The Igusa Square Root

The Borcherds determinant of $\phi_{0,1}$, its denominator algebra, and the $K3 \times E$ realization problem

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Frontispiece

The Igusa cusp form Δ_5 has five readings. The genus-two automorphic section, the denominator of a generalized Borcherds–Kac–Moody superalgebra on $\Lambda^{2,1}_{II}$, and the singular theta lift of the weak Jacobi form $\phi_{0,1}$ are theorem-level readings. The protected Pfaffian is a retained-data reading of the chosen E_3-prefactorization datum

$$\mathfrak{A}_{K3\times E} \;\in\; E_3\text{-HolFA}(K3\times E),$$

and the mirror discriminant is a conjectural period-side reading on the rank-3 Mukai-invariant $K3\times E$ component.

0.0.0.1 The five views.

① *Automorphic.* The Borcherds–Gritsenko–Nikulin section

$$\mathcal{D}_X \;=\; \Delta_5 \;\in\; H^0\big(\mathbf{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2,\; \mathcal{L}^{\otimes 5}\otimes\nu_{\Delta_5}\big)$$

of the weight-5 automorphic line on Siegel upper-half space, with multiplier ν_{Δ_5} of [9,76].

② *Borcherds–Kac–Moody denominator.* The denominator of the generalized BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$ on the rank-three Lorentzian root lattice $\Lambda^{2,1}_{II}\subset\Lambda^{2,1}=\Lambda^{(1,1)}\oplus\langle 2\rangle$ of signature $(2,1)$, the index-four sublattice with even $f_{\pm 2}$ coordinates, abstractly $U(4)\oplus\langle 2\rangle$ (discriminant -32, the index-four scaling of the hyperbolic block). The ambient lattice $\Lambda^{2,1}$ has basis (f_2,f_3,f_{-2}) with $(f_2,f_{-2})=-1$ and $(f_3,f_3)=2$; the sublattice $\Lambda^{2,1}_{II}$ is generated by $2f_2, f_3, 2f_{-2}$. The type-II Cartan matrix on simple roots $\delta_1=2f_2-f_3$, $\delta_2=2f_{-2}-f_3$, $\delta_3=f_3$ is, with $\mathbf{J}_3$ the 3×3 all-ones matrix, $(\delta_i,\delta_j)=4\mathbf{I}_3-2\mathbf{J}_3=\left(\begin{smallmatrix}2&-2&-2\\-2&2&-2\\-2&-2&2\end{smallmatrix}\right)$,

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\,\Delta_5(2Z),$$

with signed root supermultiplicities smult $\mathfrak{g}_{\Delta_5,\alpha(n,l,m)}=f(nm,l)$ on the active support $\Gamma_{\mathrm{act}}=\{(n,l,m)\in\Gamma_{\mathrm{eff}}\mid f(nm,l)\neq 0\}$, with $\alpha(n,l,m)=2nf_2-lf_3+2mf_{-2}$ [9,46–48].

③ *Compact Pfaffian.* The retained Pfaffian line of the $K3\times E$ chiral E_3-prefactorization input, presented at finite stage R by the Dirac–Igusa pro-object

$$\mathfrak{D}^{DI}_X \;=\; \lim_{\substack{\longleftarrow\\ R}}\,\big(\mathcal{A}^{E_3}_{X,R},\, F^{\mathrm{hyb}}_{X,R},\, \Gamma_{X,R},\, \Pi_X,\, \Phi_R,\, o_R,\, H_R,\, P^{\Pi}_R,\, C_{X,R},\, \Theta_{\mathrm{Kos},R},\, \mathcal{L}_{\mathrm{Pf},R},\, \mathrm{pf}_{X,R},\, \epsilon_{o,R},\, \mathrm{Rec}_R\big)$$

(the chosen E_3-prefactorization datum $\mathcal{A}^{E_3}$, the hybrid factorization carrier F^{hyb} of $\mathrm{Ran}^{\mathrm{hyb}}(E)$, the Mukai charge lattice Γ, the Gram projection Π_X, the vanishing-cycle coefficient sheaf Φ, the orientation o, the Hall datum H, the Gram-descended primitive P^{Π}, the source coalgebra C, the Koszul cobar Θ_{Kos}, the Pfaffian line $\mathcal{L}_{\mathrm{Pf}}$, the Pfaffian section pf, the orientation character ϵ_o, and the recognition map Rec; Chapter 5 specifies the finite-stage data and their residuals, with the E_3, wrapped, Koszul, and recognition lanes retained or conditional exactly as stated there). The obstruction $(D0)$ is reduced in Appendix 7 to the retained D0-HN datum; $(O1)$, $(O1)^+$, and

$(O2)$ are reduced there to retained quotient-orientation, Weyl transport, and type-II wall-atlas data. Together these data supply the Pfaffian and orientation part of the Pfaffian–Dirac theorem

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}) \;=\; \Delta_5, \qquad \epsilon_o \;=\; \nu_{\Delta_5}.$$

Here ϵ_o is the orientation character of the Pfaffian line on $W^{(2)}(\Lambda_{II}^{2,1})$, and ν_{Δ_5} is the Maass multiplier of [76] on $\mathbf{Sp}_4(\mathbb{Z})$, pulled back along the exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the inclusion $\Lambda_{II}^{2,1} \hookrightarrow \Lambda^{3,2}$ to a character on $W^{(2)}(\Lambda_{II}^{2,1})$ taking value -1 on each of the three simple type-II reflections $s_{\mathfrak{d}_i}$; see Theorem 4.1. The symbol $\mathcal{P}_X$ denotes the protected primitive Lie superalgebra $P_X^{\Pi,\mathrm{red}}$ of Chapter 5.6 (the H^0 of the protected primitives of the hybrid factorization sheaf $\mathcal{F}_X$ after Gram-projection pushforward $\overline{\Pi}^{\Theta}_{X,*}$ and Hopf-radical quotient $\overline{\mathrm{Rad}\langle\,,\,\rangle_X}$; the three notations $\mathcal{P}_X = P_X^{\Pi,\mathrm{red}} = P_X$ denote one object, abbreviated P_X where the construction data are not in play). The Lie-superalgebra recognition

$$\mathcal{P}_X \;\cong\; \mathfrak{g}_{\Delta_5}$$

is the separate primitive-recognition theorem of Chapter 7, conditional on the finite compact-source datum $(S1)$–$(S10)$.

④ *Mirror discriminant. Conjectural.* On the rank-3 Mukai-invariant $(K3 \times E)$-period component, the discriminant section is conjecturally Δ_5^{-2}; its polar divisor is twice the reduced mirror discriminant, and its reduced automorphic Heegner support is the Humbert divisor where Δ_5 vanishes (Chapter 8).

⑤ *Singular theta lift.* The theta decomposition $\phi_{0,1}^{\mathrm{Weil}}$ of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ has weight $-1/2$ for the Weil representation of $\Lambda^{3,2}$; its Borcherds regularized theta lift is an orthogonal automorphic form, and the exceptional isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies that form with the genus-two Siegel cusp form

$$\Theta^{\mathrm{Bor}}_{\mathrm{sing}}(\phi_{0,1}^{\mathrm{Weil}}) \;=\; \Delta_5.$$

This is the Borcherds–Gritsenko–Nikulin theta identity of [9, 10, 46].

0.0.0.2 The cross-level identities. The three edges ① ↔ ②, ② ↔ ⑤, ① ↔ ⑤ are theorems (Borcherds 1995; Gritsenko–Nikulin 1998); they identify the automorphic, BKM, and regularized-theta descriptions. The edge ② ↔ ③ is the conditional Pfaffian–Dirac theorem after $(D0)$, $(O1)$, $(O1)^+$, and the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$, together with the finite geometric Pfaffian datum (P_{fin}), on the $K3 \times E$ specialization in Appendix 7 (Theorems 7.7, 7.13, 7.14, 7.19), making ② ↔ ③ a theorem at the Pfaffian and orientation level on $K3 \times E$; primitive Lie-superalgebra recognition remains conditional on the cofinal finite source-recognition datum and the finite Koszul comparison datum of Definition 0.211. This edge relates the BKM denominator algebra to the Dirac–Igusa Pfaffian. The edge ① ↔ ③ is the two-step composite $\mathcal{D}_X = \Delta_5 = \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$, with the orientation match $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$ supplying the multiplier. The protected primitive Lie superalgebra $\mathcal{P}_X$ recovers the BKM target precisely when the source representatives, relations, pairings, radical quotient, PBW comparison, and full parity dimensions of $(S1)$–$(S10)$ are supplied.

0.0.0.3 The level-n scalar shadow. On the closed cobordism $K3 \times E \mapsto \mathrm{pt}$,

$$Z_{\mathrm{BPS}}^{K3\times E} \;=\; \big(\Delta_{10}\big)^{-1} \;=\; \Delta_5^{-2}, \qquad \Delta_{10} \;=\; \Delta_5^2 \;\in\; M_{10}\big(\mathbf{Sp}_4(\mathbb{Z})\big).$$

Here $\Delta_{10} = \Delta_5^2$ is the theta-normalized weight-ten cusp form on $\mathbf{Sp}_4(\mathbb{Z})$, equivalently $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$ in the unscaled Igusa convention ([51]; the multiplier ν_{Δ_5} squares to trivial [76]); the Oberdieck–Pixton-monic normalization $\chi_{10}^{\mathrm{OP}} := D_5^2 = 4096^{-1}\Delta_5^2$ (with $D_5 := 64^{-1}\Delta_5$ the Borcherds-monic form so that $[q^{1/2}r^{1/2}s^{1/2}]D_5 = 1$) gives the equivalent form $Z_{\mathrm{OP}}^{X} = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$, where $4096 = 64^2$ is the squared theta-leading constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ of the Igusa form and the minus sign is the OP scalar branch convention [91, 92]. $Z_{\mathrm{BPS}}^{K3\times E}$ is the level-n protected trace of the retained $K3 \times E$ data. A three-dimensional gravity-line interpretation with internal compact factor $K3 \times E$ requires the Hall–Borcherds residual of Vol II and is not asserted here.

0.0.0.4 The κ-tuple of $K3 \times E$. The retained $K3 \times E$ data carry the five-coordinate κ-tuple

$$\kappa_{K3\times E} \;=\; \big(\kappa_{\mathrm{cat}},\; \kappa_{\mathrm{ch}}^{\mathrm{Hodge}},\; \kappa_{\mathrm{ch}}^{\mathrm{Heis}},\; \kappa_{\mathrm{BKM}},\; \kappa_{\mathrm{fiber}}\big) \;=\; (0,\,0,\,3,\,5,\,24).$$

The categorical Euler is $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3})\,\chi(\mathcal{O}_E) = 2\cdot 0 = 0$. The Hodge supertrace $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(K3\times E) = \sum_q (-1)^q h^{0,q}(K3\times E) = 0$. The Heisenberg rank $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ is the rank of the type-II Heisenberg–Cartan obtained after the normal-ordered Gram projection Π_X to $\Lambda_{II}^{2,1}$. It is not the rank of the algebraic Mukai lattice $H^0(S,\mathbb{Z}) \oplus \mathrm{NS}(S) \oplus H^4(S,\mathbb{Z})$; for the $U \oplus \langle -2\rangle$ polarisation that lattice has rank five, while the $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$-coordinate counts only the three Gram coordinates which enter the type-II BKM Cartan. The BKM coordinate

$$\kappa_{\mathrm{BKM}}(\Phi_N) \;=\; c_N(0)/2$$

(Borcherds 1995; Gritsenko 1999) reads, at the paramodular input $N = 1$, $c_1(0) = 10$, giving $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$. The fibre coordinate $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(K3) = 24$. The additive identity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails at every $N \in \{1,2,3,4,6\}$: the Lefschetz Euler $\chi_{\mathrm{top}}(K3) = 24$ and the holomorphic Euler $\chi(\mathcal{O}_{K3}) = 2$ are distinct invariants of the K3 fibre, and κ_{BKM} sees the $c_N(0)/2$ data of the Borcherds input rather than either of them. Each subscript on κ records a distinct invariant; the unsubscripted handle leaves the categorical dimension unspecified.

Chapter 1

Pentadic Δ_5

1 The retained E_3-object and the five readings of Δ_5

The five readings attached to Δ_5 have different logical status. The automorphic, BKM, and theta-lift readings are theorems; the Pfaffian reading is relative to retained $K3 \times E$ data; the mirror reading is conjectural. The trivial-canonical threefold $K3 \times E$ supplies the ambient CY-3 input for a retained holomorphic prefactorization datum in the Kontsevich–Tamarkin and Costello–Gwilliam–Li lineage [22, 23, 63]. The notation

$$\mathfrak{A}_{K3\times E} \;\in\; E_3\text{-HolFA}(K3 \times E)$$

denotes this chosen finite-stage E_3-prefactorization datum; the unconditional functor from CY-3 dg-categories to holomorphic E_3-factorization algebras is not asserted [22, 23, 63, 65, 99]. The retained protected sector supplies a Pfaffian line on the Dirac–Igusa pro-object. The Pfaffian–Dirac theorem compares that line with the automorphic section $\mathcal{D}_X = \Delta_5$ only under the named orientation and wall data. The five readings are the automorphic section, the BKM denominator, the retained Pfaffian section, the conjectural period discriminant, and the singular theta lift.

The unconditional new mathematical content of the manuscript, beyond the classical Igusa–Borcherds–Gritsenko–Nikulin spine, is the parity-resolved root-space computation at $\delta_{123} = \delta_1 + \delta_2 + \delta_3$ –- even dimension 29 against odd dimension 93, with $29 - 93 = -64 = f(1,1)$ (Proposition 6.7) –- together with the scalar-Pfaffian monodromy proposition (Proposition 0.201). The Pfaffian–Dirac identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}) = \Delta_5$ is a recognition criterion relative to the retained data $(D0)$, $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$, (P_{fin}), which the certificate ledgers record as not supplied.

Definition 1.1 *(Pentadic Δ_5).* The five-tuple

$$\big(\mathcal{D}_X,\ \mathrm{den}(\mathfrak{g}_{\Delta_5}),\ \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}),\ \mathrm{Disc}(\mathcal{P} \to \mathbb{D}),\ \Theta^{\mathrm{Bor}}_{\mathrm{sing}}(\phi_{0,1}^{\mathrm{Weil}})\big)$$

of automorphic, BKM, retained-Pfaffian, period, and theta-lift readings is the *pentadic* Δ_5. The reading of Δ_5 is fixed by the level: at level ①, Δ_5 is $\mathcal{D}_X$; at level ②, Δ_5 is the function whose value at $2Z$ is $64\ \mathrm{den}(\mathfrak{g}_{\Delta_5})$; at level ③, Δ_5 is the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$ under $(D0)$, $(O1)$, $(O1)^+$, and the retained $(O2_{\mathrm{atlas}})$ datum; at level ④, Δ_5^{-2} is the mirror discriminant section, conjecturally; at level ⑤, Δ_5 is the Borcherds theta lift of $\phi_{0,1}^{\mathrm{Weil}}$.

2 The ① ↔ ② edge: Borcherds–Gritsenko–Nikulin

The Borcherds product expansion of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ is the genus-two cusp form

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \big(1 - q^n r^l s^m\big)^{f(nm,l)},$$

where $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right)$, $q = e^{2\pi i z_1}$, $r = e^{2\pi i z_2}$, $s = e^{2\pi i z_3}$, and $f(n,l)$ are the Fourier coefficients of $\phi_{0,1}$ (the weak Jacobi form whose discriminant decomposition records $f(0,0) = 10$, $f(0,\pm 1) = 1$, $f(1,0) = 108$,

$f(1,\pm 1) = -64, f(1,\pm 2) = 10, \dots)$ [9, 34, 46]. The leading constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64 = 2^6$ is the theta-leading coefficient (Igusa); the chamber $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ fixes the active semigroup Γ_{eff} of cusp-completed exponents.

The Gritsenko–Nikulin denominator identity converts the same product into the Borcherds–Weyl–Kac form on the type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1} = \Lambda^{(1,1)} \oplus \langle 2\rangle$:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; \sum_{w\in W^{(2)}(\Lambda^{2,1}_{II})} \mathrm{sgn}(w)\; w\Big(e^{\rho} \;-\; \sum_{a\in\Lambda^{2,1}_{II}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\, e^{\rho+a}\Big) \;=\; 64^{-1}\,\Delta_5(2Z),$$

where $\mathcal{P}_{II}$ is the type-II Weyl chamber (the intersection of the positive half-spaces of the three simple walls, Theorem 5.1) and $m(a) \in \mathbb{Z}$ are the imaginary-simple correction multiplicities read from the active support of $\phi_{0,1}$ [46, Theorem 2.1], with Weyl vector

$$\rho \;=\; \tfrac{1}{2}\big(\delta_1+\delta_2+\delta_3\big) \;=\; f_2 - \tfrac{1}{2}f_3 + f_{-2}$$

on the three simple type-II reflections $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ (Cartan matrix $(\delta_i,\delta_j) = 4\mathbf{I}_3 - 2\mathbf{J}_3 = \left(\begin{smallmatrix} 2 & -2 & -2\\ -2 & 2 & -2\\ -2 & -2 & 2\end{smallmatrix}\right)$, $\mathbf{J}_3$ the all-ones matrix); type-II reflection subgroup $W^{(2)}(\Lambda^{2,1}_{II})$ generated by these three reflections [46–48]. The imaginary-simple correction terms $-\sum_a m(a)\,e^{\rho+a}$ are the content of the Gritsenko–Nikulin theorem: the naive alternating sum $\sum_w \mathrm{sgn}(w)\,w(e^{\rho})$ alone is the denominator of the uncorrected hyperbolic Kac–Moody algebra of $W^{(2)}$ and does not equal $64^{-1}\Delta_5(2Z)$ –- the exponent $\rho + 2f_2$ carries coefficient -9 in $64^{-1}\Delta_5(2Z)$ yet lies on no Weyl translate of ρ, so it must be accounted by imaginary simple roots. The signed root supermultiplicities recover the Fourier coefficients of $\phi_{0,1}$ under the doubling $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$:

$$\mathrm{smult}\;\mathfrak{g}_{\Delta_5,\alpha(n,l,m)} \;=\; f(nm,l) \qquad \text{on } (n,l,m)\in\Gamma_{\mathrm{act}} = \{(n,l,m)\in\Gamma_{\mathrm{eff}} \mid f(nm,l)\neq 0\}.$$

The edge ① ↔ ② is the joint statement: the automorphic section $\mathcal{D}_X = \Delta_5$ of $\mathcal{L}^{\otimes 5}\otimes\nu_{\Delta_5}$ on Siegel space *equals* the BKM denominator (after the constant 64^{-1} and the variable doubling $Z \mapsto 2Z$). This is Borcherds 1995 together with the Gritsenko–Nikulin identification of the underlying lattice with $\Lambda^{2,1}_{II}$ and the Weyl-group action with $W^{(2)}(\Lambda^{2,1}_{II})$ [9, 46].

3 THE ① ↔ ⑤ AND ② ↔ ⑤ EDGES: THE SINGULAR THETA LIFT

The Borcherds product expansion of the previous section converts $\phi_{0,1}$ into Δ_5 by an arithmetic identity in Γ_{eff}; the question of *where* this identity comes from has a single answer. The exceptional isomorphism for $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies the genus-two Siegel space with the type-IV symmetric domain attached to the signature-$(3,2)$ lattice $\Lambda^{3,2} = \Lambda^{(1,1)}\oplus\Lambda^{(1,1)}\oplus\langle 2\rangle$, and it identifies the arithmetic quotient $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2$ with $\Gamma_{3,2}\backslash\mathbb{H}^{\mathbf{IV}}_{+}$, where $\Gamma_{3,2} = \wedge^2\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$. Borcherds's regularisation pairs the $\mathbf{Mp}_2(\mathbb{Z})$ Weil-representation form ϕ^{Weil} with the Siegel theta kernel of $\Lambda^{3,2}$, and produces a meromorphic Siegel modular form whose divisor is prescribed by the principal part:

$$\Theta^{\mathrm{Bor}}_{\mathrm{sing}}(\phi) \;=\; \int^{\mathrm{reg}}_{\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}} \big\langle \phi(\tau),\, \Theta_{\Lambda^{3,2}}(\tau,Z)\big\rangle\,\frac{du\,dv}{v^2}, \qquad \tau = u+iv,$$

on the Weil-representation pairing of the vector-valued weakly-holomorphic input ϕ (the theta decomposition of $\phi_{0,1}$ into the discriminant form of $\Lambda^{3,2}$) against the Siegel theta series of the signature-$(3,2)$ lattice $\Lambda^{3,2}$ — not the rank-three target lattice $\Lambda^{2,1}_{II}$, which receives the BKM root structure under doubling —

together with the antiholomorphic dependence required for absolute convergence after regularisation; the explicit construction is Chapter 5, Definition 1.9 and Theorem 1.10 [9, 10]. At the theta-decomposed input $\phi_{0,1}^{\mathrm{Weil}}$ the regularised integral computes the composite

$$\Theta^{\mathrm{Bor}}_{\mathrm{sing}} : J^{\mathrm{wk}}_{0,1} \longrightarrow M^{!}_{-1/2}(\rho_{\Lambda^{3,2}}) \longrightarrow M_5(\mathbf{Sp}_4(\mathbb{Z}),\, \nu_{\Delta_5}), \qquad \phi_{0,1} \longmapsto \phi^{\mathrm{Weil}}_{0,1} \longmapsto \Delta_5,$$

giving the edge ① ↔ ⑤. Reading the same lift through the Gritsenko–Nikulin denominator identity, $\phi_{0,1}$ is the character of the index-one Heisenberg sector and the multiplicative lift over $\Lambda^{2,1}_{II}$ is exactly $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, giving the edge ② ↔ ⑤ [9, 46–48].

4 The ② ↔ ③ edge: the Pfaffian–Dirac theorem

The theorem edges proved by Borcherds and Gritsenko–Nikulin identify the automorphic, BKM-denominator, and singular-theta readings of Δ_5; the chiral E_3-prefactorization datum of §1 is the retained operator-level source from which the View ③ datum is obtained. The structural question has two terms. The Pfaffian term asks whether the protected determinant of the $K3 \times E$ operator datum is the automorphic section Δ_5. The primitive term asks whether the compact Hall primitive sector is the BKM algebra $\mathfrak{g}_{\Delta_5}$. The first term is the Pfaffian–Dirac theorem after $(D0)$, $(O1)$, $(O1)^+$, and $(O2_{\mathrm{atlas}})$, together with (P_{fin}), in Appendix 7. The second term is the finite compact-source recognition problem $(S1)$–$(S10)$.

Theorem 4.1 *(Pfaffian–Dirac). Conditional on* $(D0)$, $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$, *and* (P_{fin})*; ambient: cosection-twisted reduced d-critical chart on $K3 \times E$ at finite cusp-truncation height R, with chain-level pro-object, Mittag–Leffler vanishing of the inverse system in R, and Pfaffian orientation o with Weyl character ϵ_o.* The protected Pfaffian of the Dirac–Igusa pro-object $\mathfrak{D}^{DI}_X = \varprojlim_R \mathfrak{D}^{DI}_{X,R}$ satisfies

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{DI}_X) \;=\; \Delta_5, \qquad \epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II}),$$

The primitive Lie-superalgebra assertion is a separate recognition statement. If the finite compact-source datum $(S1)$–$(S10)$ is supplied on a cofinal system of relation-closed degree windows, then

$$\mathcal{P}_X := H^0\ \mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)/\overline{\mathrm{Rad}\langle\,,\rangle_X} \;\cong\; \mathfrak{g}_{\Delta_5}.$$

The four obstructions and the finite geometric Pfaffian datum name the exact conditioning. $(D0)$ is the $D0$-degeneration limit of the cosection-reduced d-critical orientation theory: the limiting orientation in the $D0$-degenerate fibre is constructed and matches the orientation of the generic fibre under the cosection-twist [12, 55, 56]. $(O1)$ is the strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle: the square root of the determinant line of the cotangent complex extends across the cusp boundary as part of the retained quotient-orientation datum [16, 55, 56]. $(O1)^+$ is the Weyl-equivariant transport of the orientation along reflections in the type-II reflection subgroup $W^{(2)}(\Lambda^{2,1}_{II})$: the orientation cocycle $c_o \in Z^2(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2)$ trivializes in H^2, and the resulting character $\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ is well-defined on generators. $(O2)$ is the local Pfaffian wall normal form on type-II walls: at each simple type-II wall s_{∂_i} the rank-one normal Pfaffian crossing of Definition 0.204 is part of the retained wall-atlas datum and gives the local sign $(-1)^{N^{\mathrm{Pf}}_{\partial_i}} = -1$, and the Hall-side sign sum on the half-Hilbert orbit of the self-Ext stack of size $2^{12} = 4096$ is compared with the squared theta-leading $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ only after scalarization. The wall transport gives the character value $\epsilon_o(s_{\partial_i}) = \nu_{\Delta_5}(s_{\partial_i})$ on each generator (Appendix E). The finite geometric Pfaffian datum gives the skew perfect complexes, Pfaffian lines, finite Pfaffian sections, type-II wall divisor orders, automorphic line comparisons, support and parity rows, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility for the cofinal HN product.

Remark 4.2 *(The conditioning is named precisely).* The four obstruction classes $(D0)$, $(O1)$, $(O1)^+$, $(O2)$ and the finite geometric Pfaffian datum are the exact Pfaffian and orientation obligations; Appendix 7 reduces $(D0)$ to the retained D0-HN datum, names the retained quotient-orientation and Weyl transport data, and names $(O2_{\mathrm{atlas}})$ and (P_{fin}). At the operator level ③, $\mathcal{D}_X = \Delta_5$ is compared with the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$. After the level-$n$ protected trace, the scalar partition function is Δ_5^{-2}. The primitive algebra $\mathcal{P}_X \simeq \mathfrak{g}_{\Delta_5}$ is stronger; it requires the finite compact-source recognition datum $(S1)$–$(S10)$. The operator-level datum carries the orientation character ϵ_o that the scalar shadow has squared away, by view ③, and recovers it as ν_{Δ_5} on $W^{(2)}(\Lambda_{II}^{2,1})$ by $(O1)^+$ and $(O2)$.

5 View ④: mirror discriminant

Conjectural (ambient: variation of Hodge structure on the rank-3 Mukai-invariant $(K3 \times E)$-period component). On $\mathcal{P}^{(3)}_{K3\times E} \simeq \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})$, the mirror discriminant section is conjecturally Δ_5^{-2}. Its polar divisor is $2\mathfrak{D}_{\mathrm{mir}}$, where the reduced support of $\mathfrak{D}_{\mathrm{mir}}$ is the Humbert–Heegner divisor $\mathfrak{H}_{\mathrm{II}}$ fixed by the Borcherds product of Δ_5. This mirror-discriminant identification is conjectural and is not a hypothesis of any theorem; see Conjecture 1.1.

6 The level-n scalar shadow

The scalar partition function on the closed cobordism $K3 \times E \mapsto \mathrm{pt}$ is the Borcherds determinant of the protected K3 index, computed by Oberdieck–Pixton 2018 and Oberdieck–Pandharipande 2018 [91, 92]: the OP partition function $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is the virtual count of Hilbert schemes of $K3 \times E$ in the $(-p_{\mathrm{DT}})^n$ DT-chamber, and the BPS normalization $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ rescales by -4096^{-1}. Two normalizations of the same weight-ten cusp form on $\mathbf{Sp}_4(\mathbb{Z})$ are in use, and naming them is necessary because both appear in the formulas:

- *Theta-normalized.* $\Delta_{10}(Z) := \Delta_5(Z)^2 \in M_{10}(\mathbf{Sp}_4(\mathbb{Z}))$ (the multiplier ν_{Δ_5} squares to trivial; the leading coefficient $[q\,r\,s]\Delta_{10} = 64^2 = 4096$).
- *Oberdieck–Pixton-monic.* $\chi_{10}^{\mathrm{OP}}(Z) := D_5(Z)^2 = 64^{-2}\Delta_5(Z)^2 = 4096^{-1}\Delta_{10}(Z)$ (so $[q\,r\,s]\chi_{10}^{\mathrm{OP}} = 1$).

Oberdieck–Pixton and Oberdieck–Pandharipande prove [91, 92]

$$Z^X_{\mathrm{OP}} \;=\; -\big(\chi_{10}^{\mathrm{OP}}\big)^{-1} \;=\; -4096\,\Delta_5^{-2},$$

and the level-n protected scalar shadow is

$$Z^{K3\times E}_{\mathrm{BPS}} \;=\; \big(\Delta_{10}\big)^{-1} \;=\; \Delta_5^{-2} \;=\; -\tfrac{1}{4096}\,Z^X_{\mathrm{OP}}.$$

The factor $4096 = 64^2$ is the squared theta-leading constant of Δ_5; the OP minus sign is a scalar branch convention absorbed into the OP $(-p_{\mathrm{DT}})^n$ chamber and is independent of the orientation character ϵ_o of view ③.

$Z^{K3\times E}_{\mathrm{BPS}}$ is the level-n protected trace of the retained $K3 \times E$ data: a charge-graded virtual partition function. A three-dimensional gravity-line interpretation of the scalar shadow with internal compact factor $K3 \times E$ requires the Hall–Borcherds residual of Vol II. The universal-stage chain names the required objects: primitive factorization dg-category $\mathcal{C} \to$ boundary chart $A_b = \mathrm{End}_{\mathcal{C}}(b) \to$ bar twisting $\mathrm{Bar}(A_b) \to$ derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \to$ ambient operator algebra $\mathcal{A}$ at the boundary vacuum b of the open factorization dg-category on $(K3 \times E,\ D,\ \tau)$; this chain is not constructed here. The local witness is the formal-Darboux stalk of the mixed-holomorphic-topological strings theory [24, 25, 63]. The class $\mathrm{ob}^{\mathrm{hol}}_{\mathrm{dR}}(K3 \times E)$ is the holomorphic de Rham obstruction to gluing this local BV factorization model over $K3 \times E$; its vanishing is a necessary global condition, not the gravity-line operator algebra.

Remark 6.1 *(Primitive order).* The five views are ordered: primitive objects (the retained chiral E_3-prefactorization datum on $K3 \times E$, its four obstructions $(D0)$, $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$, and its finite Pfaffian section datum; the Cartan datum of $\mathfrak{g}_{\Delta_5}$) first; cross-level identities ((2↔3) Pfaffian–Dirac, $(\epsilon_o = \nu_{\Delta_5})$ orientation match, $(\mathcal{P}_X \cong \mathfrak{g}_{\Delta_5})$ primitive recognition) second; scalar modular forms last (the section $\mathcal{D}_X$, the BKM denominator $64^{-1}\Delta_5(2Z)$, the partition function Δ_5^{-2}). A statement is not primitive if it is true only after passing to a trace, choosing a chamber, completing a category, or installing descent data. On each first appearance the scalar reading is named: the theta-normalized form Δ_5 with leading $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$; the BKM denominator $64^{-1}\Delta_5(2Z)$ with the doubling $2Z$ on the type-II sublattice; the OP-monic form $D_5 = 64^{-1}\Delta_5$ with $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ the Oberdieck–Pixton normalization of the squared determinant section.

7 Placement in the Beilinson Chain

The pentadic Δ_5 is one identification at one specific level of the Beilinson chain. Under the chiral Koszul source datum of Definition 0.208, the finite Koszul comparison datum of Definition 0.211, and the strict Koszul Mittag–Leffler exactness of Proposition 0.214, the source coalgebra C_X maps by cobar comparison to the target chiral shadow $\mathsf{Den}_{\Delta_5,E}$, compatibly with Weyl transport, Pfaffian orientation, Hall pairing, radical quotient, and PBW filtration. Vol I's Theorem H supplies the chain-level identification $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) = \mathrm{ChirHoch}^\bullet(A_b, A_b)$ only on the retained Koszul locus, and Vol II's chiral-Hochschild Beilinson stratification (i)+(ii) supplies the CY-orientation trace pairing of degree $-\dim_{\mathbb{C}}(K3 \times E) = -3$. Relative to the retained trace datum, these inputs give the level-Z trace identity $\mathrm{Tr}^{\mathrm{bulk}}_n((-1)^F \cdot \mathrm{id})_{Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)} = \Delta_5^{-2}$ of Appendix 7, Theorem 7.23, relative to (Z_{HB}); the OP chamber scalar is -4096 times this trace. The level-A gravity-line operator algebra acting on the boundary is the open gravity-line Hall–Borcherds residual of Vol II. The Vol III κ-stratification places $\mathfrak{g}_{\Delta_5}$ at the $(d = 3$, rank 3) chamber through the identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$, where $c_N(0)$ is the constant Fourier coefficient of the input weak Jacobi or vector-valued modular form. Applied at the paramodular row $(t, N; k) = (1, 1; 5)$ of the Gritsenko–Cléry catalogue [44, 46], with Jacobi seed $\phi_{0,1}$ and Borcherds product Δ_5, the identity reads $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5$. The companion identity at input Φ_{12}, the Fake-Monster Borcherds product on the Leech-extension lattice $\mathrm{II}_{25,1}$ [8, 9], reads $\kappa_{\mathrm{BKM}}(\Phi_{12}) = c_{\Phi_{12}}(0)/2 = 12$; the two values are different rows of one universal $c_N(0)/2$ rule, indexed by different input forms and lattices, not in contradiction. The mixed holomorphic-topological strings companion identifies the formal-Darboux stalk $A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N,\ \mathrm{Kosz}([\phi_1, \phi_2]))$ of the chiral Hochschild cochain at the Dirac brane vacuum. The obstruction to globalizing this local factorization model over $K3 \times E$ is the holomorphic de Rham class $\mathrm{ob}^{\mathrm{hol}}_{\mathrm{dR}}(K3 \times E)$. A three-dimensional gravitational path-integral interpretation of the level-n scalar shadow $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ requires this obstruction to vanish and separately requires the Vol II gravity-line Hall–Borcherds operator algebra.

CONTENTS

Part I

Automorphic and BKM Data

Chapter 2

K3×E geometry, Mukai dictionary, lattice polarization, normal-ordered Gram cocycle, hybrid $\mathrm{Ran}^{\mathrm{hyb}}(E)$ carrier

The arithmetic source contains three named weight-five readings, and their normalizations are different. The automorphic chain produces the section

$$\mathcal{D}_X = \Delta_5 \in M_5\big(\mathbf{Sp}_4(\mathbb{Z}),\ \nu_{\Delta_5}\big) \qquad \text{(View ①; proved [9, 46]).}$$

The Borcherds–Kac–Moody (BKM) chain produces the denominator

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\,\Delta_5(2Z) \qquad \text{(View ②; proved [46]).}$$

The compact Pfaffian chain produces, granted the retained Do-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and the retained type-II wall-atlas datum of Theorem 7.5 of Appendix 7, the protected Pfaffian

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5 \qquad \text{(View ③; relative to the retained orientation data at the Pfaffian level).}$$

The handle "Δ_5" with no superscript is reserved for working formulas; each occurrence names which of $\mathcal{D}_X$, $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, or $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is meant. The cross-level identity ②↔③ is a theorem at the Pfaffian and orientation level on $K3 \times E$, granted (D0), the retained quotient-orientation datum for (O1), the chosen Weyl lifts for (O1$^+$), and the retained datum (O2$_{\mathrm{atlas}}$) in Theorems 7.7, 7.13, 7.14, 7.19. The primitive Lie-superalgebra recognition $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source datum ($S1$)–($S10$) of Chapter 7.

0.0.0.1 The Mukai–Gram dictionary. The Mukai–Gram dictionary on $K3 \times E$ begins with the Mukai lattice $\widetilde{H}(S,\mathbb{Z})$ of the K3 surface S, of rank 24 and signature (4, 20) [83]. The even Kunneth sector

$$\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \otimes H^{\mathrm{even}}(E,\mathbb{Z}) \cong \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$$

is the formal $K3 \times E$ charge lattice used below; the two summands are written (Q, P). For the Gram map this underlying Kunneth group is read with the two componentwise K3 Mukai pairings; the tensor Kunneth cross-pairing on $H^{\mathrm{even}}(X,\mathbb{Z})$ is a different bilinear form. The Gram-index lattice $\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$ receives the quadratic shadow

$$\Pi_X(Q,P) = \big((Q,Q)/2,\ (Q,P),\ (P,P)/2\big).$$

This map is not additive; its additivity defect is the symmetric polarization $B = -\delta\Pi_X$. The additive lattice map is the normal-ordered homomorphism

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}.$$

After this normal ordering, Γ_{gram} embeds into $\Lambda^{2,1}_{II}$ and then into the type-IV ambient lattice $\Lambda^{3,2}$ of Chapter 5. Detailed lattice computations are in Appendix 3; the dictionary fixes the $K3 \times E$ charge convention.

The geometric and lattice data feed the three established Δ_5 constructions. The compact target is fixed at $X = S \times E$ with S a smooth complex projective K3 surface and E a smooth elliptic curve over $\mathbb{C}$; this is a trivial-canonical threefold ($\dim_{\mathbb{C}} X = 3$, $K_X \cong \mathcal{O}_X$; elementary from the product formula), so $n = \chi(\mathcal{O}_Y)$ appears directly. The $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ tuple of [70] (the K3×E row of the Gritsenko–Cléry catalogue) is the categorical-dimensional anchor; every use of κ carries one of these five subscripts.

The datum has a geometric part and an arithmetic part. The geometric part is the formal dyonic charge lattice $\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$ on the Mukai sector, together with $\Gamma_{X,\sigma}^{\mathrm{eff}} \subset \Gamma_X^{\mathrm{alg}} \subset \Gamma_X^{\mathrm{form}}$. The arithmetic part is the quadratic Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}} = \mathbb{Z}^3$, its symmetric bilinear polarization $B = -\delta \Pi_X$, the normal-ordered extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}$, and the additive map $\overline{\Pi}_X(c, T) = \Pi_X(c) - T$. The fixed-lift raw-pushforward obstruction theorem forces the extension, and the projection-locality obstruction forces the wrapped stratum in $\mathrm{Ran}^{\mathrm{hyb}}(E) = \mathrm{Ran}(E)_{b=0} \sqcup \mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$.

1 Smooth K3×E threefold and Mukai lattice

1.1 The compact target

Definition 1.1 *(K3×E threefold; primitive geometric datum).* A *K3×E threefold* is a triple (X, S, E) in which S is a smooth complex projective K3 surface (so $K_S \cong \mathcal{O}_S$, $h^{1,0}(S) = 0$, $h^{2,0}(S) = 1$; [83]), E is a smooth elliptic curve over $\mathbb{C}$ with marked origin $0_E \in E$, and

$$X := S \times E.$$

The canonical bundle is trivial, $K_X \cong p_S^* K_S \otimes p_E^* K_E \cong \mathcal{O}_X$, and Hodge invariants are

$$\dim_{\mathbb{C}} X = 3, \qquad h^{1,0}(X) = h^{1,0}(E) = 1, \qquad \chi_{\mathrm{top}}(X) = \chi_{\mathrm{top}}(S) \cdot \chi_{\mathrm{top}}(E) = 24 \cdot 0 = 0, \qquad \chi(\mathcal{O}_X) = \chi(\mathcal{O}_S) \cdot \chi(\mathcal{O}_E) =$$

The threefold X is Calabi–Yau in the trivial-canonical sense used by Donaldson–Thomas theory and shifted symplectic geometry; the construction uses the chosen trivialization of K_X.

Remark 1.2 *(Distinguishing $\chi_{\mathrm{top}}(K3) = 24$ and $\chi(\mathcal{O}_S) = 2$).* The two integers enter the construction by different routes. The Borcherds weight is $f(0,0)/2 = 5$, where $f(0,0) = 10$ is the constant Fourier coefficient of the weight-zero index-one weak Jacobi form $\phi_{0,1} = r^{-1} + 10 + r + O(q)$ in the Eichler–Zagier normalization (computed, [34]). The topological Euler number $\chi_{\mathrm{top}}(K3) = 24$ enters one step removed, through the elliptic-genus specialization: at $z = 0$ the $\phi_{0,1}$ support $4n - l^2 \geq -1$ leaves only $l \in \{-1, 0, 1\}$ at $n = 0$, so the constant term is $\sum_l f(0, l) = f(0, -1) + f(0, 0) + f(0, 1) = 12$, and $Z_{K3}(\tau, 0) = 2\,\phi_{0,1}(\tau, 0)$ has constant term $2 \cdot 12 = 24 = \chi_{\mathrm{top}}(K3)$. Thus $\chi_{\mathrm{top}}(K3) = 2 \sum_l f(0, l)$, not $f(0,0)$; the Borcherds weight 5 sees only the single coefficient $f(0,0) = 10$. The integer $\chi(\mathcal{O}_S) = 2$ enters by a third route, the categorical Calabi–Yau dimension and the holomorphic part of the Mukai pairing. The two integers belong to different entries: the κ-tuple of [70] records $\kappa_{\mathrm{fiber}} = 24 = \chi_{\mathrm{top}}(K3)$, not $2 = \chi(\mathcal{O}_S)$, and the universal Borcherds identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ at $\Phi_1 = \Delta_5$ gives $c_1(0) = 10$ and $\kappa_{\mathrm{BKM}} = 5$. The expression $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_{K3}) = 0 + 2 = 2$ does not give the Borcherds weight, while $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} + \chi(\mathcal{O}_{K3}) = 3 + 2 = 5$ is only an accidental equality at the $N = 1$ row. Across the CHL frame the Borcherds weights are $c_N(0)/2 = (5, 3, 2, \frac{3}{2}, 1)$ (Jatkar–Sen [53]; Govindarajan–Krishna [41, 42]; the $N = 4$ weight is half-integral), not a fixed sum of a chiral rank and $\chi(\mathcal{O}_{K3})$ (proved, [70] counterexample). The coordinate $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight; the protected scalar shadow is written at level $n = \chi(\mathcal{O}_Y)$ for the one-dimensional subscheme $Y \subset K3 \times E$. The target $X = K3 \times E$ is a trivial-canonical threefold.

1.2 *Mukai lattice on the K3 factor*

Definition 1.3 *(Mukai lattice and Mukai pairing).* Let S be a smooth complex K3 surface. The *Mukai lattice* is the total even cohomology

$$\Lambda_S := \widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z}),$$

identified with the K-theoretic charge lattice $K_0(D^b(\mathrm{Coh}\, S))_{\mathrm{tor.\,free}}$ via the Mukai vector

$$v(\mathcal{F}) := \mathrm{ch}(\mathcal{F})\sqrt{\mathrm{Td}(S)} = \big(r(\mathcal{F}),\ c_1(\mathcal{F}),\ r(\mathcal{F}) + \mathrm{ch}_2(\mathcal{F})\big),$$

where r is the rank, c_1 is the first Chern class, and the half-shift r in the four-form term comes from $\mathrm{Td}(S) = 1 + 2[\mathrm{pt}]$ (proved, [83, 84]). The *Mukai pairing* is

$$\big\langle (r,c,s),\ (r',c',s')\big\rangle_{\mathrm{Muk}} := c\cdot c' - rs' - r's,$$

where $c \cdot c' \in \mathbb{Z}$ is the cup product on $H^2(S,\mathbb{Z})$. It is even, unimodular, of signature $(4, 20)$, and rank 24:

$$\Lambda_S \cong II_{4,20} \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2},$$

where U denotes the rank-two hyperbolic plane $U \cong \mathbb{Z}e \oplus \mathbb{Z}f$ with $e^2 = f^2 = 0$, $e \cdot f = 1$, and $E_8(-1)$ denotes the negative-definite E_8 root lattice (proved, [84, 88]).

Remark 1.4 *(Sign convention).* The signature $(4, 20)$ uses the convention that $H^0 \oplus H^4$ contributes a hyperbolic plane U via $(1, 0, 0)$ and $(0, 0, 1)$ and that $H^2(S,\mathbb{Z}) \cong II_{3,19} \cong U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$, which has positive part of dimension 3. Together, $\Lambda_S = U \oplus II_{3,19} \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}$ with positive part of dimension 4 (proved, [84]). The signature is $(4, 20)$, not $(20, 4)$: the positive direction is the smaller side. A reflection r_v across $v \in \Lambda_S$ acts by $r_v(x) = x - 2(x, v)v/(v, v)$, taking the $\mathbb{R}$-linear extension when $(v, v) \neq 0$, without implicit sign reversal.

1.3 *Künneth and the formal even Mukai sector on $X = S \times E$*

Definition 1.5 *(Formal even Mukai sector on $X = S \times E$).* Pick a basepoint $0_E \in E$ and the standard generators $1_E \in H^0(E,\mathbb{Z})$ and $\omega_E \in H^2(E,\mathbb{Z})$ with $\int_E \omega_E = 1$. Every even cohomology class on $X = S \times E$ decomposes as

$$v_X = P \otimes 1_E + Q \otimes \omega_E, \qquad P, Q \in \Lambda_S$$

(proved, Künneth on simply connected $\widetilde{H}$ tensor E-cohomology). The pair $(Q, P) \in \Lambda_S \oplus \Lambda_S$ is the *formal even charge*. The component $P \otimes 1_E$ is invariant under translation along E, while $Q \otimes \omega_E$ records the E-degree.

The *formal full-Mukai dyonic lattice* is the additive abelian group

$$\Gamma_X^{\mathrm{form}} := \Lambda_S \oplus \Lambda_S, \qquad c = (Q, P),$$

with electric component Q and magnetic component P. The symbol Γ_X^{phys} is defined here by the equality

$$\Gamma_X^{\mathrm{phys}} := \Gamma_X^{\mathrm{form}}.$$

This is not the full four-dimensional $\mathcal{N} = 4$ electric–magnetic charge lattice and not compact Hall support. It is the D6–D2–D0/OP even branch on which the Mukai Gram map is tested before imposing algebraicity, effectivity, stability, or non-emptiness of compact Hall moduli.

Definition 1.6 *(Liu product-stability datum on $S \times E$).* Fix a rational Bridgeland stability condition

$$\sigma_S = (\mathcal{A}_S, Z_S), \qquad Z_S\big(K(D^b(\mathrm{Coh}\, S))\big) \subset \mathbb{Q} + i\mathbb{Q},$$

on $D^b(\mathrm{Coh}\, S)$. Let L_E be a degree-one line bundle on the elliptic curve E, and let

$$p : S \times E \to S, \qquad q : S \times E \to E$$

be the projections. The Abramovich–Polishchuk heart used by Liu is the following heart [1, 66]:

$$\mathcal{A}_E^{\mathrm{AP}} = \big\{ M \in D^b(\mathrm{Coh}(S \times E)) \mid p_*(M \otimes q^* L_E^{\otimes n}) \in \mathcal{A}_S \text{ for } n \gg 0 \big\}.$$

For $M \in \mathcal{A}_E^{\mathrm{AP}}$, Liu's curve-factor polynomial is

$$L_M(n) := Z_S\big(p_*(M \otimes q^* L_E^{\otimes n})\big) = a(M)n + b(M) + i\big(c(M)n + d(M)\big),$$

where $a, b, c, d : K(\mathcal{A}_E^{\mathrm{AP}}) \to \mathbb{Q}$ are additive homomorphisms. For $t \in \mathbb{Q}_{>0}$, set

$$Z_t(M) = a(M)t - d(M) + i\, c(M)t, \qquad \nu_t(M) = \begin{cases} \dfrac{-a(M)t + d(M)}{c(M)t}, & c(M) \neq 0, \\ +\infty, & c(M) = 0. \end{cases}$$

Let

$$\mathcal{T}_t = \{ M \in \mathcal{A}_E^{\mathrm{AP}} \mid \nu_{t,\min}(M) > 0 \}, \qquad \mathcal{F}_t = \{ M \in \mathcal{A}_E^{\mathrm{AP}} \mid \nu_{t,\max}(M) \leq 0 \},$$

and

$$\mathcal{A}_E^t = \langle \mathcal{T}_t, \mathcal{F}_t[1] \rangle.$$

For $s \in \mathbb{Q}_{>0}$, Liu's product central charge is

$$Z_E^{s,t}(M) = c(M)s + b(M) + i\big(-a(M)t + d(M)\big).$$

The Liu product-stability condition is

$$\sigma_{s,t} := (\mathcal{A}_E^t, Z_E^{s,t}).$$

Liu proves that this is a rational Bridgeland stability condition for $s, t \in \mathbb{Q}_{>0}$, and extends the construction continuously to $(s, t) \in \mathbb{R}^2_{>0}$ [66, Theorem 4.7 and §5]. This is a source-imported stability condition. It is not a finite-type moduli construction, not a compact Hall stage, and not a Pfaffian orientation.

The packet `certificates/moduli/liu_product_stability` checks this definition. It records the source theorem, the AP heart membership formula, the linear polynomial $L_M(n)$, the four coefficient homomorphisms a, b, c, d, the weak charge Z_t, the torsion-pair tilt, and the central charge $Z_E^{s,t}$. Its verifier status is `LIU_PRODUCT_STABILITY_DEFINED`. It does not prove the separate AP-compatibility row, finite-type semistable substacks, derived enhancements, compact Hall stages, Pfaffian orientations, or protected traces.

Proposition 1.7 *(Compatibility with the Abramovich–Polishchuk heart).* [proved] With the datum of Definition 1.6, the heart of Liu's product stability condition is the torsion-pair tilt of the Abramovich–Polishchuk heart:

$$\heartsuit(\sigma_{s,t}) = \mathcal{A}_E^t = \langle \mathcal{T}_t, \mathcal{F}_t[1] \rangle,$$

where $(\mathcal{T}_t, \mathcal{F}_t)$ is the torsion pair in $\mathcal{A}_E^{\mathrm{AP}}$ defined by the slope ν_t. Consequently

$$\mathcal{A}_X^{\mathrm{AP/Liu}} = \mathcal{A}_E^t.$$

Thus the retained heart is Abramovich–Polishchuk-compatible in the finite-row sense: it is obtained from $\mathcal{A}_E^{\mathrm{AP}}$ by Liu's specified torsion-pair tilt.

Proof. The Abramovich–Polishchuk heart $\mathcal{A}_E^{\mathrm{AP}}$ is the global heart entering Liu's construction. Liu defines the weak charge Z_t on this heart, uses the resulting slope ν_t to form the torsion pair

$$\mathcal{T}_t = \{M \mid \nu_{t,\min}(M) > 0\}, \qquad \mathcal{F}_t = \{M \mid \nu_{t,\max}(M) \le 0\},$$

and then sets

$$\mathcal{A}_E^t = \langle \mathcal{T}_t, \mathcal{F}_t[1] \rangle$$

before defining

$$\sigma_{s,t} = (\mathcal{A}_E^t, Z_E^{s,t})$$

[66, §2.2 and Theorem 4.7]. Hence the heart of $\sigma_{s,t}$ is $\mathcal{A}_E^t$ by construction. The retained heart is defined as $\mathcal{A}_X^{\mathrm{AP/Liu}} := \heartsuit(\sigma_{s,t})$, so it equals $\mathcal{A}_E^t$. This proves the compatibility statement using only the definition of the tilt; it proves no noetherianity, HN-filtration, boundedness, finite-moduli, derived-enhancement, Pfaffian-orientation, or protected-trace row. □

The packet `certificates/moduli/ap_liu_compatibility` checks this heart identification. It records the AP membership formula, the weak charge Z_t, the slope ν_t, the torsion pair $(\mathcal{T}_t, \mathcal{F}_t)$, the tilted heart $\mathcal{A}_E^t$, and the equality $\mathcal{A}_X^{\mathrm{AP/Liu}} = \heartsuit(\sigma_{s,t}) = \mathcal{A}_E^t$. Its verifier status is `AP_LIU_COMPATIBILITY_VERIFIED`. It does not prove noetherianity, HN filtrations, bounded HN types, finite-type semistable substacks, derived enhancements, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.8 *(Retained Abramovich–Polishchuk/Liu heart).* Fix the Liu product-stability datum of Definition 1.6. The retained stability heart on

$$D^b(\mathrm{Coh}\, X), \qquad X = S \times E,$$

is

$$\mathcal{A}_X^{\mathrm{AP/Liu}} := \heartsuit(\sigma_{s,t}),$$

equivalently $\mathcal{A}_X^{\mathrm{AP/Liu}} = \mathcal{A}_E^t$ by Proposition 1.7. In the rest of the manuscript the symbol σ means this retained datum $\sigma_{s,t}$, and its heart is $\mathcal{A}_X^{\mathrm{AP/Liu}}$.

This definition is only the choice of the ambient heart and stability datum. Noetherianity of the retained windows, existence of Harder–Narasimhan filtrations, boundedness of HN types, finite-type semistable substacks, quasi-smooth derived enhancements, universal complexes, and compact Hall stages are separate finite rows.

The packet `certificates/moduli/stability_heart` checks this definition as a table-level datum. It records the inputs

$$(S,H),\quad (E,0_E),\quad \sigma_S,\quad L_E,\quad \mathcal{A}_X^{\mathrm{AP}},\quad \sigma_{s,t},\quad \mathcal{R}_{\mathrm{HN}},$$

and verifies the scalar firewall excluding noetherianity, HN filtrations, bounded HN types, finite-type moduli, Pfaffian orientations, and protected traces. Its verifier status is `STABILITY_HEART_DATUM_VERIFIED`.

Definition 1.9 *(Retained noetherian window).* Fix the heart $\mathcal{A}_X^{\mathrm{AP/Liu}}$. A retained finite window at height R is a full exact subcategory

$$\mathcal{A}_{X,R}^{\mathrm{ret}} \subset \mathcal{A}_X^{\mathrm{AP/Liu}}$$

with a finite set of isomorphism classes, closed under retained subobjects and retained quotients, together with a length function

$$\ell_R : \mathrm{Obj}(\mathcal{A}_{X,R}^{\mathrm{ret}}) \to \mathbb{Z}_{\geq 0}$$

such that $\ell_R(A) = 0$ exactly for the zero object and $\ell_R(A) < \ell_R(B)$ for every strict retained monomorphism $A \hookrightarrow B$.

Proposition 1.10 *(Retained finite windows are noetherian).* [proved] Every retained finite window $\mathcal{A}_{X,R}^{\mathrm{ret}}$ in the sense of Definition 1.9 is noetherian: every ascending chain of retained subobjects terminates.

Proof. Let

$$A_0 \subset A_1 \subset A_2 \subset \cdots$$

be a chain of retained subobjects of an object A in $\mathcal{A}_{X,R}^{\mathrm{ret}}$. After deleting repetitions, each inclusion is strict. The length function then gives a strictly increasing sequence

$$\ell_R(A_0) < \ell_R(A_1) < \ell_R(A_2) < \cdots .$$

All A_i lie in the finite retained window and all are subobjects of A, so $\ell_R(A_i) \leq \ell_R(A)$. A strictly increasing sequence of non-negative integers bounded above by $\ell_R(A)$ is finite. Thus the original ascending chain stabilizes. □

The packet `certificates/moduli/retained_window_noetherian` checks this finite ACC statement on a displayed retained exact window. It verifies a finite object set, subobject and quotient closure, strict increase of the length rank along each strict monomorphism, acyclicity of the strict subobject graph, and the longest-chain bound. Its verifier status is `RETAINED_WINDOW_NOETHERIAN_VERIFIED`. It does not prove global noetherianity of $\mathcal{A}_X^{\mathrm{AP/Liu}}$, Harder–Narasimhan filtrations, bounded HN types, finite-type semistable substacks, quasi-smooth moduli, Pfaffian orientations, or protected traces.

Definition 1.11 *(Retained finite Harder–Narasimhan filtration datum).* Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ be a retained finite window. A retained finite Harder–Narasimhan filtration datum consists of:

$$\Phi_R \subset \mathbb{Q}, \qquad \phi_R : \mathcal{S}_R \to \Phi_R,$$

where Φ_R is a finite totally ordered phase set and $\mathcal{S}_R$ is a class of nonzero retained semistable objects, together with, for every $A \in \mathcal{A}^{\mathrm{ret}}_{X,R}$, a finite exact filtration

$$0 = A_0 \subset A_1 \subset \cdots \subset A_m = A$$

whose retained quotients $E_i = A_i/A_{i-1}$ lie in $\mathcal{S}_R$ and satisfy

$$\phi_R(E_1) > \phi_R(E_2) > \cdots > \phi_R(E_m).$$

The zero object has the empty filtration. This is finite retained-window data. It is not a global Harder–Narasimhan theorem for $\mathcal{A}^{\mathrm{AP/Liu}}_X$, not boundedness of HN types, and not finite-type moduli.

Proposition 1.12 *(Retained finite HN filtrations exist from the datum).* [proved] If a retained finite window $\mathcal{A}^{\mathrm{ret}}_{X,R}$ carries the datum of Definition 1.11, then every object of $\mathcal{A}^{\mathrm{ret}}_{X,R}$ has a retained Harder–Narasimhan filtration.

Proof. For $A = 0$ the empty filtration is the Harder–Narasimhan filtration. For $A \neq 0$, the datum gives an exact retained chain

$$0 = A_0 \subset A_1 \subset \cdots \subset A_m = A.$$

Each quotient $E_i = A_i/A_{i-1}$ is retained and semistable, and the phase inequalities are strict. The chain is finite, hence terminates inside the exact window. These are precisely the Harder–Narasimhan conditions in the retained finite window. □

The packet `certificates/moduli/retained_window_hn_filtration` checks this finite statement on a displayed retained exact window. It verifies a finite phase set, semistable quotient factors, exact filtration chains, length additivity, and strict phase descent. Its verifier status is `RETAINED_WINDOW_HN_FILTRATION_VERIFIED`. It does not prove global Harder–Narasimhan filtrations in $\mathcal{A}^{\mathrm{AP/Liu}}_X$, bounded HN types, finite-type semistable substacks, quasi-smooth moduli, Pfaffian orientations, or protected traces.

Definition 1.13 *(Retained bounded HN type datum).* Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ be a retained finite window with a retained finite Harder–Narasimhan filtration datum. The *retained HN type* of an object A is the finite word

$$\mathrm{HNType}_R(A) = \big(([E_1], \phi_R(E_1)), \ldots, ([E_m], \phi_R(E_m))\big)$$

attached to its retained HN quotients $E_i = A_i/A_{i-1}$, with the empty word for $A = 0$. A retained bounded HN type datum is a finite set $\mathfrak{T}_R$ containing all these words, together with a finite Liu–Hilbert decoration of the semistable factor classes:

$$[E] \longmapsto \big(\widehat{c}_E, [a_E, b_E], P_E, N_E, \mathcal{F}_E\big).$$

Here $\widehat{c}_E$ is the retained numerical class, $[a_E, b_E]$ is the cohomological-amplitude interval, P_E is the retained Hilbert-polynomial label, N_E is the retained regularity bound, and $\mathcal{F}_E$ is the retained closure family. This datum is a finite bound on active-window HN types. It is not a finite-type semistable substack, not a quasi-smooth derived enhancement, and not a compact Hall stage.

Proposition 1.14 *(HN type is bounded in a retained active window).* [proved] Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ be a retained finite window with a retained finite Harder–Narasimhan filtration datum. Then the set of retained HN types occurring in $\mathcal{A}^{\mathrm{ret}}_{X,R}$ is finite. More precisely, if

$$N_R = \max\{\ell_R(A) \mid A \in \mathcal{A}^{\mathrm{ret}}_{X,R}\},$$

then every retained HN type has length at most N_R.

Proof. The retained window has finitely many isomorphism classes, hence the set of retained semistable HN factors is finite. If

$$0 = A_0 \subset A_1 \subset \cdots \subset A_m = A$$

is the retained HN filtration of A, then each inclusion $A_{i-1} \subset A_i$ is strict unless $m = 0$. The retained length rank gives

$$0 = \ell_R(A_0) < \ell_R(A_1) < \cdots < \ell_R(A_m) \leq N_R,$$

so $m \leq N_R$. Thus every retained HN type is a word of length at most N_R in a finite alphabet of semistable factor classes. The set of such words is finite, and the strict phase condition selects a finite subset. A finite Liu–Hilbert decoration of the factor classes therefore gives a finite decorated HN type set. □

The packet `certificates/moduli/retained_hn_type_bounds` checks this finite boundedness statement. It verifies the finite semistable factor-type set, the finite HN type-word set, the maximal HN height, the maximal factor count, the retained length-rank bound, and the mirror of the HN type rows in `certificates/moduli/k3e_finite_moduli/hn_type_bounds.csv`. Its verifier status is `RETAINED_HN_TYPE_BOUNDS_VERIFIED`. It does not construct finite-type semistable substacks, quasi-smooth derived enhancements, universal complexes, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.15 *(Retained finite-type semistable substack datum).* Fix a retained finite window $\mathcal{A}^{\mathrm{ret}}_{X,R}$, the retained stability condition $\sigma_{s,t}$, and the retained bounded HN type datum of Definition 1.13. A retained finite-type semistable substack datum assigns to every retained semistable factor type c a finite-type bounded Quot/Postnikov ambient stack $Q_{R,c}$, a chart

$$U_{R,c} \to Q_{R,c},$$

and a specialization-closed finite-type algebraic substack

$$\mathfrak{M}^{ss}_{R,c} \subset Q_{R,c}$$

whose objects are exactly the $\sigma_{s,t}$-semistable retained objects of type c. The datum carries two defect ranks:

$$d^{ss}_{R,c} = 0, \qquad d^{bd}_{R,c} = 0.$$

The first says that no unstable object is included and no semistable object of type c is omitted. The second says that the bounded Quot/Postnikov ambient covers the retained class. This is a finite-type semistable-substack datum. It is not a quasi-smooth derived enhancement, not a universal complex, not a rigidification, not a compact Hall correspondence, and not an orientation datum.

Proposition 1.16 *(Finite-type semistable substacks for retained classes).* [proved] Let $\mathcal{A}^{\mathrm{ret}}_{X,R}$ carry a retained bounded HN type datum and a retained finite-type semistable substack datum. Then for every retained semistable factor type c the stack $\mathfrak{M}^{ss}_{R,c}$ is of finite type over $\mathbb{C}$, is specialization-closed inside its bounded Quot/Postnikov ambient, and parametrizes exactly the retained $\sigma_{s,t}$-semistable objects of type c.

Proof. The retained HN type datum gives only finitely many semistable factor types c. For each such c, the semistable-substack datum places $\mathfrak{M}^{ss}_{R,c}$ inside a finite-type bounded Quot/Postnikov ambient $Q_{R,c}$ through the chart $U_{R,c}$, with the semistable locus represented by finitely many locally closed retained chart pieces. Finite unions of locally closed substacks of a finite-type algebraic stack are finite type. The condition $d^{bd}_{R,c} = 0$ says that the ambient covers the retained class, while $d^{ss}_{R,c} = 0$ says that the substack is precisely the $\sigma_{s,t}$-semistable locus of that class. The specialization-closed flag is part of the datum and is checked in the same finite chart. Thus each $\mathfrak{M}^{ss}_{R,c}$ is finite type and has the asserted moduli meaning. The argument uses no derived enhancement, universal complex, rigidification, finite-inertia stratification, flag correspondence, cosection atlas, transition map, Pfaffian orientation, or protected trace. □

The packet `certificates/moduli/retained_semistable_substacks` checks this finite-type row. It imports the retained HN factor types, supplies one bounded Quot/Postnikov ambient chart and one semistable substack for each retained class, verifies specialization-closedness, and checks zero semistability and boundedness defect ranks. Its verifier status is `RETAINED_SEMISTABLE_SUBSTACKS_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/semistable_substacks.csv` discharge item 167. They do not construct derived enhancements, universal complexes, scalar rigidifications, finite inertia strata, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.17 *(Retained quasi-smooth derived enhancement datum).* Fix the retained finite-type semistable substacks $\mathfrak{M}^{ss}_{R,c}$ of Definition 1.15. A retained quasi-smooth derived enhancement datum assigns to every retained semistable factor type c a derived Artin stack

$$\mathfrak{M}^{\mathrm{der}}_{R,c}$$

with a morphism $\mathfrak{M}^{\mathrm{der}}_{R,c} \to \mathrm{Perf}(X)$, an identification

$$t_0\mathfrak{M}^{\mathrm{der}}_{R,c} = \mathfrak{M}^{ss}_{R,c},$$

a perfect cotangent complex of amplitude

$$\mathbb{L}_{\mathfrak{M}^{\mathrm{der}}_{R,c}} \in D^{[-1,0]}(\mathfrak{M}^{\mathrm{der}}_{R,c}),$$

and a restricted PTVV (-1)-shifted symplectic form

$$\omega^{\mathrm{PTVV}}_{R,c}.$$

The datum carries two defect ranks

$$d^{\mathrm{Tor}}_{R,c} = 0, \qquad d^{\omega}_{R,c} = 0.$$

The first says that the cotangent-amplitude assertion has no retained Tor-amplitude defect. The second says that the PTVV form restricts to the retained derived substack without symplectic defect. This is a derived-enhancement datum. It is not a universal perfect complex, not a scalar rigidification, not an inertia stratification, not a Hall-flag correspondence, and not a cosection or orientation datum.

Proposition 1.18 *(Quasi-smooth derived Artin enhancements).* [proved] Let the retained finite-type semistable substacks carry the datum of Definition 1.17. Then every $\mathfrak{M}^{ss}_{R,c}$ has a quasi-smooth derived Artin enhancement $\mathfrak{M}^{\mathrm{der}}_{R,c}$ whose classical truncation is $\mathfrak{M}^{ss}_{R,c}$, whose cotangent complex has amplitude $[-1, 0]$, and whose (-1)-shifted symplectic form is the restricted PTVV form on the Calabi–Yau threefold $X = S \times E$.

Proof. The datum gives a derived Artin stack $\mathfrak{M}^{\mathrm{der}}_{R,c}$ with $t_0\mathfrak{M}^{\mathrm{der}}_{R,c} = \mathfrak{M}^{ss}_{R,c}$. The finite-type statement for the classical truncation is Proposition 1.16. The cotangent-amplitude condition $\mathbb{L}_{\mathfrak{M}^{\mathrm{der}}_{R,c}} \in D^{[-1,0]}$ and the vanishing $d^{\mathrm{Tor}}_{R,c} = 0$ are exactly the quasi-smoothness criterion for a derived Artin stack. Since $K_X \simeq \mathcal{O}_X$, X is a Calabi–Yau threefold, and PTVV produce the ambient (-1)-shifted symplectic form on the derived moduli of perfect complexes [95, Theorem 0.3]. The vanishing $d^{\omega}_{R,c} = 0$ says that this form restricts to the retained derived substack. Thus the retained enhancement is quasi-smooth and carries the asserted PTVV form. The proof supplies no universal complex on the rigidified stack, no scalar rigidification, no finite-inertia stratification, no extension or flag stack, no cosection atlas, no transition system, no Pfaffian orientation, and no protected trace. □

The packet `certificates/moduli/retained_derived_enhancements` checks this row. It imports the finite-type semistable substacks, records one derived Artin enhancement per retained substack, verifies cotangent amplitude $[-1, 0]$, imports the PTVV (-1)-shifted symplectic form, and checks zero Tor-amplitude and symplectic defect ranks. Its verifier status is `RETAINED_DERIVED_ENHANCEMENTS_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/derived_e` discharge item 168. They do not construct universal complexes, scalar rigidifications, finite inertia strata, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.19 *(Retained scalar rigidification and residual-inertia datum).* Fix the retained quasi-smooth derived enhancements $\mathfrak{M}^{\mathrm{der}}_{R,c}$ of Definition 1.17. A retained scalar rigidification and residual-inertia datum assigns to every retained semistable factor type c an exact sequence of stabilizer groups

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{Aut}_{R,c} \longrightarrow H_{R,c} \longrightarrow 1,$$

a scalar rigidification

$$\mathfrak{M}^{\mathrm{rig}}_{R,c} = \left[\mathfrak{M}^{\mathrm{der}}_{R,c}/\mathbb{G}_m\right],$$

and the residual inertia group $H_{R,c}$. The datum requires

$$|H_{R,c}| < \infty, \qquad d^{\mathrm{rig}}_{R,c} = 0, \qquad d^{\mathrm{in}}_{R,c} = 0.$$

Thus the scalar $\mathbb{G}_m$ has been removed, the residual inertia is finite, and the rigidification and inertia defects vanish. This is a scalar-rigidification datum for retained derived substacks. It is not a universal perfect complex, not a finite closed inertia stratification, not an extension or flag stack, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.20 *(Finite residual inertia after scalar rigidification).* [proved] Let the retained derived enhancements carry the datum of Definition 1.19. Then for each retained semistable factor type c, the scalar rigidified stack $\mathfrak{M}^{\mathrm{rig}}_{R,c}$ has finite residual inertia $H_{R,c}$. Equivalently, after quotienting the scalar $\mathbb{G}_m$-automorphisms, no positive-dimensional stabilizer remains in the retained row.

Proof. The datum gives the exact sequence

$$1 \to \mathbb{G}_m \to \mathrm{Aut}_{R,c} \to H_{R,c} \to 1.$$

The scalar rigidification is the quotient by the indicated central $\mathbb{G}_m$. Therefore the stabilizer remaining on $\mathfrak{M}^{\mathrm{rig}}_{R,c}$ is $H_{R,c}$. The condition $|H_{R,c}| < \infty$ gives finite residual inertia, and the equalities $d^{\mathrm{rig}}_{R,c} = d^{\mathrm{in}}_{R,c} = 0$ say that neither the quotient nor the inertia computation has a retained defect. The argument constructs no universal complex, finite closed inertia stratification, flag stack, cosection atlas, transition map, Pfaffian orientation, or protected trace. □

The packet `certificates/moduli/retained_rigidification_inertia` checks this row. It imports the retained derived enhancements, records one scalar rigidification for each retained substack, verifies the exact sequence

$$1 \to \mathbb{G}_m \to \mathrm{Aut}_{R,c} \to H_{R,c} \to 1,$$

checks $\mathbb{G}_m$-removal, finite residual inertia, and zero rigidification and inertia defect ranks. Its verifier status is `RETAINED_RIGIDIFICATION_INERTIA_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/scalar_rigidifications.csv` discharge item 169 and the scalar-rigidification existence row item 175. They do not construct universal complexes, finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.21 *(Retained finite class-bound datum).* Fix the retained finite-type semistable substacks $\mathfrak{M}^{ss}_{R,c}$ and the retained HN type-bound rows. A retained finite class-bound datum consists of a finite set

$$C_R = \{c_1, \ldots, c_m\}$$

of retained semistable classes, together with maps assigning to each $c \in C_R$ a retained type $\tau(c)$, a charge $\widehat{c}$, a substack $\mathfrak{M}^{ss}_{R,c}$, a finite Hilbert-polynomial label set

$$\{P_{c,i}\}_{i \in I_c}, \qquad |I_c| < \infty,$$

a cohomological-amplitude interval

$$[a_c, b_c] \subset \mathbb{Z}, \qquad a_c \leq b_c,$$

and a nonnegative regularity bound N_c. The datum requires zero class-coverage, Hilbert-bound, amplitude-bound, and regularity-bound defects. This is a retained class-bound datum. It is not a universal perfect complex, not a finite closed inertia stratification, not an extension or flag stack, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.22 *(Finite retained class bounds).* [proved] Let a retained active window carry the datum of Definition 1.21. Then the retained class set C_R is finite. For every $c \in C_R$, the Hilbert-polynomial labels $P_{c,i}$, the cohomological amplitude interval $[a_c, b_c]$, and the regularity bound N_c are finite retained data.

Proof. The datum writes $C_R = \{c_1, \ldots, c_m\}$, hence the retained class set is finite. For each c, the index set I_c of Hilbert polynomial labels is finite by hypothesis, the amplitude interval is bounded by the two integers $a_c \leq b_c$, and $N_c \in \mathbb{Z}_{\geq 0}$ is a finite integer. The vanishing of the four defect ranks says that these labels cover exactly the retained classes imported from the HN type-bound and semistable-substack rows. The argument constructs no universal family over $\mathfrak{M}^{ss}_{R,c}$, no closed inertia stratification, no flag stack, no cosection atlas, no transition map, no Pfaffian orientation, and no protected trace. □

The packet `certificates/moduli/retained_class_bounds` checks this row. It imports the retained HN type-bound rows and retained semistable substacks, records the finite class set C_R, records the Hilbert-polynomial labels $P_{c,i}$, records the amplitude intervals $[a_c, b_c]$, and records finite regularity bounds N_c. Its verifier status is `RETAINED_CLASS_BOUNDS_VERIFIED`. The mirrored class-bound data are the Hilbert-polynomial, amplitude, and regularity columns in `certificates/moduli/k3e_finite_moduli/hn_type_bounds.csv`, and they discharge items 170–173. They do not construct universal complexes, finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.23 *(Retained universal perfect-complex datum).* Fix the retained finite substacks $\mathfrak{M}^{ss}_{R,c}$, their scalar rigidifications, and the retained class-bound datum. A retained universal perfect-complex datum assigns to every $c \in C_R$ a perfect complex

$$\mathcal{U}_{R,c} \in \mathrm{Perf}\big(X \times \mathfrak{M}^{\mathrm{rig}}_{R,c}\big),$$

a base-change identification for every test morphism

$$T \longrightarrow \mathfrak{M}^{\mathrm{rig}}_{R,c}, \qquad \mathcal{U}_{R,c}|_{X\times T} \simeq E_T,$$

and a finite Tor-amplitude interval

$$\mathrm{TorAmp}(\mathcal{U}_{R,c}) \subset [a_c, b_c].$$

It carries two defect ranks

$$d^{\mathrm{desc}}_{R,c} = 0, \qquad d^{\mathrm{perf}}_{R,c} = 0.$$

The first says that the tautological complex on the bounded Quot/Postnikov chart descends across the scalar rigidification. The second says that the descended object remains perfect on the retained finite substack. This is a universal-complex datum. It is not a finite closed inertia stratification, not an extension or flag stack, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.24 *(Universal perfect complexes on retained finite substacks).* [proved] Let the retained finite substacks carry the datum of Definition 1.23. Then each retained scalar-rigidified finite substack $\mathfrak{M}^{\mathrm{rig}}_{R,c}$ carries a universal perfect complex $\mathcal{U}_{R,c}$ on $X \times \mathfrak{M}^{\mathrm{rig}}_{R,c}$, compatible with base change and of finite Tor amplitude contained in $[a_c, b_c]$.

Proof. The bounded Quot/Postnikov ambient chart carries its tautological complex. Restricting it to the retained finite substack gives the tautological complex before rigidification. The equality $d^{\mathrm{desc}}_{R,c} = 0$ is precisely the descent condition across the scalar rigidification $\mathfrak{M}^{\mathrm{der}}_{R,c} \to \mathfrak{M}^{\mathrm{rig}}_{R,c}$. The equality $d^{\mathrm{perf}}_{R,c} = 0$ says that the descended complex is perfect on $X \times \mathfrak{M}^{\mathrm{rig}}_{R,c}$. The datum also supplies the base-change identifications and the Tor-amplitude containment in $[a_c, b_c]$. Thus the retained substack carries the asserted universal perfect complex. The proof constructs no finite closed inertia stratification, extension or flag stack, cosection atlas, transition map, Pfaffian orientation, or protected trace. □

The packet `certificates/moduli/retained_universal_complexes` checks this row. It imports the retained semistable substacks, class-bound rows, derived enhancements, and scalar rigidifications, records one universal perfect complex per retained finite substack, verifies base-change compatibility, finite Tor amplitude, zero descent defect, and zero perfection defect. Its verifier status is `RETAINED_UNIVERSAL_COMPLEXES_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/universal_complexes.csv` discharge item 174. They do not construct finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.25 *(Retained E-translation rigidification datum).* Fix the retained scalar-rigidified finite substacks $\mathfrak{M}^{\mathrm{rig}}_{R,c}$ and their universal perfect complexes. A retained E-translation rigidification datum assigns to every $c \in C_R$ an action

$$a_{R,c} : E \times \mathfrak{M}^{\mathrm{rig}}_{R,c} \longrightarrow \mathfrak{M}^{\mathrm{rig}}_{R,c}$$

with identity element 0_E, free on the retained row, together with the quotient stack

$$\mathfrak{M}^{E\text{-rig}}_{R,c} = \left[\mathfrak{M}^{\mathrm{rig}}_{R,c}/E\right].$$

The datum carries a defect rank

$$d^{E\text{-rig}}_{R,c} = 0.$$

Thus the elliptic-translation direction has been removed without defect on the retained finite substack. This is an E-translation-rigidification datum. It is not a finite closed inertia stratification, not an extension or flag stack, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.26 *(E-translation rigidifications on retained finite substacks).* [proved] Let the retained scalar-rigidified finite substacks carry the datum of Definition 1.25. Then each retained finite substack has an E-translation rigidification $\mathfrak{M}^{E\text{-rig}}_{R,c}$, and the translation direction has zero retained quotient defect.

Proof. The datum gives an action of the elliptic curve E on $\mathfrak{M}^{\mathrm{rig}}_{R,c}$, with identity 0_E, and states that the action is free on the retained row. The quotient

$$\left[\mathfrak{M}^{\mathrm{rig}}_{R,c}/E\right]$$

therefore removes exactly the E-translation direction. The vanishing $d^{E\text{-rig}}_{R,c} = 0$ says that the quotient has no retained translation-rigidification defect. The proof constructs no finite closed inertia stratification, extension or flag stack, cosection atlas, transition map, Pfaffian orientation, or protected trace. ☐

The packet `certificates/moduli/retained_e_translation_rigidifications` checks this row. It imports scalar rigidifications and universal perfect complexes, records the E-translation action on each retained finite substack, verifies freeness on the retained row, constructs the quotient by E, and checks zero translation-rigidification defect. Its verifier status is `RETAINED_E_TRANSLATION_RIGIDIFICATIONS_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/` discharge item 176. They do not construct finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.27 *(Retained closed-substack datum).* Fix the retained E-translation-rigidified finite substacks $\mathfrak{M}^{E\text{-rig}}_{R,c}$. A retained closed-substack datum assigns to every $c \in C_R$ a finite family of closed substacks

$$\mathfrak{Z}_{R,c,j} \hookrightarrow \mathfrak{M}^{E\text{-rig}}_{R,c}, \qquad 1 \le j \le m_{R,c},$$

which covers the retained row. It carries the finite integers

$$m_{R,c} > 0$$

and defect ranks

$$d^{\mathrm{cov}}_{R,c} = 0, \qquad d^{\mathrm{in}}_{R,c} = 0.$$

The first defect rank says that the closed substacks cover the retained row. The second says that the finite residual inertia from the scalar rigidification remains finite on every retained closed stratum. This is a retained closed-substack datum. It is not closure under extensions, not closure under Harder–Narasimhan factors, not closure under duals, not an extension or flag stack, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.28 *(Retained closed finite substacks).* [proved] Let the retained E-translation-rigidified finite substacks carry the datum of Definition 1.27. Then each retained finite row has a finite closed cover by substacks on which the residual inertia is finite. Equivalently, the finite stratification row has positive strata count, finite residual inertia, zero coverage defect, and zero inertia defect.

Proof. The retained semistable rows are finite type and specialization-closed by Proposition 1.16. The scalar rigidification removes $\mathbb{G}_m$ and leaves finite residual inertia by Proposition 1.20. The E-translation rigidification removes the elliptic translation direction by Proposition 1.26. The closed-substack datum then supplies the closed immersions $\mathfrak{Z}_{R,c,j} \hookrightarrow \mathfrak{M}^{E\text{-rig}}_{R,c}$, their finite cover, and the equalities $d^{\mathrm{cov}}_{R,c} = d^{\mathrm{in}}_{R,c} = 0$. Thus the retained row is represented by a finite closed cover with finite residual inertia. The proof constructs no extension closure, HN-factor closure, dual closure, flag stack, cosection atlas, transition map, Pfaffian orientation, or protected trace. □

The packet `certificates/moduli/retained_closed_substacks` checks this row. It imports the retained finite-type semistable substacks, the scalar rigidifications with finite residual inertia, and the E-translation rigidifications, records one retained closed substack and one closed-cover row per retained finite row, and checks zero coverage and inertia defects. Its verifier status is `RETAINED_CLOSED_SUBSTACKS_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/stratificatio` discharge item 177. They do not prove closure under extensions, closure under HN factors, closure under duals, extension or flag stacks, cosection atlases, transitions, compact Hall stages, Pfaffian orientations, or protected traces.

Definition 1.29 *(Retained extension-closure datum).* Fix the retained finite HN word set

$$\mathcal{W}^{\mathrm{HN}}_R = \{0, s_3, s_2, s_1, e_{32}, e_{21}, f_{321}\}$$

from the bounded HN type datum. A retained extension-closure datum is a finite set of binary extension rows

$$u * v \longmapsto w, \qquad u, v, w \in \mathcal{W}^{\mathrm{HN}}_R,$$

with w retained and with defect rank

$$d^{\mathrm{ext}}_{u,v} = 0.$$

For the first retained window the nontrivial rows are

$$s_3 * s_2 = e_{32}, \qquad s_2 * s_1 = e_{21}, \qquad e_{32} * s_1 = f_{321}, \qquad s_3 * e_{21} = f_{321}.$$

This is an extension-closure datum in the retained exact window. It is not an extension stack, not a two-step flag stack, not closure under Harder–Narasimhan factors, not closure under duals, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.30 *(Retained substacks are closed under extensions).* [proved] Let the retained finite HN window carry the datum of Definition 1.29. Then every extension represented by one of the retained binary extension rows has middle term again in the retained finite HN window. The extension closure defect is zero on every such row.

Proof. The bounded HN type datum supplies the finite word set $\mathcal{W}_R^{\mathrm{HN}}$. The extension-closure datum supplies the four nonempty binary rows displayed in Definition 1.29; their middle terms are e_{32}, e_{21}, and f_{321}, all elements of $\mathcal{W}_R^{\mathrm{HN}}$. The equalities

$$d^{\mathrm{ext}}_{s_3,s_2} = d^{\mathrm{ext}}_{s_2,s_1} = d^{\mathrm{ext}}_{e_{32},s_1} = d^{\mathrm{ext}}_{s_3,e_{21}} = 0$$

say that no retained extension leaves the finite window. Thus the retained HN rows are closed under the supplied extensions. The proof constructs no extension stack, two-step flag stack, HN-factor closure, dual closure, cosection atlas, transition map, Pfaffian orientation, or protected trace. □

The packet `certificates/moduli/retained_extension_closure` checks this row. It imports the retained HN type-bound packet and retained closed-substack packet, records four nonempty binary extension rows, checks that every middle HN type lies inside the retained finite HN word set, and checks zero extension-closure defect. Its verifier status is `RETAINED_EXTENSION_CLOSURE_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/extension_closure.csv` discharge item 178. They do not construct extension or flag stacks, prove closure under HN factors, prove closure under duals, construct cosection atlases, construct transitions, construct compact Hall stages, construct Pfaffian orientations, or construct protected traces.

Definition 1.31 *(Retained HN-factor-closure datum).* Fix the retained finite HN word set $\mathcal{W}_R^{\mathrm{HN}}$ and the retained semistable factor types

$$\mathcal{T}_R = \{s_3, s_2, s_1\}.$$

A retained HN-factor-closure datum assigns to every nonzero word

$$w = (t_1, \dots, t_k) \in \mathcal{W}_R^{\mathrm{HN}}$$

its ordered factor list $t_i \in \mathcal{T}_R$, together with the retained class C_{R,t_i}, the finite-type semistable substack $\mathfrak{M}^{ss}_{R,t_i}$, and the retained closed substack $\mathfrak{Z}_{R,t_i}$ for each factor. It carries defect ranks

$$d^{\mathrm{HNfac}}_{w,i} = 0, \qquad 1 \le i \le k.$$

This is HN-factor closure in the retained finite window. It is not closure under duals, not an extension or flag stack, not subquotient closure in compactified flag stacks, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.32 *(Retained substacks are closed under HN factors).* [proved] Let the retained finite HN window carry the datum of Definition 1.31. Then every HN factor of every retained finite HN word is represented by a retained finite-type semistable substack and by a retained closed substack. All HN-factor-closure defects vanish.

Proof. The bounded HN type datum supplies the retained words

$$0,\ s_3,\ s_2,\ s_1,\ e_{32} = (s_3, s_2),\ e_{21} = (s_2, s_1),\ f_{321} = (s_3, s_2, s_1).$$

The datum assigns to each nonzero factor occurrence the corresponding retained class, finite-type semistable substack, and retained closed substack. There are ten such occurrences: three one-factor words, two two-factor words, and one three-factor word. The equalities $d^{\mathrm{HNfac}}_{w,i} = 0$ say that each factor occurrence lies in the retained finite window. Thus passing from a retained object to its HN factors does not leave the retained substack system. The proof constructs no compactified extension stack, no two-step flag stack, no subquotient closure theorem, no dual closure, no cosection atlas, no transition map, no Pfaffian orientation, and no protected trace. □

The packet `certificates/moduli/retained_hn_factor_closure` checks this row. It imports retained HN type bounds, class bounds, finite-type semistable substacks, retained closed substacks, and retained extension closure, records ten HN-factor occurrence rows, and checks that every factor occurrence is represented by a retained class, a retained finite-type semistable substack, and a retained closed substack. Its verifier status is `RETAINED_HN_FACTOR_CLOSURE_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/hn_factor_closure.csv` discharge item 179. They are not the dual-closure rows below, do not construct extension or flag stacks, do not prove subquotient closure in flag stacks, do not construct cosection atlases, do not construct transitions, do not construct compact Hall stages, do not construct Pfaffian orientations, and do not construct protected traces.

Definition 1.33 *(Retained dual-closure datum).* Fix the retained finite HN word set $\mathcal{W}^{\mathrm{HN}}_R$. A retained dual-closure datum is an involution D_R on the retained semistable factor types

$$D_R(s_3) = s_1, \qquad D_R(s_2) = s_2, \qquad D_R(s_1) = s_3,$$

together with the induced involution on HN words

$$D_R(t_1, \ldots, t_k) = (D_R(t_k), \ldots, D_R(t_1)).$$

For the first retained window this gives

$$D_R(0) = 0, \quad D_R(s_3) = s_1, \quad D_R(s_2) = s_2, \quad D_R(s_1) = s_3, \quad D_R(e_{32}) = e_{21}, \quad D_R(e_{21}) = e_{32}, \quad D_R(f_{321}) = f_{321}.$$

It carries defect ranks

$$d^{\vee}_w = 0, \qquad w \in \mathcal{W}^{\mathrm{HN}}_R.$$

This is dual closure in the retained finite window. It is not an extension or flag stack, not subquotient closure in compactified flag stacks, not a cosection atlas, not a transition system, and not an orientation datum.

Proposition 1.34 *(Retained substacks are closed under duals).* [proved] Let the retained finite HN window carry the datum of Definition 1.33. Then duality sends every retained finite HN word to a retained finite HN word, and the dual-closure defect vanishes on every retained word.

Proof. The factor duality exchanges s_3 and s_1 and fixes s_2, so it is an involution on the retained semistable factor set $\{s_3, s_2, s_1\}$. Applying it to an HN word reverses the order and dualizes each factor. Hence

$$D_R(e_{32}) = D_R(s_3, s_2) = (s_2, s_1) = e_{21}, \qquad D_R(e_{21}) = (s_3, s_2) = e_{32},$$

and

$$D_R(f_{321}) = D_R(s_3, s_2, s_1) = (s_3, s_2, s_1) = f_{321}.$$

The zero word is fixed, the one-factor words map as displayed in the definition, and all targets belong to $\mathcal{W}_R^{\mathrm{HN}}$. The equalities $d_w^\vee = 0$ say that no dualized retained word leaves the finite window. The proof constructs no compactified extension stack, no two-step flag stack, no subquotient closure theorem, no cosection atlas, no transition map, no Pfaffian orientation, and no protected trace. □

The packet `certificates/moduli/retained_dual_closure` checks this row. It imports retained HN type bounds, class bounds, retained closed substacks, and retained HN-factor closure, records the three factor dual rows and seven HN-word dual rows, verifies that type duality and HN-word duality are involutive, and checks that HN-word duality is reverse factorwise duality. Its verifier status is `RETAINED_DUAL_CLOSURE_VERIFIED`. The mirrored rows in `certificates/moduli/k3e_finite_moduli/dual_closure.csv` discharge item 180. They do not construct extension or flag stacks, prove subquotient closure in flag stacks, construct cosection atlases, construct transitions, construct compact Hall stages, construct Pfaffian orientations, or construct protected traces.

Definition 1.35 *(Algebraic/effective Hall support).* Fix a polarised K3 surface (S, H), a chosen Mukai vector lattice $\Lambda_{S,\mathrm{alg}} \subset \Lambda_S$ (the image of the algebraic K-class lattice generated by sheaves whose Mukai vector lies in $H^0 \oplus \mathrm{NS}(S) \oplus H^4$), and the retained stability datum $\sigma = \sigma_{s,t}$ with heart $\mathcal{A}_X^{\mathrm{AP/Liu}}$ from Definition 1.8. The retained-window rows above supply noetherianity, Harder–Narasimhan filtrations, bounded HN types, finite-type semistable substacks, and quasi-smooth derived enhancements, together with finite residual inertia after scalar rigidification, finite retained class-bound data, and universal perfect complexes, E-translation rigidifications, and finite closed-cover stratifications at finite height, together with extension closure, HN-factor closure, and dual closure inside the retained HN window; later rows supply flag stacks, subquotient closure in flag stacks, cosection charts, transitions, scalar firewalls, and compact Hall support. No unconditional Bridgeland stability theorem on the threefold $S \times E$ is used. Set

$$\Gamma_X^{\mathrm{alg}} := \Lambda_{S,\mathrm{alg}} \oplus \Lambda_{S,\mathrm{alg}}, \qquad \Gamma_{X,\sigma}^{\mathrm{eff}} \subset \Gamma_X^{\mathrm{alg}} \subset \Gamma_X^{\mathrm{form}},$$

where the *σ-effective semigroup* $\Gamma_{X,\sigma}^{\mathrm{eff}}$ is the additive sub-semigroup of charges supported by objects semistable for this datum and of nonzero Mukai vector. Membership in $\Gamma_{X,\sigma}^{\mathrm{eff}}$ is not implied by membership in Γ_X^{alg} or by formal lattice arithmetic (folklore).

Definition 1.36 *(Admissible Dirac–Igusa parameter datum).* An admissible parameter datum for the Dirac–Igusa construction is

$$\mathfrak{p} = (S, H, \sigma, E, 0_E, \tau_X, \mathcal{R}_{\mathrm{HN}}), \qquad X = S \times E,$$

where (S, H) is a polarized smooth complex K3 surface, E is a smooth elliptic curve with origin 0_E, $\tau_X : K_X \simeq \mathcal{O}_X$ is the chosen trivialization, σ is the retained stability datum used to define $\Gamma_{X,\sigma}^{\mathrm{eff}}$, and $\mathcal{R}_{\mathrm{HN}}$ is the cofinal finite HN index system. It includes the retained finite charge, moduli, hybrid, orientation, Hall, Pfaffian, Koszul, and primitive-recognition packets whenever those packets are invoked later. Thus H and σ are not decorations: they determine the algebraic/effective Hall support and the finite HN windows.

A morphism $\alpha : \mathfrak{p} \to \mathfrak{p}'$ is an admissible base-change isomorphism

$$\alpha = (f, u, \eta_\sigma, \eta_{\mathrm{HN}})$$

with $f : S \to S'$ an isomorphism of K3 surfaces carrying H to H', $u : E \to E'$ an isomorphism of elliptic curves with $u(o_E) = o_{E'}$, compatibility with the chosen trivializations of $K_{S\times E}$, a transport η_σ of the retained stability datum and effective Hall support, and a cofinal identification $\eta_{\mathrm{HN}} : \mathcal{R}_{\mathrm{HN}} \to \mathcal{R}'_{\mathrm{HN}}$. The induced Mukai isometry sends (Q, P) to (f_*Q, f_*P), preserves the Mukai pairing, and therefore preserves

$$\Pi_X(Q, P) = ((Q, Q)/2, (Q, P), (P, P)/2), \qquad B = -\delta\Pi_X.$$

The resulting groupoid is denoted $\mathsf{Par}^{\mathrm{DI}}$. Arbitrary non-isomorphic deformations of S, arbitrary changes of stability chamber, and arbitrary changes of elliptic origin are not morphisms of $\mathsf{Par}^{\mathrm{DI}}$ unless the corresponding wall-crossing or origin-translation datum below is supplied.

Proposition 1.37 *(Admissible base-change functoriality).* Let $\alpha : \mathfrak{p} \to \mathfrak{p}'$ be a morphism in $\mathsf{Par}^{\mathrm{DI}}$. If the finite packets attached to $\mathfrak{p}$ have zero defect rows and are transported by α, then pull-push along $f \times u : X \to X'$ induces functors at each retained height

$$\alpha_{R,*} : \mathsf{DI}^{\mathrm{fin}}_{X,R} \longrightarrow \mathsf{DI}^{\mathrm{fin}}_{X',\eta_{\mathrm{HN}}(R)}$$

preserving the components $\mathcal{A}^{E_3}$, F^{hyb}, $\widehat{\Gamma}$, Π, Φ, o, H, P^{Π}, C, Θ_{Kos}, $\mathcal{L}_{\mathrm{Pf}}$, pf, ε_o, and Rec, in the sense of the morphisms (M1)–(M8) of Definition 0.244. These functors commute with HN transition morphisms and therefore induce a functor on Dirac–Igusa pro-objects

$$\alpha_* : \mathrm{Pro}(\mathsf{DI}^{\mathrm{fin}}_X) \longrightarrow \mathrm{Pro}(\mathsf{DI}^{\mathrm{fin}}_{X'}).$$

For composable admissible morphisms, the induced functors compose up to the canonical pro-isomorphism coming from the common cofinal HN refinement. This is the base-change functoriality in S and E used by the construction.

Proof. The isomorphism f gives an isometry of Mukai lattices and carries the algebraic sublattice, the polarization, and the retained stability support by hypothesis. The isomorphism u preserves 1_E, ω_E, the origin, and the translation group law. Hence $(f \times u)_*$ preserves the formal charge lattice, the normal-ordered central extension, and the Gram map. The assumed transport of the finite packets carries the moduli stacks, correspondence stacks, orientation lines, Pfaffian lines, Hall bialgebra matrices, Koszul source complexes, and primitive-recognition matrices to their primed counterparts. Compatibility with the transition maps is exactly the zero-defect condition in the finite morphism tables. Proposition 0.246 then gives the induced pro-functor and the cofinal composition law. □

Definition 1.38 *(Polarization and elliptic-origin dependence).* The dependence on the K3 polarization is the dependence of $\Gamma^{\mathrm{eff}}_{X,\sigma}$, the HN index system, and the finite moduli stacks on (H, σ). If two choices (H, σ) and (H', σ') lie in chambers connected by a specified Kontsevich–Soibelman wall-crossing datum $\mathsf{WC}_{H,H',R}$ with zero finite defects, the resulting Dirac–Igusa pro-objects are compared by the induced pro-isomorphism. Without such a wall-crossing datum they are different parameter points, not two presentations of the same object.

Let $a \in E$ and replace the origin o_E by a. For a retained wrapped family $\mathcal{F}$, the determinant anchors of Definition 4.7 satisfy the canonical identity

$$\lambda^{\det}_a(\mathcal{F}) \simeq \lambda^{\det}_{o_E}(\mathcal{F}) \otimes \mathcal{O}_E\big(\chi(\mathcal{F})(o_E - a)\big) \quad \text{in } \mathrm{Pic}^0(E).$$

Thus changing the origin contributes the degree-zero character

$$\vartheta_{a,\mathcal{F}} := \mathcal{O}_E\big(\chi(\mathcal{F})(o_E - a)\big) \in \mathrm{Pic}^0(E),$$

and, after restriction to a finite stabilizer $H \subset E[N]$, the corresponding finite character

$$\vartheta^H_{a,\mathcal{F}} \in H^1(BH; \mathbb{C}^\times)$$

is the only permitted change in the quotient-orientation linearization. The Dirac–Igusa object is independent of the elliptic origin only after twisting the wrapped-anchor and orientation-linearization rows by this character and verifying the same finite stabilizer equations.

Proposition 1.39 *(Elliptic-origin change up to its canonical character).* Fix S, H, σ, E and the finite HN system, and let $\mathfrak{p}_0$ and $\mathfrak{p}_a$ be the admissible parameter data obtained from the origins 0_E and $a \in E$. Suppose the finite hybrid, orientation, and morphism rows for $\mathfrak{p}_a$ are obtained from those for $\mathfrak{p}_0$ by tensoring every wrapped anchor with $\vartheta_{a,\mathcal{F}} = \mathcal{O}_E(\chi(\mathcal{F})(0_E - a))$ and by adding the restricted character $\vartheta^H_{a,\mathcal{F}}$ to the finite stabilizer linearization row for each retained stabilizer $H \subset E[N]$. Then the two finite-stage systems determine canonically pro-isomorphic Dirac–Igusa pro-objects after this character twist. Before the twist, the obstruction to identifying the two origin choices is exactly the family of characters $\vartheta^H_{a,\mathcal{F}}$.

Proof. By Definition 4.7,

$$\lambda^{\det}_a(\mathcal{F}) = \det Rp_{E,*}\mathcal{F} \otimes \mathcal{O}_E(-\chi(\mathcal{F})a)$$

and

$$\lambda^{\det}_{0_E}(\mathcal{F}) = \det Rp_{E,*}\mathcal{F} \otimes \mathcal{O}_E(-\chi(\mathcal{F})0_E).$$

Their ratio is $\mathcal{O}_E(\chi(\mathcal{F})(0_E - a))$, a canonical point of $\mathrm{Pic}^0(E)$. Restriction to a finite stabilizer gives the character $\vartheta^H_{a,\mathcal{F}}$. All other components of the formal charge lattice, the Mukai–Gram map, and the K3 data are unchanged. Therefore the finite-stage morphisms are the identity on all components except the wrapped-anchor and orientation-linearization rows, where they are precisely the displayed tensor and character twists. The zero-defect assumptions give commuting transition squares; Proposition 0.246 then identifies the resulting pro-objects. If the character rows are not twisted, the displayed ratio remains as the residual finite-stabilizer linearization class, so the obstruction is exactly $\vartheta^H_{a,\mathcal{F}}$. □

Remark 1.40 *(Three-tier inclusion).* The hierarchy $\Gamma^{\mathrm{eff}}_{X,\sigma} \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$ distinguishes three discipline registers. Γ^{form}_X is the formal additive group on which Π_X is defined and the polarization identity $B = -\partial\Pi_X$ holds. Γ^{alg}_X is the K-theoretic algebraic sublattice; its rank-one extension closure inside Γ^{form}_X records only the algebraic Mukai branch (computed, [64, 84]). $\Gamma^{\mathrm{eff}}_{X,\sigma}$ is the Hall support selected by σ; effectivity and semistability are not lattice-theoretic properties and are not fibre-functorial in the K3 type. The sectorial Hall truncation of §4 works at this third level after imposing finite Harder–Narasimhan height bounds.

Lemma 1.41 *(Two orthogonal hyperbolic planes inside Λ_S).* The Mukai lattice contains two orthogonal hyperbolic planes:

$$U \oplus U \hookrightarrow \Lambda_S, \qquad U_1 = \mathbb{Z}e_1 \oplus \mathbb{Z}f_1, \quad U_2 = \mathbb{Z}e_2 \oplus \mathbb{Z}f_2,$$

with $e_i^2 = f_i^2 = 0$, $e_i \cdot f_i = 1$, and $U_1 \perp U_2$ inside the Mukai pairing (proved, [88]).

Proof. Direct from $\Lambda_S \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}$. □

2 Quadratic Gram map and lattice polarization

2.1 *Igusa variables and the Gram-index lattice*

For a Siegel period

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathcal{H}_2, \qquad q = e^{2\pi i z_1}, \quad r = e^{2\pi i z_2}, \quad s = e^{2\pi i z_3},$$

and

$$T(n,l,m) = \begin{pmatrix} n & l/2 \\ l/2 & m \end{pmatrix}, \qquad n,l,m \in \mathbb{Z},$$

the trace pairing

$$\big(T(n,l,m), Z\big)_{\mathrm{tr}} := \mathrm{tr}\big(T(n,l,m)\,Z\big) = nz_1 + lz_2 + mz_3$$

gives the Borcherds monomial $x^{(n,l,m)} = q^n r^l s^m$.

Definition 2.1 *(Gram-index lattice and effective semigroup).* The *Gram-index lattice* is

$$\Gamma_{\mathrm{gram}} := \mathbb{Z}^3, \qquad \gamma = (n,l,m).$$

The *cusp-positive semigroup* at the standard cusp $\mathfrak{c}_\infty$ with chamber convention $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ is

$$\Gamma_{\mathrm{eff}} = \{(n,l,m) : m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n,l,0) : n > 0,\ l \in \mathbb{Z}\} \cup \{(0,l,0) : l < 0\}.$$

The *Jacobi-active sublocus* $\Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{eff}}$ is the locus of nonvanishing $f(nm,l)$ in the weak Jacobi form $\phi_{0,1}$ (computed, [34]).

Proposition 2.2 *(Gram matrix convention for protected charges).* For a formal even charge $c = (Q,P) \in \Gamma_X^{\mathrm{form}}$ in the unimodular lattice Λ_S, set

$$n = \frac{(Q,Q)}{2}, \qquad l = (Q,P), \qquad m = \frac{(P,P)}{2}.$$

Then $n,l,m \in \mathbb{Z}$ (since Λ_S is even unimodular, proved),

$$\begin{pmatrix} (Q,Q) & (Q,P) \\ (P,Q) & (P,P) \end{pmatrix} = 2\,T(n,l,m), \qquad \det\big(2T(n,l,m)\big) = 4nm - l^2,$$

and $x^{(n,l,m)} = q^n r^l s^m$ is the Fourier character of $\big(T(n,l,m), Z\big)_{\mathrm{tr}}$.

Proof. The Mukai pairing is integer-valued and even on $\Lambda_S \oplus \Lambda_S$ by Definition 1.3. The matrix identity is the direct expansion

$$\begin{pmatrix} (Q,Q) & (Q,P) \\ (P,Q) & (P,P) \end{pmatrix} = \begin{pmatrix} 2n & l \\ l & 2m \end{pmatrix} = 2\,T(n,l,m).$$

The determinant follows. The Fourier character is $\exp(2\pi i\,\mathrm{tr}(TZ)) = q^n r^l s^m$. □

2.2 The Gram map and its bilinear cocycle

Definition 2.3 *(Quadratic Gram map and bilinear cocycle).* The *Gram map* is the quadratic projection

$$\Pi_X : \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \Pi_X(Q,P) = \left(\frac{(Q,Q)}{2},\ (Q,P),\ \frac{(P,P)}{2}\right).$$

The *symmetric bilinear cocycle* B associated to Π_X is

$$B : \Gamma_X^{\mathrm{form}} \times \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad B\big((Q,P),(Q',P')\big) = \big((Q,Q'),\ (Q,P') + (Q',P),\ (P,P')\big).$$

LEMMA 2.4 *(Gram additivity defect).* For $c, c' \in \Gamma_X^{\mathrm{form}}$,

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c,c').$$

The defect B is symmetric, $\mathbb{Z}$-bilinear, integer-valued, and satisfies polarization: $B(c,c) = 2\,\Pi_X(c)$. As an ordinary additive cochain identity,

$$B = -\delta\Pi_X, \qquad (\delta q)(c,c') = q(c) + q(c') - q(c+c').$$

(proved.)

Remark 2.5 *(Quadratic primitive, no linear primitive).* The equality $B = -\delta\Pi_X$ is an equality in the ordinary inhomogeneous cochain complex, where the primitive Π_X is quadratic. There is no additive homomorphism $L : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ with $B = -\delta L$. Thus the normal-ordering defect is removed by the quadratic Gram coordinate, not by a linear change of charge lattice.

Proof. Write $c = (Q,P)$, $c' = (Q',P')$; then

$$\begin{aligned}
\frac{(Q+Q')^2}{2} &= \frac{Q^2}{2} + \frac{Q'^2}{2} + Q\cdot Q', \\
(Q+Q')\cdot(P+P') &= Q\cdot P + Q'\cdot P' + Q\cdot P' + Q'\cdot P, \\
\frac{(P+P')^2}{2} &= \frac{P^2}{2} + \frac{P'^2}{2} + P\cdot P'.
\end{aligned}$$

The three cross terms are the three components of $B(c,c')$. Symmetry, bilinearity, and integrality follow from the corresponding properties of the Mukai pairing. Polarization is the case $c' = c$. The cochain identity rewrites the additivity defect. ☐

LEMMA 2.6 *(Bilinear cocycle identity).* B is normalized, $B(c,0) = B(0,c) = 0$, and the additive coboundary

$$(\delta B)(a,b,c) := B(b,c) - B(a+b,c) + B(a,b+c) - B(a,b)$$

vanishes identically on Γ_X^{form} (proved).

Proof. By bilinearity in each slot,

$$B(a+b,c) = B(a,c) + B(b,c), \qquad B(a,b+c) = B(a,b) + B(a,c).$$

Substituting,

$$(\delta B)(a,b,c) = B(b,c) - B(a,c) - B(b,c) + B(a,b) + B(a,c) - B(a,b) = 0.$$

The polarization identity $B(c,c) = 2\,\Pi_X(c)$ is the only place where B takes a non-vanishing value on the diagonal; this is structural, not a Bott-element framing. ☐

Remark 2.7 *(Polarization, not Bott element).* The identity $B(c, c) = 2\,\Pi_X(c)$ is the polarization identity for the symmetric bilinear B associated to the quadratic Π_X. It is forced by the homogeneous-of-degree-two structure of the Gram map. No external Bott-element framing or characteristic-class coincidence is involved. The full structural content is that Π_X is a homogeneous quadratic on the additive group Γ_X^{form} valued in Γ_{gram}, with associated symmetric bilinear B given by the Mukai pairing of components on each direction.

2.3 *Formal lift of every Gram triple*

Lemma 2.8 *(Formal primitive torsion-one lift of every Gram triple).* Suppose Λ_S contains two orthogonal hyperbolic planes $U_1 \oplus U_2$ with bases (e_1, f_1, e_2, f_2) as in Lemma 1.41. Then every Gram triple $(n, l, m) \in \mathbb{Z}^3$ admits a primitive saturated formal full-Mukai lift $c = (Q, P) \in \Gamma_X^{\mathrm{form}}$ with $\Pi_X(Q, P) = (n, l, m)$. One may take

$$Q = e_1 + nf_1, \qquad P = lf_1 + e_2 + mf_2.$$

For this lift, the wedge torsion invariant

$$I(Q, P) := \gcd\left\{\text{coordinates of } Q \wedge P \text{ in a } \mathbb{Z}\text{-basis of } \textstyle\bigwedge^2 \Lambda_S\right\}$$

equals 1. (proved.)

Proof. The hyperbolic-plane pairings give

$$Q^2 = 2n, \qquad P^2 = 2m, \qquad Q \cdot P = l,$$

hence $\Pi_X(Q, P) = (n, l, m)$. The submodule $\mathbb{Z}Q + \mathbb{Z}P \subset \Lambda_S$ contains the unimodular minor $\det\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$ in the rows indexed by e_1, e_2 of the coefficient matrix in (e_1, f_1, e_2, f_2), so the lift is primitive saturated. The wedge expansion

$$Q \wedge P = l\, e_1 \wedge f_1 + e_1 \wedge e_2 + m\, e_1 \wedge f_2 + n\, f_1 \wedge e_2 + nm\, f_1 \wedge f_2$$

contains the basis element $e_1 \wedge e_2$ with coefficient 1, so the gcd is 1 and $I(Q, P) = 1$. □

Remark 2.9 *(Pointwise existence is not Hall support).* Lemma 2.8 produces a formal charge plane realizing every prescribed Igusa degree. It does not prove algebraicity of (Q, P), effectivity, non-empty compact Hall moduli, duality-orbit descent, source-index descent, or any chain-level Hall strictification. The compact Hall obstruction is functorial compatibility of pointwise existence with Hall addition and the primitive bracket; only the latter is the working content of Chapter 6.

The packet `certificates/charge/formal_primitive_lift` is the finite table-level check of Lemma 2.8. It verifies, on the zero row, the three simple-root rows, the δ_{123} row, and one negative-pairing test row, that

$$\Pi_X(Q, P) = (n, l, m), \qquad I(Q, P) = 1, \qquad \alpha(\Pi_X(Q, P)) = 2nf_2 - lf_3 + 2mf_{-2}.$$

The same packet records the e_1, e_2 minor equal to 1, hence primitivity of the displayed formal lift. It is not evidence for algebraic effectivity, finite HN boundedness, compact Hall support, source Hall brackets, Pfaffian orientations, type-II wall atlases, mirror discriminants, or protected traces.

3 Normal-ordered Gram extension

3.1 Central extension and the additive normal-ordered Gram homomorphism

Definition 3.1 *(Normal-ordered extension and Gram homomorphism).* Set

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{form}} \oplus \Gamma_{\mathrm{gram}},$$

with operation

$$(c, T) \star (c', T') := \big(c + c',\ T + T' + B(c, c')\big).$$

Define the *normal-ordered Gram map*

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c, T) = \Pi_X(c) - T.$$

The *raw placement* and the *split section* are

$$i_0 : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \quad i_0(c) = (c, 0), \qquad s : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \quad s(c) = (c, \Pi_X(c)).$$

Theorem 3.2 *(Group law of the normal-ordered extension; additivity of $\overline{\Pi}_X$).* Let $\widehat{\Gamma}_X, \star, \overline{\Pi}_X$, i_0, s be as in Definition 3.1. Then:

(i) $(\widehat{\Gamma}_X, \star)$ is an abelian group with identity $(0, 0)$ and inverse $(c, T)^{-1} = (-c,\ -T + B(c, c)) = (-c,\ -T + 2\,\Pi_X(c))$.

(ii) The sequence

$$0 \to \Gamma_{\mathrm{gram}} \xrightarrow{T \mapsto (0,T)} \widehat{\Gamma}_X \xrightarrow{\mathrm{pr}_1} \Gamma_X^{\mathrm{form}} \to 0$$

is a central extension of abelian groups with cocycle B.

(iii) $\overline{\Pi}_X$ is a group homomorphism from $(\widehat{\Gamma}_X, \star)$ to the additive group Γ_{gram}.

(iv) The split section $s(c) = (c, \Pi_X(c))$ is additive, and

$$\overline{\Pi}_X(i_0(c)) = \Pi_X(c), \qquad \overline{\Pi}_X(s(c)) = 0.$$

(v) The change of variables $(c, T) \mapsto (c, T - \Pi_X(c))$ carries $\star$ to componentwise addition; thus $\widehat{\Gamma}_X \cong \Gamma_X^{\mathrm{form}} \oplus \Gamma_{\mathrm{gram}}$ as abstract abelian groups, and $[B] = 0$ in additive cohomology.

(proved.)

Proof. *(i)* Associativity follows from the cocycle identity $\delta B = 0$ (Lemma 2.6); the associator $B(a, b) + B(a + b, c) - B(a, b + c) - B(b, c)$ vanishes by bilinearity. Commutativity is symmetry of B. The unit calculation $(c, T) \star (0, 0) = (c, T + B(c, 0)) = (c, T)$ and inverse $(c, T) \star (-c, -T + B(c, c)) = (0, B(c, c) - B(c, c)) = (0, 0)$ (using $B(c, -c) = -B(c, c)$ by bilinearity) close the abelian-group structure.

(ii) Definition.

(iii) Apply Lemma 2.4:

$$\overline{\Pi}_X((c, T) \star (c', T')) = \Pi_X(c + c') - T - T' - B(c, c') = \Pi_X(c) + \Pi_X(c') - T - T' = \overline{\Pi}_X(c, T) + \overline{\Pi}_X(c', T').$$

(iv) For the section,

$$s(c) \star s(c') = (c + c',\ \Pi_X(c) + \Pi_X(c') + B(c, c')) = (c + c',\ \Pi_X(c + c')) = s(c + c').$$

The identities $\overline{\Pi}_X(i_0(c)) = \Pi_X(c) - 0 = \Pi_X(c)$ and $\overline{\Pi}_X(s(c)) = \Pi_X(c) - \Pi_X(c) = 0$ are immediate.

(v) The map $(c,T) \mapsto (c, T - \Pi_X(c))$ is a bijection with inverse $(c,T) \mapsto (c, T + \Pi_X(c))$. Direct calculation gives

$$(c, T - \Pi_X(c)) + (c', T' - \Pi_X(c')) = (c + c',\ T + T' - \Pi_X(c) - \Pi_X(c')) = (c + c',\ (T + T' + B(c,c')) - \Pi_X(c + c')),$$

matching the change of variables applied to $(c,T) \star (c',T')$. □

Remark 3.3 *(Finite charge-window verification).* The normal-ordered group law is a proved lattice identity. Its use in the Dirac–Igusa pro-object additionally requires a finite charge-window packet: active $\phi_{0,1}$-support, downward saturation, Liu HN bounds, Mukai–Gram rows, cocycle identities, normal-ordered lift orbits, target-window images, primitive formal lifts, strict transitions, and a scalar firewall. A target presentation, signed exponent list, Pfaffian product, Hilbert-scheme scalar, or source-rank table does not by itself supply this packet.

3.2 The fixed-lift raw-pushforward obstruction theorem

Definition 3.4 *(Raw homogeneous and fixed-lift Π_X-pushforwards).* Let $P = \bigoplus_{c \in \Gamma_X^{\mathrm{form}}} P_c$ be a Γ_X^{form}-graded primitive object. The *raw homogeneous Π_X-pushforward* is

$$\left(\Pi_{X,*}^{\mathrm{raw}} P\right)_\gamma := \bigoplus_{\Pi_X(c)=\gamma} P_c.$$

The *strict fixed-lift raw Π_X-pushforward* is the narrower operation obtained by choosing one lift $L(\gamma) \in \Gamma_X^{\mathrm{form}}$ for every retained γ and setting

$$\left(\Pi_{X,*}^{\mathrm{raw},L} P\right)_\gamma := P_{L(\gamma)}.$$

Theorem 3.5 *(Raw homogeneous fixed-lift Π_X-pushforward cannot realize the type-II real-root strings).* Let $P = \bigoplus_{c \in \Gamma_X^{\mathrm{form}}} P_c$ be a Γ_X^{form}-graded Lie superalgebra with $[P_c, P_{c'}] \subset P_{c+c'}$. Suppose a strict fixed-lift raw pushforward of P along the quadratic map Π_X were a Γ_{gram}-graded Lie superalgebra structure on the displayed bracket channels of chosen charge lifts. Then every nonzero bracket channel $[P_c, P_{c'}]$ contributing to that strict raw pushforward must satisfy $B(c,c') = 0$.

For the type-II real simple roots $(\delta_1, \delta_2, \delta_3)$ of the Gritsenko–Nikulin algebra $\mathfrak{g}_{\Delta_5}$ with Cartan matrix

$$A_{\mathrm{II}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

this condition is incompatible with the nonzero second real-root strings $2\delta_i + \delta_j$, $i \neq j$. The strict fixed-lift raw Π_X-descent therefore cannot carry these brackets; they require the normal-ordered extension $\widehat{\Gamma}_X$ and the additive homomorphism $\overline{\Pi}_X$. (proved, [46].)

Proof. If the raw pushforward is Γ_{gram}-graded, a nonzero bracket of classes of raw Gram degrees $\Pi_X(c)$ and $\Pi_X(c')$ must land in raw Gram degree $\Pi_X(c) + \Pi_X(c')$. But the upstairs bracket lands in physical degree $c + c'$, whose raw Gram shadow is

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c,c').$$

Hence $B(c, c') = 0$ on every nonzero bracket channel.

Take physical lifts c_i, c_j of distinct type-II real simple root degrees γ_i, γ_j, $\Pi_X(c_i) = \gamma_i$ with $\gamma_i \neq 0$. The Gritsenko–Nikulin Cartan matrix A_{II} has off-diagonal entries -2, so distinct simple roots satisfy $\langle \delta_i, \delta_j \rangle = -2$, and the real-root string through δ_j in the δ_i-direction includes the levels $\delta_i + \delta_j$ and $2\delta_i + \delta_j$ (proved, [46, §2–3]). Both correspond to nonzero brackets, hence raw pushforward forces

$$B(c_i, c_j) = 0, \qquad B(c_i, c_i + c_j) = 0.$$

Bilinearity and polarization give

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = 2\,\Pi_X(c_i) + 0 = 2\,\gamma_i,$$

which is nonzero in the torsion-free lattice Γ_{gram}. Contradiction. □

Remark 3.6 *(Scope of the no-go).* Theorem 3.5 excludes the strict *homogeneous fixed-lift* raw pushforward. It does not exclude isolated B-isotropic bracket channels and does not decide the mixed-lift fibre-summed construction in which several lifts of a single Gram degree are summed before bracketing; the latter remains an open mixed-lift question. The normal-ordered extension removes exactly the displayed obstruction because $\overline{\Pi}_X$ is additive on $\widehat{\Gamma}_X$. Chain-level Hall strictification (A_∞-compatibility, cyclic equivariance, Frobenius, multiplicative orientation) is a separate input system, named in §6 and developed in Chapter 6.

Definition 3.7 *(Gram fibres, zero fibre, and fibre-summed parity blocks).* For $\gamma \in \Gamma_{\mathrm{gram}}$, set

$$\Gamma_X(\gamma) := \Pi_X^{-1}(\gamma) \subset \Gamma_X^{\mathrm{form}}.$$

The *zero Gram fibre* is

$$K_{\Pi,0} := \Gamma_X(0,0,0) = \{(Q,P) \mid Q^2 = P^2 = Q \cdot P = 0\}.$$

It is a fibre of the quadratic map Π_X, not the kernel of a group homomorphism. If $V = \bigoplus_{c \in A} V_c$ is a finite $\mathbb{Z}/2$-graded formal source over a finite subset $A \subset \Gamma_X^{\mathrm{form}}$, the fibre-summed Gram pushforward is the $\mathbb{Z}/2$-graded vector space

$$(\Pi_{X,*}^{\mathrm{fib}} V)_{\gamma,\bar{p}} := \bigoplus_{c \in A \cap \Gamma_X(\gamma)} V_{c,\bar{p}}, \qquad \bar{p} \in \mathbb{Z}/2.$$

The signed superdimension $\dim(\Pi_{X,*}^{\mathrm{fib}} V)_{\gamma,\bar{0}} - \dim(\Pi_{X,*}^{\mathrm{fib}} V)_{\gamma,\bar{1}}$ is a scalar shadow of this two-integer datum.

Proposition 3.8 *(The zero Gram fibre is not an additive subgroup).* [proved] In the two-hyperbolic-plane chart of Lemma 2.8, the zero Gram fibre $K_{\Pi,0}$ is not closed under addition. Hence no quotient of Γ_X^{form} by $K_{\Pi,0}$ is defined at the level of abelian groups.

Proof. Let

$$k_1 = (e_1, e_2), \qquad k_2 = (f_1, -f_2).$$

Then

$$\Pi_X(k_1) = \Pi_X(k_2) = (0,0,0),$$

because e_1, e_2, f_1, f_2 are isotropic and the two hyperbolic planes are orthogonal. Their sum is

$$k_1 + k_2 = (e_1 + f_1, e_2 - f_2).$$

The first component has square 2, the second has square −2, and the two components are orthogonal. Therefore

$$\Pi_X(k_1 + k_2) = (1, 0, -1) \neq (0, 0, 0).$$

Thus $K_{\Pi,0}$ is a null-cone fibre of a quadratic map, not a subgroup. □

Remark 3.9 *(Fibre summation does not prove a Hall radical).* Definition 3.7 proves only the formal bookkeeping required before a finite Hall window can be compared with a target root window. A fibre with one even and one odd source vector has signed superdimension 0 and parity profile (1, 1); it is not the zero source degree. The radical along $K_{\Pi,0}$ is therefore a separate Hopf-radical datum inside a finite source: it must specify the pairing matrix, kernel matrix, quotient splitting, ideal/coideal identities, and transition exactness. It is not obtained by quotienting the formal charge lattice by $K_{\Pi,0}$.

The packet `certificates/charge/gram_fibre_kernel` checks these formal assertions on finite rows. It records two distinct formal charges with Gram label (1, 1, 0), two with Gram label (1, 1, 1), and two nonzero charges in $K_{\Pi,0}$ whose parity profile is (1, 1) although the signed superdimension is 0. It also records the witness

$$(e_1, e_2) + (f_1, -f_2) = (e_1 + f_1, e_2 - f_2), \qquad \Pi_X(e_1 + f_1, e_2 - f_2) = (1, 0, -1),$$

so the zero fibre is not additive. The packet does not prove finite HN boundedness, compact Hall support, source Hall brackets, source parity, a Hopf radical, exactness of a finite pushforward, Pfaffian orientations, or protected traces.

Definition 3.10 *(Zero-fibre radical on a finite formal source).* Let $V = \bigoplus_{c \in A} V_c$ be a finite Γ_X^{form}-graded $\mathbb{Q}$-vector space with $\mathbb{Z}/2$-parity, where $A \subset \Gamma_X^{\mathrm{form}}$ is finite. Let

$$V_{\Pi,0} := \bigoplus_{c \in A \cap K_{\Pi,0}} V_c.$$

For a bilinear pairing $h : V_{\Pi,0} \otimes V_{\Pi,0} \to \mathbb{Q}$, the *zero-fibre radical* is the ordinary linear-algebra kernel

$$\mathrm{Rad}_{\Pi,0}(V, h) := \ker\Big(V_{\Pi,0} \longrightarrow V_{\Pi,0}^{\vee}\Big), \qquad v \longmapsto h(v, -).$$

It is a subspace of the finite source vector space, not a quotient of Γ_X^{form} by the zero Gram fibre.

Proposition 3.11 *(Exactness of finite fibre-summed Gram pushforward).* [proved] The functor

$$\Pi_{X,*}^{\mathrm{fib}} : \mathrm{Vect}^{\mathrm{fin}}_{\Gamma_X^{\mathrm{form}}, \mathbb{Z}/2} \longrightarrow \mathrm{Vect}^{\mathrm{fin}}_{\Gamma_{\mathrm{gram}}, \mathbb{Z}/2}, \qquad (\Pi_{X,*}^{\mathrm{fib}} V)_{\gamma, \bar{p}} = \bigoplus_{\Pi_X(c) = \gamma} V_{c, \bar{p}},$$

is exact on finite-support graded $\mathbb{Q}$-vector spaces. Consequently a degreewise exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

of finite formal sources remains exact after applying $\Pi_{X,*}^{\mathrm{fib}}$.

Proof. Fix $(\gamma, \bar{p})$. The $(\gamma, \bar{p})$-component of $\Pi^{\mathrm{fib}}_{X,*}$ is the finite direct sum of the $(c, \bar{p})$-components over the fibre $\Pi_X^{-1}(\gamma) \cap A$. Finite direct sums are exact in the category of $\mathbb{Q}$-vector spaces. Taking this finite direct sum in every Gram/parity component gives the displayed exact sequence after pushforward. □

The packet `certificates/charge/fibre_pushforward_exactness` checks this finite linear algebra on six displayed Gram/parity blocks. It records matrices for $0 \to A \to B \to C \to 0$, verifies that the projection kernel equals the injection image in each block, verifies the same equality after fibre-summed pushforward, and records a zero-fibre pairing matrix whose radical is one-dimensional. The packet does not prove finite HN boundedness, compact Hall support, source Hall brackets, Hopf ideal/coideal closure of the radical, exact Hall pushforward, bar-construction commutation, Pfaffian orientations, or protected traces.

Proposition 3.12 *(Finite normal-ordered pushforward preserves parity).* [proved] Let

$$V = \bigoplus_{\hat{\gamma} \in A} \left(V_{\hat{\gamma},\bar{0}} \oplus V_{\hat{\gamma},\bar{1}} \right)$$

be a finite-support $\widehat{\Gamma}_X$-graded super vector space, and let $\overline{\Pi}_X : \widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$ be the additive normal-ordered Gram map. Define the pushed super vector space by

$$(\overline{\Pi}^{\mathrm{fib}}_{X,*} V)_{\gamma,\bar{p}} := \bigoplus_{\substack{\hat{\gamma} \in A \\ \overline{\Pi}_X(\hat{\gamma}) = \gamma}} V_{\hat{\gamma},\bar{p}}, \qquad \bar{p} \in \mathbb{Z}/2.$$

Then $\overline{\Pi}^{\mathrm{fib}}_{X,*}$ preserves the even and odd summands separately. In particular the pair

$$\left(\dim(\overline{\Pi}^{\mathrm{fib}}_{X,*} V)_{\gamma,\bar{0}}, \dim(\overline{\Pi}^{\mathrm{fib}}_{X,*} V)_{\gamma,\bar{1}} \right)$$

is the fibrewise sum of the source parity dimensions, while the signed superdimension is only its scalar shadow.

Proof. For fixed $(\gamma, \bar{p})$, the displayed definition is an ordinary finite direct sum of precisely the $\bar{p}$-parity components whose normal-ordered Gram degree is γ. No even summand is placed in an odd summand and no odd summand is placed in an even summand. Thus the pushed even block is the direct sum of source even blocks over the fibre of $\overline{\Pi}_X$, and the pushed odd block is the direct sum of source odd blocks over the same fibre. Taking the difference of the two dimensions is a later scalar trace operation and does not recover the two parity ranks. □

The packet `certificates/charge/parity_pushforward` checks this finite statement on three displayed Gram degrees. It records the two parity dimensions before and after pushforward and includes a zero-degree row with parity profile $(1, 1)$ and signed superdimension 0. It proves preservation of finite parity blocks under the additive normal-ordered pushforward. It does not prove the source parity theorem, the Gritsenko–Nikulin/Kac target parity equality, compact source representatives, primitive recognition, Pfaffian orientations, or protected traces.

Proposition 3.13 *(Finite bar construction commutes with the normal-ordered Gram pushforward).* [proved] Let $A = \mathbb{Q}\mathbf{1} \oplus \bar{A}$ be a finite-support augmented associative algebra graded by the additive group $\widehat{\Gamma}_X$, with multiplication preserving the $\widehat{\Gamma}_X$-degree. Let

$$T^c_{\leq N}(\bar{A}) = \bigoplus_{1 \leq p \leq N} \bar{A}^{\otimes p}$$

be the finite reduced tensor coalgebra with deconcatenation coproduct, graded by the sum of the $\widehat{\Gamma}_X$-degrees of the letters. Then the additive homomorphism $\overline{\Pi}_X : \widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$ induces a canonical isomorphism of finite Γ_{gram}-graded coalgebras

$$\overline{\Pi}^{\mathrm{fib}}_{X,*}\, T^c_{\leq N}(\bar{A}) \;\cong\; T^c_{\leq N}\Big(\overline{\Pi}^{\mathrm{fib}}_{X,*}\bar{A}\Big).$$

If the bar differential is defined by multiplication in A, the same isomorphism intertwines the finite bar differential.

Proof. For a word $[a_1|\cdots|a_p]$, the source degree is $\hat{\gamma}_1 + \cdots + \hat{\gamma}_p$ in the additive group $\widehat{\Gamma}_X$. Since $\overline{\Pi}_X$ is a group homomorphism,

$$\overline{\Pi}_X(\hat{\gamma}_1 + \cdots + \hat{\gamma}_p) = \overline{\Pi}_X(\hat{\gamma}_1) + \cdots + \overline{\Pi}_X(\hat{\gamma}_p).$$

Thus pushing the word degree forward after forming the tensor coalgebra and forming the tensor coalgebra after pushing the letters forward give the same finite direct sum in each Γ_{gram}-degree. Deconcatenation cuts a word into two subwords; the displayed equality is additive on the two subword degrees, so the coproduct is preserved. The internal bar differential replaces adjacent letters a_i, a_{i+1} by their product. Degree preservation of multiplication in A gives $\deg(a_i a_{i+1}) = \deg(a_i) + \deg(a_{i+1})$, hence the same Γ_{gram}-degree after $\overline{\Pi}_X$. Therefore the finite bar differential is intertwined. □

The packet `certificates/charge/bar_pushforward_commutation` checks this statement on finite words of length at most three. It records letter degrees, word degrees, deconcatenation splits, and product terms in the bar differential, all using the additive normal-ordered Gram grading. It does not certify the raw quadratic Π_X-pushforward, the chiral Koszul quasi-isomorphism, Hall bracket compatibility, Hopf pairing compatibility, Pfaffian orientations, or protected traces.

Proposition 3.14 *(Finite bracket and pairing degrees descend under normal-ordered Gram pushforward).* [proved] Let $P = \bigoplus_{\hat{\gamma}\in\widehat{\Gamma}_X} P_{\hat{\gamma}}$ be a finite-support $\widehat{\Gamma}_X$-graded Lie super vector space with

$$[P_{\hat{\gamma}}, P_{\hat{\delta}}] \subset P_{\hat{\gamma}+\hat{\delta}}.$$

Then $\overline{\Pi}^{\mathrm{fib}}_{X,*}P$ is a Γ_{gram}-graded Lie super vector space. If

$$h : P \otimes P \to \mathbb{Q}$$

is a homogeneous degree-zero bilinear form, in the sense that

$$h(P_{\hat{\gamma}}, P_{\hat{\delta}}) = 0 \quad \text{unless} \quad \hat{\gamma} + \hat{\delta} = 0,$$

then the fibre-summed form on $\overline{\Pi}^{\mathrm{fib}}_{X,*}P$ is homogeneous of degree zero for the Γ_{gram}-grading.

Proof. For $x \in P_{\hat{\gamma}}$ and $y \in P_{\hat{\delta}}$, the source bracket has degree $\hat{\gamma} + \hat{\delta}$. Applying the additive homomorphism $\overline{\Pi}_X$ gives

$$\overline{\Pi}_X(\hat{\gamma} + \hat{\delta}) = \overline{\Pi}_X(\hat{\gamma}) + \overline{\Pi}_X(\hat{\delta}),$$

so the same bracket is graded after fibre-summed pushforward. The Jacobi identity and super-skew symmetry are unchanged, since the underlying bracket is the same linear map on the same finite direct sum.

If $h(x, y) \neq 0$, then $\hat{\gamma} + \hat{\delta} = 0$. Applying $\overline{\Pi}_X$ gives $\overline{\Pi}_X(\hat{\gamma}) + \overline{\Pi}_X(\hat{\delta}) = 0$. Thus the pushed pairing can be nonzero only on opposite Gram degrees. This is degree-zero homogeneity of the pushed form. No Hopf adjointness, Frobenius cyclic identity, quotient non-degeneracy, or primitive closure is used. □

The packet `certificates/charge/hall_pairing_pushforward_compatibility` checks this finite algebraic statement on supplied bracket and pairing rows. It verifies that three bracket rows have additive normal-ordered Gram degree and that three nonzero pairing rows have total Gram degree 0. It does not construct compact Hall correspondences, prove primitive closure, prove Hopf adjointness, prove the Frobenius cyclic identity, prove quotient non-degeneracy, prove Pfaffian orientations, or prove protected traces.

Proposition 3.15 *(Finite normal-ordered pushforward preserves the PBW filtration).* [proved] Let $U = \bigoplus_{\hat{\gamma} \in A} U_{\hat{\gamma}}$ be a finite-support $\widehat{\Gamma}_X$-graded vector space equipped with an increasing PBW filtration $F_{\leq d}U$ by homogeneous subspaces:

$$F_{\leq d}U = \bigoplus_{\hat{\gamma} \in A} \big(F_{\leq d}U \cap U_{\hat{\gamma}}\big).$$

Write $F_{\leq d}U_{\hat{\gamma}} := F_{\leq d}U \cap U_{\hat{\gamma}}$. Define

$$F_{\leq d}(\overline{\Pi}^{\mathrm{fib}}_{X,*}U)_{\gamma} := \bigoplus_{\substack{\hat{\gamma} \in A \\ \overline{\Pi}_X(\hat{\gamma}) = \gamma}} F_{\leq d}U_{\hat{\gamma}}.$$

Then $\overline{\Pi}^{\mathrm{fib}}_{X,*}$ preserves the PBW filtration, and for every d

$$\mathrm{gr}^{\mathrm{PBW}}_d(\overline{\Pi}^{\mathrm{fib}}_{X,*}U)_{\gamma} \cong \bigoplus_{\substack{\hat{\gamma} \in A \\ \overline{\Pi}_X(\hat{\gamma}) = \gamma}} \mathrm{gr}^{\mathrm{PBW}}_d U_{\hat{\gamma}}.$$

If U is a filtered algebra whose multiplication has $\widehat{\Gamma}_X$-degree $\hat{\gamma} + \hat{\delta}$ and PBW degree $\leq a + b$ on $F_{\leq a}U_{\hat{\gamma}} \cdot F_{\leq b}U_{\hat{\delta}}$, the pushed multiplication has Γ_{gram}-degree $\overline{\Pi}_X(\hat{\gamma}) + \overline{\Pi}_X(\hat{\delta})$ and PBW degree $\leq a + b$.

Proof. For fixed (γ, d), the displayed formula is a finite direct sum over the fibre of $\overline{\Pi}_X$. Since $F_{\leq d}U_{\hat{\gamma}} \subset F_{\leq d+1}U_{\hat{\gamma}}$ for each $\hat{\gamma}$, the same inclusion holds after taking the finite direct sum over the fibre. Finite direct sums are exact over $\mathbb{Q}$, hence the quotient $F_{\leq d}/F_{\leq d-1}$ commutes with the displayed pushforward and gives the associated-graded isomorphism. For the algebra statement, the source multiplication sends $(\hat{\gamma}, \hat{\delta})$ to $\hat{\gamma} + \hat{\delta}$ and has PBW degree $\leq a + b$; applying the additive homomorphism $\overline{\Pi}_X$ sends the Gram degree to $\overline{\Pi}_X(\hat{\gamma}) + \overline{\Pi}_X(\hat{\delta})$ and does not change the filtration index. Thus the pushed filtered multiplication has the asserted degree and filtration bound. □

The packet `certificates/charge/pbw_pushforward` checks this finite filtered statement on three displayed Gram degrees. It records cumulative PBW dimensions, verifies monotonicity of the filtration, and recomputes the associated-graded dimensions as successive quotients. It proves only preservation of a supplied finite PBW filtration under the additive normal-ordered pushforward. It does not prove the compact-source PBW comparison, target PBW comparison, no-extra-relations theorem, primitive recognition, Pfaffian orientations, or protected traces.

Proposition 3.16 *(Finite normal-ordered pushforward preserves triangular decomposition).* [proved] Let $h : \Gamma_{\mathrm{gram}} \to \mathbb{Z}$ be an additive height functional. For a finite-support $\widehat{\Gamma}_X$-graded vector space $V = \bigoplus_{\hat{\gamma}\in A} V_{\hat{\gamma}}$, define

$$V^+ = \bigoplus_{h(\overline{\Pi}_X(\hat{\gamma}))>0} V_{\hat{\gamma}}, \qquad V^0 = \bigoplus_{h(\overline{\Pi}_X(\hat{\gamma}))=0} V_{\hat{\gamma}}, \qquad V^- = \bigoplus_{h(\overline{\Pi}_X(\hat{\gamma}))<0} V_{\hat{\gamma}}.$$

Then

$$\overline{\Pi}^{\mathrm{fib}}_{X,*}V = (\overline{\Pi}^{\mathrm{fib}}_{X,*}V)^- \oplus (\overline{\Pi}^{\mathrm{fib}}_{X,*}V)^0 \oplus (\overline{\Pi}^{\mathrm{fib}}_{X,*}V)^+$$

with each summand obtained by pushing forward the corresponding source summand over the same fibres of $\overline{\Pi}_X$. If V carries a homogeneous bracket

$$[V_{\hat{\gamma}}, V_{\hat{\delta}}] \subset V_{\hat{\gamma}+\hat{\delta}},$$

then the pushed bracket has height

$$h\Big(\overline{\Pi}_X(\hat{\gamma} + \hat{\delta})\Big) = h(\overline{\Pi}_X(\hat{\gamma})) + h(\overline{\Pi}_X(\hat{\delta})).$$

Thus positive-positive brackets remain positive, negative-negative brackets remain negative, Cartan-positive and Cartan-negative brackets remain in the corresponding sign, and positive-negative brackets land in the sign determined by the actual height sum.

Proof. The three displayed subspaces form a direct sum because every integer height is exactly one of positive, zero, or negative. For a fixed Gram degree γ, the fibre-summed pushforward takes the finite direct sum over $\overline{\Pi}_X^{-1}(\gamma)$. All source summands in this fibre have the same value $h(\gamma)$, hence the same triangular sign. Therefore no positive source summand is pushed into the Cartan or negative part, and similarly for the other two signs.

For the bracket statement, source homogeneity gives degree $\hat{\gamma} + \hat{\delta}$. Additivity of $\overline{\Pi}_X$ and of h gives the displayed height equality. The triangular sign after pushforward is therefore the sign of this integer. The four stated same-sign and Cartan-action cases are immediate; the positive-negative case is not collapsed to a universal rule, but is read from the actual height sum. □

The packet `certificates/charge/triangular_pushforward` checks this finite statement on seven displayed source blocks and five supplied bracket sign rows. It verifies that the supplied integer height agrees with the positive, Cartan, and negative labels, that the fibre-summed pushforward keeps the three direct-sum parts separate, and that the displayed bracket rows land in the triangular part determined by the pushed height. It does not prove exact triangular completion, target triangular completion, compact-source primitive recognition, no-extra-relations, PBW comparison, Pfaffian orientations, or protected traces.

Remark 3.17 *(Quadratic obstruction; no linear trivialisation).* Suppose $B = \delta L$ for some additive linear $L : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$. Setting $c' = c$, $B(c, c) = 2L(c) - L(2c)$; additivity gives $L(2c) = 2L(c)$, so $B(c, c) = 0$, contradicting $B(c, c) = 2\,\Pi_X(c) \neq 0$ for any c with $(Q, Q), (Q, P), (P, P)$ not all zero. Hence no additive linear trivialisation exists. The quadratic primitive Π_X is necessary; trivializing the additive cocycle requires a quadratic cochain, not a linear one (proved).

3.3 Normal-ordering flag formula

Lemma 3.18 *(Multiplication of raw placements).* For $c_1, \dots, c_k \in \Gamma_X^{\mathrm{form}}$,

$$i_0(c_1) \star \cdots \star i_0(c_k) = \Bigg(\sum_i c_i,\ \sum_{i<j} B(c_i, c_j)\Bigg),$$

and consequently

$$\overline{\Pi}_X(i_0(c_1) \star \cdots \star i_0(c_k)) = \sum_i \Pi_X(c_i).$$

(proved.)

Proof. Induct on k. The case $k = 2$ is the definition of $\star$. At step $k \to k+1$,

$$B\Bigg(\sum_{i \le k} c_i,\ c_{k+1}\Bigg) = \sum_{i \le k} B(c_i,\ c_{k+1})$$

by bilinearity. Applying $\overline{\Pi}_X$ subtracts the accumulated pairwise B-term from $\Pi_X(\sum_i c_i) = \sum_i \Pi_X(c_i) + \sum_{i<j} B(c_i, c_j)$. □

Remark 3.19 *(Normal-ordered primitive recognition target).* The interface with the BKM denominator algebra is the normal-ordered charge route

$$\Gamma_X^{\mathrm{form}} \hookrightarrow \widehat{\Gamma}_X \xrightarrow{\overline{\Pi}_X} \Gamma_{\mathrm{gram}} \xrightarrow{\alpha} \Lambda_{II}^{2,1},$$

where $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$ is the type-IV embedding into the BKM root lattice (proved, [46]). At chain level this becomes the descent $\overline{\Pi}^{\Theta}_{X,*}$ of Chapter 6. The primitive recognition problem is to construct a protected primitive Hall algebra whose normal-ordered descended bracket is the Borcherds–Kac bracket of $\mathfrak{g}_{\Delta_5}$. Raw Gram pushforward is not an admissible recognition target.

4 The hybrid $\mathrm{Ran}^{\mathrm{hyb}}(E)$ factorization carrier

After integration along the K3 direction, the chiral object is supported on the elliptic factor E. The ordinary finite-support carrier is the Ran space $\mathrm{Ran}(E)$ of finite subsets, but a one-dimensional effective cycle with positive elliptic degree projects onto all of E. The hybrid carrier $\mathrm{Ran}^{\mathrm{hyb}}(E)$ is a finite-stage prestack whose $b = 0$ local and $b > 0$ wrapped subprestacks are separated before the mixed local/wrapped correspondences are supplied as additional finite-stage data. At finite stage those data are represented by the tables hybrid_ran_prestack_definition, local_stratum_b0_prestack_definition, wrapped_stratum_bpositive_prestack_definition, geometric_elliptic_degree_map, geometric_elliptic_degree_additivity, geometric_borcherds_degree_separation, positive_elliptic_degree_projection, hybrid_base, local_configurations, wrapped_prequotients, local_local_correspondence_definition, mixed_local_wrapped_correspondence_ wrapped_wrapped_correspondence_definition, two_sided_mixed_correspondence_definition, correspondences, flag_atlas, higher_coloured, quotient_pseudofunctor, protected_integration, transitions, and scalar_firewall. They are not supplied by the ordinary Ran space, a quotient-first Hall product, or the Borcherds s-degree.

4.1 *Projection-locality obstruction*

LEMMA 4.1 *(Projection-locality obstruction).* Let

$$Z = \sum_i m_i [C_i] \in Z_1(S \times E), \qquad m_i > 0,$$

be an effective one-cycle. If $(p_E)_* Z = b[E]$ with $b > 0$, then $p_E(|Z|) = E$. Hence $|Z|$ is not local on $\mathrm{Ran}(E)$.

Proof. For each irreducible component C_i, the proper image $p_E(C_i)$ is either a point or a closed one-dimensional subset of the irreducible curve E. In the second case it is all of E, and $(p_E)_*[C_i] = \deg(C_i/E)[E]$. Since the effective sum $(p_E)_* Z = b[E]$ has $b > 0$, at least one component maps dominantly to E. Therefore $p_E(|Z|) = E$. □

Remark 4.2 *(Geometric versus Borcherds elliptic degree).* The lemma fixes the necessity of a wrapped stratum. The geometric elliptic-degree map b_R^{geom}, on a finite Hall stage R, records the integer $\deg_E$ of a chosen retained class. The trace exponent $m_R^{\mathrm{Bch}} := \mathrm{pr}_3 \overline{\Pi}_X$, on the same stage, records the third coordinate of the normal-ordered Gram pushforward. The two are not equal as functions on the formal lattice; the Borcherds exponent m records the magnetic Mukai pairing $(P, P)/2$, which sees Λ_S-square data, while b_R^{geom} records the E-degree of the geometric support, which sees $H^0(S) \oplus H^4(S)$ data through r and ch_2 of the K3 factor. The wrapped stratum carries $b_R^{\mathrm{geom}} > 0$ branes; the local stratum carries $b_R^{\mathrm{geom}} = 0$ branes whose magnetic component lies in the projection-finite kernel. This is the $\mathrm{Ran}(E)$-support-locality obstruction in the K3-integrated chiral shadow.

4.2 *The hybrid carrier*

Definition 4.3 *(Hybrid Ran carrier).* At finite HN height R, the *hybrid Ran carrier* is the prestack

$$\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E) : \mathrm{Aff}^{op}_{\mathbb{C}} \to \infty\text{-Gpd}$$

of Definition 0.3. A test family consists of finitely many local maps to E, retained local colour labels, finitely many wrapped colour labels, and T-families in the corresponding rigidified wrapped prequotient stacks with anchor maps to E. Its local $b = 0$ stratum is the subprestack $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ of Definition 0.4. Its wrapped $b > 0$ stratum is the subprestack $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$ of Definition 0.5. The stack $\mathrm{Ran}^{\mathrm{hyb}}_{R,\tau}(E)$ exists only after descent rows for the chosen topology τ have been supplied. The set

$$\bigsqcup_{b \geq 0} \mathrm{Ran}(E) \times \{b\}$$

is the degree shadow, not the carrier.

Remark 4.4 *(Two strata, two registers).* On the $b = 0$ stratum the standard Ran/factorization formalism applies and supplies the local chiral factorization input in $\mathrm{Fact}(\mathrm{Ran}(E))$ (folklore, [7, 35]). On the $b > 0$ stratum the support of $\mathrm{Supp}\,\mathcal{F}$ projects onto E, so the brane is global; its data live on the wrapped prequotient and after E-quotient become E-equivariant data on the elliptic curve. The hybrid carrier is the smallest carrier reflecting both strata; it is not a single Ran space.

4.3 Mixed local/wrapped correspondences

Definition 4.5 *(Mixed local/wrapped correspondences (finite Hall stage R)).* Fix a finite Hall stage R and retained Liu–Hilbert classes Ξ_A, Ξ_B, Ξ_C with $\Xi_C = \Xi_A + \Xi_B$ in the closure rules of [64]. The *retained extension correspondence*

$$\mathfrak{E}^{\mathrm{ret}}_{\Xi_A,\Xi_B;\Xi_C} \xrightarrow{p} \mathfrak{M}^{\mathrm{ss}}_{\Xi_A} \times \mathfrak{M}^{\mathrm{ss}}_{\Xi_B}, \qquad \mathfrak{E}^{\mathrm{ret}}_{\Xi_A,\Xi_B;\Xi_C} \xrightarrow{q} \mathfrak{M}^{\mathrm{ss}}_{\Xi_C}$$

is the closed derived flag-Quot/filtration stack over $\mathfrak{M}^{\mathrm{ss}}_{\Xi_C}$, with q required at this finite stage to be proper or $\mathrm{Coh}^{\mathrm{prop}}$-admissible in the compactified model (folklore). Three colours of correspondence appear, indexed by the location of Ξ_A, Ξ_B in $\mathrm{Ran}^{\mathrm{hyb}}(E)$:

(LL) *local-local*: $b_A = b_B = 0$;

(LW) *ordered local/wrapped*: $b_A = 0$, $b_B > 0$;

(WW) *wrapped-wrapped*: $b_A, b_B > 0$.

The local-local row is the ordered projection-finite diagram

$$\mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K} \longrightarrow \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\beta,R,J} \longrightarrow \mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K},$$

where $\alpha : I \to \Gamma^{\mathrm{loc}}_R$, $\beta : J \to \Gamma^{\mathrm{loc}}_R$, $\gamma = |\alpha| + |\beta|$ is retained in Γ^{loc}_R, and $K = I \sqcup J$ modulo Ran collisions. This is Definition 0.19; it has no wrapped input and no reduced E-quotient. The one-sided mixed row is the ordered pair of unreduced diagrams

$$\mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I} \longrightarrow \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

and

$$\mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

where $\alpha : I \to \Gamma^{\mathrm{loc}}_R$, $\eta \in \Gamma^{\mathrm{wr}}_R$, and $\zeta = |\alpha| + \eta$ or $\eta + |\alpha|$ is retained in Γ^{wr}_R. This is Definition 0.21; it is not the local-local row, not the wrapped-wrapped row, not the two-sided mixed row, and not a quotient-first product. The wrapped-wrapped row is the ordered unreduced diagram

$$\mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\eta_1,R} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta_2,R} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

where $\eta_1, \eta_2 \in \Gamma^{\mathrm{wr}}_R$ and $\zeta = \eta_1 + \eta_2$ is retained in Γ^{wr}_R. This is Definition 0.23; it is ordered, unreduced, and separate from symmetric descent and source/target-anchor memory. The ordered two-sided mixed row is the unreduced local–wrapped–local diagram

$$\mathfrak{E}^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+} \longrightarrow \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha_-,R,I_-} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\beta_+,R,I_+} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

where $\zeta = |\alpha_-| + \eta + |\beta_+| \in \Gamma^{\mathrm{wr}}_R$. This is Definition 0.47; it is not the full eight-word associativity atlas and not a quotient-first product. The source/target anchor-memory row is separate: for LW, WL, WW, and LWL it records the ordered wrapped source anchors and wrapped target anchor on the unreduced correspondence stack, before the reduced E-quotient, as in Definition 0.48. The local/local row has no wrapped anchor memory. This row does not construct the determinant anchor, compute its translation weight, repair the $\chi = 0$ degeneracy, or define o^{λ}_R. The arity-three shadow is the eight-word two-step binary flag atlas

$$\{L, W\}^3 = \{LLL, LLW, LWL, LWW, WLL, WLW, WWL, WWW\},$$

which records binary associativity only; higher-coloured tree configurations live on a separate coloured tree category $\mathrm{Tree}^{\mathrm{col}}_R$, supplied as additional input (folklore, [70]).

Remark 4.6 *(Eight binary words versus colored trees).* The eight binary words of Definition 4.5 exhaust the arity-three two-step compositions $\{L, W\}^3$, but they do not catalogue higher-coloured tree configurations, units, symmetry/order conventions, refinement maps, descent, or overlaps. Those data are additional finite Hall input named at §6; asserting them prematurely confuses binary associativity with the full factorization-coherence structure ([70], correspondence formalism).

4.4 Wrapped anchor and translation linearization

Definition 4.7 *(Determinant anchor on the wrapped stratum).* Fix a finite HN height R, a wrapped colour $\eta \in \Gamma_R^{\mathrm{wr}}$, and a test scheme T. Let $\mathcal{F}_{\eta,T}$ be the retained universal perfect complex on $X \times T$ pulled back from the wrapped prequotient row, and write

$$\pi_{E,T} : X \times T \longrightarrow E \times T.$$

Let $\chi_\eta = \chi(\mathcal{F}_{\eta,t})$, constant on the retained colour row. The Knudsen–Mumford determinant functor [29, 61] gives a line bundle

$$\mathscr{D}_{\eta,T} := \det R\pi_{E,T,*}\mathcal{F}_{\eta,T}$$

on $E \times T$, functorial in T as a point of the relative Picard functor. Since E has genus 1, $\deg \det C = \chi(C)$ for a perfect complex C on E. Hence $\mathscr{D}_{\eta,T}$ has relative degree χ_η, and

$$\lambda^{\det}_{\eta,R}(\mathcal{F}_{\eta,T}) := \mathscr{D}_{\eta,T} \otimes \mathrm{pr}_E^* \mathcal{O}_E(-\chi_\eta \cdot 0_E)$$

where $\mathrm{pr}_E : E \times T \to E$ is projection. This line bundle has relative degree 0. Its class defines the determinant anchor

$$\lambda^{\det}_{\eta,R} : T \longrightarrow \mathrm{Pic}^0(E) \simeq E.$$

This constructs only the normalized determinant anchor. The next propositions compute its E-translation weight and isolate the $\chi_\eta = 0$ degeneracy. The finite-cover normalization to unit weight, extra anchor data, anchor residual definition and vanishing, quotient descent, and transition compatibility are separate rows.

The packet `certificates/hybrid/determinant_anchor_construction` verifies `DETERMINANT_ANCHOR_CONSTRUCTION_VE`
the determinant-of-cohomology line is functorial in families, the normalization has relative degree zero, and the resulting map lands in $\mathrm{Pic}^0(E) \simeq E$. It does not compute the translation weight, handle the $\chi = 0$ degeneracy, add extra anchor data, define o_R^λ, or prove quotient descent.

Proposition 4.8 *(Translation weight of the determinant anchor).* [proved] Let $a : T \to E$ be a section and let $\tau_a : E \times T \to E \times T$ be translation by a. Use the support-translation convention

$$a \cdot \mathcal{F}_{\eta,T} := (\mathrm{id}_S \times \tau_a)_* \mathcal{F}_{\eta,T}$$

on $X = S \times E$. If $\Gamma_a, \Gamma_0 \subset E \times T$ are the graphs of a and 0_E, then in the relative Picard functor

$$\lambda^{\det}_{\eta,R}(a \cdot \mathcal{F}_{\eta,T}) \simeq \lambda^{\det}_{\eta,R}(\mathcal{F}_{\eta,T}) \otimes \mathcal{O}_{E\times T}\big(\chi_\eta(\Gamma_a - \Gamma_0)\big).$$

Under $\mathrm{Pic}^0(E) \simeq E$, the induced translation law is

$$\lambda^{\det}_{\eta,R}(a \cdot \mathcal{F}_{\eta,T}) = \lambda^{\det}_{\eta,R}(\mathcal{F}_{\eta,T}) + \chi_\eta a.$$

Thus the E-translation weight of the determinant anchor is χ_η. Its kernel is $E[\chi_\eta]$ when $\chi_\eta \neq 0$, and all of E when $\chi_\eta = 0$.

Proof. Set $\mathscr{D}_{\eta,T} = \det R\pi_{E,T,*}\mathcal{F}_{\eta,T}$. The Cartesian square for translation on the elliptic factor gives, in the relative Picard group,

$$\det R\pi_{E,T,*}(a \cdot \mathcal{F}_{\eta,T}) \simeq \tau_{a,*}\mathscr{D}_{\eta,T} \simeq \tau_{-a}^{*}\mathscr{D}_{\eta,T}.$$

For a relative line bundle L of degree d on $E \times T$, translation acts trivially on the $\mathrm{Pic}^0(E)$-class and shifts the degree part by

$$\tau_{-a}^{*}L \otimes L^{-1} \simeq \mathcal{O}_{E\times T}\big(d(\Gamma_a - \Gamma_0)\big) \quad \text{in } \mathrm{Pic}^0(E \times T/T).$$

Taking $L = \mathscr{D}_{\eta,T}$ and $d = \chi_\eta$, and then tensoring both sides with the fixed normalization $\mathcal{O}_{E\times T}(-\chi_\eta\Gamma_0)$, gives the displayed identity. The sign is positive because the manuscript uses pushforward by the support translation $x \mapsto x + a$; the pullback convention would reverse the sign. The kernel statement is the kernel of the multiplication homomorphism $[\chi_\eta] : E \to E$. □

The packet `certificates/hybrid/determinant_anchor_translation_weight` verifies `DETERMINANT_ANCHOR_TRANSLATION_WEIGHT_VERIFIED`: for the support-translation convention $a \cdot \mathcal{F} = (\mathrm{id}_S \times \tau_a)_*\mathcal{F}$, the determinant anchor has translation weight χ_η, with restriction $h \mapsto \chi_\eta h$ on any finite stabilizer $H \subset E[N]$. It does not choose a coherent stabilizer linearization, make the weight one by finite cover, handle the $\chi_\eta = 0$ degeneracy, add extra anchor data, define o_R^λ, prove quotient descent, or prove transition compatibility.

Proposition 4.9 *(Zero-Euler determinant-anchor degeneracy).* [proved] Assume $\chi_\eta = 0$. Then for every section $a : T \to E$,

$$\lambda_{\eta,R}^{\det}(a \cdot \mathcal{F}_{\eta,T}) = \lambda_{\eta,R}^{\det}(\mathcal{F}_{\eta,T}) \quad \text{in } \mathrm{Pic}^0(E).$$

On a retained finite row where the E-translation action is free, a geometric point $a \neq 0$ gives distinct objects $\mathcal{F}_{\eta,t}$ and $a \cdot \mathcal{F}_{\eta,t}$ with the same determinant anchor. Hence the determinant anchor has full E-orbit kernel on the $\chi_\eta = 0$ branch and has one-dimensional separation defect. A final wrapped anchor on this branch must contain extra anchor data not determined by $\lambda_{\eta,R}^{\det}$.

Proof. Proposition 4.8 gives

$$\lambda_{\eta,R}^{\det}(a \cdot \mathcal{F}_{\eta,T}) = \lambda_{\eta,R}^{\det}(\mathcal{F}_{\eta,T}) + \chi_\eta a.$$

When $\chi_\eta = 0$ this is the displayed equality for all $a \in E$. The retained E-translation-rigidification datum is free on the retained row by Proposition 1.26; hence a nonzero geometric translation gives a different point in the prequotient before the reduced quotient is taken. The two points have the same determinant anchor. Therefore determinant-anchor-only data does not separate the E-orbit. The orbit is one-dimensional, so the class-level separation defect has rank 1. Extra anchor data is therefore a necessary input for the $\chi_\eta = 0$ branch, not a consequence of determinant of cohomology. □

The packet `certificates/hybrid/determinant_anchor_chi_zero_degeneracy` verifies `DETERMINANT_ANCHOR_CHI_ZERO_DEGENERACY_VERIFIED`: on the branch $\chi_\eta = 0$, the determinant anchor is constant on the full E-translation orbit, and freeness of the retained translation action gives a rank-one separation defect for determinant-anchor-only data. It records that extra anchor data is mandatory. It does not itself construct that data, define o_R^λ, prove losslessness, prove quotient descent, choose stabilizer linearizations, or prove transition compatibility.

Proposition 4.10 *(Extra anchor data from the degree-zero Jordan–Hoelder divisor).* [proved] Let V be a semistable vector bundle of rank r and degree 0 on E. Write the associated graded object of a Jordan–Hoelder filtration as

$$\mathrm{gr}^{\mathrm{JH}} V = L_1 \oplus \cdots \oplus L_r, \qquad L_i \in \mathrm{Pic}^0(E),$$

and set

$$\mathrm{sp}^{\mathrm{JH}}(V) = [L_1] + \cdots + [L_r] \in \mathrm{Sym}^r \mathrm{Pic}^0(E).$$

This divisor is independent of the Jordan–Hoelder filtration. Its Abel sum is $\det V$. For

$$V_0 = \mathcal{O}_E^{\oplus 2}, \qquad V_L = L \oplus L^{-1}, \qquad L \neq \mathcal{O}_E,$$

one has

$$\det V_0 \simeq \det V_L \simeq \mathcal{O}_E, \qquad \mathrm{sp}^{\mathrm{JH}}(V_0) = 2[\mathcal{O}_E], \qquad \mathrm{sp}^{\mathrm{JH}}(V_L) = [L] + [L^{-1}],$$

and the two degree-two divisors are unequal in $\mathrm{Sym}^2 \mathrm{Pic}^0(E)$.

Proof. By Atiyah's classification and Tu's S-equivalence description for semistable bundles on an elliptic curve [4, 104], every stable Jordan–Hoelder factor of a degree-zero semistable vector bundle on E is a degree-zero line bundle, and the unordered collection of these factors is the S-equivalence class of V. Hence $\mathrm{sp}^{\mathrm{JH}}(V)$ is independent of the chosen Jordan–Hoelder filtration. The determinant of the associated graded object is

$$\det \mathrm{gr}^{\mathrm{JH}}(V) = L_1 \otimes \cdots \otimes L_r,$$

which equals $\det V$; under the group law on $\mathrm{Pic}^0(E)$, this is precisely the Abel sum of the divisor $\mathrm{sp}^{\mathrm{JH}}(V)$. The displayed rank-two pair has determinant $\mathcal{O}_E$ on both sides. If $2[\mathcal{O}_E] = [L] + [L^{-1}]$ in $\mathrm{Sym}^2 \mathrm{Pic}^0(E)$, then the unordered multisets $\{\mathcal{O}_E, \mathcal{O}_E\}$ and $\{L, L^{-1}\}$ are equal, so $L \simeq \mathcal{O}_E$, contrary to the hypothesis. Thus the determinant anchor agrees while the Jordan–Hoelder divisor separates the two objects. □

The packet `certificates/hybrid/determinant_anchor_extra_anchor_data` verifies `DETERMINANT_ANCHOR_EXTRA_ANCHOR_DATA_VERIFIED`: on a row where the elliptic pushforward is a degree-zero semistable vector bundle, the additional anchor datum is the Jordan–Hoelder S-equivalence divisor in $\mathrm{Sym}^r \mathrm{Pic}^0(E)$. Its Abel sum is the determinant anchor, and it separates the determinant-zero rank-two test pair $\mathcal{O}_E^{\oplus 2}$ and $L \oplus L^{-1}$ with $L \neq \mathcal{O}_E$. It does not prove global anchor losslessness, prove $o_R^\lambda = 0$, prove quotient descent, choose stabilizer linearizations, populate every finite wrapped prequotient, or prove transition compatibility.

Lemma 4.11 *(Connected E-descent: separated cohomological classes).* For the connected component E° of the identity in the E-translation action,

$$H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \qquad |u_i| = 2.$$

Hence

$$H^1(BE; \mathbb{F}_2) = 0, \qquad H^2(BE; \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2.$$

There is no degree-one connected character; there is a degree-two connected gerbe class

$$\alpha^{E,\mathrm{free}} = a_1 u_1 + a_2 u_2 \in H^2(BE; \mathbb{F}_2).$$

(proved, [88].)

Proof. $E^{\mathrm{top}} \simeq T^2$ has classifying space $BE \simeq BT^2$ and $H^*(BT^2;\mathbb{F}_2) = \mathbb{F}_2[u_1,u_2]$ with both u_i of degree 2 (mod-2 Chern classes of the universal line bundles). H^1 vanishes; H^2 is the displayed two-dimensional $\mathbb{F}_2$ vector space. Vanishing of the connected gerbe class is the equation $a_1 = a_2 = 0$ on the actual Borel/edge class of the equivariant reduced determinant complex, not a consequence of $H^1(BE) = 0$ or ordinary translation invariance. □

Lemma 4.12 *(Finite stabilizer $E[N]$-descent: gerbe and linearization classes).* For the order-two finite subgroup $E[2] \cong (\mathbb{Z}/2)^2$,

$$H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1,x_2], \qquad |x_i| = 1.$$

The degree-two gerbe class is

$$\beta^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad (b_{20},b_{11},b_{02}) = (r_1,\ r_1+r_2+r_3,\ r_2),$$

detected on cyclic subgroup restrictions by joint reduction; the $r_i \in \mathbb{F}_2$ are the three nonzero point-stabilizer parities. After $\beta^{E,E[2]} = 0$ (i.e. $r_1 = r_2 = r_3 = 0$), a separate degree-one linearization class remains:

$$\lambda^{E,E[2]} = \lambda_1x_1 + \lambda_2x_2 \in H^1(BE[2];\mathbb{F}_2), \qquad \rho_1 = \lambda_1,\ \rho_2 = \lambda_2,\ \rho_3 = \lambda_1+\lambda_2.$$

The r_i are gerbe bits in H^2; the ρ_i are linearization bits in H^1; the two live in distinct cohomological degrees and remain independent. For even $N \geq 4$ with $2^a \parallel N,\ a \geq 2$,

$$H^*\big(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2\big) = \Lambda(x_1,x_2)\otimes\mathbb{F}_2[y_1,y_2],$$

$$\beta^{E,N} = A_1^{(N)}y_1 + A_{12}^{(N)}x_1x_2 + A_2^{(N)}y_2, \qquad \lambda^{E,N} = \lambda_1^{(N)}x_1 + \lambda_2^{(N)}x_2.$$

The mixed term $A_{12}^{(N)}$ is invisible to cyclic order-two restrictions; even $N \geq 4$ tests must be checked on the full two-primary generators. For odd N, the mod-2 orientation obstruction vanishes by transfer. (proved.)

Proof. $E[2] \cong (\mathbb{Z}/2)^2$ has $H^*(B(\mathbb{Z}/2)^2;\mathbb{F}_2) = \mathbb{F}_2[x_1,x_2]$ with $|x_i| = 1$. The class $\beta^{E,E[2]}$ is the finite-stabilizer Cech–Borel edge class, not the restriction of the connected BE-class $\alpha^{E,\mathrm{free}}$. Its restriction to the three non-zero cyclic subgroups gives the three parities r_i; inversion of this three-by-three system gives $(b_{20},b_{11},b_{02}) = (r_1, r_1+r_2+r_3, r_2)$. The degree-one linearization class $\lambda^{E,E[2]}$ is the residual character of the underlying line bundle after $\beta = 0$ is imposed; it lives in H^1 and is distinct from the gerbe $\beta \in H^2$ in degree. Even $N \geq 4$ introduces the polynomial generators $y_i \in H^2(B(\mathbb{Z}/2^a))$ and the mixed term x_1x_2, which on cyclic restrictions vanishes; full two-primary detection is needed. Odd N contributes only $\mathbb{F}_p$-cohomology with p odd, on which the mod-2 orientation has zero restriction by the transfer. □

Remark 4.13 *(Translation linearization criterion, not vanishing).* For a stabilizer $H \subset E[N]$ and a determinant anchor line bundle $\lambda(\mathcal{F})$, ordinary translation invariance $g^*\lambda \simeq \lambda$ in $\mathrm{Pic}^0(E)$ does not choose coherent isomorphisms $\phi_g : \lambda \to g^*\lambda$. After the square-root gerbe is killed, the choices form a character torsor in $H^1(BH;\mathbb{F}_2)$, and quotient orientation requires the linearization class

$$\lambda^H \in H^1(BH;\mathbb{F}_2)$$

to vanish. This is a *criterion* for descent on the wrapped stratum; it is not implied by $g^*[\lambda] = [\lambda]$ in $\mathrm{Pic}(E)$. Proposition 4.8 computes the translation law for $\lambda^{\det}$ and its weight $\chi(\mathcal{F})$. The finite-cover normalisation to unit weight when $\chi(\mathcal{F}) \neq 0$ and the coherent stabilizer linearization are later anchor rows. The $\chi(\mathcal{F}) = 0$ determinant-anchor degeneracy is Proposition 4.9; the degree-zero Jordan–Hoelder divisor of Proposition 4.10 separates the determinant-zero rank-two test pair. Full anchor losslessness, finite-stage population, quotient descent, transition compatibility, and the coherent stabilizer linearization remain later rows. Until those rows are supplied, this paragraph is only the descent criterion for a chosen linearized anchor. (proved as a criterion; determinant-anchor weight proved in Proposition 4.8.)

Remark 4.14 *(Mod-2 character does not control odd-order anchor characters).* The mod-2 orientation character does not by itself control odd-order anchor characters. For anchor descent on the wrapped stratum, also compute the actual stabilizer character on the chosen anchor trivialization, valued in $\widehat{H} = \mathrm{Hom}(H, \mathbb{C}^\times)$, or define the retained stabilizer as preserving that trivialization. (unverified; supplied as part of the finite Hall datum.)

4.5 Cusp polarization and chamber transport

Definition 4.15 *(Cusp polarization on Γ_{eff}).* The chamber convention $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ at the standard cusp $\mathfrak{c}_\infty$ determines the cusp-positive semigroup Γ_{eff} of Definition 2.1. The order on triples is lexicographic: $m > 0$ first because s is smallest; if $m = 0$ then $n > 0$ because q is next; if $m = n = 0$ then $l < 0$ because $|r|^{-1} \ll 1$. This is a chamber convention for the completed product, not a microscopic charge positivity statement.

Remark 4.16 *(Chamber transport).* The pairing $\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - ll'$ is the Gram pairing on triples; the order on Γ_{eff} is a Fock polarization at one cusp. An element of the duality group $\mathbf{Sp}_4(\mathbb{Z})$ transports both the chamber and the expansion to the corresponding product chamber at the transported cusp. The arithmetic group $\mathbf{Sp}_4(\mathbb{Z})$ is the symmetry of the genus-two chemical potential Z and of the invariant Gram plane on $\Lambda^{3,2}$; it is not asserted to be the full microscopic T-/U-duality group of the K3×T^2 compactification, which is larger (folklore, [31, 49]).

Remark 4.17 *(Chamber dependence of the wrapped stratum).* The cusp chamber chooses a positive direction on Γ_{eff}. At $\mathfrak{c}_\infty$ the wrapped stratum carries $\deg_E > 0$ branes; at a transported cusp the wrapped direction is the image of $\deg_E$ under the corresponding chamber transport. Mixed Hall correspondences are chamber-equivariant when the finite Hall stage R is compatible with the duality element; otherwise the correspondence stack is the original closed flag-Quot stack restricted to the new chamber.

Definition 4.18 *(Chamber-compatible Hall descent datum).* Let σ and σ' be two chosen stability data, and let $g \in \mathbf{Sp}_4(\mathbb{Z})$ transport the standard cusp chamber to another product chamber. A chamber-compatible Hall descent datum is a Kontsevich–Soibelman wall-crossing system

$$\mathsf{WC}_{g,R} : C_{\sigma,S,\leq R} \longrightarrow C_{\sigma',S,\leq R'}$$

on the retained finite Hall stages, with transition maps in R, which preserves the normal-ordered Gram map $\overline{\Pi}_X$, transports the local/wrapped split, the quotient-after-correspondence datum, the orientation lines, and the protected integration maps. The identity $\overline{\Pi}_X(\hat{c} \star \hat{c}') = \overline{\Pi}_X(\hat{c}) + \overline{\Pi}_X(\hat{c}')$ on $\widehat{\Gamma}_X$ is a chamber-independent lattice fact; the compact Hall category and its primitive bracket are not chamber-independent without $\mathsf{WC}_{g,R}$ (folklore/β, [64]).

5 Local protected observable algebra (sectorial source datum)

The compact realization route of Chapter 6 takes as supplied input a chain-level local observable system on opens $U \subset X = S \times E$. The system is graded by the algebraic/effective Hall support, factorizes on disjoint opens, and restricts on E-projection-finite supports to a chiral object on $\mathrm{Ran}(E)_{b=0}$. The data are external assumptions named in the literature and form the source datum for the compact realization.

Definition 5.1 *(Local protected observable algebra).* Let $U \subset X = S \times E$ be open. Let $\Gamma^{\mathrm{eff}}_{X,\sigma}(U) \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$ be the supplied algebraic/effective charge support. For $c \in \Gamma^{\mathrm{eff}}_{X,\sigma}(U)$, let $\mathfrak{M}_c(U)$ be the supplied derived moduli stack of compactly supported B-branes on U of Mukai charge c. The PTVV theorem supplies the ambient (-1)-shifted symplectic structure on the CY_3 perfect-complex moduli problem (proved, [95]). Let Φ_c denote the BBDJS perverse sheaf $P^{\bullet}_{\mathfrak{M}_c,s_c}$ on an oriented d-critical truncation, when such a d-critical structure and square-root orientation are supplied (proved, [12, Theorem 6.9]). Let o_c be the orientation line, a square root of the determinant line of the cotangent complex of $\mathfrak{M}_c(U)$, supplied as part of the Hall datum.

The *local protected observable algebra of X on U* is the source object

$$\mathcal{A}_X(U) := \bigoplus_{c \in \Gamma^{\mathrm{eff}}_{X,\sigma}(U)} H^{\mathrm{BM}}_*\big(\mathfrak{M}_c(U),\ \Phi_c \otimes o_c\big),$$

graded by the supplied algebraic/effective support, not by the full formal lattice Γ^{form}_X. For pairwise disjoint $U_1, \dots, U_k \subset X$, the proposed factorization map is

$$\mathcal{A}_X(U_1) \otimes \cdots \otimes \mathcal{A}_X(U_k) \longrightarrow \mathcal{A}_X(U_1 \sqcup \cdots \sqcup U_k),$$

defined chain-level by Massey–Saito Thom–Sebastiani for vanishing cycles

$$\Phi_{F_1 \oplus \cdots \oplus F_k} \simeq \Phi_{F_1} \boxtimes \cdots \boxtimes \Phi_{F_k}$$

on disjoint d-critical charts (proved, [78, 97]), together with the Ext-vanishing $\mathrm{Ext}^*(\mathcal{F}, \mathcal{G}) = 0$ for branes $\mathcal{F}, \mathcal{G}$ with disjoint supports (folklore).

Remark 5.2 *(External source datum).* The components are external assumptions: PTVV (-1)-shifted symplectic structure on derived moduli of B-branes (proved, [95]); BBDJS vanishing-cycle perverse sheaf and its Thom–Sebastiani isomorphism (proved, [12, 78, 97]); orientation lines and Joyce–Upmeier-compatible multiplicative orientation data (ambient technology proved in [56, Theorem 4.1]; the $K3 \times E$ quotient orientation is supplied separately). The factorization map relies on disjoint-supports Ext-vanishing for compactly supported branes (folklore; absent for non-compact supports, where one recovers chiral algebra structure, not commutative factorization). Compactness on $X = S \times E$ is open because the relevant moduli of branes are typically non-quasicompact in Λ_S-charge; the sectorial truncation of §4 is the finite-stage carrier, and the compact-source recognition datum $(S1)$–$(S10)$ is the remaining finite recognition problem. At finite stage the retained-moduli assertion is represented by the tables hn_type_bounds, semistable_substacks, derived_enhancements, universal_complexes, scalar_rigidifications, stratifications, extension_flag_stacks, cosection_atlas, transitions, and scalar_firewall, not by the scalar Hilbert-scheme test. At present the packet `certificates/moduli/k3e_finite_moduli` supplies only the obstruction ledger `MODULI_OBSTRUCTION_LEDGER_VERIFIED`, with `moduli_certification=false`.

6 The four obstructions and the comparison target

The hybrid carrier and the source datum are supplied as a finite holomorphic E_3-prefactorization datum and a $K3$-to-E specialization datum on $\mathrm{Ran}^{\mathrm{hyb}}(E)$. The cross-level comparison $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ holds on $K3 \times E$ relative to four retained data, with $(O2)$ represented by the retained wall-atlas datum of Appendix 7 (Theorems 7.7, 7.13, 7.14, 7.19). They are the orientation and Pfaffian data feeding Chapter 6.

(D0) D0-degeneration limit. The retained D0-HN datum of Definition 7.6 supplies the flat D0-degeneration family, additive K3 semiregularity cosections, reduced vanishing-cycle and orientation specializations, Pfaffian-line extension, and strict HN transitions (criterion Theorem 7.7).

(O1) Strong reduced orientation. The K3×E-quotient self-Ext frame bundle carries a strong reduced orientation on the wrapped stratum, descending the BBDJS orientation line o_c across the E-quotient relative to $(O1_{\mathrm{quot}})$ (Theorem 7.13).

(O1$^+$) Weyl-equivariant transport. The orientation transports $W^{(2)}(\Lambda_{II}^{2,1})$-equivariantly along reflections relative to the chosen Weyl lifts (Theorem 7.14); after the type-II wall atlas supplies the rank-one local signs, the transported character agrees with ν_{Δ_5} on the BKM Weyl group.

(O2) Local Pfaffian wall normal form. On type-II walls of $\Lambda_{II}^{2,1}$, the local Pfaffian wall normal form is the rank-one crossing $[\mathbb{R} \xrightarrow{u} \mathbb{R}]$; the retained $(O2_{\mathrm{atlas}})$ datum transports these local signs over the half-Hilbert orbit, whose sign support has cardinality $4096 = 2^{12}$ (Theorem 7.19).

The protected scalar shadow on the closed K3×E→point cobordism is the OP/DT identity

$$Z_{\mathrm{DT}}^X = Z_{\mathrm{OP}}^X(P_{\mathrm{OP}}, Q_{\mathrm{OP}}, T_{\mathrm{OP}}) = -4096\,\Delta_5(Z)^{-2} \qquad \text{(proved, [91, 92, 101]).}$$

Equivalently $\chi_{10}^{\mathrm{OP}} = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2$ and $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$; the factor $4096 = 64^2$ is a theta-leading normalization conversion and the minus sign is the OP scalar convention. The orientation-forgetful shadow

$$Z_{\mathrm{BPS}}^{K3\times E} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$$

is the level-n protected trace of the retained $K3 \times E$ data on the closed cobordism in the unnormalized convention $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$; the inverse-square scalar cannot recover the orientation character that distinguishes Δ_5 from $-\Delta_5$. A $3d$ gravitational path-integral realization requires the Hall–Borcherds residual of [73] and is not claimed. The conditional cross-level identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ supplies the orientation/Pfaffian datum under (D0), (O1), (O1$^+$), (O2).

Remark 6.1 *(Compatibility of the κ-tuple).* The κ-tuple $(0, 0, 3, 5, 24)$ is consistent across [70] (Vol III), the Hall–Borcherds residual of [73] (Vol II), and the Beilinson–Drinfeld factorization framework of [7] via Vol I [72]. The local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ + brane completion of [71] contributes the holomorphic de Rham obstruction class on the global target; the identity $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ does not promote to the global target via formal Hamiltonian BF, and the Hall–Borcherds residual is the remaining chain-level bridge.

7 Notation and fixed data

$X = S \times E$**.** Smooth complex projective trivial-canonical threefold; S is a K3 surface, E a smooth elliptic curve. $\dim_{\mathbb{C}} X = 3$ and $h^{1,0}(X) = 1$.

Λ_S**.** Mukai lattice; even unimodular signature $(4, 20)$. $\Lambda_S \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2} \cong II_{4,20}$.

$\Gamma_X^{\mathrm{form}}, \Gamma_X^{\mathrm{phys}}$**.** Formal full-Mukai dyonic lattice $\Lambda_S \oplus \Lambda_S$. The superscript "phys" denotes this formal even branch, not the $\mathcal{N} = 4$ charge lattice.

$\Gamma_X^{\mathrm{alg}}, \Gamma_{X,\sigma}^{\mathrm{eff}}$. Algebraic K-class lattice $\subset \Gamma_X^{\mathrm{form}}$; chosen σ-effective semigroup $\subset \Gamma_X^{\mathrm{alg}}$.
$\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$. Gram-index lattice, (n, l, m).
Π_X. Quadratic Gram map, $\Pi_X(Q, P) = ((Q, Q)/2,\ (Q, P),\ (P, P)/2)$.
B. Symmetric bilinear cocycle, $B(c, c') = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P')$; satisfies $B = -\partial \Pi_X$ and $B(c, c) = 2\Pi_X(c)$.
$\widehat{\Gamma}_X$. Normal-ordered extension $\Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}$; abelian under $\star$.
$\overline{\Pi}_X$. Additive normal-ordered Gram homomorphism $\overline{\Pi}_X(c, T) = \Pi_X(c) - T$.
$\mathrm{Ran}^{\mathrm{hyb}}(E)$. Finite-stage hybrid prestack on the elliptic factor, with local $b = 0$ subprestack $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ and wrapped $b > 0$ subprestack $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$.
$\mathcal{D}_X = \Delta_5$, $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$. Three named weight-five objects: automorphic section, BKM denominator, compact Pfaffian (View 1, 2, 3).

The fixed data are: (a) the formal Mukai sector $\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$ and its three-tier support hierarchy; (b) the bilinear polarization $B = -\partial \Pi_X$ as the full structural content of the Gram defect; (c) the normal-ordered extension $\widehat{\Gamma}_X$ with additive homomorphism $\overline{\Pi}_X$, fixed by the fixed-lift raw-pushforward obstruction theorem; (d) the hybrid Ran carrier $\mathrm{Ran}^{\mathrm{hyb}}(E)$, fixed by the projection-locality obstruction; (e) the connected/finite split of the E-descent classes, with the linearization criterion separated from the gerbe class; (f) the local protected observable algebra source datum, taken as an external assumption; (g) the four obstructions (O0), (O1), (O1$^+$), (O2) conditioning the cross-level identity 2$\leftrightarrow$3.

Chapter 6 constructs from these fixed data the Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}} = \lim_R(\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P$

Chapter 3

View I: the Borcherds–Gritsenko–Nikulin section

1 The pentadic level and the Beilinson cut

The automorphic reading of Δ_5 is the first theorem-level reading. It is a holomorphic section of the weight-5 automorphic line on the genus-two Siegel modular space $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$: the Borcherds–Gritsenko–Nikulin lift of the weight-zero index-one weak Jacobi form $\phi_{0,1}$, normalized at the isotropic cusp $\mathfrak{c}_\infty$ by the theta-constant leading coefficient 64. The $\mathbf{Sp}_4(\mathbb{Z})$-equivariant data of View I are the multiplier system ν_{Δ_5}, the Borcherds chamber, the divisor support, and the weight.

1.0.0.1 Three names. The handle "Δ_5" is shared among three structurally distinct objects and must be tagged on first use:

- *the Borcherds–Gritsenko–Nikulin section* $\mathcal{D}_X = \Delta_5$ of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$, View I;
- *the BKM denominator* $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, View II, the Borcherds–Kac–Moody denominator algebra;
- *the protected Pfaffian* $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$, View III, constructed in the Dirac–Igusa chapter as a section on the Pfaffian line of the cosection-reduced compact Hall datum after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and finite geometric Pfaffian datum (P_{fin}) of Appendix 7.

The bare label Δ_5 never substitutes for any of the three: it names the automorphic section unless a reading is specified, while $\mathcal{D}_X$, $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, and $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ live at distinct categorical-dimensional levels.

1.0.0.2 Beilinson cut: scope of View I. View I is the scalar modular shadow. In the reconstitution order "primitive objects first, cross-level identities second, scalar shadows last" View I is a single holomorphic section of an automorphic line on a Shimura variety, with its multiplier system and its weight. No operator-level datum is asserted at View I. In particular, the equality $\mathcal{D}_X = \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of View I and View III is the Pfaffian–Dirac theorem of Chapter 7, relative to the retained data recorded in Appendix 7; it is not asserted in this chapter. The shadows are Δ_5, χ_{10}, the level-n BPS partition function $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$, and the Oberdieck–Pixton scalar branch $Z_{\mathrm{OP}}^{X} = -4096\,\Delta_5^{-2}$; none of them is a Hilbert space, a Hall pairing, an orientation line, or an operator product.

2 The genus-two Siegel space and the Borcherds chamber

Let $\mathbb{H}_2$ be the genus-two Siegel upper half space. An element $Z \in \mathbb{H}_2$ is a complex symmetric 2×2 matrix with positive-definite imaginary part,

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}, \qquad \Im Z > 0.$$

Set Fourier variables

$$q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$

The fixed cusp polarization at the rational isotropic boundary $\mathfrak{c}_\infty$ of $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ — the standard type-II cusp polarization — is the ordered limit

$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1.$$

Equivalently, $\Im z_3$ goes to $+\infty$ faster than $\Im z_1$ goes to $+\infty$, and $\Im z_2$ is bounded. This polarization selects a Weyl chamber on the $O(3,2)$ symmetric domain through the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge of Chapter 5, and hence selects a positive cone, the effective semigroup

$$\Gamma_{\text{eff}} = \{(n,l,m) \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n,l,0) \mid n > 0,\ l \in \mathbb{Z}\} \cup \{(0,l,0) \mid l < 0\} \subset \mathbb{Z}^3,$$

and a corresponding completed Fourier expansion ring. In this completion every Fourier coefficient of every formal product receives only finitely many contributions: the chamber walls are the inequalities defining the ordering, and the chamber-completed product is the only product considered.

2.0.0.1 Choices fixed once and for all. The cusp $\mathfrak{c}_\infty$, the Fourier variables (q,r,s), the polarization $(0 < |s| \ll |q| \ll |r|^{-1} \ll 1)$, and the effective semigroup $\Gamma_{\text{eff}} \subset \mathbb{Z}^3$ are fixed. Every View-I product expansion is the product expansion in the cusp-completed semigroup algebra of Γ_{eff}. The cusp expansion at any other rational isotropic cusp is obtained by an $\mathbf{Sp}_4(\mathbb{Z})$-translate of these data.

3 The weak Jacobi input and the K3 elliptic genus

The Borcherds input is the *weight-zero index-one weak Jacobi form* $\phi_{0,1} \in J^{\text{weak}}_{0,1}$ of Eichler–Zagier [34, §2.2], normalized so that the K3 elliptic genus is

$$Z_{K3}(\tau,z) = 2\,\phi_{0,1}(\tau,z).$$

Write the Fourier expansion as

$$\phi_{0,1}(\tau,z) = \sum_{n\geq 0,\ l\in\mathbb{Z}} f(n,l)\,q^n r^l, \qquad q = e^{2\pi i\tau}, \quad r = e^{2\pi i z}.$$

The first Fourier coefficients are

$$\begin{aligned}
&f(0,-1) = 1, && f(0,0) = 10, && f(0,1) = 1, \\
&f(1,-2) = 10, && f(1,-1) = -64, && f(1,0) = 108, && f(1,1) = -64, && f(1,2) = 10, \\
&f(2,-3) = 1, && f(2,-2) = 108, && f(2,-1) = -513, && f(2,0) = 808,
\end{aligned} \tag{3.1}$$

extending in the standard way by hyperbolic symmetry $f(n,l) = f(n,-l)$ and Eichler–Zagier support $f(n,l) = 0$ for $4n - l^2 < -1$ [34, §2.2]. The values in (3.1) are obtained by the exact rational expansion of the standard theta-quotient expression $\phi_{0,1} = 4\sum_{i=2,3,4} \theta_i(\tau,z)^2/\theta_i(\tau,0)^2$ [*computed*; cross-checked against [34, Theorem 9.2]].

3.0.0.1 Borcherds-weight handle. For a fixed Borcherds input Φ, the weight identity $\kappa_{\mathrm{BKM}}(\Phi) = c_\Phi(0)/2$ (Borcherds 1995 [9, Theorem 10.1]; Gritsenko 2018 [43]) gives, on the row $(t, N) = (1, 1)$ of the Cléry–Gritsenko catalogue [44, Theorem 1.4],

$$c_1(0) := f(0,0) = 10, \qquad \kappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = 5,$$

to be matched in Proposition 6.1 below with the geometric weight of the Borcherds lift. The notation κ_{BKM} is the BKM cusp-form-weight entry of the κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ of the $K3 \times E$ target. The bare symbol κ is not a scalar invariant; every entry carries its semantic subscript; the additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ does not define the BKM coordinate. The BKM coordinate is $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5$.

4 The theta-constant normalization

The theta-constant Igusa form Δ_5 is fixed by its leading Fourier monomial at $\mathfrak{c}_\infty$. Writing the theta-constant representation of Igusa [51, §5], [52, §2]

$$\Delta_5(Z) = \prod_{\substack{a,b \in (\mathbb{Z}/2\mathbb{Z})^2 \\ {}^t ab \equiv 0 \bmod 2}} \nu_{a,b}(Z), \qquad \nu_{a,b}(Z) = \sum_{l \in \mathbb{Z}^2} \exp\!\Big(\pi i \big(z[l + \tfrac{1}{2}a] + {}^t bl\big)\Big),$$

the four characteristics with $a = (0,0)$ start with the constant 1, the two factors with $a = (1,0)$ start with $2q^{1/8}$, the two factors with $a = (0,1)$ start with $2s^{1/8}$, and the two factors with $a = (1,1)$ start with $2q^{1/8}r^{1/4}s^{1/8}$; multiplying gives

Proposition 4.1 *(Theta-constant leading coefficient).*

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 \;=\; 2^6 \;=\; 64.$$

[*proved*; Igusa 1962 [51, Theorem 3], Gritsenko 1999 [46, Theorem 4.1]; the rational theta-product expansion gives the same leading coefficient.]

Definition 4.2 *(Monic Borcherds–Gritsenko form).* The *monic genus-two Borcherds–Gritsenko form* is

$$D_5(Z) \;:=\; 64^{-1}\Delta_5(Z).$$

By Proposition 4.1, $D_5 = q^{1/2}r^{1/2}s^{1/2} + \ldots$ at $\mathfrak{c}_\infty$.

This is the normalization used by Gritsenko–Nikulin [48, Theorem 2.1, (2.15)–(2.16)]: their Borcherds product is the monic form D_5; the theta-constant convention restores the prefactor 64.

5 The Borcherds–Gritsenko–Nikulin section

5.1 The product expansion

Borcherds's automorphic product theorem [9, Theorem 10.1], specialized to the genus-two Siegel modular variety $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ by Gritsenko–Nikulin [48, Theorem 2.1, Example 2.4] via the exceptional exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ of Chapter 5, lifts the weak Jacobi form $\phi_{0,1}$ to a meromorphic Siegel modular form of weight $f(0,0)/2 = 5$ and multiplier system ν_{Δ_5} with infinite product expansion in the cusp chamber.

THEOREM 5.1 *(Borcherds–Gritsenko–Nikulin lift, monic form).* In the cusp chamber $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$,

$$D_5(Z) \;=\; q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

the product converging absolutely on a neighborhood of $\mathfrak{c}_\infty$ and continuing meromorphically to all of $\mathbb{H}_2$ as a holomorphic Siegel modular form of weight 5 on $\mathbf{Sp}_4(\mathbb{Z})$ with character ν_{Δ_5} of order two.

[*proved*; Borcherds 1995 [9, Theorem 10.1], Gritsenko–Nikulin 1998 [48, Theorem 2.1, (2.15)–(2.16)], Gritsenko 1999 [46, Theorem 4.1].]

COROLLARY 5.2 *(Theta-normalized section).*

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The exponent $f(nm,l)$ of the (n,l,m)-factor depends only on the Hecke–Verlinde–DMVV product nm and on the Jacobi index l, reflecting the second-quantized structure of the Borcherds lift, whose combinatorial origin is the Dijkgraaf–Moore–Verlinde–Verlinde elliptic genus of symmetric products [30].

5.2 The View I datum

Definition 5.3 *(The Borcherds–Gritsenko–Nikulin section).* The *Borcherds–Gritsenko–Nikulin section* of the weight-5 automorphic line is the View I representative

$$\mathcal{D}_X \;:=\; \Delta_5$$

on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$, with Borcherds product expansion at $\mathfrak{c}_\infty$ given by Corollary 5.2.

Remark 5.4 *($\mathcal{D}_X$ is the View I name).* The handle $\mathcal{D}_X$ is the View I name of the section. It coincides as a holomorphic Siegel modular form with the theta-constant Igusa cusp form Δ_5; the renaming records the role: View I scalar shadow, the determinant of the virtual K3 index in $K_0(\mathrm{sVect}_{\mathrm{fd}})$ (the K_0-determinant identity proved at Proposition 7.2; the operator Pfaffian datum belongs to View III), and the input section of the Pfaffian–Dirac theorem of Chapter 7. The compact operator-level datum $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of View III is asserted equal to $\mathcal{D}_X$ under $(D_0), (O_1), (O_1)^+, (O_2)$; without these obstruction trivializations, only $\mathcal{D}_X$ is rigorously available.

6 WEIGHT, MULTIPLIER, AND DIVISOR

6.1 The Borcherds weight

PROPOSITION 6.1 *(Borcherds–Gritsenko weight identity at $N = 1$).*

$$\mathrm{wt}(\mathcal{D}_X) \;=\; \frac{f(0,0)}{2} \;=\; 5 \;=\; \frac{c_1(0)}{2} \;=\; \kappa_{\mathrm{BKM}}(\Delta_5).$$

The weight is integral; no metaplectic cover enters the weight factor.

[*proved*; Borcherds 1995 [9, Theorem 10.1, weight formula]; Gritsenko–Nikulin 1998 [48, Example 2.4]; Gritsenko 1999 [46, Theorem 4.1, weight]; the Borcherds weight identity $\kappa_{\mathrm{BKM}}(\Phi) = c_\Phi(0)/2$ for a fixed input form and lattice; Cléry–Gritsenko [44], Vol III.]

The row $(t, N; \mathrm{wt}, c) = (1, 1; 5, 10)$ is the leading F_j-entry of the eight-row Cléry–Gritsenko catalogue [44, Theorem 1.4]. In the F_j-order the remaining rows carry weights $(2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1)$ and constants $(4, 2, 1, 6, 4, 3, 2)$; in the input-pair order $(2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2)$ the same data read $(2, 3, 1, 2, \frac{1}{2}, \frac{3}{2}, 1)$ and $(4, 6, 2, 4, 1, 3, 2)$. Each row satisfies $\kappa_{\mathrm{BKM}}(F_j) = c_j(0)/2$. The half-integer rows are realized by $v_\eta^3 \times v_H$ and $\chi_4^{(4)} \times v_H$, respectively. The Vol III universal Borcherds-weight identity replaces the additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$: the κ-tuple records separate invariants, and the $K3 \times E$ entry is precisely $\kappa_{\mathrm{BKM}} = 5$.

6.2 *The Maass multiplier system*

The character $\nu_{\Delta_5} \colon \mathbf{Sp}_4(\mathbb{Z}) \to \mathbb{C}^\times$ of Maass [76] has order two, $\nu_{\Delta_5}^2 = 1$, and is determined on the standard generators of $\mathbf{Sp}_4(\mathbb{Z})$ by

$$\nu_{\Delta_5}\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix} = 1, \qquad \nu_{\Delta_5}\begin{pmatrix} \mathbf{I}_2 & B \\ 0 & \mathbf{I}_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3},$$

$$\nu_{\Delta_5}\begin{pmatrix} {}^t A^{-1} & 0 \\ 0 & A \end{pmatrix} = (-1)^{(1+a_1+a_4)(1+a_2+a_3)+a_1a_4},$$

with $B = \left(\begin{smallmatrix} b_1 & b_2 \\ b_2 & b_3 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} a_1 & a_2 \\ a_3 & a_4 \end{smallmatrix}\right)$ [76]. The square $\Delta_5^2 = \Delta_{10}$ is scalar. The multiplier records the obstruction to identifying Δ_5 with an ordinary modular form on $\mathbf{Sp}_4(\mathbb{Z})$: Δ_5 is a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$, where $\mathcal{L}$ is the Hodge automorphic line and ν_{Δ_5} is the order-two character; the section Δ_5 does not descend to a section of the bare Hodge line $\mathcal{L}^5$, but its square Δ_5^2 is a scalar section of $\mathcal{L}^{10}$. The finite automorphic packet

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certificates/automorphic/delta5_maass_character
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records the six chamber generator values, the relations $\nu_{\Delta_5}|_{W^{(2)}} = \det$, $\nu_{\Delta_5}|_{\mathrm{Aut}(\mathcal{P}_{II})} = 1$, and $\nu_{\Delta_5}^2 = 1$, and the line row $\Delta_5 \in H^0(\mathbb{H}_2, \mathcal{L}^5 \otimes \nu_{\Delta_5})$. The same packet excludes using ν_{Δ_5} as a Pfaffian orientation line or as an OP scalar branch.

6.3 *The diagonal divisor*

Let

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \left(\begin{smallmatrix} a & 0 \\ 0 & b \end{smallmatrix}\right) \,\middle|\, a, b \in \mathbb{H}_1 \right\} \subset \mathbb{H}_2$$

be the locus of diagonal genus-two periods. The Cléry–Gritsenko classification [44] records

$$\operatorname{supp}\operatorname{div}(\Delta_5) = \mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \mathrm{mult}_{\mathcal{H}_{\mathrm{diag}}}(\Delta_5) = 1,$$

so the divisor of Δ_5 is the simple paramodular Humbert divisor. In the cusp-chamber language, the type-II reflective walls $\mathcal{H}_{\delta_i}$, $i = 1, 2, 3$, of the lattice $\Lambda_{II}^{2,1}$ (the type-II sublattice of the Borcherds even lattice $\Lambda^{2,1}$) are the components of $\mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}$ that meet the chosen fundamental Weyl chamber $\mathcal{P}_{II}$ (Chapter 4); the corresponding simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ have Gram pairings $(\delta_i, \delta_i) = 2$ and $(\delta_i, \delta_j) = -2$ for $i \neq j$ [48, Example 2.4], and the corresponding Borcherds divisor orders are

$$d_{\delta_1}^{\mathrm{aut}} = f(0, 1) = 1, \qquad d_{\delta_2}^{\mathrm{aut}} = f(0, 1) = 1, \qquad d_{\delta_3}^{\mathrm{aut}} = f(0, -1) = 1.$$

[*proved*; Cléry–Gritsenko 2011 [44, Theorem 1.4]; Gritsenko–Nikulin 1998 [48, Example 2.4]; the Gram computation is the one displayed in Appendix 1.2.] The finite automorphic divisor packet

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records the three type-II support representatives, the simple order-one rows for Δ_5, the pole-order-two row for Δ_5^{-2}, and a separate support-versus-multiplicity row. It is not a Pfaffian wall chart, an $O2$ atlas, or a mirror-period discriminant.

6.4 *Maass-character values on the type-II walls*

LEMMA 6.2 *(Maass-character values on the type-II reflections).* With the simple type-II roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ and chamber-permuting type-I roots $\alpha_1 = f_{-2} - f_2$, $\alpha_2 = f_2 - f_3$, $\alpha_3 = f_{-2} - f_3$,

$$\nu_{\Delta_5}(s_{\delta_i}) \;=\; -1, \qquad \nu_{\Delta_5}(s_{\alpha_i}) \;=\; +1, \qquad i = 1, 2, 3.$$

Equivalently, for chosen $\mathbf{Sp}_4$-lifts and the weight-five slash action without character,

$$(\Delta_5|_5 s_{\delta_i})(Z) = -\Delta_5(Z), \qquad (\Delta_5|_5 s_{\alpha_i})(Z) = +\Delta_5(Z).$$

[*proved*; Maass [76]; Gritsenko–Nikulin [46, §3]; Gritsenko [45]. The type-II values follow from $d^{\mathrm{aut}}_{\delta_i} = 1$ and the chamber-wall cocycle calculation of Lemma 7.2, which fixes $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det$ on the three type-II generators; the type-I values follow from $d^{\mathrm{aut}}_{\alpha_i} = 0$ (vanishing of $f(0, \pm 2)$ by the support of $\phi_{0,1}$ at $q = 0$).]

The chamber-permuting reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$ act trivially on Δ_5 because the corresponding type-I roots α_i are not $\mathbf{Sp}_4$-images of timelike short roots in the cusp chamber: type-I square-2 vectors are outside the type-II sublattice $\Lambda^{2,1}_{II}$ on which the Borcherds product expansion of Δ_5 is supported, so their walls do not appear in $\mathrm{div}(\Delta_5)$ at all [34, §2.2].

7 IDENTIFICATION WITH THE VIRTUAL K_0-DETERMINANT

The Fock superdeterminant of the Borcherds-graded super vector space $\mathcal{V} = \bigoplus_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \mathbb{U}_{nm,l}$ — the K_0-class of the second-quantized Hecke–DMVV half of the K3 elliptic genus, with the cusp/Weyl boundary factors at $m = 0$ carrying the input multiplicities $f(0, l)$ and the bulk factors at $m > 0$ carrying $f(nm, l)$ on Γ_{eff} (Definition 2.1) — is

LEMMA 7.1 *(K_0-Fock superdeterminant).*

$$\mathrm{sdet}_{\mathcal{V}}(1 - x) \;=\; \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}, \qquad x^{(n,l,m)} = q^n r^l s^m.$$

[*proved*; the Fock-trace lemma is the three-way comparison $\mathrm{sdet}_{\mathcal{V}}(1 - x) = \prod(1 - x^{\gamma})^{\mathrm{sdim}\,\mathcal{V}_\gamma}$ under the Borcherds-graded second-quantization; the automorphic–orthogonal–BKM identification sits in Theorem 1.20 of Chapter 5.]

PROPOSITION 7.2 *(BGN identification of the virtual determinant).* With $v_0(Z) := 64\, q^{1/2} r^{1/2} s^{1/2}$ the cusp/Weyl monomial of Chapter 2, the normalized K_0-determinant agrees with the View I section,

$$\mathcal{D}_X(Z) \;:=\; v_0(Z)\ \mathrm{sdet}_{\mathcal{V}}(1 - x) \;=\; \Delta_5(Z),$$

in the cusp-chamber expansion at $\mathfrak{c}_\infty$ and hence as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

Proof. Combine Theorem 5.1, the theta-constant prefactor Proposition 4.1, Lemma 7.1, and Cohen's q-expansion principle for genus-two Siegel modular forms with half-integral character: a meromorphic Siegel form is determined by its expansion at one isotropic cusp. Both sides have the same Borcherds product expansion at $\mathfrak{c}_\infty$ by direct comparison of exponents, and $\mathcal{D}_X$ is holomorphic of weight 5 by Theorem 5.1. □

[*proved*; Borcherds 1995 [9, Theorem 10.1], Gritsenko–Nikulin 1998 [48, Theorem 2.1], Igusa 1962/1964 [51, 52].]

Remark 7.3 *(Scope of the BGN identification).* Proposition 7.2 is an equality of holomorphic sections of an automorphic line bundle. The carrier $\mathcal{V}$ is a class in $K_0(\Gamma_{\mathrm{eff}}$-graded super vector spaces), with no vertex, chiral, or holomorphic E_3-factorization structure. The passage from the K_0-Fock determinant to a section of an oriented Pfaffian line over the cosection-reduced compact Hall datum, and the identification of that section with $\mathcal{D}_X$, is the Pfaffian–Dirac theorem of Chapter 7, relative to the retained data recorded in Appendix 7. View I supplies only the automorphic determinant section.

8 The squared determinant section and OP normalization

The Maass character has order two, so the square $\Delta_5^2 = \Delta_{10}$ is a scalar weight-10 Igusa form on $\mathbf{Sp}_4(\mathbb{Z})$, with no multiplier system.

Proposition 8.1 *(The square is the Oberdieck–Pandharipande χ_{10}).*

$$\Delta_5^2 \;=\; \Delta_{10} \;=\; \Phi_{10}^{\mathrm{un}} \;=\; \chi_{10}.$$

[*proved*; Igusa [51, 52]; the symbol Φ_{10}^{un} is Borcherds's name for the squared theta-constant Igusa form, χ_{10} is the Oberdieck–Pandharipande [91, 92] name; both equal Δ_5^2. The naming is $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$, with the Oberdieck–Pixton OP normalization $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ the monic-on-D_5 variant.]

8.0.0.1 The OP scalar branch. With the OP normalization $\chi_{10}^{\mathrm{OP}} = D_5^2$, the Oberdieck–Pixton scalar partition function on $K3 \times E$ is [92, (0.4) and Theorem 1]

$$Z_{\mathrm{OP}}^X \;=\; -\,(\chi_{10}^{\mathrm{OP}})^{-1} \;=\; -\,4096\,\Delta_5^{-2}.$$

The factor $-4096 = -(64)^2$ records the squared theta-constant leading coefficient and the OP scalar sign convention. At View I this is the closed protected trace Δ_5^{-2}; the orientation character which distinguishes the Pfaffian section Δ_5 from its negative is lost under squaring. The operator datum entering that trace is defined at the Pfaffian and orientation level relative to the retained data of Appendix 7.

8.0.0.2 Three names for the squared form, distinguished. The notation is fixed as follows: $\Delta_5^2 = \Delta_{10}$ is the Igusa convention (theta-constant square); $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ is the Borcherds–Eisenstein convention; $\chi_{10} = \Delta_5^2$ is the Oberdieck–Pandharipande convention; and $\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2 = D_5^2$ is the Oberdieck–Pixton variant in which D_5 is the monic Borcherds form (Definition 4.2). The finite packet `certificates/normalizations/delta5_theta_leading` records these as separate convention rows and relation rows; equality of the unscaled sections is not used to identify the OP monic form or either inverse scalar branch.

9 The View II shadow: the $2Z$ denominator

The Gritsenko–Nikulin denominator algebra of the Borcherds–Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ arises as the View II shadow of Δ_5 under the doubling $Z \mapsto 2Z$:

Proposition 9.1 *(Gritsenko–Nikulin doubled denominator).*

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; D_5(2Z) \;=\; 64^{-1}\Delta_5(2Z).$$

[*proved*; Gritsenko–Nikulin 1998 [48, Theorem 2.1, (2.15)–(2.16); §5]; Gritsenko 1999 [46, Theorem 4.1].] The doubling records the embedding $\Lambda^{2,1}_{II} = \alpha(\Gamma_{\mathrm{ind}}) \subset \Lambda^{2,1}$ of index 4. Its type-II walls are primitive roots of square 2, and the substitution $Z \mapsto 2Z$ replaces $\exp(-\pi i(\rho, Z))$ by $\exp(-2\pi i(\rho, Z))$ in the Gritsenko–Nikulin denominator normalization [48, Theorem 2.1, (2.15)]. At View I this is a pointer; the Cartan, Chevalley, real-and-imaginary roots, and Weyl–Kac–Borcherds denominator identity of $\mathfrak{g}_{\Delta_5}$ sit in Chapter 4.

10 The $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge to View V

The Borcherds construction is naturally written on the orthogonal side: Δ_5 is the regularized theta lift of $\phi^{\mathrm{Weil}}_{0,1}$ on $\Lambda^{3,2}$, pulled back to $\mathbf{Sp}_4$ through the exterior-square isogeny

$$\mathbf{PSp}_4(\mathbb{R}) \;\simeq\; \mathbf{SO}(3,2)_+.$$

The corresponding spin cover is the real group with Lie algebra $\mathfrak{sp}_4(\mathbb{R})$. At the symbolic level View I (Δ_5 on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$) and View V ($\Delta_5$ as the singular theta lift on $\Gamma_{3,2}\backslash\mathbb{H}^{\mathbf{IV}}_+$) are the two sides of this bridge: the weight-5 automorphic line on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ is the pullback of the weight-5 orthogonal modular line on $\Gamma_{3,2}\backslash\mathbb{H}^{\mathbf{IV}}_+$, and the multiplier ν_{Δ_5} restricts to the orthogonal Maass character. The full theta-lift construction — Borcherds' *singular theta correspondence* [9], [10] — produces the orthogonal modular form; the automorphic section in View I is its pullback to the $\mathbf{Sp}_4$-side of the bridge.

11 Summary of View I

Theorem 11.1 *(View I datum).* The genus-two Igusa cusp form Δ_5 is the Borcherds–Gritsenko–Nikulin section

$$\mathcal{D}_X \;:=\; \Delta_5$$

of the weight-5 automorphic line $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ over $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$:

(i) At the rational isotropic cusp $\mathfrak{c}_\infty$ with cusp polarization $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$, $\mathcal{D}_X$ admits the Borcherds product expansion

$$\mathcal{D}_X(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

with leading theta-constant coefficient $[q^{1/2}r^{1/2}s^{1/2}]\mathcal{D}_X = 64$, and exponents $f(nm,l)$ the Fourier coefficients of the K3 weight-zero index-one weak Jacobi form $\phi_{0,1} = \frac{1}{2}Z_{K3}$ (Eichler–Zagier [34, §2.2]).

(ii) $\mathcal{D}_X$ has weight $\mathrm{wt}(\mathcal{D}_X) = f(0,0)/2 = 5 = c_1(0)/2 = \kappa_{\mathrm{BKM}}(\Delta_5)$, the leading entry $(t, N; \mathrm{wt}, c) = (1, 1; 5, 10)$ of the Cléry–Gritsenko catalogue [44, Theorem 1.4]. The multiplier is the Maass character ν_{Δ_5} of order two.

(iii) The divisor of $\mathcal{D}_X$ is the simple paramodular Humbert divisor $\mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}$; the type-II reflective walls $\mathcal{H}_{\delta_i}$ ($i = 1, 2, 3$) carry order $d^{\mathrm{aut}}_{\delta_i} = 1$, and $\nu_{\Delta_5}(s_{\delta_i}) = -1 = (-1)^{d^{\mathrm{aut}}_{\delta_i}}$, while the chamber-permuting roots α_i carry order 0 and $\nu_{\Delta_5}(s_{\alpha_i}) = +1$ (Lemma 6.2).

(iv) The square $\mathcal{D}_X^2 = \Delta_{10} = \Phi_{10}^{\mathrm{un}} = \chi_{10}$ is a scalar weight-10 Igusa form, and the Oberdieck–Pixton scalar branch is $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$.

(v) The View II shadow is the doubled denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 64^{-1}\Delta_5(2Z)$ (Proposition 9.1); the View III operator-level datum $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is the subject of Chapter 6 and Appendix 7.

[*proved*; Borcherds 1995 [9, Theorem 10.1], Borcherds 1998 [10], Gritsenko–Nikulin 1998 [48, Theorem 2.1, Example 2.4], Gritsenko 1999 [46, Theorem 4.1], Igusa 1962 [51, Theorem 3], Igusa 1964 [52, §2], Cléry–Gritsenko 2011 [44, Theorem 1.4], Maass 1964 [76], Eichler–Zagier 1985 [34, §2.2], Oberdieck–Pixton 2018 [92, (0.4) and Theorem 1], Oberdieck–Pandharipande 2016 [91]; the lattice and theta-product computations are displayed in Appendices 1.2 and 2.]

11.0.0.1 What View I does not assert. View I does not assert any operator-level datum on $K3 \times E$, does not construct the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ (Chapter 4), does not construct the protected first-order operator $\mathfrak{D}_X^{\mathrm{DI}}$ or the Pfaffian line $\mathscr{L}_{\mathrm{Pf},X}$ (Chapter 6), and does not commit to the $K3 \times E$ realization of the chosen E_3-prefactorization datum $\mathfrak{A}_{K3\times E}$ at any level above the scalar shadow. In the language of the Beilinson cut, View I is the bottom of the reconstitution order: a holomorphic Siegel cusp form, with its weight, multiplier, divisor, and Borcherds product, and nothing more.

Chapter 4

The Borcherds–Kac–Moody denominator algebra $\mathfrak{g}_{\Delta_5}$ on $\Lambda^{2,1}_{II}$

The Igusa cusp form Δ_5 admits a Borcherds–Kac–Moody description as a denominator. The denominator algebra is the generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ on the index-four type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$, abstractly $U(4) \oplus \langle 2 \rangle$, with three even real simple roots δ_1, δ_2, δ_3, an infinite imaginary simple-root datum read from the K3 half-index $\phi_{0,1}$, and Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ generated by the three real reflections. Its Weyl–Kac–Borcherds denominator is the weight-five Borcherds–Kac–Moody denominator

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z),$$

the doubled Igusa form normalized by the theta-constant constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$. The Borcherds-weight characteristic of the denominator is $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 10/2 = 5$, the weight of the Gritsenko paramodular form $\Delta_5 = \Phi_1$ at level 1 — equivalently $(\Phi_{10}^{\mathrm{un}})^{1/2}$ in Igusa's normalization, where $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$. The handle Φ_1 denotes Gritsenko's paramodular Φ_t at $t = 1$, distinct from Igusa's weight-ten Φ_{10}.

The denominator algebra is target-side: it is the chiral shadow of an open compact realization. No compact Hall bracket, protected Pfaffian, or microscopic state space is claimed here. The protected Pfaffian is treated in Chapter 6 relative to the retained data of Appendix 7; the compact Hall primitive recognition remains conditional on $(S1)$–$(S10)$.

The three named instances of Δ_5 are tagged distinctly. In View II the canonical handle is the Borcherds–Kac–Moody denominator,

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z).$$

The automorphic section $\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ of Chapter 3 is View I; the compact protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of Chapter 6 is View III, relative to retained data. The bare handle Δ_5 is the working symbol for the Siegel cusp form of weight five and never substitutes for any of the three.

1 The hyperbolic lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ and its type-II sublattice

Fix the rank-three Lorentzian lattice

$$\Lambda^{2,1} \;=\; \Lambda^{(1,1)} \oplus [2] \;=\; \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2},$$

where $\Lambda^{(1,1)}$ is the hyperbolic plane and $[2]$ is the rank-one lattice generated by a single vector of square $+2$. The bilinear form on the chosen basis is

$$(f_2, f_{-2}) \;=\; -1, \qquad (f_2, f_2) \;=\; (f_{-2}, f_{-2}) \;=\; 0, \qquad (f_3, f_3) \;=\; 2, \tag{4.1}$$

with all other pairings between basis vectors zero. The lattice is even, signature $(2, 1)$, and primitively embedded in the rank-five Lorentzian lattice $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ of Section 1.4.

Definition 1.1 *(Type-II sublattice).* The type-II sublattice of $\Lambda^{2,1}$ is

$$\Lambda^{2,1}_{II} \;=\; \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2\}.$$

Its dual lattice is

$$(\Lambda^{2,1}_{II})^* \;=\; \mathbb{Z}\tfrac{1}{2}f_2 \oplus \mathbb{Z}\tfrac{1}{2}f_3 \oplus \mathbb{Z}\tfrac{1}{2}f_{-2} \;=\; \tfrac{1}{2}\Lambda^{2,1}.$$

The inclusion $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ has index four, and $\Lambda^{2,1} \subset (\Lambda^{2,1}_{II})^* = \frac{1}{2}\Lambda^{2,1}$ has index eight, so the discriminant group $(\Lambda^{2,1}_{II})^*/\Lambda^{2,1}_{II}$ has order 32. Abstractly the type-II sublattice is $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2\rangle$ of discriminant -32.

Remark 1.2 *(The two divisibility classes of square-2 vectors).* A primitive element $\delta \in \Lambda^{2,1}$ with $(\delta,\delta) = 2$ has divisibility either $(\delta, \Lambda^{2,1}) = \mathbb{Z}$ (type I) or $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$ (type II). The type-I roots lie in the sublattice

$$\Lambda^{2,1}_{I} \;=\; \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m + l + n \equiv 0 \pmod 2\}$$

and the type-II roots lie in $\Lambda^{2,1}_{II}$. The split is recorded in Gritsenko–Nikulin [46, Section 1]; the underlying classification of square-2 vectors on $\Lambda^{(1,1)} \oplus [2]$ is Nikulin's [88]. The finite packet `certificates/lattice/type_ii_chamber_isotropic` records the six primitive square-two representatives

$$\delta_1, \delta_2, \delta_3, \qquad f_{-2} - f_2,\; f_2 - f_3,\; f_{-2} - f_3,$$

checks divisibility 2 for the first three and divisibility 1 for the last three, and cross-checks against the Humbert-divisor packet that only the type-II representatives occur in $\operatorname{div}(\Delta_5)$.

LEMMA 1.3 *(Reflection-group decomposition).* Let $W^{(2)}(\Lambda^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+$ be the subgroup generated by reflections in primitive square-2 vectors. Then

$$\mathbf{O}(\Lambda^{2,1})_+ \;=\; W^{(2)}(\Lambda^{2,1}),$$

and there are two normal subgroups corresponding to the two divisibility classes,

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{I})] \;=\; 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{II})] \;=\; 6. \tag{4.2}$$

Proof. The first equality is Nikulin's [88] reflection-group theorem for $\Lambda^{(1,1)} \oplus [2]$. The two indices are the two cosets of the divisibility classes in $\mathbf{O}(\Lambda^{2,1})_+$; type-II reflections generate a subgroup of index six because the three orbit representatives $\{f_{-2} - f_2,\; f_2 - f_3,\; f_{-2} - f_3\}$ of the chamber automorphism S_3 are type-I and lie outside $W^{(2)}(\Lambda^{2,1}_{II})$. □

The same packet records the three type-II Coxeter generators $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$, the pairings $(\delta_i, \delta_j) = -2$ for $i \neq j$, and the resulting absence of finite braid relations among distinct type-II generators. The finite S_3-relations belong instead to the type-I chamber automorphisms.

The change of polarization between Gram-index parametrization and the denominator-side Lorentzian root grading is the linear map

$$\alpha\colon \Gamma_{\mathrm{ind}} \longrightarrow \Lambda^{2,1}_{II}, \qquad \alpha(n,l,m) \;=\; 2nf_2 - lf_3 + 2mf_{-2}.$$

Two pairings live on $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$: the Gram-index pairing

$$\langle (n,l,m), (n',l',m') \rangle_{\mathrm{ind}} \;=\; 2(nm' + n'm) - ll',$$

and the Lorentzian pairing on $\Lambda^{2,1}_{II}$ pulled back through α,

$$(\alpha(\gamma), \alpha(\gamma')) \;=\; -2\langle \gamma, \gamma' \rangle_{\mathrm{ind}}, \qquad (\alpha(\gamma), \alpha(\gamma)) \;=\; -2(4nm - l^2). \tag{4.3}$$

The sign convention is fixed by the Borcherds product expansion and by the BKM denominator: positive root norm gives the real wall set, non-positive root norm gives the imaginary roots.

2 Real walls, the type-II Weyl chamber, and chamber automorphism

Reflections on a Lorentzian lattice

For a primitive vector $\alpha \in \Lambda^{2,1}$ with $(\alpha, \alpha) > 0$ and $(\alpha, \alpha) \mid 2(\alpha, \Lambda^{2,1})$, the Kac reflection [58, Chapter 3] is

$$s_\alpha \colon x \longmapsto x - \frac{2(x, \alpha)}{(\alpha, \alpha)}\alpha, \qquad s_\alpha(\alpha) = -\alpha, \qquad s_\alpha|_{\alpha^\perp} = \mathbf{I}.$$

For $(\alpha, \alpha) = 2$ this simplifies to

$$s_\alpha \colon x \longmapsto x - (x, \alpha)\alpha.$$

The simple type-II roots

Set

$$\delta_1 \;=\; 2f_2 - f_3, \qquad \delta_2 \;=\; 2f_{-2} - f_3, \qquad \delta_3 \;=\; f_3, \tag{4.4}$$

so $\delta_i \in \Lambda^{2,1}_{II}$, $(\delta_i, \delta_i) = 2$, and $(\delta_i, \Lambda^{2,1}) = 2\mathbb{Z}$ for $i = 1, 2, 3$. These are the Gritsenko–Nikulin type-II simple roots [46, Theorem 2.1, Section 4]; the same choice appears as case $(t = 1, II, 0)$ in [48, Section 5.1.1].

Definition 2.1 *(Type-II Weyl chamber).* For a positive-norm vector α set

$$\mathbb{H}_{\alpha,+} \;=\; \left\{\mathbb{R}_{>0}x \in C(\Lambda^{2,1})_+/\mathbb{R}_{>0} \;\middle|\; (x, \alpha) \le 0\right\}.$$

The type-II Weyl chamber is

$$\mathcal{P}_{II} \;=\; \bigcap_{i=1}^{3} \mathbb{H}_{\delta_i,+}.$$

Its primitive walls are

$$\mathcal{P}_{II,prim} \;=\; \{\delta_1, \delta_2, \delta_3\}.$$

The chamber decomposition is

$$W^{(2)}(\Lambda^{2,1}) \;\simeq\; W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}), \qquad \mathrm{Aut}(\mathcal{P}_{II}) \;\simeq\; S_3, \tag{4.5}$$

with $\mathrm{Aut}(\mathcal{P}_{II})$ generated by the three type-I reflections $s_{f_{-2}-f_2}$, $s_{f_2-f_3}$, $s_{f_{-2}-f_3}$. Compare (4.2).

The Cartan matrix is the symmetric Lorentzian Gram matrix

Lemma 2.2 *(Cartan matrix).* The Gram matrix of $(\delta_1, \delta_2, \delta_3)$ is

$$A \;=\; \big((\delta_i, \delta_j)\big)_{1 \le i,j \le 3} \;=\; \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}. \tag{4.6}$$

Proof. Direct computation from (4.1) and (4.4): the diagonal entries are $(\delta_i, \delta_i) = 2$ for $i = 1, 2, 3$ as in the type-II classification, and the off-diagonal entries are

$$(\delta_1, \delta_2) \;=\; (2f_2 - f_3, 2f_{-2} - f_3) \;=\; -4 + 2 \;=\; -2,$$

and analogously $(\delta_1, \delta_3) = (\delta_2, \delta_3) = -2$. □

The matrix A is a symmetric generalized Cartan matrix in the sense of Kac [58, Chapter 1]: the diagonal entries are positive even integers, the off-diagonal entries are non-positive integers, and $A_{ij} = 0$ if and only if $A_{ji} = 0$. The associated Kac–Moody algebra $\mathfrak{g}(A)$ is hyperbolic; it is the rank-three hyperbolic algebra $H(2, 2, 2)$ of Feingold–Frenkel.

The chamber automorphism is S_3

LEMMA 2.3 *(S_3 chamber action).* The group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ acts on the simple roots by permutation: a reflection in $f_{-2} - f_2$ exchanges δ_1 and δ_2 and fixes δ_3; a reflection in $f_2 - f_3$ exchanges δ_1 and δ_3 and fixes δ_2; a reflection in $f_{-2} - f_3$ exchanges δ_2 and δ_3 and fixes δ_1. The Cartan matrix (4.6) is S_3-symmetric.

Proof. Each chamber-permuting reflection is the type-I reflection s_α in the primitive direction $\alpha \propto \delta_i - \delta_j$ that swaps the two type-II walls. For $(\delta_1\ \delta_2)$, the difference $\delta_1 - \delta_2 = 2(f_2 - f_{-2})$ has primitive direction $f_{-2} - f_2$ of square two; direct computation gives $s_{f_{-2}-f_2}(f_2) = f_{-2}$, $s_{f_{-2}-f_2}(f_3) = f_3$, hence $s_{f_{-2}-f_2}(\delta_1) = 2f_{-2} - f_3 = \delta_2$, $s_{f_{-2}-f_2}(\delta_2) = \delta_1$, $s_{f_{-2}-f_2}(\delta_3) = \delta_3$. For $(\delta_1\ \delta_3)$, $\delta_1 - \delta_3 = 2(f_2 - f_3)$ with primitive direction $f_2 - f_3$ of square two; one checks $s_{f_2-f_3}(\delta_1) = \delta_3$, $s_{f_2-f_3}(\delta_3) = \delta_1$, $s_{f_2-f_3}(\delta_2) = \delta_2$. For $(\delta_2\ \delta_3)$, $\delta_2 - \delta_3 = 2(f_{-2} - f_3)$ with primitive direction $f_{-2} - f_3$ of square two; one checks $s_{f_{-2}-f_3}(\delta_2) = \delta_3$, $s_{f_{-2}-f_3}(\delta_3) = \delta_2$, $s_{f_{-2}-f_3}(\delta_1) = \delta_1$. The three transpositions generate $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$; the Cartan matrix (4.6) is invariant under every permutation of $\{1, 2, 3\}$, so the S_3-action extends from the simple roots to all of $\overline{\mathcal{P}_{II}}$. □

This S_3-symmetry is part of the Gritsenko–Nikulin chamber stabilizer, it fixes the Weyl vector ρ, and it permutes the three primitive isotropic rays of $\overline{\mathcal{P}_{II}}$ (compare Lemma 6.3 below).

The Weyl vector

PROPOSITION 2.4 *(Weyl vector).* The Weyl vector $\rho \in \Lambda^{2,1} \otimes \mathbb{Q}$ defined by

$$(\rho, \delta_i) \;=\; -\frac{(\delta_i, \delta_i)}{2} \;=\; -1, \qquad i = 1, 2, 3,$$

is

$$\rho \;=\; \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) \;=\; f_2 - \tfrac{1}{2}f_3 + f_{-2},$$

and lies in the open positive cone $C(\Lambda^{2,1})_+ = C(\Lambda_{II}^{2,1})_+$.

Proof. The Cartan matrix $A = 4\mathbf{I}_3 - 2\mathbf{J}_3$ acts on $(1, 1, 1)^T$ as $A(1, 1, 1)^T = (-2, -2, -2)^T$ — the all-ones vector is an eigenvector with eigenvalue -2. The orthogonal complement of $(1, 1, 1)$ is the two-plane $\sum c_i = 0$, on which $\mathbf{J}_3$ vanishes and $A = 4\mathbf{I}_3$ acts by 4. Hence the spectrum of A is $(-2, 4, 4)$ and

$$A^{-1}(1, 1, 1)^T \;=\; -\tfrac{1}{2}(1, 1, 1)^T,$$

so the unique solution of $Ac = (-1, -1, -1)^T$ is $c = (\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2})$. Substituting $\rho = \sum c_i \delta_i$ and (4.4):

$$\tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) \;=\; \tfrac{1}{2}(2f_2 - f_3 + 2f_{-2} - f_3 + f_3) \;=\; f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

Its norm is

$$(\rho, \rho) \;=\; -2 + \tfrac{1}{2} \;=\; -\tfrac{3}{2} \;<\; 0,$$

so $\rho \in C(\Lambda^{2,1})_+$. Since $\rho \in \frac{1}{2}\Lambda^{2,1}_{II}$, ρ also lies in $C(\Lambda^{2,1}_{II})_+$. The finite packet `certificates/lattice/type_ii_weyl_vector` records the same computation together with $\rho \in (\Lambda^{2,1}_{II})^\vee$ and $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2 \rangle$. □

Remark 2.5 *(Weyl vector under the S_3 chamber action).* The expression $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ is manifestly S_3-symmetric on the simple-root basis; consequently $g \cdot \rho = \rho$ for every $g \in \mathrm{Aut}(\mathcal{P}_{II})$. The Weyl vector is the only fixed line of the S_3-action up to scaling.

The dual cone and chamber containment

Lemma 2.6 *(Chamber and dual cone).* With

$$\Delta(\Lambda^{2,1}_{II})_+ \;=\; \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3,$$

$$\Delta(\Lambda^{2,1}_{II})^*_+ \;=\; \left\{x \in \Lambda^{2,1} \otimes \mathbb{R} \;\middle|\; (x, \delta_i) \leq 0,\; i = 1, 2, 3\right\},$$

one has

$$\Delta(\Lambda^{2,1}_{II})^*_+ \;\subset\; \overline{C(\Lambda^{2,1})_+} \;\subset\; \Delta(\Lambda^{2,1}_{II})_+.$$

Proof. Both inclusions are convex-geometric statements about the simplicial chamber $\mathcal{P}_{II}$. The first follows from $(\rho, \delta_i) = -1 < 0$, which places ρ in $\Delta(\Lambda^{2,1}_{II})^*_+$, and the convexity of the cone. The second follows from the explicit decomposition of any $x \in \overline{C(\Lambda^{2,1})_+}$ as a non-negative linear combination of the three simple roots after passing to the chamber $\mathcal{P}_{II}$. □

3 Real and imaginary roots; the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$

Real roots and their multiplicities

Definition 3.1 *(Real-root system).* The real simple-root set in $\mathcal{P}_{II}$ is

$$\Delta^{re}_{\mathrm{simp}} \;=\; \{\delta_1, \delta_2, \delta_3\}.$$

The full real-root set is the orbit of the simple roots under the type-II Weyl group,

$$\Delta^{re} \;=\; W^{(2)}(\Lambda^{2,1}_{II})\,\Delta^{re}_{\mathrm{simp}}.$$

The real roots have norm $(\alpha, \alpha) = 2$ on every Weyl translate because reflections preserve the bilinear form, and the multiplicity of a real root in $\mathfrak{g}_{\Delta_5}$ is one in even parity (compare Lemma 4.6 below). The real roots index the perturbative wall set

$$\mathcal{W}_{\mathrm{pert}} \;=\; \bigcup_{\delta \in \Delta^{re}_{\mathrm{simp}}} \mathbb{H}_\delta;$$

together with the chamber-automorphism walls in $\mathrm{Aut}(\mathcal{P}_{II})$, the perturbative walls of the type-I cover are $\bigcup_{\alpha \in \Delta^{re}} \mathbb{H}_\alpha$.

Imaginary roots: the K3 half-index data

The imaginary simple-root data of $\mathfrak{g}_{\Delta_5}$ are read from the K3 half-index $\phi_{0,1}$ via the Borcherds product. Recall that the weak Jacobi form $\phi_{0,1} \in J_{0,1}^{\mathrm{wk}}$ has Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0,\, l \in \mathbb{Z}} f(n, l)\, q^n r^l, \tag{4.7}$$

and, because the index is one, $f(n, l)$ is a function of the discriminant $4n - l^2$. The finite arithmetic table `certificates/jacobi/k3e_first_relation_window_coefficients` checks the normalization $\phi_{0,1} = r^{-1} + 10 + r + O(q)$, the evenness $f(n, l) = f(n, -l)$, and the first relation-closed discriminant orbits used below.

Definition 3.2 *(Imaginary simple roots).* For $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ set

$$\mu_{\bar{0}}(a) = \begin{cases} m(a), & (a, a) < 0 \text{ and } m(a) > 0, \\ \tau(a), & (a, a) = 0 \text{ and } \tau(a) > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\mu_{\bar{1}}(a) = \begin{cases} -m(a), & (a, a) < 0 \text{ and } m(a) < 0, \\ -\tau(a), & (a, a) = 0 \text{ and } \tau(a) < 0, \\ 0, & \text{otherwise.} \end{cases}$$

The functions $m(a)$ are the additive correction coefficients of Gritsenko–Nikulin [46, Theorem 2.1], read from the Weyl-alternating sum in Section 5.1 below; the functions $\tau(ta)$ on isotropic rays are the eta-product expansion exponents of Lemma 6.4 below.

The imaginary simple-root set is the multiset

$$\Delta^{im} = \Delta^{im}_{\bar{0}} \cup \Delta^{im}_{\bar{1}},$$

where $a \in \Delta^{im}_{\bar{p}}$ with multiplicity $\mu_{\bar{p}}(a)$.

Remark 3.3 *(Why the parity split appears).* The Gritsenko–Nikulin product expansion for $64^{-1}\Delta_5$ has positive and negative integer exponents. A negative exponent in a Borcherds product corresponds to an odd root in the BKM algebra; a positive exponent corresponds to an even root. The sign split of $m(a)$ and $\tau(a)$ across the imaginary support records this parity on the simple-generator fibre. The signed root supermultiplicity is the difference

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}$$

for the full root space. The simple-generator signed dimension $\mu_{\bar{0}}(a) - \mu_{\bar{1}}(a)$ need not equal $\mathrm{smult}(a)$ when the degree a is reached by brackets of lower positive degrees. On the isotropic ray $a_{ij} = \delta_i + \delta_j$, the Gritsenko–Nikulin simple fibre has signed dimension 9, while the full root space has signed multiplicity 10, the missing direction being the real bracket $[e_i, e_j]$. The equality $\mathrm{smult}(\alpha(n, l, m)) = f(nm, l)$ is therefore a denominator identity, not a statement about simple fibres.

The full root datum

Definition 3.4 *(Real and imaginary root sets).* The real-root multiset is

$$\Delta^{re}_{\bar{0}} \;=\; \Delta^{re}, \qquad \Delta^{re}_{\bar{1}} \;=\; \varnothing.$$

The imaginary-root multiset is $\Delta^{im} = \Delta^{im}_{\bar{0}} \cup \Delta^{im}_{\bar{1}}$ with multiplicities $\mu_{\bar{p}}(a)$ on the imaginary simple support and Weyl-translated copies on the non-simple imaginary support.

Lemma 3.5 *(Borcherds–Cartan inequalities on the simple support).* Let

$$\Pi \;=\; \{\delta_1, \delta_2, \delta_3\} \cup \Delta^{im}_{\mathrm{simp}}$$

be the simple support of the Borcherds–Cartan datum. For every $\alpha \in \Pi$ and every distinct pair $\alpha, \alpha' \in \Pi$,

$$(\alpha, \alpha) \;>\; 0 \text{ for } \alpha \in \{\delta_1, \delta_2, \delta_3\}, \qquad (\alpha, \alpha) \;\le\; 0 \text{ for } \alpha \in \Delta^{im}_{\mathrm{simp}}, \qquad (\alpha, \alpha') \;\le\; 0.$$

For a real simple root $\alpha = \delta_i$,

$$\frac{2(\alpha, \alpha')}{(\alpha, \alpha)} \in \mathbb{Z}.$$

The last two assertions are statements about the simple Borcherds–Cartan datum; arbitrary Weyl translates of real roots are not constrained by an off-diagonal Cartan inequality.

Proof. These are the Borcherds–Kac defining inequalities of the simple Cartan datum [58, Chapter 1], [8]. For the three real simple roots they are the entries of the Cartan matrix (4.6). If $a \in \Delta^{im}_{\mathrm{simp}} \subset \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$, then $(a, \delta_i) \le 0$ by the chamber inequalities. Two imaginary simple roots lie in the same closed Lorentzian cone and have non-positive norm; their pairing is therefore non-positive. The integrality condition follows from $(\delta_i, \delta_i) = 2$ and the integral pairing on $\Lambda^{2,1}_{II} = \mathbb{Z}\delta_1 + \mathbb{Z}\delta_2 + \mathbb{Z}\delta_3$. □

4 Generators and relations: Cartan, Chevalley, Serre, Borcherds orthogonality, super sign

The Borcherds presentation conventions of [8] and the Kac sign conventions of [58, Chapter 1] are imported intact. In the type-II chamber these conventions give the denominator identity of Section 5.

Definition 4.1 *(Borcherds–Cartan superdatum $\mathscr{B}_{\Delta_5}$).* Let I be an index set with parity map $p\colon I \to \mathbb{Z}/2\mathbb{Z}$ and root map $\mathrm{ch}\colon I \to \Delta \subset \Lambda^{2,1}_{II}$, $i \mapsto \alpha_i$. The fibres over an imaginary root a have cardinality

$$\#\{i \in I \mid \alpha_i = a,\ p(i) = p\} \;=\; \mu_{\bar{p}}(a)$$

in each parity, and the three real simple roots $\delta_1, \delta_2, \delta_3$ each have a single index with parity $p = 0$.

Definition 4.2 *(Cartan, Chevalley, Serre, orthogonality, super sign).* The denominator algebra $\mathfrak{g}_{\Delta_5}$ is generated by the even Cartan $\Lambda^{2,1}_{II} \otimes \mathbb{C}$ and by parity-homogeneous generators

$$e_i,\ f_i \qquad (i \in I), \qquad |e_i| = |f_i| = p(i), \qquad \deg e_i = \alpha_i,\ \deg f_i = -\alpha_i,$$

together with the Cartan element $h_i \in \Lambda^{2,1}_{II} \otimes \mathbb{C}$ sending $i \mapsto \alpha_i$ under the linear identification of the Cartan with the lattice. The defining relations are:
Cartan relations. For $h, h' \in \Lambda^{2,1}_{II} \otimes \mathbb{C}$ and $i, j \in I$,

$$[h, h'] \;=\; 0, \qquad [h, e_j] \;=\; (h, \alpha_j)e_j, \qquad [h, f_j] \;=\; -(h, \alpha_j)f_j.$$

Chevalley relations.

$$[e_i, f_j] \;=\; \delta_{ij} h_i.$$

Real Serre relations. For real α_i and $i \neq j$,

$$(\operatorname{ad} e_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e_j \;=\; (\operatorname{ad} f_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f_j \;=\; 0. \tag{4.8}$$

Borcherds orthogonality. If $(\alpha_i, \alpha_j) = 0$, then

$$[e_i, e_j] \;=\; [f_i, f_j] \;=\; 0.$$

Super sign rule. For homogeneous elements X, Y of parities $|X|, |Y|$,

$$[X, Y] \;=\; -(-1)^{|X||Y|}[Y, X].$$

Remark 4.3 *(The real Serre exponent is three for $i \neq j$).* For the three type-II real simple roots, $(\delta_i, \delta_j) = -2$ and $(\delta_i, \delta_i) = 2$, so

$$1 - \frac{2(\delta_i, \delta_j)}{(\delta_i, \delta_i)} \;=\; 1 - \frac{-4}{2} \;=\; 3.$$

The real Serre relations (4.8) read

$$(\operatorname{ad} e_i)^3 e_j \;=\; (\operatorname{ad} f_i)^3 f_j \;=\; 0 \qquad (i \neq j).$$

The first commutators $[e_i, e_j]$ for $i \neq j$ are *not* forced to vanish; they generate the rank-three real-root sub-Kac–Moody algebra in the standard hyperbolic-Kac–Moody fashion.

The triangular decomposition

LEMMA 4.4 *(Triangular decomposition).* With

$$Q_+ \;=\; \sum_{\alpha \in \Delta} \mathbb{Z}_{\geq 0}\alpha \;\subset\; \Lambda_{II}^{2,1},$$

the algebra $\mathfrak{g}_{\Delta_5}$ has the standard triangular decomposition

$$\mathfrak{g}_{\Delta_5} \;=\; \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_\alpha\right) \oplus (\Lambda_{II}^{2,1} \otimes \mathbb{C}) \oplus \left(\bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_{-\alpha}\right),$$

with $\mathfrak{g}_0 = \Lambda_{II}^{2,1} \otimes \mathbb{C}$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$. The positive root set is

$$\mathcal{R}_+ \;=\; \{\alpha \in Q_+ \setminus \{0\} \mid \mathfrak{g}_\alpha \neq 0\}, \qquad \mathcal{R}_- \;=\; -\mathcal{R}_+.$$

Proof. This is the standard structure of a Borcherds–Kac–Moody Lie superalgebra associated to the superdatum (4.1) and the relations of Definition 4.2; see [8, 58]. □

The signed superdimension function

Definition 4.5 *(Signed root supermultiplicity).* For $\alpha \in \Lambda^{2,1}_{II}$,

$$\mathrm{smult}(\alpha) \;=\; \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}.$$

Lemma 4.6 *(Real-root multiplicity is one).* For every $\alpha \in \Delta^{re}$,

$$\dim \mathfrak{g}_{\alpha,\bar{0}} \;=\; 1, \qquad \dim \mathfrak{g}_{\alpha,\bar{1}} \;=\; 0, \qquad \mathrm{smult}(\alpha) \;=\; 1.$$

Proof. The simple real roots δ_i are even by construction, with single generators e_i, f_i, so $\dim \mathfrak{g}_{\delta_i,\bar{0}} = 1$ and $\dim \mathfrak{g}_{\delta_i,\bar{1}} = 0$. By [58, Lemma 3.8 and Proposition 3.7], Weyl operators are automorphisms of $\mathfrak{g}_{\Delta_5}$. The $\mathfrak{g}_{\Delta_5}$ adaptation [46, Section 6, Statement 6.4] shows that, in absence of odd real simple roots, these Weyl operators are even. Hence $\mathfrak{g}_{\alpha,\overline{p}} \simeq \mathfrak{g}_{w\alpha,\overline{p}}$ for every $w \in W^{(2)}(\Lambda^{2,1}_{II})$ and every parity $\overline{p}$, and the equalities transfer from the simple roots to the full real-root orbit. □

Lemma 4.7 *(Imaginary simple-fibre signed dimension).* For an imaginary simple degree $a \in \Delta^{im}_{\mathrm{simp}}$, let

$$E^{\mathrm{simp}}_a \subset \mathfrak{g}_a$$

be the span of the Chevalley generators whose root is a. Then

$$\mathrm{sdim}\, E^{\mathrm{simp}}_a \;=\; \mu_{\bar{0}}(a) - \mu_{\bar{1}}(a).$$

If a is decomposable in the positive root monoid, the full signed root multiplicity $\mathrm{smult}(a)$ also contains the images of brackets whose total degree is a.

Proof. By construction of $\mathscr{B}_{\Delta_5}$, the imaginary simple-root fibres at a have $\mu_{\bar{0}}(a)$ generators of even parity and $\mu_{\bar{1}}(a)$ of odd parity, all of degree a. Their span is E^{simp}_a, so its signed dimension is the stated difference. The root space $\mathfrak{g}_a$ is the degree-a part of the quotient of the free Lie superalgebra by the Borcherds–Kac relations, and brackets of lower positive degrees may survive in that quotient. □

5 The denominator: Weyl alternating sum and Borcherds product

The Weyl alternating sum

Theorem 5.1 *(Weyl alternating sum for $64^{-1}\Delta_5$).* Let $W^{(2)}(\Lambda^{2,1}_{II})$ act on $\Lambda^{2,1} \otimes \mathbb{C}$ by reflection in the type-II walls, and let $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \frac{1}{2} f_3 + f_{-2}$ be the Weyl vector. Then

$$\frac{1}{64}\Delta_5(Z) \;=\; \sum_{w \in W^{(2)}(\Lambda^{2,1}_{II})} \det(w) \left(e^{-\pi i (w(\rho), z)} - \sum_{a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\, e^{-\pi i (w(\rho + a), z)} \right).$$

Proof. Write

$$\tfrac{1}{64}\Delta_5(Z) \;=\; \sum_{\lambda} c_\lambda e^{-\pi i (\lambda, z)}, \qquad \lambda \in \Lambda^{2,1}_{II} \cup (\rho + \Lambda^{2,1}_{II}).$$

The leading term has $\lambda = \rho$ with $c_\rho = 1$ by the $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$ theta-constant normalization (4.9) below. The remaining exponents λ lie in $\rho + a$ with $a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$. The Maass multiplier restriction [46, (1.3)–(1.5)], citing [76], says $\Delta_5 \,|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det$, so the Weyl translates satisfy

$$c_{w\lambda} \;=\; \det(w)\, c_\lambda.$$

If λ lies on a simple wall $\mathbb{H}_{\delta_i}$, the reflection s_{δ_i} fixes λ and forces $c_\lambda = -c_\lambda = 0$. Summing the signed Weyl translates of the remaining exponents gives the displayed alternating sum, with $m(a)$ the additive correction coefficient [46, Theorem 2.1] read from the active support of $\phi_{0,1}$. □

Borcherds product expansion of $D_5 = 64^{-1}\Delta_5$

LEMMA 5.2 *(Vacuum coefficient).*

$$[q^{1/2}r^{1/2}s^{1/2}]\,\Delta_5(Z) \;=\; 64. \tag{4.9}$$

Proof. The leading exponent of the Gritsenko additive lift of $\eta^9\vartheta(z)$ at ρ is computed in [46, Theorem 2.1 and Section 3], with the constant 64 arising from the ten-theta-constant Maass formula [46, (1.3)]; the same factor is the value of the leading Fourier coefficient $c_\Delta(1,1,1)$ in the Gritsenko cusp expansion. □

THEOREM 5.3 *(Borcherds product for D_5).* With $D_5(Z) = 64^{-1}\Delta_5(Z)$, the effective support

$$\Gamma_{\mathrm{eff}} \;=\; \{(n,l,m)\in\mathbb{Z}^3 \mid m>0,\ n\geq 0,\ l\in\mathbb{Z}\}\cup\{(n,l,0)\in\mathbb{Z}^3 \mid n>0,\ l\in\mathbb{Z}\}\cup\{(0,l,0)\in\mathbb{Z}^3 \mid l<0\},$$

and the active subset $\Gamma_{\mathrm{act}} = \{(n,l,m)\in\Gamma_{\mathrm{eff}} \mid f(nm,l)\neq 0\}$,

$$D_5(Z) \;=\; q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^nr^ls^m)^{f(nm,l)}. \tag{4.10}$$

The cusp polarization selects one of the two half-spaces under $(n,l,m)\mapsto(-n,-l,-m)$ on the type-II character lattice, namely the half-space on which the chamber $0<|s|\ll|q|\ll|r|^{-1}\ll 1$ makes $q^nr^ls^m\to 0$ ([48, Example 2.4]). The boundary triple $(0,-1,0)$ is present and gives the simple root $\delta_3 = f_3$; the inactive triples $\Gamma_{\mathrm{eff}}\setminus\Gamma_{\mathrm{act}}$ contribute trivial factors $(1-q^nr^ls^m)^0 = 1$.

Proof. This is the Borcherds product associated to the singular theta lift of the weight-zero index-one weak Jacobi form $\phi_{0,1}$, [10, Theorem 13.3] specialized to the lattice $\Lambda^{2,1}$ and the Jacobi-form input $\phi_{0,1}$, equivalently [48, Theorem 2.1 and (2.1)–(2.3)]. The exponents $f(nm,l)$ are the Fourier coefficients of $\phi_{0,1}$ (4.7), and the prefactor $q^{1/2}r^{1/2}s^{1/2}$ is the Weyl-vector vacuum $e^{-\pi i(\rho,z)}$ in the chosen polarization. The Borcherds exponent identification is (4.3). □

The denominator identity

THEOREM 5.4 *(Denominator identity, doubled form).* Let $\mathfrak{g}_{\Delta_5}$ be the Gritsenko–Nikulin denominator algebra of Definitions 4.1 and 4.2, with signed root supermultiplicity smult. Then

$$\frac{1}{64}\Delta_5(2Z) \;=\; \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})}\det(w)\left(e^{-2\pi i(w(\rho),z)} - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\,e^{-2\pi i(w(\rho+a),z)}\right) \tag{4.11}$$

on the Weyl alternating side, and on the Borcherds product side

$$\frac{1}{64}\Delta_5(2Z) \;=\; e^{-2\pi i(\rho,z)}\prod_{\alpha\in\mathcal{R}_+}(1-e^{-2\pi i(\alpha,z)})^{\mathrm{smult}(\alpha)}. \tag{4.12}$$

The two expressions are equal as the Weyl–Kac–Borcherds denominator of $\mathfrak{g}_{\Delta_5}$:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; \frac{1}{64}\Delta_5(2Z). \tag{4.13}$$

On the active support, the signed multiplicities are the K3 half-index Fourier coefficients,

$$\mathrm{smult}(\alpha(n,l,m)) \;=\; f(nm,l) \qquad ((n,l,m)\in\Gamma_{\mathrm{act}}). \tag{4.14}$$

Proof. Apply the substitution $Z \mapsto 2Z$, equivalently $z \mapsto 2z$, to Theorem 5.1: every exponent $-\pi i(w(\rho + a), z)$ becomes $-2\pi i(w(\rho + a), z)$, giving the displayed Weyl alternating sum (4.11).

The product expression (4.12) is the Borcherds product expansion of Theorem 5.3 under $Z \mapsto 2Z$ and the Lorentzian degree identification (4.3): the Gram-index exponent $q^n r^l s^m$ is $e^{-\pi i(\alpha(n,l,m),z)}$, hence $Z \mapsto 2Z$ sends it to $e^{-2\pi i(\alpha(n,l,m),z)}$. The exponents $f(nm, l)$ on the active support are the signed multiplicities $\mathrm{smult}(\alpha(n, l, m))$ of the BKM denominator, because the Borcherds product and the Weyl–Kac–Borcherds product are the same element of the completed positive-root monoid algebra. The imaginary simple-root data of Definition 3.2 determine the additive correction terms in the Weyl alternating side; the full root multiplicities are read from the denominator product.

The equality $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ identifies the Borcherds–Kac–Moody denominator of $\mathfrak{g}_{\Delta_5}$ with the doubled-Igusa product. The factor 64 is the theta-constant normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ of Lemma 5.2; it is not a separate convention choice.

The signed-multiplicity equality (4.14) on the active support is the comparison of (4.12) with the Borcherds product expansion of D_5 at $Z \mapsto 2Z$. □

Two normalizations and the doubling factor

Remark 5.5 *(The doubling $Z \mapsto 2Z$).* The denominator identity is written for $\Delta_5(2Z)$ rather than $\Delta_5(Z)$ because the BKM root grading is on $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ (index 4), while the additive lift of $\phi_{0,1}$ produces Δ_5 on the ambient $\Lambda^{2,1}$ in its primitive form. The root attached to a product monomial is

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda^{2,1}_{II}.$$

The doubling acts on the Borcherds-product variables by $q \mapsto q^2$, $r \mapsto r^2$, $s \mapsto s^2$, so $q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z))$ becomes the standard denominator character $\exp(-2\pi i(\alpha(n, l, m), z))$. No equality $\Lambda^{2,1}_{II} = 2\Lambda^{2,1}$ is used; the f_3-coordinate remains primitive. The theta-constant normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ is untouched by doubling.

Remark 5.6 *(Doubling factor as a power of two).* The constant $64 = 2^6$ is the value of the leading Fourier coefficient $c_\Delta(1, 1, 1)$ of Δ_5 in the chosen polarization. It is *not* the power 2^d associated with a doubling on a rank-d lattice or a half-Hilbert orbit count. The $2^{12} = 4096$ orbit count appearing in the type-II Pfaffian wall normal form (Chapter 6, $(O2)$) is a distinct sign-sum on the half-Hilbert orbit and is unrelated to the prefactor 64 of the denominator identity.

Proposition 5.7 *(Higher-order Fourier coefficient verification of the Borcherds product).* The Fourier coefficients $f(n, l)$ of $\phi_{0,1}$ entering the Borcherds product of Theorem 5.3 match the expansion of $D_5(Z)$ to order $\max(n) \le 4$ at the type-II cusp. The non-zero $f(n, l)$ for $0 \le n \le 4$ are:

n	$f(n, l)$ for $\lvert l\rvert \le n + 1$
0	$f(0, 0) = 10, \quad f(0, \pm 1) = 1$
1	$f(1, 0) = 108, \quad f(1, \pm 1) = -64, \quad f(1, \pm 2) = 10$
2	$f(2, 0) = 808, \quad f(2, \pm 1) = -513, \quad f(2, \pm 2) = 108, \quad f(2, \pm 3) = 1$
3	$f(3, 0) = 4016, \quad f(3, \pm 1) = -2752, \quad f(3, \pm 2) = 808, \quad f(3, \pm 3) = -64$
4	$f(4, 0) = 16524, \quad f(4, \pm 1) = -11775, \quad f(4, \pm 2) = 4016, \quad f(4, \pm 3) = -513, \quad f(4, \pm 4) = 10$

Proof. The values are read from $\phi_{0,1}(\tau, z) = h_0(\tau)\theta_{1,0}(\tau, z) + h_1(\tau)\theta_{1,1}(\tau, z)$ by the Eichler–Zagier theta decomposition (Lemma 1.12) and the standard generating-function identities for the K3 elliptic genus. All values are obtained by direct rational expansion of the theta quotient and checked against the closed-form $\eta^{-2}\theta_1^2$ expansion to depth $(n, l) = (4, 4)$. Squaring the Borcherds product doubles every exponent,

$$\Phi_{10}^{\mathrm{un}} = \Delta_5^2 = 64^2\, qrs \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{2f(nm,l)},$$

so the same coefficients $f(nm, l)$ control the weight-ten Igusa form with multiplicities $2f(nm, l)$.

The verified coefficients match the cross-check identities of [34, Proposition 5.2]: for each fixed n, the row $(f(n, l))_{|l|\leq n+1}$ is the Eichler–Zagier expansion of the n-th Fourier component of $\phi_{0,1}$ in elliptic theta-functions. At the trivial elliptic variable one has

$$Z_{\mathrm{K3}}(\tau, 0) = 2\phi_{0,1}(\tau, 0) = 24,$$

so the same normalization records the Euler characteristic of K3 before the Borcherds multiplicative lift is formed. □

Remark 5.8 *(The verified arithmetic anchors all five views).* Every arithmetic identity about Δ_5, $\mathfrak{g}_{\Delta_5}$, or the type-II Cartan matrix factors through the table of Proposition 5.7 together with the Cartan matrix, Weyl vector, and reflection computation of Appendix 1.2. The five views of the frontispiece are arithmetically reconciled: View 1 via the divisor identification $f(0, \pm 1) = 1$, $f(0, 0) = 10$; View 2 via the Borcherds-product exponents $f(nm, l)$; View 3 via the signed-multiplicity residual sdim $P^{\Pi}_{X,(n,l,m)} = f(nm, l)$; View 5 via the singular-theta lift coefficients.

6 THE ACTIVE SUPPORT: LORENTZIAN CONE AND FINITE CHECKS

Definition 6.1 *(Active Gram-index support).* The active Gram-index support is

$$\Gamma_{\mathrm{act}} \;=\; \{(n, l, m) \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}.$$

The Borcherds cone is the real cone generated by the Lorentzian images of the active support,

$$C_{\mathrm{Borch}} \;=\; \mathbb{R}_{\geq 0}\, \alpha(\Gamma_{\mathrm{act}}) \;\subset\; \Lambda_{II}^{2,1} \otimes \mathbb{R}.$$

LEMMA 6.2 *(Active support gives the type-II chamber cone).* For $(n, l, m) \in \Gamma_{\mathrm{act}}$,

$$\alpha(n, l, m) \;=\; n\delta_1 + m\delta_2 + (n + m - l)\delta_3, \tag{4.15}$$

with

$$n \;\geq\; 0, \qquad m \;\geq\; 0, \qquad n + m - l \;\geq\; 0.$$

Consequently

$$C_{\mathrm{Borch}} \;=\; \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3.$$

Proof. Substitute (4.4) into (4.15):

$$n(2f_2 - f_3) + m(2f_{-2} - f_3) + (n + m - l)f_3 \;=\; 2nf_2 - lf_3 + 2mf_{-2}.$$

The effective chamber Γ_{eff} has $n, m \geq 0$. Since $\phi_{0,1}$ is weak of index one, $f(nm, l) \neq 0$ implies $4nm - l^2 \geq -1$. If $n = 0$ or $m = 0$, this gives $l^2 \leq 1$, and the boundary cases in Γ_{eff} give $l \leq n + m$. If $n, m > 0$ and $l \geq n + m + 1$, then $l^2 \geq (n + m + 1)^2 = 4nm + (n - m)^2 + 2(n + m) + 1 > 4nm + 1$, contradicting $l^2 \leq 4nm + 1$. Hence $n + m - l \geq 0$.

The reverse inclusion $C_{\mathrm{Borch}} \supset \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3$ follows from the active triples $(1, 0, 0)$, $(0, 0, 1)$, $(0, -1, 0)$ giving $\alpha = \delta_1 + \delta_3, \delta_2 + \delta_3, \delta_3$, and $(1, 1, 0)$, $(0, 1, 1)$ giving δ_1, δ_2. □

The finite table `certificates/lattice/product_chamber_root_map` checks the chamber sectors, sector closure, the root map $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$, the decomposition (4.15), active nonnegativity, and inactive zero exponents on the displayed low rows. It is a computed verification of the target-side denominator polarization, not a source construction or Pfaffian-orientation statement.

LEMMA 6.3 *(Isotropic primitive rays, S_3-orbit).* The primitive isotropic rays of $\overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$ are generated by

$$a^{(1)} = 2f_2, \qquad a^{(2)} = 2f_{-2}, \qquad a^{(3)} = 2f_2 - 2f_3 + 2f_{-2}.$$

Equivalently, in the simple-root basis,

$$a^{(1)} = \delta_1 + \delta_3, \qquad a^{(2)} = \delta_2 + \delta_3, \qquad a^{(3)} = \delta_1 + \delta_2.$$

The chamber-automorphism group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ is transitive on the three primitive isotropic generators.

Proof. Each $a^{(j)}$ is primitive in $\Lambda_{II}^{2,1}$ and has $(a^{(j)}, a^{(j)}) = 0$ by (4.1); e.g. $(2f_2, 2f_2) = 0$. The condition $(a^{(j)}, \delta_i) \le 0$ for all i places $a^{(j)}$ in $\overline{\mathcal{P}_{II}}$. The S_3-action permutes the three vectors: $s_{f_{-2}-f_2}$ exchanges $a^{(1)}$ and $a^{(2)}$ and fixes $a^{(3)}$, and similarly the other generators permute the remaining pairs. $\square$

LEMMA 6.4 *(Eta-product expansion on isotropic rays).* For $a_0 = 2f_2$ and $t \in \mathbb{Z}_{\ge 1}$,

$$1 + \frac{1}{64}\sum_{t\ge 1} c_\Delta(1+2t,1,1)q^t = \prod_{k\ge 1}(1-q^k)^9.$$

In denominator notation,

$$1 - \sum_{t\ge 1} m(ta_0)\, q^t = \prod_{t\ge 1}(1-q^t)^{\tau(ta_0)}, \qquad \tau(ta_0) = 9 \quad (t \ge 1).$$

By S_3-symmetry, $\tau(ta) = 9$ for every primitive isotropic ray $\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$.

Proof. The product $\prod_{k\ge 1}(1-q^k)^9$ is the Gritsenko–Nikulin isotropic boundary expansion of the cusp form along the ray $\mathbb{R}_{>0}a_0$, computed via the eta-product η^9 at the relevant cusp [48, Section 5.1.1], using $\Delta_5 = \mathrm{Lift}(\eta^9\vartheta)$ in Gritsenko's additive lift. The S_3-symmetry from Lemma 6.3 transfers the value $\tau(ta) = 9$ to every primitive isotropic ray. $\square$

The packet `certificates/lattice/type_ii_chamber_isotropic` records the three active rows

$$(1,2,1), \qquad (1,0,0), \qquad (0,0,1),$$

corresponding to $\delta_1+\delta_2$, $\delta_1+\delta_3$, and $\delta_2+\delta_3$, with coefficient $f(nm,l) = 10$. It records the denominator-side split $10 = 1 + 9$: one affine $A_1^{(1)}$ real-string contribution and the Gritsenko–Nikulin isotropic exponent $\tau(a) = 9$. The rows $u_{ij,r}$, $1 \le r \le 9$, are target BKM isotropic directions on the ray a_{ij}; they are not compact-source representatives and do not split a signed multiplicity into even and odd dimensions.

Low-degree denominator brackets

PROPOSITION 6.5 *(Low-degree real-root brackets).* The denominator algebra has the following target-side low-degree checks. With

$$\alpha(n,l,m) = n\delta_1 + m\delta_2 + (n+m-l)\delta_3,$$

the active positive rows through height three are

ht β	β	$\gamma(\beta)$	(β,β)	smult(β)
1	δ_1	$(1,1,0)$	2	1
1	δ_2	$(0,1,1)$	2	1
1	δ_3	$(0,-1,0)$	2	1
2	$\delta_1+\delta_2$	$(1,2,1)$	0	10
2	$\delta_1+\delta_3$	$(1,0,0)$	0	10
2	$\delta_2+\delta_3$	$(0,0,1)$	0	10
3	$2\delta_1+\delta_2$	$(2,3,1)$	2	1
3	$\delta_1+2\delta_2$	$(1,3,2)$	2	1
3	$2\delta_1+\delta_3$	$(2,1,0)$	2	1
3	$\delta_1+2\delta_3$	$(1,-1,0)$	2	1
3	$2\delta_2+\delta_3$	$(0,1,2)$	2	1
3	$\delta_2+2\delta_3$	$(0,-1,1)$	2	1
3	$\delta_1+\delta_2+\delta_3$	$(1,1,1)$	-6	-64

(4.16)

After choosing even Chevalley generators e_i, f_i, h_i for the real simple roots and setting

$$A_{ij} = (\delta_i, \delta_j), \qquad E_{ij} := [e_i, e_j] \quad (i \neq j),$$

so $E_{ji} = -E_{ij}$, one has

$$[h_i, h_j] = 0, \qquad [h_i, e_j] = A_{ij}e_j, \qquad [h_i, f_j] = -A_{ij}f_j, \qquad [e_i, f_j] = \delta_{ij}h_i,$$

and

$$(\operatorname{ad} e_i)^3 e_j = 0 \quad (i \neq j).$$

For $\{i,j,k\} = \{1,2,3\}$,

$$[h_i, E_{ij}] = [h_j, E_{ij}] = 0, \qquad [h_k, E_{ij}] = -4E_{ij},$$

$$[f_i, E_{ij}] = 2e_j, \qquad [f_j, E_{ij}] = -2e_i, \qquad [f_k, E_{ij}] = 0.$$

For ordered $i \neq j$, put $E_{iij} := [e_i, E_{ij}]$. Then

$$[e_i, E_{iij}] = 0, \qquad [f_i, E_{iij}] = 2E_{ij}, \qquad [f_j, E_{iij}] = 0.$$

For $i < j$, the isotropic primitive degree $a_{ij} := \delta_i + \delta_j$ has

$$\text{smult}(a_{ij}) = 10 = 1 + 9:$$

one even real-real commutator direction E_{ij} and nine even isotropic simple directions $u_{ij,1}, \dots, u_{ij,9} \in \mathfrak{g}_{a_{ij}}$, $\tau(ta_{ij}) = 9$ for $t \geq 1$. The orthogonality relations give

$$[e_i, u_{ij,r}] = [e_j, u_{ij,r}] = 0, \qquad [f_i, u_{ij,r}] = [f_j, u_{ij,r}] = 0.$$

In the timelike degree $\delta_{123} := \delta_1 + \delta_2 + \delta_3$, the real bracketings

$$T_1 = [e_1, E_{23}], \qquad T_2 = [e_2, E_{31}], \qquad T_3 = [e_3, E_{12}]$$

satisfy $T_1 + T_2 + T_3 = 0$ by the graded Jacobi identity. Since $(\delta_k, a_{ij}) = -4$, the Borcherds–Kac presentation imposes

$$(\operatorname{ad} e_k)^5 u_{ij,r} = 0,$$

and analogously for the negatives. The product exponent gives

$$\operatorname{smult}(\delta_{123}) = f(1,1) = -64.$$

Proof. The table is obtained by writing $\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3$ and solving $\gamma(\beta) = (c_1, c_1 + c_2 - c_3, c_2)$. The norm is $(\beta, \beta) = -2(4nm - l^2)$, and the multiplicity is $\operatorname{smult}(\beta) = f(nm, l)$ from Theorem 5.4. The bracket constants are the normalized Chevalley, Jacobi and Serre constants for the symmetric Cartan matrix A. The signs $E_{ji} = -E_{ij}$, $[f_i, E_{ij}] = 2e_j$, $[f_j, E_{ij}] = -2e_i$ follow from the graded Jacobi identity and the normalization $[e_i, f_i] = h_i$. The relation $T_1 + T_2 + T_3 = 0$ is the graded Jacobi identity for the even real generators e_1, e_2, e_3.

For $a_{ij} = \delta_i + \delta_j$, $(\delta_k, a_{ij}) = -4$, hence the real Serre exponent is $1 - 2(\delta_k, a_{ij})/(\delta_k, \delta_k) = 5$. The equality $\tau(ta_{ij}) = 9$ on primitive isotropic rays is Lemma 6.4. □

Remark 6.6 *(The terminal stop on the affine string).* For $\alpha = \delta_k$ and $\beta = a_{ij}$ with $\{i, j, k\} = \{1, 2, 3\}$, the real-root string theorem [58, Proposition 5.1(c)] with the GN super adaptation [46, Section 6, Statement 6.4] gives $(\beta, \alpha^\vee) = -4$ and $\beta - \alpha \notin \Delta$. Hence the string $\beta + \mathbb{Z}\alpha$ stops before $a_{ij} + 5\delta_k$:

$$a_{ij},\ a_{ij} + \delta_k,\ a_{ij} + 2\delta_k,\ a_{ij} + 3\delta_k,\ a_{ij} + 4\delta_k,\ \mathfrak{g}_{a_{ij}+5\delta_k} = 0.$$

The submatrix on δ_i, δ_j is $\left(\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix}\right)$, so $a_{ij} = \delta_i + \delta_j$ is the affine null root of $A_1^{(1)}$. [58, Theorem 7.4 and Corollary 8.3] gives $\operatorname{mult}(ta_{ij}) = 1$ in the affine subalgebra, and the extra nine in $\operatorname{smult}(a_{ij}) = 10$ are the GN isotropic simple fibres of Lemma 6.4.

The first timelike presentation count

PROPOSITION 6.7 *(Timelike presentation count* 29|93 *at* δ_{123}*).* With $\delta_{123} = \delta_1 + \delta_2 + \delta_3 = 2f_2 - f_3 + 2f_{-2}$, the additive correction coefficient is

$$m(\delta_{123}) = -93,$$

and consequently

$$\operatorname{mult}_{\bar{0}}(\delta_{123}) = 29, \qquad \operatorname{mult}_{\bar{1}}(\delta_{123}) = 93, \qquad 29 - 93 = -64 = f(1,1).$$

The 29 even directions are the two-dimensional real-real-real Borcherds–Kac presentation span

$$T_1 = [e_1, [e_2, e_3]], \qquad T_2 = [e_2, [e_3, e_1]], \qquad T_3 = [e_3, [e_1, e_2]], \qquad T_1 + T_2 + T_3 = 0,$$

together with the 27 mixed real-isotropic words

$$[e_k, u_{ij,r}], \qquad \{i, j, k\} = \{1, 2, 3\},\ 1 \leq r \leq 9.$$

The 93 odd directions are the negative-norm imaginary simple generators in degree δ_{123}.

Proof. The root δ_{123} corresponds in the Weyl-sum coefficient formula to $(3-1)f_2 - \frac{3-1}{2}f_3 + (3-1)f_{-2}$. Hence $m(\delta_{123}) = -64^{-1}c_\Delta(3,3,3)$. Removing the leading monomial from $D_5 = 64^{-1}\Delta_5$ and computing $[qrs]\prod(1-q^n r^l s^m)^{f(nm,l)} = 93$ gives the integer 93: the factor $(1-qrs)^{-64}$ contributes 64, and the remaining root factors contribute 29. The signed equality $29 - 93 = -64 = f(1,1)$ is the Borcherds product identity (4.14) at $(n,l,m) = (1,1,1)$. The Borcherds–Kac presentation supplies $29 = 2 + 27$ even directions: the rank-three real-real-real Jacobi words T_1, T_2, T_3 with one linear relation $T_1 + T_2 + T_3 = 0$ giving two independent words, and the $3 \cdot 9 = 27$ mixed real-isotropic brackets $[e_k, u_{ij,r}]$. The remaining 93 odd directions are the negative-norm imaginary simple generators of Definition 3.2. □

The finite packet `certificates/targets/delta5_gn_kac/delta123_presentation_split` records this target presentation split explicitly. Its tables list the degree δ_{123}, the three Jacobi words T_1, T_2, T_3 with one relation, the 27 mixed words $[e_k, u_{ij,r}]$, and the 93 odd imaginary-simple generators $\xi_{\delta_{123},a}$. The verifier recomputes the norm -6, the signed value $f(1,1) = -64$, the monic qrs-coefficient 93, and $m(\delta_{123}) = -93$, and cross-checks the `wle3_target_parity` and `type_ii_chamber_isotropic` packets. This is a target presentation count, not a compact-source Hall bracket or Pfaffian orientation statement.

Remark 6.8 *(Determinant alone forgets the parity split).* The determinant identity $\mathrm{smult}(\delta_{123}) = -64$ records only the difference $\mathrm{mult}_{\bar{0}} - \mathrm{mult}_{\bar{1}} = 29 - 93$. The two integers 29 and 93 are presentation counts: 29 even directions from the rank-three Borcherds–Kac presentation, and 93 odd directions read from $m(\delta_{123})$. They are determined by the BKM presentation of $\mathfrak{g}_{\Delta_5}$, not by the Borcherds product.

Remark 6.9 *(Target rows in the first finite terminal window).* Let $C_{k,s} = a_{ij} + s\delta_k$ for $\{i,j,k\} = \{1,2,3\}$. The target presentation supplies

$$2a_{ij} : 10|0, \qquad C_{k,3} : 29|93, \qquad C_{k,4} : 10|0, \qquad C_{k,5} : 0|0.$$

The first row uses the $A_1^{(1)}$ subalgebra generated by δ_i, δ_j, together with the nine GN isotropic simple fibres on the ray $\mathbb{Z}_{>0}a_{ij}$. The rows $C_{k,3}$ and $C_{k,4}$ are the even real-Weyl transports of δ_{123} and a_{ij}, respectively, and $C_{k,5}$ is the terminal zero row of the real-root string [58, Proposition 5.1(c), Theorem 7.4, Corollary 8.3] [46, Sections 3–4, Statement 6.4]. In contrast,

$$C_{k,2} : \mathrm{sdim} = 108, \quad m = 90, \qquad D_k = \delta_k + 2a_{ij} : \mathrm{sdim} = -513, \quad m = 54, \qquad 2\delta_{123} : \mathrm{sdim} = 4016, \quad m = -540$$

are signed target rows only. They cannot be used as parity blocks, PBW rows, or source comparison codomains until the finite target presentation is computed in those degrees.

Definition 6.10 *(Finite target-presentation datum).* Let $W \subset \mathcal{R}_+$ be a finite downward-saturated target-root window and put $S_W = W \cup (-W) \cup \{0\}$. A finite target-presentation datum for $\mathfrak{g}_{\Delta_5}|_{S_W}$ consists of the following target objects, written in homogeneous bases:

(T1) the full parity dimensions $\dim \mathfrak{g}_{\Delta_5 \pm\beta,\bar{0}}$ and $\dim \mathfrak{g}_{\Delta_5 \pm\beta,\bar{1}}$ for every $\beta \in W$;

(T2) the real and imaginary simple-generator fibres that generate the displayed degrees, with parity and Weyl-transport or terminal-string provenance;

(T3) Cartan, Chevalley, real-Serre, Borcherds orthogonality, and super-sign relation rows for every relation whose inputs and output lie in S_W;

(T4) positive-negative invariant-pairing blocks and their target radical kernels, in parity blocks;

(T5) associated-graded PBW rows for the same finite window;

(T6) provenance for each row from the Gritsenko–Nikulin presentation, Kac real-root-string calculus, Borcherds generalized-Kac–Moody conventions, or a stated finite computation.

A signed exponent row smult(β), or the imaginary-simple integer $m(\beta)$, is only a scalar target row. It becomes a target-presentation row only after (T1)–(T6) have been supplied in that degree.

7 Maass multiplier on $W^{(2)}(\Lambda^{2,1})$

The transformation law of Δ_5 under $\mathbf{Sp}_4(\mathbb{Z})$ is

$$\Delta_5(g \cdot Z) \;=\; \nu_{\Delta_5}(g)\det(CZ+D)^5\Delta_5(Z), \qquad g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z}),$$

with multiplier system ν_{Δ_5} of order two. The scalar square is invariant of order 10:

$$\Delta_5^2(g \cdot Z) \;=\; \det(CZ+D)^{10}\Delta_5^2(Z).$$

Under the exterior-square identification of $\mathbf{PSp}_4(\mathbb{Z})$ with its projective image in $\mathbf{O}(\Lambda^{3,2})_+$ and the embedding $\mathbf{O}(\Lambda^{2,1})_+ \hookrightarrow \mathbf{O}(\Lambda^{3,2})$, the multiplier restricts to $W^{(2)}(\Lambda^{2,1})$ as a character to $\{\pm 1\}$.

Lemma 7.1 *(Reflection-class values of ν_{Δ_5}).* For the three type-II simple reflections,

$$\nu_{\Delta_5}(s_{\partial_i}) \;=\; -1, \qquad i = 1, 2, 3.$$

For the three type-I chamber-automorphism reflections,

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) \;=\; \nu_{\Delta_5}(s_{f_2-f_3}) \;=\; \nu_{\Delta_5}(s_{f_{-2}-f_3}) \;=\; +1.$$

Proof. Maass's multiplier formula for the product of the ten even theta constants is [46, (1.3)–(1.5)], citing [76]. Under the exterior-square identification of Section 1.4, Gritsenko–Nikulin compute the reflection values at [46, Proposition 2.1] and the type-II chamber restriction at [46, Proposition 2.2]. The character is -1 on the three type-II simple reflections and $+1$ on the three $\mathrm{Aut}(\mathcal{P}_{II})$-reflections. □

Lemma 7.2 *(Multiplier restriction).* Under the chamber decomposition (4.5),

$$\nu_{\Delta_5}\big|_{W^{(2)}(\Lambda^{2,1}_{II})} \;=\; \det, \qquad \nu_{\Delta_5}\big|_{\mathrm{Aut}(\mathcal{P}_{II})} \;=\; 1.$$

Equivalently, for $h = wg$ with $w \in W^{(2)}(\Lambda^{2,1}_{II})$ and $g \in \mathrm{Aut}(\mathcal{P}_{II})$, the weight-five slash action without character satisfies

$$(\Delta_5|_5 h)(Z) \;=\; \det(w)\Delta_5(Z), \qquad (\Delta_5|_5 h)(Z) := j(h,Z)^{-5}\Delta_5(hZ),$$

where $j(h,Z)$ is the automorphy factor of the chosen $\mathbf{Sp}_4$-lift of h. Thus the lemma is a statement about the multiplier character, not a raw equality of functions on Siegel space.

Proof. Lemma 7.1 provides the values on the six generators of the chamber decomposition. The first generator class generates $W^{(2)}(\Lambda^{2,1}_{II})$ as a Coxeter group on three generators of det-character -1, hence $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det$. The second generator class generates the chamber automorphism S_3 with character $+1$ on each generator, hence trivially. □

Remark 7.3 *(Why the parity restricts to determinant).* The character ν_{Δ_5} has order two and is determined by six generator values; the type-II Coxeter relations among the s_{∂_i} force the restriction $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$ to be either trivial or the determinant. Since $\nu_{\Delta_5}(s_{\partial_i}) = -1 \neq 1$, the restriction is the determinant character. This is a character computation: it does not depend on a normalization of Δ_5 and is fixed by the group theory plus the six generator values. The finite packet `certificates/automorphic/delta5_maass_character` records these six values and checks the determinant and trivial restriction rows; it does not supply an orientation line or compact source trace.

8 The Borcherds-weight characteristic $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$

The Borcherds-weight characteristic of the BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ is the weight of the paramodular form whose product expansion is the denominator. The paramodular form here is $\Delta_5 = \Phi_1$ at level 1, with weight 5.

THEOREM 8.1 *(Universal Borcherds-weight identity for $\mathfrak{g}_{\Delta_5}$).*

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = \frac{10}{2} = 5. \tag{4.17}$$

The identity is universal across the five CHL frame shapes $N \in \{1, 2, 3, 4, 6\}$ with $\varphi(N) \mid 2$: for each N in the frame, the Borcherds product Φ_N has weight $c_N(0)/2$, with $c_N(0)$ the constant Fourier coefficient of the Borcherds input of Φ_N. The ladder is uniformly in the square-root normalization: at $N = 1$ the input is the square-root seed $\phi_{0,1}$ -- half the K3 elliptic genus $\mathrm{Ell}(K3) = 2\phi_{0,1}$ -- with $c_1(0) = f(0,0) = 10$; at $N \in \{2, 3, 4, 6\}$ the inputs are the Jatkar–Sen twisted square-root seeds, with constant terms $c_N(0) = (6, 4, 3, 2)$ per Jatkar–Sen and Govindarajan–Krishna [41, 42, 53]; the weight at $N = 4$ is half-integral. The table is:

N	1	2	3	4	6
$c_N(0)$	10	6	4	3	2
$\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$	5	3	2	3/2	1

Proof. Borcherds weight formula. The Borcherds weight formula [10, Theorem 13.3] states that for a Borcherds product $\Theta(\phi)$ on the Grassmannian $\mathbb{H}^{\mathbf{IV}}_+$ associated to a signature-(3, 2) lattice $\Lambda^{3,2}$ and a weakly holomorphic vector-valued modular form ϕ of weight $-1/2$ on $\mathbf{Mp}_2(\mathbb{Z})$ with constant Fourier coefficient $c(0,0)$, the weight of $\Theta(\phi)$ on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ is $\mathrm{wt}\,\Theta(\phi) = c(0,0)/2$. Pulled back along the exterior-square map from $\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$ to the projective orthogonal group of $\Lambda^{3,2}$, this is the weight on $\mathbf{Sp}_4(\mathbb{Z})$.

The $N = 1$ case. The input form is $\phi_{0,1} \in J^{\mathrm{weak}}_{0,1}$, the weight-zero index-one weak Jacobi form, with constant Fourier coefficient $c_1(0) = f(0,0) = 10$. Hence $\mathrm{wt}\,\Delta_5 = c_1(0)/2 = 10/2 = 5$, recovering equation (4.17).

The CHL-twined values for $N \in \{2, 3, 4, 6\}$. For $g_N \in M_{24}$ of order $N \in \{2, 3, 4, 6\}$ the canonical cycle shapes on the 24-dimensional Niemeier representation are

$$1^8 2^8, \quad 1^6 3^6, \quad 1^4 2^2 4^4, \quad 1^2 2^2 3^2 6^2$$

([20, §2]; Conway–Norton). The Borcherds input of Φ_N is the Jatkar–Sen twisted square-root seed attached to g_N, with constant Fourier coefficient $(c_2(0), c_3(0), c_4(0), c_6(0)) = (6, 4, 3, 2)$ per Jatkar–Sen and, for the composite rows, Govindarajan–Krishna [41, 42, 53], giving Borcherds weights $(3, 2, \frac{3}{2}, 1)$;

the $N = 4$ weight $\frac{3}{2}$ is half-integral, and Φ_4 is a form on the genus-two metaplectic cover. The identification $c_N(0) = \chi^{g_N}(K3) = a_1$ with the g_N-Lefschetz number –- the 1-cycle count of the frame shape, which gives the once-recorded constants $(8, 6, 4, 2)$ and the integral ladder $(5, 4, 3, 2, 1)$ –- is retracted: the frame-shape half-sum is the weight of the eta product $\eta_{g_N} = \prod_d \eta(d\tau)^{a_d}$, not the Jacobi constant term of the Borcherds input. The table $(c_1(0), c_2(0), c_3(0), c_4(0), c_6(0)) = (10, 6, 4, 3, 2)$ is uniformly in the square-root normalization, with the $N = 1$ entry $c_1(0) = f(0, 0) = 10$ of $\phi_{0,1}$ –- half the constant coefficient 20 of $\mathrm{Ell}(K3) = 2\phi_{0,1}$.

Universality of the Borcherds product construction. The Borcherds singular-theta lift $\Theta(\phi_{0,1}^{(N)})$ of the g_N-twined input is well-defined for each $N \in \{1, 2, 3, 4, 6\}$ by [10, Theorem 13.3] applied to the corresponding CHL lattice $\Lambda^{(N),3,2}$; the weight is $c_N(0)/2$ on $\widetilde{\mathbf{O}}(\Lambda^{(N),3,2})_+$, pulled back to a Siegel modular form of weight $\kappa_{\mathrm{BKM}}(\Phi_N)$ (at $N = 4$, of half-integral weight $\frac{3}{2}$ on the genus-two metaplectic cover). For each $N \in \{1, 2, 3, 4, 6\}$, the table values follow from the Jatkar–Sen constant terms.

Separation from the eight-row Gritsenko–Cléry catalogue. The Gritsenko–Cléry catalogue [44, Theorem 1.4] is indexed by the eight pairs

$$(1, 1), (2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2),$$

or equivalently by the F_j-rows of Appendix 2. Its weights and constants in the input-pair order are

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1), \qquad (10, 4, 6, 2, 4, 1, 3, 2),$$

and each row satisfies $\mathrm{wt}(F_j) = c_j(0)/2$. The CHL-twining table of the present theorem is indexed by automorphisms g_N with $N \in \{1, 2, 3, 4, 6\}$ and has constants $(10, 6, 4, 3, 2)$. The common entry used by the $K3 \times E$ host is the $(t, N) = (1, 1)$ row with Jacobi seed $\phi_{0,1}$, Borcherds product Δ_5, and weight 5. $\square$

Remark 8.2 *(The additive formula fails).* The additive expression

$$\kappa_{\mathrm{BKM}}(\Phi_N) \overset{?}{=} \kappa_{\mathrm{ch}}(\mathcal{A}_X) + \chi(\mathcal{O}_{\mathrm{fiber}})$$

has no invariant meaning on the κ-tuple. On the Hodge reading at $N = 1$,

$$\kappa_{\mathrm{BKM}}(\Phi_1) = 5 \neq \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(\mathcal{A}_{K3\times E}) + \chi(\mathcal{O}_{K3}) = 0 + 2 = 2.$$

On the Heisenberg reading,

$$\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(\mathcal{A}_{K3\times E}) + \chi(\mathcal{O}_{K3}) = 3 + 2 = 5$$

matches the $N = 1$ number only accidentally: the same fixed right hand side cannot produce the CHL weights $c_N(0)/2 = (5, 3, 2, \frac{3}{2}, 1)$ for $N = 1, 2, 3, 4, 6$. The universal rule is the rowwise Borcherds weight $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$. The data live in distinct entries of the κ-tuple

$$(\kappa_{\mathrm{cat}},\ \kappa_{\mathrm{ch}}^{\mathrm{Hodge}},\ \kappa_{\mathrm{ch}}^{\mathrm{Heis}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}}) \;=\; (0, 0, 3, 5, 24),$$

matching the Vol III κ-stratification of Calabi–Yau quantum groups; the Borcherds-weight identity (4.17) is the correct universal formula on the BKM entry, and the additive formula confuses $\chi_{\mathrm{top}}(K3) = 24$ with $\chi(\mathcal{O}_{K3}) = 2$. The universal failure across the five-CHL frame $N \in \{1, 2, 3, 4, 6\}$ is the level-by-level evaluation of the κ-tuple recorded in Vol III.

Remark 8.3 *(Two Borcherds inputs and one weight rule).* The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ of [10, Theorem 13.3] applies after the input form, lattice, and Weyl chamber are fixed. The paramodular Δ_5 row, with Jacobi input $\phi_{0,1}$ and Borcherds product Δ_5 on $\Lambda_{II}^{2,1}$, reads $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 10/2 = 5$. The Vol I row records the Fake-Monster Borcherds product Φ_{12} on the Leech-extension lattice $\mathrm{II}_{25,1}$ [8, 9], reading $\kappa_{\mathrm{BKM}}(\Phi_{12}) = c_{\Phi_{12}}(0)/2 = 12$. The values 5 and 12 are different inputs to the same Borcherds-weight rule, on different lattices, not a value disagreement and not a shifted indexing of one input; each site must name its input form and lattice. The additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ is not a structural identity: the Hodge reading fails at $N = 1$, and the Heisenberg reading gives only the accidental equality $3 + 2 = 5$. The number 5 is fixed by $c_1(0)/2$; the chiral and fibre entries belong to different slots of the κ-tuple.

9 The denominator-algebra theorem and its scope

Theorem 9.1 *(Denominator-algebra identity).* The denominator algebra $\mathfrak{g}_{\Delta_5}$ associated to the Borcherds–Cartan superdatum $\mathscr{B}_{\Delta_5}$ has the following properties.

(i) *Triangular decomposition.*

$$\mathfrak{g}_{\Delta_5} = \Big(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_\alpha\Big) \oplus \mathfrak{g}_{\Delta_5 0} \oplus \Big(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_{-\alpha}\Big),$$

where $\mathfrak{g}_{\Delta_5 0} = \Lambda_{II}^{2,1} \otimes \mathbb{C}$ is the Cartan subalgebra. The chamber-interior and cusp-boundary summands decompose only the positive half:

$$\mathfrak{g}_{\Delta_5 +} = \mathfrak{g}_{\Delta_5 +}^{\mathrm{int}} \oplus \mathfrak{g}_{\Delta_5 +}^{\mathrm{cusp}}, \qquad \mathfrak{g}_{\Delta_5 +}^{\mathrm{int}} = \bigoplus_{\alpha\in\mathcal{R}_+,\, m(\alpha)>0} \mathfrak{g}_\alpha, \quad \mathfrak{g}_{\Delta_5 +}^{\mathrm{cusp}} = \bigoplus_{\alpha\in\mathcal{R}_+,\, m(\alpha)=0} \mathfrak{g}_\alpha.$$

(ii) *Bracket compatibility.*

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}.$$

(iii) *Visible signed multiplicity.* On the active support

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

(iv) *Denominator identity.*

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \frac{1}{64}\Delta_5(2Z).$$

(v) *Borcherds-weight identity.*

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = 5.$$

The theorem concerns the Borcherds denominator algebra. A microscopic state space and its operator product require separate compact realization; the operator-level pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ is defined in Chapter 6, and its protected Pfaffian is identified with Δ_5 by the Pfaffian–Dirac theorem of Chapter 7.

Proof. Parts (i) and (ii) are the standard structure of a generalized Kac–Moody Lie superalgebra associated to the superdatum $\mathscr{B}_{\Delta_5}$ and the relations of Definition 4.2; see [46, Sections 3–4] for the construction and [8, 58] for the underlying presentation theory.

Part (iii) is (4.14), comparing the Borcherds product expansion of D_5 (Theorem 5.3) with the Weyl–Kac–Borcherds denominator product (Theorem 5.4). The product exponents on the active Gram-index support Γ_{act} are the Fourier coefficients $f(nm, l)$ of $\phi_{0,1}$, and the BKM signed multiplicities smult(α) are read off equality of the two expansions in their common Lorentzian chamber.

Part (iv) is Theorem 5.4: the doubling $Z \mapsto 2Z$ identifies the type-II sublattice grading with the BKM root lattice grading, and the prefactor 64 is the theta-constant normalization (4.9).

Part (v) is Theorem 8.1, the Borcherds-weight characteristic at $N = 1$. □

Corollary 9.2 *(Real-root and isotropic multiplicities).* The signed root supermultiplicity smult takes values

$$\mathrm{smult}(\delta_i) = 1 \quad (i = 1, 2, 3), \qquad \mathrm{smult}(a_{ij}) = 10 \quad (i < j), \qquad \mathrm{smult}(\delta_{123}) = -64,$$

with the parity split at δ_{123} given by $\mathrm{mult}_{\bar{0}}(\delta_{123}) = 29$, $\mathrm{mult}_{\bar{1}}(\delta_{123}) = 93$ of Proposition 6.7.

Proof. The values follow from the active support table (4.16) and Proposition 6.7. □

10 Boundary with View I (automorphic) and View V (theta lift)

The denominator algebra is the second of the five views of Δ_5. Two cross-level identities to neighbouring views are already theorems and are recorded here as boundary statements; the third cross-level identity, View II ↔ View III, is the Pfaffian–Dirac theorem of Chapter 7, with primitive recognition separated as $(S1)$–$(S10)$.

Theorem 10.1 *(Cross-level identity* View I ↔ View II, *Borcherds 1998 / Gritsenko–Nikulin 1998).* The Borcherds denominator of the BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$ is the doubled Igusa cusp form, normalized by the theta-constant constant:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z).$$

Proof. This is Theorem 5.4 in the present chapter, equivalently [46, Sections 3–4, Proposition 3.1], with explicit product expansion in [48, Theorem 2.1, (2.1)–(2.3)] and the universal weight characteristic in [10, Theorem 13.3]. □

Theorem 10.2 *(Cross-level identity* View II ↔ View V, *Borcherds 1998).* The Borcherds denominator of $\mathfrak{g}_{\Delta_5}$ is obtained from the singular theta lift of the Weil-representation theta component $\phi_{0,1}^{\mathrm{Weil}}$ of $\phi_{0,1} \in J_{0,1}^{\mathrm{wk}}$ on the lattice $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$. The bridge between the two sides is the exceptional exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ of Section 1.4.

Proof. This is the singular theta lift theorem [10, Theorem 13.3] specialized to the lattice $\Lambda^{3,2}$ and to the Weil-representation form $\phi_{0,1}^{\mathrm{Weil}}$ obtained by theta-decomposing the Jacobi input $\phi_{0,1}$, equivalently the additive lift in Gritsenko–Nikulin [46, Section 1, Theorem 2.1] cast through the exceptional isogeny. The bridge $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the metaplectic Weil action on the input are the subject of Chapter 5. □

11 The $\mathfrak{g}_{\Delta_5}$ denominator in the κ-tuple

The denominator algebra $\mathfrak{g}_{\Delta_5}$ sits at the BKM entry of the κ-tuple

$$(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24),$$

of Vol III's CY_d-stratification. The five-tuple records: $\kappa_{\mathrm{cat}} = 0$ for the categorical-Hodge characteristic, $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 0$ for the chiral-Hodge refined supertrace, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ for the Heisenberg rank, $\kappa_{\mathrm{BKM}} = 5$ for the cusp-form weight, and $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(K3) = 24$.

The five entries are independently defined and not summed: the categorical-Hodge characteristic is the trace of the BV bracket on the K3 chiral algebra; the chiral-Hodge refined supertrace is read on the protected reduced category; the Heisenberg rank is the rank-three type-II Heisenberg–Cartan obtained from the normal-ordered Gram projection to $\Lambda_{II}^{2,1}$; the cusp-form weight is $\kappa_{\mathrm{BKM}} = c_1(0)/2 = 5$ from Theorem 8.1; the fibre Euler character $\chi_{\mathrm{top}}(K3) = 24$ records the topology of the K3 fibre.

Remark 11.1 *(Paramodular row versus Fake-Monster row).* The value $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$ is the Borcherds-weight $c_1(0)/2$ on the paramodular row with Jacobi input $\phi_{0,1}$ at $\Lambda_{II}^{2,1}$. The Fake-Monster value $\kappa_{\mathrm{BKM}}(\Phi_{12}) = 12$ is the Borcherds-weight $c_{\Phi_{12}}(0)/2$ on a different input, the Fake-Monster Borcherds product on $\mathrm{II}_{25,1}$ of [8, 9]; the two rows are independent evaluations of one universal rule (Remark 8.3). The BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ realizes the paramodular row.

12 STRUCTURAL PROPERTIES OF $\mathfrak{g}_{\Delta_5}$

Real-root sub-Kac–Moody algebra

LEMMA 12.1 *(Real rank-three sub-Kac–Moody algebra).* The subalgebra ${\mathfrak{g}_{\Delta_5}}^{re} \subset \mathfrak{g}_{\Delta_5}$ generated by $\{e_i, f_i, h_i\}_{i=1}^3$ together with the Cartan $\Lambda_{II}^{2,1} \otimes \mathbb{C}$ is the rank-three hyperbolic Kac–Moody algebra $\mathfrak{g}(A)$ associated to the symmetric Cartan matrix $A = ((\delta_i, \delta_j)) = \left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$.

Proof. The relations of Definition 4.2 restricted to real $\alpha_i \in \Delta_{\mathrm{simp}}^{re}$ are exactly the Kac–Moody relations of [58, Chapter 1] for the symmetric Cartan matrix A. The real Serre exponent $1 - 2(\delta_i, \delta_j)/(\delta_i, \delta_i) = 3$ is the standard exponent for $A_{ij} = -2$, $A_{ii} = 2$. □

Remark 12.2 *(Hyperbolic structure of ${\mathfrak{g}_{\Delta_5}}^{re}$).* The Kac–Moody algebra $\mathfrak{g}(A)$ is hyperbolic in the sense of [58, Chapter 4]: every proper connected rank-two principal submatrix is

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix},$$

the affine $A_1^{(1)}$ Cartan matrix with null root $\delta_i + \delta_j$, while the full rank-three matrix has Lorentzian signature. The full BKM algebra $\mathfrak{g}_{\Delta_5}$ is the automorphic correction of $\mathfrak{g}(A)$ in the sense of [46, Section 3], i.e. adjoining the imaginary simple roots $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ with multiplicities $\mu_{\overline{p}}(a)$ to obtain the algebra whose denominator identity matches $64^{-1}\Delta_5(2Z)$.

Bilinear pairing on $\mathfrak{g}_{\Delta_5}$

LEMMA 12.3 *(Invariant bilinear form).* There is a unique non-degenerate invariant supersymmetric bilinear form

$$(\cdot, \cdot)_{\mathfrak{g}_{\Delta_5}} : \mathfrak{g}_{\Delta_5} \otimes \mathfrak{g}_{\Delta_5} \longrightarrow \mathbb{C}$$

extending the bilinear form on $\Lambda_{II}^{2,1} \otimes \mathbb{C} \subset {\mathfrak{g}_{\Delta_5}}_0$ and satisfying the ad-invariance

$$([X, Y], Z)_{\mathfrak{g}_{\Delta_5}} = (X, [Y, Z])_{\mathfrak{g}_{\Delta_5}}.$$

The pairing $(\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha})_{\mathfrak{g}_{\Delta_5}}$ is non-degenerate; pairings between non-opposite root spaces vanish.

Proof. Standard for Borcherds–Kac–Moody algebras; see [8, Section 1] and [58, Section 2.2]. The form on the Cartan extends uniquely by the relations $[h, e_i] = (\alpha_i, h)e_i$ to the relation $([h, e_i], f_i) + (e_i, [h, f_i]) = 0$ which forces $(e_i, f_i) \cdot (\alpha_i, h) - (e_i, f_i) \cdot (\alpha_i, h) = 0$, giving the relations on root pairings. □

Weyl-equivariance of root spaces

Lemma 12.4 *(Even Weyl transport of parity).* For every $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and every root $\alpha \in \Delta$, the Weyl operator induces a parity-preserving isomorphism

$$\mathfrak{g}_{\alpha,\overline{p}} \;\simeq\; \mathfrak{g}_{w\alpha,\overline{p}}, \qquad \overline{p} \in \mathbb{Z}/2.$$

Proof. This is the Weyl-transport-of-parity statement of [58, Lemma 3.8 and Proposition 3.7], with the GN super-version of [46, Section 6, Statement 6.4]. Since the GN real simple generators are even, the reflection operators are even automorphisms of $\mathfrak{g}_{\Delta_5}$. Hence Weyl transport preserves the parity decomposition. □

13 Comparison with the chiral shadow lattice Γ_{ind}

The Gram-index parametrization $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$ of $\Lambda_{II}^{2,1}$ via $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$ provides the comparison between the BKM denominator algebra and the K3 half-index $\phi_{0,1}$. The Gram-index pairing on Γ_{ind} is

$$\langle (n, l, m), (n', l', m') \rangle_{\mathrm{ind}} \;=\; 2(nm' + n'm) - ll',$$

and the pullback to the Lorentzian pairing on $\Lambda_{II}^{2,1}$ is (4.3).

Remark 13.1 *(The denominator polarization vs. the cusp polarization).* The Gram-index lattice carries two polarizations:

- the cusp polarization, in which $\Gamma_{\mathrm{eff}} \subset \Gamma_{\mathrm{ind}}$ is the effective subset (the active support of the K3 half-index in the Borcherds product) and the order is the Fock polarization at one cusp;
- the denominator polarization, in which Γ_{ind} maps via α to $\Lambda_{II}^{2,1}$ and the order is the type-II chamber $\mathcal{P}_{II}$.

The two polarizations are the same lattice with two ordered charts; the denominator side is the Borcherds chamber with $W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$ symmetry, the cusp side is the Fock determinant at the chosen cusp. A duality element transports between the two; cf. Section 1.4 of Chapter 2.

Visible coefficients

Lemma 13.2 *(Active multiplicity equals $f(nm, l)$).* On the active support $\Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{eff}}$,

$$\mathrm{smult}(\alpha(n, l, m)) \;=\; f(nm, l).$$

For triples $(n, l, m) \in \Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$, the product exponent vanishes and no visible product factor is contributed; the BKM root space $\mathfrak{g}_{\alpha(n,l,m)}$ is not pinned by the Borcherds product alone.

Proof. The first equality is (4.14), comparing the Borcherds product expansion with the Weyl–Kac–Borcherds denominator. The second statement records that exponent zero on a triple (n, l, m) does not exclude the existence of a root space with smult = 0; the Borcherds product expansion does not see zero-superdimension root spaces. □

What the determinant determines

PROPOSITION 13.3 *(Information content of the denominator product).* The denominator identity

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z)$$

determines the signed integers $\mathrm{smult}(\alpha)$ on the active support after the BKM root system $\mathcal{R}_+$ has been fixed. It does not determine:

- invisible zero-superdimension root spaces;
- the parity decomposition $\dim \mathfrak{g}_{\alpha,\bar{0}}$, $\dim \mathfrak{g}_{\alpha,\bar{1}}$ beyond their difference;
- the structure constants of the BKM bracket $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \to \mathfrak{g}_{\alpha+\beta}$;
- simple primitive representatives, no-extra-relations theorems, or completed PBW compatibility;
- the closed radical of the positive-negative invariant pairing.

Consequently the denominator product is a necessary input for primitive recognition; it is not by itself a recognition datum.

Proof. Write $X^\alpha = e^{-2\pi i(\alpha,z)}$. In the completed positive root monoid algebra,

$$\log \prod_{\alpha\in\mathcal{R}_+} (1 - X^\alpha)^{\mathrm{smult}(\alpha)} \;=\; -\sum_{\alpha\in\mathcal{R}_+}\sum_{k\geq 1} \frac{\mathrm{smult}(\alpha)}{k} X^{k\alpha}.$$

The exponents $\mathrm{smult}(\alpha)$ are recovered recursively by height, equivalently by Mobius inversion on each primitive ray. The recovered exponent is only $\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha)$. The difference $a - b$ does not determine the pair (a, b). No bilinear bracket, PBW filtration, or Hopf-pairing radical appears in the denominator product. □

What the determinant forgets

PROPOSITION 13.4 *(Grothendieck class as the determinant invariant).* For every Γ_{eff}-graded super vector space $\mathcal{B} = \bigoplus_\gamma \mathcal{B}_\gamma$ with $\mathrm{sdim}\,\mathcal{B}_{(n,l,m)} = f(nm, l)$, the Fock determinant is

$$64q^{1/2}r^{1/2}s^{1/2}\prod_\gamma (1 - x^\gamma)^{\mathrm{sdim}\,\mathcal{B}_\gamma} \;=\; \Delta_5(Z).$$

The determinant remembers only the class of $\mathcal{B}$ in

$$K_0(\Gamma_{\mathrm{eff}}\text{-graded finite super vector spaces}).$$

A bracket $[\mathcal{B}_\gamma, \mathcal{B}_{\gamma'}] \to \mathcal{B}_{\gamma+\gamma'}$ does not appear in the determinant formula, nor do simple primitive representatives, parity dimensions, or pairing radicals.

Proof. The Fock determinant lemma gives $\mathrm{sdet}_{\mathcal{B}}(1 - x) = \prod_\gamma (1 - x^\gamma)^{\mathrm{sdim}\,\mathcal{B}_\gamma}$. Multiplying by the vacuum factor and using the hypothesis $\mathrm{sdim}\,\mathcal{B}_{(n,l,m)} = f(nm, l)$ identifies this product with the Borcherds product for $64^{-1}\Delta_5$. Replacing $\mathcal{B}$ by any other graded super vector space with the same Grothendieck class preserves the determinant; the zero bracket on such a space gives the same determinant and a non-isomorphic Lie algebra; adjoining a cancelling pair $W_\gamma \oplus \Pi W_\gamma$ in any active degree leaves the determinant unchanged while changing the parity dimensions. Hence the determinant invariant is the K_0-class. □

14 Comparison: rank-three hyperbolic vs. Borcherds correction

The denominator algebra $\mathfrak{g}_{\Delta_5}$ is the BKM correction of the rank-three hyperbolic Kac–Moody algebra $\mathfrak{g}(A)$ on the same Cartan matrix A. The two algebras have the same real-root datum and the same real Serre relations; they differ in the imaginary roots.

Table 4.1: Real-root datum of $\mathfrak{g}_{\Delta_5}$ and the rank-three hyperbolic $\mathfrak{g}(A)$.

	$\mathfrak{g}(A)$, $A = \left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$	$\mathfrak{g}_{\Delta_5}$
real simple roots	$\delta_1, \delta_2, \delta_3$	$\delta_1, \delta_2, \delta_3$
real-root mult.	1 each	1 each
real Serre exponent	3	3
imaginary simple roots	none	$\Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$
$\mu_{\bar{0}}(a)$	0	$\max(m(a), 0)$ or $\max(\tau(a), 0)$
$\mu_{\bar{1}}(a)$	0	$\max(-m(a), 0)$ or $\max(-\tau(a), 0)$
denominator	no closed product	$64^{-1}\Delta_5(2Z)$

Remark 14.1 *(Borcherds correction adjoins the imaginary simple roots).* The Gritsenko–Nikulin construction [46, Sections 3–4, Proposition 3.1] adjoins to $\mathfrak{g}(A)$ the imaginary simple generators with multiplicities $\mu_{\bar{p}}(a)$ from the Borcherds-product expansion of the additive lift of $\eta^9\vartheta$. The resulting BKM Lie superalgebra has denominator $64^{-1}\Delta_5(2Z)$, while $\mathfrak{g}(A)$ has no closed denominator product as a Kac–Moody algebra alone (its denominator is a formal power series whose closed form is unknown). The correction algebra is the central object of View II.

15 The character of the chiral shadow

The denominator algebra $\mathfrak{g}_{\Delta_5}$ is the Beilinson cut's chiral shadow at View II. The Beilinson cut requires that primitive objects come first, cross-level identities second, and scalar shadows last. Inscribed at View II:

- *Primitive object* (View III, relative to retained data): the supplied finite E_3-prefactorization input of the K3×E formal target, presented at finite stage as the pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ of Chapter 6.
- *Cross-level identity* (View II): the Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, with generators and relations as recorded above.
- *Automorphic determinant* (View I, K_0-determinant): the automorphic section $\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$, with squared determinant section $\Phi^{\mathrm{un}}_{10} = \Delta_5^2$ at weight 10.

The denominator algebra is target-side: it imports the Gritsenko–Nikulin Cartan, the Chevalley–Borcherds–Kac relations, the Weyl vector, and the Borcherds product, and supplies no compact Hall correspondence, no orientation line, no Pfaffian section, and no microscopic state space. The Pfaffian and orientation part of the cross-level identity View II $\leftrightarrow$ View III is the Pfaffian–Dirac theorem after $(D0)$, $(O1)$, $(O1)^+$, and the retained $(O2_{\mathrm{atlas}})$ datum, together with the finite geometric Pfaffian datum (P_{fin}), of Chapter 6. The primitive identification $P_X \simeq \mathfrak{g}_{\Delta_5}$ has the additional finite compact-source recognition hypothesis $(S1)$–$(S10)$ and the finite Koszul comparison datum of Definition 0.211.

Remark 15.1 *(The chiral shadow is target, not bulk).* The Beilinson cut classifies $\mathfrak{g}_{\Delta_5}$ as a chiral shadow, not a chiral bulk. The bulk object lives on the source side: at a finite retained stage the chiral bulk is $Z^{\mathrm{der}}_{\mathrm{ch}}(C_{X,R}) \simeq \mathrm{ChirHoch}^\bullet(C_{X,R}, C_{X,R})$, identified on the Koszul locus by Vol I Theorem H [72] together with Vol II's chiral-Hochschild Beilinson stratification (i)+(ii) [73], attached to a primitive

$C^{op} = C^{op}_{X,R}$ on the closed factorization datum (X, D, τ) at the boundary vacuum b. Here $X = K3 \times E$ is the formal target, D the chosen boundary divisor at infinity carrying the cusp polarization, and $\tau \in H^*(D, \mathbb{Q})$ the boundary trace class supplying the factorization gluing — the data of a closed chiral C^{op} in the sense of [73, §2]. The denominator algebra $\mathfrak{g}_{\Delta_5}$ is the chiral shadow of this bulk, not the bulk itself; the bulk object lives one categorical level above the chiral shadow, and the BKM denominator records only the shadow.

Remark 15.2 *(Beilinson chain on $K3 \times E$).* The Universal Holography master theorem of Vol II [73, § 9] identifies, on a closed factorisation datum (X, D, τ) at boundary vacuum b:

$$\text{boundary} = A_b, \qquad \text{bulk} = Z^{\mathrm{der}}_{\mathrm{ch}}(A_b), \qquad \text{interaction} = \mathrm{SC}^{\mathrm{ch,top}}\text{-brace}.$$

On $K3 \times E$, the compact-source side supplies the *boundary chart* $A_b = \mathrm{End}_C(b)$ at the boundary vacuum b, with C the open factorisation dg-category of Chapter 6, the augmented bar coalgebra $\mathrm{Bar}(A_b)$ and its MC-twisting structure of [73, §6], and the chiral shadow $\mathfrak{g}_{\Delta_5} = \mathrm{shadow}(Z^{\mathrm{der}}_{\mathrm{ch}}(A_b))$ of the bulk algebra at the BKM level — but *not* the bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$ itself as a chiral algebra on $K3 \times E$ with explicit OPE, which is the level-Z object of Vol II. The chain-level trace identification of Theorem 7.23 supplies the level-Z criterion (Definition 7.20) relative to (Z_{HB}), Vol I Theorem H, and Vol II's chiral-Hochschild Beilinson stratification; the level-A gravity-line operator-algebra facet remains open in Vol II. Equivalently, the Beilinson cut $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$ is exhibited at $\mathsf{P},\mathsf{C},\mathsf{S}$ as chiral-shadow constructions of Chapters 4 and 6, at $\mathsf{S} \to \mathsf{Z}$ as a retained trace identity on the chain-level chiral derived centre (Theorem 7.23), and at $\mathsf{Z} \to \mathsf{A}$ as the open gravity-line Hall–Borcherds residual. The K3$\times E$ threefold realisation question of Chapter 9 is the level-A open problem.

16 The denominator algebra of the chiral source-target frame

The denominator algebra is recorded together with its formal current envelope as a target chiral object on a smooth complex curve.

Definition 16.1 *(Formal current envelope).* For a smooth complex curve C, set

$$\mathrm{Cur}_C(\mathfrak{g}_{\Delta_5}) \;:=\; \mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_C$$

with the level-zero current bracket

$$[a \otimes f, b \otimes g] \;=\; [a, b] \otimes fg.$$

The chiral enveloping algebra of Beilinson–Drinfeld [7] is

$$\mathsf{Den}_{\Delta_5,C} \;:=\; U^{\mathrm{ch}}_C\big(\mathrm{Cur}_C(\mathfrak{g}_{\Delta_5})\big).$$

PROPOSITION 16.2 *(Formal current envelope realizes the denominator).* The chiral algebra $\mathsf{Den}_{\Delta_5,C}$ is the formal completed ind Beilinson–Drinfeld current envelope on C obtained from the denominator Lie superalgebra. Its constant-current Lie superalgebra is $\mathfrak{g}_{\Delta_5}$, and the denominator of that constant-current bracket is $64^{-1}\Delta_5(2Z)$.

For $C = E$, the elliptic curve in $X = K3 \times E$, the chiral algebra $\mathsf{Den}_{\Delta_5,E}$ is the target chiral object used in the $K3 \times E$ realization problem of Chapter 6. A compact E_3/Hall/chiral protected realization requires the retained Dirac–Igusa pro-object data, the Hall correspondences, the orientation line, and the Koszul comparison.

Proof. In the completed ind setting, the current construction sends a Lie superalgebra to a Lie-* algebra on a smooth curve, and the chiral enveloping construction sends a Lie-* algebra to a chiral algebra [7]. Equivalently, in vertex-algebra language, this is the universal current vertex algebra at level zero [57]. Constant sections of $\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E$ recover the original Lie superalgebra, with denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ from Theorem 9.1. □

17 The closing identity and its conditioning

The denominator identity is

$$\boxed{\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z), \qquad \kappa_{\mathrm{BKM}}(\Delta_5) \;=\; 5, \qquad \mathrm{smult}(\alpha(n,l,m)) \;=\; f(nm,l).}$$

These three are imported theorems: the first from Borcherds 1995 [9] and Gritsenko–Nikulin 1998 [48] [48, Theorem 2.1, (2.1)–(2.3)], with the algebra constructed in [46, Sections 3–4, Proposition 3.1]; the second from Borcherds 1998 [10, Theorem 13.3] with the Vol III specialization to the paramodular family; the third from the same Borcherds product expansion as the first, by comparison with the Weyl–Kac–Borcherds denominator. The conditioning of the closing identity is target-side. No compact Hall correspondence, Pfaffian section, or microscopic state space is claimed.

The cross-level identity View II ↔ View III, the Pfaffian–Dirac theorem of Chapter 7, asks for the protected Pfaffian

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) \;=\; \Delta_5,$$

the orientation character $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$, and the primitive recognition theorem $P_X \simeq \mathfrak{g}_{\Delta_5}$, conditional on the finite compact-source recognition datum $(S1)$–$(S10)$ in addition to the four orientation and Pfaffian obstructions. The BKM denominator identity is independent of those four obstructions and is imported from primary literature.

Cited identities

The denominator calculation uses only proved identities. The type-II Cartan matrix of Lemma 2.2 and the Weyl vector $\rho = f_2 - \frac{1}{2}f_3 + f_{-2}$ of Proposition 2.4 are direct lattice computations. The reflection-group decomposition is Lemma 1.3, with Nikulin's theorem as input [88]. Real-root multiplicity one, Lemma 4.6, follows from the Kac–Moody root-space theorem and the Gritsenko–Nikulin normalization [46, 58]. The imaginary simple-root parity split is Definition 3.2, read from [46, Section 3]. The Maass multiplier values and their restriction to the Weyl group are Lemmas 7.1 and 7.2, using [46, 76] and [46, Prop. 2.1, 2.2]. The Borcherds product for D_5 is Theorem 5.3, from [10, Thm. 13.3] and [48, Thm. 2.1]. The Weyl alternating sum is Theorem 5.1, from [46, Thm. 2.1]. The denominator identity (4.13) is the Gritsenko–Nikulin denominator theorem [46, Sections 3–4], [48, Thm. 2.1]. The Borcherds-weight value $\kappa_{\mathrm{BKM}} = 5$ is Theorem 8.1, from Borcherds' weight formula [10, Thm. 13.3]. The 29|93 presentation count of Proposition 6.7 is the direct finite count against the Gritsenko–Nikulin simple-root lists. The triangular decomposition and invariant bilinear form are Lemmas 4.4 and 12.3, by the general Borcherds–Kac–Moody theory [8, 58].

The closing identity is therefore proved as recorded. Its Pfaffian realization is the theorem of Chapter 7; its primitive compact-source realization remains conditional on $(S1)$–$(S10)$.

18 Arithmetic consistency of the denominator identity

The denominator-algebra theorem is constrained by the following computations; each one is fixed by the recorded primary sources and the Borcherds-weight identity.

Sign of imaginary multiplicities

The Borcherds product exponent $f(nm, l)$ on the active support determines the signed BKM multiplicity. The invariant smult is defined as the difference $\mathrm{mult}_{\bar{0}} - \mathrm{mult}_{\bar{1}}$ (Definition 4.5), and the comparison of the Borcherds product (4.10) with the Weyl–Kac–Borcherds denominator product (4.12) gives $\mathrm{smult}(\alpha(n,l,m)) = f(nm,l)$ on Γ_{act}. The parity split of smult into $(\mathrm{mult}_{\bar{0}}, \mathrm{mult}_{\bar{1}})$ requires the Gritsenko–Nikulin imaginary simple-root data $\mu_{\bar{p}}(a)$, recorded in Definition 3.2, and the Borcherds–Kac bracket calculus in the degree under consideration. The data $m(a)$ and $\tau(a)$ give the signed simple fibres; lower brackets supply the remaining summands of the full root space. Compare [46, Section 3].

Doubled vs. undoubled Δ_5

The denominator identity uses $\Delta_5(2Z)$. The additive Gritsenko lift produces the monomial product for $D_5(Z)$, and the root map $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ lands in $\Lambda_{II}^{2,1}$. The substitution $Z \mapsto 2Z$ changes $\exp(-\pi i(\alpha, z))$ into the standard denominator character $\exp(-2\pi i(\alpha, z))$; no equality of the ambient and type-II lattices is asserted. The prefactor $64 = [q^{1/2}r^{1/2}s^{1/2}]\Delta_5$ is the theta-constant normalization and is untouched by doubling. See Remark 5.5.

Doubling factor as power of two

The factor 64 is the value of the leading Fourier coefficient $c_\Delta(1,1,1)$ of Δ_5, computed in the Maass formula for the product of ten even theta constants [46, (1.3)–(1.5)]. It is not a half-Hilbert orbit count. The orbit-count factor $2^{12} = 4096$ arises in the type-II Pfaffian wall normal form $(O2)$ of Chapter 6 as the size of the half-Hilbert orbit on the local sign atlas; it is unrelated to the denominator prefactor 64. See Remark 5.6.

Simple-root system on $\Lambda_{II}^{2,1}$

Nikulin's reflection-group theorem [88] fixes $W^{(2)}(\Lambda_{II}^{2,1})$ as a normal subgroup of index 6 in $\mathbf{O}(\Lambda^{2,1})_+$, and the Gritsenko–Nikulin chamber $\mathcal{P}_{II}$ has fixed walls $\delta_1, \delta_2, \delta_3$ with Gram matrix (4.6). Any simple-root triple for the same type-II chamber is Weyl-conjugate to a permutation of $(\delta_1, \delta_2, \delta_3)$. The BKM algebra depends on the chamber Cartan datum, not on a labelling of its three walls.

Weyl vector ρ

The Weyl vector is fixed by the condition $(\rho, \delta_i) = -(\delta_i, \delta_i)/2 = -1$ for $i = 1,2,3$. The displayed value $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ satisfies this and lies in $(\Lambda_{II}^{2,1})^* \cap C(\Lambda^{2,1})_+$; $\rho' = \delta_1 + \delta_2 + \delta_3$ would have $(\rho', \delta_i) = -2$, violating the condition. The factor $\frac{1}{2}$ is forced. See Proposition 2.4.

Real-root norm convention

The lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ is even by (4.1). The type-II sublattice $\Lambda_{II}^{2,1}$ is also even (a sublattice of an even lattice). Hence $(\delta, \delta) \in 2\mathbb{Z}$ for every $\delta \in \Lambda_{II}^{2,1}$. The choice $(\delta, \delta) = 2$ is forced for primitive square-norm-2 vectors. The convention $(\delta_i, \delta_i) = 1$ does not occur.

Second Serre bracket compatibility (Gram cocycle)

The Gram cocycle $B(c, c')$ is the source-side quadratic correction on the upstairs lattice; its compatibility with the BKM bracket on $\Lambda_{II}^{2,1}$ is mediated by the normal-ordered homomorphism $\overline{\Pi}_X : \widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$ of Chapter 2, which corrects the raw quadratic shadow $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ to the additive map. The target-side Lorentzian Cartan pairing on $\Lambda_{II}^{2,1}$ needs no Gram cocycle correction. The compatibility is precisely between $\overline{\Pi}_X$ on the source and the BKM grading on the target, as established in Chapter 2.

Alternative Cartan-matrix choice

The symmetrized matrix $\left(\begin{smallmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{smallmatrix}\right)$ is not the Cartan datum of the denominator algebra. The matrix A in (4.6) has integer entries with $A_{ii} = 2$, and is the symmetrized form fixed by the even-lattice convention. A rescaled matrix $\frac{1}{2}A$ with $A_{ii} = 1$ would not be a generalized Cartan matrix in the Kac–Moody sense for the real simple roots, where the standard normalization is $A_{ii} = 2$ [58, Chapter 1]. The symmetric form A is the normalization compatible with the even lattice $\Lambda_{II}^{2,1}$.

The two values of κ_{BKM}

The values $\kappa_{\mathrm{BKM}} = 5$ and $\kappa_{\mathrm{BKM}} = 12$ index different Borcherds inputs to the same universal rule $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ of [10, Theorem 13.3]. The Δ_5 row evaluates the rule at the paramodular row with Jacobi seed $\phi_{0,1}$ and Borcherds product Δ_5 on $\Lambda_{II}^{2,1}$, with $c_1(0) = 10$ and $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$. The Vol I row evaluates the rule at the Fake-Monster row Φ_{12} on $\mathrm{II}_{25,1}$ [8, 9], with $c_{\Phi_{12}}(0) = 24$ and $\kappa_{\mathrm{BKM}}(\Phi_{12}) = 12$. Each site must name its input form and lattice; the values 5 and 12 are independent evaluations, not a single shifted quantity. See Remark 8.3 and the Vol III universal-Borcherds-weight identity (Theorem 8.1).

The determinant is a Grothendieck-class invariant

A graded super vector space with the same signed multiplicities and zero bracket gives the same denominator product but is not a Lie superalgebra. The denominator identity fixes only the signed Grothendieck class (Proposition 13.4). The bracket, the parity split, the radical, the simple primitives, and the no-extra-relations theorem are required for a recognition theorem; all of these are read on the BKM presentation of Definition 4.2, not from the denominator product alone. The recognition target $P_X \simeq \mathfrak{g}_{\Delta_5}$ requires the compact-source data of Chapter 7.

19 $\mathfrak{g}_{\Delta_5}$ AS ONE OF FIVE VIEWS

The denominator algebra $\mathfrak{g}_{\Delta_5}$ is one of five views of Δ_5: View I (automorphic section), View II (Borcherds–Kac–Moody denominator), View III (compact protected Pfaffian, conditional), View IV (mirror discriminant, conjectural), View V (singular theta lift). The cross-level identities are:

- View I $\leftrightarrow$ View II: theorem, [9, 48].
- View II $\leftrightarrow$ View V: theorem, [10].
- View I $\leftrightarrow$ View V: theorem, [10, 46].
- View II $\leftrightarrow$ View III: Pfaffian and orientation theorem on $K3 \times E$, Chapter 7; primitive recognition conditional on $(S1)$–$(S10)$.
- View II $\leftrightarrow$ View IV: *conjectural*, Chapter 8.

The denominator identity is

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z),$$

the Borcherds-weight characteristic

$$\kappa_{\mathrm{BKM}}(\Delta_5) = 5,$$

and the active multiplicity formula

$$\mathrm{smult}(\alpha(n,l,m)) = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

These three identities are the target-side data of View II.

Chapter 5

The singular theta lift

1 The singular theta lift of $\phi_{0,1}$

The automorphic, Borcherds–Kac–Moody, and singular-theta descriptions of Δ_5 are theorems of Borcherds, Gritsenko, and Nikulin. The Pfaffian description is relative to the retained Dirac–Igusa data, and the mirror-discriminant description is conjectural. The exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies the genus-two Siegel upper half-space and the type-IV symmetric domain attached to the signature-$(3,2)$ lattice into one homogeneous space; the singular theta integral of [9, 10] sends the Weil-representation theta component $\phi_{0,1}^{\mathrm{Weil}}$ of $\phi_{0,1}$ to the weight-five Borcherds product Δ_5, and the Gritsenko–Nikulin denominator identity then reads the lift as the BKM denominator on $\Lambda_{II}^{2,1}$. These identifications are theorems of [9, 10, 46, 48]; the integral lattice, arithmetic subgroup, and multiplier convention are fixed by the following normalization.

1.1 Notation and the canonical name

Three Borcherds-product handles for the weight-five Igusa cusp form are distinguished. The *automorphic section* $\mathcal{D}_X = \Delta_5$ of the line $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ is the View-I name. The *BKM denominator* $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ on $\Lambda_{II}^{2,1}$ is the View-II name. The *protected Pfaffian* $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$, defined on $K3 \times E$ relative to the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, $(O2_{\mathrm{atlas}})$, and the finite geometric Pfaffian datum (P_{fin}), is the View-III name. The View-V canonical name is

$$\Delta_5 \;=\; \Theta\big(\phi_{0,1}^{\mathrm{Weil}}\big),$$

the *singular theta lift of the Weil component* $\phi_{0,1}^{\mathrm{Weil}}$ obtained from $\phi_{0,1}$ by theta decomposition. Throughout $\phi_{0,1} \in J_{0,1}^{\mathrm{weak}}$ is the unique up to scale weight-zero index-one weak Jacobi form on the Jacobi group $\Gamma_1^J = \mathbf{SL}_2(\mathbb{Z}) \ltimes \mathbb{Z}^2$ with constant Fourier coefficient fixed by the K3 elliptic genus normalization $Z_{\mathrm{K3}} = 2\phi_{0,1}$ of [34, §2.2]; $\phi_{0,1}^{\mathrm{Weil}}$ is its weight-$-1/2$ Weil-representation theta component, and Θ is the Borcherds singular-theta integral of [9, §6], [10, §6] with the Igusa cusp trivialization $C_{\phi_{0,1}} = 64$. Its monic Borcherds product is $D_5 = 64^{-1}\Delta_5$. The bare handle Δ_5 is reserved for cross-view identifications.

1.2 The weak Jacobi form $\phi_{0,1}$

Definition 1.1 *(Weak Jacobi form, weight zero, index one).* A weak Jacobi form of weight k and index m on the Jacobi group $\Gamma_1^J = \mathbf{SL}_2(\mathbb{Z}) \ltimes \mathbb{Z}^2$ is a holomorphic function $\phi : \mathbb{H} \times \mathbb{C} \to \mathbb{C}$ satisfying, for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathbf{SL}_2(\mathbb{Z})$ and $(\lambda, \mu) \in \mathbb{Z}^2$,

$$\phi\!\left(\frac{a\tau + b}{c\tau + d},\, \frac{z}{c\tau + d}\right) = (c\tau + d)^k e^{2\pi i m c z^2/(c\tau+d)}\,\phi(\tau, z),$$
$$\phi(\tau,\, z + \lambda\tau + \mu) = e^{-2\pi i m(\lambda^2\tau + 2\lambda z)}\,\phi(\tau, z),$$

together with the cusp expansion

$$\phi(\tau, z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} f(n,l)\, q^n r^l, \qquad q = e^{2\pi i\tau},\ r = e^{2\pi i z},$$

in which the coefficient depends only on the pair $(4mn - l^2,\ l \bmod 2m)$ (Eichler–Zagier [34, Theorem 2.2]). For $m = 1$ the residue class is fixed by the parity of $4n - l^2$. The *weight-zero index-one* weak Jacobi form $\phi_{0,1}$ is fixed by $k = 0$, $m = 1$, the discriminant condition $4n - l^2 \geq -1$ and the normalization

$$\phi_{0,1}(\tau, z) = r^{-1} + 10 + r + O(q).$$

The space $J_{0,1}^{\mathrm{weak}}$ is one-dimensional over $\mathbb{C}$ (Eichler–Zagier [34, Theorem 9.4 and Table 9, §9]); $\phi_{0,1}$ generates it. The theta-quotient presentation is the input to the singular-theta integral below.

LEMMA 1.2 *(Eichler–Zagier theta-quotient presentation).* The weak Jacobi form $\phi_{0,1}$ admits the closed form

$$\phi_{0,1}(\tau, z) = 4\sum_{i=2}^{4} \frac{\theta_i(\tau, z)^2}{\theta_i(\tau, 0)^2},$$

where $\theta_2, \theta_3, \theta_4$ are the three even Jacobi theta functions in the Mumford normalization (Eichler–Zagier [34, §9, equation (9.4)]); equivalently $\phi_{0,1} = Z_{\mathrm{K3}}/2$ is one half of the K3 elliptic genus [30]. The leading Fourier coefficients are

$$f(0,0) = 10, \quad f(0,\pm1) = 1, \quad f(1,0) = 108, \quad f(1,\pm1) = -64, \quad f(1,\pm2) = 10,$$

$$f(2,0) = 808, \quad f(2,\pm1) = -513, \quad f(2,\pm2) = 108, \quad f(2,\pm3) = 1,$$

in agreement with the Mukai–Gram normalization of Appendix 3. The discriminant relation $f(n,l) = f(n',l')$ when $4n - l^2 = 4n' - (l')^2$ (Eichler–Zagier [34, Theorem 2.2]) is the index-one specialization of the $(4mn - l^2, l \bmod 2m)$ rule and organises these coefficients into the equivalence classes

$$\{f(1,0),\, f(2,\pm2)\} = 108\ (\text{disc } 4), \quad \{f(0,0),\, f(1,\pm2)\} = 10\ (\text{disc } 0), \quad \{f(0,\pm1),\, f(2,\pm3)\} = 1\ (\text{disc } -1),$$

each independently consistent with the verified expansion. The finite arithmetic table

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certificates/jacobi/k3e_first_relation_window_coefficients
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records the computed rows $\phi_{0,1} = r^{-1} + 10 + r + O(q)$, the finite evenness checks $f(n,l) = f(n,-l)$, and the finite discriminant-dependence checks $f(n,l) = c(4n - l^2)$ for the first relation-closed window. This table checks the displayed Fourier window; the all-coefficient statement is the Eichler–Zagier theorem cited above.

Proof. The closed form is standard in Eichler–Zagier [34, §§9–10]; the K3 elliptic-genus identity $Z_{\mathrm{K3}} = 2\phi_{0,1}$ is [34, §2.2] together with the sigma-model right-moving Witten index of [30]. The constant term $f(0,0) = 10$ is the right-moving-localized character of the K3 Ramond sector at $L_0 = c/24$; the remaining values follow from the discriminant relation $f(n,l) = f(n',l')$ when $4n - l^2 = 4n' - (l')^2$ (Eichler–Zagier [34, Theorem 2.2]) together with explicit theta-quotient evaluation, which fixes the disc-3 class ($f(1,\pm1) = -64$) and the disc-4 class ($f(1,0) = f(2,\pm2) = 108$) by direct computation from the theta-quotient of Lemma 1.2. The squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2 = 64^2 = 4096$ of the Borcherds-product expansion in §3 is the next arithmetic check on these coefficients via the cusp-chamber product $\Delta_5(Z) = 64q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1 - q^n r^l s^m)^{f(nm,l)}$. $\square$

1.3 The signature-$(3,2)$ lattice and the type-IV domain

Definition 1.3 *(The signature-$(3,2)$ lattice).* The lattice $\Lambda^{3,2}$ is the orthogonal complement of the alternating Pfaffian element $q_0 = e_1 \wedge e_3 + e_2 \wedge e_4 \in \wedge^2\Lambda^4$ inside the signature-$(3,3)$ lattice $\big(\wedge^2\Lambda^4,\ (\ ,\ )\big) \simeq \mathrm{II}_{3,3}$, where $\Lambda^4 = \bigoplus_{i=1}^4 \mathbb{Z}e_i$ and the bilinear form (u,v) is the Pfaffian pairing $u \wedge v = (u,v)\, e_1 \wedge e_2 \wedge e_3 \wedge e_4$. The vector q_0 has square -2. In the BKM sign convention it is the even discriminant-two lattice

$$\Lambda^{3,2} \ \simeq\ \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2], \qquad \Lambda^{(1,1)} = \left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right),$$

written in the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ of the exterior-square model section.

Remark 1.4 *(Signature conventions).* The lattice $\Lambda^{3,2}$ carries three positive and two negative directions in the (u,v)-pairing of Definition 1.3. The *primitive hyperbolic sublattice* $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ on which the BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ is graded has signature $(2,1)$ in the fixed convention: the $[2]$ factor and one hyperbolic direction are positive, and the other hyperbolic direction is negative. The decomposition $\Lambda^{3,2} = \Lambda^{2,1} \oplus \Lambda^{(1,1)}$ splits an extra hyperbolic plane on which the singular theta integral lives; this is the $\mathrm{II}_{2,2}$-fibre at the standard cusp under the exceptional isomorphism of §1.4. No $\Lambda^{2,1} \oplus \langle -2 \rangle$ presentation is invoked; the fixed convention is the $\Lambda^{(1,1)} \oplus [2]$ Nikulin form.

Let

$$\Gamma_{3,2} := \wedge^2\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$$

be the projective exterior-square arithmetic group. The type-IV symmetric domain attached to $\Lambda^{3,2}$ is

$$\mathbb{H}^{\mathbf{IV}} = \big\{ Z \in \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C}) \ \big|\ (Z,Z) = 0,\ (Z,\overline{Z}) < 0 \big\},$$

with positive component $\mathbb{H}^{\mathbf{IV}}_+$ singled out by $\mathrm{Im}\, z_1 > 0$ in the parametrization ${}^t(z_2^2 - z_1z_3, z_3, z_2, z_1, 1)z_0$ of the exterior-square model section. Its arithmetic quotient $\Gamma_{3,2}\backslash\mathbb{H}^{\mathbf{IV}}_+$ is the moduli space on which the Borcherds singular-theta lift of [9] produces automorphic forms with prescribed singularities along rational quadratic divisors.

1.4 The exceptional isogeny $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$

Theorem 1.5 *(Exterior-square exceptional isomorphism).* The exterior-square representation $\wedge^2 : \mathbf{SL}_4 \to \mathrm{Aut}(\wedge^2\Lambda^4, (\ ,\ ))$ restricts to a homomorphism

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \ \xrightarrow{\ \simeq\ }\ \Gamma_{3,2},$$

realizing the exceptional isomorphism of adjoint real Lie groups $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$, together with its spin lift on the covering groups. Under this identification the genus-two Siegel upper half-space $\mathbb{H}_2$ embeds onto the positive component $\mathbb{H}^{\mathbf{IV}}_+$ by

$$\left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right) \ \longmapsto\ {}^t(z_2^2 - z_1z_3,\ z_3,\ z_2,\ z_1,\ 1)z_0,$$

intertwining the $\mathbf{Sp}_4(\mathbb{Z})$ and $\Gamma_{3,2}$ actions.

Proof. The Pfaffian form $u \wedge v = (u,v)e_1 \wedge e_2 \wedge e_3 \wedge e_4$ is the canonical $\mathbf{SL}_4$-invariant symmetric bilinear form on $\wedge^2\Lambda^4$; the subgroup of $\mathbf{SL}_4$ fixing $q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$ is the symplectic group $\mathbf{Sp}_4$ of the alternating form B_{q_0}. The orthogonal complement $\Lambda^{3,2} = q_0^\perp$ is preserved, and the kernel of $\wedge^2|_{\mathbf{Sp}_4}$ on $q_0^\perp$ is $\{\pm\mathbf{I}_4\}$ (the exterior-square model section). This is the integral form of the classical exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ of [60, Chapter II.6]; the spin cover supplies the corresponding lift with Lie algebra $\mathfrak{sp}_4(\mathbb{R})$. This cover is distinct from the metaplectic cover on the $\mathbf{Mp}_2(\mathbb{Z})$ input. The displayed substitution into the type-IV domain produces the commutative square of the exterior-square model section. The integral-Igusa form of the identification $\mathbb{H}_2 \simeq \mathbb{H}_+^{\mathbf{IV}}$ is the content of [51, §2]; see also [36, Chapter II] and [105, §6]. □

Remark 1.6 *(Three guises of one homogeneous space).* The same homogeneous space carries three names: the genus-two Siegel domain $\mathbf{Sp}_4(\mathbb{R})/\mathbf{U}(2)$, the type-IV symmetric domain $\mathbf{SO}(3,2)_+/(\mathbf{SO}(3) \times \mathbf{SO}(2))$, and the spin double cover $\mathbf{Spin}(3,2)/(\mathbf{U}(2)\text{-cover})$. The regularized theta integral below uses the metaplectic cover $\mathbf{Mp}_2(\mathbb{Z})$ on the input variable τ through the Weil representation of the discriminant form of $\Lambda^{3,2}$. The integrality of $\mathrm{wt}(\Delta_5) = 5$ places the resulting weight on the ordinary automorphic line; the multiplier system ν_{Δ_5} records the residual reflection character on $W^{(2)}(\Lambda^{2,1}_{II})$ (Lemma 7.2).

1.5 The Borcherds theta correspondence and the exceptional isomorphism

Theorem 1.7 *(Regularized theta correspondence on $\Lambda^{3,2}$).* Let $L = \Lambda^{3,2}$. Borcherds's regularized theta integral pairs a weakly holomorphic form $\phi \in M^!_{-1/2}(\rho_L)$ for $\mathbf{Mp}_2(\mathbb{Z})$ with the non-holomorphic Siegel theta kernel of L, and produces an automorphic form on $\widetilde{\mathbf{O}}(L)_+$ of weight $c_\phi(0,0)/2$. Its rational-quadratic divisor is determined by the negative-index part of ϕ, and its Weyl-chamber product uses the Fourier coefficients of ϕ in all effective product degrees. For $\phi = \phi^{\mathrm{Weil}}_{0,1}$, the pullback along

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \longrightarrow \Gamma_{3,2} \subset \mathbf{O}(\Lambda^{3,2})$$

is the Igusa cusp form Δ_5.

[proved; [9, 10, 46, 51]]

Remark 1.8 *(The two groups in the lift).* The metaplectic group in the construction is $\mathbf{Mp}_2(\mathbb{Z})$, acting on the input form through the Weil representation $\rho_{\Lambda^{3,2}}$. The genus-two group $\mathbf{Sp}_4$ enters after the orthogonal automorphic form has been constructed, through the exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the integral exterior-square map of Theorem 1.5. Thus the theta input, the orthogonal automorphic form, and the Siegel modular form are three successive incarnations of one Borcherds product.

1.6 The Borcherds singular-theta integral

Borcherds's correspondence assigns to a Weil-representation form obtained from a Jacobi input a meromorphic orthogonal modular form by integration against a vector-valued Siegel theta series. The integral is divergent at the lift's input weight; the Borcherds regularization produces a meromorphic Siegel modular form, after the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ pullback, with explicit divisor.

Definition 1.9 *(Siegel theta-series of $\Lambda^{3,2}$).* For the lattice $\Lambda^{3,2}$ of Definition 1.3, the Siegel theta-series at $Z \in \mathbb{H}_+^{\mathbf{IV}}$ and $\tau \in \mathbb{H}$ is

$$\Theta_{\Lambda^{3,2}}(\tau, Z) = \sum_{\gamma \in A_{\Lambda^{3,2}}} e_\gamma \sum_{\lambda \in \Lambda^{3,2}+\gamma} \exp\bigl(\pi i\,\overline{\tau}\,(\lambda,\lambda)_{Z^+} + \pi i\,\tau\,(\lambda,\lambda)_{Z^-}\bigr),$$

where $(\ ,\ )_{Z^\pm}$ are the positive and negative parts of the (u,v)-pairing under the orthogonal decomposition determined by Z (Borcherds [9, §4], [10, §4]). The theta-series is a non-holomorphic modular form of weight $(3/2, 1)$ on $\mathbf{Mp}_2(\mathbb{Z})$ valued in the Weil representation of $\Lambda^{3,2}$ on $\mathbb{C}[A_{\Lambda^{3,2}}]$. The scalar sum over $\Lambda^{3,2}$ is only the $\gamma = 0$ component.

Theorem 1.10 *(Singular theta lift* [9, Theorem 13.3], [10, Theorem 6.2]*).* Let ϕ be a weakly holomorphic vector-valued modular form for the dual Weil representation of $\Lambda^{3,2}$ on $\mathbf{Mp}_2(\mathbb{Z})$ of weight $-1/2$ with integral Fourier coefficients $c_\phi(n,\lambda) \in \mathbb{Z}$ at the cusp. The regularized integral

$$\Theta(\phi)(Z) = \int^{\mathrm{reg}}_{\mathbf{SL}_2(\mathbb{Z})\backslash\mathbb{H}} \big\langle \phi(\tau), \Theta_{\Lambda^{3,2}}(\tau, Z)\big\rangle\, v\,\frac{du\,dv}{v^2}, \qquad \tau = u + iv,$$

in the regularization sense of [10, §6] produces a meromorphic function on $\mathbb{H}^{\mathbf{IV}}_{+}$ with the following properties.

(1) *Modular invariance.* $\Theta(\phi)$ is an automorphic form of weight $c_\phi(0,0)/2$ on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ with multiplier system determined by the Weil action on $\Lambda^{3,2*}/\Lambda^{3,2}$.

(2) *Borcherds product.* On the cone of definition, $\Theta(\phi)$ admits the Borcherds product expansion

$$\Theta(\phi)(Z) = C_\phi e^{-2\pi i(\rho,Z)} \prod_{\substack{\alpha\in\Lambda^{3,2*}\\ (\alpha,W)>0}} \left(1 - e^{-2\pi i(\alpha,Z)}\right)^{c_\phi(-(\alpha,\alpha)/2,\ \alpha \bmod \Lambda^{3,2})},$$

with $C_\phi \in \mathbb{C}^\times$ fixed by the chosen cusp trivialization, Weyl vector ρ determined by ϕ, and a chamber W of the positive cone.

(3) *Divisor.* The divisor of $\Theta(\phi)$ is the linear combination of rational quadratic divisors $\lambda^\perp$ weighted by $c_\phi(-(\lambda,\lambda)/2, \lambda \bmod \Lambda^{3,2})$ over primitive $\lambda \in \Lambda^{3,2*}$ with negative discriminant.

The factor v is the invariant normalization for signature $(3,2)$: the input has holomorphic weight $-1/2$, the theta kernel has weight $(3/2, 1)$, and $v\langle\phi, \Theta_{\Lambda^{3,2}}\rangle$ has weight $(0,0)$.

[proved; [9, 10]]

Remark 1.11 *(Two regularization conventions).* The Borcherds regularization [10, §6] subtracts the divergent $v \to \infty$ contribution by analytic continuation in an auxiliary parameter, and is the convention used here. The Harvey–Moore regularization [49] differs from the Borcherds regularization by a finite renormalization of the Weyl vector and the constant term, with no effect on the product expansion's exponents or divisor. The two conventions agree on quotient divisors $\Theta(\phi_1)/\Theta(\phi_2)$, on the BKM denominator identity, and on the weight identity $\operatorname{wt}\Theta(\phi) = c_\phi(0,0)/2$.

1.7 From $\phi_{0,1}$ to a vector-valued Weil-representation form

The singular-theta integral takes a weakly holomorphic vector-valued form on $\mathbf{Mp}_2(\mathbb{Z})$; the input $\phi_{0,1}$ is a scalar weak Jacobi form of index one. Theta-decomposition reduces the latter to the former.

Lemma 1.12 *(Theta-decomposition of $\phi_{0,1}$).* Let $\phi_{0,1} \in J^{\mathrm{weak}}_{0,1}$ and decompose along the elliptic theta-functions of index one,

$$\phi_{0,1}(\tau,z) = h_0(\tau)\theta_{1,0}(\tau,z) + h_1(\tau)\theta_{1,1}(\tau,z),$$

where $\theta_{1,\mu}(\tau,z) = \sum_{\ell\equiv\mu\,(2)} q^{\ell^2/4} r^\ell$ for $\mu \in \{0,1\}$ (Eichler–Zagier [34, §5]). Then $h = (h_0, h_1)$ is a weakly holomorphic vector-valued modular form of weight $-1/2$ for the dual Weil representation $\rho^\vee_{[2]}$, equivalently for the Weil representation of the opposite discriminant form $[-2]$, on $\mathbf{Mp}_2(\mathbb{Z})$. Under the finite quadratic-module identification

$$A_{[2]} \;\simeq\; A_{\Lambda^{(1,1)}\oplus[2]} \;\simeq\; A_{\Lambda^{3,2}},$$

it pairs with the $\rho_{\Lambda^{3,2}}$-valued theta kernel of Definition 1.9. Its Fourier coefficients are exactly

$$[q^{D/4}]h_\mu(\tau) = c(D), \qquad c(4n - l^2) = f(n,l), \qquad D \equiv -\mu^2 \pmod 4,$$

or equivalently $[q^{n-l^2/4}]h_{l \bmod 2} = f(n,l)$. In the index-one normalization the first coefficients are $[q^0]h_0 = 10$ and $[q^{-1/4}]h_1 = 1$ reproducing $\phi_{0,1} = r^{-1} + 10 + r + O(q)$. Since $\theta_{1,1}(\tau+1,z) = i\,\theta_{1,1}(\tau,z)$, invariance of $\phi_{0,1}$ under T forces $h_1(\tau+1) = -i\,h_1(\tau)$, the dual eigenvalue. The vector h is the weakly holomorphic input $\phi_{0,1}^{\mathrm{Weil}}$ of the weight and representation type required by Theorem 1.10.

[proved; [9, 34, 46]]

Proof. Theta-decomposition along the lattice $\Lambda_{\mathrm{ind}} = \mathbb{Z}$ inside the Jacobi-elliptic variable is Eichler–Zagier [34, Theorem 5.1]. The two-component vector h is weakly holomorphic of weight $-1/2$ on $\mathbf{Mp}_2(\mathbb{Z})$ with the Weil representation of the rank-one lattice $\langle 2\rangle$. Comparing the coefficient of r^ℓ gives $[q^{n-\ell^2/4}]h_{\ell \bmod 2} = f(n,\ell)$, hence the discriminant formula above. The hyperbolic plane $\Lambda^{(1,1)}$ is unimodular, so adjoining one or two copies of $\Lambda^{(1,1)}$ does not change the discriminant form. Thus

$$(\Lambda^{(1,1)} \oplus [2])^*/(\Lambda^{(1,1)} \oplus [2]) \simeq [2]^*/[2] \simeq (\Lambda^{3,2})^*/\Lambda^{3,2},$$

with the same finite quadratic form. This discriminant-form isometry, is the transport of h to the required weakly holomorphic vector-valued form for the Weil representation of $\Lambda^{3,2}$. The constant Fourier coefficient $c_{\phi_{0,1}^{\mathrm{Weil}}}(0,0) = f(0,0) = 10$ is preserved under this discriminant-form identification because the trivial coset $0 \in \Lambda^{3,2*}/\Lambda^{3,2}$ matches the trivial coset on the rank-one factor. Borcherds [9, §15] records the explicit computation for the $\mathbf{Sp}_4$-case in the equivalent paramodular language. □

Remark 1.13 *(Index-one normalization of $c_1(0)$).* The constant Fourier coefficient $c_1(0) := f(0,0) = 10$ of $\phi_{0,1}$ is the Eichler–Zagier index-one normalization of the K3 elliptic genus. The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ of [45] names N as the paramodular level of the input denominator; at $N = 1$ the input form is $\phi_{0,1}$ and $c_1(0) = 10$, giving

$$\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5.$$

The integer-versus-half-integer ambiguity for the metaplectic weight factor is killed by the integrality of $c_1(0)/2 = 5$, and the Eichler–Zagier normalization is the row F_1 normalization in the Cléry–Gritsenko catalogue [44]. On the other seven rows the symbol $c(0)$ is indexed by the pair (t,N), not by a CHL twining order alone. No $c_N(0) = 2k_N + 1$ half-integer convention enters at $N = 1$.

1.8 Theta lift identifies Δ_5

The lift produces an automorphic form on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$, pulled back along $\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \xrightarrow{\simeq} \Gamma_{3,2}$ of Theorem 1.5 to a Siegel modular form on $\mathbf{Sp}_4(\mathbb{Z})$.

THEOREM 1.14 *(Δ_5 as singular theta lift).* The singular-theta lift of the weak Jacobi form $\phi_{0,1}$ is the weight-five Igusa cusp form:

$$\Theta(\phi_{0,1}^{\mathrm{Weil}})\big|_{\mathbb{H}_2} = \Delta_5,$$

the right-hand side written in the convention $\mathcal{D}_X = \Delta_5$ of §1.1 and the left-hand side regularized in the Borcherds sense of [10, §6]. Equivalently, in the BKM normalization,

$$\Theta(\phi_{0,1}^{\mathrm{Weil}}) = 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z \mapsto Z/2},$$

i.e. the lift is Δ_5 on $\mathbb{H}_2$ and $64^{-1}\Delta_5(2Z)$ on $\Lambda_{II}^{2,1}$-coordinates of $\mathbb{H}_+^{\mathbf{IV}}$.

[proved; [9, 10, 46, 48]]

Proof. The Borcherds singular-theta lift in Theorem 1.10 produces a meromorphic automorphic form $\Theta(\phi_{0,1}^{\mathrm{Weil}})$ on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ of weight $c_1(0)/2 = 5$. Its divisor is the linear combination of rational quadratic divisors weighted by $c_{\phi_{0,1}^{\mathrm{Weil}}}(-(\lambda,\lambda)/2, \lambda \bmod \Lambda^{3,2})$. For $\phi_{0,1}$, the negative-discriminant coefficients $f(0,\pm 1) = 1$ give the simple reflective Humbert divisor of Δ_5, while $f(0,0) = 10$ gives the weight 5. The finite packet `certificates/automorphic/delta5_humbert_divisor` records this divisor support and order-one multiplicity separately from the pole-order-two statement for Δ_5^{-2}. The Borcherds-product expansion of Theorem 1.10(2), specialized to $\phi_{0,1}^{\mathrm{Weil}}$, reads

$$\Theta(\phi_{0,1}^{\mathrm{Weil}}) = C_{\phi_{0,1}} q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}$$

in the cusp chamber, with Γ_{eff} the cusp-positive semigroup and Weyl vector $\rho_0 = (1/2, 1/2, 1/2)$. The monic Borcherds product is $D_5(Z)$ of §1.1; the Igusa theta-constant cusp trivialization fixes $C_{\phi_{0,1}} = 64$, hence $\Theta(\phi_{0,1}^{\mathrm{Weil}}) = \Delta_5$. The pullback to $\mathbb{H}_2$ along $\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \to \Gamma_{3,2}$ of Theorem 1.5 identifies $\Theta(\phi_{0,1}^{\mathrm{Weil}})\big|_{\mathbb{H}_2}$ with the $\mathbf{Sp}_4(\mathbb{Z})$-cusp form whose Borcherds product is exactly the Igusa cusp form Δ_5 of weight five. The BKM normalization $64^{-1}\Delta_5(2Z)$ on $\Lambda_{II}^{2,1}$ is Gritsenko–Nikulin [46, Theorem 2.1], [48, Theorem 2.1, Section 5.1.1]: the doubled lattice argument $Z \mapsto 2Z$ corresponds to the polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda_{II}^{2,1}$ of §1.9. Both sides have the same Borcherds-product expansion, and their leading coefficients agree by the theta-constant $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$. □

Corollary 1.15 *(Borcherds–Gritsenko–Nikulin pentadic identity).* The theorem-level automorphic, BKM-denominator, and theta-lift readings coincide:

$$\begin{aligned}(\text{View I})\ \mathcal{D}_X &= (\text{View II})\ 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z\mapsto Z/2} \\ &= (\text{View V})\ \Theta\big(\phi_{0,1}^{\mathrm{Weil}}\big).\end{aligned}$$

The Pfaffian identification $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ of View III holds on $K3 \times E$ at the Pfaffian and orientation level, granted the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and retained finite geometric Pfaffian datum (P_{fin}) of Theorem 7.5 of Appendix 7. The lift does not provide the level-`A` gravity-line operator algebra; a 3d-gravity path-integral realization requires the level-`A` Hall–Borcherds residual, while Definition 7.20 records only the level-`Z` trace comparison.

[proved for Views I, II, V; conditional for View III]

1.9 The multiplier system and the Weyl vector

Theorem 1.16 *(Multiplier system of the lift).* The lift $\Theta(\phi_{0,1}^{\mathrm{Weil}})$ carries the $\mathbf{Sp}_4(\mathbb{Z})$-multiplier system ν_{Δ_5} of Maass [76], equivalently described as the character $v_\eta^{12} \times v_H$ on the Jacobi parabolic and as the Gritsenko character on the BKM side [46, Proposition 2.1], [48, §5.1.1, case $(t = 1, \mathrm{II}, 0)$]. On the type-II generators of the Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$,

$$\nu_{\Delta_5}(s_{\hat{\delta}_i}) = -1, \qquad i = 1, 2, 3,$$

and on the chamber automorphism factor $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ the multiplier is trivial:

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_{-2}-f_3}) = +1.$$

This is exactly the multiplier of Lemma 7.2. The finite automorphic packet `certificates/automorphic/delta5_maass_charac` checks these generator values, the order-two relation, and the automorphic-line placement $\mathcal{L}^5 \otimes \nu_{\Delta_5}$; the packet is not an orientation construction.

[proved; [46, 48, 76]]

Remark 1.17 *(Singular-theta multiplier vs Maass multiplier).* The Borcherds singular-theta lift produces a multiplier system on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ determined by the Weil action on $\Lambda^{3,2*}/\Lambda^{3,2}$ (Borcherds [9, Theorem 13.3]). For the present $\Lambda^{3,2}$, the discriminant group has order two, generated by $f_3/2$ in the convention of Remark 1.38. The finite Weil module is the two-dimensional module attached to this discriminant form. The Borcherds multiplier determined by that module and by the Weyl vector restricts on the Borcherds chamber to the same reflection character. Pulled back along $\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \to \Gamma_{3,2}$ it matches Maass's multiplier formula for the product of the ten even theta constants [46, (1.3)–(1.5)], [76]: the -1-eigenvalues on the three type-II simple reflections and the $+1$-eigenvalues on the three chamber automorphisms. The two descriptions are not independent — they are the two sides of the same character — and their agreement is part of the regularized theta correspondence of Theorem 1.7.

Theorem 1.18 *(Weyl vector of the lift).* The Weyl vector of the Borcherds product $\Theta(\phi_{0,1}^{\mathrm{Weil}})$ in the type-II chamber $\mathcal{P}_{II} \subset \Lambda_{II}^{2,1} \otimes \mathbb{R}$ is

$$\rho = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_2 + \tfrac{1}{2}\delta_3 = f_2 - \tfrac{1}{2}f_3 + f_{-2},$$

satisfying $(\rho, \delta_i) = -1$ for $i = 1, 2, 3$, with simple-real-root Cartan matrix

$$A = \big((\delta_i, \delta_j)\big) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

This is the Weyl vector of $\mathfrak{g}_{\Delta_5}$ on $\Lambda_{II}^{2,1}$ of the Weyl-chamber section, identified with the Weyl vector of the Borcherds product by [46, Theorem 4.1]. The finite lattice packet `certificates/lattice/type_ii_weyl_vector` checks the Cartan matrix, discriminant -32, dual-lattice membership of ρ, and the displayed pairings.

[proved; [46, 48]]

Proof. The Weyl vector of this Borcherds product is determined by the q^0-row of ϕ at the cusp. For $\phi_{0,1} = r^{-1} + 10 + r + O(q)$, the negative-discriminant coefficients are $f(0, \pm 1) = 1$, and the constant coefficient is $f(0, 0) = 10$; the Weyl-vector formula of [9, Theorem 13.3, equation (13.10)] yields $\rho = (1/2, 1/2, 1/2)$ in cusp coordinates, transported to $\Lambda_{II}^{2,1}$ by the polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda_{II}^{2,1}$ of the Weyl-chamber section. This recovers $\rho = f_2 - \frac{1}{2}f_3 + f_{-2}$; the Cartan matrix is the Gram matrix of $\{\delta_1, \delta_2, \delta_3\}$ computed in the Weyl-chamber section. □

1.10 Compatibility with View II: lift's image is the BKM denominator

Theorem 1.19 *(Lift–denominator compatibility* [46, Proposition 3.1]*).* The image of the singular-theta lift, $\Theta(\phi_{0,1}^{\mathrm{Weil}}) = \Delta_5$, satisfies the Weyl–Kac–Borcherds denominator identity for the Lorentzian BKM superalgebra $\mathfrak{g}_{\Delta_5}$ on $\Lambda_{II}^{2,1}$:

$$64^{-1}\Delta_5(2Z) = e^{-2\pi i(\rho, Z)} \prod_{\alpha \in \mathcal{R}_+} \big(1 - e^{-2\pi i(\alpha, Z)}\big)^{\mathrm{smult}(\alpha)},$$

with positive-root system $\mathcal{R}_+$, Weyl vector ρ of Theorem 1.18, and signed root supermultiplicities

$$\mathrm{smult}\big(\alpha(n,l,m)\big) = f(nm,l), \qquad (n,l,m) \in \Gamma_{\mathrm{act}}.$$

The simple imaginary-root data are the Gritsenko–Nikulin coefficients $m(a)$ on negative-norm rays and $\tau(a)$ on isotropic rays of the Weyl-chamber section.

[proved; [46, 48]]

Proof. The Borcherds-product expansion $\Delta_5 = 64\, q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ of Theorem 1.14 is the BKM denominator product for $\mathfrak{g}_{\Delta_5}$ after the polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda^{2,1}_{II}$: the exponent $f(nm,l)$ is the signed multiplicity of the root $\alpha(n,l,m)$, and the doubled argument $2Z$ converts $\exp(-\pi i(\alpha, z))$ into the standard denominator character $\exp(-2\pi i(\alpha, z))$. The finite packet `certificates/lattice/product_chamber_root_map` checks these two target-side steps: the row-wise root map into $\Lambda^{2,1}_{II}$ and the $Z \mapsto 2Z$ character conversion. The algebra $\mathfrak{g}_{\Delta_5}$ is the Gritsenko–Nikulin algebra of [46, §§3–4, Proposition 3.1]; the denominator identity is read off the product expansion as in Chapter 4. Equivalently, the imaginary-simple-root characters $m(a)$ and $\tau(a)$ are the additive data of [48, Theorem 2.1, equations (2.1)–(2.3)]. □

1.11 Compatibility with View I and the cross-level identities

Theorem 1.20 *(Borcherds–Gritsenko–Nikulin three-way agreement).* The three identifications

$\Theta(\phi^{\mathrm{Weil}}_{0,1}) = \Delta_5$	(Borcherds 1995, 1998),
$64^{-1}\Delta_5(2Z) = \mathrm{den}(\mathfrak{g}_{\Delta_5})$	(Gritsenko–Nikulin 1998),
$\mathcal{D}_X = \Delta_5$	(Igusa 1962)

are mutually consistent: $\Theta(\phi^{\mathrm{Weil}}_{0,1}) = \mathcal{D}_X = 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z \mapsto Z/2}$ on $\mathbb{H}^{\mathbf{IV}}_+$, and the multiplier system ν_{Δ_5} of Theorem 1.16 is common.

[proved; [9, 10, 46, 48, 51]]

Proof. Theorem 1.14 is identification (1). Theorem 1.19 is identification (2). Identification (3) is the Igusa identification $\mathcal{D}_X = \Delta_5$ of [51, Theorem of §2], transported through the exterior-square model section. All three sides have the Borcherds-product expansion $\Delta_5 = 64\, q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ in the cusp chamber and pull back along the exceptional isomorphism of Theorem 1.5 to the same automorphic form on $\mathbf{Sp}_4(\mathbb{Z})$. The multiplier-system agreement is Theorem 1.16. □

1.12 The weight-five universal Borcherds identity at $N = 1$

Theorem 1.21 *(Universal Borcherds weight identity, $N = 1$).* At paramodular level $N = 1$ the Vol III universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ specializes to

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = \frac{f(0,0)}{2} = \frac{10}{2} = 5.$$

This is the row ($N = 1$, wt $= 5$, $c_1(0) = 10$) of the Cléry–Gritsenko catalogue [44], and is the value entered in the K3×E $\kappa_\bullet$-spectrum $\{\kappa_{\mathrm{cat}} = 0, \kappa^{\mathrm{Hodge}}_{\mathrm{ch}} = 0, \kappa^{\mathrm{Heis}}_{\mathrm{ch}} = 3, \kappa_{\mathrm{BKM}} = 5, \kappa_{\mathrm{fiber}} = 24\}$ of Vol III.

[proved; [9, 44]; Vol III universal identity]

Proof. The lift's weight is $c_1(0)/2$ by Theorem 1.10(1), and $c_1(0) = f(0,0) = 10$ by Lemma 1.2 and Lemma 1.12. The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is [45, §§2–3], generalized in Vol III across the eight Cléry–Gritsenko rows; at $N = 1$ it is exactly the Borcherds weight formula of Theorem 1.10(1). $\square$

Remark 1.22 *(Subscript discipline).* The bare symbol κ does not appear in any theorem-level identity. Every occurrence carries the subscript fixing its construction reading (κ_{cat}, $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$, κ_{BKM}, κ_{fiber}). The additive expression $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{K3})$ is a mixed-reading collision. With $\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}^{\mathrm{Heis}}$ it gives the accidental equality $3 + 2 = 5$ at $N = 1$, but it has no rowwise extension to $N \in \{2, 3, 4, 6\}$. The universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the invariant statement; at $N = 1$ the right hand side is $c_1(0)/2 = 5$.

1.13 Beilinson cut: the lift is the structural content

The five views of §1 lie on the Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$ of Vol II. View V's primitive content is the singular-theta integral $\Theta : J_{0,1}^{\mathrm{weak,Weil}} \to M_*(\widetilde{\mathbf{O}}(\Lambda^{3,2})_+)$ of Theorem 1.10, sending Jacobi forms to Borcherds products. Its value at $\phi_{0,1}$ is the automorphic section Δ_5, the level-S image of the lift.

Remark 1.23 *(Beilinson level).* The integral operator Θ lives at level C of the universal chain: it is a chart-dependent passage from a primitive Jacobi-form input to its Borcherds product, dependent on the Weil representation of $\Lambda^{3,2}$ and on the regularization of [10, §6]. The scalar Δ_5 lives at level S: it is the automorphic value of the lift on $\phi_{0,1}$. The retained $\mathfrak{D}_X^{\mathrm{DI}}$ datum of View III is the conditional Pfaffian construction whose protected Pfaffian recovers Δ_5; the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ of View II lives at level S under the trace lane of Vol II. The primitive object in this passage is the integral operator Θ, and the automorphic, BKM, Pfaffian, and theta-lift names of Δ_5 are readings, not primitive objects.

1.14 Consistency of the lattice conventions

Remark 1.24 *($\Lambda^{3,2}$ versus $\Lambda^{2,1} \oplus \langle -2\rangle$).* The signature-$(3,2)$ lattice $\Lambda^{3,2}$ of Definition 1.3 decomposes as $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ on the integral level. The sublattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ is primitive in $\Lambda^{3,2}$; its orthogonal complement is the second hyperbolic plane $\Lambda^{(1,1)}$, not the rank-one form $\langle -2\rangle$ (which would have wrong signature). The polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda_{II}^{2,1}$ sends $(n, l, m) \mapsto 2nf_2 - lf_3 + 2mf_{-2}$, so the BKM grading lattice is $\Lambda_{II}^{2,1}$, the index-four sublattice of $\Lambda^{2,1}$ with primitive f_3-coordinate. The doubled argument $Z \mapsto 2Z$ in $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ of Gritsenko–Nikulin changes the exponential convention from $\exp(-\pi i(\alpha, z))$ to $\exp(-2\pi i(\alpha, z))$. The $\mathrm{II}_{2,2}$ presentation of $\Lambda^{3,2}$ used in some references (Borcherds 1995 [9, §15], Borcherds 1998 [10, Example 16.4]) is the same lattice expressed via two hyperbolic planes plus the [2]-form, and produces the same Borcherds product Δ_5 under the same lift. No $\mathrm{II}_{3,2}$ versus $\mathrm{II}_{2,1} \oplus \langle -2\rangle$ ambiguity remains.

Remark 1.25 *(Multiplier and metaplectic accounting).* The Borcherds singular-theta lift produces an automorphic form on the integral form $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$, pulled back to $\mathbf{Sp}_4(\mathbb{Z})$ by Theorem 1.5. Since $c_1(0)/2 = 5$ is integral, the resulting automorphy factor is defined on $\mathbf{Sp}_4(\mathbb{Z})$. The metaplectic cover enters on the $\mathbf{Mp}_2(\mathbb{Z})$ input side through $\rho_{\Lambda^{3,2}}$; after restriction to integral weight its central sign does not alter the $\mathbf{Sp}_4(\mathbb{Z})$ automorphy factor. The residual reflection character is exactly the multiplier ν_{Δ_5} of Theorem 1.16. For the CHL constants at $N \in \{1, 2, 3, 6\}$, the numbers $c_N(0)/2 = (5, 3, 2, 1)$ are integral, and the metaplectic cover belongs to the input Weil representation while the resulting automorphic forms live on the corresponding integral orthogonal or Siegel arithmetic groups. At $N = 4$ the constant $c_4(0) = 3$ is odd and the weight $c_4(0)/2 = \frac{3}{2}$ is half-integral (Govindarajan–Krishna [42]); Φ_4 is a genuine form on the genus-two metaplectic cover.

Remark 1.26 *(The eta-multiplier v_η^{12}).* The character of Δ_5 on the Jacobi parabolic of $\mathbf{Sp}_4(\mathbb{Z})$ is described in [45] as $v_\eta^{12} \times v_H$, where v_η is the eta-multiplier on $\mathbf{SL}_2(\mathbb{Z})$ and v_H is the Heisenberg multiplier. The first four Cléry–Gritsenko rows carry eta exponents 12, 6, 4, 3 for $N = 1, 2, 3, 4$, respectively [44, Theorem 1.4]; the present form is the $N = 1$ row. Equivalently, after the lift through the exceptional isomorphism, v_η^{12} remains the eta multiplier in the Jacobi–Heisenberg presentation; it is not a discriminant character of $\mathrm{II}_{2,2}$, since $\mathrm{II}_{2,2}$ is unimodular. No v_η^N inconsistency between [45], [46], and [44] arises at $N = 1$; the three computations agree.

1.15 Theta-lift boundary

The singular theta lift is one realization of Δ_5 among five. The lift's primitive content is the Borcherds regularized theta operator $\Theta : \phi^{\mathrm{Weil}} \mapsto \Theta(\phi^{\mathrm{Weil}})$ of Theorem 1.10; its value at $\phi_{0,1}^{\mathrm{Weil}}$ is the weight-five Igusa cusp form. The five-view identification $\mathcal{D}_X = 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z\mapsto Z/2} = \Theta(\phi_{0,1}^{\mathrm{Weil}})$ is the theorem-level automorphic–BKM–theta identity; its Pfaffian extension to View III is conditional on the retained $K3 \times E$ data of Appendix 7. The mirror-discriminant View IV is the remaining conjectural cross-level identity, recorded in the mirror-discriminant chapter (Chapter 8) and not used here.

Borcherds's theta-correspondence formulation makes two structural facts visible. First, the weight-five integrality is forced by $c_1(0) = 10$ and the integrality of $c_1(0)/2$; no metaplectic cover is needed at the weight level. Second, the BKM-denominator structure of View II is not an extra postulate but a theorem about the lift's image: the Gritsenko–Nikulin denominator identity reads the lift's Borcherds product as the denominator of $\mathfrak{g}_{\Delta_5}$ on $\Lambda_{II}^{2,1}$ (Theorem 1.19), and the Borcherds–Gritsenko–Nikulin three-way agreement (Theorem 1.20) crystallizes the lift's value in three convertible normalizations. The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is, at $N = 1$, precisely the Borcherds-weight formula $\mathrm{wt}\,\Theta(\phi) = c_\phi(0,0)/2$ of Theorem 1.10(1) applied to $\phi = \phi_{0,1}$.

Part II

Dirac–Igusa Pro-Object

Chapter 6

The Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ on $K3 \times E$ (View ③)

The Igusa cusp form Δ_5 of weight 5 on $\mathrm{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2$ has three established identities: the Borcherds–Gritsenko–Nikulin section (View ①), the denominator of a generalized Borcherds–Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ on the root lattice $\Lambda_{II}^{2,1}$ (View ②), and the singular theta lift of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ (View ⑤). The View ③ datum is the protected Pfaffian line and compact source comparison for a finite E_3-prefactorization algebra on the $K3 \times E$ formal target, presented at finite stage as the Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \big(\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\big)$$

with eight finite constructions and four named obstructions.

Let S be a smooth complex K3 surface and let E be a smooth elliptic curve. Singular K3 limits, nodal or cuspidal elliptic degenerations, imprimitive curve-class contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures belong to separate boundary data. The standing finite-stage hypotheses are [K3×E-fin] (finite-type semistable substacks at bounded HN height), [K3×E-atlas] (finite d-critical/cosection Hall atlas), and [ML] (Mittag–Leffler descent for the HN inverse system). Two further standing functorial inputs are [E_3-HolFA], a chosen finite holomorphic E_3-prefactorization model for $\mathcal{A}_{X,R}^{E_3}$, including the BV quantum master equation, anomaly cancellation, locality, and framing/formality data needed to make the E_3-operations act on the retained finite stage, and [$K3{\to}E$-sp], a chosen chain-level specialization from the $K3$-direction to the hybrid chiral carrier on E. At finite stage these two inputs are the table system

formal_target_charts, field_complexes, e3_operations, bv_qme, anomaly_framing, compact_support, factoriz

The scalar firewall excludes the automorphic section, the BKM denominator, the target current envelope, the protected trace, and the hybrid carrier as replacements for the compact E_3-source. The finite E_3-source packet `certificates/e3_source/k3e_e3_holfa` currently supplies only the obstruction ledger `E3_SOURCE_OBSTRUCTION_LEDGER_VERIFIED`, with `e3_source_certification=false`. Thus [E_3-HolFA] and [K3→E-sp] are retained inputs until the formal-target, field-complex, E_3-operation, BV-QME, anomaly, compact-support, descent, specialization, transition, and scalar-firewall rows are supplied. The finite charge-window hypotheses are row-level data:

active_support, window_saturation, liu_hn_window, gram_map, cocycle_identities, normal_ordered_lifts, t

The current finite charge-window packet is empty-blocked and supplies none of these rows. The formal Mukai–Gram cocycle packet `certificates/charge/mukai_gram_cocycle` supplies only the chart-level identities for Π_X, B, the normal-ordered section, and $\overline{\Pi}_X$. It does not supply finite active support, Liu bounds, target-window images, primitive formal lifts, strict transitions, Mittag–Leffler rows, or the scalar firewall required for the finite charge window. The formal primitive-lift packet `certificates/charge/formal_primitive_lift` supplies only pointwise lattice lifts and the root grading $\alpha(\Pi_X(Q,P)) = 2nf_2 - lf_3 + 2mf_{-2}$ on the supplied formal rows. It is not a replacement for finite charge-window rows, effectivity, compact support, source Hall brackets, Pfaffian orientations, or type-II wall atlases. The formal Gram-fibre packet `certificates/charge/gram_fibre_kernel` supplies only same-Gram fibre grouping, separate even/odd summation, the definition of $K_{\Pi,0} = \Pi_X^{-1}(0,0,0)$, and the explicit fact that $K_{\Pi,0}$ is not an additive subgroup. It is not a source parity theorem, a Hopf radical, or an exactness theorem for a finite pushforward. The finite exactness packet `certificates/charge/fibre_pushforward_exactness` supplies only finite vector-space exactness of the fibre-summed Gram pushforward and a zero-fibre radical as a pairing kernel. It is not the exact Hall pushforward, Hopf radical ideal/coideal closure, or bar-commutation datum required by the compact source theorem. The finite bar-pushforward packet `certificates/charge/bar_pushforward_commutation` supplies only tensor-coalgebra compatibility for the additive normal-ordered Gram grading. It is not the raw quadratic Π_X-pushforward, the chiral Koszul quasi-isomorphism, Hall bracket compatibility, or Hopf pairing compatibility. The finite bracket-pairing pushforward packet `certificates/charge/hall_pairing_pushforward_compatibility` supplies only degree compatibility for a supplied bracket and supplied homogeneous pairing after normal-ordered Gram pushforward. It is not compact Hall geometry, primitive closure, Hopf adjointness, Frobenius cyclicity, quotient non-degeneracy, Pfaffian orientation, or protected trace data. The finite parity-pushforward packet `certificates/charge/parity_pushforward` supplies only the super-vector-space identity that the additive normal-ordered Gram pushforward sums even and odd finite blocks separately. Its signed superdimension rows are scalar shadows of the two parity ranks. It is not a source parity theorem, the Gritsenko–Nikulin/Kac parity equality, compact source data, primitive recognition, Pfaffian orientation, or protected trace data. The finite PBW-pushforward packet `certificates/charge/pbw_pushforward` supplies only the filtered-vector-space identity that the additive normal-ordered Gram pushforward preserves a supplied finite PBW filtration and its associated graded. It is not the compact-source PBW comparison, target PBW comparison, no-extra-relations theorem, primitive recognition, Pfaffian orientation, or protected trace data. The finite triangular-pushforward packet `certificates/charge/triangular_pushforward` supplies only the direct-sum identity that a supplied positive/Cartan/negative decomposition, checked by an integer height on Gram degrees, remains triangular after additive normal-ordered Gram pushforward. It is not exact triangular completion, target triangular completion, compact-source primitive recognition, no-extra-relations, Pfaffian orientation, or protected trace data. The retained stability-heart packet `certificates/moduli/stability_heart` supplies only the chosen Abramovich–Polishchuk/Liu heart $\mathcal{A}_X^{\mathrm{AP/Liu}} \subset D^b(\mathrm{Coh}(S \times E))$ and its input data. It verifies `STABILITY_HEART_DATUM_VERIFIED`, but it is not noetherianity, HN existence, bounded HN type data, finite-type semistable moduli, derived enhancements, Pfaffian orientation, or protected trace data. The Liu product-stability packet `certificates/moduli/liu_product_stability` supplies the precise source-imported definition of $\sigma_{s,t}$: the AP heart, the linear polynomial $L_M(n)$, the coefficient homomorphisms a, b, c, d, the weak charge Z_t, the tilt by $(\mathcal{T}_t, \mathcal{F}_t)$, and the central charge $Z_E^{s,t}$. It verifies `LIU_PRODUCT_STABILITY_DEFINED`, but it is not the AP-compatibility theorem, finite-type semistable moduli, derived enhancements, Pfaffian orientation, or protected trace data. The AP–Liu compatibility packet `certificates/moduli/ap_liu_compatibility` supplies Proposition 1.7: the retained heart $\mathcal{A}_X^{\mathrm{AP/Liu}}$ is the Liu tilt $\mathcal{A}_E^t = \langle \mathcal{T}_t, \mathcal{F}_t[1] \rangle$ of the Abramovich–Polishchuk heart $\mathcal{A}_E^{\mathrm{AP}}$. It verifies `AP_LIU_COMPATIBILITY_VERIFIED`, but it is not noetherianity, HN existence, bounded HN type data, finite-type semistable moduli, derived enhancements, Pfaffian orientation, or protected trace data. The retained-window noetherian packet `certificates/moduli/retained_window_noetherian` supplies only the finite ACC theorem for a supplied subobject-closed and quotient-closed retained exact window with a strict length rank. It verifies `RETAINED_WINDOW_NOETHERIAN_VERIFIED`, but it is not global noetherianity of the AP/Liu heart, HN existence, bounded HN type data, finite-type moduli, derived enhancements, Pfaffian orientation, or protected trace data. The retained-window HN-filtration packet `certificates/moduli/retained_window_hn_filtration` supplies only finite exact HN filtrations in the retained window: the quotient factors are semistable, the phases strictly decrease, and lengths add across the exact steps. It verifies `RETAINED_WINDOW_HN_FILTRATION_VERIFIED`, but it is not a global HN theorem for the AP/Liu heart, bounded HN type data, finite-type moduli, derived enhancements, Pfaffian orientation, or protected trace data. The retained HN type-bound packet `certificates/moduli/retained_hn_type_bounds` supplies only bounded HN type data in the active finite window. It verifies `RETAINED_HN_TYPE_BOUNDS_VERIFIED`: the semistable factor alphabet is finite, HN words have bounded length, and the finite-moduli HN type rows are mirrored from the certified packet. It is not finite-type semistable moduli, derived enhancements, universal complexes, Pfaffian orientation, or protected trace data. The retained semistable-substack packet `certificates/moduli/retained_semistable_substacks` supplies Proposition 1.16: each retained semistable HN factor type has a specialization-closed finite-type substack inside a bounded Quot/Postnikov ambient. It verifies `RETAINED_SEMISTABLE_SUBSTACKS_VERIFIED`, but it is not a derived enhancement, universal complex, rigidification, finite-inertia stratification, flag stack, cosection atlas, transition system, Pfaffian orientation, or protected trace datum. The retained derived-enhancement packet `certificates/moduli/retained_derived_enhancements` supplies Proposition 1.18: each retained semistable substack has a quasi-smooth derived Artin enhancement with cotangent amplitude $[-1, 0]$ and the restricted PTVV (-1)-shifted symplectic form. It verifies `RETAINED_DERIVED_ENHANCEMENTS_VERIFIED`, but it is not a universal complex, rigidification, finite-inertia stratification, flag stack, cosection atlas, transition system, Pfaffian orientation, or protected trace datum. The retained rigidification-inertia packet `certificates/moduli/retained_rigidification_inertia` supplies Proposition 1.20: the scalar $\mathbb{G}_m$ is removed from every retained derived substack, and the residual inertia group is finite with zero rigidification and inertia defects. It verifies `RETAINED_RIGIDIFICATION_INERTIA_VERIFIED`, but it is not a universal complex, finite closed inertia stratification, flag stack, cosection atlas, transition system, Pfaffian orientation, or protected trace datum. The retained class-bound packet `certificates/moduli/retained_class_bounds` supplies Proposition 1.22: the retained class set C_R, the Hilbert-polynomial labels $P_{c,i}$, the cohomological-amplitude intervals $[a_c, b_c]$, and the regularity bounds N_c are finite row data. It verifies `RETAINED_CLASS_BOUNDS_VERIFIED`, but it is not a universal complex, finite closed inertia stratification, flag stack, cosection atlas, transition system, Pfaffian orientation, or protected trace datum. The retained universal-complex packet `certificates/moduli/retained_universal_complexes` supplies Proposition 1.24: each retained scalar-rigidified finite substack carries a universal perfect complex with base-change compatibility, finite Tor amplitude, zero descent defect, and zero perfection defect. It verifies `RETAINED_UNIVERSAL_COMPLEXES_VERIFIED`, but it is not a finite closed inertia stratification, flag stack, cosection atlas, transition system, Pfaffian orientation, or protected trace datum. The retained E-translation-rigidification packet `certificates/moduli/retained_e_translation_rigidifications` supplies Proposition 1.26: the elliptic-translation action is removed on every retained finite row with zero translation-rigidification defect. It verifies `RETAINED_E_TRANSLATION_RIGIDIFICATIONS_VERIFIED`, but it is not a finite closed inertia stratification, flag stack, cosection atlas, transition system, Pfaffian orientation, or protected trace datum. The retained closed-substack packet `certificates/moduli/retained_closed_substacks` supplies Proposition 1.28: every retained finite row has a finite closed cover with finite residual inertia and zero coverage and inertia defects. It verifies `RETAINED_CLOSED_SUBSTACKS_VERIFIED`, but it is not closure under extensions, closure under HN factors, closure under duals, a flag stack, a cosection atlas, a transition system, a Pfaffian orientation, or a protected trace datum. The retained extension-closure packet `certificates/moduli/retained_extension_closure` supplies Proposition 1.30: every supplied binary extension row has middle term in the retained finite HN word set with zero extension-closure defect. It verifies `RETAINED_EXTENSION_CLOSURE_VERIFIED`, but it is not an extension stack, a two-step flag stack, closure under HN factors, closure under duals, a cosection atlas, a transition system, a Pfaffian orientation, or a protected trace datum. The retained HN-factor-closure packet `certificates/moduli/retained_hn_factor_closure` supplies Proposition 1.32: every HN factor of a retained finite HN word is represented by a retained class, a finite-type semistable substack, and a retained closed substack. It verifies `RETAINED_HN_FACTOR_CLOSURE_VERIFIED`, but it is not a dual-closure datum, not an extension stack, not a two-step flag stack, not subquotient closure in flag stacks, not a cosection atlas, not a transition system, not a Pfaffian orientation, and not a protected trace datum. The retained dual-closure packet `certificates/moduli/retained_dual_closure` supplies Proposition 1.34: the finite duality involution $s_3 \leftrightarrow s_1$, $s_2 \mapsto s_2$, together with reversal of HN words, preserves the retained finite HN word set with zero defect. It verifies `RETAINED_DUAL_CLOSURE_VERIFIED`, but it is not an extension stack, not a two-step flag stack, not subquotient closure in flag stacks, not a cosection atlas, not a transition system, not a Pfaffian orientation, and not a protected trace datum. The finite-moduli and atlas hypotheses are likewise row-level data:

hn_type_bounds, semistable_substacks, derived_enhancements, universal_complexes, scalar_rigidifications, stra

These rows record the retained bounded Quot/Postnikov slices, quasi-smooth derived enhancements, universal complexes, scalar rigidifications, finite residual inertia, retained closed stratifications, retained extension closure, retained HN-factor closure, retained dual closure, compactified flag correspondences, d-critical/cosection charts, and strict transitions. Liu stability without the finite charge-window rows, a target charge window, or a Hilbert-scheme scalar test does not supply $[K3 \times E\text{-fin}]$ or $[K3 \times E\text{-atlas}]$. The finite-moduli packet `certificates/moduli/k3e_finite_moduli` currently records the HN-bound, semistable-substack, derived-enhancement, and rigidification-inertia rows, together with the finite class-bound rows, universal-complex rows, E-translation-rigidification rows, retained closed-substack rows, retained extension-closure rows, retained HN-factor-closure rows, and retained dual-closure rows. The remaining rows stay in the obstruction ledger `MODULI_OBSTRUCTION_LEDGER_VERIFIED`, with `moduli_certification=false`. Thus the full finite-type moduli and d-critical/cosection atlas inputs remain open until the flag-stack, cosection-atlas, transition, and scalar-firewall rows are supplied.

The bar coalgebra is MC twisting, not the level-Z bulk. The imported Borcherds–Kac–Moody current envelope $\mathsf{Den}_{\Delta_5,E}$ is the target chiral shadow, and the cobar comparison $\Theta_{\mathrm{Kos},R}$ is the source-to-target map on the hybrid source. Its invertibility is the Koszul source datum, not a target-internal counit.

Construction theorem for the Dirac–Igusa package

Theorem 0.1 *(Construction of the Dirac–Igusa pro-object on $K3 \times E$).* Granted the first five components of the retained obstruction datum $\mathfrak{R}_X^{\mathrm{DI}}$ of Definition 7.4, as used in Theorem 7.5, and the standing finite-stage hypotheses [K3×E-fin], [K3×E-atlas], [ML], the standing hypotheses [E_3-HolFA] and [$K3{\to}E$-sp] through the finite compact-source proof tables above, the chiral Koszul source datum of Definition 0.208, its finite Koszul comparison datum of Definition 0.211, its strict Mittag–Leffler exactness of Proposition 0.214, and the cofinal primitive-recognition datum of Definition 0.217, together with the finite-stage morphism proof tables of §0.7 and the finite theorem-dependency ledger

beilinson_levels, objects, morphisms, theorem_dependencies, dependency_edges, forbidden_edges, status_sep

verified only as the schema-complete theorem-dependency ledger and not as a certificate for the retained construction packets, the finite stages define a Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_{R\in\mathcal{R}_{\mathrm{HN}}} \left(\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P^{\Pi}_R, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\right)$$

on the moduli stack $\mathfrak{M}_c^{\mathrm{red},+}(X)$ of effective dyonic-class objects in the chosen stability heart of Chapter 2. More precisely, the finite stages form a functor

$$\mathcal{R}_{\mathrm{HN}}^{\mathrm{op}} \longrightarrow \mathsf{DI}_X^{\mathrm{fin}}$$

in the finite-stage category of Definition 0.244, and the displayed inverse limit is notation for the associated object of $\mathrm{Pro}_{\mathcal{R}_{\mathrm{HN}}}(\mathsf{DI}_X^{\mathrm{fin}})$. Componentwise inverse limits are asserted only for components whose strict Mittag–Leffler and $R^1 \varprojlim = 0$ rows are supplied. The universal property and cofinal presentation invariance are those of Proposition 0.246. The dependence on the admissible parameter datum (S, H, σ, E, o_E), base-change functoriality in S and E, and the elliptic-origin character correction are those of Definitions 1.36, 1.38 and Propositions 1.37 and 1.39. The typed finite stages are:

(L1) (*Hybrid carrier*, §0.1.) $F^{\mathrm{hyb}}_{X,R}$ is the hybrid wrapped factorization carrier on $\mathrm{Ran}^{\mathrm{hyb}}(E)$ over the elliptic factor, with local/wrapped/mixed atlas matching the type-II wall structure $\{\gamma(\delta_1), \gamma(\delta_2), \gamma(\delta_3)\} = \{(1,1,0), (0,1,1), (0,-1,0)\}$ on $\Lambda^{2,1}_{II}$, as computed in Proposition 2.1. Compact Mukai-chart representatives for these three triples are constructed in Proposition 3.16; retained wall objects, wall charts, and protected integration remain O2 and hybrid inputs.

(L2) (*Reduced d-critical orientation*, §0.2.) o_R is the reduced orientation datum: a square root of the reduced determinant line, Thom–Sebastiani transport on retained extension strata, null-trivialisations of the reduced and equivariant degree-two orientation gerbes, and zero finite-stabilizer linearization characters, as in Definition 0.86.

(L3) (*Hall correspondences*, §0.3.) H_R is the finite protected Hall datum on $H_c^\bullet(\mathfrak{M}^{\mathrm{red},+}_{c,R}(X), \Phi_W \otimes o_R)$: product, collision coproduct, primitive projection, and Hopf pairing on the cosection-reduced heart. The comparison with the Drinfeld double $G(X) = D(Y^+(X))$ requires the Hall pairing, integral form, stable-envelope transport, completion, and primitive-recognition data.

(L4) (*First-order Dirac block and Pfaffian line*, §0.4.) $\mathscr{L}_{\mathrm{Pf},R}$ is the Pfaffian line bundle on $\mathfrak{Q}_R = (\mathfrak{M}^{\mathrm{red},+}_{c,R}(X)/E)^{\mathrm{red}}$ with finite Pfaffian sections $\mathrm{pf}_{X,R}$ carried into the inverse limit by Mittag–Leffler descent.

(L5) (*Chiral Koszul source coalgebra*, §0.5.) $C_{X,R}$ is the bar of the augmented binary/two-step hybrid correspondence algebra; it becomes the bar of a hybrid factorization algebra only after the higher-coloured residual vanishes. The chiral Koszul cobar map $\Theta_{\mathrm{Kos},R} : \Omega^{\mathrm{ch}}_E C_X \longrightarrow \mathsf{Den}_{\Delta_5,E}$ is the source-to-target comparison and is a quasi-isomorphism under the chiral Koszul source datum and its Mittag–Leffler exactness. Its compatibility with Weyl transport, Pfaffian orientation, Hall pairing, radical quotient, and PBW filtration is the finite Koszul comparison datum of Definition 0.211.

(L6) (*Primitive recognition*, §4.) Rec_R is the cofinal-window primitive recognition criterion (S1)–(S10) of Definition 0.217. Under that finite compact-source datum, it identifies $P_X^{\Pi,\mathrm{red}}$ with $\mathfrak{g}_{\Delta_5}$.

(L7) (*Eight finite constructions*, §0.7.) The pro-object is the inverse limit of the eight finite constructions of Appendix 6; its transition maps are the morphisms $\rho_{RR'}$ of Definition 0.244, and each component carries its obstruction class.

(L8) (*Obstruction classes*, §1.2.) The obstruction $(D0)$ is reduced on $K3 \times E$ to the retained D0-HN datum, $(O1)$ and $(O1)^+$ are reduced to retained orientation data, and $(O2)$ is reduced to $(O2_{\mathrm{atlas}})$, by Theorem 7.5.

Proof. The pro-object is represented by the functor of finite-stage data on the HN system $\mathcal{R}_{\mathrm{HN}}$. The Pfaffian, orientation, Hall, and carrier entries are obtained from the finite constructions of Appendix 6, and the functorial transition system is supplied by the fixed finite-stage data [K3×E-fin], [K3×E-atlas], [ML] together with the projections $(D0_{\mathrm{HN}}, O1_{\mathrm{quot}}, \mathcal{T}_W, O2_{\mathrm{atlas}})$ of $\mathfrak{R}_X^{\mathrm{DI}}$. The Koszul entry $\Theta_{\mathrm{Kos},R}$ is supplied by the chiral Koszul source datum and its exact inverse-system condition. The recognition entry Rec_R is supplied by the finite compact-source recognition datum. Thus (L1)–(L8) are finite components with separate source, orientation, Hall, Koszul, Pfaffian, and recognition hypotheses, and their transition morphisms satisfy (M1)–(M8) by the finite morphism proof tables; no component is inferred from another. □

Entries (L1)–(L8) are the finite constituents of $\mathfrak{D}_X^{\mathrm{DI}}$. Each constituent has its finite-stage morphism and obstruction class. A section equality, denominator product, or scalar trace does not determine an isomorphism of Dirac–Igusa pro-objects unless the morphisms of Definition 0.244 and the pro-isomorphism of Definition 0.245 are supplied with cofinality, component-map, composition-law, Mittag–Leffler, pro-isomorphism, and scalar-firewall rows. The finite formulas are in Appendix 6; the retained obstruction data are in Appendix 7. The current finite morphism packet `certificates/morphisms/k3e_dirac_igusa_pro_object` verifies only the obstruction ledger `MORPHISM_OBSTRUCTION_LEDGER_VERIFIED`, with `pro_object_certification=false`. Thus the present checked data record the missing pro-object morphism rows; they do not construct the transition functor or a pro-isomorphism.

0.1 The hybrid factorization carrier $\mathrm{Ran}^{\mathrm{hyb}}(E)$

The first entry of the Dirac–Igusa pro-object is the carrier on which chiral operations live. An ordinary Beilinson–Drinfeld factorization algebra on $\mathrm{Ran}(E)$ [7, §3.1] sees only a finite multiset of points of E and the operations attached to their collisions; this is the projection-finite stratum. The Igusa product Δ_5 carries a Borcherds variable s whose positive powers record positive elliptic-degree wrapping. An effective one-cycle on $K3 \times E$ of positive elliptic degree has a horizontal component and therefore projects *onto* E. Locality on $\mathrm{Ran}(E)$ alone has no mechanism for it.

PROPOSITION 0.2 *(Positive elliptic degree projects onto E; proved).* Let

$$Z = \sum_i m_i [C_i] \in Z_1(K3 \times E), \qquad m_i > 0,$$

be an effective one-cycle with irreducible curve components C_i. If

$$(p_E)_*Z = b[E], \qquad b > 0,$$

then $p_E(|Z|) = E$. Equivalently, at least one component C_i maps dominantly to E. If a retained object A has a support realization row identifying $\mathrm{cyc}_1(A) = Z$ with the effective fundamental one-cycle of $\mathrm{Supp}_1 A$, then

$$b_R^{\mathrm{geom}}(A) > 0 \quad \Longrightarrow \quad p_E(\mathrm{Supp}_1 A) = E.$$

Thus such an object is not localized at finitely many points of E.

Proof. For each irreducible component C_i, the proper image $p_E(C_i)$ is either a point or all of the irreducible curve E. In the first case $(p_E)_*[C_i] = 0$; in the second case $(p_E)_*[C_i] = \deg(C_i/E)[E]$ with positive degree. Since

$$(p_E)_*Z = \sum_i m_i(p_E)_*[C_i] = b[E]$$

and $b > 0$, at least one component has positive degree over E. That component maps dominantly onto E, hence

$$E = p_E(C_i) \subset p_E(|Z|) \subset E.$$

The support statement follows when the support-realization row identifies Z with the effective fundamental cycle of the one-dimensional support of A. □

The packet

`certificates/hybrid/positive_elliptic_degree_projection`

verifies `POSITIVE_ELLIPTIC_DEGREE_PROJECTION_VERIFIED`: the effective-cycle theorem is proved, while the retained object row, effective support-cycle realization row, positive degree row, object projection row, and transition row are recorded as missing open obligations.

The hybrid carrier replaces the Ran space by a two-stratum object that records, simultaneously, finite local supports and wrapped elliptic legs. At every finite Harder–Narasimhan height R the datum is a finite-coloured correspondence base, not a Ran space with a scalar attached.

Definition 0.3 *(Finite-stage hybrid Ran prestack).* Fix a finite HN height R. The hybrid carrier at height R is first a prestack

$$\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E) : \mathrm{Aff}^{op}_{\mathbb{C}} \longrightarrow \infty\text{-Gpd}.$$

For a test affine T, a T-point is represented by the following finite data:

(i) a finite local index set I and maps $x_i : T \to E$, $i \in I$;

(ii) retained local colour labels $\alpha_i \in \Gamma_R^{\mathrm{loc}}$;

(iii) a finite wrapped index set J, retained wrapped colour labels $\eta_j \in \Gamma_R^{\mathrm{wr}}$, and T-families

$$W_j \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta_j,R}(T)$$

in the rigidified wrapped prequotient stacks, equipped with their anchor maps

$$\lambda_{\eta_j,R}(W_j) : T \longrightarrow E.$$

Two representatives are identified by finite relabelling of I and J, and by the usual Ran collision maps on the local coordinates. Equivalently,

$$\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E) = \operatorname*{colim}_{(I,J,\alpha,\eta)\in \mathsf{Fin}^{\mathrm{hyb}}_R} \left(E^I \times \prod_{j\in J} \mathcal{M}^{\mathrm{wr,rig}}_{\eta_j,R} \right),$$

where $\mathsf{Fin}^{\mathrm{hyb}}_R$ is the finite indexing category generated by relabellings and local collision maps, and where $\alpha : I \to \Gamma^{\mathrm{loc}}_R$, $\eta : J \to \Gamma^{\mathrm{wr}}_R$. This is a prestack definition. For a topology τ, the stack

$$\mathrm{Ran}^{\mathrm{hyb}}_{R,\tau}(E) := a_\tau \, \mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$$

is used only after the descent rows for a_τ have been supplied. The scalar object

$$\bigsqcup_{b\geq 0} \mathrm{Ran}(E) \times \{b\}$$

is only the degree shadow; it is not the hybrid prestack.

The packet

`certificates/hybrid/hybrid_ran_prestack_definition`

verifies `HYBRID_RAN_PRESTACK_DEFINITION_VERIFIED`: the finite-stage carrier is defined as a prestack functor, while the local-stratum component rows, wrapped-stratum component rows, descent rows, stackification rows, and transition rows are recorded as missing open obligations.

Definition 0.4 *(Local $b = 0$ stratum as a subprestack).* At finite HN height R, the local stratum is the full subprestack

$$\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E) \subset \mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$$

cut out by the conditions $J = \varnothing$ and $\alpha : I \to \Gamma^{\mathrm{loc}}_R$, where

$$\Gamma^{\mathrm{loc}}_R = \{\gamma \in \Gamma^{\mathrm{HN}}_{\sigma,S}(R) \mid b^{\mathrm{geom}}_R(\gamma) = 0\}.$$

Thus, for a test affine T,

$$\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)(T) = \operatorname*{colim}_{(I,\alpha)} \mathrm{Map}(T, E)^I, \qquad \alpha : I \to \Gamma^{\mathrm{loc}}_R,$$

where the colimit is over finite sets, relabellings, and collision maps. The forgetful map to the ordinary Ran prestack discards the colour map α. This definition contains no wrapped family and no anchor map. The local closed-configuration incidence stacks $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$ and the sheaves $\mathcal{F}^{\mathrm{loc}}_{\alpha,R}$ are coefficient data over this carrier, not the carrier itself.

The packet

`certificates/hybrid/local_stratum_b0_prestack_definition`

verifies `LOCAL_STRATUM_B0_PRESTACK_DEFINITION_VERIFIED`: the $b = 0$ local stratum is defined as a subprestack of the hybrid prestack, while component rows, closed-configuration rows, local sheaf rows, descent rows, and transition rows are recorded as missing open obligations.

Definition 0.5 *(Wrapped $b > 0$ stratum as a subprestack).* At finite HN height R, the wrapped stratum is the full subprestack

$$\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E) \subset \mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$$

cut out by the conditions $I = \varnothing$ and $\eta : J \to \Gamma^{\mathrm{wr}}_R$, where

$$\Gamma^{\mathrm{wr}}_R = \{\gamma \in \Gamma^{\mathrm{HN}}_{\sigma,S}(R) \mid b^{\mathrm{geom}}_R(\gamma) > 0\}.$$

Thus, for a test affine T,

$$\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)(T) = \operatorname*{colim}_{(J,\eta)} \prod_{j \in J} \mathcal{M}^{\mathrm{wr,rig}}_{\eta_j,R}(T), \qquad \eta : J \to \Gamma^{\mathrm{wr}}_R,$$

where the colimit is over finite sets and relabellings. The anchor maps

$$\lambda_{\eta_j,R}(W_j) : T \longrightarrow E$$

are retained as part of each wrapped prestack point. This definition contains no finite local support maps $x_i : T \to E$ and no local colour labels. The wrapped coefficient objects $\mathcal{G}^{\mathrm{wr}}_{\eta,R}$, anchor residuals, and quotient-after-correspondence data are coefficient and correspondence data over this carrier, not the carrier itself.

The packet

`certificates/hybrid/wrapped_stratum_bpositive_prestack_definition`

verifies `WRAPPED_STRATUM_BPOSITIVE_PRESTACK_DEFINITION_VERIFIED`: the $b > 0$ wrapped stratum is defined as a subprestack of the hybrid prestack, while component rows, populated wrapped prequotient rows, anchor rows, support-realization rows, descent rows, and transition rows are recorded as missing open obligations.

Definition 0.6 *(E-equivariant wrapped prequotient stacks).* Fix a finite HN height R and a wrapped colour $\eta \in \Gamma^{\mathrm{wr}}_R$. The wrapped prequotient row is the scalar-rigidified retained wrapped stack

$$\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \subset \mathfrak{M}^{\mathrm{rig}}_{\eta,R}$$

whose objects have geometric elliptic degree

$$b^{\mathrm{geom}}_R(\eta) > 0.$$

The superscript rig means that the scalar automorphisms have been rigidified. It does not mean that the elliptic translation direction has been divided out. The map

$$\rho_{\eta,R} : \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \longrightarrow \mathcal{M}^{\mathrm{wr}}_{\eta,R} \subset \mathfrak{M}_{\eta,R}$$

forgets this scalar rigidification and the finite wrapped bookkeeping. The reduced quotient, when later used, is written

$$\mathcal{M}^{\mathrm{wr},E}_{\eta,R} := \left[\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}/E\right],$$

and belongs to the quotient-descent rows, not to the prequotient definition.

For a test affine T and a section $a : T \to E$, let $\tau_a : E \times T \to E \times T$ be relative translation by a. An object $A \in \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}(T)$, represented by the universal perfect complex on $K3 \times E \times T$, is acted on by

$$a \cdot A := \left(1_{K3} \times \tau_a\right)_* A.$$

The scalar rigidification and the universal complex are transported by this automorphism. The origin 0_E is fixed once and for all and acts as the identity. This defines the E-equivariant wrapped prequotient stack. It does not prove finite type, properness, final anchor population, anchor losslessness, quotient descent, or transition compatibility.

LEMMA 0.7 *(E-action law on the wrapped prequotient; proved).* The operation of Definition 0.6 is an E-action on $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$. It preserves the wrapped colour η and the condition $b_R^{\mathrm{geom}}(\eta) > 0$.

Proof. Relative translations satisfy $\tau_{0_E} = \mathrm{id}$ and $\tau_a \circ \tau_b = \tau_{a+b}$. Functoriality of pushforward along automorphisms gives

$$0_E \cdot A = A, \qquad a \cdot (b \cdot A) = (a + b) \cdot A.$$

Translation on the elliptic factor acts trivially on the numerical Mukai class and carries effective one-cycles of a fixed p_E-degree to effective one-cycles of the same p_E-degree. Hence the retained HN colour and b_R^{geom} are unchanged. The freeness and zero translation-rigidification defect on retained rows are exactly the input of Proposition 1.26; they are imported here only for the E-action row and do not supply finite type, properness, quotient descent, anchors, or transitions. □

The packet

`certificates/hybrid/equivariant_wrapped_prequotient_definition`

verifies `E_EQUIVARIANT_WRAPPED_PREQUOTIENT_DEFINITION_VERIFIED`: the scalar-rigidified wrapped prequotient $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$ is defined before the reduced E-quotient, the translation action with origin 0_E is defined, and the action law and wrapped-colour preservation have zero defect. Finite type, properness, final anchor population, anchor losslessness, quotient descent, transition compatibility, and aggregate hybrid-carrier population are recorded as missing open obligations.

PROPOSITION 0.8 *(Finite type of wrapped prequotients).* [proved at finite HN height] Assume the retained finite class-bound datum of Definition 1.21, the retained finite-type semistable substack datum of Definition 1.15, and the scalar-rigidification datum of Definition 1.19. Then, for every wrapped colour $\eta \in \Gamma_R^{\mathrm{wr}}$, the prequotient stack

$$\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$$

of Definition 0.6 is of finite type over $\mathbb{C}$. This is a finite-stage statement before the reduced E-quotient is taken.

Proof. Let

$$C_R(\eta) = \{c \in C_R \mid \widehat{c} = \eta\}.$$

Proposition 1.22 makes C_R finite; hence $C_R(\eta)$ is finite. For each $c \in C_R(\eta)$, Proposition 1.16 gives a finite-type retained semistable stack $\mathfrak{M}_{R,c}^{ss}$. The scalar subgroup $\mathbb{G}_m$ is flat, finitely presented, and central in the retained stabilizer sequence. The rigidification

$$\mathfrak{M}_{R,c}^{ss} \longrightarrow \mathfrak{M}_{R,c}^{\mathrm{rig}} = \mathfrak{M}_{R,c}^{ss}/\mathbb{G}_m$$

is therefore an fppf gerbe by the rigidification theorem [100, Tag 04V2]; finite type descends across such an fppf cover by fpqc descent for finite type [100, Tag 041K, Lemma 74.11.11]. Hence every $\mathfrak{M}^{\mathrm{rig}}_{R,c}$ with $c \in C_R(\eta)$ is finite type. The wrapped prequotient is the finite union of these scalar-rigidified retained class pieces:

$$\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} = \bigcup_{c \in C_R(\eta)} \mathfrak{M}^{\mathrm{rig}}_{R,c}.$$

Finite unions of finite-type substacks are finite type. The condition $b^{\mathrm{geom}}_R(\eta) > 0$ is numerical on the retained class and does not introduce an infinite support parameter. The proof does not establish properness, quotient descent, final anchor population, anchor losslessness, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/wrapped_prequotient_finite_type`

verifies `WRAPPED_PREQUOTIENT_FINITE_TYPE_VERIFIED`: finite type of $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ follows from the finite retained class set, finite-type retained semistable substacks, and scalar rigidification by $\mathbb{G}_m$. Properness, final anchor population, anchor losslessness, quotient descent, transition compatibility, and aggregate hybrid-carrier population are still missing open obligations.

Corollary 0.9 *(Finite-support Ran locality misses the geometric elliptic-degree direction; proved).* A support-local factorization algebra on $\mathrm{Ran}(E)$, defined only by separating supports over disjoint opens of E, sees the projection-finite sector. It does not realize the sectors of positive geometric elliptic degree.

Proof. The obstruction uses the geometric degree $b^{\mathrm{geom}}_R = \deg_E$, not the Borcherds coordinate $m = \mathrm{pr}_3 \overline{\Pi}_X$. On the rank-one Oberdieck–Pandharipande/Donaldson–Thomas branch one has the branch-local equality $m_{\mathrm{Bch}} = d_E - 1$ [90]; it is not a definition of the wrapped stratum. By Proposition 0.2, an effective support cycle with $b^{\mathrm{geom}}_R > 0$ projects onto all of E. The object-level statement requires the support-realization row named in that proposition. Such wrapped supports cannot be separated by disjoint opens. Therefore ordinary Ran-locality has no locality mechanism for the wrapped sector. □

Definition 0.10 *(Geometric elliptic-degree map).* Fix a finite HN height R and a retained HN numerical class $\gamma \in \Gamma^{\mathrm{HN}}_{\sigma,S}(R)$. Let

$$\mathrm{cyc}_1(\gamma) \in N_1(K3 \times E)$$

be the numerical one-cycle obtained from the codimension-two Chern character component, equivalently the class of $\mathrm{ch}_2(A) \cap [K3 \times E]$ for any retained semistable object A of numerical class γ. Since $N_1(E) = \mathbb{Z}[E]$, there is a unique integer $b^{\mathrm{geom}}_R(\gamma)$ such that

$$(p_E)_* \,\mathrm{cyc}_1(\gamma) = b^{\mathrm{geom}}_R(\gamma)[E].$$

On the effective retained one-dimensional sector this integer is nonnegative. The local and wrapped charge windows are

$$\Gamma^{\mathrm{loc}}_R = \{\gamma \mid b^{\mathrm{geom}}_R(\gamma) = 0\}, \qquad \Gamma^{\mathrm{wr}}_R = \{\gamma \mid b^{\mathrm{geom}}_R(\gamma) > 0\}.$$

The map b^{geom}_R is not the Borcherds trace coordinate

$$m^{\mathrm{Bch}}_R = \mathrm{pr}_3 \overline{\Pi}_X.$$

The latter is a normal-ordered Gram coordinate; the former is the proper-pushforward degree of the geometric support along $p_E : K3 \times E \to E$.

The packet

`certificates/hybrid/geometric_elliptic_degree_map`

verifies `GEOMETRIC_ELLIPTIC_DEGREE_MAP_OBSTRUCTION_VERIFIED`: the definition is fixed, while the retained HN domain rows, one-cycle class rows, class-constancy rows, pushforward rows, integer degree rows, nonnegativity rows, local/wrapped split rows, transition rows, and retained-extension additivity rows are not supplied. The last item is the separate row-202 packet below.

Definition 0.11 *(Geometric elliptic-degree additivity datum).* At finite HN height R, a geometric elliptic-degree additivity datum consists, for every retained extension row

$$0 \longrightarrow A_{\gamma'} \longrightarrow A_\gamma \longrightarrow A_{\gamma''} \longrightarrow 0$$

in the finite Hall stage, of the following rows:

(i) the retained extension-pair row, with $\gamma', \gamma'', \gamma \in \Gamma^{\mathrm{HN}}_{\sigma,S}(R)$;

(ii) the one-cycle equality

$$\mathrm{cyc}_1(\gamma) = \mathrm{cyc}_1(\gamma') + \mathrm{cyc}_1(\gamma'') \quad \text{in } N_1(K3 \times E);$$

(iii) the pushed-forward equality

$$(p_E)_* \,\mathrm{cyc}_1(\gamma) = (p_E)_* \,\mathrm{cyc}_1(\gamma') + (p_E)_* \,\mathrm{cyc}_1(\gamma'') \quad \text{in } N_1(E);$$

(iv) the integer degree equality

$$b_R^{\mathrm{geom}}(\gamma) = b_R^{\mathrm{geom}}(\gamma') + b_R^{\mathrm{geom}}(\gamma'');$$

(v) compatibility of these rows with the retained transition maps $R \le R'$.

The same definition applies to a distinguished triangle representing the Hall extension correspondence, with the corresponding equality in $K_0(K3 \times E)$.

PROPOSITION 0.12 *(Additivity criterion for the geometric elliptic degree).* Assume a finite stage carries the datum of Definition 0.11. Then

$$b_R^{\mathrm{geom}}(\gamma) = b_R^{\mathrm{geom}}(\gamma') + b_R^{\mathrm{geom}}(\gamma'')$$

for every retained extension row

$$0 \to A_{\gamma'} \to A_\gamma \to A_{\gamma''} \to 0.$$

Proof. In $K_0(K3 \times E)$ the retained extension gives $[A_\gamma] = [A_{\gamma'}] + [A_{\gamma''}]$. The Chern character is additive on K_0, hence the codimension-two component gives the one-cycle equality in $N_1(K3 \times E)$. Proper pushforward along p_E is linear on numerical one-cycles, so the equality remains true in $N_1(E)$. Since $N_1(E) = \mathbb{Z}[E]$, equality of the pushed-forward cycles is equality of the coefficients of $[E]$, which are precisely the integers b_R^{geom} of Definition 0.10. The transition rows state that this coefficient equality is preserved under $R \le R'$. □

The packet

`certificates/hybrid/geometric_elliptic_degree_additivity`

verifies `GEOMETRIC_ELLIPTIC_DEGREE_ADDITIVITY_OBSTRUCTION_VERIFIED`: the additivity criterion is fixed, the degree-map definition packet and the retained-extension-closure packet are imported, and the missing retained extension-pair rows, one-cycle additivity rows, pushforward additivity rows, integer degree additivity rows, and transition rows are recorded fail-closed. Retained word closure by itself proves only that the middle HN word is retained; it does not prove additivity of $(p_E)_* \operatorname{cyc}_1$.

Definition 0.13 *(Geometric/Borcherds degree comparison datum).* At finite HN height R, set

$$m_R^{\mathrm{Bch}}(\widehat{\gamma}) := \mathrm{pr}_3 \overline{\Pi}_X(\widehat{\gamma}), \qquad \overline{\Pi}_X(\widehat{\gamma}) = (n, l, m).$$

A comparison datum between b_R^{geom} and m_R^{Bch} consists of:

(i) a specified branch of retained HN classes;
(ii) a lift of each retained class on that branch to a charge $\widehat{\gamma}$ in the domain of $\overline{\Pi}_X$;
(iii) populated geometric degree rows computing b_R^{geom} from $(p_E)_* \operatorname{cyc}_1$;
(iv) populated Gram-coordinate rows computing $\mathrm{pr}_3 \overline{\Pi}_X(\widehat{\gamma})$;
(v) a declared equality or affine comparison formula with zero defect on the branch;
(vi) compatibility of the comparison with retained transitions $R \le R'$.

Proposition 0.14 *(Geometric/Borcherds degree separation).* The finite hybrid datum distinguishes the support degree

$$b_R^{\mathrm{geom}}(\gamma) = \operatorname{coeff}_{[E]}(p_E)_* \operatorname{cyc}_1(\gamma)$$

from the Borcherds trace coordinate

$$m_R^{\mathrm{Bch}}(\widehat{\gamma}) = \mathrm{pr}_3 \overline{\Pi}_X(\widehat{\gamma}).$$

The local/wrapped split is made by b_R^{geom}. The protected trace is graded by m_R^{Bch}. No formula identifying the two coordinates, or replacing one by the other, is part of the hybrid datum unless a comparison datum of Definition 0.13 is supplied.

Proof. The two maps have different constructions. The map b_R^{geom} is obtained from the numerical one-cycle class of a retained object by proper pushforward to $N_1(E) = \mathbb{Z}[E]$; it is defined before the local/wrapped factorization carrier is formed. The coordinate m_R^{Bch} is the third coordinate of the normal-ordered Gram homomorphism $\overline{\Pi}_X$; it is used after the Gram descent to grade protected integration in the Borcherds variables. A branch-local identity, such as the rank-one Oberdieck–Pandharipande comparison $m_{\mathrm{Bch}} = d_E - 1$ [90], becomes a theorem only after the branch, lift, geometric degree rows, Gram-coordinate rows, and transition compatibility are supplied. In the absence of these rows there is no typed morphism that turns $\mathrm{pr}_3 \overline{\Pi}_X$ into $\operatorname{coeff}_{[E]}(p_E)_* \operatorname{cyc}_1$. □

The packet

`certificates/hybrid/geometric_borcherds_degree_separation`

verifies `GEOMETRIC_BORCHERDS_DEGREE_SEPARATION_OBSTRUCTION_VERIFIED`: the two coordinates are separated, substitution of $\mathrm{pr}_3 \overline{\Pi}_X$ for b_R^{geom} is excluded, and branch-comparison rows are recorded as missing open obligations.

PROPOSITION 0.15 *(Middle type-II wall routes through wrapped and mixed geometry).* [routing proved; retained wall object not constructed] Let

$$\gamma_{\mathrm{wall}}(\delta_2) = (0, 1, 1)$$

be the middle simple type-II wall image of Proposition 2.1. On the type-II wall branch where the geometric elliptic degree is written in the OP/Borcherds coordinate as $d = m + 1$, this wall has $d = 2 > 0$. Consequently, any retained finite-stage class η with a supplied branch-comparison row

$$\overline{\Pi}_X(\eta) = (0, 1, 1), \qquad b_R^{\mathrm{geom}}(\eta) = d = 2,$$

lies in Γ_R^{wr}, not in Γ_R^{loc}. Its one-wall carrier is the wrapped prequotient side $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$ of Definition 0.6. Any Hall operation inserting projection-finite local colours against this wall is typed by the one-sided LW/WL mixed correspondences of Definition 0.21, and by the ordered two-sided mixed correspondence of Definition 0.47 when local insertions occur on both sides. It is not an LL operation on the ordinary Ran stratum.

This is a routing theorem. It does not supply the retained wall object $x_{\delta_2,R}$, an effective support-realization row, a populated wrapped anchor, quotient descent, the O2 wall atlas, or protected integration.

Proof. Proposition 2.1 gives the middle wall triple $(n, l, m) = (0, 1, 1)$. The wall-branch convention $d = m + 1$ gives $d = 2$. After the retained branch-comparison row identifies this integer with $b_R^{\mathrm{geom}}(\eta)$, the definition

$$\Gamma_R^{\mathrm{loc}} = \{b_R^{\mathrm{geom}} = 0\}, \qquad \Gamma_R^{\mathrm{wr}} = \{b_R^{\mathrm{geom}} > 0\}$$

puts η in the wrapped window and excludes it from the local window. By Proposition 0.2, if a retained object realizing η is supplied together with its effective support-cycle row, its one-dimensional support projects onto all of E, so finite-support Ran locality cannot contain it. Definition 0.5 therefore places the one-wall carrier in the wrapped $b > 0$ subprestack, with the prequotient stack of Definition 0.6. The local/wrapped and two-sided mixed correspondence definitions are exactly the before-E-quotient operations whose source contains one wrapped input and projection-finite local inputs. The final sentence records the rows still absent from this routing calculation. □

The packet

```
certificates/hybrid/middle_type_ii_wall_wrapped_mixed
```

verifies `MIDDLE_TYPE_II_WALL_WRAPPED_MIXED_ROUTING_VERIFIED`: the middle wall image has $m = 1$, the wall-branch degree $d = m + 1 = 2$ is positive, the routing is to the wrapped $b > 0$ side, and local insertions are typed by mixed correspondences. Its obstruction rows record that retained branch-comparison, wall-object, support, anchor, O2 atlas, quotient, protected-integration, aggregate-carrier, and transition rows are still missing.

PROPOSITION 0.16 *(Hybrid operation profiles recover the three simple type-II walls).* [operation-profile recovery proved; carrier population not constructed] Let

$$\gamma_{\mathrm{wall}}(\delta_1) = (1, 1, 0), \qquad \gamma_{\mathrm{wall}}(\delta_2) = (0, 1, 1), \qquad \gamma_{\mathrm{wall}}(\delta_3) = (0, -1, 0)$$

be the three simple type-II wall images of Proposition 2.1. On the same type-II wall branch $d = m + 1$, their elliptic degrees are

$$d_{\delta_1} = 1, \qquad d_{\delta_2} = 2, \qquad d_{\delta_3} = 1.$$

Thus, after retained branch-comparison rows identify

$$b_R^{\mathrm{geom}}(\eta_i) = d_{\partial_i}, \qquad \overline{\Pi}_X(\eta_i) = \gamma_{\mathrm{wall}}(\partial_i),$$

the three simple type-II walls are recovered by the hybrid vocabulary as the three singleton wrapped colour profiles

$$\eta_i \in \Gamma_R^{\mathrm{wr}}, \qquad i = 1, 2, 3.$$

For each i, a finite local insertion on the left or right of this wall is typed by the LW or WL mixed correspondence, and local insertions on both sides are typed by the ordered LWL correspondence. No LL-only local operation recovers a type-II wall profile.

This is recovery of operation *types*. It does not populate the wrapped prequotient stacks, mixed extension stacks, anchors, quotient descent, O2 wall atlas, protected integration, or transition system.

Proof. The triples are those of Proposition 2.1. Applying the wall-branch formula $d = m + 1$ gives

$$(1, 1, 0) \mapsto 1, \qquad (0, 1, 1) \mapsto 2, \qquad (0, -1, 0) \mapsto 1.$$

All three integers are positive. Once a retained branch-comparison row identifies these integers with b_R^{geom}, the definition

$$\Gamma_R^{\mathrm{wr}} = \{b_R^{\mathrm{geom}} > 0\}, \qquad \Gamma_R^{\mathrm{loc}} = \{b_R^{\mathrm{geom}} = 0\}$$

places all three η_i in the wrapped window and none in the local window. Definition 0.5 supplies the singleton wrapped carrier profile. Definitions 0.21 and 0.47 are exactly the before E-quotient operation profiles with one wrapped input and local inputs on one or both sides. Since an LL correspondence has only local inputs and local target colour, it cannot have target colour $\eta_i \in \Gamma_R^{\mathrm{wr}}$. The final sentence separates operation-profile recovery from the missing populated finite-stage rows. □

The packet

```
certificates/hybrid/type_ii_wall_hybrid_operation_recovery
```

verifies `TYPE_II_WALL_HYBRID_OPERATION_RECOVERY_VERIFIED`: the three imported wall triples are distinct, have positive wall-branch degrees 1, 2, 1, are routed to singleton wrapped colour profiles, and carry exactly the LW, WL, and LWL mixed operation profiles. Its obstruction rows record that retained branch-comparison, wall-object, support, anchor, mixed-stack, O2 atlas, E-quotient losslessness, quotient, protected-integration, aggregate-carrier, and transition rows are still missing.

Proposition 0.17 *(Type-II wall profiles survive the E-quotient assignment).* [profile-level survival proved; object-level nonvanishing not proved] Let η_i, $i = 1, 2, 3$, be the singleton wrapped type-II wall profiles of Proposition 0.16, and assume the finite quotient-after-correspondence datum of Definition 0.50. Then the quotient assignment sends each retained unreduced wrapped profile to

$$Q_{E,R}^{\mathrm{corr}}(\mathcal{M}_{\eta_i,R}^{\mathrm{wr,rig}}) = [\mathcal{M}_{\eta_i,R}^{\mathrm{wr,rig}}/E].$$

For every retained finite local insertion on one or both sides of η_i, the associated LW, WL, or LWL operation profile is sent to the quotient of the already formed unreduced correspondence stack. Hence none of the three simple type-II wall profiles is deleted by the reduced E-quotient at the quotient-after-correspondence assignment level.

This does not prove that $\mathcal{M}_{\eta_i,R}^{\mathrm{wr,rig}}$ is nonempty, that its quotient cycle has nonzero Borel–Moore class, that vanishing-cycle coefficients descend nontrivially, or that the reduced object enters the O2 wall atlas, protected Pfaffian, aggregate hybrid carrier, or transition system.

Proof. By Proposition 0.16, the three simple type-II wall images have positive wall-branch elliptic degrees 1, 2, 1. After the retained branch-comparison rows they are therefore the singleton wrapped colours $\eta_i \in \Gamma_R^{\mathrm{wr}}$. The same proposition identifies the finite local insertions on these wall profiles with the one-sided LW and WL mixed operation profiles and the two-sided LWL operation profile, all before the reduced E-quotient.

Definition 0.50 assigns $[Y/E]$ to every retained object Y of $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, including wrapped rigidified prequotients. Taking $Y = \mathcal{M}_{\eta_i,R}^{\mathrm{wr,rig}}$ gives the displayed quotient object for each i. The same definition assigns to every generating unreduced correspondence

$$e = (Y_0 \xleftarrow{p_e} \mathfrak{E}_e \xrightarrow{q_e} Y_1)$$

the already-formed quotient correspondence

$$\overline{e} = (\overline{Y}_0 \xleftarrow{\overline{p}_e} [\mathfrak{E}_e/E] \xrightarrow{\overline{q}_e} \overline{Y}_1).$$

Applying this to the LW, WL, and LWL operation profiles therefore preserves the operation labels as quotient-after-correspondence arrows. Proposition 0.54 records that this is not a quotient-first reconstruction from reduced object stacks, and Proposition 0.60 keeps any supplied transition row in the same quotient-presented form. The final paragraph states the unsupplied rows and prevents this profile-level assignment theorem from being read as object-level nonvanishing, O2 atlas construction, or protected integration. □

The packet

```
certificates/hybrid/type_ii_wall_e_quotient_survival
```

verifies `TYPE_II_WALL_E_QUOTIENT_SURVIVAL_VERIFIED`: the three wrapped type-II wall profiles are assigned quotient objects, their nine LW, WL, and LWL operation profiles are assigned quotient-after-correspondence arrows, quotient-first reconstruction is excluded, and HN transition compatibility is imported. Its firewall rows record that object-level nonzero survival, coefficient descent, orientation descent, O2 wall-atlas insertion, protected integration, and aggregate hybrid-carrier population remain open.

PROPOSITION 0.18 *(No type-I wall enters the type-II hybrid wall inlet).* [profile-level exclusion proved; aggregate carrier not populated] Let

$$\vartheta_{12} = f_{-2} - f_2, \qquad \vartheta_{13} = f_2 - f_3, \qquad \vartheta_{23} = f_{-2} - f_3$$

be the three type-I chamber-automorphism roots of Lemma 2.3. At finite HN height R, let the type-II wall-profile inlet be the finite set of retained profiles

$$\Gamma_R^{\mathrm{wall},II} = \left\{ \eta_i \;\middle|\; \overline{\Pi}_X(\eta_i) = \gamma_{\mathrm{wall}}(\partial_i), b_R^{\mathrm{geom}}(\eta_i) = d_{\partial_i}, i = 1, 2, 3 \right\}$$

appearing in Propositions 0.16 and 0.17. Then every element of $\Gamma_R^{\mathrm{wall},II}$ is labelled by one of the three type-II divisor roots $\partial_1, \partial_2, \partial_3$. No type-I root $\vartheta_{12}, \vartheta_{13}, \vartheta_{23}$ supplies a wall profile in $\Gamma_R^{\mathrm{wall},II}$, and none is created by the quotient-after-correspondence assignment.

This is an exclusion theorem for the wall-profile inlet. It is not a global exhaustion of all retained hybrid colours, not a population theorem for the aggregate hybrid carrier, and not an O2 wall-atlas type-I exclusion row.

Proof. Remark 1.2 and the finite `type_ii_chamber_isotropic` packet record the two square-two divisibility classes. The roots $\delta_1, \delta_2, \delta_3$ have ambient divisibility 2, lie in the type-II sublattice, and occur in $\operatorname{div}(\Delta_5)$ with order 1. The roots $\vartheta_{12}, \vartheta_{13}, \vartheta_{23}$ have ambient divisibility 1, generate the S_3 chamber automorphism action of Lemma 2.3, and have zero Δ_5-divisor order.

Proposition 0.16 imports only the three type-II wall images

$$\gamma_{\mathrm{wall}}(\delta_1), \qquad \gamma_{\mathrm{wall}}(\delta_2), \qquad \gamma_{\mathrm{wall}}(\delta_3)$$

and routes them to singleton wrapped profiles with positive wall-branch degrees 1, 2, 1. Hence the wall-profile inlet just defined is indexed by the divisor-supported type-II roots and by no other square-two root. The type-I reflections act only by permuting the three δ_i; they do not add a fourth wall label, and they do not turn a zero-order type-I chamber wall into a Δ_5-divisor wall.

Proposition 0.17 then applies $Q^{\mathrm{corr}}_{E,R}$ to the same three already formed wall profiles and to their LW, WL, and LWL operation profiles. Since quotient-after-correspondence assigns quotient objects and quotient arrows to retained inputs, it preserves this three-element indexing set and introduces no new type-I profile. The final paragraph separates this profile-level firewall from the unsupplied global carrier and O2-atlas rows. □

The packet

`certificates/hybrid/type_i_false_wall_exclusion`

verifies `TYPE_I_FALSE_WALL_EXCLUSION_VERIFIED`: the admitted wall-profile rows are exactly the three type-II profiles $\delta_1, \delta_2, \delta_3$, the three type-I chamber roots have divisibility 1, zero Δ_5-divisor support, and zero Δ_5-divisor order, and the E-quotient survival row does not resurrect a type-I wall profile. Its firewall rows record that global colour exhaustion, aggregate hybrid-carrier population, retained wall objects, coefficient descent, O2 wall-atlas insertion, protected integration, and transition to O2 remain open.

Definition 0.19 *(Local/local correspondence datum).* Fix a finite HN height R. Let $\alpha : I \to \Gamma^{\mathrm{loc}}_R$ and $\beta : J \to \Gamma^{\mathrm{loc}}_R$ be finite local colour maps, and write $|\alpha| = \sum_{i\in I} \alpha_i$, $|\beta| = \sum_{j \in J} \beta_j$. A local/local correspondence row is defined only for a retained target colour $\gamma \in \Gamma^{\mathrm{loc}}_R$ satisfying

$$\gamma = |\alpha| + |\beta|.$$

The target index set K is $I \sqcup J$, modulo the ordinary Ran collision maps. The ordered local/local correspondence is the projection-finite diagram

$$\mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K} \xrightarrow{p^{\mathrm{LL}}} \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\beta,R,J}, \qquad \mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K} \xrightarrow{q^{\mathrm{LL}}} \mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K},$$

classifying retained extensions

$$0 \longrightarrow A_{\beta,J} \longrightarrow B_{\gamma,K} \longrightarrow A_{\alpha,I} \longrightarrow 0.$$

This diagram lives over $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E) \times \mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ and targets $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$. It has no wrapped input and no reduced E-quotient. Symmetric descent, collision compatibility, properness of the local extension stack, Thom–Sebastiani transport, and transition compatibility are separate rows. The diagram is not a mixed correspondence, not a wrapped/wrapped correspondence, not a two-sided mixed correspondence, and not an ordinary uncoloured Ran product.

The packet

```
certificates/hybrid/local_local_correspondence_definition
```

verifies `LOCAL_LOCAL_CORRESPONDENCE_DEFINITION_VERIFIED`: the ordered LL correspondence type is defined, while extension-stack rows, source/target-map rows, descent rows, properness rows, Thom–Sebastiani rows, collision rows, and transition rows are not supplied by that definition packet.

PROPOSITION 0.20 *(Target properness for local/local extensions).* [proved for the Hall target map] Assume the local closed-incidence stacks $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$, $\mathfrak{M}^{\mathrm{loc,cl}}_{\beta,R,J}$, and $\mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K}$ carry the universal perfect complexes and the retained extension-closure datum of Definition 1.29. Then the target map

$$q^{\mathrm{LL}} : \mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K} \longrightarrow \mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K}$$

is proper. The source map p^{LL} is not asserted to be proper.

Proof. Let $B_{\gamma,K}$ be the universal target object on $\mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K}$. The stack $\mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K}$ is the closed substack of the relative Quot scheme over $\mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K}$ parametrising quotients

$$B_{\gamma,K} \twoheadrightarrow A_{\alpha,I}$$

whose kernel is an object $A_{\beta,J}$, with exact sequence

$$0 \longrightarrow A_{\beta,J} \longrightarrow B_{\gamma,K} \longrightarrow A_{\alpha,I} \longrightarrow 0.$$

The retained extension-closure datum says that the middle term stays inside the retained finite HN window. The local support conditions over E^I, E^J, and E^K, and the retained numerical colour conditions α, β, γ, are closed conditions inside the relative Quot scheme. Grothendieck projectivity of Quot [50, Theorem 2.2.4] makes the relative Quot scheme projective over $\mathfrak{M}^{\mathrm{loc,cl}}_{\gamma,R,K}$, hence proper. A closed substack of a proper stack over the same base is proper, so q^{LL} is proper.

The source map has a different fibre. Over fixed endpoints $(A_{\alpha,I}, A_{\beta,J})$, the extension parameters contain $\mathrm{Ext}^1(A_{\alpha,I}, A_{\beta,J})$ modulo the usual Hom-automorphisms. When this Ext group is nonzero, the fibre contains affine directions and is not proper in general. Thus the properness needed for the Hall pushforward is the target-properness of q^{LL}, not source-properness of p^{LL}. □

The packet

```
certificates/hybrid/local_local_extension_properness
```

verifies `LOCAL_LOCAL_TARGET_PROPERNESS_VERIFIED`: the local/local extension stack is represented as a closed relative Quot locus over the target local incidence stack, the target map q^{LL} is proper, and the false source-properness substitute is excluded. Descent, collision compatibility, Thom–Sebastiani transport, transition compatibility, and aggregate hybrid-carrier population remain separate obligations. The compact-support exceptional pushforward model is supplied separately by Definition 0.25.

Definition 0.21 *(One-sided mixed local/wrapped correspondence datum).* Fix a finite HN height R. Let $\alpha : I \to \Gamma^{\mathrm{loc}}_R$ be a finite local colour map and write $|\alpha| = \sum_{i \in I} \alpha_i$. Let $\eta \in \Gamma^{\mathrm{wr}}_R$. A one-sided mixed correspondence row is defined only for a retained target colour $\zeta \in \Gamma^{\mathrm{wr}}_R$ satisfying the typed equality

$$\zeta = |\alpha| + \eta \quad \text{or} \quad \zeta = \eta + |\alpha|,$$

with the positive wrapped condition read through b_R^{geom}. The two ordered mixed diagrams are the unreduced correspondence stacks

$$\mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I} \xrightarrow{p^{\mathrm{LW}}} \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}, \qquad \mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I} \xrightarrow{q^{\mathrm{LW}}} \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

classifying retained extensions

$$0 \longrightarrow W_\eta \longrightarrow B_\zeta \longrightarrow A_\alpha \longrightarrow 0,$$

and

$$\mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I} \xrightarrow{p^{\mathrm{WL}}} \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}, \qquad \mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I} \xrightarrow{q^{\mathrm{WL}}} \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

classifying retained extensions

$$0 \longrightarrow A_\alpha \longrightarrow B_\zeta \longrightarrow W_\eta \longrightarrow 0.$$

The symbols $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$ and $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ are the local closed incidence stack and wrapped rigidified prequotient supplied by the finite hybrid datum. These diagrams live before the reduced E-quotient. They are not local-local correspondences, not wrapped-wrapped correspondences, not ordered two-sided mixed correspondences, and not a proof of admissibility, base change, projection formula, Thom–Sebastiani transport, or source/target-anchor losslessness.

The packet

`certificates/hybrid/mixed_local_wrapped_correspondence_definition`

verifies `MIXED_LOCAL_WRAPPED_CORRESPONDENCE_DEFINITION_VERIFIED`: the one-sided LW and WL mixed correspondence types are defined, while extension-stack rows, source/target-map rows, populated anchor-memory rows, admissibility rows, base-change rows, projection-formula rows, Thom–Sebastiani rows, and transition rows are recorded as missing open obligations in that definition packet.

Proposition 0.22 *(Target admissibility for mixed local/wrapped extensions).* [proved for the Hall target maps] Assume the local closed-incidence stacks $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$, the wrapped prequotients $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ and $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$, the universal perfect complexes, the retained closed-substack datum, and the retained extension-closure datum are supplied at the fixed finite HN height R. Then the target maps in the one-sided unreduced mixed correspondences

$$q^{\mathrm{LW}} : \mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}, \qquad q^{\mathrm{WL}} : \mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$$

are proper. Hence the target leg needed for the mixed Hall pushforward is admissible as a proper geometric map before the reduced E-quotient. The source maps p^{LW} and p^{WL} are not asserted to be proper.

Proof. Let B_ζ be the universal target object on $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$. For the LW order, the stack $\mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I}$ is the closed substack of the relative Quot scheme over $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$ parametrising quotients

$$B_\zeta \twoheadrightarrow A_{\alpha,I}$$

whose kernel is an object W_η, with exact sequence

$$0 \longrightarrow W_\eta \longrightarrow B_\zeta \longrightarrow A_{\alpha,I} \longrightarrow 0.$$

The quotient is required to lie in the local closed-incidence stack, and the kernel is required to lie in the retained wrapped prequotient. The retained closed-substack datum makes these endpoint conditions closed on the relative Quot scheme. The retained colour condition $\zeta = |\alpha| + \eta$ and the finite HN window condition are fixed row conditions; the retained extension-closure datum records that the allowed middle term remains in the finite window. Grothendieck projectivity of Quot [50, Theorem 2.2.4] makes the relative Quot scheme projective, hence proper, over $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$. A closed substack of this relative Quot scheme is proper over the same target, so q^{LW} is proper.

For the WL order one uses the relative Quot scheme over the same target, but now with quotient

$$B_\zeta \twoheadrightarrow W_\eta$$

and kernel $A_{\alpha,I}$:

$$0 \longrightarrow A_{\alpha,I} \longrightarrow B_\zeta \longrightarrow W_\eta \longrightarrow 0.$$

The wrapped quotient condition, the local-kernel condition, and the retained colour condition $\zeta = \eta + |\alpha|$ are closed row conditions by the same retained closed-substack and extension-closure inputs. The relative Quot scheme is projective over $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$, and the WL extension stack is a closed substack of it. Hence q^{WL} is proper.

The source maps have the usual Hall extension fibres. Over fixed endpoints in the LW order one sees extension classes in $\mathrm{Ext}^1(A_{\alpha,I}, W_\eta)$ modulo Hom-automorphisms; in the WL order one sees $\mathrm{Ext}^1(W_\eta, A_{\alpha,I})$ modulo the same automorphism relation. These fibres contain affine directions when the corresponding Ext^1-group is nonzero, so source properness is not a consequence of the argument and is not claimed. The compact-support exceptional pushforward model, base change, projection formula, Thom–Sebastiani transport, anchor losslessness, quotient descent, and transition compatibility remain separate rows. □

The packet

`certificates/hybrid/mixed_local_wrapped_extension_admissibility`

verifies `MIXED_LOCAL_WRAPPED_ADMISSIBILITY_VERIFIED`: the one-sided mixed extension stacks are represented as closed relative Quot loci over the wrapped target prequotient, the target maps q^{LW} and q^{WL} are proper, and the false source-properness substitute is excluded. The compact-support exceptional pushforward model is supplied separately by Definition 0.25. Populated anchor-memory rows, base change, projection formula, Thom–Sebastiani transport, transition compatibility, quotient descent, and aggregate hybrid-carrier population remain separate obligations.

Definition 0.23 *(Wrapped/wrapped correspondence datum).* Fix a finite HN height R. Let $\eta_1, \eta_2 \in \Gamma_R^{\mathrm{wr}}$. A wrapped/wrapped correspondence row is defined only for a retained target colour $\zeta \in \Gamma_R^{\mathrm{wr}}$ satisfying

$$\zeta = \eta_1 + \eta_2,$$

with the positive wrapped condition read through b_R^{geom}. The ordered wrapped/wrapped correspondence is the unreduced diagram

$$\mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R} \xrightarrow{p^{\mathrm{WW}}} \mathcal{M}^{\mathrm{wr,rig}}_{\eta_1,R} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta_2,R}, \qquad \mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R} \xrightarrow{q^{\mathrm{WW}}} \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

classifying retained extensions

$$0 \longrightarrow W_{\eta_2} \longrightarrow B_\zeta \longrightarrow W_{\eta_1} \longrightarrow 0.$$

This diagram lives over $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E) \times \mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$ and targets $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$, before the reduced E-quotient. The source pair is ordered; symmetric descent for swapping η_1 and η_2 is a separate row. Source and target anchor memory, admissibility, Thom–Sebastiani transport, and transition compatibility are also separate rows. The diagram is not a mixed correspondence, not a local-local correspondence, not a two-sided mixed correspondence, and not a scalar or quotient-first replacement.

The packet

```
certificates/hybrid/wrapped_wrapped_correspondence_definition
```

verifies `WRAPPED_WRAPPED_CORRESPONDENCE_DEFINITION_VERIFIED`: the ordered WW correspondence type is defined, while extension-stack rows, source/target-map rows, populated anchor-memory rows, admissibility rows, Thom–Sebastiani rows, symmetric-descent rows, and transition rows are recorded as missing open obligations in that definition packet.

Proposition 0.24 *(Target admissibility for wrapped/wrapped extensions).* [proved for the Hall target map] Assume the wrapped prequotients $\mathcal{M}^{\mathrm{wr,rig}}_{\eta_1,R}$, $\mathcal{M}^{\mathrm{wr,rig}}_{\eta_2,R}$, and $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$, the universal perfect complexes, the retained closed-substack datum, and the retained extension-closure datum are supplied at the fixed finite HN height R. Then the target map in the ordered unreduced wrapped/wrapped correspondence

$$q^{\mathrm{WW}} : \mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R} \longrightarrow \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$$

is proper. Hence the target leg needed for the wrapped/wrapped Hall pushforward is admissible as a proper geometric map before the reduced E-quotient. The source map p^{WW} is not asserted to be proper.

Proof. Let B_ζ be the universal target object on $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$. The stack $\mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R}$ is the closed substack of the relative Quot scheme over $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$ parametrising quotients

$$B_\zeta \twoheadrightarrow W_{\eta_1}$$

whose kernel is an object W_{η_2}, with exact sequence

$$0 \longrightarrow W_{\eta_2} \longrightarrow B_\zeta \longrightarrow W_{\eta_1} \longrightarrow 0.$$

The quotient and kernel are required to lie in the retained wrapped prequotients of colours η_1 and η_2. The retained closed-substack datum makes these endpoint conditions closed on the relative Quot scheme. The retained colour condition $\zeta = \eta_1 + \eta_2$ and the finite HN window condition are fixed row conditions; the retained extension-closure datum records that the allowed middle term remains in the finite window. Grothendieck projectivity of Quot [50, Theorem 2.2.4] makes the relative Quot scheme projective, hence proper, over $\mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$. A closed substack of this relative Quot scheme is proper over the same target, so q^{WW} is proper.

The source map has the usual Hall extension fibre. Over fixed endpoints (W_{η_1}, W_{η_2}), the extension parameters contain $\mathrm{Ext}^1(W_{\eta_1}, W_{\eta_2})$ modulo the usual Hom-automorphisms. When this Ext^1-group is nonzero, the fibre contains affine directions and is not proper in general. Thus the properness supplied here is target properness of q^{WW}, not source properness of p^{WW}. The compact-support exceptional pushforward model, symmetric descent, populated anchor-memory rows, Thom–Sebastiani transport, quotient descent, transition compatibility, and aggregate hybrid-carrier population remain separate rows. □

The packet

`certificates/hybrid/wrapped_wrapped_extension_admissibility`

verifies `WRAPPED_WRAPPED_ADMISSIBILITY_VERIFIED`: the ordered wrapped/wrapped extension stack is represented as a closed relative Quot locus over the wrapped target prequotient, the target map q^{WW} is proper, and the false source-properness substitute is excluded. The compact-support exceptional pushforward model is supplied separately by Definition 0.25. Symmetric descent, populated anchor-memory rows, Thom–Sebastiani transport, quotient descent, transition compatibility, and aggregate hybrid-carrier population remain separate obligations.

Definition 0.25 *(Compact-support exceptional pushforward model).* Let

$$Y \xleftarrow{p} \mathfrak{E} \xrightarrow{f} Z$$

be one of the retained finite correspondence maps in the local, mixed, wrapped, or flag correspondence systems at HN height R, and let

$$K_{\mathfrak{E}} \in D_c^b(\mathfrak{E})$$

be the constructible vanishing-cycle complex with orientation line on the source of f. A compact-support exceptional pushforward model for $(f, K_{\mathfrak{E}})$ is a tuple

$$\mathfrak{Q}^{\mathrm{cs}}(f, K_{\mathfrak{E}}) = (j_f, \overline{\mathfrak{E}}_f, \overline{f}, K_{\mathfrak{E}}, f_!^{\mathrm{cs}} K_{\mathfrak{E}})$$

such that:

(i) $j_f : \mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_f$ is an open immersion into a finite-type Artin stack with finite residual inertia on the retained strata;

(ii) $\overline{f} : \overline{\mathfrak{E}}_f \to Z$ is proper and
$$f = \overline{f} \circ j_f;$$

(iii) $j_{f,!} K_{\mathfrak{E}}$ exists in the chosen constructible six-functor theory on the retained d-critical stack, and
$$f_!^{\mathrm{cs}} K_{\mathfrak{E}} := \overline{f}_* \, j_{f,!} K_{\mathfrak{E}} \in D_c^b(Z)$$
is the compact-support exceptional pushforward along f;

(iv) if f is proper, the identity compactification
$$j_f = \mathrm{id}_{\mathfrak{E}}, \qquad \overline{\mathfrak{E}}_f = \mathfrak{E}, \qquad \overline{f} = f$$
is allowed, and then $f_!^{\mathrm{cs}} = f_*$.

The datum is a chosen compact-support model, not a theorem that two compactifications give canonically identical functors. It does not prove base change, the projection formula, Thom–Sebastiani compatibility, quotient descent, transition compatibility, symmetric descent, or associativity on the two-step flag atlas. Those are the separate functorial rows required before

$$q_!\,\mathrm{TS}^{\mathrm{red}}\,p^*$$

is a coherent Hall multiplication on the finite hybrid carrier.

The packet

```
certificates/hybrid/compact_support_exceptional_pushforward_model
```

verifies `COMPACT_SUPPORT_EXCEPTIONAL_PUSHFORWARD_MODEL_DEFINED`: Definition 0.25 defines the compactification, extension-by-zero, and proper pushforward formula for a retained finite correspondence map, and it records the proper-map specialization $f_!^{\mathrm{cs}} = f_*$. It does not populate all correspondence maps, prove independence of the compactification, prove base change, prove the projection formula, transport Thom–Sebastiani, descend through the E-quotient, prove transition compatibility, or populate the aggregate hybrid carrier.

Proposition 0.26 *(Base change for one-sided mixed correspondences).* [proved for Cartesian compact-support squares] Let $o \in \{\mathrm{LW}, \mathrm{WL}\}$, and let

$$q^o : \mathfrak{E}^o_{\alpha,\eta;\zeta,R,I} \longrightarrow Z^o_{\zeta,R} := \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$$

be the target map of the corresponding one-sided mixed correspondence. Let $K^o \in D^b_c(\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I})$ be the constructible vanishing-cycle complex with orientation line. Suppose that a retained target morphism $g : Z' \to Z^o_{\zeta,R}$ is supplied, and that the mixed correspondence and the compact-support model are pulled back by Cartesian squares

$$\begin{tikzcd}
\mathfrak{E}^{o\prime} \arrow[r, "\widetilde{g}"] \arrow[d, "j_o'"'] & \mathfrak{E}^o_{\alpha,\eta;\zeta,R,I} \arrow[d, "j_o"] \\
\overline{\mathfrak{E}}^{o\prime} \arrow[r, "\overline{g}"] \arrow[d, "\overline{q}^{o\prime}"'] & \overline{\mathfrak{E}}^o \arrow[d, "\overline{q}^o"] \\
Z' \arrow[r, "g"'] & Z^o_{\zeta,R}
\end{tikzcd}$$

with $\overline{q}^o$ and $\overline{q}^{o\prime}$ proper, and with a coefficient identification

$$K^{o\prime} \simeq \widetilde{g}^* K^o.$$

Then the compact-support exceptional pushforward of Definition 0.25 carries a canonical base-change isomorphism

$$g^*(q^o)^{\mathrm{cs}}_! K^o \xrightarrow{\ \sim\ } (q^{o\prime})^{\mathrm{cs}}_! K^{o\prime}.$$

For the proper target maps of Proposition 0.22, the identity compactification gives the ordinary proper-base-change isomorphism for q^{LW} and q^{WL}.

Proof. By definition,

$$(q^o)_!^{\mathrm{cs}} K^o = \overline{q}^o_* \, j_{o,!} K^o.$$

The lower square is Cartesian and $\overline{q}^o$ is proper, so the proper base-change isomorphism in the chosen constructible six-functor theory gives

$$g^* \overline{q}^o_* \, j_{o,!} K^o \simeq \overline{q}^{o\prime}_* \, \overline{g}^* \, j_{o,!} K^o.$$

The upper square is Cartesian and j_o is an open immersion; hence extension by zero commutes with this base change:

$$\overline{g}^* \, j_{o,!} K^o \simeq j'_{o,!} \, \widetilde{g}^* K^o.$$

Using the supplied coefficient identification $K^{o\prime} \simeq \widetilde{g}^* K^o$, the last term becomes

$$\overline{q}^{o\prime}_* \, j'_{o,!} K^{o\prime} = (q^{o\prime})_!^{\mathrm{cs}} K^{o\prime}.$$

This proves the displayed base-change isomorphism. The argument proves only the Cartesian compact-support base-change row for the chosen model. It does not prove the projection formula, Thom–Sebastiani compatibility, quotient descent, transition compatibility, choice independence for compactifications, or associativity on the two-step flag atlas. □

The packet

`certificates/hybrid/mixed_correspondence_base_change`

verifies `MIXED_CORRESPONDENCE_BASE_CHANGE_VERIFIED`: Proposition 0.26 supplies the Cartesian compact-support base-change isomorphisms for the LW and WL target maps, including the proper-target specialization. It does not prove the projection formula, Thom–Sebastiani transport, quotient descent, transition compatibility, compactification-choice independence, or aggregate hybrid-carrier population.

PROPOSITION 0.27 *(Projection formula for one-sided mixed correspondences).* [proved for retained finite-Tor target coefficients] Let $o \in \{\mathrm{LW}, \mathrm{WL}\}$, and let

$$q^o : \mathfrak{E}^o_{\alpha,\eta;\zeta,R,I} \longrightarrow Z^o_{\zeta,R} := \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R}$$

be the target map of the corresponding one-sided mixed correspondence, equipped with a chosen compact-support model

$$\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I} \xrightarrow{j_o} \overline{\mathfrak{E}}^o \xrightarrow{\overline{q}^o} Z^o_{\zeta,R}, \qquad q^o = \overline{q}^o \circ j_o,$$

where j_o is an open immersion and $\overline{q}^o$ is proper. Let $K^o \in D^b_c(\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I})$ be the constructible vanishing-cycle complex with orientation line. Let $L \in D^b_c(Z^o_{\zeta,R})$ be a retained target coefficient of finite Tor-amplitude, so that the open-immersion and proper projection formulas hold in the chosen constructible six-functor theory. Then there is a canonical isomorphism

$$(q^o)_!^{\mathrm{cs}}\big(K^o \otimes^{\mathbf{L}} (q^o)^* L\big) \xrightarrow{\sim} (q^o)_!^{\mathrm{cs}} K^o \otimes^{\mathbf{L}} L.$$

For the proper target maps of Proposition 0.22, the identity compactification specializes this to the ordinary proper projection formula for q^{LW} and q^{WL}.

Proof. Since $q^o = \overline{q}^o \circ j_o$, one has

$$(q^o)^* L = j_o^* (\overline{q}^o)^* L.$$

Expanding the compact-support model gives

$$(q^o)_!^{\mathrm{cs}}\big(K^o \otimes^{\mathbf{L}} (q^o)^* L\big) = \overline{q}^o_* \, j_{o,!}\big(K^o \otimes^{\mathbf{L}} j_o^* (\overline{q}^o)^* L\big).$$

The projection formula for the open immersion j_o identifies the right-hand side with

$$\overline{q}^o_*\big(j_{o,!} K^o \otimes^{\mathbf{L}} (\overline{q}^o)^* L\big).$$

The map $\overline{q}^o$ is proper; the proper projection formula therefore gives

$$\overline{q}^o_*\big(j_{o,!} K^o \otimes^{\mathbf{L}} (\overline{q}^o)^* L\big) \simeq \overline{q}^o_* j_{o,!} K^o \otimes^{\mathbf{L}} L = (q^o)_!^{\mathrm{cs}} K^o \otimes^{\mathbf{L}} L.$$

This proves the stated isomorphism. The proof uses the chosen compactification and the retained finite-Tor target coefficient. It does not prove compactification-choice independence, Thom–Sebastiani compatibility, quotient descent, transition compatibility, or associativity on the two-step flag atlas. □

The packet

```
certificates/hybrid/mixed_correspondence_projection_formula
```

verifies `MIXED_CORRESPONDENCE_PROJECTION_FORMULA_VERIFIED`: Proposition 0.27 supplies the compact-support projection formula for the LW and WL target maps with retained finite-Tor target coefficients. It does not prove Thom–Sebastiani transport, quotient descent, transition compatibility, compactification-choice independence, or aggregate hybrid-carrier population.

Proposition 0.28 *(Thom–Sebastiani transport for one-sided mixed correspondences).* [proved for supplied reduced additive d-critical charts] Let $o \in \{\mathrm{LW}, \mathrm{WL}\}$, and let

$$Y^o_{\mathrm{src}} = \begin{cases} \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}, & o = \mathrm{LW}, \\ \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}, & o = \mathrm{WL} \end{cases}$$

be the ordered source product of the one-sided mixed correspondence with source map $p^o : \mathfrak{E}^o_{\alpha,\eta;\zeta,R,I} \to Y^o_{\mathrm{src}}$. Write

$$K^o_{\mathrm{src}} = \Phi^{\mathrm{red}}_{Y^o_{\mathrm{src}}} \otimes o^{\mathrm{red}}_{Y^o_{\mathrm{src}}}, \qquad K^o_{\mathfrak{E}} = \Phi^{\mathrm{red}}_{\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I}} \otimes o^{\mathrm{red}}_{\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I}}.$$

Assume that the retained mixed extension stack is covered by Joyce–BBDJS d-critical charts for which:

(i) the extension potential is the pullback of the ordered sum of the two source potentials, up to the standard nondegenerate quadratic stabilization;
(ii) the K3 semiregularity cosection is additive along the mixed extension correspondence, so the reduced obstruction theories and reduced vanishing-cycle complexes are pulled from the same quotient of the ordinary obstruction theory;
(iii) the Joyce–Upmeier orientation line on $\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I}$ is identified with $(p^o)^* o^{\mathrm{red}}_{Y^o_{\mathrm{src}}}$, and the orientation Thom–Sebastiani defect is zero on chart overlaps.

Then the mixed correspondence carries a canonical reduced Thom–Sebastiani transport isomorphism

$$\mathrm{TS}^{\mathrm{red},o}_{\alpha,\eta;R,I} : (p^o)^* K^o_{\mathrm{src}} \xrightarrow{\sim} K^o_{\mathfrak{E}}$$

in $D^b_c(\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I})$. The construction is ordered: the LW and WL isomorphisms differ by the Koszul symmetry of the source product and are recorded as separate rows.

Proof. On a retained d-critical chart, condition (1) puts the mixed extension potential in the BBDJS Thom–Sebastiani normal form for the ordered sum of the two source potentials, after harmless quadratic stabilization [12]. Hence the BBDJS Thom–Sebastiani theorem gives an isomorphism between the pullback of the exterior product of the two source vanishing-cycle complexes and the vanishing-cycle complex of the extension chart. Condition (2) identifies the ordinary and cosection-reduced sides of this isomorphism after quotienting by the common additive semiregularity cosection. Condition (3) tensors the vanishing-cycle isomorphism with the Joyce–Upmeier orientation transport and makes it independent of the chosen chart overlap. The descent data therefore glue the chartwise isomorphisms to the displayed object of $D^b_c(\mathfrak{E}^o_{\alpha,\eta;\zeta,R,I})$.

The proof supplies only the binary mixed Thom–Sebastiani transport for the chosen reduced d-critical and orientation atlas. It does not construct the orientation line or discharge $(O1)$. The quotient-after-correspondence descent of this transport is Proposition 0.53. Transition compatibility and the eight-word two-step flag pentagon remain separate obligations. □

The packet

`certificates/hybrid/mixed_correspondence_thom_sebastiani`

verifies `MIXED_THOM_SEBASTIANI_TRANSPORT_CONDITIONAL_VERIFIED`: Proposition 0.28 supplies the binary mixed Thom–Sebastiani transport for LW and WL from supplied additive reduced d-critical charts and zero-defect orientation transport. It does not prove $(O1)$, quotient descent, transition compatibility, two-step flag associativity, or aggregate hybrid-carrier population.

Definition 0.29 *(Eight local/wrapped binary words).* At a fixed finite HN height R, let

$$\mathbb{A}_{\mathrm{hyb}} = \{\mathrm{L}, \mathrm{W}\}$$

be the ordered hybrid alphabet, where L denotes a projection-finite local input of elliptic degree 0, and W denotes a rigidified wrapped input of positive elliptic degree. The binary type product is

$$\mathrm{L} \star \mathrm{L} = \mathrm{L}, \qquad \mathrm{L} \star \mathrm{W} = \mathrm{W}, \qquad \mathrm{W} \star \mathrm{L} = \mathrm{W}, \qquad \mathrm{W} \star \mathrm{W} = \mathrm{W}.$$

Thus a length-three word

$$w = \epsilon_1\epsilon_2\epsilon_3 \in \mathbb{A}^3_{\mathrm{hyb}}$$

has left and right intermediate types

$$\epsilon_{12} = \epsilon_1 \star \epsilon_2, \qquad \epsilon_{23} = \epsilon_2 \star \epsilon_3,$$

and final type

$$\epsilon_{123} = (\epsilon_1 \star \epsilon_2) \star \epsilon_3 = \epsilon_1 \star (\epsilon_2 \star \epsilon_3).$$

The eight ordered words are

$$\mathcal{W}^{(3)}_{\mathrm{hyb}} = \{\mathrm{LLL}, \mathrm{LLW}, \mathrm{LWL}, \mathrm{WLL}, \mathrm{LWW}, \mathrm{WLW}, \mathrm{WWL}, \mathrm{WWW}\}.$$

For compatible ordered charges (ξ_1, ξ_2, ξ_3), the two parenthesization symbols attached to w are

$$((\epsilon_1\epsilon_2)\epsilon_3) : (\xi_1, \xi_2, \xi_3) \mapsto ((\xi_1 + \xi_2), \xi_3),$$

$$(\epsilon_1(\epsilon_2\epsilon_3)) : (\xi_1, \xi_2, \xi_3) \mapsto (\xi_1, (\xi_2 + \xi_3)).$$

This is only the word and type convention. It does not construct the two-step flag stacks, prove that the stacks are retained, or prove associativity, pentagon coherence, quotient descent, transition compatibility, or aggregate hybrid-carrier population.

The packet

`certificates/hybrid/eight_word_binary_vocabulary`

verifies `EIGHT_WORD_BINARY_VOCABULARY_DEFINED`: Definition 0.29 fixes the ordered alphabet, the type product, the eight length-three words, and the two parenthesization symbols. It does not construct the two-step flag stacks or prove associativity.

Proposition 0.30 *(Eight two-step local/wrapped flag stacks).* [constructed as retained flag-Quot substacks] Assume the retained local and wrapped object stacks, the four ordered binary correspondence stacks LL, LW, WL, WW, the retained universal complexes, closed endpoint conditions, and retained extension closure at height R. For every word

$$w = \epsilon_1\epsilon_2\epsilon_3 \in \mathcal{W}^{(3)}_{\mathrm{hyb}}$$

and every compatible ordered colour triple (ξ_1, ξ_2, ξ_3) of those types, there is a retained finite-type Artin stack

$$\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}$$

classifying two-step filtrations

$$0 = C_0 \subset C_1 \subset C_2 \subset C_3$$

in the finite HN heart such that

$$C_i/C_{i-1} \in \mathfrak{M}^{\epsilon_i}_{\xi_i,R} \qquad (i = 1, 2, 3),$$

the intermediate objects C_2/C_0, C_3/C_1, and C_3/C_0 have the types

$$\epsilon_{12}, \qquad \epsilon_{23}, \qquad \epsilon_{123}$$

of Definition 0.29, and every intermediate colour is retained. If one of these types is W, the corresponding object carries the fixed wrapped rigidification and the before-quotient anchor memory.

Let

$$\mathfrak{E}^{\epsilon\delta}_{\xi,v;\xi+v,R}$$

denote the ordered binary correspondence stack of type $\epsilon\delta$, interpreted as LL, LW, WL, or WW by Definitions 0.19, 0.21, and 0.23. The flag stack carries canonical comparison morphisms

$$\lambda^w : \mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R} \longrightarrow \mathfrak{E}^{\epsilon_{12}\epsilon_3}_{\xi_1+\xi_2,\xi_3;\xi_1+\xi_2+\xi_3,R} \times_{\mathfrak{M}^{\epsilon_{12}}_{\xi_1+\xi_2,R}} \mathfrak{E}^{\epsilon_1\epsilon_2}_{\xi_1,\xi_2;\xi_1+\xi_2,R}$$

and

$$\rho^w : \mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R} \longrightarrow \mathfrak{E}^{\epsilon_1\epsilon_{23}}_{\xi_1,\xi_2+\xi_3;\xi_1+\xi_2+\xi_3,R} \times_{\mathfrak{M}^{\epsilon_{23}}_{\xi_2+\xi_3,R}} \mathfrak{E}^{\epsilon_2\epsilon_3}_{\xi_2,\xi_3;\xi_2+\xi_3,R}.$$

The maps are defined by the two forgetful operations on the filtration: λ^w remembers $C_1 \subset C_2 \subset C_3$ as $((\xi_1, \xi_2), \xi_3)$, while ρ^w remembers it as $(\xi_1, (\xi_2, \xi_3))$.

Proof. Fix w and (ξ_1, ξ_2, ξ_3). Over the retained terminal object stack of colour $\xi_1 + \xi_2 + \xi_3$, take the relative two-step flag-Quot stack parametrising quotients

$$C_3 \twoheadrightarrow C_3/C_2, \qquad C_2 \twoheadrightarrow C_2/C_1$$

or equivalently subobjects $C_1 \subset C_2 \subset C_3$. Grothendieck projectivity for relative Quot schemes and its flag version give finite type over the retained terminal stack [50, Theorem 2.2.4]. The conditions that the three graded pieces lie in the prescribed local or wrapped retained substacks, that the intermediate colours equal the displayed sums, and that the wrapped pieces carry the fixed rigidifications and anchor memory are closed retained row conditions. The retained extension-closure packet ensures that C_2/C_0, C_3/C_1, and C_3/C_0 remain in the finite HN window. Hence the required $\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}$ is a retained finite-type closed substack of the flag-Quot stack.

Forgetting the first or the second extension step gives the two displayed morphisms to the iterated correspondence fibre products. The construction is carried out before the reduced E-quotient and only constructs the eight flag stacks and their comparison maps. It does not prove that λ^w and ρ^w induce the same Hall operation, does not prove associativity defects vanish, and does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/eight_word_two_step_flag_stacks`

verifies `EIGHT_WORD_TWO_STEP_FLAG_STACKS_CONSTRUCTED`: Proposition 0.30 constructs the retained two-step flag stack and the two comparison maps for each of the eight words. It does not prove associativity, pentagon coherence, quotient descent, transition compatibility, or aggregate population.

PROPOSITION 0.31 *(Associativity for the LLL word).* [proved for the pure local two-step flag stack] Let α, β, δ be retained local colours at height R, and write

$$\gamma_{12} = \alpha + \beta, \qquad \gamma_{23} = \beta + \delta, \qquad \gamma_{123} = \alpha + \beta + \delta.$$

Assume the LLL flag stack $\mathfrak{F}^{(2),\mathrm{LLL}}_{\alpha,\beta,\delta,R}$ of Proposition 0.30, the local/local target-proper correspondence rows of Proposition 0.20, the chosen compact-support pushforward model, and the local base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the LLL flag square. Then the pure local Hall product satisfies

$$m^{\mathrm{LL}}_{\gamma_{12},\delta,R} \circ \left(m^{\mathrm{LL}}_{\alpha,\beta,R} \otimes 1\right) = m^{\mathrm{LL}}_{\alpha,\gamma_{23},R} \circ \left(1 \otimes m^{\mathrm{LL}}_{\beta,\delta,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the three local inputs. Equivalently, the LLL associativity defect class

$$\mathfrak{d}^{\mathrm{assoc}}_{\mathrm{LLL},R}$$

vanishes.

Proof. Both composites are pull-push operations attached to the same two-step local filtration stack

$$0 \subset C_1 \subset C_2 \subset C_3$$

with graded pieces of colours α, β, δ. The left parenthesization uses the comparison map λ^{LLL} to first form the LL extension of α by β, then the extension of γ_{12} by δ. The right parenthesization uses ρ^{LLL} to first form the extension of β by δ, then the extension of α by γ_{23}. These are two descriptions of the same flag object, so their source and target maps factor through the same retained flag stack.

Proper base change identifies the two iterated target pushforwards after pullback to the flag stack; the projection formula moves the source coefficients through the same compact-support pushforward; and the reduced Thom–Sebastiani identification on the LLL flag square identifies the two tensor products of vanishing-cycle and orientation lines. Hence the two pull-push composites agree. This proves $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LLL},R} = 0$.

The argument is purely local. It proves no associativity statement for LLW, LWL, WLL, LWW, WLW, WWL, or WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/lll_associativity`

verifies `LLL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.31 proves the LLL word associativity defect vanishes under the stated local flag-stack, base-change, projection-formula, and reduced Thom–Sebastiani coefficient rows. It does not prove the seven remaining word associativity rows or the four-input pentagon.

Proposition 0.32 *(Associativity for the LLW word).* [proved for the local/local–mixed flag stack] Let α, β be retained local colours and let η be a retained wrapped colour at height R. Write

$$\gamma_{12} = \alpha + \beta, \qquad \zeta_{23} = \beta + \eta, \qquad \zeta_{123} = \alpha + \beta + \eta.$$

Assume the LLW flag stack $\mathfrak{F}^{(2),\mathrm{LLW}}_{\alpha,\beta,\eta,R}$ of Proposition 0.30, the LL and LW target-admissible correspondence rows, the chosen compact-support pushforward model, and the required local and mixed base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the LLW flag square. Then

$$m^{\mathrm{LW}}_{\gamma_{12},\eta,R} \circ \left(m^{\mathrm{LL}}_{\alpha,\beta,R} \otimes 1\right) = m^{\mathrm{LW}}_{\alpha,\zeta_{23},R} \circ \left(1 \otimes m^{\mathrm{LW}}_{\beta,\eta,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered inputs. Equivalently,

$$\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LLW},R} = 0.$$

Proof. Both composites are represented by the same retained flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in L_\alpha, \quad C_2/C_1 \in L_\beta, \quad C_3/C_2 \in W_\eta.$$

The left comparison map λ^{LLW} records first the local/local extension of α by β, and then the local/wrapped extension of γ_{12} by η. The right comparison map ρ^{LLW} records first the local/wrapped extension of β by η, and then the local/wrapped extension of α by ζ_{23}. These two iterated descriptions have the same underlying filtered object.

The local and mixed base-change isomorphisms identify the two iterated compact-support pushforwards after pullback to $\mathfrak{F}^{(2),\mathrm{LLW}}_{\alpha,\beta,\eta,R}$. The projection formulas move the external source coefficients through the same target pushforward, and the reduced Thom–Sebastiani transports identify the two coefficient systems on the common flag stack. Hence the two composites agree, so $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LLW},R} = 0$.

The proof uses no wrapped/wrapped binary product. It proves no associativity statement for LWL, WLL, LWW, WLW, WWL, or WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. ☐

The packet

`certificates/hybrid/llw_associativity`

verifies `LLW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.32 proves the LLW word associativity defect vanishes under the stated LL/LW flag-stack, base-change, projection-formula, and reduced Thom–Sebastiani coefficient rows. It does not prove the six remaining word associativity rows or the four-input pentagon.

PROPOSITION 0.33 *(Associativity for the LWL word).* [proved for the mixed–mixed flag stack] Let α, β be retained local colours and let η be a retained wrapped colour at height R, ordered as (α, η, β). Write

$$\zeta_{12} = \alpha + \eta, \qquad \zeta_{23} = \eta + \beta, \qquad \zeta_{123} = \alpha + \eta + \beta.$$

Assume the LWL flag stack $\mathfrak{F}^{(2),\mathrm{LWL}}_{\alpha,\eta,\beta,R}$ of Proposition 0.30, the LW and WL target-admissible correspondence rows, the chosen compact-support pushforward model, and the required mixed base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the LWL flag square. Then

$$m^{\mathrm{WL}}_{\zeta_{12},\beta,R} \circ \left(m^{\mathrm{LW}}_{\alpha,\eta,R} \otimes 1\right) = m^{\mathrm{LW}}_{\alpha,\zeta_{23},R} \circ \left(1 \otimes m^{\mathrm{WL}}_{\eta,\beta,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered inputs. Equivalently,

$$\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LWL},R} = 0.$$

Proof. The common refinement is the retained LWL flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in L_\alpha, \quad C_2/C_1 \in W_\eta, \quad C_3/C_2 \in L_\beta.$$

The left comparison λ^{LWL} reads this filtration as an LW extension followed by a WL extension. The right comparison ρ^{LWL} reads it as a WL extension followed by an LW extension. Both descriptions have the same terminal filtered object of wrapped colour ζ_{123}.

The mixed base-change isomorphisms identify the two iterated compact-support pushforwards on the common flag stack. The mixed projection formula moves the ordered source coefficients through the same pushforward, and the mixed reduced Thom–Sebastiani transports identify the two coefficient systems. Therefore the two composites agree, and $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LWL},R} = 0$.

The proof is the ordered LWL case only. It proves no associativity statement for WLL, LWW, WLW, WWL, or WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. ☐

The packet

`certificates/hybrid/lwl_associativity`

verifies `LWL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.33 proves the LWL word associativity defect vanishes under the stated LW/WL flag-stack, base-change, projection-formula, and reduced Thom–Sebastiani coefficient rows. It does not prove the five remaining word associativity rows or the four-input pentagon.

Proposition 0.34 *(Associativity for the WLL word).* [proved for the mixed–local/local flag stack] Let α, β be retained local colours and let η be a retained wrapped colour at height R, ordered as (η, α, β). Write

$$\zeta_{12} = \eta + \alpha, \qquad \gamma_{23} = \alpha + \beta, \qquad \zeta_{123} = \eta + \alpha + \beta.$$

Assume the WLL flag stack $\mathfrak{F}^{(2),\mathrm{WLL}}_{\eta,\alpha,\beta,R}$ of Proposition 0.30, the WL and LL target-admissible correspondence rows, the chosen compact-support pushforward model, and the required local and mixed base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the WLL flag square. Then

$$m^{\mathrm{WL}}_{\zeta_{12},\beta,R} \circ \left(m^{\mathrm{WL}}_{\eta,\alpha,R} \otimes 1\right) = m^{\mathrm{WL}}_{\eta,\gamma_{23},R} \circ \left(1 \otimes m^{\mathrm{LL}}_{\alpha,\beta,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered inputs. Equivalently,

$$\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{WLL},R} = 0.$$

Proof. The common refinement is the retained WLL flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in W_\eta, \quad C_2/C_1 \in L_\alpha, \quad C_3/C_2 \in L_\beta.$$

The left comparison λ^{WLL} records first the wrapped/local extension of η by α, then the wrapped/local extension of ζ_{12} by β. The right comparison ρ^{WLL} records first the local/local extension of α by β, then the wrapped/local extension of η by γ_{23}. Both descriptions have terminal wrapped colour ζ_{123}.

The mixed and local base-change isomorphisms identify the two iterated compact-support pushforwards on the common flag stack. The projection formulas move the ordered source coefficients through the same pushforward, and the reduced Thom–Sebastiani transports identify the two coefficient systems. Therefore the two composites agree, and $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{WLL},R} = 0$.

The proof uses no wrapped/wrapped binary product. It proves no associativity statement for LWW, WLW, WWL, or WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/wll_associativity`

verifies `WLL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.34 proves the WLL word associativity defect vanishes under the stated WL/LL flag-stack, base-change, projection-formula, and reduced Thom–Sebastiani coefficient rows. It does not prove the four remaining word associativity rows or the four-input pentagon.

Proposition 0.35 *(Associativity for the LWW word).* [proved conditionally for the mixed–wrapped/wrapped flag stack] Let α be a retained local colour and let η_1, η_2 be retained wrapped colours at height R, ordered as (α, η_1, η_2). Write

$$\zeta_{12} = \alpha + \eta_1, \qquad \zeta_{23} = \eta_1 + \eta_2, \qquad \zeta_{123} = \alpha + \eta_1 + \eta_2.$$

Assume the LWW flag stack $\mathfrak{F}^{(2),\mathrm{LWW}}_{\alpha,\eta_1,\eta_2,R}$ of Proposition 0.30, the LW and WW target-admissible correspondence rows, the chosen compact-support pushforward model, and the required mixed and wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the LWW flag square. Then

$$m^{\mathrm{WW}}_{\zeta_{12},\eta_2,R} \circ \left(m^{\mathrm{LW}}_{\alpha,\eta_1,R} \otimes 1\right) = m^{\mathrm{LW}}_{\alpha,\zeta_{23},R} \circ \left(1 \otimes m^{\mathrm{WW}}_{\eta_1,\eta_2,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered inputs. Equivalently,

$$\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LWW},R} = 0.$$

The wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani rows are hypotheses of this proposition; they are not supplied by the wrapped/wrapped target-admissibility theorem.

Proof. The common refinement is the retained LWW flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in L_\alpha, \quad C_2/C_1 \in W_{\eta_1}, \quad C_3/C_2 \in W_{\eta_2}.$$

The left comparison λ^{LWW} records first the local/wrapped extension of α by η_1, then the wrapped/wrapped extension of ζ_{12} by η_2. The right comparison ρ^{LWW} records first the wrapped/wrapped extension of η_1 by η_2, then the local/wrapped extension of α by ζ_{23}. Both descriptions have terminal wrapped colour ζ_{123}.

The mixed and wrapped/wrapped base-change isomorphisms identify the two iterated compact-support pushforwards on the common flag stack. The corresponding projection formulas move the ordered source coefficients through the same pushforward, and the reduced Thom–Sebastiani transports identify the two coefficient systems. Therefore the two composites agree, and $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LWW},R} = 0$.

The proof is wordwise and conditional on the wrapped/wrapped functorial coefficient rows. It proves no associativity statement for WLW, WWL, or WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

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certificates/hybrid/lww_associativity
```

verifies `LWW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.35 proves the LWW word associativity defect vanishes under the stated LW/WW flag-stack, mixed functorial rows, and wrapped/wrapped functorial hypotheses. It does not prove the three remaining word associativity rows or the four-input pentagon.

PROPOSITION 0.36 *(Associativity for the WLW word).* [proved conditionally for the mixed–wrapped/wrapped flag stack] Let α be a retained local colour and let η_1, η_2 be retained wrapped colours at height R, ordered as (η_1, α, η_2). Write

$$\zeta_{12} = \eta_1 + \alpha, \qquad \zeta_{23} = \alpha + \eta_2, \qquad \zeta_{123} = \eta_1 + \alpha + \eta_2.$$

Assume the WLW flag stack $\mathfrak{F}^{(2),\mathrm{WLW}}_{\eta_1,\alpha,\eta_2,R}$ of Proposition 0.30, the WL, LW, and WW target-admissible correspondence rows, the chosen compact-support pushforward model, and the required mixed and wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the WLW flag square. Then

$$m^{\mathrm{WW}}_{\zeta_{12},\eta_2,R} \circ \left(m^{\mathrm{WL}}_{\eta_1,\alpha,R} \otimes 1\right) = m^{\mathrm{WW}}_{\eta_1,\zeta_{23},R} \circ \left(1 \otimes m^{\mathrm{LW}}_{\alpha,\eta_2,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered inputs. Equivalently,

$$\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{WLW},R} = 0.$$

The wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani rows are hypotheses of this proposition; they are not supplied by the wrapped/wrapped target-admissibility theorem.

Proof. The common refinement is the retained WLW flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in W_{\eta_1}, \quad C_2/C_1 \in L_\alpha, \quad C_3/C_2 \in W_{\eta_2}.$$

The left comparison λ^{WLW} records first the wrapped/local extension of η_1 by α, then the wrapped/wrapped extension of ζ_{12} by η_2. The right comparison ρ^{WLW} records first the local/wrapped extension of α by η_2, then the wrapped/wrapped extension of η_1 by ζ_{23}. Both descriptions have terminal wrapped colour ζ_{123}.

The mixed and wrapped/wrapped base-change isomorphisms identify the two iterated compact-support pushforwards on the common flag stack. The corresponding projection formulas move the ordered source coefficients through the same pushforward, and the reduced Thom–Sebastiani transports identify the two coefficient systems. Therefore the two composites agree, and $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{WLW},R} = 0$.

The proof is wordwise and conditional on the wrapped/wrapped functorial coefficient rows. It proves no associativity statement for WWL or WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/wlw_associativity`

verifies `WLW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.36 proves the WLW word associativity defect vanishes under the stated $WL/LW/WW$ flag-stack, mixed functorial rows, and wrapped/wrapped functorial hypotheses. It does not prove the two remaining word associativity rows or the four-input pentagon.

Proposition 0.37 *(Associativity for the WWL word)*. [proved conditionally for the wrapped/wrapped–mixed flag stack] Let α be a retained local colour and let η_1, η_2 be retained wrapped colours at height R, ordered as (η_1, η_2, α). Write

$$\zeta_{12} = \eta_1 + \eta_2, \qquad \zeta_{23} = \eta_2 + \alpha, \qquad \zeta_{123} = \eta_1 + \eta_2 + \alpha.$$

Assume the WWL flag stack $\mathfrak{F}^{(2),\mathrm{WWL}}_{\eta_1,\eta_2,\alpha,R}$ of Proposition 0.30, the WW and WL target-admissible correspondence rows, the chosen compact-support pushforward model, and the required mixed and wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the WWL flag square. Then

$$m^{\mathrm{WL}}_{\zeta_{12},\alpha,R} \circ \left(m^{\mathrm{WW}}_{\eta_1,\eta_2,R} \otimes 1\right) = m^{\mathrm{WW}}_{\eta_1,\zeta_{23},R} \circ \left(1 \otimes m^{\mathrm{WL}}_{\eta_2,\alpha,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered inputs. Equivalently,

$$\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{WWL},R} = 0.$$

The wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani rows are hypotheses of this proposition; they are not supplied by the wrapped/wrapped target-admissibility theorem.

Proof. The common refinement is the retained WWL flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in W_{\eta_1}, \quad C_2/C_1 \in W_{\eta_2}, \quad C_3/C_2 \in L_\alpha.$$

The left comparison λ^{WWL} records first the wrapped/wrapped extension of η_1 by η_2, then the wrapped/local extension of ζ_{12} by α. The right comparison ρ^{WWL} records first the wrapped/local extension of η_2 by α, then the wrapped/wrapped extension of η_1 by ζ_{23}. Both descriptions have terminal wrapped colour ζ_{123}.

The wrapped/wrapped and mixed base-change isomorphisms identify the two iterated compact-support pushforwards on the common flag stack. The corresponding projection formulas move the ordered source coefficients through the same pushforward, and the reduced Thom–Sebastiani transports identify the two coefficient systems. Therefore the two composites agree, and $\mathfrak{d}^{\mathrm{assoc}}_{\mathrm{WWL},R} = 0$.

The proof is wordwise and conditional on the wrapped/wrapped functorial coefficient rows. It proves no associativity statement for WWW, and it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

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certificates/hybrid/wwl_associativity
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verifies `WWL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.37 proves the WWL word associativity defect vanishes under the stated WW/WL flag-stack, mixed functorial rows, and wrapped/wrapped functorial hypotheses. It does not prove the remaining WWW word associativity row or the four-input pentagon.

PROPOSITION 0.38 *(Associativity for the WWW word).* [proved conditionally for the wrapped/wrapped flag stack] Let η_1, η_2, η_3 be retained wrapped colours at height R, ordered as (η_1, η_2, η_3). Write

$$\zeta_{12} = \eta_1 + \eta_2, \qquad \zeta_{23} = \eta_2 + \eta_3, \qquad \zeta_{123} = \eta_1 + \eta_2 + \eta_3.$$

Assume the WWW flag stack $\mathfrak{F}^{(2),\mathrm{WWW}}_{\eta_1,\eta_2,\eta_3,R}$ of Proposition 0.30, the WW target-admissible correspondence rows, the chosen compact-support pushforward model, and the required wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications on the WWW flag square. Then

$$m^{\mathrm{WW}}_{\zeta_{12},\eta_3,R} \circ \left(m^{\mathrm{WW}}_{\eta_1,\eta_2,R} \otimes 1\right) = m^{\mathrm{WW}}_{\eta_1,\zeta_{23},R} \circ \left(1 \otimes m^{\mathrm{WW}}_{\eta_2,\eta_3,R}\right)$$

as morphisms on the retained compactly supported reduced vanishing-cycle complexes of the ordered wrapped inputs. Equivalently,

$$\mathfrak{d}^{\mathrm{assoc}}_{\mathrm{WWW},R} = 0.$$

The wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani rows are hypotheses of this proposition; they are not supplied by the wrapped/wrapped target-admissibility theorem.

Proof. The common refinement is the retained WWW flag stack

$$0 \subset C_1 \subset C_2 \subset C_3, \qquad C_1/C_0 \in W_{\eta_1}, \quad C_2/C_1 \in W_{\eta_2}, \quad C_3/C_2 \in W_{\eta_3}.$$

The left comparison λ^{WWW} records first the wrapped/wrapped extension of η_1 by η_2, then the wrapped/wrapped extension of ζ_{12} by η_3. The right comparison ρ^{WWW} records first the wrapped/wrapped extension of η_2 by η_3, then the wrapped/wrapped extension of η_1 by ζ_{23}. Both descriptions have terminal wrapped colour ζ_{123}.

The wrapped/wrapped base-change isomorphisms identify the two iterated compact-support pushforwards on the common flag stack. The wrapped/wrapped projection formulas move the ordered source coefficients through the same pushforward, and the wrapped/wrapped reduced Thom–Sebastiani transports identify the two coefficient systems. Therefore the two composites agree, and $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{WWW},R} = 0$.

The proof is wordwise and conditional on the wrapped/wrapped functorial coefficient rows. It proves the last length-three wordwise associativity defect, but it does not prove the four-input pentagon, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/www_associativity`

verifies `WWW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.38 proves the WWW word associativity defect vanishes under the stated WW/WW flag-stack and wrapped/wrapped functorial hypotheses. It completes the length-three wordwise associativity list, but it does not prove the four-input pentagon.

Proposition 0.39 *(Four-input pentagon coherence)*. [proved conditionally before the reduced E-quotient] Let

$$u = \epsilon_1\epsilon_2\epsilon_3\epsilon_4 \in \{\mathrm{L}, \mathrm{W}\}^4$$

and let $(\xi_1, \xi_2, \xi_3, \xi_4)$ be retained ordered colours of these types at height R. Assume the binary correspondence rows for LL, LW, WL, WW, the compact-support pushforward model, the length-three associativity rows of Propositions 0.31–0.38, and the required local, mixed, and wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani coefficient identifications. Assume also the BBDJS–Joyce–Upmeier Thom–Sebastiani orientation pentagon on the retained four-step d-critical charts. Then there is a retained finite-type flag-Quot stack

$$\mathfrak{F}^{(3),u}_{\xi_1,\xi_2,\xi_3,\xi_4,R}$$

classifying filtrations

$$0 = C_0 \subset C_1 \subset C_2 \subset C_3 \subset C_4$$

with C_i/C_{i-1} of type ϵ_i and colour ξ_i, all intermediate colours retained, and wrapped rigidifications and anchors remembered before the E-quotient. The five planar binary parenthesizations of the four inputs give five iterated correspondence fibre products receiving canonical maps from $\mathfrak{F}^{(3),u}_{\xi_1,\xi_2,\xi_3,\xi_4,R}$. The two composites around the Mac Lane pentagon of associators agree:

$$a_{\xi_1,\xi_2,\xi_3+\xi_4,R}\, a_{\xi_1+\xi_2,\xi_3,\xi_4,R} = (1 \otimes a_{\xi_2,\xi_3,\xi_4,R})\, a_{\xi_1,\xi_2+\xi_3,\xi_4,R}\, (a_{\xi_1,\xi_2,\xi_3,R} \otimes 1).$$

Equivalently, for every $u \in \{\mathrm{L}, \mathrm{W}\}^4$,

$$\mathfrak{o}^{\mathrm{pent}}_{u,R} = 0.$$

Proof. Fix u and the ordered colours. Over the retained terminal object stack of colour $\xi_1 + \xi_2 + \xi_3 + \xi_4$, take the relative three-step flag-Quot stack parametrising the flag

$$0 = C_0 \subset C_1 \subset C_2 \subset C_3 \subset C_4$$

with the prescribed four graded pieces. The retained closed endpoint conditions, universal complexes, and extension-closure datum cut out a finite-type retained substack. Every intermediate object C_j/C_i carries the type forced by the product rule $\mathrm{L} \cdot \mathrm{L} = \mathrm{L}$ and otherwise W; if the type is W, the wrapped rigidification and anchor memory are part of the retained datum. This constructs $\mathfrak{F}^{(3),u}_{\xi_1,\xi_2,\xi_3,\xi_4,R}$ before quotienting by E.

Each planar binary tree on four leaves forgets the same flag in a different order and therefore gives a map from the common four-step flag stack to the corresponding iterated binary correspondence fibre product. The associator attached to an internal triple is the two-step flag comparison of Propositions 0.31–0.38 pulled back along the appropriate contraction of the four-step flag. Thus both paths around the pentagon are represented on the same source stack $\mathfrak{F}^{(3),u}_{\xi_1,\xi_2,\xi_3,\xi_4,R}$, with the same terminal target colour.

Proper base change for the retained compact-support model identifies the target pushforwards after pullback to the common four-step flag stack. The projection formula moves the four ordered source coefficients through the same pushforward. The reduced Thom–Sebastiani coefficient systems on the two paths are the two parenthesizations of the same fourfold external product. The BBDJS Thom–Sebastiani pentagon [12, Theorem 5.10(iv)], together with the Joyce–Upmeier multiplicative orientation cocycle [56, §4], identifies these two coefficient systems. Hence the two pull-push functors around the pentagon agree, and $\mathfrak{o}^{\mathrm{pent}}_{u,R} = 0$.

The proof is before the reduced E-quotient and at fixed height R. It proves no quotient descent, transition compatibility, unit compatibility, higher-coloured tree category, or aggregate hybrid-carrier population. Whenever a WW binary correspondence appears in the four-input word, the wrapped/wrapped base-change, projection-formula, and reduced Thom–Sebastiani rows remain hypotheses, as in Propositions 0.35–0.38. □

The packet

`certificates/hybrid/four_input_pentagon_coherence`

verifies `FOUR_INPUT_PENTAGON_CONDITIONAL_VERIFIED`: Proposition 0.39 constructs the retained four-step flag stack for each word in $\{\mathrm{L}, \mathrm{W}\}^4$ and proves the associator pentagon on that common refinement under the stated functorial hypotheses. It does not define the higher-coloured tree category or prove quotient descent, transition compatibility, or aggregate population.

Definition 0.40 *(Higher-coloured hybrid tree category).* Fix R. The nonunital higher-coloured hybrid tree category

$$\mathsf{Tree}_R^{E,\mathrm{hyb}}$$

is the free coloured category generated by the retained local and wrapped profiles below.

An object is a finite ordered hybrid profile

$$\underline{\xi} = ((\epsilon_i, \xi_i, \mathfrak{s}_i))_{i=1}^n, \qquad \epsilon_i \in \{\mathrm{L}, \mathrm{W}\},$$

where ξ_i is a retained colour of the indicated type. If $\epsilon_i = \mathrm{L}$, the symbol $\mathfrak{s}_i$ is a retained closed local configuration chart, including its finite indexing set and diagonal-refinement datum. If $\epsilon_i = \mathrm{W}$, the symbol $\mathfrak{s}_i$ is the retained wrapped rigidification with source anchor memory before the E-quotient. The output colour is

$$|\underline{\xi}| = \sum_i \xi_i,$$

with output type L only for an all-local profile and W otherwise.

A generating morphism
$$T : \underline{\xi} \longrightarrow |\underline{\xi}|$$
is a planar rooted tree with leaves ordered by $\underline{\xi}$. Every internal edge carries the retained intermediate colour given by the sum of the leaves below that edge and the type rule $\mathrm{L} \cdot \mathrm{L} = \mathrm{L}$, otherwise W. Every internal vertex of arity $k \geq 2$ is decorated by the retained extension correspondence or flag-Quot stack for the ordered subprofile below that vertex. Local vertices remember closed-configuration refinements and symmetric relabellings; mixed and wrapped vertices remember source, intermediate, and target anchors before quotient. Composition is grafting of planar rooted trees followed by contraction of unary identity edges.

The defect of realizing this category by compactly supported reduced vanishing-cycle pull-push functors is the higher-coloured residual
$$\mathfrak{o}_R^{\mathrm{col}} = \left(\mathfrak{o}_R^{\mathrm{tree}},\ \mathfrak{o}_R^{\mathrm{unit}},\ \mathfrak{o}_R^{\mathrm{sym}},\ \mathfrak{o}_R^{\mathrm{ref}},\ \mathfrak{o}_R^{\mathrm{des}},\ \mathfrak{o}_R^{\mathrm{ov}}\right).$$
The first component measures compatibility of tree contractions beyond the four-input pentagon. The remaining components measure, respectively, vacuum units, symmetric relabelling of local configurations, refinement of closed local charts, descent from ordered charts, and overlap compatibility of retained local and wrapped charts. This definition supplies the category and the names of its residual components; it does not assert $\mathfrak{o}_R^{\mathrm{col}} = 0$.

The packet

```
certificates/hybrid/higher_coloured_tree_category_definition
```

verifies `HIGHER_COLOURED_TREE_CATEGORY_DEFINED`: Definition 0.40 defines the nonunital higher-coloured tree category and the six components of $\mathfrak{o}_R^{\mathrm{col}}$. It does not prove the residual vanishes, and it does not supply unit compatibility, symmetric descent, wrapped order conventions, quotient descent, transition compatibility, or aggregate population.

Proposition 0.41 *(Higher-coloured residual vanishing criterion).* [conditional finite-stage criterion] At fixed height R, assume Definition 0.40 and Proposition 0.39. Suppose the following finite rows are supplied in the retained E-equivariant correspondence category before quotient:

(i) vacuum object and unit correspondences with left and right unit defect zero for local and wrapped inputs;
(ii) symmetric relabelling descent for projection-finite local configuration charts;
(iii) refinement compatibility for closed local configuration charts;
(iv) descent from ordered local charts to the symmetric finite-colour colimit;
(v) wrapped insertion order and braid-convention compatibility for ordered wrapped profiles;
(vi) overlap compatibility for retained local and wrapped charts on common refinements.

Then the higher-coloured residual of Definition 0.40 vanishes:
$$\mathfrak{o}_R^{\mathrm{col}} = 0.$$
Equivalently, all six components
$$\mathfrak{o}_R^{\mathrm{tree}},\quad \mathfrak{o}_R^{\mathrm{unit}},\quad \mathfrak{o}_R^{\mathrm{sym}},\quad \mathfrak{o}_R^{\mathrm{ref}},\quad \mathfrak{o}_R^{\mathrm{des}},\quad \mathfrak{o}_R^{\mathrm{ov}}$$
have defect rank zero.

Proof. The tree-contraction component is the coherence problem for binary parenthesizations in $\mathsf{Tree}_R^{E,\mathrm{hyb}}$. The length-three associators are the rows of Propositions 0.31–0.38, and their four-input pentagon is Proposition 0.39. Hence the standard coherence theorem for parenthesized products identifies any two binary contraction paths with the same planar leaves by a composite of four-input pentagons. Pulling those pentagons to the retained common flag-Quot refinement gives $\mathfrak{o}_R^{\mathrm{tree}} = 0$.

The remaining five components are not consequences of the pentagon. By the supplied rows in the statement, the left and right vacuum correspondences act as units on local and wrapped inputs, so $\mathfrak{o}_R^{\mathrm{unit}} = 0$. Symmetric relabelling descent sets $\mathfrak{o}_R^{\mathrm{sym}} = 0$. Refinement compatibility of closed local charts sets $\mathfrak{o}_R^{\mathrm{ref}} = 0$. Descent from ordered charts to the symmetric finite-colour colimit sets $\mathfrak{o}_R^{\mathrm{des}} = 0$. Overlap compatibility on common local/wrapped refinements sets $\mathfrak{o}_R^{\mathrm{ov}} = 0$. The residual is the direct tuple of these six components; therefore $\mathfrak{o}_R^{\mathrm{col}} = 0$.

This criterion does not construct the non-tree input rows. In the present correction ledger the unit, symmetric-relabelling, and wrapped order rows are supplied separately; refinement, ordered-chart descent, and overlap rows remain separate obligations needed for aggregate population. □

The packet

`certificates/hybrid/higher_coloured_residual_vanishing_criterion`

verifies `HIGHER_COLOURED_RESIDUAL_CONDITIONAL_CRITERION_VERIFIED`: Proposition 0.41 proves the finite-stage implication from the six named component rows to $\mathfrak{o}_R^{\mathrm{col}} = 0$. It proves the tree-contraction component from the associativity rows and four-input pentagon, but it does not supply the refinement, ordered-chart descent, or overlap rows required to invoke the criterion unconditionally. The unit, symmetric-relabelling, and wrapped order rows are supplied by separate packets.

Definition 0.42 *(Hybrid unit object and vacuum correspondences).* [definition only] Fix a finite HN height R. A hybrid unit datum consists of the following retained rows before the reduced E-quotient.

First, the zero object of the retained Hall heart determines a local zero colour

$$0_R \in \Gamma_R^{\mathrm{loc}}, \qquad b_R^{\mathrm{geom}}(0_R) = 0,$$

with empty local support chart $\varnothing$. The corresponding unit stack is the one-point closed local incidence stack

$$\mathbf{1}_R^{\mathrm{hyb}} := \mathfrak{M}^{\mathrm{loc,cl}}_{0_R,R,\varnothing}.$$

The same symbol $0_R : \varnothing \to \Gamma_R^{\mathrm{loc}}$ denotes the empty local colour profile whose total colour is 0_R. There is no wrapped zero colour: the unit is local because the wrapped sector is defined by $b_R^{\mathrm{geom}} > 0$.

Second, for every finite local colour map $\alpha : I \to \Gamma_R^{\mathrm{loc}}$, the datum contains the split local unit substacks

$$\mathfrak{U}^{\mathrm{LL},\ell}_{0_R,\alpha;\alpha,R,I} \subset \mathfrak{E}^{\mathrm{LL}}_{0_R,\alpha;\alpha,R,\varnothing,I,I}, \qquad \mathfrak{U}^{\mathrm{LL},r}_{\alpha,0_R;\alpha,R,I} \subset \mathfrak{E}^{\mathrm{LL}}_{\alpha,0_R;\alpha,R,I,\varnothing,I}.$$

They classify, respectively, the split exact sequences

$$0 \longrightarrow A_{\alpha,I} \longrightarrow A_{\alpha,I} \longrightarrow 0 \longrightarrow 0, \qquad 0 \longrightarrow 0 \longrightarrow A_{\alpha,I} \longrightarrow A_{\alpha,I} \longrightarrow 0.$$

The source maps remember the ordered pair $(\mathbf{1}_R^{\mathrm{hyb}}, A_{\alpha,I})$ or $(A_{\alpha,I}, \mathbf{1}_R^{\mathrm{hyb}})$, and the target maps send the split sequence to $A_{\alpha,I}$.

Third, for every wrapped colour $\eta \in \Gamma_R^{\mathrm{wr}}$, the datum contains the split mixed unit substacks

$$\mathfrak{U}^{\mathrm{LW},\ell}_{0_R,\eta;\eta,R} \subset \mathfrak{E}^{\mathrm{LW}}_{0_R,\eta;\eta,R,\varnothing}, \qquad \mathfrak{U}^{\mathrm{WL},r}_{\eta,0_R;\eta,R} \subset \mathfrak{E}^{\mathrm{WL}}_{\eta,0_R;\eta,R,\varnothing}.$$

They classify the split exact sequences

$$0 \longrightarrow W_\eta \longrightarrow W_\eta \longrightarrow 0 \longrightarrow 0, \qquad 0 \longrightarrow 0 \longrightarrow W_\eta \longrightarrow W_\eta \longrightarrow 0.$$

The wrapped rigidification and source/target anchor memory are those of W_η; the zero local factor carries the empty support chart and no anchor. These rows are the unit correspondences used by $\mathfrak{d}_R^{\mathrm{unit}}$. This definition does not prove that the induced compact-support pull-push maps are the identity on local or wrapped coefficients; those are the local and wrapped unit compatibility rows.

The packet

`certificates/hybrid/hybrid_unit_object_and_vacuum_correspondences`

verifies `HYBRID_UNIT_OBJECT_AND_VACUUM_CORRESPONDENCES_DEFINED`: Definition 0.42 defines the local zero unit object and the four split vacuum correspondence types LL_ℓ, LL_r, LW_ℓ, and WL_r. It does not prove the local or wrapped unit identity laws, quotient descent, transition compatibility, or aggregate population.

Proposition 0.43 *(Local unit compatibility for the hybrid product).* [proved before quotient at finite height] Assume the hybrid unit datum of Definition 0.42, the local/local correspondence datum of Definition 0.19, the target-properness row of Proposition 0.20, and the compact-support model of Definition 0.25. Let

$$K^{\mathrm{loc}}_{\alpha,R,I} := \Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I}$$

be the reduced local coefficient on $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$, and fix the standard zero-object normalization

$$K^{\mathrm{loc}}_{0_R,R,\varnothing} \simeq \mathbb{C}_{\mathbf{1}_R^{\mathrm{hyb}}}.$$

Let $p^{\mathrm{LL},\ell}$, $q^{\mathrm{LL},\ell}$ and $p^{\mathrm{LL},r}$, $q^{\mathrm{LL},r}$ denote the restrictions of the LL source and target maps to $\mathfrak{U}^{\mathrm{LL},\ell}_{0_R,\alpha;\alpha,R,I}$ and $\mathfrak{U}^{\mathrm{LL},r}_{\alpha,0_R;\alpha,R,I}$. Then the compact-support pull-push maps attached to the split local unit correspondences satisfy

$$\mu^{\mathrm{LL},\ell}_{0_R,\alpha;R,I} = (q^{\mathrm{LL},\ell})^{\mathrm{cs}}_! \, \mathrm{TS}^{\mathrm{red}}_{0_R,\alpha} \, (p^{\mathrm{LL},\ell})^* = \mathrm{id}_{K^{\mathrm{loc}}_{\alpha,R,I}},$$

and

$$\mu^{\mathrm{LL},r}_{\alpha,0_R;R,I} = (q^{\mathrm{LL},r})^{\mathrm{cs}}_! \, \mathrm{TS}^{\mathrm{red}}_{\alpha,0_R} \, (p^{\mathrm{LL},r})^* = \mathrm{id}_{K^{\mathrm{loc}}_{\alpha,R,I}}.$$

Equivalently, the local unit defect component has defect rank zero:

$$\mathfrak{d}_R^{\mathrm{unit,loc}} = 0.$$

Proof. For the left unit row, the split substack

$$\mathfrak{U}^{\mathrm{LL},\ell}_{0_R,\alpha;\alpha,R,I}$$

is isomorphic to $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$. The isomorphism sends $A_{\alpha,I}$ to the split sequence

$$0 \longrightarrow A_{\alpha,I} \longrightarrow A_{\alpha,I} \longrightarrow 0 \longrightarrow 0,$$

and its inverse takes the target object of the sequence. Under this identification the source map is the closed insertion

$$A_{\alpha,I} \longmapsto (\mathbf{1}_R^{\mathrm{hyb}}, A_{\alpha,I}),$$

and the target map is the identity on $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$. Since the target map is proper, the compact-support model uses the identity compactification and $(q^{\mathrm{LL},\ell})^{\mathrm{cs}}_!$ is ordinary pushforward along the identity.

The pullback of the ordered source coefficient is

$$(p^{\mathrm{LL},\ell})^*\big(K^{\mathrm{loc}}_{0_R,R,\varnothing} \boxtimes K^{\mathrm{loc}}_{\alpha,R,I}\big) \simeq K^{\mathrm{loc}}_{\alpha,R,I}.$$

The zero-object normalization identifies the first factor with the constant unit complex on the point. The BBDJS Thom–Sebastiani isomorphism for the sum of a zero potential and the local potential [12, Theorem 5.10], tensored with the Joyce–Upmeier orientation unit [56, §4], is the identity on this coefficient. Hence

$$(q^{\mathrm{LL},\ell})^{\mathrm{cs}}_! \ \mathrm{TS}^{\mathrm{red}}_{0_R,\alpha}\ (p^{\mathrm{LL},\ell})^* = \mathrm{id}_{K^{\mathrm{loc}}_{\alpha,R,I}}.$$

The right unit row is identical with the factors reversed. The substack $\mathfrak{U}^{\mathrm{LL},r}_{\alpha,0_R;\alpha,R,I}$ is the graph of the split sequence

$$0 \longrightarrow 0 \longrightarrow A_{\alpha,I} \longrightarrow A_{\alpha,I} \longrightarrow 0,$$

so its source map inserts $(A_{\alpha,I}, \mathbf{1}_R^{\mathrm{hyb}})$ and its target map is again the identity. The ordered Thom–Sebastiani isomorphism for the local potential plus the zero potential is the right-unit form of the same zero-factor identity. Thus the right pull-push is $\mathrm{id}_{K^{\mathrm{loc}}_{\alpha,R,I}}$.

The proof is before the reduced E-quotient and at fixed height R. It proves no wrapped unit identity, quotient descent, transition compatibility, symmetric descent, refinement compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/local_unit_compatibility`

verifies `LOCAL_UNIT_COMPATIBILITY_VERIFIED`: Proposition 0.43 proves the left and right LL vacuum pull-push maps are the identity on local coefficients. It does not prove wrapped unit compatibility, quotient descent, transition compatibility, or aggregate population.

PROPOSITION 0.44 *(Wrapped unit compatibility for the hybrid product).* [proved before quotient at finite height] Assume the hybrid unit datum of Definition 0.42, the one-sided mixed correspondence datum of Definition 0.21, the mixed target-admissibility row of Proposition 0.22, the compact-support model of Definition 0.25, and the mixed Thom–Sebastiani transport row of Proposition 0.28. Let

$$K^{\mathrm{wr}}_{\eta,R} := \Phi^{\mathrm{wr}}_{\eta,R} \otimes o^{\mathrm{wr}}_{\eta,R}$$

be the reduced wrapped coefficient on $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$, and keep the zero-object normalization

$$K^{\mathrm{loc}}_{0_R,R,\varnothing} \simeq \mathbb{C}_{\mathbf{1}_R^{\mathrm{hyb}}}.$$

Let $p^{\mathrm{LW},\ell}, q^{\mathrm{LW},\ell}$ and $p^{\mathrm{WL},r}, q^{\mathrm{WL},r}$ denote the restrictions of the mixed source and target maps to $\mathfrak{U}^{\mathrm{LW},\ell}_{0_R,\eta;\eta,R}$ and $\mathfrak{U}^{\mathrm{WL},r}_{\eta,0_R;\eta,R}$. Then the compact-support pull-push maps attached to the split wrapped unit correspondences satisfy

$$\mu^{\mathrm{LW},\ell}_{0_R,\eta;R} = (q^{\mathrm{LW},\ell})^{\mathrm{cs}}_! \ \mathrm{TS}^{\mathrm{red,LW}}_{0_R,\eta}\ (p^{\mathrm{LW},\ell})^* = \mathrm{id}_{K^{\mathrm{wr}}_{\eta,R}},$$

and

$$\mu^{\mathrm{WL},r}_{\eta,0_R;R} = (q^{\mathrm{WL},r})^{\mathrm{cs}}_! \ \mathrm{TS}^{\mathrm{red},\mathrm{WL}}_{\eta,0_R} \ (p^{\mathrm{WL},r})^* = \mathrm{id}_{K^{\mathrm{wr}}_{\eta,R}} .$$

Equivalently, the wrapped unit defect component has defect rank zero:

$$\mathfrak{d}^{\mathrm{unit},\mathrm{wr}}_R = 0.$$

Proof. For the left wrapped unit row, the split substack

$$\mathfrak{U}^{\mathrm{LW},\ell}_{0_R,\eta;\eta,R}$$

is isomorphic to $\mathcal{M}^{\mathrm{wr},\mathrm{rig}}_{\eta,R}$. The isomorphism sends W_η to the split sequence

$$0 \longrightarrow W_\eta \longrightarrow W_\eta \longrightarrow 0 \longrightarrow 0,$$

and its inverse takes the target wrapped object. Under this identification the source map is the insertion

$$W_\eta \longmapsto (\mathbf{1}^{\mathrm{hyb}}_R, W_\eta),$$

and the target map is the identity on $\mathcal{M}^{\mathrm{wr},\mathrm{rig}}_{\eta,R}$. The unit definition records that the zero local factor carries no anchor; hence the source and target anchor memory on the split row is exactly the anchor memory of W_η. No E-quotient is taken. Since the target map is proper, the compact-support model uses the identity compactification and $(q^{\mathrm{LW},\ell})^{\mathrm{cs}}_!$ is ordinary pushforward along the identity.

The pullback of the ordered source coefficient is

$$(p^{\mathrm{LW},\ell})^*\big(K^{\mathrm{loc}}_{0_R,R,\varnothing} \boxtimes K^{\mathrm{wr}}_{\eta,R}\big) \simeq K^{\mathrm{wr}}_{\eta,R}.$$

The zero-object normalization identifies the first factor with the constant unit complex on the point. The mixed Thom–Sebastiani transport of Proposition 0.28 specializes, for the zero local potential plus the wrapped potential, to the BBDJS zero-factor Thom–Sebastiani identity [12, Theorem 5.10], tensored with the Joyce–Upmeier orientation unit [56, §4]. Therefore

$$(q^{\mathrm{LW},\ell})^{\mathrm{cs}}_! \ \mathrm{TS}^{\mathrm{red},\mathrm{LW}}_{0_R,\eta} \ (p^{\mathrm{LW},\ell})^* = \mathrm{id}_{K^{\mathrm{wr}}_{\eta,R}} .$$

The right wrapped unit row is the same argument with the factors reversed. The split substack $\mathfrak{U}^{\mathrm{WL},r}_{\eta,0_R;\eta,R}$ is the graph of

$$0 \longrightarrow 0 \longrightarrow W_\eta \longrightarrow W_\eta \longrightarrow 0,$$

the source map inserts $(W_\eta, \mathbf{1}^{\mathrm{hyb}}_R)$, the target map is the identity on $\mathcal{M}^{\mathrm{wr},\mathrm{rig}}_{\eta,R}$, and the wrapped anchor memory is unchanged. The ordered WL Thom–Sebastiani transport is the right zero-factor identity. Hence the right pull-push is $\mathrm{id}_{K^{\mathrm{wr}}_{\eta,R}}$.

The proof is before the reduced E-quotient and at fixed height R. It proves no quotient descent, transition compatibility, symmetric descent, refinement compatibility, overlap compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/wrapped_unit_compatibility`

verifies `WRAPPED_UNIT_COMPATIBILITY_VERIFIED`: Proposition 0.44 proves the left LW and right WL vacuum pull-push maps are the identity on wrapped coefficients, with wrapped anchor memory unchanged before the reduced E-quotient. It does not prove quotient descent, transition compatibility, or aggregate population.

PROPOSITION 0.45 *(Symmetric descent for local configuration charts).* [proved before quotient at finite height] Assume the local $b = 0$ carrier of Definition 0.4 and the projection-finite closed local configuration sector of the finite stage. Let $\alpha : I \to \Gamma_R^{\mathrm{loc}}$ be a finite local colour map and let $\sigma : I \xrightarrow{\sim} I'$ be a bijection. Define

$$(\sigma_*\alpha)_{i'} = \alpha_{\sigma^{-1}(i')}, \qquad \rho_\sigma : E^I \xrightarrow{\sim} E^{I'}, \qquad (\rho_\sigma(\mathbf{e}))_{i'} = e_{\sigma^{-1}(i')}.$$

Then

$$(\mathrm{id}_E \times \rho_\sigma)(Z_I) = Z_{I'}$$

and ρ_σ induces an isomorphism of closed local configuration charts

$$r_\sigma : \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \xrightarrow{\sim} \mathfrak{M}^{\mathrm{loc,cl}}_{\sigma_*\alpha,R,I'}, \qquad (A, \mathbf{e}) \longmapsto (A, \rho_\sigma(\mathbf{e})),$$

with

$$\pi_{\sigma_*\alpha,R,I'} \circ r_\sigma = \rho_\sigma \circ \pi_{\alpha,R,I}.$$

For the reduced local coefficient

$$K^{\mathrm{loc}}_{\alpha,R,I} := \Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I},$$

there is a functorial relabelling isomorphism

$$r_\sigma^* K^{\mathrm{loc}}_{\sigma_*\alpha,R,I'} \simeq K^{\mathrm{loc}}_{\alpha,R,I},$$

and these isomorphisms satisfy the cocycle identity for composable bijections. Hence the stabilizer

$$\mathfrak{S}_\alpha = \{\sigma \in \mathrm{Aut}(I) \mid \sigma_*\alpha = \alpha\}$$

acts on $(\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}, K^{\mathrm{loc}}_{\alpha,R,I})$, and the coefficient descends effectively to the quotient stack

$$[\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}/\mathfrak{S}_\alpha].$$

Equivalently, the fixed-chart symmetric relabelling component of the higher-coloured residual has defect rank zero:

$$\mathfrak{d}_R^{\mathrm{sym}} = 0$$

on projection-finite local configuration charts before quotient.

Proof. The incidence locus $Z_I \subset E \times E^I$ is the union of the graphs $e = e_i$. Under $\mathrm{id}_E \times \rho_\sigma$ the graph indexed by $i \in I$ is sent to the graph indexed by $\sigma(i) \in I'$; hence $(\mathrm{id}_E \times \rho_\sigma)(Z_I) = Z_{I'}$. Thus the condition

$$p_E(\mathrm{Supp}\, A) \subset Z_I(\mathbf{e})$$

is equivalent to

$$p_E(\mathrm{Supp}\, A) \subset Z_{I'}(\rho_\sigma(\mathbf{e})).$$

Relabelling the finite colour profile by σ changes no underlying sheaf A and preserves the total retained local colour. Therefore

$$(A, \mathbf{e}) \longmapsto (A, \rho_\sigma(\mathbf{e}))$$

defines the stated isomorphism r_σ, with inverse $r_{\sigma^{-1}}$, and the displayed identity with the configuration maps follows from the definition of ρ_σ.

The local d-critical chart, its potential, and its cosection-reduced vanishing-cycle complex are attached to the object A together with the ordered support coordinates. The map r_σ is an isomorphism of these charts: it only relabels the support coordinates and the matching colour labels. Functoriality of vanishing cycles under isomorphism, tensored with the transported Joyce–Upmeier orientation line, gives

$$r_\sigma^*(\Phi^{\mathrm{loc}}_{\sigma_*\alpha,R,I'} \otimes o^{\mathrm{loc}}_{\sigma_*\alpha,R,I'}) \simeq \Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I}.$$

Since $r_\tau r_\sigma = r_{\tau\sigma}$, these pullback isomorphisms satisfy the cocycle identity.

For $\sigma \in \mathfrak{S}_\alpha$ the target profile is again α, so the preceding isomorphisms define a finite group action on the stack and an equivariant structure on the coefficient complex. Constructible complexes on a quotient stack by a finite group are the same data as equivariant constructible complexes on the source stack; therefore the descent to $[\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}/\mathfrak{S}_\alpha]$ is effective. This proves the symmetric relabelling component for a fixed projection-finite local chart.

The proof does not prove collision compatibility, refinement of closed local charts, descent from ordered charts to the symmetric finite-colour colimit, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

```
certificates/hybrid/local_configuration_symmetric_descent
```

verifies `LOCAL_CONFIGURATION_SYMMETRIC_DESCENT_VERIFIED`: Proposition 0.45 proves finite symmetric-group descent for each projection-finite closed local configuration chart and its reduced coefficient before the reduced E-quotient. It does not prove collision compatibility, refinement compatibility, descent from ordered charts to the symmetric finite-colour colimit, quotient descent, transition compatibility, or aggregate population.

Proposition 0.46 *(Wrapped insertion order conventions)*. [proved before quotient at finite height] Fix R and the ordered hybrid tree category of Definition 0.40. For an ordered hybrid profile

$$\underline{\xi} = ((\epsilon_i, \xi_i, \mathfrak{s}_i))_{i=1}^n$$

write

$$\mathrm{Wr}(\underline{\xi}) = (\xi_{i_1}, \ldots, \xi_{i_m}), \qquad i_1 < \cdots < i_m, \qquad \epsilon_{i_a} = \mathrm{W},$$

for the ordered subsequence of wrapped leaves. The wrapped insertion convention is the following.

(i) At every binary vertex the wrapped order of the output is the concatenation of the wrapped orders of the ordered inputs.

(ii) For a WW vertex with ordered source pair (η_1, η_2), the Hall correspondence is

$$0 \longrightarrow W_{\eta_2} \longrightarrow B_{\eta_1+\eta_2} \longrightarrow W_{\eta_1} \longrightarrow 0,$$

and the ordered wrapped source list is (η_1, η_2): first the quotient colour, then the subobject colour. The target wrapped colour remembers this source order as the anchor-memory order before the reduced E-quotient.

(iii) For LW, WL, and LWL vertices, local factors do not change the order of the wrapped subsequence.

(iv) The comparison maps for the eight two-step flag stacks of Proposition 0.30 preserve the ordered wrapped subsequence Wr for each of the words

$$LLL,\ LLW,\ LWL,\ WLL,\ LWW,\ WLW,\ WWL,\ WWW.$$

Therefore parenthesization changes in $\mathsf{Tree}_R^{E,\mathrm{hyb}}$ use the planar leaf order and introduce no braid operator and no permutation of wrapped inputs. The wrapped-order input row in Proposition 0.41 has defect rank zero at fixed height before quotient.

Proof. The object set of $\mathsf{Tree}_R^{E,\mathrm{hyb}}$ is ordered by definition, and a generating morphism is a planar rooted tree whose leaves are ordered by the input profile. Thus every internal vertex sees an ordered subprofile. The rule for the output colour is additive, and the type rule sends an output to the wrapped sector as soon as one input below the vertex is wrapped. This gives a canonical ordered wrapped subsequence: scan the leaves below the vertex from left to right and discard the local leaves.

For an LW or WL vertex there is exactly one wrapped input, so the ordered wrapped subsequence is unchanged. For an LWL vertex the two local leaves are discarded by the scan, and the wrapped subsequence is again the single middle wrapped colour. For a WW vertex the ordered source pair in Definition 0.23 is (η_1, η_2), while the Hall extension convention is

$$0 \to W_{\eta_2} \to B_{\eta_1+\eta_2} \to W_{\eta_1} \to 0.$$

Hence the first source factor is the quotient and the second source factor is the subobject. This fixes the ordered source-anchor list as (η_1, η_2). No transposition is applied when the target wrapped colour is formed.

It remains to check parenthesization changes. The two-step flag stack for a word $w = \epsilon_1\epsilon_2\epsilon_3$ records a filtration whose successive quotients are the three ordered leaves. The left comparison map contracts the first two leaves and then the resulting intermediate object with the third leaf. The right comparison map contracts the last two leaves and then the first leaf with the resulting intermediate object. In both cases the ordered wrapped subsequence is obtained by scanning the same ordered list $(\epsilon_1, \epsilon_2, \epsilon_3)$ and discarding local entries. Thus the comparison maps preserve Wr for all eight words.

Consequently every contraction in the planar tree category preserves the wrapped subsequence by induction on the number of vertices. Since the category has no braid generator and no symmetric identification of wrapped leaves, a braid word is not a hidden morphism in the realization. The only braid compatible with the convention is the identity braid, which acts by the identity on the ordered source coefficient. Hence the wrapped insertion order row has defect rank zero.

The proof does not prove symmetric descent for unordered wrapped pairs, closed-chart refinement, descent from ordered charts to the symmetric finite-colour colimit, overlap compatibility, quotient descent, transition compatibility, or aggregate hybrid-carrier population. □

The packet

`certificates/hybrid/wrapped_insertion_order_conventions`

verifies `WRAPPED_INSERTION_ORDER_CONVENTIONS_VERIFIED`: Proposition 0.46 fixes the planar wrapped-source order for mixed and wrapped vertices, checks the WW quotient/subobject convention, and proves that the eight two-step flag comparisons preserve the same ordered wrapped subsequence before the reduced E-quotient. It does not prove unordered wrapped-pair symmetric descent, ordered-chart descent, overlap compatibility, quotient descent, transition compatibility, or aggregate population.

Definition 0.47 *(Ordered two-sided mixed correspondence datum).* Fix a finite HN height R. Let $\alpha_- : I_- \to \Gamma_R^{\mathrm{loc}}$ and $\beta_+ : I_+ \to \Gamma_R^{\mathrm{loc}}$ be finite local colour maps, and let $\eta \in \Gamma_R^{\mathrm{wr}}$. Write $|\alpha_-| = \sum_{i \in I_-} \alpha_{-,i}$ and $|\beta_+| = \sum_{j \in I_+} \beta_{+,j}$. A two-sided mixed correspondence row is defined only for a retained wrapped target colour $\zeta \in \Gamma_R^{\mathrm{wr}}$ satisfying

$$\zeta = |\alpha_-| + \eta + |\beta_+|.$$

The ordered local–wrapped–local correspondence is the unreduced diagram

$$\mathfrak{E}^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+} \xrightarrow{p^{\mathrm{LWL}}} \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha_-,R,I_-} \times \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \times \mathfrak{M}^{\mathrm{loc,cl}}_{\beta_+,R,I_+},$$

$$\mathfrak{E}^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+} \xrightarrow{q^{\mathrm{LWL}}} \mathcal{M}^{\mathrm{wr,rig}}_{\zeta,R},$$

classifying retained flags

$$0 \subset A_{\beta_+} \subset B_{\eta+\beta_+} \subset C_\zeta$$

with associated graded pieces

$$A_{\beta_+}, \qquad W_\eta, \qquad A_{\alpha_-}$$

from bottom to top. Equivalently,

$$0 \to A_{\beta_+} \to B_{\eta+\beta_+} \to W_\eta \to 0, \qquad 0 \to B_{\eta+\beta_+} \to C_\zeta \to A_{\alpha_-} \to 0.$$

The diagram lives before the reduced E-quotient. It is a three-input correspondence type, not the full eight-word two-step flag atlas and not an associativity theorem. Source and target anchor memory, admissibility, base change, projection formula, Thom–Sebastiani transport, and transition compatibility are separate rows.

The packet

```
certificates/hybrid/two_sided_mixed_correspondence_definition
```

verifies `TWO_SIDED_MIXED_CORRESPONDENCE_DEFINITION_VERIFIED`: the ordered LWL two-sided mixed correspondence type is defined, while flag-stack rows, source/target-map rows, populated anchor-memory rows, admissibility rows, base-change rows, projection-formula rows, Thom–Sebastiani rows, and transition rows are recorded as missing open obligations.

Definition 0.48 *(Source and target anchors before quotient).* Fix a finite HN height R. Assume the wrapped prequotient sector has the prequotient stacks of Definition 0.6 and has the abstract anchors

$$\lambda_{\eta,R} : \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \longrightarrow E$$

of Definition 0.69. For every unreduced correspondence with a wrapped input or wrapped target, the source/target anchor memory is part of the correspondence datum before the reduced E-quotient is applied.

For the one-sided mixed rows of Definition 0.21, it is

$$a^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I} : \mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I} \longrightarrow E^2, \qquad \xi \longmapsto \big(\lambda_{\eta,R}(W_\eta), \lambda_{\zeta,R}(B_\zeta)\big),$$

and

$$a^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I} : \mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I} \longrightarrow E^2, \qquad \xi \longmapsto \big(\lambda_{\eta,R}(W_\eta), \lambda_{\zeta,R}(B_\zeta)\big).$$

For the wrapped/wrapped row of Definition 0.23, it is

$$a^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R} : \mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R} \longrightarrow E^3, \qquad \xi \longmapsto \big(\lambda_{\eta_1,R}(W_1), \lambda_{\eta_2,R}(W_2), \lambda_{\zeta,R}(B_\zeta)\big).$$

For the ordered two-sided mixed row of Definition 0.47, it is

$$a^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+} : \mathfrak{E}^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+} \longrightarrow E^2, \qquad \xi \longmapsto \big(\lambda_{\eta,R}(W_\eta), \lambda_{\zeta,R}(C_\zeta)\big).$$

These maps record ordered source anchors and target anchors on the unreduced stack. The local/local correspondence has no wrapped input and no wrapped anchor-memory row. The determinant-anchor construction and translation-weight computation are dedicated later packets; the $\chi = 0$ degeneracy and the degree-zero Jordan–Hoelder extra anchor are isolated by their own packets. The anchor residual definition o^λ_R, its vanishing, anchor losslessness, quotient descent, stabilizer linearization, and transition compatibility remain separate rows. Replacing the tuple above by a scalar Fock factor, a Borcherds s-degree, or a post-quotient class is not this datum.

The packet

`certificates/hybrid/source_target_anchor_memory_definition`

verifies `SOURCE_TARGET_ANCHOR_MEMORY_DEFINITION_VERIFIED`: the LW, WL, WW, and LWL source/target anchor-memory maps are defined before quotient. The determinant anchor is constructed by the next packet, and its translation weight is computed by the following packet. The $\chi = 0$ degeneracy is isolated by the chi-zero packet. The degree-zero Jordan–Hoelder extra-anchor datum is recorded by `certificates/hybrid/determinant_anchor_extra_anchor_data`; anchor-residual definition and vanishing, full losslessness, quotient-descent, stabilizer linearization, populated finite-stack, and transition rows remain separate obligations.

The packet

`certificates/hybrid/determinant_anchor_construction`

verifies `DETERMINANT_ANCHOR_CONSTRUCTION_VERIFIED`: Definition 4.7 constructs the normalized determinant-of-cohomology anchor

$$\lambda^{\det}_{\eta,R}(\mathcal{F}_{\eta,T}) = \det R\pi_{E,T,*}\mathcal{F}_{\eta,T} \otimes \mathrm{pr}_E^* \mathcal{O}_E(-\chi_\eta \cdot o_E) \in \mathrm{Pic}^0(E)$$

as a functorial map to $\mathrm{Pic}^0(E) \simeq E$. The packet checks the degree-zero normalization and determinant-functor base-change; unit-weight normalization, the Jordan–Hoelder extra-anchor datum, the residual definition o^λ_R, residual vanishing, losslessness, quotient descent, populated wrapped-prequotient rows, and transition rows remain separate. The $\chi = 0$ degeneracy is isolated by the chi-zero packet below. The translation-weight packet

`certificates/hybrid/determinant_anchor_translation_weight`

verifies `DETERMINANT_ANCHOR_TRANSLATION_WEIGHT_VERIFIED`: for the support-translation convention $a \cdot \mathcal{F} = (\mathrm{id}_S \times \tau_a)_* \mathcal{F}$, Proposition 4.8 gives

$$\lambda^{\det}_{\eta,R}(a \cdot \mathcal{F}) = \lambda^{\det}_{\eta,R}(\mathcal{F}) + \chi_\eta a \quad \text{in } \mathrm{Pic}^0(E) \simeq E.$$

The packet records the finite-stabilizer restriction $h \mapsto \chi_\eta h$. It does not choose coherent stabilizer linearizations, make the weight one, handle the $\chi_\eta = 0$ degeneracy, identify the Jordan–Hoelder extra anchor, prove losslessness, descend through the quotient, or prove transition compatibility. The chi-zero degeneracy packet

`certificates/hybrid/determinant_anchor_chi_zero_degeneracy`

verifies `DETERMINANT_ANCHOR_CHI_ZERO_DEGENERACY_VERIFIED`: Proposition 4.9 proves that when $\chi_\eta = 0$, the determinant anchor is constant on the full E-translation orbit. Freeness of the retained E-translation action gives a rank-one separation defect for determinant-anchor-only data. The packet records the necessity of extra anchor data; it does not itself construct that data, define o_R^λ, prove losslessness, descend through the quotient, or prove transition compatibility. The determinant-anchor extra-anchor-data packet

`certificates/hybrid/determinant_anchor_extra_anchor_data`

verifies `DETERMINANT_ANCHOR_EXTRA_ANCHOR_DATA_VERIFIED`: Proposition 4.10 adds the degree-zero Jordan–Hoelder divisor of the elliptic pushforward as anchor data. The Abel sum is the determinant anchor, and the rank-two pair $\mathcal{O}_E^{\oplus 2}$ and $L \oplus L^{-1}$, $L \neq \mathcal{O}_E$, is separated despite equal determinant. This packet does not prove global anchor losslessness, prove $o_R^\lambda = 0$, descend the repaired anchor through the quotient, choose stabilizer linearizations, populate all finite wrapped-prequotient rows, or prove transition compatibility.

Definition 0.49 *(Finite anchor residual).* Fix a finite HN height R. For $\eta \in \Gamma_R^{\mathrm{wr}}$, let

$$\mathcal{A}_{\eta,R} := \mathrm{Sym}^{r_\eta} \operatorname{Pic}^0(E), \qquad \mathrm{Ab}_{\eta,R} : \mathcal{A}_{\eta,R} \longrightarrow \operatorname{Pic}^0(E) \simeq E$$

be the Abel-sum morphism. A repaired wrapped anchor over the prequotient row is a functorial map

$$\Lambda_{\eta,R} : \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \longrightarrow \mathcal{A}_{\eta,R}$$

such that

$$\mathrm{Ab}_{\eta,R} \circ \Lambda_{\eta,R} = \lambda_{\eta,R}^{\det},$$

and whose restriction to the degree-zero vector-bundle row is the Jordan–Hoelder divisor of Proposition 4.10. For an exact sequence of semistable degree-zero elliptic pushforwards, the anchor target has the divisor-union operation

$$\mu_{\eta_1,\eta_2} : \mathcal{A}_{\eta_1,R} \times \mathcal{A}_{\eta_2,R} \longrightarrow \mathcal{A}_{\eta_1+\eta_2,R}.$$

The finite anchor residual is the five-component defect

$$\mathfrak{o}_R^\lambda = \left(\mathfrak{o}_R^{\lambda,\mathrm{ex}},\ \mathfrak{o}_R^{\lambda,\mathrm{unit}},\ \mathfrak{o}_R^{\lambda,\mathrm{loss}},\ \mathfrak{o}_R^{\lambda,\mathrm{multi}},\ \mathfrak{o}_{R'R}^{\lambda,\mathrm{tr}}\right),$$

with the following zero conditions. The E-valued source/target anchors of Definition 0.48 are the Abel-sum shadows $\mathrm{Ab}_{\eta,R} \circ \Lambda_{\eta,R}$; they do not replace the repaired symmetric-product anchor before the residual vanishes.

$$\mathfrak{o}_R^{\lambda,\mathrm{ex}} = 0$$

if and only if every $\eta \in \Gamma_R^{\mathrm{wr}}$ carries such a functorial repaired anchor $\Lambda_{\eta,R}$, including the semistable vector-bundle locus needed for the Jordan–Hoelder divisor. The component $\mathfrak{o}_R^{\lambda,\mathrm{unit}}$ vanishes if and only if the chosen repaired anchors admit a finite-cover normalization with translation weight one for the retained E-action. The component $\mathfrak{o}_R^{\lambda,\mathrm{loss}}$ vanishes if and only if the kernel pair of $\Lambda_{\eta,R}$ on each retained wrapped row is the retained equivalence relation, so that no extra wrapped position is identified by the anchor. The component $\mathfrak{o}_R^{\lambda,\mathrm{multi}}$ vanishes if and only if, on every LW, WL, WW, and LWL unreduced correspondence, the target repaired anchor equals the divisor-union operation applied to the ordered wrapped source anchors. The component $\mathfrak{o}_{R'R}^{\lambda,\mathrm{tr}}$ vanishes if and only if, for every $R \le R'$, the repaired anchors commute with the HN restriction $R' \to R$. This definition supplies the residual and its zero semantics. It does not prove any component vanishes, does not descend the anchor through the reduced E-quotient, and does not choose stabilizer linearizations.

The packet

`certificates/hybrid/anchor_residual_definition`

verifies `ANCHOR_RESIDUAL_DEFINITION_VERIFIED`: Definition 0.49 defines the repaired anchor target, its Abel-sum projection, and the five residual components ex, unit, loss, multi, and tr. The packet does not prove first-window vanishing, quotient descent, stabilizer linearization, finite-stage population, or transition compatibility.

Definition 0.50 *(Quotient-after-correspondence pseudofunctor).* [definition only] Fix a finite HN height R. Let $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ be the unreduced E-equivariant hybrid correspondence category generated by the retained local, mixed, wrapped, two-sided mixed, and flag correspondence stacks before the reduced E-quotient. A quotient-after-correspondence datum consists of the following finite rows.

For every retained object Y of $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, including local closed configuration charts, wrapped rigidified prequotients, correspondence stacks, and flag stacks, choose the quotient stack

$$\overline{Y} = [Y/E]$$

for the retained diagonal translation action and the quotient map

$$q_Y : Y \longrightarrow \overline{Y}.$$

For every generating unreduced correspondence

$$e = (Y_0 \xleftarrow{p_e} \mathfrak{C}_e \xrightarrow{q_e} Y_1)$$

whose maps are E-equivariant, choose the reduced correspondence

$$\overline{e} = (\overline{Y}_0 \xleftarrow{\overline{p}_e} [\mathfrak{C}_e/E] \xrightarrow{\overline{q}_e} \overline{Y}_1)$$

induced after the same correspondence stack has been formed. The assignment

$$Q_{E,R}^{\mathrm{corr}}(Y) = \overline{Y}, \qquad Q_{E,R}^{\mathrm{corr}}(e) = \overline{e}$$

is the quotient-after-correspondence pseudofunctor at the level of objects and generating arrows.

For a reduced vanishing-cycle coefficient K_Y on Y, the datum also specifies the coefficient-descent interface

$$\overline{K}_Y \in D_c^b(\overline{Y}), \qquad q_Y^* \overline{K}_Y \simeq K_Y,$$

whenever the relevant stabilizer linearization and orientation descent rows are supplied. The induced chain functor

$$Q_{E,R} : \mathsf{BM}_R^{E,\mathrm{hyb}} \longrightarrow \mathsf{BM}_R^{\mathrm{red},\mathrm{hyb}}$$

is not part of this definition; it is the later Borel–Moore chain row built from these quotient and coefficient-descent interfaces.

The finite quotient residual attached to this datum is

$$\mathfrak{o}_R^Q = \left(\mathfrak{o}_R^{Q,\mathrm{form}}, \mathfrak{o}_R^{Q,\mathrm{comp}}, \mathfrak{o}_R^{Q,\mathrm{bc}}, \mathfrak{o}_R^{Q,\mathrm{TS}}, \mathfrak{o}_R^{Q,\mathrm{flag}}, \mathfrak{o}_R^{Q,\mathrm{BM}}, \mathfrak{o}_R^{Q,\theta}, \mathfrak{o}_{R'R}^{Q,\mathrm{tr}}\right).$$

The first component says that objects, arrows, quotient maps, and coefficient-descent interfaces are defined. The remaining components measure, respectively, preservation of composition, base-change, Thom–Sebastiani transport, flag-stack coherences, Borel–Moore chain functoriality, the comparison maps $\theta^Q_{\mu,R}$, and compatibility with $R' \to R$ transitions.

This is a quotient-after-correspondence definition: the correspondence stack $\mathfrak{E}_e$ is formed before applying $[-/E]$. It is not a quotient-first Hall product on already reduced object stacks.

The packet

```
certificates/hybrid/quotient_after_correspondence_pseudofunctor_definition
```

verifies `QUOTIENT_AFTER_CORRESPONDENCE_PSEUDOFUNCTOR_DEFINED`: Definition 0.50 defines the object assignment, arrow assignment, coefficient-descent interface, and eight quotient residual components of the quotient-after-correspondence pseudofunctor before Borel–Moore chains. The composition component is supplied by Proposition 0.51. The base-change square component is supplied by Proposition 0.52. The Thom–Sebastiani descent component is supplied by Proposition 0.53. The quotient-first exclusion theorem is supplied by Proposition 0.54. The Borel–Moore chain functor is supplied by Proposition 0.55. The operation-comparison maps $\theta^Q_{\mu,R}$ are supplied by Proposition 0.56. The quotient two-step flag associativity component is supplied by Proposition 0.57. It does not prove compact-support pushforward base change, transition compatibility, or aggregate population.

Proposition 0.51 *(Quotient-after-correspondence preserves composition).* [proved for finite-height composition of spans] Fix R and the quotient-after-correspondence datum of Definition 0.50. Let

$$e = (Y_0 \xleftarrow{p_e} \mathfrak{E}_e \xrightarrow{q_e} Y_1), \qquad f = (Y_1 \xleftarrow{p_f} \mathfrak{E}_f \xrightarrow{q_f} Y_2)$$

be composable retained E-equivariant hybrid correspondences. The unreduced composite has middle stack

$$\mathfrak{E}_e \times_{Y_1} \mathfrak{E}_f$$

with the diagonal E-action. There is a canonical isomorphism of reduced correspondences

$$\Xi_{f,e} \colon Q^{\mathrm{corr}}_{E,R}(f \circ e) \xrightarrow{\sim} Q^{\mathrm{corr}}_{E,R}(f) \circ Q^{\mathrm{corr}}_{E,R}(e),$$

whose middle-stack component is

$$\left[(\mathfrak{E}_e \times_{Y_1} \mathfrak{E}_f)/E\right] \simeq [\mathfrak{E}_e/E] \times_{[Y_1/E]} [\mathfrak{E}_f/E].$$

The isomorphisms $\Xi_{f,e}$ satisfy the associativity cocycle for three composable correspondences. Hence

$$\mathfrak{o}_R^{Q,\mathrm{comp}} = 0.$$

This proves no base-change theorem for compact-support pushforwards, no Thom–Sebastiani compatibility, no Borel–Moore chain functor $Q_{E,R}$, no comparison map $\theta^{Q}_{\mu,R}$, no quotient-first exclusion theorem, and no transition or aggregate statement.

Proof. The maps q_e and p_f are E-equivariant, so the fibre product $\mathfrak{E}_e \times_{Y_1} \mathfrak{E}_f$ carries the diagonal E-action. For a test scheme T, an object of $[\mathfrak{E}_e/E] \times_{[Y_1/E]} [\mathfrak{E}_f/E]$ over T consists of E-torsors $P_e, P_f \to T$, equivariant maps

$$u : P_e \to \mathfrak{E}_e, \qquad v : P_f \to \mathfrak{E}_f,$$

and an isomorphism $\alpha : P_e \xrightarrow{\sim} P_f$ over the common image in $[Y_1/E]$; equivalently $p_f v\alpha = q_e u$ as maps $P_e \to Y_1$. Transporting v along α gives one torsor $P_e \to T$ with compatible equivariant maps to $\mathfrak{E}_e$ and $\mathfrak{E}_f$. This is precisely an E-equivariant map

$$P_e \longrightarrow \mathfrak{E}_e \times_{Y_1} \mathfrak{E}_f,$$

or equivalently an object of $[(\mathfrak{E}_e \times_{Y_1} \mathfrak{E}_f)/E]$ over T. The converse sends one torsor with a map to the fibre product to the two quotient-stack objects with the identity torsor isomorphism. Morphisms are transported by the same torsor isomorphisms. The two constructions are inverse and functorial in T, giving the displayed isomorphism of quotient stacks.

The source and target maps to $[Y_0/E]$ and $[Y_2/E]$ are induced from p_e and q_f, so the middle-stack isomorphism identifies the quotient of the unreduced composite with the composite of the two reduced correspondences. For three composable correspondences both iterated comparisons are obtained from the same quotient of the triple fibre product, and the two maps agree by the associativity of 2-fibre products and the universal property of quotient stacks. Thus the composition defect has rank zero, while all coefficient, base-change, Thom–Sebastiani, Borel–Moore, transition, and aggregate claims remain outside this proposition. $\square$

The packet

`certificates/hybrid/quotient_after_correspondence_composition`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_COMPOSITION_VERIFIED`: Proposition 0.51 proves that quotient-after-correspondence commutes with finite-height composition of retained E-equivariant spans and that the associativity cocycle has defect rank zero. The base-change component is supplied by Proposition 0.52. The Thom–Sebastiani descent component is supplied by Proposition 0.53. The quotient-first exclusion theorem is supplied by Proposition 0.54. The Borel–Moore chain functor is supplied by Proposition 0.55. The operation-comparison maps $\theta^{Q}_{\mu,R}$ are supplied by Proposition 0.56. The quotient two-step flag associativity component is supplied by Proposition 0.57. The HN-transition compatibility of $Q_{E,R}$ is supplied by Proposition 0.60. It does not prove aggregate population.

PROPOSITION 0.52 *(Quotient-after-correspondence preserves base-change squares).* [proved for finite-height cartesian squares of stacks] Fix R and the quotient-after-correspondence datum of Definition 0.50. Let

$$\begin{array}{ccc} X' & \xrightarrow{\tilde{g}} & X \\ h'\downarrow & & \downarrow h \\ Y' & \xrightarrow{g} & Y \end{array} \qquad (\square)$$

be a retained E-equivariant cartesian square of stacks appearing as a source, target, open, or compactified square of the finite hybrid correspondence calculus. Then the quotient square

$$\begin{array}{ccc} [X'/E] & \xrightarrow{[\widetilde{g}/E]} & [X/E] \\ {\scriptstyle [h'/E]}\downarrow & & \downarrow{\scriptstyle [h/E]} \\ [Y'/E] & \xrightarrow{[g/E]} & [Y/E] \end{array}$$

is cartesian. Equivalently, the canonical morphism

$$[X'/E] \longrightarrow [X/E] \times_{[Y/E]} [Y'/E]$$

is an isomorphism. Hence quotient-after-correspondence preserves the retained base-change squares and

$$\mathfrak{d}_R^{Q,\mathrm{bc}} = 0.$$

This proves only the stack-level cartesian-square component of the quotient residual. It does not prove compact-support pushforward base change, coefficient pullback for vanishing cycles, the projection formula, Thom–Sebastiani compatibility, the chain functor $Q_{E,R}$, the comparison maps $\theta^Q_{\mu,R}$, transition compatibility, or aggregate population.

Proof. Since the square (□) is E-equivariant and cartesian, there is an E-equivariant isomorphism

$$X' \simeq X \times_Y Y'.$$

For a test scheme T, an object of $[X/E] \times_{[Y/E]} [Y'/E]$ over T consists of E-torsors $P_X, P_{Y'} \to T$, equivariant maps

$$u : P_X \to X, \qquad v : P_{Y'} \to Y',$$

and an isomorphism $\alpha : P_X \xrightarrow{\sim} P_{Y'}$ over the common image in $[Y/E]$. After transporting v along α, this is one E-torsor $P_X \to T$ with maps $u : P_X \to X$ and $v\alpha : P_X \to Y'$ satisfying

$$h\,u = g\,v\,\alpha.$$

By the cartesian identity $X' \simeq X \times_Y Y'$, this is the same as an E-equivariant map $P_X \to X'$, hence an object of $[X'/E]$ over T. The converse sends an equivariant map $P \to X'$ to its two projections with the identity torsor isomorphism. Morphisms are carried by the same torsor isomorphisms, and the two constructions are inverse and functorial in T. Therefore the quotient square is cartesian.

Applying this argument to each retained source, target, open, and compactified square in the finite hybrid correspondence calculus gives zero defect for preservation of base-change squares. The proof has not applied $h_!$, h^*, reduced Thom–Sebastiani transport, or Borel–Moore chains, so the corresponding functorial and chain-level claims remain separate obligations. □

The packet

`certificates/hybrid/quotient_after_correspondence_base_change`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_BASE_CHANGE_VERIFIED`: Proposition 0.52 proves that quotient-after-correspondence preserves retained cartesian base-change squares of finite-height

E-equivariant stacks. It does not prove compact-support pushforward base change, coefficient pullback, or aggregate population. The Thom–Sebastiani descent component is supplied by Proposition 0.53. The quotient-first exclusion theorem is supplied by Proposition 0.54. The Borel–Moore chain functor is supplied by Proposition 0.55. The operation-comparison maps $\theta^{Q}_{\mu,R}$ are supplied by Proposition 0.56. The quotient two-step flag associativity component is supplied by Proposition 0.57. The HN-transition compatibility of $Q_{E,R}$ is supplied by Proposition 0.60.

PROPOSITION 0.53 *(Quotient-after-correspondence preserves Thom–Sebastiani transport).* [proved for equivariant finite-height Thom–Sebastiani rows] Fix R and the quotient-after-correspondence datum of Definition 0.50. Let

$$e = (Y_{\mathrm{src}} \xleftarrow{p_e} \mathfrak{E}_e \xrightarrow{q_e} Y_{\mathrm{tar}})$$

be a retained E-equivariant correspondence whose unreduced reduced vanishing-cycle coefficients carry an E-equivariant Thom–Sebastiani transport

$$\mathrm{TS}_e^{\mathrm{red}} : p_e^* K_{Y_{\mathrm{src}}} \xrightarrow{\sim} K_{\mathfrak{E}_e}.$$

Assume the coefficient-descent interfaces

$$q_{Y_{\mathrm{src}}}^* \overline{K}_{Y_{\mathrm{src}}} \simeq K_{Y_{\mathrm{src}}}, \qquad q_{\mathfrak{E}_e}^* \overline{K}_{\mathfrak{E}_e} \simeq K_{\mathfrak{E}_e}$$

are supplied, including the stabilizer linearizations and orientation descent needed for the reduced vanishing-cycle complexes. Then there is a unique reduced Thom–Sebastiani transport

$$\overline{\mathrm{TS}}_{\overline{e}}^{\mathrm{red}} : \overline{p}_e^* \overline{K}_{Y_{\mathrm{src}}} \xrightarrow{\sim} \overline{K}_{\mathfrak{E}_e}$$

on the quotient correspondence

$$\overline{e} = ([Y_{\mathrm{src}}/E] \xleftarrow{\overline{p}_e} [\mathfrak{E}_e/E] \xrightarrow{\overline{q}_e} [Y_{\mathrm{tar}}/E])$$

whose pullback along $q_{\mathfrak{E}_e}$ is $\mathrm{TS}_e^{\mathrm{red}}$ under the descent identifications and the cartesian quotient square of Proposition 0.52. Hence

$$\mathfrak{o}_R^{Q,\mathrm{TS}} = 0.$$

This proves only descent of the reduced Thom–Sebastiani coefficient transport through the quotient-after-correspondence construction. It does not prove the existence of the unreduced BBDJS Thom–Sebastiani row, the Joyce–Upmeier orientation row, compact-support pushforward base change, flag-stack coherences, the Borel–Moore chain functor $Q_{E,R}$, the comparison maps $\theta^{Q}_{\mu,R}$, transition compatibility, or aggregate population.

Proof. The quotient square for the source map is cartesian by Proposition 0.52:

$$\begin{array}{ccc} \mathfrak{E}_e & \xrightarrow{q_{\mathfrak{E}_e}} & [\mathfrak{E}_e/E] \\ p_e\downarrow & & \downarrow\overline{p}_e \\ Y_{\mathrm{src}} & \xrightarrow{q_{Y_{\mathrm{src}}}} & [Y_{\mathrm{src}}/E]. \end{array}$$

Therefore

$$q_{\mathfrak{E}_e}^* \overline{p}_e^* \overline{K}_{Y_{\mathrm{src}}} \simeq p_e^* q_{Y_{\mathrm{src}}}^* \overline{K}_{Y_{\mathrm{src}}} \simeq p_e^* K_{Y_{\mathrm{src}}}.$$

The second descent interface gives

$$q_{\mathfrak{E}_e}^* \overline{K}_{\mathfrak{E}_e} \simeq K_{\mathfrak{E}_e}.$$

Thus the unreduced Thom–Sebastiani isomorphism is a morphism between the pullbacks of the two reduced coefficient objects on $[\mathfrak{E}_e/E]$.

Constructible sheaves on the quotient stack $[\mathfrak{E}_e/E]$ are equivalent to E-equivariant constructible sheaves on $\mathfrak{E}_e$, with the stated stabilizer linearizations. Under this equivalence, morphisms on the quotient stack are exactly equivariant morphisms on $\mathfrak{E}_e$. Since $\mathrm{TS}_e^{\mathrm{red}}$ is E-equivariant by hypothesis, it descends uniquely to a morphism

$$\overline{\mathrm{TS}}_{\overline{e}}^{\mathrm{red}} : \overline{p}_e^* \overline{K}_{Y_{\mathrm{src}}} \longrightarrow \overline{K}_{\mathfrak{E}_e}.$$

Its pullback is an isomorphism, hence the descended morphism is an isomorphism. The same equivalence of quotient sheaves proves uniqueness and functoriality for every retained finite-height Thom–Sebastiani row. No pushforward, Borel–Moore chain functor, or flag comparison has been applied, so those coherences remain outside this proposition. □

The packet

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certificates/hybrid/quotient_after_correspondence_thom_sebastiani
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verifies `QUOTIENT_AFTER_CORRESPONDENCE_THOM_SEBASTIANI_VERIFIED`: Proposition 0.53 proves that an equivariant reduced Thom–Sebastiani transport descends uniquely through the quotient-after-correspondence construction once the coefficient-descent interfaces are supplied. It does not prove aggregate population. The operation-comparison maps $\theta_{\mu,R}^{Q}$ are supplied by Proposition 0.56. The quotient-first exclusion theorem is Proposition 0.54. The Borel–Moore chain functor is Proposition 0.55. The quotient two-step flag associativity component is Proposition 0.57. The HN-transition compatibility of $Q_{E,R}$ is Proposition 0.60.

Proposition 0.54 *(Quotient-first construction is excluded).* [proved as a finite-height image characterization] Fix R and the quotient-after-correspondence datum of Definition 0.50. Every generating arrow in the image of $\boldsymbol{Q}_{E,R}^{\mathrm{corr}}$ is represented by a retained unreduced E-equivariant correspondence

$$e = (Y_0 \xleftarrow{p_e} \mathfrak{E}_e \xrightarrow{q_e} Y_1)$$

together with the quotient presentation

$$\overline{e} = ([Y_0/E] \xleftarrow{\overline{p}_e} [\mathfrak{E}_e/E] \xrightarrow{\overline{q}_e} [Y_1/E]).$$

Equivalently, the two squares

$$\begin{array}{ccccccc} \mathfrak{E}_e & \longrightarrow & [\mathfrak{E}_e/E] & \qquad & \mathfrak{E}_e & \longrightarrow & [\mathfrak{E}_e/E] \\ {\scriptstyle p_e}\downarrow & & \downarrow{\scriptstyle \overline{p}_e} & & {\scriptstyle q_e}\downarrow & & \downarrow{\scriptstyle \overline{q}_e} \\ Y_0 & \longrightarrow & [Y_0/E] & & Y_1 & \longrightarrow & [Y_1/E] \end{array}$$

are part of the arrow data. A quotient-first span, obtained by first passing to $[Y_0/E]$ and $[Y_1/E]$ and then choosing an independent middle stack over their product, is not an arrow of $\boldsymbol{Q}_{E,R}^{\mathrm{corr}}$ unless it is equipped with a specified isomorphism to $[\mathfrak{E}_e/E]$ compatible with these two squares. In that exceptional case it is the same quotient-after- correspondence arrow after transport along the specified isomorphism. Thus no quotient-first Hall product is used in the finite quotient correspondence calculus, and the quotient-first exclusion row has defect rank zero.

Proof. The object assignment of Definition 0.50 sends an unreduced retained object Y to $[Y/E]$. Its arrow assignment does not choose a new reduced extension stack. It sends the already formed unreduced correspondence

$$Y_0 \xleftarrow{p_e} \mathfrak{E}_e \xrightarrow{q_e} Y_1$$

to the reduced correspondence with middle stack $[\mathfrak{E}_e/E]$. Hence the image of $Q^{\mathrm{corr}}_{E,R}$ is the full finite-height family of reduced spans carrying such an unreduced quotient presentation.

Let

$$[Y_0/E] \xleftarrow{a} Z \xrightarrow{b} [Y_1/E]$$

be a quotient-first candidate obtained after quotienting the two object stacks. If Z is not supplied with an isomorphism

$$Z \simeq [\mathfrak{E}_e/E]$$

for a retained unreduced E-equivariant correspondence e, and if a, b are not the transported maps $\overline{p}_e, \overline{q}_e$, then this candidate lacks the quotient map $\mathfrak{E}_e \to [\mathfrak{E}_e/E]$ used in the definition. It is therefore not in the image of $Q^{\mathrm{corr}}_{E,R}$. If such an isomorphism is supplied, transporting the span structure along it identifies the candidate with the quotient-after-correspondence span assigned to e; no independent quotient-first construction remains.

The composition comparison of Proposition 0.51 uses the quotient of the unreduced fibre product $\mathfrak{E}_e \times_{Y_1} \mathfrak{E}_f$. The base-change comparison of Proposition 0.52 uses quotient presentations of the unreduced cartesian squares. The Thom–Sebastiani descent of Proposition 0.53 pulls coefficients back along $\mathfrak{E}_e \to [\mathfrak{E}_e/E]$. None of these rows can be interpreted on an independent quotient-first middle stack unless it has first been identified with the quotient of the unreduced middle stack. The exclusion is therefore a statement about the finite correspondence calculus itself, not a Borel–Moore, theta-comparison, transition, or aggregate-population claim. □

The packet

`certificates/hybrid/quotient_after_correspondence_quotient_first_exclusion`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_QUOTIENT_FIRST_EXCLUSION_VERIFIED`: Proposition 0.54 proves that every reduced generating arrow used by $Q^{\mathrm{corr}}_{E,R}$ carries a quotient presentation from a preformed unreduced E-equivariant correspondence. An object-quotient-first Hall product is excluded unless it is identified with that quotient presentation. This packet does not prove transition compatibility for Hall products or aggregate population. The Borel–Moore chain functor is supplied by Proposition 0.55. The operation-comparison maps $\theta^{Q}_{\mu,R}$ are supplied by Proposition 0.56. The quotient two-step flag associativity component is supplied by Proposition 0.57. The HN-transition compatibility of $Q_{E,R}$ is supplied by Proposition 0.60.

PROPOSITION 0.55 *(The quotient E-descent functor on Borel–Moore chains).* [proved for quotient-presented finite-height coefficient objects] Fix R and the quotient-after-correspondence datum of Definition 0.50. For every retained object Y of $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, suppose the reduced vanishing-cycle coefficient $K_Y \in D^b_{c,E}(Y)$ is supplied with the descent interface

$$\overline{K}_Y \in D^b_c([Y/E]), \qquad q_Y^*\overline{K}_Y \simeq K_Y.$$

Let

$$C_*^{\mathrm{BM}}(Z, L) := R\Gamma(Z, \mathbb{D}_Z L)$$

denote Borel–Moore chains with constructible coefficient L, with $\mathbb{D}_Z$ the Verdier duality functor on the retained finite stack Z. Define

$$C_*^{\mathrm{BM},E}(Y, K_Y) := C_*^{\mathrm{BM}}([Y/E], \overline{K}_Y).$$

Then the assignments

$$(Y, K_Y) \longmapsto ([Y/E], \overline{K}_Y), \qquad C_*^{\mathrm{BM},E}(Y, K_Y) \longmapsto C_*^{\mathrm{BM}}([Y/E], \overline{K}_Y)$$

extend to a functor

$$Q_{E,R} : \mathsf{BM}_R^{E,\mathrm{hyb}} \longrightarrow \mathsf{BM}_R^{\mathrm{red},\mathrm{hyb}}.$$

On an E-equivariant coefficient morphism $\alpha : K_Y \to L_Y$, the functor sends α to the descended morphism

$$\overline{\alpha} : \overline{K}_Y \longrightarrow \overline{L}_Y$$

under the quotient-stack descent equivalence, and then applies $C_*^{\mathrm{BM}}([Y/E], -)$. Hence the Borel–Moore chain component of the quotient residual vanishes:

$$\mathfrak{d}_R^{Q,\mathrm{BM}} = 0.$$

This proves only the chain-level descent functor. It does not prove flag coherences, compact-support pushforward base change, the comparison isomorphisms $\theta^Q_{\mu,R}$, compatibility with products or coproducts, transition compatibility, protected-integration compatibility, or aggregate population.

Proof. The coefficient-descent interface identifies the supplied E-equivariant coefficient K_Y with the pullback of a constructible coefficient on the quotient stack. The standard descent equivalence

$$D^b_c([Y/E]) \simeq D^b_{c,E}(Y)$$

is fully faithful on morphisms: an equivariant morphism $\alpha : K_Y \to L_Y$ is uniquely the pullback of a morphism $\overline{\alpha} : \overline{K}_Y \to \overline{L}_Y$ whenever both coefficients carry the prescribed descent data. Identities and composites descend to identities and composites because this is an equivalence of categories.

Borel–Moore chains are functorial for coefficient morphisms on the fixed stack:

$$\overline{\alpha} : \overline{K}_Y \to \overline{L}_Y \quad \Longrightarrow \quad R\Gamma([Y/E], \mathbb{D}\overline{\alpha}) : C_*^{\mathrm{BM}}([Y/E], \overline{K}_Y) \to C_*^{\mathrm{BM}}([Y/E], \overline{L}_Y).$$

The object assignment $Y \mapsto [Y/E]$ and the coefficient assignment $K_Y \mapsto \overline{K}_Y$ therefore define a functor on the finite additive chain category generated by the retained quotient-presented coefficient objects. This is $Q_{E,R}$.

No map of the form $q_!\,\mathrm{TS}\,p^*$ has been compared with its reduced counterpart. The present functor supplies the chain objects and descended coefficient morphisms on which the later $\theta^Q_{\mu,R}$ comparison maps may act; it does not construct those maps. $\square$

The packet

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certificates/hybrid/quotient_after_correspondence_bm_chain_functor
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verifies `QUOTIENT_AFTER_CORRESPONDENCE_BM_CHAIN_FUNCTOR_VERIFIED`: Proposition 0.55 constructs $Q_{E,R}$ on quotient-presented Borel–Moore chain objects and proves $\mathfrak{o}_R^{Q,\mathrm{BM}} = 0$. This packet does not prove compact-support pushforward base change, protected integration, or aggregate population. The operation-comparison maps $\theta^Q_{\mu,R}$ are supplied by Proposition 0.56. The quotient two-step flag associativity component is supplied by Proposition 0.57. The HN-transition compatibility of $Q_{E,R}$ is supplied by Proposition 0.60.

PROPOSITION 0.56 *(Quotient comparison maps for retained operations).* [proved for quotient-presented finite-height operation rows] Fix R and the quotient-after-correspondence datum of Definition 0.50. Let

$$e = \left(Y_1 \times \cdots \times Y_k \xleftarrow{p_e} \mathfrak{E}_e \xrightarrow{q_e} Y_0\right)$$

be a retained k-ary local, mixed, wrapped, or flag operation row. Assume the input, middle, and target coefficient objects carry quotient-descent interfaces, the external product coefficient descends to

$$\overline{K}_1 \boxtimes \cdots \boxtimes \overline{K}_k \quad \text{on} \quad [Y_1/E] \times \cdots \times [Y_k/E],$$

the reduced Thom–Sebastiani transport descends as in Proposition 0.53, and the chosen compact-support pushforward row for q_e is quotient-presented by an isomorphism

$$\delta_{q,e} \colon \overline{q^{\mathrm{cs}}_{e,!} L_e} \xrightarrow{\sim} \overline{q}^{\mathrm{cs}}_{e,!} \overline{L}_e,$$

where

$$L_e = \mathrm{TS}^{\mathrm{red}}_e \, p_e^*(K_1 \boxtimes \cdots \boxtimes K_k), \qquad \overline{L}_e = \overline{\mathrm{TS}}^{\mathrm{red}}_{\overline{e}} \, \overline{p}_e^{\,*}(\overline{K}_1 \boxtimes \cdots \boxtimes \overline{K}_k).$$

Then there is a canonical chain isomorphism

$$\theta^Q_{\mu,e,R} \colon Q_{E,R} \, q^{\mathrm{cs}}_{e,!} \, \mathrm{TS}^{\mathrm{red}}_e \, p_e^* \xrightarrow{\sim} \overline{q}^{\mathrm{cs}}_{e,!} \, \overline{\mathrm{TS}}^{\mathrm{red}}_{\overline{e}} \, \overline{p}_e^{\,*}(Q_{E,R}^{\boxtimes k}).$$

For every retained operation row with the displayed descent data,

$$\mathfrak{o}_R^{Q,\theta} = 0.$$

This proves only the existence of the operation-comparison isomorphisms $\theta^Q_{\mu,R}$. It does not prove their compatibility with two-step flag associativity, coproducts, primitive projection, $R' \to R$ transitions, protected integration, or aggregate hybrid-carrier population.

Proof. By Proposition 0.52, the quotient square for p_e is cartesian. Hence the pullback of the descended external product along the quotient middle stack is the descent of

$$p_e^*(K_1 \boxtimes \cdots \boxtimes K_k).$$

Proposition 0.53 then identifies the descended middle coefficient with

$$\overline{L}_e = \overline{\mathrm{TS}}^{\mathrm{red}}_{\overline{e}} \, \overline{p}_e^{\,*}(\overline{K}_1 \boxtimes \cdots \boxtimes \overline{K}_k).$$

The quotient-presented compact-support row for the target map gives the specified isomorphism

$$\delta_{q,e} : \overline{q^{\mathrm{cs}}_{e,!} L_e} \xrightarrow{\sim} \overline{q}^{\mathrm{cs}}_{e,!} \overline{L}_e$$

on $[Y_0/E]$.

Applying the Borel–Moore chain functor of Proposition 0.55 to $\delta_{q,e}$ gives the displayed chain isomorphism. Its source is, by definition,

$$Q_{E,R}\, q^{\mathrm{cs}}_{e,!}\, \mathrm{TS}^{\mathrm{red}}_{e}\, p^{*}_{e}(K_1, \ldots, K_k),$$

and its target is the reduced operation

$$\overline{q}^{\mathrm{cs}}_{e,!}\, \overline{\mathrm{TS}}^{\mathrm{red}}_{\overline{e}}\, \overline{p}^{*}_{e}(Q_{E,R}K_1, \ldots, Q_{E,R}K_k).$$

Uniqueness follows from full faithfulness of constructible descent on quotient stacks. The proof compares one retained operation with its reduced quotient operation; it does not compare two parenthesized operations, a product with a coproduct, a primitive projection, or two HN heights. □

The packet

`certificates/hybrid/quotient_after_correspondence_theta_mu_comparison`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_THETA_MU_COMPARISON_VERIFIED`: Proposition 0.56 constructs the operation-comparison isomorphisms $\theta^{Q}_{\mu,R}$ for quotient-presented finite-height operation rows and proves $\mathfrak{o}^{Q,\theta}_{R} = 0$ on those rows. This packet imports associativity compatibility from Proposition 0.57. It imports coproduct compatibility from Proposition 0.58. It imports primitive-projection compatibility from Proposition 0.59. It imports the HN-transition compatibility of $Q_{E,R}$ from Proposition 0.60. It does not prove Hall-operation transition compatibility, protected integration, or aggregate population.

Proposition 0.57 *(Associativity compatibility of the quotient comparison maps).* [proved for the eight two-step flag associativity rows] Fix R, and let

$$w = \epsilon_1\epsilon_2\epsilon_3 \in \{\mathrm{L}, \mathrm{W}\}^3$$

be one of the eight local/wrapped words with retained ordered colours (ξ_1, ξ_2, ξ_3). Assume the two-step flag stack $\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}$ and the wordwise associativity row of Propositions 0.31–0.38. Let

$$A_w = \mu_{\xi_1+\xi_2,\xi_3,R} \circ (\mu_{\xi_1,\xi_2,R} \otimes 1), \qquad B_w = \mu_{\xi_1,\xi_2+\xi_3,R} \circ (1 \otimes \mu_{\xi_2,\xi_3,R})$$

be the two unreduced parenthesized operations, and let

$$a_w : A_w \xrightarrow{\sim} B_w$$

be the associativity isomorphism represented by the common flag stack. Let $\overline{A}_w$, $\overline{B}_w$, and $\overline{a}_w : \overline{A}_w \xrightarrow{\sim} \overline{B}_w$, be the corresponding reduced operations after quotient-after- correspondence. If every binary and composite operation occurring in A_w and B_w carries the quotient-presented data required in Proposition 0.56, then the square

$$\begin{tikzcd} Q_{E,R} A_w \arrow[r, "\Theta_w^L"] \arrow[d, "Q_{E,R}(a_w)"'] & \overline{A}_w\,(Q_{E,R}^{\boxtimes 3}) \arrow[d, "\overline{a}_w"] \\ Q_{E,R} B_w \arrow[r, "\Theta_w^R"'] & \overline{B}_w\,(Q_{E,R}^{\boxtimes 3}) \end{tikzcd}$$

commutes. Here Θ_w^L and Θ_w^R are obtained by composing the binary $\theta^Q_{\mu,R}$ maps of Proposition 0.56 with the composition comparison of Proposition 0.51. Consequently the quotient two-step flag associativity component vanishes:

$$\mathfrak{o}_R^{Q,\mathrm{flag}} = 0.$$

This proves only compatibility of $\theta^Q_{\mu,R}$ with the eight length-three associativity rows. It does not prove coproduct compatibility, primitive compatibility, $R' \to R$ transition compatibility, protected-integration compatibility, or aggregate hybrid-carrier population.

Proof. For a fixed w, both parenthesizations are represented before quotient by the same retained two-step flag stack

$$\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}.$$

The maps λ^w and ρ^w of Proposition 0.30 identify this flag stack with the two iterated correspondence fibre products used by A_w and B_w. The wordwise associativity propositions state that the two pull-push operations obtained from this common flag stack agree after the required base-change, projection-formula, and reduced Thom–Sebastiani identifications.

Apply the quotient-after-correspondence construction to the same diagram. Proposition 0.51 identifies the quotient of each iterated correspondence fibre product with the corresponding iterated quotient correspondence. Proposition 0.52 identifies the quotient of the common flag square with the reduced cartesian square, and Proposition 0.53 descends the coefficient transports. Thus the reduced associator $\overline{a}_w$ is the image of a_w under quotient descent.

The maps Θ_w^L and Θ_w^R are induced by the same quotient-presented compact-support pushforward descent isomorphisms $\delta_{q,e}$ used in Proposition 0.56, pulled back to the common quotient flag stack. Full faithfulness of constructible descent on quotient stacks makes the two morphisms in the displayed square equal after pullback to $\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}$. Since descent is faithful, the square commutes on the quotient stack. The argument is wordwise for the eight length-three words; it does not involve the Hall coproduct, primitive projection, finite-height transitions, or protected integration. □

The packet

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certificates/hybrid/quotient_after_correspondence_theta_mu_associativity
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verifies `QUOTIENT_AFTER_CORRESPONDENCE_THETA_MU_ASSOCIATIVITY_VERIFIED`: Proposition 0.57 proves that the quotient comparison maps $\theta^Q_{\mu,R}$ commute with the eight wordwise two-step flag associativity rows, hence $\mathfrak{o}_R^{Q,\mathrm{flag}} = 0$ for those rows. This packet imports coproduct compatibility from Proposition 0.58. It imports primitive-projection compatibility from Proposition 0.59. It imports the HN-transition compatibility of $Q_{E,R}$ from Proposition 0.60. It does not prove Hall-operation transition compatibility, protected integration, or aggregate population.

Proposition 0.58 *(Coproduct compatibility of the quotient comparison maps).* [proved for supplied quotient-presented Hall bialgebra rows] Fix R. Let μ_R be a retained binary Hall operation row, and let

$$\Delta_R : \mathcal{H}_R^{E,\mathrm{hyb}} \longrightarrow \mathcal{H}_R^{E,\mathrm{hyb}} \boxtimes \mathcal{H}_R^{E,\mathrm{hyb}}$$

be a retained collision coproduct row. Assume the product correspondence, the splitting correspondence defining Δ_R, and the four-leg bialgebra comparison correspondence carry the quotient-presented compact-support, base-change, and reduced Thom–Sebastiani data required in Proposition 0.56. Assume also that the source finite Hall bialgebra identity

$$\Delta_R\mu_R = (\mu_R \boxtimes \mu_R)(1 \boxtimes \tau \boxtimes 1)(\Delta_R \boxtimes \Delta_R)$$

has defect rank zero on that retained correspondence, with Koszul sign braiding τ. Let $\overline{\mu}_R$, $\overline{\Delta}_R$, and the reduced bialgebra identity be the quotient-after-correspondence images of these rows.

Then the two morphisms

$$(Q_{E,R} \boxtimes Q_{E,R})\Delta_R\mu_R \rightrightarrows (\overline{\mu}_R \boxtimes \overline{\mu}_R)(1 \boxtimes \tau \boxtimes 1)(\overline{\Delta}_R \boxtimes \overline{\Delta}_R)Q_{E,R}^{\boxtimes 2}$$

coincide. The first morphism applies the quotient comparison for the coproduct row, then $\overline{\Delta}_R(\theta^Q_{\mu,R})$, and then the reduced bialgebra identity. The second applies the source bialgebra identity, then the two coproduct comparison maps and the two product comparison maps $\theta^Q_{\mu,R}$. Consequently the quotient product-coproduct compatibility defect vanishes:

$$\mathfrak{d}_R^{Q,\Delta} = 0.$$

This proves only compatibility of $\theta^Q_{\mu,R}$ with supplied finite coproduct/bialgebra rows. It does not construct the compact Hall coproduct in the aggregate carrier, prove primitive compatibility, $R' \to R$ transition compatibility, protected integration, or aggregate hybrid-carrier population.

Proof. The source bialgebra identity is represented by the retained four-leg extension-splitting correspondence: both $\Delta_R\mu_R$ and $(\mu_R \boxtimes \mu_R)(1 \boxtimes \tau \boxtimes 1)(\Delta_R \boxtimes \Delta_R)$ are its two pull-push descriptions. The zero-defect source row asserts equality after the listed base-change, projection-formula, compact-support, and Thom–Sebastiani identifications.

Apply quotient-after-correspondence to the same correspondence. Proposition 0.51 identifies the quotient of the composite correspondence with the composite of the quotient correspondences. Proposition 0.52 identifies the quotient of the bialgebra comparison squares with the reduced cartesian squares, and Proposition 0.53 descends the product and inverse-coproduct coefficient transports. Thus the reduced bialgebra identity is the quotient image of the source identity.

The comparison maps for the product and coproduct rows are induced by the same quotient-presented compact-support descent isomorphisms $\delta_{q,e}$ used in Proposition 0.56, with source and target legs interchanged for the coproduct correspondence. After pullback to the common equivariant four-leg correspondence, the two displayed composites are the same morphism because the source bialgebra square has defect rank zero. Full faithfulness of constructible descent on quotient stacks then identifies the two descended morphisms on the quotient correspondence. The argument uses only the supplied finite product-coproduct square; it does not form primitive kernels, finite-height transition maps, or the protected integration map. □

The packet

`certificates/hybrid/quotient_after_correspondence_theta_mu_coproduct`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_THETA_MU_COPRODUCT_VERIFIED`: Proposition 0.58 proves that the quotient comparison maps $\theta^{Q}_{\mu,R}$ commute with supplied finite Hall coproduct/bialgebra rows. This packet imports primitive-projection compatibility from Proposition 0.59. It imports the HN-transition compatibility of $Q_{E,R}$ from Proposition 0.60. It does not prove Hall-coproduct transition compatibility, protected integration, or aggregate population.

PROPOSITION 0.59 *(Primitive-projection compatibility of the quotient comparison maps).* [proved for supplied finite primitive kernel and projection rows] Fix R. Let

$$\widetilde{\Delta}_R = \Delta_R - \eta_R\epsilon_R \boxtimes \mathrm{id} - \mathrm{id} \boxtimes \eta_R\epsilon_R, \qquad \widetilde{\overline{\Delta}}_R = \overline{\Delta}_R - \overline{\eta}_R\overline{\epsilon}_R \boxtimes \mathrm{id} - \mathrm{id} \boxtimes \overline{\eta}_R\overline{\epsilon}_R$$

be the reduced coproducts attached to the retained finite Hall bialgebra rows. Assume the finite primitive rows supply the kernel subobjects and idempotent projections

$$P^E_R = \ker(\widetilde{\Delta}_R) \cap \ker(\epsilon_R), \qquad \overline{P}_R = \ker(\widetilde{\overline{\Delta}}_R) \cap \ker(\overline{\epsilon}_R),$$
$$\pi^{\mathrm{Prim}}_R : \mathcal{H}^{E,\mathrm{hyb}}_R \to P^E_R, \qquad \overline{\pi}^{\mathrm{Prim}}_R : \mathcal{H}^{\mathrm{red},\mathrm{hyb}}_R \to \overline{P}_R,$$

and assume that the quotient-after-correspondence descent of the unit, counit, and reduced coproduct rows identifies

$$Q_{E,R}P^E_R \simeq \overline{P}_R, \qquad Q_{E,R}\pi^{\mathrm{Prim}}_R = \overline{\pi}^{\mathrm{Prim}}_R Q_{E,R}$$

on the supplied primitive kernel/projection row. For a retained binary Hall product row μ_R, let $\theta^{Q}_{\mu,R}$ be the comparison map of Proposition 0.56, and assume the coproduct compatibility of Proposition 0.58 for the same product-coproduct row.

Then $\theta^{Q}_{\mu,R}$ commutes with the primitive projection on the output Hall object:

$$\overline{\pi}^{\mathrm{Prim}}_R\, \overline{\mu}_R\, Q^{\boxtimes 2}_{E,R}\, \theta^{Q}_{\mu,R} = \theta^{Q,\mathrm{Prim}}_{\mu,R}\, Q_{E,R}\pi^{\mathrm{Prim}}_R\, \mu_R,$$

where $\theta^{Q,\mathrm{Prim}}_{\mu,R} : Q_{E,R}P^E_R \to \overline{P}_R$ is the quotient identification on the primitive summand. Equivalently, the square

$$\begin{tikzcd} Q_{E,R}\mu_R \arrow[r, "\theta^{Q}_{\mu,R}"] \arrow[d, "Q_{E,R}\pi^{\mathrm{Prim}}_R"'] & \overline{\mu}_R Q^{\boxtimes 2}_{E,R} \arrow[d, "\overline{\pi}^{\mathrm{Prim}}_R"] \\ Q_{E,R}P^E_R \arrow[r, "\theta^{Q,\mathrm{Prim}}_{\mu,R}"'] & \overline{P}_R \end{tikzcd}$$

commutes. Consequently the quotient primitive-projection defect vanishes on the supplied finite rows:

$$\mathfrak{d}^{Q,\mathrm{Prim}}_R = 0.$$

This proves only compatibility of $\theta^{Q}_{\mu,R}$ with the supplied primitive kernel/projection rows. It does not construct source compact Hall primitive matrices, prove $R' \to R$ transition compatibility, prove protected-integration compatibility, or populate the aggregate hybrid carrier.

Proof. The primitive subobject is the intersection of two kernels: the kernel of the reduced coproduct and the kernel of the counit. By Proposition 0.58, the quotient comparison sends the source reduced-coproduct equation to the reduced quotient equation. The primitive row additionally supplies the unit and counit descent, so $Q_{E,R}$ sends $\ker(\widetilde{\Delta}_R) \cap \ker(\epsilon_R)$ into $\ker(\widetilde{\overline{\Delta}}_R) \cap \ker(\overline{\epsilon}_R)$. This gives the kernel identification $Q_{E,R} P_R^E \simeq \overline{P}_R$ recorded in the hypothesis.

The two composites in the displayed square are morphisms obtained from the same quotient-presented binary Hall correspondence after adjoining the supplied primitive idempotent on the output. On the coproduct kernel their reduced-coproduct components agree by the row-254 coproduct square; on the augmentation kernel their counit components agree by the supplied counit descent. Full faithfulness of constructible descent on the quotient stack identifies the two idempotent-split morphisms. Hence the primitive projection square commutes and the finite primitive defect has rank zero. $\square$

The packet

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certificates/hybrid/quotient_after_correspondence_theta_mu_primitives
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verifies `QUOTIENT_AFTER_CORRESPONDENCE_THETA_MU_PRIMITIVES_VERIFIED`: Proposition 0.59 proves that the quotient comparison maps $\theta^{Q}_{\mu,R}$ commute with the supplied finite primitive kernel and projection rows. This packet imports the comparison, associativity, and coproduct packets. It imports the HN-transition compatibility of $Q_{E,R}$ from Proposition 0.60. It does not prove HN-transition preservation of primitive subspaces, protected-integration compatibility, or aggregate population.

Proposition 0.60 *(HN-transition compatibility of the quotient descent functor).* [proved for supplied finite HN transition rows] Let $R \leq R'$. Suppose the finite quotient-after-correspondence data at heights R' and R are supplied, and suppose every retained object row $Y_{R'} \to Y_R$ in the HN transition system is E-equivariant. Write

$$T^E_{R'R} : \mathsf{BM}^{E,\mathrm{hyb}}_{R'} \longrightarrow \mathsf{BM}^{E,\mathrm{hyb}}_{R}, \qquad \overline{T}_{R'R} : \mathsf{BM}^{\mathrm{red},\mathrm{hyb}}_{R'} \longrightarrow \mathsf{BM}^{\mathrm{red},\mathrm{hyb}}_{R}$$

for the supplied source and reduced Borel–Moore transition functors. Assume that each transition row is quotient-presented: the square

$$\begin{array}{ccc} Y_{R'} & \longrightarrow & Y_R \\ \downarrow & & \downarrow \\ [Y_{R'}/E] & \longrightarrow & [Y_R/E] \end{array}$$

commutes, the lower map is the quotient of the upper E-equivariant map, and the vanishing-cycle coefficient transport at height $R' \to R$ descends to the quotient coefficient transport. Equivalently, the source transition chain map and the reduced transition chain map are induced from the same equivariant transition correspondence.

Then the quotient descent functors commute with the HN transition:

$$\overline{T}_{R'R}\, Q_{E,R'} = Q_{E,R}\, T^E_{R'R}$$

as functors on the quotient-presented finite Borel–Moore chain category. Consequently the transition component of the quotient residual vanishes on the supplied rows:

$$\mathfrak{o}^{Q,\mathrm{tr}}_{R'R} = 0.$$

This proves only the HN-transition compatibility of $Q_{E,R}$ on quotient-presented finite transition rows. It does not prove that HN transitions preserve the Hall product, Hall coproduct, primitive subspaces, protected-integration compatibility, or aggregate hybrid-carrier population.

Proof. For a retained transition object $Y_{R'} \to Y_R$, the map is E-equivariant. The quotient-stack universal property therefore gives a unique reduced transition map

$$[Y_{R'}/E] \longrightarrow [Y_R/E]$$

for which the quotient square commutes. The coefficient transport is assumed to be an E-equivariant morphism between quotient-presented vanishing-cycle coefficients, and hence descends uniquely to the corresponding coefficient transport on the two quotient stacks.

Apply the Borel–Moore chain construction of Proposition 0.55. The composite $\overline{T}_{R'R}Q_{E,R'}$ first replaces the source object by the quotient stack $[Y_{R'}/E]$ and then applies the descended transition map to $[Y_R/E]$. The composite $Q_{E,R}T^E_{R'R}$ first applies the equivariant source transition and then descends the result to the same quotient stack $[Y_R/E]$. Both composites are therefore the same functor

$$C_*^{\mathrm{BM}}([Y_{R'}/E], \overline{K}_{R'}) \longrightarrow C_*^{\mathrm{BM}}([Y_R/E], \overline{K}_R)$$

with the same descended coefficient morphism. Full faithfulness of constructible descent on quotient stacks identifies the two chain maps and their composites over successive height restrictions. Thus the finite transition defect of $Q_{E,R}$ has rank zero. □

The packet

`certificates/hybrid/quotient_after_correspondence_hn_transition_compatibility`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_HN_TRANSITION_COMPATIBILITY_VERIFIED`: Proposition 0.60 proves that $Q_{E,R}$ commutes with supplied finite HN transition rows. It does not prove Hall product, Hall coproduct, primitive-subspace, protected-integration, or aggregate transition compatibility.

Definition 0.61 *(Finite protected integration datum).* Fix R, the reduced Borel–Moore chain object

$$\mathcal{H}_R^{\mathrm{red,hyb}} := Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R},$$

and its finite charge decomposition

$$\mathcal{H}_R^{\mathrm{red,hyb}} = \bigoplus_{\gamma \in \Gamma_R} \mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}}.$$

A finite protected integration datum at height R consists of:

(i) a finite commutative monoid T_R and labels $\omega_R(\gamma) \in \mathsf{T}_R$ for the retained charges;

(ii) the finite subspace
$$\mathbb{C}[\mathsf{T}_R]_{\le R} := \sum_{\gamma \in \Gamma_R} \mathbb{C}e_{\omega_R(\gamma)} \subset \mathbb{C}[\mathsf{T}_R];$$

(iii) for every $\gamma \in \Gamma_R$, a degree-zero chain map
$$\int_{\gamma,R}^{\mathrm{prot}} : \mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}} \longrightarrow \mathbb{C}e_{\omega_R(\gamma)}.$$

The associated protected integration map is the finite direct sum

$$I_R^{\mathrm{prot}} := \bigoplus_{\gamma\in\Gamma_R} \int_{\gamma,R}^{\mathrm{prot}} : \mathcal{H}_R^{\mathrm{red,hyb}} \longrightarrow \mathbb{C}[\mathsf{T}_R]_{\le R}.$$

Thus, for $x \in \mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}}$,

$$I_R^{\mathrm{prot}}(x) \in \mathbb{C}e_{\omega_R(\gamma)}.$$

The label $\omega_R(\gamma)$ is not yet identified with $\overline{\Pi}_X(\gamma)$, and $e_{\omega_R(\gamma)}$ is not yet identified with $q^n r^l s^m$. Those are the Gram-degree and $\overline{\Pi}_X$-separation rows. This datum also does not assert multiplicativity, comultiplicativity, primitive-projection compatibility, HN-transition compatibility, the closed $K3 \times E \to \mathrm{pt}$ protected trace, or a Pfaffian line.

Proposition 0.62 *(Definition row for protected integration).* [proved for supplied finite chargewise integration maps] The datum of Definition 0.61 defines a degree-zero chain map

$$I_R^{\mathrm{prot}} : Q_{E,R}\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}} \longrightarrow \mathbb{C}[\mathsf{T}_R]_{\le R}.$$

Equivalently, the chain-map component of the protected-integration residual vanishes on the supplied finite rows:

$$\mathfrak{o}_R^{I,\mathrm{ch}} = 0.$$

This proves only the definition and well-typedness of I_R^{prot}. It does not prove product compatibility, coproduct compatibility, primitive-projection compatibility, HN-transition compatibility, the Gram exponent law, the use of $\overline{\Pi}_X$ rather than b_R^{geom}, or aggregate hybrid-carrier population.

Proof. The charge set Γ_R is finite at height R. Each summand $\int_{\gamma,R}^{\mathrm{prot}}$ is a degree-zero chain map to the one-dimensional complex $\mathbb{C}e_{\omega_R(\gamma)}$. The finite direct sum of these chain maps is therefore a degree-zero chain map from the finite direct sum $\bigoplus_{\gamma\in\Gamma_R}\mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}}$ to the finite subspace of the monoid algebra spanned by the labels $e_{\omega_R(\gamma)}$. This map is exactly I_R^{prot}. No Hall operation, coproduct, primitive idempotent, transition functor, Gram-coordinate comparison, or closed trace functor enters the construction. □

The packet

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certificates/hybrid/protected_integration_definition
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verifies `PROTECTED_INTEGRATION_DEFINITION_VERIFIED`: Proposition 0.62 defines I_R^{prot} as the finite chargewise direct sum of supplied degree-zero protected integration maps. Product compatibility is supplied separately by Proposition 0.63. Coproduct compatibility is supplied separately by Proposition 0.64. Primitive-projection compatibility is supplied separately by Proposition 0.65. Gram degree is supplied separately by Proposition 0.66. The normal-ordered label comparison is supplied separately by Proposition 0.67. The packet records transition compatibility as a separate open row.

Proposition 0.63 *(Product compatibility of protected integration).* [proved for supplied finite binary product rows] Fix R and the finite protected integration datum of Definition 0.61. Let

$$w = \epsilon_1\epsilon_2 \in \{\mathrm{L}, \mathrm{W}\}^2$$

be a retained binary local/wrapped word, and let

$$\overline{\mu}_{w,\gamma_1,\gamma_2,R} : \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1,R} \otimes \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_2,R} \longrightarrow \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1+\gamma_2,R}$$

be the corresponding reduced Hall product row after quotient-after-correspondence and the comparison map $\theta^{Q}_{\mu,R}$. Assume the target monoid labels satisfy the supplied product row

$$m_{\mathsf{T}_R}\left(e_{\omega_R(\gamma_1)} \otimes e_{\omega_R(\gamma_2)}\right) = e_{\omega_R(\gamma_1+\gamma_2)}$$

with defect rank zero. Assume also that the chargewise protected integration functionals satisfy the supplied square

$$\begin{tikzcd}
\mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1,R} \otimes \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_2,R} \arrow[r, "\overline{\mu}_{w,\gamma_1,\gamma_2,R}"] \arrow[d, "\int^{\mathrm{prot}}_{\gamma_1,R} \otimes \int^{\mathrm{prot}}_{\gamma_2,R}"'] & \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1+\gamma_2,R} \arrow[d, "\int^{\mathrm{prot}}_{\gamma_1+\gamma_2,R}"] \\
\mathbb{C}e_{\omega_R(\gamma_1)} \otimes \mathbb{C}e_{\omega_R(\gamma_2)} \arrow[r, "m_{\mathsf{T}_R}"'] & \mathbb{C}e_{\omega_R(\gamma_1+\gamma_2)}
\end{tikzcd}$$

with defect rank zero. Then the protected integration map is multiplicative on this product row:

$$I^{\mathrm{prot}}_R\,\overline{\mu}_{w,\gamma_1,\gamma_2,R} = m_{\mathsf{T}_R}\,(I^{\mathrm{prot}}_R \otimes I^{\mathrm{prot}}_R)$$

on $\mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1,R} \otimes \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_2,R}$. Consequently

$$\mathfrak{d}^{I,\mathrm{prod}}_{w,R} = 0$$

for every supplied finite binary product row of word w. This proves only product compatibility of I^{prot}_R. It does not prove coproduct compatibility, primitive-projection compatibility, HN-transition compatibility, the Gram exponent law, the use of $\overline{\Pi}_X$ rather than b^{geom}_R, the level-Z protected trace, or a Pfaffian line.

Proof. Restrict I^{prot}_R to the two source charge summands and to the target charge summand. By Definition 0.61, these restrictions are exactly

$$\int^{\mathrm{prot}}_{\gamma_1,R}, \qquad \int^{\mathrm{prot}}_{\gamma_2,R}, \qquad \int^{\mathrm{prot}}_{\gamma_1+\gamma_2,R}.$$

The displayed component square is therefore precisely the restriction of the global multiplicativity square for I^{prot}_R to the summand $\mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1,R} \otimes \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_2,R}$. Since Γ_R is finite, the global product square is the finite direct sum of these component squares. Each component has defect rank zero by hypothesis, and the target label multiplication identifies the one-dimensional target line with $\mathbb{C}e_{\omega_R(\gamma_1+\gamma_2)}$. Hence the product defect of protected integration vanishes on the supplied row. The proof uses no coproduct, primitive projection, transition functor, Gram-coordinate identification, closed trace functor, or Pfaffian line. □

The packet

`certificates/hybrid/protected_integration_product_compatibility`

verifies `PROTECTED_INTEGRATION_PRODUCT_COMPATIBILITY_VERIFIED`: Proposition 0.63 proves that I^{prot}_R respects every supplied finite binary local/wrapped product row. Coproduct compatibility is supplied separately by Proposition 0.64. Primitive-projection compatibility is supplied separately by Proposition 0.65. Gram degree is supplied separately by Proposition 0.66. The normal-ordered label comparison is supplied separately by Proposition 0.67. The packet records transition compatibility as a separate open row.

Proposition 0.64 *(Coproduct compatibility of protected integration).* [proved for supplied finite binary coproduct rows] Fix R and the finite protected integration datum of Definition 0.61. Let

$$w = \epsilon_1\epsilon_2 \in \{\mathrm{L}, \mathrm{W}\}^2$$

be a retained binary local/wrapped output word, and let

$$\overline{\Delta}_{w,\gamma_1,\gamma_2,R} \colon \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1+\gamma_2,R} \longrightarrow \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1,R} \otimes \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_2,R}$$

be the corresponding reduced Hall coproduct row after quotient-after-correspondence and the coproduct comparison row $\theta^{Q}_{\Delta,R}$. Assume the target label coalgebra has the supplied component row

$$\Delta_{\mathsf{T}_R,w}\big(e_{\omega_R(\gamma_1+\gamma_2)}\big) = e_{\omega_R(\gamma_1)} \otimes e_{\omega_R(\gamma_2)}$$

with defect rank zero. Assume also that the chargewise protected integration functionals satisfy the supplied square

$$\begin{array}{ccc}
\mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1+\gamma_2,R} & \xrightarrow{\overline{\Delta}_{w,\gamma_1,\gamma_2,R}} & \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1,R} \otimes \mathcal{H}^{\mathrm{red,hyb}}_{\gamma_2,R} \\
{\scriptstyle \int^{\mathrm{prot}}_{\gamma_1+\gamma_2,R}}\Big\downarrow & & \Big\downarrow{\scriptstyle \int^{\mathrm{prot}}_{\gamma_1,R} \otimes \int^{\mathrm{prot}}_{\gamma_2,R}} \\
\mathbb{C}e_{\omega_R(\gamma_1+\gamma_2)} & \xrightarrow[\Delta_{\mathsf{T}_R,w}]{} & \mathbb{C}e_{\omega_R(\gamma_1)} \otimes \mathbb{C}e_{\omega_R(\gamma_2)}
\end{array}$$

with defect rank zero. Then the protected integration map is comultiplicative on this coproduct row:

$$(I^{\mathrm{prot}}_R \otimes I^{\mathrm{prot}}_R)\,\overline{\Delta}_{w,\gamma_1,\gamma_2,R} = \Delta_{\mathsf{T}_R,w}\, I^{\mathrm{prot}}_R$$

on $\mathcal{H}^{\mathrm{red,hyb}}_{\gamma_1+\gamma_2,R}$. Consequently

$$\mathfrak{d}^{I,\mathrm{co}}_{w,R} = 0$$

for every supplied finite binary coproduct row of word w. This proves only coproduct compatibility of I^{prot}_R. It does not prove primitive-projection compatibility, HN-transition compatibility, the Gram exponent law, the use of $\overline{\Pi}_X$ rather than b^{geom}_R, the level-Z protected trace, or a Pfaffian line.

Proof. Restrict I^{prot}_R to the source charge summand and to the two target charge summands. By Definition 0.61, these restrictions are

$$\int^{\mathrm{prot}}_{\gamma_1+\gamma_2,R}, \qquad \int^{\mathrm{prot}}_{\gamma_1,R}, \qquad \int^{\mathrm{prot}}_{\gamma_2,R}.$$

The displayed component square is therefore the restriction of the global coproduct square for I^{prot}_R to the charge $\gamma_1 + \gamma_2$ and the output split (γ_1, γ_2). Since Γ_R is finite, the supplied global coproduct square is the finite direct sum of these component squares over the retained split rows. Each component has defect rank zero by hypothesis, and the target label coproduct identifies the one-dimensional source label line with $\mathbb{C}e_{\omega_R(\gamma_1)} \otimes \mathbb{C}e_{\omega_R(\gamma_2)}$. Hence the coproduct defect of protected integration vanishes on the supplied row. The proof uses no primitive projection, transition functor, Gram-coordinate identification, closed trace functor, or Pfaffian line. □

The packet

```
certificates/hybrid/protected_integration_coproduct_compatibility
```

verifies `PROTECTED_INTEGRATION_COPRODUCT_COMPATIBILITY_VERIFIED`: Proposition 0.64 proves that I_R^{prot} respects every supplied finite binary local/wrapped coproduct row. Primitive-projection compatibility is supplied separately by Proposition 0.65. Gram degree is supplied separately by Proposition 0.66. The normal-ordered label comparison is supplied separately by Proposition 0.67. The packet records transition compatibility as a separate open row.

PROPOSITION 0.65 *(Primitive-projection compatibility of protected integration).* [proved for supplied finite primitive projection rows] Fix R and the finite protected integration datum of Definition 0.61. Assume the quotient primitive-projection row of Proposition 0.59 has supplied the reduced primitive subobject

$$\overline{P}_R = \ker(\overline{\widetilde{\Delta}}_R) \cap \ker(\overline{\epsilon}_R) \subset \mathcal{H}_R^{\mathrm{red,hyb}}$$

and an idempotent projection

$$\overline{\pi}_R^{\mathrm{Prim}} : \mathcal{H}_R^{\mathrm{red,hyb}} \longrightarrow \overline{P}_R.$$

Assume also that the target label coalgebra has a supplied primitive projector

$$\pi_{\mathsf{T}_R}^{\mathrm{Prim}} : \mathbb{C}[\mathsf{T}_R]_{\le R} \longrightarrow \mathbb{C}[\mathsf{T}_R]_{\le R}^{\mathrm{Prim}}$$

with defect rank zero, and that the protected integration functionals satisfy the supplied square

$$\begin{tikzcd}
\mathcal{H}_R^{\mathrm{red,hyb}} \arrow[r, "\overline{\pi}_R^{\mathrm{Prim}}"] \arrow[d, "I_R^{\mathrm{prot}}"'] & \overline{P}_R \arrow[d, "I_R^{\mathrm{prot}}|_{\overline{P}_R}"] \\
\mathbb{C}[\mathsf{T}_R]_{\le R} \arrow[r, "\pi_{\mathsf{T}_R}^{\mathrm{Prim}}"'] & \mathbb{C}[\mathsf{T}_R]_{\le R}^{\mathrm{Prim}}
\end{tikzcd}$$

with defect rank zero. Then I_R^{prot} respects the primitive projection:

$$I_R^{\mathrm{prot}}\, \overline{\pi}_R^{\mathrm{Prim}} = \pi_{\mathsf{T}_R}^{\mathrm{Prim}}\, I_R^{\mathrm{prot}}$$

on $\mathcal{H}_R^{\mathrm{red,hyb}}$, after identifying $I_R^{\mathrm{prot}}|_{\overline{P}_R}$ with its target primitive subspace. Consequently

$$\mathfrak{d}_R^{I,\mathrm{Prim}} = 0$$

on the supplied finite primitive projection row. This proves only primitive-projection compatibility of I_R^{prot}. It does not prove HN-transition compatibility, the Gram exponent law, the use of $\overline{\Pi}_X$ rather than b_R^{geom}, the level-Z protected trace, or a Pfaffian line.

Proof. The reduced primitive subobject $\overline{P}_R$ is already the idempotent-split summand supplied by Proposition 0.59; the present statement does not reconstruct it. Restrict the chargewise direct sum I_R^{prot} to this summand and compose with the target primitive projector. The displayed square is exactly the assertion that these two composites agree. Its defect rank is zero by hypothesis, hence the primitive-projection component of the protected-integration residual vanishes on the supplied finite row. No transition functor, Gram-coordinate identification, closed trace functor, or Pfaffian line enters the argument. □

The packet

```
certificates/hybrid/protected_integration_primitive_compatibility
```

verifies `PROTECTED_INTEGRATION_PRIMITIVE_COMPATIBILITY_VERIFIED`: Proposition 0.65 proves that I_R^{prot} respects the supplied finite primitive projection row. Gram degree is supplied separately by Proposition 0.66. The normal-ordered label comparison is supplied separately by Proposition 0.67. The packet records transition compatibility as a separate open row.

Proposition 0.66 *(Gram degree of protected integration).* [proved for supplied finite Gram-monomial label rows] Fix R and the finite protected integration datum of Definition 0.61. Assume a finite Gram-monomial target row supplies a map

$$g_R : \mathsf{T}_R \longrightarrow \mathbb{Z}^3, \qquad g_R(\omega_R(\gamma)) = (n_R(\gamma), l_R(\gamma), m_R(\gamma)),$$

and a monomialization map

$$\chi_R : \mathbb{C}[\mathsf{T}_R]_{\leq R} \longrightarrow \mathbb{C}[q^{\pm 1}, r^{\pm 1}, s^{\pm 1}]$$

such that

$$\chi_R\big(e_{\omega_R(\gamma)}\big) = q^{n_R(\gamma)} r^{l_R(\gamma)} s^{m_R(\gamma)}$$

on every retained charge label, with defect rank zero. Then, for every

$$x \in \mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}},$$

one has

$$(\chi_R I_R^{\mathrm{prot}})(x) \in \mathbb{C}\, q^{n_R(\gamma)} r^{l_R(\gamma)} s^{m_R(\gamma)}.$$

Equivalently, the s-degree of the protected integration image is

$$\deg_s(\chi_R I_R^{\mathrm{prot}}(x)) = m_R(\gamma)$$

whenever the scalar coefficient is nonzero. Consequently

$$\mathfrak{d}_R^{I,\deg} = 0$$

on the supplied finite Gram-monomial rows. This proves only the target Gram-degree statement for protected integration. It does not prove that $g_R\omega_R$ is the normal-ordered charge map $\overline{\Pi}_X$, does not prove that the third coordinate is different from b_R^{geom}, does not prove HN-transition compatibility, and does not construct the level-Z protected trace or a Pfaffian line.

Proof. By Definition 0.61,

$$I_R^{\mathrm{prot}}(x) \in \mathbb{C}e_{\omega_R(\gamma)}$$

for $x \in \mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}}$. Applying the supplied monomialization map χ_R sends the one-dimensional target line $\mathbb{C}e_{\omega_R(\gamma)}$ to the one-dimensional monomial line

$$\mathbb{C}q^{n_R(\gamma)} r^{l_R(\gamma)} s^{m_R(\gamma)}.$$

Thus the image has the displayed q, r, s-degree, and in particular s-degree $m_R(\gamma)$ when the coefficient is nonzero. The argument uses only the target label row and the chargewise definition of I_R^{prot}; it does not use a comparison between ω_R and $\overline{\Pi}_X$, does not compare with b_R^{geom}, and does not use transition or trace data. $\square$

The packet

`certificates/hybrid/protected_integration_gram_degree`

verifies `PROTECTED_INTEGRATION_GRAM_DEGREE_VERIFIED`: Proposition 0.66 proves that the supplied finite monomialization sends the protected integration image of charge γ to the line $\mathbb{C}q^{n_R(\gamma)}r^{l_R(\gamma)}s^{m_R(\gamma)}$. The packet records transition compatibility and leaves the normal-ordered label comparison to Proposition 0.67.

PROPOSITION 0.67 *(Normal-ordered Gram label of protected integration).* [proved for supplied finite normal-ordered label comparison rows] Fix R and the finite protected integration datum of Definition 0.61. Assume the Gram-degree row of Proposition 0.66. Assume also that a finite protected-label comparison row supplies a lift

$$\widehat{\omega}_R : \Gamma_R \longrightarrow \widehat{\Gamma}_R$$

to the domain of the normal-ordered Gram map and a zero-defect equality

$$g_R\omega_R = \overline{\Pi}_X\widehat{\omega}_R : \Gamma_R \longrightarrow \mathbb{Z}^3.$$

Write

$$\overline{\Pi}_X(\widehat{\omega}_R(\gamma)) = \big(n_R^{\Pi}(\gamma), l_R^{\Pi}(\gamma), m_R^{\Pi}(\gamma)\big).$$

Then, for every

$$x \in \mathcal{H}_{\gamma,R}^{\mathrm{red,hyb}},$$

one has

$$(\chi_R I_R^{\mathrm{prot}})(x) \in \mathbb{C}q^{n_R^{\Pi}(\gamma)}r^{l_R^{\Pi}(\gamma)}s^{m_R^{\Pi}(\gamma)}$$

and

$$\deg_s(\chi_R I_R^{\mathrm{prot}}(x)) = m_R^{\Pi}(\gamma) = \mathrm{pr}_3\,\overline{\Pi}_X(\widehat{\omega}_R(\gamma))$$

whenever the scalar coefficient is nonzero. The exponent of protected integration is therefore the normal-ordered Gram coordinate

$$m_R^{\mathrm{Bch}} := \mathrm{pr}_3\,\overline{\Pi}_X\widehat{\omega}_R,$$

not the geometric support degree b_R^{geom}. Equivalently, the protected-integration label residual satisfies

$$\mathfrak{d}_R^{I,\Pi} = 0$$

on the supplied finite comparison rows. This proves only the $\overline{\Pi}_X$-label comparison for protected integration. It does not prove HN-transition compatibility, a branch formula relating m_R^{Bch} to b_R^{geom}, the level-`Z` protected trace, or a Pfaffian line.

Proof. By Proposition 0.66, the image of a homogeneous class of charge γ lies in the monomial line

$$\mathbb{C}q^{n_R(\gamma)}r^{l_R(\gamma)}s^{m_R(\gamma)}$$

where

$$g_R(\omega_R(\gamma)) = (n_R(\gamma), l_R(\gamma), m_R(\gamma)).$$

The supplied comparison row identifies this triple with

$$\overline{\Pi}_X(\widehat{\omega}_R(\gamma)) = \big(n_R^{\Pi}(\gamma), l_R^{\Pi}(\gamma), m_R^{\Pi}(\gamma)\big).$$

Substitution gives the displayed monomial line and the displayed s-degree. Proposition 0.14 separates the support map

$$b_R^{\mathrm{geom}}(\gamma) = \mathrm{coeff}_{[E]}(p_E)_* \,\mathrm{cyc}_1(\gamma)$$

from the Borcherds coordinate

$$\mathrm{pr}_3\, \overline{\Pi}_X(\widehat{\omega}_R(\gamma)).$$

Thus b_R^{geom} controls the local/wrapped support split, whereas the protected-integration monomial is labelled by the normal-ordered Gram coordinate. No equality or affine relation between these two integers is used; such a relation would require the separate branch-comparison datum of Definition 0.13. □

The packet

`certificates/hybrid/protected_integration_pi_x_separation`

verifies `PROTECTED_INTEGRATION_PI_X_SEPARATION_VERIFIED`: Proposition 0.67 proves that the protected-integration monomial exponent is $\overline{\Pi}_X(\widehat{\omega}_R(\gamma))$, with s-degree $\mathrm{pr}_3\, \overline{\Pi}_X(\widehat{\omega}_R(\gamma))$, and not the geometric support degree $b_R^{\mathrm{geom}}(\gamma)$. The packet records transition compatibility as a separate open row.

Proposition 0.68 *(First-window anchor residual vanishing criterion).* [proved as a criterion; unverified for the present first-window scaffold] Fix a first relation-closed finite window W_1 and a height R whose wrapped colours lie in W_1. The restriction

$$\mathfrak{o}_R^{\lambda}|_{W_1} = 0$$

is equivalent to the simultaneous vanishing on W_1 of the five component defects in Definition 0.49: existence of repaired anchors $\Lambda_{\eta,R}$, unit translation weight after the chosen finite cover, kernel-pair losslessness, multiplicativity on the LW, WL, WW, and LWL unreduced correspondences, and compatibility with the $R' \to R$ HN-transition maps. In the current first-window scaffold

`certificates/first_window/k3e_relation_closed_window,`

the compact-source window rows, target-source representative rows, correspondence theorem rows, and strict transition rows are empty. Consequently the present data do not prove $\mathfrak{o}_R^{\lambda}|_{W_1} = 0$.

Proof. The equivalence is the product-zero condition for the five residual components in Definition 0.49. Each component is defined by an equality of finite data on the retained prequotient, correspondence, or transition row; the product residual vanishes exactly when all five equalities hold. The first-window obstruction ledger records no compact-source `window_closure.csv` rows, no `target_source_representatives.csv` rows, no theorem rows in `first_window_theorems.csv`, and no strict transition rows. These missing tables contain the required first-window existence, losslessness, multiplicativity, and transition data. Hence the criterion is defined, but its hypotheses are not supplied by the current first-window scaffold. □

The packet

```
certificates/hybrid/anchor_residual_first_window_obstruction
```

verifies `ANCHOR_RESIDUAL_FIRST_WINDOW_OBSTRUCTION_VERIFIED`: Proposition 0.68 gives the exact first-window zero criterion and records that the present first-window scaffold does not contain the rows needed to prove it. This packet is not a proof of row 218; it is the obstruction ledger for that proof.

FINITE HYBRID LOCAL/WRAPPED FACTORIZATION DATUM

The full carrier at finite stage is the following supplied chain datum, with named obstruction classes that must be trivialized.

Definition 0.69 *(Finite hybrid local/wrapped factorization datum).* Fix a Bridgeland-style stability sector (σ, S) in the K3-surface lineage [5, 13] and a finite HN height R. A finite-stage hybrid datum is

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} = \big(C_{\sigma,S,\leq R},\ \mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E),\ b^{\mathrm{geom}}_R,\ \mathcal{F}^{\mathrm{loc}}_R,\ \mathcal{G}^{\mathrm{wr}}_R,\ \mathfrak{E}^{\mathrm{loc}}_R,\ \mathfrak{E}^{\mathrm{hyb}}_R,\ \mathfrak{E}^{\mathrm{wr}}_R,\ \mathfrak{Flag}^{\mathrm{hyb}}_R,\ \mathsf{Corr}^{E,\mathrm{hyb}}_R,\ \lambda_R,\ Q_{E,R},\ \theta^Q_R,\ o_R,\ I^{\mathrm{prot}}_R\big)$$

with the seven clauses below. Every stack, sheaf, operation, and coherence lives at the fixed height R; the inverse limit appears only after all finite stages and transition maps have been supplied.

(i) *Finite Hall stage, hybrid prestack, and elliptic degree.* $C_{\sigma,S,\leq R}$ is a supplied extension-closed charge-support HN quotient of the candidate compact Hall category under the finite-type hypothesis [K3×E-fin], with finite-type substacks $\mathfrak{M}_{\gamma,R}$ for $\gamma \in \Gamma^{\mathrm{HN}}_{\sigma,S}(R)$, vanishing-cycle coefficients $\Phi_{\gamma,R}$, and orientation lines $o_{\gamma,R}$. The carrier prestack is $\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$ of Definition 0.3; its stackification is not used unless the descent rows of that definition are supplied. The geometric elliptic-degree component
$$b^{\mathrm{geom}}_R : \Gamma^{\mathrm{HN}}_{\sigma,S}(R) \longrightarrow \mathbb{Z}_{\geq 0}$$
is the map of Definition 0.10; it splits the finite charge support into local and wrapped pieces:
$$\Gamma^{\mathrm{loc}}_R = \{b^{\mathrm{geom}}_R = 0\}, \qquad \Gamma^{\mathrm{wr}}_R = \{b^{\mathrm{geom}}_R > 0\}. \tag{6.1}$$
The $b = 0$ carrier is $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ of Definition 0.4; it is the $J = \varnothing$ subprestack of the hybrid carrier, not the later local coefficient sheaf. The $b > 0$ carrier is $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$ of Definition 0.5; it is the $I = \varnothing$ subprestack whose points are wrapped prequotient families with anchors, not the later wrapped coefficient object $\mathcal{G}^{\mathrm{wr}}_R$. Additivity on retained extensions is the datum and criterion of Definition 0.11 and Proposition 0.12; terms of height $> R$ do not enter the retained quotient. The Borcherds trace coordinate $m^{\mathrm{Bch}}_R = \mathrm{pr}_3\, \overline{\Pi}_X$ is not used to form the split in (6.1); any relation between m^{Bch}_R and b^{geom}_R requires the comparison datum of Definition 0.13.

(ii) *Projection-finite local sector through closed configurations.* Over the local carrier $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$, and for a finite set I, set
$$Z_I = \bigcup_{i\in I} \{(e, (e_j)_j) \in E \times E^I \mid e = e_i\}.$$
For $\alpha \in \Gamma^{\mathrm{loc}}_R$ the stage supplies a closed-configuration incidence stack
$$\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} = \{(A, \mathbf{e}) \in \mathfrak{M}_{\alpha,R} \times E^I \mid p_E(\mathrm{Supp}\, A) \subset Z_I(\mathbf{e})\}$$

compatible with diagonals $E^J \to E^I$, symmetric-group descent, and the d-critical/vanishing-cycle structure. Open-support locality is recovered as a limit

$$\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U) \simeq \operatorname*{colim}_I \big(\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times_{E^I} U^I\big),$$

and the configuration map $\pi_{\alpha,R,I}$ produces

$$\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I} = (\pi_{\alpha,R,I})_!(\Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I}).$$

For an open $U \subset E$,

$$\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U) = \bigoplus_{I/S_I} R\Gamma_c\Big(U^I, j^!_{U,I}\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I}\Big),$$

and $\mathcal{F}^{\mathrm{loc}}_R(U)$ is the direct sum over Γ^{loc}_R. Locality is !-restriction on closed configuration spaces; the open-support Borel–Moore formula is a theorem to be derived from this atlas, not the primary datum.

(iii) *Wrapped prequotient sector and rigidification.* Over $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$, and for $\eta \in \Gamma^{\mathrm{wr}}_R$, the scalar-rigidified prequotient stack of Definition 0.6

$$\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \xrightarrow{\rho_{\eta,R}} \mathcal{M}^{\mathrm{wr}}_{\eta,R} \subset \mathfrak{M}_{\eta,R}$$

is supplied with a fixed origin $o_E \in E$. The map $\rho_{\eta,R}$ forgets the scalar rigidification; the reduced E-quotient is not taken in this clause. The normalized determinant construction of Definition 4.7 gives the candidate anchor

$$\lambda^{\det}_{\eta,R}(A) = \det Rp_{E,*}A \otimes \mathcal{O}_E(-\chi(A)\cdot o_E) \in \mathrm{Pic}^0(E) \simeq E.$$

Its translation weight is $\chi(A)$ by Proposition 4.8. If $\chi(A) = 0$, Proposition 4.9 shows that this determinant anchor is constant along the retained E-translation orbit and therefore has rank-one separation defect. The degree-zero Jordan–Hoelder divisor of Proposition 4.10 separates the determinant-zero rank-two test pair $\mathcal{O}_E^{\oplus 2}$ and $L \oplus L^{-1}$. Unit-weight finite-cover normalization, global anchor losslessness, quotient descent, stabilizer linearization, transition compatibility, and finite-stage population are separate rows before it becomes the final wrapped anchor $\lambda_{\eta,R} : \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \to E$. The finite anchor residual $\mathfrak{o}^\lambda_R$ is Definition 0.49; its vanishing, not its definition, is required before the repaired anchors become the final wrapped anchors. After the final wrapped anchor rows are supplied, set

$$\mathcal{G}^{\mathrm{wr}}_{\eta,R} = H^{\mathrm{BM}}_*\Big(\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}, \Phi^{\mathrm{wr}}_{\eta,R} \otimes o^{\mathrm{wr}}_{\eta,R}\Big), \qquad \mathcal{G}^{\mathrm{wr}}_R = \bigoplus_\eta \mathcal{G}^{\mathrm{wr}}_{\eta,R}.$$

(iv) *Local, mixed, and wrapped extension correspondences.* For disjoint opens $(U_i) \subset E$, local charges α_i, and wrapped charges η, η_1, η_2, the finite stage contains the correspondence stacks

$$\mathfrak{E}^{\mathrm{loc}}_{\alpha_1,\dots,\alpha_r,R}(U_1,\dots,U_r), \quad \mathfrak{E}^{\mathrm{hyb}}_{\alpha_1,\dots,\alpha_r;\eta,R}, \quad \mathfrak{E}^{\mathrm{wr}}_{\eta_1,\eta_2,R},$$

where the local symbol is the LL correspondence datum of Definition 0.19. The one-sided mixed symbol is the LW/WL correspondence datum of Definition 0.21 after collecting the finite local colours into a local colour map $\alpha : I \to \Gamma^{\mathrm{loc}}_R$. It is still before the reduced E-quotient. The wrapped symbol is the ordered WW correspondence datum of Definition 0.23. These stacks are supplied together with the ordered *two-sided* mixed stacks of Definition 0.47,

$$\mathfrak{E}^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}((U_i),(V_j))$$

classifying ordered insertions of finite local supports on both sides of one wrapped input. The defects of closed-configuration finite-type properness, pull-push admissibility, and Thom–Sebastiani transport are recorded by

$$\mathfrak{o}_R^{\mathrm{prop,LL}},\ \mathfrak{o}_R^{\mathrm{prop,mix}},\ \mathfrak{o}_R^{\mathrm{prop,WW}},$$

and aggregate residuals $\mathfrak{o}_R^{\mathrm{conf}}$, $\mathfrak{o}_R^{\mathrm{conf-prop}}$, $\mathfrak{o}_R^{\mathrm{adm}}$, $\mathfrak{o}_R^{\mathrm{TS}}$. Mixed and wrapped stacks remember source and target anchors before quotienting by the maps a^{LW}, a^{WL}, a^{WW}, and a^{LWL} of Definition 0.48. Replacing this memory by a scalar Fock factor, a Hecke factor, a Borcherds trace degree, or a post-quotient class is not the hybrid datum. These stacks generate a finite E-equivariant correspondence category $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ with operation maps $\mu_R^\bullet = q_! \circ \mathrm{TS}_R \circ p^*$.

(v) *Eight-word two-step binary flag atlas.* Write L for a projection-finite local input and W for a rigidified wrapped input. For every word $w = (\epsilon_1, \epsilon_2, \epsilon_3) \in \{\mathrm{L}, \mathrm{W}\}^3$ and every compatible ordered triple (ξ_1, ξ_2, ξ_3) of charges and local opens, the datum supplies a two-step flag stack $\mathfrak{Flag}^w_{\xi_1,\xi_2,\xi_3,R}$ classifying filtrations with the prescribed associated graded pieces, with comparison maps to the iterated correspondence fibre products. The eight words

$$\mathrm{LLL},\ \mathrm{LLW},\ \mathrm{LWL},\ \mathrm{WLL},\ \mathrm{LWW},\ \mathrm{WLW},\ \mathrm{WWL},\ \mathrm{WWW}$$

are all part of the datum; in mixed and wrapped words the flag stack retains initial, intermediate, and terminal anchors before E-quotient. The two parenthesizations give the same map in the finite quotient,

$$\mu_{\xi_1+\xi_2,\xi_3,R} \circ (\mu_{\xi_1,\xi_2,R} \otimes 1) = \mu_{\xi_1,\xi_2+\xi_3,R} \circ (1 \otimes \mu_{\xi_2,\xi_3,R}),$$

with defect classes $\mathfrak{o}_{w,R}^{\mathrm{assoc}}$ required to vanish for all eight words; the chosen 2-morphisms must satisfy the four-input pentagon on the four-step flag-of-flags stacks. The higher-coloured residual $\mathfrak{o}_R^{\mathrm{col}}$ of Definition 0.40 records the remaining coloured configuration data, units, symmetry conventions, refinement maps, descent, and overlap coherences for all finite HN windows. The eight-word atlas gives only binary associativity; full hybrid factorization requires the higher-coloured residual to vanish in addition.

(vi) *Quotient-after-correspondence functor and reduced E-quotient.* All stacks and correspondences above are E-equivariant before quotienting. The quotient datum supplies a pseudofunctor

$$\mathsf{Corr}_R^{E,\mathrm{hyb}} \longrightarrow \mathsf{Corr}_R^{\mathrm{red,hyb}}$$

as in Definition 0.50. Proposition 0.51 supplies the coherent 2-morphism for composition. The later Proposition 0.52 supplies preservation of the stack-level cartesian base-change squares. The later quotient rows supply compact-support pushforward base change, Thom–Sebastiani transport, and chain-level comparison; applying vanishing-cycle Borel–Moore chains gives the finite correspondence module categories $\mathsf{BM}_R^{E,\mathrm{hyb}}$ and $\mathsf{BM}_R^{\mathrm{red,hyb}}$, and the induced functor $Q_{E,R}$ between them. For each operation $\mu_R^\bullet = q_!\,\mathrm{TS}\,p^*$ the datum carries a comparison isomorphism

$$\theta^Q_{\mu,R} : Q_{E,R}\, q_!\, \mathrm{TS}\, p^* \xrightarrow{\sim} \overline{q}_!\, \overline{\mathrm{TS}}\, \overline{p}^*(Q_{E,R}^{\boxtimes k}),$$

compatible with the flag-stack coherences. The quotient residual lanes are refined by

$$\mathfrak{o}_R^Q = (\mathfrak{o}_R^{Q,\mathrm{form}}, \mathfrak{o}_R^{Q,\mathrm{comp}}, \mathfrak{o}_R^{Q,\mathrm{bc}}, \mathfrak{o}_R^{Q,\mathrm{TS}}, \mathfrak{o}_R^{Q,\mathrm{flag}}, \mathfrak{o}_R^{Q,\mathrm{BM}}, \mathfrak{o}_R^{Q,\theta}, \mathfrak{o}_{R'R}^{Q,\mathrm{tr}}),$$

and these components must vanish before the reduced hybrid product is used. A quotient-first construction (object-stack quotient followed by a new Hall product) is *not* this datum.

(vii) *Protected integration with Gram-trace degree.* The finite stage carries a protected integration datum of Definition 0.61 and hence a chain map

$$I_R^{\mathrm{prot}} : Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \longrightarrow \mathbb{C}[\mathsf{T}_R]_{\le R}.$$

For homogeneous x of charge γ,

$$I_R^{\mathrm{prot}}(Q_{E,R}x) \in \mathbb{C}e_{\omega_R(\gamma)}.$$

The finite Gram-degree row identifies

$$e_{\omega_R(\gamma)} = q^n r^l s^m, \qquad g_R\omega_R(\gamma) = (n, l, m),$$

in the OP/Borcherds variables of Proposition 1.1. The normal-ordered label comparison of Proposition 0.67 identifies $g_R\omega_R$ with $\overline{\Pi}_X\widehat{\omega}_R$ and excludes b_R^{geom} as protected-integration exponent. The integration defects

$$\mathfrak{o}_R^{I,\mathrm{ch}}, \{\mathfrak{o}_{w,R}^{I,\mathrm{prod}}\}_w, \mathfrak{o}_R^{I,\mathrm{co}}, \mathfrak{o}_R^{I,\mathrm{Prim}}, \mathfrak{o}_{R'R}^{I,\mathrm{tr}}, \mathfrak{o}_R^{I,\mathrm{deg}}, \mathfrak{o}_R^{I,\Pi}$$

record chain-map well-typedness, multiplicativity on each word, comultiplicativity, primitive compatibility, HN transition, the Gram-degree law, and the exclusion of b_R^{geom} as the Borcherds exponent.

The aggregate finite obstruction tuple of the hybrid lane is

$$\mathfrak{O}_{\mathrm{hyb},R} = (\mathfrak{o}_R^{\mathrm{stage}}, \mathfrak{o}_R^{\lambda}, \mathfrak{o}_R^{\mathrm{corr,LL}}, \mathfrak{o}_R^{\mathrm{corr,mix}}, \mathfrak{o}_R^{\mathrm{corr,WW}}, \mathfrak{o}_R^{\mathrm{conf}}, \{\mathfrak{o}_{w,R}^{\mathrm{assoc}}\}_w, \mathfrak{o}_R^{\mathrm{col}}, \mathfrak{o}_R^{Q,\bullet}, \mathfrak{o}_R^{I,\bullet}, \mathfrak{o}_{\mathrm{hyb},R'R}^{\mathrm{tr}}).$$

At table level the hybrid datum is supplied by

hybrid_ran_prestack_definition, local_stratum_bo_prestack_definition, wrapped_stratum_bpositive_prestack_definiti

The scalar firewall excludes ordinary Ran locality alone, determinant anchors alone, quotient-first Hall products, Fock traces, Borcherds s-degree alone, and scalar Pfaffian products as hybrid carrier data.

Proposition 0.70 *(Finite consequences of the hybrid datum; conditional on $\mathfrak{O}_{\mathrm{hyb},R} = 0$).* Assume the data of Definition 0.69 are supplied for all R with compatible transition maps, and $\mathfrak{O}_{\mathrm{hyb},R} = 0$. Then:

(i) the $b = 0$ restriction is an ordinary support-local factorization algebra on $\mathrm{Ran}(E)$ in the sense of Beilinson–Drinfeld [7, §3.4], equivalently a Costello–Gwilliam factorization algebra over the smooth real curve E_{top} [23, Vol. II §10];

(ii) every operation with a wrapped input lands in a wrapped stratum;

(iii) the ordered mixed maps, including the two-sided $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$, are the only Hall actions of projection-finite insertions on wrapped sectors in $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ before reduced E-quotient;

(iv) the eight-word atlas plus pentagon coherence yield an associative finite-stage reduced Hall product after vanishing cycles, orientations, HN quotienting, the quotient-after-correspondence pseudofunctor, and the $\theta^{Q}_{\mu,R}$;

(v) for homogeneous x of charge γ, the protected integration map is chargewise typed by

$$I_R^{\mathrm{prot}}(Q_{E,R}x) \in \mathbb{C}e_{\omega_R(\gamma)};$$

the Gram-degree and $\overline{\Pi}_X$-separation rows give $e_{\omega_R(\gamma)} = q^n r^l s^m$ for $\overline{\Pi}_X(\widehat{\omega}_R(\gamma)) = (n, l, m)$, while the local/wrapped type of x is still measured by $b_R^{\mathrm{geom}}(\gamma)$.

Proof. Items (i) and (ii) are part of the elliptic-degree stratification and Proposition 0.2. Item (iii) is the definition of the mixed operations: before reduced E-quotient they are the only displayed operations remembering relative E-position between finite local supports and a wrapped leg. Items (iv) and (v) are the eight-word coherence and the protected integration clause. □

COROLLARY 0.71 *(No quotient-first universal hybrid repair; proved).* A candidate finite correspondence category for the wrapped sector gives the source object $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ only if it contains the unreduced local, wrapped, and ordered mixed source stacks of Definition 0.69, their anchors, the eight-word flag atlas, the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$ after Borel–Moore chains, the comparison maps $\theta^{Q}_{\mu,R}$, and protected integration with $\mathfrak{D}_{\mathrm{hyb},R} = 0$. A quotient of object stacks formed *before* the mixed and wrapped correspondences, or a determinant/Fock category carrying only the trace degree $s^{m_R^{\mathrm{Bch}}}$, does not define the hybrid source.

Proof. By Proposition 0.2, a positive elliptic-degree sector with an effective support-realization row is not a finite-support Ran sector; only the rigidified wrapped prequotient with anchor, together with the unreduced mixed and wrapped correspondences, retains the relative E-position used by the source data. A quotient-first construction loses these anchors. □

COMPARISON WITH THE TWO-STAGE CHIRAL CONSTRUCTION

The hybrid carrier is the finite $K3 \times E$ specialization datum attached to the two-stage construction CY_d-Cat $\to$ E_d-HolFA(X) $\to$ ChirAlg(C) of Vol III Calabi–Yau quantum groups, read here as a chosen specialization datum rather than a constructed functor. The projection-finite local stratum is the retained holomorphic E_3-prefactorization input restricted to the smooth elliptic factor, and the wrapped stratum is the chain-level $K3$-fusion datum that the local stratum cannot see. Beilinson and Drinfeld's Ran construction [7, §3.4–§3.5] provides the $b = 0$ part; the $b > 0$ part is the Hall–Borcherds residual. The finite E_3-source and specialization rows formal_target_charts, field_complexes, e3_operations, bv_qme, anomaly_framing, compact_support, factorization_descent, k3_to_e_specialization, transitions, and scalar_firewall are upstream of this hybrid carrier. They are not supplied by the hybrid carrier itself. The eight-word two-step binary flag atlas $w \in \{\mathrm{L}, \mathrm{W}\}^3$ is the smallest closed coherence calculus that detects the local/wrapped split through binary associativity; it is independent of the higher-coloured residual $\mathfrak{o}_R^{\mathrm{col}}$ which closes higher symmetric arities, units, and overlaps.

The $D0$ specialization

The finite-stage components of $\mathfrak{o}_R^{\mathrm{stage}}$ and $\mathfrak{o}_R^{\lambda}$ are constructed in Appendix 7.2. The wrapped-leg construction and the eight-word atlas at finite height remain conditional on the named hybrid obstruction tuple; the $b = 0$ stratum is the classical projection-finite Ran sector of [7]. The hybrid carrier is the projection-finite Ran stratum on the $b = 0$ component and the wrapped stratum relative to the named finite obstruction tuple $\mathfrak{O}_{\mathrm{hyb},R}$.

Chiral operations on $\mathrm{Ran}^{\mathrm{hyb}}(E)$ require an orientation line on the cosection-reduced d-critical heart. On $K3 \times E$ this orientation descends through the reduced E-quotient only after the retained quotient-orientation datum is supplied; on rank-2 E-strata the Klein-four calculation gives the edge criterion and leaves the degree-one linearization character to be set to zero. Entry (L2), constructed in §0.2, is this orientation datum.

0.2 *Cosection-twisted reduced d-critical orientation*

The full reduced d-critical orientation entry of the compact Hall sector on $K3 \times E$ is the second entry of the pro-object. Its completed form consists of a Joyce d-critical structure, a vanishing-cycle coefficient system, a Joyce–Upmeier–type orientation line, and a quotient-descent datum for the reduced E-quotient. The rows below separate these inputs. Before any Pfaffian wall or Weyl transport can be discussed, the orientation line must descend through the cosection-reduced moduli.

The compact Hall heart is determined by the chosen stability datum (σ, S) of Chapter 2; its K3-surface component lies in the Bridgeland–Bayer–Macri lineage [5, 13]. The first finite orientation input is the PTVV (-1)-shifted symplectic form on the retained derived substacks. The Joyce d-critical truncation, the K3 semiregularity cosection, cosection surjectivity, the reduced self-Ext complex, the determinant line, and its square root are separate rows.

Proposition 0.72 *(PTVV form on retained finite substacks).* [proved for retained derived enhancements; d-critical truncation not constructed] Let R be a finite HN height and let $\mathfrak{M}^{ss}_{R,c}$ be one of the retained finite-type semistable substacks carrying the quasi-smooth derived enhancement

$$\mathfrak{M}^{\mathrm{der}}_{R,c}$$

of Proposition 1.18. Then $\mathfrak{M}^{\mathrm{der}}_{R,c}$ carries the restricted PTVV (-1)-shifted symplectic form

$$\omega^{\mathrm{PTVV}}_{R,c}.$$

Its cotangent amplitude is $[-1, 0]$, and its symplectic restriction defect is zero on every retained row.

This is the unreduced derived symplectic input. It does not construct the Joyce d-critical truncation, the K3 semiregularity cosection, cosection surjectivity, $\mathrm{RHom}_{\mathrm{red}}$, the determinant line, a square-root orientation, quotient orientation, an O2 wall atlas, or protected integration.

Proof. The retained derived-enhancement datum of Definition 1.17 assigns to each retained finite-type semistable substack a derived Artin enhancement with cotangent amplitude $[-1, 0]$. Since

$$K_{K3\times E} \simeq \mathcal{O}_{K3\times E},$$

the ambient perfect-complex moduli problem is Calabi–Yau of dimension three. Pantev–Toën–Vaquié–Vezzosi [95, Theorem 0.3] therefore supplies a (-1)-shifted symplectic form on the derived moduli of perfect complexes. Proposition 1.18 is exactly the retained finite-row restriction of this form to $\mathfrak{M}^{\mathrm{der}}_{R,c}$, with zero Tor-amplitude and symplectic restriction defects. The final paragraph lists the later orientation rows not supplied by this PTVV input. □

The packet

`certificates/orientation/ptvv_finite_substack_symplectic`

verifies `PTVV_FINITE_SUBSTACK_SYMPLECTIC_VERIFIED`: it imports the retained derived-enhancement packet, checks the three retained finite substacks, verifies shifted degree -1, cotangent amplitude $[-1, 0]$, and zero symplectic defect, and records that d-critical truncation, K3 cosection, reduced self-Ext, determinant, orientation square root, quotient orientation, O2, and protected integration rows remain open.

PROPOSITION 0.73 *(Joyce d-critical truncation of retained finite substacks).* [proved for classical truncations; vanishing cycles not constructed] Let $\mathfrak{M}^{\mathrm{der}}_{R,c}$ be a retained finite quasi-smooth derived enhancement carrying the PTVV form $\omega^{\mathrm{PTVV}}_{R,c}$ of Proposition 0.72. Then its classical truncation

$$\mathfrak{M}^{\mathrm{cl}}_{R,c} := t_0\left(\mathfrak{M}^{\mathrm{der}}_{R,c}\right)$$

carries the Joyce d-critical structure

$$s^{\mathrm{Joyce}}_{R,c} \in H^0\left(\mathfrak{M}^{\mathrm{cl}}_{R,c}, \mathcal{S}^0_{\mathfrak{M}^{\mathrm{cl}}_{R,c}}\right)$$

canonically associated to $\omega^{\mathrm{PTVV}}_{R,c}$. The d-critical truncation defect is zero on every retained finite row.

This row constructs only the d-critical structure on the classical truncation. It does not construct the BBDJS vanishing-cycle complex, the K3 semiregularity cosection, cosection surjectivity, $\mathrm{RHom}_{\mathrm{red}}$, determinant lines, square-root orientations, quotient orientation, transition compatibility, or protected integration.

Proof. The input of Proposition 0.72 is a retained quasi-smooth derived enhancement equipped with a (-1)-shifted symplectic form. Brav–Bussi–Dupont–Joyce–Szendrői [12, Theorem 6.9] associate to the classical truncation of a (-1)-shifted symplectic derived scheme or Artin chart a canonical Joyce d-critical structure, functorial on overlaps. Applying this theorem to each retained finite derived row gives $s^{\mathrm{Joyce}}_{R,c}$ on $t_0(\mathfrak{M}^{\mathrm{der}}_{R,c})$. The retained packet records zero defect for these three truncation rows. The final paragraph separates the d-critical truncation from the later coefficient, cosection, reduced-obstruction, orientation, quotient, and integration rows. □

The packet

`certificates/orientation/joyce_dcritical_truncation`

verifies `JOYCE_DCRITICAL_TRUNCATION_VERIFIED`: it imports the PTVV finite-substack packet, applies the BBDJS d-critical truncation theorem to the three retained finite derived substacks, and records zero d-critical defect. Its firewall rows record that vanishing cycles, K3 cosections, reduced self-Ext complexes, determinant lines, square-root orientations, quotient orientation, transition compatibility, and protected integration remain open.

The holomorphic symplectic two-form on the K3 fibre makes the unreduced Donaldson–Thomas theory of $K3 \times E$ vanish identically: it generates trivial first-order deformations that obstruct the virtual class. The standard remedy, as in reduced K3 enumerative geometry [80], is to quotient these out by a semiregularity cosection.

Proposition 0.74 *(K3 semiregularity cosection on retained finite substacks).* [proved as an unreduced cosection; surjectivity not proved] Let $X = S \times E$, where S is a K3 surface with holomorphic two-form σ_S, and put $\sigma = p_S^* \sigma_S$. For each retained finite classical truncation $\mathfrak{M}^{\mathrm{cl}}_{R,c}$ of Proposition 0.73, the obstruction sheaf $\mathrm{Ob}_{R,c}$ carries an $\mathcal{O}_{\mathfrak{M}^{\mathrm{cl}}_{R,c}}$-linear cosection

$$\mathrm{CS}_{R,c} : \mathrm{Ob}_{R,c} \longrightarrow \mathcal{O}_{\mathfrak{M}^{\mathrm{cl}}_{R,c}}.$$

At a point represented by a perfect complex F on X, and for $\xi \in \mathrm{Ext}^2_X(F, F)$, the fibre map is

$$\mathrm{CS}_{R,c}|_F(\xi) = \int_X \sigma \wedge \mathrm{Tr}\big((\xi \otimes 1_{\Omega^1_X}) \circ \mathrm{At}(F)\big),$$

where $\mathrm{At}(F) \in \mathrm{Ext}^1_X(F, F \otimes \Omega^1_X)$ is the Atiyah class. The construction defect is zero on every retained finite row.

This row constructs only the unreduced K3 semiregularity cosection. It does not prove cosection surjectivity on retained strata, filtered Hall additivity, the reduced self-Ext complex, perfectness of the reduced obstruction theory, determinant lines, square-root orientations, quotient orientation, transition compatibility, vanishing cycles, or protected integration.

Proof. The obstruction sheaf of the retained quasi-smooth derived enhancement has fibre $\mathrm{Ext}^2_X(F, F)$ on the chart represented by the universal perfect complex F. The Atiyah class and trace are functorial under restriction to etale charts. Hence the assignment displayed above glues from charts to an $\mathcal{O}$-linear morphism of obstruction sheaves. The wedge product lies in

$$H^3(X, \Omega^3_X) = H^3(X, K_X) \simeq \mathbb{C}$$

because $\sigma \in H^0(X, \Omega^2_X)$. This is the Buchweitz–Flenner–Bloch semiregularity map in the $K3 \times E$ form used by Maulik–Toda [81, Lemma 2.6, Theorem 2.7] and by the Kiem–Li cosection formalism [59]. The packet checks the same formula on the three retained truncation rows. No nonvanishing or surjectivity assertion enters this construction. □

The packet

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certificates/orientation/k3_semiregularity_cosection
```

verifies `K3_SEMIREGULARITY_COSECTION_VERIFIED`: it imports the Joyce d-critical truncation packet, writes the semiregularity formula above on the three retained finite rows, and records zero construction defect. Its firewall rows record that surjectivity, filtered Hall additivity, reduced self-Ext complexes, determinant lines, square-root orientations, quotient orientation, transition compatibility, vanishing cycles, and protected integration remain open.

Proposition 0.75 *(Cosection surjectivity criterion on retained K3-class strata).* [criterion proved; retained nonzero K3-class witnesses not supplied] Let $\mathfrak{M}^{\mathrm{cl}}_{R,c}$ be a retained finite classical truncation equipped with the cosection $\mathrm{CS}_{R,c}$ of Proposition 0.74. Suppose the retained row carries a K3-class witness

$$\beta_{R,c} \in H_2(S, \mathbb{Z}), \qquad \beta_{R,c} \neq 0,$$

recorded by the support or charge map for every geometric point of $\mathfrak{M}^{\mathrm{cl}}_{R,c}$, and suppose the row lies on the reduced DT/PT branch on which the Maulik–Toda semiregularity theorem applies. Then

$$\mathrm{CS}_{R,c} : \mathrm{Ob}_{R,c} \twoheadrightarrow \mathcal{O}_{\mathfrak{M}^{\mathrm{cl}}_{R,c}}$$

is surjective, equivalently its cokernel has rank zero.

The current retained finite packets do not supply the witness $\beta_{R,c} \neq 0$, a branch-identification row, or a cokernel-rank row. Therefore row 274 is not discharged here.

Proof. Surjectivity of a cosection of the obstruction sheaf is local on the classical moduli stack. At a geometric point F, the fibre map is the semiregularity map displayed in Proposition 0.74. On the reduced nonzero K3-class DT/PT branch, Maulik–Toda [81, Lemma 2.6, Theorem 2.7] identify this map with the standard K3 semiregularity cosection and prove surjectivity; the same surjective branch is used in the reduced stable-pair theory of Oberdieck [90, Theorems 1 and 3]. If the retained row records a nonzero K3 class $\beta_{R,c}$ at every geometric point and identifies the row with that branch, these pointwise surjections glue to a surjection of sheaves. The last assertion is a table-level audit: the retained class, charge, and Do-HN semiregularity tables have no row recording $\beta_{R,c} \neq 0$ and no row recording cokernel rank zero. □

The packet

`certificates/orientation/k3_cosection_surjectivity`

verifies `K3_COSECTION_SURJECTIVITY_OBSTRUCTION_VERIFIED`: it records the criterion above, checks that the current retained finite class tables lack a K3-class witness column or row, checks that the Do semiregularity table is empty, and keeps the surjectivity and cokernel tables empty. This status is not a proof of surjectivity; it is the fail-closed ledger for row 274.

After a future retained-stratum packet supplies the nonzero K3-class witness and the cokernel-rank-zero row, the later additivity row checks the cosection on extension strata. On the extension stack $\mathfrak{E}_{A,B}$ of short exact sequences $0 \to A \to C \to B \to 0$ — with projections p_1, p_2 to the substacks $\mathfrak{M}_A, \mathfrak{M}_B$ of sub- and quotient-object and q to $\mathfrak{M}_C$, the correspondence set up in §0.3 — the later additivity row is the filtered Hall identity

$$q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B,$$

the compatibility hypothesis in Kiem–Li cosection localization [54, 56].

Definition 0.76 *(Cosection-reduced self-Ext cone).* Let F be the universal object over a retained finite classical truncation $\mathfrak{M}^{\mathrm{cl}}_{R,c}$. Assume the row carries the cosection $\mathrm{CS}_{R,c}$ of Proposition 0.74. Define the cosection-reduced self-Ext cone by

$$\mathrm{RHom}_{\mathrm{red}}(F,F) := \mathrm{fib}\big(\Theta_F : \mathrm{RHom}_X(F,F) \longrightarrow R\Gamma(X,\mathcal{O}_X) \oplus H^2(S,\mathcal{O}_S)[-1]\big),$$

where

$$\Theta_F = (\mathrm{tr}, \mathrm{CS}_F),$$

and equivalently

$$\mathrm{RHom}_{\mathrm{red}}(F,F) = \mathrm{Cone}(\Theta_F)[-1].$$

The first component removes the scalar trace direction; the second is the K3 semiregularity cosection direction. If the surjectivity witness of Proposition 0.75 is also supplied, this cone is the complex entering the Kiem–Li reduced obstruction theory on the retained branch. This definition does not prove surjectivity, does not prove that the cone is perfect, does not identify its determinant line, does not construct a square root, and does not supply quotient orientation, transition compatibility, vanishing cycles, or protected integration.

The packet

`certificates/orientation/rhomred_definition`

verifies `RHOMRED_DEFINITION_VERIFIED`: it imports the semiregularity-cosection packet and the surjectivity-obstruction packet, records the cone formula above for the three retained finite rows, and records zero definition defect. Its firewall rows record that surjectivity, determinant lines, square-root orientations, quotient orientation, transition compatibility, vanishing cycles, and protected integration are not supplied by the cone-definition row.

Proposition 0.77 *(Perfectness of the cosection-reduced self-Ext cone).* [proved for the cone complex; determinant line not yet defined] Let $F = \mathcal{U}_{R,c}$ be the universal perfect complex on $X \times \mathfrak{M}^{\mathrm{rig}}_{R,c}$ supplied by Proposition 1.24, and let Θ_F be the trace–cosection morphism of Definition 0.76. Then

$$\mathrm{RHom}_{\mathrm{red}}(F, F) = \mathrm{fib}(\Theta_F)$$

is a perfect complex on the retained finite row. The perfectness defect is zero for each of the three retained substacks $Mss_{R0,s3}, Mss_{R0,s2}, Mss_{R0,s1}$.

This proves row 276 at the level of the reduced self-Ext cone. It does not prove cosection surjectivity, does not construct the determinant line $\det \mathrm{RHom}_{\mathrm{red}}(F, F)$, does not construct a square-root orientation, and does not supply quotient orientation, transition compatibility, vanishing cycles, or protected integration.

Proof. Let

$$p : X \times \mathfrak{M}^{\mathrm{rig}}_{R,c} \longrightarrow \mathfrak{M}^{\mathrm{rig}}_{R,c}$$

be the projection. Proposition 1.24 gives $F = \mathcal{U}_{R,c} \in \mathrm{Perf}(X \times \mathfrak{M}^{\mathrm{rig}}_{R,c})$, with finite Tor amplitude and zero perfection defect. Hence

$$R\mathcal{H}om(F, F)$$

is perfect on $X \times \mathfrak{M}^{\mathrm{rig}}_{R,c}$. The morphism p is smooth and proper because $X = K3 \times E$ is smooth projective. Therefore $Rp_* R\mathcal{H}om(F, F)$, written fibrewise as $\mathrm{RHom}_X(F, F)$, is perfect by the Stacks Project [100, Tag 0685, Lemma 37.61.13]. The target

$$R\Gamma(X, \mathcal{O}_X) \oplus H^2(S, \mathcal{O}_S)[-1]$$

is a finite-dimensional constant complex, hence perfect. The stable ∞-category of perfect complexes is closed under fibres and cofibres. Thus

$$\mathrm{fib}\Big(\mathrm{RHom}_X(F, F) \xrightarrow{\Theta_F} R\Gamma(X, \mathcal{O}_X) \oplus H^2(S, \mathcal{O}_S)[-1]\Big)$$

is perfect. The row 276 packet checks this argument on the three retained rows by importing the universal-perfect-complex packet and the cone-definition packet. □

The packet

`certificates/orientation/rhomred_perfectness`

verifies `RHOMRED_PERFECTNESS_VERIFIED`: it imports the retained universal-perfect-complex packet, the retained derived-enhancement packet, and the cone-definition packet; records perfect source and perfect target complexes; checks closure of perfect complexes under fibres for the three retained rows; and records zero perfectness defect. Its firewall rows record that determinant lines, square-root orientations, quotient orientation, transition compatibility, vanishing cycles, and protected integration are not supplied by the perfectness row.

Definition 0.78 *(Determinant line of the reduced self-Ext complex).* Let $F = \mathcal{U}_{R,c}$ be a retained universal perfect complex and put

$$K_F^{\mathrm{red}} := \mathrm{RHom}_{\mathrm{red}}(F, F).$$

By Proposition 0.77, K_F^{red} is perfect. The reduced determinant line is the Knudsen–Mumford determinant

$$\mathcal{L}_F^{\det} := \det K_F^{\mathrm{red}} := \mathrm{Det}_{\mathrm{KM}}\left(K_F^{\mathrm{red}}\right) \in \mathrm{Pic}\left(\mathfrak{M}_{R,c}^{\mathrm{rig}}\right)$$

[29, 61]. On a local perfect representative $K^\bullet$, this line is

$$\det K_F^{\mathrm{red}} = \bigotimes_i \det(K^i)^{(-1)^i},$$

with the Knudsen–Mumford transition signs. It is invariant under quasi-isomorphism and functorial for pullback of the retained row. The determinant-defect rank is zero for

$$Mss_{R0,s3}, \qquad Mss_{R0,s2}, \qquad Mss_{R0,s1}.$$

This definition supplies row 277. It does not choose a square root of $\mathcal{L}_F^{\det}$, does not compute $w_2(\mathcal{L}_F^{\det})$, does not construct quotient orientation or Weyl transport, and does not supply vanishing cycles or protected integration.

The packet

`certificates/orientation/rhomred_determinant`

verifies `RHOMRED_DETERMINANT_VERIFIED`: it imports the perfectness packet, applies the Knudsen–Mumford determinant functor to the three retained perfect reduced self-Ext complexes, records the three determinant lines, and records zero determinant defect. Its firewall rows record that square-root orientations, w_2-classes, quotient orientation, transition compatibility, vanishing cycles, and protected integration are not supplied by the determinant row.

PROPOSITION 0.79 *(Square-root criterion for the reduced determinant line).* [criterion proved; square-root rows not supplied] Let $\mathcal{L}_F^{\det}$ be the determinant line of Definition 0.78. A retained row constructs the reduced orientation square root precisely when it supplies a line

$$o_F \in \mathrm{Pic}\left(\mathfrak{M}_{R,c}^{\mathrm{rig}}\right)$$

and an isomorphism

$$\epsilon_F : o_F^{\otimes 2} \xrightarrow{\sim} \mathcal{L}_F^{\det}.$$

Equivalently, $\mathcal{L}_F^{\det}$ lies in the essential image of the squaring morphism

$$[2] : \mathrm{Pic}\left(\mathfrak{M}_{R,c}^{\mathrm{rig}}\right) \longrightarrow \mathrm{Pic}\left(\mathfrak{M}_{R,c}^{\mathrm{rig}}\right).$$

When it exists, the set of square roots is a torsor under the two-torsion subgroup $\mathrm{Pic}(\mathfrak{M}_{R,c}^{\mathrm{rig}})[2]$.

The current retained orientation packet supplies no line o_F, no isomorphism ϵ_F, and no zero square-defect row for $Mss_{R0,s3}$, $Mss_{R0,s2}$, $Mss_{R0,s1}$. Therefore row 278 is not discharged by the present finite tables. The class

$$w_2(\mathcal{L}_F^{\det})$$

is the mod-2 obstruction entering row 279 and the quotient orientation gerbes; its vanishing is necessary for a topological square root, but the algebraic square root itself is the Picard-groupoid datum (o_F, ϵ_F).

Proof. The assertion is the definition of a square root of a line bundle in the Picard groupoid. Passing to isomorphism classes gives the squaring map on Pic; the fibre of this map over $\mathcal{L}_F^{\det}$ is either empty or a torsor under the kernel of squaring, namely the two-torsion subgroup. The finite-table assertion is a direct inspection of `certificates/orientation/k3e_reduced_orientation`: its `orientation_lines.csv` table has no square-root rows and its obstruction ledger records the missing square-root and orientation-class vanishing rows. □

The packet

`certificates/orientation/rhomred_square_root`

verifies `RHOMRED_SQUARE_ROOT_OBSTRUCTION_VERIFIED`: it imports the determinant packet and the reduced-orientation obstruction ledger, records the Picard-groupoid square-root criterion, checks that the three determinant lines have no supplied square roots in the current orientation table, and keeps w_2, quotient orientation, transition compatibility, vanishing cycles, and protected integration as open obligations.

Proposition 0.80 *(Stiefel–Whitney formula for the reduced determinant line).* [formula proved; retained w_2-values not supplied] Let $\mathcal{L}_F^{\det}$ be the determinant line of Definition 0.78, regarded as a complex line bundle on the retained analytic stack. Its second Stiefel–Whitney class is

$$w_2\Big((\mathcal{L}_F^{\det})_{\mathbb{R}}\Big) = c_1(\mathcal{L}_F^{\det}) \bmod 2 \in H^2(\mathfrak{M}^{\mathrm{rig}}_{R,c}; \mathbb{F}_2).$$

If K_F^{red} is represented by a perfect kernel $\mathcal{K}_F^{\mathrm{red}}$ on $X \times \mathfrak{M}^{\mathrm{rig}}_{R,c}$ and $\pi : X \times \mathfrak{M}^{\mathrm{rig}}_{R,c} \to \mathfrak{M}^{\mathrm{rig}}_{R,c}$, then the determinant-of-cohomology calculation required for row 279 is the degree-one part of Grothendieck–Riemann–Roch:

$$c_1(\mathcal{L}_F^{\det}) = \Big[\pi_*\Big(\mathrm{ch}(\mathcal{K}_F^{\mathrm{red}})\ \mathrm{td}(T_\pi)\Big)\Big]_1.$$

Thus row 279 is discharged only by supplying, for each retained substack, an integral c_1-row or an equivalent GRR expansion for $\mathcal{K}_F^{\mathrm{red}}$, together with its mod-2 reduction.

The present determinant and square-root packets supply none of these rows for $Mss_{R0,s3}$, $Mss_{R0,s2}$, $Mss_{R0,s1}$. They define the three determinant lines and record the absence of square roots, but they do not compute $c_1(\mathcal{L}_F^{\det})$, do not reduce it modulo 2, and do not verify $w_2 = 0$.

Proof. For a complex vector bundle E, the Stiefel–Whitney classes of the underlying real bundle satisfy $w_{2i}(E_{\mathbb{R}}) = c_i(E) \bmod 2$ and $w_{2i+1}(E_{\mathbb{R}}) = 0$ [82, Theorem 14.10]; the case of a complex line gives the displayed formula. The determinant line is the Knudsen–Mumford determinant of the perfect complex [29,61]; its first Chern class is computed from determinant of cohomology by the displayed Grothendieck–Riemann–Roch expression [102]. Inspection of `certificates/orientation/rhomred_determinant` and `certificates/orientation/rhomred_square_root` shows that the retained rows have no c_1-payload, no GRR expansion row, no mod-2 reduction row, and no w_2-value row. □

The packet

`certificates/orientation/rhomred_w2`

verifies `RHOMRED_W2_FORMULA_OBSTRUCTION_VERIFIED`: it imports the determinant and square-root packets, records the $w_2 = c_1 \bmod 2$ and determinant-of-cohomology formulae, checks the three retained determinant lines, and keeps the actual c_1, GRR, mod-2 reduction, w_2-vanishing, square-root, quotient orientation, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.81 *(Vanishing criterion for the reduced orientation gerbe).* [criterion proved; $\alpha^{\mathrm{red}} = 0$ rows not supplied] For a retained charge stratum C, define the reduced orientation gerbe class

$$\alpha_C^{\mathrm{red}} = w_2(\det K_C^{\mathrm{red}}) + \mathrm{CS}_C^* \, w_2(\det \operatorname{coker} \mathrm{CS}_C) \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2).$$

A retained row proves $\alpha_C^{\mathrm{red}} = 0$ precisely when it supplies:

$$u_C = w_2(\det K_C^{\mathrm{red}}), \qquad v_C = w_2(\det \operatorname{coker} \mathrm{CS}_C),$$

the pullback $\mathrm{CS}_C^* \, v_C$, and the equality

$$u_C + \mathrm{CS}_C^* \, v_C = 0$$

in $H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$. If the cosection-cokernel row proves $\operatorname{coker} \mathrm{CS}_C = 0$, then the second summand is zero and row 280 reduces to the w_2-vanishing row of Proposition 0.80. If the cokernel is not proved zero, its determinant line and mod-2 class are separate inputs.

The present finite tables do not prove $\alpha^{\mathrm{red}} = 0$ for $Mss_{R0,s3}$, $Mss_{R0,s2}$, $Mss_{R0,s1}$. They do not supply retained-stratum $w_2(\det K^{\mathrm{red}})$-values, do not supply cosection-cokernel rank-zero rows, do not supply $w_2(\det \operatorname{coker} \mathrm{CS})$-rows, and do not supply a zero-sum row for α^{red}. The null-homotopy of this class is a stronger datum and belongs to row 281.

Proof. The displayed formula is the definition of the reduced degree-two orientation gerbe in the quotient-orientation tower: the first summand comes from the determinant line of the reduced self-Ext complex, and the second is the cosection-cokernel correction pulled back along the cosection map. Since the coefficient group is $\mathbb{F}_2$, vanishing of the class is exactly the equality of the two summands with sum zero. If the cokernel sheaf is zero, its determinant line is the trivial line, so its second Stiefel–Whitney class is zero. Inspection of `certificates/orientation/rhomred_w2`, `certificates/orientation/k3_cosection_surjectivity`, and `certificates/orientation/k3e_reduced_orientation` shows that the determinant w_2-values, cokernel rank-zero rows, cokernel w_2-rows, quotient-Borel α^{red}-rows, and null-homotopy rows are absent. □

The packet

`certificates/orientation/rhomred_alpha_red`

verifies `RHOMRED_ALPHA_RED_OBSTRUCTION_VERIFIED`: it imports the row 279 w_2 packet, the cosection-surjectivity obstruction packet, and the reduced-orientation obstruction ledger; records the formula and zero-sum criterion for α^{red}; checks the three retained strata; and keeps the determinant w_2, cokernel rank, cokernel w_2, zero-sum, null-homotopy, quotient orientation, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.82 *(Null-homotopy criterion for the reduced gerbe).* [criterion proved; null-homotopy rows not supplied] Fix a retained Cech–Borel cover of the reduced stratum $\mathfrak{M}_C^{\mathrm{red}}$. Let

$$a_C^{\mathrm{red}} \in Z^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$$

be a cocycle representative of α_C^{red}. A row constructs a null-homotopy of α_C^{red} precisely when it supplies a 1-cochain

$$h_C^{\mathrm{red}} \in C^1(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$$

and verifies the coboundary equation

$$\delta h_C^{\mathrm{red}} = a_C^{\mathrm{red}}.$$

Thus row 281 is stronger than row 280: the equality $\alpha_C^{\mathrm{red}} = 0$ proves only that such a cochain exists after choosing a representative; row 281 records the representative, the cochain, and the zero coboundary-defect row. When nonempty, the set of null-homotopies is a torsor under $H^1(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$.

The present finite tables do not construct a null-homotopy for $Mss_{R0,s3}$, $Mss_{R0,s2}$, $Mss_{R0,s1}$. They do not supply a cocycle representative of α^{red}, do not prove the class zero, do not supply a 1-cochain, and do not verify the coboundary equation. No quotient-orientation descent row follows from an empty quotient-Borel table.

Proof. This is the definition of a null-homotopy in the cochain model for degree-two $\mathbb{F}_2$-valued Cech–Borel cohomology. A degree-two cocycle is null-homotopic precisely when it is a coboundary; the possible primitives differ by 1-cocycles, and changing a primitive by a 1-coboundary gives the same trivialization. Hence the set of choices is a torsor under H^1. Inspection of `certificates/orientation/rhomred_alpha_red` and `certificates/orientation/k3e_reduced_orientation` shows that no cocycle representative, no zero-class row, no 1-cochain, and no coboundary-defect-zero row is supplied. □

The packet

`certificates/orientation/rhomred_alpha_red_nullhomotopy`

verifies `RHOMRED_ALPHA_RED_NULLHOMOTOPY_OBSTRUCTION_VERIFIED`: it imports the row 280 packet and the reduced-orientation obstruction ledger, records the cochain criterion $\delta h^{\mathrm{red}} = a^{\mathrm{red}}$, checks the three retained strata, and keeps the cocycle representative, zero-class, 1-cochain, coboundary-defect, quotient orientation, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.83 *(Computation criterion for the connected free E-Borel class).* [criterion proved; a_1, a_2 rows not supplied] Let $E^{\mathrm{top}} \simeq T^2$, and use the basis

$$H^2(BE; \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2$$

of Lemma 4.11. For a retained stratum C with free E-translation action, the connected-free quotient-orientation class is the $E_\infty^{2,0}$-edge component of the equivariant degree-two Borel class of the reduced determinant complex:

$$\alpha_C^{E,\mathrm{free}} = \mathrm{edge}_{2,0}\big(w_2^E(\mathscr{D}_C)\big) = a_1(C)u_1 + a_2(C)u_2.$$

A row computes $\alpha_C^{E,\mathrm{free}}$ precisely when it supplies the equivariant Borel representative $w_2^E(\mathscr{D}_C)$, the spectral-sequence edge calculation including any incoming $d_2 : E_2^{0,1} \to E_2^{2,0}$ contribution, and the two coefficients

$$a_1(C), a_2(C) \in \mathbb{F}_2.$$

The vanishing equation $a_1 = a_2 = 0$ is row 283, and a chosen null-trivialization of that zero class is row 284.

The present finite tables do not compute $\alpha^{E,\mathrm{free}}$ for $Mss_{R0,s3}$, $Mss_{R0,s2}$, $Mss_{R0,s1}$. The retained E-translation packet supplies free actions and quotient stacks, but the reduced-orientation packet supplies no `free_E` quotient-Borel rows, no Borel spectral-sequence edge rows, and no a_1, a_2 coefficient rows. The equality $H^1(BE; \mathbb{F}_2) = 0$ rules out connected degree-one characters; it does not compute the degree-two class above.

Proof. The cohomology of the connected torus action is $H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2]$, $|u_i| = 2$, by Lemma 4.11. Hence every connected-free degree-two edge class has a unique expansion $a_1u_1 + a_2u_2$. The Borel fibration spectral sequence computes the equivariant class from $H_E^*(C; \mathbb{F}_2)$, and the edge component in bidegree $(2, 0)$ is well-defined only after the relevant incoming differential has been killed or included in the reported coefficient row. Inspection of `certificates/moduli/retained_e_translation_rigidifications` shows that the free actions are supplied, while inspection of `certificates/orientation/k3e_reduced_orientation` shows that the quotient-Borel table has no `free_E` row and no coefficient or edge-reduction payload. □

The packet

`certificates/orientation/rhomred_free_e_borel`

verifies `RHOMRED_FREE_E_BOREL_OBSTRUCTION_VERIFIED`: it imports the retained E-translation rigidification packet and the reduced-orientation obstruction ledger, records the basis and edge-class criterion for $\alpha^{E,\mathrm{free}} = a_1u_1 + a_2u_2$, checks the three retained free E-actions, and keeps the equivariant Borel representative, edge-reduction, a_1, a_2 coefficient, vanishing, null-trivialization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.84 *(Vanishing criterion for the connected free E-Borel class).* [criterion proved; $\alpha^{E,\mathrm{free}} = 0$ rows not supplied] Let C be one of the retained free E-translation strata, and suppose row 282 has supplied the connected-free edge class

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2) = \mathbb{F}_2u_1 \oplus \mathbb{F}_2u_2.$$

Then row 283 proves $\alpha_C^{E,\mathrm{free}} = 0$ precisely by supplying the same coefficient row together with the two equalities

$$a_1(C) = 0, \qquad a_2(C) = 0$$

in $\mathbb{F}_2$. The vanishing is not implied by $H^1(BE; \mathbb{F}_2) = 0$, by freeness of the E-action, by the existence of the quotient stack, by a finite-stabilizer calculation, or by any scalar trace or character value. A null-trivialization of this zero class is the separate row 284.

The present finite tables do not prove $\alpha^{E,\mathrm{free}} = 0$. The row 282 packet has no a_1, a_2 values; the reduced-orientation quotient-Borel table has no `free_E` row; and no zero-coefficient row is supplied for $Mss_{R0,s3}$, $Mss_{R0,s2}$, $Mss_{R0,s1}$.

Proof. The classes u_1, u_2 form a basis of $H^2(BE;\mathbb{F}_2)$ by Lemma 4.11. Therefore the expansion of $\alpha_C^{E,\mathrm{free}}$ in Proposition 0.83 is unique, and the class is zero if and only if both coordinates are zero. This is a basis calculation after the Borel edge class has been computed; it is not a replacement for that computation. The packet `certificates/orientation/rhomred_free_e_borel` records the absence of the coefficient row, and `certificates/orientation/k3e_reduced_orientation` records the empty quotient-Borel table, so the current finite evidence supplies the criterion but not its hypotheses. □

The packet

`certificates/orientation/rhomred_free_e_vanishing`

verifies `RHOMRED_FREE_E_VANISHING_OBSTRUCTION_VERIFIED`: it imports the row 282 packet and the reduced-orientation obstruction ledger, records the basis-zero criterion $\alpha^{E,\mathrm{free}} = 0 \iff a_1 = a_2 = 0$, checks the three retained free E-actions, and keeps the a_1, a_2 coefficient, zero-coefficient, null-trivialization, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.85 *(Null-trivialization criterion for the connected free E-Borel class).* [criterion proved; null-trivialization rows not supplied] Fix a cochain model for the connected BE-edge component used in Proposition 0.83. Let

$$a_C^{E,\mathrm{free}} \in Z^2(BE;\mathbb{F}_2)$$

be a cocycle representative of the connected-free edge class $\alpha_C^{E,\mathrm{free}}$. Row 284 constructs a null-trivialization of $\alpha_C^{E,\mathrm{free}}$ precisely when it supplies a 1-cochain

$$h_C^{E,\mathrm{free}} \in C^1(BE;\mathbb{F}_2)$$

and verifies the coboundary equation

$$\delta h_C^{E,\mathrm{free}} = a_C^{E,\mathrm{free}}.$$

Thus row 284 is stronger than row 283: the vanishing $\alpha_C^{E,\mathrm{free}} = 0$ supplies the cohomology condition, while row 284 records a representative, a primitive, and a defect-zero coboundary calculation. Since $H^1(BE;\mathbb{F}_2) = 0$, the connected BE-edge homotopy class of such a trivialization has no H^1-torsor ambiguity after the cochain model is fixed; this uniqueness does not replace the cochain-level construction.

The present finite tables do not construct this null-trivialization. They do not supply a cocycle representative of $\alpha^{E,\mathrm{free}}$, do not prove the class zero, do not supply a 1-cochain, and do not verify a defect-zero coboundary equation for $Mss_{R0,s3}, Mss_{R0,s2}, Mss_{R0,s1}$.

Proof. A null-trivialization of a degree-two $\mathbb{F}_2$-valued Borel class is, by definition, a primitive for a chosen degree-two cocycle representative. Hence the datum is exactly a cochain h with $\delta h = a$, plus the verification that the coboundary defect is zero. The equality $H^1(BE;\mathbb{F}_2) = 0$ from Lemma 4.11 says that two connected-edge trivializations of the same cocycle are homotopic after fixing the cochain model; it does not give the cocycle representative, the primitive, or the coboundary calculation. Inspection of `certificates/orientation/rhomred_free_e_vanishing` and `certificates/orientation/k3e_reduced_orientation` shows that none of these rows is supplied. □

The packet

`certificates/orientation/rhomred_free_e_nulltrivialization` verifies `RHOMRED_FREE_E_NULLTRIVIALIZATION_OBSTRUCTION_VERIFIED`: it imports the row 283 packet and the reduced-orientation obstruction ledger, records the cochain criterion $\delta h^{E,\mathrm{free}} = a^{E,\mathrm{free}}$, checks the three retained free E-actions, and keeps the cocycle representative, zero-class, 1-cochain, coboundary-defect, quotient orientation, transition, vanishing-cycle, and protected-integration rows open.

The reduced determinant complex

The following notation belongs to the later orientation rows. After the cosection, cone-definition, perfectness, and determinant rows have been supplied, for an object C in the chosen finite stage write

$$K_C^{\mathrm{red}} = \mathrm{RHom}_{\mathrm{red}}(C, C),$$

the cosection-reduced self-Ext complex of Definition 0.76; it is perfect by Proposition 0.77. Its determinant line is

$$\mathcal{L}_C^{\det} = \det K_C^{\mathrm{red}}$$

by Definition 0.78. The later orientation row, when supplied, equips the reduced orientation line with a chosen square root,

$$o_C \in \mathrm{Pic}(\mathfrak{M}_X^{\mathrm{or}}), \qquad o_C^{\otimes 2} \simeq \det K_C^{\mathrm{red}},$$

in the sense of [56], and is multiplicative under the reduced extension correspondences: $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$ compatibly with the Hall additivity above. This is the data on which Thom–Sebastiani transport is supplied for the vanishing-cycle coefficient sheaves.

(O1) Strong reduced orientation through the quotient-orientation tower

The compact Hall product on $X = K3 \times E$ exists only after the reduced E-quotient. The orientation line must descend through this quotient, and descent is detected by a tower of mod-2 Borel-equivariant gerbes. For a charge stratum C:

(i) the reduced gerbe class

$$\alpha_C^{\mathrm{red}} = w_2(\det K_C^{\mathrm{red}}) + \mathrm{CS}_C^* \, w_2(\det \mathrm{coker}\, \mathrm{CS}_C) \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2);$$

(ii) the free E-class

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2);$$

(iii) for each finite-stabilizer stratum S with stabilizer $H \subset E_{\mathrm{tors}}$, the equivariant class $\widetilde{\beta}_{C,S}^H \in H_H^2(S; \mathbb{F}_2)$, whose point-stabilizer restriction $\beta_{C,S}^H \in H^2(BH; \mathbb{F}_2)$ is read only after the mixed Borel terms, the stabilizer action on $H^\bullet(S; \mathbb{F}_2)$, and the spectral-sequence differentials have vanished.

Definition 0.86 *((O1) Strong multiplicative reduced orientation).* The (O1) datum at finite HN height R consists of: the reduced orientation line o_C with chosen square root; its multiplicativity under reduced extension correspondences with chosen Thom–Sebastiani transport; null-trivializations of α_C^{red}, $\alpha_C^{E,\mathrm{free}}$, $\widetilde{\beta}_{C,S}^H$ and $\beta_{C,S}^H$ on every charge stratum used by the Hall correspondences; and a choice of zero $H^1(BH; \mathbb{F}_2)$-linearization character $\lambda_{C,S}^H = 0$ for each finite stabilizer.

The full-level specialization at $H = E[2]$ is the Klein-four condition

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \quad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2),$$

$$\mathrm{res}_{\langle e_i \rangle} \beta_{C,S}^{E,2} = r_i \tau_i^2,$$

with vanishing $r_1 = r_2 = r_3 = 0$ on every retained two-torsion stratum. For full-level $H = E[N]$ with $2^a \parallel N$, $a \geq 2$, the two-primary residual takes value in $H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle$ and contains the mixed coefficient $A_{12}^{(N)}$ *not* detected by cyclic restrictions.

(O1) is therefore not a single Picard-line assertion: it is the ordinary square root, the null-trivialization of the reduced and equivariant degree-two gerbes, and the choice of zero degree-one linearization characters for the actual finite stabilizers. At finite HN height this is the cocycle system of [56] restricted to $K3 \times E$.

After finite trivialisations, the $(O1)$ datum is recorded by the orientation proof tables

orientation_lines, ts_multiplicativity, quotient_borel, finite_stabilizers.

The first table records the determinant square root and the vanishing orientation class. The second records Thom–Sebastiani multiplicativity and flag-pentagon defects. The third records the four quotient Cech–Borel classes α^{red}, $\alpha^{E,\mathrm{free}}$, $\widetilde{\beta}^H$, and λ^H, with chosen null-trivialisations and edge-reduction status. The fourth records the finite-stabilizer edge calculation, including the Klein-four coefficients, the two-primary coefficients $(A_1^{(H)}, A_{12}^{(H)}, A_2^{(H)})$, and the two degree-one linearization coefficients. All ranks and coefficients in these rows must vanish. A row using only cyclic restrictions is not an orientation row.

Proposition 0.87 *(Computation criterion for the finite-stabilizer Borel class).* [criterion proved; $\beta^H_{C,S}$ rows not supplied] Let S be a retained stratum with residual finite stabilizer $H \subset E_{\mathrm{tors}}$. Row 285 computes the finite-stabilizer class

$$\beta^H_{C,S} \in H^2(BH; \mathbb{F}_2)$$

precisely when it supplies:

(i) a finite-stabilizer stratum row identifying H and its two-primary type;

(ii) a Cech–Borel representative

$$\widetilde{\beta}^H_{C,S} = w_2^H(\mathscr{D}_C|_S) \in H^2_H(S; \mathbb{F}_2);$$

(iii) an edge-reduction row killing or accounting for the ordinary term in $H^2(S; \mathbb{F}_2)^H$, the mixed term in $H^1(BH; H^1(S; \mathbb{F}_2))$, and the incoming Borel spectral-sequence differentials;

(iv) the classifying-space coordinates of the resulting edge class.

After these rows are supplied, the computation is the following group cohomology calculation. If H has odd order, and also for $H = 1$, positive-degree $H^*(BH; \mathbb{F}_2)$ vanishes. If $H \simeq E[2]$, then

$$\beta^H_{C,S} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2.$$

If the two-primary part is $(\mathbb{Z}/2^a)^2$, $a \geq 2$, then

$$\beta^H_{C,S} = A_1^{(H)} y_1 + A_{12}^{(H)} x_1x_2 + A_2^{(H)} y_2.$$

The coefficient $A_{12}^{(H)}$ is a rank-two stabilizer coefficient; it is not read from cyclic order-two restrictions. Row 286, not row 285, is the assertion that the computed class vanishes for every retained H. Rows 288–289 concern the independent degree-one linearization character.

The present finite tables do not compute $\beta^H_{C,S}$. The retained rigidification-inertia packet records finite residual groups $H_{R0,s3}$, $H_{R0,s2}$, $H_{R0,s1}$ of order 1, but it does not supply finite closed inertia stratifications, Cech–Borel representatives, edge-reduction rows, or finite-stabilizer orientation rows. The reduced-orientation packet has an empty `finite_stabilizers` table and an empty `quotient_borel` table.

Proof. The Borel spectral sequence for $S \to EH \times_H S \to BH$ filters $H^2_H(S; \mathbb{F}_2)$ by the classifying-space edge $H^2(BH; \mathbb{F}_2)$, the mixed term $H^1(BH; H^1(S; \mathbb{F}_2))$, and the ordinary term $H^2(S; \mathbb{F}_2)^H$, with the relevant incoming differentials. Only after the non-edge terms are killed or explicitly accounted for does the restriction of $\widetilde{\beta}^H$ define $\beta^H \in H^2(BH; \mathbb{F}_2)$. Lemma 4.12 and Lemmas 7.10–7.11 give the displayed coordinates: transfer kills odd stabilizers, $(\mathbb{Z}/2)^2$ gives the quadratic basis x_1^2, x_1x_2, x_2^2, and $(\mathbb{Z}/2^a)^2$, $a \geq 2$, gives the basis y_1, x_1x_2, y_2. Inspection of `certificates/moduli/retained_rigidification_inertia` and `certificates/orientation/k3e_reduced_orientation` shows that only residual finite group input is present; the edge and coefficient rows are absent. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_beta`

verifies `RHOMRED_FINITE_STABILIZER_BETA_OBSTRUCTION_VERIFIED`: it imports the residual-inertia packet and the reduced-orientation obstruction ledger, records the odd, Klein-four, and two-primary coordinate criteria for row 285, checks the three retained residual groups, and keeps the finite closed inertia stratum, Cech–Borel representative, edge-reduction, β^H-coefficient, vanishing, null-trivialization, linearization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.88 *(Vanishing criterion for the finite-stabilizer Borel class).* [criterion proved; $\beta^H_{C,S} = 0$ rows not supplied] Assume row 285 has computed

$$\beta^H_{C,S} \in H^2(BH; \mathbb{F}_2)$$

for every retained finite-stabilizer stratum. Row 286 proves

$$\beta^H_{C,S} = 0$$

for every retained H precisely when it verifies the corresponding zero condition in the classifying-space basis for every retained row:

$$H = 1 \text{ or } |H| \text{ odd}: \quad H^2(BH; \mathbb{F}_2) = 0 \quad \text{after edge reduction},$$

$$H \simeq E[2]: \quad b_{20} = b_{11} = b_{02} = 0,$$

and, for two-primary rank two stabilizers,

$$H_{(2)} \simeq (\mathbb{Z}/2^a)^2,\ a \geq 2: \quad A_1^{(H)} = A_{12}^{(H)} = A_2^{(H)} = 0.$$

The condition is imposed on every retained finite-stabilizer row. A cyclic-only calculation does not prove the two-primary case, because it does not see $A_{12}^{(H)}$. Row 287 is the subsequent choice of compatible null-trivializations, and rows 288–289 concern the independent $H^1(BH; \mathbb{F}_2)$-linearization character.

The present finite tables do not prove the displayed vanishing. The row 285 packet records residual finite groups of order 1, but it records no finite-stabilizer orientation row, no edge-reduction row, no β^H-value, and no zero row. The group-cohomology fact $H^2(B1; \mathbb{F}_2) = 0$ is a zero template for a supplied edge row; it is not itself the missing finite-stabilizer orientation datum.

Proof. After the edge-reduction hypotheses of Proposition 0.87, the class is an element of $H^2(BH;\mathbb{F}_2)$. Transfer kills this group for odd H, and it is zero for $H = 1$. Lemma 7.10 identifies the $E[2]$ basis x_1^2, x_1x_2, x_2^2, so vanishing is equivalent to the three quadratic coefficients being zero. Lemma 7.11 identifies the two-primary rank-two basis y_1, x_1x_2, y_2, so vanishing is equivalent to the three displayed coefficients being zero. The coefficient $A_{12}^{(H)}$ is independent of cyclic order-two restrictions. Inspection of `certificates/orientation/rhomred_finite_stabilizer_beta` and `certificates/orientation/k3e_reduced_orientation` shows that the current finite evidence supplies the criterion but not its hypotheses. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_beta_vanishing`

verifies `RHOMRED_FINITE_STABILIZER_BETA_VANISHING_OBSTRUCTION_VERIFIED`: it imports the row 285 packet and the reduced-orientation obstruction ledger, records the odd/trivial, Klein-four, and two-primary zero criteria for row 286, checks the three retained residual groups, and keeps the β^H-value, zero-coordinate, null-trivialization, linearization, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.89 *(Compatible null-trivialization criterion for finite-stabilizer Borel classes).* [criterion proved; compatible null-trivialization rows not supplied] Assume rows 285–286 have supplied, for every retained finite-stabilizer stratum S with residual stabilizer H, a Cech–Borel cocycle

$$a^H_{C,S} \in Z^2(BH;\mathbb{F}_2)$$

representing $\beta^H_{C,S}$, and have verified $\beta^H_{C,S} = 0$. Row 287 is the following stronger cochain-level datum. It must supply a 1-cochain

$$h^H_{C,S} \in C^1(BH;\mathbb{F}_2)$$

with

$$\delta h^H_{C,S} = a^H_{C,S}.$$

For every retained subgroup inclusion $\iota\colon K \hookrightarrow H$ it must also verify

$$\mathrm{res}_\iota\, a^H_{C,S} = a^K_{C,S}$$

and supply a comparison 0-cochain

$$q^\iota_{C,S} \in C^0(BK;\mathbb{F}_2)$$

such that

$$\mathrm{res}_\iota\, h^H_{C,S} - h^K_{C,S} = \delta q^\iota_{C,S}.$$

For every chain $L \xrightarrow{\jmath} K \xrightarrow{\iota} H$, the chosen comparisons must have zero defect

$$q^{\iota\circ\jmath}_{C,S} = q^\jmath_{C,S} + \mathrm{res}_\jmath\, q^\iota_{C,S}$$

in the chosen Cech–Borel cochain model. This is the subgroup restriction compatibility demanded in row 287. It is not the linearization character of rows 288–289, and it is not implied by a scalar trace, by the residual group order, or by the abstract vanishing of $H^2(BH;\mathbb{F}_2)$ unless the representative $a^H_{C,S}$, the cochain $h^H_{C,S}$, and all restriction comparisons are supplied.

The present finite tables do not construct row 287. They record three residual groups of order 1, but they record no finite-stabilizer orientation row, no Cech–Borel representative $a^H_{C,S}$, no verified zero row from row 286, no 1-cochain $h^H_{C,S}$, no retained subgroup-inclusion table, no comparison 0-cochain, and no chain-coherence row. Thus row 287 remains an open cochain-level obstruction, even in the trivial residual-group template.

Proof. A null-trivialization of the representative cocycle $a^H_{C,S}$ is, by definition, a degree-one cochain whose coboundary is that representative. Therefore the cohomological statement $\beta^H_{C,S} = 0$ is only the existence statement; row 287 asks for a chosen primitive. Restriction along $\iota\colon K \hookrightarrow H$ is a cochain map, so $\delta(\mathrm{res}_\iota h^H - h^K) = 0$ once $\mathrm{res}_\iota a^H = a^K$. Compatibility requires this closed 1-cochain to be killed by a chosen 0-cochain q^ι, and functoriality over a two-step chain is exactly the displayed defect-zero equation. These are cochain equalities, not statements about the final scalar shadow. Inspection of `certificates/orientation/rhomred_finite_stabilizer_beta_vanishing`, `certificates/orientation/k3e_reduced_orientation`, and `certificates/orientation/transition_orientation_pre` shows that the required cochains and subgroup-restriction rows are not present. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_beta_nulltrivialization`

verifies `RHOMRED_FINITE_STABILIZER_BETA_NULLTRIVIALIZATION_OBSTRUCTION_VERIFIED`: it imports the row 286 packet and the reduced-orientation obstruction ledger, records the cochain and subgroup-restriction criteria for row 287, checks the three retained residual groups, and keeps the β^H-cocycle, zero row, 1-cochain, coboundary-defect, subgroup-inclusion, comparison 0-cochain, chain-coherence, linearization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.90 *(Computation criterion for the finite-stabilizer linearization character).* [criterion proved; linearization-character rows not supplied] Fix a retained finite-stabilizer stratum S with residual stabilizer H. After the row 287 null-trivialization has killed the degree-two Borel gerbe, the remaining finite-stabilizer descent datum is a character

$$\lambda^H_{C,S} \in H^1(BH;\mathbb{F}_2) = \mathrm{Hom}(H,\mathbb{F}_2).$$

Row 288 computes this class precisely when it supplies the orientation-line action of H on the already trivialized Pfaffian orientation line, an ordered generator basis for the two-primary part of H, and the sign values of this action on the chosen generators. For $H = 1$, and more generally for odd H, the target group $H^1(BH;\mathbb{F}_2)$ is zero after the retained stabilizer row and orientation-line action have been supplied. For $H \simeq E[2] \simeq (\mathbb{Z}/2)^2$, with basis x_1, x_2 dual to the chosen generators, the computation is

$$\lambda^H_{C,S} = \lambda_1 x_1 + \lambda_2 x_2, \qquad \rho_1 = \lambda_1, \quad \rho_2 = \lambda_2, \quad \rho_3 = \lambda_1 + \lambda_2$$

on the three non-zero cyclic subgroups. For $H_{(2)} \simeq (\mathbb{Z}/2^a)^2$, $a \geq 2$, it is

$$\lambda^H_{C,S} = \lambda_1^{(H)} x_1 + \lambda_2^{(H)} x_2 \in H^1(BH;\mathbb{F}_2),$$

where x_i are the mod-2 degree-one classes dual to the chosen two-primary generators. Row 288 is only this computation of $\lambda^H_{C,S}$; row 289 is the separate assertion that the computed character is zero.

The present finite tables do not compute row 288. The determinant anchor translation-weight packet records the class-level restriction $h \mapsto \chi_\eta h$, but it marks coherent stabilizer linearization as not certified. The reduced-orientation packet has no finite-stabilizer rows, no quotient Borel rows, and no orientation-line action rows. The row 287 packet has no null-trivializations and records `linearization_computed = false` on every retained residual group. Thus the equality $H^1(B1; \mathbb{F}_2) = 0$, cyclic quadratic restrictions, and translation-weight formulas are templates or separate data; none is a computed value of $\lambda^H_{C,S}$ for row 288.

Proof. Once the degree-two gerbe has been trivialized, two finite-stabilizer linearizations of the same orientation line differ by a sign character of H. This torsor is $H^1(BH; \mathbb{F}_2)$, so a computation of the residual class is exactly a computation of the sign character on generators. Transfer gives $H^1(BH; \mathbb{F}_2) = 0$ for odd H, and the trivial group has no positive cohomology. Lemma 7.10 identifies the $E[2]$ basis and the three cyclic restrictions ρ_1, ρ_2, ρ_3. Lemma 7.11 identifies the degree-one basis x_1, x_2 for the two-primary rank two case. These are coordinate computations for a supplied action on the orientation line. Inspection of `certificates/orientation/rhomred_finite_stabilizer_beta_nulltrivialization`, `certificates/orientation/k3e_reduced_orientation`, and `certificates/hybrid/determinant_anchor_translation_` shows that the action rows, generator-basis rows, and character values are not present. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_linearization_character`

verifies `RHOMRED_FINITE_STABILIZER_LINEARIZATION_CHARACTER_OBSTRUCTION_VERIFIED`: it imports the row 287 packet, the reduced-orientation obstruction ledger, and the determinant-anchor translation-weight packet, records the $H^1(BH; \mathbb{F}_2)$ coordinate criterion for row 288, checks the three retained residual groups, and keeps the orientation-line action, generator-basis, character-value, H^1-coordinate, row 289 vanishing, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.91 *(Vanishing criterion for the finite-stabilizer linearization character).* [criterion proved; zero-linearization rows not supplied] Assume row 288 has computed

$$\lambda^H_{C,S} \in H^1(BH; \mathbb{F}_2)$$

for every retained finite-stabilizer stratum. Row 289 proves

$$\lambda^H_{C,S} = 0$$

for every retained H precisely when it verifies the corresponding zero coordinate row in the $H^1(BH; \mathbb{F}_2)$-basis:

$$H = 1 \text{ or } |H| \text{ odd}: \quad H^1(BH; \mathbb{F}_2) = 0 \quad \text{after the retained action row is supplied,}$$

$$H \simeq E[2]: \quad \lambda_1 = \lambda_2 = 0 \quad \text{equivalently} \quad \rho_1 = \rho_2 = \rho_3 = 0$$

with $\rho_3 = \lambda_1 + \lambda_2$, and, for two-primary rank two stabilizers,

$$H_{(2)} \simeq (\mathbb{Z}/2^a)^2, \ a \geq 2: \quad \lambda_1^{(H)} = \lambda_2^{(H)} = 0.$$

This is a degree-one vanishing statement. It is not implied by $\beta^H_{C,S} = 0$, by a row 287 null-trivialization of the degree-two gerbe, by the determinant-anchor translation-weight formula, or by class invariance in Pic. Row 290 records this independence as a separate firewall.

The present finite tables do not prove row 289. They record no orientation-line action, no generator-basis row, no character values, and no computed H^1-coordinate for $\lambda^H_{C,S}$. The row 288 packet records three order-one residual group templates, but the corresponding λ-values are `missing`, with `lambda_character_computed = false` and `lambda_vanishing_verified = false`. Hence the zero $H^1(B1;\mathbb{F}_2)$ template cannot be promoted to a zero-linearization row for the retained orientation problem.

Proof. Once row 288 has produced an element of $H^1(BH;\mathbb{F}_2)$, vanishing is a coordinate calculation. The groups $H = 1$ and odd H have zero mod-2 degree-one cohomology. Lemma 7.10 identifies the $E[2]$ character as $\lambda_1 x_1 + \lambda_2 x_2$ and the cyclic restrictions as $\rho_1 = \lambda_1$, $\rho_2 = \lambda_2$, $\rho_3 = \lambda_1 + \lambda_2$; all three cyclic restrictions vanish if and only if $\lambda_1 = \lambda_2 = 0$. Lemma 7.11 identifies the two-primary degree-one basis x_1, x_2, so vanishing is equivalent to the two displayed coefficients being zero. These statements require the computed character from row 288. Inspection of `certificates/orientation/rhomred_finite_stabilizer_linearization_character` and `certificates/orientation/k3e_reduced_orientation` shows that no such character value or zero-coordinate row is present. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_linearization_vanishing`

verifies `RHOMRED_FINITE_STABILIZER_LINEARIZATION_VANISHING_OBSTRUCTION_VERIFIED`: it imports the row 288 packet and the reduced-orientation obstruction ledger, records the $H^1(BH;\mathbb{F}_2)$ zero criteria for row 289, checks the three retained residual groups, and keeps the computed-character, zero-coordinate, all-retained-coverage, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.92 *(Gerbe vanishing does not imply zero finite-stabilizer linearization).* [firewall proved] The implication

$$\beta^H_{C,S} = 0 \implies \lambda^H_{C,S} = 0$$

is false as a finite-stabilizer descent principle. For $H \simeq E[2]$, the equations

$$b_{20} = b_{11} = b_{02} = 0$$

kill the degree-two class $\beta^H_{C,S} \in H^2(BH;\mathbb{F}_2)$. The degree-one class

$$\lambda^H_{C,S} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BH;\mathbb{F}_2)$$

is independent; for example $(\lambda_1, \lambda_2) = (1, 0)$ gives

$$\rho_1 = 1, \qquad \rho_2 = 0, \qquad \rho_3 = 1,$$

so the gerbe vanishes and the linearization does not. For $H_{(2)} \simeq (\mathbb{Z}/2^a)^2$, $a \geq 2$, the same separation is

$$A_1^{(H)} = A_{12}^{(H)} = A_2^{(H)} = 0, \qquad (\lambda_1^{(H)}, \lambda_2^{(H)}) = (1, 0).$$

Thus rows 286–287 cannot replace rows 288–289. If $H = 1$ or $|H|$ is odd, then $H^1(BH;\mathbb{F}_2) = 0$; this is a separate group-cohomological zero target, not a consequence of $\beta^H_{C,S} = 0$.

The present finite tables obey this firewall. The row 289 packet records `beta_vanishing_implies_lambda_vanishing = false` and `nulltrivialization_implies_lambda_vanishing = false`; it also records missing λ-values and no zero-linearization rows. Hence no scalar trace, determinant-anchor translation weight, gerbe vanishing row, or null-trivialization row is used as a substitute for the degree-one linearization calculation.

Proof. The cohomological degrees already separate the two obstructions. Lemma 7.10 gives

$$H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \qquad |x_i| = 1.$$

The finite gerbe lives in degree 2, with coordinates (b_{20}, b_{11}, b_{02}), while the residual linearization lives in degree 1, with coordinates (λ_1, λ_2). These are coordinates on different graded pieces, so the zero locus of the degree-two coordinates contains every value of the degree-one coordinates. The displayed choice $(\lambda_1, \lambda_2) = (1, 0)$ is an explicit counterexample to the implication. Lemma 7.11 gives the same degree separation for $(\mathbb{Z}/2^a)^2$: the gerbe coordinates are $(A_1^{(H)}, A_{12}^{(H)}, A_2^{(H)})$ in degree 2, and the linearization coordinates are $(\lambda_1^{(H)}, \lambda_2^{(H)})$ in degree 1. The trivial and odd cases have zero degree-one target by transfer or by triviality; that is not an implication from the degree-two class. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_gerbe_linearization_independence`

verifies `RHOMRED_FINITE_STABILIZER_GERBE_LINEARIZATION_INDEPENDENCE_VERIFIED`: it imports the row 289 packet and the reduced-orientation obstruction ledger, records explicit $E[2]$ and two-primary counterexamples to the false implication, records the trivial/odd zero-target exception, and checks that no current orientation packet uses a gerbe, scalar, or determinant-anchor row as a zero-linearization row.

Proposition 0.93 *($E[2]$ Klein-four finite-stabilizer class).* [local computation proved; retained $E[2]$-edge row not supplied] Let $H = E[2] \simeq (\mathbb{Z}/2)^2$, choose generators e_1, e_2, and let $x_1, x_2 \in H^1(BH;\mathbb{F}_2)$ be the dual basis. After the finite-stabilizer Cech–Borel class on a retained $BE[2]$-stratum has been reduced to the classifying-space edge, its degree-two part is uniquely of the form

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2 \in H^2(BE[2];\mathbb{F}_2).$$

This is row 291. The symbols b_{20}, b_{11}, b_{02} are the coordinates of the supplied edge class in the ordered basis

$$x_1^2, \quad x_1x_2, \quad x_2^2.$$

Rows 292–294 are the subsequent restriction computations which express these coordinates through the three cyclic restrictions r_1, r_2, r_3. Row 291 does not by itself supply a retained $E[2]$-stratum, an equivariant Cech–Borel representative, an edge-reduction row, or the values of the three coefficients.

Proof. For a complex elliptic curve, $E[2] \simeq (\mathbb{Z}/2)^2$. The classifying-space cohomology is

$$H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \qquad |x_i| = 1.$$

Therefore its degree-two part has basis

$$H^2(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle.$$

An edge-reduced finite-stabilizer Borel class is an element of this three-dimensional vector space, hence has the unique displayed coordinate expansion. The proof is a computation in $H^*(B(\mathbb{Z}/2)^2;\mathbb{F}_2)$; it does not assert that the current finite tables have supplied the geometric edge class whose coordinates are b_{20}, b_{11}, b_{02}. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_klein_four_class`

verifies `RHOMRED_FINITE_STABILIZER_KLEIN_FOUR_CLASS_VERIFIED`: it imports the finite-stabilizer beta packet, the row 290 firewall, and the reduced-orientation obstruction ledger, records the $H^2(BE[2];\mathbb{F}_2)$ basis and the displayed Klein-four edge class, and keeps the retained $E[2]$-stratum, edge-reduction, coefficient-value, cyclic-restriction, vanishing, linearization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.94 *(First cyclic restriction of the $E[2]$ Klein-four class).* [local computation proved; first cyclic restriction row not supplied] Keep the notation of Proposition 0.93. Let $\iota_1\colon \langle e_1\rangle \hookrightarrow E[2]$ be the first cyclic subgroup and let

$$t \in H^1(B\langle e_1\rangle;\mathbb{F}_2)$$

be its degree-one generator. The restriction map satisfies

$$\iota_1^*(x_1) = t, \qquad \iota_1^*(x_2) = 0.$$

Therefore

$$\iota_1^*\beta_{C,S}^{E,2} = b_{20}t^2.$$

If the first cyclic restriction is written

$$\iota_1^*\beta_{C,S}^{E,2} = r_1 t^2,$$

then

$$b_{20} = r_1.$$

This is row 292. It is the first coefficient extraction from the row 291 class. It does not supply the retained $E[2]$-edge row, the first cyclic restriction row, or the value of r_1.

Proof. The inclusion ι_1 sends the generator of $\langle e_1\rangle$ to e_1. Hence the dual class x_1 restricts to the generator t, while x_2 restricts to zero. Applying ι_1^* to

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$$

gives $b_{20}t^2$. Since $H^2(B\langle e_1\rangle;\mathbb{F}_2) = \mathbb{F}_2\langle t^2\rangle$, equality with r_1t^2 is equivalent to equality of coefficients, namely $b_{20} = r_1$. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_klein_four_b20`

verifies `RHOMRED_FINITE_STABILIZER_KLEIN_FOUR_B20_VERIFIED`: it imports the row 291 Klein-four class packet, records the first cyclic restriction map $x_1 \mapsto t,\ x_2 \mapsto 0$, verifies the coefficient identity $b_{20} = r_1$, and keeps the retained edge row, the first cyclic restriction value, the b_{11} and b_{02} restriction computations, vanishing, linearization, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.95 *(Diagonal cyclic restriction of the $E[2]$ Klein-four class).* [local computation proved; cyclic restriction rows not supplied] Keep the notation of Propositions 0.93 and 0.94. Let

$$\iota_2\colon \langle e_2\rangle \hookrightarrow E[2], \qquad \iota_3\colon \langle e_1+e_2\rangle \hookrightarrow E[2]$$

be the second and diagonal cyclic subgroups, and let t denote the degree-one generator on either cyclic subgroup. The restriction maps satisfy

$$\iota_2^*(x_1) = 0, \qquad \iota_2^*(x_2) = t,$$

and

$$\iota_3^*(x_1) = t, \qquad \iota_3^*(x_2) = t.$$

Thus

$$\iota_2^*\beta_{C,S}^{E,2} = b_{02}t^2, \qquad \iota_3^*\beta_{C,S}^{E,2} = (b_{20}+b_{11}+b_{02})t^2.$$

If the three cyclic restrictions are written

$$\iota_i^*\beta_{C,S}^{E,2} = r_i t^2, \qquad i = 1, 2, 3,$$

where $i = 3$ denotes the diagonal subgroup, then

$$b_{11} = r_1 + r_2 + r_3$$

in $\mathbb{F}_2$. This is row 293. It does not supply the retained $E[2]$-edge row, the three cyclic restriction rows, the values of r_1, r_2, r_3, or the value of b_{11}. Row 294 records the second-square extraction $b_{02} = r_2$ as its own ledger row.

Proof. The inclusion ι_2 kills the dual class x_1 and sends x_2 to the cyclic generator t. The inclusion ι_3 sends the cyclic generator to $e_1 + e_2$, so both dual classes restrict to t. Applying these restrictions to

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$$

gives the two displayed formulae. The first cyclic restriction gives $b_{20} = r_1$ by Proposition 0.94; the second restriction gives $b_{02} = r_2$; and the diagonal restriction gives

$$b_{20} + b_{11} + b_{02} = r_3.$$

Substitution yields $r_1 + b_{11} + r_2 = r_3$. Since the coefficient field is $\mathbb{F}_2$, this is equivalent to

$$b_{11} = r_1 + r_2 + r_3.$$

All equalities here are identities in the cohomology of cyclic classifying spaces. They do not assert that the current finite tables have supplied the geometric cyclic restrictions or their values. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_klein_four_b11`

verifies `RHOMRED_FINITE_STABILIZER_KLEIN_FOUR_B11_VERIFIED`: it imports the row 291 and row 292 packets, records the second and diagonal cyclic restriction maps, verifies the mixed coefficient identity $b_{11} = r_1 + r_2 + r_3$, and keeps the retained edge row, the cyclic restriction values, the standalone b_{02} row, vanishing, linearization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.96 *(Second cyclic restriction of the $E[2]$ Klein-four class).* [local computation proved; second cyclic restriction row not supplied] Keep the notation of Proposition 0.93. Let $\iota_2 \colon \langle e_2 \rangle \hookrightarrow E[2]$ be the second cyclic subgroup and let

$$t \in H^1(B\langle e_2 \rangle; \mathbb{F}_2)$$

be its degree-one generator. The restriction map satisfies

$$\iota_2^*(x_1) = 0, \qquad \iota_2^*(x_2) = t.$$

Therefore

$$\iota_2^* \beta_{C,S}^{E,2} = b_{02} t^2.$$

If the second cyclic restriction is written

$$\iota_2^* \beta_{C,S}^{E,2} = r_2 t^2,$$

then

$$b_{02} = r_2.$$

This is row 294. It is the standalone second-square coefficient extraction from the row 291 class. It does not supply the retained $E[2]$-edge row, the second cyclic restriction row, or the value of r_2.

Proof. The inclusion ι_2 sends the generator of $\langle e_2 \rangle$ to e_2. Hence x_1 restricts to zero and x_2 restricts to the generator t. Applying ι_2^* to

$$\beta_{C,S}^{E,2} = b_{20} x_1^2 + b_{11} x_1 x_2 + b_{02} x_2^2$$

gives $b_{02} t^2$. Since $H^2(B\langle e_2 \rangle; \mathbb{F}_2) = \mathbb{F}_2\langle t^2 \rangle$, equality with $r_2 t^2$ is equivalent to $b_{02} = r_2$. This is a classifying-space calculation only; it does not assert that the finite-stabilizer tables have supplied the retained cyclic restriction or its value. □

The packet

```
certificates/orientation/rhomred_finite_stabilizer_klein_four_b02
```

verifies `RHOMRED_FINITE_STABILIZER_KLEIN_FOUR_B02_VERIFIED`: it imports the row 291 and row 293 packets, records the second cyclic restriction map $x_1 \mapsto 0$, $x_2 \mapsto t$, verifies the coefficient identity $b_{02} = r_2$, and keeps the retained edge row, the second cyclic restriction value, vanishing, linearization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.97 *(Klein-four zero-restriction criterion).* [criterion proved; zero restriction rows not supplied] Keep the notation of Propositions 0.93, 0.94, 0.95, and 0.96. Let

$$\iota_i^* \beta_{C,S}^{E,2} = r_i t^2, \qquad i = 1, 2, 3,$$

where $i = 1$ is the subgroup $\langle e_1 \rangle$, $i = 2$ is $\langle e_2 \rangle$, and $i = 3$ is $\langle e_1 + e_2 \rangle$. Then

$$\beta_{C,S}^{E,2} = 0 \quad \Longleftrightarrow \quad r_1 = r_2 = r_3 = 0.$$

Equivalently, the three cyclic zero restrictions are necessary and sufficient for the vanishing of the classifying-space edge of the $E[2]$-finite-stabilizer Borel class. This is the algebraic content of row 295. The present retained finite-stabilizer tables do not yet supply the three cyclic restriction rows or the values r_1, r_2, r_3, so this proposition is not a geometric proof that those values vanish on the retained $K3 \times E$ strata.

Proof. Rows 292–294 give

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3$$

in $\mathbb{F}_2$. If $r_1 = r_2 = r_3 = 0$, then $b_{20} = b_{02} = b_{11} = 0$, hence

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$$

vanishes in $H^2(BE[2]; \mathbb{F}_2)$. Conversely, if $\beta_{C,S}^{E,2} = 0$, then the coefficients in the basis x_1^2, x_1x_2, x_2^2 vanish. Thus $r_1 = b_{20} = 0$, $r_2 = b_{02} = 0$, and

$$r_3 = b_{20} + b_{11} + b_{02} = 0.$$

The implication is a classifying-space calculation. A proof of the geometric assertion $r_1 = r_2 = r_3 = 0$ still requires retained cyclic restriction rows, or an equivalent edge-reduced vanishing row, in the finite-stabilizer orientation tables. □

The packet

```
certificates/orientation/rhomred_finite_stabilizer_klein_four_zero_restrictions
```

verifies `RHOMRED_FINITE_STABILIZER_KLEIN_FOUR_ZERO_RESTRICTIONS_OBSTRUCTION_VERIFIED`: it imports the row 291–294 packets, proves the equivalence between the three zero cyclic restrictions and vanishing of the Klein-four classifying-space edge, and records that the retained cyclic zero rows, the r_i-values, finite-stabilizer beta vanishing, quotient orientation, transition compatibility, vanishing-cycle, and protected-integration rows remain open.

Proposition 0.98 *(Even two-primary finite-stabilizer class).* [local computation proved; retained two-primary row not supplied] Let $N \geq 4$ be even and write $2^a \parallel N$, $a \geq 2$. The two-primary part of the N-torsion stabilizer of a complex elliptic curve is

$$E[N]_{(2)} \simeq (\mathbb{Z}/2^a)^2.$$

Choose generators and let

$$x_1, x_2 \in H^1(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2), \qquad y_1, y_2 \in H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2)$$

be the corresponding exterior and polynomial generators. Then

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2,$$

and

$$H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

After the finite-stabilizer Cech–Borel class on a retained two-primary stratum has been reduced to the classifying-space edge, its degree-two part is uniquely of the form

$$\beta_{C,S}^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1x_2 + A_2^{(N)} y_2.$$

This is row 296. The coefficients $A_1^{(N)}, A_{12}^{(N)}, A_2^{(N)}$ are coordinates of the supplied edge class in the ordered basis y_1, x_1x_2, y_2. Row 296 does not supply a retained two-primary stratum, an equivariant Cech–Borel representative, an edge-reduction row, or the values of the three coefficients. Row 297 is the separate mixed-term detection check for $A_{12}^{(N)}$.

Proof. For a complex elliptic curve, $E[N] \simeq (\mathbb{Z}/N)^2$. Primary decomposition gives the displayed two-primary subgroup. For $a \geq 2$,

$$H^*(B\mathbb{Z}/2^a; \mathbb{F}_2) = \Lambda(x) \otimes \mathbb{F}_2[y], \qquad |x| = 1, \quad |y| = 2.$$

The Künneth formula gives

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2].$$

Its degree-two part has basis y_1, x_1x_2, y_2. Therefore an edge-reduced two-primary finite-stabilizer Borel class is a unique linear combination of these three basis vectors. The argument is only a classifying-space computation; it does not assert that the current finite tables have supplied the retained edge class whose coordinates are $A_1^{(N)}, A_{12}^{(N)}, A_2^{(N)}$. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_two_primary_class`

verifies `RHOMRED_FINITE_STABILIZER_TWO_PRIMARY_CLASS_VERIFIED`: it imports the finite-stabilizer beta packet and the reduced orientation obstruction ledger, records the $\Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2]$ cohomology ring, the degree-two basis y_1, x_1x_2, y_2, and the displayed two-primary edge-class form, and keeps the retained stratum, edge-reduction, coefficient-value, mixed-term detection, vanishing, linearization, transition, vanishing-cycle, and protected-integration rows open.

PROPOSITION 0.99 *(Detection of the two-primary mixed term).* [local detection proved; retained A_{12}-value not supplied] Keep the notation of Proposition 0.98. Define the coefficient functional

$$\pi_{12} : H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) \longrightarrow \mathbb{F}_2$$

by

$$\pi_{12}(y_1) = 0, \qquad \pi_{12}(x_1x_2) = 1, \qquad \pi_{12}(y_2) = 0.$$

Then

$$\pi_{12}\Big(\beta_{C,S}^{E,N}\Big) = A_{12}^{(N)}.$$

For every cyclic subgroup $C \subset (\mathbb{Z}/2^a)^2$, the restriction map satisfies

$$\mathrm{res}_C^{(\mathbb{Z}/2^a)^2}(x_1x_2) = 0.$$

Thus the mixed coefficient $A_{12}^{(N)}$ is detected only by the rank-two classifying-space edge, or by an equivalent rank-two coefficient row; it is not determined by cyclic restrictions. This is row 297. It does not supply the retained two-primary edge row or the value of $A_{12}^{(N)}$.

Proof. The first assertion is the definition of the coordinate functional in the ordered basis y_1, x_1x_2, y_2. It remains to check the cyclic restriction statement. Let $C \subset (\mathbb{Z}/2^a)^2$ be cyclic. If $|C| = 2$, then $C \subset 2(\mathbb{Z}/2^a)^2$ because $a \geq 2$; therefore the two mod-2 characters x_1, x_2 restrict trivially to C. If $|C| \geq 4$, then

$$H^*(BC; \mathbb{F}_2) = \Lambda(u) \otimes \mathbb{F}_2[v], \qquad |u| = 1, \quad |v| = 2,$$

and both x_1 and x_2 restrict to scalar multiples of u. Hence their product restricts to a scalar multiple of $u^2 = 0$. Thus x_1x_2 restricts to zero on every cyclic subgroup. The coefficient $A_{12}^{(N)}$ can therefore not be recovered from cyclic restrictions of the degree-two class. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_two_primary_mixed_term_detection`

verifies `RHOMRED_FINITE_STABILIZER_TWO_PRIMARY_MIXED_TERM_DETECTION_VERIFIED`: it imports the row 296 two-primary class packet, records the rank-two coefficient functional π_{12}, proves that cyclic subgroup restrictions annihilate x_1x_2, and keeps the retained edge row, $A_{12}^{(N)}$-value, vanishing, linearization, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.100 *(Odd-order finite-stabilizer transfer).* [local transfer proved; retained odd edge row not supplied] Let $H \subset E[N]$ be a finite stabilizer of odd order. Then

$$H^i(BH;\mathbb{F}_2) = 0, \qquad i > 0.$$

Consequently, after the finite-stabilizer Cech–Borel class on a retained odd-order stratum has been reduced to the classifying-space edge,

$$\beta_{C,S}^H \in H^2(BH;\mathbb{F}_2)$$

vanishes. The residual degree-one stabilizer character also vanishes after edge reduction, since $H^1(BH;\mathbb{F}_2) = 0$. This is row 298. It proves the odd-order target calculation; it does not supply a retained odd-order stratum, an equivariant Cech–Borel representative, an edge-reduction row, an all-retained-stabilizer coverage row, or the compatible null-trivialisations required for (O1).

Proof. Since $|H|$ is odd, multiplication by $|H|$ is the identity on $\mathbb{F}_2$. The restriction-transfer identity in group cohomology gives multiplication by $|H|$ on $H^i(BH;\mathbb{F}_2)$, while restriction to the point is zero in positive degree. Thus every positive-degree class in $H^*(BH;\mathbb{F}_2)$ is zero. Equivalently, the averaging idempotent makes the invariants functor exact because $|H|$ is invertible in $\mathbb{F}_2$. In particular, the degree-two classifying-space edge of the finite-stabilizer Borel class and the degree-one residual character both vanish once the geometric data have been reduced to those targets. □

The packet

`certificates/orientation/rhomred_finite_stabilizer_odd_transfer`

verifies `RHOMRED_FINITE_STABILIZER_ODD_TRANSFER_VERIFIED`: it imports the finite-stabilizer beta and beta-vanishing obstruction packets, records the odd-order transfer calculation $H^{>0}(BH;\mathbb{F}_2) = 0$, and keeps the retained odd stratum, edge-reduction, all-retained coverage, null-trivialisation, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.101 *(Finite-stabilizer orientation detection; proved after edge reduction).* Let S be a retained finite-stabilizer stratum with stabilizer $H \subset E_{\mathrm{tors}}$. Suppose the equivariant Cech–Borel orientation class $\widetilde{\beta}_{C,S}^H \in H_H^2(S;\mathbb{F}_2)$ has been reduced to its classifying-space edge, i.e. its ordinary component in $H^2(S;\mathbb{F}_2)^H$, mixed component in $H^1(BH;H^1(S;\mathbb{F}_2))$, and incoming spectral-sequence differential terms vanish with chosen null-trivialisations. Then the finite stabilizer part of (O1) is exactly the following finite calculation.

If H has odd order, the mod-2 classifying-space edge vanishes. If $H \simeq E[2] \simeq (\mathbb{Z}/2)^2$, write

$$\beta_{C,S}^H = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad \lambda_{C,S}^H = \lambda_1x_1 + \lambda_2x_2.$$

Then quotient orientation on this stratum is equivalent to

$$b_{20} = b_{11} = b_{02} = \lambda_1 = \lambda_2 = 0,$$

or equivalently to the three cyclic quadratic restrictions and the three cyclic linearization restrictions all vanishing, as in Lemma 7.10. If the two-primary part of H is $(\mathbb{Z}/2^a)^2$, $a \geq 2$, write

$$\beta^H_{C,S} = A_1^{(H)} y_1 + A_{12}^{(H)} x_1 x_2 + A_2^{(H)} y_2, \qquad \lambda^H_{C,S} = \lambda_1^{(H)} x_1 + \lambda_2^{(H)} x_2.$$

Then quotient orientation is equivalent to the five equations

$$A_1^{(H)} = A_{12}^{(H)} = A_2^{(H)} = \lambda_1^{(H)} = \lambda_2^{(H)} = 0.$$

The mixed coefficient $A_{12}^{(H)}$ is not detected by cyclic order-two restrictions. Therefore a finite-stabilizer row which lists only cyclic restrictions is not an (O1) quotient-orientation row unless the edge-reduction and mixed-coefficient computations are supplied.

Proof. The edge-reduction hypothesis identifies the relevant obstruction with an element of $H^2(BH;\mathbb{F}_2)$, and the residual linearization with an element of $H^1(BH;\mathbb{F}_2)$. For odd H, transfer kills positive degree cohomology with $\mathbb{F}_2$-coefficients. For $E[2]$, the coordinate calculation is Lemma 7.10: the three nonzero cyclic restrictions invert the quadratic coefficients, and the linearization character is an independent H^1-torsor. For $(\mathbb{Z}/2^a)^2$, $a \geq 2$, Lemma 7.11 gives

$$H^2(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

The term x_1x_2 pulls back trivially to cyclic order-two subgroups, so it must be computed on the rank-two stabilizer. These are exactly the equations in the statement. □

PROPOSITION 0.102 *(Direct-sum orientation multiplicativity criterion).* [Picard-groupoid criterion proved; direct-sum Thom–Sebastiani row not supplied] Let $C, \mathcal{D}$ be retained objects at a fixed finite HN height. Suppose that the cosection-reduced self-Ext construction has been supplied on the direct-sum chart and is compatible with the block decomposition

$$K^{\mathrm{red}}_{C\oplus\mathcal{D}} \simeq K^{\mathrm{red}}_{C} \oplus K^{\mathrm{red}}_{\mathcal{D}} \oplus K^{\mathrm{red}}_{C,\mathcal{D}} \oplus K^{\mathrm{red}}_{\mathcal{D},C}.$$

Then the determinant functor gives

$$\mathcal{L}^{\det}_{C\oplus\mathcal{D}} \simeq \mathcal{L}^{\det}_{C} \otimes \mathcal{L}^{\det}_{\mathcal{D}} \otimes \det K^{\mathrm{red}}_{C,\mathcal{D}} \otimes \det K^{\mathrm{red}}_{\mathcal{D},C}.$$

Direct-sum multiplicativity of orientations is therefore not the displayed determinant identity. It is the additional Picard-groupoid datum consisting of square roots $o_C^{\otimes 2} \simeq \mathcal{L}^{\det}_C$, $o_{\mathcal{D}}^{\otimes 2} \simeq \mathcal{L}^{\det}_{\mathcal{D}}$, $o_{C\oplus\mathcal{D}}^{\otimes 2} \simeq \mathcal{L}^{\det}_{C\oplus\mathcal{D}}$, a chosen Calabi–Yau hyperbolic trivialization

$$\det K^{\mathrm{red}}_{C,\mathcal{D}} \otimes \det K^{\mathrm{red}}_{\mathcal{D},C} \simeq q_{C,\mathcal{D}}^{\otimes 2},$$

and an isomorphism

$$m_{\oplus,C,\mathcal{D}} : o_{C\oplus\mathcal{D}} \xrightarrow{\sim} o_C \otimes o_{\mathcal{D}} \otimes q_{C,\mathcal{D}}$$

which is symmetric in $(C, \mathcal{D})$, functorial under pullback to the retained direct-sum and two-step flag charts, and has zero Thom–Sebastiani pentagon defect. Row 299 is proved for a retained pair exactly when this datum is supplied for that pair and recorded as a zero-defect row in ts_multiplicativity.

In the present finite packet the table orientation_lines has no square-root rows and ts_multiplicativity has no direct-sum isomorphism row. Thus the current status of row 299 is the criterion above together with an open obstruction, not a proof of direct-sum multiplicativity.

Proof. The first displayed isomorphism is the standard block decomposition of derived endomorphisms of a direct sum, after restricting to a chart where the reduced cone construction has been made compatible with direct sum. The Knudsen–Mumford determinant functor is multiplicative for finite direct sums of perfect complexes, hence gives the determinant-line formula. The off-diagonal factors are not killed by this formula; they must be paired by the Calabi–Yau duality and trivialized as a square before one can compare square roots.

A reduced orientation is a square root in the Picard groupoid, and a multiplicative reduced orientation is therefore the isomorphism $m_{\oplus,\mathcal{C},\mathcal{D}}$ together with the symmetry, pullback, and pentagon coherences stated above. These are exactly the entries carried by a direct-sum Thom–Sebastiani row. The current orientation ledger records the square-root row and the Thom–Sebastiani multiplicativity row as missing open obligations, and the corresponding tables are empty. Hence the proposition proves the criterion and identifies the missing data; it does not assert that the missing data have been constructed. $\square$

The packet

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certificates/orientation/rhomred_orientation_direct_sum_multiplicativity
```

verifies `RHOMRED_ORIENTATION_DIRECT_SUM_MULTIPLICATIVITY_OBSTRUCTION_VERIFIED`: it imports the reduced determinant, square-root obstruction, and reduced-orientation ledger packets, records the direct-sum Picard-groupoid criterion, checks that orientation_lines and ts_multiplicativity are empty, and keeps the square-root, cross-term hyperbolic trivialization, direct-sum Thom–Sebastiani isomorphism, pentagon, extension, quotient, transition, vanishing-cycle, and protected-integration rows open.

Proposition 0.103 *(Extension orientation multiplicativity criterion).* [criterion proved; extension Thom–Sebastiani row not supplied] Let

$$\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}'} \xleftarrow{p=(p_1,p_2)} \overline{\mathfrak{E}}^{\mathrm{red}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q} \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}+\widehat{c}'}$$

be a retained compactified extension correspondence. Row 300 is an orientation row only if it supplies a Joyce–Upmeier multiplicativity isomorphism

$$\mathrm{TS}^{o}_{\mathfrak{E}} : p_1^* o_{R,\widehat{c}} \otimes p_2^* o_{R,\widehat{c}'} \xrightarrow{\sim} q^* o_{R,\widehat{c}+\widehat{c}'}$$

whose square is the cosection-reduced determinant comparison on $\overline{\mathfrak{E}}^{\mathrm{red}}_{R,\widehat{c},\widehat{c}'}$. Equivalently, after tensoring with the BBDJS reduced Thom–Sebastiani isomorphism for vanishing cycles, it gives the coefficient-system isomorphism

$$p^*\big((\Phi^{\mathrm{red}}_{R,\widehat{c}} \otimes o_{R,\widehat{c}}) \boxtimes (\Phi^{\mathrm{red}}_{R,\widehat{c}'} \otimes o_{R,\widehat{c}'})\big) \xrightarrow{\sim} q^*(\Phi^{\mathrm{red}}_{R,\widehat{c}+\widehat{c}'} \otimes o_{R,\widehat{c}+\widehat{c}'}).$$

It must be functorial on overlaps, compatible with quotient descent, and its pullbacks to every retained two-step flag stack must have zero Thom–Sebastiani pentagon defect. Thus row 300 is proved exactly by a zero-defect row in ts_multiplicativity with

(extension_stack_id, left_line_id, right_line_id, target_line_id, ts_isomorphism_id).

The retained extension-closure rows record only that certain middle HN types stay inside the finite window. The present orientation table ts_multiplicativity is empty, and the mixed Thom–Sebastiani packet is conditional on supplied orientation transport. Consequently row 300 is presently a criterion with an open extension-orientation obstruction.

Proof. The Hall product pulls the external coefficient system on $\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}'}$ to the extension stack and pushes it along q. The BBDJS Thom–Sebastiani theorem identifies the reduced vanishing-cycle factors after the reduced d-critical charts and the extension potential comparison have been supplied. That theorem does not choose square roots of the orientation gerbe. Joyce–Upmeier multiplicativity is the additional Picard-groupoid isomorphism $\mathrm{TS}^{o}_{\mathfrak{E}}$ whose square is the determinant comparison for the exact-triangle self-Ext complex.

Associativity of the oriented Hall product requires the same isomorphism to satisfy the two-step flag pentagon; quotient descent and HN transition compatibility are separate rows. The current certificate `certificates/orientation/k3e_reduced_orientation` has no rows in orientation_lines or ts_multiplicativity, and its obstruction ledger records both ts_multiplicativity and ts_pentagon as missing open obligations. Retained extension closure and conditional mixed Thom–Sebastiani transport therefore do not prove row 300. □

The packet

`certificates/orientation/rhomred_orientation_extension_multiplicativity`

verifies `RHOMRED_ORIENTATION_EXTENSION_MULTIPLICATIVITY_OBSTRUCTION_VERIFIED`: it imports retained extension closure, the reduced-orientation ledger, the square-root obstruction packet, the direct-sum multiplicativity criterion, and the mixed Thom–Sebastiani packet; it checks that the extension-closure rows do not construct extension stacks or orientations, that the mixed Thom–Sebastiani packet is conditional on orientation transport, and that orientation_lines and ts_multiplicativity are empty.

PROPOSITION 0.104 *(Thom–Sebastiani compatibility for orientation lines).* [compatibility criterion proved; orientation-line Thom–Sebastiani row not supplied] Let $\mathrm{TS}^{o}_{\mathfrak{E}}$ be a candidate orientation-line isomorphism on a retained extension correspondence as in Proposition 0.103. It is compatible with Thom–Sebastiani if and only if the following finite row is supplied.

(i) The square-root diagram commutes, equivalently

$$q^*\sigma_{\widehat{c}+\widehat{c}'} \circ (\mathrm{TS}^{o}_{\mathfrak{E}})^{\otimes 2} = \det(\mathrm{TS}^{\mathrm{red}}_{\mathfrak{E}}) \circ (\sigma_{\widehat{c}} \otimes \sigma_{\widehat{c}'}),$$

as morphisms from $\left(p_1^* o_{R,\widehat{c}} \otimes p_2^* o_{R,\widehat{c}'}\right)^{\otimes 2}$ to $q^*\mathcal{L}^{\det}_{R,\widehat{c}+\widehat{c}'}$.

(ii) The oriented coefficient-system map obtained by tensoring $\mathrm{TS}^{o}_{\mathfrak{E}}$ with the BBDJS reduced Thom–Sebastiani isomorphism is the row used by the Hall product:

$$\mathrm{TS}^{\Phi\otimes o}_{\mathfrak{E}} = \mathrm{TS}^{\Phi,\mathrm{red}}_{\mathfrak{E}} \otimes \mathrm{TS}^{o}_{\mathfrak{E}}\,.$$

(iii) The chart-overlap cocycle, quotient descent cocycle, and two-step flag pentagon defect of this oriented Thom–Sebastiani row vanish.

Thus row 301 is not a new scalar identity and not merely the BBDJS vanishing-cycle Thom–Sebastiani theorem. It is the zero-defect compatibility row saying that the Joyce–Upmeier orientation isomorphism is the square root of the determinant Thom–Sebastiani comparison and that its tensor with the vanishing-cycle Thom–Sebastiani map is the coefficient-system transport.

The present packets do not supply such a row. The hybrid Thom–Sebastiani packets are conditional on supplied orientation transport, the four-input pentagon packet treats the orientation pentagon as a functorial input, and orientation_lines and ts_multiplicativity are empty.

Proof. An orientation line is a square root of the determinant line in the Picard groupoid. Therefore compatibility of an orientation Thom–Sebastiani isomorphism is exactly compatibility after squaring: the square of $\mathrm{TS}^{o}_{\mathfrak{E}}$ must be the determinant-line comparison induced by the reduced exact-triangle self-Ext complex. This proves the first condition.

BBDJS Thom–Sebastiani identifies the vanishing-cycle factors after the reduced d-critical chart and potential comparison have been supplied. It does not choose the Joyce–Upmeier square root. The oriented Hall coefficient system is $\Phi^{\mathrm{red}} \otimes o$, so its Thom–Sebastiani transport is the tensor product of the vanishing-cycle transport and the orientation-line transport. This gives the second condition.

The same row must be invariant under chart changes, descend through the quotient, and satisfy the two-step flag pentagon; otherwise the oriented Hall product depends on choices or fails associativity. The current orientation ledger records the relevant orientation-line, Thom–Sebastiani, and pentagon rows as missing open obligations. The mixed and four-input Thom–Sebastiani packets use orientation compatibility as a hypothesis. Hence the proposition proves the criterion and records the missing data, not an unconditional compatibility theorem. □

The packet

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certificates/orientation/rhomred_orientation_thom_sebastiani_compatibility
```

verifies `RHOMRED_ORIENTATION_THOM_SEBASTIANI_COMPATIBILITY_OBSTRUCTION_VERIFIED`: it imports the row 300 extension-multiplicativity packet, the reduced-orientation ledger, the mixed Thom–Sebastiani packet, and the four-input pentagon packet; it records that the BBDJS Thom–Sebastiani rows and conditional pentagon rows do not construct the missing orientation-line compatibility row.

Proposition 0.105 *(Orientation descent through the E-quotient).* [descent criterion proved; quotient-orientation rows not supplied] Let o_C be a reduced orientation line on a retained stratum of the $K3 \times E$ reduced moduli stack. The line descends through the quotient by E precisely when the following finite Picard-groupoid datum is supplied.

(i) a square-root row for $o_C^{\otimes 2} \simeq \det K_C^{\mathrm{red}}$;
(ii) a null-trivialization of the ordinary reduced gerbe α_C^{red};
(iii) a Cech–Borel representative of the connected free E-class

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2)$$

with $a_1 = a_2 = 0$ and a chosen null-trivialization;
(iv) for every retained finite-stabilizer stratum S, an edge-reduced equivariant class

$$\widetilde{\beta}^H_{C,S} \rightsquigarrow \beta^H_{C,S} \in H^2(BH; \mathbb{F}_2),$$

a zero row for $\beta^H_{C,S}$, and compatible null-trivialisations under retained subgroup restrictions;
(v) a computed residual linearization character

$$\lambda^H_{C,S} \in H^1(BH; \mathbb{F}_2)$$

with $\lambda^H_{C,S} = 0$ for every retained stabilizer;
(vi) compatibility of these choices on overlaps and on the quotient presentation of the retained extension and flag charts.

Equivalently, row 302 is proved by a populated quotient_borel table and a populated finite_stabilizers table whose obstruction ranks and degree-one characters vanish. The descent object is the descended line o_C/E on the quotient stack together with the chosen quotient cocycle

$$\mathfrak{q}_C = (\alpha_C^{\mathrm{red}}, \alpha_C^{E,\mathrm{free}}, \{(\beta_{C,S}^H, \lambda_{C,S}^H)\}_{S,H}) \simeq 0.$$

The present packets do not supply this datum. They record criteria for the reduced gerbe, connected free E-class, finite-stabilizer classes, and residual linearization characters, but orientation_lines, quotient_borel, and finite_stabilizers are empty. Thus row 302 is currently a descent criterion with an open quotient-orientation obstruction, not a descended orientation.

Proof. For a line on an E-stack to descend to the quotient stack, its equivariant gerbe and stabilizer linearizations must be trivialized. In the reduced orientation problem the ordinary component is α_C^{red}. The connected part of the E-action contributes the $H^2(BE;\mathbb{F}_2)$ edge class $\alpha_C^{E,\mathrm{free}}$. Finite stabilizers contribute the Borel classes $\widetilde{\beta}_{C,S}^H$, whose classifying-space edges $\beta_{C,S}^H$ are legitimate targets only after the mixed Borel and differential terms have been killed. After the degree-two gerbes vanish, a residual degree-one stabilizer character $\lambda_{C,S}^H$ remains; nonzero λ^H changes the descended orientation by a sign character and therefore prevents descent.

The six listed rows are exactly the data required to trivialize these obstructions and make the equivariant orientation line descend. The compatibility clauses are needed because descent is a statement in the Picard groupoid on the quotient presentation, not a collection of unrelated pointwise vanishings. The current reduced-orientation fixture records all quotient-Borel and finite-stabilizer rows as missing open obligations, and the specialized packets for α^{red}, $\alpha^{E,\mathrm{free}}$, β^H, and λ^H are obstruction ledgers. Hence the criterion is proved, while the descended orientation is not constructed. □

The packet

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certificates/orientation/rhomred_orientation_e_quotient_descent
```

verifies `RHOMRED_ORIENTATION_E_QUOTIENT_DESCENT_OBSTRUCTION_VERIFIED`: it imports the square-root, reduced-gerbe nullhomotopy, connected free E-nulltrivialization, finite-stabilizer nulltrivialization, finite-stabilizer linearization-vanishing, and reduced-orientation ledger packets; it checks that the quotient tables are empty and records the exact missing descent rows.

PROPOSITION 0.106 *(Orientation descent and the Hall product).* [product-descent criterion proved; Hall-product orientation rows not supplied] Fix a finite height R and a retained compactified extension correspondence

$$\mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'} \xleftarrow{p} \overline{\mathfrak{E}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q} \mathfrak{M}_{R,\widehat{c}+\widehat{c}'}.$$

Put $K_{R,\widehat{c}} := \Phi_{R,\widehat{c}}^{\mathrm{red}} \otimes o_{R,\widehat{c}}$. The statement that quotient-orientation descent commutes with the Hall product is the commutative square

$$\begin{tikzcd}
Q_{E,R}\, q_!\, \mathrm{TS}_R^{\Phi\otimes o}\, p^* \arrow[r, "\theta^Q_{\mu,R}"] \arrow[d, "Q_{E,R} m_R"'] & \overline{q}_!\, \overline{\mathrm{TS}_R}^{\Phi\otimes o}\, \overline{p}^*(Q_{E,R} \boxtimes Q_{E,R}) \arrow[d, "\overline{m}_R"] \\
Q_{E,R} K_{R,\widehat{c}+\widehat{c}'} \arrow[r, "\theta^Q_{K,R}"'] & \overline{K}_{R,\widehat{c}+\widehat{c}'}
\end{tikzcd}$$

in the Borel–Moore correspondence category on the quotient stack, where $\theta^{Q}_{\mu,R}$ is the quotient-after-correspondence operation comparison and $\theta^{Q}_{K,R} : Q_{E,R}K_{R,\widehat{c}+\widehat{c}'} \simeq \overline{K}_{R,\widehat{c}+\widehat{c}'}$ is the descended coefficient-system identification.

This square is proved by supplying the following finite datum.

(i) quotient-orientation rows for the two input strata, the extension stack, and the output stratum, with the quotient cocycles

$$p^*(\mathfrak{q}_{R,\widehat{c}} \boxtimes \mathfrak{q}_{R,\widehat{c}'}) \quad \text{and} \quad q^*\mathfrak{q}_{R,\widehat{c}+\widehat{c}'}$$

identified by a zero Picard-groupoid defect row;

(ii) the oriented Thom–Sebastiani row

$$\mathrm{TS}^{\Phi\otimes o}_{R} = \mathrm{TS}^{\Phi,\mathrm{red}}_{R} \otimes \mathrm{TS}^{o}_{R}$$

whose square is the reduced determinant Thom–Sebastiani comparison and whose quotient-descent defect is zero;

(iii) external-product descent

$$Q_{E,R}(K_{R,\widehat{c}} \boxtimes K_{R,\widehat{c}'}) \simeq \overline{K}_{R,\widehat{c}} \boxtimes \overline{K}_{R,\widehat{c}'}$$

on the product quotient;

(iv) compact-support pushforward descent for $q_!$, with proper base-change and projection-formula defect ranks zero;

(v) the quotient-after-correspondence comparison $\theta^{Q}_{\mu,R}$ for the supplied binary Hall product row;

(vi) compatibility of the preceding rows with the two-step flag pentagon, so that product descent is associative after quotient.

The current packets do not supply this datum. Row 302 supplies no quotient-orientation rows, row 301 supplies no oriented Thom–Sebastiani row, and the compact Hall source packet supplies no geometric Hall product matrix. The quotient-after-correspondence packet constructs $\theta^{Q}_{\mu,R}$ only for operation rows with descent interfaces already supplied. Thus row 303 is presently a product-descent criterion, not a proof that orientation descent commutes with the Hall product.

Proof. The Hall product is the pull–transport–push composite

$$m_R = q_!\,\mathrm{TS}^{\Phi\otimes o}_{R}\,p^*.$$

A quotient comparison for this composite is an isomorphism of composites, not a scalar identity. Therefore it decomposes into three functorial requirements: descent of p^* and of the external product coefficient system, descent of the oriented Thom–Sebastiani transport, and descent of $q_!$ through compact supports. The first requirement is exactly the Picard-groupoid equality of the pulled-back quotient cocycles on the extension correspondence. The second is the orientation-line row of Proposition 0.104 tensored with the reduced BBDJS Thom–Sebastiani transport. The third is the compact-support pushforward comparison used by Proposition 0.56.

If these rows are supplied, the naturality of pullback, the descended Thom–Sebastiani isomorphism, and compact-support pushforward descent identify

$$Q_{E,R}q_!\,\mathrm{TS}^{\Phi\otimes o}_{R}\,p^* \quad \text{with} \quad \overline{q}_!\overline{\mathrm{TS}}^{\Phi\otimes o}_{R}\,\overline{p}^*(Q_{E,R} \boxtimes Q_{E,R}),$$

which is the displayed square. Associativity after quotient is the same statement on the retained two-step flag stack. The present tables fail before this proof can be applied: the quotient-orientation tables are empty, the oriented Thom–Sebastiani row is absent, and the source Hall product rows are not populated. Hence only the criterion is proved. □

The packet

`certificates/orientation/rhomred_orientation_hall_product_descent`

verifies `RHOMRED_ORIENTATION_HALL_PRODUCT_DESCENT_OBSTRUCTION_VERIFIED`: it imports the row 302 quotient-orientation descent packet, the row 301 oriented Thom–Sebastiani packet, the quotient-after-correspondence θ_μ^Q packet, and the transition Hall-product obstruction packet; it records that the current tree has comparison maps for already supplied operation rows but no quotient-orientation Hall-product descent row.

PROPOSITION 0.107 *(Orientation descent and the Hall coproduct).* [coproduct-descent criterion proved; Hall-coproduct orientation rows not supplied] Fix a finite height R and the same retained compactified correspondence, read in the splitting direction

$$\mathfrak{M}_{R,\widehat{c}+\widehat{c}'} \xleftarrow{q} \overline{\mathfrak{E}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{p} \mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'}.$$

With $K_{R,\widehat{c}} = \Phi^{\mathrm{red}}_{R,\widehat{c}} \otimes o_{R,\widehat{c}}$, quotient-orientation descent commutes with the Hall coproduct precisely when the pull–inverse-transport–push comparison

$$\begin{tikzcd}
Q_{E,R}\, p_!(\mathrm{TS}_R^{\Phi\otimes o})^{-1} q^* \arrow[r, "\theta^Q_{\Delta,R}"] \arrow[d, "Q_{E,R}\Delta_R"'] & \overline{p}_!\, (\overline{\mathrm{TS}}_R^{\Phi\otimes o})^{-1} \overline{q}^* Q_{E,R} \arrow[d, "\overline{\Delta}_R"] \\
Q_{E,R}(K_{R,\widehat{c}} \boxtimes K_{R,\widehat{c}'}) \arrow[r, "\theta^Q_{K\boxtimes K,R}"'] & \overline{K}_{R,\widehat{c}} \boxtimes \overline{K}_{R,\widehat{c}'}
\end{tikzcd}$$

commutes on the quotient Borel–Moore correspondence category.

This square is proved by supplying the following finite datum.

(i) quotient-orientation rows for the source stratum, the splitting stack, and the two output strata, with the cocycle comparison

$$q^* \mathfrak{q}_{R,\widehat{c}+\widehat{c}'} = p^*(\mathfrak{q}_{R,\widehat{c}} \boxtimes \mathfrak{q}_{R,\widehat{c}'})$$

in the Picard groupoid;

(ii) the inverse oriented Thom–Sebastiani row

$$(\mathrm{TS}_R^{\Phi\otimes o})^{-1} = (\mathrm{TS}_R^{\Phi,\mathrm{red}})^{-1} \otimes (\mathrm{TS}_R^{o})^{-1}$$

with determinant-square and quotient-descent defects zero;

(iii) external-product descent for the two-output coefficient system $K_{R,\widehat{c}} \boxtimes K_{R,\widehat{c}'}$;

(iv) compact-support pushforward descent for $p_!$, with proper base-change and projection-formula defect ranks zero;

(v) a supplied collision-coproduct row Δ_R, finite coproduct matrix D_R, counit row, and zero coassociativity and counit defects;

(vi) the quotient-after-correspondence comparison $\theta^Q_{\Delta,R}$ and compatibility with the bialgebra square relating Δ_R and m_R.

The current packets do not supply this datum. Row 302 supplies no quotient-orientation rows, row 301 supplies no oriented Thom–Sebastiani row, the compact Hall source packet supplies no D-entries or counit rows, and the quotient-after-correspondence coproduct packet applies only to already supplied quotient-presented coproduct and bialgebra rows. Thus row 304 is presently a coproduct-descent criterion, not a proof that orientation descent commutes with the Hall coproduct.

Proof. The collision coproduct is the pull–inverse-transport–push composite

$$\Delta_R = p_!(\mathrm{TS}_R^{\Phi\otimes o})^{-1}q^*.$$

For this composite to descend, the quotient cocycle on the source object must agree, after pullback to the splitting correspondence, with the external product of the two output quotient cocycles. This is the Picard-groupoid equality displayed above. The coefficient transport is then the inverse of the oriented Thom–Sebastiani isomorphism, not the product orientation row alone. Finally the exceptional pushforward is along p, so the six-functor comparison is the $p_!$-descent row for the splitting correspondence.

Given those rows, functoriality of pullback, inverse Thom–Sebastiani descent, and compact-support descent identify

$$Q_{E,R}\,p_!(\mathrm{TS}_R^{\Phi\otimes o})^{-1}q^* \quad \text{with} \quad \overline{p}_!(\overline{\mathrm{TS}}_R^{\Phi\otimes o})^{-1}\overline{q}^*Q_{E,R},$$

which is the displayed square. Coassociativity and counitality after quotient are the same statement on the retained two-step flag and zero-object rows. The present tables fail before the criterion can be applied: the quotient-orientation tables are empty, the oriented Thom–Sebastiani row is absent, and the source coproduct and counit rows are not populated. Hence only the criterion is proved. □

The packet

```
certificates/orientation/rhomred_orientation_hall_coproduct_descent
```

verifies `RHOMRED_ORIENTATION_HALL_COPRODUCT_DESCENT_OBSTRUCTION_VERIFIED`: it imports the row 302 quotient-orientation descent packet, the row 301 oriented Thom–Sebastiani packet, the row 303 product-descent packet, the quotient-after-correspondence θ^Q_Δ packet, and the transition Hall-coproduct obstruction packet; it records that the current tree has conditional coproduct comparison rows but no quotient-orientation Hall-coproduct descent row.

Proposition 0.108 *(Orientation descent and primitive projection).* [primitive-projection descent criterion proved; primitive orientation rows not supplied] Fix R. Let

$$K_R = \Phi_R^{\mathrm{red}} \otimes o_R$$

be the oriented vanishing-cycle coefficient system on the retained finite Hall object $\mathcal{H}_R(K_R)$. Write

$$\widetilde{\Delta}_R = \Delta_R - \eta_R\epsilon_R \boxtimes \mathrm{id} - \mathrm{id} \boxtimes \eta_R\epsilon_R, \qquad P_R(K_R) = \ker\widetilde{\Delta}_R \cap \ker\epsilon_R$$

and let $\pi_R^{\mathrm{Prim}}\colon \mathcal{H}_R(K_R) \to P_R(K_R)$ be the supplied primitive idempotent. Denote the quotient objects by $\overline{\mathcal{H}}_R(\overline{K}_R)$, $\overline{P}_R(\overline{K}_R)$, and $\overline{\pi}_R^{\mathrm{Prim}}$. Orientation descent commutes with primitive projection precisely when the idempotent square

$$\begin{tikzcd}
Q_{E,R}\mathcal{H}_R(K_R) \arrow[r, "\theta^{Q}_{H,R}"] \arrow[d, "Q_{E,R}\pi_R^{\mathrm{Prim}}"'] & \overline{\mathcal{H}}_R(\overline{K}_R) \arrow[d, "\overline{\pi}_R^{\mathrm{Prim}}"] \\
Q_{E,R}P_R(K_R) \arrow[r, "\theta^{Q,\mathrm{Prim}}_{o,R}"'] & \overline{P}_R(\overline{K}_R)
\end{tikzcd}$$

commutes in the idempotent-complete quotient Borel–Moore correspondence category.

This square is proved by supplying the following finite datum.

(i) quotient-orientation rows for the ambient Hall object, the zero-object stratum, and every splitting correspondence entering $\widetilde{\Delta}_R$;

(ii) the Hall-coproduct orientation-descent datum of Proposition 0.107, together with unit and counit orientation descent;

(iii) finite source and quotient primitive kernel rows

$$P_R(K_R) = \ker(\widetilde{\Delta}_R, \epsilon_R), \qquad \overline{P}_R(\overline{K}_R) = \ker(\widetilde{\overline{\Delta}}_R, \overline{\epsilon}_R),$$

with inclusion and projection matrices for the idempotents π_R^{Prim} and $\overline{\pi}_R^{\mathrm{Prim}}$;

(iv) the quotient-after-correspondence primitive comparison

$$\theta_R^{Q,\mathrm{Prim}}\colon Q_{E,R}P_R(K_R) \xrightarrow{\sim} \overline{P}_R(\overline{K}_R)$$

with zero reduced-coproduct and counit kernel-intertwining defects;

(v) the projection-commutator identity

$$\overline{\pi}_R^{\mathrm{Prim}}\theta^{Q}_{H,R} = \theta^{Q,\mathrm{Prim}}_{o,R} Q_{E,R}\pi_R^{\mathrm{Prim}};$$

(vi) for the pro-object, transition-compatible primitive matrices and zero composition defects for $R' \to R$.

The present packets do not supply this datum. The quotient primitive-projection packet proves compatibility of $\theta^{Q}_{\mu,R}$ only for already supplied finite primitive kernel and projection rows. The compact Hall source packet supplies no D-entries, counit rows, primitive kernels, or primitive projection matrices; the transition primitive-subspace packet records empty transition tables. Hence row 305 is a primitive-projection descent criterion, not a proof that orientation descent commutes with primitive projection.

Proof. The primitive subobject is the kernel of the map

$$(\widetilde{\Delta}_R, \epsilon_R)\colon \mathcal{H}_R(K_R) \longrightarrow \mathcal{H}_R(K_R) \boxtimes \mathcal{H}_R(K_R) \oplus \mathbf{1}.$$

If the coproduct, unit, counit, and orientation coefficient systems descend through $Q_{E,R}$, then applying $Q_{E,R}$ carries this kernel diagram to the quotient kernel diagram. The supplied kernel rows make this statement finite: the reduced-coproduct and counit matrices intertwine with the quotient comparison, so $Q_{E,R}P_R(K_R)$ identifies with $\overline{P}_R(\overline{K}_R)$.

The primitive projection is then an idempotent splitting of this kernel inside the idempotent-complete correspondence category. Two morphisms from $Q_{E,R}\mathcal{H}_R(K_R)$ to $\overline{P}_R(\overline{K}_R)$ are equal once their compositions with the inclusion into $\overline{\mathcal{H}}_R(\overline{K}_R)$ agree. That equality is exactly the kernel-intertwining identity for $\widetilde{\Delta}_R$ and ϵ_R, together with the projection-commutator row. Transition compatibility is a separate strictness condition needed only after passing from one finite height to the pro-object.

The available data stop before this criterion can be applied. Row 304 does not provide an actual oriented Hall-coproduct descent row, the compact source has no coproduct or counit matrices, and the primitive transition packet has no primitive kernel or projection matrices. Thus only the criterion is proved. □

The packet

```
certificates/orientation/rhomred_orientation_primitive_projection_descent
```

verifies `RHOMRED_ORIENTATION_PRIMITIVE_PROJECTION_DESCENT_OBSTRUCTION_VERIFIED`: it imports row 302, row 304, the quotient-after-correspondence primitive-projection packet, the transition primitive-subspace obstruction packet, and the compact Hall source ledger; it records that the present tree has a conditional $\theta^{Q,\mathrm{Prim}}$ comparison but no compact Hall primitive matrices and no orientation primitive-projection descent row.

(O1)$^+$ Weyl-equivariant transport along $W^{(2)}(\Lambda^{2,1}_{II})$

The previous criterion isolates the data needed for descent through the reduced E-quotient. Once such quotient-orientation data are supplied, the Igusa structure still requires compatibility with the Weyl group of the BKM target. For each of the three type-II simple reflections s_{∂_i} ($i = 1, 2, 3$), the $K3 \times E$ side carries a wall transport

$$\tau_i : o_C \longrightarrow s^*_{\partial_i} o_C.$$

The word *coherent* is a hypothesis. At finite height, the first object is the finite partial action groupoid $\mathcal{G}_R$ whose objects are retained oriented quotient strata and whose arrows are the retained type-II wall transports; a choice of determinant-line lifts gives a 2-cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$. Only on a complete $W^{(2)}(\Lambda^{2,1}_{II})$-orbit with constant coefficients does this reduce to the group cohomology class

$$[c_o] \in H^2\big(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2\big).$$

Definition 0.109 *((O1)$^+$ Weyl-equivariant determinant-line transport).* The (O1)$^+$ datum is the chosen set of lifts $\{\tau_i\}$ satisfying $[c_o] = 0$ with chosen 1-cochain killing c_o, the normalization $\tau_i^2 = 1$, and the transport of the quotient-orientation cocycle

$$\mathfrak{q}_C = \big(\alpha^{\mathrm{red}}_C, \alpha^{E,\mathrm{free}}_C, \{(\beta^H_C, \lambda^H_C)\}_{H\in\mathcal{H}(C)}\big)$$

along τ_i between charge strata over Weyl-related Gram degrees. The torsor defect of this transport is

$$\omega_{i,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{s_{\partial_i}C}; \mathbb{F}_2) \oplus \bigoplus_{H\in\mathcal{H}(C)} H^1(BH; \mathbb{F}_2),$$

required to vanish. The Coxeter presentation of $W^{(2)}(\Lambda^{2,1}_{II})$ then assembles these lifts into an honest action on the oriented Pfaffian line over the wall-complement of the reduced compact Hall sector.

$(\mathrm{O1})^+$ is a determinant-line monodromy datum. No scalar formula enters; the transports must move the entire null-trivialization datum, not only the underlying line. The $(\mathrm{O1})^+$ data comprise $[c_o] = 0$, the chosen cochain, the unitarity $\tau_i^2 = 1$, and the torsor vanishing $\omega_{i,C} = 0$ on every retained orbit. In finite coordinates this is recorded by

weyl_lifts, coxeter_cocycles, transitions, scalar_firewall.

The Weyl-lift rows carry $\tau_i^2 = 1$, torsor-defect zero, and quotient-cocycle transport zero. The Coxeter rows carry the vanishing projective cocycle, the cochain killing it, and the type-II Coxeter defect. Transition rows carry strictness and $R^1 \varprojlim = 0$ for orientation lines, quotient data, and Weyl lifts. The scalar firewall excludes scalar trace, squared determinant, OP scalar branch, and Maass-character values as orientation data.

PROPOSITION 0.110 *(Weyl wall transport maps).* [wall-transport construction criterion proved; τ_i rows not supplied] Fix R, a retained quotient stratum S, and a type-II simple reflection $s_{\hat{\partial}_i}$, $i \in \{1, 2, 3\}$. A Weyl wall transport

$$\tau_{i,R,S} \colon o_{R,S} \longrightarrow s_{\hat{\partial}_i}^* o_{R,s_{\hat{\partial}_i} S}$$

is constructed by a finite `weyl_lifts` row with the following payload.

(i) a verified type-II generator row for $s_{\hat{\partial}_i}$ and a retained wall correspondence carrying S to $s_{\hat{\partial}_i} S$;

(ii) source and target Joyce–Upmeier orientation lines

$$o_{R,S}^{\otimes 2} \simeq \det K_{R,S}^{\mathrm{red}}, \qquad o_{R,s_{\hat{\partial}_i} S}^{\otimes 2} \simeq \det K_{R,s_{\hat{\partial}_i} S}^{\mathrm{red}};$$

(iii) a Picard-groupoid lift matrix for $\tau_{i,R,S}$, whose square with the determinant transport commutes:

$$(\tau_{i,R,S})^{\otimes 2} \colon o_{R,S}^{\otimes 2} \longrightarrow s_{\hat{\partial}_i}^* o_{R,s_{\hat{\partial}_i} S}^{\otimes 2}$$

agrees with the determinant transport $\det K_{R,S}^{\mathrm{red}} \to s_{\hat{\partial}_i}^* \det K_{R,s_{\hat{\partial}_i} S}^{\mathrm{red}}$;

(iv) transport of the quotient-orientation cocycle $\mathfrak{q}_{R,S}$ to $s_{\hat{\partial}_i}^* \mathfrak{q}_{R,s_{\hat{\partial}_i} S}$, including the finite-stabilizer Borel classes and the zero linearization characters;

(v) vanishing torsor and quotient-cocycle transport defects

$$\omega_{i,R,S} = 0, \qquad \mathrm{defect}_{q,i,R,S} = 0.$$

The condition $\tau_{i,R,S}^2 = 1$ is a separate involutivity row; it is not part of this construction criterion.

The current data prove only the target and profile prerequisites: the type-II chamber packet supplies the three simple reflections, and the quotient-after-correspondence wall-profile packet proves that the three profiles survive the E-quotient. The reduced-orientation packet has no orientation-line rows and no `weyl_lifts` rows, and the wall-atlas packet has no retained wall-object rows. Thus row 306 is presently a Weyl-wall transport criterion, not a construction of the maps τ_i.

Proof. The map $\tau_{i,R,S}$ is a Picard-groupoid isomorphism, not a scalar sign. Hence the first input is an actual source and target orientation line over Weyl-related quotient strata. The determinant square-root condition forces the displayed square to commute, so the lift preserves the Joyce–Upmeier square root of the reduced determinant complex. The quotient-orientation datum contains the reduced gerbe, the free E-Borel class, the finite-stabilizer Borel classes, and the linearization characters; transporting only the underlying line would lose this data. The vanishing of $\omega_{i,R,S}$ and of the quotient-cocycle defect is exactly the assertion that the entire quotient-orientation package moves across the wall.

The type-II generator and profile-survival rows do not construct this Picard isomorphism. They identify which reflections and quotient profiles must be used. The construction itself starts only when the orientation-line rows, retained wall-object rows, and Weyl-lift rows are populated. Those tables are empty in the current packets, so the criterion is the strongest proved statement. □

The packet

```
certificates/orientation/rhomred_weyl_wall_transport_tau
```

verifies `RHOMRED_WEYL_WALL_TRANSPORT_TAU_OBSTRUCTION_VERIFIED`: it imports the blocked reduced-orientation ledger, the type-II chamber generator packet, the quotient wall-profile survival packet, and the blocked wall-atlas ledger; it records that the three target reflections and quotient profiles are available, but no source orientation-line, wall-object, or τ_i-transport rows are supplied.

Proposition 0.111 *(Involutivity of the Weyl wall transport).* [involutivity criterion proved; τ_i^2 rows not supplied] Fix R, a retained quotient stratum S, and a supplied Weyl wall transport

$$\tau_{i,R,S} \colon o_{R,S} \longrightarrow s_{\delta_i}^* o_{R,s_{\delta_i}S}.$$

The identity $\tau_{i,R,S}^2 = 1$ is the commutativity of the Picard-groupoid square

$$\begin{tikzcd}
o_{R,S} \arrow[r, "\tau_{i,R,S}"] \arrow[dr, "\mathrm{id}"'] & s_{\delta_i}^* o_{R,s_{\delta_i}S} \arrow[d, "s_{\delta_i}^* \tau_{i,R,s_{\delta_i}S}"] \\
& s_{\delta_i}^* s_{\delta_i}^* o_{R,S} \arrow[d, "\kappa_{i,R,S}"] \\
& o_{R,S}
\end{tikzcd}$$

where $\kappa_{i,R,S}$ is the supplied identification of the double-reflected orientation line with $o_{R,S}$ over the equality $s_{\delta_i}^2 = 1$ on the retained wall correspondence.

It is proved by a finite `weyl_lifts` involutivity row with the following data:

(i) a row constructing both lifts $\tau_{i,R,S}$ and $\tau_{i,R,s_{\delta_i}S}$ as in Proposition 0.110;

(ii) the target Coxeter row $s_{\delta_i}^2 = 1$ for the type-II reflection, and the corresponding double-wall source correspondence;

(iii) a determinant-square compatibility row showing that $(s_{\delta_i}^* \tau_{i,R,s_{\delta_i}S}) \circ \tau_{i,R,S}$ squares to the identity transport of $\det K_{R,S}^{\mathrm{red}}$;

(iv) transport of the quotient-orientation cocycle around the two-edge loop with zero reduced-gerbe, free-E, finite-stabilizer, and linearization defects;

(v) a finite matrix row

$$\mathrm{defect}_{\tau^2,i,R,S} = 0$$

recording that the composite equals $\mathrm{id}_{o_{R,S}}$, not only its square on the determinant line.

The present data do not supply these rows. Row 306 records only the criterion for constructing τ_i. The type-II chamber packet records the target relation $s_{\partial_i}^2 = 1$, and the Maass character packet records $\nu_{\Delta_5}(s_{\partial_i})^2 = 1$; neither constructs an orientation-line lift nor proves its square is the identity. Thus row 307 is presently an involutivity criterion, not a proof of $\tau_i^2 = 1$.

Proof. The reflection identity $s_{\partial_i}^2 = 1$ is an identity in the target type-II Coxeter group. The lift $\tau_{i,R,S}$ lives in a Picard groupoid over the retained source stratum. Therefore involutivity is a statement about the two-edge source loop

$$S \longrightarrow s_{\partial_i} S \longrightarrow S$$

and about the induced automorphism of the orientation line over that loop.

The determinant-square row removes the first possible defect: after squaring, the composite lift acts trivially on the determinant complex. This is not enough, because a square root can still differ from the identity by the residual sign automorphism of the orientation line and by the quotient-orientation torsor. The quotient-cocycle transport rows remove the Borel and finite-stabilizer defects. The final τ^2-defect row asserts that the remaining line automorphism is the identity. These are finite matrix and Picard-groupoid checks, not consequences of the scalar character value.

Since the current `weyl_lifts` table is empty, none of the composites in the displayed square is supplied. The available target and automorphic rows identify the relation to be lifted; they do not lift it. Hence only the criterion is proved. □

The packet

`certificates/orientation/rhomred_weyl_tau_square`

verifies `RHOMRED_WEYL_TAU_SQUARE_OBSTRUCTION_VERIFIED`: it imports the row 306 transport criterion, the blocked reduced-orientation ledger, the type-II chamber packet, and the Maass character packet; it records that the target reflection relation and the scalar square value are available, but no τ_i-lift or τ_i^2-defect row is supplied.

Definition 0.112 *(Finite partial action groupoid).* [groupoid datum specified; finite groupoid rows not supplied] At height R, the finite partial action groupoid $\mathcal{G}_R$ is the following finite datum.

(i) The objects are retained quotient-orientation strata S for which the orientation line $o_{R,S}$, quotient cocycle $\mathfrak{q}_{R,S}$, and retained support condition are supplied.

(ii) The generating arrows are the retained type-II wall transports

$$g_{i,R,S} : S \longrightarrow s_{\partial_i} S$$

for $i \in \{1, 2, 3\}$, defined only when S and $s_{\partial_i} S$ both lie in the retained finite window and the row $\tau_{i,R,S}$ of Proposition 0.110 is supplied.

(iii) The identity arrow id_S, inverse arrow

$$g_{i,R,S}^{-1} = g_{i,R,s_{\partial_i} S},$$

and partial compositions

$$g_{i_k,R,S_{k-1}} \circ \cdots \circ g_{i_1,R,S_0}$$

are included exactly when every intermediate stratum S_j is retained and every wall-transport row in the word is supplied.

(iv) The structure tables record source, target, identity, inverse, and associativity defects; all are required to have defect rank zero.

The coefficient system for the orientation cocycle on $\mathcal{G}_R$ is not forced to be constant. It is the local system whose fibre at S is the automorphism group of the complete quotient-orientation package over S. Only on a complete retained orbit with identified fibres may one replace it by the constant $\underline{\mathbb{F}}_2$-system.

The present data do not populate $\mathcal{G}_R$. The type-II chamber packet gives the three target generators and the no-finite-braid relation among distinct generators, and the quotient wall-profile packet gives three surviving profiles. The reduced-orientation packet has no retained object rows, no Weyl-lift rows, and no `coxeter_cocycles` rows. Thus row 308 specifies the finite partial action groupoid datum, but the current tree contains no finite $\mathcal{G}_R$ object, arrow, inverse, or composition tables.

The packet

`certificates/orientation/rhomred_weyl_partial_action_groupoid`

verifies `RHOMRED_WEYL_PARTIAL_ACTION_GROUPOID_OBSTRUCTION_VERIFIED`: it imports the target type-II chamber packet, the row 306 and row 307 Weyl-lift criteria, the quotient wall-profile survival packet, and the blocked reduced-orientation ledger; it records that the target generators are available but the finite groupoid object, arrow, inverse, and composition rows are not supplied.

Definition 0.113 *(Orientation projective cocycle).* [cocycle formula specified; finite $c_{o,R}$ rows not supplied] Assume the finite partial action groupoid $\mathcal{G}_R$ of Definition 0.112 has been populated, and that every arrow $g : S \to T$ in $\mathcal{G}_R$ is equipped with a determinant-line lift

$$\tau_g : o_{R,S} \longrightarrow g^* o_{R,T}$$

transporting the full quotient-orientation package. For every composable pair

$$S \xrightarrow{g} T \xrightarrow{h} U$$

with composite hg, the orientation projective cocycle is the unique element

$$c_{o,R}(h, g; S) \in \mathrm{Aut}_{\mathfrak{Q}^{\mathrm{or}}_{R,S}}(o_{R,S}) \simeq \mathbb{F}_2$$

defined by

$$g^* \tau_h \circ \tau_g = (-1)^{c_{o,R}(h,g;S)}\, \tau_{hg}.$$

Equivalently, $c_{o,R}$ is the finite table measuring the defect between composition of chosen Weyl determinant-line lifts and the chosen lift of the partial product in $\mathcal{G}_R$.

The row is valid only after the following data are supplied:

(i) object, arrow, inverse, and composition rows for $\mathcal{G}_R$;

(ii) determinant-line lifts for g, h, and hg, including the quotient-cocycle transport rows;

(iii) an identified coefficient fibre $\mathrm{Aut}_{\mathfrak{Q}^{\mathrm{or}}_{R,S}}(o_{R,S})$ at the source object;

(iv) a cocycle matrix row recording $c_{o,R}(h, g; S)$ for every composable pair;

(v) a normalization row

$$c_{o,R}(g, \mathrm{id}_S; S) = c_{o,R}(\mathrm{id}_T, g; S) = 0;$$

(vi) a cocycle-defect row

$$dc_{o,R}(k,h,g;S) = 0$$

for every composable triple in the partial groupoid.

The present packets do not provide these rows. The row 308 groupoid packet is empty, the `weyl_lifts` table is empty, and the `coxeter_cocycles` table is empty. Thus row 309 defines the orientation 2-cocycle formula, but no finite cocycle table is currently populated.

The packet

`certificates/orientation/rhomred_weyl_orientation_cocycle`

verifies `RHOMRED_WEYL_ORIENTATION_COCYCLE_OBSTRUCTION_VERIFIED`: it imports the row 308 partial-action-groupoid packet, the row 306 and row 307 lift criteria, and the blocked reduced-orientation ledger; it records that the formula for $c_{o,R}$ is fixed but that the finite composable-pair, lift, and cocycle-defect rows are absent.

PROPOSITION 0.114 *(Vanishing criterion for the orientation cocycle class).* [criterion proved; cohomology-vanishing rows not supplied] Let $\mathcal{G}_R$, the coefficient system $\underline{\mathbb{F}}_{2,o}$, and the normalized cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o})$ be supplied as in Definition 0.113. Choose ordered finite bases for the normalized groupoid cochain spaces

$$C^1_R = C^1(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o}), \qquad C^2_R = C^2(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o}),$$

write the coboundary as a matrix

$$d^1_R : C^1_R \longrightarrow C^2_R,$$

and write the cocycle as a column vector $v(c_{o,R}) \in C^2_R$. Then

$$[c_{o,R}] = 0 \in H^2(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o})$$

if and only if

$$\operatorname{rank}_{\mathbb{F}_2}(d^1_R) = \operatorname{rank}_{\mathbb{F}_2}\left[d^1_R \mid v(c_{o,R})\right].$$

Equivalently, the augmented linear system

$$d^1_R b = v(c_{o,R})$$

is soluble over $\mathbb{F}_2$. This is the row 310 criterion. A supplied rank equality proves the vanishing of the cohomology class; row 311 is the subsequent choice of a particular solution b.

The criterion is not triggered by the target Coxeter graph, by the Maass character, by the relation $s^2_{\delta_i} = 1$, or by an abstract calculation of $H^2(W^{(2)}; \mathbb{F}_2)$ before the retained groupoid, coefficient system, finite orbit, and cocycle vector have been supplied. The present finite tables contain no values of $c_{o,R}$, no normalized cochain-complex bases, no coboundary matrix, and no rank certificate. Hence row 310 remains an open cohomological obstruction.

Proof. For any finite groupoid with coefficients in a local system of $\mathbb{F}_2$-lines, the normalized cochain complex is a finite cochain complex of $\mathbb{F}_2$-vector spaces. A closed 2-cochain represents zero in H^2 exactly when it lies in $\operatorname{im} d^1_R$. After ordered bases are fixed, membership in the image of a matrix is equivalent to equality of the rank of the matrix and the rank of the matrix augmented by the target vector. This proves the displayed criterion. Inspection of `certificates/orientation/rhomred_weyl_orientation_cocycle` and `certificates/orientation/k3e_reduced_orientation/coxeter_cocycles.csv` shows that the required cocycle, complex, coboundary, and rank rows are not present. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_cocycle_class_vanishing
```

verifies `RHOMRED_WEYL_ORIENTATION_COCYCLE_CLASS_VANISHING_OBSTRUCTION_VERIFIED`: it imports the row 309 cocycle packet and the blocked reduced-orientation ledger, records the finite rank criterion for $[c_{o,R}] = 0$, and keeps the cocycle-value, closedness, cochain-complex, coboundary-matrix, rank-certificate, and witness cochain rows open.

Proposition 0.115 *(Cochain trivialization criterion for the orientation cocycle).* [criterion proved; killing 1-cochain not supplied] Assume the data of Proposition 0.114 are supplied and that row 310 has produced a rank certificate for $[c_{o,R}] = 0$. A row 311 cochain trivialization is a named vector

$$b_R \in C^1(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o})$$

such that, in the ordered bases used for row 310,

$$d^1_R b_R = v(c_{o,R}).$$

Equivalently, it is a chosen lift of $v(c_{o,R})$ through the coboundary map d^1_R, together with a defect row

$$\mathrm{def}(b_R) := d^1_R b_R - v(c_{o,R}) = 0$$

over $\mathbb{F}_2$. The class vanishing in row 310 proves that such b_R exists; row 311 chooses one, records its coordinates, and verifies the displayed equation. Compatibility of these choices under $R \to R'$ is row 312, not part of row 311.

The present finite tables do not construct row 311. The row 309 cocycle values are absent, the row 310 rank-certificate rows are absent, and no vector b_R, gauge normalization, or coboundary-defect row is supplied.

Proof. Once ordered bases are fixed, a 1-cochain is a column vector in the finite $\mathbb{F}_2$-vector space C^1_R. The condition that it kill the projective cocycle is the single matrix equation $d^1_R b_R = v(c_{o,R})$. Therefore the construction of b_R is stronger than the rank equality of row 310: it supplies an actual preimage, not only the statement that a preimage exists. The residual freedom is the addition of a 1-cocycle in $\ker d^1_R$, so a complete finite row must also record either a gauge convention or the torsor in which the choice lives. Inspection of the row 309 and row 310 packets shows that the input cocycle, the coboundary matrix, the rank certificate, and the witness vector are not present. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_cocycle_cochain_trivialization
```

verifies `RHOMRED_WEYL_ORIENTATION_COCYCLE_COCHAIN_TRIVIALIZATION_OBSTRUCTION_VERIFIED`: it imports the row 309 cocycle packet and the row 310 class-vanishing packet, records the equation defining the killing 1-cochain, and keeps the cochain-vector, gauge-normalization, and coboundary-defect rows open.

PROPOSITION 0.116 *(Transition compatibility for the Coxeter cochain).* [criterion proved; transition rows not supplied] Let $R' \le R$. Suppose rows 309–311 have supplied $\mathcal{G}_R, \mathcal{G}_{R'}$, the coefficient systems, cocycles, cochain complexes, and chosen cochains b_R and $b_{R'}$. A row 312 transition compatibility datum consists of cochain maps

$$\rho^i_{RR'} : C^i(\mathcal{G}_{R'}; \underline{\mathbb{F}}_{2,o}) \longrightarrow C^i(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o}), \qquad i = 1, 2,$$

induced by the finite-stage transition $t_{RR'}$, such that

$$d^1_R \rho^1_{RR'} = \rho^2_{RR'} d^1_{R'}, \qquad \rho^2_{RR'} v(c_{o,R'}) = v(c_{o,R}),$$

and

$$\rho^1_{RR'}(b_{R'}) = b_R.$$

Equivalently, the transition defect

$$\Delta^b_{RR'} := b_R - \rho^1_{RR'}(b_{R'}) \in C^1(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o})$$

must be zero in the chosen gauge. For a chain $R'' \le R' \le R$, the cochain maps must also satisfy

$$\rho^1_{RR''} = \rho^1_{RR'} \rho^1_{R'R''}, \qquad \rho^2_{RR''} = \rho^2_{RR'} \rho^2_{R'R''}$$

on the retained cochain bases. Thus the b_R form a strict inverse system of cochain trivializations. Compatibility only up to an unrecorded 1-cocycle is not enough; the comparison cocycle or the gauge fixing must be part of the finite row.

The present tables do not prove row 312. No row 311 cochain vectors are supplied, the transition-orientation packet has no Coxeter cochain transition row, and the reduced-orientation transition table is empty.

Proof. The first displayed equation says that the transition maps form a cochain map. The second says that the projective cocycle at level R' pulls back to the projective cocycle at level R. If $d^1_{R'} b_{R'} = v(c_{o,R'})$, these two equations imply

$$d^1_R \rho^1_{RR'}(b_{R'}) = \rho^2_{RR'} d^1_{R'} b_{R'} = \rho^2_{RR'} v(c_{o,R'}) = v(c_{o,R}).$$

Thus both $\rho^1_{RR'}(b_{R'})$ and b_R kill $c_{o,R}$. Row 312 asks that the chosen lifts agree, not merely that both exist. The defect $\Delta^b_{RR'}$ is therefore a closed 1-cochain; strict compatibility is exactly the assertion $\Delta^b_{RR'} = 0$ in the fixed gauge, together with the two-step composition equations for the ρ^i. Inspection of the row 311 packet, the transition-orientation packet, and `certificates/orientation/k3e_reduced_orientation/transitions.csv` shows that none of these finite rows are present. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_cocycle_cochain_transition
```

verifies `RHOMRED_WEYL_ORIENTATION_COCYCLE_COCHAIN_TRANSITION_OBSTRUCTION_VERIFIED`: it imports the row 311 cochain packet and the transition-orientation ledger, records the equations defining compatibility under $R \to R'$, and keeps the transition cochain-map, cocycle-pullback, cochain-defect, gauge, and two-step coherence rows open.

Proposition 0.117 *(Computation criterion for the Weyl torsor defects).* [criterion proved; torsor-defect rows not supplied] Fix R, a retained quotient stratum S, and a simple type-II reflection s_{∂_i}. Suppose the row 306 lift $\tau_{i,R,S}$ and the quotient-orientation packages

$$\mathfrak{q}_{R,S}, \qquad \mathfrak{q}_{R,s_{\partial_i}S}$$

are supplied by explicit Cech–Borel representatives, null trivialisations, finite-stabilizer edge representatives, and linearization characters. Let

$$\Theta^1_{i,R,S} : \boldsymbol{Q}^1_{R,S} \longrightarrow \boldsymbol{Q}^1_{R,s_{\partial_i}S}$$

be the degree-one cochain map induced by $\tau_{i,R,S}$ on the quotient-orientation complex

$$\boldsymbol{Q}^\bullet_{R,S} := C^\bullet(\mathfrak{M}^{\mathrm{red}}_{R,S}; \mathbb{F}_2) \oplus \bigoplus_{H \in \mathcal{H}(R,S)} C^\bullet(BH; \mathbb{F}_2),$$

with the connected E-free summand already killed by its row 284 null-trivialization. If $\eta_{R,S} \in \boldsymbol{Q}^1_{R,S}$ and $\eta_{R,s_{\partial_i}S} \in \boldsymbol{Q}^1_{R,s_{\partial_i}S}$ denote the chosen quotient-orientation null trivialisations, the torsor-defect cochain is

$$\Delta^\omega_{i,R,S} := \Theta^1_{i,R,S}(\eta_{R,S}) - \eta_{R,s_{\partial_i}S}.$$

After the degree-two quotient representatives have been transported with zero defect, $\Delta^\omega_{i,R,S}$ is closed. The row 313 torsor defect is the cohomology class

$$\omega_{i,R,S} := [\Delta^\omega_{i,R,S}] \in H^1(\mathfrak{M}^{\mathrm{red}}_{R,s_{\partial_i}S}; \mathbb{F}_2) \oplus \bigoplus_{H \in \mathcal{H}(R,s_{\partial_i}S)} H^1(BH; \mathbb{F}_2),$$

recorded in a finite basis for the displayed H^1-space. Thus row 313 computes the defect vector; row 314 is the separate assertion that this vector is zero for every retained i, R, S.

The present finite tables do not compute row 313. They contain no row 306 Weyl lifts, no quotient Borel representatives, no finite-stabilizer transport rows, no linearization transport rows, no $\boldsymbol{Q}^\bullet$-basis rows, and no torsor-defect vector rows.

Proof. The quotient orientation is not a single line: it is the orientation line together with null trivialisations of the reduced gerbe, the free-E Borel class, the finite-stabilizer Borel classes, and the zero linearization characters. Transport along $\tau_{i,R,S}$ therefore gives cochain maps on the Cech–Borel complexes carrying these representatives. If the degree-two representatives are transported with zero defect, the difference between the transported source null-trivialization and the target null-trivialization has zero coboundary; hence it defines the displayed H^1-class. Coordinates of this class in a chosen finite basis are exactly the torsor-defect computation. A target Weyl relation, Maass-character value, or local Pfaffian sign has no such cochain representative and cannot compute $\omega_{i,R,S}$. Inspection of the reduced-orientation and row 306 packets shows that the required rows are absent. □

The packet

`certificates/orientation/rhomred_weyl_orientation_torsor_defects`

verifies `RHOMRED_WEYL_ORIENTATION_TORSOR_DEFECTS_OBSTRUCTION_VERIFIED`: it imports the row 306 Weyl-wall transport criterion, the row 312 cochain-transition packet, and the blocked reduced-orientation ledger; it records the finite cochain formula for $\omega_{i,R,S}$ and keeps the quotient-orientation representative, transport, H^1-basis, and defect-vector rows open.

PROPOSITION 0.118 *(Vanishing criterion for the Weyl torsor defects).* [criterion proved; zero-defect rows not supplied] Assume the row 313 data of Proposition 0.117 have been supplied for every retained triple (i, R, S). Thus each $\Delta^{\omega}_{i,R,S}$ is a closed cochain in the target quotient-orientation complex and has coordinates

$$v(\omega_{i,R,S}) \in H^1(\mathfrak{M}^{\mathrm{red}}_{R,s_{\partial_i}S}; \mathbb{F}_2) \oplus \bigoplus_{H \in \mathcal{H}(R,s_{\partial_i}S)} H^1(BH; \mathbb{F}_2)$$

in a fixed finite basis. Row 314 is proved exactly when the finite vanishing table contains, for every retained triple (i, R, S), the following four rows:

$$d\Delta^{\omega}_{i,R,S} = 0, \qquad v(\omega_{i,R,S}) = 0, \qquad \operatorname{rank}\big(v(\omega_{i,R,S})\big) = 0, \qquad \operatorname{cov}\{(i, R, S)\} = \operatorname{cov}_{\mathrm{ret}}.$$

Equivalently, the row 313 defect vectors must be present, closed, computed in the displayed H^1-basis, and zero with complete coverage over the retained Weyl-wall atlas. Under these hypotheses the Weyl lift $\tau_{i,R,S}$ preserves the quotient-orientation torsor, and $\omega_{i,R,S} = 0$ for all retained i, R, S.

The present finite tables do not prove row 314. The row 313 packet contains no torsor-defect vector rows, no H^1-basis rows, no zero-vector certificates, and no coverage rows for retained triples. Therefore the vanishing of $\omega_{i,R,S}$ remains an open orientation obligation.

Proof. In the fixed H^1-basis, a closed defect cochain represents the zero torsor class if and only if its coordinate vector is zero. Thus the four displayed finite checks are precisely the statement that each class $\omega_{i,R,S}$ vanishes, together with the assertion that no retained type-II wall has been omitted. This is the needed source-side orientation statement: it says that the transported null trivialisation equals the target null trivialisation in the quotient orientation torsor. A target Weyl relation, a Maass character, a Pfaffian wall sign, or an empty defect table is not a coordinate calculation in the displayed H^1-space. Inspection of the row 313 packet shows that the required defect vectors and zero certificates are not supplied. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_torsor_defect_vanishing
```

verifies `RHOMRED_WEYL_ORIENTATION_TORSOR_DEFECT_VANISHING_OBSTRUCTION_VERIFIED`: it imports the row 313 torsor-defect computation criterion, records the finite zero-vector criterion for row 314, and keeps the computed defect-vector, closedness, zero-certificate, and retained coverage rows open.

PROPOSITION 0.119 *(Weyl preservation criterion for the reduced gerbe).* [criterion proved; preservation rows not supplied] Fix R, a retained quotient stratum S, and a simple type-II reflection s_{∂_i}. Suppose the row 306 lift $\tau_{i,R,S}$ is supplied, and suppose row 280 supplies Cech–Borel cocycles

$$a^{\mathrm{red}}_{R,S} \in Z^2(\mathcal{B}_{R,S}; \mathbb{F}_2), \qquad a^{\mathrm{red}}_{R,s_{\partial_i}S} \in Z^2(\mathcal{B}_{R,s_{\partial_i}S}; \mathbb{F}_2)$$

representing $\alpha^{\mathrm{red}}_{R,S}$ and $\alpha^{\mathrm{red}}_{R,s_{\partial_i}S}$. Let

$$\Theta^{\bullet,\mathrm{red}}_{i,R,S} \colon C^\bullet(\mathcal{B}_{R,S}; \mathbb{F}_2) \longrightarrow C^\bullet(\mathcal{B}_{R,s_{\partial_i}S}; \mathbb{F}_2)$$

be the cochain map induced by the Weyl transport on the reduced Cech–Borel groupoids. The row 315 preservation defect is

$$\Delta^{\mathrm{red}}_{i,R,S} := \Theta^{2,\mathrm{red}}_{i,R,S}(a^{\mathrm{red}}_{R,S}) - a^{\mathrm{red}}_{R,s_{\partial_i}S}.$$

Weyl transport preserves the reduced gerbe on this retained wall exactly when there is a finite cochain

$$u^{\mathrm{red}}_{i,R,S} \in C^1(\mathcal{B}_{R,s_{\partial_i}S}; \mathbb{F}_2)$$

with

$$du^{\mathrm{red}}_{i,R,S} = \Delta^{\mathrm{red}}_{i,R,S};$$

equivalently,

$$[\Delta^{\mathrm{red}}_{i,R,S}] = 0 \in H^2(\mathcal{B}_{R,s_{\partial_i}S}; \mathbb{F}_2).$$

The finite row proving row 315 must record the source and target α^{red}-cocycles, the $\Theta^{2,\mathrm{red}}$ matrix, the transported defect vector, the coboundary witness $u^{\mathrm{red}}_{i,R,S}$, a zero class-rank certificate, and coverage of every retained triple (i, R, S). If strict representative preservation is imposed as the chosen gauge, the same row must record the stronger condition $\Delta^{\mathrm{red}}_{i,R,S} = 0$.

The present finite tables do not prove row 315. The row 280 packet contains no computed α^{red}-values, the row 281 packet contains no Cech–Borel null-homotopies, the row 306 packet contains no Weyl-lift cochain maps, and row 314 has not proved vanishing of the quotient-orientation torsor defects.

Proof. Because $\Theta^{\bullet,\mathrm{red}}_{i,R,S}$ is required to be a cochain map, the displayed defect is closed whenever the source and target representatives are closed and the degree-three transport defect is zero. The transported reduced gerbe class equals the target reduced gerbe class precisely when this closed defect is a coboundary. A chosen cochain $u^{\mathrm{red}}_{i,R,S}$ with $du^{\mathrm{red}}_{i,R,S} = \Delta^{\mathrm{red}}_{i,R,S}$ is the finite certificate of equality in H^2. Complete retained-wall coverage is part of the statement because $O1^+$ is a Weyl-equivariant orientation assertion, not a single-wall computation. A target Weyl relation, a scalar multiplier, a Pfaffian sign, or the uncomputed vanishing of the full torsor defect is not a cochain calculation for α^{red}. Inspection of the row 280, row 281, row 306, and row 314 packets shows that the needed rows are absent. $\square$

The packet

```
certificates/orientation/rhomred_weyl_alpha_red_preservation
```

verifies `RHOMRED_WEYL_ALPHA_RED_PRESERVATION_OBSTRUCTION_VERIFIED`: it imports the reduced-gerbe, reduced-gerbe null-homotopy, Weyl-wall transport, and torsor-defect vanishing packets; it records the finite cochain criterion for row 315; and it keeps the source-target α^{red}-cocycle, transport matrix, coboundary witness, zero class-rank, and retained coverage rows open.

Proposition 0.120 *(Weyl preservation criterion for the connected free E-Borel class).* [criterion proved; preservation rows not supplied] Fix R, a retained free E-quotient stratum S, and a simple type-II reflection s_{∂_i}. Suppose the row 306 lift $\tau_{i,R,S}$ is supplied and suppose row 282 supplies connected BE-edge representatives

$$a^{E,\mathrm{free}}_{R,S}, \qquad a^{E,\mathrm{free}}_{R,s_{\partial_i}S} \in Z^2(BE; \mathbb{F}_2)$$

with coordinates

$$\alpha^{E,\mathrm{free}}_{R,S} = a_1(R,S)u_1 + a_2(R,S)u_2, \qquad \alpha^{E,\mathrm{free}}_{R,s_{\partial_i}S} = a_1(R,s_{\partial_i}S)u_1 + a_2(R,s_{\partial_i}S)u_2$$

in the basis of Lemma 4.11. Let

$$M^{E}_{i,R,S} \in \mathrm{GL}_2(\mathbb{F}_2)$$

be the induced map of the Weyl transport on $H^2(BE;\mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2$, computed from the cochain-level transport of the connected E-translation rigidification. The row 316 preservation defect is the coordinate vector

$$\Delta^{E,\mathrm{free}}_{i,R,S} := M^E_{i,R,S}\begin{pmatrix} a_1(R,S) \\ a_2(R,S)\end{pmatrix} - \begin{pmatrix} a_1(R,s_{\partial_i}S) \\ a_2(R,s_{\partial_i}S)\end{pmatrix} \in \mathbb{F}_2^2.$$

Weyl transport preserves the connected free E-Borel class on this retained wall exactly when

$$\Delta^{E,\mathrm{free}}_{i,R,S} = 0.$$

Equivalently, in a cochain model one may record a degree-one witness for the difference between the transported source representative and the target representative; after passing to the connected BE-edge, this is the same as the displayed zero coordinate vector, since u_1, u_2 form a basis of $H^2(BE;\mathbb{F}_2)$.

The finite row proving row 316 must therefore contain the source and target a_1, a_2 coefficient rows, the induced matrix $M^E_{i,R,S}$, the transported coordinate vector, the zero-vector certificate, the underlying cochain-transport defect if strict representatives are used, and coverage of every retained triple (i, R, S). The present finite tables do not prove row 316: rows 282, 283, and 284 supply no connected BE-edge coefficients, zero checks, or null-trivialisations; row 306 supplies no Weyl cochain map; and row 315 has not proved preservation of α^{red}.

Proof. The connected part of the quotient orientation is controlled by the degree-two edge of $H^*_E(-;\mathbb{F}_2)$. In degree two the connected BE-edge has the fixed basis u_1, u_2. Therefore a Weyl transport preserves $\alpha^{E,\mathrm{free}}$ if and only if the transported coordinate vector equals the target coordinate vector. The displayed equation is exactly this equality in $\mathbb{F}_2^2$. If the comparison is made before passage to the edge, the difference of cocycle representatives must be recorded as a coboundary; its edge coordinate is the same defect vector. Freeness of the E-action, $H^1(BE;\mathbb{F}_2) = 0$, target Weyl relations, Maass-character values, and Pfaffian signs do not compute the $H^2(BE)$ coordinate transport. Inspection of the row 282, row 283, row 284, row 306, and row 315 packets shows that the required finite rows are absent. □

The packet

```
certificates/orientation/rhomred_weyl_alpha_e_free_preservation
```

verifies `RHOMRED_WEYL_ALPHA_E_FREE_PRESERVATION_OBSTRUCTION_VERIFIED`: it imports the connected free E-Borel computation, vanishing, null-trivialization, Weyl-wall transport, and reduced-gerbe preservation packets; it records the finite coordinate criterion for row 316; and it keeps the source-target a_1, a_2 rows, induced $H^2(BE)$-matrix, zero-vector certificate, cochain-transport defect, and retained coverage rows open.

PROPOSITION 0.121 *(Weyl preservation criterion for the finite-stabilizer Borel classes).* [criterion proved; preservation rows not supplied] Fix R, a retained finite-stabilizer stratum S, and a simple type-II reflection s_{∂_i}. Suppose the row 306 lift $\tau_{i,R,S}$ is supplied and identifies the source and target finite stabilizers by an isomorphism

$$\varphi_{i,R,S}\colon H_{R,S} \xrightarrow{\sim} H_{R,s_{\partial_i}S}.$$

Suppose row 285 has supplied classifying-space edge representatives

$$\beta^{H_{R,S}}_{R,S} \in H^2(BH_{R,S};\mathbb{F}_2), \qquad \beta^{H_{R,s_{\partial_i}S}}_{R,s_{\partial_i}S} \in H^2(BH_{R,s_{\partial_i}S};\mathbb{F}_2),$$

with coordinates in the appropriate odd, Klein-four, or two-primary basis of Proposition 0.87. Let

$$M^H_{i,R,S}\colon H^2(BH_{R,S};\mathbb{F}_2) \longrightarrow H^2(BH_{R,s_{\partial_i}S};\mathbb{F}_2)$$

be the matrix induced by $B\varphi_{i,R,S}$ on the chosen classifying-space bases. The row 317 preservation defect is

$$\Delta^{\beta}_{i,R,S} := M^H_{i,R,S}\, v(\beta^{H_{R,S}}_{R,S}) - v(\beta^{H_{R,s_{\partial_i}S}}_{R,s_{\partial_i}S}).$$

Weyl transport preserves the finite-stabilizer Borel class on this retained wall exactly when

$$\Delta^{\beta}_{i,R,S} = 0$$

for every retained triple (i, R, S). If the comparison is made at the Cech–Borel representative level, the row must also record the transport of $\widetilde{\beta}^H$, the edge-reduction compatibility, and a zero cochain-level transport defect before taking the displayed classifying-space coordinate vector.

Thus a row proving 317 must contain the source and target finite-stabilizer orientation rows, the stabilizer isomorphism $\varphi_{i,R,S}$, the induced $H^2(BH)$-matrix, the transported β^H-coordinate vector, a zero-vector certificate, edge-reduction compatibility, and complete retained-wall coverage. The present finite tables do not prove row 317: row 285 has no finite-stabilizer orientation or edge rows, row 286 has no $\beta^H = 0$ rows, row 287 has no compatible null-trivializations, row 306 has no Weyl lift on stabilizers, and row 316 has not proved preservation of $\alpha^{E,\mathrm{free}}$.

Proof. The finite-stabilizer component of the quotient orientation is the classifying-space edge of the H-equivariant Borel class $\widetilde{\beta}^H$. A Weyl lift can preserve this component only after it identifies the source and target stabilizer groups and carries the source edge representative to the target edge representative. In the fixed group-cohomology bases this condition is precisely the vanishing of the displayed coordinate defect. For odd or trivial stabilizers the target space is zero only after the finite-stabilizer edge row has been supplied; for rank-two two-primary stabilizers the mixed coefficient is part of the basis and cannot be read from cyclic restrictions. A residual group order, a scalar trace, a Maass character, a Pfaffian sign, or preservation of the connected free E-component is not a finite-stabilizer edge transport calculation. Inspection of the row 285, row 286, row 287, row 306, and row 316 packets shows that the required finite rows are absent. □

The packet

```
certificates/orientation/rhomred_weyl_finite_stabilizer_beta_preservation
```

verifies `RHOMRED_WEYL_FINITE_STABILIZER_BETA_PRESERVATION_OBSTRUCTION_VERIFIED`: it imports the finite-stabilizer Borel computation, vanishing, null-trivialization, Weyl-wall transport, and connected free E-preservation packets; it records the finite edge-coordinate criterion for row 317; and it keeps the stabilizer isomorphism, source-target β^H-coordinates, $H^2(BH)$-matrix, edge-transport defect, zero-vector certificate, and retained coverage rows open.

Proposition 0.122 *(Weyl preservation criterion for the zero finite-stabilizer linearization).* [criterion proved; linearization-transport rows not supplied] Fix R, a retained finite-stabilizer stratum S, and a simple type-II reflection s_{∂_i}. Suppose the row 306 lift $\tau_{i,R,S}$ is supplied and identifies the source and target finite stabilizers by an isomorphism

$$\varphi_{i,R,S}\colon H_{R,S} \xrightarrow{\sim} H_{R,s_{\partial_i}S}.$$

Suppose row 288 has supplied the finite-stabilizer linearization characters

$$\lambda_{R,S}^{H_{R,S}} \in H^1(BH_{R,S};\mathbb{F}_2), \qquad \lambda_{R,s_{\partial_i}S}^{H_{R,s_{\partial_i}S}} \in H^1(BH_{R,s_{\partial_i}S};\mathbb{F}_2),$$

with coordinates in the bases of Proposition 0.90, and suppose row 289 has supplied the corresponding zero rows. Let

$$N_{i,R,S}^H \colon H^1(BH_{R,S};\mathbb{F}_2) \longrightarrow H^1(BH_{R,s_{\partial_i}S};\mathbb{F}_2)$$

be the matrix induced by $B\varphi_{i,R,S}$ on the chosen degree-one classifying-space bases. The row 318 linearization-transport defect is

$$\Delta_{i,R,S}^{\lambda} := N_{i,R,S}^H\, v(\lambda_{R,S}^{H_{R,S}}) - v(\lambda_{R,s_{\partial_i}S}^{H_{R,s_{\partial_i}S}}).$$

Weyl transport preserves the zero finite-stabilizer linearization on this retained wall exactly when

$$\Delta_{i,R,S}^{\lambda} = 0$$

for every retained triple (i, R, S). If the comparison is made at the orientation-line action level, the row must also record the transport of the sign action of $H_{R,S}$ to $H_{R,s_{\partial_i}S}$ and a zero action-transport defect before passing to $H^1(BH;\mathbb{F}_2)$.

Thus a row proving 318 must contain the source and target λ^H-coordinate rows, the stabilizer isomorphism $\varphi_{i,R,S}$, the induced $H^1(BH)$-matrix, the transported coordinate vector, a rank-zero defect row, orientation-line action transport, and complete retained-wall coverage. The present finite tables do not prove row 318: row 288 has no orientation-line action rows, generator-basis rows, character values, or computed λ^H-coordinates; row 289 has no zero-coordinate rows; row 306 has no Weyl lift on finite stabilizers; and row 317 has not proved preservation of the degree-two finite-stabilizer Borel classes.

Proof. After the degree-two finite-stabilizer gerbe has been trivialized, the remaining descent ambiguity of the orientation line is the sign character

$$\lambda^H \in H^1(BH;\mathbb{F}_2) = \mathrm{Hom}(H,\mathbb{F}_2).$$

A Weyl lift can preserve the zero character only after it identifies the source and target stabilizer groups and transports the source sign action to the target sign action. In the chosen $H^1(BH)$-bases, this condition is exactly the vanishing of the displayed coordinate defect. If rows 288 and 289 have supplied zero characters on both sides, the class-level equality follows from functoriality of group cohomology, but the transport still has no mathematical meaning without the stabilizer isomorphism, the induced degree-one matrix, and the action-level comparison. The identity $H^1(B1;\mathbb{F}_2) = 0$, a vanishing β^H-class, a null-trivialization of β^H, a connected E-Borel calculation, a Maass-character value, a Pfaffian wall sign, or a scalar trace does not compute this degree-one transport. Inspection of the row 288, row 289, row 306, and row 317 packets shows that the required finite rows are absent. □

The packet

`certificates/orientation/rhomred_weyl_finite_stabilizer_lambda_zero_preservation`

verifies `RHOMRED_WEYL_FINITE_STABILIZER_LAMBDA_ZERO_PRESERVATION_OBSTRUCTION_VERIFIED`: it imports the finite-stabilizer linearization-character, linearization-vanishing, Weyl-wall transport, and finite-stabilizer Borel-preservation packets; it records the degree-one coordinate criterion for row 318; and it keeps the stabilizer isomorphism, source-target λ^H-coordinates, $H^1(BH)$-matrix, orientation-line action transport, coordinate defect, rank-zero row, and retained coverage rows open.

Definition 0.123 *(Orientation character of the strictified Weyl transport).* [definition specified; finite character rows not supplied] Assume rows 306–318 have supplied a strictified $W^{(2)}(\Lambda^{2,1}_{II})$-transport of the complete quotient-oriented determinant-line package on a retained finite height R: the finite partial action groupoid $\mathcal{G}_R$, the lifts $\tau_{i,R,S}$, the cochain b_R killing the projective cocycle $c_{o,R}$, transition compatibility of b_R, vanishing torsor defects, and preservation of α^{red}, $\alpha^{E,\mathrm{free}}$, β^H, and $\lambda^H = 0$. For each simple type-II reflection s_{δ_i} and retained source stratum S, let

$$\sigma^o_{i,R,S} \in \{\pm 1\}$$

be the residual sign of the strictified lift on the oriented one-dimensional Pfaffian square-root line after the target line and the complete quotient-orientation package have been identified with the transported source package. A row defining the finite-stage orientation character must prove that $\sigma^o_{i,R,S}$ is independent of S on every retained connected wall component and is compatible under $R' \le R$. Write the resulting sign as $\sigma^o_{i,R}$.

The finite-stage orientation character is then the homomorphism

$$\epsilon_{o,R} : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}$$

defined on a reduced word

$$w = s_{\delta_{i_1}} \cdots s_{\delta_{i_m}}$$

by

$$\epsilon_{o,R}(w) := \prod_{a=1}^{m} \sigma^o_{i_a,R}.$$

The type-II Coxeter presentation has only the relations

$$s^2_{\delta_i} = 1, \qquad i = 1, 2, 3,$$

and no braid relations between distinct simple reflections. Hence the displayed formula is well-defined once the generator signs are supplied, since $(\sigma^o_{i,R})^2 = 1$. A compatible inverse system of finite characters defines

$$\epsilon_o = \varprojlim_R \epsilon_{o,R}.$$

This is a definition from source-side orientation transport. It is not the automorphic Maass character ν_{Δ_5}, not the determinant character of the target Coxeter group, not the OP scalar branch, and not a consequence of the scalar trace. Row 320 computes the generator values $\epsilon_o(s_{\delta_i})$; row 321 compares the resulting character with the determinant character; row 322 compares it with ν_{Δ_5}. The present finite tables do not define $\epsilon_{o,R}$: row 312 has no compatible cochain b_R, row 314 has no zero torsor-defect rows, rows 315–318 do not prove preservation of the four quotient-orientation components, and no generator-sign or transition-compatible character row is supplied.

The packet

```
certificates/orientation/rhomred_weyl_orientation_character_definition
```

verifies `RHOMRED_WEYL_ORIENTATION_CHARACTER_DEFINITION_OBSTRUCTION_VERIFIED`: it imports the Coxeter-cochain transition, torsor-defect vanishing, and four component-preservation packets; it records the word-evaluation definition of $\epsilon_{o,R}$; it imports the automorphic Maass character only as a firewall; and it keeps the strict Weyl action, generator-sign, stratum-independence, transition-compatibility, finite-character, and inverse-limit character rows open.

PROPOSITION 0.124 *(Simple-wall sign criterion for the orientation character).* [local sign formula proved; finite simple-wall sign rows not supplied] Assume the finite-stage orientation character of Definition 0.123 has been supplied. Fix a simple type-II reflection s_{∂_i}, a retained generic wall stratum $S \subset \mathbb{H}_{\partial_i}$, and a retained ($O2_{\mathrm{atlas}}$) wall chart with real normal coordinate u_i. Suppose the chart supplies the rank-one normal Pfaffian crossing

$$K^{\mathrm{nor}}_{\partial_i,R,S} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}], \qquad \mathrm{Pf}(K^{\mathrm{nor}}_{\partial_i,R,S}) = v_{i,R,S}u_i,$$

with

$$s_{\partial_i}(u_i) = -u_i, \qquad s_{\partial_i}(v_{i,R,S}) = v_{i,R,S}, \qquad N^{\mathrm{Pf}}_{\partial_i,R,S} = 1.$$

Then the source-side generator sign of row 320 is

$$\sigma^{o}_{i,R,S} = \frac{s^{*}_{\partial_i}(v_{i,R,S}u_i)}{v_{i,R,S}u_i} = -1 = (-1)^{N^{\mathrm{Pf}}_{\partial_i,R,S}}.$$

If the row also supplies stratum-independence and transition compatibility for these signs, then

$$\epsilon_{o,R}(s_{\partial_i}) = -1, \qquad i = 1, 2, 3,$$

and the compatible inverse-limit character satisfies

$$\epsilon_{o}(s_{\partial_i}) = -1, \qquad i = 1, 2, 3.$$

Thus a row proving 320 must contain, for all three simple type-II walls, retained wall-object rows, rank-one wall-chart rows, the normal coordinate flip $u_i \mapsto -u_i$, invariant tangent Pfaffian units, normal rank 1, divisor order 1, local sign rows, the row 319 orientation-character row, stratum-independence rows, and transition compatibility rows. The present finite tables do not prove row 320: the row 319 packet has no constructed orientation character or generator-sign rows, the ($O2_{\mathrm{atlas}}$) packet has no wall-object, rank-one chart, coordinate-flip, invariant-unit, or local-sign rows, and the finite Pfaffian packet has no type-II wall chart or sign rows.

Proof. The local calculation is the rank-one Pfaffian crossing of Proposition 5.2. On the retained chart the tangent Pfaffian unit is invariant and the reflection changes only the normal coordinate:

$$s^{*}_{\partial_i}(v_{i,R,S}u_i) = v_{i,R,S}(-u_i) = -v_{i,R,S}u_i.$$

The normal complex contains one odd crossing, so $N^{\mathrm{Pf}}_{\partial_i,R,S} = 1$ and the sign is $(-1)^1 = -1$. Squaring removes this sign:

$$s^{*}_{\partial_i}((v_{i,R,S}u_i)^2) = (v_{i,R,S}u_i)^2.$$

Hence squared determinants, scalar traces, and OP scalar branches cannot recover the orientation character. The Maass character value $\nu_{\Delta_5}(s_{\partial_i}) = -1$ is an automorphic target value; it does not supply the source wall chart, invariant unit, normal-rank row, or generator-sign row. Inspection of the row 319, ($O2_{\mathrm{atlas}}$), and finite Pfaffian packets shows that those finite rows are absent. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_simple_sign_computation
```

verifies `RHOMRED_WEYL_ORIENTATION_SIMPLE_SIGN_COMPUTATION_OBSTRUCTION_VERIFIED`: it imports the orientation-character definition, the $(O2_{\mathrm{atlas}})$ wall-atlas ledger, the finite Pfaffian ledger, and the automorphic Maass character as a firewall; it records the rank-one formula for row 320; and it keeps the wall-object, rank-one chart, invariant-unit, coordinate-flip, local-sign, generator-sign, stratum-independence, and transition-compatibility rows open.

Proposition 0.125 *(Determinant criterion for the orientation character).* [group-theoretic criterion proved; source generator signs not supplied] Let

$$\det_W : W^{(2)}(\Lambda_{II}^{2,1}) \longrightarrow \{\pm 1\}$$

be the determinant character of the reflection representation on $\Lambda_{II}^{2,1} \otimes_{\mathbb{Z}} \mathbb{R}$. Suppose the source-side finite-stage character $\epsilon_{o,R}$ of Definition 0.123 has been constructed and the row 320 finite sign rows prove

$$\epsilon_{o,R}(s_{\partial_i}) = -1, \qquad i = 1, 2, 3.$$

Then

$$\epsilon_{o,R} = \det_W \quad \text{on} \quad W^{(2)}(\Lambda_{II}^{2,1}).$$

If the finite characters are transition-compatible, then the inverse limit character satisfies

$$\epsilon_o = \det_W \quad \text{on} \quad W^{(2)}(\Lambda_{II}^{2,1}).$$

The present finite data do not prove row 321. The row 319 packet does not construct $\epsilon_{o,R}$, and the row 320 packet contains no generator-sign rows. The automorphic packet records $\nu_{\Delta_5}|_{W^{(2)}} = \det_W$, but that is a target Maass character computation, not a source orientation-character computation.

Proof. Each s_{∂_i} is a real reflection in the retained rank-three Lorentzian reflection representation, hence

$$\det_W(s_{\partial_i}) = -1, \qquad i = 1, 2, 3.$$

The type-II Coxeter presentation used in Definition 0.123 has generators $s_{\partial_1}, s_{\partial_2}, s_{\partial_3}$, relations $s_{\partial_i}^2 = 1$, and no braid relation between distinct generators. Therefore a $\{\pm 1\}$-valued character is determined by its three generator values. If row 320 supplies the three source values $\epsilon_{o,R}(s_{\partial_i}) = -1$, then $\epsilon_{o,R}$ and $\det_W$ agree on the generators and hence on all words in $W^{(2)}(\Lambda_{II}^{2,1})$. The same argument passes to the inverse limit once the transition-compatibility rows for $\epsilon_{o,R}$ are supplied. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_determinant_comparison
```

verifies `RHOMRED_WEYL_ORIENTATION_DETERMINANT_COMPARISON_OBSTRUCTION_VERIFIED`: it imports the row 319 orientation-character definition, the row 320 simple-sign criterion, and the automorphic determinant comparison as a firewall; it records the determinant criterion for row 321; and it keeps the source character, source generator signs, equality-on-generators, transition compatibility, and inverse-limit comparison rows open.

PROPOSITION 0.126 *(Maass-character criterion for the orientation character).* [comparison criterion proved; source determinant comparison not supplied] Assume the row 321 source comparison has been supplied:

$$\epsilon_o = \det_W \quad \text{on} \quad W^{(2)}(\Lambda_{II}^{2,1}).$$

Assume also the automorphic Maass character computation for the Borcherds–Gritsenko–Nikulin square root,

$$\nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})} = \det_W.$$

Then

$$\epsilon_o = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The present finite data do not prove row 322. The Maass character packet proves the automorphic equality $\nu_{\Delta_5}|_{W^{(2)}} = \det_W$, but the row 321 packet does not prove the source equality $\epsilon_o = \det_W$. Therefore the Maass character cannot be used as a substitute for the missing source-side orientation character or the missing type-II wall sign rows.

Proof. The assertion is the transitivity of equality of characters on $W^{(2)}(\Lambda_{II}^{2,1})$. If the source orientation character has already been identified with $\det_W$, and the automorphic Maass character restricts to the same determinant character, then both characters have identical values on every element of $W^{(2)}(\Lambda_{II}^{2,1})$. The second equality is automorphic target data. The first equality is the source-side obstruction in row 321 and is not supplied by the current packets. □

The packet

```
certificates/orientation/rhomred_weyl_orientation_maass_comparison
```

verifies `RHOMRED_WEYL_ORIENTATION_MAASS_COMPARISON_OBSTRUCTION_VERIFIED`: it imports the row 321 determinant-comparison criterion and the automorphic Maass-character packet; it records the transitivity criterion for row 322; and it keeps the source determinant comparison, inverse-limit source character, and orientation-versus-Maass comparison rows open.

PROPOSITION 0.127 *(S_3-chamber automorphisms have trivial Maass character).* [proved; automorphic character only] The restriction of the Maass character of the Igusa square root to the type-I chamber automorphism group is trivial:

$$\nu_{\Delta_5}\big|_{\mathrm{Aut}(\mathcal{P}_{II})} = 1, \qquad \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3.$$

Equivalently,

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_{-2}-f_3}) = +1.$$

This is an automorphic multiplier statement. It does not decide whether ϵ_o extends from $W^{(2)}(\Lambda_{II}^{2,1})$ to $W^{(2)}(\Lambda_{II}^{2,1}) \rtimes S_3$; that is a separate source-side orientation-transport problem.

Proof. Lemma 2.3 identifies $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ and shows that the three type-I reflections

$$s_{f_{-2}-f_2}, \qquad s_{f_2-f_3}, \qquad s_{f_{-2}-f_3}$$

act as the three transpositions of the type-II walls. These transpositions generate S_3. The Maass multiplier computation of Lemma 5.3, equivalently the automorphic packet `certificates/automorphic/delta5_maass_character`, gives value +1 on each of the three type-I chamber reflections. Since ν_{Δ_5} is a character, it is +1 on every product of these generators, hence trivial on $\mathrm{Aut}(\mathcal{P}_{II})$. □

The packet

```
certificates/automorphic/delta5_s3_chamber_maass_triviality
```

verifies `DELTA5_S3_CHAMBER_MAASS_TRIVIALITY_VERIFIED`: it imports the Maass-character packet, records the three type-I chamber transpositions and their +1 values, checks the chamber-automorphism relation row, and excludes any conclusion about the S_3-extension of the compact orientation character.

Proposition 0.128 *(Semidirect extension decision for the orientation character).* [abstract character criterion proved; source semidirect Hall lift not supplied] Let $W = W^{(2)}(\Lambda^{2,1}_{II})$ and $\Sigma = \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$. The group Σ acts on the simple reflections of W by permutation. Suppose the source-side orientation character $\epsilon_o : W \to \{\pm 1\}$ has been constructed and satisfies

$$\epsilon_o(s_{\eth_1}) = \epsilon_o(s_{\eth_2}) = \epsilon_o(s_{\eth_3}) = -1.$$

Then ϵ_o is Σ-invariant and therefore extends as an abstract character to $W \rtimes \Sigma$. The extensions are

$$\widetilde{\epsilon}_\chi(w, \sigma) = \epsilon_o(w)\chi(\sigma), \qquad \chi \in \mathrm{Hom}(\Sigma, \{\pm 1\}) = \{1, \mathrm{sgn}\}.$$

The unique extension whose restriction to Σ equals the Maass character of row 323 is the trivial-Σ extension

$$\widetilde{\epsilon}_{\mathrm{Maass}}(w, \sigma) = \epsilon_o(w).$$

This is only the abstract character decision. A compact-source extension of the orientation character requires semidirect Hall lifts of the chamber-permuting type-I reflections, their action on the orientation line, determinant square-root compatibility, preservation of the quotient-orientation null-trivialisations, and the semidirect relations. The present finite packets do not supply those rows.

Proof. The chamber action of Lemma 2.3 sends $s_{\eth_i}$ to $s_{\eth_{\sigma(i)}}$. Hence, under the displayed hypothesis,

$$\epsilon_o(\sigma s_{\eth_i} \sigma^{-1}) = \epsilon_o(s_{\eth_{\sigma(i)}}) = -1 = \epsilon_o(s_{\eth_i})$$

for every generator $s_{\eth_i}$ and every $\sigma \in \Sigma$. Since the type-II Coxeter presentation has only the involutive relations $s_{\eth_i}^2 = 1$ and no braid relations, this equality on the generators gives Σ-invariance of ϵ_o. A character on W extends to the semidirect product precisely after this invariance condition holds; any two extensions differ by a character of Σ. Since $\mathrm{Hom}(S_3, \{\pm 1\}) = \{1, \mathrm{sgn}\}$, there are exactly two abstract extensions. Proposition 0.127 shows that ν_{Δ_5} is trivial on Σ, so the Maass-compatible abstract extension is the one with $\chi = 1$. $\square$

The packet

```
certificates/orientation/rhomred_weyl_orientation_s3_extension_decision
```

verifies `RHOMRED_WEYL_ORIENTATION_S3_EXTENSION_DECISION_OBSTRUCTION_VERIFIED`: it imports the determinant-comparison criterion and the S_3 Maass triviality packet; it records the two abstract semidirect character extensions and the Maass-compatible choice; and it keeps the source-side semidirect Hall lifts, orientation-line transport, square-root compatibility, quotient-orientation transport, semidirect relations, and compact-source extension rows open.

PROPOSITION 0.129 *(Semidirect Hall-lift construction criterion).* [criterion proved; semidirect Hall lifts not supplied] Let $W = W^{(2)}(\Lambda_{II}^{2,1})$ and $\Sigma = \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$. A compact-source lift of the abstract extension of Proposition 0.128 is a finite-stage action of $W \rtimes \Sigma$ on the retained compact Hall correspondence package. It consists of:

(i) the populated type-II partial action groupoid for W, with orientation-line arrows $\tau_{i,R,S}$ and zero projective Coxeter defects;

(ii) for the three chamber transpositions $\sigma \in \Sigma$, Hall correspondences

$$\mathfrak{H}_{\sigma,R,S} : \mathcal{M}_{R,S} \longleftarrow \mathcal{Z}_{\sigma,R,S} \longrightarrow \mathcal{M}_{R,\sigma S}$$

over the retained quotient stacks, together with source and target maps on the Hall product, Hall coproduct, unit, counit, primitive subspace, pairing, radical, and PBW filtration rows;

(iii) Picard-groupoid isomorphisms

$$\theta^{o}_{\sigma,R,S} : o_{R,S} \xrightarrow{\sim} \sigma^* o_{R,\sigma S}$$

whose squares are the determinant-line transports;

(iv) transport of α^{red}, $\alpha^{E,\mathrm{free}}$, $\widetilde{\beta}^H$, β^H, and $\lambda^H = 0$ through the same chamber correspondences;

(v) zero-defect rows for the S_3 relations and for the semidirect conjugation relations

$$\theta^{o}_{\sigma,R,\delta_i}\, \tau_{i,R,S}\, (\theta^{o}_{\sigma,R,\delta_i})^{-1} = \tau_{\sigma(i),R,\sigma S};$$

(vi) transition compatibility in R and $R^1 \varprojlim = 0$ for the semidirect Hall and Picard-groupoid data.

If these rows are supplied, the trivial-S_3 abstract extension of row 324 is represented by compact-source orientation transport on $W \rtimes S_3$. The present finite packets do not supply these rows: the compact Hall source packet is empty blocked, the type-II partial action groupoid has no object or arrow rows, the τ_i packet has no orientation-line transport rows, and the transition orientation packet has no Picard-groupoid transition rows.

Proof. The listed data are exactly the data of an action of the semidirect product in the oriented Hall correspondence 2-category. Items 1 and 2 give the underlying source correspondences on the retained Hall stacks. Item 3 lifts the action from line bundles to Joyce–Upmeier orientations by enforcing the determinant square-root diagram. Item 4 says that the quotient-orientation structure is transported with the orientation line. Item 5 is the group law: the S_3 relations define the chamber part, while the conjugation relations identify the chamber transport of the type-II Weyl lifts. Item 6 is the inverse-system condition needed to pass from finite height to the pro-object. Without these rows there is only an abstract character extension, not a compact-source Hall action. □

The packet

`certificates/orientation/rhomred_weyl_semidirect_hall_lifts`

verifies `RHOMRED_WEYL_SEMIDIRECT_HALL_LIFTS_OBSTRUCTION_VERIFIED`: it imports the row 324 extension decision, the compact Hall source obstruction ledger, the type-II partial-action-groupoid ledger, the τ_i-transport ledger, and the transition-orientation ledger; it records the semidirect Hall-lift construction criterion; and it keeps the source Hall correspondences, orientation transports, determinant square-root rows, quotient-orientation transports, semidirect relation rows, transition rows, and inverse-limit rows open.

Proposition 0.130 *(Imaginary divisor monodromy is not Weyl monodromy).* [firewall proved; local divisor monodromy not yet defined] The orientation character

$$\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}$$

is a real-root reflection character. Its domain is generated by the three real simple reflections

$$s_{\delta_1}, \qquad s_{\delta_2}, \qquad s_{\delta_3}.$$

An imaginary simple root of the Borcherds–Kac–Moody denominator algebra contributes a multiplicity in the denominator formula; it does not define an element of $W^{(2)}(\Lambda^{2,1}_{II})$. Hence monodromy around a divisor component indexed by an imaginary root, if such monodromy is constructed in the compact Hall theory, is not a Weyl character value and is not a value of ϵ_o.

Thus no imaginary-divisor monodromy is being called Weyl monodromy in the orientation-character construction. Row 327 defines local divisor monodromy as a separate datum; row 328 proves that this separate datum is irrelevant to the Maass character ν_{Δ_5} on $W^{(2)}(\Lambda^{2,1}_{II})$.

Proof. Definition 3.1 defines the real-root system as the $W^{(2)}(\Lambda^{2,1}_{II})$-orbit of $\{\delta_1, \delta_2, \delta_3\}$, and the type-II Weyl group is generated by the associated real simple reflections. The imaginary simple roots are the Borcherds correction recorded in Table 4.1: they are absent from the rank-three Kac–Moody real-root datum and enter $\mathfrak{g}_{\Delta_5}$ by the multiplicities $m(a)$ and $\tau(a)$ in the denominator formula. A Weyl reflection is attached to a real root of norm 2; an imaginary simple root contributes a denominator factor and a root space, not a reflection generator. Therefore the notation $\epsilon_o(w)$ is meaningful only for $w \in W^{(2)}(\Lambda^{2,1}_{II})$, and no loop around an imaginary divisor component can be read as such a value without adding a new definition. That new definition is isolated in row 327. □

The packet

```
certificates/orientation/rhomred_weyl_imaginary_divisor_firewall
```

verifies `RHOMRED_WEYL_IMAGINARY_DIVISOR_FIREWALL_VERIFIED`: it imports the real-root definition, the BKM real/imaginary comparison, the Chapter 6 imaginary-root firewall, and the row 325 lift criterion; it records that real-root reflections are the only Weyl generators; and it keeps local divisor monodromy outside the Weyl orientation character until row 327 defines it.

Definition 0.131 *(Local divisor monodromy datum).* [definition specified; local divisor rows not supplied] Fix a finite height R and a retained compact Hall stratum $\mathcal{M}_{R,S}$. A local divisor monodromy datum with label η consists of:

(i) a retained effective Cartier divisor or normal-crossing component

$$\mathcal{D}_{R,S,\eta} \subset \mathcal{M}_{R,S}$$

together with the row identifying the source of the label η; if η is an imaginary Borcherds label, the label is only a divisor or multiplicity label and not a Weyl-reflection generator;

(ii) an analytic or formal normal tube $N_{R,S,\eta}$ of $\mathcal{D}_{R,S,\eta}$ in $\mathcal{M}_{R,S}$, its punctured tube

$$N^{\times}_{R,S,\eta} := N_{R,S,\eta} \setminus \mathcal{D}_{R,S,\eta},$$

and a base point $x_{R,S,\eta} \in N^{\times}_{R,S,\eta}$;

(iii) a positively oriented meridian class

$$\gamma_{R,S,\eta} \in \pi_1(N^{\times}_{R,S,\eta}, x_{R,S,\eta});$$

(iv) the oriented reduced coefficient system

$$K_R := \Phi_R^{\mathrm{red}} \otimes o_R$$

restricted to $N^{\times}_{R,S,\eta}$, and, when the Pfaffian line is separately supplied, the corresponding rank-one Pfaffian-line factor;

(v) the monodromy representation of this local system and the operator

$$T_{R,S,\eta} := \rho_{K_R}(\gamma_{R,S,\eta}) \in \mathrm{Aut}\big((K_R)_{x_{R,S,\eta}}\big).$$

When a rank-one Pfaffian-line summand is supplied and preserved by the meridian action, its sign is denoted $\mu_{R,S,\eta} \in \{\pm 1\}$;

(vi) for every successor $R' \leq R$, transition rows identifying the divisor components, normal tubes, meridians, coefficient systems, and monodromy operators.

The finite-stage local divisor monodromy is the partially defined assignment

$$(R, S, \eta) \longmapsto T_{R,S,\eta}$$

on the supplied rows. Its domain is a set of local compact Hall divisor data, not $W^{(2)}(\Lambda^{2,1}_{II})$. It is not a value of ϵ_0, not a value of the Maass character ν_{Δ_5}, and not determined by the Borcherds multiplicity attached to η. Without the divisor component, normal tube, meridian, oriented coefficient system, and transition rows above, no local divisor monodromy value is defined.

The packet

```
certificates/orientation/rhomred_local_divisor_monodromy_definition
```

verifies `RHOMRED_LOCAL_DIVISOR_MONODROMY_DEFINITION_OBSTRUCTION_VERIFIED`: it imports the row 326 firewall, the compact Hall obstruction ledger, the reduced-orientation obstruction ledger, the reduced vanishing-cycle transition ledger, and the finite Pfaffian obstruction ledger; it records Definition 0.131; and it keeps the divisor-component, normal-tube, meridian, coefficient-system, Pfaffian-line, monodromy-operator, and transition rows open.

PROPOSITION 0.132 *(Local divisor monodromy is irrelevant to the Maass character).* [proved as a domain-separation theorem] Let ν_{Δ_5} be the Maass character of the automorphic Igusa square root, with values on the type-II Weyl generators and on the S_3 chamber automorphisms as recorded by the automorphic Maass-character packet. Let $(R, S, \eta) \mapsto T_{R,S,\eta}$ be any supplied local divisor monodromy datum in the sense of Definition 0.131. Then the values of ν_{Δ_5} on

$$W^{(2)}(\Lambda^{2,1}_{II}) \quad \text{and} \quad \mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$$

are independent of the operators $T_{R,S,\eta}$.

More precisely, there is no expression

$$\nu_{\Delta_5}(\gamma_{R,S,\eta})$$

in the present data. Such an expression would require an additional comparison map from the local meridian groupoid of the compact Hall divisor rows to the automorphic or Weyl group on which ν_{Δ_5} is defined. No such comparison map is part of Definition 0.131, and no such row is supplied by the current packets.

Proof. The Maass character is automorphic target data. Its recorded inputs are the type-II real-root reflections s_{∂_i} and the chamber automorphisms in $\mathrm{Aut}(\mathcal{P}_{II})$; the packet `certificates/automorphic/delta5_maass_character` records these values and contains no compact Hall source rows. By Definition 0.131, local divisor monodromy is instead a meridian action on the restricted oriented coefficient system $K_R = \Phi_R^{\mathrm{red}} \otimes o_R$ over a punctured normal tube of a retained compact Hall divisor component. Its input is the meridian class $\gamma_{R,S,\eta}$, not an element of $W^{(2)}(\Lambda_{II}^{2,1})$ and not a chamber automorphism.

Therefore the Maass-character computation has no argument on which a local divisor monodromy value can be evaluated. Changing or supplying operators $T_{R,S,\eta}$ while keeping the automorphic Maass packet fixed changes only source-side local-system data; it does not change the automorphic multiplier rows defining ν_{Δ_5}. A comparison would require a new meridian-to-automorphic map and a proof that the Maass character pulls back along it. Those rows are absent. □

The packet

`certificates/orientation/rhomred_local_divisor_monodromy_maass_irrelevance`

verifies `RHOMRED_LOCAL_DIVISOR_MONODROMY_MAASS_IRRELEVANCE_VERIFIED`: it imports Definition 0.131, the row 322 Maass-comparison criterion, and the automorphic Maass-character packet; it records the domain-separation theorem; and it keeps any meridian-to-automorphic comparison map, local monodromy operator rows, and row 329 simple-wall orientation computations separate.

Proposition 0.133 *(Orientation on the three simple wall charts).* [chart-level computation criterion proved; finite chart rows not supplied] For $i = 1, 2, 3$, let a retained $(O2_{\mathrm{atlas}})$ chart over the simple type-II wall $\mathbb{H}_{\partial_i}$ supply the data

$$(U_{i,R,S}, T_{i,R,S}, u_i, \mathcal{L}^{\tan}_{i,R,S}, v_{i,R,S}, o_{i,R,S})$$

with

$$U_{i,R,S} = T_{i,R,S} \times (-\varepsilon, \varepsilon), \qquad s_{\partial_i}(t, u_i) = (t, -u_i),$$

rank-one normal Pfaffian block

$$K^{\mathrm{cross}}_{i,R,S} = [\mathbb{R} \xrightarrow{u_i} \mathbb{R}], \qquad \mathrm{pf}^{\mathrm{loc}}_{i,R,S} = v_{i,R,S} u_i,$$

divisor order 1, and invariant tangent Pfaffian unit

$$s_{\partial_i}(v_{i,R,S}) = v_{i,R,S}.$$

Then the source orientation sign on each of the three supplied simple wall charts is

$$\omega^{\mathrm{chart}}_{i,R,S} := \frac{s^*_{\partial_i}\, \mathrm{pf}^{\mathrm{loc}}_{i,R,S}}{\mathrm{pf}^{\mathrm{loc}}_{i,R,S}} = -1, \qquad i = 1, 2, 3.$$

Equivalently, the three-chart orientation vector is

$$(\omega^{\mathrm{chart}}_{1,R,S}, \omega^{\mathrm{chart}}_{2,R,S}, \omega^{\mathrm{chart}}_{3,R,S}) = (-1, -1, -1)$$

on the supplied rows.

The present finite packets do not supply the three rows to which this calculation applies. The $(O2_{\mathrm{atlas}})$ packet has no retained wall objects and no wall-chart rows; the finite Pfaffian packet has no type-II wall chart, Pfaffian line, invariant unit, or local sign row; and the row 320 packet contains only the abstract rank-one sign criterion, not populated chart-orientation rows.

Proof. The computation is componentwise. On the i-th supplied rank-one chart the reflection fixes the tangent factor and sends the normal coordinate to its negative. Hence

$$s_{\delta_i}^*(v_{i,R,S}u_i) = v_{i,R,S}(-u_i) = -v_{i,R,S}u_i.$$

The quotient of the transported local Pfaffian section by the original section is therefore -1. The same calculation holds for $i = 1, 2, 3$. The assertion is a source chart-orientation computation: it uses the normal coordinate, invariant tangent unit, divisor order, and retained quotient orientation on the chart. By Proposition 0.132, no Maass-character value or local divisor monodromy datum supplies these rows. □

The packet

`certificates/orientation/rhomred_three_simple_wall_chart_orientation`

verifies `RHOMRED_THREE_SIMPLE_WALL_CHART_ORIENTATION_OBSTRUCTION_VERIFIED`: it imports the row 320 simple-wall sign criterion, the $(O2_{\mathrm{atlas}})$ wall-atlas ledger, the finite Pfaffian ledger, and the row 328 Maass-irrelevance firewall; it records the three-chart orientation computation criterion for row 329; and it keeps the three retained chart rows, invariant-unit rows, quotient orientation rows, local sign rows, and transition rows open.

PROPOSITION 0.134 *(Orientation on the isotropic a_{ij}-charts).* [chart-level parity criterion proved; source charts not supplied] For $1 \le i < j \le 3$, let $a_{ij} := \delta_i + \delta_j$ be the primitive isotropic boundary ray of the type-II chamber. Suppose a retained compact Hall chart $\mathcal{U}_{ij,R,S}$ in degree a_{ij} supplies:

(i) a source decomposition of the oriented reduced coefficient block into 10 even and 0 odd one-dimensional summands, compatible with the target $a_{ij} : 10|0$ row;
(ii) the quotient-orientation line and its determinant-square row on $\mathcal{U}_{ij,R,S}$;
(iii) identification of the single affine $A_1^{(1)}$ null-root direction and the nine Gritsenko–Nikulin isotropic directions $u_{ij,1}, \ldots, u_{ij,9}$;
(iv) transition compatibility for $R' \le R$.

Then the local isotropic chart orientation parity is even and the chart sign is

$$\omega_{ij,R,S}^{\mathrm{iso}} = (-1)^{\dim K_{1,ij,R,S}^{\mathrm{iso}}} = (-1)^0 = +1.$$

Thus the three supplied isotropic chart rows would give

$$(\omega_{12,R,S}^{\mathrm{iso}}, \omega_{13,R,S}^{\mathrm{iso}}, \omega_{23,R,S}^{\mathrm{iso}}) = (+1, +1, +1).$$

The present packets do not supply these source chart rows. The `type_ii_chamber_isotropic` and `wle3_target_parity` packets certify the target arithmetic $a_{ij} : 10|0$, the decomposition $10 = 1 + 9$, and the three isotropic rays a_{12}, a_{13}, a_{23}; they do not construct compact Hall source charts, determinant-square orientation rows, or transition rows. Consequently row 330 is presently a chart-orientation criterion, not an unconditional source orientation computation.

Proof. The target $W_{\leq 3}$ parity table records

$$a_{ij} : 10|0, \qquad i < j,$$

and the chamber-isotropic packet records the decomposition of each primitive ray into the affine null-root contribution 1 and the Gritsenko–Nikulin isotropic contribution 9. If a source compact Hall chart realizes these ten directions as even oriented summands and supplies no odd summand, the parity exponent of the local determinant orientation is the odd rank. This rank is 0. Hence the sign is $(-1)^0 = +1$ for each of the three isotropic charts. The computation uses source chart and orientation rows; the target parity table alone does not provide them. □

The packet

```
certificates/orientation/rhomred_isotropic_aij_chart_orientation
```

verifies `RHOMRED_ISOTROPIC_AIJ_CHART_ORIENTATION_OBSTRUCTION_VERIFIED`: it imports the type-II chamber-isotropic packet, the $W_{\leq 3}$ target-parity packet, the row 329 simple-wall chart-orientation packet, and the compact Hall source obstruction ledger; it records the isotropic chart-orientation criterion for row 330; and it keeps the compact source chart, quotient-orientation, determinant-square, isotropic-source parity, and transition rows open.

Proposition 0.135 *(Orientation on the δ_{123}-chart).* [chart-level parity criterion proved; source chart not supplied] Let

$$\delta_{123} := \delta_1 + \delta_2 + \delta_3.$$

Suppose a retained compact Hall chart $\mathcal{U}_{123,R,S}$ in degree δ_{123} supplies:

(i) a source decomposition of the oriented reduced coefficient block into 29 even and 93 odd one-dimensional summands, compatible with the target $\delta_{123} : 29|93$ row;
(ii) source rows realizing the two real–real–real directions, the 27 mixed real–isotropic directions, and the 93 odd imaginary-simple directions of Proposition 6.7;
(iii) the quotient-orientation line and its determinant-square row on $\mathcal{U}_{123,R,S}$;
(iv) transition compatibility for $R' \leq R$.

Then the local first-timelike chart orientation parity is odd and the chart sign is

$$\omega^{\mathrm{time}}_{123,R,S} = (-1)^{\dim K^{\mathrm{time}}_{1,123,R,S}} = (-1)^{93} = -1.$$

The present packets do not supply the source chart to which this calculation applies. The `delta123_presentation_split` and `wle3_target_parity` packets certify the target presentation count $29|93$, with $29 = 2 + 27$ and $29 - 93 = -64$; they do not construct a compact Hall source chart, a source parity decomposition, a quotient orientation, a determinant-square row, or transition rows. Consequently row 331 is presently a chart-orientation criterion, not an unconditional source orientation computation.

Proof. Proposition 6.7 and the finite target packet `certificates/targets/delta5_gn_kac/delta123_presentation_split` compute the first timelike target presentation:

$$\mathrm{mult}_{\bar{0}}(\delta_{123}) = 29, \qquad \mathrm{mult}_{\bar{1}}(\delta_{123}) = 93, \qquad 29 - 93 = -64.$$

The even part is the direct presentation count $2 + 27$: two real–real–real directions after the Jacobi relation and twenty-seven mixed real–isotropic brackets. The odd part consists of the ninety-three negative-norm imaginary-simple directions. If a compact Hall chart realizes this decomposition as the source reduced coefficient block and supplies the determinant-square orientation row, the parity exponent of the local determinant orientation is the odd rank. That rank is 93. Hence the chart sign is $(-1)^{93} = -1$. The target presentation count alone is not a source orientation row. □

The packet

```
certificates/orientation/rhomred_delta123_chart_orientation
```

verifies `RHOMRED_DELTA123_CHART_ORIENTATION_OBSTRUCTION_VERIFIED`: it imports the target ∂_{123} presentation-split packet, the $W_{\leq 3}$ target-parity packet, the compact Hall source obstruction ledger, the finite Pfaffian obstruction ledger, and the row 330 isotropic chart-orientation firewall; it records the ∂_{123} chart-orientation criterion for row 331; and it keeps the compact source chart, source parity, source presentation decomposition, quotient-orientation, determinant-square, and transition rows open.

PROPOSITION 0.136 *(Orientation-line compatibility with the* 29|93 *parity split).* [line-level compatibility criterion proved; source parity and orientation line not supplied] Let $\mathcal{U}_{123,R,S}$ be a retained compact Hall chart in degree ∂_{123}, and let

$$K^{\mathrm{red}}_{123,R,S}$$

be the corresponding reduced coefficient block. Suppose the following source data are supplied:

(i) a parity involution $J_{123,R,S}$ on $K^{\mathrm{red}}_{123,R,S}$, with eigenspace decomposition

$$K^{\mathrm{red}}_{123,R,S} = K_{\bar{0},123,R,S} \oplus K_{\bar{1},123,R,S}, \qquad \dim K_{\bar{0},123,R,S} = 29, \quad \dim K_{\bar{1},123,R,S} = 93;$$

(ii) source comparison rows identifying the even summand with the two real–real–real directions and the 27 mixed real–isotropic directions, and identifying the odd summand with the 93 odd imaginary-simple directions of the target ∂_{123}-presentation;

(iii) a Joyce–Upmeier orientation line $o_{123,R,S}$ and a commuting determinant-square diagram

$$o^{\otimes 2}_{123,R,S} \xrightarrow{\sim} \det K^{\mathrm{red}}_{123,R,S} \xrightarrow{\sim} \det K_{\bar{0},123,R,S} \otimes \left(\det K_{\bar{1},123,R,S}\right)^{-1};$$

(iv) quotient-orientation null-trivialisations on the chart, compatible with the parity splitting;

(v) transition compatibility: for every $R' \leq R$, the transition map commutes with J_{123}, preserves the determinant-square diagram above, and has zero Picard-groupoid, parity-commutator, off-diagonal, composition, and $R^1 \varprojlim$ defects.

Then the orientation line is compatible with the 29|93 parity split. In this case the determinant parity exponent is the source odd rank 93, so the local orientation sign on the first timelike chart is

$$(-1)^{\dim K_{\bar{1},123,R,S}} = (-1)^{93} = -1.$$

The present packets do not supply these hypotheses. The target $\partial_{123} : 29|93$ presentation is verified, and Proposition 0.135 records the local chart criterion, but the compact Hall source packet has no parity block rows, the reduced-orientation packet has no orientation-line square-root rows, the transition-parity packet has no parity involution or block-diagonal transition rows, and the transition-orientation packet has no Picard-groupoid transport rows. Hence row 332 is presently a compatibility criterion and obstruction ledger, not a proof that the orientation line is compatible with the 29|93 split.

Proof. The determinant functor on a $\mathbb{Z}/2$-graded finite block gives the Berezinian factorisation

$$\det K^{\mathrm{red}}_{123,R,S} \simeq \det K_{\bar{0},123,R,S} \otimes \left(\det K_{\bar{1},123,R,S}\right)^{-1}.$$

A Joyce–Upmeier orientation on the chart is a square root of this determinant line in the Picard groupoid, not a scalar dimension. If the displayed square diagram commutes, the chosen orientation line is an orientation of the parity-split determinant line itself. The target presentation count 29|93 identifies the required ranks; the source comparison rows in item 2 assert that these target ranks are the actual source parity ranks on $\mathcal{U}_{123,R,S}$. The local sign is then the parity of the odd determinant factor, namely $(-1)^{93} = -1$. The transition rows in item 5 make the statement independent of the finite HN height and prevent a parity split at one stage from being read as a pro-object orientation. □

The packet

```
certificates/orientation/rhomred_delta123_parity_compatibility
```

verifies `RHOMRED_DELTA123_PARITY_COMPATIBILITY_OBSTRUCTION_VERIFIED`: it imports the row 331 ∂_{123}-chart criterion, the reduced determinant-line packet, the square-root obstruction ledger, the compact Hall source obstruction ledger, the transition-parity obstruction ledger, and the transition-orientation obstruction ledger; it records the line-level compatibility criterion for row 332; and it keeps the source parity block, parity involution, source-to-target comparison, orientation square-root, determinant-square, quotient-orientation, parity-transition, and Picard-groupoid transition rows open.

Proposition 0.137 *(Orientation-line compatibility with negative roots).* [negative-root compatibility criterion proved; source pairing and orientation transport not supplied] Let γ be a retained positive degree whose target row carries a formal negative-dual block, and let $-\gamma$ denote the corresponding negative root degree. Suppose the compact Hall source supplies:

(i) retained source charts $\mathcal{U}_{\gamma,R,S}$ and $\mathcal{U}_{-\gamma,R,S}$ and a compact-source duality morphism

$$\mathbb{D}_{\gamma,R,S} : \mathcal{U}_{\gamma,R,S} \longrightarrow \mathcal{U}_{-\gamma,R,S}$$

lifting the retained finite-window dual closure;

(ii) parity decompositions of

$$K^{\mathrm{red}}_{\gamma,R,S}, \qquad K^{\mathrm{red}}_{-\gamma,R,S}$$

with equal even ranks and equal odd ranks, identified by the positive–negative target pairing block for γ;

(iii) a nondegenerate source Hopf-pairing matrix

$$G_{\gamma,R,S} : K^{\mathrm{red}}_{\gamma,R,S} \otimes K^{\mathrm{red}}_{-\gamma,R,S} \longrightarrow \mathbb{C}$$

whose off-parity blocks vanish and whose radical is the recorded source radical;

(iv) orientation square roots $o_{\gamma,R,S}$ and $o_{-\gamma,R,S}$, with determinant-square diagrams commuting with the duality isomorphism

$$\det K^{\mathrm{red}}_{-\gamma,R,S} \simeq \left(\det K^{\mathrm{red}}_{\gamma,R,S}\right)^{-1} \otimes \varepsilon^{\mathrm{CY}_3}_{\gamma,R,S},$$

where $\varepsilon^{\mathrm{CY}_3}_{\gamma,R,S}$ is the supplied Calabi–Yau threefold Serre-pairing sign line;

(v) quotient-orientation null-trivialisations and transition rows preserving the duality morphism, the pairing matrix, the parity decompositions, and the two orientation square roots.

Then the reduced orientation line is compatible with the negative root $-\gamma$: the negative-root orientation is the Serre-dual Picard-groupoid transport of the positive-root orientation, and the positive–negative pairing is even and nondegenerate modulo the recorded radical.

The present packets do not supply these hypotheses. The retained dual-closure packet proves only that the finite HN window is closed under duals. The target presentation packet records formal positive–negative blocks, but every such row has `source_pairing=false`. The compact Hall source packet has no G-entries, Hopf-pairing identities, radical rows, source negative-root parity blocks, or Chevalley anti-involution rows; the orientation packets have no square-root row, no quotient-orientation transport row, and no Picard-groupoid transition row. Hence row 333 is presently a negative-root compatibility criterion and obstruction ledger, not a proof of negative-root orientation compatibility.

Proof. The proof is formal once the listed source data are supplied. The pairing matrix $G_{\gamma,R,S}$ identifies the negative source block with the dual of the positive source block after quotienting by the recorded radical. Since its off-parity blocks vanish, the identification respects the even and odd determinant factors. The displayed determinant-duality isomorphism then identifies the determinant line of the negative block with the dual determinant line of the positive block, up to the supplied Calabi–Yau Serre sign line. Commutativity of the square-root diagrams lifts this determinant identity to the Joyce–Upmeier orientation lines. The quotient null-trivialisations make the statement one on the reduced quotient chart, and the transition rows make the positive–negative identification a statement in the pro-object rather than at a single finite height. No target formal pairing block supplies the source pairing matrix, radical, or orientation-line transport. □

The packet

```
certificates/orientation/rhomred_negative_root_orientation_compatibility
```

verifies `RHOMRED_NEGATIVE_ROOT_ORIENTATION_COMPATIBILITY_OBSTRUCTION_VERIFIED`: it imports the retained dual-closure packet, the target presentation packet, the formal Hall-pairing pushforward packet, the compact Hall source obstruction ledger, the row 332 parity-compatibility packet, and the transition-orientation obstruction ledger; it records the negative-root orientation-compatibility criterion for row 333; and it keeps the compact source negative-root charts, source positive–negative pairing matrices, Hopf-pairing identities, radical quotient rows, source parity blocks, orientation square roots, Serre-duality sign line, quotient-orientation transport, and Picard-groupoid transition rows open.

PROPOSITION 0.138 *(Chevalley anti-involution on orientation data).* [anti-involution criterion proved; source maps not supplied] Let $P^{\mathrm{red}}_{R,S}$ be the reduced compact Hall primitive quotient at finite stage (R, S), with retained positive and negative degree pieces and Joyce–Upmeier orientation lines $o_{\gamma,R,S}$. Suppose the compact Hall source supplies:

(i) source simple representatives e_i, f_i, and h_i, including the imaginary simple generators and the recorded parity blocks, with e_i and f_i lying in opposite retained degrees and h_i in the Cartan part;

(ii) an even involutive linear map
$$\vartheta_{R,S} : P^{\mathrm{red}}_{R,S} \longrightarrow P^{\mathrm{red}}_{R,S}$$
which sends $P^{\mathrm{red}}_{\gamma,R,S}$ to $P^{\mathrm{red}}_{-\gamma,R,S}$, fixes the Cartan part, sends $e_i \leftrightarrow f_i$, and preserves the source parity decomposition;

(iii) source bracket and relation rows showing

$$\vartheta_{R,S}([x,y]) = [\vartheta_{R,S}(y), \vartheta_{R,S}(x)]$$

for homogeneous primitive classes, together with zero-defect Cartan, Chevalley, real-Serre, Borcherds-orthogonality, and super-sign relation matrices after quotienting by the recorded radical;

(iv) a source positive–negative Hopf pairing for which $\vartheta_{R,S}$ is adjoint, the radical is stable under $\vartheta_{R,S}$, and the quotient pairing is nondegenerate and parity preserving;

(v) orientation-line isomorphisms in the Picard groupoid

$$\vartheta^{o}_{\gamma,R,S} : o_{\gamma,R,S} \xrightarrow{\sim} o^{\vee}_{-\gamma,R,S} \otimes \varepsilon^{\mathrm{CY}_3}_{\gamma,R,S}$$

whose determinant-square diagrams commute with the source duality map, the Hopf pairing, and the Calabi–Yau threefold Serre sign line;

(vi) quotient-orientation null-trivialisations and transition rows preserving $\vartheta_{R,S}$, $\vartheta^{o}_{\gamma,R,S}$, the radical quotient, the parity decomposition, and the determinant-square diagrams.

Then $(\vartheta_{R,S}, \vartheta^{o}_{R,S})$ is a Chevalley anti-involution on the oriented reduced compact Hall primitive package. It exchanges the positive and negative simple root data, fixes the Cartan orientation, is anti-multiplicative for the Lie superbracket, is adjoint for the Hopf pairing on the radical quotient, and transports Joyce–Upmeier orientations by the Serre-dual Picard-groupoid isomorphism.

The present packets do not supply these hypotheses. The target presentation contains Chevalley and Serre language only at the formal target level. The compact Hall source packet has no simple representative rows, no bracket B-entries, no Chevalley relation rows, no G-entries, no radical rows, and no parity blocks. The negative-root compatibility packet records the dual-orientation criterion but supplies no source anti-involution map. The orientation-transition packet has no Picard-groupoid transition map. Hence row 334 is presently an anti-involution criterion and obstruction ledger, not a construction of a Chevalley anti-involution on orientation data.

Proof. All assertions follow from the listed source identities. The linear map $\vartheta_{R,S}$ is involutive and exchanges the positive and negative primitive source pieces by item 2. Item 3 is exactly the anti-automorphism identity for the even Lie-superalgebra anti-involution, with the Cartan, Chevalley, Serre, Borcherds, and super-sign relations checked after passage to the radical quotient. Item 4 makes this anti-involution adjoint for the source Hopf pairing and prevents a target formal pairing block from replacing the compact source pairing. Item 5 lifts the determinant duality to the Joyce–Upmeier orientation lines and records the Calabi–Yau Serre sign line. Item 6 makes the construction compatible with the quotient null-trivialisations and with finite-stage transitions. These are precisely the data of a Chevalley anti-involution on orientation data. □

The packet

```
certificates/orientation/rhomred_orientation_chevalley_antiinvolution
```

verifies `RHOMRED_ORIENTATION_CHEVALLEY_ANTIINVOLUTION_OBSTRUCTION_VERIFIED`: it imports the row 333 negative-root compatibility packet, the compact Hall source obstruction ledger, the target presentation packet, the retained dual-closure packet, the formal Hall-pairing pushforward packet, and the transition-orientation obstruction ledger; it records the source Chevalley anti-involution criterion for row 334; and it keeps the source simple representatives, positive–negative source charts, bracket matrices, Chevalley relation rows, Cartan orientation rows, Hopf-pairing matrices, radical quotient, parity preservation, orientation square roots, determinant-duality square, Serre sign line, quotient-orientation transport, Picard-groupoid transition, involutivity identity, and anti-multiplicativity identity open.

PROPOSITION 0.139 *(Frobenius property of the orientation pairing).* [Frobenius criterion proved; source cyclicity not supplied] Let $P^{\mathrm{red}}_{R,S}$ be the oriented reduced compact Hall primitive quotient at finite stage (R,S). Suppose the compact Hall source supplies:

(i) Hall product and coproduct matrices $M_{R,S}$ and $D_{R,S}$, unit and counit rows, and zero-defect associativity, coassociativity, bialgebra-compatibility, and primitive-closure rows;

(ii) a source Hopf pairing
$$\langle -,-\rangle^{o}_{R,S} : P^{\mathrm{red}}_{R,S} \otimes P^{\mathrm{red}}_{R,S} \longrightarrow \ell_{R,S}$$
with orientation-line factors included in the target line $\ell_{R,S}$, together with a source trace functional whose degree is recorded in the next row;

(iii) zero-defect Hopf-adjointness rows
$$\langle xy, z\rangle^{o}_{R,S} = \langle x \otimes y, \Delta z\rangle^{o}_{R,S}$$
with the Koszul signs and Thom–Sebastiani orientation signs recorded at the source;

(iv) zero-defect primitive Frobenius-cyclicity rows
$$\langle [x,y], z\rangle^{o}_{R,S} = \langle x, [y,z]\rangle^{o}_{R,S}$$
for homogeneous primitive classes, with the parity signs and the Chevalley anti-involution of Proposition 0.138 acting by adjunction;

(v) radical-kernel, quotient-splitting, radical ideal/coideal, and quotient nondegeneracy rows, so that the pairing descends to a nondegenerate form on the reduced radical quotient;

(vi) Picard-groupoid orientation pairing squares, Calabi–Yau Serre sign-line rows, radical-transition rows, pairing-kernel Mittag–Leffler rows, and orientation-transition rows making the cyclicity identities independent of the finite HN height.

Then $\langle -,-\rangle^{o}_{R,S}$ is a Frobenius pairing on the oriented reduced compact Hall primitive quotient: it is nondegenerate after quotienting by the recorded radical, it is adjoint for the Hall product and coproduct, and its restriction to primitives is cyclic for the Lie superbracket with the recorded orientation signs.

The present packets do not supply these hypotheses. The formal Hall-pairing pushforward packet proves only degree compatibility for a supplied pairing; it explicitly does not prove Hopf adjointness, Frobenius cyclicity, or quotient nondegeneracy. The compact Hall source packet has no M-, D-, B-, G-, K-, or Q-entries, no unit or counit rows, no Hall-bialgebra identities, no Hopf-pairing identity rows, and no radical ideal/coideal rows. The row 334 packet records the Chevalley anti-involution only as a criterion. Hence row 335 is presently a Frobenius-pairing criterion and obstruction ledger, not a proof that the orientation pairing is Frobenius.

Proof. The argument is the standard finite-dimensional Frobenius argument once the listed source data are present. The product, coproduct, unit, counit, and bialgebra identities make the finite-stage Hall package a bialgebra on the retained source quotient. Hopf adjointness identifies left multiplication with the adjoint of the coproduct. Restricting to primitive classes turns this adjointness into the displayed cyclic identity for the Lie superbracket; the Koszul signs are exactly the source parity signs, and the orientation signs are the Thom–Sebastiani and Serre signs recorded in the Picard groupoid. Radical ideal/coideal stability and quotient nondegeneracy descend the pairing to the reduced radical quotient. The transition and Mittag–Leffler rows make the construction independent of the finite height. No scalar trace, target pairing block, or formal degree compatibility row supplies these source identities. □

The packet

```
certificates/orientation/rhomred_orientation_frobenius_pairing
```

verifies `RHOMRED_ORIENTATION_FROBENIUS_PAIRING_OBSTRUCTION_VERIFIED`: it imports the row 334 Chevalley anti-involution criterion, the compact Hall source obstruction ledger, the formal Hall-pairing pushforward packet, the transition-radical obstruction ledger, the pairing-kernel $R^1\varprojlim$ obstruction ledger, and the transition-orientation obstruction ledger; it records the Frobenius pairing criterion for row 335; and it keeps the Hall product and coproduct matrices, bialgebra identities, source Hopf-pairing matrix, Hopf-adjointness row, Frobenius-cyclicity row, quotient nondegeneracy row, radical kernel, quotient splitting, radical ideal/coideal, orientation pairing line, orientation cyclic signs, Serre sign line, radical transition rows, pairing-kernel Mittag–Leffler rows, orientation-transition rows, and row 336 trace-degree computation open.

Proposition 0.140 *(Degree of the Frobenius trace).* [trace-degree criterion proved; source trace not supplied] Let $P^{\mathrm{red}}_{R,S}$ be the oriented reduced compact Hall primitive quotient, and suppose the Frobenius pairing of Proposition 0.139 has been supplied on the radical quotient. Suppose in addition that the compact Hall source supplies:

(i) a source trace functional
$$\mathrm{tr}^{o}_{R,S} : P^{\mathrm{red}}_{R,S,\mathrm{rad}} \longrightarrow \ell^{\mathrm{tr}}_{R,S}[-3]$$
on the reduced radical quotient, with the target trace line $\ell^{\mathrm{tr}}_{R,S}$ recorded in the orientation Picard groupoid;

(ii) a source grading convention row for cohomological degree, parity, orientation-line degree, and Calabi–Yau threefold Serre shift, with the trace-degree identity
$$\deg(\mathrm{tr}^{o}_{R,S}) = -3$$
having defect rank zero;

(iii) zero-defect trace–pairing rows
$$\langle x, y\rangle^{o}_{R,S} = \mathrm{tr}^{o}_{R,S}(xy)$$
for homogeneous classes x, y, including the Koszul and Thom–Sebastiani orientation signs;

(iv) zero-defect trace cyclicity rows
$$\mathrm{tr}^{o}_{R,S}(xy) = (-1)^{|x||y|}\,\mathrm{tr}^{o}_{R,S}(yx)$$
and compatibility with the Chevalley anti-involution and the Calabi–Yau Serre sign line;

(v) radical-quotient, quotient-orientation, transition, and Mittag–Leffler rows preserving the trace functional, the trace degree, and the trace–pairing identity in the pro-object.

Then the Frobenius trace has degree −3. Equivalently, the orientation pairing has cohomological degree −3:

$$\deg\langle x, y\rangle^{o}_{R,S} = \deg x + \deg y - 3$$

for homogeneous classes on the reduced radical quotient.

The present packets do not supply these hypotheses. The row 335 packet records the Frobenius pairing only as a criterion and explicitly keeps the source trace functional open. The compact Hall source packet has no source trace rows, no product or pairing matrices, no Hopf-pairing identities, and no radical quotient rows. The formal Hall-pairing pushforward packet proves only homogeneous degree-zero compatibility for supplied pairing rows. The protected-integration Gram-degree packet records the scalar target monomial s-degree m_R of I_R^{prot}, not the cohomological degree of a source Frobenius trace. The level-Z protected trace packet is an empty obstruction ledger. Hence row 336 is presently a trace-degree criterion and obstruction ledger, not a proof that the Frobenius trace has degree −3.

Proof. The statement follows immediately from the supplied source trace data. The trace-degree row fixes the target shift of $\mathrm{tr}^{o}_{R,S}$ to $[-3]$. The trace–pairing identity then identifies the Frobenius pairing with the composite $(x, y) \mapsto \mathrm{tr}^{o}_{R,S}(xy)$, so a homogeneous product of degree $\deg x + \deg y$ is sent to the trace line in degree $\deg x + \deg y - 3$. Trace cyclicity and the Chevalley/Serre compatibility rows show that this degree is the Frobenius trace degree, not only a degree of one chosen product expression. Radical and transition rows descend the identity to the reduced pro-object. A scalar protected-integration monomial degree does not provide the source trace functional or the cohomological shift. □

The packet

`certificates/orientation/rhomred_orientation_frobenius_trace_degree`

verifies `RHOMRED_ORIENTATION_FROBENIUS_TRACE_DEGREE_OBSTRUCTION_VERIFIED`: it imports the row 335 Frobenius-pairing criterion, the compact Hall source obstruction ledger, the formal Hall-pairing pushforward packet, the protected-integration Gram-degree packet, the level-Z protected trace obstruction ledger, and the transition-orientation obstruction ledger; it records the trace-degree criterion for row 336; and it keeps the source trace functional, trace-degree row, trace–pairing identity, trace cyclicity, orientation trace line, Calabi–Yau Serre shift, radical quotient, quotient-orientation transport, transition compatibility, and row 337 Serre-sign computation open.

PROPOSITION 0.141 *(Calabi–Yau threefold Serre signs).* [Serre-sign criterion proved; source sign rows not supplied] Let $K^{\mathrm{red}}_{\gamma,R,S}$ and $K^{\mathrm{red}}_{-\gamma,R,S}$ be the reduced oriented compact Hall complexes in opposite retained degrees. Suppose the compact Hall source supplies:

(i) a reduced Serre-duality isomorphism

$$S_{\gamma,R,S} : K^{\mathrm{red}}_{\gamma,R,S} \xrightarrow{\sim} \left(K^{\mathrm{red}}_{-\gamma,R,S}\right)^{\vee}[-3]$$

on the radical quotient, compatible with the source Ext-pairing and with the quotient null-trivialisations;

(ii) source rows fixing cohomological degree, parity, determinant-line ordering, and the Koszul sign convention for the shift $[-3]$, with a sign-exponent row $\sigma^{\mathrm{CY}_3}_{\gamma,R,S}$ of defect rank zero;

(iii) a Picard-groupoid sign line $\varepsilon^{\mathrm{CY}_3}_{\gamma,R,S}$ and a commuting determinant-duality square

$$\det K^{\mathrm{red}}_{-\gamma,R,S} \simeq \left(\det K^{\mathrm{red}}_{\gamma,R,S}\right)^{-1} \otimes \varepsilon^{\mathrm{CY}_3}_{\gamma,R,S};$$

(iv) zero-defect comparison rows identifying the Thom–Sebastiani sign, the Chevalley anti-involution sign, the Frobenius-cyclicity sign, and the trace-cyclicity sign with $\varepsilon^{\mathrm{CY}_3}_{\gamma,R,S}$;

(v) radical-quotient, quotient-orientation, finite-stage transition, and Mittag–Leffler rows preserving $S_{\gamma,R,S}$, the sign exponent, and the determinant-duality square.

Then the orientation signs agree with Calabi–Yau threefold Serre duality: every sign used in negative-root transport, Chevalley adjunction, Frobenius cyclicity, and trace cyclicity is the sign obtained from the displayed Serre-duality isomorphism and the recorded source grading convention.

The present packets do not supply these hypotheses. The PTVV packet imports the ambient (-1)-shifted symplectic form on retained finite derived substacks, but it explicitly does not construct the Joyce d-critical truncation, reduced RHom, determinant line, orientation square root, or quotient orientation. The row 333, 334, 335, and 336 packets all keep the Calabi–Yau Serre sign line or shift open. The compact Hall source packet has no source pairing matrices, Hopf-pairing identities, parity blocks, radical rows, determinant-duality squares, or Serre sign rows. The formal Hall-pairing pushforward and target negative-dual blocks are not source sign computations. Hence row 337 is presently a Serre-sign criterion and obstruction ledger, not a proof that the orientation signs agree with Calabi–Yau threefold Serre duality.

Proof. Once the listed source rows are present, the assertion is a determinant calculation. The reduced Serre-duality isomorphism identifies the negative complex with the shifted dual of the positive complex. The source degree and parity rows determine the Koszul exponent attached to the shift $[-3]$; the determinant-ordering row transports this exponent to the determinant line. The determinant-duality square is therefore the Picard-groupoid form of the Calabi–Yau threefold Serre sign line. The comparison rows force the signs used in Thom–Sebastiani gluing, Chevalley adjunction, Frobenius cyclicity, and trace cyclicity to be this same line. Radical and transition rows make the equality one on the reduced pro-object. An ambient shifted symplectic form or a target pairing block gives no source determinant ordering and therefore no Serre-sign computation. □

The packet

```
certificates/orientation/rhomred_orientation_cy3_serre_signs
```

verifies `RHOMRED_ORIENTATION_CY3_SERRE_SIGNS_OBSTRUCTION_VERIFIED`: it imports the row 333 negative-root compatibility criterion, the row 334 Chevalley anti-involution criterion, the row 335 Frobenius-pairing criterion, the row 336 trace-degree criterion, the PTVV finite-substack shifted-symplectic packet, the compact Hall source obstruction ledger, and the transition-orientation obstruction ledger; it records the Calabi–Yau Serre-sign criterion for row 337; and it keeps the reduced Serre-duality isomorphism, source sign-exponent table, orientation sign line, determinant-duality square, Thom–Sebastiani sign comparison, Chevalley sign comparison, Frobenius-cyclicity sign comparison, trace-cyclicity sign comparison, radical quotient, quotient orientation, and transition compatibility open.

Proposition 0.142 *(Reduced Calabi–Yau pairing modulo the radical).* [nondegeneracy criterion proved; quotient matrices not supplied] Let $P^{\mathrm{red}}_{\gamma,R,S}$ and $P^{\mathrm{red}}_{-\gamma,R,S}$ be the retained opposite-degree primitive source spaces at finite stage (R, S). Suppose the compact Hall source supplies:

(i) a Calabi–Yau threefold Hopf-pairing matrix

$$G^{\mathrm{CY}_3}_{\gamma,R,S} : P^{\mathrm{red}}_{\gamma,R,S} \otimes P^{\mathrm{red}}_{-\gamma,R,S} \longrightarrow \ell^{\mathrm{CY}_3}_{\gamma,R,S}$$

with the source Serre sign line of Proposition 0.141 included in the target line;

(ii) left- and right-kernel rows

$$K^L_{\gamma,R,S} = \ker G^{\mathrm{CY}_3}_{\gamma,R,S}, \qquad K^R_{\gamma,R,S} = \ker(G^{\mathrm{CY}_3}_{\gamma,R,S})^t$$

and zero-defect rows identifying these kernels with the recorded Hopf-pairing radical in the positive and negative source spaces;

(iii) quotient splitting matrices

$$Q_{\gamma,R,S} : P^{\mathrm{red}}_{\gamma,R,S} \twoheadrightarrow \overline{P}_{\gamma,R,S}, \qquad Q_{-\gamma,R,S} : P^{\mathrm{red}}_{-\gamma,R,S} \twoheadrightarrow \overline{P}_{-\gamma,R,S}$$

whose kernels are exactly the recorded radicals;

(iv) a quotient-pairing matrix

$$\overline{G}_{\gamma,R,S} = Q_{\gamma,R,S}\, G^{\mathrm{CY}_3}_{\gamma,R,S}\, Q^t_{-\gamma,R,S}$$

with full left rank and full right rank, equivalently determinant or Smith defect rank zero after choosing finite quotient bases;

(v) radical ideal/coideal rows, transition-radical rows, and pairing-kernel Mittag–Leffler rows proving that the quotient pairing and its nondegeneracy are independent of the finite HN height.

Then the reduced Calabi–Yau threefold pairing is nondegenerate modulo the radical:

$$\overline{G}_{\gamma,R,S} : \overline{P}_{\gamma,R,S} \otimes \overline{P}_{-\gamma,R,S} \longrightarrow \ell^{\mathrm{CY}_3}_{\gamma,R,S}$$

has zero left and right kernels, with the orientation sign line fixed by Calabi–Yau threefold Serre duality.

The present packets do not supply these hypotheses. The compact Hall source packet has no G-entries, kernel matrices, quotient splittings, quotient-nondegeneracy identities, radical ideal/coideal rows, or quotient transition rows. The formal Hall-pairing pushforward packet proves only homogeneous degree-zero compatibility for three supplied formal pairings and explicitly does not prove quotient nondegeneracy. The row 333, 335, and 337 packets record negative-root compatibility, Frobenius pairing, and Serre signs only as criteria. The transition-radical and pairing-kernel Mittag–Leffler packets are obstruction ledgers with empty transition and kernel tables. Hence row 338 is presently a nondegeneracy-modulo-radical criterion and obstruction ledger, not a proof that the reduced Calabi–Yau pairing is nondegenerate modulo the radical.

Proof. The assertion is the elementary quotient-pairing argument once the listed finite matrices exist. By the kernel rows, the left and right radicals of $G^{\mathrm{CY}_3}_{\gamma,R,S}$ are exactly the kernels of $Q_{\gamma,R,S}$ and $Q_{-\gamma,R,S}$. Therefore $G^{\mathrm{CY}_3}_{\gamma,R,S}$ factors through $\overline{P}_{\gamma,R,S} \otimes \overline{P}_{-\gamma,R,S}$. The full-rank row for $Q_{\gamma,R,S}G^{\mathrm{CY}_3}_{\gamma,R,S}Q^t_{-\gamma,R,S}$ states precisely that the induced form has no left or right kernel. The radical ideal/coideal rows make the quotient compatible with the Hall algebra operations; the transition and Mittag–Leffler rows make the finite-stage statement stable in the pro-object. The Serre-sign line fixes the orientation of the target line but does not by itself imply full rank. □

The packet

```
certificates/orientation/rhomred_orientation_cy3_pairing_nondegeneracy
```

verifies RHOMRED_ORIENTATION_CY3_PAIRING_NONDEGENERACY_OBSTRUCTION_VERIFIED: it imports the row 333 negative-root compatibility criterion, the row 335 Frobenius-pairing criterion, the row 337 Serre-sign criterion, the compact Hall source obstruction ledger, the formal Hall-pairing pushforward packet, the transition-radical obstruction ledger, and the pairing-kernel Mittag–Leffler obstruction ledger; it records the reduced Calabi–Yau pairing nondegeneracy criterion for row 338; and it keeps the source pairing matrix, left and right kernel rows, radical identification, quotient splitting matrices, quotient-pairing full-rank row, radical ideal/coideal rows, radical transition rows, pairing-kernel Mittag–Leffler rows, and inverse-limit nondegeneracy open.

Definition 0.143 *(Orientation-preserving transition datum).* Let $R' \leq R$, and suppose the finite-moduli transition geometry of Definition 0.242 is supplied. An orientation-preserving transition datum consists of:

(i) a Picard-groupoid isomorphism over the retained transition stack

$$\theta^{o}_{RR'} : (t_{RR'})^{*} o_{R'} \xrightarrow{\sim} o_{R};$$

(ii) commutativity of the determinant square-root diagram

$$\big((t_{RR'})^{*} o_{R'}\big)^{\otimes 2} \longrightarrow (t_{RR'})^{*} \det K^{\mathrm{red}}_{R'} \longrightarrow \det K^{\mathrm{red}}_{R},$$

with the square $o_R^{\otimes 2} \simeq \det K^{\mathrm{red}}_{R}$;

(iii) transport of the null-trivialisations of α^{red}, $\alpha^{E,\mathrm{free}}$, $\widetilde{\beta}^{H}$, β^{H}, and $\lambda^{H} = 0$ for every retained finite stabilizer stratum;

(iv) compatibility of $\theta^{o}_{RR'}$ with the Thom–Sebastiani orientation isomorphisms on extension and two-step flag stacks;

(v) compatibility with the Weyl lifts τ_i, the cochain killing the projective Coxeter cocycle, and the quotient-cocycle transport rows of $(O1)^{+}$;

(vi) zero Picard-groupoid, square-root, null-trivialization, Thom–Sebastiani, Weyl, and composition defect ranks, together with $R^1 \varprojlim = 0$ for the orientation Picard data.

Proper or closed stack transitions alone do not define this datum. Scalar traces, squared determinants, OP scalar branches, Maass-character values, Pfaffian products, and target root windows are not orientation transport data.

Proposition 0.144 *(Transition preservation of Joyce–Upmeier orientations; proved).* Given an orientation-preserving transition datum of Definition 0.143, the transition $R \to R'$ preserves the reduced Joyce–Upmeier orientation line, the quotient-orientation null-trivialisations, the Thom–Sebastiani multiplicative structure, and the $(O1)^{+}$ Weyl-equivariant determinant-line transport.

Proof. The isomorphism $\theta^{o}_{RR'}$ identifies the pulled-back target orientation line with the source orientation line in the Picard groupoid. The square-root diagram in item 2 shows that this identification is an identification of Joyce–Upmeier orientations, not only of underlying line bundles. Item 3 transports the quotient Cech–Borel null-trivialisations and the zero finite-stabilizer linearization characters. Item 4 makes the Hall Thom–Sebastiani orientation isomorphisms commute with transition, and item 5 makes the type-II Weyl lifts and Coxeter cochain commute with transition. The zero-defect and $R^1 \varprojlim$ rows in item 6 give strict composition and inverse-limit exactness for the orientation Picard data. □

The packet `certificates/orientation/transition_orientation_preservation` records the present state of this datum. Its verifier status is `TRANSITION_ORIENTATION_PRESERVATION_OBSTRUCTION_VERIFIED`: the stack-transition input, orientation-line pullback isomorphisms, determinant square compatibility, quotient Cech–Borel null-trivialization transport rows, finite-stabilizer and linearization transport rows, Thom–Sebastiani transition rows, Weyl-lift transition rows, Coxeter-cochain transition rows, strict Picard-groupoid transition rows, and orientation Mittag–Leffler rows are not yet supplied. Hence transition preservation of orientations remains open.

PROPOSITION 0.145 *(Survival of orientation classes in the inverse limit).* [inverse-limit survival criterion proved; orientation tower not supplied] Let $\mathcal{R}$ be the retained cofinal HN index category, and for each $R \in \mathcal{R}$ let o_R be a Joyce–Upmeier orientation line on the reduced compact Hall frame at stage R. Suppose the compact orientation source supplies:

(i) finite-stage orientation rows

$$o_R^{\otimes 2} \simeq \det K_R^{\mathrm{red}}, \qquad \mathrm{cl}(o_R) = 0$$

with zero square-root and orientation-class defect ranks for every retained stratum;

(ii) for every successor $R' \leq R$, Picard-groupoid transition isomorphisms

$$\theta^o_{RR'} : (t_{RR'})^* o_{R'} \xrightarrow{\sim} o_R$$

whose determinant-square, quotient-orientation, Thom–Sebastiani, Weyl, and Coxeter transition defects are zero;

(iii) strict identity and composition rows for the transitions $\theta^o_{RR'}$, so that the o_R form an inverse system in the orientation Picard groupoid;

(iv) an orientation Mittag–Leffler row with

$$R^1 \varprojlim_R \pi_0 \operatorname{Pic}_R^{or} = 0$$

and compatible image-stabilization witnesses for the orientation classes;

(v) a limit row identifying the pro-orientation class

$$o_\infty = \varprojlim_R (o_R, \theta^o_{RR'})$$

with the class used by the Dirac–Igusa pro-object.

Then the finite-stage orientation classes survive the inverse limit: the compatible system $(o_R, \theta^o_{RR'})$ determines an orientation class o_∞ on the pro-object, and every finite projection of o_∞ is the prescribed o_R.

The present packets do not supply these hypotheses. The reduced orientation ledger has no orientation-line rows, no square-root rows, no orientation-class rows, and no transition rows. The square-root packet records three missing square-root rows and has no orientation class zero row. The transition-orientation packet has no Picard-groupoid pullback maps, no determinant-square transition rows, no strict composition rows, and no orientation Mittag–Leffler row. The row 338 nondegeneracy packet keeps inverse-limit nondegeneracy open and does not construct orientation classes. Hence row 339 is presently an inverse-limit survival criterion and obstruction ledger, not a proof that orientation classes survive the inverse limit.

Proof. The claim is the standard exactness statement for an inverse system in Picard groupoids once the listed rows are present. The finite-stage rows give objects o_R with the prescribed determinant square and vanishing orientation obstruction. The transition rows make these objects compatible under pullback. Strict identity and composition rows replace weak pairwise compatibility by an inverse system. The Mittag–Leffler row kills the $R^1 \varprojlim$ obstruction to lifting the compatible finite classes to a pro-class. The limit row then identifies the resulting object with the orientation datum used in the Dirac–Igusa pro-object. This proves survival of the prescribed classes. It does not prove that every orientation class at the limit comes from a finite-stage class; that is the separate row 340 obligation. □

The packet

```
certificates/orientation/rhomred_orientation_class_limit_survival
```

verifies `RHOMRED_ORIENTATION_CLASS_LIMIT_SURVIVAL_OBSTRUCTION_VERIFIED`: it imports the reduced-orientation obstruction ledger, the square-root obstruction packet, the transition-orientation obstruction ledger, and the row 338 nondegeneracy criterion; it records the inverse-limit survival criterion for row 339; and it keeps the finite orientation line rows, square-root rows, orientation-class zero rows, Picard transition maps, strict composition rows, orientation Mittag–Leffler rows, limit class row, and row 340 no-new-class statement open.

Proposition 0.146 *(No new orientation class at the limit).* [no-phantom criterion proved; pro-Picard comparison not supplied] Let Pic^{or}_∞ be the orientation Picard groupoid of the Dirac–Igusa pro-object, and let Pic^{or}_R be the finite-stage orientation Picard groupoid at HN height R. Suppose the compact orientation source supplies:

(i) a pro-Picard comparison functor

$$\Xi : \mathrm{Pic}^{or}_\infty \longrightarrow \varprojlim_R \mathrm{Pic}^{or}_R$$

whose object and automorphism components are defined by the finite stack transitions;

(ii) finite-stage effectivity rows showing that every orientation line on the pro-object restricts to the recorded finite-stage orientation line, square-root, quotient-orientation, Thom–Sebastiani, Weyl, and Coxeter data;

(iii) separatedness rows proving that a pro-orientation class whose finite restrictions are all trivial is itself trivial;

(iv) automorphism Mittag–Leffler rows with

$$R^1 \varprojlim_R \pi_1 \mathrm{Pic}^{or}_R = 0$$

and object-class Mittag–Leffler rows with zero $R^1 \varprojlim$ defect for the relevant orientation-class tower;

(v) a zero-defect comparison row

$$\pi_0 \mathrm{Pic}^{or}_\infty \xrightarrow{\sim} \varprojlim_R \pi_0 \mathrm{Pic}^{or}_R$$

on the subfunctor cut out by the determinant square, quotient orientation, Thom–Sebastiani, Weyl, and Coxeter conditions.

Then no new orientation class appears at the limit: every Dirac–Igusa pro-orientation class is detected by its compatible finite-stage restrictions, and the class is the image of the finite tower under Proposition 0.145.

The present packets do not supply these hypotheses. The row 339 packet records only the survival criterion and has no limit-survival rows. The reduced-orientation ledger has no finite orientation-line or transition rows. The transition-orientation packet has no Picard-groupoid transition maps, no strict composition rows, no orientation Mittag–Leffler rows, and no pro-Picard comparison rows. The square-root packet records missing finite square roots and no orientation-class zero rows. Hence row 340 is presently a no-phantom-class criterion and obstruction ledger, not a proof that no new orientation class appears at the limit.

Proof. Assume the listed comparison and exactness rows. The functor Ξ restricts a pro-orientation class to its finite-stage orientation tower. The effectivity rows identify this tower with the same finite-stage orientation data used in the Dirac–Igusa construction. Separatedness says that the kernel of $\pi_0(\Xi)$ is zero: a class with trivial finite restrictions is trivial at the pro-level. Vanishing of $R^1 \varprojlim$ on automorphisms and object classes removes the usual derived-limit ambiguity in passing between Picard groupoids and their towers. The displayed zero-defect comparison row therefore identifies pro-orientation classes with compatible finite orientation classes. By Proposition 0.145, each compatible finite tower gives the prescribed limit class. No additional class can occur at the limit. □

The packet

```
certificates/orientation/rhomred_orientation_no_new_limit_class
```

verifies `RHOMRED_ORIENTATION_NO_NEW_LIMIT_CLASS_OBSTRUCTION_VERIFIED`: it imports the row 339 inverse-limit survival criterion, the reduced-orientation obstruction ledger, the square-root obstruction packet, and the transition-orientation obstruction ledger; it records the no-new-limit-class criterion for row 340; and it keeps the pro-Picard comparison functor, finite-stage effectivity rows, separatedness rows, automorphism Mittag–Leffler rows, object-class Mittag–Leffler rows, zero-defect comparison row, and pro-orientation class table open.

Definition 0.147 *(Vanishing-cycle-preserving transition datum).* Let $R' \leq R$. Suppose the finite-moduli transition geometry of Definition 0.242 and the orientation-preserving transition datum of Definition 0.143 are supplied. A vanishing-cycle-preserving transition datum consists of:

(i) transition maps of oriented d-critical charts, with transition graph $\mathfrak{G}_{RR'}$ and projections

$$p_R : \mathfrak{G}_{RR'} \to \mathfrak{Z}_R, \qquad p_{R'} : \mathfrak{G}_{RR'} \to \mathfrak{Z}_{R'};$$

(ii) potential compatibility on charts:

$$f_R = p_{R'}^* f_{R'} + q_{RR'}$$

after pullback to the transition graph, where $q_{RR'}$ is the recorded non-degenerate quadratic stabilization and its parity shift is recorded;

(iii) pullback compatibility of the K3 semiregularity cosections and a morphism of reduced obstruction complexes with zero cosection and reduced-complex defect ranks;

(iv) a constructible-complex isomorphism

$$\theta^{\Phi}_{RR'} : p_R^* \Phi^{\mathrm{red}}_R \xrightarrow{\sim} p_{R'}^* \Phi^{\mathrm{red}}_{R'}$$

on $\mathfrak{G}_{RR'}$, with BBDJS perverse normalization and support defects zero;

(v) compatibility of $\theta^{\Phi}_{RR'}$ with the Thom–Sebastiani vanishing-cycle isomorphisms on extension and two-step flag stacks and with proper base change on retained support;

(vi) zero d-critical, potential, cosection, vanishing-cycle transport, Thom–Sebastiani, base-change, and composition defect ranks, together with $R^1 \varprojlim = 0$ for the reduced vanishing-cycle complexes.

Scalar traces, Euler characteristics, Behrend-weighted numbers, Pfaffian products, orientation lines alone, target root windows, D0 scalar tests, and signed multiplicities are not vanishing-cycle transition data.

Proposition 0.148 *(Transition preservation of reduced vanishing cycles; proved).* Given a vanishing-cycle-preserving transition datum of Definition 0.147, the transition $R \to R'$ preserves the reduced BBDJS perverse sheaves of vanishing cycles, their cosection reduction, their Thom–Sebastiani products, and their compact-support base-change domains.

Proof. The d-critical chart transition and potential compatibility identify the local vanishing-cycle functors after the recorded quadratic stabilization. The cosection compatibility identifies the reduced obstruction theories, so the reduced vanishing-cycle complexes lie on the same reduced critical locus after pullback to $\mathfrak{G}_{RR'}$. The isomorphism $\theta^{\Phi}_{RR'}$ identifies the two BBDJS complexes in $D^b_c(\mathfrak{G}_{RR'})$, and the perverse-normalization row records the shift introduced by stabilization. The Thom–Sebastiani and proper-base-change rows make this identification compatible with extension and two-step flag correspondences. The zero-defect and $R^1 \varprojlim$ rows give strict composition and inverse-limit exactness for the reduced vanishing-cycle system. □

The packet `certificates/vanishing_cycles/transition_vanishing_cycle_preservation` records the present state of this datum. Its verifier status is `TRANSITION_VANISHING_CYCLE_PRESERVATION_OBSTRUCTION_VERIFI` the stack-transition input, orientation-transition input, d-critical chart transition rows, potential-compatibility rows, quadratic stabilization rows, cosection-pullback rows, reduced obstruction complex maps, graph-pullback vanishing-cycle isomorphisms, perverse normalization rows, support and base-change rows, Thom–Sebastiani vanishing-cycle compatibility rows, strict composition rows, and vanishing-cycle Mittag–Leffler rows are not yet supplied. Hence transition preservation of reduced vanishing cycles remains open.

Cosection localization and the cosection-reduced heart

After the surjectivity, kernel, and perfectness rows have been supplied, the K3 semiregularity cosection CS_C selects the reduced obstruction theory. Only then does the corresponding Behrend constructible function ν^{red} and the reduced vanishing-cycle complex Φ^{red} enter the Hall product [6, 54]. The filtered Hall additivity $q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B$ on extension stacks is the later cosection-localization compatibility required by the Hall correspondence; the Joyce–Upmeier orientation lines are required to be Thom–Sebastiani-coherent on the resulting two-step and four-input flag stacks.

Proposition 0.149 *(Finite d-critical Hall product; conditional on the (O1) and atlas data).* Fix R and closed-configuration substacks $\mathfrak{M}^{\mathrm{loc,cl}}_{c,R,I}$ in a noetherian abelian Hall heart, and assume:

(i) finite-type quasi-smooth derived Artin enhancements of the selected object, extension, and two-step flag stacks, with perfect obstruction theories;

(ii) finite residual inertia after scalar, E-translation, and wrapped rigidifications;

(iii) retained closed flag-Quot/filtration extension and flag stacks, or an explicitly specified compact-support six-functor model, with admissible p^* and $q_!$;

(iv) PTVV (-1)-shifted symplectic structures restricting to these substacks, classical truncations carrying Joyce d-critical structures [12, 95];

(v) surjective K3 semiregularity cosections with filtered Hall additivity;

(vi) Joyce–Upmeier orientation lines and orientation transports [56] with all finite-stabilizer characters trivialized;

(vii) Thom–Sebastiani transport for the reduced vanishing-cycle sheaves satisfying two-step flag pentagons and higher-coloured configuration coherences.

Then the finite reduced local-local product

$$m_R^{\mathrm{red}} = q_! \circ \mathrm{TS}_R^{\mathrm{red}} \circ p^*$$

is an honest morphism on $H_c^\bullet(\mathfrak{M}_{c,R,I}^{\mathrm{loc,cl}}, \Phi_{c,R,I}^{\mathrm{red}} \otimes o_{c,R,I}^{\mathrm{red}})$ and is associative on the finite stage.

The first obstruction to removing these hypotheses is finite inertia: proper finite-type correspondences do not by themselves eliminate residual $\mathbb{G}_m$, direct-sum, or parabolic automorphisms, and a residual $\mathbb{G}_m$ already contributes $H^\bullet(B\mathbb{G}_m) = \mathbb{C}[u]$ to protected cohomology. The phrase *finite stage* here means finite charge support, finite-type stacks, and finite-dimensional protected cohomology after the stated quotients and admissibility hypotheses; it does not mean that the Hall correspondence morphisms are themselves finite morphisms. The raw exact-triangle stack is not a proper Hall correspondence; the proper object is the retained compactified filtration stack, or else the compact-support functor $q_!$ is an extra datum.

Quotient-after-correspondence ordering

The E-quotient is taken *after* the extension correspondences, not before. A pseudofunctor $\mathsf{Corr}_R^{E,\mathrm{hyb}} \to \mathsf{Corr}_R^{\mathrm{red,hyb}}$ supplies the descent; applying the vanishing-cycle Borel–Moore functor with orientation lines included gives $Q_{E,R}$ between the corresponding finite correspondence module categories. For each operation $\mu_R^\bullet = q_!\,\mathrm{TS}\,p^*$ a chosen comparison isomorphism

$$\theta_{\mu,R}^{Q} : Q_{E,R}\, q_!\, \mathrm{TS}\, p^* \xrightarrow{\sim} \overline{q}_! \overline{\mathrm{TS}}\, \overline{p}^*(Q_{E,R}^{\boxtimes k})$$

is part of the datum. Its compatibility with the eight-word flag 2-morphisms and HN transitions is the (O1) descent calculus. A quotient-first construction (object-stack quotient followed by a new Hall product) is not the same datum.

The orientation obstruction class

The orientation lane carries the residual

$$\mathfrak{O}_{\mathrm{or},R} = \left(\mathfrak{o}_C^{\mathrm{red}}, \mathfrak{o}_C^{E,\mathrm{free}}, \{\mathfrak{o}_{C,S}^{H}\}_{H,S}, \{\mathfrak{o}_{C,S}^{\lambda^H}\}_{H,S}, [c_o]_R,\ \xi_{c_o,R},\ \{\mathfrak{o}_{i,R}^{\tau,2}\}_i,\ \{\omega_{i,C,R}\}_i\right)$$

required to vanish at every finite HN height, with chosen null-trivializations. Theorems 7.13 and 7.14 of Appendix 7 are the quotient-orientation and Weyl-transport criteria relative to that retained datum. The packet

`certificates/orientation/k3e_reduced_orientation/blocked_obligations.csv` records the missing finite equations: determinant square-root and orientation-class rows, Thom–Sebastiani and pentagon rows, the four quotient Cech–Borel class rows, finite-stabilizer edge reduction, odd-transfer, Klein-four and two-primary coefficient rows, Weyl lift involutivity, torsor-defect, quotient-cocycle transport, projective-cocycle, cochain, Coxeter, transition, Mittag–Leffler, and scalar-firewall rows. Its verifier returns `ORIENTATION_OBSTRUCTION_LEDGER_VERIFIED`, with `orientation_certification=false` and `mathematical_certification=false`. It is the finite list of equations still required for $(O1)$ and $(O1)^{+}$, not a proof of those equations. The Klein-four and two-primary finite-stabilizer lemmas there give the edge coordinates of (O1). Proposition 0.101 is the finite-site test: the Borel spectral sequence must first reduce the equivariant class to the classifying-space edge, and for two-primary rank two stabilizers the mixed coefficient must be computed directly. The wrapped-stratum Weyl-equivariant transport along $W^{(2)}(\Lambda^{2,1}_{II})$ is reduced to the no-braid type-II Coxeter presentation and the Maass-character match.

Comparison with the Calabi–Yau quantum-group stratum

Vol III's Calabi–Yau quantum-groups stratification observes that $K3 \times E$ is not presented by a global toric/quiver-variety NCCR atlas of the Schiffmann–Vasserot type, so the orientation lane cannot be supplied by such an NCCR-induced heart; the cosection-twisted reduced d-critical structure is forced on the chosen stability heart instead. The displayed Borel calculation at $E[2]$ is the smallest finite stabilizer detected by the cosection-reduced determinant; even $N \geq 4$ stabilizers carry the mixed coefficient $A^{(N)}_{12}$, which the displayed cyclic restrictions miss, and which (O1) requires to be null-trivialized as part of the full degree-two gerbe.

The orientation o_R trivializes the Joyce–Upmeier obstruction to a compatible Hall product on the cosection-twisted heart. Once o_R is in place, the Hall product, the collision coproduct, and the protected primitive extraction are unambiguous. Entry (L3) is the Hall apparatus of §0.3.

0.3 Hall correspondences: product, coproduct, primitives

The third entry of the Dirac–Igusa pro-object is the Hall structure on the cosection-reduced d-critical heart. It is the algebraic content that the orientation lane equips and the hybrid carrier indexes. Three operations live here: the Hall product m^{red}_R, the collision coproduct Δ^{ch}_R, and the primitive projection P^{Prim}_R. After normal-ordered Gram descent these form at most a graded bialgebra-and-primitive datum. The word "Hopf" is reserved for the additional antipode data of Definition 0.161; recognition with the Borcherds–Kac–Moody target is the §5.6 obligation only after that data and the radical quotient data have been supplied.

The Kontsevich–Soibelman cohomological Hall algebra of a 3-Calabi–Yau category [64] is the prototype; on $K3 \times E$ the cosection reduction [54, 56] replaces the unreduced critical CoHA, and the wrapped sector is the residual the hybrid carrier of §0.1 retains. The Maulik–Okounkov stable-envelope construction and the Schiffmann–Vasserot theorems on Drinfeld doubles of CoHA operate on the same input; their integral form is the Hall side of the present construction.

Definition 0.150 *(Finite compact Hall object).* Fix a finite HN height R and a finite retained charge set $\widehat{\Gamma}_R$. A finite compact Hall object at height R is the $\widehat{\Gamma}_R$-graded object

$$\mathcal{H}^{\mathrm{geom}}_R = \bigoplus_{\widehat{c}\in\widehat{\Gamma}_R} H^{\mathrm{BM}}_*\Big(\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}}, \Phi^{\mathrm{red}}_{R,\widehat{c}} \otimes o_{R,\widehat{c}}\Big),$$

where the summand is defined only after the following data are fixed: the finite-type retained rigidified stack $\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}}$ with finite residual inertia, the cosection-reduced BBDJS vanishing-cycle complex $\Phi^{\mathrm{red}}_{R,\widehat{c}}$, the

Joyce–Upmeier orientation line $o_{R,\widehat{c}}$, and an admissible compact-support Borel–Moore six-functor model giving finite-rank groups. The parity is the cohomological parity of these Borel–Moore groups together with the orientation parity recorded in the source basis provenance.

The object $\mathcal{H}_R^{\mathrm{geom}}$ is only the finite coefficient object. It does not include the Hall product, coproduct, unit, counit, Hopf pairing, radical quotient, PBW filtration, primitive subspace, normal-ordered Gram descent, or transition maps. Those are additional source rows. A scalar trace, Euler characteristic, signed root multiplicity, Pfaffian product, target root window, or denominator product is not a substitute for the displayed Borel–Moore direct sum.

The packet `certificates/hall/finite_geometric_hall_object` verifies `FINITE_GEOMETRIC_HALL_OBJECT_DEFINITION`; it records Definition 0.150 as the row 341 definition and imports the retained finite-moduli packet, the reduced-orientation obstruction ledger, the transition vanishing-cycle obstruction ledger, and the compact Hall source obstruction ledger. The present source packets do not yet populate $\mathcal{H}_R^{\mathrm{geom}}$: no orientation-line rows, vanishing-cycle coefficient rows, object-component rows, source-degree rows, homogeneous basis provenance rows, or finite-rank rows are supplied. Hence row 341 is a definition of the finite compact Hall object, not a proof that the compact $K3 \times E$ source stage has been populated.

The compact Hall product

Definition 0.151 *(Compact Hall product on the cosection-reduced heart).* Let $C_{\sigma,S,\le R}$ be the finite HN quotient of §0.1. Assume the retained Liu–Hilbert compactified correspondence and the compact-support pushforward of Proposition 0.149, including finite residual inertia, base-change, projection-formula, Thom–Sebastiani, and quotient-after-correspondence coherences. The compact Hall product at height R is the operation

$$m_R^{\mathrm{red}} = q_{c,c'!} \circ \mathrm{TS}_{c,c'}^{\mathrm{red}} \circ p_{c,c'}^* : H_*^{\mathrm{BM}}\big(\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma, \Phi_c^{\mathrm{red}} \boxtimes \Phi_{c'}^{\mathrm{red}} \otimes o_c \boxtimes o_{c'}\big) \to H_*^{\mathrm{BM}}\big(\mathfrak{M}_{c+c'}^\sigma, \Phi_{c+c'}^{\mathrm{red}} \otimes o_{c+c'}\big)$$

attached to the retained compactified extension correspondence

$$\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma \xleftarrow{p_{c,c'}} \overline{\mathfrak{E}}_{c,c'}^{\mathrm{ret},\sigma} \xrightarrow{q_{c,c'}} \mathfrak{M}_{c+c'}^\sigma,$$

in the cosection-reduced d-critical heart, with vanishing-cycle coefficients Φ^{red}, Joyce–Upmeier orientation lines o, and Thom–Sebastiani transport $\mathrm{TS}_{c,c'}^{\mathrm{red}}$. This definition names the operator once the three functorial pieces p^*, $\mathrm{TS}^{\mathrm{red}}$, and $q_!$ are supplied in the chosen six-functor model. It does not prove that p^* exists, that $q_!$ exists for the retained correspondence, or that $q_!$ is independent of a compactification; these are separate functorial obligations. It also does not supply product matrices, associativity, transition compatibility, or primitive closure. Associativity on the finite stage is the eight-word two-step binary flag pentagon together with the quotient-after-correspondence and HN-transition coherences of §0.1.

The packet `certificates/hall/finite_hall_product_operator` verifies `FINITE_HALL_PRODUCT_OPERATOR_DEFINITION`; it records the row 342 definition

$$m_R^{\mathrm{red}} = q_!\, \mathrm{TS}^{\mathrm{red}}\, p^*$$

and imports the row 341 finite compact Hall object, the compact-support exceptional pushforward model, the compact Hall source obstruction ledger, and the transition-Hall-product obstruction ledger. The present packets give only the formal product expression and the chosen compact-support notation. They do not populate retained extension-correspondence rows, p^* rows, $q_!$ rows, reduced Thom–Sebastiani rows on the Hall correspondence, product matrix rows, base-change or projection-formula rows, compactification-independence rows, associativity rows, or product-transition rows.

Proposition 0.152 *(Pullback leg of the finite Hall product).* [six-functor criterion proved; retained p-row not supplied] Let

$$p_{c,c'} : \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \longrightarrow \mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'}$$

be a supplied retained extension-correspondence morphism at finite HN height R. Assume that $p_{c,c'}$ is a finite-type morphism in the retained Artin-stack category on which the chosen constructible six-functor formalism is installed, and that the input coefficient

$$K_{R,c,c'} = (\Phi^{\mathrm{red}}_{R,c} \otimes o_{R,c}) \boxtimes (\Phi^{\mathrm{red}}_{R,c'} \otimes o_{R,c'})$$

is a bounded constructible complex on the input product stack. Then the pullback

$$p^{*}_{c,c'} K_{R,c,c'} \in D^{b}_{c}(\overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'})$$

exists in the same six-functor formalism. This is the p^*-leg of the finite Hall product m^{red}_{R}.

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall product. The finite-moduli packet has no extension-stack rows, finite-type flag rows, source-map rows, or d-critical coefficient rows on the extension correspondence. The row 341 finite-Hall-object packet has no object-component or coefficient rows. The row 342 product-operator packet has no populated p^* rows. The compact-support packet gives notation for compact-support pushforwards, not a populated pullback leg. Hence row 343 is presently this pullback-existence criterion and obstruction ledger, not a proof that the retained Hall product has a populated p^*-leg.

Proof. In a constructible six-functor theory on the retained finite-type Artin-stack category, every morphism p in the category carries a pullback functor

$$p^{*} : D^{b}_{c}(Y) \longrightarrow D^{b}_{c}(X).$$

The finite-type and constructibility hypotheses ensure that bounded constructible coefficients remain in the retained coefficient category after pullback. Applying this to the supplied extension morphism $p_{c,c'}$ and to $K_{R,c,c'}$ gives the displayed object. The argument uses only the pullback functor; it does not construct $q_!$, prove Thom–Sebastiani compatibility, produce product matrices, or prove associativity. □

The packet `certificates/hall/finite_hall_product_pullback_functor` verifies `FINITE_HALL_PRODUCT_PULLBACK_FU` it imports the row 342 product-operator packet, the finite-moduli obstruction ledger, the row 341 finite-Hall-object packet, and the compact-support model; it records the pullback-existence criterion; and it keeps the retained extension correspondence, finite-type p-map, constructible input coefficient, populated p^*-row, and product matrix rows open.

Proposition 0.153 *(Exceptional pushforward leg of the finite Hall product).* [compact-support criterion proved; retained q-row not supplied] Let

$$q_{c,c'} : \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \longrightarrow \mathfrak{M}^{\mathrm{red}}_{R,c+c'}$$

be a supplied retained extension-correspondence morphism at finite HN height R, and let

$$K_{\mathfrak{E}} \in D^{b}_{c}(\overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'})$$

be the bounded constructible coefficient obtained after the preceding pullback and reduced Thom–Sebastiani transport. Assume that $q_{c,c'}$ lies in the chosen retained finite-type Artin-stack category with finite residual inertia and that one of the following two pieces of functorial data is supplied:

(i) $q_{c,c'}$ is proper in that category; or

(ii) a compact-support model

$$j_q : \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \hookrightarrow \overline{\mathfrak{E}}_q, \qquad \overline{q} : \overline{\mathfrak{E}}_q \to \mathfrak{M}^{\mathrm{red}}_{R,c+c'}, \qquad q_{c,c'} = \overline{q} \circ j_q$$

is supplied as in Definition 0.25, with $j_{q,!}K_{\mathfrak{E}}$ and $\overline{q}_* j_{q,!}K_{\mathfrak{E}}$ in the same constructible six-functor formalism.

Then the exceptional pushforward leg

$$q^{\mathrm{cs}}_{c,c',!}K_{\mathfrak{E}} := \overline{q}_* j_{q,!}K_{\mathfrak{E}} \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c+c'})$$

exists; in the proper case it is the ordinary proper pushforward $q_{c,c',*}K_{\mathfrak{E}}$. This is the $q_!$-leg of the finite Hall product m^{red}_R.

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall product. The row 342 product operator records $q_!$ only as notation. The row 343 packet proves only the pullback criterion. The finite-moduli ledger has no extension flag-stack row, no retained q-map row, no properness row for q, and no d-critical coefficient row on the extension correspondence. The compact-support model defines $\overline{q}_* j_{q,!}$, but its map-population and choice-independence tables are empty. The compact Hall source packet has no product matrix rows. Hence row 344 is this exceptional pushforward criterion and obstruction ledger, not a proof that the retained Hall product has a populated $q_!$-leg.

Proof. In the proper case, the six-functor formalism supplies the proper direct image

$$q_{c,c',*} : D^b_c(\overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'}) \longrightarrow D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c+c'}),$$

and properness preserves constructibility. In the compact-support case, the supplied open immersion gives extension by zero

$$j_{q,!}K_{\mathfrak{E}} \in D^b_c(\overline{\mathfrak{E}}_q),$$

and the supplied proper map $\overline{q}$ gives the proper direct image

$$\overline{q}_* j_{q,!}K_{\mathfrak{E}} \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c+c'}).$$

This composite is exactly the compact-support exceptional pushforward of Definition 0.25. The argument does not compare two compactifications, prove base change or the projection formula, construct the reduced Thom–Sebastiani row, produce product matrices, or prove associativity. □

The packet `certificates/hall/finite_hall_product_pushforward_functor` verifies `FINITE_HALL_PRODUCT_PUSHFORWARD_FUNCTOR_OBSTRUCTION_VERIFIED`: it imports the row 342 product-operator packet, the row 343 pullback packet, the finite-moduli obstruction ledger, the row 341 finite-Hall-object packet, the compact-support model, and the compact Hall source obstruction ledger; it records the compact-support pushforward criterion; and it keeps the retained extension correspondence, retained q-map, properness or compactification row, constructible coefficient row, populated $q_!$-row, compactification independence, and product matrix rows open.

Proposition 0.154 *(Compactification independence criterion for the Hall $q_!$-leg).* [common-refinement criterion proved; compactification pair not supplied] Let

$$q : \mathfrak{E} \longrightarrow Z$$

be a supplied retained Hall target morphism at finite HN height R, and let $K_{\mathfrak{E}} \in D^b_c(\mathfrak{E})$ be the bounded constructible coefficient on the correspondence source. Suppose two compact-support models for $(q, K_{\mathfrak{E}})$ are supplied:

$$j_i : \mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_i, \qquad \overline{q}_i : \overline{\mathfrak{E}}_i \to Z, \qquad q = \overline{q}_i \circ j_i, \qquad i = 1, 2,$$

with $\overline{q}_i$ proper. Assume further that a common proper refinement is supplied:

$$j_{12} : \mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_{12}, \qquad r_i : \overline{\mathfrak{E}}_{12} \to \overline{\mathfrak{E}}_i, \qquad \overline{q}_{12} : \overline{\mathfrak{E}}_{12} \to Z,$$

where the maps r_i and $\overline{q}_{12}$ are proper and satisfy

$$r_i \circ j_{12} = j_i, \qquad \overline{q}_i \circ r_i = \overline{q}_{12}.$$

Finally assume that the extension-by-zero comparison morphisms

$$r_{i,*} j_{12,!} K_{\mathfrak{E}} \xrightarrow{\sim} j_{i,!} K_{\mathfrak{E}}, \qquad i = 1, 2,$$

are isomorphisms in the chosen constructible six-functor formalism. Then the two compact-support pushforwards are canonically identified:

$$\overline{q}_{1,*} j_{1,!} K_{\mathfrak{E}} \cong \overline{q}_{12,*} j_{12,!} K_{\mathfrak{E}} \cong \overline{q}_{2,*} j_{2,!} K_{\mathfrak{E}}.$$

Thus $q_!^{\mathrm{cs}} K_{\mathfrak{E}}$ is independent of the two compactifications whenever the common-refinement datum and the two comparison isomorphisms are supplied.

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall product. The compact-support model has only the abstract formula $\overline{q}_* j_{q,!}$; its choice-independence table is empty. The row 344 packet records the existence criterion for a supplied compactification, not a comparison between two choices. The finite-moduli ledger has no retained q-map row, no compactification pair row, and no common-refinement row. Hence row 345 is this common-refinement criterion and obstruction ledger, not a proof that the actual retained Hall $q_!$-leg is already independent of compactification.

Proof. For $i = 1, 2$, proper functoriality gives

$$\overline{q}_{i,*} r_{i,*} = (\overline{q}_i \circ r_i)_* = \overline{q}_{12,*}.$$

Composing this equality with the supplied comparison isomorphism

$$r_{i,*} j_{12,!} K_{\mathfrak{E}} \simeq j_{i,!} K_{\mathfrak{E}}$$

gives

$$\overline{q}_{i,*} j_{i,!} K_{\mathfrak{E}} \simeq \overline{q}_{i,*} r_{i,*} j_{12,!} K_{\mathfrak{E}} = \overline{q}_{12,*} j_{12,!} K_{\mathfrak{E}}.$$

The two choices $i = 1, 2$ are therefore identified through the same middle object. The proof uses only proper functoriality and the supplied extension-by-zero comparison maps; it does not construct the common refinement, prove base change or the projection formula, produce product matrices, or prove associativity. $\square$

The packet `certificates/hall/finite_hall_product_compactification_independence` verifies `FINITE_HALL_PRODUCT_COMPACTIFICATION_INDEPENDENCE_OBSTRUCTION_VERIFIED`: it imports the row 342 product-operator packet, the row 344 pushforward packet, the finite-moduli obstruction ledger, the compact-support model, and the compact Hall source obstruction ledger; it records the common-refinement criterion; and it keeps the retained q-map, compactification-pair row, common-refinement row, comparison-isomorphism row, populated choice-independence row, and product matrix rows open.

PROPOSITION 0.155 *(Associativity criterion via retained two-step flags).* [two-step-flag criterion proved; populated m_R-rows not supplied] Let $c_1, c_2, c_3 \in \widehat{\Gamma}_R$. Suppose the finite Hall product legs

$$m^{\mathrm{red}}_{a,b} = q^{\mathrm{cs}}_{a,b,!}\, \mathrm{TS}^{\mathrm{red}}_{a,b}\, p^*_{a,b}$$

are supplied, choice-independent, and constructible for the four pairs

$$(c_1, c_2), \quad (c_1 + c_2, c_3), \quad (c_2, c_3), \quad (c_1, c_2 + c_3)$$

that occur in the two parenthesizations. Assume also that a retained two-step extension flag stack

$$\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$$

is supplied, with comparison morphisms to the two iterated correspondence fibre products

$$\lambda : \mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3} \to \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_1+c_2,c_3} \times_{\mathfrak{M}^{\mathrm{red}}_{R,c_1+c_2}} \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_1,c_2},$$

$$\rho : \mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3} \to \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_1,c_2+c_3} \times_{\mathfrak{M}^{\mathrm{red}}_{R,c_2+c_3}} \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_2,c_3}.$$

Assume the squares defining λ and ρ satisfy the base-change isomorphisms for the relevant p^*- and $q^{cs}_!$ legs, that the projection formula holds for the coefficient objects, and that the reduced Thom–Sebastiani identifications on the two parenthesizations agree on $\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$. Finally assume that the two compact-support target pushforwards from the flag stack to $\mathfrak{M}^{\mathrm{red}}_{R,c_1+c_2+c_3}$ are identified by the compactification-independence criterion of Proposition 0.154.

Then for all supplied constructible inputs

$$K_i \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c_i}), \qquad i = 1, 2, 3,$$

there is a canonical associativity isomorphism

$$m^{\mathrm{red}}_{c_1+c_2,c_3}\big(m^{\mathrm{red}}_{c_1,c_2}(K_1, K_2), K_3\big) \cong m^{\mathrm{red}}_{c_1,c_2+c_3}\big(K_1, m^{\mathrm{red}}_{c_2,c_3}(K_2, K_3)\big).$$

After Borel–Moore homology and after choosing homogeneous bases, this is the rowwise associativity identity for the finite product matrix M_R.

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall product. Rows 342–345 define the operator and record criteria for p^*, $q_!$, and compactification independence, but their populated product, pushforward, and independence tables are empty. The hybrid two-step flag packet constructs the eight local/wrapped flag stacks and comparison maps before the reduced quotient; the wordwise associativity packets are conditional on functorial rows and do not populate the compact Hall carrier. The compact Hall source packet has no product matrix entries and no Hall-bialgebra associativity rows. Hence row 346 is this associativity criterion and obstruction ledger, not a proof that the actual retained m_R is already associative as a populated finite matrix.

Proof. Expand the left parenthesization by the definition of m^{red}. It is the compact-support pushforward along the left iterated correspondence after pulling back $K_1 \boxtimes K_2 \boxtimes K_3$ and applying reduced Thom–Sebastiani twice. The right parenthesization is the analogous pushforward along the right iterated correspondence. Pull both iterated correspondences back to $\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$ by λ and ρ. The supplied base-change and projection-formula isomorphisms identify the two iterated pull–push expressions with compact-support pushforwards from this common flag stack. The supplied Thom–Sebastiani associativity isomorphism identifies the two coefficient complexes on the flag stack. The supplied compactification-independence comparison identifies the two final compact-support pushforwards to $\mathfrak{M}^{\mathrm{red}}_{R,c_1+c_2+c_3}$. The resulting isomorphism is the displayed associativity isomorphism. Passing to Borel–Moore homology and to homogeneous bases gives the matrix identity whenever the product matrix rows are supplied. □

The packet `certificates/hall/finite_hall_product_associativity_two_step_flags` verifies `FINITE_HALL_PRODUCT_ASSOCIATIVITY_TWO_STEP_FLAGS_OBSTRUCTION_VERIFIED`: it imports the row 342 product operator, rows 344–345, the finite-moduli obstruction ledger, the compact-support model, the eight-word two-step flag-stack packet, the conditional wordwise associativity packets, and the compact Hall source obstruction ledger; it records the two-step-flag associativity criterion; and it keeps the populated compact product rows, compact Hall two-step flag rows, functorial base-change/projection/Thom–Sebastiani rows, quotient descent, transition rows, product matrices, and Hall-bialgebra associativity rows open.

For local-only inputs and ordered mixed and wrapped inputs, the same construction produces the three operation families $\mu_R^{\mathrm{loc}}, \mu_R^{\mathrm{hyb}}, \mu_R^{\mathrm{wr}}$ of §0.1. Existence of the product as a genuine morphism on Borel–Moore chains requires the admissibility/properness residuals of that section to vanish.

The collision coproduct on the source coalgebra

The collision coproduct is defined on the same correspondence geometry, read in the opposite direction: instead of contracting an extension to a single middle term, one expands an object into the pairs of its sub/quotient extensions. At finite stage, this is the 2-step flag stack already supplied as part of the hybrid datum.

Definition 0.156 *(Collision coproduct).* The finite collision coproduct

$$\Delta_R^{\mathrm{ch}} : C_{X,R} \longrightarrow C_{X,R} \otimes C_{X,R}$$

is the chain-level pull-push along the 2-step flag stack $\mathfrak{Flag}^{(2)}_{c,c',R}$ of subobjects of charge c and quotients of charge c' in $\mathfrak{M}^{\sigma}_{c+c',R}$, computed in the cosection-reduced d-critical heart, before the reduced E-quotient. If the higher-coloured residual, unit maps, refinement coherences, and quotient-after-correspondence residuals of §0.1 vanish, then after $Q_{E,R}$ it satisfies the coalgebra identities

$$(\Delta_R^{\mathrm{ch}} \otimes 1)\Delta_R^{\mathrm{ch}} = (1 \otimes \Delta_R^{\mathrm{ch}})\Delta_R^{\mathrm{ch}}, \qquad (\varepsilon_R^C \otimes 1)\Delta_R^{\mathrm{ch}} = (1 \otimes \varepsilon_R^C)\Delta_R^{\mathrm{ch}} = \mathrm{id},$$

which are the dual of the Hall pentagon on the same flag stack. For $R' \geq R$ the transition map $\rho^C_{R'R}$ commutes with Δ^{ch}.

Definition 0.157 *(Compact Hall coproduct on the cosection-reduced heart).* Let

$$\mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'} \xleftarrow{p_{c,c'}} \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \xrightarrow{q_{c,c'}} \mathfrak{M}^{\mathrm{red}}_{R,c+c'}$$

be the retained compactified extension correspondence at finite HN height R, with the same reduced vanishing-cycle and orientation coefficients as in Definition 0.151. The compact Hall coproduct operator is the formal pull–inverse Thom–Sebastiani–push operator

$$\Delta_R^{\mathrm{red}} = p_{c,c',!}^{\mathrm{cs}} \circ (\mathrm{TS}_{c,c'}^{\mathrm{red}})^{-1} \circ q_{c,c'}^{*}.$$

It sends a coefficient on $\mathfrak{M}_{R,c+c'}^{\mathrm{red}}$ to a coefficient on $\mathfrak{M}_{R,c}^{\mathrm{red}} \times \mathfrak{M}_{R,c'}^{\mathrm{red}}$, and after Borel–Moore Kunneth identification it is read as

$$\Delta_R^{\mathrm{red}} : \mathcal{H}_{R,c+c'}^{\mathrm{geom}} \longrightarrow \mathcal{H}_{R,c}^{\mathrm{geom}} \otimes \mathcal{H}_{R,c'}^{\mathrm{geom}}.$$

This definition names the operator once the three functorial pieces q^*, $(\mathrm{TS}^{\mathrm{red}})^{-1}$, and $p_!^{\mathrm{cs}}$ are supplied in the chosen six-functor model. It does not prove that q^* exists, that $p_!$ exists for the reversed correspondence, that $p_!$ is independent of a compactification, or that the coproduct is coassociative. It also does not supply counit maps, coproduct matrices, transition compatibility, bialgebra compatibility, or primitive closure.

The packet `certificates/hall/finite_hall_coproduct_operator` verifies `FINITE_HALL_COPRODUCT_OPERATOR_DEFINI` it records the row 347 definition

$$\Delta_R^{\mathrm{red}} = p_!(\mathrm{TS}^{\mathrm{red}})^{-1} q^*$$

and imports the row 341 finite compact Hall object, the compact-support exceptional pushforward model, the compact Hall source obstruction ledger, and the transition-Hall-coproduct obstruction ledger. The present packets give only the formal coproduct expression and the chosen compact-support notation. They do not populate retained splitting-correspondence rows, q^* rows, $p_!$ rows, inverse Thom–Sebastiani rows, compactification-independence rows, coproduct matrix rows, counit rows, coassociativity rows, bialgebra rows, or coproduct-transition rows.

PROPOSITION 0.158 *(Exceptional pushforward leg of the finite Hall coproduct).* [compact-support criterion proved; retained p-row not supplied] Let

$$p_{c,c'} : \overline{\mathfrak{E}}_{R,c,c'}^{\mathrm{ret}} \longrightarrow \mathfrak{M}_{R,c}^{\mathrm{red}} \times \mathfrak{M}_{R,c'}^{\mathrm{red}}$$

be a supplied retained splitting-correspondence morphism at finite HN height R, and let

$$K_{\mathfrak{E}} \in D_c^b(\overline{\mathfrak{E}}_{R,c,c'}^{\mathrm{ret}})$$

be the bounded constructible coefficient obtained after the preceding pullback $q_{c,c'}^*$ and inverse reduced Thom–Sebastiani transport. Assume that $p_{c,c'}$ lies in the chosen retained finite-type Artin-stack category with finite residual inertia and that one of the following two pieces of functorial data is supplied:

(i) $p_{c,c'}$ is proper in that category; or

(ii) a compact-support model

$$j_p : \overline{\mathfrak{E}}_{R,c,c'}^{\mathrm{ret}} \hookrightarrow \overline{\mathfrak{E}}_p, \qquad \overline{p} : \overline{\mathfrak{E}}_p \to \mathfrak{M}_{R,c}^{\mathrm{red}} \times \mathfrak{M}_{R,c'}^{\mathrm{red}}, \qquad p_{c,c'} = \overline{p} \circ j_p$$

is supplied as in Definition 0.25, with $j_{p,!}K_{\mathfrak{E}}$ and $\overline{p}_* j_{p,!}K_{\mathfrak{E}}$ in the same constructible six-functor formalism.

Then the exceptional pushforward leg

$$p^{\mathrm{cs}}_{c,c',!}K_{\mathfrak{E}} := \overline{p}_* j_{p,!}K_{\mathfrak{E}} \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'})$$

exists; in the proper case it is the ordinary proper pushforward $p_{c,c',*}K_{\mathfrak{E}}$. This is the $p_!$-leg of the finite Hall coproduct Δ^{red}_R.

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall coproduct. The row 347 coproduct operator records $p_!$ only as notation. The finite-moduli ledger has no extension flag-stack row, no retained splitting p-map row, no properness row for p, and no d-critical coefficient row on the splitting correspondence. The compact-support model defines $\overline{p}_* j_{p,!}$, but its map-population and choice-independence tables are empty. The compact Hall source packet has no coproduct matrix rows, and the transition-Hall-coproduct packet has no coproduct-transport rows. Hence row 348 is this exceptional pushforward criterion and obstruction ledger, not a proof that the retained Hall coproduct has a populated $p_!$-leg.

Proof. In the proper case, the six-functor formalism supplies the proper direct image

$$p_{c,c',*} : D^b_c(\overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'}) \longrightarrow D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'}),$$

and properness preserves constructibility. In the compact-support case, the supplied open immersion gives extension by zero

$$j_{p,!}K_{\mathfrak{E}} \in D^b_c(\overline{\mathfrak{E}}_p),$$

and the supplied proper map $\overline{p}$ gives the proper direct image

$$\overline{p}_* j_{p,!}K_{\mathfrak{E}} \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'}).$$

This composite is exactly the compact-support exceptional pushforward of Definition 0.25. The argument does not compare two compactifications, prove base change or the projection formula, construct the q^* row, construct the inverse reduced Thom–Sebastiani row, produce coproduct matrices, or prove coassociativity. □

The packet `certificates/hall/finite_hall_coproduct_pushforward_functor` verifies `FINITE_HALL_COPRODUCT_PUSHFORWARD_FUNCTOR_OBSTRUCTION_VERIFIED`: it imports the row 347 coproduct-operator packet, the finite-moduli obstruction ledger, the row 341 finite-Hall-object packet, the compact-support model, the compact Hall source obstruction ledger, and the transition-Hall-coproduct obstruction ledger; it records the compact-support pushforward criterion; and it keeps the retained splitting correspondence, retained p-map, properness or compactification row, constructible coefficient row, inverse Thom–Sebastiani row, populated $p_!$-row, compactification independence, coproduct matrix rows, and transition-coproduct rows open.

Proposition 0.159 *(Coassociativity criterion via retained two-step splitting flags).* [two-step-splitting-flag criterion proved; populated Δ_R-rows not supplied] Let $c_1, c_2, c_3 \in \widehat{\Gamma}_R$. Suppose the finite Hall coproduct legs

$$\Delta^{\mathrm{red}}_{a,b} = p^{\mathrm{cs}}_{a,b,!}(\mathrm{TS}^{\mathrm{red}}_{a,b})^{-1}q^*_{a,b}$$

are supplied, choice-independent, and constructible for the four pairs

$$(c_1 + c_2, c_3), \quad (c_1, c_2), \quad (c_1, c_2 + c_3), \quad (c_2, c_3)$$

that occur in the two coproduct parenthesizations. Assume also that a retained two-step splitting flag stack

$$\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$$

is supplied, with comparison morphisms to the two iterated splitting correspondence fibre products

$$\lambda : \mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3} \to \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_1+c_2,c_3} \times_{\mathfrak{M}^{\mathrm{red}}_{R,c_1+c_2}} \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_1,c_2},$$

$$\rho : \mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3} \to \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_1,c_2+c_3} \times_{\mathfrak{M}^{\mathrm{red}}_{R,c_2+c_3}} \overline{\mathfrak{E}}^{\mathrm{ret}}_{R;c_2,c_3}.$$

Assume the squares defining λ and ρ satisfy the base-change isomorphisms for the relevant q^*- and $p_!^{\mathrm{cs}}$-legs, that the projection formula holds for the coefficient objects, and that the inverse reduced Thom–Sebastiani identifications on the two parenthesizations agree on $\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$. Finally assume that the two compact-support pushforwards from the flag stack to

$$\mathfrak{M}^{\mathrm{red}}_{R,c_1} \times \mathfrak{M}^{\mathrm{red}}_{R,c_2} \times \mathfrak{M}^{\mathrm{red}}_{R,c_3}$$

are identified by a supplied compactification-independence comparison for the $p_!^{\mathrm{cs}}$-legs.

Then for every supplied constructible input

$$K \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,c_1+c_2+c_3})$$

there is a canonical coassociativity isomorphism

$$(\Delta^{\mathrm{red}}_{c_1,c_2} \otimes 1)\Delta^{\mathrm{red}}_{c_1+c_2,c_3}(K) \cong (1 \otimes \Delta^{\mathrm{red}}_{c_2,c_3})\Delta^{\mathrm{red}}_{c_1,c_2+c_3}(K).$$

After Borel–Moore homology and after choosing homogeneous bases, this is the rowwise coassociativity identity for the finite coproduct matrix D_R.

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall coproduct. Row 347 names the coproduct operator but has no populated Δ_R-rows. Row 348 proves only the $p_!$-existence criterion for a supplied retained splitting map, and its pushforward table is empty. The finite-moduli ledger has no retained two-step splitting flag stack. The compact-support model has no base-change, projection-formula, Thom–Sebastiani, or choice-independence rows. The hybrid two-step flag packet constructs local/wrapped flag stacks before the reduced quotient and does not populate the compact Hall carrier. The compact Hall source packet has no coproduct matrix entries and no Hall-bialgebra coassociativity rows. Hence row 349 is this coassociativity criterion and obstruction ledger, not a proof that the actual retained Δ_R is already coassociative as a populated finite matrix.

Proof. Expand the left parenthesization by the definition of Δ^{red}. It is the compact-support pushforward to $\mathfrak{M}^{\mathrm{red}}_{R,c_1} \times \mathfrak{M}^{\mathrm{red}}_{R,c_2} \times \mathfrak{M}^{\mathrm{red}}_{R,c_3}$ obtained by pulling K back along the two relevant q-maps, applying inverse reduced Thom–Sebastiani twice, and pushing along the two relevant p-maps. The right parenthesization is the analogous expression for the other iterated splitting correspondence. Pull both iterated correspondences back to $\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$ by λ and ρ. The supplied base-change and projection-formula isomorphisms identify the two iterated pull–inverse-TS–push expressions with compact-support pushforwards from this common flag stack. The supplied inverse Thom–Sebastiani coassociativity isomorphism identifies the two coefficient complexes on the flag stack. The supplied compactification-independence comparison identifies the two compact-support pushforwards to the triple product. The resulting isomorphism is the displayed coassociativity isomorphism. Passing to Borel–Moore homology and to homogeneous bases gives the matrix identity whenever the coproduct matrix rows are supplied. $\square$

The packet `certificates/hall/finite_hall_coproduct_coassociativity_two_step_flags` verifies `FINITE_HALL_COPRODUCT_COASSOCIATIVITY_TWO_STEP_FLAGS_OBSTRUCTION_VERIFIED`: it imports the row 347 coproduct operator, row 348, the finite-moduli obstruction ledger, the compact-support model, the eight-word two-step flag-stack packet, the compact Hall source obstruction ledger, and the transition-Hall-coproduct obstruction ledger; it records the two-step-splitting-flag coassociativity criterion; and it keeps the populated compact coproduct rows, compact Hall two-step splitting flag rows, functorial base-change/projection/inverse-Thom–Sebastiani rows, compactification-independence rows, quotient descent, transition rows, coproduct matrices, and Hall-bialgebra coassociativity rows open.

Proposition 0.160 *(Bialgebra compatibility criterion for the finite Hall product and coproduct).* [product–coproduct square criterion proved; populated bialgebra rows not supplied] Let $a, b, u, v \in \widehat{\Gamma}_R$ with $u + v = a + b$, and set

$$\mathsf{S}(a, b; u, v) = \{(a_1, a_2, b_1, b_2) \mid a = a_1 + a_2,\ b = b_1 + b_2,\ u = a_1 + b_1,\ v = a_2 + b_2\}.$$

The set $\mathsf{S}(a, b; u, v)$ is finite at finite HN height. Suppose the product legs $m^{\mathrm{red}}_{a,b}$, $m^{\mathrm{red}}_{a_1,b_1}$, $m^{\mathrm{red}}_{a_2,b_2}$ and the coproduct legs $\Delta^{\mathrm{red}}_{u,v}$, $\Delta^{\mathrm{red}}_{a_1,a_2}$, $\Delta^{\mathrm{red}}_{b_1,b_2}$ are supplied, constructible, and choice-independent for every $(a_1, a_2, b_1, b_2) \in \mathsf{S}(a, b; u, v)$. Assume also that a retained product–splitting square stack

$$\mathfrak{Q}^{\mathrm{ret}}_{R;a,b}{}^{u,v}$$

is supplied, with comparison morphisms to the iterated correspondence for $\Delta_{u,v} m_{a,b}$ and to the finite disjoint union, indexed by $\mathsf{S}(a, b; u, v)$, of the iterated correspondences for

$$(m^{\mathrm{red}}_{a_1,b_1} \otimes m^{\mathrm{red}}_{a_2,b_2})(1 \otimes \tau \otimes 1)(\Delta^{\mathrm{red}}_{a_1,a_2} \otimes \Delta^{\mathrm{red}}_{b_1,b_2}).$$

Assume the comparison squares satisfy proper base change for the relevant q^*, p^*, $q^{\mathrm{cs}}_!$, and $p^{\mathrm{cs}}_!$ legs; the projection formula for the coefficient objects; the product–coproduct Thom–Sebastiani coherence identifying the product-side and coproduct-side coefficient complexes on $\mathfrak{Q}^{\mathrm{ret}}_{R;a,b}{}^{u,v}$; the Koszul sign convention for the middle braiding τ; and compact-support choice-independence for all final pushforwards to $\mathfrak{M}^{\mathrm{red}}_{R,u} \times \mathfrak{M}^{\mathrm{red}}_{R,v}$.

Then for all supplied constructible inputs

$$K_a \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,a}), \qquad K_b \in D^b_c(\mathfrak{M}^{\mathrm{red}}_{R,b}),$$

there is a canonical bialgebra-compatibility isomorphism

$$\Delta^{\mathrm{red}}_{u,v} m^{\mathrm{red}}_{a,b}(K_a, K_b) \cong \bigoplus_{\mathsf{S}(a,b;u,v)} (m^{\mathrm{red}}_{a_1,b_1} \otimes m^{\mathrm{red}}_{a_2,b_2})(1 \otimes \tau \otimes 1)(\Delta^{\mathrm{red}}_{a_1,a_2} K_a \otimes \Delta^{\mathrm{red}}_{b_1,b_2} K_b).$$

After Borel–Moore homology and after choosing homogeneous bases, this is the rowwise matrix identity

$$D_R^{a+b;u,v} M_R^{a,b} = \sum_{\mathsf{S}(a,b;u,v)} (M_R^{a_1,b_1} \otimes M_R^{a_2,b_2})(1 \otimes \tau \otimes 1)(D_R^{a;a_1,a_2} \otimes D_R^{b;b_1,b_2}).$$

The present packets do not supply the hypotheses for the actual compact $K3 \times E$ Hall bialgebra. Row 342 names the product operator and row 347 names the coproduct operator; neither packet has populated operation rows. Rows 346 and 349 record the associativity and coassociativity criteria, but their identity tables are empty. The finite-moduli ledger has no retained product–splitting square stack. The compact-support model has no base-change, projection-formula, Thom–Sebastiani, or choice-independence rows for the mixed product–coproduct square. The compact Hall source packet has no M-entries, D-entries, or Hall-bialgebra identity rows. Hence row 350 is this bialgebra compatibility criterion and obstruction ledger, not a proof that the actual retained finite source already satisfies $\Delta m = (m \otimes m)\Delta$.

Proof. Expand the left side by the definitions of m^{red} and Δ^{red}. It is a compact-support pull–TS–push along the product correspondence followed by a pull–inverse-TS–push along the splitting correspondence. Expand the right side by first splitting the two inputs, applying the Koszul braiding to exchange the two middle factors, and multiplying the two output pairs. Pull the two iterated correspondences to $\mathfrak{Q}^{\mathrm{ret}}_{R;a,b}{}^{u,v}$. Proper base change and the projection formula identify the two iterated pull-push functors on this common square stack. The supplied product–coproduct Thom–Sebastiani coherence identifies the coefficient complexes; the middle exchange contributes exactly the Koszul sign τ. The compact-support choice-independence comparisons identify the final pushforwards to $\mathfrak{M}^{\mathrm{red}}_{R,u} \times \mathfrak{M}^{\mathrm{red}}_{R,v}$. The resulting isomorphism is the displayed bialgebra-compatibility isomorphism. Applying Borel–Moore homology, Kunneth, and homogeneous compact-source bases gives the displayed finite matrix identity whenever M_R, D_R, and the bialgebra identity rows are supplied. □

The packet `certificates/hall/finite_hall_bialgebra_compatibility` verifies `FINITE_HALL_BIALGEBRA_COMPATIBIL` it imports the product and coproduct operator packets, the row 346 associativity packet, the row 349 coassociativity packet, the finite-moduli obstruction ledger, the compact-support model, the compact Hall source obstruction ledger, and the product- and coproduct-transition obstruction ledgers; it records the product–coproduct square criterion; and it keeps the populated product rows, populated coproduct rows, retained product–splitting square rows, base-change/projection/Thom–Sebastiani rows, compactification-independence rows, M- and D-matrices, transition rows, and Hall-bialgebra compatibility rows open.

Definition 0.161 *(Finite Hall antipode datum).* A finite Hall antipode datum at HN height R is extra structure on a supplied finite Hall bialgebra

$$(\mathcal{H}_R, m_R, \Delta_R, \eta_R, \epsilon_R)$$

whose unit, counit, associativity, coassociativity, and product–coproduct compatibility rows have zero defect. It consists of either:

(i) explicit homogeneous antipode matrices

$$S_{R,\gamma} : \mathcal{H}_{R,\gamma} \longrightarrow \mathcal{H}_{R,\iota(\gamma)}$$

for a supplied charge involution ι preserving the retained finite window; or

(ii) a supplied connected conilpotent filtration for which the standard recursive antipode

$$S_R(1) = 1, \qquad S_R(x) = -x - \sum S_R(x')\,x'' \quad \big(\widetilde{\Delta}_R x = \sum x' \otimes x''\big)$$

terminates in every retained homogeneous block.

In either presentation the convolution identities

$$m_R(S_R \otimes \mathrm{id})\Delta_R = \eta_R\epsilon_R, \qquad m_R(\mathrm{id} \otimes S_R)\Delta_R = \eta_R\epsilon_R$$

must be supplied as zero-defect rows, and S_R must commute with the HN transition maps on every block where the pro-object is formed. A finite Hall bialgebra becomes a finite Hall Hopf algebra only after this antipode datum is supplied.

The present compact $K3 \times E$ packets do not supply an antipode datum. The source packet has no antipode matrix rows, no charge involution row, no connected conilpotent recursion row, and no left or right convolution identity row. The bialgebra-compatibility packet records only the criterion for $\Delta m = (m \otimes m)(1 \otimes \tau \otimes 1)(\Delta \otimes \Delta)$. Hopf-pairing rows, Drinfeld double comparisons, bar-coalgebra deconcatenation, primitive closure, scalar traces, and signed multiplicities are not antipode data.

The packet `certificates/hall/finite_hall_antipode_datum` verifies `FINITE_HALL_ANTIPODE_DATUM_OBSTRUCTION_VE` it imports the row 350 bialgebra-compatibility packet, the compact Hall source obstruction ledger, and the product- and coproduct-transition obstruction ledgers; it records Definition 0.161; and it keeps the antipode matrices, charge involution or connected-conilpotent filtration, unit and counit rows, bialgebra rows, convolution-identity rows, and transition-antipode rows open. Thus row 351 defines the antipode data required by any finite Hall Hopf-algebra claim; it does not assert that the current compact Hall source is a Hopf algebra.

Proposition 0.162 *(No Hopf algebra without an antipode; proved).* Let

$$(\mathcal{H}_R, m_R, \Delta_R, \eta_R, \epsilon_R)$$

be a supplied finite Hall bialgebra stage. If the finite Hall antipode datum of Definition 0.161 is not supplied, then this stage is not a finite Hall Hopf algebra in the verified source package of the manuscript. It may be used only as a bialgebra-level input. Every downstream occurrence of a Hopf algebra claim is therefore conditional on the antipode matrices, or on the terminating connected-conilpotent recursion, together with the two convolution identities and transition compatibility.

This is not a non-existence theorem for antipodes on the underlying graded vector space. It is the source-data boundary used in this manuscript: absent S_R and its convolution identities, the Hopf algebra name is unavailable.

Proof. A Hopf algebra is a bialgebra equipped with an antipode. In the finite Hall presentation this means exactly the datum S_R and the two identities

$$m_R(S_R \otimes \mathrm{id})\Delta_R = \eta_R\epsilon_R, \qquad m_R(\mathrm{id} \otimes S_R)\Delta_R = \eta_R\epsilon_R.$$

If no such row is supplied, no convolution inverse for the identity endomorphism of $\mathcal{H}_R$ has been constructed. Bialgebra compatibility, Hopf-pairing adjointness, a target Drinfeld double, a bar-coalgebra model, primitive closure, and scalar trace data do not define that convolution inverse. The verified source package therefore remains bialgebra-level until the antipode datum is added. □

The packet `certificates/hall/finite_hall_nonhopf_boundary` verifies `FINITE_HALL_NONHOPF_BOUNDARY_VERIFIED`: it imports the row 351 antipode-datum packet, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger; it records Proposition 0.162; it checks that the current source has no antipode rows, no convolution identity rows, and no finite Hall Hopf-algebra claim; and it records explicitly that no non-existence theorem for future antipodes is being asserted.

This is the source coalgebra of §0.5; its identification with the bar construction $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X)$ is the bar comparison there.

Primitive projection and the bialgebra datum

The primitive subspace is a bialgebra notion. It uses the augmentation, unit, counit, and coproduct; its closure under the supercommutator uses bialgebra compatibility. It does not make the source object Hopf. Hopf-level recognition is available only after the antipode datum named in Proposition 0.162 is supplied.

Definition 0.163 *(Finite primitive Hall subspace).* Let $(\mathcal{H}_R, \Delta_R, \eta_R, \epsilon_R)$ be a finite counital Hall coalgebra stage with supplied homogeneous finite bases. Put

$$I_R = \ker \epsilon_R.$$

The reduced coproduct is the map

$$\bar{\Delta}_R : I_R \longrightarrow I_R \otimes I_R, \qquad \bar{\Delta}_R(x) = \Delta_R x - \eta_R(1) \otimes x - x \otimes \eta_R(1),$$

equivalently the projection of $\Delta_R|_{I_R}$ to $I_R \otimes I_R$. The finite primitive Hall subspace is

$$P_R = \mathrm{Prim}(\mathcal{H}_R) = \ker\big(\bar{\Delta}_R : I_R \to I_R \otimes I_R\big).$$

In finite coordinates, row 353 consists of the counit matrix, the augmentation-ideal basis, the reduced-coproduct matrix $\bar{D}_R$, and kernel rows giving a basis, inclusion, and projection for P_R. It is a definition of a subspace of the supplied finite coalgebra. It does not assert that the compact $K3 \times E$ source has supplied those rows, and it does not assert primitive closure under the Hall commutator.

The packet `certificates/hall/finite_hall_primitive_subspace_definition` verifies `FINITE_HALL_PRIMITIVE_SUBSPACE_DEFINITION_OBSTRUCTION_VERIFIED`: it imports the row 352 non-Hopf boundary, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger; it records Definition 0.163; and it keeps the source coproduct matrix, unit and counit rows, augmentation-ideal basis, reduced-coproduct matrix, primitive-kernel rows, primitive basis, inclusion, projection, and degree–parity split open. Thus row 353 defines $P_R = \ker \bar{\Delta}_R$ for supplied finite data; it does not populate P_R in the present compact source.

Definition 0.164 *(Primitive Hall sub-bracket).* Write

$$\widetilde{P}_X = H^0\, \mathrm{Prim}_{\mathrm{prot}}(\mathcal{F}_X) = \ker\big(\widetilde{\Delta} : \mathcal{F}_X \to \mathcal{F}_X \otimes \mathcal{F}_X\big)$$

for the primitive layer of the protected Hall bialgebra object, where $\widetilde{\Delta} = \Delta - (\eta\varepsilon \otimes \mathrm{id} + \mathrm{id} \otimes \eta\varepsilon)$. The Hall product restricts to a $\widehat{\Gamma}_X$-graded bracket

$$[\,\cdot\,,\,\cdot\,]_X : \widetilde{P}_{X,\widehat{c}} \otimes \widetilde{P}_{X,\widehat{c}'} \longrightarrow \widetilde{P}_{X,\widehat{c}\star\widehat{c}'},$$

the protected primitive Hall bracket. After normal-ordered Gram descent (§0.1) and the Hopf-radical quotient, the descended bracket is Γ_{gram}-graded and the descended primitive layer is the candidate $P_X^{\Pi,\mathrm{red}}$ to be identified with $\mathfrak{g}_{\Delta_5}$.

The finite packet `certificates/charge/hall_pairing_pushforward_compatibility` supplies only the grading calculation behind this sentence: a supplied $\widehat{\Gamma}_X$-graded bracket and a supplied homogeneous degree-zero pairing remain graded after pushforward by the additive normal-ordered map $\overline{\Pi}_X$. The compact Hall correspondence, primitive closure, Hopf adjointness, Frobenius cyclic identity, quotient non-degeneracy, and protected trace remain separate data.

Proposition 0.165 *(Product-only Hall data do not define primitives; proved).* Let $\mathcal{H}_R$ be a finite HN stage carrying the Hall product m_R and unit η_R. Without a counital coproduct Δ_R and the bialgebra compatibility of m_R with Δ_R, there is no defined primitive subspace

$$\mathrm{Prim}(\mathcal{H}_R) = \ker\big(\Delta_R - \eta_R\varepsilon_R \otimes \mathrm{id} - \mathrm{id} \otimes \eta_R\varepsilon_R\big),$$

and the supercommutator of m_R does not define a Lie bracket on primitives. Thus a compact Hall product, even if associative and oriented, is not a primitive-recognition datum.

Proof. The displayed formula is the definition of primitives in a counital bialgebra. If Δ_R or ε_R is absent, the kernel is not an object. If Δ_R is present but not compatible with m_R, the bialgebra identity

$$\Delta_R([x, y]) = [\Delta_R x, \Delta_R y]$$

in the graded tensor-product algebra does not hold. Hence $\Delta_R x = x \otimes 1 + 1 \otimes x$ and $\Delta_R y = y \otimes 1 + 1 \otimes y$ no longer imply $\Delta_R([x, y]) = [x, y] \otimes 1 + 1 \otimes [x, y]$.

The product does not determine the coproduct. Over a characteristic zero field, the graded algebra $A = k[x, y]$, $|y| = 2|x|$, has the primitive coproduct $\Delta_0 x = x \otimes 1 + 1 \otimes x$, $\Delta_0 y = y \otimes 1 + 1 \otimes y$, and also the counital coassociative algebra coproduct

$$\Delta_1 x = x \otimes 1 + 1 \otimes x, \qquad \Delta_1 y = y \otimes 1 + 1 \otimes y + x \otimes x.$$

The multiplication is the same in both bialgebras, but $\mathrm{Prim}(A, \Delta_0)$ and $\mathrm{Prim}(A, \Delta_1)$ are different subspaces: y is primitive for Δ_0, while $y - \frac{1}{2}x^2$ is primitive for Δ_1. Hence the Hall product alone cannot determine P_R^{Prim}, the Hopf radical, or the PBW comparison used in primitive recognition. □

Definition 0.166 *(Finite Hall bialgebra defect tuple).* Fix homogeneous finite bases on a retained HN stage and write (M, D, η, ϵ) for the matrices of the Hall product, coproduct, unit, and counit. The finite Hall bialgebra defect tuple is

$$\mathfrak{D}_{\mathrm{Hbialg},R} = \big(\mathfrak{d}_R^{\mathrm{unit},L}, \mathfrak{d}_R^{\mathrm{unit},R}, \mathfrak{d}_R^{\mathrm{counit},L}, \mathfrak{d}_R^{\mathrm{counit},R}, \mathfrak{d}_R^{\mathrm{assoc}}, \mathfrak{d}_R^{\mathrm{coassoc}}, \mathfrak{d}_R^{\mathrm{bialg}}, \mathfrak{d}_R^{\mathrm{prim}}\big),$$

where

$$\mathfrak{d}_R^{\mathrm{assoc}} = m_R(m_R \otimes 1) - m_R(1 \otimes m_R), \qquad \mathfrak{d}_R^{\mathrm{coassoc}} = (\Delta_R \otimes 1)\Delta_R - (1 \otimes \Delta_R)\Delta_R,$$

$$\mathfrak{d}_R^{\mathrm{bialg}} = \Delta_R m_R - (m_R \otimes m_R)(1 \otimes \tau \otimes 1)(\Delta_R \otimes \Delta_R),$$

with τ the Koszul sign braiding, and

$$\mathfrak{d}_R^{\mathrm{prim}}(x, y) = \widetilde{\Delta}_R\big(m_R(x, y) - (-1)^{|x||y|} m_R(y, x)\big)$$

for $x, y \in \ker \widetilde{\Delta}_R \cap \ker \epsilon_R$. The unit and counit components are the four usual left/right defects

$$m_R(\eta_R \otimes 1) - 1, \quad m_R(1 \otimes \eta_R) - 1, \quad (\epsilon_R \otimes 1)\Delta_R - 1, \quad (1 \otimes \epsilon_R)\Delta_R - 1.$$

Proposition 0.167 *(Hall primitives are formed only after the bialgebra defects vanish; proved).* At a finite HN stage, the matrices (M, D, η, ϵ) define a counital graded bialgebra whose supercommutator restricts to a bracket on the primitive subspace if and only if the first seven components of $\mathfrak{D}_{\mathrm{Hbialg},R}$ vanish; then $\mathfrak{d}_R^{\mathrm{prim}} = 0$ follows. Conversely, a finite source proof table in which any component of $\mathfrak{D}_{\mathrm{Hbialg},R}$ is absent or non-zero does not define the primitive Hall Lie superalgebra used in Theorem 0.215.

Proof. The unit, counit, associativity, coassociativity, and bialgebra compatibility equations are exactly the first seven displayed defects. When they vanish, the standard calculation in a graded bialgebra gives, for primitive x, y,

$$\Delta_R([x,y]) = [x,y] \otimes 1 + 1 \otimes [x,y], \qquad \epsilon_R([x,y]) = 0,$$

because Δ_R is multiplicative and ϵ_R is an algebra homomorphism. Hence the supercommutator lies in the primitive subspace, which is the vanishing of $\mathfrak{o}_R^{\mathrm{prim}}$. If one of the first seven equations is missing or nonzero, the object is not a counital bialgebra on the chosen finite stage, and the displayed primitive-bracket calculation is not available. □

The packet `certificates/hall/finite_hall_primitive_commutator_closure` verifies `FINITE_HALL_PRIMITIVE_COMMUTATOR_CLOSURE_OBSTRUCTION_VERIFIED`: it imports the row 353 primitive-subspace definition, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger; it records Proposition 0.167 as the row 354 relative proof; and it keeps the product matrix, coproduct matrix, unit and counit rows, primitive-kernel basis, bialgebra identity rows, primitive-closure defect rows, and supercommutator matrix rows open. Hence the proposition proves the closure theorem for supplied finite bialgebra data; the present compact $K3 \times E$ source still has no populated primitive Hall bracket.

PROPOSITION 0.168 *(The primitive Hall supercommutator satisfies graded Jacobi; proved).* Let A be an associative $\mathbb{Z}/2$-graded algebra and let

$$[x,y] = xy - (-1)^{|x||y|} yx$$

for homogeneous $x, y \in A$. Then the bracket is graded skew-symmetric and satisfies the graded Jacobi identity

$$(-1)^{|x||z|}[x,[y,z]] + (-1)^{|y||x|}[y,[z,x]] + (-1)^{|z||y|}[z,[x,y]] = 0.$$

Consequently, if the finite Hall bialgebra data of Proposition 0.167 are supplied so that P_R is closed under the supercommutator, then P_R is a finite graded Lie algebra object under the Hall bracket.

Proof. Graded skew-symmetry is immediate from the definition. For Jacobi, expand the three displayed summands in the associative algebra A. The twelve monomials $xyz, xzy, yxz, yzx, zxy, zyx$, each with its Koszul sign, occur in cancelling pairs after using only associativity and the parity rule $|xy| = |x| + |y|$. Thus the supercommutator on any associative graded algebra is a graded Lie bracket. If $x, y, z \in P_R$, Proposition 0.167 keeps each inner and outer commutator in P_R, so the same identity holds in the primitive Hall subspace. □

The packet `certificates/hall/finite_hall_graded_jacobi` verifies `FINITE_HALL_GRADED_JACOBI_OBSTRUCTION_VERI` it imports the row 354 primitive-commutator closure packet, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger; it records Proposition 0.168; and it keeps the source product matrix, primitive bracket matrix, associativity rows, primitive-closure rows, Jacobi defect rows, parity rows, and normal-ordered bracket matrices open. Hence row 355 proves graded Jacobi for supplied associative finite Hall data; it does not populate a compact $K3 \times E$ primitive bracket.

PROPOSITION 0.169 *(The Hall bracket preserves normal-ordered Gram degree; proved for supplied homogeneous data).* Let $P_R = \bigoplus_{\hat{c} \in \hat{\Gamma}_R} P_{R,\hat{c}}$ be a finite primitive Hall stage whose bracket is obtained from a homogeneous Hall product:

$$[P_{R,\hat{c}}, P_{R,\hat{c}'}] \subset P_{R,\hat{c}+\hat{c}'}.$$

Let

$$\overline{\Pi}_X : \widehat{\Gamma}_R \longrightarrow \Gamma_R^{\Pi}$$

be the additive normal-ordered Gram homomorphism. Define the descended fibre-summed pieces

$$P_{R,\gamma}^{\Pi} = \bigoplus_{\overline{\Pi}_X(\widehat{c})=\gamma} P_{R,\widehat{c}}.$$

Then

$$[P_{R,\gamma}^{\Pi}, P_{R,\gamma'}^{\Pi}] \subset P_{R,\gamma+\gamma'}^{\Pi}.$$

Equivalently, every non-zero finite bracket row has zero normal-ordered Gram-degree defect

$$\overline{\Pi}_X(\widehat{e}) - \overline{\Pi}_X(\widehat{c}) - \overline{\Pi}_X(\widehat{c}') = 0$$

for its source degrees $\widehat{c}, \widehat{c}'$ and target degree $\widehat{e} = \widehat{c} + \widehat{c}'$.

Proof. The Hall product is homogeneous by hypothesis, so both ordered products xy and yx in the supercommutator have source degree $\widehat{c} + \widehat{c}'$. Subtracting them with the Koszul sign cannot change the charge degree. Applying the additive normal-ordered homomorphism gives

$$\overline{\Pi}_X(\widehat{c} + \widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}').$$

If several source charges have the same descended Gram label, the definition of $P_{R,\gamma}^{\Pi}$ is the direct sum over that fibre; the same additivity sends the whole bracket of the two fibres into the fibre over $\gamma + \gamma'$. This proves the displayed inclusion. □

The packet `certificates/hall/finite_hall_normal_ordered_gram_degree` verifies `FINITE_HALL_NORMAL_ORDERED_GRAM_DEGREE_OBSTRUCTION_VERIFIED`: it imports the row 355 graded-Jacobi packet, the finite `hall_pairing_pushforward_compatibility` packet, the empty $K3 \times E$ charge-window fixture, and the compact Hall source obstruction ledger; it records Proposition 0.169; and it keeps the current source degree rows, normal-ordered lift rows, Hall product rows, primitive bracket rows, and bracket-degree defect rows open. Hence row 356 proves the Gram-degree statement for supplied homogeneous normal-ordered Hall data; it does not populate the compact $K3 \times E$ bracket or replace the row 357 parity check.

Proposition 0.170 *(The Hall bracket respects parity; proved for supplied homogeneous data).* Let $A = A_{\bar{0}} \oplus A_{\bar{1}}$ be a $\mathbb{Z}/2$-graded associative algebra whose product is even:

$$A_p A_q \subset A_{p+q} \qquad (p, q \in \mathbb{Z}/2).$$

For homogeneous $x \in A_p$ and $y \in A_q$, set

$$[x, y] = xy - (-1)^{pq} yx.$$

Then $[x, y] \in A_{p+q}$. Consequently, if the finite primitive Hall stage P_R is supplied as a parity-graded subspace closed under this supercommutator, then

$$[P_{R,\bar{p}}, P_{R,\bar{q}}] \subset P_{R,\bar{p}+\bar{q}}.$$

After normal-ordered Gram descent, the same statement holds fibrewise:

$$[P_{R,\gamma,\bar{p}}^{\Pi}, P_{R,\gamma',\bar{q}}^{\Pi}] \subset P_{R,\gamma+\gamma',\bar{p}+\bar{q}}^{\Pi}$$

whenever the row 356 degree hypotheses and the parity block rows are supplied.

Proof. Since the multiplication is even, both xy and yx lie in the same parity summand A_{p+q}. Their signed difference therefore also lies in A_{p+q}. Restricting to P_R uses only the supplied closure of P_R under the supercommutator and the supplied parity decomposition of P_R. The normal-ordered statement is obtained by combining this parity calculation with Proposition 0.169: the Gram label adds independently of the parity label, while the parity label adds in $\mathbb{Z}/2$. □

The packet `certificates/hall/finite_hall_bracket_parity` verifies `FINITE_HALL_BRACKET_PARITY_OBSTRUCTION_VE` it imports the row 355 graded-Jacobi packet, the row 356 normal-ordered Gram-degree packet, the finite `hall_pairing_pushforward_compatibility` bracket rows, the finite `parity_pushforward` packet, and the compact Hall source obstruction ledger; it records Proposition 0.170; and it keeps the current source parity blocks, homogeneous product rows, primitive bracket rows, and bracket-parity defect rows open. Hence row 357 proves parity compatibility for supplied homogeneous finite Hall data; it does not prove full $K3 \times E$ source parity or the GN–Kac parity comparison required later in primitive recognition.

Definition 0.171 *(Compact geometric Hall bialgebra datum).* At a finite HN height R, a compact geometric Hall bialgebra datum is the following finite correspondence package on the cosection-reduced d-critical heart:

(i) finite-type retained substacks $\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}}$ for all $\widehat{c} \in \widehat{\Gamma}_R$, with finite residual inertia, reduced vanishing-cycle sheaves $\Phi^{\mathrm{red}}_{R,\widehat{c}}$, and Joyce–Upmeier orientation lines $o_{R,\widehat{c}}$;

(ii) compactified extension correspondences

$$\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}'} \xleftarrow{\ p\ } \overline{\mathfrak{E}}^{\mathrm{red}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{\ q\ } \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}+\widehat{c}'}$$

with admissible p^*, $q_!$ for the product and admissible q^*, $p_!$ for the collision coproduct, in either a retained compactified flag-Quot model or an explicitly supplied compact-support six-functor model;

(iii) reduced Thom–Sebastiani isomorphisms $\mathrm{TS}^{\mathrm{red}}_{\widehat{c},\widehat{c}'}$ and their inverse transports, compatible with the multiplicative orientation cocycle;

(iv) vacuum correspondences for the zero object, giving the unit η_R and counit ϵ_R;

(v) three-step and four-step retained flag stacks carrying the associativity, coassociativity, and bialgebra-compatibility diagrams, with base-change, projection-formula, Koszul sign, and Thom–Sebastiani coherence witnesses;

(vi) reduced E-quotient and HN-transition compatibilities for all the preceding correspondences.

The product and coproduct attached to this datum are

$$m_R = q_!\, \mathrm{TS}^{\mathrm{red}}\, p^*, \qquad \Delta_R = p_!(\mathrm{TS}^{\mathrm{red}})^{-1} q^*.$$

Definition 0.172 *(Hall-product-preserving transition datum).* Let $R' \le R$. Suppose the finite-moduli transition geometry of Definition 0.242, the orientation-preserving transition datum of Definition 0.143, and the vanishing-cycle-preserving transition datum of Definition 0.147 are supplied. A Hall-product-preserving transition datum consists of:

(i) a transition cube for the retained compactified extension correspondences

$$\mathfrak{M}_R \times \mathfrak{M}_R \xleftarrow{\ p_R\ } \overline{\mathfrak{E}}_R \xrightarrow{\ q_R\ } \mathfrak{M}_R, \qquad \mathfrak{M}_{R'} \times \mathfrak{M}_{R'} \xleftarrow{\ p_{R'}\ } \overline{\mathfrak{E}}_{R'} \xrightarrow{\ q_{R'}\ } \mathfrak{M}_{R'},$$

whose input and output squares commute with the stage transitions;

(ii) transport of the coefficient system $\Phi_R^{\mathrm{red}} \otimes o_R$ on the extension graph, including the Thom–Sebastiani product isomorphism

$$\mathrm{TS}_R^{\mathrm{red}} : p_R^*(\Phi_R^{\mathrm{red}} \boxtimes \Phi_R^{\mathrm{red}}) \xrightarrow{\sim} q_R^* \Phi_R^{\mathrm{red}};$$

(iii) proper base-change, projection-formula, and compact-support pushforward isomorphisms comparing the two pull-push functors

$$q_{R,!}\, \mathrm{TS}_R^{\mathrm{red}}\, p_R^* \quad \text{and} \quad q_{R',!}\, \mathrm{TS}_{R'}^{\mathrm{red}}\, p_{R'}^*$$

after applying the transition maps;

(iv) finite product matrices $M_R, M_{R'}$ in compact source bases and transition matrices $T_R, T_{R'}$ satisfying the product intertwining identity

$$T_{\alpha+\beta}\, M_R^{\alpha,\beta} = M_{R'}^{\alpha,\beta}(T_\alpha \otimes T_\beta)$$

in every retained degree and parity block;

(v) compatibility with the associativity square on the retained two-step flag stack;

(vi) zero correspondence, coefficient-system, base-change, projection-formula, product-intertwining, associativity-transition, and composition defect ranks, together with $R^1 \varprojlim = 0$ for the Hall product maps.

Scalar traces, denominator products, signed multiplicities, target BKM products, Pfaffian products, orientation transport alone, and vanishing-cycle transport alone are not Hall-product transition data.

Proposition 0.173 *(Transition preservation of compact Hall products; proved).* Given a Hall-product-preserving transition datum of Definition 0.172, the finite-stage transition $R \to R'$ preserves the compact Hall product: on every retained finite degree block,

$$T_R \circ m_R = m_{R'} \circ (T_R \otimes T_R).$$

Proof. The transition cube identifies the input pullbacks p_R^* and $p_{R'}^*$ after passage to the transition graph. The coefficient transport identifies $\Phi^{\mathrm{red}} \otimes o$ and the Thom–Sebastiani product isomorphisms on the two sides. Proper base change, the projection formula, and compact-support pushforward compatibility then identify the two exceptional pushforwards along q_R and $q_{R'}$. Hence the two pull-push functors defining m_R and $m_{R'}$ commute with transition. In finite compact source bases this is exactly the displayed matrix identity. The associativity-transition and $R^1 \varprojlim$ rows give strict composition and inverse-limit exactness for the product maps. □

The packet `certificates/hall/transition_hall_product_preservation` records the present state of this datum. Its verifier status is `TRANSITION_HALL_PRODUCT_PRESERVATION_OBSTRUCTION_VERIFIED`: the transition-geometry input, orientation-transition input, vanishing-cycle transition input, source product matrices, extension correspondence transition squares, input and output product-stack squares, coefficient-system transport rows, Thom–Sebastiani product transport rows, proper base-change rows, projection-formula rows, compact-support pushforward rows, product-matrix intertwining rows, associativity-transition rows, and Hall-product Mittag–Leffler rows are not yet supplied. Hence transition preservation of compact Hall products remains open.

Definition 0.174 *(Hall-coproduct-preserving transition datum).* Let $R' \leq R$. Suppose the finite-moduli transition geometry of Definition 0.242, the orientation-preserving transition datum of Definition 0.143, and the vanishing-cycle-preserving transition datum of Definition 0.147 are supplied. A Hall-coproduct-preserving transition datum consists of:

(i) a transition cube for the same retained compactified extension correspondences, read in the splitting direction

$$\mathfrak{M}_R \xleftarrow{q_R} \overline{\mathfrak{E}}_R \xrightarrow{p_R} \mathfrak{M}_R \times \mathfrak{M}_R, \qquad \mathfrak{M}_{R'} \xleftarrow{q_{R'}} \overline{\mathfrak{E}}_{R'} \xrightarrow{p_{R'}} \mathfrak{M}_{R'} \times \mathfrak{M}_{R'},$$

whose q-source square and p-two-output square commute with the stage transitions;

(ii) transport of the coefficient system $\Phi_R^{\mathrm{red}} \otimes o_R$ on the splitting graph, including the inverse Thom–Sebastiani coproduct isomorphism

$$(\mathrm{TS}_R^{\mathrm{red}})^{-1} : q_R^* \Phi_R^{\mathrm{red}} \xrightarrow{\sim} p_R^*(\Phi_R^{\mathrm{red}} \boxtimes \Phi_R^{\mathrm{red}});$$

(iii) proper base-change, projection-formula, and compact-support pushforward isomorphisms comparing the two pull-push functors

$$p_{R,!}(\mathrm{TS}_R^{\mathrm{red}})^{-1} q_R^* \quad \text{and} \quad p_{R',!}(\mathrm{TS}_{R'}^{\mathrm{red}})^{-1} q_{R'}^*$$

after applying the transition maps;

(iv) finite coproduct matrices $D_R, D_{R'}$ in compact source bases, counit matrices $\epsilon_R, \epsilon_{R'}$, and transition matrices T_γ satisfying the coproduct intertwining identity

$$(T_\mu \otimes T_\nu)\, D_R^{\rho;\mu,\nu} = D_{R'}^{\rho;\mu,\nu}\, T_\rho$$

in every retained degree and parity block with $\rho = \mu + \nu$;

(v) compatibility with the coassociativity square on the retained two-step flag stack and with the zero-object counit square;

(vi) zero correspondence, coefficient-system, base-change, projection-formula, coproduct-intertwining, coassociativity-transition, counit-transition, and composition defect ranks, together with $R^1 \varprojlim = 0$ for the Hall coproduct maps.

Scalar traces, denominator products, signed multiplicities, target BKM coalgebra data, Pfaffian operations, bar-coalgebra deconcatenation, Hall-product preservation, orientation transport alone, and vanishing-cycle transport alone are not Hall-coproduct transition data.

PROPOSITION 0.175 *(Transition preservation of compact Hall coproducts; proved).* Given a Hall-coproduct-preserving transition datum of Definition 0.174, the finite-stage transition $R \to R'$ preserves the compact Hall coproduct: on every retained finite degree block with $\rho = \mu + \nu$,

$$(T_\mu \otimes T_\nu) \circ \Delta_R^{\rho;\mu,\nu} = \Delta_{R'}^{\rho;\mu,\nu} \circ T_\rho.$$

Proof. The transition cube identifies the source pullbacks q_R^* and $q_{R'}^*$ after passage to the transition graph. The coefficient transport identifies $\Phi^{\mathrm{red}} \otimes o$ and the inverse Thom–Sebastiani isomorphisms on the two sides. Proper base change, the projection formula, and compact-support pushforward compatibility then identify the two exceptional pushforwards along p_R and $p_{R'}$. Hence the two pull-push functors defining Δ_R and $\Delta_{R'}$ commute with transition. In finite compact source bases this is exactly the displayed matrix identity. The coassociativity-transition, counit-transition, and $R^1 \varprojlim$ rows give strict composition, counital compatibility, and inverse-limit exactness for the coproduct maps. □

The packet `certificates/hall/transition_hall_coproduct_preservation` records the present state of this datum. Its verifier status is `TRANSITION_HALL_COPRODUCT_PRESERVATION_OBSTRUCTION_VERIFIED`: the transition-geometry input, orientation-transition input, vanishing-cycle transition input, source coproduct matrices, source counit maps, splitting correspondence transition squares, source object and target product stack squares, coefficient-system transport rows, inverse Thom–Sebastiani coproduct transport rows, proper base-change rows, projection-formula rows, compact-support pushforward rows, coproduct-matrix intertwining rows, coassociativity-transition rows, counit-transition rows, and Hall-coproduct Mittag–Leffler rows are not yet supplied. Hence transition preservation of compact Hall coproducts remains open.

Definition 0.176 *(Primitive-subspace-preserving transition datum).* Let $R' \leq R$, and suppose the finite Hall stages carry counital coproducts (Δ_R, ϵ_R) and $(\Delta_{R'}, \epsilon_{R'})$. Put

$$\widetilde{\Delta}_R = \Delta_R - \eta_R\epsilon_R \otimes \mathrm{id} - \mathrm{id} \otimes \eta_R\epsilon_R, \qquad P_R = \ker \widetilde{\Delta}_R \cap \ker \epsilon_R.$$

A primitive-subspace-preserving transition datum consists of:

(i) finite compact-source matrices for $\Delta_R, \Delta_{R'}, \epsilon_R, \epsilon_{R'}$, the associated reduced coproducts $\widetilde{\Delta}_R, \widetilde{\Delta}_{R'}$, and the ambient Hall transition $T_R : \mathcal{H}_R \to \mathcal{H}_{R'}$;

(ii) source and target primitive kernel rows, including inclusions $\iota_R : P_R \hookrightarrow \mathcal{H}_R$, $\iota_{R'} : P_{R'} \hookrightarrow \mathcal{H}_{R'}$ and finite primitive projection matrices;

(iii) the reduced-coproduct and counit intertwining identities

$$\widetilde{\Delta}_{R'} T_R = (T_R \otimes T_R)\widetilde{\Delta}_R, \qquad \epsilon_{R'} T_R = \epsilon_R;$$

(iv) a restricted transition matrix

$$T_R^{\mathrm{Prim}} : P_R \longrightarrow P_{R'}$$

satisfying $\iota_{R'} T_R^{\mathrm{Prim}} = T_R \iota_R$, together with zero primitive-image, projection-commutator, and composition defect ranks.

Hall product preservation, scalar trace data, denominator products, target root spaces, bar-coalgebra primitives, normal-ordered degree maps, and primitive ranks without inclusions are not primitive-subspace transition data.

Proposition 0.177 *(Transition preservation of primitive Hall subspaces; proved).* Given a primitive-subspace-preserving transition datum of Definition 0.176, the ambient Hall transition carries primitives to primitives:

$$T_R(P_R) \subseteq P_{R'}.$$

Equivalently, $T_R|_{P_R}$ factors through the displayed $T_R^{\mathrm{Prim}} : P_R \to P_{R'}$.

Proof. Let $x \in P_R$. Then $\widetilde{\Delta}_R x = 0$ and $\epsilon_R x = 0$. The two intertwining identities give

$$\widetilde{\Delta}_{R'}(T_R x) = (T_R \otimes T_R)\widetilde{\Delta}_R x = 0, \qquad \epsilon_{R'}(T_R x) = \epsilon_R x = 0.$$

Thus $T_R x \in \ker \widetilde{\Delta}_{R'} \cap \ker \epsilon_{R'} = P_{R'}$. The finite restricted matrix T_R^{Prim} is the same statement in the chosen primitive bases. The projection and composition defect rows assert that this factorization is compatible with the finite primitive summands and with successive stage transitions. $\square$

The packet `certificates/hall/transition_primitive_subspace_preservation` records the present state of this datum. Its verifier status is `TRANSITION_PRIMITIVE_SUBSPACE_PRESERVATION_OBSTRUCTION_VERIFIED`: the source and target coproduct matrices, source and target counit maps, reduced-coproduct matrices, primitive kernel rows, ambient Hall transition matrices, Hall-coproduct transition input, counit-transition compatibility, kernel-intertwining rows, primitive projection matrices, restricted primitive transition matrices, and primitive-transition composition rows are not yet supplied. Hence transition preservation of primitive Hall subspaces remains open.

PROPOSITION 0.178 *(Geometric Hall data give the finite bialgebra defect rows; proved relative to the datum).* Let the compact geometric Hall bialgebra datum of Definition 0.171 be given at height R. Then the matrices of $(m_R, \Delta_R, \eta_R, \epsilon_R)$ satisfy the unit, counit, associativity, coassociativity, and bialgebra-compatibility equations of Definition 0.166. Hence the first seven components of $\mathfrak{O}_{\mathrm{Hbialg},R}$ vanish, and Proposition 0.167 forms the primitive Hall bracket. Conversely, finite matrices (M, D, η, ϵ) whose bialgebra defect ranks vanish but which are not obtained from such a compact geometric Hall bialgebra datum are not compact $K3 \times E$ source rows; they are only an abstract finite bialgebra.

Proof. The left and right unit identities are the pull-push identities for the vacuum extension correspondences. The counit identities are their collision analogues after applying the augmentation to the zero-object factor. Associativity of m_R is the equality of the two pull-push composites on the retained three-step extension flag stack; the base-change isomorphism, projection formula, and triple Thom–Sebastiani coherence identify the two composites. The same argument applied to the reversed correspondence gives coassociativity of Δ_R.

Bialgebra compatibility is the comparison between first multiplying and then splitting an extension, and first splitting the two inputs and then multiplying the four resulting pieces. The retained four-step flag-of-flags stack is the common refinement of these two operations. Base change on this common refinement gives the equality of the two pull-push functors; the middle interchange contributes exactly the Koszul sign braiding τ, and the multiplicative Thom–Sebastiani orientation transport identifies the coefficient systems. This is the matrix identity

$$\Delta_R m_R = (m_R \otimes m_R)(1 \otimes \tau \otimes 1)(\Delta_R \otimes \Delta_R).$$

Thus all first seven defect matrices have rank zero. The primitive closure is then the algebraic calculation of Proposition 0.167.

The converse sentence is the Beilinson cut: an abstract finite bialgebra with the same matrices is not a $K3 \times E$ source object unless its entries are the pull-push matrices of the retained compact correspondences with the stated orientation, quotient, and transition data. ☐

LEMMA 0.179 *(Primitive bracket lifting; conditional on the Frobenius defect $\mathfrak{o}_F = 0$).* Let $\langle \cdot, \cdot \rangle_X$ be the protected Hopf pairing on $\widetilde{P}_X$, non-degenerate on the radical quotient. The trilinear Frobenius defect

$$\mathfrak{o}_F(x, y, z) := \langle [x, y], z \rangle + (-1)^{|x| \cdot |y|} \langle y, [x, z] \rangle, \quad x, y, z \in \widetilde{P}_X, \quad c_x + c_y + c_z = 0,$$

viewed as a class in $C^3_{\mathrm{cyc}}(\widetilde{P}_X; k)$ after the reduced E-quotient and radical quotient, vanishes precisely when the bracket is graded-symmetric for the Hopf pairing. Vanishing is the cyclic-CY-3 Serre-functor identity of Theorem 4.1. Non-degeneracy of the Hopf pairing alone does *not* force $\mathfrak{o}_F = 0$; it only detects the vanishing once supplied.

Definition 0.180 *(Finite positive-negative Hall pairing).* Let

$$P_R^{\Pi} = \bigoplus_{\gamma \in \Gamma_R^{\Pi},\ \bar{p} \in \mathbb{Z}/2} P_{R,\gamma,\bar{p}}^{\Pi}$$

be a finite normal-ordered primitive Hall stage, equipped with the descended product m_R^{Θ} and a degree-zero trace

$$\mathrm{tr}_{0,R} : H_{R,0,\bar{0}}^{\Pi} \longrightarrow k.$$

A finite positive-negative Hall pairing is the collection of bilinear forms

$$\langle \cdot, \cdot \rangle_{R,\gamma,\bar{p}} : P_{R,\gamma,\bar{p}}^{\Pi} \otimes P_{R,-\gamma,\bar{p}}^{\Pi} \longrightarrow k$$

defined, in homogeneous finite bases, by the matrix

$$G_{\gamma,\bar{p};ab} = \mathrm{tr}_{0,R}\big(m_R^{\Theta}(e_{\gamma,\bar{p},a}, e_{-\gamma,\bar{p},b})\big).$$

Equivalently, $G_{\gamma,\bar{p};ab}$ is the pull-push coefficient of the compact positive-negative pairing correspondence

$$\mathfrak{P}_{R,\gamma,-\gamma;0}^{\mathrm{red}} \longrightarrow \mathfrak{M}_{R,0}^{\mathrm{red}}$$

with reduced Thom–Sebastiani and orientation transport, followed by $\mathrm{tr}_{0,R}$. A finite row of type G consists of the degree γ, the parity $\bar{p}$, the positive and negative primitive basis labels, the scalar coefficient, the pairing correspondence, and the geometric source reference.

This is a definition of the source matrix G. It does not assert that G is homogeneous, graded-symmetric, invariant under the Hall bracket, compatible with the coproduct, or non-degenerate after the radical quotient. Those are the separate defect rows of Definition 0.183.

The packet `certificates/hall/finite_hall_positive_negative_pairing` verifies `FINITE_HALL_POSITIVE_NEGATIVE_` it imports the formal degree-zero pairing rows of `hall_pairing_pushforward_compatibility`, the row 356 normal-ordered Gram-degree packet, the row 357 parity packet, and the compact Hall source obstruction ledger; it records Definition 0.180; and it keeps the current source G-entries, degree-zero trace row, positive-negative pairing correspondences, orientation transport rows, and Hopf-pairing identity rows open. Hence row 358 defines the finite positive-negative Hall pairing matrix; it does not prove any of the row 359–362 pairing identities.

Proposition 0.181 *(The finite Hall pairing is homogeneous; proved for supplied source data).* Assume the finite positive-negative Hall pairing of Definition 0.180 is supplied from a homogeneous normal-ordered Hall product and a degree-zero trace. Extend it by zero to all pairs

$$P_{R,\gamma,\bar{p}}^{\Pi} \otimes P_{R,\gamma',\bar{q}}^{\Pi}.$$

Then

$$\langle P_{R,\gamma,\bar{p}}^{\Pi}, P_{R,\gamma',\bar{q}}^{\Pi} \rangle_R = 0 \quad \text{unless} \quad \gamma + \gamma' = 0 \text{ and } \bar{p} = \bar{q}.$$

Equivalently, every non-zero finite pairing row has total normal-ordered Gram degree 0 and belongs to one parity block.

Proof. For homogeneous inputs $x \in P^{\Pi}_{R,\gamma,\bar{p}}$ and $y \in P^{\Pi}_{R,\gamma',\bar{q}}$, the row 356 degree calculation gives

$$m_R^{\Theta}(x, y) \in H^{\Pi}_{R,\gamma+\gamma',\bar{p}+\bar{q}}.$$

The trace in Definition 0.180 is defined on the degree-zero block. Hence $\mathrm{tr}_{0,R}(m_R^{\Theta}(x, y))$ can contribute only when $\gamma + \gamma' = 0$. The positive-negative pairing matrix G is defined blockwise on $P^{\Pi}_{R,\gamma,\bar{p}} \otimes P^{\Pi}_{R,-\gamma,\bar{p}}$; its extension by zero kills the off-parity blocks. Thus every non-zero entry has total normal-ordered Gram degree 0 and one fixed parity label. □

The packet `certificates/hall/finite_hall_pairing_homogeneity` verifies `FINITE_HALL_PAIRING_HOMOGENEITY_OBST` it imports the row 358 positive-negative pairing definition, the row 356 normal-ordered Gram-degree packet, the formal degree-zero pairing rows of `hall_pairing_pushforward_compatibility`, and the compact Hall source obstruction ledger; it records Proposition 0.181; and it keeps the current source G-entries, trace row, pairing correspondence rows, and homogeneity defect rows open. Hence row 359 proves the homogeneity criterion for supplied finite source data; it does not prove graded supersymmetry, Frobenius invariance, coproduct adjointness, or quotient non-degeneracy.

PROPOSITION 0.182 *(The finite Hall pairing is supersymmetric; proved for supplied signed-transpose rows).* Assume the finite positive-negative Hall pairing of Definition 0.180 is homogeneous in the sense of Proposition 0.181. Suppose that for every retained degree γ, parity $\bar{p}$, and pair of finite basis labels a, b, the signed-transpose row

$$G_{\gamma,\bar{p};ab} - (-1)^{\bar{p}} G_{-\gamma,\bar{p};ba} = 0$$

is supplied. Then the extended finite pairing is supersymmetric:

$$\langle x, y\rangle_R = (-1)^{|x||y|}\langle y, x\rangle_R$$

for all homogeneous $x, y \in P^{\Pi}_R$.

Conversely, a compact-source proof of this identity requires the source G-entries on both signs, source parity blocks, signed-transpose defect rows, and the reduced Calabi–Yau threefold Serre-sign and orientation-transport data. Formal degree-zero pairing rows, target positive-negative blocks, and scalar traces do not supply these signs.

Proof. Write $x \in P^{\Pi}_{R,\gamma,\bar{p}}$ and $y \in P^{\Pi}_{R,\gamma',\bar{q}}$. By Proposition 0.181, both $\langle x, y\rangle_R$ and $\langle y, x\rangle_R$ vanish unless $\gamma' = -\gamma$ and $\bar{q} = \bar{p}$. On a non-zero block one therefore has $|x| = |y| = \bar{p}$, so $(-1)^{|x||y|} = (-1)^{\bar{p}}$. Expanding in the chosen finite bases, the asserted supersymmetry equation is exactly

$$G_{\gamma,\bar{p};ab} = (-1)^{\bar{p}} G_{-\gamma,\bar{p};ba}.$$

Linearity gives the identity for arbitrary homogeneous vectors. This is a finite signed-transpose criterion; the geometric computation of the signs is the source Serre/orientation problem named in the final sentence of the statement. □

The packet `certificates/hall/finite_hall_pairing_supersymmetry` verifies `FINITE_HALL_PAIRING_SUPERSYMMETRY_` it imports the row 359 homogeneity packet, the row 358 positive-negative pairing definition, the reduced Calabi–Yau threefold Serre-sign criterion packet, and the compact Hall source obstruction ledger; it records Proposition 0.182; and it keeps the current source G-entries, source parity blocks, signed-transpose rows, Serre-sign rows, and orientation-transport rows open. Hence row 360 proves the finite supersymmetry criterion for supplied source data; it does not prove Frobenius invariance, coproduct adjointness, or quotient non-degeneracy.

Definition 0.183 *(Finite Hopf-pairing identity tuple).* Fix homogeneous finite bases on a retained primitive stage and write (M, B, D, G, K, Q) for the product, bracket, coproduct, Hopf-pairing, radical, and quotient-splitting matrices. The finite Hopf-pairing identity tuple is

$$\mathfrak{O}_{\mathrm{Hpair},R} = \left(\mathfrak{o}_R^{\mathrm{adj}},\, \mathfrak{o}_R^{\mathrm{Frob}},\, \mathfrak{o}_R^{\mathrm{qnd}}\right),$$

where

$$\mathfrak{o}_R^{\mathrm{adj}}(x, y, z) = \langle m_R^{\Theta}(x, y), z\rangle_R - \langle x \otimes y, \Delta_R^{\Theta} z\rangle_R,$$
$$\mathfrak{o}_R^{\mathrm{Frob}}(x, y, z) = \langle [x, y]_\Theta, z\rangle_R + (-1)^{|x||y|}\langle y, [x, z]_\Theta\rangle_R,$$

and $\mathfrak{o}_R^{\mathrm{qnd}}$ is the finite rank defect of the pairing induced by G on $(P_R^\Pi/\mathrm{Rad}_R^\Pi)_\gamma \otimes (P_R^\Pi/\mathrm{Rad}_R^\Pi)_{-\gamma}$ in every retained parity block. In matrix form these are the adjointness, Frobenius cyclic, and quotient non-degeneracy rows of the finite source proof tables.

Proposition 0.184 *(The finite Hall pairing is invariant; proved for supplied Frobenius rows).* Assume the finite primitive Hall bracket is the graded-skew supercommutator and is parity-compatible, the finite Hall pairing is supersymmetric in the sense of Proposition 0.182, and the Frobenius cyclic component of Definition 0.183 vanishes:

$$\mathfrak{o}_R^{\mathrm{Frob}}(x, y, z) = 0$$

for every homogeneous retained triple. Then the finite pairing is invariant under the Hall bracket:

$$\langle [x, y]_\Theta, z\rangle_R = \langle x, [y, z]_\Theta\rangle_R$$

for all homogeneous $x, y, z \in P_R^\Pi$.

Conversely, a compact-source proof of this identity requires source primitive bracket rows, source G-entries, parity blocks, Frobenius defect rows, cyclic three-point pairing correspondences, and the reduced Calabi–Yau threefold Serre-sign/orientation transport data. The graded Jacobi identity, pairing supersymmetry, or target invariant form does not supply these source rows.

Proof. Put $p = |x|$, $q = |y|$, $r = |z|$. Apply the assumed vanishing of $\mathfrak{o}_R^{\mathrm{Frob}}$ to the triple (y, z, x):

$$0 = \langle [y, z]_\Theta, x\rangle_R + (-1)^{qr}\langle z, [y, x]_\Theta\rangle_R.$$

The bracket parity row gives $|[y, z]_\Theta| = q + r$ and $|[x, y]_\Theta| = p + q$. Supersymmetry of the pairing gives

$$\langle [y, z]_\Theta, x\rangle_R = (-1)^{p(q+r)}\langle x, [y, z]_\Theta\rangle_R.$$

Using graded skew-symmetry $[y, x]_\Theta = -(-1)^{pq}[x, y]_\Theta$ and supersymmetry again,

$$\langle z, [y, x]_\Theta\rangle_R = -(-1)^{pq+(p+q)r}\langle [x, y]_\Theta, z\rangle_R.$$

The two exponents $p(q + r)$ and $qr + pq + (p + q)r$ agree in $\mathbb{Z}/2$. After removing the common sign, the displayed vanishing becomes

$$\langle x, [y, z]_\Theta\rangle_R - \langle [x, y]_\Theta, z\rangle_R = 0,$$

which is the asserted invariance. □

The packet `certificates/hall/finite_hall_pairing_invariance` verifies `FINITE_HALL_PAIRING_INVARIANCE_OBSTRU` it imports the row 360 supersymmetry packet, the row 357 bracket-parity packet, the row 355 graded-Jacobi packet, the reduced Calabi–Yau threefold Serre-sign criterion packet, and the compact Hall source obstruction ledger; it records Proposition 0.184; and it keeps the current source B-entries, G-entries, parity blocks, Frobenius defect rows, cyclic three-point correspondences, Serre-sign rows, and orientation-transport rows open. Hence row 361 proves the finite invariance criterion for supplied source data; it does not prove coproduct adjointness or quotient non-degeneracy.

Proposition 0.185 *(The finite Hall product and coproduct are adjoint; proved for supplied Hopf-adjointness rows).* Assume the finite Hall product m_R^{Θ}, coproduct Δ_R^{Θ}, and positive-negative pairing G are supplied in homogeneous finite bases. Equip $P_R^{\Pi} \otimes P_R^{\Pi}$ with the Koszul tensor product pairing

$$\langle x_1 \otimes x_2, u_1 \otimes u_2 \rangle_{R \otimes R} = (-1)^{|x_2||u_1|} \langle x_1, u_1 \rangle_R \langle x_2, u_2 \rangle_R.$$

If every adjointness component of Definition 0.183 vanishes,

$$\mathfrak{o}_R^{\mathrm{adj}}(x, y, z) = 0,$$

then

$$\langle m_R^{\Theta}(x, y), z \rangle_R = \langle x \otimes y, \Delta_R^{\Theta} z \rangle_{R \otimes R}$$

for all homogeneous retained finite vectors x, y, z in the domain of the Hopf-pairing tuple.

In coordinates, for homogeneous basis vectors e_a, e_b, e_d, this is the rank-zero matrix identity

$$\sum_c M_{ab}^c G_{cd} - \sum_{u,v} D_d^{uv} (-1)^{|e_b||e_u|} G_{au} G_{bv} = 0.$$

A compact-source proof therefore requires source M-entries, D-entries, G-entries, tensor-pairing sign rows, Hopf adjointness defect rows, and the product/coproduct correspondence orientation data. Bialgebra compatibility, pairing invariance, target Hopf forms, and scalar traces do not supply this adjointness matrix.

Proof. Write

$$m_R^{\Theta}(e_a, e_b) = \sum_c M_{ab}^c e_c, \qquad \Delta_R^{\Theta}(e_d) = \sum_{u,v} D_d^{uv} e_u \otimes e_v.$$

By bilinearity of G,

$$\langle m_R^{\Theta}(e_a, e_b), e_d \rangle_R = \sum_c M_{ab}^c G_{cd}.$$

By the Koszul tensor product rule,

$$\langle e_a \otimes e_b, \Delta_R^{\Theta}(e_d) \rangle_{R \otimes R} = \sum_{u,v} D_d^{uv} (-1)^{|e_b||e_u|} G_{au} G_{bv}.$$

Thus the displayed coordinate defect is exactly $\mathfrak{o}_R^{\mathrm{adj}}(e_a, e_b, e_d)$. Its vanishing on the finite homogeneous basis gives the identity for basis vectors, and linearity gives it for all homogeneous vectors. ☐

The packet `certificates/hall/finite_hall_pairing_coproduct_adjointness` verifies `FINITE_HALL_PAIRING_COPRODUCT_ADJOINTNESS_OBSTRUCTION_VERIFIED`: it imports the row 361 pairing-invariance packet, the row 342 product-operator packet, the row 347 coproduct-operator packet, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger; it records Proposition 0.185; and it keeps the current source M-entries, D-entries, G-entries, tensor-pairing sign rows, Hopf-adjointness defect rows, product/coproduct correspondence rows, and orientation-transport rows open. Hence row 362 proves the coproduct-adjointness criterion for supplied finite source data; it does not prove quotient non-degeneracy.

Proposition 0.186 *(The finite quotient Hopf pairing is non-degenerate; proved for supplied quotient-rank rows).* Let $G_{\gamma,\bar{p}}$ be a supplied homogeneous finite positive-negative pairing matrix on

$$P^{\Pi}_{R,\gamma,\bar{p}} \otimes P^{\Pi}_{R,-\gamma,\bar{p}}.$$

Let $K_{\gamma,\bar{p}} \subset P^{\Pi}_{R,\gamma,\bar{p}}$ and $K_{-\gamma,\bar{p}} \subset P^{\Pi}_{R,-\gamma,\bar{p}}$ be supplied left and right radical bases for $G_{\gamma,\bar{p}}$, and let

$$Q_{\gamma,\bar{p}} : L_{\gamma,\bar{p}} \longrightarrow P^{\Pi}_{R,\gamma,\bar{p}}, \qquad Q_{-\gamma,\bar{p}} : L_{-\gamma,\bar{p}} \longrightarrow P^{\Pi}_{R,-\gamma,\bar{p}}$$

be quotient splittings. Write

$$\overline{G}_{\gamma,\bar{p}} = Q^{t}_{\gamma,\bar{p}} G_{\gamma,\bar{p}} Q_{-\gamma,\bar{p}}$$

for the induced quotient pairing matrix. If the finite rank rows give

$$\operatorname{rank} \overline{G}_{\gamma,\bar{p}} = \dim L_{\gamma,\bar{p}} = \dim L_{-\gamma,\bar{p}}$$

for every retained $(\gamma, \bar{p})$, then the induced pairing

$$L_{\gamma,\bar{p}} \otimes L_{-\gamma,\bar{p}} \longrightarrow k$$

is non-degenerate. This is the quotient non-degeneracy component $\mathfrak{o}^{\mathrm{qnd}}_{R} = 0$ of Definition 0.183.

Conversely, a compact-source proof requires the source G-entries, left and right radical bases $K_{\pm\gamma,\bar{p}}$, quotient splittings $Q_{\pm\gamma,\bar{p}}$, quotient-pairing matrices, and rank or determinant rows proving full rank. Hopf adjointness, Frobenius invariance, target radical tables, rank equality alone, and signed dimensions do not supply quotient non-degeneracy.

Proof. Because $K_{\gamma,\bar{p}}$ and $K_{-\gamma,\bar{p}}$ are supplied as the left and right radicals of $G_{\gamma,\bar{p}}$, the pairing descends to

$$P^{\Pi}_{R,\gamma,\bar{p}}/K_{\gamma,\bar{p}} \quad \text{and} \quad P^{\Pi}_{R,-\gamma,\bar{p}}/K_{-\gamma,\bar{p}}.$$

The splittings $Q_{\gamma,\bar{p}}$ and $Q_{-\gamma,\bar{p}}$ give finite bases of these quotients, and the matrix of the descended pairing in these bases is precisely $\overline{G}_{\gamma,\bar{p}} = Q^{t}_{\gamma,\bar{p}} G_{\gamma,\bar{p}} Q_{-\gamma,\bar{p}}$. Full rank equal to both quotient dimensions says that the two adjoint maps

$$L_{\gamma,\bar{p}} \longrightarrow L^{*}_{-\gamma,\bar{p}}, \qquad L_{-\gamma,\bar{p}} \longrightarrow L^{*}_{\gamma,\bar{p}}$$

are isomorphisms. This is non-degeneracy of the quotient pairing. □

The packet `certificates/hall/finite_hall_quotient_pairing_nondegeneracy` verifies `FINITE_HALL_QUOTIENT_PAIRING_NONDEGENERACY_OBSTRUCTION_VERIFIED`: it imports the row 362 coproduct-adjointness packet, the row 360 supersymmetry packet, the row 358 positive-negative pairing definition, the pairing-kernel $\varprojlim^{1}$ obstruction packet, the transition-radical obstruction packet, and the compact Hall source obstruction ledger; it records Proposition 0.186; and it keeps the current source G-entries, K-entries, Q-entries, quotient-pairing matrices, quotient-rank rows, determinant rows, and radical-identification rows open. Hence row 363 proves the quotient non-degeneracy criterion for supplied finite source data; it does not prove transition preservation of radicals or $\varprojlim^{1}$-vanishing for the radical tower.

Definition 0.187 *(Compact geometric Hopf-pairing datum).* At finite HN height R, assume the compact geometric Hall bialgebra datum of Definition 0.171. A compact geometric Hopf-pairing datum on the descended primitive stage is the following additional source data:

(i) a degree-zero protected trace

$$\mathrm{tr}_{0,R} : H_c^\bullet(\mathfrak{M}^{\mathrm{red}}_{R,0}, \Phi^{\mathrm{red}}_{R,0} \otimes o_{R,0}) \longrightarrow k$$

compatible with the vacuum counit and with HN transitions;

(ii) positive-negative compact pairing correspondences $\mathfrak{P}^{\mathrm{red}}_{R,\gamma,-\gamma;0}$ obtained by restricting the Hall product to total descended Gram degree 0, with orientation and Thom–Sebastiani transport inherited from the Hall datum;

(iii) homogeneous pairing matrices

$$G_{\gamma,ab} = \mathrm{tr}_{0,R}\Big(m_R(e_{\gamma,a}, e_{-\gamma,b})\Big)$$

on every retained parity block, together with kernel matrices $K_{\gamma,\bar{p}}$ and quotient splittings $Q_{\gamma,\bar{p}}$;

(iv) triple pairing correspondences for charges $\gamma_1 + \gamma_2 + \gamma_3 = 0$, with cyclic permutation isomorphisms induced by the 3-Calabi–Yau Serre pairing and by Joyce–Upmeier orientation transport;

(v) finite rank-zero witnesses for Hopf adjointness and the primitive Frobenius cyclic identity, and finite rank witnesses that the induced pairing on $(P_R^\Pi/\mathrm{Rad}_R^\Pi)_\gamma \otimes (P_R^\Pi/\mathrm{Rad}_R^\Pi)_{-\gamma}$ is non-degenerate in every retained parity block.

The radical is

$$\mathrm{Rad}^\Pi_{R,\gamma,\bar{p}} = \{x \in P^\Pi_{R,\gamma,\bar{p}} \mid \langle x, y\rangle_R = 0 \text{ for all } y \in P^\Pi_{R,-\gamma,\bar{p}}\},$$

with both signs included. The datum is source-side: the target Gritsenko–Nikulin radical may be used only as a comparison codomain after the source radical and quotient have been formed.

The packet `certificates/hall/finite_hall_compact_geometric_hopf_pairing_datum` verifies `FINITE_HALL_COMPACT_GEOMETRIC_HOPF_PAIRING_DATUM_OBSTRUCTION_VERIFIED`: it imports the row 363 quotient non-degeneracy packet, the row 362 coproduct-adjointness packet, the row 361 Frobenius invariance packet, the reduced Calabi–Yau threefold Serre-sign criterion packet, the level-`Z` protected-trace obstruction ledger as a firewall, and the compact Hall source obstruction ledger; it records Definition 0.187; and it keeps the source degree-zero trace, positive-negative pairing correspondences, orientation and Thom–Sebastiani transport, G-entries, K-entries, Q-entries, cyclic three-point correspondences, Hopf-adjointness witnesses, Frobenius cyclic witnesses, and quotient non-degeneracy witnesses open. Hence row 364 records the compact geometric Hopf-pairing datum as a source-level datum; it does not construct it, and the level-`Z` protected trace does not substitute for the source degree-zero trace.

Definition 0.188 *(Radical-preserving transition datum).* Let $R' \leq R$. Suppose the primitive-subspace transition datum of Definition 0.176 is given after normal-ordered descent, so that

$$T_R^{\mathrm{Prim}} : P_R^\Pi \longrightarrow P_{R'}^\Pi$$

is defined. A radical-preserving transition datum consists of:

(i) source and target Hopf-pairing matrices $G_{R,\gamma,\bar{p}}$ and $G_{R',\gamma,\bar{p}}$, with the Hopf adjointness, Frobenius cyclic, and quotient non-degeneracy rows of Definition 0.183;

(ii) source and target radical kernel rows

$$K_{R,\gamma,\bar{p}} = \mathrm{Rad}^{\Pi}_{R,\gamma,\bar{p}}, \qquad K_{R',\gamma,\bar{p}} = \mathrm{Rad}^{\Pi}_{R',\gamma,\bar{p}},$$

together with quotient splittings $Q_{R,\gamma,\bar{p}}$ and $Q_{R',\gamma,\bar{p}}$;

(iii) for every target opposite primitive $y' \in P^{\Pi}_{R',-\gamma,\bar{p}}$, a lifted source opposite primitive $L^{-\gamma}_{R'R} y' \in P^{\Pi}_{R,-\gamma,\bar{p}}$ with $T^{\mathrm{Prim},-\gamma}_{R} L^{-\gamma}_{R'R} y' = y'$;

(iv) the pairing-transition identity

$$\langle T^{\mathrm{Prim},\gamma}_{R} x, y' \rangle_{R'} = \langle x, L^{-\gamma}_{R'R} y' \rangle_{R}$$

for every $x \in P^{\Pi}_{R,\gamma,\bar{p}}$ and $y' \in P^{\Pi}_{R',-\gamma,\bar{p}}$;

(v) a restricted radical transition matrix

$$T^{\mathrm{Rad}}_{R} : K_{R,\gamma,\bar{p}} \longrightarrow K_{R',\gamma,\bar{p}}$$

and a descended quotient transition matrix

$$\bar{T}_{R} : P^{\Pi}_{R,\gamma,\bar{p}} / K_{R,\gamma,\bar{p}} \longrightarrow P^{\Pi}_{R',\gamma,\bar{p}} / K_{R',\gamma,\bar{p}},$$

with zero radical-image, quotient well-definedness, splitting compatibility, and composition defect ranks.

Scalar traces, denominator products, signed multiplicities, target radical tables, pairing matrices without Hopf identities, rank equalities, primitive transition alone, and normal-ordered degree maps are not radical-transition data.

Proposition 0.189 *(Transition preservation of Hopf radicals; proved).* Given a radical-preserving transition datum of Definition 0.188, the primitive transition carries the Hopf-pairing radical into the Hopf-pairing radical:

$$T^{\mathrm{Prim},\gamma}_{R} \big(\mathrm{Rad}^{\Pi}_{R,\gamma,\bar{p}} \big) \subseteq \mathrm{Rad}^{\Pi}_{R',\gamma,\bar{p}} .$$

Consequently T^{Prim}_{R} descends to the radical quotients through the displayed $\bar{T}_{R}$.

Proof. Let $x \in \mathrm{Rad}^{\Pi}_{R,\gamma,\bar{p}}$, and let $y' \in P^{\Pi}_{R',-\gamma,\bar{p}}$. Choose the supplied lift $L^{-\gamma}_{R'R} y' \in P^{\Pi}_{R,-\gamma,\bar{p}}$. The pairing-transition identity gives

$$\langle T^{\mathrm{Prim},\gamma}_{R} x, y' \rangle_{R'} = \langle x, L^{-\gamma}_{R'R} y' \rangle_{R} = 0,$$

because x pairs trivially with every source opposite primitive. Since this holds for every y', the vector $T^{\mathrm{Prim},\gamma}_{R} x$ lies in $\mathrm{Rad}^{\Pi}_{R',\gamma,\bar{p}}$. The finite restricted matrix T^{Rad}_{R} records this inclusion in radical bases, and the quotient well-definedness row is precisely the statement that T^{Prim}_{R} factors through $\bar{T}_{R}$. The splitting and composition rows make the quotient transition independent of the chosen finite representatives and compatible with successive stages. □

The packet `certificates/hall/transition_radical_preservation` records the present state of this datum. Its verifier status is `TRANSITION_RADICAL_PRESERVATION_OBSTRUCTION_VERIFIED`: the primitive-transition input, source and target Hopf-pairing matrices, source and target Hopf-pairing identity rows, source and target radical kernels, opposite-primitive lift rows, pairing-transition identities, radical-image identities, restricted radical transition matrices, source and target quotient splittings, descended quotient transition matrices, and radical-transition composition rows are not yet supplied. Hence transition preservation of Hopf radicals remains open.

PROPOSITION 0.190 *(Geometric Hopf-pairing data give the finite Hopf-pairing defect rows; proved relative to the datum).* Let the compact geometric Hall bialgebra datum and the compact geometric Hopf-pairing datum be given at height R. Then

$$\mathfrak{D}_{\mathrm{Hpair},R} = 0.$$

Consequently the radical of the source pairing is a coproduct coideal; if the cyclic Frobenius component of the datum is imposed on primitives, it is also a Lie ideal, and the quotient carries the descended primitive Lie bracket and Hopf pairing. Conversely, a matrix G, a kernel matrix K, and a quotient matrix Q with the right ranks are not a compact source Hopf-pairing datum unless their entries are obtained from the protected trace, pairing correspondences, and cyclic three-point coherences above.

Proof. Hopf adjointness is the equality of two pull-push evaluations on the same three-point extension/collision correspondence. On one side one multiplies x and y, then pairs with z by $\mathrm{tr}_{0,R}$; on the other one splits z by Δ_R, pairs the two resulting factors with x and y, and then applies the tensor trace. The common flag refinement and the projection formula identify the two Borel–Moore chain maps, while the Thom–Sebastiani and orientation transports identify the coefficient systems. This is

$$\langle m_R^{\Theta}(x, y), z\rangle_R = \langle x \otimes y, \Delta_R^{\Theta} z\rangle_R.$$

The Frobenius cyclic identity is the same three-point trace after a cyclic permutation of the inputs. The 3-Calabi–Yau Serre pairing changes the order by the Koszul sign; the supercommutator therefore gives

$$\langle [x, y]_{\Theta}, z\rangle_R + (-1)^{|x||y|}\langle y, [x, z]_{\Theta}\rangle_R = 0.$$

The quotient non-degeneracy component is not a consequence of the first two equations; it is the finite rank condition in the datum after dividing by $K_{\gamma,\bar{p}}$. Hence the three components of $\mathfrak{D}_{\mathrm{Hpair},R}$ vanish.

The coideal and Lie-ideal conclusions are Lemma 0.192. The converse sentence is a source/target firewall: a finite bilinear form can have the same kernel rank as the target radical without being the protected trace pairing of the compact $K3 \times E$ Hall source. □

The packet `certificates/hall/finite_hall_hopf_pairing_defect_rows` verifies `FINITE_HALL_HOPF_PAIRING_DEFECT` it imports the row 364 compact geometric Hopf-pairing datum packet, the row 350 Hall bialgebra compatibility packet, the row 362 Hopf-adjointness criterion, the row 361 Frobenius cyclic criterion, the row 363 quotient non-degeneracy criterion, and the compact Hall source obstruction ledger; it records Proposition 0.190; and it keeps the present compact source Hopf-adjointness rows, Frobenius cyclic rows, quotient non-degeneracy rows, M-, D-, G-, K-, and Q-entries, radical ideal/coideal rows, degree-zero source trace, positive-negative pairing correspondences, and cyclic three-point coherences open. Hence row 365 proves only the relative implication from supplied compact geometric Hopf-pairing data to $\mathfrak{D}_{\mathrm{Hpair},R} = 0$; it does not construct the datum and does not prove Hopf-radical descent for the current compact source.

Proposition 0.191 *(A pairing matrix is not a Hopf-pairing proof; proved).* At a finite HN stage, the matrix G, its kernel matrix K, and a quotient splitting Q do not by themselves prove that the radical is the Hopf radical used in primitive recognition. The finite stage proves Hopf-radical descent only after $\mathfrak{D}_{\mathrm{Hpair},R} = 0$: Hopf adjointness, the primitive Frobenius cyclic identity, and non-degeneracy of the induced quotient pairing. Under these three vanishings, the radical is a coproduct coideal and a Lie ideal, hence the quotient carries the descended primitive Lie bracket and Hopf pairing.

Proof. The matrix G defines a bilinear form and $K = \ker G$ records its linear radical. Coideal stability is not a linear-algebra consequence of being a kernel: it follows from Hopf adjointness $\langle m(x, y), r\rangle = \langle x \otimes y, \Delta r\rangle$ and non-degeneracy of the quotient tensor pairing. Lie-ideal stability is not a consequence of kernel membership either; it follows from the Frobenius identity $\langle [x, r], z\rangle = -(-1)^{|x||r|}\langle r, [x, z]\rangle$. These are exactly the first two defects in $\mathfrak{D}_{\mathrm{Hpair},R}$, and quotient non-degeneracy is the third. With all three vanishing, Lemma 0.192 applies. If any one is absent, K is only the radical of a finite matrix, not the Hopf radical required by Theorem 0.215. □

The packet `certificates/hall/finite_hall_pairing_matrix_firewall` verifies `FINITE_HALL_PAIRING_MATRIX_FIREW` it imports the row 365 Hopf-pairing defect packet, the row 362 Hopf-adjointness criterion, the row 361 Frobenius cyclic criterion, the row 363 quotient non-degeneracy criterion, and the compact Hall source obstruction ledger; it records Proposition 0.191; and it keeps the current source G-, K-, and Q-entries, Hopf-adjointness rows, Frobenius cyclic rows, quotient non-degeneracy rows, radical coideal rows, radical Lie-ideal rows, and quotient Hopf-pairing rows open. Hence row 366 is a firewall theorem: G, K, Q and their ranks can identify a finite bilinear kernel, but they do not prove the Hopf radical used in primitive recognition. Hopf-radical descent begins only after $\mathfrak{D}_{\mathrm{Hpair},R} = 0$.

Lemma 0.192 *(Hopf radical is a Lie ideal and a coideal; conditional on $\mathfrak{o}_F = 0$ and Hopf adjointness).* Let P^{Π}_R be a finite normal-ordered primitive stage with transported product m^{Θ}_R, coproduct Δ^{Θ}_R, graded symmetric Hopf pairing $\langle\cdot,\cdot\rangle_R$, and radical Rad^{Π}_R. Assume Hopf adjointness $\langle m^{\Theta}_R(x, y), z\rangle_R = \langle x \otimes y, \Delta^{\Theta}_R z\rangle_R$ and non-degeneracy on $P^{\Pi}_R/\mathrm{Rad}^{\Pi}_R \otimes P^{\Pi}_R/\mathrm{Rad}^{\Pi}_R$. Then

$$(\bar{q}_{\Pi,R} \otimes \bar{q}_{\Pi,R})\Delta^{\Theta}_R(r) = 0 \quad (r \in \mathrm{Rad}^{\Pi}_R),$$

i.e. Rad^{Π}_R is a coideal. If, in addition, $\mathfrak{o}_{F,R} = 0$, then Rad^{Π}_R is a Lie ideal.

Proof. For $r \in \mathrm{Rad}^{\Pi}_R$ and $x, y \in P^{\Pi}_R$, Hopf adjointness gives $\langle x \otimes y, \Delta^{\Theta}_R(r)\rangle_R = \langle m^{\Theta}_R(x, y), r\rangle_R = 0$; non-degeneracy of the quotient tensor pairing then kills the class. The Lie-ideal half is direct: if $\mathfrak{o}_{F,R} = 0$ then $\langle [x, r]_\Theta, z\rangle_R = -(-1)^{|x||r|}\langle r, [x, z]_\Theta\rangle_R = 0$ for every z, so $[x, r]_\Theta \in \mathrm{Rad}^{\Pi}_R$. □

The packet `certificates/hall/finite_hall_hopf_radical_ideal_coideal` verifies `FINITE_HALL_HOPF_RADICAL_IDEAL_COIDEAL_OBSTRUCTION_VERIFIED`: it imports the row 366 pairing-matrix firewall, the row 365 Hopf-pairing defect packet, the row 362 Hopf-adjointness criterion, the row 361 Frobenius cyclic criterion, the row 363 quotient non-degeneracy criterion, and the compact Hall source obstruction ledger; it records Lemma 0.192; and it keeps the present source Hopf-adjointness rows, Frobenius cyclic rows, quotient tensor non-degeneracy rows, radical coideal rows, radical Lie-ideal rows, G-, K-, and Q-entries open. Hence row 367 proves the conditional algebraic criterion for a supplied finite Hopf pairing; it does not prove the radical ideal/coideal rows for the current compact $K3 \times E$ source.

INTEGRAL FORM AND STABLE-ENVELOPE TRANSPORT

Two parallel constructions act on the same compact Hall heart:

(i) the integral-form comparison modeled on the Schiffmann–Vasserot elliptic Hall algebra; a compact $K3 \times E$ specialization would require a non-toric Maulik–Okounkov-type stable-envelope transport;

(ii) the proposed Drinfeld double $\mathcal{D}_\hbar(\mathrm{CoHA}_{K3\times E})$ of Vol III, whose positive half must be the protected primitive Hopf algebra above and whose Drinfeld pairing must be the Hopf pairing.

The compatibility datum between these two constructions is a comparison with Calabi–Yau quantum groups. A different constant or sign would be a distinct mathematical claim; the Drinfeld double is not asserted here as an input theorem.

Definition 0.193 *(Integral-form stable-envelope comparison datum).* At finite height R, an integral-form stable-envelope comparison datum for the compact $K3 \times E$ Hall heart consists of the following finite rows:

(i) an integral lattice $(Y^+_{R,\mathbb{Z}}(X), m_R, \Delta_R, \langle -, - \rangle_R)$ inside the protected Hall bialgebra, stable under product, coproduct, and the Hopf pairing;

(ii) a separated and complete topology on $Y^+_R(X)$, compatible with the HN transition maps and the Hopf radical quotient;

(iii) a completed Drinfeld double $\mathcal{D}_\hbar(Y^+_R(X))$ with cross-relations whose structure constants are the finite Hall pairing rows;

(iv) a Cartan part in the double, identified with $\Lambda^{2,1}_{II} \otimes \mathbb{C}$, and rows proving that this Cartan survives the reduced primitive quotient;

(v) a compact non-toric stable-envelope correspondence satisfying support, normalization, degree, and chamber-cocycle identities on the retained $K3 \times E$ moduli;

(vi) comparison constants, signs, and degree shifts identifying the preceding double with the Calabi–Yau quantum-group normalization.

The datum is source-level. The target BKM denominator, the Schiffmann–Vasserot affine A_0 theorem, and the Maulik–Okounkov quiver-variety stable envelope do not supply these finite compact rows.

PROPOSITION 0.194 *(Stable-envelope and Drinfeld-double scope firewall; proved).* The Maulik–Okounkov stable-envelope theorem and the Schiffmann–Vasserot elliptic-Hall comparison may be used here only in their stated scope. They do not construct the integral form, separated completion, compact non-toric stable envelope, Drinfeld double cross-relations, or Cartan identification required for the compact $K3 \times E$ Hall heart. Any assertion

$$\mathcal{D}_\hbar(Y^+(K3 \times E))^{\mathrm{int}}_{\mathrm{compl}} \simeq Y_\hbar(\widehat{\mathfrak{g}}_{\Delta_5})$$

is therefore conditional on the datum of Definition 0.193.

Proof. Appendix 4 records the precise imported theorems. Maulik–Okounkov stable envelopes are constructed for smooth symplectic varieties in the quiver-variety setting with a torus action and chamber data; their theorem gives the geometric R-matrix in that category. Schiffmann–Vasserot identify the $\mathbb{C}^3$ cohomological Hall algebra with the positive half in the affine A_0 elliptic-Hall setting. Neither theorem contains a compact non-toric $K3 \times E$ moduli atlas, a retained D0-HN comparison, an $O2$ wall atlas, or the Hopf-radical quotient rows above.

Thus the cited theorems fix the target shape and the normalization problem, but they do not supply the finite source rows of Definition 0.193. The compact comparison is a separate datum: without those rows the symbol $\mathcal{D}_\hbar(Y^+(K3\times E))^{\mathrm{int}}_{\mathrm{compl}}$ is a requested construction, not an established input theorem. $\square$

The packet `certificates/hall/finite_hall_integral_form_stable_envelope_transport` verifies `FINITE_HALL_INTEGRAL_FORM_STABLE_ENVELOPE_TRANSPORT_OBSTRUCTION_VERIFIED`: it imports the row 367 Hopf-radical ideal/coideal criterion, the compact Hall source ledger, the Do-HN obstruction ledger, and the $(O2)$ wall-atlas ledger; it records Definition 0.193 and Proposition 0.194; and it keeps the present compact source integral-form rows, product stability rows, coproduct stability rows, pairing stability rows, separated-completion rows, complete-completion rows, Drinfeld-double rows, cross-relation rows, stable-envelope rows, Cartan identification rows, primitive Cartan survival rows, compact non-toric replacement theorem, and Calabi–Yau quantum-group comparison rows open. Hence row 368 proves only the scope firewall and the required finite datum; it does not construct the compact $K3\times E$ integral form, the compact Drinfeld double, or stable-envelope transport.

Conilpotent source coalgebra and Hopf radical quotient

Definition 0.195 *(Finite conilpotent source-coalgebra datum).* A finite conilpotent source-coalgebra datum at HN height R is a source-side package

$$(C_{X,R},\eta^C_R,\varepsilon^C_R,F^{\mathrm{co}}_{R,\bullet},\Delta^{\mathrm{ch}}_R,\rho^C_{R'R})$$

obtained from the retained collision correspondences on $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$, after the reduced E-quotient, with the following finite rows:

(i) $C_{X,R}$ is a coaugmented chiral coalgebra on E, with coaugmentation η^C_R, counit ε^C_R, and coalgebra identities for Δ^{ch}_R;

(ii) $C_{X,R}$ is filtered by a finite increasing source charge filtration

$$0=F^{\mathrm{co}}_{R,-1}\subset F^{\mathrm{co}}_{R,0}\subset\cdots\subset F^{\mathrm{co}}_{R,N_R}=C_{X,R}$$

whose associated graded pieces are finite dimensional in each Γ^Π_R-degree, where $\Gamma^\Pi_R:=\overline{\Pi}_X(\widehat{\Gamma}_R)$ is the image, under the *normal-ordered* Gram homomorphism $\overline{\Pi}_X$ of Chapter 2, of the retained normal-ordered charge window $\widehat{\Gamma}_R$ at Harder–Narasimhan height R (§0.1);

(iii) the additive $\overline{\Pi}_X$, not the raw quadratic Π_X, is the grading map;

(iv) the reduced iterated coproduct is nilpotent:

$$(\overline{\Delta}^{\mathrm{ch}}_R)^{N_R+1}=0;$$

(v) for every successor $R'\le R$, the transition map $\rho^C_{R'R}$ preserves the filtration, counit, coaugmentation, and collision coproduct.

Only after these transition rows and the strict Mittag–Leffler rows are supplied does the inverse limit define the source coalgebra

$$C_X=\varprojlim_R C_{X,R}.$$

PROPOSITION 0.196 *(Conilpotent source-coalgebra firewall; proved).* The notation $C_{X,R}$, the formal bar-coalgebra construction, the target BKM denominator, the stable-envelope comparison, and the Drinfeld-double comparison do not construct the finite conilpotent source-coalgebra datum of Definition 0.195. A compact $K3 \times E$ source coalgebra is available only after the coaugmentation, counit, finite filtration, collision-coproduct, coalgebra-identity, conilpotence, quotient-after-correspondence, transition, and Mittag–Leffler rows are supplied.

Proof. A conilpotent coalgebra is not determined by its name. It requires a coalgebra object, counit and coaugmentation, a coproduct satisfying coassociativity and counit identities, and a filtration for which the reduced iterated coproduct terminates. In this manuscript those structures must be produced by retained compact collision correspondences and their quotient-after-correspondence identities. The target denominator and the later Drinfeld-double comparison live on the comparison side; stable-envelope transport is a separate normalization problem. None of them supplies the source chain maps or zero-defect rows listed in Definition 0.195. The inverse-limit object C_X additionally requires transition compatibility and strict Mittag–Leffler exactness, since otherwise $R^1 \varprojlim$ remains an obstruction. □

The packet `certificates/hall/finite_hall_conilpotent_source_coalgebra` verifies `FINITE_HALL_CONILPOTENT_SOURCE_COALGEBRA_OBSTRUCTION_VERIFIED`: it imports the row 368 stable-envelope/Drinfeld-double firewall, the hybrid carrier ledger, the compact Hall source obstruction ledger, the Do-HN obstruction ledger, the Hall product and coproduct transition ledgers, and the primitive-space and pairing-kernel $\varprojlim^1$ ledgers; it records Definition 0.195 and Proposition 0.196; and it keeps the present compact source coalgebra rows, coaugmentation rows, counit rows, finite filtration rows, collision-coproduct rows, coalgebra-identity rows, conilpotence rows, quotient coalgebra identity rows, transition coalgebra rows, inverse-limit Mittag–Leffler rows, and bar-cobar comparison rows open. Hence row 369 proves only the source-data boundary for $C_{X,R}$ and C_X; it does not construct the compact $K3 \times E$ conilpotent source coalgebra.

PROPOSITION 0.197 *(Sectorial Hall truncation and inverse limit; conditional on* [K3×E-fin]*,* [K3×E-atlas]*, and* [ML]*).* Under the three standing hypotheses [K3×E-fin] (finite-type semistable substacks at bounded HN height), [K3×E-atlas] (finite d-critical/cosection Hall atlas with Joyce–Upmeier orientation lines), and [ML] (Mittag–Leffler descent for the HN inverse system), the finite Hall product, coproduct, primitive projection, and Hopf pairing are compatible with the transition maps $\rho_{R'R}$, and their inverse limits define the completed Hall Hopf datum $\widetilde{P}_X, C_X$. The Mittag–Leffler hypothesis controls the $R^1 \varprojlim$ obstruction on every finite-window slice; without it, only a compatible pro-object is obtained.

The hypothesis [K3×E-fin] is not a consequence of fibrewise K3 boundedness alone. A precise sufficient input is product-stability boundedness in the Abramovich–Polishchuk/Liu sense [1, 66]: for a rational K3 stability condition $\sigma_S = (\mathcal{A}_S, Z_S)$ and Liu's product stability $\sigma_{s,t}$ on $D^b(\mathrm{Coh}(S \times E))$, fixed-class boundedness

$$\mathfrak{M}^{ss}_{\sigma_{s,t}}(\gamma) \text{ is finite type over } \mathbb{C}$$

together with Liu's support property would imply [K3×E-fin] for bounded h_S-windows. Piyaratne–Toda boundedness for threefold Bridgeland/BMT [96] does not automatically supply this theorem; fibrewise Bayer–Macri boundedness alone does not either.

Wall-crossing and chamber dependence

The lattice cocycle $B(c, c')$ of the dyonic Mukai construction is chamber-independent. The compact Hall side $D_0(X; \sigma, S)$ is chamber-dependent: finite-type semistable substacks change under wall-crossing, and the Kontsevich–Soibelman wall-crossing formula [64] relates Hall counts in adjacent chambers. If the datum $D_0(X; \sigma, S)$ is supplied in chamber σ, then $\overline{\Pi}_X$ remains a homomorphism and the recognition target is $\mathfrak{g}_{\Delta_5}$ provided the chamber identifications hold for the OP/DT scalar import; that the chosen σ lies in the OP/DT chamber, or that the wall-crossing transport is compatible with $\overline{\Pi}_X$, is a separate conjectural input. No chamber-independence of the recognition target is asserted.

The Hall obstruction class

The Hall lane carries the residuals

$$\mathfrak{O}_{\mathrm{Hall},R} = (\mathfrak{o}_R^{\mathrm{stage}},\ \mathfrak{o}_R^{\mathrm{conf}},\ \mathfrak{o}_R^{\mathrm{prop}},\ \mathfrak{o}_R^{\mathrm{TS}},\ \mathfrak{O}_{\mathrm{Hbialg},R},\ \mathfrak{O}_{\mathrm{Hpair},R},\ \mathfrak{o}_{R'R}^{\mathrm{ML}}).$$

The first four entries assert the existence of the compact geometric Hall bialgebra datum of Definition 0.171: finite retained substacks, compact-support functoriality, and coherent Thom–Sebastiani orientation transport. The tuple $\mathfrak{O}_{\mathrm{Hbialg},R}$ is then the algebraic defect tuple of Definition 0.166; it is not defined before product, coproduct, unit, and counit matrices have been obtained from the compact correspondences. The tuple $\mathfrak{O}_{\mathrm{Hpair},R}$ is the Hopf-pairing identity tuple of Definition 0.183; it is independent of the bialgebra equations and is not defined before the compact geometric Hopf-pairing datum of Definition 0.187: protected trace, positive-negative pairing correspondences, cyclic three-point coherences, source radical, quotient splitting, and quotient non-degeneracy. On the projection-finite stratum the wrapped product is the classical d-critical CoHA product after cosection reduction; on the wrapped stratum the construction is open and governed by the (D0) obstruction. PBW compatibility remains a separate recognition datum on the descended primitive layer.

Comparison with Drinfeld doubling

Vol III's Drinfeld doubling $Y^+(X) \to G(X)$ is the comparison target for the compact source side; the integral form is the chiral lift of the Schiffmann–Vasserot elliptic Hall algebra after the stable-envelope and completion data are fixed. Maulik–Okounkov stable-envelope transport supplies the integral form on the projection-finite Ran stratum; the wrapped stratum is the corresponding stratum-completion. No black-box equality CoHA $= W_\infty$ is asserted: $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\mathfrak{gl}_1)$ in Vol III, and $W_{1+\infty}$ appears only after Drinfeld doubling, centre, vacuum evaluation. The same discipline applies here: the protected primitive Hopf algebra of $K3 \times E$ becomes the BKM target only after the Hopf radical quotient, the Frobenius identity, and the no-extra-relations theorem of §0.6.

Under $\mathfrak{O}_{\mathrm{Hall},R} = 0$ and the Mittag–Leffler transition datum, the Hall correspondences give a finite-stage protected primitive Hopf datum on the cosection-twisted heart. This datum lives at the operator level and has no automatic comparison to the scalar denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ imported in Chapter 4. Closing the cross-level identity ②↔③ requires a chain-level scalar incarnation of the primitive structure: a Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ on the reduced moduli, a Dirac block decomposition, and a first-order Pfaffian section whose $\mathbf{Sp}_4(\mathbb{Z})$-pullback recovers Δ_5. This is the fourth entry (L4), constructed in §0.4.

0.4 *First-order Dirac block and protected Pfaffian*

The fourth entry of the Dirac–Igusa pro-object is the protected Pfaffian line and its first-order Dirac block decomposition. The Pfaffian, not the determinant, is the object of interest for a precise reason: the automorphic target Δ_5 is the square root of the scalar weight-ten form $\Delta_5^2 = \chi_{10}$, and the square root carries the order-two Maass multiplier ν_{Δ_5} that the square sends to the trivial character. Geometrically the determinant line $\mathscr{L}_{\det}$ is canonically trivialisable and forgets orientation; its square root, the Pfaffian line, retains the orientation character ϵ_o, and it is precisely the match $\epsilon_o = \nu_{\Delta_5}$ that distinguishes the first-order object Δ_5 from its orientation-blind square Δ_5^2. Two definitions of this Pfaffian meet. The geometric definition starts from the cosection-reduced d-critical heart, carries a Joyce–Upmeier orientation, descends to the reduced quotient, and associates a Pfaffian line to the determinant of the reduced obstruction theory in the Brav–Bussi–Dupont–Joyce–Szendrői framework [12]. The algebraic construction starts from the imported Borcherds–Kac–Moody denominator algebra, fixes a positive window in Γ_{gram}, and writes the algebraic super-Pfaffian

$$\mathrm{sPf}_R^{\mathrm{alg}}(D) = \prod_{\gamma \in A_R} (1 - x^\gamma)^{\mathrm{sdim}\, V_\gamma}$$

of the trivial first-order block decomposition D_γ on the target side. The Pfaffian–Dirac theorem of Chapter 6 is the identification of these two Pfaffian sections after the chosen line-comparison ι_{aut} and the four-obstruction datum of Appendix 7.

Pfaffian line and squaring isomorphism

Definition 0.198 *(Pfaffian line on the cosection-reduced quotient).* Let K_C^{red} denote the cosection-reduced self-Ext complex on a charge stratum, with reduced determinant $\det K_C^{\mathrm{red}}$ and Joyce–Upmeier orientation o_C. The Pfaffian line at finite stage is

$$\mathscr{L}_{\mathrm{Pf},R} = \bigotimes_C \mathrm{Pf}(K_{C,R}^{\mathrm{red}})$$

on the reduced moduli, with squaring isomorphism

$$\mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \simeq \mathscr{L}_{\det,R}$$

in the Brav–Bussi–Dupont–Joyce–Szendrői sense [12]. The successor maps $\rho_{R^+R}^{\mathrm{Pf}}$ commute with these squaring isomorphisms, identifying the inverse limit with a Pfaffian line $\mathscr{L}_{\mathrm{Pf},X}$ on the inverse-limit object. A normalized section $\mathrm{pf}_{X,R}$ at finite stage and $\mathrm{pf}_X = \varprojlim_R \mathrm{pf}_{X,R}$ are part of the datum.

Pfaffian-to-automorphic comparison

The recognition statement is read on the automorphic line. A section-level identification $\mathrm{pf}_X = \Delta_5$ requires a comparison of $\mathscr{L}_{\mathrm{Pf},X}$ with the pullback of the automorphic weight-five line to the moduli base.

Definition 0.199 *(Pfaffian-to-automorphic line comparison).* A section-level Pfaffian-to-automorphic comparison is a chosen isomorphism of line bundles

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$$

to the pullback of the automorphic weight-five line, twisted by the multiplier ν_{Δ_5}, compatible with cusp trivialization at $\mathfrak{c}_\infty$ and with the squaring map $\mathscr{L}_{\mathrm{Pf},X}^{\otimes 2} \simeq \mathscr{L}_{\det,X}$.

Definition 0.200 *(Finite geometric Pfaffian datum (P_{fin})).* At finite HN height R, a finite geometric Pfaffian datum is the following system on the reduced compact quotient $\mathfrak{Q}_R = (\mathfrak{M}_R/E)^{\mathrm{red}}$.

(P1) For every retained stratum C, a cosection-reduced perfect self-Ext complex $K^{\mathrm{red}}_{C,R}$ with odd skew self-duality

$$\vartheta_{C,R} : K^{\mathrm{red}}_{C,R} \xrightarrow{\sim} (K^{\mathrm{red}}_{C,R})^\vee[1],$$

compatible with the reduced obstruction theory and the quotient by E.

(P2) A Joyce–Upmeier orientation $o_{C,R}$ and the corresponding Pfaffian line

$$\mathscr{L}_{\mathrm{Pf},R} = \bigotimes_C \mathrm{Pf}(K^{\mathrm{red}}_{C,R}), \qquad \mathscr{L}^{\otimes 2}_{\mathrm{Pf},R} \simeq \mathscr{L}_{\det,R}.$$

(P3) A finite-rank skew first-order block $D_{R,\gamma}$ on each retained primitive degree $\gamma \in \Gamma^{\Pi,+}_R$, with Pfaffian section

$$\mathrm{pf}_{X,R} \in H^0(\mathfrak{Q}_R, \mathscr{L}_{\mathrm{Pf},R}).$$

In a cusp frame this section has support exactly on the retained active degrees, full even/odd parity dimensions $p^{\bar 0}_{R,\gamma}, p^{\bar 1}_{R,\gamma}$, and signed exponent $p^{\bar 0}_{R,\gamma} - p^{\bar 1}_{R,\gamma} = a_\Delta(\gamma)$.

(P4) For every retained simple type-II wall chart, with wall coordinate $u_{\mathfrak{d}_i,R}$, a local factorization

$$\mathrm{pf}_{X,R} = v_{\mathfrak{d}_i,R}\, u_{\mathfrak{d}_i,R} \cdot \mathrm{pf}^{\tan}_{i,R}, \qquad \mathrm{ord}_{u_{\mathfrak{d}_i,R}=0}(\mathrm{pf}_{X,R}) = 1,$$

where the unit $v_{\mathfrak{d}_i,R}$ is invariant under the chosen reflection lift and the normal Pfaffian rank is $N^{\mathrm{Pf}}_{\mathfrak{d}_i,R} = 1$. No type-I wall is allowed in this datum.

(P5) A finite-stage comparison $\iota_{\mathrm{aut},R} : \mathscr{L}_{\mathrm{Pf},R} \to \mathcal{L}^5 \otimes \nu_{\Delta_5}$ on the automorphic chart, compatible with the cusp trivialization, the leading coefficient $\mathrm{lc}_{\mathfrak{c}_\infty}(\mathrm{pf}_{X,R}) = 64$, and squaring to the determinant line.

(P6) Strict transition isomorphisms $\rho^{\mathrm{Pf}}_{R^+R} : \mathscr{L}_{\mathrm{Pf},R^+}|_{\mathfrak{Q}_R} \xrightarrow{\sim} \mathscr{L}_{\mathrm{Pf},R}$ carrying pf_{X,R^+} to $\mathrm{pf}_{X,R}$, with closed images and vanishing $\lim^1$ for the line and section systems.

The scalar Borcherds product, the squared determinant Δ_5^2, and the OP scalar $-4096\Delta_5^{-2}$ are not entries of (P_{fin}). They are shadows obtained after choosing the cusp frame and, in the OP case, after squaring and inverting the Pfaffian section.

After finite homogeneous trivialisations, (P_{fin}) is recorded by the Pfaffian proof tables

strata, skew_complexes, pfaffian_lines, pfaffian_sections, wall_charts, automorphic_comparisons, transitions,

The rows are not scalar product rows. They record retained strata, cosection-reduced skew perfect complexes, Pfaffian lines and their squares, finite Pfaffian sections, type-II rank-one wall charts, Pfaffian-to-automorphic line comparisons, strict transition maps, and the exclusion of scalar Borcherds products, squared determinants, OP scalar traces, and exponent tables as entries of (P_{fin}). Defect ranks, exponent residuals, leading residuals, and $R^1 \varprojlim$-ranks must be zero; the type-II normal Pfaffian rank and divisor order must be one; the simple-reflection sign is -1; the automorphic comparison has leading coefficient 64, weight 5, and character ν_{Δ_5}. Every row must carry geometric source provenance and a proof reference.

Proposition 0.201 *(Scalar Pfaffian data do not determine a Pfaffian section; proved).* Let U be a quotient chart in the type-II wall complement, let $\mathcal{L}^5$ be the automorphic weight-five line on U, and let T_χ be a non-trivial order-two flat line with character $\chi : \pi_1(U) \to \{\pm 1\}$. A divisor, a leading coefficient, a scalar Borcherds product written after choosing a cusp trivialization of $\mathcal{L}^5$, or the square of that product in $\mathcal{L}^{10}$ does not determine a section of $\mathcal{L}^5 \otimes T_\chi$. Thus a section-level Pfaffian identity

$$\mathrm{pf}_X = \Delta_5$$

requires, in addition to the scalar product data, the Pfaffian line $\mathscr{L}_{\mathrm{Pf},X}$, the section pf_X, the comparison ι_{aut}, the order-two character $\epsilon_o = \nu_{\Delta_5}$, and the inverse-limit transition data for these objects. In particular, the exponent table $f(nm, l)$, the squared identity $\Delta_5^2 = \chi_{10}$, and the OP scalar $-4096\,\Delta_5^{-2}$ do not construct $\mathrm{pf}_{X,R}$ or the orientation character $\epsilon_{o,R}$.

Proof. Choose a simply connected open set $V \subset U$ and a flat frame τ of $T_\chi|_V$. For a local section s of $\mathcal{L}^5$, the section $s \otimes \tau$ of $\mathcal{L}^5 \otimes T_\chi$ has the same local scalar expression as s after the frame τ is set equal to 1. Under the canonical trivialization $T_\chi^{\otimes 2} \simeq O_U$, the squares of s and $s \otimes \tau$ give the same section s^2 of $\mathcal{L}^{10}$. Their divisors and leading coefficients in the chosen cusp trivialization are therefore identical.

The two objects are nevertheless different globally. Along a loop ℓ with $\chi(\ell) = -1$, analytic continuation fixes s and sends $s \otimes \tau$ to $-s \otimes \tau$. This monodromy is exactly the information carried by the order-two orientation character. Hence the square, the local product, and the scalar trace forget the data needed to define the Pfaffian section as a section of a twisted weight-five line. The line, section, line comparison, character, and transition compatibilities are therefore separate data, not consequences of the scalar product. $\square$

FIRST-ORDER DIRAC BLOCK DECOMPOSITION

Definition 0.202 *(First-order Dirac block).* For $\gamma \in \Gamma_R^{\Pi,+}$, write $P_{R,\gamma}^{\Pi,+}$ for the cusp-positive primitive piece in degree γ and put $x^\gamma = q^n r^l s^m$ for $\gamma = (n, l, m)$. The two-term Pfaffian/Koszul block on $P_{R,\gamma}^{\Pi,+} \oplus (P_{R,\gamma}^{\Pi,+})^\vee[1]$ is the skew operator

$$J_\gamma = \begin{pmatrix} 0 & 1 - x^\gamma \\ -(1 - x^\gamma) & 0 \end{pmatrix}, \qquad \mathrm{Pf}(J_\gamma) = (1 - x^\gamma)^{\operatorname{sdim} P_{R,\gamma}^{\Pi,+}}.$$

For $\gamma \notin \Gamma_R^{\Pi,+}$, set $P_{R,\gamma}^{\Pi,+} = 0$. The first-order block of $\mathfrak{D}_X^{D_0}$ at height R is the direct sum

$$\mathfrak{D}_{X,R}^{D_0} = \bigoplus_{\gamma \in \Gamma_R^{\Pi,+} \setminus \{0\}} J_\gamma,$$

acting on $\bigoplus_\gamma P_{R,\gamma}^{\Pi,+} \oplus (P_{R,\gamma}^{\Pi,+})^\vee[1]$.

The principal symbol of $\mathfrak{D}_{X,R}^{D_0}$ with respect to the finite Chevalley–Eilenberg/Koszul filtration is the normal-ordered Hall bracket of §0.3: the Pfaffian line is the determinant detector, and the symbol carries the Lie data. The phrase *first-order* means the Pfaffian/primitive object before scalar squaring; no analytic Dirac operator on a constructed state space is asserted.

THE PFAFFIAN PRODUCT AT FINITE STAGE

PROPOSITION 0.203 *(Pfaffian product at finite stage; conditional on (P_{fin})).* Suppose the orientation lane (O1, O1$^+$) has descended to the reduced quotient, the finite geometric Pfaffian datum (P_{fin}) of Definition 0.200 is supplied, and the support and parity residuals

$$\mathfrak{o}_{R,\gamma}^{\mathrm{supp}} = \begin{cases} P_{R,\gamma}^{\Pi,+}, & a_\Delta(\gamma) = 0 \\ 0, & a_\Delta(\gamma) \neq 0 \end{cases}, \qquad \mathfrak{o}_{R,\gamma}^{\mathrm{par}} = p_{R,\gamma}^{\bar{0}} - p_{R,\gamma}^{\bar{1}} - a_\Delta(\gamma) \in \mathbb{Z},$$

together with the leading-coefficient residual $\mathfrak{o}_R^{\mathrm{lead}} = \mathrm{lc}_{\mathfrak{c}_\infty}(\mathrm{pf}_{X,R}) - 64$, all vanish on the test window, where $a_\Delta(\gamma)$ is the Borcherds exponent and $p_{R,\gamma}^{\bar{p}} = \dim P_{R,\gamma,\bar{p}}^{\Pi,+}$. Then the Pfaffian section at finite stage has expansion

$$\mathrm{pf}_{X,R}|_{\mathfrak{c}_\infty} = 64q^{1/2}r^{1/2}s^{1/2} \prod_{\substack{\gamma\in\Gamma_R^{\Pi,+}\\ \gamma\neq 0}} (1 - x^\gamma)^{\mathrm{sdim}\, P_{R,\gamma}^{\Pi,+}}.$$

The cusp-positive Pfaffian half is recorded; negative root representatives and the Cartan/imaginary completion enter the Lie recognition statement of §0.6.

The support residual is a super-vector-space residual, not a K_0 class: its vanishing excludes invisible even–odd cancelling primitive pieces in target-zero degrees. The parity residual is a signed superdimension residual. These finite-stage residuals on the test window do not by themselves prove compact source coverage; the inclusion $A_R \subset \Gamma_R^{\Pi,+}$ of active target degrees in the source image is itself a residual condition, not a consequence of the Borcherds product.

Local Pfaffian wall on type-II walls

The (O2) obstruction concerns the local Pfaffian normal form on the three simple type-II walls $\mathbb{H}_{\partial_i}$, $i = 1, 2, 3$, of the Borcherds–Cartan datum $\mathscr{B}_{\Delta_5}$.

Definition 0.204 *(Rank-one type-II Pfaffian crossing).* For a simple type-II root ∂, let T_∂ be a real analytic tangent chart with trivial s_∂-action, set $U_\partial = T_\partial \times (-\varepsilon, \varepsilon)$ with $s_\partial(t, u) = (t, -u)$, and let $\mathscr{L}_\partial^{\tan}$ be an oriented tangent Pfaffian line with s_∂-invariant unit v_∂. The rank-one odd Pfaffian crossing is the skew family

$$D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad K_\partial^{\mathrm{cross}} = [\mathbb{R} \xrightarrow{u} \mathbb{R}],$$

in odd Pfaffian parity. The total local Pfaffian section is

$$\mathrm{pf}_\partial = v_\partial\, u, \qquad s_\partial^* \,\mathrm{pf}_\partial = -\,\mathrm{pf}_\partial, \qquad s_\partial^*(\mathrm{pf}_\partial^2) = \mathrm{pf}_\partial^2\,.$$

Proposition 0.205 *(Rank-one crossing gives the type-II local sign; proved).* For $i = 1, 2, 3$, the rank-one crossing of Definition 0.204 forms the finite generic type-II local-sign datum with $N_{\partial_i}^{\mathrm{Pf}} = 1$, and at finite HN height R, with successor maps preserving the tangent charts, the wall coordinate $u_{\partial_i,R}$, and the invariant units, the (O2) residual entries

$$\mathfrak{o}_{\partial_i,R}^{\mathrm{chart}},\ \mathfrak{o}_{\partial_i,R}^{\mathrm{wall}},\ \mathfrak{o}_{\partial_i,R}^{\mathrm{split}},\ \mathfrak{o}_{\partial_i,R}^{\mathrm{unit}},\ \mathfrak{o}_{\partial_i,R}^{\mathrm{rank}},\ \mathfrak{o}_{\partial_i,R'R}^{\mathrm{tr}}$$

vanish for these local models, and $\varepsilon_{\partial_i,R}^{\mathrm{loc}} = -1$.

Proof. The fixed locus of $s_\partial(t, u) = (t, -u)$ is $u = 0$, so the wall and the tangent fixed locus agree. The determinant factor is split: the tangent line is $\mathscr{L}_\partial^{\tan}$, and the normal factor is $K_\partial^{\mathrm{cross}}$. The unit v_∂ is invariant by construction. The skew matrix has Pfaffian u, so the normal-mode rank is one. Reflection sends $u \mapsto -u$ and fixes v_∂; hence the local Pfaffian section changes sign and its square is fixed. The transition hypothesis is exactly preservation of these data under successor maps. □

HALF-HILBERT ORBIT AND THE TYPE-II SIGN SUM

The (O2) obstruction also requires a wall *atlas*: at finite HN height R, each retained component and boundary stratum of $\mathbb{H}_{\partial_i}$ needs a chart of the rank-one form, and the normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps must agree on chart overlaps. A single generic chart computes only a local sign.

The finite O2 wall-atlas proof packet is the table system

wall_objects, wall_charts, overlaps, orbit_transport, half_hilbert_orbit, compatibilities, scalar_separation.

Its rows must carry the retained wall objects, rank-one chart coordinates, Cech/cochain overlap equations, Weyl transport, the twelve-coordinate half-Hilbert orbit, and the component, boundary, Thom–Sebastiani, HN-transition, quotient-orientation, and protected integration identities. The scalar-separation rows record $4096 = 2^{12} = 64^2$ without identifying the OP scalar normalization or the Maass character value with wall-atlas data.

The half-Hilbert orbit of the type-II reflection group on the relevant retained stratum carries a sign sum of size $4096 = 2^{12}$, counting the supplied finite-window normal Pfaffian signs across the twelve retained components and boundary strata of the type-II wall configuration; this sum is the signature of the (O2) wall normal-form datum. Local rank-one charts contribute the local factors; the chosen Weyl lifts convert the retained global system into the orientation character on the $W^{(2)}(\Lambda_{II}^{2,1})$-orbit. The full normal-form theorem with overlap, boundary, and component coherence is relative to the retained $(O2_{\mathrm{atlas}})$ datum and Theorem 7.19 of Appendix 7.5; the local model is the finite rank-one wall factor.

PROPOSITION 0.206 *(Generic local sign and the orientation character).* Suppose $(\mathrm{O1})^+$ supplies a normalized lift $\tau_{i,R,S} : o_{R,S} \to s_{\partial_i}^* o_{R,S}$ of the simple type-II reflection on every retained orbit, and the (O2)/hybrid residual for ∂_i vanishes. In a local chart with $K_{\partial_i,R}^{\mathrm{nor}} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$, with normal Pfaffian unit v_i and $s_{\partial_i}(v_i) = v_i$, let $\ell_{\partial_i,R,S}$ be the path in the reduced quotient wall-complement carrying $u_i = \epsilon$ to $u_i = -\epsilon$, closed by $\tau_{i,R,S}$. Then

$$M_{\ell_{\partial_i,R,S}}(v_i u_i) = v_i(-u_i) = -v_i u_i, \qquad \epsilon_{o,R}(s_{\partial_i}) = (-1)^{N^{\mathrm{Pf}}_{\partial_i,R}}.$$

For the rank-one type-II model $\epsilon_{o,R}(s_{\partial_i}) = -1 = \nu_{\Delta_5}(s_{\partial_i})$. The global statement $\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$ is the chamber-permuting half of the Pfaffian–Dirac orientation identity, conditional on the (O2) wall atlas.

PFAFFIAN NORMALIZATION AND TRACE

The supplied Pfaffian-to-automorphic comparison ι_{aut} then gives, on the imported Borcherds product side,

$$\iota_{\mathrm{aut}}\big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)\big) = \mathcal{D}_X = \Delta_5,$$

with cusp-side expansion

$$\mathrm{pf}_X\,|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

matching the imported Borcherds–Gritsenko–Nikulin product. The scalar branch obtained after taking the protected trace of the inverse squared Pfaffian section is the Oberdieck–Pixton normalization

$$Z_{\mathrm{OP}}^X = -4096\, \Delta_5^{-2},$$

where the negative sign is the OP scalar branch and $4096 = 2^{12} = 64^2$ is the scalar equality between the retained half-Hilbert sign count and the squared theta-leading coefficient. Neither number defines the orientation character.

Comparison with the Pfaffian identity

The Vol II primitive-to-shadow comparison singles out the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5$ as the target identity for the retained Pfaffian datum; the $K3 \times E$ side supplies the Pfaffian line and scalar branch data. BBDJS [12] provides the orientation-line and Pfaffian calculus on the cosection-reduced d-critical heart; Joyce–Upmeier [56] provides multiplicativity under reduced extension correspondences; Joyce–Tanaka–Upmeier [55] extends the framework to 4-fold analogues but is not used directly here. The constant $4096 = 2^{12}$ is fixed at the orientation level by the retained half-Hilbert wall atlas and at the scalar level by 64^2; the local sign -1 is fixed by the rank-one Pfaffian wall computation. The Oberdieck–Pixton minus sign is a scalar branch convention, not the source of the orientation character. No identification of the Pfaffian Δ_5 with a compact microscopic state space is asserted: Δ_5 is the protected Pfaffian section, the scalar trace is Δ_5^{-2}, and the operator-level datum is the pro-object $\mathfrak{D}_X^{\mathrm{DI}}$, defined at the Pfaffian and orientation level relative to the retained data of Appendix 7.

The Pfaffian recovers Δ_5 at the target scalar level, but the cross-level recognition that identifies the primitive Hopf algebra of §0.3 with the Borcherds–Kac–Moody target $\mathfrak{g}_{\Delta_5}$ requires a separately constructed source coalgebra and an explicit chiral-Koszul cobar comparison. The source coalgebra C_X is the bar of the augmented binary/two-step hybrid correspondence algebra; it becomes the bar of a hybrid factorization algebra only after the higher-coloured residual vanishes. The cobar map $\Theta_{\mathrm{Kos},R}$ is the quasi-isomorphism to the imported current envelope $\mathsf{Den}_{\Delta_5,E}$. This Koszul lane (L5) is the content of §0.5.

0.5 *Koszul source coalgebra and chiral MC twisting*

The fifth entry of the Dirac–Igusa pro-object is the Koszul lane: the source coalgebra C_X, the bar comparison b_X, and the cobar quasi-isomorphism Θ_{Kos} to the imported current envelope $\mathsf{Den}_{\Delta_5,E} = U_E^{\mathrm{ch}}(\mathrm{Cur}_E(\mathfrak{g}_{\Delta_5}))$, the chiral enveloping algebra of the level-zero current algebra of the BKM denominator algebra over E (Definition 16.1). Source–target separation is structural. The target counit

$$\varepsilon_{\Delta,R} : \Omega_E^{\mathrm{ch}} \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\leq R}) \longrightarrow \mathsf{Den}_{\Delta_5,E,\leq R}$$

is internal to the imported denominator current envelope and produces no source data. The compact Koszul comparison $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$ is a source-to-target statement on the supplied hybrid source, conditional on the chiral Koszul source datum, the finite Koszul comparison datum, and strict Koszul Mittag–Leffler exactness.

The Beilinson cut applies in its purest form on this lane. $\bar{B}^{\mathrm{ch}}$ is MC twisting / coupling coalgebra; it is not the level-Z bulk. The target of the lane is $\mathsf{Den}_{\Delta_5,E}$, the chiral shadow of the BKM denominator algebra; the source coalgebra C_X is the supplied compact Hall data after collision coproduct, conilpotent filtration, and bar-length truncation, and the cobar map is the comparison.

Source–target separation

Lemma 0.207 *(Source–target separation for the Koszul lane; proved).* Fix a finite height R. The target counit $\varepsilon_{\Delta,R}$ above is a morphism internal to the imported current envelope. A source Koszul datum is different data: it must first construct

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, \qquad C_{X,R}, \qquad F^{\mathrm{co}}_{R,\bullet}, \qquad \Delta^{\mathrm{ch}}_R$$

from the finite hybrid correspondence category $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$, orientations, the eight-word two-step binary flag atlas, and the separate higher-coloured configuration residual; only then may it compare

$$b_{X,R} : C_{X,R} \simeq \bar{B}_E^{\mathrm{ch}}(\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}, \varepsilon_{X,R}^{\mathrm{hyb}}), \quad \Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \simeq \mathsf{Den}_{\Delta_5,E,\le R}.$$

Setting $C_{X,R} = \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\le R})$, or choosing a homotopy inverse to $\varepsilon_{\Delta,R}$, does not produce a source datum.

Proof. The domain of $\varepsilon_{\Delta,R}$ contains only target augmentation, target charge filtration, and target chiral operations; it does not contain rigidified wrapped prequotients, anchors, mixed local/wrapped correspondences, quotient-after-correspondence pseudofunctor, orientation transport, protected integration, or primitive representatives. These are the data from which the source coalgebra and collision coproduct are defined. □

CHIRAL KOSZUL SOURCE DATUM

Definition 0.208 *(Chiral Koszul source datum).* Fix (σ, S, R) and the Gram image Γ_R^{Π}. A finite-stage chiral Koszul source datum is

$$\mathfrak{K}_{X,R}^{\mathrm{Kos}} = (\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}, C_{X,R}, F_{R,\bullet}^{\mathrm{co}}, \Delta_R^{\mathrm{ch}}, b_{X,R}, \Theta_{\mathrm{Kos},R})$$

with the following clauses; every entry is source-side finite data except for the terminal comparison with the imported target.

(i) *Source chiral object.* $\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}$ is the finite hybrid object of §0.1, with $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and with a supplied wrapped bar-cobar domain datum. The wrapped sector is not made Koszul by the projection-finite $b = 0$ Ran sector.

(ii) *Conilpotent chiral coalgebra.* $C_{X,R}$ is a coaugmented conilpotent chiral coalgebra on E, filtered by a finite increasing source charge filtration $F_{R,\bullet}^{\mathrm{co}}$; reduced iterated coproduct nilpotent on each finite stage.

(iii) *Collision coproduct.* Δ_R^{ch} is the collision coproduct from the finite Hall correspondences (local-local, ordered local-wrapped, wrapped-wrapped, and eight-word flag stacks), defined before $Q_{E,R}$; the quotient coalgebra identities are the vanishing of the higher-coloured, unit, refinement, and quotient residuals. Transition maps $\rho_{R'R}^{C}$ commute with Δ^{ch}.

(iv) *Bar comparison.* $b_{X,R} : C_{X,R} \simeq \bar{B}_E^{\mathrm{ch}}(\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}, \varepsilon_{X,R}^{\mathrm{hyb}})$ is a charge-preserving quasi-isomorphism of conilpotent chiral coalgebras compatible with primitive projection, the completed Hopf pairing, the reduced E-quotient, and the normal-ordered descent $\overline{\Pi}_{X,*}^{\Theta}$, separately on the projection-finite and wrapped summands and on their mixed words.

(v) *Cobar comparison with the target envelope.* $\Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \simeq \mathsf{Den}_{\Delta_5,E,\le R}$ is a charge-preserving quasi-isomorphism of augmented chiral algebras. Restricted to constant-current primitives after the Hopf-radical quotient it gives the primitive recognition morphism

$$\iota_R^{\mathrm{Prim}} : H^0 \,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}_{X,*}^{\Theta} \mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}) \longrightarrow \mathfrak{g}_{\Delta_5,\le R}.$$

The exact finite Koszul obstruction tuple is

$$\mathfrak{O}_{\mathrm{Kos},R} = (\mathfrak{o}_R^{C}, \mathfrak{o}_R^{\mathrm{fil}}, \mathfrak{o}_R^{\mathrm{conil}}, \mathfrak{o}_R^{\Delta,\mathrm{ch}}, \{\mathfrak{o}_{R'R}^{C,\mathrm{tr}}\}_{R'\ge R}, \mathfrak{o}_R^{b}, \mathfrak{o}_R^{\Omega}, \mathfrak{o}_R^{\mathrm{wrKos}}, \mathfrak{o}_R^{\mathrm{hyb}}, \mathfrak{o}_R^{\Pi}, \mathfrak{o}_R^{\mathrm{sep}}, \mathfrak{o}_R^{\mathrm{Prim}}).$$

Canonical finite source-bar coalgebra

For a hybrid source with bounded-length augmentation ideal, a canonical model exists.

Proposition 0.209 *(Canonical finite source-bar coalgebra; conditional on $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and bounded length).* Suppose $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and there is an integer N_R such that every nonzero ordered word of non-vacuum hybrid inputs of total HN height $\le R$ has length at most N_R, with HN height positive on every non-vacuum colour surviving in the reduced bar construction. Suppose an augmentation $\varepsilon^{\mathrm{hyb}}_{X,R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \to k\mathbf{1}$ is supplied. Then the canonical source-bar coalgebra

$$C^{\mathrm{bar}}_{X,R} = \bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R}) = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_{E}(\overline{\mathcal{F}}_{X,R}),$$

filtered by bar-length, with deconcatenation collision coproduct

$$\Delta^{\mathrm{bar}}_{R}([x_1|\cdots|x_p]) = \mathbf{1}\otimes[x_1|\cdots|x_p] + [x_1|\cdots|x_p]\otimes\mathbf{1} + \sum_{i=1}^{p-1}[x_1|\cdots|x_i]\otimes[x_{i+1}|\cdots|x_p],$$

realizes the coalgebra and source-bar part of a chiral Koszul source datum with $\mathfrak{o}^{C}_{R} = \mathfrak{o}^{\mathrm{fil}}_{R} = \mathfrak{o}^{\mathrm{conil}}_{R} = \mathfrak{o}^{\Delta,\mathrm{ch}}_{R} = 0$ and $b_{X,R} = \mathrm{id}$.

Proof. Apply the Beilinson–Drinfeld chiral bar construction [7, §3.7] to the supplied hybrid object. Since the finite HN quotient kills charges of height $> R$ and non-vacuum words have length $\le N_R$, only finitely many bar lengths occur. The deconcatenation coproduct on each ordered word is realized by cutting the corresponding ordered collision flag in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R}$; admissibility of pull-push and Thom–Sebastiani transport are part of $\mathfrak{O}_{\mathrm{hyb},R} = 0$. The eight-word flag pentagon gives binary coassociativity; the higher-coloured, unit, refinement, and quotient coherences contained in $\mathfrak{O}_{\mathrm{hyb},R} = 0$ give the chiral coalgebra identities. After at most $p \le N_R$ reduced iterations, every branch has length one and the next reduced coproduct is zero, so $(\overline{\Delta}^{\mathrm{bar}}_{R})^{N_R+1} = 0$. □

The finite packet `certificates/charge/bar_pushforward_commutation` supplies only the grading fact used in the preceding construction: for the additive normal-ordered map $\overline{\Pi}_X : \widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$, finite bar words, deconcatenation, and multiplication terms have the same Gram degree before and after pushforward. It does not supply $\mathfrak{O}_{\mathrm{hyb},R} = 0$, the wrapped bar-cobar domain, the source-internal counit, or the source-to-target chiral quasi-isomorphism.

Cobar obstruction

The canonical source gives a conilpotent chiral bar coalgebra. A cobar quasi-isomorphism to the denominator chiral algebra is an additional datum, consisting of a source-internal bar-cobar counit and a source-to-target chiral algebra quasi-isomorphism.

Corollary 0.210 *(Source-to-target cobar obstruction; proved).* Under the hypotheses of Proposition 0.209, a cobar quasi-isomorphism $\Theta_{\mathrm{Kos},R}$ requires:

(i) *Source-internal bar-cobar counit.* The supplied hybrid source, including the wrapped summands and mixed local/wrapped words, must lie in the Beilinson–Drinfeld/Francis–Gaitsgory completed bar-cobar domain [35, Theorem 6.2.2], with the source-internal counit

$$\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_{E} C^{\mathrm{bar}}_{X,R} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$$

a quasi-isomorphism in the chosen completed/coderived conilpotent category. Defects: $\mathfrak{o}_R^{\varepsilon,\mathrm{src}} = (\mathfrak{o}_R^{\mathrm{aug}}, \mathfrak{o}_R^{\mathrm{FG}}, \mathfrak{o}_R^{\mathrm{wrKos}}, H^\bullet \operatorname{Cone} \varepsilon_{\mathrm{bar},R}^{\mathrm{src}})$.

(ii) *Source-to-target chiral algebra quasi-isomorphism.* $\vartheta_R : \mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\leq R}$ preserving differential, product, unit, transition maps, and primitive restriction. Defects:

$$\mathfrak{o}_R^{\vartheta} = \big(d_\Delta \vartheta_R - \vartheta_R d_{\mathrm{hyb}}, \vartheta_R \mu_R^{\mathrm{hyb}} - \mu_{\Delta,R}(\vartheta_R \otimes \vartheta_R), \vartheta_R \eta_R^{\mathrm{hyb}} - \eta_{\Delta,R}, \vartheta_R \circ \tau_{R'R}^{E,\mathrm{hyb}} - \pi_{R'R}^{\Delta} \circ \vartheta_{R'}, H^\bullet \operatorname{Cone} \vartheta_R, \vartheta_R|_{\mathrm{Prim}} - \iota$$

Then $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon_{\mathrm{bar},R}^{\mathrm{src}}$, and the surviving cobar residual is $\mathfrak{o}_R^{\Omega} = (\mathfrak{o}_R^{\varepsilon,\mathrm{src}}, \mathfrak{o}_R^{\vartheta})$. No homotopy inverse to the target counit $\varepsilon_{\Delta,R}$ enters this construction.

Definition 0.211 *(Finite Koszul comparison datum).* Fix a finite HN height R, and suppose the finite-stage chiral Koszul source datum $\mathfrak{K}_{X,R}^{\mathrm{Kos}}$ of Definition 0.208 is fixed. A finite Koszul comparison datum is

$$\mathfrak{K}_{X,R}^{\mathrm{cmp}} = \big(\varepsilon_{\mathrm{bar},R}^{\mathrm{src}}, \vartheta_R, H_R^W, H_R^{\mathrm{Pf}}, H_R^{\mathrm{Hall}}, H_R^{\mathrm{rad}}, H_R^{\mathrm{PBW}}, H_{R'R}^{\mathrm{tr}}\big)$$

with the following finite chain-level clauses.

(K0) *Two quasi-isomorphism cones.* The source-internal bar-cobar counit

$$\varepsilon_{\mathrm{bar},R}^{\mathrm{src}} : \Omega_E^{\mathrm{ch}} C_{X,R} \longrightarrow \mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}$$

and the source-to-target chiral algebra morphism

$$\vartheta_R : \mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}} \longrightarrow \mathsf{Den}_{\Delta_5,E,\leq R}$$

are supplied with finite filtrations by HN height, bar length, local/wrapped word, and parity. Their cone cohomology groups vanish in each filtered piece.

(K1) *Weyl compatibility.* For every retained type-II simple reflection s_{∂_i} and chosen lift $\tau_{i,R}$, the square comparing the source Weyl transport on $C_{X,R}$ with the target Weyl action on $\mathsf{Den}_{\Delta_5,E,\leq R}$ commutes up to a specified homotopy $H_{i,R}^W$. The Coxeter and chamber-overlap defect complexes of these homotopies are acyclic.

(K2) *Pfaffian orientation compatibility.* The determinant line of the finite cobar comparison is identified with the Pfaffian-line transport of $(O1)^+$ and the $(O2_{\mathrm{atlas}})$ wall datum. On each type-II wall chart the induced sign is the rank-one Pfaffian sign $\epsilon_{o,R}(s_{\partial_i})$; after the target comparison it is $\nu_{\Delta_5}(s_{\partial_i})$. The difference is the finite defect of H_R^{Pf}, required to vanish.

(K3) *Hall product, coproduct, and pairing compatibility.* On primitive classes, $\vartheta_R \circ \varepsilon_{\mathrm{bar},R}^{\mathrm{src}}$ intertwines the Hall supercommutator, collision coproduct, unit, counit, and Hopf pairing with the corresponding operations in the finite target presentation. Equivalently the finite matrices $M, D, \eta, \epsilon, B, G$ of Definition 0.236 are carried to the target product, coproduct, bracket, and invariant pairing blocks.

(K4) *Radical quotient compatibility.* The primitive restriction maps the finite source Hopf radical onto the finite GN/Kac radical block and induces the comparison on radical quotients:

$$\mathrm{Rad}_{X,R}^{\Pi} \xrightarrow{\Theta_{\mathrm{Kos},R}|_{\mathrm{Prim}}} \mathrm{Rad}_{\mathrm{GN},R}, \qquad P_{X,R}^{\Pi,\mathrm{red}} \xrightarrow{\iota_R^{\mathrm{Prim}}} \mathfrak{g}_{\Delta_5,\leq R}.$$

The kernel and cokernel defect complexes of this quotient square are zero.

(K5) *PBW and primitive-recognition compatibility.* The primitive restriction of $\Theta_{\mathrm{Kos},R}$ is the same map ι_R^{Prim} used in primitive recognition, and it preserves the PBW filtrations on the source and target enveloping algebras. The associated-graded comparison agrees with the PBW rows of Definition 0.236.

(K6) *Strict transitions.* For $R' \geq R$, the transition homotopy $H^{\mathrm{tr}}_{R',R}$ identifies the two composites obtained by first applying $\Theta_{\mathrm{Kos},R'}$ and then truncating, or first truncating the source data and then applying $\Theta_{\mathrm{Kos},R}$. The same strictness holds for the Weyl, Pfaffian, Hall-pairing, radical, and PBW compatibility complexes.

(K7) *Koszul Mittag–Leffler exactness with compatibility.* The inverse systems of the two quasi-isomorphism cones and of all defect complexes in $(K1)$–$(K6)$ are strict Mittag–Leffler systems; $R^1 \varprojlim = 0$ for each of them.

The finite Koszul comparison residual

$$\mathfrak{D}_{\mathrm{Kcmp},R} = \left(H^\bullet C\varepsilon^{\mathrm{src}}_{\mathrm{bar},R}, H^\bullet C\vartheta_R, \mathfrak{d}_R^W, \mathfrak{d}_R^{\mathrm{Pf}}, \mathfrak{d}_R^{\mathrm{Hall}}, \mathfrak{d}_R^{\mathrm{rad}}, \mathfrak{d}_R^{\mathrm{PBW}}, \mathfrak{d}^{\mathrm{tr}}_{R',R}, \mathfrak{d}^{\mathrm{ML}}_{\mathrm{Kcmp}}\right)$$

vanishes exactly when $(K0)$–$(K7)$ hold. A target-internal bar-cobar counit, a primitive restriction, or a scalar denominator identity supplies none of these compatibility homotopies.

After choosing finite homogeneous bases, the datum is recorded by three source tables:

`koszul_cones`, `koszul_comparison_identities`, `koszul_transition_ml`.

The first table records the two acyclic cones `source_bar_cobar_counit` and `source_to_target_quasi_isomorphism`, with zero cohomology rank in every HN, bar-length, word, and parity block. The second table records zero finite defects for

`weyl_action`, `pfaffian_orientation`, `hall_product`, `hall_coproduct`, `hopf_pairing`, `radical_quotient`,

The third table records strict transition and Mittag–Leffler vanishing for

`source_cone`, `target_cone`, `weyl_action`, `pfaffian_orientation`, `hall_pairing`, `radical_quotient`, `pbw`,

with $R^1 \varprojlim$-rank zero. These rows require geometric source provenance and proof references, just as the Hall and primitive-recognition tables do.

Proposition 0.212 *(Finite Koszul comparison compatibility; conditional on $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$).* If the finite Koszul comparison datum $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ is supplied and $\mathfrak{D}_{\mathrm{Kcmp},R} = 0$, then

$$\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E C_{X,R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\leq R}$$

is a quasi-isomorphism of augmented chiral algebras whose primitive restriction respects the Weyl action, the Pfaffian orientation character, the Hall product/coproduct, the Hopf pairing, the radical quotient, and the PBW associated graded on the finite window.

Proof. Clause (K0) gives the quasi-isomorphism by acyclicity of the two cones. Clauses (K1)–(K2) identify the Weyl and Pfaffian-orientation squares, with zero Coxeter and wall-sign defects. Clause (K3) says that the source Hall operations and Hopf pairing are carried to the finite target blocks, so the primitive bracket and pairing used by recognition are the restrictions of the same cobar comparison. Clause (K4) identifies the radical quotient, and (K5) identifies the PBW associated gradeds. The conclusion is therefore a finite diagram chase in the common homogeneous bases. No scalar Borcherds product or target-internal counit participates in these diagrams. □

PROPOSITION 0.213 *(Primitive restriction is not a Koszul quasi-isomorphism; proved).* Let

$$\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} \longrightarrow \mathsf{Den}_{\Delta_5,E,\leq R}$$

be a morphism of augmented chiral algebras whose restriction to the constant-current primitive summand agrees with ι_R^{Prim}. This primitive restriction does not imply that ϑ_R is a quasi-isomorphism. The required finite obstruction is the full cohomology

$$H^\bullet \operatorname{Cone}(\vartheta_R)$$

on every HN weight, bar length, local/wrapped word, and parity block. Consequently the chiral Koszul comparison $\Theta_{\mathrm{Kos},R}$ is proved only after the source-internal counit cone and the source-to-target cone are acyclic, compatibly in R.

Proof. The assertion already fails for ordinary augmented dg algebras, hence also for constant chiral algebras on E. Let B be any augmented dg algebra and let M be a non-zero dg B-bimodule placed in a nonprimitive HN weight, with zero differential. The square-zero extension $A = B \oplus M$ has multiplication

$$(b, m)(b', m') = (bb', bm' + mb'), \qquad MM = 0,$$

and the projection $p : A \to B$ is a morphism of augmented dg algebras. On the chosen primitive summand of $B \subset A$, p is the identity. But

$$H^\bullet \operatorname{Cone}(p) \cong H^\bullet(M)[1] \neq 0.$$

Thus agreement on primitives detects only the selected weight-one summand. It does not see square-zero, wrapped, mixed, higher bar-length, or parity-cancelling classes in the source. In the Dirac–Igusa notation these missing classes are exactly the cone terms in $\mathfrak{o}_R^{\varepsilon,\mathrm{src}}$ and $\mathfrak{o}_R^{\vartheta}$, together with the Koszul Mittag–Leffler obstruction in the inverse system. □

CLASS M COMPLETION AND CHIRAL KOSZULNESS

The bar/cobar comparison takes place in the completed ambient category. Class M holds in the completed ambient: analytic HS-sewing, coderived BV = bar, weight-completed, pro-, J-adic. On chain-level ordinary complexes, the chiral bar/cobar pair is the comparison input, not the closed Koszulness theorem; chiral Koszulness in the homotopy setting is a separate theorem of Francis–Gaitsgory [35].

KOSZUL MITTAG–LEFFLER EXACTNESS

The completed bar/cobar inverse system is controlled by a finite-slice Mittag–Leffler condition on every relevant tower.

PROPOSITION 0.214 *(Consequence of the chiral Koszul source datum; conditional on Koszul Mittag–Leffler).* Assume $\mathfrak{K}^{\mathrm{Kos}}_{X,R}$ is supplied for all R with compatible transition maps and $\mathfrak{o}^{C,\mathrm{tr}}_{R'R} = 0$, and that the Koszul Mittag–Leffler condition holds: $R^1 \varprojlim = 0$ for the inverse systems of source coalgebras, completed chiral bar/cobar constructions, target truncations, and the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ [106, §3.5]. Then $C_X = \varprojlim_R C_{X,R}$ is a completed pro-conilpotent chiral coalgebra attached to the hybrid compact source, conilpotent on every finite HN quotient, and the maps $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ induce

$$b_X : C_X \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X), \qquad \Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

If, in addition, the finite Koszul comparison data $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ of Definition 0.211 are supplied for a cofinal HN subsystem and satisfy the strict compatibility Mittag–Leffler condition $(K7)$, then the inverse-limit map Θ_{Kos} respects the Weyl action, Pfaffian orientation character, Hall product/coproduct, Hopf pairing, radical quotient, and PBW associated graded. This identifies the supplied source cobar algebra with the formal target current envelope; it does not construct the E_3-source, the hybrid wrapped Hall correspondences, the orientation line, the primitive recognition data, or C_X itself from the target-internal counit.

Proof. Inverse-limit existence follows from transition compatibility, the bounded exhaustive finite charge filtration, finite coassociative and counital collision coproduct, and continuity hypotheses. The coradical filtration is finite on every Γ^{Π}_{R}-truncation; the Koszul Mittag–Leffler hypothesis kills $R^1 \varprojlim$ on the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$, giving completed quasi-isomorphisms. If the finite comparison data are present, the same argument applied to the Weyl, Pfaffian, Hall-pairing, radical, and PBW defect complexes in $(K1)$–$(K7)$ passes the zero finite defects to the inverse limit. Without that exactness one obtains only a compatible pro-quasi-isomorphism, with no theorem-level control of these recognition compatibilities. □

Chiral MC twisting / coupling coalgebra

Bar is twisting; bar is not the level-Z bulk. The reduced bar coalgebra $\bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X})$ is the Maurer–Cartan coupling coalgebra of the supplied hybrid chiral algebra: its primitive elements are the chain-level inputs of MC twisting in the Beilinson–Drinfeld/Francis–Gaitsgory framework, and its grouplike completion records the homotopy-coherent twisting data. The level-Z bulk in Vol II is $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ for the relevant boundary algebra A; the visible target is its BKM chiral shadow $\mathsf{Den}_{\Delta_5,E}$. The Koszul comparison maps C_X, an instance of MC coupling, into that target chiral algebra by the cobar construction.

The Koszul obstruction class

The Koszul lane carries the residuals

$$\mathfrak{O}_{\mathrm{Kos},R}, \qquad \mathfrak{O}_{\mathrm{Kcmp},R}, \qquad \mathfrak{o}^{\mathrm{ML,Kos}}_{R' R}.$$

With $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and the canonical source-bar construction, the coalgebra defects vanish; only the cobar residual $\mathfrak{o}^{\Omega}_{R} = (\mathfrak{o}^{\varepsilon,\mathrm{src}}_{R}, \mathfrak{o}^{\vartheta}_{R})$ remains, with the wrapped Koszul defect $\mathfrak{o}^{\mathrm{wrKos}}_{R}$ included in $\mathfrak{o}^{\varepsilon,\mathrm{src}}_{R}$. Vanishing of the source-internal counit defect and construction of the source-to-target chiral algebra quasi-isomorphism operates in the completed Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar domain. The comparison residual $\mathfrak{O}_{\mathrm{Kcmp},R}$ is separate: it records whether this quasi-isomorphism respects the Weyl action, Pfaffian orientation, Hall pairing, radical quotient, and PBW filtration used by primitive recognition. The Koszul Mittag–Leffler hypothesis controls the inverse-system passage for both the quasi-isomorphism cones and these compatibility complexes.

Source-target separation

The source–target separation is the $K3 \times E$-side instance of the Vol II Beilinson cut: bar is twisting, bar is not bulk; the boundary algebra is $A = \mathcal{F}^{\mathrm{hyb}}_{X}$ at finite stage, and the bulk is $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ which receives the cobar of the bar coalgebra after MC coupling. Vol I master patterns prohibit the identification of the chiral bar with the bulk chiral algebra and require the Koszul comparison to be a separate theorem in the homotopy setting; the same separation is enforced above. The compact Koszul comparison and the chiral Koszulness theorem are not identical; the present construction lives on the comparison side, with chiral Koszulness in the Francis–Gaitsgory sense as the abstract input.

Under the Koszul source datum and the finite Koszul comparison datum, $\Theta_{\mathrm{Kos},R}$ identifies the cobar of the source coalgebra with the imported current envelope of $\mathfrak{g}_{\Delta_5}$, compatibly with the Weyl, Pfaffian, Hall-pairing, radical, and PBW structures. Under the recognition datum $(S1)$–$(S10)$, the source-side protected primitive $P_X^{\Pi,\mathrm{red}}$, extracted from C_X via the Hopf radical quotient, the Frobenius identity, and the no-extra-relations criterion, is identified with the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. Entry (L6) is this recognition apparatus, in §0.6.

0.6 *Primitive recognition apparatus*

The sixth entry of the Dirac–Igusa pro-object is the primitive recognition criterion. Its conclusion, after the compact source data are supplied, is an isomorphism of Gram-graded Lie superalgebras

$$P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$$

between the cosection-twisted reduced Hall primitive layer of $K3 \times E$ and the imported Borcherds–Kac–Moody denominator algebra. Signed superdimensions alone do not suffice for this theorem. The Borcherds product fixes $\operatorname{sdim} P_{X,\gamma}^{\Pi,\mathrm{red}} = f(nm, l)$ for $\gamma = (n, l, m)$; this is one integer in each degree. A source–target identification fixes two integers in every degree (the parity dimensions), the relation ideal, the radical equality, the no-extra-relations theorem, the completed PBW filtration, and the Hopf-radical coideal stability.

The determinant admits the zero-bracket graded super vector space with the same signed dimensions: the denominator is identical, while the Chevalley diagonal, the non-orthogonal brackets, and the Hopf pairing datum have disappeared. Recognition therefore requires representatives, relations, pairings, kernels, and PBW data, not numerics.

BORCHERDS–CARTAN TARGET DATUM

The target side imports the Borcherds–Kac superdatum $\mathscr{B}_{\Delta_5} = (H, I, p, \alpha_i, (\ ,\ ))$ of Gritsenko–Nikulin [46–48] after the [8, 58] presentation conventions. The Cartan is $H = \Lambda_{II}^{2,1} \otimes \mathbb{C}$, the three real simple roots $\delta_1, \delta_2, \delta_3$ form a hyperbolic triangle with Gram matrix

$$((\delta_i, \delta_j))_{i,j} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

and the imaginary fibres of $I \to H^*$ carry the Gritsenko–Nikulin parity split $\mu_{\bar{0}}(a), \mu_{\bar{1}}(a)$ for imaginary simple roots. The degree map is

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

with $\widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})$ the completed free Lie superalgebra on the parity-homogeneous generators $e_i, f_i, i \in I$, and J_{BK} the closed Borcherds–Kac ideal generated by Cartan, Chevalley, real Serre, orthogonality, and super sign relations. The target denominator algebra is

$$G(\mathscr{B}_{\Delta_5}) = \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) / \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}} = \mathscr{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}.$$

SOURCE-SIDE DATA

THEOREM 0.215 *(Primitive recognition from finite source representatives; conditional on (S1)–(S10)).* Let a compact Hall/factorization construction give a completed $\widehat{\Gamma}_X$-graded primitive Hall Lie superalgebra $\widetilde{P}_X$ and a completed Hopf pairing. Its normal-ordered Gram descent

$$P_X^{\Pi,\mathrm{desc}} = H^\circ \operatorname{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*} \mathscr{F}_X) = \overline{\Pi}^{\Theta}_{X,*} \widetilde{P}_X, \quad P_X^{\Pi,\mathrm{red}} = P_X^{\Pi,\mathrm{desc}} / \overline{\mathrm{Rad}\langle\ ,\ \rangle_X}.$$

Suppose the following ten source-side data are constructed.

(S1) *Charge and normal-ordered descent.* The chain-level descent $\overline{\Pi}^{\Theta}_{X,*}$ categorifies the homomorphism $\overline{\Pi}_X(c,T) = \Pi_X(c) - T$, so that the descended bracket is Γ_{gram}-graded. Each descended homogeneous parity piece is finite dimensional.

(S2) *Cartan and root-degree matching.* An even Cartan identification $\iota_H : P^{\Pi,\mathrm{red}}_{X,0} \xrightarrow{\sim} \Lambda^{2,1}_{II} \otimes \mathbb{C}$ aligning the descended Gram grading with the GN root grading via $\alpha(n,l,m)$.

(S3) *Simple primitive representatives and parities.* For every simple-root index $i \in I$, choose $\gamma_i \in \Gamma_{\mathrm{gram}}$ with $\alpha(\gamma_i) = \alpha_i$. The source supplies parity-homogeneous representatives e^X_i, f^X_i, h^X_i in $P^{\Pi,\mathrm{red}}_{X,\gamma_i}$, $P^{\Pi,\mathrm{red}}_{X,-\gamma_i}$, and $P^{\Pi,\mathrm{red}}_{X,0}$, with $|e^X_i| = |f^X_i| = p(i)$, $h^X_i = \iota_H^{-1}(\alpha_i)$, and the GN multiplicity fibres $\mu_{\bar 0}(a)$, $\mu_{\bar 1}(a)$ for imaginary simple roots. Real representatives for $\delta_1, \delta_2, \delta_3$ are even.

(S4) *Borcherds–Kac relations in the Hall bracket.* In the protected Hall/factorization bracket the representatives satisfy:

$$[h,h'] = 0, \quad [h,e^X_i] = (h,\alpha_i)e^X_i, \quad [h,f^X_i] = -(h,\alpha_i)f^X_i, \quad [e^X_i, f^X_j] = \delta_{ij}h^X_i,$$

the real Serre relations

$$(\operatorname{ad} e^X_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} e^X_j = 0, \quad (\operatorname{ad} f^X_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)} f^X_j = 0 \qquad (\text{real } i,\ i \neq j),$$

the Borcherds orthogonality relations $(\alpha_i,\alpha_j) = 0 \Rightarrow [e^X_i, e^X_j] = [f^X_i, f^X_j] = 0$, and the super sign rule. In the rank-three real subsystem, the Serre exponent is 3 ($(\delta_i,\delta_j) = -2$); first brackets $[e^X_i, e^X_j]$ are not forced to vanish. For the primitive isotropic degrees $a_{ij} = \delta_i + \delta_j$ ($i < j$), choose $\gamma_{ij} \in \Gamma_{\mathrm{gram}}$ with $\alpha(\gamma_{ij}) = a_{ij}$. The source supplies the nine Gritsenko–Nikulin isotropic simple representatives $u^{\pm,X}_{ij,r} \in P^{\Pi,\mathrm{red}}_{X,\pm\gamma_{ij}}$, $1 \le r \le 9$, satisfying orthogonality and Chevalley zero with adjacent real roots, with real-string relations $(\operatorname{ad} e^X_k)^5 u^{+,X}_{ij,r} = 0$ for the complementary $\{i,j,k\} = \{1,2,3\}$. The tenth even vector in $\mathfrak{g}_{a_{ij}}$ is the real bracket $[e^X_i, e^X_j]$.

(S5) *Counital Hall bialgebra, Hopf pairing, and radical quotient.* A continuous completed Hall product, unit, coproduct, and counit satisfy associativity, coassociativity, the unit and counit identities, and the bialgebra compatibility

$$\Delta_X m_X = (m_X \otimes m_X)(1 \otimes \tau \otimes 1)(\Delta_X \otimes \Delta_X)$$

with the Koszul sign braiding τ. The bracket is the supercommutator of this product restricted to primitives. The Hopf pairing is continuous, homogeneous, invariant for bracket and coproduct, has vanishing Frobenius cyclic defect, is non-degenerate after the radical quotient, and has closed homogeneous radical $\mathrm{Rad}_X = \{x \mid \langle x, y\rangle_X = 0\ \forall y\}$ a Lie ideal and coideal. After quotienting by $\overline{\mathrm{Rad}_X}$, the source radical quotient is identified with the GN/Kac quotient $\overline{\mathrm{Rad}_{\mathrm{GN}}}$. At every finite stage the Lie-ideal and coideal stability hold and the inverse-limit kernel is exactly Rad_X.

(S6) *No extra relations.* If $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P^{\Pi,\mathrm{red}}_X$ is the continuous map determined by the representatives, $\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$. Equivalently, $G(\mathscr{B}_{\Delta_5}) \to P^{\Pi,\mathrm{red}}_X$ is injective. This is checked finite stage by finite stage on saturated windows, with kernel identifications commuting with HN transitions.

(S7) *Generation.* $P^{\Pi,\mathrm{red}}_X$ is topologically generated by $P^{\Pi,\mathrm{red}}_{X,0}$ and the simple primitive representatives.

(S8) *Completed PBW.* PBW filtration compatibility:

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P_X^{\Pi,\mathrm{red}})_W \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_W$$

on every saturated finite window W, with strict transition maps and inverse limit in the HN topology of §0.3.

(S9) *Exact completion.* Inverse limit exact on the saturated finite root windows: transition maps strict, images closed, and passage to the completed radical quotient commuting with finite-window kernels, cokernels, PBW associated gradeds, and parity pieces. Equivalently, no new relation, generator, radical element, or parity-cancelling summand appears only after completion.

(S10) *Full parity dimensions.* For every nonzero γ with $\alpha(\gamma) \in \mathcal{R}$,

$$\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}} = \mathrm{mult}_{\overline{p}}(\alpha(\gamma)), \qquad p = 0, 1.$$

The same equality is required in degree $-\gamma$, with the target parity dimensions transported by the Borcherds–Kac Chevalley anti-involution; it is not inferred from the positive half or from the signed denominator. On the first timelike channel $\gamma = (1, 1, 1)$,

$$d^{\mathrm{GN}}_{(1,1,1),\overline{0}} = 29, \qquad d^{\mathrm{GN}}_{(1,1,1),\overline{1}} = 93.$$

These are presentation-level target dimensions, not an inference from $29 - 93 = -64$.

Then $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ as completed Lie superalgebras graded by Γ_{gram} through α, with denominator $64^{-1}\Delta_5(2Z)$. If in addition the chiral Koszul source datum and the finite Koszul comparison datum of §0.5 are constructed, including the strict Koszul Mittag–Leffler exactness for the compatibility complexes, then $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$ through a map respecting the Weyl action, Pfaffian orientation character, Hall pairing, radical quotient, and PBW associated graded.

Proof. The simple primitive representatives define a continuous homomorphism $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$. (S1) and (S2) make π a homomorphism of the descended Gram-graded completions. (S3) fixes the real and imaginary simple generators with their parities and multiplicity fibres. (S4) puts J_{BK} in the kernel. (S5) identifies the compact Hopf radical with the GN/Kac radical, so π descends to $G(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$. (S7) gives surjectivity; (S6) gives injectivity after the completed radical quotient. (S8) gives PBW-compatible finite-height identifications; (S9), in the form of Proposition 0.235, gives exactness in the inverse limit; (S10) rules out hidden zero-superdimension root spaces and cancelling ideals invisible to the determinant. The chiral statement uses C_X, b_X, Θ_{Kos}, the finite Koszul comparison datum $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$, and the Koszul Mittag–Leffler exactness of §0.5. □

COFINAL FINITE-WINDOW PRIMITIVE RECOGNITION

The Borcherds product fixes signed target superdimensions; the Gritsenko–Nikulin/Kac presentation fixes full target parity dimensions only in the degrees where the presentation has been computed. A compact $K3 \times E$ source is identified with the denominator algebra only by source representatives, pairings, radicals, relation ideals, PBW, and exact transition maps.

The imported target on the first window $W_{\le 3}$ has parity table

$$\delta_i : 1|0, \quad a_{ij} : 10|0, \quad 2\delta_i + \delta_j : 1|0, \quad \delta_{123} : 29|93,$$

so $29 - 93 = -64 = f(1,1)$. The row a_{ij} : $10|0$ is the real bracket $[e_i, e_j]$ together with the nine Gritsenko–Nikulin isotropic simple directions. These are target data; they do not construct compact source primitives, source pairings, radical kernels, or source bracket constants. The finite target packet `wle3_target_parity` verifies this table from the theta-series coefficients and the Gritsenko–Nikulin/Kac presentation: a_{ij} is isotropic, the real-string exponent for $2\delta_i + \delta_j$ is 3, and δ_{123} splits as $29 = 2 + 27$ even directions and 93 odd imaginary-simple directions. It is a target-reference table only.

On the first base terminal and saturation-extension target rows the coefficient table `k3e_first_relation_window_coefficie` computes

$$f(4,4) = 10, \quad f(2,2) = 108, \quad f(2,1) = f(4,3) = -513, \quad f(3,3) = -64, \quad f(4,2) = 4016,$$

together with their discriminant-orbit translates and the terminal zero rows $f(5,5) = f(1,-3) = 0$. These equalities match the signed dimensions in the A071 target presentation. They remain one-integer target arithmetic; they do not supply the two parity dimensions required by (R3), or any compact-source row in (R0)–(R8). The A071 target-presentation check verifies the target rows

$$2a_{ij} : 10|0, \qquad C_{k,3} : 29|93, \qquad C_{k,4} : 10|0, \qquad C_{k,5} : 0|0,$$

and verifies that $C_{k,2}$, $D_k = \delta_k + 2a_{ij}$, and $2\delta_{123}$ are signed-only: they have no target basis, positive-negative pairing block, radical block, or PBW row. This is a target-reference computation, not a compact-source theorem.

The first relation-closed compact-source window is therefore a separate finite computation. Its row packet is

window_closure, target_source_representatives, chevalley_serre_matrices, hall_boundary_complex, spectral_seque

These rows must prove downward saturation and relation closure, geometric source representatives with full parity ranks, Cartan, Chevalley, real-Serre, Borcherds-orthogonality, and super-sign relations, the Hall–Chevalley boundary d_1, strong convergence of the finite-window spectral sequence, computed kernel bases, no-extra kernel equality, PBW associated-graded equality, and strict Mittag–Leffler transition. Target labels, signed dimensions, scalar traces, Pfaffian products, denominator products, target PBW rows, arbitrary matrices, and status-only rows are excluded by the scalar firewall.

Proposition 0.216 *(Target-presentation rows are a prerequisite; proved).* Let $W \subset \mathcal{R}_+$ be a finite downward-saturated target-root window. If W is used in a primitive-recognition datum satisfying (R0)–(R8), then the target restriction $\mathfrak{g}_{\Delta_5}|_{S_W}$ must be represented by a finite target-presentation datum in the sense of Definition 6.10. Signed rows $\mathrm{smult}(\beta)$ and imaginary-simple integers $m(\beta)$ alone are not admissible codomains for the comparison maps $A_{\beta,\overline{p}}$.

Proof. Clause (R2) compares source relation rows with the Borcherds–Kac target relations, so the Cartan, Chevalley, real-Serre, orthogonality, and super-sign rows must be present in the target window. Clause (R3) compares two parity dimensions in each sign, not the signed difference. Clause (R4) compares the source positive-negative Hopf pairing and radical quotient with the target invariant-pairing kernel. Clause (R7) compares PBW associated gradeds. Thus the target side must carry the parity, relation, pairing, radical, and PBW rows of Definition 6.10. A scalar row $\mathrm{smult}(\beta)$ or $m(\beta)$ supplies none of the pairing, radical, relation, or PBW codomains, and supplies only one integer where (R3) requires two. □

Definition 0.217 *(Cofinal finite-window primitive-recognition datum).* A finite set $W \subset \mathcal{R}_+$ is *downward saturated* if $\beta \in W$, $\beta' \in \mathcal{R}_+$, $0 < \beta' \leq \beta$, and $\beta - \beta' \in Q_+$ imply $\beta' \in W$. The canonical height-window convention is

$$W_\nu = \{\beta = b_1\delta_1 + b_2\delta_2 + b_3\delta_3 \in \mathcal{R}_+ \mid b_1 + b_2 + b_3 \leq \nu\}.$$

A cofinal finite-window primitive-recognition datum for $P_X^{\Pi,\mathrm{red}}$ is an increasing sequence of finite downward-saturated target-root windows $W_1 \subset W_2 \subset \cdots$ with $\bigcup_\nu W_\nu = \mathcal{R}_+$, together with the following finite data on every W_ν, compatible under $W_\nu \subset W_{\nu+1}$. Put

$$\Gamma(W_\nu) := \{\gamma \in \Gamma_{\mathrm{gram}} \mid \alpha(\gamma) \in W_\nu\}.$$

(R0) *Compact source labels.* Every source basis vector of degree in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$ satisfies the compact source admissibility condition. Target labels from $G(\mathscr{B}_{\Delta_5})$ appear only in target rows or as codomains of comparison maps.

(R1) *Representatives and parities.* Simple-root representatives e_i^X, f_i^X, h_i^X whose target roots lie in W_ν are supplied in source degrees $\pm\gamma_i$ with $\alpha(\gamma_i) = \alpha_i$, with the GN parity fibres for imaginary simple roots and even real representatives for $\delta_1, \delta_2, \delta_3$.

(R2) *Relations.* Cartan, Chevalley, real Serre, orthogonality, and super sign relations checked for every relation whose displayed target roots lie in $W_\nu \cup (-W_\nu) \cup \{0\}$ and whose source lifts lie in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$.

(R3) *Full finite parity dimensions.* For every $\gamma \in \Gamma(W_\nu)$ and for its negative $-\gamma$, $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}} = \mathrm{mult}^{\mathrm{GN}}_{\overline{p}}(\alpha(\gamma))$ ($p = 0, 1$), with the negative-root parity dimensions carried by the Borcherds–Kac Chevalley anti-involution, and with zero on source degrees whose image is not a target root. Two-integer equality, not signed-superdimension.

(R4) *Hall bialgebra, Hopf pairing, and radical.* The finite product M, coproduct D, unit η, and counit ϵ satisfy associativity, coassociativity, unit, counit, and bialgebra compatibility identities on the window; the supercommutator bracket sends primitives to primitives. Positive-negative pairing matrices are written in parity blocks; kernels agree with GN/Kac invariant-pairing kernels; Hopf adjointness, the primitive Frobenius cyclic identity, and non-degeneracy on the radical quotient are verified by finite rank-zero defect rows; stable under finite brackets in the window; coideal for finite coproduct; carried to the next window by transitions; inverse-limit radical the closure of finite kernels.

(R5) *No extra relations.* Finite kernel equality $\ker \pi_\nu = \overline{(J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}})}_{\leq W_\nu}$.

(R6) *Generation.* Source primitive spaces over $\Gamma(W_\nu) \cup (-\Gamma(W_\nu))$ are generated by Cartan and supplied simple representatives through brackets whose intermediate target roots stay in W_ν.

(R7) *Completed PBW compatibility.* Strict PBW filtration preservation under transitions, inverse-limit PBW compatibility.

(R8) *Exact triangular completion.* For any source stage R assigned to W_ν, words whose total Gram degree lies in the finite target test window Γ_R^{test} of Definition 0.249 only through positive-negative cancellation are not added as extra finite generators; transition maps strict, images closed, $\lim^1$ vanishing on the triangular finite-window systems.

The datum is finite at each ν, but not numerical: it contains representatives, relation verifications, pairing matrices, kernel equalities, and PBW transition data. The Borcherds product and the OP scalar square provide none of these source-side objects. The further assertion that every relation-closed target degree occurs at some finite HN stage is the finite-R exhaustion datum of Definition 0.250; it is not a consequence of target relation closure alone.

Proposition 0.218 *(Height root windows and their Gram preimages; proved).* The sets W_ν of Definition 0.217 are finite, downward saturated, and cofinal in $\mathcal{R}_+$. If $\beta = b_1\delta_1 + b_2\delta_2 + b_3\delta_3 \in W_\nu$, then

$$\gamma_\beta = (b_1,\ b_1 + b_2 - b_3,\ b_2) \in \Gamma_{\mathrm{gram}}$$

is the unique element with

$$\alpha(\gamma_\beta) = b_1\delta_1 + b_2\delta_2 + b_3\delta_3.$$

Hence

$$\Gamma(W_\nu) = \{(b_1, b_1 + b_2 - b_3, b_2) \mid b_1\delta_1 + b_2\delta_2 + b_3\delta_3 \in W_\nu\}.$$

Moreover γ_β lies in the type-II product chamber:

$$b_2 > 0 \Rightarrow m > 0, \qquad b_2 = 0,\ b_1 > 0 \Rightarrow m = 0,\ n > 0, \qquad b_1 = b_2 = 0,\ b_3 > 0 \Rightarrow m = n = 0,\ l < 0.$$

Proof. Finiteness follows because $b_1 + b_2 + b_3 \leq \nu$ gives only finitely many triples in Q_+. If $\beta' \in \mathcal{R}_+$, $0 < \beta' \leq \beta$, and $\beta \in W_\nu$, then $\mathrm{ht}(\beta') \leq \mathrm{ht}(\beta) \leq \nu$; hence $\beta' \in W_\nu$. Cofinality follows because every positive root has finite height.

For $\gamma = (n, l, m)$, the product-chamber root map is

$$\alpha(n, l, m) = n\delta_1 + m\delta_2 + (n + m - l)\delta_3.$$

Solving $\alpha(n, l, m) = b_1\delta_1 + b_2\delta_2 + b_3\delta_3$ gives $n = b_1$, $m = b_2$, and $l = b_1 + b_2 - b_3$, proving uniqueness and the displayed formula for $\Gamma(W_\nu)$. The three chamber cases are then immediate from $b_i \geq 0$ and $\beta \neq 0$. □

The packet `certificates/lattice/downward_root_windows` records Proposition 0.218. It verifies `DOWNWARD_ROOT_WINDOWS_VERIFIED`: the height-window formula, the inverse formula for $\Gamma(W_\nu)$, the product-chamber sector rule, and the ambient positive-cone audit for $1 \leq \nu \leq 7$. It is a target-side root-window convention, not a compact-source recognition packet, not a Pfaffian orientation, and not a protected trace.

Proposition 0.219 *(Active height windows exhaust the positive roots; proved).* Let $W_\nu^{\mathrm{act}} := W_\nu$. Then

$$\bigcup_{\nu \geq 1} W_\nu^{\mathrm{act}} = \mathcal{R}_+.$$

Equivalently, every positive root $\beta = b_1\delta_1 + b_2\delta_2 + b_3\delta_3$ is witnessed in the finite active window $W_{\mathrm{ht}(\beta)}$, where $\mathrm{ht}(\beta) = b_1 + b_2 + b_3$.

Proof. By definition $W_\nu \subset \mathcal{R}_+$ for every ν. Conversely, if $\beta \in \mathcal{R}_+$, then its coefficients in the positive root monoid are non-negative and have finite sum $\mathrm{ht}(\beta)$. Hence $\beta \in W^{\mathrm{act}}_{\mathrm{ht}(\beta)}$. The two inclusions give the displayed equality. □

The packet `certificates/lattice/active_root_window_exhaustion` records Proposition 0.219. It verifies `ACTIVE_ROOT_WINDOW_EXHAUSTION_VERIFIED`: the height-witness formula $\beta \in \mathcal{R}_+ \Rightarrow \beta \in W_{\mathrm{ht}(\beta)}$, the low active membership samples, the inverse γ_β formula, and the type-II chamber sector rule. It is not the compact-source support exhaustion required before moduli transitions, orientation transport, vanishing-cycle transport, Hall operations, radical transport, or PBW transport.

Definition 0.220 *(Source-window compact-support exhaustion datum).* A source-window compact-support exhaustion datum consists of:

(i) a set $\mathscr{S}_X^{\mathrm{cpt}}$ of compact-source support rows, each carrying a source degree, a compact geometric provenance row, proper support, finite stabilizer data, and the orientation row used by the source Hall construction;

(ii) a height function
$$h_X : \mathscr{S}_X^{\mathrm{cpt}} \longrightarrow \mathbb{Z}_{\geq 0}$$
with finite fibres;

(iii) finite source windows
$$\mathscr{S}_{X,\leq R}^{\mathrm{cpt}} = \{s \in \mathscr{S}_X^{\mathrm{cpt}} \mid h_X(s) \leq R\};$$

(iv) a cofinality witness proving that every $s \in \mathscr{S}_X^{\mathrm{cpt}}$ lies in $\mathscr{S}_{X,\leq h_X(s)}^{\mathrm{cpt}}$;

(v) transition rows $\mathscr{S}_{X,\leq R}^{\mathrm{cpt}} \hookrightarrow \mathscr{S}_{X,\leq R'}^{\mathrm{cpt}}$ for $R \leq R'$, later refined to closed embeddings or proper maps before orientation, vanishing-cycle, Hall, radical, and PBW transport is asserted.

Target root labels, signed multiplicities, Pfaffian products, and protected traces are not entries of this datum.

PROPOSITION 0.221 *(Source-window compact-support exhaustion criterion; proved).* If a datum of Definition 0.220 is supplied, then the finite source windows exhaust compact support:
$$\bigcup_{R\geq 0} \mathscr{S}_{X,\leq R}^{\mathrm{cpt}} = \mathscr{S}_X^{\mathrm{cpt}}.$$

Proof. The inclusion from left to right is part of the definition of the finite source windows. Conversely, for $s \in \mathscr{S}_X^{\mathrm{cpt}}$, the cofinality witness gives $s \in \mathscr{S}_{X,\leq h_X(s)}^{\mathrm{cpt}}$. Thus every compact source support row occurs in one finite source window. □

The packet `certificates/sources/source_window_compact_support` records the present state of this datum. Its verifier status is `SOURCE_WINDOW_COMPACT_SUPPORT_OBSTRUCTION_VERIFIED`: the obstruction ledger lists the missing compact support index set, source height function, finite source windows, compact provenance rows, cofinality witness, source-window transitions, and target-source compatibility rows. Its core source tables are empty, so the compact-source support-exhaustion theorem remains open. This is the firewall separating the target-root exhaustion of Proposition 0.219 from compact-source support exhaustion.

Definition 0.222 *(Target/source window compatibility datum).* Let W_ν be the target height window and $\mathscr{S}_{X,\leq R}^{\mathrm{cpt}}$ a finite compact-source window. A target/source window compatibility datum consists of:

(i) a degree map
$$d_R : \mathscr{S}_{X,\leq R}^{\mathrm{cpt}} \longrightarrow \Gamma_{\mathrm{gram}}$$
recording the normal-ordered Gram degree of every compact support row;

(ii) a target index $\nu(R)$ such that
$$d_R(s) \in \Gamma(W_{\nu(R)}) \qquad \text{for every } s \in \mathscr{S}_{X,\leq R}^{\mathrm{cpt}};$$

(iii) the row-wise equality
$$\alpha(d_R(s)) \in W_{\nu(R)}$$
with zero degree-map defect;

(iv) compatibility with compact provenance: the support row, its source degree, its compact geometric provenance, and its Gram degree all carry the same support identifier;

(v) compatibility under transitions

$$\mathscr{S}^{\mathrm{cpt}}_{X,\le R} \hookrightarrow \mathscr{S}^{\mathrm{cpt}}_{X,\le R'}, \qquad W_{\nu(R)} \subset W_{\nu(R')},$$

so that the source degree map commutes with passage to larger windows.

Proposition 0.223 *(Compatibility criterion for target and source windows; proved).* Given a datum of Definition 0.222, every compact source support row in $\mathscr{S}^{\mathrm{cpt}}_{X,\le R}$ has a well-defined target codomain in $\Gamma(W_{\nu(R)}) \cup (-\Gamma(W_{\nu(R)})) \cup \{0\}$, and the finite comparison maps on that source window are degree-compatible with the target window $W_{\nu(R)}$.

Proof. For a positive source degree the codomain is $\alpha(d_R(s)) \in W_{\nu(R)}$ by the compatibility datum. The negative degree is carried to $-\Gamma(W_{\nu(R)})$, and the Cartan degree is 0. The zero-defect equality makes the target degree independent of notation, and the transition condition shows that enlarging R does not move a previously defined source support row outside the enlarged target window. Thus every finite comparison row has a target codomain in the chosen window. □

The packet `certificates/sources/target_source_window_compatibility` records the current state of this datum. Its verifier status is `TARGET_SOURCE_WINDOW_COMPATIBILITY_OBSTRUCTION_VERIFIED`: the source degree maps, $\Gamma(W_\nu)$-membership rows, target-window matching rows, compact-provenance compatibility rows, cofinal-index compatibility rows, transition compatibility rows, and zero compatibility-defect rows are not yet supplied. Hence target/source window compatibility remains open.

Proposition 0.224 *(Finite-window presentation recognition; proved).* Fix a downward-saturated finite target-root window W and put $S_W = W \cup (-W) \cup \{0\}$. For a graded Lie superalgebra L, write

$$L|_{S_W} := \bigoplus_{\beta \in S_W} L_\beta$$

with the bracket retained only on pairs whose total degree remains in S_W. Suppose a compact $K3 \times E$ source supplies (R0)–(R8) on W. Then the finite comparison matrices $A_{\beta,\tilde{p}}$ determine an isomorphism of truncated Gram-graded Lie superalgebras

$$A_W : P^{\Pi,\mathrm{red}}_X|_{\alpha^{-1}(S_W)} \xrightarrow{\sim} \mathfrak{g}_{\Delta_5}|_{S_W},$$

compatible with the Cartan identification, the simple-root representatives, the Hopf-pairing radical quotient, and the associated-graded PBW filtration on W. Conversely, such an isomorphism, together with compact-source basis provenance, parity blocks, simple representatives, product, coproduct, unit, counit, Hopf-pairing, radical, quotient, and transition maps from which its matrices are obtained, gives (R0)–(R8).

Proof. The representatives of (R1) define a homomorphism from the free Borcherds–Kac presentation to the source truncation. The relation rows of (R2) put every Cartan, Chevalley, real-Serre, orthogonality, and super-sign relation whose degree lies in S_W in the kernel. The Hopf bialgebra and pairing data of (R4) make the source quotient a truncated Lie superalgebra: product and coproduct are compatible, the supercommutator closes on primitives, and the radical is a Lie ideal and a coproduct coideal. The no-extra relation row (R5) is exactly the finite equality

$$\ker \pi_W = \big(\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}\big)|_{S_W}.$$

Thus the presentation map descends injectively from $\mathfrak{g}_{\Delta_5}|_{S_W}$ to the source radical quotient. Generation (R6) gives surjectivity on every source degree in $\alpha^{-1}(S_W)$. The parity dimensions of (R3) identify the even and odd dimensions degree by degree in both signs, so no zero-superdimension summand or parity-cancelling kernel can remain. PBW compatibility (R7) identifies the associated gradeds of the universal enveloping algebras on W. Exact triangular completion (R8) says that the finite truncation is stable under passage to the next window and that no relation is created only by completion. The matrix A_W is therefore a degree-preserving bijection preserving all brackets, pairing quotients, and PBW associated gradeds present in the finite window.

Conversely, a truncated isomorphism with specified compact-source basis provenance, parity blocks, simple representatives, product, coproduct, unit, counit, Hopf pairing, radical, quotient, and transition maps yields the relation rows, radical identities, no-extra equality, generation spans, PBW rows, and transition rows by writing those finite maps in homogeneous bases. Without the named compact-source maps one has only a linear isomorphism of graded vector spaces, not (R0)–(R8). □

Definition 0.225 *(PBW-transition datum).* Let $W_\nu \subset W_{\nu+1}$ be two downward-saturated finite windows. Put

$$L_\nu = P^{\Pi,\mathrm{red}}_{X,\nu}/\mathrm{Rad}^{\Pi}_{X,\nu}, \qquad L_{\nu+1} = P^{\Pi,\mathrm{red}}_{X,\nu+1}/\mathrm{Rad}^{\Pi}_{X,\nu+1}.$$

A PBW-transition datum for $L_{\nu+1} \to L_\nu$ consists of the following finite rows:

(i) radical-quotient transition matrices $T^{Q}_{\nu+1,\nu}$ in homogeneous quotient bases;

(ii) ordered PBW word bases for $F^{\mathrm{PBW}}_{\le d} U(L_\nu)$ and $F^{\mathrm{PBW}}_{\le d} U(L_{\nu+1})$, with their associated-graded ranks;

(iii) source and target relation-rewrite rows which identify the normal forms in the enveloping algebras;

(iv) the multiplicative word transition induced by $T^{Q}_{\nu+1,\nu}$, together with the zero-defect relation-rewrite identity saying that it descends to

$$U(T^{Q}_{\nu+1,\nu}) : U(L_{\nu+1}) \longrightarrow U(L_\nu);$$

(v) for every retained d, a zero-defect filtered-image row

$$U(T^{Q}_{\nu+1,\nu})\big(F^{\mathrm{PBW}}_{\le d} U(L_{\nu+1})\big) \subset F^{\mathrm{PBW}}_{\le d} U(L_\nu);$$

(vi) the induced matrices on $\mathrm{gr}^{\mathrm{PBW}}_d\, U(L_{\nu+1}) \to \mathrm{gr}^{\mathrm{PBW}}_d\, U(L_\nu)$, the PBW-symbol intertwining identity, the A_β-PBW transition square, and the composition law for three consecutive windows.

The datum is absent if only primitive transitions, radical transitions, PBW rank equalities, target PBW rows, Hilbert series, scalar traces, or the formal normal-ordered PBW-pushforward packet are supplied.

PROPOSITION 0.226 *(Transition preservation of PBW filtrations; proved).* A PBW-transition datum in the sense of Definition 0.225 determines a filtered algebra homomorphism

$$U(T^{Q}_{\nu+1,\nu}) : \big(U(L_{\nu+1}), F^{\mathrm{PBW}}_{\le\bullet}\big) \longrightarrow \big(U(L_\nu), F^{\mathrm{PBW}}_{\le\bullet}\big)$$

and hence maps

$$\mathrm{gr}^{\mathrm{PBW}}_d\, U(L_{\nu+1}) \longrightarrow \mathrm{gr}^{\mathrm{PBW}}_d\, U(L_\nu)$$

for every retained PBW degree d. These associated-graded maps are compatible with the finite comparison matrices A_β, and they compose functorially on triples of windows. Without the rows of Definition 0.225, transition preservation of PBW filtrations is not established.

Proof. The quotient transition matrix sends degree-one primitive generators of $L_{\nu+1}$ to degree-one elements of L_ν. Multiplication gives a word-level map from the tensor algebra on $L_{\nu+1}$ to the tensor algebra on L_ν. The relation-rewrite zero-defect row is exactly the statement that the image of every source enveloping relation is a target enveloping relation; hence the word-level map descends to the universal enveloping quotients.

The filtered-image row says, in the chosen finite PBW bases, that the matrix of the descended map has no component from words of length $\leq d$ to target normal forms of length $> d$. This is precisely $U(T^{Q}_{\nu+1,\nu})(F^{\mathrm{PBW}}_{\leq d}) \subset F^{\mathrm{PBW}}_{\leq d}$. Therefore the standard quotient $F_{\leq d}/F_{\leq d-1}$ produces the displayed associated-graded matrix. The PBW-symbol row identifies this quotient map with the symbol of the degree-one transition, and the A_β-square is the same matrix identity after applying the finite source-to-target comparison. The composition row gives equality of the two matrix products for three windows, so the associated graded maps form an inverse system. No scalar or rank-only datum contains these matrix identities. □

The packet

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certificates/hall/transition_pbw_filtration_preservation
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records the current state of this datum. Its verifier returns `TRANSITION_PBW_FILTRATION_PRESERVATION_OBSTRUCTION_VERIFIE` the radical-quotient transition input, source and target PBW filtration rows, associated-graded rank rows, ordered word bases, relation-rewrite systems, quotient transition matrices, multiplicative word transitions, relation-rewrite compatibility rows, filtered-image identities, restricted filtered transition matrices, associated-graded transition matrices, PBW-symbol identities, A_β-PBW transition squares, and PBW-transition composition rows are not yet supplied.

Definition 0.227 *(Parity-transition datum).* Let $V_{\nu,\gamma}$ and $V_{\nu+1,\gamma}$ be the retained finite primitive source spaces, after the finite radical quotient if recognition is being tested after (R4). A parity-transition datum for

$$T_{\nu+1,\nu,\gamma} : V_{\nu+1,\gamma} \longrightarrow V_{\nu,\gamma}$$

consists of parity-homogeneous bases

$$V_{\nu,\gamma} = V_{\nu,\gamma,\bar{0}} \oplus V_{\nu,\gamma,\bar{1}}, \qquad V_{\nu+1,\gamma} = V_{\nu+1,\gamma,\bar{0}} \oplus V_{\nu+1,\gamma,\bar{1}},$$

the finite fermion-parity involutions

$$J_{\nu,\gamma} = +1 \text{ on } V_{\nu,\gamma,\bar{0}}, \qquad J_{\nu,\gamma} = -1 \text{ on } V_{\nu,\gamma,\bar{1}},$$

and the transition matrix of $T_{\nu+1,\nu,\gamma}$, with the following zero-defect rows:

$$J_{\nu,\gamma} T_{\nu+1,\nu,\gamma} = T_{\nu+1,\nu,\gamma} J_{\nu+1,\gamma},$$

equivalently

$$T_{\overline{0}\overline{1}} = 0, \qquad T_{\overline{1}\overline{0}} = 0.$$

The datum also includes the restricted matrices

$$T_{\overline{p}} : V_{\nu+1,\gamma,\overline{p}} \longrightarrow V_{\nu,\gamma,\overline{p}}, \qquad p = 0, 1,$$

the same rows for $-\gamma$ transported by the Borcherds–Kac Chevalley anti-involution, the comparison square with the finite A_β-matrices on each parity piece, and the composition law for three consecutive windows. Signed superdimension, target parity tables, parity ranks without transition matrices, primitive/radical/PBW transition rows, and the formal parity-pushforward packet are not such a datum.

PROPOSITION 0.228 *(Transition preservation of parity decompositions; proved).* A parity-transition datum in the sense of Definition 0.227 makes each finite transition $T_{\nu+1,\nu,\gamma}$ an even linear map. Hence

$$T_{\nu+1,\nu,\gamma}\big(V_{\nu+1,\gamma,\overline{p}}\big) \subset V_{\nu,\gamma,\overline{p}}, \qquad p = 0, 1,$$

and the even and odd retained root-space towers are defined separately. The same conclusion holds for the negative-root spaces and is compatible with the finite comparison maps A_β. Without the rows of Definition 0.227, transition preservation of parity decompositions is not established.

Proof. The commutator identity $J_{\nu,\gamma} T_{\nu+1,\nu,\gamma} = T_{\nu+1,\nu,\gamma} J_{\nu+1,\gamma}$ is the coordinate-free statement that the transition is even. If $x \in V_{\nu+1,\gamma,\overline{p}}$, then

$$J_{\nu,\gamma} T_{\nu+1,\nu,\gamma} x = T_{\nu+1,\nu,\gamma} J_{\nu+1,\gamma} x = (-1)^p T_{\nu+1,\nu,\gamma} x.$$

Thus $T_{\nu+1,\nu,\gamma} x$ lies in the $(-1)^p$-eigenspace of $J_{\nu,\gamma}$, namely $V_{\nu,\gamma,\overline{p}}$. In parity-homogeneous bases this is equivalent to the two off-diagonal block identities $T_{\overline{01}} = T_{\overline{10}} = 0$. The restricted matrices are the diagonal blocks. The Chevalley anti-involution and the A_β-comparison square are additional finite matrix identities in the datum, so the same parity preservation holds on negative roots and after comparison with the target presentation. The composition row gives functoriality of the even and odd restricted maps. □

The packet

`certificates/hall/transition_parity_decomposition_preservation`

records the current state of this datum. Its verifier returns `TRANSITION_PARITY_DECOMPOSITION_PRESERVATION_OBSTRUCTION_V` the primitive-transition input, source and target-stage parity blocks, parity-homogeneous bases, source and target fermion-parity involutions, ambient transition matrices, parity-commutator identities, off-diagonal zero identities, even and odd restricted transition matrices, negative-root parity transport rows, comparison parity squares, and parity-transition composition rows are not yet supplied.

Definition 0.229 *(Primitive-space Mittag–Leffler datum).* Fix a cofinal sequence $W_1 \subset W_2 \subset \cdots$ as in Definition 0.217. For each retained Gram degree γ with $\alpha(\gamma) \in W_\nu \cup (-W_\nu)$, and for each parity $\overline{p}$, let

$$P_{\nu,\gamma,\overline{p}} := P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}}\big|_{\alpha^{-1}(W_\nu \cup (-W_\nu))}$$

denote the finite primitive Hall space after normal-ordered descent and before quotienting by the Hopf radical. When $W_\nu \subset W_{\nu'}$, transition preservation of primitive subspaces gives a restricted map

$$t_{\nu'\nu,\gamma,\overline{p}} : P_{\nu',\gamma,\overline{p}} \longrightarrow P_{\nu,\gamma,\overline{p}}.$$

A primitive-space Mittag–Leffler datum consists of:

(i) primitive-space rows for every $(\nu,\gamma,\overline{p})$, including a finite basis, parity, sign, target root, and finite rank;
(ii) restricted primitive transition matrices $t_{\nu'\nu}$, together with identity and composition rows;
(iii) coverage rows for every retained positive and negative root degree, for both parities, on the cofinal window sequence;
(iv) for every base window W_{ν_0}, every retained $(\gamma,\overline{p})$, and every later tower containing that degree, an image-stabilization witness: there is $\nu_1 \geq \nu_0$ such that for all $\nu \geq \nu_1$,

$$\operatorname{im}\big(P_{\nu,\gamma,\overline{p}} \to P_{\nu_0,\gamma,\overline{p}}\big) = \operatorname{im}\big(P_{\nu_1,\gamma,\overline{p}} \to P_{\nu_0,\gamma,\overline{p}}\big),$$

with the stabilized image written in the chosen basis of $P_{\nu_0,\gamma,\overline{p}}$;
(v) the defect rows

$$\text{strict} = 1, \qquad \text{ML} = 1, \qquad \operatorname{rank} R^1 \varprojlim_{\nu} P_{\nu,\gamma,\overline{p}} = 0$$

for every retained primitive-space tower, and compatibility rows for the A_β-comparison maps.

Transition preservation of primitive subspaces, finite rank, equality of signed multiplicities, target-window containment, radical-transition data, PBW-transition data, and denominator-product identities are not a primitive-space Mittag–Leffler datum.

Proposition 0.230 *(Vanishing of $R^1\varprojlim$ on primitive spaces; proved).* For every tower $P_{\nu,\gamma,\overline{p}}$ carrying the datum of Definition 0.229, the inverse system of primitive spaces is Mittag–Leffler. Consequently

$$R^1 \varprojlim_{\nu} P_{\nu,\gamma,\overline{p}} = 0$$

for every retained Gram degree, target root, sign, and parity. Without the rows of Definition 0.229, the passage from finite primitive Hall spaces to the completed primitive source is not exact.

Proof. The primitive-space rows and restricted transition matrices define an inverse system of finite-rank vector spaces once the identity and composition rows have zero defect. The image-stabilization row is precisely the Mittag–Leffler condition: for each fixed base window, the images of all sufficiently large windows inside the base primitive space are constant. The standard derived inverse-limit criterion for inverse systems of vector spaces therefore gives $R^1\varprojlim = 0$ on each primitive-space tower. The conclusion is degreewise and paritywise. It cannot be inferred from pointwise primitive kernels, dimension counts, signed Borcherds multiplicities, or transition preservation without the stabilized-image rows. □

The packet

`certificates/hall/primitive_space_lim1_vanishing`

records the current state of this datum. Its verifier returns `PRIMITIVE_SPACE_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal root-window sequence, primitive-space rows, primitive bases, finite-rank rows, parity and sign coverage rows, restricted transition matrices, transition-functoriality rows, image subspace rows, image-stabilization witnesses, strict Mittag–Leffler rows, zero $R^1\varprojlim$ rows, degreewise coverage rows, and comparison-tower compatibility rows are not yet supplied.

Definition 0.231 *(Pairing-kernel Mittag–Leffler datum).* Assume the primitive-space rows of Definition 0.229 have been supplied. For each retained $(\nu,\gamma,\overline{p})$, a finite Hopf-pairing row is a matrix for

$$\lambda_{\nu,\gamma,\overline{p}} : P_{\nu,\gamma,\overline{p}} \longrightarrow P^{\vee}_{\nu,-\gamma,\overline{p}}, \qquad x \longmapsto \langle x, - \rangle_{\nu},$$

together with its opposite transpose

$$\rho_{\nu,\gamma,\overline{p}} : P_{\nu,-\gamma,\overline{p}} \longrightarrow P^{\vee}_{\nu,\gamma,\overline{p}}, \qquad y \longmapsto \langle -, y\rangle_\nu .$$

Put

$$K^{\ell}_{\nu,\gamma,\overline{p}} := \ker \lambda_{\nu,\gamma,\overline{p}}, \qquad K^{r}_{\nu,\gamma,\overline{p}} := \ker \rho_{\nu,\gamma,\overline{p}}.$$

The Hopf radical is formed later from these pairing kernels and the radical-intersection/quotient data; a pairing-kernel datum by itself is not a radical quotient.

A pairing-kernel Mittag–Leffler datum consists of:

(i) finite Hopf-pairing matrices $\lambda_{\nu,\gamma,\overline{p}}$ and $\rho_{\nu,\gamma,\overline{p}}$, with Hopf adjointness, Frobenius-cyclicity, and quotient-nondegeneracy identity rows marking that these are the pairings used in recognition;
(ii) finite bases and ranks for every left kernel $K^{\ell}_{\nu,\gamma,\overline{p}}$ and right kernel $K^{r}_{\nu,\gamma,\overline{p}}$;
(iii) transition maps between the left kernels and between the right kernels, induced by the primitive transition matrices and the pairing-transition identity, with identity and composition rows;
(iv) coverage rows for every retained positive and negative root degree, both parities, and both kernel sides;
(v) for every base window W_{ν_0}, retained $(\gamma, \overline{p})$, and side $s \in \{\ell, r\}$, an image-stabilization witness: there is $\nu_1 \geq \nu_0$ such that for all $\nu \geq \nu_1$,

$$\mathrm{im}\big(K^{s}_{\nu,\gamma,\overline{p}} \to K^{s}_{\nu_0,\gamma,\overline{p}}\big) = \mathrm{im}\big(K^{s}_{\nu_1,\gamma,\overline{p}} \to K^{s}_{\nu_0,\gamma,\overline{p}}\big),$$

with the stabilized image written in a finite kernel basis;
(vi) the defect rows

$$\mathrm{strict} = 1, \qquad \mathrm{ML} = 1, \qquad \mathrm{rank}\, R^1 \varprojlim_{\nu} K^{s}_{\nu,\gamma,\overline{p}} = 0$$

for every retained pairing-kernel tower, and compatibility rows with the A_β-comparison to the target invariant-pairing kernels.

Hopf-pairing matrices, rank equalities, quotient non-degeneracy, primitive-space Mittag–Leffler exactness, radical-transition preservation, target invariant kernels, and denominator identities are not a pairing-kernel Mittag–Leffler datum.

PROPOSITION 0.232 *(Vanishing of $R^1 \varprojlim$ on pairing kernels; proved).* For every tower $K^{s}_{\nu,\gamma,\overline{p}}$, $s \in \{\ell, r\}$, carrying the datum of Definition 0.231, the inverse system of pairing kernels is Mittag–Leffler. Consequently

$$R^1 \varprojlim_{\nu} K^{s}_{\nu,\gamma,\overline{p}} = 0$$

for every retained Gram degree, target root, sign, parity, and kernel side. Without the rows of Definition 0.231, kernel formation for the Hopf pairing need not commute with passage from finite windows to the completed primitive source.

Proof. The pairing-kernel rows and restricted transition matrices define an inverse system of finite-rank vector spaces once identity and composition defects vanish. The image-stabilization row is exactly the Mittag–Leffler condition for this kernel system. The standard derived inverse-limit criterion for inverse systems of vector spaces gives $R^1 \varprojlim = 0$ for each left and right pairing-kernel tower. This is a statement about the kernels as subspaces with their restricted transition maps. It does not follow from the existence of a Hopf pairing matrix, from rank equality, from non-degeneracy after quotient, or from Mittag–Leffler exactness of the ambient primitive spaces without strict kernel-transition and stabilized-image rows. □

The packet

`certificates/hall/pairing_kernel_lim1_vanishing`

records the current state of this datum. Its verifier returns `PAIRING_KERNEL_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal root-window sequence, primitive-space input, Hopf-pairing matrix rows, Hopf-pairing identity rows, left and right kernel rows, kernel bases, finite-rank rows, pairing-transition input, restricted kernel transition matrices, transition-functoriality rows, image subspace rows, image-stabilization witnesses, strict Mittag–Leffler rows, zero $R^1 \varprojlim$ rows, degreewise coverage rows, and comparison-kernel compatibility rows are not yet supplied.

Definition 0.233 *(PBW associated-graded Mittag–Leffler datum).* Assume the radical quotient transition data and the PBW-transition datum of Definition 0.225 have been supplied. Let

$$L_\nu := P_X^{\Pi,\mathrm{red}}|_{\alpha^{-1}(S_{W_\nu})}/\mathrm{Rad}^{\Pi}_{X,\nu}$$

be the finite primitive radical quotient on the window W_ν. For each PBW degree d, retained Gram degree χ, and parity $\overline{p}$, put

$$G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}} := \left(\mathrm{gr}^{\mathrm{PBW}}_d U(L_\nu)\right)_{\chi,\overline{p}}.$$

The associated-graded transition maps of Proposition 0.226 give maps

$$g_{\nu'\nu,d,\chi,\overline{p}} : G^{\mathrm{PBW}}_{\nu',d,\chi,\overline{p}} \longrightarrow G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}} \qquad (\nu \le \nu').$$

A PBW associated-graded Mittag–Leffler datum consists of:

(i) finite PBW filtration rows $F^{\mathrm{PBW}}_{\le d} U(L_\nu)$, lower filtration rows $F^{\mathrm{PBW}}_{\le d-1} U(L_\nu)$, and quotient rows for every $G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}}$;

(ii) ordered word bases, relation-rewrite rows, associated-graded bases, and associated-graded rank rows;

(iii) associated-graded transition matrices $g_{\nu'\nu}$, together with PBW-symbol intertwining, A_β-PBW comparison squares, identity rows, and composition rows;

(iv) coverage rows for every retained PBW degree, Gram degree, parity, side, and cofinal window;

(v) for every base window W_{ν_0}, every retained $(d, \chi, \overline{p})$, and every side, an image-stabilization witness: there is $\nu_1 \ge \nu_0$ such that for all $\nu \ge \nu_1$,

$$\mathrm{im}\big(G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}} \to G^{\mathrm{PBW}}_{\nu_0,d,\chi,\overline{p}}\big) = \mathrm{im}\big(G^{\mathrm{PBW}}_{\nu_1,d,\chi,\overline{p}} \to G^{\mathrm{PBW}}_{\nu_0,d,\chi,\overline{p}}\big),$$

with the stabilized image written in a finite associated-graded basis;

(vi) the defect rows

$$\mathrm{strict} = 1, \qquad \mathrm{ML} = 1, \qquad \mathrm{rank}\, R^1 \varprojlim_{\nu} G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}} = 0$$

for every retained PBW associated-graded tower.

PBW filteredness, associated-graded rank equality, target PBW rows, Hilbert series, formal PBW push-forward, denominator identities, and primitive or pairing-kernel Mittag–Leffler exactness are not a PBW associated-graded Mittag–Leffler datum.

PROPOSITION 0.234 *(Vanishing of $R^1 \varprojlim$ on PBW associated gradeds; proved).* For every tower $G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}}$ carrying the datum of Definition 0.233, the inverse system of PBW associated-graded pieces is Mittag–Leffler. Consequently

$$R^1 \varprojlim_{\nu} G^{\mathrm{PBW}}_{\nu,d,\chi,\overline{p}} = 0$$

for every retained PBW degree, Gram degree, parity, and side. Without the rows of Definition 0.233, PBW associated gradeds need not commute with passage from finite windows to the completed enveloping algebra.

Proof. The associated-graded piece rows and associated-graded transition matrices define an inverse system of finite-rank vector spaces once the identity and composition rows have zero defect. The image-stabilization row is precisely the Mittag–Leffler condition on that inverse system. The standard derived inverse-limit criterion for inverse systems of vector spaces gives $R^1 \varprojlim = 0$ on every retained PBW associated-graded tower. Filteredness alone gives maps of filtered pieces; it does not imply that the images on the quotients $F_{\le d}/F_{\le d-1}$ stabilize. Rank equality and Hilbert-series agreement likewise do not compute the derived inverse limit. □

The packet

```
certificates/hall/pbw_associated_graded_lim1_vanishing
```

records the current state of this datum. Its verifier returns `PBW_ASSOCIATED_GRADED_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal root-window sequence, radical-quotient input, PBW-transition input, PBW filtration rows, ordered word bases, relation-rewrite rows, associated-graded piece rows, graded bases, associated-graded rank rows, associated-graded transition maps, PBW-symbol intertwining rows, A_β-PBW squares, transition-functoriality rows, image subspace rows, image-stabilization witnesses, strict Mittag–Leffler rows, zero $R^1 \varprojlim$ rows, degreewise coverage rows, and comparison-graded compatibility rows are not yet supplied.

PROPOSITION 0.235 *(Strict Mittag–Leffler completion for primitive recognition; proved).* Let $\{W_\nu\}_{\nu \ge 1}$ be a cofinal increasing sequence of downward-saturated finite root windows. For each ν, let

$$A_\nu : P_X^{\Pi,\mathrm{red}}|_{\alpha^{-1}(S_{W_\nu})} \xrightarrow{\sim} \mathfrak{g}_{\Delta_5}|_{S_{W_\nu}}$$

be the finite-window isomorphism of Proposition 0.224. Suppose the transition maps for $W_{\nu+1} \to W_\nu$ satisfy the following strict Mittag–Leffler conditions:

(i) the source primitive-space towers of Definition 0.229 and the target root-space parity towers are Mittag–Leffler, and A_ν commutes with the transition maps on each parity piece;

(ii) the pairing-kernel towers of Definition 0.231, the kernel towers of the finite presentation maps, the Hopf radical intersection towers, and the relation-radical quotient towers are strict subsystems: transition maps carry each subspace onto the corresponding image and induce the quotient transition maps;

(iii) the PBW filtrations are strict under transition, and the induced towers on the associated graded pieces carry the Mittag–Leffler datum of Definition 0.233;

(iv) the triangular finite-window towers for positive-negative cancellation have $R^1 \varprojlim = 0$.

Then inverse limit is exact on the finite recognition diagram. In particular,

$$\varprojlim_{\nu} \big(P_X^{\Pi,\mathrm{red}}|_{\alpha^{-1}(S_{W_\nu})} / \mathrm{Rad}^{\Pi}_{X,\nu} \big) \cong \big(\varprojlim_{\nu} P_X^{\Pi,\mathrm{red}}|_{\alpha^{-1}(S_{W_\nu})} \big) / \overline{\varprojlim_{\nu} \mathrm{Rad}^{\Pi}_{X,\nu}},$$

the completed kernel is the closure of the finite kernels, the completed PBW associated graded is the inverse limit of the finite PBW associated gradeds, and the maps A_ν assemble to an isomorphism of completed Gram-graded Lie superalgebras

$$\varprojlim_{\nu} P_X^{\Pi,\mathrm{red}}|_{\alpha^{-1}(S_{W_\nu})} \xrightarrow{\sim} \varprojlim_{\nu} \mathfrak{g}_{\Delta_5}|_{S_{W_\nu}}.$$

Without these strict Mittag–Leffler conditions, finite-window isomorphisms do not by themselves identify the completed primitive source with the completed BKM target.

Proof. For inverse systems of vector spaces, the derived exact sequence for $\varprojlim$ shows that exactness of

$$0 \to K_\nu \to V_\nu \to Q_\nu \to 0$$

passes to inverse limits exactly when $R^1 \varprojlim K_\nu = 0$, or more generally when the tower is Mittag–Leffler. Apply this to the presentation-kernel tower, the Hopf-radical tower, the quotient tower, and each even/odd root-space tower. Strictness says that taking subobjects, quotients, and finite PBW filtered pieces commutes with the transition maps; hence the inverse-limit quotient is the quotient by the closure of the finite radicals, and the inverse-limit kernel is the closure of the finite kernels. The strict filtered statement gives

$$\mathrm{gr}_{\mathrm{PBW}} \varprojlim_{\nu} U(P_\nu) \cong \varprojlim_{\nu} \mathrm{gr}_{\mathrm{PBW}}\, U(P_\nu)$$

on every retained degree.

The maps A_ν commute with the transition maps, so they form an isomorphism of inverse systems. Since inverse limit is a functor and the kernel and cokernel towers of A_ν are zero, the inverse-limit map has zero kernel and cokernel. If the Mittag–Leffler hypotheses are removed, the $R^1 \varprojlim$ term can contribute a hidden kernel, cokernel, radical element, PBW class, or parity-cancelling summand which is invisible on each fixed finite window. This is exactly the completion failure prohibited by (R8). ☐

Definition 0.236 *(Finite source proof tables).* After choosing homogeneous bases on a finite window W_ν, the finite-window primitive-recognition datum of Definition 0.217 is equivalent to the following source proof tables:

degrees, parity_blocks, basis_provenance, simple_representatives, M, D, unit_counit, B, G, hopf_pairing_identities, K
hall_bialgebra_identities, radical_ideal_coideal, relation_rows, no_extra, generation, pbw, transitions, A_β_comparison_

When the same packet is used to assert the current-envelope statement $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$, it must also contain the finite Koszul comparison tables

koszul_cones, koszul_comparison_identities, koszul_transition_ml of Definition 0.211. They are not implied by the A_β-comparison maps: A_β records the primitive radical-quotient comparison, whereas the Koszul tables record the acyclic cones, compatibility homotopies, and strict Mittag–Leffler systems of the chiral cobar comparison.

Their mathematical meanings are fixed as follows. The degree table records the Gram labels γ, their signs, window membership, and compact source-admissibility. The provenance table records the retained charge lift, source stratum, vanishing-cycle or intersection-complex summand, reduced orientation, quotient orientation, Thom–Sebastiani transport, finite-stabilizer linearization, protected integration, and transition compatibility of every basis vector. The parity table records the two ranks in each parity, not only their difference. The simple-representative table names the source basis vectors realizing e_i^X, f_i^X, h_i^X, together with their parity and target reference. The matrices M, D, B record, respectively, the source Hall product, source coproduct, and supercommutator bracket on the chosen source bases. The unit_counit table records the unit and counit maps. The Hall bialgebra identity table records the vanishing of the finite defects for associativity, coassociativity, left and right unit, left and right counit, product-coproduct compatibility, and primitive closure of the supercommutator. The matrix G records the Hopf pairing. The Hopf-pairing identity table records the finite defects for Hopf adjointness $\langle m(x,y), z\rangle = \langle x \otimes y, \Delta z\rangle$, the Frobenius cyclic identity for the primitive bracket, and non-degeneracy of the induced pairing on the radical quotient. The matrices K, Q record the radical kernel and a quotient splitting. The matrix A records the comparison from the source radical quotient to the imported target basis; target labels are allowed only in this codomain.

Each row is *geometrically generated*. It carries a source identifier for the compact correspondence, reduced stratum, vanishing- cycle or intersection-complex summand, orientation transport, Hopf-pairing construction, radical computation, quotient splitting, comparison map, or transition morphism from which the row is obtained, together with a proof reference for the relevant theorem, equation, finite computation, or source certificate. A row whose matrices satisfy the displayed rank identities but are not obtained from these compact-source data is not a row of the finite source proof tables. In particular, arbitrary matrices with the correct dimensions, target-only rows, signed-only rows, and status-only rows do not define the compact source primitive-recognition datum.

The remaining tables record identities. The radical_ideal_coideal rows state that K is stable as a Lie ideal and as a coproduct coideal. The relation_rows table records the Cartan, Chevalley, real Serre, Borcherds orthogonality, and super sign relations. The no_extra table records the finite rank identity $\ker \pi_\nu = (J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}})_{\le W_\nu}$. The generation table records, for each source basis vector in the radical quotient, a bracket-word span by the Cartan and simple primitive representatives with all intermediate degrees remaining in W_ν. The pbw table records the associated-graded PBW rank comparison. The transitions table records strictness and Mittag–Leffler exactness under $W_{\nu+1} \to W_\nu$. The A_β-comparison table records that the comparison matrix A intertwines the source bracket, coproduct, pairing, radical quotient, and PBW filtration with the target blocks. Admissibility of these finite tables is numerical: every defect-rank row above must have rank 0, and every rank-comparison row must have equal ranks on the two sides and on their combined span. A non-zero defect rank or a failed rank equality is a failed finite-window recognition datum. The identity rows are also exhaustive. In the finite source proof tables the canonical labels are

`unit_left`, `unit_right`, `counit_left`, `counit_right`, `associativity`, `coassociativity`, `bialgebra_compatibility`, `prim`

`hopf_adjointness`, `frobenius_cyclic`, `quotient_nondegenerate`, `lie_ideal`, `coproduct_coideal`,

`cartan`, `chevalley`, `real_serre`, `borcherds_orthogonality`, `super_sign`, `bracket`, `coproduct`, `pairing`, `radical_qu`

Omitting one family is a failed finite-window recognition datum. If the current-envelope comparison is also asserted, the Koszul table families listed above are part of the same finite packet; omitting one of them is a failed finite Koszul comparison datum, even if (R0)–(R8) hold algebraically.

For the compact $K3 \times E$ Hall source packet used in this manuscript, the companion obstruction ledger

`certificates/sources/k3e_compact_hall/blocked_obligations.csv`

records precisely the missing source artifacts. Its verifier returns `COMPACT_HALL_OBSTRUCTION_LEDGER_VERIFIED`, while still recording `compact_source_recognition=false` and `mathematical_certification=false`. The ledger is therefore not evidence for primitive recognition. It is the finite list of objects which must be supplied before $P_X^{\Pi,\mathrm{red}}$ exists as a constructed compact-source primitive Hall Lie superalgebra: the coproduct D, unit and counit, bialgebra compatibility, primitive closure, Hopf pairing, radical ideal/coideal, no-extra kernel equality, PBW comparison, strict Mittag–Leffler transition, the A_β-intertwining rows, and the finite Koszul comparison rows.

Proposition 0.237 *(The finite source proof tables are exactly the finite recognition datum; proved).* For a fixed downward-saturated finite window W_ν, a populated system of finite source proof tables in the sense of Definition 0.236, with all stated rank and intertwining identities verified over the chosen coefficient ring and with every row geometrically generated from the compact-source correspondences and orientation/Hopf data named there, is equivalent to clauses (R0)–(R8) of Definition 0.217 after a choice of homogeneous bases. In particular, a header-only table system, or a table system whose rows contain status text but no mathematical payload, or a table system of arbitrary matrices without compact-source provenance, does not define a finite-window primitive-recognition datum.

Proof. Choosing homogeneous bases turns each finite-dimensional source primitive space, pairing block, radical, quotient, and comparison map into matrices, but only after the corresponding finite Hall correspondence, reduced orientation, Hopf pairing, radical quotient, and transition maps have been constructed. The row-level geometric source and proof reference record this construction before coordinates are chosen. The degree and provenance tables are precisely (R0). The simple-representative table is (R1). The parity table is (R3). The representative rows together with the relation table are (R1)–(R2). The M, D, η, ϵ, B matrices and the Hall bialgebra identity table are the product-coproduct and primitive-closure part of (R4). The G matrix and the Hopf-pairing identity table are the pairing adjointness, Frobenius, and quotient non-degeneracy part of (R4). The K, Q matrices and the radical_ideal_coideal identities are the radical part of (R4). The no_extra rank identity is (R5). The generation span table is (R6). The PBW table is (R7), and the transition table is (R8). The A_β-comparison rows assert that the source quotient is compared to the same target blocks in every operation, so the matrix statements are independent of the chosen coordinates. Conversely, any finite-window datum satisfying (R0)–(R8) yields these tables after choosing bases and writing the finite maps and identities in matrix form, with the geometric source and proof reference inherited from the finite maps themselves. Empty rows have no source vectors, maps, ranks, or identities; arbitrary matrices with no compact-source origin have no Hall correspondence, orientation datum, pairing construction, or transition map. They satisfy none of the clauses. □

Proposition 0.238 *(Global recognition is exactly the cofinal datum; proved).* Assume (S1)–(S4) of Theorem 0.215. Then the global source-side assertions (S5)–(S10) are equivalent to a cofinal finite-window primitive-recognition datum $\{W_\nu, \text{(R0)–(R8)}\}$ in the sense of Definition 0.217, and conversely.

Proof. The two sides record the same data: each W_ν is a saturated finite window in (S5)–(S9), and (R3) is exactly the two-integer parity equality of (S10) on W_ν. The Hopf pairing matrices, kernel equalities, PBW filtrations, no-extra-relations relations, and transition strictness are exactly the cofinal contents of (S5)–(S9), and the cusp-positive arithmetic side of $f(nm, l)$ is recovered from the parity equalities (R3) on the active target degrees. Conversely, given (R0)–(R8) on a cofinal system, the strict Mittag–Leffler completion theorem Proposition 0.235 gives the global statements (S5)–(S9), and (R3) on a cofinal exhaustion gives (S10). □

Recognition criterion at the window $W_{\leq 3}$

For a finite window W with $S_W = W \cup (-W) \cup \{0\}$, a compact source datum consists of compact-source bases $b^X_{R,\gamma_\beta,\overline{p},a}$, parity pieces, product, coproduct, bracket, pairing, radical, and quotient matrices M, D, B, G, K, Q, comparison maps $A_{\beta,\overline{p}}$, relation identities, the no-extra kernel equality, PBW associated gradeds, and strict Mittag–Leffler transition maps. The target labels e_i, E_{ij}, $u_{ij,r}$, T_i, $M_{ij,r}$, w_s occur only in target blocks or as codomain coordinates of $A_{\beta,\overline{p}}$.

The window $W_{\leq 3}$ is the first arithmetic window, not a relation-closed degree window. A relation-closed degree test must include real-Serre terminal rows, doubled isotropic rows, δ_{123}-real strings, odd–odd δ_{123} rows, and complementary height-seven strings, or pass to the downward-saturated closure. In the target-reference packet `wle3_target_parity`, $W_{\leq 3}$ has the verified rows $\delta_i : 1|0, a_{ij} : 10|0, 2\delta_i + \delta_j : 1|0$, and $\delta_{123} : 29|93$; its scalar firewall excludes source parity, Hall brackets, pairing radicals, PBW comparison, and primitive recognition. The finite packet `parity_pushforward` verifies a different formal statement: once finite source parity blocks are supplied, the additive normal-ordered Gram pushforward preserves the even and odd blocks separately. It supplies neither the source parity dimensions required in (R3) nor the Gritsenko–Nikulin/Kac target parity equality; those remain primitive-recognition data. The finite packet `pbw_pushforward` is likewise formal: once a finite PBW filtration is supplied, additive normal-ordered Gram pushforward preserves its filtered pieces and associated graded. It does not supply the compact-source PBW comparison required in (R7), the target PBW rows, or the no-extra-relations theorem. The finite packet `triangular_pushforward` is the analogous formal statement for a supplied positive/Cartan/negative splitting: the normal-ordered pushforward preserves the three direct-sum parts and the displayed bracket signs. It does not supply the exact triangular completion required in (R8), target triangular completion, or the Mittag–Leffler exactness of triangular finite-window systems. The finite target degree-closure packet `wrel_degree_closure` verifies that the enlarged target degree set has exactly the 35 rows

$$W^{act}_{\leq 3} \cup R_4 \cup I_4 \cup C_{\leq 7} \cup \{2\delta_{123}\},$$

with R_4 and $C_{k,5}$ terminal zero rows. Its downward-saturation audit finds exactly three nonzero proper subdegree defects, namely

$$D_k = \delta_k + 2a_{ij}, \qquad \{i, j, k\} = \{1, 2, 3\},$$

below $2\delta_{123}$, each with signed multiplicity -513. This is still only the target degree closure. In the larger target degree window, Remark 6.9 supplies full parity only for $2a_{ij}$, $C_{k,3}$, $C_{k,4}$, and $C_{k,5}$. The rows $C_{k,2}$, D_k, and $2\delta_{123}$ remain signed target rows; they cannot serve as source comparison codomains until the finite target presentation is computed in those degrees. The base-string terminal degree set is therefore not yet a primitive-recognition window in the sense of (R0)–(R8). The A071 target-presentation verifier records this boundary: it checks the $2a_{ij}$, $C_{k,3}$, $C_{k,4}$, and $C_{k,5}$ parity rows, and rejects $C_{k,2}$, D_k, and $2\delta_{123}$ as target basis, pairing, radical, or PBW rows. The failure is structural, not a missing scalar coefficient.

The minimal nonzero downward-saturation extension $W^+_{\mathrm{rel}} \cup \{D_1, D_2, D_3\}$ is now target-recorded; a primitive-recognition window must still supply the missing target-presentation and compact-source data. The post-saturation real-string audit records 21 terminal codomains outside the saturated extension, 10 with nonzero signed multiplicity. The terminal-layer audit groups these codomains into 21 distinct terminal degrees; adjoining them creates 114 nonzero subdegree-defect incidences across 26 distinct subdegrees. The next target-degree computation adjoins exactly these 26 rows and gives an 85-row degree set with zero remaining nonzero proper-subdegree defect. This closes downward saturation for the displayed terminal layer, but full finite relation closure remains a further target closure problem: if these 26 rows are promoted to target-generator rows, the real-root strings force 55 terminal checks, 54 terminal codomains outside the 85-row set, and 12 missing codomains with nonzero signed multiplicity. Adjoining those 55 terminal codomains would create 546 nonzero proper-subdegree defect incidences across 98 distinct subdegrees. Adjoining those 98 rows gives a 237-row target degree set with zero remaining nonzero proper-subdegree defect for that layer. If this 98-row layer is promoted to target-generator rows, the real-root strings force 206 terminal checks, 182 new terminal degrees, and 24 missing nonzero terminal degrees. The exact terminal-layer table records all 206 codomains, and the exact saturation-extension table records the 50 distinct nonzero proper subdegrees created by adjoining them, with 624 incidences. Adjoining both layers gives a 469-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the newly adjoined 50-row layer forces 96 real-string terminal checks, 90 new terminal degrees, no missing nonzero terminal degree, and a 12-row nonzero saturation extension with 44 incidences. Adjoining these rows gives a 571-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the newly adjoined 12-row layer forces 20 real-string terminal checks, all with zero signed terminal multiplicity, and a 6-row nonzero saturation extension with 8 incidences. Adjoining these rows gives a 597-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the newly adjoined 6-row layer forces 10 real-string terminal checks, all with zero signed terminal multiplicity, and a 6-row nonzero saturation extension with 18 incidences. Adjoining these rows gives a 613-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the next newly adjoined 6-row layer forces 10 real-string terminal checks, all with zero signed terminal multiplicity, and a 6-row nonzero saturation extension with 8 incidences. Adjoining these rows gives a 629-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the next newly adjoined 6-row layer forces 10 real-string terminal checks, all with zero signed terminal multiplicity, and a 6-row nonzero saturation extension with 18 incidences. Adjoining these rows gives a 645-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. The tail is not a finite accident. For every even $N \geq 14$, put

$$
\begin{aligned}
A_N &= \{(0, N, N-1), (0, N, N), (0, N+1, N), (N, 0, N-1), (N, 0, N), (N+1, 0, N)\},\\
B_N &= \{(0, N, N+1), (0, N+1, N+1), (0, N+1, N+2), (N, 0, N+1), (N+1, 0, N+1), (N+1, 0, N+2)\}.
\end{aligned}
$$

The verifier `wrel_tail_staircase` proves symbolically, using the Lorentzian pairing and the q^0-term $y^{-1} + 10 + y$ of $\phi_{0,1}$, that the real-root terminal audit and downward saturation force

$$A_N \Longrightarrow B_N, \qquad B_N \Longrightarrow A_{N+2}.$$

Its finite-anchor table checks the six rows of A_{14} inside the `wrel_degree_closure` packet. Hence the target-degree audit contains an infinite six-row staircase. Its `unbounded_tail.csv` table records the subsequences A_{14+2k} and B_{14+2k} with a strictly increasing coordinate. The source comparison can therefore be formulated only through finite presentation data, not by closing a finite target-degree enumeration. To promote this target audit to the first relation-closed compact-source theorem, one must populate the first-window packet

window_closure, target_source_representatives, chevalley_serre_matrices, hall_boundary_complex, spectral_seque

The theorem rows include the first-window instances of $O1$, $O1^+$, $O2$, P_{fin}, Hall product, Hall coproduct, Hopf pairing, Koszul comparison, PBW comparison, primitive recognition, and Pfaffian equality. A target relation-closed degree set without these compact-source rows is not a source theorem. At present the packet `certificates/first_window/k3e_relation_closed_window` verifies only the obstruction ledger `FIRST_WINDOW_OBSTRUCTION_LEDGER_VERIFIED`, with `first_window_certification=false`. Thus the first relation-closed source theorem remains open until these finite rows are supplied.

PROPOSITION 0.239 *(Target-admissibility obstruction for the height-seven closure; proved).* Let W be a downward-saturated finite target-root window containing the height-7 closure W_{rel}^+ of Chapter 7. If W is a primitive-recognition window satisfying (R0)–(R8), then the target restriction must contain a finite target-presentation datum. In particular, for every row $C_{k,2}$, $D_k = \delta_k + 2a_{ij}$, and $2\delta_{123}$, it must supply the full even/odd parity dimensions, the positive-negative pairing block, the target radical block, and the PBW associated-graded row. The signed target identities

$$\mathrm{sdim}(C_{k,2}) = 108, \quad m(C_{k,2}) = 90, \qquad \mathrm{sdim}(D_k) = -513, \quad m(D_k) = 54, \qquad \mathrm{sdim}(2\delta_{123}) = 4016, \quad m(2\delta_{123}) = -540$$

do not supply these data. Hence the presently available target data do not make the saturated extension of W_{rel}^+ a primitive-recognition window.

Proof. The rows $C_{k,2}$ and $2\delta_{123}$ lie in W_{rel}^+, and D_k lies in every downward-saturated window containing W_{rel}^+, since

$$0 < D_k = \delta_k + 2a_{ij} < 2\delta_{123}.$$

By Proposition 0.216, they must be target-presentation rows. Clause (R3) requires the two integers $\mathrm{mult}_{\bar{0}}^{\mathrm{GN}}(\beta)$ and $\mathrm{mult}_{\bar{1}}^{\mathrm{GN}}(\beta)$ for every $\beta \in W$, and also in the negative degree by the Borcherds–Kac Chevalley anti-involution. A signed multiplicity gives only their difference. The integer $m(\beta)$ records the Gritsenko–Nikulin imaginary-simple contribution in the denominator presentation; it is not the full quotient root-space parity split and does not determine an invariant positive-negative pairing.

Clauses (R4), (R5), and (R7) then require the target pairing kernel, radical block, and PBW associated-graded comparison in the same degrees. Without those target blocks the source-to-target matrices $A_{\beta,\bar{p}}$ have no defined codomain in these rows, and the equations expressing Hopf radical equality, no-extra-relations, and PBW compatibility are not well-formed. Thus signed data in $C_{k,2}$, D_k, and $2\delta_{123}$ are not admissible target data for (R0)–(R8). □

WHAT SIGNED DIMENSIONS MISS

LEMMA 0.240 *(Signed dimensions are not recognition; proved).* Let $V_\bullet$ be a $\widehat{\Gamma}_X$-graded super vector space with zero bracket and zero coproduct, with the same signed superdimensions $\mathrm{sdim}\, V_\gamma = f(nm, l)$ as $P_X^{\Pi,\mathrm{red}}$. Then $V_\bullet$ has the same Borcherds denominator $64^{-1}\Delta_5(2Z)$ as $\mathfrak{g}_{\Delta_5}$ but is not isomorphic to it as a Lie superalgebra.

Proof. The denominator depends only on signed superdimensions; the bracket data, no-extra-relations theorem, Hopf-radical coideal stability, and PBW filtration are independent. If the Cartan is mapped nontrivially, the Chevalley relation $[e_i, f_i] = h_i$ fails in $V_\bullet$. If the Cartan is killed, the root-degree Cartan identification fails. In either case the kernel of the free presentation map contains all non-Cartan brackets, hence is strictly larger than $\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$ in degrees where the BKM bracket is nonzero. With the zero Hopf pairing the radical is all of $V_\bullet$, so the radical quotient is zero. □

This lemma is the counterexample that prohibits the determinant route to recognition. It is also the structural reason the ten clauses (S1)–(S10) do not collapse to their signed-dimension shadow: the signed dimensions are necessary and not sufficient; (S1)–(S9) supply the source presentation, and (S10) supplies the missing parity split.

Lemma 0.241 *(Finite-window scalar exclusions; proved).* Let W be a finite target-root window. None of the following data implies a finite-window primitive-recognition datum on W in the sense of Definition 0.217: target labels without compact representatives, signed dimensions, the scalar trace $Z_{BPS}^{K3\times E}$, the Pfaffian product, the Borcherds denominator product, target PBW rows without source PBW strictness, arbitrary matrices with the right ranks but no compact provenance, or rows whose only content is a verification status.

Proof. Clauses (R0)–(R8) require source objects before comparison: compact basis vectors, source representatives, Hall product and coproduct, unit and counit, Hopf pairing, radical and quotient matrices, relation rows, no-extra kernel equality, generation spans, PBW strictness, and Mittag–Leffler transition maps. Target labels give only codomain coordinates for $A_{\beta,\bar{p}}$, not the source vectors mapped to them. Signed dimensions and the denominator product give one integer per degree; (R3) requires the two parity dimensions, while (R4)–(R8) require operations and kernels. Lemma 0.240 exhibits a zero-bracket object with the same denominator and no recognition isomorphism. The scalar trace and the Pfaffian product live after taking a protected scalar shadow; they forget the bracket, coproduct, Hopf radical, PBW filtration, and finite transition system. Target PBW rows test only the imported target presentation and do not give source PBW strictness or a source-to-target intertwining matrix. Arbitrary matrices with correct ranks do not supply a Hall correspondence, reduced orientation, compact cycle, pairing construction, or transition morphism, so they fail the geometric provenance part of (R0) and the construction clauses of (R4)–(R8). Status-only rows contain no vector space, map, matrix, kernel, rank, or identity. Hence each listed substitute omits at least one structural component of (R0)–(R8), and none defines a finite-window primitive-recognition datum. □

Obstruction classes

The recognition lane carries the residuals

$$\mathfrak{O}_{\mathrm{Rec},\nu} = \left(\mathfrak{o}_\nu^{(\mathrm{R1})}, \dots, \mathfrak{o}_\nu^{(\mathrm{R8})}, \mathfrak{o}_{\nu+1,\nu}^{\mathrm{tr}}\right),$$

required to vanish on every cofinal window. The first arithmetic window $W_{\leq 3}$ is achievable from the GN target side; the first relation-closed degree window includes the real-Serre terminal rows and the δ_{123} odd–odd rows, but some added target rows are still only signed target rows by Remark 6.9. The finite-stage components of the (R0)–(R8) criterion, including source-radical/Hopf-coideal stability, remain source-recognition data. Appendix 7 supplies the orientation lane; it does not supply the source matrices M, D, B, G, K, Q, the comparison maps $A_{\beta,\bar{p}}$, the kernel equality, the PBW associated-graded comparison, or the finite Koszul comparison residual $\mathfrak{O}_{\mathrm{Kcmp},R}$. Construction of the source-internal recognition criterion at any one relation-closed degree window beyond $W_{\leq 3}$, together with the Koszul compatibility checks of Definition 0.211, is the unresolved compact-source datum of Chapter 10.

Comparison with the denominator algebra

The recognition theorem is the source side of the Vol III Drinfeld double construction. The target presentation uses the generalized Kac–Moody convention of Borcherds [8] and Kac [58]; the automorphic correction algebra is the Gritsenko–Nikulin algebra of [46, Sections 3–4, Proposition 3.1], and the product-lift data are those of [48, Theorem 2.1, §5.1.1] together with Borcherds' automorphic product [9,10]. The identity $\mathrm{CoHA}(\mathbb{C}^3) = W_{1+\infty}$ belongs to the local toric case; the present recognition criterion is at the level of the protected primitive Hall algebra, before Drinfeld doubling; after the compact source datum is supplied, it identifies that algebra with $\mathfrak{g}_{\Delta_5}$, not with the centred or vacuum-evaluated double.

Entries (L1)–(L6) name the constructed objects. The finite HN system in §0.7 supplies the charge windows, Hall correspondences, orientation transports, Pfaffian lines, Koszul source coalgebras, primitive-recognition data, and their local obstruction classes.

0.7 Eight finite constructions

The seventh entry of the Dirac–Igusa pro-object is the finite HN system of charge windows, Hall correspondences, orientation transports, Pfaffian lines, Koszul source coalgebras, and primitive-recognition data. At HN height R these data are finite-type or finite-dimensional, and their obstruction classes vanish exactly when the transition system defines the inverse limit $\mathfrak{D}_X^{\mathrm{DI}}$.

At fixed HN height all objects lie on finite-type stacks and finite-dimensional Borel–Moore chain spaces; the inverse limit is taken after the finite stages have compatible transition maps. For $R \in \mathcal{R}_{\mathrm{HN}}$, the constructions assume: [K3×E-fin] (finite-type semistable substacks at bounded HN height), [K3×E-atlas] (finite d-critical/cosection Hall atlas with Joyce–Upmeier orientation lines), and [ML] (Mittag–Leffler descent).

At finite height the first two hypotheses are the table system

hn_type_bounds, semistable_substacks, derived_enhancements, universal_complexes, scalar_rigidifications, stra

The scalar firewall excludes Liu stability alone, the formal charge window, a target window, a Hilbert-scheme scalar test, or a Pfaffian product as finite-moduli data. The finite-moduli packet `certificates/moduli/k3e_finite_moduli` currently records the HN-bound, semistable-substack, derived-enhancement, and rigidification-inertia rows, together with the finite class-bound and universal-complex rows and the E-translation-rigidification rows, retained closed-substack rows, and retained extension-closure rows, retained HN-factor-closure rows, and retained dual-closure rows. The remaining rows stay in the obstruction ledger `MODULI_OBSTRUCTION_LEDGER_VERIFIED`, with `moduli_certification=false`. It is still not a construction of the full $[K3 \times E\text{-fin}]$ package or $[K3 \times E\text{-atlas}]$.

Definition 0.242 *(Finite-moduli transition geometry datum).* Let $R' \le R$. A finite-moduli transition geometry datum from height R to height R' consists of the following rows.

(i) Transition morphisms

$$t^{\mathrm{obj}}_{RR'} : \mathfrak{M}^{\mathrm{obj}}_R \to \mathfrak{M}^{\mathrm{obj}}_{R'}, \qquad t^{\mathrm{ext}}_{RR'} : \mathfrak{M}^{\mathrm{ext}}_R \to \mathfrak{M}^{\mathrm{ext}}_{R'}, \qquad t^{\mathrm{fl}}_{RR'} : \mathfrak{M}^{\mathrm{fl}}_R \to \mathfrak{M}^{\mathrm{fl}}_{R'}$$

on the retained object, extension, and two-step flag stacks.

(ii) Retained closed substacks

$$\mathfrak{Z}^a_R \subset \mathfrak{M}^a_R, \qquad \mathfrak{Z}^a_{R'} \subset \mathfrak{M}^a_{R'}, \qquad a \in \{\mathrm{obj}, \mathrm{ext}, \mathrm{fl}\},$$

such that $t^a_{RR'}(\mathfrak{Z}^a_R) \subset \mathfrak{Z}^a_{R'}$.

(iii) For each a, the restricted morphism

$$t^{a}_{RR'}|\mathfrak{Z}^{a}_{R} : \mathfrak{Z}^{a}_{R} \to \mathfrak{Z}^{a}_{R'}$$

is either proper or a closed immersion of finite-type stacks. In the closed-immersion case the image, equivalently the graph, is recorded as a closed substack.

(iv) The universal complexes, scalar rigidifications, stabilizer stratifications, cosections, vanishing-cycle domains, and orientation domains pull back to the corresponding rows on the source stage.

(v) The base-change, retained-locus, stabilizer, orientation-domain, vanishing-cycle-domain, and composition defect ranks are zero.

Root windows, source-window indices, signed multiplicities, Pfaffian products, and scalar traces are not transition morphisms of moduli stacks.

Proposition 0.243 *(Proper or closed finite-moduli transitions; proved).* Given a finite-moduli transition geometry datum of Definition 0.242, the (M2) component of Definition 0.244 is a system of proper or closed transition morphisms on the retained object, extension, and two-step flag stacks, compatible with the cosection, vanishing-cycle, and Joyce–Upmeier orientation domains.

Proof. The datum supplies the three stack morphisms and their restrictions to the retained closed substacks. Item 3 says that each restricted component is proper or a closed immersion of finite-type stacks. The closed-locus condition in item 2 makes these restrictions the transition morphisms of the retained finite stages. Item 4 gives the required pullback compatibility for universal complexes, cosections, vanishing-cycle domains, and orientation domains. The zero-defect rows in item 5 give strict base-change and composition. Thus the moduli part of the finite-stage transition is exactly the (M2) datum. □

The packet `certificates/moduli/finite_moduli_transition_geometry` records the present state of this datum. Its verifier status is `FINITE_MODULI_TRANSITION_GEOMETRY_OBSTRUCTION_VERIFIED`: the object-stack, extension-stack, and two-step flag-stack transition morphisms, the proper-or-closed component rows, the finite-type source-target rows, the closed image or graph rows, and the base-change, retained-locus, composition, orientation-domain, and vanishing-cycle-domain zero-defect rows are not yet supplied. Hence the finite-moduli transition-geometry theorem remains open.

Definition 0.244 *(Finite-stage Dirac–Igusa category).* For $R \in \mathcal{R}_{\mathrm{HN}}$, a finite-stage Dirac–Igusa datum is the tuple

$$\mathfrak{D}^{\mathrm{DI}}_{X,R} = \big(\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P^{\Pi}_R, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\big)$$

with the finite data and residual equations specified in §0.7. The category $\mathsf{DI}^{\mathrm{fin}}_{X,R}$ has these tuples as objects and the morphisms below as arrows. The finite-stage category $\mathsf{DI}^{\mathrm{fin}}_{X}$ is the disjoint union over retained heights, together with transition arrows for comparable heights. For $R' \le R$, a transition morphism

$$\rho_{RR'} : \mathfrak{D}^{\mathrm{DI}}_{X,R} \longrightarrow \mathfrak{D}^{\mathrm{DI}}_{X,R'}$$

is the following compatible system.

(M1) A restriction of charge windows and central extensions $\widehat{\Gamma}_R \to \widehat{\Gamma}_{R'}$ preserving Π_X, the normal-ordering cocycle B, and the target Gram labels.

(M2) Proper or closed transition morphisms of the object, extension, and two-step flag stacks, with the corresponding cosection, vanishing-cycle, and orientation domains, as in Proposition 0.243, and with reduced vanishing-cycle transport as in Proposition 0.148.

(M3) A morphism of hybrid local/wrapped factorization carriers over $\mathrm{Ran}^{\mathrm{hyb}}(E)$, compatible with the eight-word flag atlas, quotient-after-correspondence pseudofunctor, and protected integration.

(M4) Isomorphisms of quotient-orientation data, Weyl lifts, and type-II wall charts preserving the orientation cocycle, normal Pfaffian units, wall coordinates, and the local rank $N_{\partial_i}^{\mathrm{Pf}} = 1$, with orientation transport as in Proposition 0.144.

(M5) A morphism of finite Hall bialgebra data preserving product, coproduct, unit, counit, Hopf pairing, primitive projection, radical quotient, and normal-ordered Gram descent, with product preservation as in Proposition 0.173 and coproduct preservation as in Proposition 0.175, and primitive-subspace preservation as in Proposition 0.177, and radical preservation as in Proposition 0.189.

(M6) A Pfaffian-line transition $\rho_{RR'}^{\mathrm{Pf}} : \mathscr{L}_{\mathrm{Pf},R}|_{\mathfrak{Q}_{R'}} \xrightarrow{\sim} \mathscr{L}_{\mathrm{Pf},R'}$ carrying $\mathrm{pf}_{X,R}$ to $\mathrm{pf}_{X,R'}$, compatible with squaring and with the automorphic comparison.

(M7) A filtered coalgebra morphism $C_{X,R} \to C_{X,R'}$ preserving the bar differential, augmentation, Koszul source datum, and the comparison Θ_{Kos}.

(M8) A morphism of primitive-recognition tables preserving representatives, relation rows, parity blocks, product and coproduct matrices, Hopf-pairing matrices, radical and quotient splittings, comparison maps, no-extra rows, PBW filtrations, and Mittag–Leffler transition rows. PBW-filtration preservation is Proposition 0.226; parity-decomposition preservation is Proposition 0.228; primitive-space $R^1 \varprojlim$-vanishing is Proposition 0.230; pairing-kernel $R^1 \varprojlim$-vanishing is Proposition 0.232; and PBW associated-graded $R^1 \varprojlim$-vanishing is Proposition 0.234.

All diagrams expressing these compatibilities commute in the relevant derived, Borel–Moore, filtered, or completed category. Composition is componentwise composition. Thus the identity and associativity conditions for $\mathsf{DI}_X^{\mathrm{fin}}$ are precisely the zero-defect rows in cofinal_subsystems, stage_morphisms, and composition_laws. An isomorphism of finite stages is a transition morphism whose components are equivalences of stacks where applicable, quasi-isomorphisms on chain and coalgebra objects, isomorphisms on orientation and Pfaffian lines, isomorphisms of Hall bialgebra and recognition matrices on the retained windows, and equality of the orientation characters.

Definition 0.245 *(Dirac–Igusa pro-object and pro-isomorphism).* A Dirac–Igusa pro-object is an object of

$$\mathrm{Pro}_{\mathcal{R}_{\mathrm{HN}}}(\mathsf{DI}_X^{\mathrm{fin}}),$$

represented by a functor

$$\mathfrak{D}_X^{\mathrm{DI}} : \mathcal{R}_{\mathrm{HN}}^{\mathrm{op}} \longrightarrow \mathsf{DI}_X^{\mathrm{fin}}$$

from a cofinal HN subsystem to the finite-stage category of Definition 0.244. The symbol

$$\varprojlim_R \mathfrak{D}_{X,R}^{\mathrm{DI}}$$

means this pro-object. It is not a componentwise inverse limit unless the corresponding ml_exactness rows give strict Mittag–Leffler towers and $R^1 \varprojlim = 0$. When a component is realized as a completed module or complex, its topology is the projection topology: a fundamental system of neighbourhoods of zero is given by the kernels of the projections to finite retained heights. For Picard-groupoid components, this means compatible line bundles and trivialisations on a cofinal subsystem. For completed Lie, Hall, and Koszul components, it means the completed filtered object determined by the finite quotients.

A pro-isomorphism between two Dirac–Igusa pro-objects is an isomorphism in this pro-category: after passing to a common cofinal subsystem, it is represented by finite-stage isomorphisms commuting with all transition morphisms and with two-sided inverse defects zero. Equality of scalar Pfaffian products, equality of squared determinants, or equality of signed denominator exponents is not a pro-isomorphism unless the morphism data (M1)–(M8) are supplied.

Proposition 0.246 *(Universal property and cofinal presentation invariance).* Let $\mathsf{C}_X = \mathsf{DI}_X^{\mathrm{fin}}$, and suppose the finite morphism rows of Definition 0.244 give an inverse system

$$\mathfrak{D} = (\mathfrak{D}_R, \rho_{RR'}) : \mathcal{R}_{\mathrm{HN}}^{\mathrm{op}} \longrightarrow \mathsf{C}_X.$$

For a finite-stage object $T \in \mathsf{C}_X$, regarded as a constant pro-object, the Dirac–Igusa pro-object satisfies

$$\mathrm{Hom}_{\mathrm{Pro}(\mathsf{C}_X)}(T, \mathfrak{D}) \cong \varprojlim_{R \in \mathcal{R}_{\mathrm{HN}}} \mathrm{Hom}_{\mathsf{C}_X}(T, \mathfrak{D}_R).$$

For another Dirac–Igusa pro-object $\mathfrak{E} = (\mathfrak{E}_S)_{S \in \mathcal{S}}$,

$$\mathrm{Hom}_{\mathrm{Pro}(\mathsf{C}_X)}(\mathfrak{E}, \mathfrak{D}) \cong \varprojlim_{R \in \mathcal{R}_{\mathrm{HN}}} \varinjlim_{S \in \mathcal{S}} \mathrm{Hom}_{\mathsf{C}_X}(\mathfrak{E}_S, \mathfrak{D}_R).$$

These two formulas are the universal property of $\mathfrak{D}_X^{\mathrm{DI}}$. They are statements in the pro-category; they do not assert componentwise inverse limits of complexes, line bundles, Hall objects, or Lie superalgebras.

If $i : \mathcal{R}'_{\mathrm{HN}} \hookrightarrow \mathcal{R}_{\mathrm{HN}}$ is a cofinal subsystem, the restriction $i^*\mathfrak{D}$ defines a canonically isomorphic pro-object. More generally, two finite presentations of the Dirac–Igusa datum over cofinal HN subsystems define the same pro-object exactly when, after passage to a common cofinal refinement, the finite-stage isomorphisms commute with (M1)–(M8) and have zero two-sided inverse defects. This is the independence of finite presentation choices used in the notation $\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \mathfrak{D}_{X,R}^{\mathrm{DI}}$.

Proof. The first displayed identity is the special case of the standard pro-category formula with the source represented by a constant system. The second is the defining Hom formula in $\mathrm{Pro}(\mathsf{C}_X)$: morphisms $(\mathfrak{E}_S)_S \to (\mathfrak{D}_R)_R$ are compatible families of finite-stage morphisms after allowing a cofinal refinement of the source index. Cofinal restriction leaves these Hom functors unchanged, hence gives the canonical pro-isomorphism by Yoneda. The finite morphism tables supply the only non-formal input: they ensure that the stage maps exist, compose strictly, and have zero inverse defects on the common refinement. The Mittag–Leffler rows enter only when one replaces a pro-object by an actual completed component; they are not part of the pro-object universal property itself. □

Definition 0.247 *(Moduli-cohomology Mittag–Leffler datum).* Fix a cofinal sequence in the retained HN system. For each retained object, extension, mixed, wrapped, and two-step flag stack $\mathfrak{M}_{R,a}$, for each coefficient system $\mathcal{K}_{R,a}$ used in the finite Hall or reduced vanishing-cycle construction, and for each cohomological degree q, put

$$C^q_{R,a,\mathcal{K}} := H^q_c(\mathfrak{M}_{R,a}, \mathcal{K}_{R,a})$$

or the corresponding Borel–Moore cohomology group, according to the six-functor convention of the finite Hall operation under discussion. A moduli-cohomology Mittag–Leffler datum consists of:

(i) finite-rank rows for every $C^q_{R,a,\mathcal{K}}$, with the stack kind, coefficient system, support block, and topology recorded;

(ii) transition maps
$$u^q_{RR',a,\mathcal{K}} : C^q_{R,a,\mathcal{K}} \longrightarrow C^q_{R',a,\mathcal{K}}$$
induced by the proper pushforward, closed restriction, or specified six-functor operation along the retained moduli transition;

(iii) identity and composition rows for the maps $u^q_{RR'}$;

(iv) for every base stage R_0, stack kind a, coefficient $\mathcal{K}$, and degree q, an image-stabilization witness: there is $R_1 \geq R_0$ such that for all $R \geq R_1$,
$$\mathrm{im}\big(C^q_{R,a,\mathcal{K}} \to C^q_{R_0,a,\mathcal{K}}\big) = \mathrm{im}\big(C^q_{R_1,a,\mathcal{K}} \to C^q_{R_0,a,\mathcal{K}}\big),$$
with the stabilized image written in a finite basis;

(v) the defect rows
$$\text{strict} = 1, \qquad \mathrm{ML} = 1, \qquad \operatorname{rank} R^1 \varprojlim_R C^q_{R,a,\mathcal{K}} = 0$$
for every retained tower.

Finite type of the stacks, properness or closedness of transition morphisms, bounded HN types, Euler characteristics, protected traces, and target-root windows are not a moduli-cohomology Mittag–Leffler datum.

PROPOSITION 0.248 *(Vanishing of $R^1 \varprojlim$ on moduli cohomology; proved).* For every tower $C^\bullet_{R,a,\mathcal{K}}$ carrying the datum of Definition 0.247, the inverse system is Mittag–Leffler in each cohomological degree. Consequently
$$R^1 \varprojlim_R C^q_{R,a,\mathcal{K}} = 0$$
for every retained stack kind a, coefficient system $\mathcal{K}$, support block, and degree q. Without the rows of Definition 0.247, the passage from finite retained moduli cohomology to componentwise inverse-limit cohomology is not established.

Proof. The identity and composition rows make $\{C^q_{R,a,\mathcal{K}}, u^q_{RR'}\}$ an inverse system of finite-rank modules. The image-stabilization row is exactly the Mittag–Leffler condition: for every fixed base stage R_0, the images inside $C^q_{R_0,a,\mathcal{K}}$ are eventually constant. The standard derived-inverse-limit criterion for inverse systems of modules then gives $R^1 \varprojlim = 0$ on each such tower. The vanishing is degreewise and coefficientwise; it cannot be inferred from finite type, from a scalar Euler characteristic, or from the existence of proper or closed transition morphisms without the stabilized-image rows. □

The packet

`certificates/moduli/moduli_cohomology_lim1_vanishing`

records the current state of this datum. Its verifier returns `MODULI_COHOMOLOGY_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal HN sequence, retained moduli stack inputs, transition geometry input, coefficient systems, finite cohomology groups, cohomology transition maps, functoriality rows, image subspace rows, image-stabilization witnesses, strict Mittag–Leffler status rows, zero $R^1 \varprojlim$ rows, and degreewise coverage rows are not yet supplied.

0.7.0.1 Finite proof tables for morphisms and pro-isomorphisms. The finite-stage functor is supplied by the tables

cofinal_subsystems, stage_objects, stage_morphisms, component_maps, composition_laws, ml_exactness, pr

The table stage_objects contains one row for each of

$$\mathcal{A}^{E_3},\ F^{\mathrm{hyb}},\ \Gamma,\ \Pi,\ \Phi,\ o,\ H,\ P^{\Pi},\ C,\ \Theta_{\mathrm{Kos}},\ \mathscr{L}_{\mathrm{Pf}},\ \mathrm{pf},\ \varepsilon_o,\ \mathrm{Rec}$$

at each retained height. The table component_maps contains rows for (M1)–(M8), with zero compatibility, kernel, and cokernel ranks in the corresponding derived, Borel–Moore, filtered, or completed category. The tables cofinal_subsystems, stage_morphisms, and composition_laws record directed cofinality, strict transition morphisms, identity laws, and $\rho_{R'R''}\rho_{RR'} = \rho_{RR''}$. The table ml_exactness records $R^1 \varprojlim = 0$ for the component towers. The table pro_isomorphisms records finite-stage isomorphism families over a common cofinal subsystem, with commuting squares and two-sided inverse defects zero. The table scalar_firewall excludes scalar Pfaffian products, denominator products, squared determinants, signed exponent tables, and protected traces as morphism or pro-isomorphism data. The current packet `certificates/morphisms/k3e_dirac_igusa_pro_object` verifies only the obstruction ledger `MORPHISM_OBSTRUCTION_LEDGER_VERIFIED`, with `pro_object_certification=false`. It records the missing cofinal subsystem, stage-object, transition, component-map, composition-law, Mittag–Leffler, and pro-isomorphism rows; it does not construct the functor $\mathcal{R}_{\mathrm{HN}}^{\mathrm{op}} \to \mathsf{DI}_X^{\mathrm{fin}}$.

0.7.0.2 Finite proof tables for the compact E_3-source. The inputs [E_3-HolFA] and [K3→E-sp] are supplied at finite height by

formal_target_charts, field_complexes, e3_operations, bv_qme, anomaly_framing, compact_support, factoriz

These rows record the finite formal target charts, field complexes, holomorphic E_3-operations, BV quantum master equation, anomaly cancellation, framing and formality, compact retained support, prefactorization descent, $K3$-to-E chain specialization, strict transitions, and $R^1 \varprojlim = 0$. The scalar firewall excludes the automorphic section, BKM denominator, target current envelope, protected trace, and hybrid-carrier-only data as entries of $\mathcal{A}^{E_3}_{X,R}$. The current finite E_3-source packet records these rows only by the obstruction ledger `E3_SOURCE_OBSTRUCTION_LEDGER_VERIFIED`, with `e3_source_certification=false`. It is an absence ledger, not a construction of $\mathcal{A}^{E_3}_{X,R}$.

Charge window

Definition 0.249 (Finite target test window). Fix an HN height R. Let $\Gamma_R^{HN} \subset F_{\sigma,S}(R) \subset \Gamma_X^{\mathrm{form}}$ be the retained finite HN charge set. For $c \in \Gamma_R^{HN}$, set

$$\mathcal{T}_R(c) = \Big\{ \sum_{i<j} B(c_i, c_j) \,\Big|\, c = \sum_i c_i,\ c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \le R \Big\} \subset \Gamma_{\mathrm{gram}},$$

and

$$\widehat{\Gamma}_R := \{(c,T) \mid c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\} \subset \widehat{\Gamma}_X.$$

The finite target test window is

$$\Gamma_R^{\mathrm{test}} := \overline{\Pi}_X(\widehat{\Gamma}_R) = \{\Pi_X(c) + T \mid c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\} \subset \Gamma_{\mathrm{gram}}.$$

It is finite once Γ_R^{HN} and the translate sets $\mathcal{T}_R(c)$ are finite. For $R \le R'$, compatible inclusions of HN charge sets and translate sets induce $\Gamma_R^{\mathrm{test}} \subset \Gamma_{R'}^{\mathrm{test}}$. This is a target window in Γ_{gram}, not a compact source window and not the (O1) orientation datum.

Hypotheses. Stability datum (σ, S), finite HN height R, and a retained HN charge set Γ_R^{HN}.

Finite datum. Finite saturated downward window $F_{\sigma,S}(R) \subset \Gamma_X^{\mathrm{HN}}(R)$, the dyonic formal Mukai/effective-charge support, the Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$, the cocycle B, the finite normal-ordered lift $\widehat{\Gamma}_R$, and the target test window $\Gamma_R^{\mathrm{test}} \subset \Gamma_{\mathrm{gram}}$ of Definition 0.249.

Construction.

(i) Form the Liu charge $L_F(n) = Z_S(R p_{S,*}(F \otimes p_E^* H^n))$ [66] for σ_S a rational K3 stability condition and a degree-one line bundle H on E. Restrict to charges with bounded h_S-height $\le R$; the resulting set is $F_{\sigma,S}(R)$.

(ii) Compute the Gram map and cocycle $B(c, c')$ on $F_{\sigma,S}(R) \times F_{\sigma,S}(R)$ via the Mukai dictionary of §3.3; check the symmetric bilinear 2-cocycle condition.

(iii) Form the finite normal-ordered lift set $\widehat{\Gamma}_R$ and its image $\Gamma_R^{\mathrm{test}} = \overline{\Pi}_X(\widehat{\Gamma}_R)$.

At table level this input would be supplied by

finite_test_window_definition, active_support, window_saturation, liu_hn_window, gram_map, cocycle_ident

When supplied, these rows verify the active $\phi_{0,1}$-support, downward saturation, Liu height cutoff, Mukai–Gram map, cocycle identities, normal-ordered lift orbit, target test-window image, primitive formal lift, strict transition, and $R^1 \varprojlim = 0$. The finite-test-window definition row is separated in

`certificates/charge/finite_test_window`.

Its verifier returns `FINITE_TEST_WINDOW_DEFINITION_VERIFIED`, with populated test-window certification and relation-closed target exhaustion both false. It supplies the formula for Γ_R^{test}, not the HN charge rows, normal-ordered translate rows, image-membership rows, finiteness witnesses, transition inclusions, or the assertion that every relation-closed target degree appears at finite R. The scalar firewall rejects target-presentation rows, signed exponent lists, Pfaffian products, Hilbert-scheme scalar shadows, orientation data, and source-rank-only data as charge-window evidence.

Definition 0.250 *(Relation-closed target finite-R exhaustion datum).* Let $\mathcal{D}_{\mathrm{rel}} \subset \Gamma_{\mathrm{gram}}$ be the target-degree set on which relation closure is asserted. In the present target audit this includes the finite W_{rel}-cascade recorded by `certificates/targets/delta5_gn_kac/wrel_degree_closure` and the symbolic tail $A_N \Rightarrow B_N \Rightarrow A_{N+2}$ for even $N \ge 14$ recorded by `certificates/targets/delta5_gn_kac/wrel_tail_staircase`. A finite-R exhaustion datum for $\mathcal{D}_{\mathrm{rel}}$ consists of:

(i) a target-scope table listing the finite prefix and the symbolic tail families;

(ii) for every finite-prefix degree γ, a finite height $R(\gamma)$, a charge $c_\gamma \in \Gamma_{R(\gamma)}^{HN}$, and a translate $T_\gamma \in \mathcal{T}_{R(\gamma)}(c_\gamma)$;

(iii) for each symbolic tail family, formulae R_N, c_N, and T_N valid for every allowed parameter N;

(iv) the image identities

$$\Pi_X(c_\gamma) + T_\gamma = \gamma, \qquad \Pi_X(c_N) + T_N = \gamma_N,$$

with zero defect in Γ_{gram};

(v) compatibility rows for $R \leq R'$, and a zero coverage-defect row for the finite prefix and every symbolic family.

Target relation closure, the infinite tail staircase, active-root cofinality, signed multiplicities, and the formula for Γ_R^{test} are not finite-R exhaustion data.

Proposition 0.251 *(Finite-R realization of relation-closed target degrees; proved).* If $\mathcal{D}_{\mathrm{rel}}$ carries the datum of Definition 0.250, then for every $\gamma \in \mathcal{D}_{\mathrm{rel}}$ there exists a finite HN height R such that

$$\gamma \in \Gamma_R^{\mathrm{test}}.$$

Proof. For a finite-prefix degree γ, the datum gives $R(\gamma)$, $c_\gamma \in \Gamma_{R(\gamma)}^{HN}$, and $T_\gamma \in \mathcal{T}_{R(\gamma)}(c_\gamma)$ with $\Pi_X(c_\gamma) + T_\gamma = \gamma$. By Definition 0.249, $\gamma \in \Gamma_{R(\gamma)}^{\mathrm{test}}$. The same argument applies to a symbolic tail degree γ_N using R_N, c_N, T_N. The zero coverage-defect row says that these two cases exhaust $\mathcal{D}_{\mathrm{rel}}$. No target-only table supplies the charges c or translates T, so target relation closure alone does not prove the proposition. □

The packet

```
certificates/charge/relation_closed_target_finite_R_exhaustion
```

verifies `RELATION_CLOSED_TARGET_FINITE_R_EXHAUSTION_OBSTRUCTION_VERIFIED`: the finite prefix degree-to-R rows, symbolic tail R_N-formulae, HN charge witnesses, normal-ordered translate witnesses, image-equality rows, finite-R bounds, transition compatibility rows, compact source compatibility rows, and zero coverage-defect row are not supplied.

Obstruction classes. [K3×E-fin] on the Liu charge bound; B cocycle property.

Finite Hall stage and atlas

Hypotheses. The charge window, plus the cosection-twisted reduced d-critical heart of §0.2.

Finite datum. The finite Hall stage $C_{\sigma,S,\leq R}$, finite-type derived Artin enhancements $\mathfrak{M}_{\widehat{c},R}, \mathfrak{E}_{\widehat{c},\widehat{c}',R}, \mathfrak{F}^{(2)}_{\widehat{c},\widehat{c}',\widehat{c}'',R}$ of object, extension, and two-step flag stacks; reduced cosections; reduced vanishing cycles Φ^{red}; orientation lines o^{red}. At table level this datum is supplied by hn_type_bounds, semistable_substacks, derived_enhancements, universal_complexes, scalar_rigidifications, stratifications, extension_flag_stacks, cosection_atlas, transitions, and scalar_firewall. The current packet verifies only `MODULI_OBSTRUCTION_LEDGER_VERIFIED`, with `moduli_certification=false`; no finite Hall stage is thereby constructed.

Construction.

(i) Restrict the chosen stability heart to charges in $F_{\sigma,S}(R)$; form the finite-type derived enhancements with PTVV (-1)-shifted symplectic structures [95] on the truncations; descend to Joyce d-critical structures [12].

(ii) Compute the K3 semiregularity cosection CS_C on each retained charge stratum. Surjectivity on reduced strata and filtered Hall additivity on extension stacks are separate rows.

(iii) Choose Joyce–Upmeier orientation lines [56] with chosen square roots, multiplicative under reduced extension correspondences.

Obstruction classes. [K3×E-atlas]; finite closed inertia stratifications after the wrapped quotient steps, together with closure under extensions, HN factors, and duals where needed.

HYBRID WRAPPED FACTORIZATION DATUM

Hypotheses. The finite Hall stage.

Finite datum. The geometric elliptic-degree map b_R^{geom} of Definition 0.10; the hybrid factorization carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$ of §0.1: the finite-stage prestack $\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$; rigidified wrapped prequotients $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ with anchors $\lambda_{\eta,R}$; closed-configuration local sectors $\mathcal{F}_R^{\mathrm{loc}}$; wrapped sectors $\mathcal{G}_R^{\mathrm{wr}}$; the eight-word two-step binary flag atlas; the higher-coloured residual $\mathfrak{o}_R^{\mathrm{col}}$; the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$ and comparison isomorphisms $\theta^Q_{\mu,R}$; protected integration I_R^{prot}. At table level this datum would be supplied by

hybrid_ran_prestack_definition, local_stratum_bo_prestack_definition, geometric_elliptic_degree_map, geometric_

When supplied, these rows record the hybrid prestack definition, the $b = 0$ local subprestack, the $b > 0$ wrapped subprestack, elliptic-degree map, additivity criterion, Borcherds-coordinate separation, positive-degree projection criterion, and split, local closed configuration stacks, wrapped anchors, local/mixed/wrapped correspondences, source/target anchor memory before quotient, all eight words in $\{\mathrm{L},\mathrm{W}\}^3$, determinant anchors, determinant-anchor weights, hypersurface $\chi = 0$ determinant-anchor degeneracy, hypersurface $\chi = 0$ Jordan–Hoelder extra-anchor data, higher-coloured coherences, quotient-after-correspondence, protected integration, and strict HN transitions. The degree-map definition is isolated in

`certificates/hybrid/geometric_elliptic_degree_map.`

Its verifier returns `GEOMETRIC_ELLIPTIC_DEGREE_MAP_OBSTRUCTION_VERIFIED`: b_R^{geom} is the coefficient of $[E]$ in $(p_E)_*\,\mathrm{cyc}_1(\gamma)$, not $\mathrm{pr}_3\,\overline{\Pi}_X$. The retained HN class rows, one-cycle rows, pushforward rows, local/wrapped split rows, and transition rows are not supplied. The additivity criterion is isolated in

`certificates/hybrid/geometric_elliptic_degree_additivity.`

Its verifier returns `GEOMETRIC_ELLIPTIC_DEGREE_ADDITIVITY_OBSTRUCTION_VERIFIED`: Definition 0.11 and Proposition 0.12 give the finite-stage criterion, while extension-pair, one-cycle additivity, pushforward additivity, integer degree additivity, and transition rows are not supplied. The retained-extension-closure packet does not by itself prove additivity of b_R^{geom}. The geometric/Borcherds separation packet is

`certificates/hybrid/geometric_borcherds_degree_separation.`

Its verifier returns `GEOMETRIC_BORCHERDS_DEGREE_SEPARATION_OBSTRUCTION_VERIFIED`: Proposition 0.14 separates the support degree b_R^{geom} from the trace coordinate $m_R^{\mathrm{Bch}} = \mathrm{pr}_3\,\overline{\Pi}_X$; branch comparison rows are not supplied. The positive elliptic-degree projection packet is

`certificates/hybrid/positive_elliptic_degree_projection.`

Its verifier returns `POSITIVE_ELLIPTIC_DEGREE_PROJECTION_VERIFIED`: Proposition 0.2 proves that an effective one-cycle with positive E-degree projects onto all of E, while retained object support-realization rows are not supplied. The hybrid Ran prestack definition packet is

`certificates/hybrid/hybrid_ran_prestack_definition.`

Its verifier returns `HYBRID_RAN_PRESTACK_DEFINITION_VERIFIED`: Definition 0.3 defines the finite-stage carrier as a prestack functor; local component, wrapped component, descent, stackification, and transition rows are not supplied. The local $b = 0$ stratum packet is

`certificates/hybrid/local_stratum_b0_prestack_definition`.

Its verifier returns `LOCAL_STRATUM_B0_PRESTACK_DEFINITION_VERIFIED`: Definition 0.4 defines the full $J = \varnothing$ subprestack with colours in Γ_R^{loc}; closed-configuration, local sheaf, descent, and transition rows are not supplied. The wrapped $b > 0$ stratum packet is

`certificates/hybrid/wrapped_stratum_bpositive_prestack_definition`.

Its verifier returns `WRAPPED_STRATUM_BPOSITIVE_PRESTACK_DEFINITION_VERIFIED`: Definition 0.5 defines the full $I = \varnothing$ subprestack with colours in Γ_R^{wr}; the E-equivariant prequotient definition is separated in the next packet, while anchor, support-realization, descent, and transition rows are not supplied. The E-equivariant wrapped prequotient packet is

`certificates/hybrid/equivariant_wrapped_prequotient_definition`.

Its verifier returns `E_EQUIVARIANT_WRAPPED_PREQUOTIENT_DEFINITION_VERIFIED`: Definition 0.6 defines $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$ before the reduced E-quotient and the translation action with origin 0_E; finite type is separated in the next packet, while properness, anchor population, anchor losslessness, quotient descent, aggregate hybrid-carrier population, and transition rows are not supplied. The wrapped-prequotient finite-type packet is

`certificates/hybrid/wrapped_prequotient_finite_type`.

Its verifier returns `WRAPPED_PREQUOTIENT_FINITE_TYPE_VERIFIED`: Proposition 0.8 proves finite type for $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}$ at fixed R from the retained finite class set, finite-type semistable substacks, and scalar rigidification; properness, anchor population, quotient descent, aggregate hybrid-carrier population, and transition rows are not supplied. The local/local correspondence packet is

`certificates/hybrid/local_local_correspondence_definition`.

Its verifier returns `LOCAL_LOCAL_CORRESPONDENCE_DEFINITION_VERIFIED`: Definition 0.19 defines the ordered LL projection-finite diagram; extension stack, source/target map, descent, properness, Thom–Sebastiani, collision, and transition rows are not supplied by that definition packet. The local/local target-properness packet is

`certificates/hybrid/local_local_extension_properness`.

Its verifier returns `LOCAL_LOCAL_TARGET_PROPERNESS_VERIFIED`: Proposition 0.20 realizes the LL extension stack as a closed relative Quot locus and proves properness of the target map q^{LL}. It does not assert properness of the source map p^{LL}, and descent, collision compatibility, Thom–Sebastiani transport, transition rows, and aggregate hybrid-carrier population are not supplied. The mixed local/wrapped correspondence packet is

`certificates/hybrid/mixed_local_wrapped_correspondence_definition`.

Its verifier returns `MIXED_LOCAL_WRAPPED_CORRESPONDENCE_DEFINITION_VERIFIED`: Definition 0.21 defines the one-sided LW and WL unreduced mixed diagrams; extension stack, source/target map, populated anchor-memory, admissibility, base-change, projection-formula, Thom–Sebastiani, and transition rows are not supplied by that definition packet. The mixed local/wrapped target-admissibility packet is

`certificates/hybrid/mixed_local_wrapped_extension_admissibility`.

Its verifier returns `MIXED_LOCAL_WRAPPED_ADMISSIBILITY_VERIFIED`: Proposition 0.22 realizes the LW and WL extension stacks as closed relative Quot loci over the wrapped target prequotient and proves properness of the target maps q^{LW} and q^{WL}. It does not assert properness of p^{LW} or p^{WL}, and compact-support pushforward, populated anchor-memory rows, base change, projection formula, Thom–Sebastiani transport, quotient descent, transition rows, and aggregate hybrid-carrier population are not supplied. The wrapped/wrapped correspondence packet is

`certificates/hybrid/wrapped_wrapped_correspondence_definition`.

Its verifier returns `WRAPPED_WRAPPED_CORRESPONDENCE_DEFINITION_VERIFIED`: Definition 0.23 defines the ordered WW unreduced diagram; extension stack, source/target map, populated anchor-memory, admissibility, Thom–Sebastiani, symmetric-descent, and transition rows are not supplied by that definition packet. The wrapped/wrapped target-admissibility packet is

`certificates/hybrid/wrapped_wrapped_extension_admissibility`.

Its verifier returns `WRAPPED_WRAPPED_ADMISSIBILITY_VERIFIED`: Proposition 0.24 realizes the WW extension stack as a closed relative Quot locus over the wrapped target prequotient and proves properness of the target map q^{WW}. It does not assert properness of p^{WW}, and compact-support pushforward, populated anchor-memory rows, symmetric descent, Thom–Sebastiani transport, quotient descent, transition rows, and aggregate hybrid-carrier population are not supplied. The compact-support exceptional pushforward model packet is

`certificates/hybrid/compact_support_exceptional_pushforward_model`.

Its verifier returns `COMPACT_SUPPORT_EXCEPTIONAL_PUSHFORWARD_MODEL_DEFINED`: Definition 0.25 defines a chosen compactification $\mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_f \to Z$ and the formula $f_!^{\mathrm{cs}}K = \overline{f}_* j_{f,!}K$, with the proper-map specialization $f_!^{\mathrm{cs}} = f_*$. It does not prove choice independence, base change, projection formula, Thom–Sebastiani transport, symmetric descent, quotient descent, transition compatibility, associativity, or aggregate hybrid-carrier population. The mixed-correspondence base-change packet is

`certificates/hybrid/mixed_correspondence_base_change`.

Its verifier returns `MIXED_CORRESPONDENCE_BASE_CHANGE_VERIFIED`: Proposition 0.26 proves the Cartesian compact-support base-change isomorphisms for the LW and WL target maps. It does not prove compactification-choice independence, projection formula, Thom–Sebastiani transport, quotient descent, transition compatibility, associativity, or aggregate hybrid-carrier population. The mixed-correspondence projection-formula packet is

`certificates/hybrid/mixed_correspondence_projection_formula`.

Its verifier returns MIXED_CORRESPONDENCE_PROJECTION_FORMULA_VERIFIED: Proposition 0.27 proves the compact-support projection formula for the LW and WL target maps with retained finite-Tor target coefficients. It does not prove compactification-choice independence, Thom–Sebastiani transport, quotient descent, transition compatibility, associativity, or aggregate hybrid-carrier population. The mixed-correspondence Thom–Sebastiani packet is

certificates/hybrid/mixed_correspondence_thom_sebastiani.

Its verifier returns MIXED_THOM_SEBASTIANI_TRANSPORT_CONDITIONAL_VERIFIED: Proposition 0.28 proves the binary LW/WL reduced Thom–Sebastiani transport from supplied additive d-critical charts and zero-defect orientation transport. It does not prove $(O1)$, quotient descent, transition compatibility, two-step flag associativity, or aggregate hybrid-carrier population. The eight-word binary vocabulary packet is

certificates/hybrid/eight_word_binary_vocabulary.

Its verifier returns EIGHT_WORD_BINARY_VOCABULARY_DEFINED: Definition 0.29 fixes the alphabet {L, W}, the binary type product, the eight length-three words, and their two parenthesization symbols. It does not construct the two-step flag stacks or prove associativity. The eight-word two-step flag-stack packet is

certificates/hybrid/eight_word_two_step_flag_stacks.

Its verifier returns EIGHT_WORD_TWO_STEP_FLAG_STACKS_CONSTRUCTED: Proposition 0.30 constructs the retained two-step flag stacks and the two comparison maps for all eight words. It does not prove associativity, pentagon coherence, quotient descent, transition compatibility, or aggregate population. The LLL associativity packet is

certificates/hybrid/lll_associativity.

Its verifier returns LLL_ASSOCIATIVITY_CONDITIONAL_VERIFIED: Proposition 0.31 proves the pure local associativity defect vanishes on the LLL two-step flag stack, under the stated local base-change, projection-formula, and reduced Thom–Sebastiani coefficient rows. It does not prove the seven mixed or wrapped word associativity rows or the four-input pentagon. The LLW associativity packet is

certificates/hybrid/llw_associativity.

Its verifier returns LLW_ASSOCIATIVITY_CONDITIONAL_VERIFIED: Proposition 0.32 proves the LLW associativity defect vanishes on the local/local–mixed two-step flag stack, under the stated local and mixed functorial coefficient rows. It does not prove the six remaining word associativity rows or the four-input pentagon. The LWL associativity packet is

certificates/hybrid/lwl_associativity.

Its verifier returns LWL_ASSOCIATIVITY_CONDITIONAL_VERIFIED: Proposition 0.33 proves the LWL associativity defect vanishes on the ordered mixed–mixed two-step flag stack, under the stated mixed functorial coefficient rows. It does not prove the five remaining word associativity rows or the four-input pentagon. The WLL associativity packet is

certificates/hybrid/wll_associativity.

Its verifier returns `WLL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.34 proves the WLL associativity defect vanishes on the mixed–local/local two-step flag stack, under the stated local and mixed functorial coefficient rows. It does not prove the four WW-dependent word associativity rows or the four-input pentagon. The LWW associativity packet is

`certificates/hybrid/lww_associativity`.

Its verifier returns `LWW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.35 proves the LWW associativity defect vanishes on the mixed–wrapped/wrapped two-step flag stack, under the stated mixed functorial rows and wrapped/wrapped functorial hypotheses. It does not prove the three remaining WW-dependent word associativity rows or the four-input pentagon. The WLW associativity packet is

`certificates/hybrid/wlw_associativity`.

Its verifier returns `WLW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.36 proves the WLW associativity defect vanishes on the mixed–wrapped/wrapped two-step flag stack, under the stated mixed functorial rows and wrapped/wrapped functorial hypotheses. It does not prove the two remaining WW-dependent word associativity rows or the four-input pentagon. The WWL associativity packet is

`certificates/hybrid/wwl_associativity`.

Its verifier returns `WWL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.37 proves the WWL associativity defect vanishes on the wrapped/wrapped–mixed two-step flag stack, under the stated mixed functorial rows and wrapped/wrapped functorial hypotheses. It does not prove the remaining WWW word associativity row or the four-input pentagon. The WWW associativity packet is

`certificates/hybrid/www_associativity`.

Its verifier returns `WWW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.38 proves the WWW associativity defect vanishes on the wrapped/wrapped two-step flag stack, under the stated wrapped/wrapped functorial hypotheses. The eight length-three wordwise associativity defects are then zero under the stated functorial rows. The four-input pentagon remains separate. The four-input pentagon packet is

`certificates/hybrid/four_input_pentagon_coherence`.

Its verifier returns `FOUR_INPUT_PENTAGON_CONDITIONAL_VERIFIED`: Proposition 0.39 constructs the retained four-step flag stack for every word in $\{\mathrm{L}, \mathrm{W}\}^4$ and proves the Mac Lane pentagon for the binary hybrid product on that common refinement, under the stated functorial hypotheses. It does not define the higher-coloured tree category or prove quotient descent, transition compatibility, or aggregate population. The higher-coloured tree-category packet is

`certificates/hybrid/higher_coloured_tree_category_definition`.

Its verifier returns `HIGHER_COLOURED_TREE_CATEGORY_DEFINED`: Definition 0.40 defines the nonunital higher-coloured hybrid tree category and the six components of $\mathfrak{o}_R^{\mathrm{col}}$. It does not prove $\mathfrak{o}_R^{\mathrm{col}} = 0$, unit compatibility, symmetric descent, wrapped order conventions, quotient descent, transition compatibility, or aggregate population. The higher-coloured residual criterion packet is

`certificates/hybrid/higher_coloured_residual_vanishing_criterion`.

Its verifier returns `HIGHER_COLOURED_RESIDUAL_CONDITIONAL_CRITERION_VERIFIED`: Proposition 0.41 proves the finite-stage implication from the six named component rows to $\mathfrak{o}_R^{\mathrm{col}} = 0$. The tree component is discharged by the eight associativity rows and the four-input pentagon. The unit and symmetric-relabelling rows are supplied by separate packets. The full non-tree tuple remains incomplete: refinement, ordered-chart descent, overlap, quotient, transition, and aggregate rows are not supplied. The hybrid unit packet is

`certificates/hybrid/hybrid_unit_object_and_vacuum_correspondences`.

Its verifier returns `HYBRID_UNIT_OBJECT_AND_VACUUM_CORRESPONDENCES_DEFINED`: Definition 0.42 defines the local zero unit object and the four split vacuum correspondences for local and wrapped inputs. It does not prove the local or wrapped unit identity laws, quotient descent, transition compatibility, or aggregate population. The local unit compatibility packet is

`certificates/hybrid/local_unit_compatibility`.

Its verifier returns `LOCAL_UNIT_COMPATIBILITY_VERIFIED`: Proposition 0.43 proves that the left and right LL split vacuum correspondences act as identity pull-push maps on local coefficients. It does not prove wrapped unit compatibility, quotient descent, transition compatibility, or aggregate population. The wrapped unit compatibility packet is

`certificates/hybrid/wrapped_unit_compatibility`.

Its verifier returns `WRAPPED_UNIT_COMPATIBILITY_VERIFIED`: Proposition 0.44 proves that the left LW and right WL split vacuum correspondences act as identity pull-push maps on wrapped coefficients, with wrapped anchor memory unchanged before the reduced E-quotient. It does not prove quotient descent, transition compatibility, or aggregate population. The local configuration symmetric descent packet is

`certificates/hybrid/local_configuration_symmetric_descent`.

Its verifier returns `LOCAL_CONFIGURATION_SYMMETRIC_DESCENT_VERIFIED`: Proposition 0.45 proves finite symmetric-group descent for projection-finite closed local configuration charts and their reduced coefficients before the reduced E-quotient. It does not prove collision compatibility, closed-chart refinement, ordered-chart descent to the symmetric finite-colour colimit, quotient descent, transition compatibility, or aggregate population. The wrapped insertion order convention packet is

`certificates/hybrid/wrapped_insertion_order_conventions`.

Its verifier returns `WRAPPED_INSERTION_ORDER_CONVENTIONS_VERIFIED`: Proposition 0.46 fixes the planar wrapped source order, the WW quotient/subobject convention, and the order preservation of the eight two-step flag comparison maps before the reduced E-quotient. It does not prove unordered wrapped-pair symmetric descent, ordered-chart descent, overlap compatibility, quotient descent, transition compatibility, or aggregate population. The ordered two-sided mixed correspondence packet is

`certificates/hybrid/two_sided_mixed_correspondence_definition`.

Its verifier returns `TWO_SIDED_MIXED_CORRESPONDENCE_DEFINITION_VERIFIED`: Definition 0.47 defines the ordered LWL unreduced flag diagram; flag-stack, source/target map, populated anchor-memory, admissibility, base-change, projection-formula, Thom–Sebastiani, and transition rows are not supplied. The source/target anchor-memory packet is

`certificates/hybrid/source_target_anchor_memory_definition`.

Its verifier returns `SOURCE_TARGET_ANCHOR_MEMORY_DEFINITION_VERIFIED`: Definition 0.48 defines the LW, WL, WW, and LWL anchor-memory maps on unreduced correspondence stacks before the E-quotient. The determinant anchor is constructed by the next packet and its translation weight by the following packet. The $\chi = 0$ degeneracy is isolated by the chi-zero packet. The degree-zero Jordan–Hoelder extra-anchor datum is recorded by the extra-anchor-data packet; anchor-residual, full losslessness, quotient-descent, stabilizer linearization, populated finite-stack, and transition rows are not supplied by the source/target-memory packet. The determinant-anchor construction packet is

`certificates/hybrid/determinant_anchor_construction`.

Its verifier returns `DETERMINANT_ANCHOR_CONSTRUCTION_VERIFIED`: Definition 4.7 constructs $\lambda_{\eta,R}^{\det}$ as the normalized determinant of cohomology, checks that it lands in $\mathrm{Pic}^0(E) \simeq E$, and records determinant-functor base-change; unit-weight, $\chi = 0$ degeneracy, Jordan–Hoelder extra-anchor data, anchor-residual, losslessness, quotient-descent, populated wrapped-prequotient, and transition rows are not supplied. The determinant-anchor translation-weight packet

`certificates/hybrid/determinant_anchor_translation_weight`

returns `DETERMINANT_ANCHOR_TRANSLATION_WEIGHT_VERIFIED`: under the support-translation convention it computes $\lambda_{\eta,R}^{\det}(a \cdot \mathcal{F}) = \lambda_{\eta,R}^{\det}(\mathcal{F}) + \chi_\eta a$. The determinant-anchor chi-zero degeneracy packet

`certificates/hybrid/determinant_anchor_chi_zero_degeneracy`

returns `DETERMINANT_ANCHOR_CHI_ZERO_DEGENERACY_VERIFIED`: on the branch $\chi_\eta = 0$, the determinant anchor is constant on the full retained E-translation orbit and has rank-one separation defect. The determinant-anchor extra-anchor-data packet

`certificates/hybrid/determinant_anchor_extra_anchor_data`

returns `DETERMINANT_ANCHOR_EXTRA_ANCHOR_DATA_VERIFIED`: the degree-zero Jordan–Hoelder divisor refines the determinant anchor and separates the rank-two pair $O_E^{\oplus 2}$ and $L \oplus L^{-1}$, $L \neq O_E$. Losslessness, anchor-residual, quotient-descent, stabilizer linearization, populated wrapped-prequotient, and transition rows are not supplied. The quotient-after-correspondence pseudofunctor definition packet is

`certificates/hybrid/quotient_after_correspondence_pseudofunctor_definition`.

Its verifier returns `QUOTIENT_AFTER_CORRESPONDENCE_PSEUDOFUNCTOR_DEFINED`: Definition 0.50 defines the object assignment $Y \mapsto [Y/E]$, the generating-arrow assignment $(Y_0 \leftarrow \mathfrak{E}_e \rightarrow Y_1) \mapsto ([Y_0/E] \leftarrow [\mathfrak{E}_e/E] \rightarrow [Y_1/E])$, the coefficient-descent interface, and the quotient residual components. The composition component is supplied by

`certificates/hybrid/quotient_after_correspondence_composition`.

Proposition 0.51 proves preservation of finite-height composition of retained spans. The base-change square component is supplied by

`certificates/hybrid/quotient_after_correspondence_base_change.`

Proposition 0.52 proves preservation of retained finite-height cartesian squares after quotient. The Thom–Sebastiani descent component is supplied by

`certificates/hybrid/quotient_after_correspondence_thom_sebastiani.`

Proposition 0.53 proves descent of equivariant reduced Thom–Sebastiani transport through the quotient correspondence. The quotient-first exclusion component is supplied by

`certificates/hybrid/quotient_after_correspondence_quotient_first_exclusion.`

Proposition 0.54 proves that every reduced generating arrow used by $\boldsymbol{Q}^{\mathrm{corr}}_{E,R}$ is the quotient of a preformed unreduced E-equivariant correspondence. The Borel–Moore chain functor component is supplied by

`certificates/hybrid/quotient_after_correspondence_bm_chain_functor.`

Proposition 0.55 constructs $Q_{E,R}$ on quotient-presented Borel–Moore chain objects. The operation-comparison maps are supplied by

`certificates/hybrid/quotient_after_correspondence_theta_mu_comparison.`

Proposition 0.56 constructs $\theta^{Q}_{\mu,R}$ for quotient-presented operation rows. The quotient two-step flag associativity component is supplied by Proposition 0.57. The quotient product-coproduct compatibility component is supplied by Proposition 0.58. The quotient primitive-projection compatibility component is supplied by Proposition 0.59. The HN-transition compatibility of $Q_{E,R}$ is supplied by Proposition 0.60. Compact-support pushforward base change, protected-integration transition compatibility, and aggregate population are not supplied. Hall product, Hall coproduct, and primitive-subspace transition preservation are not supplied by row 256. The definition of the protected integration map is supplied separately by Proposition 0.62. Its product compatibility is supplied separately by Proposition 0.63. Its coproduct compatibility is supplied separately by Proposition 0.64. Its primitive-projection compatibility is supplied separately by Proposition 0.65. Its Gram degree is supplied separately by Proposition 0.66. Its normal-ordered label comparison is supplied separately by Proposition 0.67.

Construction.

(i) Split Γ_R into Γ_R^{loc} ($b_R^{\mathrm{geom}} = 0$) and Γ_R^{wr} ($b_R^{\mathrm{geom}} > 0$).

(ii) Form the local carrier $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ as the $J = \varnothing$ subprestack of $\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$, with local colour labels in Γ_R^{loc}.

(iii) Form the wrapped carrier $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$ by imposing $I = \varnothing$ in the hybrid prestack and allowing only colour maps $\eta : J \to \Gamma_R^{\mathrm{wr}}$, with wrapped prequotient families and their anchors retained.

(iv) For $\alpha \in \Gamma_R^{\mathrm{loc}}$: form the closed-configuration incidence stacks $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$ and the configuration maps $\pi_{\alpha,R,I} : \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \to E^I$, with vanishing-cycle/orientation pushforwards $\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I}$. Define $\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U)$ by symmetric-group descent on E^I.

(v) Retain the ordered LL local/local correspondence diagram of Definition 0.19; its target-proper extension stack is Proposition 0.20. Source properness is not claimed. Descent, collision compatibility, Thom–Sebastiani, compact-support pushforward, aggregate population, and transition rows remain separate obligations.

(vi) For $\eta \in \Gamma_R^{\mathrm{wr}}$: form the rigidified prequotient $\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \to \mathcal{M}_{\eta,R}^{\mathrm{wr}}$ of Definition 0.6, with its E-translation action before the reduced quotient is taken. The finite-type assertion is Proposition 0.8. Then attach the determinant anchor $\lambda_{\eta,R}^{\det}$ of Definition 4.7. Its translation weight, computed in Proposition 4.8, is χ_η. On the $\chi_\eta = 0$ branch, Proposition 4.9 gives a rank-one determinant-anchor separation defect. Attach the degree-zero Jordan–Hoelder divisor of Proposition 4.10 as extra anchor data; it separates the determinant-zero rank-two test pair but is not full anchor losslessness. Unit-weight normalization, losslessness, quotient descent, stabilizer linearization, transition compatibility, and finite-stage population remain separate rows. The finite anchor residual $\mathfrak{o}_R^{\lambda}$ is Definition 0.49; its vanishing is a later row.

(vii) Retain the one-sided LW and WL mixed local/wrapped correspondence diagrams of Definition 0.21; their target-admissible extension-stack and target-map rows are supplied by Proposition 0.22. Their populated anchor-memory, compact-support pushforward, base-change, projection-formula, Thom–Sebastiani, quotient-descent, and transition rows remain separate obligations.

(viii) Retain the ordered WW wrapped/wrapped correspondence diagram of Definition 0.23; its target-admissible extension-stack and target-map rows are supplied by Proposition 0.24. Its populated anchor-memory, compact-support pushforward, symmetric descent, Thom–Sebastiani, quotient-descent, and transition rows remain separate obligations.

(ix) Retain the compact-support exceptional pushforward model of Definition 0.25. It supplies the chosen formula $f_!^{\mathrm{cs}} = \overline{f}_* j_!$ and the proper-map specialization. It does not populate every correspondence map with a compactification and does not prove base change, projection formula, Thom–Sebastiani, symmetric descent, quotient descent, transition compatibility, or associativity.

(x) Retain the LW/WL mixed base-change isomorphisms of Proposition 0.26. They apply to Cartesian compact-support squares with pulled coefficient complexes. Projection formula, Thom–Sebastiani, quotient descent, transitions, associativity, and aggregate population remain separate rows.

(xi) Retain the LW/WL mixed projection-formula isomorphisms of Proposition 0.27. They apply to retained finite-Tor target coefficients in the chosen compact-support model. Thom–Sebastiani, quotient descent, transitions, associativity, and aggregate population remain separate rows.

(xii) Retain the LW/WL mixed Thom–Sebastiani transports of Proposition 0.28. They apply on the supplied reduced additive d-critical charts with zero-defect Joyce–Upmeier orientation transport. Quotient descent, transitions, two-step flag associativity, and aggregate population remain separate rows.

(xiii) Retain the eight-word binary vocabulary of Definition 0.29. This fixes {LLL, LLW, LWL, WLL, LWW, WLW, WWL, W the L/W type product, and the two parenthesization symbols. The actual two-step flag stacks, associativity defects, pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xiv) Retain the eight two-step flag stacks and their left/right comparison maps from Proposition 0.30. Associativity defects, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xv) Retain the LLL associativity identity of Proposition 0.31. The seven remaining word associativity rows, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xvi) Retain the LLW associativity identity of Proposition 0.32. The six remaining word associativity rows, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xvii) Retain the LWL associativity identity of Proposition 0.33. The five remaining word associativity rows, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xviii) Retain the WLL associativity identity of Proposition 0.34. The four remaining WW-dependent associativity rows, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xix) Retain the LWW associativity identity of Proposition 0.35. The three remaining WW-dependent associativity rows, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xx) Retain the WLW associativity identity of Proposition 0.36. The two remaining WW-dependent associativity rows, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xxi) Retain the WWL associativity identity of Proposition 0.37. The remaining WWW associativity row, the four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xxii) Retain the WWW associativity identity of Proposition 0.38. This completes the length-three wordwise associativity list. The four-input pentagon, quotient descent, transitions, and aggregate population remain separate rows.

(xxiii) Retain the four-input pentagon coherence of Proposition 0.39. The higher-coloured tree category, remaining unit compatibilities, quotient descent, transitions, and aggregate population remain separate rows.

(xxiv) Retain the nonunital higher-coloured tree category of Definition 0.40. The vanishing of $\mathfrak{o}_R^{\mathrm{col}}$, remaining unit compatibilities, quotient descent, transitions, and aggregate population remain separate rows.

(xxv) Retain the conditional higher-coloured residual criterion of Proposition 0.41. The tree component is discharged by associativity and pentagon coherence. The unit and symmetric-relabelling rows are supplied by separate proof packets. Refinement, ordered-chart descent, overlap, quotient descent, transitions, and aggregate population rows remain separate rows.

(xxvi) Retain the hybrid unit object and split vacuum correspondences of Definition 0.42. The local and wrapped unit identity laws are supplied by the next two proof rows; quotient descent, transitions, and aggregate population remain separate rows.

(xxvii) Retain the local unit compatibility theorem of Proposition 0.43. The wrapped unit identity law, quotient descent, transitions, and aggregate population remain separate rows.

(xxviii) Retain the wrapped unit compatibility theorem of Proposition 0.44. The unit component of the higher-coloured residual is now discharged at fixed height before quotient. Quotient descent, transitions, and aggregate population remain separate rows.

(xxix) Retain the local configuration symmetric descent theorem of Proposition 0.45. The symmetry component of the higher-coloured residual is now discharged for projection-finite local configuration charts at fixed height before quotient. Collision compatibility, refinement, ordered-chart descent, overlap, quotient descent, transitions, and aggregate population remain separate rows.

(xxx) Retain the wrapped insertion order convention theorem of Proposition 0.46. The wrapped order input row of the higher-coloured residual criterion is now supplied at fixed height before quotient. Unordered wrapped-pair symmetric descent, ordered-chart descent, overlap, quotient descent, transitions, and aggregate population remain separate rows.

(xxxi) Retain the ordered LWL two-sided mixed correspondence diagram of Definition 0.47; its flag-stack, source/target-map, populated anchor-memory, admissibility, base-change, projection-formula, Thom–Sebastiani, and transition rows remain separate obligations.

(xxxii) Retain the source/target anchor-memory maps a^{LW}, a^{WL}, a^{WW}, and a^{LWL} of Definition 0.48 on the unreduced mixed and wrapped correspondence types. This step assumes the abstract wrapped anchors $\lambda_{\eta,R}$. The determinant-anchor construction and translation-weight computation are separate packets, and the $\chi = 0$ determinant-anchor degeneracy is handled by its own packet. The degree-zero Jordan–Hoelder extra-anchor datum and the residual definition o_R^{λ} are separate packets; residual vanishing, losslessness, quotient descent, and transition rows remain separate obligations.

(xxxiii) Retain the quotient-after-correspondence pseudofunctor definition of Definition 0.50. This supplies the object, arrow, and coefficient-descent interface rows before Borel–Moore chains.

(xxxiv) Retain the composition comparison of Proposition 0.51. This supplies $\mathfrak{o}_R^{Q,\mathrm{comp}} = 0$ for finite-height retained spans.

(xxxv) Retain the base-change square comparison of Proposition 0.52. This supplies $\mathfrak{o}_R^{Q,\mathrm{bc}} = 0$ for finite-height retained cartesian squares. Compact-support pushforward base change and aggregate population remain separate rows; the quotient flag component is treated below.

(xxxvi) Retain the Thom–Sebastiani descent comparison of Proposition 0.53. This supplies $\mathfrak{o}_R^{Q,\mathrm{TS}} = 0$ for equivariant finite-height reduced Thom–Sebastiani rows.

(xxxvii) Retain the quotient-first exclusion of Proposition 0.54. This supplies the row-250 firewall: an independent object-quotient- first Hall product is not used unless it is identified with $[\mathfrak{E}_e/E]$ for a preformed unreduced correspondence.

(xxxviii) Retain the Borel–Moore chain functor of Proposition 0.55. This supplies $\mathfrak{o}_R^{Q,\mathrm{BM}} = 0$ on quotient-presented coefficient objects.

(xxxix) Retain the operation-comparison maps of Proposition 0.56. This supplies $\mathfrak{o}_R^{Q,\theta} = 0$ for quotient-presented finite-height operation rows.

(xl) Retain the associativity compatibility of Proposition 0.57. This supplies $\mathfrak{o}_R^{Q,\mathrm{flag}} = 0$ for the eight wordwise two-step flag associativity rows.

(xli) Retain the coproduct compatibility of Proposition 0.58. This supplies $\mathfrak{o}_R^{Q,\Delta} = 0$ for supplied finite product-coproduct rows.

(xlii) Retain the primitive-projection compatibility of Proposition 0.59. This supplies $\mathfrak{o}_R^{Q,\mathrm{Prim}} = 0$ for supplied finite primitive kernel and projection rows.

(xliii) Retain the HN-transition compatibility of $Q_{E,R}$ from Proposition 0.60. This supplies $\mathfrak{o}_{R',R}^{Q,\mathrm{tr}} = 0$ for supplied quotient-presented finite HN object and coefficient transition rows. Protected integration and aggregate population remain separate rows. Hall product, Hall coproduct, and primitive-subspace transition preservation remain outside row 256.

(xliv) Retain the eight-word vocabulary $\mathcal{W}_{\mathrm{hyb}}^{(3)}$ of Definition 0.29 and the two-step binary flag stacks $\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}$ of Proposition 0.30. The vanishing of $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LLL},R}$ is supplied by Proposition 0.31, and $\mathfrak{o}^{\mathrm{assoc}}_{\mathrm{LLW},R}$ is supplied by Proposition 0.32; the LWL associativity defect is zero by Proposition 0.33; and the WLL associativity defect is zero by Proposition 0.34. The LWW associativity defect is zero by Proposition 0.35. The WLW associativity defect is zero by Proposition 0.36. The WWL associativity defect is zero by Proposition 0.37; and the WWW associativity defect is zero by Proposition 0.38. The four-input pentagon is supplied by Proposition 0.39. The unit, refinement, symmetry, descent, and higher-coloured tree data remain in $\mathfrak{o}_R^{\mathrm{col}}$.

(xlv) Use the supplied quotient-after-correspondence pseudofunctor definition $\mathsf{Corr}_R^{E,\mathrm{hyb}} \to \mathsf{Corr}_R^{\mathrm{red,hyb}}$, then supply the separate compact-support pushforward base-change and transition rows before treating the quotient residual as zero.

(xlvi) Define I_R^{prot} on the reduced functor by Proposition 0.62. Its product compatibility is supplied by Proposition 0.63. Its coproduct compatibility is supplied by Proposition 0.64. Its primitive-projection compatibility is supplied by Proposition 0.65. Its Gram degree is supplied by Proposition 0.66. The normal-ordered label comparison is supplied by Proposition 0.67. The transition compatibility row remains separate.

Obstruction classes. $\mathfrak{D}_{\mathrm{hyb},R}$ (§0.1).

Orientation descent and Weyl transport

Hypotheses. The hybrid wrapped factorization datum.

Finite datum. The (O1) and (O1)$^+$ data: null-trivializations of the reduced and equivariant orientation gerbes, $E[N]$-linearizations with vanishing degree-one characters, and Weyl-equivariant lifts τ_i ($i = 1, 2, 3$) compatible with the quotient-orientation cocycle $\mathfrak{q}_C$.

Construction.

(i) For each charge stratum C: compute the reduced gerbe α_C^{red}, free E-class $\alpha_C^{E,\mathrm{free}}$, and equivariant finite-stabilizer classes $\widetilde{\beta}^H_{C,S}$ for the actual stabilizers $H \subset E_{\mathrm{tors}}$ on finite-stabilizer strata.

(ii) Verify the Klein-four condition at $H = E[2]$: $(b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2)$ with $r_1 = r_2 = r_3 = 0$; for full-level even $N \geq 4$, verify the two-primary residual on $E[N]_{(2)}$ including the mixed coefficient $A_{12}^{(N)}$.

(iii) Choose zero $H^1(BH; \mathbb{F}_2)$-linearization characters $\lambda^H_{C,S} = 0$.

(iv) For each simple type-II reflection s_{δ_i}, supply the lift $\tau_i : o_C \to s_{\delta_i}^* o_C$ on the finite partial action groupoid $\mathcal{G}_R$; compute the 2-cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$ and verify $[c_{o,R}] = 0$ with chosen 1-cochain.

(v) Verify $\tau_i^2 = 1$ and the Coxeter relations on the $W^{(2)}(\Lambda^{2,1}_{II})$-orbit; verify $\omega_{i,C} = 0$ for each torsor defect of the quotient-orientation transport.

Obstruction classes. $\mathfrak{D}_{\mathrm{or},R}$ (§0.2); the (O2) wall-atlas enters the Pfaffian wall normal form.

HALL PRODUCT, COPRODUCT, AND PRIMITIVE PROJECTION

Hypotheses. The finite Hall atlas, hybrid carrier, and orientation transport.

Finite datum. The compact Hall product m_R^{red}, collision coproduct Δ_R^{ch}, primitive projection P_R^{Prim}, and Hopf pairing $\langle\, , \,\rangle_R$ on the cosection-reduced d-critical heart, with normal-ordered Gram descent $\overline{\Pi}^{\Theta}_{X,*}$. The product alone is not a primitive datum by Proposition 0.165.

Construction.

(i) Form $m_R^{\mathrm{red}} = q_! \circ \mathrm{TS}_R^{\mathrm{red}} \circ p^*$ on each binary correspondence (LL, LW, WL, WW); verify two-step pentagon coherence on the eight-word atlas. This operation is defined only in the retained compactified correspondence model, or in an explicitly supplied exceptional compact-support model as in Definition 0.25; for LW/WL Cartesian target refinements, base change is Proposition 0.26; for LW/WL retained finite-Tor target coefficients, the projection formula is Proposition 0.27; for LW/WL binary coefficient transport, Thom–Sebastiani is Proposition 0.28.

(ii) Form Δ_R^{ch} by pull-push on the 2-step flag stack; verify coassociativity and counitality after $Q_{E,R}$ using the higher-coloured, unit, refinement, quotient, and transition coherences.

(iii) Form $\langle\, , \,\rangle_R$ on the primitive layer; verify Hopf adjointness with the coproduct on the radical quotient.

(iv) Compute the Frobenius defect $\mathfrak{o}_{F,R}$ via the CY-3 Serre-functor identity [64]; verify $\mathfrak{o}_{F,R} = 0$.

(v) Compute the radical Rad_R^{Π}; verify Hopf-coideal and Lie-ideal stability.

(vi) Form the chain-level normal-ordered Gram descent $\overline{\Pi}^{\Theta}_{X,*}$ via the negative-cyclic Loday–Quillen–Tsygan route or the PTVV–Joyce orientation route [67, 68, 95, 103]; verify the triple homotopy compatibility $d_{\mathrm{Hoch}}\Theta_{\Pi} = B_{\mathrm{ch}}$ and the $\widehat{\Gamma}_R$-grading absorption.

Obstruction classes. $\mathfrak{O}_{\mathrm{Hall},R}$, including $\mathfrak{o}_{F,R}$, the Hopf-coideal residual, and the Θ-pentagon residual.

DIRAC BLOCK, PFAFFIAN LINE, AND TYPE-II WALL ATLAS

Hypotheses. The orientation transport and Hall product.

Finite datum. The first-order Dirac block decomposition $\mathfrak{D}_{X,R}^{D_0} = \bigoplus_\gamma J_\gamma$, the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ with squaring isomorphism, the Pfaffian-to-automorphic comparison $\iota_{\mathrm{aut},R}$, the normalized section $\mathrm{pf}_{X,R}$, and the type-II wall atlas (rank-one normal model on every retained component, boundary stratum, and overlap).

Construction.

(i) Form the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R} = \bigotimes_C \mathrm{Pf}(K_{C,R}^{\mathrm{red}})$ on the cosection-reduced quotient [12]; verify the squaring isomorphism.

(ii) Form J_γ on $P_{R,\gamma}^{\Pi,+} \oplus (P_{R,\gamma}^{\Pi,+})^\vee[1]$ for each $\gamma \in \Gamma_R^{\Pi,+} \setminus \{0\}$; verify Pfaffian contribution $(1 - x^\gamma)^{\mathrm{sdim}\, P_{R,\gamma}^{\Pi,+}}$.

(iii) Verify the support, parity, and leading-coefficient residuals $\mathfrak{o}_{R,\gamma}^{\mathrm{supp}} = \mathfrak{o}_{R,\gamma}^{\mathrm{par}} = \mathfrak{o}_R^{\mathrm{lead}} = 0$ on the test window.

(iv) For each simple type-II wall $\mathbb{H}_{\delta_i}$ ($i = 1, 2, 3$): take the rank-one Pfaffian crossing chart on every retained component and boundary stratum; verify $N_{\delta_i}^{\mathrm{Pf}} = 1$ and the local sign $\varepsilon_{\delta_i,R}^{\mathrm{loc}} = -1$; verify chart-overlap agreement of the normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps; verify the half-Hilbert orbit sign sum (size $4096 = 2^{12}$).

(v) Supply the Pfaffian-to-automorphic comparison $\iota_{\mathrm{aut},R} : \mathscr{L}_{\mathrm{Pf},R} \to \mathcal{L}^5 \otimes \nu_{\Delta_5}$ compatible with cusp trivialization and squaring.

Obstruction classes. (D0-Pf), (O2)/hybrid compatibility residuals, and the Pfaffian-to-automorphic comparison residuals.

Koszul source coalgebra and cobar comparison

Hypotheses. The hybrid carrier and Hall product, an augmentation of the hybrid source, and a wrapped bar-cobar domain datum for the wrapped summands and mixed local/wrapped words.

Finite datum. The conilpotent chiral source coalgebra $C_{X,R}$, finite filtration $F^{\mathrm{co}}_{R,\bullet}$, collision coproduct Δ^{ch}_{R}, bar comparison $b_{X,R}$, and cobar comparison $\Theta_{\mathrm{Kos},R}$.

Construction.

(i) Choose an augmentation $\varepsilon^{\mathrm{hyb}}_{X,R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} \to k\mathbf{1}$ and verify bounded length N_R.

(ii) Form the canonical source-bar coalgebra $C^{\mathrm{bar}}_{X,R} = \bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R})$ [7, §3.7] with bar-length filtration and deconcatenation collision coproduct.

(iii) Verify binary coassociativity on the eight-word atlas, the higher-coloured and unit coherences, bar-length conilpotence, and transition compatibility; set $b_{X,R} = \mathrm{id}$.

(iv) Verify that the projection-finite summands, wrapped summands, and mixed local/wrapped words lie in the chosen completed Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar domain. Under that wrapped Koszul datum, verify the source-internal counit $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_{E} C^{\mathrm{bar}}_{X,R} \to \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ is a quasi-isomorphism in the chosen completed/coderived category [35, Theorem 6.2.2].

(v) Construct the source-to-target chiral algebra quasi-isomorphism $\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\leq R}$ preserving differential, product, unit, transition maps, and primitive restriction.

(vi) Supply the finite Koszul comparison datum $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ of Definition 0.211: verify the Weyl transport square, Pfaffian-orientation sign square, Hall product/coproduct and Hopf-pairing square, radical quotient square, PBW associated-graded square, and their transition homotopies.

(vii) Set $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$; verify the Koszul Mittag–Leffler condition on the cones of b and Θ_{Kos} and on the comparison-defect complexes of $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$.

Obstruction classes. $\mathfrak{D}_{\mathrm{Kos},R}$, including $\mathfrak{o}^{\Omega}_{R} = (\mathfrak{o}^{\varepsilon,\mathrm{src}}_{R}, \mathfrak{o}^{\vartheta}_{R})$ and the wrapped Koszul defect $\mathfrak{o}^{\mathrm{wrKos}}_{R}$, together with $\mathfrak{D}_{\mathrm{Kcmp},R}$ for the Weyl, Pfaffian, Hall-pairing, radical, PBW, transition, and Mittag–Leffler comparison defects.

Primitive recognition in cofinal finite windows

Hypotheses. The Hall product, Pfaffian datum, and Koszul source coalgebra, plus the imported Borcherds–Kac target datum $\mathscr{B}_{\Delta_5}$.

Conditional datum. For every cofinal saturated finite window W_ν, the recognition data (R0)–(R8); after these data are supplied on a cofinal relation-closed system, the inverse-limit isomorphism $P^{\Pi,\mathrm{red}}_{X} \cong \mathfrak{g}_{\Delta_5}$, and after the Koszul source datum is also supplied, the chiral identification $\Omega^{\mathrm{ch}}_{E} C_X \simeq \mathsf{Den}_{\Delta_5,E}$.

Construction.

(i) For each ν, identify the saturated downward target-root window $W_\nu = \{\beta \in \mathcal{R}_+ \mid \mathrm{ht}\,\beta \leq \nu\}$ and the source window $\Gamma(W_\nu) = \{\gamma \in \Gamma_{\mathrm{gram}} \mid \alpha(\gamma) \in W_\nu\}$. In coordinates $\beta = b_1\delta_1 + b_2\delta_2 + b_3\delta_3$, this means $\gamma_\beta = (b_1, b_1 + b_2 - b_3, b_2)$. The packet `certificates/lattice/downward_root_windows` verifies `DOWNWARD_ROOT_WINDOWS_VERIFIED`; it does not supply compact source representatives or the recognition rows below. The packet `certificates/lattice/active_root_window_exhaustion` verifies `ACTIVE_ROOT_WINDOW_EXHAUSTION_VERIFIED`:

$\bigcup_\nu W_\nu = \mathcal{R}_+$ by the height witness $\beta \in W_{\mathrm{ht}(\beta)}$. This is target-root exhaustion, not compact-source support exhaustion. The packet `certificates/sources/source_window_compact_support` verifies `SOURCE_WINDOW_COMPACT_SUPPORT_OBSTRUCTION_VERIFIED`: the compact support index set, source height function, finite source windows, compact provenance rows, cofinality witness, transitions, and target-source compatibility rows are still absent. The packet `certificates/sources/target_source_window_compatibility` verifies `TARGET_SOURCE_WINDOW_COMPATIBILITY_OBSTRUCTION_VERIFIED`: source degree maps into $\Gamma(W_\nu)$, target-window matching, compact-provenance compatibility, cofinal-index compatibility, transition compatibility, and zero compatibility-defect rows are still absent.

(ii) Choose the simple-root representatives e_i^X, f_i^X, h_i^X in source degrees $\pm\gamma_i$, with $\alpha(\gamma_i) = \alpha_i \in W_\nu$; verify parity assignments via the GN multiplicity fibres.

(iii) Verify the Borcherds–Kac relations (Cartan, Chevalley, real Serre, orthogonality, super sign) on every relation whose source and target degrees lie in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$ and $W_\nu \cup (-W_\nu) \cup \{0\}$, respectively.

(iv) Compute the full finite parity dimensions $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\bar{p}}$ $(p = 0, 1)$ for $\gamma \in \Gamma(W_\nu)$; verify two-integer equality with $\mathrm{mult}^{\mathrm{GN}}_{\bar{p}}(\alpha(\gamma))$.

(v) Compute the Hopf pairing matrices in parity blocks; verify kernel agreement with GN/Kac, coideal stability, and transition compatibility.

(vi) Verify the no-extra-relations equality $\ker \pi_\nu = (\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}})_{\le W_\nu}$.

(vii) Verify generation by Cartan and simple representatives; compute PBW associated gradeds and verify strict transition preservation.

(viii) Verify exact triangular completion: $\lim^1 = 0$ on the triangular finite-window systems.

Obstruction classes. $\mathfrak{O}_{\mathrm{Rec},\nu}$ (§0.6); for the chiral identification, the finite Koszul comparison residual $\mathfrak{O}_{\mathrm{Kcmp},R}$ and the inverse-limit Koszul Mittag–Leffler condition of §0.5.

FINITE HN ORDER

At HN height R the finite datum has the following order:

$$
\begin{aligned}
(\sigma, S, R) &\longmapsto F_{\sigma,S}(R), \widehat{\Gamma}_R \\
&\longmapsto C_{\sigma,S,\le R}, \Phi^{\mathrm{red}}, o^{\mathrm{red}} \\
&\longmapsto \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \\
&\longmapsto (\mathrm{O1}), (\mathrm{O1})^+ \\
&\longmapsto m_R^{\mathrm{red}}, \Delta_R^{\mathrm{ch}}, \langle\, , \,\rangle_R, \overline{\Pi}^{\Theta}_{X,*} \\
&\longmapsto \mathfrak{D}^{D_0}_{X,R}, \mathcal{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, (\mathrm{O2}) \\
&\longmapsto C_{X,R}, b_{X,R}, \Theta_{\mathrm{Kos},R} \\
&\longmapsto (\mathrm{R0})\text{–}(\mathrm{R8}) \text{ on } W_\nu
\end{aligned}
$$

The pro-object

$$
\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \left(\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathcal{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\right)
$$

is the inverse limit when every finite obstruction class vanishes compatibly.

Finite-stage theorems

Theorem 0.252 *(Finite-stage Dirac–Igusa pro-object; conditional on the finite obstruction classes).* Suppose the finite charge window, Hall atlas, hybrid carrier, orientation transport, Hall product, Pfaffian datum, Koszul source coalgebra, and primitive-recognition datum are defined at heights $R \in \mathcal{R}_{\mathrm{HN}}$, all finite obstruction classes vanish, the HN transitions are strict, and the finite morphism proof tables above have zero defects. Then these data determine a functor

$$R \longmapsto \mathfrak{D}^{\mathrm{DI}}_{X,R}$$

from $\mathcal{R}^{\mathrm{op}}_{\mathrm{HN}}$ to the finite-stage category of Definition 0.244. Equivalently, for every $R' \leq R$ there is a morphism $\rho_{RR'}$ satisfying (M1)–(M8), and the composition law $\rho_{R'R''}\rho_{RR'} = \rho_{RR''}$ holds for $R'' \leq R' \leq R$. The inverse limit $\mathfrak{D}^{\mathrm{DI}}_X = \varprojlim_R \mathfrak{D}^{\mathrm{DI}}_{X,R}$ is the resulting Dirac–Igusa pro-object of Definition 0.245.

Theorem 0.253 *(Pfaffian–Dirac identification at finite stage; with local orientation and wall data at R).* Suppose the finite geometric Pfaffian datum, orientation transport, type-II wall atlas, and primitive-recognition datum have vanishing obstruction classes at height R. Then on the cusp expansion

$$\mathrm{pf}_{X,R}|_{\mathfrak{c}_\infty} = 64q^{1/2}r^{1/2}s^{1/2} \prod_{\gamma \in \Gamma^{\Pi,+}_R \setminus \{0\}} (1 - x^\gamma)^{\operatorname{sdim} P^{\Pi,+}_{R,\gamma}},$$

the Pfaffian section matches the Borcherds product on the test window, with the orientation character $\varepsilon_{o,R}|_{W^{(2),\leq R}} = \nu_{\Delta_5}|_{W^{(2),\leq R}}$ on the type-II reflection truncation. The recognition map $\iota^{\mathrm{Prim}}_R : H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}) \to \mathfrak{g}_{\Delta_5,\leq R}$ is an isomorphism on the saturated finite window.

Relative theorem on $K3 \times E$

The finite Hall atlas, hybrid carrier, and orientation transport at the test window use the retained D0-HN datum through Theorem 7.7; the orientation descent at (O1) by the retained quotient-orientation datum and Theorem 7.13; these data at $(\mathrm{O1})^+$ by the chosen Weyl lifts and Theorem 7.14; the Pfaffian wall atlas at (O2) by the retained $(O2_{\mathrm{atlas}})$ datum and Theorem 7.19; the cofinal recognition criterion beyond the base terminal and saturated target audits is the unresolved compact-source recognition datum of Chapter 10.

The eight finite constructions give the Dirac–Igusa pro-object only when the obstruction classes vanish. Entry (L8) names $(D0)$, $(O1)$, $(O1)^+$, and $(O2_{\mathrm{atlas}})$, locates each in the construction, and carries the retained data and the wall-atlas transport through Appendix 7 in §0.8.

0.8 *Four named obstructions*

The eighth entry of the Dirac–Igusa pro-object is the obstruction theory. The cross-level identity ②↔③ between the imported Borcherds–Kac–Moody denominator algebra and the compact protected Pfaffian passes through four obstruction classes and the finite geometric Pfaffian datum. The first obstruction is reduced on $K3 \times E$ to the retained D0-HN datum in Appendix 7 (Theorem 7.7); $(O1)$ and $(O1)^+$ are reduced there to the retained quotient-orientation and Weyl-transport data (Theorems 7.13, 7.14); the fourth is reduced to the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$ by Theorem 7.19. The finite geometric Pfaffian datum (P_{fin}) of Definition 0.200 supplies the cosection-reduced skew perfect complexes, Pfaffian line, finite Pfaffian section, type-II wall divisor order, automorphic line-comparison, support and parity rows, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility for the cofinal HN product. The principal Pfaffian and orientation theorem

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) = \Delta_5, \qquad \varepsilon_o = \nu_{\Delta_5}$$

holds relative to the first five components of the retained obstruction datum $\mathfrak{R}_X^{\mathrm{DI}}$, namely $(D0_{\mathrm{HN}}, O1_{\mathrm{quot}}, \mathcal{T}_W, O2_{\mathrm{atlas}}, P_{\mathrm{fin}})$. The Lie-superalgebra recognition $P_X \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source recognition datum $(S1)$–$(S10)$ of §0.6. On $K3 \times E$ the D0 criterion, the quotient-orientation criteria, and the conditional wall-atlas transport are stated in Appendix 7; the compact-source recognition datum is a separate hypothesis. The retained obstruction data are the components of the tuple $\mathfrak{R}_X^{\mathrm{DI}}$ of Definition 7.4. Thus "retained datum" in this section means a projection of that tuple, not a scalar normalization or a theorem label.

The fourth obstruction is the type-II wall-atlas datum; the Appendix G theorem transports that datum along the Weyl orbit and compares its character with ν_{Δ_5}.

(D0) D0-DEGENERATION LIMIT OF THE COSECTION-REDUCED D-CRITICAL ORIENTATION THEORY

Obstruction. At every finite HN height R and every retained charge stratum $C \in C_{\sigma,S,\leq R}$, the cosection-reduced d-critical structure on the K3-fibre direction must extend through a retained flat $D0$-degeneration family to the zero-dimensional sector on $K3 \times E$. Equivalently, the datum $(D0_{\mathrm{HN}})$ of Definition 7.6 supplies the finite derived degeneration family, the HN transition morphisms, the reduced obstruction theory, the additive K3 semiregularity cosection $\mathrm{CS}_{C,R}$, the reduced vanishing-cycle specialization, the Joyce–Upmeier orientation specialization, the Pfaffian-line and protected-integration specialization, and strict Mittag–Leffler exactness. The filtered Hall additivity equation

$$q^* \,\mathrm{CS}_C = p_1^* \,\mathrm{CS}_A + p_2^* \,\mathrm{CS}_B$$

is part of this datum on extension stacks throughout the limit. The local/wrapped boundary $b_R^{\mathrm{geom}} \to 0$ belongs to the hybrid Ran carrier; it enters $(D0)$ only through compatibility of the retained Hall correspondences with the zero-dimensional limit.

Finite site. The finite Hall atlas and hybrid carrier of §0.7. Failure of (D0) prevents construction of the reduced Hall product m_R^{red}, the orientation gerbe descent of (O1), and the Pfaffian line of (D0-Pf) on the wrapped sector.

Criterion. Criterion on $K3 \times E$ in Theorem 7.7, relative to the retained D0-HN datum of Definition 7.6. The fibrewise K3 framework of [54, 56] supplies the orientation technology on the projection-finite stratum; the wrapped stratum limit is part of the retained datum of Appendix 7.2. The finite D0 proof tables of Definition 7.6 are the checkable form of this datum. A Hilbert-scheme scalar specialization, a K3 scalar test, or a scalar trace row is not a D0-HN transition row.

(O1) STRONG REDUCED ORIENTATION ON THE $K3 \times E$-QUOTIENT SELF-EXT FRAME BUNDLE

Obstruction. On the cosection-reduced moduli $\mathfrak{M}_{C,R}^{\mathrm{red}}$ at every finite HN height R, the orientation line o_C has chosen square root $o_C^{\otimes 2} \simeq \det K_C^{\mathrm{red}}$ [56], multiplicative under reduced extension correspondences, with the quotient-descent equations: vanishing of the reduced gerbe class

$$\alpha_C^{\mathrm{red}} = w_2(\det K_C^{\mathrm{red}}) + \mathrm{cs}^* w_2(\det \operatorname{coker} \mathrm{cs}) \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2),$$

the free E-class $\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2)$, and, for each finite-stabilizer stratum S with stabilizer $H \subset E_{\mathrm{tors}}$, the equivariant gerbe class $\widetilde{\beta}_{C,S}^H \in H_H^2(S; \mathbb{F}_2)$ together with the choice of zero $H^1(BH; \mathbb{F}_2)$-linearization character $\lambda_{C,S}^H = 0$. The statement applies on every charge stratum used by the Hall correspondences and is compatible with HN transitions.

Finite site. The orientation descent datum of §0.7; (O1) datum of §0.2. Failure of (O1) prevents quotient-after-correspondence descent and obstructs the reduced Hall product, the Pfaffian line on the reduced quotient, and the orientation character.

Criterion. Theorem 7.13 is a quotient-orientation criterion relative to the retained ($O1_{\mathrm{quot}}$) datum. The Klein-four condition at $H = E[2]$ reduces the edge class to $r_1 = r_2 = r_3 = 0$, but quotient orientation also requires the zero degree-one linearization character. For even $N \geq 4$ the full two-primary residual includes the mixed coefficient $A_{12}^{(N)}$, which cyclic restrictions do not detect.

(O1)$^+$ Weyl-equivariant transport along $W^{(2)}(\Lambda_{II}^{2,1})$

Obstruction. The orientation line of (O1) carries Weyl-equivariant lifts of the three type-II wall transports $\tau_i : o_C \to s_{\delta_i}^* o_C$ ($i = 1, 2, 3$) such that:

(i) the 2-cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}_2})$ on the finite partial action groupoid $\mathcal{G}_R$ has vanishing cohomology class $[c_{o,R}] = 0$ in $H^2(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2)$ on each complete orbit, with chosen 1-cochain killing $c_{o,R}$;
(ii) $\tau_i^2 = 1$ for $i = 1, 2, 3$ and the Coxeter relations of $W^{(2)}(\Lambda_{II}^{2,1})$ hold on the lifts;
(iii) the quotient-orientation cocycle $\mathfrak{q}_C$ transports along τ_i between charge strata over Weyl-related Gram degrees, with vanishing torsor defect $\omega_{i,C} = 0$.

Finite site. The Weyl-transport datum of §0.7; (O1)$^+$ datum of §0.2. Failure of (O1)$^+$ obstructs the orientation character ε_o and prevents recognition of the Weyl orientation pattern as $\nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$.

Conditional theorem. The equivariance part of Theorem 7.14 is conditional on the retained quotient-orientation datum and chosen Weyl lifts. The local rank-one Pfaffian crossing supplies the local sign $\varepsilon_{\delta_i,R}^{\mathrm{loc}} = -1$ on a generic chart; the character comparison also uses the retained ($O2_{\mathrm{atlas}}$) datum. The global Coxeter coherence on $W^{(2)}(\Lambda_{II}^{2,1})$ orbits is established by the no-braid type-II Coxeter presentation and Maass-character match of Appendix 7.4.

(O2) Local Pfaffian wall normal form on type-II walls

Obstruction. At every finite HN height R and every retained component, boundary stratum, and chart overlap of the three simple type-II walls $\mathbb{H}_{\delta_i}$, $i = 1, 2, 3$, the cosection-reduced self-Ext complex splits as

$$K_{\delta_i,R}^{\mathrm{red}} \simeq K_{\delta_i,R}^{\tan} \oplus K_{\delta_i,R}^{\mathrm{nor}},$$

with s_{δ_i}-equivariant tangent factor and the rank-one normal form

$$K_{\delta_i,R}^{\mathrm{nor}} \simeq [\mathbb{R} \xrightarrow{u_{\delta_i,R}} \mathbb{R}], \qquad N_{\delta_i,R}^{\mathrm{Pf}} = 1,$$

with normal Pfaffian unit $\mathrm{Pf}(K_{\delta_i,R}^{\mathrm{nor}}) = v_{\delta_i,R} u_{\delta_i,R}$, $s_{\delta_i}(v_{\delta_i,R}) = v_{\delta_i,R}$, $s_{\delta_i}(u_{\delta_i,R}) = -u_{\delta_i,R}$. The normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps agree on chart overlaps. The half-Hilbert orbit sign sum of size $4096 = 2^{12}$ on the retained components and boundary strata of the $W^{(2)}(\Lambda_{II}^{2,1})$-orbit reproduces the reflection character $\nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$.

Finite site. The Pfaffian wall-atlas datum of §0.7; rank-one Pfaffian crossing of §0.4. The finite proof surface is wall_objects, wall_charts, overlaps, orbit_transport, half_hilbert_orbit, compatibilities, and scalar_separation; these rows distinguish the rank-one local sign, the global half-Hilbert atlas, and the scalar equality $2^{12} = 64^2$. Failure of (O2) prevents the local Pfaffian section $\mathrm{pf}_{\delta_i,R} = v_{\delta_i,R} u_{\delta_i,R}$ from descending to the orientation character and obstructs the section-level identification $\mathrm{pf}_X = \Delta_5$ on the wall complement.

Conditional theorem. Conditional on $(O2_{\mathrm{atlas}})$ in Definition 7.1, Theorem 7.19. The local rank-one crossing model is established (Proposition 0.205); component, boundary, and overlap compatibility on the full half-Hilbert orbit is part of the retained wall-atlas datum and is transported by Theorem 7.19. The $4096 = 2^{12}$ sign count is the scalar shadow of that retained datum, as separated in Appendix 7.5. The full normal-form theorem with this sign count is relative to Appendix 5.

Joint theorem with recognition datum

Theorem 0.254 *(Pfaffian–Dirac identification with primitive recognition datum).* On $K3 \times E$, suppose (D0), the retained $(O1_{\mathrm{quot}})$ datum, the $(O1)^+$ Weyl lifts, and $(O2_{\mathrm{atlas}})$ hold at every finite HN height R with strict HN transitions as in Appendix 7. Suppose the finite geometric Pfaffian datum (P_{fin}) of Definition 0.200 is supplied on a cofinal HN system. Suppose the recognition data (R0)–(R8) of §0.6 are supplied on a cofinal sequence of saturated downward finite windows. Suppose the chiral Koszul source datum of §0.5 is supplied, together with the finite Koszul comparison datum of Definition 0.211 and the strict Koszul Mittag–Leffler exactness hypothesis. Then

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5, \qquad \varepsilon_o = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}, \qquad P_X \cong \mathfrak{g}_{\Delta_5},$$

and $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$.

The eight finite constructions of §0.7, the recognition theorem of §0.6, and the Koszul comparison of §0.5 give the displayed implication. The Pfaffian and orientation conclusion holds by the retained D0-HN datum, the retained $(O1_{\mathrm{quot}})$ datum, the chosen Weyl lifts, and the retained $(O2_{\mathrm{atlas}})$ datum; the primitive Lie-superalgebra and current-envelope conclusions require the cofinal recognition datum, the Koszul source datum, and the finite Koszul comparison datum. The current-envelope conclusion uses the finite comparison compatibility of Proposition 0.212; without that datum, the source cobar quasi-isomorphism need not preserve the Weyl action, Pfaffian orientation character, Hall pairing, radical quotient, or PBW filtration used in recognition.

The 4096 sign sum and the (O2) wall atlas

The constant $4096 = 2^{12}$ is the size of the half-Hilbert orbit of the type-II reflection group on the retained finite-window components and boundary strata of the type-II wall configuration. At the local rank-one chart, each retained component contributes a sign $\varepsilon^{\mathrm{loc}}_{\partial_i,R} = -1$; the global sign system is the retained (O2) wall atlas of Appendix E transported along the chosen Weyl lifts, and its character is $\nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$. The Oberdieck–Pixton branch $Z_{\mathrm{OP}}^X = -4096\,\Delta_5^{-2}$ is a scalar convention and is not the source of the orientation character. The equality $\varepsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$ is relative to the retained quotient orientation, the chosen Weyl lifts, and the (O2) wall atlas; the local rank-one Pfaffian computation gives only the seed sign.

Compatibility with the companion volumes

The four obstruction classes are source-side geometric data. Their automorphic shadow is smaller: the Gritsenko–Cléry catalogue fixes the Borcherds products, weights, characters, and denominator rows; it does not record the $K3 \times E$ quotient orientation, the Weyl-equivariant determinant-line lifts, or the Pfaffian wall atlas. Thus (D0) is the cosection-reduced $K3$-fibre degeneration datum, (O1) is the reduced orientation gerbe with finite-stabilizer descent, $(O1)^+$ is the Weyl-equivariant determinant-line transport, and (O2) is the retained type-II Pfaffian wall atlas. The relation-closed window tests compatibility of these source data with the target BKM relations; it is not an equality theorem between geometric obstruction classes and automorphic catalogue rows.

The inverse limit on the HN system $\mathcal{R}_{\mathrm{HN}}$ contains the hybrid carrier, the cosection-twisted orientation, the Hall correspondences, the protected Pfaffian, the Koszul source coalgebra, the primitive recognition apparatus, the finite constructions, and the retained orientation and wall-atlas data into

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R (\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \ldots, \mathrm{Rec}_R).$$

The Pfaffian and orientation part of ② $\leftrightarrow$ ③ is then the retained-data theorem of Chapter 7; the Lie-superalgebra recognition remains conditional on $(S1)$–$(S10)$.

Part III

Pfaffian–Dirac Identity

Chapter 7

The cross-level identity ② ↔ ③: Pfaffian–Dirac, orientation character, primitive recognition

The Igusa cusp form Δ_5 of weight five on $\mathbf{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2$ carries three names, each rooted in a distinct categorical-dimensional level:

- $\mathcal{D}_X = \Delta_5$ — the Borcherds–Gritsenko–Nikulin section of the weight-5 automorphic line over the genus-two Siegel space (View ①);
- $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ — the denominator of the generalized Borcherds–Kac–Moody (BKM) superalgebra $\mathfrak{g}_{\Delta_5}$ on $\Lambda^{2,1}_{II}$ (View ②);
- $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ — the protected Pfaffian of the Dirac–Igusa pro-object on $X = K3 \times E$ (View ③).

The cross-level identity ①↔② is Borcherds 1995 and Gritsenko–Nikulin 1998. The edge ②↔③ has two pieces. The four obstruction classes and the finite geometric Pfaffian datum (P_{fin}) give the Pfaffian; the orientation classes give the orientation character on $K3 \times E$. The Lie-superalgebra recognition has the additional finite compact-source datum $(S1)$–$(S10)$ and the finite Koszul comparison datum of Definition 0.211. The three statements have separate hypotheses:

(i) the *Pfaffian–Dirac theorem*: under (D0), (O1), (O1)$^+$, (O2), and the finite geometric Pfaffian datum (P_{fin}) on a cofinal HN system,

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) \;=\; \Delta_5$$

as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$;

(ii) the *orientation character identity*: under (O1)+(O1)$^+$+(O2),

$$\epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II});$$

(iii) the *primitive recognition theorem*: under the finite compact-source recognition datum (Cartan, simple representatives, Chevalley, real Serre, imaginary orthogonality, generation, completed PBW, Hopf radical equality, no-extra-relations, full parity dimensions),

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)\big/\overline{\mathrm{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X} \;\cong\; \mathfrak{g}_{\Delta_5}$$

after passage to the inverse limit over the cofinal root windows, with the chiral cobar $\Omega^{\mathrm{ch}}_E C_X$ identified as the formal current envelope $\mathtt{Den}_{\Delta_5,E}$ only after the chiral Koszul source datum is also fixed.

The theorem-dependency ledger for these three assertions is

beilinson_levels, objects, morphisms, theorem_dependencies, dependency_edges, forbidden_edges, status_sep

It is the finite graph recording which retained packets enter each proof and which edges are forbidden. In particular, the Pfaffian and orientation proofs do not use the mirror conjecture, the gravity-line residual, the scalar trace, or the OP scalar branch; primitive recognition does not use the scalar trace, the denominator product, a target counit, or target labels as source representatives. Claim statuses do not collapse: imported BKM and automorphic identities, retained-data Pfaffian identities, conditional recognition, mirror conjectures, scalar trace identities, and open gravity-line residuals occupy distinct rows. The definition-separation rows assert non-circularity at the level of finite data: the Pfaffian line is not defined by the automorphic section or by its desired value, the orientation character is not defined by scalar signs, the chiral Koszul source is not the target counit, the primitive source is not defined by target multiplicities, and the Dirac–Igusa pro-object is not defined by a scalar shadow. The obstruction-independence rows assert that $(D0)$, $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$, P_{fin}, the Koszul source comparison, and primitive recognition are separate finite data; typing dependencies are not logical implications between vanishing statements. The certificate `certificates/dependencies/k3e_theorem_dependencies` is schema-complete only: its verifier returns `SCHEMA_COMPLETE_SCHEMA_ONLY` and records `dependency_certification=false` and `mathematical_certification=false`. It verifies that the finite ledger is typed, populated, separated by status, and guarded by forbidden-edge, definition-separation, and obstruction-independence rows; it does not prove the retained Pfaffian, orientation, Koszul, recognition, mirror, scalar-trace, or gravity-line claims.

Proposition 0.1 *(Hypothesis-deletion residues).* The theorem lane has the following exact residues when one named datum is omitted.

(i) Without the finite compact-source recognition datum $(S1)$–$(S10)$, the Pfaffian–Dirac section identity and the orientation character identity remain under their own hypotheses, but there is no theorem identifying
$$P_X^{\Pi,\mathrm{red}} \quad \text{with} \quad \mathfrak{g}_{\Delta_5}.$$
The surviving primitive object is only the retained Hall primitive object with its supplied product, coproduct, pairing, radical, and PBW rows; denominator exponents or signed multiplicities do not determine its Lie bracket.

(ii) Without the chiral Koszul source datum, the finite Koszul comparison datum, and strict Koszul Mittag–Leffler exactness, the source coalgebra C_X is not identified with the target current envelope. The theorem no longer supplies a quasi-isomorphism
$$\Omega_E^{\mathrm{ch}} C_X \longrightarrow \mathsf{Den}_{\Delta_5,E}$$
with acyclic cone. The exact residue is the compact Hall primitive source, if supplied, before the source-to-target chiral comparison.

(iii) Without the finite geometric Pfaffian datum (P_{fin}), the obstruction data may still define orientation and local wall signs, but there is no global object
$$\mathrm{pf}_{X,R} \in H^0(\mathfrak{Q}_R, \mathscr{L}_{\mathrm{Pf},R})$$
whose inverse limit can be compared with $\mathcal{D}_X = \Delta_5$. The residue is a reduced orientation/wall datum, not a Pfaffian section identity.

(iv) Without $(O2_{\mathrm{atlas}})$, the retained D0-HN datum, quotient orientation, and Weyl-lift torsor do not compute the local rank-one Pfaffian signs. The residue is a Weyl-transported orientation line on the retained quotient, but no theorem gives $\epsilon_o(s_{\partial_i}) = -1$, no divisor order along the type-II walls is computed, and the Pfaffian section is not identified with Δ_5.

(v) Without $(O1)^{+}$, the quotient orientation may exist, but it is not transported by a coherent action of $W^{(2)}(\Lambda^{2,1}_{II})$. The residue is an oriented reduced quotient on the retained finite stages; there is no global character

$$\epsilon_{o} : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}.$$

(vi) Without $(O1)$, the quotient square root and reduced orientation line on the $K3 \times E$-quotient self-Ext frame bundle are absent. The residue is the finite d-critical/cosection and D0-specialization data before orientation; Hall products and Pfaffian sections are unsigned and do not define the protected oriented Pfaffian.

(vii) Without $(D0)$, the cosection-reduced orientation theory has no D0-degeneration endpoint on the cofinal HN pro-system. The residue is the formal Mukai–Gram lattice, the imported automorphic/BKM/theta triangle, and any finite stage objects supplied independently; it is not a compact $K3 \times E$ Pfaffian–Dirac theorem.

Each residue is a positive statement about the surviving object. None of the rows permits a scalar trace, an automorphic target, or a denominator product to replace the deleted datum.

Proof. The conclusion is a direct reading of the source and target of each comparison. Primitive recognition needs the source representatives, relations, radical, pairing, parity, and PBW rows; without them there is no Lie-superalgebra isomorphism. The Koszul comparison is the only map from the source coalgebra to the target current envelope. The finite geometric Pfaffian datum supplies the Pfaffian line and section, so deleting it removes the domain of the section identity. The $(O2_{\text{atlas}})$ datum supplies the local rank-one signs at the type-II walls; deleting it removes the generator values of the orientation character. The $(O1)^{+}$ datum supplies coherent Weyl transport; deleting it removes the homomorphism from the Weyl group. The $(O1)$ datum supplies the quotient orientation line; deleting it removes the oriented Pfaffian input. The $(D0)$ datum supplies the D0-degeneration endpoint and transition-compatible reduced orientation theory; deleting it removes the compact pro-limit in which the Pfaffian–Dirac comparison lives. These are object-level failures in the Beilinson chain, not failures of notation. □

PROPOSITION 0.2 *(Implication and non-redundancy of obstruction rows).* Let

$$\mathsf{Obs}^{\mathrm{DI}}_{\mathrm{fin}} = \mathsf{D0} \times \mathsf{O1} \times \mathsf{O1}^{+} \times \mathsf{O2} \times \mathsf{Pfin} \times \mathsf{Kos} \times \mathsf{Rec}$$

be the finite theorem-dependency product whose seven factors are the row categories for $D0_{\mathrm{HN}}$, $O1_{\mathrm{quot}}$, $\mathcal{T}_{W}$, $O2_{\mathrm{atlas}}$, P_{fin}, the chiral Koszul comparison datum, and the compact-source primitive recognition datum. Each vanishing predicate is a predicate on its own factor. The only implication used by the Pfaffian–Dirac lane is the conjunction of the relevant vanishing predicates into the corresponding theorem:

$$D0 \wedge O1 \wedge O1^{+} \wedge O2 \wedge P_{\mathrm{fin}} \Longrightarrow \text{Pfaffian section and orientation character},$$

and, separately,

$$\text{compact source} \wedge \text{Koszul comparison} \wedge \text{primitive recognition} \Longrightarrow P^{\Pi,\mathrm{red}}_{X} \cong \mathfrak{g}_{\Delta_5}.$$

There is no theorem-level implication from any one named vanishing predicate to any other. The type dependencies recorded in `obstruction_independence.csv` say only that a later datum is not meaningful until its earlier input object exists; they do not prove the earlier datum.

For each of the seven rows, deleting that row and leaving the other row slots formally zero gives a finite theorem-dependency witness to non-redundancy. The residual theorem in the deleted row is exactly the corresponding item of Proposition 0.1. These are counterexamples to illegitimate implications inside the finite dependency calculus. They are not asserted to be geometric $K3 \times E$ families with all other analytic hypotheses realized.

Proof. In the product category $\mathtt{Obs}^{\mathrm{DI}}_{\mathrm{fin}}$, a vanishing predicate is pulled back from one projection. A predicate pulled back from one projection cannot force a predicate pulled back from another projection unless such an implication is added as a row of the dependency ledger. The table `obstruction_independence.csv` requires `logical_implications=none` for the seven named data and records the type dependencies separately. Thus the row $O1^+$ being typed over $O1$ means that Weyl transport needs an orientation line; it is not a proof that the orientation line exists. The same applies to $O2_{\mathrm{atlas}}$, P_{fin}, the Koszul comparison, and primitive recognition.

Deleting one row removes the source, target, or comparison named in Proposition 0.1, while leaving the remaining row slots available as formal data. This supplies the finite-row countermodel to every attempted proof that the deleted row is redundant. Since each deleted row removes a different object or comparison map, no two rows have the same residue; the obstruction classes are non-redundant in the theorem-dependency calculus. □

The eight finite constructions of §5.7 define, at every finite stage R in the Harder–Narasimhan inverse system, the data $\mathcal{A}^{E_3}_{X,R}$, $F^{\mathrm{hyb}}_{X,R}$, $\Gamma_{X,R}$, Π_X, Φ_R, o_R, H_R, P^{Π}_R, $C_{X,R}$, $\Theta_{\mathrm{Kos},R}$, $\mathscr{L}_{\mathrm{Pf},R}$, $\mathrm{pf}_{X,R}$, $\varepsilon_{o,R}$, and Rec_R of the pro-object $\mathfrak{D}^{\mathrm{DI}}_X$; the four obstruction classes of §5.8 are exactly (D0), (O1), $(\mathrm{O1})^+$, (O2). These four classes act on the Pfaffian line and the orientation character; the assertion that Rec_R is eventually an isomorphism is the separate compact-source recognition datum. The relevant geometric data are the open factorization chart of the vacuum b, the protected Pfaffian functor, the Hall–Drinfeld correspondence, the chosen $\mathrm{Sp}^{\mathrm{ch}}$ specialization, the pro-object ambient, the D0-degeneration endpoint, the Bussi–Brav–Dupont–Joyce–Szendrői orientation, the non-degeneracy of the Pfaffian line, and the type-II wall normal form.

The Pfaffian and orientation theorems are retained-data theorems on $K3 \times E$, as in Appendix 7; the primitive recognition theorem is conditional on the compact-source datum $(S1)$–$(S10)$. The BKM denominator identity (View ②) is *proved*, imported from [8, 9, 46, 48, 58]; the automorphic identification (View ①) is *proved*, imported from [51, 52]; the operator-level Pfaffian datum $\mathfrak{D}^{\mathrm{DI}}_X$ is *proved on $K3 \times E$ relative to the retained D0-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts,* $(O2_{\mathrm{atlas}})$*, and the retained finite geometric Pfaffian datum* (P_{fin}) of Appendix 7; the chiral cobar identification is *conditional* on the Koszul source datum of Definition 0.208. The κ-tuple of the Calabi–Yau quantum-groups companion, $(\kappa_{\mathrm{cat}}, \kappa^{\mathrm{Hodge}}_{\mathrm{ch}}, \kappa^{\mathrm{Heis}}_{\mathrm{ch}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$, fixes the level of Δ_5: the cusp-form weight is $\kappa_{\mathrm{BKM}} = 5$ and the elliptic genus weight is $\chi_{\mathrm{top}}(K3) = \kappa_{\mathrm{fiber}} = 24$; the additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ has no invariant meaning on the κ-tuple.

1 Data and Obstructions

1.1 *The pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ and its eight finite-stage components*

Three names of Δ_5: the section $\mathcal{D}_X = \Delta_5$ (View ①); the BKM denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View ②); the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ (View ③, relative to retained data). The handle "Δ_5" is reserved for the working symbol; theorem-level sentences specify the named view.

The Dirac–Igusa pro-object is

$$\mathfrak{D}_X^{\mathrm{DI}} = \lim_R \big(\mathcal{A}^{E_3}_{X,R},\ F^{\mathrm{hyb}}_{X,R},\ \Gamma_{X,R},\ \Pi_X,\ \Phi_R,\ o_R,\ H_R,\ P_R^{\Pi},\ C_{X,R},\ \Theta_{\mathrm{Kos},R},\ \mathscr{L}_{\mathrm{Pf},R},\ \mathrm{pf}_{X,R},\ \varepsilon_{o,R},\ \mathrm{Rec}_R\big).$$

The components are the values of the eight finite constructions of §5.7, with the ambient category the cofinal HN inverse system of Definition 3.14; R ranges over a chosen cofinal subsystem $\mathcal{R}_{\mathrm{HN}} \subset \Lambda_{\mathrm{ample}}$. The transition maps are the finite-stage morphisms of Definition 0.244; isomorphism of two such pro-objects means pro-isomorphism in the sense of Definition 0.245. At finite stage, these transition and pro-isomorphism claims are the cofinal_subsystems, stage_objects, stage_morphisms, component_maps, composition_laws, ml_exactness, pro_isomorphisms, and scalar_firewall tables of §0.7. The label γ is the pro-object ambient. The universal property of this pro-object is the Hom formula of Proposition 0.246; in particular, changing the finite HN presentation by a cofinal subsystem does not change $\mathfrak{D}_X^{\mathrm{DI}}$ in the pro-category.

Remark 1.1 *(The three Δ_5-names).* The equality $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X$ is a cross-level identity, not a tautology. The right-hand side is an automorphic section; the left-hand side is the inverse limit of finite Pfaffian sections $\mathrm{pf}_{X,R}$ of the line bundles $\mathscr{L}_{\mathrm{Pf},R}$ on the reduced quotient stacks $\mathfrak{Q}_R$; the two are related by the fixed comparison $\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$. The denominator name $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is the same global section $\mathcal{D}_X$ written in the type-II chamber polarization that exposes the Weyl–Kac–Borcherds product expansion; the prefactor 64^{-1} and the doubling $2Z$ of the modular variable enter from the theta normalization of the Borcherds product on the index-4 sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$. The source side $P_X^{\Pi,\mathrm{red}}$ identifies with $\mathfrak{g}_{\Delta_5}$ only after the recognition criterion of §4 is established.

1.2 The four obstruction classes: (D0), (O1), (O1)$^+$, (O2)

The four classes match the eight finite constructions one-to-one only on the orientation lane. They are the obstruction data of Theorem 0.253. On $K3 \times E$, Appendix 7 reduces $(D0)$ to the retained D0-HN datum by Theorem 7.7; reduces $(O1)$ to the retained quotient-orientation datum by Theorem 7.13; reduces $(O1)^+$ to the chosen Weyl lifts by Theorem 7.14; and reduces $(O2)$ to the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$ by Theorem 7.19. The $(R1)$–$(R2)$ scalar separation for the type-II middle-wall constant $4096 = 2^{12} = 64^2$ inside $(O2)$ is Theorem 7.17. The finite D0 packet `certificates/d0/k3e_d0_hn` is presently verified only through its obstruction ledger: `D0_OBSTRUCTION_LEDGER_VERIFIED`, with `d0_certification=false`. Hence the D0-dependent assertions in this chapter are relative theorems until the flat degeneration, retained-substack, HN-transition, derived-enhancement, semiregularity-cosection, vanishing-cycle, orientation, Pfaffian, protected-integration, and Mittag–Leffler rows are supplied. The finite geometric Pfaffian packet `certificates/pfaffian/k3e_finite_pfaffian` is likewise verified only through its obstruction ledger: `PFAFFIAN_OBSTRUCTION_LEDGER_VERIFIED`, with `pfaffian_certification=false`. Hence the protected Pfaffian identity below is a relative section identity until the skew perfect complexes, Pfaffian lines, sections, wall charts, automorphic-comparison rows, and Pfaffian-line transition rows are constructed. The finite O2 wall-atlas packet `certificates/wall_atlas/k3e_o2_atlas` is also verified only through its obstruction ledger: `O2_OBSTRUCTION_LEDGER_VERIFIED`, with `o2_certification=false`. Hence every assertion using $(O2_{\mathrm{atlas}})$ is relative until the simple-wall coverage, wall-object, wall-chart, overlap, Weyl-transport, half-Hilbert-orbit, compatibility, type-I-exclusion, D0-limit, and scalar-separation rows are supplied. The finite orientation packet `certificates/orientation/k3e_reduced_orientation` is presently verified only through its obstruction ledger: `ORIENTATION_OBSTRUCTION_LEDGER_VERIFIED`, with `orientation_certification=false`. Thus the orientation-dependent assertions in this chapter are relative theorems until the square-root, quotient Cech–Borel, finite-stabilizer, Weyl-lift, Coxeter, and transition rows are supplied.

1.2.0.1 (Do) — Do-degeneration endpoint of the cosection-reduced d-critical orientation theory. At every R, the retained Do_{HN} datum of Definition 7.6 is fixed: a flat Do-degeneration family, retained finite-type object/extension/flag stacks, proper or closed HN transition morphisms, quasi-smooth d-critical derived enhancements, surjective additive K3 semiregularity cosections, reduced vanishing-cycle specialization, Joyce–Upmeier orientation specialization, Pfaffian-line and protected-integration specialization, and strict Mittag–Leffler exactness. Compatibly in R, these clauses give the HN-completed state object and the finite first-order block $\mathfrak{D}_X$. The finite first-order block also carries the exponent-matching clause

$$\operatorname{sdim} P^{\Pi}_{X,(n,l,m)} = f(nm, l) \qquad ((n, l, m) \in \Gamma_{\mathrm{eff}}),$$

where $f(n, l)$ is the Fourier coefficient of the weak Jacobi form $\phi_{0,1}$ of weight zero and index one. This clause is not a consequence of the reduced orientation alone; it is part of the retained finite-stage Dirac block. The cosection contraction of the Pantev–Toen–Vaquie–Vezzosi (PTVV) (−1)-shifted symplectic form by the K3 semiregularity cosection σ_{K3} is the Maulik–Toda / Oberdieck reduced obstruction theory [81, 90]; this absorbs the parity flip $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ into the reduced orientation line as the $\mathbb{Z}/2$-grading shift $o^{\mathrm{red}}_C = o^{\mathrm{std}}_C \otimes \Pi$.

1.2.0.2 (O1) — strong multiplicative reduced orientation datum on the reduced $K3 \times E$-quotient self-Ext frame bundle. A Joyce–Upmeier-type orientation line $o_C \in \operatorname{Pic}(\mathfrak{M}^{or}_X)$ with $o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C)$, extension-multiplicative under $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$, and equipped with the full quotient-orientation datum

$$\theta_C = \left(\theta^{\mathrm{red}}_C, \theta^{E,\mathrm{free}}_C, \{(\theta^H_C, \lambda^H_C = 0)\}_{H\in\mathcal{H}(C)}\right)$$

on every retained charge stratum. The vanishings $\alpha^{\mathrm{red}}_C = 0$, $\alpha^{E,\mathrm{free}}_C = 0$, $\widetilde{\beta}^H_C = 0$, $\lambda^H_C = 0$ are the orientation-line, the Borel-equivariant free-action, the residual finite-stabilizer Borel, and the residual finite-stabilizer linearization conditions; for $N = 2$, the Klein-four calculation $\beta^{E,E[2]}_C = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$ specializes the residual. For $2^a \parallel N$, $a \geq 2$, the rank-two coefficient $A^{(N)}_{12}x_1x_2$ is not detected by cyclic restrictions and is part of the finite-stabilizer orientation test of Proposition 0.101. The orientation lift technology is BBDJS [12] on the unreduced side and Joyce–Upmeier [55, 56] on the multiplicative side; the reduced $K3 \times E$-quotient theorem is separate.

1.2.0.3 (O1)⁺ — Weyl-equivariant determinant-line transport along $W^{(2)}(\Lambda^{2,1}_{II})$ reflections. The chosen lifts $\tau_i : o_C \to s^*_{\delta_i} o_C$, $i = 1, 2, 3$, of the three type-II wall transports satisfy $[c_o] = 0$ in $H^2(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2)$ (equivalently, in the finite action-groupoid cohomology $H^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$ at every R), $\widetilde{\tau}_i^2 = 1$, and the quotient-orientation transport defects $\omega_{i,C} = 0$ for every simple reflection. Then $\widetilde{\tau}_i$ define an honest Weyl action on the reduced quotient orientation; the passage from the three generator signs to a character on $W^{(2)}(\Lambda^{2,1}_{II})$ is the group-theoretic part of Definition 0.109.

1.2.0.4 (O2) — Pfaffian local model and wall atlas on type-II walls. At a generic point of each $\mathbb{H}_{\delta_i}$, the chosen real local equation $u_i = u_{\delta_i}$ with $s_{\delta_i}(u_i) = -u_i$, the normal determinant-complex splitting

$$K^{\mathrm{nor}}_{\delta_i,R} \simeq \left[\mathbb{R} \xrightarrow{u_i} \mathbb{R}\right], \qquad \operatorname{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = v_{\delta_i,R}u_i, \qquad s_{\delta_i}(v_{\delta_i,R}) = v_{\delta_i,R},$$

and the rank-one normal-form rank $N^{\mathrm{Pf}}_{\eth_i} = 1$ hold. The wall charts cover every retained component and boundary stratum of $\mathbb{H}_{\eth_i}$; on overlaps the tangent/normal splittings, normal Pfaffian units, quotient orientations, and protected integration maps agree after a zero Čech sign class. The retained atlas datum must contain, for the middle wall $\eth_2 \leftrightarrow (0,1,1)$, a wrapped wall object $x_{\eth_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ with $b^{\mathrm{geom}}_R(\eta) > 0$ and $\overline{\Pi}_X(\eta) = (0,1,1)$. Proposition 5.2 supplies only the rank-one local Pfaffian sign on such a chart; the component, boundary, overlap, quotient, and protected-integration compatibilities are part of $(O2_{\mathrm{atlas}})$.

1.3 The OP minus sign is a scalar branch convention

The Oberdieck–Pixton (OP) signed scalar is $C_{\mathrm{OP},X} = -4096$. Its absolute square-normalization is $C_{\square,X} = 4096 = 64^2$, and its sign is the OP scalar branch recorded by the fermionic shift $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber. It is not the source of the orientation character ϵ_o; orientation lines are killed by squaring. The orientation character is computed by monodromy of the Pfaffian section around the type-II walls (Lemma 5.2); equivalently, by the $(-1)^{N^{\mathrm{Pf}}_{\eth_i}}$ values from (O2).

2 The Pfaffian–Dirac theorem

The protected Pfaffian of the Dirac–Igusa pro-object is compared with the automorphic section.

Theorem 2.1 *(Pfaffian–Dirac section identity).* Granted the four obstruction classes of Section 1.2, with $(O2)$ understood as the retained wall-atlas datum $(O2_{\mathrm{atlas}})$, and granted the finite geometric Pfaffian datum (P_{fin}) of Definition 0.200 on a cofinal HN system, including skew perfect complexes, Pfaffian line and section, support, parity, active-support coverage, type-II wall divisor order, leading coefficient, Pfaffian-line comparison, and Pfaffian-line Mittag–Leffler compatibilities, the protected Pfaffian of the Dirac–Igusa pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ equals the automorphic section $\mathcal{D}_X = \Delta_5$:

$$\boxed{\iota_{\mathrm{aut}}\Big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)\Big) \;=\; \mathcal{D}_X \;=\; \Delta_5}$$

as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ over the genus-two Siegel space $\mathbf{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2$, where

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$$

is the fixed Pfaffian-to-automorphic line comparison, compatible with the cusp trivialization at $\mathfrak{c}_\infty$ and with squaring to $\mathcal{L}^{10}$. Equivalently, in the type-II Borcherds chamber,

$$\iota_{\mathrm{aut}}\Big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)\Big)\Big|_{\mathfrak{c}_\infty} = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

After squaring, the orientation forgets and the OP scalar branch identifies the level-n protected scalar shadow with the inverse squared Pfaffian:

$$Z^X_{\mathrm{OP}} \;=\; -64^2\,(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X))^{-2} \;=\; -4096\,\Delta_5^{-2},$$

in agreement with [92], where $4096 = 64^2 = ([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ is the squared theta-normalization and the leading minus sign is the OP scalar branch convention. The BPS normalization rescales by -4096^{-1}: $Z^{K3\times E}_{\mathrm{BPS}} = -4096^{-1} Z^X_{\mathrm{OP}} = \Delta_5^{-2}$.

Proof. The inverse-limit assembly consists of a Pfaffian product, a leading coefficient identification with a known weight-five Siegel form, and a q-expansion principle on a punctured neighbourhood of the type-II cusp $\mathfrak{c}_\infty$.

Finite-stage Pfaffian product ($\delta + \varepsilon$). The retained Do-HN datum supplies the cosection-reduced d-critical pro-limit on the retained HN system. The finite geometric Pfaffian datum (P_{fin}) supplies, at every cofinal R, the cosection-reduced skew perfect complexes $K^{\mathrm{red}}_{C,R}$, the first-order block $\mathfrak{D}_{X,R}$ on the reduced compact quotient $\mathfrak{Q}_R = (\mathfrak{M}_R/E)^{\mathrm{red}}$, the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$, the section $\mathrm{pf}_{X,R}$, active-support coverage, the support and parity residual vanishings, the type-II wall divisor order, and the normalized leading coefficient $\mathfrak{o}^{\mathrm{lead}}_R = 0$. Hypothesis (O1) supplies the orientation datum whose square-root line o_C satisfies $o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C, C)$ and extension-multiplicativity $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$, descended under the reduced E-quotient. The protected Pfaffian functor of (β) is then the graded Pfaffian product of the determinant of $\mathrm{RHom}_{\mathrm{red}}$ over the retained charge strata. The exponent-matching clause of the retained finite-stage Dirac block gives

$$\mathrm{sdim}\, P^{\Pi}_{X,(n,l,m)} = f(nm, l) \quad ((n, l, m) \in \Gamma_{\mathrm{eff}}),$$

so the graded Pfaffian product, expanded in the type-II cusp polarization $(q, r, s) = (e^{2\pi i z_1}, e^{2\pi i z_2}, e^{2\pi i z_3})$, is

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_{X,R}) = v_0 \prod_{(n,l,m)\in\Gamma_{\mathrm{eff,R}}} (1 - q^n r^l s^m)^{f(nm,l)}, \qquad v_0 = 64 q^{1/2} r^{1/2} s^{1/2},$$

where $\Gamma_{\mathrm{eff,R}}$ is the active support cut by HN height R. The Weyl monomial v_0 is the theta normalization $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$ of [46, Theorem 4.1]. The finite normalization table `delta5_theta_leading` records the scalar equalities $64^2 = 4096$, $D_5 = 64^{-1}\Delta_5$, and $\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_{10}$, together with separate convention rows for Δ_{10}, χ_{10}, Φ_{10}^{un}, the OP monic form, and the inverse scalar branches; it has no Pfaffian-line, orientation-character, or O2 wall-atlas content. This product is the cusp expansion of the supplied Pfaffian section, not a replacement for that section: by Proposition 0.201, the scalar product, its square, and the OP scalar do not determine $\mathscr{L}_{\mathrm{Pf},R}$, $\mathrm{pf}_{X,R}$, or $\epsilon_{o,R}$ without the Pfaffian-line and line-comparison data.

Inverse limit (γ). The pro-object ambient is the cofinal HN system; the retained Do-HN datum and finite geometric Pfaffian datum supply strict transition maps with vanishing successor residuals $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = \mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$; the finite Pfaffian sections $\mathrm{pf}_{X,R}$ form a Mittag-Leffler system on the active support, and the inverse limit is the formal Borcherds product

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})\big|_{\mathfrak{c}_\infty} = \lim_R \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_{X,R}) = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Class M completion (chain-level after weight-completion, in the sense of Vol I's class M global triangle) is exactly the ambient in which $\lim_R$ commutes with the graded Pfaffian product on the active support; this is the γ clause.

Comparison with the imported Borcherds product (β). By Proposition 7.2, the formal product $64\, q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ equals the Fourier expansion of the automorphic section $\mathcal{D}_X = \Delta_5$ at $\mathfrak{c}_\infty$; this is the Borcherds–Gritsenko–Nikulin theorem [9, 46, 48]. The fixed $\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$ of (β), compatible with cusp trivialization and squaring, transports the inverse-limit Pfaffian to a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

q-expansion principle (α). The chart α is the open factorization category C on (X, D, τ) with boundary vacuum b: on the punctured neighbourhood of $\mathfrak{c}_\infty$ inside $\mathrm{Sp}(2, \mathbb{Z})\backslash\mathcal{H}_2$ the chart algebra

$A_b = \mathrm{End}_C(b)$ supplies the local factorization-algebra structure within which the Pfaffian section is holomorphic. Two holomorphic genus-two Siegel modular forms with the same Maass character ν_{Δ_5} and the same Fourier expansion at $\mathfrak{c}_\infty$ are equal: this is the q-expansion principle for holomorphic Siegel modular forms [51, 52]. Both sides have weight 5 and Maass character ν_{Δ_5}: the right-hand side by the Borcherds–Gritsenko–Nikulin theorem; the left-hand side because the orientation character of the retained $\mathfrak{D}_X^{\mathrm{DI}}$, under (O1)+(O1)$^+$+(O2), is exactly ν_{Δ_5} on $W^{(2)}(\Lambda_{II}^{2,1})$ (this is Theorem 3.1). Hence $\iota_{\mathrm{aut}}(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})) = \Delta_5$ as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

Squared scalar (β). Squaring forgets orientation: the squared line $\left(\mathcal{L}^5 \otimes \nu_{\Delta_5}\right)^{\otimes 2} = \mathcal{L}^{10}$ is trivially charactered, and $\Delta_5^2 = \Delta_{10}$ is scalar. The OP scalar normalization of Convention 1.1 fixes the partition function $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$, with $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ and $D_5 = 64^{-1}\Delta_5$. Composition of the squared Pfaffian section with the OP scalar branch gives the OP chamber scalar representative

$$Z_{\mathrm{OP}}^X \;=\; -\,(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}))^{-2} \cdot C_{\square,X}, \qquad C_{\square,X} = 4096,$$

identifying $Z_{\mathrm{OP}}^X = -4096\,\Delta_5^{-2}$; the minus sign is the OP scalar branch convention, recorded by $(-p_{\mathrm{DT}})^n$, and is not a feature of the orientation character. The protected scalar trace is $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$. $\square$

Remark 2.2 *(Why (D0)+(O1)+(O1)$^+$+(O2)+(P_{fin})).* Each named datum is essential. (D0) gives the existence and the finite first-order block; without it there is no $\mathfrak{D}_{X,R}$ and no Pfaffian product to take. (O1) gives the orientation line and the reduced E-quotient descent; without the line the Pfaffian section has no domain. (O1)$^+$ gives the honest Weyl action on the reduced quotient orientation; without it the type-II reflection signs are projective and the orientation character is projective, not a homomorphism. (O2) gives the local Pfaffian normal form; without it the rank $N_{\delta_i}^{\mathrm{Pf}}$ is undetermined and the local sign cannot be computed. The finite geometric Pfaffian datum (P_{fin}) gives the skew perfect complexes, Pfaffian line, section, support, parity, active-support coverage, wall divisor order, leading coefficient, line comparison, and Pfaffian-line Mittag–Leffler compatibility; without it the inverse-limit product has no section-level domain and no determinant-line object. If one of these data is omitted, one of the proof steps loses its object.

3 The orientation character identity

The second theorem identifies the geometric orientation monodromy character ϵ_o of the Dirac–Igusa pro-object with the automorphic Maass character ν_{Δ_5} on the type-II Weyl group. The proof splits into a local Pfaffian wall computation, a group-theoretic extension, and an independent automorphic check.

Theorem 3.1 *(Orientation character).* Recall the type-II simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ on $\Lambda_{II}^{2,1} \subset \Lambda^{2,1} = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$ (Equation (4.4)). Granted the orientation classes (O1), (O1)$^+$, (O2), the geometric orientation monodromy character of the Dirac–Igusa pro-object equals the automorphic Maass character on the type-II Weyl group:

$$\boxed{\epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda_{II}^{2,1})}\,.$$

The character takes the value -1 on each of the three type-II simple reflections $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$, and the identity is reduced to these three values by the abelianization $W^{(2)}(\Lambda_{II}^{2,1})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^3$ (Definition 0.109).

The identity is a comparison of two characters defined independently: ϵ_o is computed by transport of the Pfaffian section under the chosen Weyl action of (O1)$^+$; ν_{Δ_5} is the Maass multiplier of the holomorphic

Siegel form $\mathcal{D}_X = \Delta_5$ computed from the multiplier system of the corresponding theta lift [45, 46, 76]. No scalar branch convention enters this sign comparison: replacing Δ_5 by $c\Delta_5$, $c \in \mathbb{C}^\times$, changes neither the Maass character nor the divisor orders. The factor 64 names the monic Borcherds product $D_5 = 64^{-1}\Delta_5$; the OP/DT scalar -4096 is unused.

Proof. Three inputs combine: Proposition 5.2 computes the local Pfaffian signs on the three simple type-II walls; Definition 0.109 extends a generator sign-tuple $(\epsilon_1, \epsilon_2, \epsilon_3) \in \{\pm 1\}^3$ to a unique character on $W^{(2)}(\Lambda^{2,1}_{II})$; and Lemma 7.2 computes the automorphic target values $\nu_{\Delta_5}(s_{\delta_i}) = -1$ directly from the Maass multiplier and the Borcherds divisor order $d^{\mathrm{aut}}_{\delta_i} = 1$.

Local Pfaffian signs from $(O2)$. By (O2), at a generic point of each $\mathbb{H}_{\delta_i}$, $i = 1, 2, 3$, the chosen local equation u_i satisfies $s_{\delta_i}(u_i) = -u_i$ and the normal determinant-complex splits as $K^{\mathrm{nor}}_{\delta_i} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$ with $\mathrm{Pf} = v_{\delta_i} u_i$ and $s_{\delta_i}(v_{\delta_i}) = v_{\delta_i}$. Transport along the closed path ℓ_{δ_i} (the path in the reduced quotient wall-complement carrying $u_i = \epsilon$ to $u_i = -\epsilon$, closed by the $(\mathrm{O1})^+$ lift $\widetilde{\tau}_i$) sends $v_{\delta_i} u_i$ to $-v_{\delta_i} u_i$, giving

$$\epsilon_o(s_{\delta_i}) = (-1)^{N^{\mathrm{Pf}}_{\delta_i}} = -1, \qquad i = 1, 2, 3.$$

This is Proposition 5.2. Squaring eliminates the sign and hence eliminates the OP scalar branch ambiguity: the squared Pfaffian $\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i})^2 = u_i^2$ is fixed by s_{δ_i}.

Group-theoretic extension via $(O1)^+$. The type-II Weyl group has Coxeter presentation

$$W^{(2)}(\Lambda^{2,1}_{II}) \simeq \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s_{\delta_1}^2 = s_{\delta_2}^2 = s_{\delta_3}^2 = 1 \rangle,$$

with no braid relations — the off-diagonal entries $(\delta_i, \delta_j) = -2$ for $i \neq j$ give Coxeter exponents $m_{ij} = \infty$. Hence $W^{(2)}(\Lambda^{2,1}_{II})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^3$ with one generator per s_{δ_i}, and any $\{\pm 1\}$-valued assignment on the generators extends uniquely to a homomorphism. By $(\mathrm{O1})^+$, $[c_o] = 0$, $\widetilde{\tau}_i^2 = 1$, and $\omega_{i,C} = 0$; the lifts $\widetilde{\tau}_i$ define an honest action on the reduced quotient orientation, and the induced character has $\epsilon_o(s_{\delta_i}) = -1$ for $i = 1, 2, 3$. This is Definition 0.109.

Independent automorphic computation of ν_{Δ_5}. The Maass multiplier ν_{Δ_5} of the singular theta lift of $\phi^{\mathrm{Weil}}_{0,1}$ is computed from the multiplier system of the Borcherds product (Borcherds [9], Gritsenko–Nikulin [46, Section 3],[45], Maass [76]). On the type-II simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$, the Borcherds divisor orders are

$$d^{\mathrm{aut}}_{\delta_i} := \mathrm{ord}_{\mathbb{H}_{\delta_i}}(\Delta_5) = f(0, \pm 1) = 1, \qquad i = 1, 2, 3,$$

so $\nu_{\Delta_5}(s_{\delta_i}) = (-1)^{d^{\mathrm{aut}}_{\delta_i}} = -1$ for $i = 1, 2, 3$. This is Lemma 7.2. The chamber-permuting roots $f_{-2} - f_2, f_2 - f_3, f_{-2} - f_3$ lie in $\Lambda^{2,1} \setminus \Lambda^{2,1}_{II}$ and are not type-II walls; their Maass values are $+1$ but no Hall orientation sign is asserted on them unless a semidirect-product Hall lift is constructed (Remark 3.4).

The comparison $\epsilon_o = \nu_{\Delta_5}$. Two $\{\pm 1\}$-valued characters of $W^{(2)}(\Lambda^{2,1}_{II})$ that agree on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ agree everywhere by the Coxeter presentation above. The local Pfaffian computation and the automorphic computation give that both ϵ_o and ν_{Δ_5} take the value -1 on each of the three simple reflections. Therefore $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$. □

Remark 3.2 *(The OP minus sign is not the source of* ϵ_o*).* The local Pfaffian computation and the automorphic multiplier computation are independent. The former uses no automorphic data; the latter uses no Pfaffian local model. In particular, the OP scalar $C_{\mathrm{OP},X} = -4096 = (-1) \cdot C_{\square,X}$ does not source the orientation character: squaring forgets orientation, so $C_{\mathrm{OP},X}$ carries no information about ϵ_0. The minus sign $\mathrm{sign}(C_{\mathrm{OP},X}) = -1$ is the OP scalar branch convention of [92], recorded by the fermionic shift $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber. The orientation character is a *geometric* monodromy datum, computed by transport of the Pfaffian line around type-II walls; it is fixed by the data of (O1)+(O1)$^+$+(O2), and is independently checked against the automorphic target by the Maass multiplier computation.

Remark 3.3 *(Imaginary roots are not Weyl reflections).* The Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ of the type-II BKM datum is generated by real simple reflections only. The imaginary simple roots contribute to the Weyl–Kac–Borcherds denominator through the multiplicity coefficients $m(a)$ and $\tau(a)$ of Gritsenko–Nikulin [48, Theorem 2.1],[46, Section 3]; they do not define reflections in $W^{(2)}(\Lambda^{2,1}_{II})$. Hence $\epsilon_0 : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ is a real-root reflection datum. Local divisor monodromy of a compact Hall theory around imaginary components of the Borcherds divisor is not a Weyl character and is not asserted here.

Remark 3.4 *(Extension to $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$).* The chamber-permuting symmetry group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ acts on the Coxeter generators $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ by permutation; its three reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$ have $\nu_{\Delta_5} = +1$. An extension of ϵ_0 to $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$ would require a separate Hall semidirect-product lift of the chamber-permuting reflections.

4 The primitive recognition theorem

The third theorem is a criterion for identifying the source-side primitive Lie superalgebra of the Dirac–Igusa pro-object with the BKM denominator algebra $\mathfrak{g}_{\Delta_5}$. The conditions form the *recognition criterion*: ten finite-stage data carrying compact geometric provenance, none of which is inferable from the Borcherds product or signed determinant alone.

Theorem 4.1 *(Primitive recognition).* Granted the recognition criterion (S1)–(S10) of Theorem 0.215, the cofinal finite-window datum of Definition 0.217, the chiral Koszul source datum of Definition 0.208, the finite Koszul comparison datum of Definition 0.211, and the retained orientation data $(O1_{\mathrm{quot}})$, the chosen Weyl lifts, and $(O2_{\mathrm{atlas}})$, suppose the compact $K3 \times E$ Hall/factorization construction supplies, on a cofinal sequence of downward-saturated finite root windows $W_1 \subset W_2 \subset \cdots$ with $\bigcup_\nu W_\nu = \mathcal{R}_+$:

- *(S1) Chain-level descent endpoint.* The (D0)-degenerated source coalgebra $(C_{X,W_\nu}, Q_{X,W_\nu})$ of Definition 0.208, descended chain-level to its primitive part on every finite window W_ν, carrying both the orientation lane and the Frobenius–Serre primitive structure feeding the recognition window.
- *(S2) Cartan and root-degree matching.* An even Cartan identification $\iota_H : P^{\Pi,\mathrm{red}}_{X,0} \xrightarrow{\sim} \Lambda^{2,1}_{II} \otimes \mathbb{C}$, with Gram-grading $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$.
- *(S3) Simple primitive representatives.* Parity-homogeneous e_i^X, f_i^X, h_i^X for every simple-root index $i \in I$ of $\mathscr{B}_{\Delta_5}$, with Gritsenko–Nikulin parity dimensions $\mu_{\bar{0}}(a)$, $\mu_{\bar{1}}(a)$ on imaginary fibres.
- *(S4) Borcherds–Kac relations.* Cartan, Chevalley ($[e_i^X, f_j^X] = \delta_{ij} h_i^X$), real Serre, Borcherds orthogonality, super sign, and isotropic-string relations satisfied in the protected Hall bracket.
- *(S5) Completed Hopf radical equality.* The completed Hopf pairing has closed homogeneous radical Rad_X which is a Lie ideal and a coideal, and identifies under the radical quotient with $\mathrm{Rad}_{\mathrm{GN}}$.
- *(S6) No-extra-relations.* The presentation map $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P^{\Pi,\mathrm{red}}_X$ has $\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$, on every finite window separately.
- *(S7) Generation.* $P^{\Pi,\mathrm{red}}_X$ is topologically generated by $P^{\Pi,\mathrm{red}}_{X,0}$ and $\{e_i^X, f_i^X\}_{i \in I}$.

- *(S8) Completed PBW.* The associated graded $\mathrm{gr}_{\mathrm{PBW}}\, U(P_X^{\Pi,\mathrm{red}})_W \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_W$ is an isomorphism on every finite window, with strict transitions.
- *(S9) Exact triangular completion.* The HN inverse limit is exact on the saturated finite root windows: $\lim^1$ vanishes for the kernels, radicals, PBW associated gradeds, and root-space parity pieces.
- *(S10) Full parity dimensions.* Two integers per nonzero root degree, in both signs: $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}} = \mathrm{mult}^{\mathrm{GN}}_{\overline{p}}(\alpha(\gamma))$, $p = 0, 1$, and the same equality in degree $-\gamma$ through the Borcherds–Kac Chevalley anti-involution; not only their signed difference.

Then the recognition datum determines an isomorphism of completed Γ_{gram}-graded Lie superalgebras

$$\boxed{H^0\,\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)\big/\overline{\mathrm{Rad}\langle\cdot,\cdot\rangle_X}\;\cong\;\mathfrak{g}_{\Delta_5}}$$

and the Weyl–Kac–Borcherds denominator of $P_X^{\Pi,\mathrm{red}}$ is $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$. If in addition the chiral Koszul source datum of Definition 0.208 is fixed, including the finite Koszul comparison datum of Definition 0.211 and the strict Koszul Mittag–Leffler exactness hypothesis of Proposition 0.214, then the chiral cobar of the source coalgebra C_X is identified with the formal current envelope compatibly with the Weyl action, Pfaffian orientation character, Hall pairing, radical quotient, and PBW associated graded:

$$\Omega^{\mathrm{ch}}_E C_X \;\simeq\; \mathsf{Den}_{\Delta_5,E}.$$

Proof. The argument is the inverse-limit assembly of (S1)–(S10) on each finite window, together with the chiral cobar identification. The window-wise theorem is Theorem 0.215, with the algebraic finite-window step isolated in Proposition 0.224; the additional input is the chiral-cobar comparison $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$, with the compatibility statement supplied by Proposition 0.212 and its Mittag–Leffler passage in Proposition 0.214.

Presentation map and kernel ($\alpha + \beta$). (S2) and (S3) define the continuous Lie-superalgebra homomorphism

$$\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \longrightarrow P_X^{\Pi,\mathrm{red}}$$

on the descended Gram-graded completion. (S4) puts the Borcherds–Kac ideal J_{BK} in the kernel. (S6) gives the kernel equality $\ker\pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$ on every finite window, hence injectivity of the descended map $G(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$. (S7) gives surjectivity. Thus the descended map is an isomorphism on every finite window by Proposition 0.224.

Radical descent (ε). (S5) identifies the source Hopf radical $\overline{\mathrm{Rad}_X}$ with the target radical $\overline{\mathrm{Rad}_{\mathrm{GN}}}$. The required Hopf adjointness, ideal/coideal property, quotient-tensor non-degeneracy, and closedness on the inverse system are part of (S5), equivalently the finite matrix conditions (R4)–(R5). The orientation lane supplies the trace pairing and Frobenius formalism on which those equations are written; it does not prove the source matrices M, D, B, G, K, Q, the kernel equality, or the PBW comparison.

Full parity dimensions (δ). (S10) is the two-integer equality $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}} = \mathrm{mult}^{\mathrm{GN}}_{\overline{p}}(\alpha(\gamma))$ for $p = 0, 1$, not just the signed difference. This rules out hidden zero-superdimension source summands, wrong parity lifts invisible to the determinant, and parity-cancelling ideals. A putative compact source matching every Hall bracket template of $\mathfrak{g}_{\Delta_5}$ on a finite window can still fail (S5): a kernel element $k \in K_{\gamma_\rho}$ with $(Q_{\gamma_\mu} \otimes Q_{\gamma_\nu}) D^{\mu,\nu}_\rho(k) \neq 0$ is invisible to the quotient target table yet breaks the radical coideal property, so the Hopf quotient is not defined (Theorem 0.215). This is the finite tensor-Hopf obstruction hidden by signed determinant data; (S5)+(S10) together block it.

PBW comparison and exact completion (γ). (S8) is the associated-graded isomorphism on every finite window, with strict transitions. (S9) is the exact-completion clause: $\lim^1$ vanishes for kernels, radicals, PBW associated gradeds, and root-space parity pieces. Equivalently, no new generator, relation, radical element, or parity-cancelling summand appears only after completion. Proposition 0.235 is the homological algebra that passes from the finite-window isomorphisms to the completed source and target. Class M completion is the ambient: chain-level identifications hold after weight-completion; this is the γ clause.

Passage to the inverse limit. Since the windows are cofinal ($\bigcup_\nu W_\nu = \mathcal{R}_+$) and the radical in (S5) is the closure of the finite kernels, the finite isomorphisms of the finite-window clauses determine the inverse-limit isomorphism

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)\big/\overline{\mathrm{Rad}\langle\cdot\,,\,\cdot\rangle_X} = P_X^{\Pi,\mathrm{red}} \xrightarrow{\sim} \mathfrak{g}_{\Delta_5}.$$

The Weyl–Kac–Borcherds denominator of $P_X^{\Pi,\mathrm{red}}$ is therefore $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, by Theorem 9.1.

Chiral cobar identification. Suppose the chiral Koszul source datum of Definition 0.208 is fixed: $(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, C_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_R, b_{X,R}, \Theta_{\mathrm{Kos},R})$ with vanishing residual $\mathfrak{O}_{\mathrm{Kos},R} = 0$, together with the finite Koszul comparison datum $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ and $\mathfrak{O}_{\mathrm{Kcmp},R} = 0$, compatibly in R and with the strict Mittag–Leffler exactness hypothesis of Proposition 0.214. The target side is the formal current envelope $\mathsf{Den}_{\Delta_5,E} = U^{\mathrm{ch}}_E\big(\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}})\big)$ of Proposition 16.2. The cobar comparison of Propositions 0.212 and 0.214 identifies

$$\Omega^{\mathrm{ch}}_E C_X \;\simeq\; \mathsf{Den}_{\Delta_5,E},$$

through the source-internal bar/cobar admissibility (the residual $\mathfrak{o}_R^{\varepsilon,\mathrm{src}} = 0$) and the source-to-target quasi-isomorphism ($\mathfrak{o}_R^{\vartheta} = 0$), and it preserves the Weyl, Pfaffian-orientation, Hall-pairing, radical-quotient, and PBW structures used above. Agreement on the primitive summand is not enough by Proposition 0.213; the full cone of ϑ_R and all compatibility defect complexes must be acyclic. The construction avoids the target-internal bar/cobar counit; this is the β-separation between observables and states. □

Remark 4.2 *(The zero-bracket graded super-vector-space counterexample).* The criterion (S1)–(S10) blocks the trivial counterexample of a graded super vector space $V_\bullet$ with all brackets zero and parity dimensions matching $\mathrm{mult}^{\mathrm{GN}}_{\overline{p}}$ on every window. Such a $V_\bullet$ may satisfy the target parity table and the signed-superdimension residual on the active support. It fails (S4): the Chevalley relation $[e_i^X, f_j^X] = \delta_{ij}h_i^X$ fails on the diagonal. It fails (S6): the kernel of the presentation map $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to V_\bullet$ is not $\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$ but contains every commutator $[e_i, e_j]$ with $(\alpha_i, \alpha_j) \neq 0$, so the induced map $G(\mathscr{B}_{\Delta_5}) \to V_\bullet$ is not injective. It fails (S5) for the zero pairing because the radical is all of $V_\bullet$, so the radical quotient is zero and not the BKM target. If one installs a nonzero pairing by hand, the denominator still does not supply its invariant Hopf adjointness, coideal radical, or equality with $\mathrm{Rad}_{\mathrm{GN}}$. The counterexample shows that signed-superdimension matching alone is not recognition; the bracket-level data of (S4)+(S5)+(S6) is the mathematical content.

Remark 4.3 *(Recognition without (D0)+(O1)+(O1)$^+$+(O2)).* The recognition criterion of Theorem 4.1 nominally lists no orientation hypothesis among (S1)–(S10). The orientation classes enter through (S1) — the chain-level descent $\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X$ is conditional on the orientation lane via Definition 0.86 and the Frobenius identity from the CY-3 Serre functor — and through (S5), where the Hopf radical's coideal property is fixed by the same Frobenius identity (Theorem 4.1). In the chapter's combined statement (Theorem 2.1+Theorem 3.1+Theorem 4.1) the orientation classes enter all three theorems.

5 Compatibility with chiral Koszul duality and Calabi–Yau quantum groups

The three theorems sit inside chiral Koszul duality and Calabi–Yau quantum groups as the View ② / View ③ face of the Calabi–Yau-to-chiral comparison. Three compatibilities are required.

5.1 *Primitive-to-shadow comparison*

The primitive-to-shadow comparison is the cross-level identity ②↔③ applied to the primitive-to-shadow part of the chain. The Vol II primitive datum on (X, D, τ) is the bicoloured nine-tuple

$$\big(C\big|_{(X,D,\tau)},\ \mathcal{D}^{\mathrm{cl}}\big|_X;\ b,\ A_b,\ F^{\mathrm{cl}};\ \sigma_{\mathrm{half}},\ \mathrm{tr}^{\mathrm{cl}}_X;\ \mathrm{Tr}_C\big).$$

The chart of vacuum b is the α-datum; the chart algebra $A_b = \mathrm{End}_C(b)$ is the open primitive algebra; the closed-colour vacuum factorization algebra F^{cl} gives the closed-side trace $\mathrm{tr}^{\mathrm{cl}}_X$ which evaluates at the type-II cusp $\mathfrak{c}_\infty$ to the weight-5 Borcherds–Gritsenko–Nikulin section $\mathcal{D}_X = \Delta_5$ (View ①). The comparison $A_b = \Phi_d^{(\Sigma_{d-1},C)}(C_X)$ is the chain-fusion datum of chiral Koszul duality and Calabi–Yau quantum groups at level 2: under that datum the Calabi–Yau-side chiral algebra A_X is compared with the open-side chart algebra A_b on the same curve C with boundary vacuum induced by (C_X, Σ_{d-1}, C). In the present setting, the comparison concerns the chiral algebra on E attached to the Hall source of $K3 \times E$; conditional on the chiral Koszul source datum, the finite Koszul comparison datum, and strict Koszul Mittag–Leffler exactness, the chiral cobar $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$ of Theorem 4.1 is the source-side chiral expression at level 2, and its constant-current Lie superalgebra is $\mathfrak{g}_{\Delta_5}$ (View ②). The protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ is the pre-trace Pfaffian section. After squaring and applying the retained level-Z datum, Vol I Theorem H (chiral-Hochschild concentration on the Koszul locus) and Vol II's cyclic-Hochschild Beilinson stratification (i)+(ii) identify the derived-centre trace pairing of degree $-\dim_{\mathbb{C}}(K3 \times E)$ whose scalar is Δ_5^{-2} (Appendix 7, Theorem 7.23). The level-A gravity-line operator algebra acting on the boundary is the open Hall–Borcherds residual of Vol II and is not constructed by the level-Z trace criterion.

5.2 *Vol III chain fusion at level* 2

The Vol III κ-stratification supplies the κ-tuple $(\kappa_{\mathrm{cat}}, \kappa^{\mathrm{Hodge}}_{\mathrm{ch}}, \kappa^{\mathrm{Heis}}_{\mathrm{ch}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ at $K3 \times E$; the row $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight realized by Δ_5, and the row $\kappa_{\mathrm{fiber}} = 24$ is $\chi_{\mathrm{top}}(K3) = 24$. The additive expression $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ has no invariant meaning: the Hodge reading gives $0 + 2 = 2$ at $N = 1$, and the Heisenberg reading gives the accidental equality $3 + 2 = 5$ only on that row. The CHL frame has Borcherds weights $(5, 3, 2, \frac{3}{2}, 1)$, fixed by $c_N(0)/2$ with $c_N(0) = (10, 6, 4, 3, 2)$ per Jatkar–Sen and Govindarajan–Krishna [41, 42, 53], not by a fixed additive formula. Thus κ_{fiber} is the Euler characteristic of the topological fibre and not of $\mathcal{O}_{\mathrm{fiber}}$; for K3, $\chi_{\mathrm{top}}(K3) = 24$ but $\chi(\mathcal{O}_{K3}) = 2$. Under the Vol III chain-fusion datum at level 2, the chiral algebra A_X on the CY face is compared with the chart algebra A_b on the Vol II open face. In the $K3 \times E$ specialization this comparison becomes $A_X = A_b = \Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$ only under the hypotheses of Theorem 4.1.

Remark 5.1 *(Coefficient projection is not a Hall comparison).* A Hall–Borcherds comparison map cannot be defined by coefficient projection alone, and a coefficient projection respecting charge addition because both products use the same cone is not a recognition theorem. A coefficient projection is not a Hall product/bracket matrix comparison, and does not prove Borcherds–Serre rows, radical descent, no-extra-relations, or PBW. The recognition criterion (S4)–(S10) of Theorem 4.1 is the mathematical statement:

the source-side bracket matrices, Hopf radical, kernel equality, and PBW associated-graded comparison are independent data, none of which is inferable from a coefficient projection.

5.3 BCOV / mixed holomorphic-topological strings

The mixed holomorphic-topological strings companion provides the local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ with brane completion as one realization of the factorization-algebra datum Φ^{FA}_d; the global obstruction is the holomorphic de Rham obstruction class. The level-n scalar shadow $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ on the closed $K3 \times E \mapsto$ point cobordism is the BPS partition function (Chapter 9). A three-dimensional gravity-line interpretation with internal compact factor $K3 \times E$ requires the Hall–Borcherds residual of Vol II and is not claimed by Theorem 2.1. The identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) = \mathcal{D}_X = \Delta_5$ is a protected Pfaffian identity before trace; its inverse square is the level-n protected scalar trace. The Pfaffian and orientation parts of the operator-level datum $\mathfrak{D}^{\mathrm{DI}}_X$ are defined on $K3 \times E$ relative to the retained data of Appendix 7; the primitive Lie-superalgebra part is the compact-source recognition problem.

6 Residual obstructions and open subroutes

The Pfaffian and orientation theorems are retained-data theorems on $K3 \times E$, relative to Appendix 7. The residuals in this chapter are the compact-source recognition datum $(S1)$–$(S10)$ and the gravity-line Hall–Borcherds operator algebra.

6.0.0.1 (D0). Theorem 7.7 is the criterion saying that the retained D0-HN datum supplies the cosection-twisted reduced d-critical orientation through the D0-degeneration pro-limit and the cofinal HN inverse system. The D_0-state, D_0-quotient, D_0-observable, D_0-descent, and D_0-Pfaffian rows are therefore part of that datum.

6.0.0.2 (O1). Theorem 7.13 supplies the strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle relative to the retained quotient-orientation datum. The finite-stabilizer linearization characters and the connected BE-edge class are part of that datum, not consequences of the primitive recognition theorem.

6.0.0.3 (O1)$^+$. The equivariance part of Theorem 7.14 is relative to the chosen Weyl lifts, the vanishing projective cocycle, and the zero torsor defects on the retained groupoid. It gives the honest $W^{(2)}(\Lambda^{2,1}_{II})$-transport of the orientation character. The comparison with ν_{Δ_5} also uses the rank-one wall signs from $(O2_{\mathrm{atlas}})$. Neither statement supplies the relation-closed source matrices of the recognition datum.

6.0.0.4 (O2). Relative to the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$, the rank-one normal-form rank $N^{\mathrm{Pf}}_{\delta_i} = 1$ holds for $i = 1, 2, 3$ (Proposition 5.2). The type-II middle-wall constant $4096 = 2^{12} = 64^2$ admits two readings: as the squared theta-leading on the type-II Borcherds product (where the $64 = 2^6$ prefactor of $D_5 = 64^{-1}\Delta_5$ doubles by $\Delta_5^2 = \Delta_{10}$) and as the 2^{12} sign assignments over the twelve binary wall coordinates of the half-Hilbert orbit. The equality of these two readings is scalar only, by Theorem 7.17. The OP scalar branch sign -1 is recorded by $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber, not by the orientation line. The middle wall $\delta_2 \leftrightarrow (0, 1, 1)$ is represented by a wrapped wall object with positive geometric b-degree only after the retained $(O2_{\mathrm{atlas}})$ datum supplies that object. Proposition 5.2 is the rank-one local sign calculation; the component, boundary, overlap, quotient, and protected-integration compatibilities are the global content of $(O2_{\mathrm{atlas}})$. Higher-rank wall atlases are not used in the proof of Theorem 2.1.

6.0.0.5 Recognition criterion (S1)–(S10). The window $W_{\leq 3}$ is the first arithmetic window but is not relation-closed. Put $a_{ij} = \partial_i + \partial_j$, $\partial_{123} = \partial_1 + \partial_2 + \partial_3$, and $C_{k,s} = a_{ij} + s\partial_k$ for $\{i, j, k\} = \{1, 2, 3\}$. The first base-string terminal degree window is the height-7 closure

$$W_{\mathrm{rel}}^{+} = W_{\leq 3}^{\mathrm{act}} \cup R_4 \cup I_4 \cup C_{\leq 7} \cup \{2\partial_{123}\},$$

where $R_4 = \{3\partial_i + \partial_j : i \neq j\}$, $I_4 = \{2a_{ij} : i < j\}$, and $C_{\leq 7} = \{C_{k,s} : 2 \leq s \leq 5\}$. It adds the real-Serre terminal rows, doubled isotropic rows, ∂_{123}-real strings, odd–odd ∂_{123} rows, and complementary height-seven strings. In this enlarged window, the finite target degree-closure packet `wrel_degree_closure` checks the 35 rows, their ∂-basis coordinates, heights, norms, Jacobi signed coefficients, six real-Serre terminal zero rows, and the three complementary-string terminal zero rows. Its downward-saturation audit proves the precise defect: W_{rel}^{+} is not downward saturated, and the only nonzero proper subdegrees outside the 35-row set are

$$D_k = \partial_k + 2a_{ij}, \qquad \{i, j, k\} = \{1, 2, 3\},$$

all below $2\partial_{123}$, with signed multiplicity $\mathrm{smult}(D_k) = -513$. The same packet verifies that adjoining these three rows gives the minimal nonzero downward-saturation extension

$$W_{\mathrm{sat}}^{+} = W_{\mathrm{rel}}^{+} \cup \{D_1, D_2, D_3\},$$

with zero remaining nonzero proper-subdegree defect. Remark 6.9 supplies full target parity for $I_4 = \{2a_{ij}\}$, $C_{k,3}$, $C_{k,4}$, and $C_{k,5}$, while $C_{k,2}$, D_k, and $2\partial_{123}$ remain signed target rows until the finite target presentation is computed. The finite coefficient table `k3e_first_relation_window_coefficients` checks the Jacobi arithmetic for these rows:

$$f(4,4) = 10, \qquad f(2,2) = 108, \qquad f(2,1) = f(4,3) = -513, \qquad f(3,3) = -64, \qquad f(4,2) = 4016,$$

with the terminal rows $f(5,5) = f(1,-3) = 0$. This is a target signed-dimension check, not a parity split or a source comparison. The A071 target-presentation verifier separately checks the target parity rows $2a_{ij} : 10|0$, $C_{k,3} : 29|93$, $C_{k,4} : 10|0$, and $C_{k,5} : 0|0$, and records $C_{k,2}$, D_k, and $2\partial_{123}$ as signed-only rows with no target basis, pairing, radical, or PBW blocks. Proposition 0.239 proves that these signed rows have no admissible parity, pairing, radical, or PBW target blocks for (R0)–(R8). Thus W_{rel}^{+} is a base-string terminal degree set with a three-dimensional saturation defect, and W_{sat}^{+} is a target-recorded saturated extension with signed-only rows. The post-saturation real-string audit then finds 21 terminal codomains outside W_{sat}^{+}, ten with nonzero signed multiplicity. Full finite relation closure therefore starts beyond W_{sat}^{+}. The terminal-layer audit groups these codomains into 21 distinct terminal degrees; adjoining them creates 114 nonzero subdegree-defect incidences across 26 distinct subdegrees. The next target-degree computation adjoins exactly these 26 rows and gives an 85-row degree set with zero remaining nonzero proper-subdegree defect. This closes downward saturation for the displayed terminal layer, not the real-string relation audit and not the compact-source comparison. If these 26 rows are promoted to target-generator rows, the real-root strings force 55 terminal checks; 54 terminal codomains lie outside the 85-row set, and 12 of the missing codomains have nonzero signed multiplicity. If those 55 terminal codomains are adjoined, the next downward-saturation audit already has 546 nonzero proper-subdegree defect incidences across 98 distinct subdegrees. Adjoining those 98 rows gives a 237-row target degree set with zero remaining nonzero proper-subdegree defect for that layer. If the 98-row layer is promoted to target-generator rows, the real-root strings force 206 terminal checks, 182 new terminal degrees, and 24 missing nonzero terminal degrees. The exact terminal-layer table records those 206 codomains, and the exact saturation-extension table records 50 nonzero proper subdegrees with 624 incidences; adjoining both gives a 469-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the 50-row layer forces 96 real-string checks, 90 new terminal degrees, no missing nonzero terminal degree, and a 12-row saturation extension with 44 incidences; adjoining both gives a 571-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the 12-row layer forces 20 zero terminal checks and a 6-row saturation extension with 8 incidences; adjoining both gives a 597-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the newly adjoined 6-row layer forces 10 zero terminal checks and a 6-row saturation extension with 18 incidences; adjoining both gives a 613-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the next newly adjoined 6-row layer forces 10 zero terminal checks and a 6-row saturation extension with 8 incidences; adjoining both gives a 629-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the next newly adjoined 6-row layer forces 10 zero terminal checks and a 6-row saturation extension with 18 incidences; adjoining both gives a 645-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. The `wrel_tail_staircase` verifier then proves the symbolic tail $A_N \Rightarrow B_N \Rightarrow A_{N+2}$ for every even $N \geq 14$, and its finite-anchor table checks A_{14} inside the `wrel_degree_closure` packet. Hence the target-degree audit has an infinite six-row staircase; its unbounded-tail table records a strictly increasing coordinate subsequence. No finite prefix of this target-only audit is a primitive-recognition theorem. Neither displayed target window is yet a primitive-recognition window satisfying (R0)–(R8). At $W_{\leq 3}$ the target parity table $\delta_i : 1|0$, $a_{ij} : 10|0$, $2\delta_i + \delta_j : 1|0$, $\delta_{123} : 29|93$ is computed; the row $a_{ij} : 10|0$ is the real bracket $[e_i, e_j]$ together with the nine Gritsenko–Nikulin isotropic simple directions. The `wle3_target_parity` verifier checks this target table and its scalar firewall; a compact source implementing (S3)–(S10) on $W_{\leq 3}$ is open, and the source identification at W^{+}_{rel} requires a finite source-and-target presentation computation. The first relation-closed source theorem is exactly the finite packet

window_closure, target_source_representatives, chevalley_serre_matrices, hall_boundary_complex, spectral_seque

It must compute the compact representatives, Chevalley and Serre matrices, Hall–Chevalley boundary, kernel bases, no-extra equality, PBW associated gradeds, and first-window theorem rows. It is not provided by the target parity table, the signed Borcherds exponents, the Pfaffian product, or the scalar trace. The current packet `certificates/first_window/k3e_relation_closed_window` records these missing rows by the obstruction ledger `FIRST_WINDOW_OBSTRUCTION_LEDGER_VERIFIED`, with `first_window_certification=false`. Hence Chapter 7 uses no first-window primitive-recognition theorem beyond this checked absence status.

6.0.0.6 Vol III Hall–Drinfeld–Borcherds spectral sequence on compact non-toric CY3. The Hall–Drinfeld–Borcherds spectral sequence abutting from Vol III's CoHA-type algebra to the BKM denominator on $K3 \times E$ is open on the compact non-toric trivial-canonical threefold, even after (D0), (O1), $(\mathrm{O1})^{+}$, and (O2). The finite-stage Hall source kernel of Theorem 0.252 is the finite-stage substitute.

6.0.0.7 Protected determinant and character level. The Pfaffian and orientation theorems identify the 2↔3 edge at the protected determinant and character level on $K3 \times E$. The primitive Lie-superalgebra recognition remains the finite compact-source datum $(S1)$–$(S10)$. The cross-level identities 1↔2 and

②↔⑤ are imported from Borcherds–Gritsenko–Nikulin and the regularized theta lift. The operator-level datum $\mathfrak{D}_X^{\mathrm{DI}}$ is defined at the Pfaffian and orientation level relative to the retained data of Appendix 7; the scalar shadow $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ is the level-n protected trace. The compact microscopic state-space interpretation requires the compact-source recognition datum; the $3d$ gravitational path integral requires the Hall–Borcherds residual; the Calabi–Yau realization requires a construction external to the Pfaffian theorem.

Part IV

Mirror Discriminant and BPS Trace

Chapter 8

The mirror discriminant

1 The mirror discriminant

The automorphic, BKM, Pfaffian, and theta-lift constructions place Δ_5 on chains that begin with explicit modular or representation-theoretic data: the Borcherds–Gritsenko–Nikulin section of the weight-5 automorphic line over $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2$ (View ①), the denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ of the Borcherds–Kac–Moody Lie superalgebra on $\Lambda^{2,1}_{II}$ (View ②), the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ of the Dirac–Igusa pro-object (View ③), and the singular theta lift of the weak Jacobi form $\phi_{0,1}$, namely the Borcherds regularized theta integral on the $\mathrm{O}(3,2)$ type-IV domain, with $\mathbf{Mp}_2$-Weil input and the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ exterior-square bridge (View ⑤). The period-side assertion is geometric. Let $\mathcal{P}^{(3)}_{K3\times E}$ be the rank-3 Mukai-invariant period quotient transported to $\mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})$. Let $\mathfrak{H}_{\mathrm{II}}$ denote the type-II Heegner divisor on this quotient. The automorphic calculation gives the divisor equality

$$[\,\Delta_5^{-2}\,] \;=\; 2[\mathfrak{H}_{\mathrm{II}}] \qquad \text{(equality of divisor classes on the } (K3\times E)\text{-component)}, \tag{8.1}$$

where $[\Delta_5^{-2}]$ is the polar divisor of the meromorphic section Δ_5^{-2} of the automorphic line. The conjectural content is the existence of a rank-3 Mukai–Hodge quotient of a mirror period family whose discriminant support is $\mathfrak{H}_{\mathrm{II}}$, together with the comparison of the automorphic line with the period Hodge line.

This identification is conjectural. The cross-level identities ④↔①, ④↔②, ④↔⑤ are open. The automorphic identity, the BKM denominator identity, the conditional Pfaffian identity, and the singular theta-lift identity do not use the period-discriminant conjecture.

The handle Δ_5 splits into three names. The automorphic section over $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2$ is denoted $\mathcal{D}_X = \Delta_5$ (View ①); the BKM denominator is $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View ②); the protected Pfaffian is $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ (View ③). The mirror conjecture concerns the first: the divisor identification (8.1) is a statement about $\mathcal{D}_X$ as an automorphic section, transported to the period side.

1.1 *The K3 mirror-period family*

The lattice-polarised K3 mirror is fixed by Dolgachev [32] and tracked, throughout, through the Mukai lattice. Let

$$\Lambda_{K3} \;=\; 3U \oplus 2E_8(-1), \qquad \widetilde{\Lambda}_{K3} \;=\; U \oplus \Lambda_{K3} \;\simeq\; 4U \oplus 2E_8(-1),$$

the K3 lattice of signature $(3,19)$ and the Mukai lattice of signature $(4,20)$ (Mukai [83, 84]). A *lattice polarisation* of a complex K3 surface S is a primitive embedding $\iota\colon M \hookrightarrow \mathrm{Pic}(S)$ of an even hyperbolic lattice M, of signature $(1,t)$, such that $M^\perp \subset \Lambda_{K3}$ admits a primitive embedding of the hyperbolic plane U; the pair (S,ι) is then *Nikulin-admissible*, in the convention of Gritsenko–Nikulin [48] § 3 and the rank-3 sublattice constructions of Borcherds [9] § 15. Here

$$t = \mathrm{rk}\, M - 1$$

is the negative index of M, so that $\operatorname{rk} M^\perp = 21 - t$ and $\operatorname{sign}(M^\perp) = (2, 19 - t)$. The convention is $\operatorname{sign}(L) = (\#\text{positive}, \#\text{negative})$.

The associated period domain

$$\Omega_M \;=\; \big\{[\omega] \in \mathbb{P}(M^\perp \otimes \mathbb{C}) \;:\; \langle \omega, \omega \rangle = 0,\ \langle \omega, \bar{\omega} \rangle > 0\big\}^+$$

is the type-IV symmetric domain of signature $(2, 19 - t)$ and dimension $19 - t$, with arithmetic group $\Gamma_M = \mathrm{O}^+(M^\perp)$ acting properly discontinuously. The *Dolgachev mirror lattice* is the sub-lattice

$$\check{M} \;=\; M^\perp \cap (U)^\perp \;\subset\; \Lambda_{K3}, \qquad M^\perp \;=\; U \oplus \check{M},$$

so that $\check{M}$ carries signature $(1, 18 - t)$ and is hyperbolic. The mirror K3 surface $S^\vee$ is lattice-polarised by $\check{M}$. Dolgachev mirror symmetry gives a correspondence between the M-polarised and $\check{M}$-polarised moduli quotients; it is an involution on a single quotient only in the self-mirror cases.

The Aspinwall–Morrison form of the same construction [3] uses the Mukai lattice to compare complex and complexified Kähler directions at large-volume and large-complex-structure cusps. The full K3 sigma-model moduli space is larger than Ω_M / Γ_M; the lattice correspondence is the only input needed for the Δ_5 discriminant statement. The type-IV lattice of the mirror quotient is

$$\widetilde{M} \;=\; U_{\mathrm{mir}} \oplus M \;\simeq\; \check{M}^\perp \;\subset\; \Lambda_{K3}, \qquad \operatorname{sign}(\widetilde{M}) = (2, t + 1).$$

Here U_{mir} is the hyperbolic plane split from $M^\perp = U_{\mathrm{mir}} \oplus \check{M}$. In the full Mukai lattice $\widetilde{\Lambda}_{K3} = U_{\mathrm{Muk}} \oplus \Lambda_{K3}$, the orthogonal complement of $U_{\mathrm{Muk}} \oplus M$ is $M^\perp = U_{\mathrm{mir}} \oplus \check{M}$; the extra Mukai plane is contained in the augmented sublattice. No canonical action on $\widetilde{\Lambda}_{K3}$ is needed here; only the primitive decomposition and the induced period domains enter the argument.

1.1.0.1 The rank-three case. The polarisation relevant to View 4 is

$$M \;=\; U \oplus \langle -2 \rangle, \qquad \operatorname{rank}(M) = 3, \qquad \operatorname{sign}(M) = (1, 2),$$

with Dolgachev mirror lattice

$$\check{M} \;=\; \big(M^\perp \cap U^\perp\big)_{\Lambda_{K3}} \;\simeq\; U \oplus E_7(-1) \oplus E_8(-1),$$

of rank 17 and signature $(1, 16)$ inside Λ_{K3}; the convention is sign $= (\#\text{positive}, \#\text{negative})$ on the bilinear form of the lattice. The decomposition follows from the embedding $\langle -2 \rangle \hookrightarrow E_8(-1)$ on a single root, after which the orthogonal complement of the root inside $E_8(-1)$ is $E_7(-1)$. The rank-five mirror-domain lattice is

$$\widetilde{M} \;=\; U \oplus M \;=\; 2U \oplus \langle -2 \rangle, \qquad \operatorname{rank}(\widetilde{M}) = 5, \qquad \operatorname{sign}(\widetilde{M}) = (2, 3).$$

Under the exterior-square convention (the model of $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3, 2)_+$ at § 1.4) the same lattice is written $\Lambda^{3,2} = U \oplus U \oplus \langle 2 \rangle$, with the hyperbolic-plane summand $\Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ carrying signature $(1, 1)$ and the rank-1 summand carrying form value $+2$ on a generator. The two writings are the same lattice expressed in opposite sign conventions on the bilinear form: the convention $(p, q) = (3, 2)$ is signature "three positive, two negative" for the fixed quadratic form, while the Dolgachev convention writes the same signature "two positive, three negative" on the form with reversed sign. The $(+2)$-vectors in the fixed convention are the (-2)-vectors of the Dolgachev convention; the Heegner divisors and the period domain are independent of the sign choice.

In the Dolgachev sign on $\widetilde{M}$, the period domain attached to $\widetilde{M}$ is the type-IV symmetric domain

$$\Omega_{\widetilde{M}} \;=\; \left\{[Z] \in \mathbb{P}\big(\widetilde{M} \otimes \mathbb{C}\big) \;:\; (Z, Z) = 0,\; (Z, \bar{Z}) > 0\right\}^{+}, \qquad \dim_{\mathbb{C}} \Omega_{\widetilde{M}} = 3,$$

whereas the exterior-square sign of $\Lambda^{3,2}$ writes the same domain with $(Z, \bar{Z}) < 0$. The two inequalities define the same oriented two-plane after multiplying the bilinear form by -1. The arithmetic group is

$$\Gamma_{\widetilde{M}} := \wedge^{2}\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \subset \mathrm{PO}^{+}(\widetilde{M})$$

acting properly discontinuously, where the component is fixed by the positive cone $C(\Lambda^{2,1})_{+}$ (§ 1.4). The exceptional isogeny

$$\mathbf{PSp}_4(\mathbb{R}) \;\simeq\; \mathbf{SO}(3,2)_{+}, \qquad \mathbf{Sp}_4(\mathbb{Z})/\{\pm 1\} \;\simeq\; \Gamma_{\widetilde{M}},$$

imported as View ⑤ (the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_{+}$ bridge), gives the identification

$$\Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}} \;\;\simeq\;\; \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}). \tag{8.2}$$

The form $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ of View ①, viewed as a section on the right-hand side of (8.2), is the Borcherds lift of the weak Jacobi form $\phi_{0,1}$ under the data $(\widetilde{M}, \phi_{0,1})$ on the left-hand side (Borcherds [10, Thm. 13.3] applied to signature $(2,3)$; Gritsenko–Nikulin [48, Thm. 1.1] for the identification with the additive Gritsenko lift; the rank-3 row of the Vol III lattice-polarised $\mathfrak{g}_L$ catalogue at $L = U \oplus \langle -2 \rangle$).

The K3-side period domain on the rank-3 polarisation is a different, larger object:

$$\Omega_{M} \;=\; \left\{[\omega] \in \mathbb{P}\big(M^{\perp} \otimes \mathbb{C}\big) \;:\; \langle \omega, \omega \rangle = 0,\; \langle \omega, \bar{\omega} \rangle > 0\right\}^{+}, \qquad \dim_{\mathbb{C}} \Omega_{M} = 17,$$

where $M^{\perp} \subset \Lambda_{K3}$ has signature $(2,17)$ and parametrises the *transcendental* Hodge filtration of a lattice-polarised K3 surface. The Borcherds lift used in View ④ is on $\Omega_{\widetilde{M}}$, not on Ω_M: the difference is the adjoining of one Mukai hyperbolic plane to the rank-3 K3 lattice, not an identification of the even elliptic cohomology with the elliptic period. The two period spaces are not interchangeable: $\Omega_{\widetilde{M}}$ of dimension 3 is the genus-two Siegel space and supports Δ_5 directly via the exceptional isogeny, while Ω_M of dimension 17 is the K3-only period space, on which the rank-3 lattice-polarised lift of $\phi_{0,1}$ lands only after the additional Mukai-U augmentation.

1.2 The K3 × E Mukai–Hodge quotient

The full Calabi–Yau-threefold variation of Hodge structure on $X = S \times E_\tau$ is

$$H^{3}(X, \mathbb{Z}) \;=\; H^{2}(S, \mathbb{Z}) \otimes H^{1}(E_\tau, \mathbb{Z}),$$

of rank 44. The elliptic modulus τ belongs to $H^1(E_\tau)$. It does not belong to the even elliptic lattice $H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z})$, whose Hodge type is constant.

The Mukai lattice $\widetilde{H}(S, \mathbb{Z}) = H^{*}(S, \mathbb{Z})$ of signature $(4,20)$ carries the Mukai pairing $\langle (r, c, s), (r', c', s') \rangle_{\mathrm{Muk}} = c \cdot c' - rs' - r's$ (Definition 1.3). The even cohomology of the elliptic curve is the hyperbolic plane $U_E = H^{\mathrm{even}}(E, \mathbb{Z}) = H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z})$ with the standard hyperbolic form. Since $H^{\mathrm{odd}}(S) = 0$, the even integral cohomology is, as an integral group with its Mukai tensor pairing,

$$H^{\mathrm{even}}(K3 \times E, \mathbb{Z}) \;=\; \widetilde{H}(S, \mathbb{Z}) \otimes U_E \;=\; \widetilde{H}(S, \mathbb{Z}) \otimes U,$$

a lattice of rank 48. This even Mukai lattice is not the Calabi–Yau-threefold period lattice; it is the lattice from which the rank-3 Mukai-invariant quotient is cut.

For $(K3\times E)$-pairs (S, E_τ) with S lattice-polarised by $M = U \oplus \langle -2\rangle$, the retained rank-3 Mukai–Hodge quotient is

$$\mathcal{P}^{(3)}_{K3\times E} \;:=\; \Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}} \;\overset{(8.2)}{\simeq}\; \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}), \tag{8.3}$$

where the augmented lattice $\widetilde{M} = U_{\mathrm{mir}} \oplus M$ is a rank-5 type-IV lattice. The summand U_{mir} is an abstract Mukai hyperbolic plane in the period quotient. It is not the constant even elliptic lattice U_E. The elliptic period enters through $H^1(E_\tau)$ and then through the exterior-square $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ coordinate system on $\Omega_{\widetilde{M}}$. Equation (8.3) is the period quotient on which View 4 is stated. The quotient is the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ model; the assertion that it is the base of the compact mirror discriminant is Conjecture 1.1.

The K3-only period domain Ω_M of dimension 17 does not enter (8.3): its role is to parametrise the transcendental Hodge filtration of S alone. The rank-3 $(K3\times E)$ Mukai–Hodge quotient is on $\Omega_{\widetilde{M}}$, of dimension 3. The relation $\widetilde{M} = U_{\mathrm{mir}} \oplus M$ is the rank-five type-IV bridge, not an equality with the full $H^3(K3\times E)$ period domain.

1.3 *The discriminant locus*

The candidate period quotient carries a natural Heegner divisor. It becomes the support of the discriminant divisor of a mirror period family only after the comparison asserted in Conjecture 1.1. On the type-IV symmetric domain $\Omega_{\widetilde{M}}$, the Heegner divisors are

$$\mathfrak{H}_{(\delta)} \;=\; \bigl\{[Z]\in\Omega_{\widetilde{M}} \;:\; (Z,\delta)=0\bigr\}, \qquad \delta\in\widetilde{M},\;\; (\delta,\delta)=2,$$

indexed by primitive square-2 vectors of the fixed form on $\widetilde{M} = \Lambda^{3,2}$ (equivalently, square-(-2) vectors of the Dolgachev form after the sign flip recorded above). Such a hyperplane records an extra algebraic Mukai vector; on a K3 slice its H^2-projection is the ordinary nodal (-2)-class when the projection is nonzero. The type-II Heegner divisor is

$$\mathfrak{H}_{\mathrm{II}} \;=\; \bigcup_{\substack{\delta\in\widetilde{M}\\ (\delta,\delta)=2\\ (\delta,\widetilde{M})=2\mathbb{Z}}} \mathfrak{H}_{(\delta)}/\Gamma_{\widetilde{M}} \;\subset\; \Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}}. \tag{8.4}$$

The $\Gamma_{\widetilde{M}}$-orbit structure on primitive square-2 vectors of $\widetilde{M}$ breaks into type-I and type-II classes ($\delta\cdot\widetilde{M}=\mathbb{Z}$ versus $\delta\cdot\widetilde{M}=2\mathbb{Z}$, the dichotomy already imported as the type-II versus type-I distinction at the $W^{(2)}(\Lambda^{2,1}_{II})$-orbit level); the union in (8.4) runs over the type-II orbit, which is the orbit relevant to the $\mathbf{Sp}_4(\mathbb{Z})$-image of the exceptional isogeny. The image in the quotient is one irreducible Heegner-type divisor (Bruinier–Funke [14, § 2] for the general lattice-Heegner formulation in signature $(2,3)$; the index-1 case is implicit in Borcherds [10, Thm. 10.1]). The same divisor, transported across the bridge (8.2), is the *Humbert surface*

$$\mathcal{H}_2^{(\mathrm{Hum})} \;=\; \bigl\{[Z]\in\mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}) \;:\; Z=\begin{pmatrix}\tau_1 & 0\\ 0 & \tau_2\end{pmatrix} \text{ up to } \mathbf{Sp}_4(\mathbb{Z})\bigr\},$$

the diagonal locus on which the principally polarised abelian surface $\mathbb{C}^2/(\mathbb{Z}^2+Z\mathbb{Z}^2)$ decomposes as a product of two elliptic curves. Under the lattice identification (8.2), this Humbert surface is the Heegner support of a norm-(-2) class in the rank-3 Mukai-invariant period domain. The assertion that it is the full discriminant support of a compact mirror family is part of Conjecture 1.1.

The form Δ_5 of View ① vanishes precisely along this divisor: $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ has a simple zero on $\mathcal{H}_2^{(\mathrm{Hum})}$ [47, Thm. 3.1][48, § 3], with multiplier ν_{Δ_5} accounting for the half-integral shift. Concretely, the Borcherds product expansion of View ① gives

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

and the automorphic zero divisor is exactly $\mathcal{H}_2^{(\mathrm{Hum})}$, counted with multiplicity 1. Thus the theorem-level divisor identity is

$$[\,\mathrm{div}(\Delta_5)\,] \quad = \quad [\,\mathcal{H}_2^{(\mathrm{Hum})}\,] \quad (\text{divisor classes on } \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})). \tag{8.5}$$

The additional equality with the mirror discriminant $\mathfrak{D}_{\mathrm{mir}}$ is not contained in the automorphic theorem; it is the support part of the mirror-discriminant conjecture.

1.4 The conjecture

The discriminant of a period quotient carries multiplicities, not just a support. Let $\mathfrak{D}_{\mathrm{mir}}$ denote the conjectural reduced discriminant divisor of that quotient, and let $\lambda_{\mathrm{per}} = \mathcal{F}^2$ denote the descended K3 period Hodge line on the same quotient. In the rank-3 Mukai-invariant component the expected section of $\lambda_{\mathrm{per}}^{-10}$ has polar order 2 along the type-II Heegner divisor: locally this is the Picard–Lefschetz contribution of a nodal K3 vanishing cycle to the Hodge filtration, and automorphically it is the square of the simple zero of Δ_5. This multiplicity comparison is part of the conjecture until the rank-3 quotient and the comparison line have been constructed.

Conjecture 1.1 *(The mirror discriminant identity).* On the rank-3 $(K3 \times E)$ Mukai–Hodge quotient parametrised by (8.3), the meromorphic section

$$\Delta_5^{-2} \;\in\; \Gamma\Big(\mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}),\;\; \mathcal{L}_{\mathrm{aut}}^{-10}\Big)$$

of the inverse 10th tensor power of the automorphic line bundle $\mathcal{L}_{\mathrm{aut}}$ coincides, as a meromorphic section of $\lambda_{\mathrm{per}}^{-10}$ on $\mathcal{P}^{(3)}_{K3\times E}$, with the discriminant section of the rank-3 quotient. Equivalently, the polar divisor of Δ_5^{-2} is twice the quotient discriminant $\mathfrak{D}_{\mathrm{mir}}$, the support of $\mathfrak{D}_{\mathrm{mir}}$ is $\mathfrak{H}_{\mathrm{II}}$, and the comparison map

$$\mathcal{L}_{\mathrm{aut}}^{-10} \;\xrightarrow{\;\sim\;}\; \lambda_{\mathrm{per}}^{-10}$$

on the open complement $\mathcal{P}^{\circ}_{K3\times E} = \mathcal{P}^{(3)}_{K3\times E} \setminus \mathfrak{D}_{\mathrm{mir}}$ extends to a global identification with Δ_5^{-2} as a section of $\lambda_{\mathrm{per}}^{-10}$.

The conjecture is, at the level of divisor classes,

$$[\,\Delta_5^{-2}\,] \quad = \quad 2\,[\,\mathfrak{D}_{\mathrm{mir}}\,] \quad = \quad 2\,[\,\mathfrak{H}_{\mathrm{II}}\,] \quad = \quad 2\,[\,\mathcal{H}_2^{(\mathrm{Hum})}\,];$$

the identity (8.5) fixes the right-hand side, and the conjectural content is the matching of multiplicities and the trivialisation of the comparison line bundle.

1.4.0.1 Evidence. The genus-two Bryan–Pandharipande generating function for disconnected genus-two BPS states on $K3 \times E$ predicts the cusp branch governed by the reciprocal Igusa form [15, Conj. 1]; Oberdieck–Pixton prove the normalized OP/DT/PT scalar representative

$$Z_{\mathrm{OP}}^{X} = -4096\, \Delta_5^{-2}$$

[92, Thm. 1]. In the trace convention of this manuscript

$$Z_{\mathrm{BPS}}^{K3\times E} := -4096^{-1} Z_{\mathrm{OP}}^{X} = \Delta_5^{-2}.$$

This is the input to View ① on the curve-counting side. The local Picard–Lefschetz model and Saito's Hodge-filtration formalism predict the same order-two degeneration as the automorphic divisor $[\Delta_5^{-2}]$. The unproved point is the global comparison: the line $\mathcal{L}_{\mathrm{aut}}^{-10} \otimes \lambda_{\mathrm{per}}^{10}$ must be trivialised and the identification of the two sides must extend across the discriminant divisor.

A separate piece of evidence is the rank-4 precedent set by Gritsenko–Nikulin [48, Thm. 4.1] in the lattice-polarised setting at the polarisation L producing a Borcherds product identified with the quasi-pull-back of the discriminant section of the universal lattice-polarised Hodge bundle on Ω_L/Γ_L. Conjecture 1.1 is the rank-3 analogue of the same identification, evaluated at the type-II rank-3 Mukai sublattice $M = U \oplus \langle -2 \rangle$ of $\Lambda^{2,1}$ (see Appendix 3); the Fake-Monster Borcherds product Φ_{12} on the Leech-extension lattice $\mathrm{II}_{25,1}$ of [8, 9] is a distinct, signature-$(26, 2)$ object and is not invoked here.

1.4.0.2 Epistemic strength. The identification claim of Conjecture 1.1 is conjectured. The comparison line bundle is a standard Hodge-theoretic object; the automorphic Heegner identity (8.5) is a theorem; the equality $\mathrm{Supp}(\mathfrak{D}_{\mathrm{mir}}) = \mathfrak{H}_{\mathrm{II}}$, the multiplicity 2, and the global extension across the discriminant divisor are conjectural.

1.4.0.3 Consequences. If proved, View ④ embeds the Δ_5 identity inside mirror symmetry on $K3 \times E$ at the level of period-side discriminants. The identification would give a period-side derivation of Δ_5 independent of the Borcherds product (View ①), the BKM denominator (View ②), the Pfaffian of the Dirac–Igusa pro-object (View ③), and the singular theta lift (View ⑤). It is a conjectural fifth chain.

A proof of Conjecture 1.1 would identify Δ_5^{-2} with the section of $\lambda_{\mathrm{per}}^{-10}$ on the rank-3 $(K3 \times E)$ Mukai–Hodge quotient; together with the Bryan–Oberdieck–Pixton theorem [92, Thm. 1] that $Z_{\mathrm{OP}}^{X} = -4096\, \Delta_5^{-2}$, this would give a period-side interpretation of the unscaled trace $Z_{\mathrm{BPS}}^{K3\times E} := -4096^{-1} Z_{\mathrm{OP}}^{X}$.

1.5 Range of the period conjecture

The Beilinson cut is in force: a statement is not primitive if it holds only after passing to a trace, choosing a chamber, completing a category, or installing descent data. The conjectural identification is on the period side: it concerns the section Δ_5^{-2} of an automorphic line bundle and the discriminant divisor of the rank-3 $K3 \times E$ mirror period component. It is not a claim about the protected operator-level datum $\mathfrak{D}_X^{\mathrm{DI}}$ of View ③; that datum is established at the Pfaffian and orientation level only relative to the retained data in the View ②↔③ chapters and Appendix 7. View ④ does not enter the operator chain.

In the trace convention of Chapter 9, the level-n scalar partition function $Z_{\mathrm{BPS}}^{K3\times E} := \Delta_5^{-2}$ is the orientation-forgetful trace of the Pfaffian datum; its three-dimensional gravity-line realization requires the Hall–Borcherds residual (Vol II, [73]). The mirror conjecture above says nothing about the gravitational lift; it addresses only the geometric origin of the section Δ_5^{-2} on the rank-3 mirror period component.

The Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is on the algebraic side; it is a theorem of Gritsenko–Nikulin [48, § 3, Thm. 3.1]. The mirror conjecture is on the period side and does not enter the algebraic chain.

1.6 Lattice-convention check

The lattice convention adopted above is the Dolgachev convention $\Lambda_{K3} = 3U \oplus 2E_8(-1)$, $M = U \oplus \langle -2\rangle$, mirror lattice $\check{M} \simeq U \oplus E_7(-1) \oplus E_8(-1)$ (rank 17, signature $(1,16)$). Aspinwall–Morrison's original mirror-pair construction [3] adopts the same lattice up to the choice of which hyperbolic plane in Λ_{K3} is split off; the two conventions agree on the K3 mirror lattice. The Mukai-lattice form $\widetilde{M} = U \oplus M = 2U \oplus \langle -2\rangle$ of rank 5 and signature $(2,3)$ is the convention adopted in Vol III [70, Prop. 6.4]; it is the convention relevant to the period-domain bridge (8.2). The lattice $\widetilde{M}$ is identified with $\Lambda^{3,2}$ of § 1.4 by the sign-flip on the rank-1 summand recorded in § 1.1. Disagreement on the lattice convention appears only as a sign convention on the Mukai pairing; it does not affect the discriminant divisor.

The K3 vs. $(K3 \times E)$ period-space matching (8.3) adjoins the distinguished hyperbolic summand U_{mir} to the rank-3 K3 lattice and lets the elliptic variation enter through $H^1(E_\tau)$. It does not identify U_{mir} with the constant even elliptic lattice U_E, and it is not an abstract equality between the K3-mirror period family and the $(K3 \times E)$-mirror period family. The possible discrepancy is exactly the comparison between the $\mathbf{Sp}_4$-side and the $\mathrm{O}^+(2,3)$-side; the isomorphism (8.2) is exactly the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge of View 5.

The word discriminant names a section, not only a support, once multiplicities are present. There are two natural divisors on the period side: the Heegner support $\mathfrak{H}_{\mathrm{II}}$, and the polar divisor of the section of $\lambda_{\mathrm{per}}^{-10}$. The conjecture concerns the second: Δ_5^{-2} is conjectured to be that section, with polar divisor twice the support divisor $\mathfrak{H}_{\mathrm{II}}$. The factor of 2 is predicted by the local Picard–Lefschetz model and is matched, on the automorphic side, by the square of Δ_5; a mismatch at the level of multiplicity would falsify the conjecture by a factor that the Borcherds product does not allow.

The relation to the Borcherds–Gritsenko–Nikulin quasi-pull-back theorems is the cleanest comparison. There the Borcherds form Φ_{12} on the even unimodular lattice of signature $(2,26)$ is pulled back to lattice-polarised type-IV domains, and its divisor is read as a discriminant section of the corresponding Hodge bundle [48, Thm. 4.1]. The rank-3 lattice $L = U \oplus \langle -2\rangle$ requires a separate comparison line and a separate Heegner-divisor analysis; the Φ_{12} theorem is evidence, not a proof.

1.7 Components of the period conjecture

(M1) *Trivialisation of the comparison line.* The comparison line bundle $\mathcal{L}_{\mathrm{aut}}^{-10} \otimes \lambda_{\mathrm{per}}^{10}$ on $\mathcal{P}^\circ_{K3\times E}$ must be trivialised. Since $\nu_{\Delta_5}^2 = 1$, the inverse-square section has no residual character; the obstruction is an ordinary Hodge line class. A trivialisation of this line, together with the divisor and boundary conditions below, strengthens the divisor comparison to an equality of meromorphic sections.

(M2) *Extension and multiplicity.* The section comparison is first defined on $\mathcal{P}^\circ_{K3\times E}$, where the Heegner divisor has been removed. The period condition is the extension across $\mathfrak{H}_{\mathrm{II}}$ with polar order 2, together with the extension to the rank-2 cusp with boundary exponent governed by the Borcherds singular weight $c_1(0)/2 = 5$ ([9, § 15]). Thus the divisor of the extended section, not the open part itself, is compared with $[\Delta_5^{-2}]$.

(M3) *The mirror correspondence on the section.* Dolgachev mirror symmetry gives a correspondence between the M-polarised and $\check{M}$-polarised period quotients, not a canonical involution of $\Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}}$. The period condition is that a chosen correspondence identifies the period Hodge line with the automorphic line and carries the type-II Heegner divisor to the type-II Heegner divisor. After this comparison, Δ_5^{-2} is required to be the same meromorphic section of the compared inverse Hodge line.

(M4) *Rank* ≥ 5 *lattice compatibility.* The Vol III rank-≥ 5 lattice-polarised $\mathfrak{g}_L$ family [69] is the conjectural extension of Conjecture 1.1 to higher Picard rank. At rank ≥ 5 the discriminant identity is conditional on the Gritsenko–Cléry 2018 admissibility conjecture (Conj. 5.1 of [44]); the rank-3 statement is the corresponding conjectural statement for the K3-Mukai-invariant lattice and is the rank relevant to the Δ_5 row.

The four period conditions (M1)–(M4) are distinct from the four obstruction classes (D0), (O1), (O1$^+$), (O2) of View ③; the latter govern the operator-level chain and are the conditioning for ②↔③, while the former govern the period-side chain and are the conditioning for ④↔①. The two systems lie on different chains: (M1)–(M4) are period-line and Heegner-divisor conditions, whereas (D0), (O1), (O1$^+$), and (O2) are Pfaffian-line and orientation conditions.

1.8 Compatibility with the Vol III lattice convention

The lattice convention $M = U \oplus \langle -2 \rangle$, $\check{M} \simeq U \oplus E_7(-1) \oplus E_8(-1)$, $\widetilde{M} = U \oplus M$ of § 1.1 is the convention of Vol III [70, § 6.4 + Prop. 6.4 + Thm. 6.7]; on the $(K3 \times E)$-component the Vol III κ-tuple $\left(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}\right) = (0, 0, 3, 5, 24)$ is recorded with each entry sourced from a distinct construction. The mirror discriminant of Conjecture 1.1 is on the $\kappa_{\mathrm{BKM}} = 5$ row of the κ-tuple; the period-side datum is the discriminant divisor on the lattice-polarised period stratum of the Vol III chart classification (Vol III [70, § V]: the "lattice-polarised period domain" equivariance stratum). The $(K3 \times E)$-fiber rank $\kappa_{\mathrm{fiber}} = 24$ is the Mukai-lattice rank of $\widetilde{\Lambda}_{K3}$; it is the rank of the Heisenberg algebra of the Mukai lattice, not a discriminant invariant.

The mixed-holomorphic-topological-strings companion volume [71] records that the modular line bundle on $\overline{\mathcal{A}}_g$ at level $g = 2$ is $\Omega_{\mathrm{central}}$, and that $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$ is the genus-2 section. The mirror-discriminant conjecture is consistent with that statement: the polar divisor of $\Delta_5^{-2} = (\Phi_{10}^{\mathrm{un}})^{-1}$ on $\overline{\mathcal{A}}_2$ is twice the divisor of Δ_5, and the mirror discriminant identification is precisely the geometric origin of that section.

The Vol II Hall–Borcherds residual is silent on View ④: the period-side discriminant identification is independent of the gravitational-lift bridge, and the conjecture above does not require the residual. Solving the Hall–Borcherds residual on the $(K3 \times E)$-component would not by itself prove Conjecture 1.1; conversely, a proof of Conjecture 1.1 would not automatically construct the Hall–Borcherds residual.

View ④ remains conjectural. The automorphic, BKM, Pfaffian, and theta-lift views form the theorem-level quadrilateral at the stated categorical dimensions. The mirror-discriminant identity is the remaining period-side equality.

Chapter 9

The level-n protected scalar shadow and the $K3 \times E$ realization question

The denominator algebra and the protected Pfaffian sit at the primitive and chart levels of the Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$. After taking the protected trace, the same Δ_5 survives, squared, as the partition function on the closed $K3 \times E \to \mathrm{pt}$ cobordism, equivalently the level-n modular-functor scalar trace of the realization formalism of Vol IV [74]. The scalar shadow is exactly the object allowed by the Beilinson cut: an automorphic identity, not an algebra; a trace, not an operator; an orientation-forgetful invariant, not an orientation lift. The orientation-level object is $\mathfrak{D}_X^{\mathrm{DI}}$ of Chapter 5; the scalar identity is the level-n shadow it casts.

Three distinct meanings of Δ_5 coexist and are tagged on first occurrence: the automorphic section $\mathcal{D}_X = \Delta_5$ (View I –- Borcherds–Gritsenko–Nikulin); the Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View II –- Gritsenko–Nikulin); the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ (View III –- Chapter 5). The handle "Δ_5" alone never substitutes for any of the three.

1 The level-n protected scalar shadow on $K3 \times E \to \mathrm{pt}$

1.1 The Oberdieck–Pixton scalar identity

Let $X = S \times E$, with S a smooth complex K3 surface and E a smooth elliptic curve. The reduced Donaldson–Thomas / stable-pair / Gromov–Witten correspondence on X, in the primitive K3-class chamber, gives the normalized scalar representative

$$Z^X_{\mathrm{OP/DT/PT}} = -4096\,\Delta_5^{-2}$$

on the genus-2 Siegel upper half-space $\mathfrak{H}_2$. The unscaled level-n trace retained below is $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$.

Convention 1.1 *(OP chamber, sign branch, primitive class).* The variables $(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}})$ are the Oberdieck–Pixton variables of [92, §0]: T_{OP} is the K3-surface degree, Q_{OP} the K3-curve class, and P_{OP} the elliptic-genus variable (equivalently $P_{\mathrm{OP}} = -p_{\mathrm{DT}}$, giving the $(-p_{\mathrm{DT}})^n$ DT-chamber). The K3 curve class β_h is taken *primitive* of self-intersection $\beta_h \cdot \beta_h = 2h - 2$ ([80]); the identity (9.1) is the primitive-class branch. The leading -1 of $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1}$ is the OP scalar convention; the squared theta-leading $4096 = 64^2$ is the normalization conversion $\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2$. None of these is the orientation character.

The scalar identity (9.1) is an identity in the OP chamber. If the retained compact Hall source lies in a different stability chamber, comparison with (9.1) requires the chamber-compatible Hall descent datum of Definition 4.18; the normal-ordered lattice map alone does not transport Hall brackets.

Under Convention 1.1 and [92, Theorem 1], [91, Proposition 5], [80],

$$Z^X_{\mathrm{DT}} \;=\; Z^X_{\mathrm{OP}}(P_{\mathrm{OP}}, Q_{\mathrm{OP}}, T_{\mathrm{OP}}) \;=\; -\,4096\,\Delta_5(Z)^{-2}, \qquad Z = \Omega \in \mathfrak{H}_2. \tag{9.1}$$

The constant $4096 = 64^2$ is the squared theta-leading coefficient $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5$ of the automorphic section, and the leading sign is the Oberdieck–Pixton scalar convention $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1}$. Both are normalization data of the imported reduced theory.

Theorem 1.2 *(Oberdieck–Pixton scalar branch and unscaled Igusa shadow).* With variables $(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}})$ normalized in the Oberdieck–Pixton chamber of the Oberdieck–Pixton normalization [92, Theorem 1], the proved reduced DT/PT scalar branch is

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = Z^X_{\mathrm{PT}} = -4096\,\Delta_5(Z)^{-2}.$$

In the unscaled trace convention of this manuscript, the level-n protected scalar shadow on the closed $K3 \times E \to \mathrm{pt}$ cobordism is defined by

$$Z^{K3\times E}_{\mathrm{BPS}}(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}}) \;=\; \big(\Phi^{\mathrm{un}}_{10}(Z)\big)^{-1} \;=\; \Delta_5(Z)^{-2}. \tag{9.2}$$

Equivalently, in Oberdieck–Pixton normalization $\chi^{\mathrm{OP}}_{10} = 2^{-12}\Delta_5^2$, $\Phi^{\mathrm{un}}_{10} = \chi_{10} = \Delta_5^2$, and the unscaled form $(\Phi^{\mathrm{un}}_{10})^{-1}$ is the retained level-n trace convention on each primitive closed K3-class fibre in the OP chamber. The normalization packet `certificates/normalizations/delta5_theta_leading` checks the named convention rows Δ_{10}, χ_{10}, Φ^{un}_{10}, χ^{OP}_{10}, $(\Phi^{\mathrm{un}}_{10})^{-1}$, $(\chi^{\mathrm{OP}}_{10})^{-1}$, and Z^X_{OP} separately; the scalar equalities below are relations among those rows, not a collapse of the conventions. The divisor packet `certificates/automorphic/delta5_humbert_divisor` checks that the polar divisor of the inverse square Δ_5^{-2} has order 2 on the same Humbert support on which Δ_5 has order 1; this is automorphic divisor arithmetic, not the mirror-discriminant assertion of Chapter 8. The Behrend-weighted reduced quotient integration of Oberdieck's reduced PT/DT scalar-quotient integration [90] gives the OP chamber branch

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = Z^X_{\mathrm{PT}} = -4096\, Z^{K3\times E}_{\mathrm{BPS}}.$$

Thus the OP branch is a normalized scalar representative of the same unscaled trace, not the definition of $Z^{K3\times E}_{\mathrm{BPS}}$.

Proof. Equation (9.2) is the dictionary between the Oberdieck–Pixton primitive series and the unscaled Igusa form Φ^{un}_{10}: the Oberdieck–Pixton normalization $\chi^{\mathrm{OP}}_{10} = D_5^2 = 64^{-2}\Delta_5^2 = 2^{-12}\Delta_5^2$ of [92, §0] together with the Igusa identification $\Phi^{\mathrm{un}}_{10} = \chi_{10} = \Delta_5^2$ of [51, 52] gives $(\Phi^{\mathrm{un}}_{10})^{-1} = \Delta_5^{-2}$. Multiplying by $2^{12} = 4096$ and the OP scalar sign restores (9.1). The reduced DT/PT comparison is [90, Theorem 3(i)], the reduced PT/GW comparison on primitive K3 classes is [91, Proposition 5], and the OP product identity is [92, Theorem 1]; their composition gives the displayed equality in the OP chamber. □

1.2 Three readings of the same exponent

The integers $f(n,l)$ of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ drive both sides of the OP/Borcherds dictionary. At the primitive datum they are virtual K_0-superdimensions of $\mathbb{U}_{n,l}$, and after taking scalar shadows they appear twice: once as exponents of the Borcherds product on the automorphic side, once as multiplicities in the Mukai dimension count of the Hilbert-scheme stratification on the geometric side. In $\mathfrak{g}_{\Delta_5}$, the same $f(nm,l)$ is the signed root supermultiplicity at $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$.

The variable map of the Oberdieck–Pixton normalization [92, Theorem 1] encodes this triple reading. A factor $(1 - q^n r^l s^m)^{f(nm,l)}$ of the Borcherds product on the automorphic side is, on the Oberdieck–Pixton side, a factor $(1 - p^k_{\mathrm{OP}} q^h_{\mathrm{OP}} \tilde{q}^d_{\mathrm{OP}})^{c(4hd-k^2)}$ of χ^{OP}_{10}, the K3-elliptic-genus exponents $c(4hd - k^2) =$

$2f(hd,k)$ recording the squared automorphic line. The multiplicities $2f$ are exactly the K3 elliptic-genus coefficients of $Z_{\mathrm{K3}} = 2\phi_{0,1}$; the squaring is part of the identification $\chi_{10}^{\mathrm{OP}} = D_5^2$ of the Oberdieck–Pixton/Pandharipande scalar identity [92, Theorem 1].

Remark 1.3 *(Constants are normalization, not orientation).* Three constants intervene in (9.1). The leading $64 = [q^{1/2}r^{1/2}s^{1/2}]\Delta_5$ is the theta-constant normalization of the Igusa section; the monic Borcherds product is $D_5 = 64^{-1}\Delta_5$. The factor $4096 = 64^2$ is the squared theta-leading coefficient produced by the OP convention $\chi_{10}^{\mathrm{OP}} = D_5^2$. The leading minus sign is the OP scalar branch $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$. None of the three is the orientation character. The protected orientation character ϵ_o and the automorphic reflection character ν_{Δ_5} are determined elsewhere: ν_{Δ_5} by the Maass transformation law of Δ_5, ϵ_o by the rank-one type-II Pfaffian signs together with the retained half-Hilbert wall atlas. The number $4096 = 2^{12}$ occurs in two different structures, not as one orientation datum: the Pfaffian-side 2^{12} is the cardinality of the retained sign orbit, while the OP-side 2^{12} is the squared theta-leading coefficient in the imported scalar identity.

1.3 Worked Fourier-coefficient check

The leading Fourier coefficients of Δ_5 and Δ_5^2 provide the elementary verification that the constants in (9.1) are normalization, not enumerative content. Recall $\phi_{0,1}$ has leading expansion

$$\phi_{0,1}(\tau,z) \;=\; (r^{-1}+10+r) + q\big(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2\big) + O(q^2),$$

with coefficients $f(0,-1) = f(0,1) = 1$, $f(0,0) = 10$, $f(1,1) = f(1,-1) = -64$, $f(1,0) = 108$, $f(1,2) = f(1,-2) = 10$, $f(1,3) = f(1,-3) = 0$. The Borcherds–Gritsenko–Nikulin product

$$\Delta_5(Z) \;=\; 64\, q^{1/2}r^{1/2}s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1-q^n r^l s^m)^{f(nm,l)} \tag{9.3}$$

opens with the theta-leading term $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$; the next non-trivial timelike coefficient belongs to the monic product after the leading monomial is removed:

$$[qrs]\, D_5/(qrs)^{1/2} = 93, \qquad D_5 = 64^{-1}\Delta_5.$$

The isolated factor $(1-qrs)^{f(1,1)} = (1-qrs)^{-64}$ contributes $64qrs$, and the remaining root factors contribute the other $29qrs$, as in Appendix 2. Squaring (9.3),

$$\Delta_5^2(Z) \;=\; 4096\, q\, r\, s \prod (1-q^n r^l s^m)^{2f(nm,l)}, \tag{9.4}$$

with theta-leading $64^2 = 4096$. The OP product $\chi_{10}^{\mathrm{OP}} = p\, q\, \tilde{q} \prod (1 - p^k q^h \tilde{q}^d)^{c(4hd-k^2)}$, with K3-elliptic exponents $c(4hd-k^2) = 2f(hd,k)$, is exactly the right-hand side of (9.4) divided by 4096. Thus

$$\chi_{10}^{\mathrm{OP}} = 4096^{-1}\,\Delta_5^2, \qquad [pq\tilde{q}]\,\chi_{10}^{\mathrm{OP}} = 1, \qquad [q^{1/2}r^{1/2}s^{1/2}]\,(64^{-1}\Delta_5) = 1.$$

The squared theta-leading coefficient 4096 is therefore the single explicit normalization constant in the OP/DT scalar identity $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$; the sign -1 is OP convention; together they produce the constant -4096 of (9.1). The Fourier coefficient at $q^{1/2}r^{1/2}s^{1/2}$ identifies the scalar 4096 as $\big([q^{1/2}r^{1/2}s^{1/2}]\Delta_5\big)^2$, a leading-coefficient square, not an enumerative invariant.

1.4 Cobordism positioning and the closed $K3 \times E \to \mathrm{pt}$ cobordism

The cobordism $K3 \times E \to \mathrm{pt}$ is the closed Calabi–Yau-threefold target viewed in the retained trace convention. The trace is the Laurent expansion of the meromorphic automorphic section Δ_5^{-2}; evaluation at a point of $\mathfrak{H}_2$ outside the polar divisor gives a complex number. The "level-n" label is the curve-counting integer

$$n = \chi(O_Y)$$

for a one-dimensional subscheme $Y \subset K3 \times E$ of class (β_h, d). The target has

$$\chi(O_{K3\times E}) = \chi(O_{K3})\chi(O_E) = 2 \cdot 0 = 0,$$

and this target Euler characteristic is not the exponent n. The Laurent variables record n, the elliptic degree d, and the K3 norm $h - 1$. The dual notation $Z_{\mathrm{BPS}}^{K3\times E}(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}})$ stores the elliptic and K3 gradings as a Laurent expansion of the inverse Igusa cusp form on $\mathfrak{H}_2$.

The $K3 \times E$ target is a threefold; the perfect obstruction theories and reduced virtual fundamental classes live on the stable-pair and DT moduli spaces $P_n(X, (\beta_h, d))$ and $I_n(X, (\beta_h, d))$, not on the target itself and not on any auxiliary fourfold. The closed K3-class fibre at primitive curve class β_h of norm $2h - 2$ supplies the K3 component of the coefficient indexed by (h, n, d), not a new target dimension.

2 Spin L-factor normalization for Δ_5^2

2.1 The Saito–Kurokawa eigenline

The unscaled Igusa form $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ is the weight-10 Saito–Kurokawa lift of the unique normalized cusp form $g = \Delta E_6 \in S_{18}(\mathbf{SL}_2(\mathbb{Z}))$; the Hecke eigenline is the one-dimensional $\mathbb{C}\Delta_{10}^{\theta} = \mathbb{C}\chi_{10}^{\mathrm{OP}}$, independent of the scalar normalization of its generator.

Proposition 2.1 *(Hecke eigenvalues on the Saito–Kurokawa eigenline).* For each prime p, let $T_1(p)$ and $T_2(p^2)$ denote the degree-one and degree-two genus-two Hecke operators on $S_{10}(\mathbf{Sp}_4(\mathbb{Z}))$ of [2, §4]. The Saito–Kurokawa form $\Delta_{10}^{\theta} = \Delta_5^2$ is a simultaneous eigenform with eigenvalues

$$T_1(p)\,\Delta_{10}^{\theta} \;=\; \lambda_1^{\mathrm{SK}}(p)\,\Delta_{10}^{\theta}, \qquad T_2(p^2)\,\Delta_{10}^{\theta} \;=\; \lambda_2^{\mathrm{SK}}(p^2)\,\Delta_{10}^{\theta},$$

where the eigenvalues are determined by the Saito–Kurokawa correspondence from the elliptic Hecke eigenvalues of $g = \Delta E_6$:

$$\lambda_1^{\mathrm{SK}}(p) \;=\; \lambda_g(p) + p^8 + p^9, \qquad \lambda_2^{\mathrm{SK}}(p^2) \;=\; \lambda_g(p)^2 + \lambda_g(p)\big(p^8 + p^9\big) + p^{18},$$

read off the X- and X^2-coefficients of the local Euler factor associated with (9.6), with $\lambda_g(p)$ the normalized p-th Hecke eigenvalue of $g = \Delta E_6$, the unique normalized cusp form in $S_{18}(\mathbf{SL}_2(\mathbb{Z}))$: from $\Delta E_6 = q - 528q^2 - 4284q^3 + \cdots$, $\lambda_g(2) = -528$, $\lambda_g(3) = -4284$ [107, Table 1]. Concretely $\lambda_1^{\mathrm{SK}}(2) = -528 + 2^8 + 2^9 = 240$ and $\lambda_1^{\mathrm{SK}}(3) = -4284 + 3^8 + 3^9 = 21960$, while $\lambda_2^{\mathrm{SK}}(4) = 135424$ and $\lambda_2^{\mathrm{SK}}(9) = 293343849$; the Ramanujan bound is violated ($|\lambda_1^{\mathrm{SK}}(p)| \sim p^9$, whereas tempered classical eigenvalues at weight 10 have size $p^{17/2}$), the signature of the Saito–Kurokawa CAP form.

Proof. The Saito–Kurokawa correspondence [107, §1] produces, for each weight $2k$ elliptic newform $g \in S_{2k}(\mathbf{SL}_2(\mathbb{Z}))$, a weight-$k+1$ Siegel cusp form $F_g \in S_{k+1}(\mathbf{Sp}_4(\mathbb{Z}))$ with spinor L-function $L_{\mathrm{spin}}(s, F_g) = \zeta(s-k+1)\zeta(s-k)L(s,g)$ (which is exactly (9.6) at $k = 9$). The Euler product factorisation determines the Hecke eigenvalues at each prime. In Andrianov's polynomial the

Siegel weight is $K = k + 1$, so $p^{2K-4} = p^{2k-2}$, $p^{2K-3} = p^{2k-1}$, and $p^{4K-6} = p^{4k-2}$. Writing $L_{\mathrm{spin}}(s, F_g) = \prod_p L_p(p^{-s}, F_g)^{-1}$ with

$$L_p(X, F_g) = (1 - p^{k-1}X)(1 - p^k X)(1 - \lambda_g(p)X + p^{2k-1}X^2)$$

and comparing with the Andrianov factorisation [2, §4.5]

$$L_p(X, F_g) = 1 - \lambda_1^{\mathrm{SK}}(p)X + (\lambda_1^{\mathrm{SK}}(p)^2 - \lambda_2^{\mathrm{SK}}(p^2) - p^{2k-2})X^2 - \lambda_1^{\mathrm{SK}}(p)p^{2k-1}X^3 + p^{4k-2}X^4,$$

the X-coefficient gives $\lambda_1^{\mathrm{SK}}(p) = \lambda_g(p) + p^{k-1} + p^k$. The X^2-coefficient of the Euler product is

$$2p^{2k-1} + (p^{k-1} + p^k)\lambda_g(p).$$

Since the Andrianov coefficient of X^2 is $\lambda_1^{\mathrm{SK}}(p)^2 - \lambda_2^{\mathrm{SK}}(p^2) - p^{2k-2}$, one obtains

$$\begin{aligned}\lambda_2^{\mathrm{SK}}(p^2) &= (\lambda_g(p) + p^{k-1} + p^k)^2 - p^{2k-2} - 2p^{2k-1} - (p^{k-1} + p^k)\lambda_g(p)\\ &= \lambda_g(p)^2 + \lambda_g(p)(p^{k-1} + p^k) + p^{2k}.\end{aligned}$$

At $k = 9$ this is the displayed formula. The dimension formula $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$ forces $g = \Delta E_6$ at $k = 9$, and the eigenvalues at $p = 2, 3$ are tabulated in [107, Table 1]. □

Remark 2.2 *(The SK-correspondence onto the full-level Maaß line).* The SK-correspondence $g \mapsto F_g$ is a Hecke-equivariant injection $S_{2k}(\mathbf{SL}_2(\mathbb{Z})) \hookrightarrow S_{k+1}(\mathbf{Sp}_4(\mathbb{Z}))$ whose image is the Maaß subspace $\mathrm{Maass}_{k+1} \subset S_{k+1}(\mathbf{Sp}_4(\mathbb{Z}))$. At $k = 9$, $\dim \mathrm{Maass}_{10} = \dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$ and $\mathrm{Maass}_{10} = \mathbb{C}\Delta_5^2$; the Hecke eigenvalues λ_i^{SK} of Proposition 2.1 record the explicit arithmetic translation of the full-level elliptic Hecke action on $g = \Delta E_6$ under the SK-correspondence. No new modular forms or arithmetic data enter; the Saito–Kurokawa eigenline of Δ_5^2 is fully determined by the full-level arithmetic of ΔE_6 and the SK-correspondence.

The OP monic normalization is

$$\chi_{10}^{\mathrm{OP}} = 2^{-12}\,\Delta_{10}^{\theta} = 4096^{-1}\,\Delta_5^2, \tag{9.5}$$

and the level-n shadow (9.2) is the inverse of the *theta-normalized* Hecke eigenline element $\Delta_{10}^{\theta} = \Delta_5^2$.

Theorem 2.3 *(Andrianov factorization of the spinor L-function).* For the Saito–Kurokawa representation generated by $\Delta_{10}^{\theta} = \Delta_5^2$,

$$L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) = \zeta(s-8)\,\zeta(s-9)\,L(s,g), \qquad g = \Delta E_6. \tag{9.6}$$

Equivalently, for Andrianov's local polynomial

$$Q_{p,F}(t) = 1 - \lambda_F(p)t + \big(\lambda_F(p)^2 - \lambda_F(p^2) - p^{16}\big)t^2 - \lambda_F(p)p^{17}t^3 + p^{34}t^4$$

at weight 10, the Euler product $\prod_p Q_{p,F}(p^{-s})^{-1}$ is the left-hand side of (9.6). Andrianov's completed function is

$$\Psi_F(s) = (2\pi)^{-s}\Gamma(s)\Gamma(s-8)L_{\mathrm{spin}}(s,F).$$

This is Andrianov's spin L-factor factorization [2, Theorem 3.1.1]; the polynomial $Q_{p,F}$ is Andrianov's genus-two Hecke polynomial in weight 10, with coefficient $\lambda_F(p)^2 - \lambda_F(p^2) - p^{2\cdot 10-4}$ at t^2. The proof uses Andrianov's spinor L-factor formula for the degree-2 Saito–Kurokawa lift attached to the weight-18 elliptic cusp form g [2], together with the dimension count $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$ which forces the source form to be $g = \Delta E_6$ [107].

Proposition 2.4 *(Polar coefficient at $s = 10$).* With $\Lambda(g,s) = (2\pi)^{-s}\Gamma(s)L(s,g)$, the functional equation $\Lambda(g,s) = -\Lambda(g,18-s)$ gives $L(9,g) = 0$, so the elliptic central zero cancels the possible $\zeta(s-8)$-pole in the spin product at $s = 9$. At $s = 10$ the polar coefficient is

$$\operatorname*{Res}_{s=10} L_{\mathrm{spin}}(s,\Delta_{10}^{\theta}) \;=\; \zeta(2)\,L(10,g) \;=\; \frac{\pi^2}{6}\,L(10,g),$$

with

$$L(10,g)/(2\pi i)^{10}\Omega^{+}(g) \in \mathbb{Q}$$

by Manin–Deligne periods. The period theorem fixes the normalisation of the critical value; it is not a non-vanishing theorem. Thus $L_{\mathrm{spin}}(s,\Delta_{10}^{\theta})$ has a simple pole at $s = 10$ exactly when $L(10,g) \neq 0$.

This is the Saito–Kurokawa critical-value identity [107] together with the Manin–Deligne period decomposition [28, 77]; the proof uses the Eichler–Shimura–Manin period theorem for the rational newform g [62, 77], in Deligne's period normalization [28].

Remark 2.5 *(Hecke eigenline is normalization-independent).* The spinor L-function depends on the Hecke eigenline $\mathbb{C}\Delta_{10}^{\theta} = \mathbb{C}\chi_{10}^{\mathrm{OP}}$, not on the scalar used to label its generator. Theorem 2.3 and Proposition 2.4 are statements about the eigenline. The OP scalar constant 2^{-12} of (9.5) relabels the generator from Δ_5^2 to χ_{10}^{OP}; it does not change L_{spin}, the residue, or the cancellation of the elliptic central zero against the zeta pole at $s = 9$.

2.2 *Independence from the Dirac–Igusa construction*

The Saito–Kurokawa factorization of $L_{\mathrm{spin}}(s,\Delta_5^2)$ and the OP/DT scalar identity (9.1) are imported results: the first from Andrianov [2], the second from Oberdieck–Pixton [92] and Oberdieck [90]. Neither uses the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}$ of Chapter 5, the orientation character ϵ_o of (O1)/(O1$^+$), the type-II wall normal form of (O2), or the primitive recognition $P_X \cong \mathfrak{g}_{\Delta_5}$ problem of Chapter 6. Conversely, the construction of $\mathfrak{D}_X^{\mathrm{DI}}$ and the Pfaffian–Dirac theorem $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ do not use the spinor L-factor or the scalar inverse Igusa form. The two routes meet at the squared section $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$ and at no earlier point.

3 What the scalar shadow does not supply

3.1 Z_{BPS} *is the level-n protected trace, not a path integral*

The squared section $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$ is the determinant section, and its inverse $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ is the protected scalar trace in the Beilinson chain, after the primitive object at level P (the open factorization dg-category, the boundary vacuum b, and $A_b = \mathrm{End}_{\mathcal{C}}(b)$) and after the chart object at level C (the disk algebra and its bar / cobar / chiral Hochschild data). The scalar partition function is the *trace* of the protected data; it does not by itself construct the level-A acting algebra.

At finite height the protected trace is a separate row-level datum:

trace_categories, protected_trace_functors, trace_operators, scalar_normalizations, trace_identities, forgetful_m

These rows record the level-Z trace category, the closed $K3 \times E \to \mathrm{pt}$ trace functor, the operator whose determinant square is Φ_{10}^{un}, the constants 64, 4096, and the OP sign -1, the identity $\Delta_5^{-2} = (\Phi_{10}^{\mathrm{un}})^{-1}$, and the forgetful maps killing orientation, primitive bracket, Hopf pairing, PBW, and parity data. The gravity-residual rows must mark the Hall–Borcherds level-A operator algebra as conjectural, not constructed, and unused in the trace proof. The scalar firewall excludes orientation characters, Pfaffian lines, primitive brackets, Hopf pairings, PBW bases, parity splittings, source Koszul maps, and gravitational path integrals as data recovered from the scalar trace.

Scope 3.1 *(No scalar trace is a 3D gravitational path integral).* The identity (9.1), proved through OP/DT/PT primary literature, is a closed automorphic equation between scalar power series. Neither this scalar identity, nor the automorphic section of Chapter 3, nor the protected Pfaffian of Chapter 5 identifies $Z_{\mathrm{BPS}}^{K3\times E}$ to a 3-dimensional gravitational path integral on $\mathrm{AdS}_3 \times S^3 \times K3$, to a metric Euclidean path-integral over the full conformal class of the boundary, or to a holographic dual partition function in any sense beyond the boundary-CFT reading recorded by Vol II. The equality $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ gives the determinant section of a candidate gravity-line theory; the partition character would be the reciprocal section Δ_5^{-2}. The construction of the gravity-line operator algebra is open.

Remark 3.2 *(Protected trace versus path integral).* The assertion that Z_{BPS} is a $3d$ gravitational path integral identifies a protected scalar trace with a level-A acting theory. The precise statement is that $Z_{\mathrm{BPS}}^{K3\times E}$ is the level-n protected trace of the retained $K3 \times E$ data; the gravitational path integral, if such a thing exists at this boundary, requires the Hall–Borcherds residual of Vol II (§3.2). The residual is open; the path-integral realization is not claimed. In particular, no microscopic state space $\mathcal{H}_{K3\times E}$ is asserted to support $\dim \mathcal{H}_{K3\times E,\gamma} = f(\gamma)$ outside the charge-graded virtual reading recorded in Chapter 2.

3.2 The Hall–Borcherds residual

The passage from a scalar partition function to a chain-level gravity-line operator algebra requires a comparison datum that lives neither in the automorphic / OP reduced theory nor in the compact Pfaffian datum. Vol II names this comparison.

Conjecture 3.3 *(Hall–Borcherds residual; gravity-line comparison).* Let $X = K3 \times E$ and assume the oriented critical Hall datum used for the Vol III positive-half comparison

$$\Theta_{\mathrm{Hall}\to\mathrm{Borch}} : \widehat{\mathrm{CoHA}}^{\mathrm{red}}_{\Lambda^{2,1}_{II}}(X) \longrightarrow U^{\mathrm{ch}}\big(\widehat{\mathfrak{n}_+(\mathfrak{g}_{\Delta_5})}\big). \tag{9.7}$$

The Hall–Borcherds residual is the conjectural extension of (9.7), after Drinfeld doubling, current enveloping along E, and weight-completion, to a filtered $\mathsf{SC}^{\mathrm{ch,top}}$ morphism from the $K3 \times E$ Hall boundary object to the gravitational boundary line algebra, whose derived-centre trace has scalar character $(\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$.

The residual is the Vol II gravity-line Hall–Borcherds residual [73]: it asks for the gravity-line operator algebra acting on the boundary. It is not claimed here; its existence and the construction of its witnessing morphism are open.

Remark 3.4 *(Compatibility with the κ-stratum).* The κ-tuple of the K3×E witness consists of the structural data behind the scalar shadow:

$$\big(\kappa_{\mathrm{cat}},\ \kappa_{\mathrm{ch}}^{\mathrm{Hodge}},\ \kappa_{\mathrm{ch}}^{\mathrm{Heis}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}}\big)_{K3\times E} = (0,\ 0,\ 3,\ 5,\ 24).$$

The fourth entry $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight $\mathrm{wt}(\Delta_5)$, the Δ_5 row; the fifth entry $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(K3) = 24$ is the topological Euler characteristic of the K3 fibre, distinct from

$\chi(\mathcal{O}_{K3}) = 2$. The additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ confuses these two: with $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 0$ and $\chi(\mathcal{O}_{K3}) = 2$, it predicts $\kappa_{\mathrm{BKM}} = 2$, which fails because the universal Borcherds weight at $N = 1$ is $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 10/2 = 5$ (Borcherds 1995, Gritsenko 1999). The five entries of the tuple are five *distinct* constructions; they cannot be combined additively. The Vol III counterexample at $N = 1$ is the controlling instance.

3.3 Local-to-global in mixed holomorphic–topological language

A second route from local protected data to a global compact factorization algebra runs through the mixed holomorphic–topological model of Costello and the holomorphic de Rham obstruction class. Formal local Hamiltonian BF data imply such a global object only after the global holomorphic de Rham obstruction class $[\mathrm{obs}_{HT,\mathrm{glob}}] \in H^1(X, \mathcal{O}_X)$, plus descent, quantum master equation, anomaly, and locality data, with the Hamiltonian condition supplied by the local model. For $X = K3\times E$ the global obstruction group is $H^1(K3\times E, \mathcal{O}_{K3\times E}) \cong \mathbb{C}$ (non-zero), so the formal-local-Hamiltonian-BF route does not by itself identify $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ to a global compact factorization algebra. Both bridges –- Hall–Borcherds for the Vol II gravity line, mixed-HT for the Costello local-to-global ascent –- are open; the only established object here is the scalar shadow.

4 The $K3\times E$ threefold realization question

The square root in the title is meant in Dirac's sense. The OP/DT scalar branch $Z^X_{\mathrm{OP/DT}} = -4096\,\Delta_5^{-2}$ is orientation-forgetful: the sign character is lost under squaring, and the inverse-square scalar cannot recover the sign by which a Pfaffian-line section distinguishes Δ_5 from $-\Delta_5$. The missing first-order datum is the protected Pfaffian, whose square is the inverse of the protected scalar shadow.

4.1 From the scalar trace to the retained operator datum

The Dirac square-root pattern compares the chiral-shadow Pfaffian with the scalar trace. Before taking the trace one has the protected Pfaffian of View 3; after the trace one has the inverse square as scalar shadow. The Pfaffian carries the orientation data; inverting its square forgets it and gives the orientation-forgetful scalar invariant.

Theorem 4.1 *(Orientation-forgetting scalar consequence of the Pfaffian).* On $K3\times E$, after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum, together with the retained finite geometric Pfaffian datum (P_{fin}), of Appendix 7, the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of Chapter 5 is the automorphic section $\mathcal{D}_X = \Delta_5$ of Chapter 3, and its inverse square is the level-n protected scalar shadow:

$$\big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})\big)^{-2} = \Delta_5(Z)^{-2} = \big(\Phi_{10}^{\mathrm{un}}(Z)\big)^{-1} = Z_{\mathrm{BPS}}^{K3\times E}(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}}), \tag{9.8}$$

and the OP/DT scalar branch is $-4096\, Z_{\mathrm{BPS}}^{K3\times E}$.

Proof. The Pfaffian–Dirac theorem of Chapter 6, with (D0), (O1), (O1$^+$), and $(O2_{\mathrm{atlas}})$ represented by the retained data of Theorem 7.5, and with the retained finite geometric Pfaffian datum (P_{fin}), asserts $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$. Squaring, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})^2 = \Delta_5^2 = \Phi_{10}^{\mathrm{un}}$; inverting, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})^{-2} = (\Phi_{10}^{\mathrm{un}})^{-1}$. By Theorem 1.2 the right-hand side equals the level-n protected scalar shadow. The Oberdieck–Pixton/Pandharipande scalar identity [92, Theorem 1] says that the OP chamber representative is -4096 times this trace; the factor is the OP convention, the squared theta-leading coefficient, and the sign branch of (9.1). $\square$

4.2 The orientation datum the inverse-square scalar cannot recover

The square-root sign is invisible to the scalar. An orientation-line sign $\varepsilon \in \{\pm 1\}$ acting by $\Delta_5 \mapsto \varepsilon\Delta_5$ satisfies $\varepsilon^2 = 1$; the scalar Δ_5^{-2} is therefore ε-invariant. Selecting the sign requires a pre-trace Pfaffian-orientation datum of View ③: the protected orientation character ϵ_o of the type-II Pfaffian wall normal form (O2) and its restriction to the type-II Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$, matched against the automorphic reflection character ν_{Δ_5}. Equality

$$\epsilon_o \,\big|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5} \,\big|_{W^{(2)}(\Lambda_{II}^{2,1})}$$

is the orientation-monodromy-character identity of Chapter 7; Appendix 7 states the retained-data criteria for (Do), (O1), (O1$^+$), and (O2) on $K3 \times E$. No constant appearing in (9.1) –- not the leading 64, not the squared 4096, not the OP scalar minus sign –- supplies ϵ_o. The local sign is checked on a rank-one Pfaffian wall; the global character requires the retained quotient orientation and the chosen Weyl transport, not a normalization constant.

4.3 The four obstructions and the realization question

Let $\mathfrak{D}_X^{\mathrm{DI}}$ denote the Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \lim_R \big(\mathcal{A}_{X,R}^{E_3},\, F_{X,R}^{\mathrm{hyb}},\, \Gamma_{X,R},\, \Pi_X,\, \Phi_R,\, o_R,\, H_R,\, P_R^{\Pi},\, C_{X,R},\, \Theta_{\mathrm{Kos},R},\, \mathcal{L}_{\mathrm{Pf},R},\, \mathrm{pf}_{X,R},\, \varepsilon_{o,R},\, \mathrm{Rec}_R\big)$$

of Chapter 5: the inverse limit, over finite root windows R, of fourteen pieces of data including the chosen finite E_3-prefactorization algebra $\mathcal{A}_{X,R}^{E_3}$, the hybrid $\mathrm{Ran}^{\mathrm{hyb}}(E)$ carrier $F_{X,R}^{\mathrm{hyb}}$, the cosection-twisted reduced d-critical orientation o_R, the Hall correspondence datum H_R, the Pfaffian line $\mathcal{L}_{\mathrm{Pf},R}$, the Pfaffian section $\mathrm{pf}_{X,R}$, and the recognition map Rec_R. The retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ datum, and retained finite geometric Pfaffian datum (P_{fin}) are the Pfaffian and orientation data of Appendix 7 for Theorem 4.1.

4.4 The threefold target, not a fourfold

The K3×E target is a complex threefold: $\dim_{\mathbb{C}} K3 \times E = 3$. The integer n in the "level-n shadow" is

$$n = \chi(\mathcal{O}_Y)$$

for the one-dimensional subscheme $Y \subset K3 \times E$, whereas $\chi(\mathcal{O}_{K3\times E}) = 2 \cdot 0 = 0$ is the target holomorphic Euler characteristic. The perfect obstruction theories in Chapter 5 are threefold theories; their virtual classes are reduced virtual classes on the moduli of objects on the threefold; the Pfaffian $\mathrm{Pf}_{\mathrm{prot}}$ of Chapter 5 is the Pfaffian of the threefold's protected operator data. The Borcherds product formula and the Gritsenko–Nikulin denominator identity give the Mukai-vector dictionary on the threefold by projection along the elliptic factor; the OP/DT/PT comparison gives the scalar shadow on the threefold; the protected Pfaffian gives the square root. No fourfold target intervenes anywhere in the chain.

4.5 Comparison with the determinant-only construction

The virtual K_0-determinant identity $\mathcal{D}_X(Z) = \Delta_5(Z)$ is a Grothendieck-class identity: it sees only the signed superdimensions of the underlying virtual super vector spaces. It does not see brackets, not parities of representatives, not relation constants, not completed PBW data, not Hopf-pairing radicals, not orientation gerbes, not the Pfaffian line $\mathcal{L}_{\mathrm{Pf}}$. Squaring the determinant gives Δ_5^2; inverting gives the scalar shadow. All the information about the retained operator datum $\mathfrak{D}_X^{\mathrm{DI}}$ that distinguishes the

Realization question (the K3×E threefold) — retained scalar-trace level. The square-root identity (9.8) is a theorem on $K3 \times E$ at the Pfaffian and scalar-trace level relative to the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and retained finite geometric Pfaffian datum (P_{fin}) in Appendix 7:

$(D0)$ *Do-degeneration limit of the cosection-reduced d-critical orientation theory.* **Criterion**: Theorem 7.7, relative to the retained Do-HN datum of Definition 7.6: flat Do-degeneration family, additive K3 semiregularity cosections, vanishing-cycle and orientation specialization, Pfaffian-line extension, protected-integration compatibility, and strict HN transitions.

$(O1)$ *Strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle.* **Relative to** $(O1_{\mathrm{quot}})$: Theorem 7.13.

$(O1^{+})$ *Weyl-equivariant transport of the orientation along $W^{(2)}(\Lambda^{2,1}_{II})$ reflections.* **Relative to the chosen Weyl lifts**: Theorem 7.14.

$(O2)$ *Local Pfaffian wall normal form on type-II walls; the $2^{12} = 4096$ sign sum on the half-Hilbert orbit, with the $(R1)$–$(R2)$ scalar separation.* **Relative to** $(O2_{\mathrm{atlas}})$: Theorems 7.19, 7.17.

(P_{fin}) *Finite geometric Pfaffian datum on a cofinal HN system: skew perfect complexes, Pfaffian lines and sections, type-II wall divisor orders, automorphic line comparisons, support and parity rows, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility.* **Relative to** (P_{fin}): Definition 7.3.

The level-Z Hall–Borcherds trace comparison is relative to the retained (Z_{HB}) datum and Theorem 7.23: the retained trace datum places the compact source in the level-Z chiral-Hochschild domain, and Vol I Theorem H [72] together with Vol II [73, chiral-Hochschild Beilinson stratification] clauses (i) and (ii) give the chain-level chiral-Hochschild trace $\mathrm{Tr}^{\mathrm{bulk}}_{n} = \Delta_5^{-2}$ on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$. The $\mathrm{Z}_{\mathrm{BPS}}$ square-root identity is therefore a retained-datum theorem on $K3 \times E$ at level Z, granted (Z_{HB}) and those two cited theorems; the level-A gravity-line operator algebra remains open in Vol II. The finite packet `certificates/trace/k3e_protected_trace` currently records (Z_{HB}) only by the obstruction ledger `TRACE_OBSTRUCTION_LEDGER_VERIFIED`, with `protected_trace_certification=false`.

retained-data identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_{X}) = \Delta_5$ from a non-Pfaffian square-root choice $\Delta_5 \mapsto -\Delta_5$ lives in the Pfaffian datum and the orientation character. The inverse-square scalar shadow (9.2), whose OP representative is (9.1), is the maximal information visible after the trace; the Pfaffian and the orientation are the stronger pre-trace data.

5 Open structures

5.1 The Hall–Borcherds residual revisited

Conjecture 3.3 asks for the filtered $\mathsf{SC}^{\mathrm{ch,top}}$ morphism from the $K3 \times E$ Hall boundary object to the gravitational boundary line algebra of Vol II, whose derived-centre trace has scalar character Δ_5^{-2}. Its trace part is the bridge from the level-S scalar identity $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ to a level-Z chiral-Hochschild trace datum. Its acting part is the further level-A gravity-line operator algebra.

The residual consists of three completions. The Drinfeld doubling $\widehat{\mathrm{CoHA}}^{\mathrm{red}} \to D(\widehat{\mathrm{CoHA}}^{\mathrm{red}})$ of the Vol III positive-half pairing: the pairing, completion, integral form, and stable-envelope transport that convert the Hall positive half $\mathsf{Y}^{+}(K3 \times E)$ into the full quantum group $G(K3 \times E)$. The current enveloping along E and weight-completion: the chiral specialization $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$ along the reference curve $C = E$, together with the weight-completed ambient required to host the formal generators of $\mathfrak{g}_{\Delta_5}$. The filtered $\mathsf{SC}^{\mathrm{ch,top}}$ morphism property: the resulting boundary object must be a filtered morphism in the two-coloured Swiss-cheese chiral-topological operad, not merely a graded-vector-space comparison. Their existence is the Vol II Hall–Borcherds residual conjecture.

5.2 *The gravity-line operator algebra and the Pentagon-face determinant*

The Vol II gravity-line problem asks for a chain-level operator algebra $\mathfrak{Op}^{K3\times E}_{\mathrm{grav}}$ whose Pentagon-face determinant section is $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$. If such an algebra is constructed, its partition character is the reciprocal section $(\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$. The Brown–Henneaux dictionary $c = 3\ell/(2G_N)$ fixes the Virasoro boundary datum at Vir_c; it does not construct the $K3 \times E$ gravity-line algebra and does not evaluate the Pentagon-face determinant. A K3-Heisenberg witness would have to specialize the Virasoro λ-bracket $\{T_\lambda T\} = \partial T + 2T\lambda + (c/12)\lambda^3$ to the closed K3 boundary at $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$ and identify the resulting determinant with $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$. Existence of the gravity-line operator algebra, agreement of its reciprocal partition character with the unscaled trace (9.2), equivalently with the OP representative (9.1) after OP normalization on each primitive closed K3-class fibre, and factorisation through the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) = \mathcal{D}_X = \Delta_5$ of Chapter 5 are open level-A conditions.

5.3 *Mathieu and umbral moonshine adjacency*

The K3 elliptic genus $Z_{\mathrm{K3}} = 2\phi_{0,1}$ admits the $N = 4$ superconformal-character decomposition of Eguchi–Ooguri–Tachikawa [33]: after the massless characters are removed, the holomorphic mock-modular series

$$\Sigma(\tau) = -2q^{-1/8} + \sum_{n\geq 1} A_n q^{n-1/8}$$

has coefficients A_n which are dimensions of, and after twining characters of, M_{24}-modules [37, 39]. The decomposition is the infinite massive-sector decomposition of the $N = 4$ algebra. The coefficients $f(n,l)$ of $\phi_{0,1}$ which drive the Borcherds product are the untwined Jacobi coefficients; an M_{24}-twining would replace $\phi_{0,1}$ by the corresponding twined K3 elliptic genus before the singular theta lift. The umbral moonshine extension of Cheng–Duncan–Harvey [19] is indexed by the 23 Niemeier root systems with non-empty root system, with M_{24} the A_1^{24} case. The interaction between this sporadic-group structure on the K3 elliptic genus and the automorphic / BKM denominator structure on the $K3 \times E$ Igusa cusp form is the Mathieu / umbral moonshine adjacency.

5.4 *The closed-cobordism partition reads as one*

The scalar shadow is the trace, the Pfaffian is the pre-trace operator-level section, and the primitive input is the retained E_3-prefactorization input whose Pfaffian line is presented by $\mathfrak{D}^{\mathrm{DI}}_X$. The closed $K3 \times E \to \mathrm{pt}$ cobordism sees only the orientation-forgetful scalar trace. At that level the inverse of the squared Pfaffian section is the Laurent expansion $\Delta_5^{-2} = (\Phi_{10}^{\mathrm{un}})^{-1}$ of the inverse Igusa cusp form, and the orientation information that distinguishes Δ_5 from $-\Delta_5$ lives in the Pfaffian datum at level Z. The closed-cobordism partition function reads as one because Δ_5 is the same section throughout: the Borcherds–Gritsenko–Nikulin section of the weight-5 automorphic line, the Borcherds–Kac–Moody denominator on the root lattice $\Lambda^{2,1}_{II}$, and the protected Pfaffian of the Dirac–Igusa pro-object on $K3 \times E$, relative to the retained orientation and wall-atlas data. The squared shadow of this same section is the protected partition function on the closed cobordism $K3 \times E \to \mathrm{pt}$; the Hall–Borcherds residual of Vol II, when constructed, will lift it to the gravity-line operator algebra at level A; the threefold realization question of

§4 is the retained-data construction of the Dirac–Igusa pro-object and the remaining compact-source recognition problem.

Bibliographic notes

The OP/DT scalar identity is Oberdieck–Pixton [92], Oberdieck–Pandharipande [91], and Oberdieck [90]; the variable conventions follow Bryan [15]. The Oberdieck–Pixton/Pandharipande scalar identity [92, Theorem 1] is the consolidated form used here. The Andrianov factorization of the spinor L-function is [2]; the Saito–Kurokawa correspondence at weight 10 is [107]; the Manin–Deligne period theorem is [77], [62], [28]. The dyon-counting interpretation in CHL backgrounds is Dijkgraaf–Verlinde–Verlinde [31] and Borcherds [10]; these inputs are the boundary-row physical hosts and are kept separate from the $K3 \times E$ protected scalar shadow throughout.

The Hall–Borcherds residual conjecture in §3.2 is Vol II's Construction Problem 2 [73], the gravity-line Hall–Borcherds comparison. The protected D-brane index conjecture $\Omega_X(h, d, n) \overset{\text{phys}}{=} \mathbf{DT}^X_{n,(\beta_h,d)}$ is recorded as the protected D-brane-index conjecture of Chapter 5; it is not used in the OP/DT scalar proof. The Mukai dictionary, the K3-class chamber convention, and the $\Lambda^{2,1}_{II}$ lattice data are imported from Chapter 2. The protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}$ and the Dirac–Igusa pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ are specified in Chapter 5; their Pfaffian and orientation parts are established relative to the retained data of Appendix 7; the Pfaffian–Dirac theorem is Chapter 6; the four obstruction classes (D0), (O1), (O1$^+$), (O2) are defined and tracked in §5.8 of Chapter 5.

The κ-tuple $(0, 0, 3, 5, 24)$ and its non-additivity at $N = 1$ (the controlling instance for Δ_5) are recorded in Vol III; the Hall positive half $\Upsilon^+(K3 \times E)$ and its Drinfeld doubling datum toward $G(K3 \times E)$, as well as the 6d holomorphic Chern–Simons quartic obstruction $\int_X \mathrm{Tr}_{\mathrm{ad}}(A(F_A)^3)$, are also Vol III. The local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ and the holomorphic de Rham obstruction class are mixed-HT-strings. None of these supply a 3D gravitational path integral.

Chapter 10

Open obstruction classes

1 Open obstruction classes

The cross-level identity ② ↔ ③ has two logically distinct parts. The Pfaffian and orientation part, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ and $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$, holds on $K3 \times E$ relative to the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum, together with the retained finite geometric Pfaffian datum (P_{fin}), of Appendix 7. The primitive recognition $P_X \cong \mathfrak{g}_{\Delta_5}$ is conditional on the finite compact-source recognition datum $(S1)$–$(S10)$. The exact theorem that remains after deleting any one of these data is Proposition 0.1; no deletion is filled by a scalar trace, automorphic section, or denominator product. The implication and non-redundancy statement is Proposition 0.2. The finite Pfaffian packet `certificates/pfaffian/k3e_finite_pfaffian` currently records (P_{fin}) only by its obstruction ledger `blocked_obligations.csv`: skew perfect complexes, Pfaffian lines, finite sections, wall charts, automorphic line comparisons, transitions, and scalar firewalls remain missing, and the verifier records `pfaffian_certification=false`. The finite O2 wall-atlas packet `certificates/wall_atlas/k3e_o2_atlas` likewise records $(O2_{\mathrm{atlas}})$ only by its obstruction ledger: simple-wall coverage, wall objects, rank-one charts, overlaps, Weyl transports, half-Hilbert-orbit rows, compatibility rows, type-I exclusion, Do-limit extension, and scalar firewalls remain missing, and the verifier records `o2_certification=false`. The automorphic identity ① ↔ ② is theorem [9, 46]; the singular-theta-lift identity ② ↔ ⑤ is theorem [9]; the BKM denominator computation $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is theorem; the Lie-superalgebra recognition remains conditional from the finite-stage compact $K3 \times E$ source to the closed denominator algebra.

Each of the four obstructions sits at a definite categorical-dimensional level on the universal chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$. The remaining structures beyond the theorem are the Hall–Borcherds residual comparing the level-n protected scalar shadow with a chain-level trace datum, the chain-fusion conjecture for compact non-toric $K3 \times E$, and the gravity-line operator algebra required to have Pentagon-face determinant section Δ_5^2. The Mathieu / umbral moonshine adjacency stands between the compact source and these structures: its connection to $\mathfrak{g}_{\Delta_5}$ is conjectural. The conjecture asks for the twined denominator algebra and its reciprocal partition character. The finite protected-trace packet `certificates/trace/k3e_protected_trace` presently records the level-Z trace datum only by its obstruction ledger: trace categories, protected trace functors, operators, scalar normalizations, identities, forgetful maps, gravity-residual separation, transitions, and scalar firewalls remain missing, and the verifier records `protected_trace_certification=false`. Beilinson cut separates the retained-data Pfaffian theorem from the primitive source recognition and the gravity-line operator algebra.

The three names of Δ_5 are kept distinct on first occurrence: the automorphic section $\mathcal{D}_X = \Delta_5 \in H^0(\mathcal{H}_2, \mathcal{L}^5 \otimes \nu_{\Delta_5})$ of view ①; the BKM denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ of view ②; and the compact protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of view ③, relative to the retained data of Appendix 7.

1.1 *The four named obstructions*

The four obstructions act on distinct mathematical objects before any scalar trace is taken. The compact target is the Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R (\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathcal{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}$ The level discipline is *primitive object first, cross-level identity second, scalar shadow last*. The four obstructions and (P_{fin}) are pre-trace Pfaffian-orientation data of View ③; the level-Z object is the derived-centre trace pairing. They do not identify the compact primitive Lie superalgebra; that is the separate datum $(S1)$–$(S10)$.

1.1.1 (D0): D0-DEGENERATION LIMIT OF THE COSECTION-REDUCED D-CRITICAL ORIENTATION THEORY

1.1.1.1 D0-degeneration. The cosection-reduced d-critical orientation theory on the retained Hall stack $\mathcal{H}_R^{\mathrm{or}}$ of finite HN height R extends through the typed D0-HN datum of Definition 7.6: a flat D0-degeneration family, retained object/extension/flag stacks, proper or closed HN transitions, additive K3 semiregularity cosections, reduced vanishing-cycle specialization, Joyce–Upmeier orientation specialization, Pfaffian-line and protected-integration compatibility, and strict Mittag–Leffler exactness. In particular, the orientation line of the reduced determinant complex

$$\mathscr{D}_{R,C}^{\mathrm{red}} = \det \mathrm{RHom}_{\mathrm{red}}(C, C)$$

admits a Brav–Bussi–Dupont–Joyce–Szendrői [12] square-root gerbe section over the closure $\overline{\mathfrak{M}_{R,C}^{\mathrm{st}}} \subset \overline{\mathcal{H}_R^{\mathrm{or}}}$ including the D0-degeneration stratum, compatible with the cosection contraction inherited from $K3 \times E$. The retained Hall datum supplies an oriented critical finite-stage Pandharipande–Thomas/Gholampour–Sheshmani inverse system $\widehat{\mathcal{H}}_{\sigma,S}^{\mathrm{or}} = \varprojlim_R \mathsf{F}_R \mathcal{H}_{\sigma,S}^{\mathrm{or}}$ on the finite HN quotients of Proposition 0.197; (D0) asks that this system carry a coherent orientation through the D0-degeneration limit. The finite packet `certificates/d0/k3e_d0_hn` records the missing rows in `blocked_obligations.csv`: flat degeneration families, retained substacks, HN transitions, derived and d-critical charts, semiregularity cosections, vanishing-cycle and orientation specializations, Pfaffian/protected-integration specializations, Mittag–Leffler rows, and scalar firewalls. Its verifier returns `D0_OBSTRUCTION_LEDGER_VERIFIED`, with `d0_certification=false`. This is the finite D0-HN obstruction list, not a D0-HN construction.

1.1.1.2 Criterion. The retained D0-HN datum of Definition 7.6 supplies the D0-degeneration orientation through the retained inverse system via Theorem 7.7, and Theorems 7.13, 7.14, and 7.19 transport it to the type-II Weyl character. The Maass character ν_{Δ_5} and the geometric orientation ϵ_o then agree on $W^{(2)}(\Lambda_{II}^{2,1})$. BBDJS [12] vanishing-cycle data require an oriented d-critical locus, not boundedness alone, and Joyce–Upmeier [56] square-root data on CY_3 moduli do not automatically extend to the reduced $K3 \times E$ quotient: the cosection-induced shift in the determinant complex must be shown compatible with descent. The connected BE Borel/edge class $\alpha^{E,\mathrm{free}} = a_1 u_1 + a_2 u_2 \in H^2(BE; \mathbb{F}_2)$ is separated from finite $BE[N]$ Borel terms in the Borel spectral sequence; the quotient-orientation descent across object, extension, mixed, wrapped, two-step flag, overlap, Thom–Sebastiani, and $R' \to R$ transition strata is conditional, with overlap/correspondence/flag/transition compatibilities named separately.

1.1.1.3 Argument. The retained Hall boundedness datum [K3×E-fin], motivated by [66, 96], supplies a finite-type retained slice; the cited boundedness results do not by themselves prove this slice for the compact $K3 \times E$ Hall sector. The datum also does not give quasi-smoothness, a d-critical chart, BBDJS coefficients, an orientation line, or compact-support six-functor admissibility through the D0 limit. The finite rows required for this assertion are hn_type_bounds, semistable_substacks, derived_enhancements, universal_complexes, scalar_rigidifications, stratifications, extension_flag_stacks, cosection_atlas, transitions, and scalar_firewall. The finite-moduli packet `certificates/moduli/k3e_finite_moduli` presently records the HN-bound, semistable-substack, derived-enhancement, and rigidification-inertia rows, together with the finite class-bound and universal-complex rows and the E-translation-rigidification rows, retained closed-substack rows, and retained extension-closure rows, retained HN-factor-closure rows, and retained dual-closure rows, while its obstruction ledger keeps the flag stacks, cosection charts, transitions, and scalar firewalls open; the verifier records `moduli_certification=false`. Pantev–Toën–Vaquié–Vezzosi [95] shifted symplectic structures cover ambient derived moduli under Calabi–Yau hypotheses; the retained atlas is not produced by PTVV alone. Joyce–Upmeier [56] square-root orientation on Calabi–Yau$_3$ moduli requires exact-sequence/direct-sum compatibility; the reduced $K3 \times E$-quotient inherits this only after the cosection-induced shift is shown compatible with descent.

1.1.1.4 Quotient-orientation classes. The retained quotient-orientation datum consists of $\alpha^{\mathrm{red}}_{R,C} = w_2(\mathscr{D}^{\mathrm{red}}_{R,C}) \in H^2(S;\mathbb{F}_2)$, $\alpha^{E,\mathrm{free}}_{R,C}, \widetilde{\beta}^H_{R,C} \in H^2_H(S;\mathbb{F}_2)$, and $\lambda^H_{R,C} \in H^1(BH;\mathbb{F}_2)$ on every retained object, extension, mixed, wrapped, and flag stratum C, with killing of the Borel mixed terms and edge-spectral-sequence differentials at every node of the descent diagram. On the rank-one D6–D2–D0 stratum $C = (1, \beta, n)$, with β the K3-fibre primitive class, the BBDJS line carries the cosection contraction explicitly, and the four obstruction classes $(\alpha^{\mathrm{red}}, \alpha^{E,\mathrm{free}}, \widetilde{\beta}^H, \lambda^H)$ are the retained quotient-orientation datum of Appendix 7.3. The remaining source-side problems are this finite-stabilizer quotient orientation and the finite compact-source recognition datum for the Hall primitive algebra. The finite orientation packet `certificates/orientation/k3e_reduced_orientation` records this as an obstruction ledger, not as an orientation proof. Its `blocked_obligations.csv` table lists the missing square-root, Thom–Sebastiani, quotient Borel, finite-stabilizer, Weyl-lift, Coxeter, transition, and scalar-firewall rows, and the verifier reports `ORIENTATION_OBSTRUCTION_LEDGER_VERIFIED` with `orientation_certification=false`.

1.1.2 (O1): strong reduced orientation on the K3×E-quotient self-Ext frame bundle

1.1.2.1 Reduced quotient orientation. The BBDJS [12] orientation gerbe on the moduli of stable sheaves on $K3 \times E$ descends to the $K3 \times E$-quotient self-Ext frame bundle, with reduced orientation gauge: on the stable stratum $\mathfrak{M}^{\mathrm{st}}_{R,C} \subset \mathcal{H}^{\mathrm{or}}_R$ the obstruction tuple

$$\mathfrak{o}^{\mathrm{or}}_{R,C} = \big(\alpha^{\mathrm{red}}_{R,C}, \alpha^{E,\mathrm{free}}_{R,C}, \{(\widetilde{\beta}^H_{R,C}, \lambda^H_{R,C})\}_H, \mu^{\mathrm{or}}_{R,e}, \dots\big)$$

vanishes simultaneously, with all overlap, correspondence, flag, and $R' \to R$ transition data trivialized coherently. (O1) demands the strong reduced descent: the connected-BE class $\alpha^{E,\mathrm{free}}$, the finite-$BE[N]$ gerbe classes $\widetilde{\beta}^H$, and the residual finite linearization characters λ^H all vanish, separately, on every retained stratum, and the spectral-sequence differentials and Borel mixed terms are killed.

1.1.2.2 Criterion. The Hall well-formedness criterion is prior to recognition: by Proposition 0.165, an associative Hall product without a counital coproduct, bialgebra compatibility, Hopf pairing, radical, quotient, and strict transition data has no defined primitive source object. The primitive recognition criterion then reads: target arithmetic on $W_{\leq 3}$ is target Borcherds–Gritsenko–Nikulin–Kac data only, and recognition requires an actual compact source datum supplying $M, D, B, G, K, Q, A_\beta$, relation rows, Hopf radical/coideal kernel equality, PBW associated-graded comparison, and strict transition/Mittag–Leffler rows. The Koszul comparison adds the finite comparison residual $\mathfrak{O}_{\mathrm{Kcmp},R}$: by Proposition 0.213, agreement of the source-to-target map on constant-current primitives does not imply the quasi-isomorphism $\Omega^{\mathrm{ch}}_E C_X \simeq \mathtt{Den}_{\Delta_5,E}$, and by Definition 0.211 even that quasi-isomorphism must still respect Weyl transport, Pfaffian orientation, Hall pairing, radical quotient, PBW filtration, and HN transitions. The pro-object comparison also requires the finite morphism rows cofinal_subsystems, stage_objects, stage_morphisms, component_maps, composition_laws, ml_exactness, pro_isomorphisms, and scalar_firewall; without them a scalar Pfaffian product or denominator identity does not define a Dirac–Igusa pro-object. Theorem 7.13 of Appendix 7 states what the retained (O1) datum carries to the recognition theorem. The connected E and finite $E[N]$ cohomologies are *distinct* obstruction spaces: $H^*(BE;\mathbb{F}_2) = \mathbb{F}_2[u_1,u_2]$ with $|u_i| = 2$, so $H^1(BE;\mathbb{F}_2) = 0$; for $E[2] \cong (\mathbb{Z}/2)^2$, $H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1,x_2]$ with $|x_i| = 1$, and the residual linearization $\lambda = \lambda_1 x_1 + \lambda_2 x_2$ is independent data after the degree-two gerbe class is killed.

1.1.2.3 Argument. Translation invariance under E-action does not produce orientation descent: a fixed underlying line may carry a nontrivial finite character, and ordinary translation invariance is *not* a substitute for vanishing of λ^H on full two-primary generators of $H^*(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \Lambda(x_1,x_2)\otimes\mathbb{F}_2[y_1,y_2]$ at $N = 2^a$. The quotient-orientation criterion is finite Čech–Borel descent on the retained groupoid, not a vanishing theorem.

1.1.2.4 Quotient-orientation theorem. Theorem 7.13 is the quotient-orientation criterion relative to the retained $(O1_{\mathrm{quot}})$ datum. That datum contains the BBDJS gerbe sections on the self-Ext frame bundle, their null classes on the rank-one D6–D2–D0 stratum and its extension, mixed, wrapped, and two-step flag completions, and the overlap and correspondence transitions on the cofinal HN system. The data $M, D, B, G, K, Q, A_\beta$, the relation rows, the Hopf radical equality, and the PBW comparison required by primitive recognition are still separate compact-source data.

1.1.3 (O1)$^+$: Weyl-equivariant transport of the orientation along $W^{(2)}(\Lambda^{2,1}_{II})$ reflections

1.1.3.1 Weyl transport. The orientation character on the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ is Weyl-equivariant: the orientation line of (O1) is equipped with coherent lifts $\tau_i : o_C \to s^*_{\delta_i} o_C$ of the three type-II simple reflections, each acting with sign -1, and the projective determinant-line cocycle $c_o \in Z^2(W^{(2)}(\Lambda^{2,1}_{II});\mathbb{F}_2)$ of the lift is a coboundary, with quotient torsor defects $\omega_{i,C} = 0$ on every retained groupoid component. At finite height R, the partial action groupoid $\mathcal{G}_R$ carries $c_{o,R} \in Z^2(\mathcal{G}_R;\underline{\mathbb{F}_2})$, and (O1)$^+$ asks that $[c_{o,R}] = 0$, $\tau_i^2 = 1$, and $\omega_{i,C} = 0$ hold coherently on every stratum and at every transition height.

1.1.3.2 Criterion. The local sign-extraction criterion reads: only after the unit character is killed and $N^{\mathrm{Pf}}_\delta = 1$ is computed from the reduced normal self-Ext chart does the local sign equal -1, and Maass values $\nu_{\Delta_5}(s_{\delta_i}) = -1$ plus divisor orders $d^{\mathrm{aut}}_{\delta_i} = f(0,\pm1) = 1$ supply target data only –- they do not compute N^{Pf}_δ, χ_v, or ϵ_o. The Coxeter presentation

$$W^{(2)}(\Lambda^{2,1}_{II}) = \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s^2_{\delta_i} = 1\rangle$$

implies that any orientation character on $W^{(2)}(\Lambda^{2,1}_{II})$ is determined by three generator signs once (O1)$^+$ gives the honest Weyl lift; the three rank-one signs established by $(O2_{\mathrm{atlas}})$ are -1, and then propagate to the full character. The Weyl-equivariant transport is a separate identity, distinct from the no-extra-relations and PBW identities, established relative to the Weyl lifts in Theorem 7.14.

1.1.3.3 Argument. Three local signs on the simple reflections imply a character on the full type-II Weyl group only after $[c_{o,R}] = 0$ and $\omega_{i,C} = 0$: the Weyl determinant-line cocycle measures whether the lift is honest, and the quotient torsor defects measure whether the descent commutes with the reflection action on the retained groupoid. The cocycle reduces to the classical group-cohomology class only on a complete $W^{(2)}(\Lambda^{2,1}_{II})$-orbit with constant coefficients; finite height R is a partial action groupoid whose component data must be identified on intersections.

1.1.3.4 Conditional theorem. Theorem 7.14 is the criterion for finite-height lifts $\tau_{i,R}$ whose squares are one, whose projective cocycle class $[c_{o,R}]$ is zero, and whose torsor defects $\omega_{i,C}$ vanish on the retained groupoid components. These data give Weyl-equivariant orientation transport once supplied. In the current finite orientation packet they are exactly the missing Weyl-lift and Coxeter rows in the obstruction ledger. They do not supply the height-seven source matrices required for the relation-closed primitive recognition window.

1.1.4 (O2): local Pfaffian wall normal form on type-II walls

1.1.4.1 Type-II wall normal form. On the three type-II walls $\mathbb{H}_{\delta_i}$ for $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$, the retained wall-atlas datum has half-Hilbert sign support of size $4096 = 2^{12}$. The local computation in Appendix E is only the rank-one crossing. Concretely, in an s_δ-equivariant real-analytic d-critical/Kuranishi chart $(U_\delta, x_\delta, u_\delta, K^{\mathrm{red}}_\delta)$, with K^{red}_δ the real form of the cosection-reduced self-Ext complex $\mathrm{RHom}_{\mathrm{red}}(C_\delta, C_\delta)$, the splitting $K^{\mathrm{red}}_\delta \simeq K^{\mathrm{tan}}_\delta \oplus K^{\mathrm{nor}}_\delta$ is equipped with the extension class $\eta_\delta \in \mathrm{Ext}^1_{D^{s_\delta}(U_\delta)}(K^{\mathrm{nor}}_\delta, K^{\mathrm{tan}}_\delta)$ trivial after the reduced E-quotient and quotient-orientation trivializations, and the normal Pfaffian admits the rank-one form $\mathrm{Pf}(K^{\mathrm{nor}}_\delta) = v_\delta\, u_\delta^{N^{\mathrm{Pf}}_\delta}$ with $v_\delta(x_\delta) \neq 0$, $s_\delta(v_\delta) = v_\delta$, and $N^{\mathrm{Pf}}_\delta = 1$.

1.1.4.2 The 4096 sign sum. The integer $4096 = 2^{12}$ is the half-Hilbert sign count retained in $(O2_{\mathrm{atlas}})$: twelve sign degrees of freedom, each $\mathbb{Z}/2$, give $2^{12} = 4096$ signed states. The type-II wall locus realizes this count only after the component, boundary, overlap, quotient, and protected-integration compatibilities of the retained atlas hold. The arithmetic match $-4096 = -(64)^2$ with the OP/DT scalar $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is the squared-theta normalization, fixed by Borcherds normalization and the OP scalar sign convention; it is *not* a derivation of ϵ_o.

1.1.4.3 Criterion. The local model and the exact separation from OP and Maass scalars are the following: scalar constants $D_5 = 64^{-1}\Delta_5$, $\chi^{\mathrm{OP}}_{10} = 4096^{-1}\Delta_5^2$, $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ are scalar normalizations and do not define ϵ_o; the Pfaffian normal form is the content. Proposition 0.201 strengthens this separation: the scalar product, its square, and the OP trace also do not determine $\mathrm{pf}_{X,R}$ as a section of the Pfaffian line, because the order-two monodromy is invisible after squaring. Theorem 7.19 is relative to $(O2_{\mathrm{atlas}})$, whose matrices over $\mathbb{Z}$ or $\mathbb{Q}$ record the Pfaffian-rank identity $N^{\mathrm{Pf}}_\delta = 1$, the splitting $\eta_\delta = 0$, the unit v_δ with s_δ-character trivial, and overlap/transition compatibility across the cofinal HN system. The packet `certificates/wall_atlas/k3e_o2_atlas` presently supplies only the obstruction ledger

`O2_OBSTRUCTION_LEDGER_VERIFIED`, with `o2_certification=false`. It records the missing atlas data; it does not supply the wall atlas.

1.1.4.4 Argument. The condition $(\delta, \delta) = 2$, the automorphic divisor order $d^{\mathrm{aut}}_{\delta_i} = 1$, and the Maass character $\nu_{\Delta_5}(s_{\delta_i}) = -1$ are target data that do not imply the Pfaffian rank assertion $N^{\mathrm{Pf}}_{\delta} = 1$. Graph-isogeny atom candidates $B_2 = i_{\Gamma_\varphi,*}L$ are Koszul-Ext-local; they are *candidates*, not constructions, until semistability, full $\widehat{\Gamma}_X$ charge match, reduced normal Ext splitting, invariant unit, and atlas overlap compatibility are established. The "higher terms do not alter" phrase in the rank-one Ext criterion requires the explicit equivariant real/parametric Morse lemma and unit check before it operates as a theorem.

1.1.4.5 Wall-atlas transport. Definition 7.1 gives the retained type-II wall-atlas datum. Theorem 7.19 transports that datum over the half-Hilbert orbit; it does not construct the graph-isogeny or reducible-curve atoms. Theorem 7.17 identifies the 2^{12} sign count with the squared theta-leading normalization only after scalarization. The remaining relation-closed source comparison is the recognition datum $(S1)$–$(S10)$, not a type-II wall normal-form problem.

1.2 Mathieu and umbral moonshine adjacency

The K3 elliptic genus has a precise representation-theoretic decomposition. Eguchi, Ooguri, and Tachikawa [33] observed in 2010 that the elliptic genus $Z_{\mathrm{K3}}(\tau, z) = 2\phi_{0,1}(\tau, z)$ admits an M_{24}-graded character expansion: the mock-modular completion $\Sigma(\tau)$ of the elliptic genus is a graded M_{24}-module, and the multiplicities of irreducible M_{24}-representations in the first several Fourier coefficients of the holomorphic part match the dimensions of irreducible M_{24}-representations on the nose. Gannon [39] proves the arithmetic identity; the geometric origin via a sigma-model action of M_{24} on the K3 elliptic-genus state space remains open outside the classes accounted for by known K3 sigma-model symmetries. The M_{24} action visible at the level of the elliptic genus does not lift to a global geometric symmetry of $D^b(\mathrm{Coh}(K3))$; the work of Gaberdiel–Hohenegger–Volpato [37] on Mathieu twining characters identifies the precise refinement at the level of supersymmetric sigma-model conformal field theory data.

The umbral moonshine of Cheng–Duncan–Harvey [19] extends Mathieu moonshine to a family of 23 cases indexed by Niemeier root systems, one case per Niemeier lattice having a non-empty root system. For a Niemeier lattice N^X with root system X, the finite group is the umbral group

$$G^X = \mathrm{Aut}(N^X)/W^X,$$

not the discriminant group. Mathieu moonshine is the $X = A_1^{24}$ case. The Leech lattice is the 24th Niemeier lattice and has empty root system; it belongs to Conway moonshine, not to the 23-case umbral list. The umbral moonshine conjecture of Cheng–Duncan–Harvey [19], in the module-existence form proved by Gannon [39], asserts that each pair (X, G^X) carries a graded module whose graded character is the prescribed vector-valued mock modular form.

1.2.0.1 The conjectural connection to $\mathfrak{g}_{\Delta_5}$. The Mathieu moonshine module lies on the $N = 4$ superconformal character side of the K3 elliptic genus. The Eguchi–Ooguri–Tachikawa module is graded by the massive level n; the z-dependence and the l-index are carried by the $N = 4$ character, not by a primitive (n, l)-grading on the M_{24}-module. The Beilinson-cut statement is therefore a conjectural extension of the twined K3 elliptic genus to the Borcherds–Kac–Moody datum on $\Lambda^{2,1}_{II}$, producing a

g-twined refinement of $\mathfrak{g}_{\Delta_5}$ with denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ for each M_{24}-conjugacy class g. The umbral cases of [19, 20] extend the conjecture to a family of 23 Borcherds–Kac–Moody-style algebras associated to the 23 Niemeier root systems with non-empty root sets.

1.2.0.2 Precise conjecture. For each conjugacy class $g \in M_{24}$, let $\phi_{0,1}^{(g)}(\tau, z)$ denote the g-twined K3 elliptic genus of [37]. The Cheng–Duncan–Harvey–Gannon umbral moonshine conjecture, refined to the BKM stratum of $\Lambda_{II}^{2,1}$, asserts the existence of a g-twined Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}^{(g)}$ with denominator

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)}) = \Theta(\phi_{0,1}^{(g),\mathrm{Weil}})\big|_{\mathbb{H}_2},$$

the singular theta lift of the g-twined Weil-representation extension of $\phi_{0,1}^{(g)}$, with the chamber and Weyl-vector data determined by the relevant umbral pair (X, G^X), when such a pair is part of the twining datum. No map from all M_{24}-classes to Niemeier root systems is asserted here.

1.2.0.3 The order-seven and order-fifteen classes. The four classes $\{7A, 7B, 15A, 15B\}$ lie outside the diagonal Gritsenko–Cléry/Borcherds frame used here, namely the rows selected by $\varphi(N) \mid 2$; this does not exclude their occurrence in other CHL dyon towers. Their cyclotomic fields have degrees 6 and 8. Their cycle shapes on the standard 24-dimensional permutation representation are

$$7A : 1^3 \cdot 7^3, \qquad 7B : 1^3 \cdot 7^3, \qquad 15A : 1 \cdot 3 \cdot 5 \cdot 15, \qquad 15B : 1 \cdot 3 \cdot 5 \cdot 15,$$

the $7A/7B$ pair distinguished by Galois conjugation under $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ and the $15A/15B$ pair by $\mathrm{Gal}(\mathbb{Q}(\zeta_{15})/\mathbb{Q})$; the rational character on each pair is integer-valued, but individual class characters take values in the cyclotomic field of order equal to the class order. These cycle shapes have different geometric status: $7A$ and $7B$ have six orbits and pass the Mukai–Kondo orbit test, while $15A$ and $15B$ have four orbits and are non-geometric in that test [38, §2]. The orders $|g| \in \{7, 15\}$ satisfy $\varphi(|g|) \nmid 2$ ($\varphi(7) = 6$, $\varphi(15) = 8$), placing them outside the Gritsenko–Cléry frame $N \in \{1, 2, 3, 4, 6\}$ of κ_{BKM}-universality selected by $\varphi(N) \mid 2$ of Theorem 8.1. A Borcherds–Kac–Moody construction for these twined genera therefore requires data beyond that denominator frame.

Conjecture 1.1 *(Order-seven and order-fifteen twined denominators).* For each $g \in \{7A, 7B, 15A, 15B\}$:

- (M★7A) The Gaberdiel–Hohenegger–Volpato $7A$-twined K3 elliptic genus $\phi_{0,1}^{(7A)}$ is a weight-0, index-1 weak Jacobi form with multiplier for the appropriate congruence subgroup attached to the $7A$ class. Its singular theta lift, if the Weil extension and Weyl chamber data are fixed, has Borcherds weight $c_{7A}(0)/2$ and multiplier, and equals $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(7A)})$ for a conjectural $7A$-twined Borcherds–Kac–Moody algebra.
- (M★7B) Identical statement for the Galois conjugate class $7B$, with $\phi_{0,1}^{(7B)} = \phi_{0,1}^{(7A)} \circ \sigma_{\zeta_7 \to \zeta_7^3}$ the Galois-conjugate character.
- (M★15A) The $15A$-twined character $\phi_{0,1}^{(15A)}$ is a weight-0, index-1 weak Jacobi form with multiplier for the appropriate congruence subgroup attached to the $15A$ class. Its singular theta lift, if the Weil extension and Weyl chamber data are fixed, has Borcherds weight $c_{15A}(0)/2$ and multiplier, and equals $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(15A)})$.
- (M★15B) Identical statement for the Galois conjugate class $15B$.

The four clauses have finite necessary conditions in their first Fourier coefficients, but no finite truncation proves the denominator identity.

Remark 1.2 *(Fourier conditions and the parity obstruction).* Three Fourier conditions constrain each clause ($\mathrm{M}^{\star}g$) at finite truncation depth.

(i) The leading Fourier coefficient $c_g(0) = [q^0 r^0]\phi_{0,1}^{(g)}$ is the Gaberdiel–Hohenegger–Volpato McKay–Thompson value of g at the $q^0 r^0$ position of the K3 elliptic genus, tabulated in [37, Table 1] for all 26 conjugacy classes. Outside the CHL frame $\{1A, 2A, 3A, 4A, 6A\}$, this leading coefficient need not be an even integer in the Borcherds lattice normalisation; for the classes $\{7A, 7B, 15A, 15B\}$ it is not one of the values $c_N(0) \in \{10, 6, 4, 3, 2\}$ of Theorem 8.1 (within the frame the $N = 4$ entry $c_4(0) = 3$ is already odd, with half-integral weight $\frac{3}{2}$).

(ii) The Borcherds weight formula gives $\kappa_{\mathrm{BKM}}(\Phi^{(g)}) = c_g(0)/2$ only after the twined Borcherds datum and its lattice have been fixed. For the order-seven and order-fifteen classes this weight need not be an integer; the lift must therefore be treated as a multiplier-bearing automorphic product, not as a weight-5 paramodular form. This is the arithmetic obstruction to extending the level-Z denominator construction beyond the CHL frame.

(iii) Mathieu-moonshine consistency with the EOT M_{24}-multiplicity predictions through the first eight Fourier coefficients of $\Sigma(\tau)$ imposes a finite necessary condition on each clause; proving ($\mathrm{M}^{\star}g$) requires identifying the geometric origin of the M_{24}-action on the K3 elliptic-genus state space and its umbral refinement in the sense of Cheng–Harrison [20].

1.2.0.4 Independence from the four Pfaffian–Dirac obstructions. The Pfaffian theorem 2.1 and the orientation-character theorem 3.1 are retained-data theorems on $K3 \times E$, after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, ($O2_{\mathrm{atlas}}$), and the retained finite geometric Pfaffian datum (P_{fin}) of Theorem 7.5. The primitive recognition theorem remains conditional on ($S1$)–($S10$), and the level-A gravity-line operator algebra remains conditional on the Vol II Hall–Borcherds residual. The moonshine adjacency is independent of both: it lives at level Z (centre/denominator) and at level A (acting object on the K3 elliptic-genus state space). A proven identification of the moonshine module as a g-twined character of $\mathfrak{g}_{\Delta_5}$ does not by itself solve the Vol II Hall–Borcherds residual; conversely the Hall–Borcherds residual does not prove the moonshine adjacency.

1.2.0.5 Twined denominator problem. The g-twining operation would produce $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ from $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ and an M_{24}-action on a denominator-compatible lift of the Mathieu module. The ordinary massive-level grading of the M_{24}-module supplies the mock modular coefficients; the Borcherds product requires the additional Jacobi variable and $\Lambda_{II}^{2,1}$-root grading. That lift, and the denominator-compatible action on it, are not part of the EOT theorem.

1.2.0.6 Compatibility with Vol III. The Vol III dimension-stratified BKM catalogue [70] places $\mathfrak{g}_{\Delta_5}$ as the $d = 3$ sibling, with companions at $d = 3$ (Borcherds Monster $V^{\natural}$ on $\mathrm{II}_{1,1}$) and $d = 5$ (Fake Monster on $\mathrm{II}_{25,1}$ via $K3 \times K3 \times E$). The Mathieu/umbral moonshine adjacency to $\mathfrak{g}_{\Delta_5}$ is a level-Z cross-stratum conjecture: the denominator-compatible M_{24} action on the K3 elliptic-genus state space for $\{7A, 7B, 15A, 15B\}$ is the missing piece on the K3-fibre side; the g-twined denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ is the missing piece on the BKM side.

1.2.0.7 Twined denominator identity. For each of the 26 conjugacy classes of M_{24}, the expected object is a g-twined Borcherds product on $\Lambda^{2,1}_{II}$. Its specialization to the $N = 4$ character expansion must recover the twined McKay–Thompson coefficients of EOT–Gannon. A coefficientwise refinement of the EOT multiplicities by BKM denominator coefficients is therefore a consequence only after the denominator-compatible Jacobi lift and the $\Lambda^{2,1}_{II}$-root grading have been fixed. The required identity is the consistency of the g-twined denominator with the OP scalar $Z^{X}_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ under the g-twining construction, including the squared sign convention.

1.3 *Residual operator structures*

The Beilinson cut separates the remaining operator data by the object on which the missing structure must live: the Hall–Borcherds operator lift, the compact non-toric chain-fusion equivalence, and the gravity-line algebra.

1.3.1 The Hall–Borcherds residual

1.3.1.1 Hall–Borcherds residual. The level-n protected scalar shadow $Z^{K3\times E}_{\mathrm{BPS}} = (\Phi^{\mathrm{un}}_{10})^{-1} = \Delta_5^{-2}$ on the closed $K3 \times E \mapsto$ point cobordism is the BPS partition function. A 3d gravitational path-integral interpretation with a genuine operator-level lift requires the Hall–Borcherds residual: an operator-valued completion of the scalar trace whose trace recovers the level-n shadow and whose algebraic content carries the metric degrees of freedom. This residual remains open in Vol II.

1.3.1.2 Comparison with Vol II. Vol II [73] names the gravity-line operator algebra required to have Pentagon-face determinant section $\Phi^{\mathrm{un}}_{10} = \Delta_5^2$, whose partition character is $(\Phi^{\mathrm{un}}_{10})^{-1} = \Delta_5^{-2}$, and identifies the Hall–Borcherds residual as the comparison arrow from the bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$ to the line-operator algebra. The Vol II residual asks for the construction of the gravity-line operator algebra with Pentagon-face determinant $\det_{\mathrm{Pentagon}} = \Phi^{\mathrm{un}}_{10} = \Delta_5^2$ on the closed cobordism, whose primitive square-root Borcherds weight is $\omega_{\mathrm{Borcherds}}(\Delta_5) = c_1(0)/2 = 5$ at the K3-Heisenberg witness. The Igusa-side datum is the same automorphic determinant identity $\Delta_5^2 = \Phi^{\mathrm{un}}_{10}$, with the Appendix G Pfaffian datum $\mathfrak{D}^{\mathrm{DI}}_X$ carrying the retained orientation and Weyl-transport data on the chiral side.

1.3.1.3 Argument. The protected graded trace

$$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F q^{L_0} r^{J_0} s^{\widetilde{L}_0}\big) = \Delta_5^{-2}$$

forgets orientation; the OP chamber representative is -4096 times this trace. The sign-character ϵ_0 is invisible in the squared form and is part of the Pfaffian–Dirac orientation theorem, not of the gravity-line residual. The Hall–Borcherds residual asks instead for an acting operator algebra whose trace recovers the scalar shadow. The realization as a three-dimensional gravitational path integral is conditional on Brown–Henneaux, modular invariance, vacuum dominance, and saddle dominance [49]; no such realization is asserted.

1.3.1.4 Scalar non-faithfulness. The structural reason Δ_5^{-2} does not by itself recover the operator-level gravity-line algebra is the scalar non-faithfulness of the graded Euler character on filtered $SC^{\mathrm{ch,top}}$-objects, proved as a lemma in Vol III [70]: an equality $\chi(C_1) = \chi(C_2)$ between filtered $SC^{\mathrm{ch,top}}$-objects does not determine a quasi-isomorphism, an $SC^{\mathrm{ch,top}}$-morphism, a Hall product, a Drinfeld pairing, a current-envelope locality structure, or a derived-centre trace –- adding a filtered acyclic summand

K with $\chi(K) = 0$ leaves the scalar unchanged while admitting non-trivial extensions of the operadic action, and a filtered differential may be perturbed within the associated graded without altering χ. The missing chain-level content underneath Δ_5^{-2} sits under Vol III's Hall–Borcherds datum: a positive-half Hall–Borcherds bialgebra morphism on E, a completed Drinfeld-double extension, a current-envelope morphism along E, and trace compatibility with Δ_5^{-2} [70]. The Igusa scalar is only the trace shadow of this datum. The Hall–Borcherds residual is the missing chain-level lift together with its finite-window identities.

1.3.1.5 Pentagon-face cyclic trace. The required cyclic trace is $\mathrm{Tr}^{\mathrm{cl}}_{\mathrm{Pentagon}}$ on the closed-colour factorization algebra F^{cl} at the $K3 \times E$-Heisenberg witness. Its evaluation must recover $c_1(0)/2 = 5$ on the universal Borcherds-weight identity, and its operator-level lift to the open-colour boundary algebra $A_b = \mathrm{End}_C(b)$ must factor through the chain-fusion datum below. The compatibility condition is the consistency of the Pentagon-face scalar with the unscaled trace Δ_5^{-2} under the squared-trace operation. The OP chamber representative is -4096 times this trace.

1.3.2 Chain fusion for compact non-toric K3×E

1.3.2.1 Chain-fusion datum. For a Calabi–Yau$_d$ category C_X and a chart datum (Σ_{d-1}, C), a chain-fusion datum consists of a boundary vacuum $b(X, \Sigma, C)$ in an open factorization dg-category on (C, D_C, τ_C), a comparison morphism

$$\Phi_d^{(\Sigma_{d-1}, C)}(C_X) \longrightarrow \mathrm{End}_C(b(X, \Sigma, C)),$$

and the compatibility of this morphism with clutching, trace, and the Calabi–Yau orientation. The chain-fusion conjecture asserts that this morphism is a quasi-isomorphism for the admissible compact chart. The affine toric examples $\mathbb{C}^3$, local $\mathbb{P}^2$, and the conifold furnish local comparison models; they do not construct the compact non-toric boundary vacuum for $K3 \times E$. At $d = 3$, the retained Dirac–Igusa data supply the protected scalar shadow on $K3 \times E$; the boundary-vacuum construction required for chain fusion remains open.

1.3.2.2 Comparison with Vol III. Vol III's catalogue [70] names this conjecture in the two-stage factorization frame and identifies, after chain fusion, the level-Z Drinfeld double obligation $G(X) = D(Y^+(X))$ for compact non-toric CY_3, together with the conditional TCFT/framing/anomaly/completion datum needed for compact non-formal CY_3 in general. Chain fusion remains a conjecture, not a theorem; the assertion of Δ_5 as a protected Pfaffian section on $K3 \times E$ is the retained-data identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ of view ③, not an underlying microscopic state space.

1.3.2.3 Argument. The compact $K3 \times E$ case at $d = 3$ is a candidate instance of chain fusion via the rectified finite $K3$-to-E specialization (Definition 0.86). The proposed chiral specialization, if constructed from the retained finite-stage datum, has target form $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\mathfrak{A}_{K3\times E}) = U^{\mathrm{ch}}(\mathfrak{heis}_{\mathrm{Muk}}) \otimes U^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{BPS}}_{K3})$ on $\mathrm{Ran}(E)$. The map $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$ is the chosen $[K3{\rightarrow}E\text{-sp}]$ datum, not a canonical functor. The finite proof rows for that datum are formal_target_charts, field_complexes, e3_operations, bv_qme, anomaly_framing, compact_support, factorization_descent, k3_to_e_specialization, transitions, and scalar_firewall; without them the compact source is only a named input. The finite packet `certificates/e3_source/k3e_e3_holfa` currently records these rows only by the obstruction ledger

`E3_SOURCE_OBSTRUCTION_LEDGER_VERIFIED`, with `e3_source_certification=false`. The boundary algebra $A_b = \mathrm{End}_C(b)$ on the open factorization side can coincide with this target only after the chain-fusion boundary vacuum $b(K3 \times E, \Sigma_2, E)$ exists. Compact non-toric $K3 \times E$ generalizations to other compact non-toric CY_3-data (e.g. Borcea–Voisin threefolds, Schoen's small resolutions) require independent boundary vacua.

1.3.2.4 Boundary vacuum condition. The missing object is $b(K3 \times E, \Sigma_2, E)$, a boundary vacuum in an open factorization dg-category on (E, D_E, τ_E) with D_E at the Igusa cusp $\mathfrak{c}_\infty$ of the Siegel space. The chain-fusion isomorphism on $W_{\leq 3}$ at finite HN height $R = 3$ must identify the chiral algebra with $A_{b(K3\times E,\Sigma_2,E)}$. The necessary compatibility condition is the consistency of the chain-fusion isomorphism with the unscaled trace Δ_5^{-2}, whose OP representative is $-4096\,\Delta_5^{-2}$, and with the universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 5$; see the κ-tuple discussion below.

1.3.3 The gravity-line operator algebra

1.3.3.1 Gravity-line algebra. A gravity-line operator algebra, if constructed, is an operator algebra on the closed-colour factorization algebra F^{cl} on $K3 \times E$ whose Pentagon-face determinant section is required to be $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$, a weight-10 automorphic section. Its partition character is the reciprocal section Δ_5^{-2}. The primitive square-root row has Borcherds weight

$$\omega_{\mathrm{Borcherds}}(\Delta_5) = \kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5.$$

This is the same Hall–Borcherds residual and remains open in Vol II.

1.3.3.2 Comparison with Vol II. Vol II [73] names the gravity-line Hall–Borcherds comparison conjecture and the Pentagon-face determinant; Vol III [70] names $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ as the universal Borcherds-weight identity [9], with $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 5$ at the paramodular Δ_5 datum.

1.3.3.3 The κ-tuple. The five-fold κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ at $K3 \times E$ records five separate invariants, distinct in construction and meaning, never additively combined. Here $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3}) \cdot \chi(\mathcal{O}_E) = 2 \cdot 0 = 0$ is the Künneth-multiplicative categorical Euler; $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = \sum_q (-1)^q h^{0,q}(K3 \times E) = 0$ by Serre duality on the compact total space; $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ is the rank of the Heisenberg–Cartan in the chiral Hall specialization on $\mathrm{Ran}(E)$ and equals the Cartan rank of $\mathfrak{g}_{\Delta_5}$; $\kappa_{\mathrm{BKM}} = c_1(0)/2 = 5$ is the universal Borcherds-weight identity at the Gritsenko–Cléry row whose Jacobi seed is $\phi_{0,1}$ and whose Borcherds product is Δ_5, with $c_1(0) = 10$; $\kappa_{\mathrm{fiber}} = 24$ is the Mukai-lattice rank $\mathrm{rk}\,\widetilde{H}^*(K3, \mathbb{Z}) = 24$ of the K3 fibre, equal to $\chi_{\mathrm{top}}(K3)$. The additive identity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails because it confuses $\chi_{\mathrm{top}}(K3) = 24$ with $\chi(\mathcal{O}_{K3}) = 2$; the universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the correct frame.

1.3.3.4 Argument. The Pentagon-face determinant is the modularity-witness of the open-side cyclic trace plus clutching; modularity is a property of the open trace, not of the closed algebra. On the Δ_5 row, $c_1(0)/2 = 5$ is the weight of the primitive Borcherds factor. The determinant required by chain fusion is its square $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$, hence a form of total weight 10 on the closed cobordism; the level-n protected scalar shadow is the reciprocal Laurent expansion Δ_5^{-2}.

1.3.3.5 Gravity-line trace condition. The required object is the gravity-line operator algebra $\mathrm{GravLine}_{K3\times E}$ on the open factorization dg-category on (E, D_E, τ_E) at the Igusa cusp, with Pentagon-face determinant identity $\det^{\mathrm{cl}}_{\mathrm{Pentagon}}(\mathrm{GravLine}_{K3\times E}) = \Delta_5^2$ and compatibility with the universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_1) = 5$ and with the κ-tuple $(0, 0, 3, 5, 24)$ at the K3-Heisenberg witness. The compatibility condition is the existence of chain fusion at $b(K3 \times E, \Sigma_2, E)$ and the Hall–Borcherds residual lifting the protected trace $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ toward the acting operator-algebra level.

The finite compact-source recognition datum $(S1)$–$(S10)$ on the first relation-closed window is the packet

window_closure, target_source_representatives, chevalley_serre_matrices, hall_boundary_complex, spectral_seque

The compact Hall source packet presently has a verified obstruction ledger,

`certificates/sources/k3e_compact_hall/blocked_obligations.csv`,

which lists the missing $M, D, \eta, \epsilon, B, G, K, Q, A$ matrices, the Hall bialgebra identity rows, Hopf-pairing identity rows, radical ideal/coideal rows, relation rows, no-extra equality, generation, PBW, strict transition, A_β-intertwining, and finite Koszul comparison rows. Its verifier records `compact_source_recognition=false`. This ledger is the finite obstruction list for the source theorem, not a substitute for the source theorem. The first relation-closed window has its own obstruction ledger,

`certificates/first_window/k3e_relation_closed_window/blocked_obligations.csv`,

with status `FIRST_WINDOW_OBSTRUCTION_LEDGER_VERIFIED` and `first_window_certification=false`. It records the missing window, representative, relation, Hall-boundary, spectral, kernel, PBW, theorem, and transition rows for the first finite primitive-recognition theorem. The Dirac–Igusa pro-object morphism packet has its own obstruction ledger,

`certificates/morphisms/k3e_dirac_igusa_pro_object/blocked_obligations.csv`,

with status `MORPHISM_OBSTRUCTION_LEDGER_VERIFIED` and `pro_object_certification=false`. It records the missing cofinal HN subsystem, fourteen finite-stage component rows, strict transition morphisms, component maps for $(M1)$–$(M8)$, composition laws, Mittag–Leffler exactness, and pro-isomorphism rows. The theorem-dependency ledger

beilinson_levels, objects, morphisms, theorem_dependencies, dependency_edges, forbidden_edges, status_sep

is now a populated schema-complete ledger. Its verifier checks Beilinson-level sites, typed objects, morphisms, theorem rows, allowed proof edges, forbidden proof edges, claim-status separation, Beilinson-level equality scopes, equality-site rows, transition rows, definition-separation rows, obstruction-independence rows, and scalar-firewall coverage; it still records `dependency_certification=false` and `mathematical_certification=false`. Thus the proof-level exclusions are finite rows, but the retained construction packets, recognition matrices, mirror comparison, and gravity-line operator algebra remain separate mathematical obligations. Together with these packets, the chain-fusion boundary vacuum $b(K3 \times E, \Sigma_2, E)$, the Hall–Borcherds residual lifting $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$, the gravity-line operator algebra with Pentagon-face determinant Δ_5^2, and the Mathieu-twined denominators $\mathrm{den}(\mathfrak{g}^{(g)}_{\Delta_5})$ on the 26 M_{24}-conjugacy classes are the unresolved data. The four obstructions concern, respectively, the D0-degeneration of the reduced orientation theory, quotient orientation, Weyl transport of the orientation line, and the type-II Pfaffian wall atlas. The remaining operator obstructions concern the Hall–Borcherds operator lift, compact non-toric chain fusion, and the gravity-line algebra. The Beilinson cut separates these objects before any scalar trace is taken.

1 Lattice computations

The seven integral lattices are recorded here in one place: the hyperbolic plane $U = \mathrm{II}_{1,1}$, the Lorentzian root lattice $\Lambda^{2,1}_{II}$ of the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$, the K3 cohomology lattice $\Lambda_{K3} = \mathrm{II}_{3,19}$, the lattice-polarized K3 sublattice Λ_{pol}, the Mukai lattice $\widetilde{H}(S,\mathbb{Z}) \simeq \mathrm{II}_{4,20}$, the formal dyonic charge lattice $\Gamma^{\mathrm{form}}_X = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$, and the signature-(3, 2) lattice $\Lambda^{3,2}$ that bridges $\mathbf{Sp}_4(\mathbb{Z})$ to the type-IV orthogonal group via the projective exterior-square image. Each is given by its Gram matrix or root data, signature, discriminant, and automorphism group, with primary citations (Conway–Sloane 1999 §2.5, §4, §16; Nikulin 1980 Theorem 1.1.2 and Theorem 1.14.4; Mukai 1984; Gritsenko–Nikulin 1998 Part II §2, §5). Ranks, signatures, and the Cartan matrix of the type-II simple roots are checked by the rational computations displayed below.

The identities about Δ_5, $\mathfrak{g}_{\Delta_5}$, the orientation character ν_{Δ_5}, and the Pfaffian wall normal form on $W^{(2)}(\Lambda^{2,1}_{II})$ use the lattice data fixed here, not a sign or scaling chosen later.

Convention 1.1 *(Sign of the bilinear form; orientation of the time-like cone).* The standard mathematical convention for the hyperbolic plane U is $U = (\mathbb{Z}^2, ((0,1),(1,0)))$, even unimodular of signature $(1,1)$ and discriminant -1 (Conway–Sloane 1999 §2.5). This convention is primary.

The Borcherds–Gritsenko–Nikulin packaging used in §1.4 of Chapter 5 and the exterior-square model of $\mathbf{Sp}_4(\mathbb{Z})$ employs the negated form

$$\Lambda^{(1,1)} = \left(\mathbb{Z}^2, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\right),$$

so that the time-like cone $C(\Lambda^{2,1})_+ \subset \Lambda^{2,1} \otimes \mathbb{R}$ has *negative* square $(x,x) < 0$. The relation is $\Lambda^{(1,1)} = U(-1)$. Every Gram matrix below is given in the standard convention; a sign flip transports it to the BKM convention without changing rank, discriminant absolute value, or automorphism group.

1.1 *The hyperbolic plane $U = \mathrm{II}_{1,1}$*

The hyperbolic plane is the unique even unimodular indefinite lattice of signature $(1,1)$ (Conway–Sloane 1999 §2.5; Nikulin 1980 Theorem 1.1.2).

Definition 1.2 *($U = \mathrm{II}_{1,1}$).* [definitional] Let $U := \mathbb{Z}e \oplus \mathbb{Z}f$ with Gram matrix

$$G_U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad e^2 = f^2 = 0, \quad e \cdot f = 1.$$

Verification 1.3 *(Rank, signature, discriminant of U).* [computed] $\mathrm{rk}\, U = 2$. The eigenvalues of G_U are ± 1, so the signature is $(1,1)$. The discriminant is

$$\mathrm{disc}\, U = \det G_U = -1.$$

U is even (every G_U-diagonal entry is $0 \equiv 0 \pmod 2$) and unimodular ($|\det G_U| = 1$). These are the defining invariants of $\mathrm{II}_{1,1}$ in the Conway–Sloane classification (Conway–Sloane 1999 §2.5, Theorem 2.5.1).

PROPOSITION 1.4 *(*$\mathrm{Aut}(U)$*).* [proved] The orthogonal group of U is

$$\mathrm{O}(U) = \{\pm\mathbf{I}\} \times \langle\sigma\rangle \simeq (\mathbb{Z}/2)^2, \qquad \sigma : e \leftrightarrow f.$$

The proper subgroup $\mathrm{O}^+(U) \simeq \mathbb{Z}/2$ swaps e and f but fixes the spinor norm.

Proof. An isometry $\phi \in \mathrm{O}(U)$ preserves the isotropic vectors of U. In rank 2 over $\mathbb{Z}$, the isotropic primitive vectors are exactly $\pm e, \pm f$, so ϕ permutes these four vectors. The permutations preserving the bilinear form give the four-element group above (Conway–Sloane 1999 §4.1). □

Remark 1.5 *(Standard sign convention versus BKM convention).* The BKM convention uses $\Lambda^{(1,1)} = U(-1)$ in the cusp polarization of §1.4 and in the exterior-square model of the automorphic group, where the time-like cone of $\Lambda^{2,1}$ has negative square. The map $U \to U(-1)$, $(e, f) \mapsto (e, f)$ is a bijection of underlying $\mathbb{Z}$-modules; only the bilinear form flips sign. The lattices $\Lambda^{2,1}$ and $\Lambda^{3,2}$ are presented in BKM coordinates; even-unimodular ambient lattices (Λ_{K3}, $\widetilde{H}(S, \mathbb{Z})$) are presented in the standard Conway–Sloane convention.

1.2 The Lorentzian root lattice $\Lambda^{2,1}_{II}$

The Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ of [48, Theorem 2.1, Section 5.1.1, case $(t = 1, II, \bar{0})$] is graded by an integral lattice of signature $(2, 1)$. This is the lattice on which the type-II simple-root Cartan matrix is $2(2\mathbf{I}_3 - \mathbf{J}_3)$ where $\mathbf{J}_3$ is the all-ones matrix.

Definition 1.6 *(Ambient* $\Lambda^{2,1} = U(-1) \oplus \langle 2\rangle$*).* [definitional] Set

$$\Lambda^{2,1} := U(-1) \oplus \langle 2\rangle = \mathbb{Z}e \oplus \mathbb{Z}f \oplus \mathbb{Z}h, \qquad e \cdot f = -1,\ e^2 = f^2 = 0,\ h^2 = 2.$$

The Gram matrix in the basis (e, f, h) is

$$G_{2,1} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad \det G_{2,1} = -2, \qquad \operatorname{disc} \Lambda^{2,1} = -2.$$

The BKM lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ in (f_2, f_3, f_{-2})-coordinates uses exactly this Gram, with $\Lambda^{(1,1)} = U(-1)$ on the (f_2, f_{-2})-block and $[2] = \langle 2\rangle$ on the f_3-axis.

Definition 1.7 *(*$\Lambda^{2,1}_{II}$ *as the index-four type-II sublattice of* $\Lambda^{2,1}$*).* [definitional] The type-II sublattice

$$\Lambda^{2,1}_{II} := \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv 0\,(2),\ n \equiv 0\,(2)\} \subset \Lambda^{2,1}$$

is the index-4 sublattice on which the type-II Cartan matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$ is the Gram of the simple roots $(\delta_1, \delta_2, \delta_3)$ of Proposition 1.10. Abstractly it is the lattice

$$\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2\rangle, \qquad \operatorname{disc} \Lambda^{2,1}_{II} = -32,$$

matching the terminology of Chapter 1 and the type-II sublattice convention of Chapter 4.

Verification 1.8 *(Rank, signature, discriminant of $\Lambda^{2,1}_{II}$).* [computed] $\operatorname{rk}\Lambda^{2,1}_{II} = 3$. The Gram of $\Lambda^{2,1}_{II}$ in $(\delta_1, \delta_2, \delta_3)$ of Proposition 1.10 is the Cartan matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$ of equation (4.6); its determinant is

$$\det\bigl(2(2\mathbf{I}_3 - \mathbf{J}_3)\bigr) = -32.$$

The hyperbolic-plus-rank-one block of the ambient $\Lambda^{2,1}$ has signature $(2,1)$; the index-4 sublattice $\Lambda^{2,1}_{II}$ inherits this signature, so

$$\sigma(\Lambda^{2,1}_{II}) = (2,1).$$

The discriminant group $(\Lambda^{2,1}_{II})^*/\Lambda^{2,1}_{II}$ has order 32 as required. The lattice $\Lambda^{2,1}_{II}$ is even because every $\delta_i^2 = 2$ and the off-diagonal couplings $(\delta_i, \delta_j) = -2$ are integral. Changing the sign of one hyperbolic basis vector gives the standard hyperbolic-plane convention $U = ((0,1),(1,0))$ and presents $\Lambda^{2,1} \simeq U \oplus \langle 2\rangle$ and $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2\rangle$; rank, signature, discriminant group order, automorphism group, and Cartan datum are unchanged.

Remark 1.9 *(Identification with $\Lambda^{2,1}$).* The BKM convention uses $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$ with Gram in (f_2, f_3, f_{-2})

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

matching $G_{2,1}$ of Definition 1.6 up to the basis swap $e \mapsto f_2, f \mapsto f_{-2}, h \mapsto f_3$. The type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ of Definition 1.7 is the index-4 sublattice on which the simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ of Proposition 1.10 live; it is the even Lorentzian root lattice of $\mathfrak{g}_{\Delta_5}$.

Proposition 1.10 *(Type-II simple-root Cartan).* [proved] In the coordinates (f_2, f_3, f_{-2}) fixed by the Weyl chamber, set

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Then $\delta_i^2 = 2$ for $i = 1, 2, 3$ and

$$\bigl((\delta_i, \delta_j)\bigr)_{i,j=1,2,3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix} = 2(2\mathbf{I}_3 - \mathbf{J}_3),$$

where $\mathbf{J}_3$ is the 3×3 all-ones matrix. Each δ_i is type-II in the sense that $(\delta_i, \Lambda^{2,1}) = 2\mathbb{Z}$.

Proof. Direct computation in the BKM Gram gives the matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$, equal to the displayed Cartan matrix. The type-II divisibility statement follows because each δ_i pairs against $\Lambda^{2,1}$ with even values: $(\delta_i, f_2), (\delta_i, f_3), (\delta_i, f_{-2}) \in 2\mathbb{Z}$ for each i. This is the GN root datum of [48, Section 5.1.1, case $(1, II, \bar{0})$]. □

Proposition 1.11 *(Weyl vector ρ).* [proved] The Weyl vector

$$\rho := \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2} \in \Lambda^{2,1} \otimes \mathbb{Q}$$

satisfies $(\rho, \delta_i) = -1$ for $i = 1, 2, 3$ and

$$(\rho, \rho) = -\tfrac{3}{2}, \qquad \rho \in C(\Lambda^{2,1})_+.$$

In particular ρ lies in the time-like cone, so the chamber $\mathcal{P}_{II} = \{x \mid (x, \delta_i) \le 0,\ i = 1, 2, 3\}$ has finite hyperbolic volume.

Proof. The simple-root condition $(\rho, \delta_i) = -(\delta_i^2)/2 = -1$ follows from Kac [58, §2.5, Lemma 2.5] for any choice of simple roots of square 2; for the GN type-II datum see [48, (2.7)]. The rational computation gives $\rho = (1, -1/2, 1)$ in (f_2, f_3, f_{-2}) coordinates and $(\rho, \delta_i) = -1$. Direct computation in the BKM Gram of Definition 1.7 gives

$$(\rho, \rho) = 2(f_2, f_{-2}) + \tfrac{1}{4}(f_3, f_3) = 2(-1) + \tfrac{1}{4}(2) = -\tfrac{3}{2}.$$

Since $(\rho, \rho) < 0$, $\rho \in C(\Lambda^{2,1})_+$. □

The finite lattice packet

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certificates/lattice/type_ii_weyl_vector
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records the simple-root coordinates, the Cartan matrix $4\mathbf{I}_3 - 2\mathbf{J}_3$, the discriminant -32, the explicit $U(4) \oplus \langle 2 \rangle$ basis $\delta_1 + \delta_3, \delta_2 + \delta_3, \delta_3$, and the checks $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$, $(\rho, \delta_i) = -1$, $(\rho, \rho) = -3/2$, and $\rho \in (\Lambda^{2,1}_{II})^\vee$. The packet is target-side lattice arithmetic only.

Proposition 1.12 *(Reflection group; Nikulin's theorem).* [proved; Nikulin 1980 Theorem 1.4.1] Let $W^{(2)}(\Lambda^{2,1})$ be the subgroup of $\mathbf{O}(\Lambda^{2,1})$ generated by reflections in primitive roots of square 2. Then

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where the subscript + denotes the index-two subgroup preserving the time-like cone $C(\Lambda^{2,1})_+$. Equivalently, every isometry of $\Lambda^{2,1}$ preserving the chosen positive cone is a product of square-2 reflections.

Proof. Nikulin [88, Theorem 1.4.1] establishes for the rank-three hyperbolic lattice $\Lambda^{(1,1)} \oplus [2]$ the equality $\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1})$; see also Gritsenko–Nikulin [46, Proposition 1.5]. □

Verification 1.13 *(Type-II reflection orbit).* [computed] For each $\delta \in \{\delta_1, \delta_2, \delta_3\}$, the reflection $s_\delta : x \mapsto x - (x, \delta)\delta$ satisfies $s_\delta(\delta) = -\delta$ and preserves the Cartan matrix on the δ-orbit. Concretely, $s_{\delta_1}(\delta_1) = -\delta_1$, $s_{\delta_1}(\delta_2) = \delta_2 + 2\delta_1$, etc.; the bilinear form on $\{\delta_1, \delta_2, \delta_3\}$ is reflection-invariant by direct matrix computation.

Remark 1.14 *(The two type-2 divisibility classes).* Each $\delta \in \Lambda^{2,1}$ with $\delta^2 = 2$ has divisibility $(\delta, \Lambda^{2,1}) \in \{\mathbb{Z}, 2\mathbb{Z}\}$. The two cases are recorded as *type I* ($(\delta, \Lambda^{2,1}) = \mathbb{Z}$) and *type II* ($(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$); both classes generate index-finite normal subgroups of $W^{(2)}(\Lambda^{2,1})$:

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_I)] = 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{II})] = 6,$$

following [46, Sections 4–5]. The denominator algebra $\mathfrak{g}_{\Delta_5}$ uses the type-II reflection chamber.

1.3 The K3 lattice $\Lambda_{K3} = \mathrm{II}_{3,19}$

For S a complex K3 surface, the integral cohomology $H^2(S, \mathbb{Z})$ with cup-product pairing is the unique even unimodular indefinite lattice of signature $(3, 19)$ (Conway–Sloane 1999 §16.4; Nikulin 1980 Theorem 1.1.4).

Definition 1.15 *($\Lambda_{K3} = \mathrm{II}_{3,19}$).* [definitional] The K3 cohomology lattice is

$$\Lambda_{K3} := H^2(S, \mathbb{Z}) \simeq -E_8 \oplus -E_8 \oplus U \oplus U \oplus U \equiv -2E_8 \oplus 3U,$$

where E_8 is the rank-8 root lattice with the standard positive- definite Cartan pairing (Conway–Sloane 1999 §4.8) and $-E_8$ denotes E_8 with bilinear form negated.

Verification 1.16 *(Rank, signature, discriminant of Λ_{K3}).* [computed]

$$\operatorname{rk}\Lambda_{K3} = 8 + 8 + 2 + 2 + 2 = 22, \qquad \sigma(\Lambda_{K3}) = (3, 19).$$

The three U-summands contribute signature $(3, 3)$ and the two $-E_8$-summands contribute signature $(0, 16)$; total $(3, 19)$. The discriminant is

$$\operatorname{disc}\Lambda_{K3} = (\det -E_8)^2 \cdot (\det U)^3 = 1 \cdot (-1)^3 = -1,$$

so $|\operatorname{disc}| = 1$ and the lattice is unimodular (Conway–Sloane 1999 §16.4, Theorem 16.4.1). Even unimodularity follows because each summand is even unimodular.

Proposition 1.17 *(Aut and Hodge data on Λ_{K3}).* [proved] The orthogonal group $\mathbf{O}(\Lambda_{K3})$ acts transitively on the set of primitive vectors of any fixed square (Eichler 1952; Nikulin [87, Theorem 1.14.4]). For a polarized K3 with period $\omega \in H^{2,0}(S) \subset \Lambda_{K3} \otimes \mathbb{C}$, the transcendental and algebraic sublattices fit into the orthogonal splitting

$$\Lambda_{K3} = \mathrm{NS}(S) \oplus T(S),$$

where $\mathrm{NS}(S) \subset H^{1,1}(S, \mathbb{Z})$ is the Néron–Severi (algebraic) sublattice of signature $(1, \rho - 1)$ and $T(S)$ is the transcendental lattice of signature $(2, 20 - \rho)$, with $\rho = \operatorname{rk}\mathrm{NS}(S)$ the Picard rank (Beauville–Bourguignon–Demazure 1985 §6; [87, §2]).

Proof. The transitivity statement is Eichler's theorem on indefinite unimodular lattices, sharpened by Nikulin [87, Theorem 1.14.4]: for an even unimodular indefinite lattice L of rank ≥ 3, $\mathbf{O}(L)$ acts transitively on primitive vectors of fixed square in each squared-class. Decomposition by Hodge type partitions $\Lambda_{K3} \otimes \mathbb{C}$ into $H^{2,0} \oplus H^{1,1} \oplus H^{0,2}$; intersecting with Λ_{K3} gives the transcendental–algebraic splitting at the integral level. □

1.4 *The lattice-polarized K3 sublattice* Λ_{pol}

A lattice polarization is a primitive embedding of an integral lattice Λ_{pol} into Λ_{K3} whose orthogonal complement carries period structure.

Definition 1.18 *(Lattice polarization $\Lambda_{\mathrm{pol}} \hookrightarrow \Lambda_{K3}$).* [definitional] A *lattice polarization* of signature $(1, \rho - 1)$ on a K3 surface S is a primitive embedding

$$\Lambda_{\mathrm{pol}} \hookrightarrow \Lambda_{K3} = \mathrm{II}_{3,19}, \qquad \sigma(\Lambda_{\mathrm{pol}}) = (1, \rho - 1), \quad 1 \leq \rho \leq 19,$$

where $\Lambda_{\mathrm{pol}} \subset H^{1,1}(S, \mathbb{Z})$ contains an ample class. The orthogonal complement

$$\Lambda_{\mathrm{pol}}^{\perp} \subset \Lambda_{K3}, \qquad \sigma(\Lambda_{\mathrm{pol}}^{\perp}) = (2, 20 - \rho),$$

is the lattice on which the period map of the family $\mathcal{F}_{\Lambda_{\mathrm{pol}}} \to \mathcal{D}_{\Lambda_{\mathrm{pol}}^{\perp}}$ is defined (Dolgachev 1996; Nikulin 1980 Proposition 1.6.1).

Verification 1.19 *(Numerics of Λ_{pol} and $\Lambda_{\mathrm{pol}}^{\perp}$).* [computed]

$$\operatorname{rk}\Lambda_{\mathrm{pol}} = \rho, \quad \sigma(\Lambda_{\mathrm{pol}}) = (1, \rho - 1), \quad \operatorname{rk}\Lambda_{\mathrm{pol}}^{\perp} = 22 - \rho, \quad \sigma(\Lambda_{\mathrm{pol}}^{\perp}) = (2, 20 - \rho).$$

Since Λ_{K3} is unimodular and $\Lambda_{\mathrm{pol}} \subset \Lambda_{K3}$ is primitive, Nikulin's gluing theorem gives an anti-isometry of finite quadratic forms

$$(A_{\Lambda_{\mathrm{pol}}}, q_{\Lambda_{\mathrm{pol}}}) \simeq (A_{\Lambda_{\mathrm{pol}}^{\perp}}, -q_{\Lambda_{\mathrm{pol}}^{\perp}}),$$

hence

$$|A_{\Lambda_{\rm pol}}| = |A_{\Lambda_{\rm pol}^\perp}|, \qquad |\det \Lambda_{\rm pol}| = |\det \Lambda_{\rm pol}^\perp|.$$

The quotient $\Lambda_{K3}/(\Lambda_{\rm pol} \oplus \Lambda_{\rm pol}^\perp)$ is the isotropic gluing graph inside $A_{\Lambda_{\rm pol}} \oplus A_{\Lambda_{\rm pol}^\perp}$; there is no quotient $\Lambda_{\rm pol}^\perp/\Lambda_{\rm pol}^*$.

Proposition 1.20 *(Nikulin embedding criterion).* [proved; Nikulin 1980 Theorem 1.14.4] A primitive embedding $\Lambda_{\rm pol} \hookrightarrow \Lambda_{K3}$ exists if and only if there is an isomorphism

$$(A_{\Lambda_{\rm pol}}, q_{\Lambda_{\rm pol}}) \simeq (A_{\Lambda_{\rm pol}^\perp}, -q_{\Lambda_{\rm pol}^\perp})$$

of finite quadratic forms, where $A_L := L^*/L$ is the discriminant group and q_L its quadratic form (Nikulin 1980 Definition 1.3.1 and Theorem 1.14.4; Conway–Sloane 1999 §15.5). The embedding is unique up to $\mathbf{O}(\Lambda_{K3})$ when

$$\operatorname{rk}\Lambda_{\rm pol} + \ell(A_{\Lambda_{\rm pol}}) \le 22 - \operatorname{rk}\Lambda_{\rm pol}$$

and the discriminant form satisfies the Nikulin uniqueness criterion (Nikulin 1980 Theorem 1.14.4).

Proof. This is the gluing-via-discriminant-form theorem ([87, §1.14]; [21, §15.5]). An embedding into Λ_{K3} exists iff the discriminant form on $\Lambda_{\rm pol}^*/\Lambda_{\rm pol}$ is realized by the orthogonal complement; uniqueness is the Nikulin orbit-uniqueness statement under the corank inequality. □

Remark 1.21 *(Mirror lattice convention).* Choose a primitive hyperbolic plane $U_{\rm mir} \subset \Lambda_{\rm pol}^\perp$. The Aspinwall–Morrison–Dolgachev mirror to a $\Lambda_{\rm pol}$-polarized K3 family is polarized by the orthogonal complement

$$\Lambda_{\rm pol}^\perp = U_{\rm mir} \oplus \check\Lambda_{\rm pol}, \qquad \check\Lambda_{\rm pol} = U_{\rm mir}^\perp \cap \Lambda_{\rm pol}^\perp \simeq \Lambda_{\rm pol}^\perp \ominus U_{\rm mir},$$

the splitting depending on the cusp $U_{\rm mir} \hookrightarrow \Lambda_{\rm pol}^\perp$ (Dolgachev 1996 §5; Aspinwall 1997). Thus $\operatorname{rk}\check\Lambda_{\rm pol} = 20 - \operatorname{rk}\Lambda_{\rm pol}$. For $\Lambda_{\rm pol} = U \oplus \langle -2\rangle$ one obtains $\check\Lambda_{\rm pol} \simeq U \oplus E_7(-1) \oplus E_8(-1)$, of signature $(1, 16)$. The hyperbolic plane $U_{\rm mir}$ re-enters only in the Mukai/type-IV comparison, where the period Hodge line is compared after adjoining the standard Mukai U-summand. The assertion that Δ_5^{-2} is the corresponding mirror section of the inverse Hodge line with reduced type-II Heegner support (View ④) remains conjectural and is independent of Chapter 6.

1.5 The Mukai lattice $\widetilde{H}(S, \mathbb{Z}) = \mathrm{II}_{4,20}$

For S a complex K3 surface, the full integral cohomology $H^*(S, \mathbb{Z})$ with the Mukai pairing is an even unimodular lattice of signature $(4, 20)$ (Mukai 1984 Theorem 1.5; Mukai 1987).

Definition 1.22 *(Mukai lattice $\widetilde{H}(S, \mathbb{Z})$).* [definitional] The Mukai lattice of S is

$$\widetilde{H}(S, \mathbb{Z}) := H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}), \qquad \widetilde{H}(S, \mathbb{Z}) \simeq U \oplus U \oplus U \oplus U \oplus -2E_8 \equiv 4U \oplus 2(-E_8),$$

graded by even cohomological degree. The Mukai pairing is

$$\langle (r, D, s), (r', D', s') \rangle_{\rm Muk} := D \cdot D' - rs' - r's$$

for $(r, D, s), (r', D', s') \in \mathbb{Z} \oplus H^2(S, \mathbb{Z}) \oplus \mathbb{Z}$ (Mukai 1984 Theorem 1.5). The pairing of an H^0-class with an H^4-class carries a minus sign relative to the cup product, so that the rank-4 component of $H^*(S, \mathbb{Z}) \otimes H^*(S, \mathbb{Z})$ contributes $-rs' - r's$.

Verification 1.23 *(Rank, signature, discriminant of $\widetilde{H}(S,\mathbb{Z})$).* [computed]

$$\operatorname{rk}\widetilde{H}(S,\mathbb{Z}) = 1 + 22 + 1 = 24, \qquad \sigma(\widetilde{H}(S,\mathbb{Z})) = (4,20).$$

The four U-summands give signature $(4,4)$ and the $-2E_8$ pieces give signature $(0,16)$, total $(4,20)$. Discriminant

$$\operatorname{disc}\widetilde{H}(S,\mathbb{Z}) = (\det U)^4 \cdot (\det -E_8)^2 = (-1)^4 \cdot 1 = 1,$$

so $\widetilde{H}(S,\mathbb{Z})$ is even unimodular, denoted $\mathrm{II}_{4,20}$ (Conway–Sloane 1999 §16.4). The lattice-polarization symbol is Λ_{pol}; the full Mukai lattice is written $\widetilde{H}(S,\mathbb{Z})$.

Proposition 1.24 *(Mukai vector).* [proved; Mukai 1987] For F a coherent sheaf on S, the Mukai vector is

$$v(F) := \operatorname{ch}(F)\sqrt{\operatorname{td}(S)} = (\operatorname{rk}F,\ c_1(F),\ \operatorname{ch}_2(F) + \operatorname{rk}F\cdot[\mathrm{pt}]) \in \widetilde{H}(S,\mathbb{Z}),$$

where $\sqrt{\operatorname{td}(S)} = (1,0,1)$ in Mukai notation on a K3 surface ([84, Theorem 3]). The Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$ recovers the Euler characteristic of Hom:

$$\langle v(F), v(G)\rangle_{\mathrm{Muk}} = -\chi(F,G) = -\sum_i (-1)^i \dim \operatorname{Ext}^i(F,G).$$

Proof. Mukai [84, Theorem 3] computes $\sqrt{\operatorname{td}(S)}$ for K3 surfaces. The Hirzebruch–Riemann–Roch formula gives $\chi(F,G) = \int_S \operatorname{ch}(F^\vee)\operatorname{ch}(G)\operatorname{td}(S) = -\langle v(F), v(G)\rangle_{\mathrm{Muk}}$ once the sign of the H^0–H^4 pairing is taken into account; see Mukai 1984 §2. □

Remark 1.25 *(Sign of the Mukai pairing on rank-0 components).* The Mukai pairing $\langle (r,D,s),(r',D',s')\rangle_{\mathrm{Muk}} = D\cdot D' - rs' - r's$ restricts to the cup-product pairing on the middle H^2-component

$$\langle (0,D,0),(0,D',0)\rangle_{\mathrm{Muk}} = D\cdot D',$$

and to the negative cup-product on the rank–Euler component

$$\langle (r,0,s),(r',0,s')\rangle_{\mathrm{Muk}} = -rs' - r's.$$

The relative minus sign is forced by Hirzebruch–Riemann–Roch [50, Theorem 5.1.1]: $-\chi(F,G)$ contains a $-\operatorname{rk}F\cdot\chi(\mathcal{O}_S,G) - \operatorname{rk}G\cdot\chi(\mathcal{O}_S,F)$ term coming from $\sqrt{\operatorname{td}(S)} = (1,0,1)$. Negating the H^0–H^4 pairing or dropping the minus sign converts the Mukai lattice to an isotropic-rank-2 unimodular form not of signature $(4,20)$, in disagreement with the Mukai 1984 classification ([83, Theorem 1.5]) and the unique even unimodular type $\mathrm{II}_{4,20}$. The sign as stated is forced by these requirements.

Proposition 1.26 *(Aut of the Mukai lattice).* [proved] $\mathbf{O}(\widetilde{H}(S,\mathbb{Z}))$ acts transitively on the set of primitive vectors of any fixed square (Eichler 1952; Nikulin 1980 Theorem 1.14.4). In particular, the Mukai vector $v = (1,0,1)$ (corresponding to the structure sheaf $\mathcal{O}_S$) and the generator $(0,0,1)$ of $H^4(S,\mathbb{Z}) \cap \widetilde{H}(S,\mathbb{Z})$ are $\mathbf{O}(\widetilde{H}(S,\mathbb{Z}))$-conjugate to any other primitive vector of the same square. The derived autoequivalence group of $D^b(S)$ maps to $\mathbf{O}(\widetilde{H}(S,\mathbb{Z}))$ via the action on Mukai vectors (Mukai 1987 §2; Orlov 1997).

Proof. Eichler's theorem ([87, Theorem 1.14.4]) gives the transitivity statement. Mukai [84, §2] constructs the representation of derived autoequivalences on $\widetilde{H}(S,\mathbb{Z})$ via Fourier–Mukai kernels. □

1.6 The formal dyonic charge lattice $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$

The formal additive dyonic lattice on $X = S \times E$ encodes the electric–magnetic charge data on which the Gram-shadow projection Π_X is defined. Its two factors are full Mukai lattices, not lattice-polarization sublattices of $H^2(S,\mathbb{Z})$.

Definition 1.27 *(Γ_X^{form} and its Gram pairing).* [definitional; from Definition 1.3] For $X = S \times E$ with S a K3 surface and E an elliptic curve,

$$\Gamma_X^{\mathrm{form}} := \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z}), \qquad c = (Q,P) \in \Gamma_X^{\mathrm{form}}, \quad Q,P \in \widetilde{H}(S,\mathbb{Z}).$$

The Gram pairing $\langle\, , \,\rangle_\Gamma : \Gamma_X^{\mathrm{form}} \times \Gamma_X^{\mathrm{form}} \longrightarrow \mathbb{Z}$ is inherited from the Mukai pairing on each factor:

$$\langle (Q,P),(Q',P')\rangle_\Gamma := \langle Q,Q'\rangle_{\mathrm{Muk}} + \langle P,P'\rangle_{\mathrm{Muk}}.$$

The Gram-index projection

$$\Pi_X : \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}} = \mathbb{Z}^3, \qquad (Q,P) \mapsto \left(\tfrac{Q^2}{2},\, Q\cdot P,\, \tfrac{P^2}{2}\right),$$

records the genus-two Gram triple of the dyonic class. This map is quadratic, not additive: the obstruction is the symmetric bilinear 2-cocycle B of Lemma 2.4.

VERIFICATION 1.28 *(Rank, signature of Γ_X^{form}).* [computed]

$$\mathrm{rk}\,\Gamma_X^{\mathrm{form}} = 2\cdot \mathrm{rk}\,\widetilde{H}(S,\mathbb{Z}) = 48, \qquad \sigma(\Gamma_X^{\mathrm{form}}) = 2\cdot(4,20) = (8,40).$$

The discriminant is

$$\mathrm{disc}\,\Gamma_X^{\mathrm{form}} = (\mathrm{disc}\,\widetilde{H}(S,\mathbb{Z}))^2 = 1,$$

so Γ_X^{form} is even unimodular. In Conway–Sloane notation it is $\mathrm{II}_{8,40}$ (Conway–Sloane 1999 §16.4).

Remark 1.29 *(Relation to the K3×E Mukai geometry).* Under the Künneth decomposition for $X = S \times E$, the integral even cohomology decomposes as

$$H^{\mathrm{ev}}(X,\mathbb{Z}) = H^*(S,\mathbb{Z}) \otimes H^{\mathrm{even}}(E,\mathbb{Z}) = H^*(S,\mathbb{Z}) \otimes (\mathbb{Z}[1] \oplus \mathbb{Z}[\omega_E]),$$

where $\omega_E \in H^2(E,\mathbb{Z})$ is the cohomological point class with $\int_E \omega_E = 1$. Every Mukai vector on X in the formal even sector decomposes as $v_X = P \otimes 1_E + Q \otimes \omega_E$ with $Q,P \in \widetilde{H}(S,\mathbb{Z})$, and $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$ is the additive lattice on which the (P,Q)-grading lives (Definition 1.5). The tensor Künneth pairing on the formal even sector pairs against the elliptic U-summand $H^0(E,\mathbb{Z}) \oplus H^2(E,\mathbb{Z}) \simeq U$, giving

$$\langle v_X, v'_X\rangle_{\mathrm{Kun}} = Q\cdot P' + Q'\cdot P,$$

where each summand is the Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$. The direct-sum Gram pairing on Γ_X^{form} keeps Q^2, $Q\cdot P$, and P^2 separately; the resulting genus-two Gram triple $(Q^2/2, Q\cdot P, P^2/2)$ is the Gram-index data $\Pi_X(Q,P)$ used in the Borcherds product (Theorem 3.9).

LEMMA 1.30 *(Two orthogonal hyperbolic planes inside $\widetilde{H}(S,\mathbb{Z})$).* [proved] $\widetilde{H}(S,\mathbb{Z}) \simeq 4U \oplus 2(-E_8)$ contains four mutually orthogonal copies of U. In particular, two of them give

$$U_1 \oplus U_2 \subset \widetilde{H}(S,\mathbb{Z}), \qquad U_i \simeq U,\ U_1 \perp U_2,$$

which is the hypothesis of the formal primitive lift Lemma 2.8: every Gram triple $(n,l,m) \in \mathbb{Z}^3$ has a primitive saturated formal lift $c = (Q,P) \in \Gamma_X^{\mathrm{form}}$ with $\Pi_X(Q,P) = (n,l,m)$.

Proof. By Definition 1.3, $\widetilde{H}(S,\mathbb{Z}) \simeq 4U \oplus 2(-E_8)$ contains four mutually orthogonal U-summands. Selecting any two of them gives the displayed sublattice, which is itself isomorphic to $U \oplus U \simeq \mathrm{II}_{2,2}$ as a sublattice of $\widetilde{H}(S,\mathbb{Z})$. The Gram-triple lift then follows from Lemma 2.8: take $Q = e_1 + nf_1$, $P = lf_1 + e_2 + mf_2$ in $U_1 \oplus U_2$ to obtain $\Pi_X(Q,P) = (n,l,m)$. □

Verification 1.31 *(Formal primitive-lift packet).* [computed] The packet `certificates/charge/formal_primitive_li` records the same two-hyperbolic-plane formula. Its verifier checks $\Pi_X(Q,P) = (n,l,m)$, the e_1, e_2 primitive minor, wedge torsion one, and

$$\alpha(\Pi_X(Q,P)) = 2nf_2 - lf_3 + 2mf_{-2}$$

for the listed formal rows. The packet is lattice arithmetic in $\widetilde{H}(S,\mathbb{Z})$; it supplies neither algebraic effectivity nor a finite HN charge window.

1.7 The signature-$(3,2)$ *lattice* $\Lambda^{3,2}$ *and the* $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ *bridge*

The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_+(3,2)$ identifies the Siegel domain with the type-IV orthogonal domain of an integral lattice $\Lambda^{3,2}$ of signature $(3,2)$, and it identifies $\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$ with the exterior-square arithmetic group $\Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$ ([46, Section 3]; Borcherds 1995 §5). The Igusa cusp form Δ_5 lives on the Siegel side; the Borcherds product expansion lives on the orthogonal side.

Definition 1.32 *($\Lambda^{3,2}$ in the exterior-square model).* [definitional] Let $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$ and consider the exterior square $\wedge^2\Lambda^4$ with the symmetric pairing

$$u \wedge v \cdot u' \wedge v' = [\det(u,v,u',v')]_{e_1\wedge e_2\wedge e_3\wedge e_4} \in \mathbb{Z}.$$

Then $(\wedge^2\Lambda^4, (\ ,\ )) \simeq \mathrm{II}_{3,3}$ as an even unimodular lattice of signature $(3,3)$. Fix the primitive vector $q_0 := e_1 \wedge e_3 + e_2 \wedge e_4 \in \wedge^2\Lambda^4$, for which $q_0^2 = -2$, and define

$$\Lambda^{3,2} := q_0^{\perp} \subset \wedge^2\Lambda^4.$$

Verification 1.33 *(Rank, signature, discriminant of* $\Lambda^{3,2}$*).* [computed] $\operatorname{rk} \wedge^2\Lambda^4 = \binom{4}{2} = 6$. Taking the orthogonal complement of the primitive vector q_0 gives $\operatorname{rk}\Lambda^{3,2} = 5$. The explicit Gram matrix of $\Lambda^{3,2}$ in the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ is

$$G_{3,2} = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where $f_1 = e_1 \wedge e_2$, $f_2 = e_2 \wedge e_3$, $f_3 = e_1 \wedge e_3 - e_2 \wedge e_4$, $f_{-2} = e_4 \wedge e_1$, $f_{-1} = e_4 \wedge e_3$. Direct computation: the matrix is antidiagonal except for the central $h^2 = 2$ entry, with off-diagonal entries $-1, -1, -1, -1$. Expansion gives

$$\det G_{3,2} = 2 \cdot (-1)^4 \cdot \operatorname{sgn}(1,2,3,4 \mapsto 4,3,2,1) = 2 \cdot 1 \cdot (+1) = +2,$$

since the permutation $(1,2,3,4) \mapsto (4,3,2,1)$ has six inversions and even parity. Therefore

$$\sigma(\Lambda^{3,2}) = (3,2), \qquad \operatorname{disc}\Lambda^{3,2} = 2.$$

This is a rational computation in the exterior-square lattice.

Proposition 1.34 *($\Lambda^{3,2} \simeq U \oplus U \oplus \langle 2\rangle$ in the fixed convention; $\Lambda^{3,2} \simeq U(-1) \oplus U(-1) \oplus \langle 2\rangle$ in the standard convention).* [proved] In the BKM convention of the exterior-square model,

$$\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2],$$

with $\Lambda^{(1,1)} = U(-1)$ and $[2]$ the rank-1 lattice with Gram (2). The Gram presentation displayed in Verification 1.33 arises from the f-basis: (f_1, f_{-1}) pairs as a copy of $\Lambda^{(1,1)}$ ($(f_1, f_{-1}) = -1$), (f_2, f_{-2}) pairs as a second copy of $\Lambda^{(1,1)}$ ($(f_2, f_{-2}) = -1$), and f_3 contributes $(f_3, f_3) = 2$.

Proof. The pairings $(f_1, f_{-1}) = -1$, $(f_2, f_{-2}) = -1$, $(f_3, f_3) = 2$ are read off from $G_{3,2}$. The orthogonal decomposition is direct: $\mathbb{Z}f_1 + \mathbb{Z}f_{-1} \perp \mathbb{Z}f_2 + \mathbb{Z}f_{-2} \perp \mathbb{Z}f_3$ inside $\Lambda^{3,2}$ because every cross-pair vanishes in $G_{3,2}$. The identification with $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ follows because each rank-2 block has Gram $((0,-1),(-1,0)) = \Lambda^{(1,1)}$ and the rank-1 block has Gram $(2) = [2]$. □

Proposition 1.35 *(The projective exterior-square arithmetic group).* [proved; exterior-square model of $\mathbf{Sp}_4(\mathbb{Z})$] The exterior-square representation $\wedge^2$ induces an isomorphism

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\} \xrightarrow{\sim} \Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+, \qquad \Gamma_{3,2} := \wedge^2 \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\}.$$

Concretely, the integral generators

$$g_0 = \begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix}, \quad g_M = \begin{pmatrix} \mathbf{I}_2 & M \\ 0 & \mathbf{I}_2 \end{pmatrix}, \quad g_A = \begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix} \quad (A \in \mathbf{GL}_2(\mathbb{Z}))$$

of $\mathbf{Sp}_4(\mathbb{Z})$ map under $\wedge^2$ to integral isometries of $\Lambda^{3,2}$, displayed as the three explicit 5×5 matrices $\wedge^2(g_0)$, $\wedge^2(g_M)$, $\wedge^2(g_A)$ in the exterior-square model. The image is the arithmetic group through which the Siegel modular group acts on the type-IV model. The Siegel space $\mathbb{H}_2$ embeds into the type-IV upper half-space $\mathbb{H}_+^{\mathbf{IV}} \subset \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C})$ via

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \mapsto {}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1) z_0,$$

and the action of $\mathbf{Sp}_4(\mathbb{Z})$ on $\mathbb{H}_2$ corresponds to the action of $\Gamma_{3,2}$ on $\mathbb{H}_+^{\mathbf{IV}}$.

Proof. The Pfaffian pairing on $\wedge^2\Lambda^4$ is preserved by $\wedge^2\mathbf{SL}_4$. The subgroup fixing q_0 is $\mathbf{Sp}_4(\mathbb{Z})$ by definition of the symplectic form $B_{q_0}(x,y) e_1 \wedge e_2 \wedge e_3 \wedge e_4 = -x \wedge y \wedge q_0$. The kernel on $q_0^\perp = \Lambda^{3,2}$ is $\{\pm \mathbf{I}_4\}$. The exceptional Lie-algebra isomorphism $\mathfrak{sp}_4(\mathbb{R}) \simeq \mathfrak{so}(3,2)$ extends to the integral form on the displayed generators. The discriminant group of $\Lambda^{3,2}$ has one non-zero element; hence the stable and full projective components coincide in the Gritsenko–Nikulin normalization. This is the content of [46, Section 3] and the exterior-square model of the automorphic group. □

Remark 1.36 *(Embedding $\Lambda_{II}^{2,1} \subset \Lambda^{3,2}$).* The primitive sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

of the Weyl-chamber section is the BKM root lattice: discarding the (f_1, f_{-1})-block of $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ leaves $\Lambda^{(1,1)} \oplus [2] \simeq \Lambda^{2,1}$. The type-II sublattice $\Lambda_{II}^{2,1} = \{mf_2 + lf_3 + nf_{-2} \mid m, n \equiv 0 \pmod 2\}$ is the lattice of even integral type-II classes, on which the simple-root Cartan

$$\big((\delta_i, \delta_j)\big) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

of Proposition 1.10 is realized via $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$. Every isometry of $\Lambda^{2,1}$ extends to an isometry of $\Lambda^{3,2}$ by the identity on $(\Lambda^{2,1})^\perp \subset \Lambda^{3,2}$.

1.8 Cross-references and consistency check

The lattices recorded above enter as follows.

- $U = \mathrm{II}_{1,1}$: every U-summand of $\Lambda^{3,2}$, Λ_{K3}, and $\widetilde{H}(S,\mathbb{Z})$. The relation $\Lambda^{(1,1)} = U(-1)$ records the BKM negated convention.
- $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2\rangle$ (with $U(4)$ written as $U(-4)$ in the BKM sign convention): the BKM root lattice on which the type-II simple-root Cartan $2(2\mathbf{I}_3 - \mathbf{J}_3)$ and the Weyl vector $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ are realized. The denominator algebra $\mathfrak{g}_{\Delta_5} = \mathfrak{g}_{\Delta_5}$ is graded by $\Lambda^{2,1}_{II}$ ([48, Theorem 2.1]).
- $\Lambda_{K3} = \mathrm{II}_{3,19}$: K3 cohomology with intersection pairing. Polarized K3 sublattices Λ_{pol} embed primitively into Λ_{K3}.
- Λ_{pol}: signature $(1, \rho - 1)$ for $\rho \le 19$ (the algebraic Picard rank). It is the lattice-polarization sublattice of Λ_{K3}, not the full Mukai lattice.
- $\widetilde{H}(S,\mathbb{Z}) = \mathrm{II}_{4,20}$: the Mukai lattice on which the D6–D2–D0 Mukai vectors $v_X(\mathcal{I}_Y)$ live; the Gram-index map Π_X is computed via this pairing (Theorem 3.9).
- $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$: the formal additive dyonic charge lattice of $X = S \times E$, on which the quadratic Gram-shadow projection Π_X and the symmetric bilinear 2-cocycle B are defined (Definition 1.3).
- $\Lambda^{3,2} \simeq U(-1) \oplus U(-1) \oplus \langle 2\rangle$: the signature-$(3,2)$ orthogonal lattice that bridges $\mathbf{PSp}_4(\mathbb{Z})$ to the type-IV orthogonal group via the exterior-square representation; on this lattice the Borcherds product expansion of Δ_5 is written.

Verification 1.37 *(End-to-end computation check).* [computed] The lattice data satisfy, in one self-contained $\mathbb{Q}$-arithmetic computation: (i) the rank-5 Gram matrix $G_{3,2}$ in the $(f_1, f_2, f_3, f_{-2}, f_{-1})$-basis; (ii) the orthogonality of the symplectic invariant $q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$ to the basis; (iii) the type-II simple-root Cartan $2(2\mathbf{I}_3 - \mathbf{J}_3)$; (iv) the Weyl vector $\rho = (1, -1/2, 1)$ in (f_2, f_3, f_{-2}) coordinates and the $(\rho, \delta_i) = -1$ relation; (v) the type-II reflection action and Cartan invariance under each s_{δ_i}. Thus the Gram, Cartan, and Weyl-vector data of §1.2 and §1.7 agree in the common rational lattice.

Remark 1.38 *(Discriminant convention; sign of $\Lambda^{3,2}$).* In the BKM convention $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$, both hyperbolic-plane summands have Gram determinant -1 and the $[2]$ summand contributes $+2$, so $\operatorname{disc}\Lambda^{3,2} = (-1)^2 \cdot 2 = +2$. In the standard convention $U \oplus U \oplus \langle -2\rangle$, the two U-summands contribute determinant -1 each and $\langle -2\rangle$ contributes -2, so $\operatorname{disc}\Lambda^{3,2} = (-1)^2 \cdot (-2) = -2$. The two presentations differ by an overall sign on the rank-2 hyperbolic blocks; both carry an order-2 discriminant group with single nontrivial generator $f_3/2 \in (\Lambda^{3,2})^*/\Lambda^{3,2}$. The same analysis applies to the ambient $\Lambda^{2,1}$: discriminant -2 in the BKM convention, discriminant $+2$ in the standard convention, and discriminant group $\mathbb{Z}/2$ generated by $h/2$. The type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ of index four has $|\operatorname{disc}\Lambda^{2,1}_{II}| = 32$, with discriminant group $(\Lambda^{2,1}_{II})^*/\Lambda^{2,1}_{II}$ of order 32 (Verification 1.8); the type-II divisibility statement $(\delta, \Lambda^{2,1}) \in \{\mathbb{Z}, 2\mathbb{Z}\}$ depends on the order 2 of the ambient $\Lambda^{2,1}$ discriminant group, not on the sign of any determinant.

Remark 1.39 *(Cited primary literature).* The primary references behind every lattice claim above are: Conway–Sloane 1999 §2.5, §4.8, §15.5, §16.4 (classification of even unimodular indefinite lattices, E_8 root system, discriminant-form gluing, K3 lattice as $\mathrm{II}_{3,19}$); Nikulin 1980 Theorem 1.1.2, Theorem 1.4.1 (=

[88] Theorem 1.4.1 in the K3-paper version), Theorem 1.6.1, Theorem 1.14.4 (signature classification, reflection group of square-2 vectors, primitive embeddings, discriminant-form uniqueness); Mukai 1984 Theorem 1.5 (Mukai pairing, signature $(4, 20)$), Mukai 1987 Theorem 3 ($\sqrt{\mathrm{td}(S)} = (1, 0, 1)$ on K3); Gritsenko–Nikulin 1998 Part II Theorem 2.1, Section 5.1.1, case $(t = 1, II, \overline{0})$ (BKM root datum for $\mathfrak{g}_{\Delta_5}$, Weyl vector, type-II simple-root Cartan).

2 Borcherds-product expansion: $\phi_{0,1}$, the Δ_5 infinite product, and the Gritsenko–Cléry eight-row catalogue

The Fourier expansion of the weight-zero index-one weak Jacobi form $\phi_{0,1}$, the Borcherds–Gritsenko–Nikulin product expansion of the Igusa cusp form Δ_5 in the type-II cusp chamber, the theta-constant normalization constant $64 = 2^6$, the doubling identity that turns the monic Borcherds product into the denominator of the Lorentzian Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$, and the full eight-row Gritsenko–Cléry catalogue of diagonal-divisor Siegel modular forms are the automorphic data at the type-II cusp. Every displayed numerical Fourier coefficient is produced by exact rational expansion of the theta quotient; every identity is attributed to author, year, and theorem of the primary literature. Orientation data, Pfaffian wall normal forms, and compact Hall input do not enter these automorphic identities; they live entirely on the automorphic / target side, prior to the geometric content of (D0), (O1), $(\mathrm{O1})^+$, and (O2).

Three names for Δ_5 appear and are kept distinct: the theta-constant automorphic section $\mathcal{D}_X = \Delta_5$ (View 1); the Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View 2); and the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of the pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ (View 3, after the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and finite geometric Pfaffian datum (P_{fin}) of Appendix 7). The first and third sections coincide as weight-five automorphic sections; the second coincides with their Borcherds denominator after the normalization $D_5 = 64^{-1}\Delta_5$ and the pullback $Z \mapsto 2Z$. The orientation character that distinguishes Δ_5 from $-\Delta_5$ enters only at View 3.

B.1 The weak Jacobi form $\phi_{0,1}$

The K3 elliptic genus is the modular input. Eichler–Zagier [34, §2.2 and Theorem 9.3] fix the weight-zero index-one weak-Jacobi normalization

$$\phi_{0,1}(\tau, z) \in J^{\mathrm{w}}_{0,1}, \qquad \phi_{0,1}(\tau, z) = \sum_{n \geq 0,\ l \in \mathbb{Z}} f(n, l)\, q^n r^l, \qquad q = e^{2\pi i \tau},\ r = e^{2\pi i z}.$$

The relation to the K3 right-moving elliptic genus is $Z_{\mathrm{K3}}(\tau, z) = 2\phi_{0,1}(\tau, z)$ [34, §2.2]; the half is the Borcherds-lift representative for the Borcherds product. The cusp normalization is

$$\phi_{0,1}(\tau, z) = r^{-1} + 10 + r + O(q).$$

The Jacobi weak-form support is $f(n, l) = 0$ whenever $4n - l^2 < -1$, with isolated extreme entries $f(0, \pm 1) = 1$ on the discriminant hyperbola $4n - l^2 = -1$. *(proved;* [34, §2.2, Theorem 9.3]*;* [48, Example 2.4].*)*

2.0.0.1 Verified Fourier table. The exact coefficients $f(n,l)$ for $n \le 2$ are the rational integers computed by the theta-quotient identity $\phi_{0,1} = 4\sum_{i=2,3,4} \theta_i(\tau,z)^2/\theta_i(\tau,0)^2$ of [34, §2.2] by rational arithmetic:

	$l=-3$	$l=-2$	$l=-1$	$l=0$	$l=1$	$l=2$	$l=3$
$n=0$	0	0	1	10	1	0	0
$n=1$	0	10	-64	108	-64	10	0
$n=2$	1	108	-513	808	-513	108	1

(computed; cross-checked with [48, Example 2.4] *and* [46, Theorem 4.1].*)* The finite table `certificates/jacobi/k3e_first_relation_window_coefficients` records this leading normalization together with the checked evenness and discriminant-dependence rows used in the first relation-closed window. The negative entries reflect the index-one Jacobi-form theta-decomposition $\phi_{0,1} = \theta_{1,0}h_0 + \theta_{1,1}h_1$ with theta-coefficient expansions $h_0 = 2(45E_4E_{4,1} - 50E_6E_{6,1})/\eta^{24} + \cdots$ in [34, §2.2]; the integer values are not target signs but signed multiplicities of the Borcherds source.

2.0.0.2 Fourier identities.

$$\begin{aligned} &f(0,0)=10, && f(0,\pm1)=1, && f(0,l)=0\ \ (|l|\ge 2),\\ &f(1,0)=108, && f(1,\pm1)=-64, && f(1,\pm2)=10,\\ &f(2,0)=808, && f(2,\pm1)=-513, && f(2,\pm2)=108,\\ & && f(2,\pm3)=1, && f(1,\pm3)=0. \end{aligned}$$

The diagonal entry $f(0,0)=10$ controls the Borcherds-lift weight $\mathrm{wt}(\Delta_5) = f(0,0)/2 = 5$; the entries $f(0,\pm1) = 1$ control the simple type-II divisor orders $d^{\mathrm{aut}}_{\delta_i} = 1$; the entry $f(1,1) = -64$ controls the multiplicity of the first timelike imaginary root, computed and used below.

B.2 The cusp chamber and the effective semigroup

Set $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right) \in \mathbb{H}_2$, $q = e^{2\pi i z_1}$, $r = e^{2\pi i z_2}$, $s = e^{2\pi i z_3}$. The standard type-II cusp polarization is the ordered limit

$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1,$$

which determines the effective semigroup

$$\Gamma_{\mathrm{eff}} = \{(n,l,m) \in \mathbb{Z}^3 \mid m > 0,\ n \ge 0\} \cup \{(n,l,0) \mid n > 0\} \cup \{(0,l,0) \mid l < 0\}.$$

Inside the Gram-index lattice $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$, the *active support*

$$\Gamma_{\mathrm{act}} := \{(n,l,m) \in \Gamma_{\mathrm{eff}} \mid f(nm,l) \neq 0\}$$

is the set of triples that contribute a visible product factor; the remaining triples in Γ_{eff} have zero exponent and fix ordering only. The signed Gram pairing $(\alpha(n,l,m), \alpha(n,l,m)) = -2(4nm - l^2)$ under

$$\alpha : (n,l,m) \longmapsto 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda^{2,1}_{II}$$

restricts to the Lorentzian root lattice of $\mathfrak{g}_{\Delta_5}$. The finite packet `certificates/lattice/product_chamber_root_map` gives a computed check of the three sector definitions of Γ_{eff}, their pairwise closure under addition, the active rows selected by $f(nm,l) \neq 0$, the inactive zero-exponent rows, and the identity

$$\alpha(n,l,m) = n\delta_1 + m\delta_2 + (n+m-l)\delta_3$$

used in the type-II chamber. This is target-side lattice arithmetic; it does not construct a compact $K3 \times E$ source, a Pfaffian orientation, an $O2$ wall atlas, a mirror discriminant, or a protected trace.

PROPOSITION 2.1 *(Images of the three simple type-II walls).* [computed] Let

$$\gamma_{\text{wall}} : \{\partial_1, \partial_2, \partial_3\} \longrightarrow \Gamma_{\text{eff}} \subset \mathbb{Z}^3$$

be the inverse of the product-chamber root map

$$\alpha(n, l, m) = n\partial_1 + m\partial_2 + (n + m - l)\partial_3$$

on the three simple type-II roots. Then

$$\gamma_{\text{wall}}(\partial_1) = (1, 1, 0), \qquad \gamma_{\text{wall}}(\partial_2) = (0, 1, 1), \qquad \gamma_{\text{wall}}(\partial_3) = (0, -1, 0).$$

Equivalently, under

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2},$$

these triples map respectively to

$$2f_2 - f_3, \qquad 2f_{-2} - f_3, \qquad f_3.$$

This is a target-side wall-image computation. It does not construct compact $K3 \times E$ representatives, retained type-II wall objects, wall charts, Pfaffian signs, protected traces, or a Hall carrier.

Proof. For a triple (n, l, m), the coefficient of $(\partial_1, \partial_2, \partial_3)$ in the type-II chamber is

$$(n,\ m,\ n + m - l).$$

For ∂_1 this coefficient vector must be $(1, 0, 0)$, hence

$$n = 1, \qquad m = 0, \qquad 1 - l = 0,$$

so $l = 1$ and $\gamma_{\text{wall}}(\partial_1) = (1, 1, 0)$. For ∂_2 the coefficient vector is $(0, 1, 0)$, hence

$$n = 0, \qquad m = 1, \qquad 1 - l = 0,$$

so $l = 1$ and $\gamma_{\text{wall}}(\partial_2) = (0, 1, 1)$. For ∂_3 the coefficient vector is $(0, 0, 1)$, hence

$$n = 0, \qquad m = 0, \qquad -l = 1,$$

so $l = -1$ and $\gamma_{\text{wall}}(\partial_3) = (0, -1, 0)$. Substituting the three triples into $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$ gives

$$2f_2 - f_3, \qquad -f_3 + 2f_{-2}, \qquad f_3,$$

which are precisely $\partial_1, \partial_2, \partial_3$. □

The finite packet

`certificates/lattice/type_ii_wall_images`

verifies `TYPE_II_WALL_IMAGES_VERIFIED`: the three inverse images above have zero root-map defect and lie in the product chamber sectors. The packet records compact source representatives, retained wall objects, wall charts, Pfaffian signs, protected traces, and Hall carrier data as excluded scalar substitutes.

B.3 The Borcherds product formula for Δ_5

Theorem 2.2 *(Borcherds–Gritsenko–Nikulin product expansion).* On the type-II cusp polarization $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$, the weight-five Igusa cusp form admits the absolutely convergent infinite product

$$\Delta_5(Z) = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

with theta-leading coefficient

$$[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 \;=\; 64 \;=\; 2^6.$$

The monic product is $D_5 := 64^{-1}\Delta_5$; the constant 64 is the theta-constant Igusa normalization, not a separately chosen scalar. *(proved;* [9, Theorem 10.1], [48, Theorem 2.1, Example 2.4, equations (2.15)–(2.16)], [46, Theorem 4.1].*)*

Proof. The general Borcherds singular theta-lift sends a weakly holomorphic vector-valued modular form of negative weight for the Weil representation of a lattice of signature $(2, n)$ to a meromorphic orthogonal automorphic form realized as an infinite product [9, Theorem 10.1]; the regularisation and convergence on divisors with singularities in higher rank is in [10, §§6–13]. The genus-two specialization at the Siegel cusp uses the exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3, 2)_+$ and the Eichler–Zagier dictionary $J^{\mathrm{w}}_{0,1} \leftrightarrow$ vector-valued Mp_2-modular forms [34, §5]; the ordered limit $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ selects the type-II cusp [48, §2, Example 2.4]. Gritsenko–Nikulin [48, (2.15)–(2.16)] write the resulting product in the unnormalized form

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

and identify D_5 with 64^{-1} times the theta-constant cusp form Δ_5 of Igusa [46, Theorem 4.1], [51], [52]; the leading constant $2^6 = 64$ is the explicit theta-product coefficient. □

2.0.0.3 The constant 2^6 as a theta-product leading coefficient. Igusa's classical product formula displays Δ_5 as the product of the ten even theta constants of genus two,

$$\Delta_5(Z) = \prod_{\substack{a,b\in(\mathbb{Z}/2\mathbb{Z})^2\\ {}^t ab\equiv 0 \pmod 2}} \nu_{a,b}(Z), \qquad \nu_{a,b}(Z) = \sum_{l\in\mathbb{Z}^2} \exp\!\big(\pi i (Z[l + \tfrac{1}{2}a] + {}^t b l)\big),$$

[51], [46, Theorem 4.1], [36, §6]. In the type-II cusp polarization, with z_2 the off-diagonal modulus, the four characteristics $a = (0, 0)$ start with 1, the two characteristics $a = (1, 0)$ start with $2q^{1/8}$, the two characteristics $a = (0, 1)$ start with $2s^{1/8}$, and the two characteristics $a = (1, 1)$ start with $2q^{1/8} r^{1/4} s^{1/8}$; the total leading monomial is $q^{1/2} r^{1/2} s^{1/2}$ with constant $1 \cdot 2^2 \cdot 2^2 \cdot 2^2 = 2^6 = 64$ *(proved;* [51], [46, Theorem 4.1].*)* This is not a normalization choice: the constant 64 is read off the theta product, with no scalar freedom. Replacing Δ_5 by $c\Delta_5$, $c \in \mathbb{C}^\times$, would change neither the Maass character ν_{Δ_5} nor the divisor orders $d^{\mathrm{aut}}_{\partial}$ of [46, Theorem 4.1].

B.4 Sample exponent verifications

The exponent $f(nm, l)$ in the product formula is read directly from the Jacobi-coefficient table of B.1. The following sample is verified exactly by the displayed coefficient table:

factor (n, l, m)	Lorentzian Gram label $\alpha = 2nf_2 - lf_3 + 2mf_{-2}$	exponent $f(nm, l)$
$(1, 1, 0)$	$\delta_1 = 2f_2 - f_3$	$f(0, 1) = 1$
$(0, 1, 1)$	$\delta_2 = 2f_{-2} - f_3$	$f(0, 1) = 1$
$(0, -1, 0)$	$\delta_3 = f_3$	$f(0, -1) = 1$
$(0, 0, m)$ $(m \geq 1)$	cusp ray $2mf_{-2}$	$f(0, 0) = 10$
$(1, 0, 1)$	$2f_2 + 2f_{-2}$ (timelike, $(\alpha, \alpha) = -8$)	$f(1, 0) = 108$
$(1, 1, 1)$	$2f_2 - f_3 + 2f_{-2}$ (timelike, $(\alpha, \alpha) = -6$)	$f(1, 1) = -64$
$(1, 2, 1)$	$2f_2 - 2f_3 + 2f_{-2}$ (isotropic, $(\alpha, \alpha) = 0$)	$f(1, 2) = 10$
$(1, 3, 1)$	$2f_2 - 3f_3 + 2f_{-2}$ (spacelike, $(\alpha, \alpha) = 10$)	$f(1, 3) = 0$
$(2, 1, 1)$	$4f_2 - f_3 + 2f_{-2}$ ($(\alpha, \alpha) = -14$)	$f(2, 1) = -513$
$(1, 2, 2)$	$2f_2 - 2f_3 + 4f_{-2}$ ($(\alpha, \alpha) = -12$)	$f(2, 2) = 108$

The triple $(1, 1, 1)$ gives the first timelike imaginary root: its exponent $f(1, 1) = -64$ is the signed superdimension of the root space $\mathfrak{g}_{\Delta_5, \alpha(1,1,1)}$ on the active support, and is the content of the lattice cross-check $f(1, 1) = 29 - 93 = -64$ from the free-Lie/imaginary-simple split computed by the Gritsenko–Nikulin target presentation. The triple $(1, 3, 1)$ has exponent zero because $f(1, 3) = 0$: this triple lies in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ and contributes ordering, not a visible factor. These are signed superdimensions, not positive dimensions; negative entries record an odd excess in the Borcherds source, and only after the (O1), (O1)$^+$, (O2) orientation data become available do they translate to operator parities.

2.0.0.4 Theta-leading coefficient. The coefficient 64 is the coefficient of the leading monomial of Δ_5:

$$\Delta_5(Z) = 64\, q^{1/2} r^{1/2} s^{1/2} \big(1 + \cdots\big).$$

After the leading monomial is divided out, the first qrs-coefficient of $D_5 = 64^{-1}\Delta_5$ is 93, not 64. The factor $(1 - qrs)^{-64}$ contributes $64\, qrs$, and the remaining root factors contribute the other $29\, qrs$; this is the identity

$$[qrs]\, D_5/(qrs)^{1/2} = 93, \qquad 29 - 93 = -64 = f(1, 1).$$

(computed by exact rational expansion of the Borcherds product.)

B.5 The doubling identity $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$

Lemma 2.3 *(Two exponential normalizations).* The product chamber convention writes monomials as $q^n r^l s^m$ with $q^n r^l s^m = \exp(-\pi i(\alpha(n, l, m), z))$, while the standard denominator-side convention of a Borcherds–Kac–Moody algebra writes formal characters as $\exp(-2\pi i(\alpha, z))$. Replacing Z by $2Z$ converts the first character to the second, so

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; D_5(2Z) \;=\; 64^{-1}\Delta_5(2Z).$$

(proved; [46, Proposition 3.1], [48, Section 5.1.1] *together with Theorem 2.2.)*

Proof. The product side of Theorem 2.2 gives $D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ with leading monomial $q^{1/2} r^{1/2} s^{1/2}$. Doubling $Z \mapsto 2Z$ replaces $q^n r^l s^m$ by $q^{2n} r^{2l} s^{2m}$ and hence $\exp(-\pi i(\alpha, z))$ by $\exp(-2\pi i(\alpha, z))$. The Lorentzian Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ of Gritsenko–Nikulin [46, Sections 3–4, Proposition 3.1] carries the denominator identity

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \sum_{w \in W^{(2)}(\Lambda_{II}^{2,1})} \det(w) \left(e^{-2\pi i(w\rho, z)} - \sum_a m(a)\, e^{-2\pi i(w(\rho + a), z)}\right)$$

on the imaginary-positive cone, where $m(a)$ is determined by the negative-norm additive coefficients and by $1 - \sum_{t\geq 1} m(ta)q^t = \prod_{t\geq 1}(1-q^t)^{\tau(ta)}$ on isotropic rays. The right-hand side is equal to $D_5(2Z)$ in the doubled exponential. Equivalently $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$. □

The factor 64 in $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is the same theta-constant 2^6 from the leading-coefficient calculation, transported through $Z \mapsto 2Z$; it is not a separate normalization, an integration constant, or a free scalar. The three identifications $\mathcal{D}_X = \Delta_5$, $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$ are kept distinct; the doubling lemma supplies the denominator-character normalization.

B.6 The eight-row Gritsenko–Cléry catalogue

Gritsenko and Cléry [44, Theorem 1.4] classify all Siegel modular forms of genus two, with character or multiplier system on a Hecke-type congruence subgroup $\Gamma_t(N) \subset \mathrm{Sp}(4,\mathbb{Q})$, whose divisor is supported on the $\Gamma_t(N)$-translates of the diagonal Humbert surface $\mathcal{H}_{\mathrm{diag}} = \{Z = \mathrm{diag}(a,b) : a, b \in \mathbb{H}_1\}$ with multiplicity one. The classification produces exactly eight rows. *(proved;* [44, Theorem 1.4 and Theorem 3.1]*.)*

The rows are:

j	F_j	(t,N)	Γ_j	$\mathrm{wt}(F_j)$	multiplier ν_j	$c_j(0)$
1	Δ_5	$(1,1)$	$\Gamma_1 = \mathrm{Sp}(4,\mathbb{Z})$	5	$v_\eta^{12} \times v_H$	10
2	Δ_2	$(2,1)$	Γ_2^+	2	$v_\eta^{6} \times v_H$	4
3	Δ_1	$(3,1)$	Γ_3^+	1	$v_\eta^{4} \times v_H$	2
4	$\Delta_{1/2}$	$(4,1)$	Γ_4^+	$\frac{1}{2}$	$v_\eta^{3} \times v_H$ (degree 8)	1
5	∇_3	$(1,2)$	$\Gamma_0^{(2)}(2)$	3	$\chi_2^{(2)} \times v_H$	6
6	∇_2	$(1,3)$	$\Gamma_0^{(2)}(3)$	2	$\chi_2^{(3)} \times v_H$	4
7	$\nabla_{3/2}$	$(1,4)$	$\Gamma_0^{(2)}(4)$	$\frac{3}{2}$	$\chi_4^{(4)} \times v_H$ (order 4)	3
8	Q_1	$(2,2)$	$\Gamma_2(2)$	1	$\chi_4^{(2)} \times v_H$	2

In the displayed F_j-order the weights and constant Fourier coefficients are

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1), \qquad (10, 4, 2, 1, 6, 4, 3, 2).$$

In the input-pair order

$$(1,1), (2,1), (1,2), (3,1), (1,3), (4,1), (1,4), (2,2)$$

the same data read

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1), \qquad (10, 4, 6, 2, 4, 1, 3, 2).$$

Each row satisfies the Borcherds–Gritsenko universal weight identity

$$\mathrm{wt}(F_j) = \frac{c_j(0)}{2} = \kappa_{\mathrm{BKM}}(F_j), \qquad j = 1, \ldots, 8.$$

(proved; [9], [44, Theorem 3.1]*; the κ_{BKM} relabelling is the universal Borcherds-weight identity used in the Calabi–Yau quantum-groups κ-stratification.)*

The half-integer weights $\frac{1}{2}$ (row 4) and $\frac{3}{2}$ (row 7) are realized by multiplier systems $v_\eta^k \times v_H$ on the ambient paramodular cover, not by passing to a metaplectic double cover. The multiplier v_η is the Dedekind eta multiplier on $\mathrm{SL}_2(\mathbb{Z})$; the multiplier v_H is the Heisenberg multiplier on the Heisenberg lift of $\mathrm{Sp}(4,\mathbb{Z})$; and $v_\eta^3 \times v_H$ has order 8 on the row-4 cover, while $\chi_4^{(4)} \times v_H$ has order 4 on the row-7 cover. *(proved;* [44, Theorem 3.1]*.)*

2.0.0.5 Denominator content of the rows. Rows 1–4 carry Gritsenko–Nikulin denominator theorems [48, Theorem 2.1, equations (2.11), (2.20), (2.21)]; in the $e_t^{(n,l,m)} = q^n r^l s^{tm}$ exponential normalization of [48, §5.1.1], the products are

$$\Delta_2(Z) = q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_2^{\mathrm{prod}}} (1 - q^n r^l s^{2m})^{f_2(nm,l)},$$
$$\Delta_1(Z) = q^{1/6} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_3^{\mathrm{prod}}} (1 - q^n r^l s^{3m})^{f_3(nm,l)},$$
$$\Delta_{1/2}(Z) = q^{1/8} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_4^{\mathrm{prod}}} (1 - q^n r^l s^{4m})^{f_4(nm,l)},$$

with cusp Jacobi inputs $\phi_{0,2}, \phi_{0,3}, \phi_{0,4}$ supplied explicitly by [48, §5]. Rows 5–8 carry Cléry–Gritsenko products on the corresponding congruence covers, with cusp Jacobi inputs ϕ_j as in [44, Theorem 3.1], and the additional denominator data (Weyl chamber, simple real roots, reflection character, Weyl alternating sum) supplied by Govindarajan's twisted-CHL theorem [40], [41], [18] for $F_5 = \nabla_3$, $F_6 = \nabla_2$, $F_7 = \nabla_{3/2}$, $F_8 = Q_1$. For F_7 the denominator theorem lives on the order-four multiplier cover $\Gamma_0^{(2)}(4)\langle \chi_4^{(4)} \times v_H \rangle$. *(proved;* [40], [48, Theorem 2.1], [44, Theorem 3.1].*)*

2.0.0.6 Level separation. The eight-row catalogue is a list of *automorphic products plus denominator theorems on the rows where the data have been supplied*. It is not a list of compact Hall sources, reduced curve-counting backgrounds, or chiral hosts. For a row j, the automorphic product F_j acquires a compact physical host only after separate choices of an automorphic line, a denominator algebra, a scalar trace, a chiral source, and a Hall realization; these choices are row-dependent. The row $j = 1$ ($F_1 = \Delta_5$) is the only row with a proved reduced CY/DT scalar host (Oberdieck–Pixton, Oberdieck–Pandharipande: $Z_{\mathrm{OP}}^X = -4096\Delta_5^{-2}$); even row 1 has open compact Hall/chiral host at the primitive-recognition level, governed by $(S1)$–$(S10)$.

B.7 The Borcherds-weight coordinate $\kappa_{\mathrm{BKM}} = 5$

The universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the row-by-row specialization of the Borcherds singular-theta-lift weight formula [9, Theorem 10.1] together with the Gritsenko–Nikulin unification [46, §2], [45]. The relevant row is $j = 1$, with

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \mathrm{wt}(\Delta_5) = 5 = \frac{c_1(0)}{2} = \frac{f(0,0)}{2} = \frac{10}{2}.$$

(proved; [9, Theorem 10.1], [34, §2.2], [44, Theorem 3.1].*)*

This is the Borcherds-weight invariant: the same integer $\kappa_{\mathrm{BKM}} = 5$ identifies the Δ_5 row inside the companion Calabi–Yau quantum-groups κ-stratification, and it is the entry that distinguishes $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5)$ (the Igusa cusp-form weight) from the unrelated quantities $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 0$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$, $\kappa_{\mathrm{cat}} = 0$, $\kappa_{\mathrm{fiber}} = 24 = \chi_{\mathrm{top}}(\mathrm{K3})$. The additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ would identify $24 = \chi_{\mathrm{top}}(\mathrm{K3})$ with $2 = \chi(\mathcal{O}_{\mathrm{K3}})$. The Heisenberg reading gives an accidental equality at $N = 1$, but it does not extend to the CHL weights $c_N(0)/2 = (5, 3, 2, \frac{3}{2}, 1)$; the appendix asserts $\kappa_{\mathrm{BKM}} = 5$ as the row entry of the Borcherds-weight identity, not as the value of any additive formula.

B.8 Normalizations and residual obstructions

The following normalizations fix the Borcherds product without residual scalar, chamber, multiplier, or coefficient ambiguity.

- *Theta-constant normalization.* The constant 64 is not a free scalar on Δ_5. The identity $64 = 2^6$ is the explicit theta-product leading coefficient [51], [46, Theorem 4.1]; with the theta normalization fixed, no residual scalar freedom remains. Cross-checked against the consistency $f(1,1) = -64$ and the qrs-coefficient identity $c_\Delta(1,1,1) = 64$ [46, Theorem 4.1].
- *Cusp chamber definition* Γ_{eff}. The semigroup Γ_{eff} is the ordered-limit semigroup of the type-II cusp polarization $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ [48, Example 2.4]; not a free choice. Triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ have zero exponent and therefore contribute no visible product factor.
- *Multiplier system for half-integer weights.* Half-integer weights $\frac{1}{2}$ and $\frac{3}{2}$ are realized by $v_\eta^k \times v_H$, not by the Weil-representation metaplectic double cover; the integer weight 5 of Δ_5 uses the abelian multiplier $v_\eta^{12} \times v_H$ (which is automorphic-trivial modulo the Maass character ν_{Δ_5} on the type-II Weyl generators). No metaplectic factor enters the Borcherds weight identity. See [44, Theorem 3.1] for the full classification.
- *Doubling factor* $64^{-1}\Delta_5(2Z)$. The factor 64^{-1} in $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is the same theta-constant 2^6 from Theorem 2.2, transported through $Z \mapsto 2Z$; no new constant. See Lemma 2.3.
- *The* $\kappa_{\mathrm{BKM}} = 5$ *row.* The integer $\kappa_{\mathrm{BKM}} = 5$ is the row entry of the Borcherds-weight identity $\mathrm{wt}(F_j) = c_N(0)/2$ in the catalogue of [44, Theorem 3.1]; it agrees with $\mathrm{wt}(\Delta_5) = f(0,0)/2 = 5$ from [34, §2.2]; it agrees with the entry $(c_1(0), \mathrm{wt}) = (10, 5)$ in the companion Calabi–Yau quantum-groups κ-stratification.
- *Eichler–Zagier vs Gritsenko–Nikulin convention.* The present appendix uses the Eichler–Zagier convention $\phi_{0,1} = Z_{\mathrm{K3}}/2$ [34, §2.2]; this matches the Gritsenko–Nikulin Jacobi convention in [48, Example 2.4] and [48, §5.1.1] on the (n,l)-coefficient table given in B.1, and matches the singular theta-lift normalization [9, Theorem 10.1] after the Mp_2-vector-valued bridge. No residual factor of 2 enters the exponent $f(nm,l)$ of the Borcherds product.

2.0.0.7 Residual obstructions. The four obstruction classes (D0), (O1), $(\mathrm{O1})^+$, (O2) are not part of the Borcherds product calculation; they are geometric content on the source side of Δ_5. Within the automorphic calculation, no residual Borcherds-product obstruction remains. The universal identity $\mathrm{wt}(F_j) = c_N(0)/2$ fixes the row $j = 1$, $\kappa_{\mathrm{BKM}} = 5$, and any disagreement with the Vol III catalogue, the Borcherds singular theta-lift weight formula, or the Gritsenko–Cléry classification would be a contradiction among the stated comparison theorems.

2.0.0.8 Computations. Fourier coefficients are obtained from the rational theta-quotient expansion; lattice and Gram-pairing identities are the computations of Appendix 1.2; the type-II cusp chamber, active support, and Borcherds product are in Proposition 7.2; the doubling identity is Remark 5.5; the catalogue tables are in Appendix 6.

3 Mukai–Gram dictionary on $K3 \times E$

Let $X = S \times E$ be the smooth trivial-canonical threefold, with S a smooth complex K3 surface and E a smooth complex elliptic curve. The Mukai vector and Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$, together with the Künneth decomposition on $H^*(X,\mathbb{Z})$, produce the Gram-degree projection $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ and its normal-ordered companion $\overline{\Pi}_X$ on the central extension $\widehat{\Gamma}_X$. The Borcherds expansion variables $q^n r^l s^m$ of an Igusa cusp expansion record Gram-degree shadows in Γ_{gram}. When the shadow is carried by an actual one-dimensional subscheme $Y \subset X$, its dyonic charge $c = (Q_Y, P_Y)$ satisfies the formula below. The existence of such a subscheme is the algebraic-effective Hall-support condition, not a formal consequence of the Fourier monomial. For Y of class $[Y] = (\beta_h, d)$ and $\chi(\mathcal{O}_Y) = n$,

$$\Pi_X(Q_Y, P_Y) = (h - 1,\ n,\ d - 1).$$

The threefold arithmetic Euler characteristic $n = \chi(\mathcal{O}_Y)$ is the direct structure-sheaf invariant; no "Todd correction" enters, and no ambient fourfold appears.

The Mukai–Gram construction sits at Beilinson chart layer α (intrinsic charge data) and γ (ambient Mukai/Gram lattice). The κ-tuple, orientation character ϵ_o, Pfaffian section, and recognition theorem live at higher Beilinson levels and depend on this chart datum.

3.1 Mukai vector and Mukai pairing on a K3 surface

Definition 3.1 *(Mukai lattice and Mukai vector).* Let S be a smooth complex K3 surface. The Mukai lattice is the total cohomology

$$\widetilde{H}(S, \mathbb{Z}) := H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

graded by even cohomological degree, an even unimodular lattice of signature $(4, 20)$ and rank 24 when equipped with the Mukai pairing below. For a coherent sheaf $F \in \mathrm{Coh}(S)$, respectively for an object $F \in D^b(S)$, the Mukai vector is

$$v_S(F) := \mathrm{ch}(F) \cdot \sqrt{\mathrm{td}(S)} = \big(\mathrm{rk}\, F,\ c_1(F),\ \mathrm{ch}_2(F) + \mathrm{rk}\, F \cdot [\mathrm{pt}]\big) \ \in\ \widetilde{H}(S, \mathbb{Z}).$$

The square root $\sqrt{\mathrm{td}(S)} = (1, 0, 1)$ on the K3 surface uses the Todd class $\mathrm{td}(S) = (1, 0, 2[\mathrm{pt}])$, so $\sqrt{\mathrm{td}(S)} = (1, 0, [\mathrm{pt}])$; identifying $[\mathrm{pt}]$ with the integer 1 in the displayed coordinates gives the displayed vector. The triple $v_S(F) = (r, D, s)$ records the rank $r \in H^0$, the first Chern class $D \in H^2$, and the Mukai-corrected second component $s = \mathrm{ch}_2(F) + r \in H^4$. Mukai introduced this lattice and pairing in [84, Theorem 1.5], with the symplectic structure of the moduli space of sheaves established in [83].

PROPOSITION 3.2 *(Mukai pairing on the K3 surface).* The Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ is the symmetric $\mathbb{Z}$-bilinear form

$$\big\langle (r, D, s), (r', D', s') \big\rangle_{\mathrm{Muk}} = D \cdot D' - rs' - r's,$$

equivalently

$$\langle v, w \rangle_{\mathrm{Muk}} = -\int_S v^\vee \cup w, \qquad v^\vee := (r, -D, s),$$

where $v_2 \cup w_2$ is the K3 cup product on $H^2(S, \mathbb{Z})$ and $v_0 \cup w_4, v_4 \cup w_0$ record the cross-degree pairings. The form is symmetric, even, unimodular, of signature $(4, 20)$; its hyperbolic-plane summands are

$$\widetilde{H}(S, \mathbb{Z}) \simeq U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}$$

as integral lattices. For a polarized K3 of Picard rank ρ, the Néron–Severi lattice $\mathrm{NS}(S)$ has signature $(1, \rho - 1)$, while the algebraic Mukai lattice

$$\widetilde{H}^{1,1}_{\mathrm{alg}}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus \mathrm{NS}(S) \oplus H^4(S, \mathbb{Z})$$

has signature $(2, \rho)$. This is Mukai [84, Theorem 1.5(ii)]; see also Huybrechts–Lehn [50, Chapter 6].

Proof. The bilinear identity is the definition; its symmetry is immediate from the symmetry of the K3 cup product and the identification of degree-zero and degree-four components. Even unimodularity follows from the K3 intersection lattice on H^2 (even unimodular of signature $(3, 19)$) and the unimodular pairing $H^0 \times H^4 \to \mathbb{Z}$ under $rs' \mapsto rs'$. The signature $(4, 20)$ follows by Hodge index plus the hyperbolic factor on $H^0 \oplus H^4$. The same hyperbolic factor changes $\mathrm{NS}(S)$ of signature $(1, \rho - 1)$ into $\widetilde{H}^{1,1}_{\mathrm{alg}}(S, \mathbb{Z})$ of signature $(2, \rho)$. The lattice isomorphism is [84, Theorem 1.5(ii)]. □

Remark 3.3 *(Mukai pairing is symmetric on both parts).* The Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ is symmetric in both the algebraic and transcendental parts. There is no orthosymplectic $\mathfrak{osp}$ refinement at the level of the lattice. The Hodge $\mathbb{Z}/2$-super-extension $(\mathbb{C}, \omega_S) \mapsto \mathbb{C} \oplus \omega_S$ used in Mukai self-mirror language imposes a separate $\mathbb{Z}/2$-grading on a Yangian-type quantum group constructed from this lattice; that grading does not change the underlying lattice pairing, which remains symmetric. The classical Lie limit of the Mukai-form Yangian is the orthogonal Lie algebra $\mathfrak{so}(4, 20)$, not $\mathfrak{osp}(4 \mid 20)$; see [98, §2] for the Mukai-side Yangian classical limit and [64] for the cohomological Hall algebra realization.

Example 3.4 *(Mukai vector of a point and of an ideal sheaf).* $v_S(\mathcal{O}_p) = (0, 0, 1)$ for the structure sheaf of a point $p \in S$; $v_S(\mathcal{O}_S) = (1, 0, 1)$ for the trivial line bundle; $v_S(L) = (1, c_1(L), c_1(L)^2/2 + 1)$ for a line bundle L. For an ideal sheaf $\mathcal{I}_C$ of a smooth curve $C \subset S$ of genus g, the adjunction formula on the K3 surface gives $C^2 = c_1(\mathcal{O}_S(C))^2 = 2g - 2$, and

$$v_S(\mathcal{I}_C) = (1, -c_1(\mathcal{O}_S(C)), \mathrm{ch}_2(\mathcal{I}_C) + 1).$$

The Mukai pairing $\langle v_S(\mathcal{O}_p), v_S(\mathcal{O}_p)\rangle_{\mathrm{Muk}} = 0 - 0 \cdot 1 - 0 \cdot 1 = 0$, recording that points have zero self-pairing in the Mukai form. The pairing $\langle v_S(\mathcal{O}_S), v_S(\mathcal{O}_p)\rangle_{\mathrm{Muk}} = 0 - 1 \cdot 1 - 0 \cdot 0 = -1$ records the Euler characteristic $\chi(\mathcal{O}_S, \mathcal{O}_p) = 1$ up to the Mukai sign convention $\chi(F, G) = -\langle v_S(F), v_S(G)\rangle_{\mathrm{Muk}}$ of [84, Proposition 2.2].

3.2 Künneth on $K3 \times E$ and the rank-3 Gram lattice

Proposition 3.5 *(Künneth for $X = K3 \times E$).* Let S be a smooth complex K3 surface and E a smooth complex elliptic curve. The even integral cohomology of $X = S \times E$ splits as

$$H^{\mathrm{even}}(X, \mathbb{Z}) \simeq H^*(S, \mathbb{Z}) \otimes_{\mathbb{Z}} H^{\mathrm{even}}(E, \mathbb{Z}),$$

the odd–odd Künneth term $H^{\mathrm{odd}}(S) \otimes H^{\mathrm{odd}}(E)$ absent because S carries no odd cohomology, both factors free $\mathbb{Z}$-modules. Fixing a generator $\omega_E \in H^2(E, \mathbb{Z})$ with $\int_E \omega_E = 1$, $H^{\mathrm{even}}(E, \mathbb{Z}) = \mathbb{Z} \cdot 1_E \oplus \mathbb{Z} \cdot \omega_E$, and every even class on X decomposes uniquely as

$$v_X = v^{(0)} \otimes 1_E + v^{(\omega)} \otimes \omega_E, \qquad v^{(0)}, v^{(\omega)} \in H^*(S, \mathbb{Z}).$$

The formal Künneth lattice

$$H^{\mathrm{even}}(X, \mathbb{Z}) \simeq \widetilde{H}(S, \mathbb{Z}) \otimes U_E, \qquad U_E := H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z}),$$

carries the tensor Künneth pairing of rank 48 and signature $(24, 24)$, obtained from the $(4, 20)$-Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ and the $(1, 1)$-hyperbolic pairing on U_E. This cross-pairing is distinct from the direct-sum componentwise Gram pairing on Γ_X^{form} used by Π_X. The odd Künneth term $H^{\mathrm{odd}}(S) \otimes H^{\mathrm{odd}}(E)$ is absent because the K3 surface S carries no odd cohomology. See Huybrechts [50, Chapter 3].

Proof. E and S have torsion-free integral cohomology, so Künneth holds integrally and produces the displayed tensor decomposition. The tensor-pairing factorization is bilinearity of the cup product over the Künneth isomorphism. The lattice signature comes from $(4, 20)$ on $\widetilde{H}(S, \mathbb{Z})$ (Proposition 3.2) tensored with $(1, 1)$ on U_E. $\square$

Definition 3.6 *(Gram-degree projection on $X = K3 \times E$).* The formal additive even-Mukai sector is

$$\Gamma_X^{\mathrm{form}} := \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z}), \qquad c = (Q, P) \in \Gamma_X^{\mathrm{form}},$$

with $Q, P \in \widetilde{H}(S, \mathbb{Z})$. The Gram-index lattice is

$$\Gamma_{\mathrm{gram}} := \mathbb{Z}^3,$$

and the Gram-degree projection is the quadratic map

$$\Pi_X : \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad (Q, P) \longmapsto \left(\tfrac{(Q,Q)_{\mathrm{Muk}}}{2},\ (Q, P)_{\mathrm{Muk}},\ \tfrac{(P,P)_{\mathrm{Muk}}}{2}\right).$$

The triple $(n, l, m) = \Pi_X(Q, P)$ is the Gram-degree shadow; it is the target of the Igusa Fourier expansion at the standard cusp. It is *not* an additive Hall grading: the additivity defect is the symmetric bilinear cocycle B of §3.4 below.

3.3 D6–D2–D0 dictionary

The Künneth decomposition of $v_X(\mathcal{I}_Y)$ on the threefold $X = K3 \times E$ produces the D6–D2–D0 dictionary $\Pi_X(Q_Y, P_Y) = (h - 1, n, d - 1)$ of Theorem 3.9 as a chart-data identity. The Mukai-vector calculation isolates the K3-fibre rank-one component $P_Y = (1, 0, 1 - d)$ on the 1_E-summand and the curve-class component $Q_Y = (0, -\beta^\vee, -n)$ on the ω_E-summand, with $\beta^\vee \in H^2(S, \mathbb{Z})$ the Poincaré dual of β; the Mukai pairing on the K3 components delivers the Gram-degree shadow.

THEOREM 3.7 *(Mukai vector of an ideal sheaf on the threefold $K3 \times E$).* Let $X = S \times E$ with S a smooth complex K3 surface and E a smooth complex elliptic curve, so $K_X \simeq \mathcal{O}_X$ and X is a smooth trivial-canonical threefold. Let $Y \subset X$ be a one-dimensional closed subscheme of class $[Y] = (\beta, d) \in H_2(S, \mathbb{Z}) \oplus H_2(E, \mathbb{Z})$, write $\beta^\vee \in H^2(S, \mathbb{Z})$ for the Poincaré dual of β, and let $n = \chi(\mathcal{O}_Y) \in \mathbb{Z}$. The Mukai vector of the ideal sheaf $\mathcal{I}_Y \in D^b(X)$ decomposes against the Künneth basis as

$$v_X(\mathcal{I}_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta^\vee, -n) \otimes \omega_E.$$

Proof. The trivial-canonical threefold condition gives $\sqrt{\mathrm{td}(X)} = \pi_S^* \sqrt{\mathrm{td}(S)}\, \pi_E^* \sqrt{\mathrm{td}(E)} = (1, 0, 1) \otimes 1$, since the elliptic curve is one-dimensional with trivial Todd class $\mathrm{td}(E) = 1$. The ideal-sheaf identity is $\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y)$. For the structure sheaf of a one-dimensional subscheme of a trivial-canonical threefold, $\mathrm{ch}_0(\mathcal{O}_Y) = 0$, $\mathrm{ch}_1(\mathcal{O}_Y) = 0$, $\mathrm{ch}_2(\mathcal{O}_Y) = \mathrm{PD}[Y]$, and $\int_X \mathrm{ch}_3(\mathcal{O}_Y) = \chi(\mathcal{O}_Y) = n$. Decomposing $[Y] = (\beta, d)$ on the Künneth basis, $\mathrm{PD}[Y] = \beta^\vee \otimes \omega_E + d[\mathrm{pt}_S] \otimes 1_E$, where $[\mathrm{pt}_S]$ is the K3 point class; the fibrewise elliptic-degree component $d[\mathrm{pt}_S] \otimes 1_E$ records the elliptic degree. Hence

$$\mathrm{ch}(\mathcal{O}_Y) = (0, 0, d) \otimes 1_E + (0, \beta^\vee, n) \otimes \omega_E,$$

and $1 - \mathrm{ch}(\mathcal{O}_Y) = (1, 0, -d) \otimes 1_E + (0, -\beta^\vee, -n) \otimes \omega_E$. Multiplying by $\sqrt{\mathrm{td}(X)} = (1, 0, 1) \otimes 1_E$ using the K3 cup product on each Künneth summand, $(1, 0, 1) \cup (1, 0, -d) = (1, 0, 1 - d)$ since $[\mathrm{pt}_S] \cup 1 = [\mathrm{pt}_S]$, and $(1, 0, 1) \cup (0, -\beta^\vee, -n) = (0, -\beta^\vee, -n)$ since $\beta^\vee \cup [\mathrm{pt}_S] = 0$ in $H^*(S, \mathbb{Z})$, yields the displayed Mukai vector $v_X(\mathcal{I}_Y) = (1, 0, 1 - d) \otimes 1_E + (0, -\beta^\vee, -n) \otimes \omega_E$. $\square$

Remark 3.8 *(Threefold Euler characteristic, no Todd correction).* The integer $n = \chi(\mathcal{O}_Y) \in \mathbb{Z}$ is the holomorphic Euler characteristic of the structure sheaf $\mathcal{O}_Y$ on the threefold $X = S \times E$; the trivial-canonical threefold condition makes $\sqrt{\mathrm{td}(X)}$ act as identity on the third Chern character so that $\int_X \mathrm{ch}_3(\mathcal{O}_Y) = n$ directly. There is no auxiliary "Todd correction" to install on top of n; the Mukai vector $(0, -\beta^\vee, -n) \otimes \omega_E$ records n in the K3 H^4-coordinate on the elliptic ω_E-summand. The sign is fixed by $\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y)$ and the Künneth orientation $\int_E \omega_E = +1$.

Theorem 3.9 *(D6–D2–D0 Mukai–Gram dictionary, restated as chart data).* With the standing data of Theorem 3.7, set

$$P_Y := (1, 0, 1 - d), \qquad Q_Y := (0, -\beta^\vee, -n) \in \widetilde{H}(S, \mathbb{Z}),$$

the Künneth components of $v_X(\mathcal{I}_Y)$. Then the Gram-degree projection of Definition 3.6 reads

$$\Pi_X(Q_Y, P_Y) = \left(\tfrac{(Q_Y, Q_Y)_{\mathrm{Muk}}}{2},\ (Q_Y, P_Y)_{\mathrm{Muk}},\ \tfrac{(P_Y, P_Y)_{\mathrm{Muk}}}{2}\right) = \left(\tfrac{(\beta^\vee)^2}{2},\ n,\ d - 1\right).$$

For the elliptically fibered K3 surface $S \to \mathbb{P}^1$ with section s_1 and curve class $\beta = \beta_h = s_1 + hF$ ($h \in \mathbb{Z}_{\geq 0}$, F the fibre class), the self-intersection $\beta_h^2 = 2h - 2$ gives

$$\Pi_X(Q_Y, P_Y) = (h - 1,\ n,\ d - 1).$$

Proof. Apply the Mukai pairing of Proposition 3.2 component by component:

$$\begin{aligned}
(Q_Y, Q_Y)_{\mathrm{Muk}} &= (-\beta^\vee) \cdot (-\beta^\vee) - 0 \cdot (-n) - 0 \cdot (-n) = (\beta^\vee)^2, \\
(Q_Y, P_Y)_{\mathrm{Muk}} &= (-\beta^\vee) \cdot 0 - 0 \cdot (1 - d) - 1 \cdot (-n) = n, \\
(P_Y, P_Y)_{\mathrm{Muk}} &= 0 \cdot 0 - 1 \cdot (1 - d) - 1 \cdot (1 - d) = 2(d - 1),
\end{aligned}$$

giving the half-pairings $((\beta^\vee)^2/2, n, d - 1)$. For the elliptic class $\beta_h = s_1 + hF$, identified with its Poincaré dual, the K3 intersection $s_1^2 = -2$ on a section of an elliptic K3 surface (genus formula $2g(s_1) - 2 = s_1^2 + s_1 \cdot K_S = -2 + 0$, giving $g(s_1) = 0$), $s_1 \cdot F = 1$, $F^2 = 0$, so $\beta_h^2 = s_1^2 + 2h(s_1 \cdot F) + h^2F^2 = -2 + 2h = 2h - 2$. The Gram-degree shadow follows. The cross-level identity with the OP/DT scalar normalization is the $Q_Y^2/2 = h - 1$ line of Theorem 3.9. $\square$

Remark 3.10 *(Algebraic vs primitive D2 charges).* The K3 curve class $\beta \in H_2(S, \mathbb{Z})$ appearing through its Poincaré dual in $Q_Y = (0, -\beta^\vee, -n)$ is algebraic: $\beta^\vee \in \mathrm{NS}(S)$, since $Y \subset X$ is a closed subscheme. Primitivity of β — that β is not a non-trivial integer multiple of another effective class — is a separate condition; the dictionary holds for any algebraic β, not only primitive ones. The Mukai-pairing formula $(Q_Y, Q_Y)_{\mathrm{Muk}} = (\beta^\vee)^2$ records the $\mathrm{NS}(S)$-self-intersection regardless of primitivity. The Bryan–Oberdieck–Pandharipande–Yin/Maulik–Pandharipande conventions [15, 81, 91] take β algebraic; the elliptic degree $d \in \mathbb{Z}_{\geq 0}$ is unconstrained beyond non-negativity in the effective semigroup.

Remark 3.11 *(Chart-data layer; not the additive Hall grading).* The triple $(n, l, m) = \Pi_X(Q_Y, P_Y)$ records the Gram-degree invariants of the dyonic charge $c = (Q, P)$. It is the target of the Igusa Fourier expansion and the Borcherds product variables $q^n r^l s^m$. It is *not* the additive Hall grading on a hypothetical compact $K3 \times E$ Hall sector; the additive grading is the upstairs charge $c \in \Gamma_X^{\mathrm{form}}$ (or its normal-ordered lift $\widehat{\Gamma}_X$ below), and Π_X is a quadratic map onto Gram invariants. The polarization defect that obstructs the upstairs additive grading from descending directly is the symmetric bilinear cocycle B of §3.4.

3.4 Symmetric bilinear Gram cocycle

LEMMA 3.12 *(Gram additivity defect, restated as chart structure).* For $c_i = (Q_i, P_i) \in \Gamma_X^{\mathrm{form}}$, $i = 1, 2$,

$$\Pi_X(c_1 + c_2) = \Pi_X(c_1) + \Pi_X(c_2) + B(c_1, c_2),$$

where the symmetric bilinear cocycle B is

$$B(c, c') := \big((Q, Q')_{\mathrm{Muk}},\ (Q, P')_{\mathrm{Muk}} + (Q', P)_{\mathrm{Muk}},\ (P, P')_{\mathrm{Muk}}\big).$$

The map B is symmetric and $\mathbb{Z}$-bilinear with values in $\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$, satisfies the polarization identity

$$B(c, c) = 2\Pi_X(c),$$

and equals $-\delta\Pi_X$ as an additive coboundary, $(\delta\Pi_X)(c, c') := \Pi_X(c) + \Pi_X(c') - \Pi_X(c + c')$. Thus the ordinary additive cohomology class $[B] = 0$; the obstruction recorded here is the impossibility of an *additive linear* trivialization, not a non-zero additive cohomology class.

Proof. This is Lemma 2.4. Symmetry, bilinearity, integrality, polarization, and δ-cocycle property are Lemma 3.12(i)–(iii). The splitting $B = -\delta\Pi_X$ by the quadratic refinement Π_X is clause (vi) of the same lemma; clause (vii) shows no additive linear $L : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ can satisfy $B = \delta L$, since $B(c, c) = 2\Pi_X(c)$ is non-zero on charges with non-trivial Mukai self-pairings. $\square$

Remark 3.13 *(Cleanly stated polarization identity).* The polarization identity $B(c, c) = 2\Pi_X(c)$ is the symmetric companion to $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$; evaluated at $c = c'$ the latter gives $\Pi_X(2c) = 2\Pi_X(c) + B(c, c)$, and $\Pi_X(2c) = 4\Pi_X(c)$ by quadraticity of Π_X, hence $B(c, c) = 2\Pi_X(c)$. This is purely chart-data structure; no Heisenberg or central extension hypothesis on the source is invoked.

3.5 Normal-ordered Gram map

Definition 3.14 *(Normal-ordered charge lattice).* Set

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}},$$

the underlying set $\Gamma_X^{\mathrm{form}} \times \Gamma_{\mathrm{gram}}$ equipped with the abelian-group law

$$(c, T) \star (c', T') := \big(c + c',\ T + T' + B(c, c')\big).$$

The identity is $(0, 0)$; the inverse of (c, T) is $(-c,\ -T + B(c, c)) = (-c,\ -T + 2\Pi_X(c))$. The split section

$$s : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \qquad s(c) := (c, \Pi_X(c)),$$

is a homomorphism of abelian groups; the raw placement

$$i_0 : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \qquad i_0(c) := (c, 0),$$

is not a homomorphism unless $B \equiv 0$.

Proposition 3.15 *(Normal-ordered Gram homomorphism).* The map

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) := \Pi_X(c) - T,$$

is a homomorphism of abelian groups, $\overline{\Pi}_X : (\widehat{\Gamma}_X, \star) \to (\Gamma_{\mathrm{gram}}, +)$; equivalently,

$$\overline{\Pi}_X\big((c,T) \star (c',T')\big) = \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T').$$

On the split section, $\overline{\Pi}_X(s(c)) = 0$; on the raw placement, $\overline{\Pi}_X(i_0(c)) = \Pi_X(c)$.

Proof. Direct calculation,

$$\begin{aligned}
\overline{\Pi}_X((c,T) \star (c',T')) &= \overline{\Pi}_X(c + c', T + T' + B(c,c')) \\
&= \Pi_X(c + c') - T - T' - B(c,c') \\
&= \big(\Pi_X(c) + \Pi_X(c') + B(c,c')\big) - T - T' - B(c,c') \\
&= \big(\Pi_X(c) - T\big) + \big(\Pi_X(c') - T'\big) \\
&= \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T'),
\end{aligned}$$

using Lemma 3.12. The split-section and raw-placement identities are the definitions. □

The finite formal packet `certificates/charge/mukai_gram_cocycle` records explicit rank-one Mukai-slice rows for the three simple Gram shadows δ_1, δ_2, δ_3 and one test charge. Its verifier checks the fixed Mukai sign convention, integrality of Π_X, symmetry and bilinearity of B, the identity

$$B(c,c') = \Pi_X(c + c') - \Pi_X(c) - \Pi_X(c'),$$

the polarization identity $B(c,c) = 2\Pi_X(c)$, additivity of the split section $s(c) = (c, \Pi_X(c))$, non-additivity of the raw placement $(c, 0)$, and the homomorphism property of $\overline{\Pi}_X$. This packet is formal chart arithmetic; it is not a finite HN charge window, a compact source construction, a Hall bracket computation, a Pfaffian orientation, an $O2$ wall atlas, a mirror discriminant, or a protected trace.

Proposition 3.16 *(Compact Mukai representatives for the three simple wall charges).* [constructed in the compact $K3$ Mukai chart] Let S be a smooth projective $K3$ surface. Choose a closed point $p \in S$ and a zero-dimensional closed subscheme $Z \subset S$ of length 2. With the convention

$$v(E) = \mathrm{ch}(E)\sqrt{\mathrm{td}(S)}, \qquad \sqrt{\mathrm{td}(S)} = (1,0,1),$$

one has

$$v(\mathcal{I}_Z) = (1,0,-1), \qquad v(\mathcal{I}_p) = (1,0,0), \qquad v(\mathcal{O}_p) = (0,0,1).$$

Define compact Mukai-pair representatives in $\widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$ by

$$x_{\delta_1} := (\mathcal{I}_Z, \mathcal{I}_p), \qquad x_{\delta_2} := (\mathcal{I}_p, \mathcal{I}_Z), \qquad x_{\delta_3} := (\mathcal{I}_p, \mathcal{O}_p).$$

Then

$$\Pi_X(v(x_{\delta_1})) = (1,1,0), \qquad \Pi_X(v(x_{\delta_2})) = (0,1,1), \qquad \Pi_X(v(x_{\delta_3})) = (0,-1,0).$$

Thus the three triples of Proposition 2.1 have compact representatives at the Mukai-chart level. This construction does not supply retained HN semistability, a compact Hall source basis, reduced orientations, vanishing-cycle summands, E-quotient data, O2 wall charts, Pfaffian signs, or protected integration.

Proof. For a zero-dimensional closed subscheme $W \subset S$ of length r, $\mathrm{ch}(\mathcal{I}_W) = 1 - r[\mathrm{pt}]$, hence

$$v(\mathcal{I}_W) = (1, 0, 1 - r).$$

For $r = 2$ this gives $v(\mathcal{I}_Z) = (1, 0, -1)$, and for $r = 1$ it gives $v(\mathcal{I}_p) = (1, 0, 0)$. The skyscraper sheaf has $\mathrm{ch}(\mathcal{O}_p) = (0, 0, 1)$, hence $v(\mathcal{O}_p) = (0, 0, 1)$.

Use the rank-one Mukai pairing

$$\big((r, d, s), (r', d', s')\big) = 2dd' - rs' - r's.$$

The three pair representatives have Mukai vectors

	Q	P
x_{∂_1}	$(1, 0, -1)$	$(1, 0, 0)$
x_{∂_2}	$(1, 0, 0)$	$(1, 0, -1)$
x_{∂_3}	$(1, 0, 0)$	$(0, 0, 1)$.

Therefore

$$\Pi_X(Q, P) = \Big(\frac{(Q, Q)}{2}, (Q, P), \frac{(P, P)}{2}\Big)$$

gives respectively

$$\Big(\frac{2}{2}, 1, \frac{0}{2}\Big) = (1, 1, 0), \qquad \Big(\frac{0}{2}, 1, \frac{2}{2}\Big) = (0, 1, 1), \qquad \Big(\frac{0}{2}, -1, \frac{0}{2}\Big) = (0, -1, 0).$$

All three sheaves are coherent sheaves on the projective surface S, so their supports are proper over $\mathbb{C}$. The final sentence records the extra structure not produced by the Mukai-vector calculation. □

The finite packet

`certificates/charge/type_ii_compact_mukai_representatives`

verifies `TYPE_II_COMPACT_MUKAI_REPRESENTATIVES_VERIFIED`: the three compact Mukai-pair representatives above have the required Gram-degree shadows. Its scalar firewall excludes reading these rows as compact Hall source basis vectors or retained O2 wall objects.

Remark 3.17 *(Polarization defect).* The composite $\overline{\Pi}_X \circ i_0 = \Pi_X$ records the quadratic Gram-degree shadow of the bare upstairs charge; the composite $\overline{\Pi}_X \circ s \equiv 0$ records the split-section convention that absorbs the entire Gram-degree into the central coordinate T. The choice of section is the choice of normal ordering: bare placements collect cross-pairing $B(c_i, c_j)$ in the central coordinate (Lemma 3.12), while s-placements record $\overline{\Pi}_X = 0$ on every component and the Gram-degree information lives entirely in the upstairs $\Pi_X(c)$. The two presentations agree on the homomorphism $\overline{\Pi}_X$. This is the additive group law on which a hypothetical compact Hall product can descend without a chain-level associator at the lattice level.

Remark 3.18 *(Chart data, not Hall support).* The lattice $\widehat{\Gamma}_X$ is the formal additive group on which charges and their Gram-degree shadows are tracked. It is not the algebraic effective semigroup Γ_X^{eff} of curve classes representable by closed subschemes $Y \subset X$; membership in the effective sub-semigroup is a separate algebraic-effective condition (Definition 1.35). The Beilinson chart layer α (charge data) and γ (Mukai lattice) license $\widehat{\Gamma}_X$ as ambient; the Hall-support restriction is part of the categorical-input layer at higher Beilinson level, not the Mukai–Gram chart datum.

3.6 Three Igusa names and the Mukai–Gram chart

Three Igusa names are distinct mathematical objects and remain distinct under the Mukai–Gram chart datum:

- the automorphic section $\mathcal{D}_X = \Delta_5$, a section of the weight-5 automorphic line over the genus-2 Siegel space with Maass character ν_{Δ_5};
- the BKM denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ on the root lattice $\Lambda^{2,1}_{II}$; and
- the compact protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$, a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on the conditional pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ of §5.

The Mukai–Gram dictionary supplies the Gram-degree shadow $\Pi_X : \Gamma^{\mathrm{form}}_X \to \Gamma_{\mathrm{gram}}$ on which the cusp-Fourier expansion of Δ_5, the root-degree map of $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, and the wall data of $\mathrm{Pf}_{\mathrm{prot}}$ all read. The Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ is symmetric in both algebraic and transcendental parts; the ungraded Mukai-form classical limit of an associated Yangian-type construction is the orthogonal Lie algebra $\mathfrak{so}(4, 20)$, not the orthosymplectic $\mathfrak{osp}(4 \mid 20)$. The Hodge $\mathbb{Z}/2$-super-extension that appears in Yangian conventions imposes a separate $\mathbb{Z}/2$-grading on the algebraic side; it does not enter the lattice pairing above. The Bryan/Oberdieck–Pandharipande/ Maulik–Toda conventions [15, 81, 91] take β algebraic and d elliptic-degree, agreeing with Theorem 3.9.

The Mukai–Gram chart datum does not touch the orientation character ϵ_o of the Weyl-transport condition 0.109, the Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$, or the recognition theorem of Theorem 0.253; those structures live at higher Beilinson levels and depend on the chart datum.

4 Hall correspondences on $\mathfrak{M}^{+}_{\mathrm{eff}}(K3 \times E)$

The Hall side of the Dirac–Igusa construction rests on the following imported theorems and on their $K3 \times E$ specialization. The Hall correspondences are the convolution correspondences through which the Beilinson–Drinfeld factorization structure on the elliptic carrier acts on the retained $K3 \times E$ Hall stack [7]. Equivalently, they comprise the chart-augmented A_b data for the open factorization dg-category (X, D, τ) at a chosen boundary vacuum b, with chiral shadow on the elliptic carrier supplied separately by § 9.

The cited theorems fix the exact hypotheses needed for the $K3 \times E$ specialization: finite retained substacks, proper extension correspondences, compact-support functoriality, vanishing cycles, Thom–Sebastiani maps, and orientation coefficients.

The standing data are: S is a smooth complex projective $K3$ surface, E is a smooth elliptic curve, $X = S \times E$ is the trivial-canonical threefold, $\sigma_S \in H^0(S, K_S)$ is the holomorphic two-form trivializing K_S, and $\sigma = \mathrm{pr}_S^*\sigma_S \in H^0(X, p_S^* K_S)$ is the inflated cosection of §9. The algebraic Mukai lattice used by the Hall support is

$$\Lambda_{S,\mathrm{alg}} = H^0(S, \mathbb{Z}) \oplus \mathrm{NS}(S) \oplus H^4(S, \mathbb{Z}),$$

the algebraic sublattice of the full Mukai lattice $\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$, endowed with the restricted Mukai pairing of [83]. Γ_{eff} is the algebraic/effective Hall support of Definition 1.35; and $\mathfrak{M}^{+}_{\mathrm{eff}}(X)$ is the moduli stack of perfect complexes of effective dyonic class semistable for the supplied stability datum on X.

D.1 The Hall product on the moduli of stable sheaves

The first imported theorem is the existence and associativity of a Hall product on the equivariant cohomology of the moduli stack of semistable sheaves, with vanishing-cycle coefficients, after Joyce and Song.

THEOREM 4.1 *(Joyce–Song generalised* DT *Hall product; γ, proved on* CY_3 *projective; conditional on (O1), (O1)$^+$ on $K3 \times E$).* Let Y be a smooth complex projective Calabi–Yau threefold. The moduli stack $\mathfrak{M}(Y)$ of compactly supported coherent sheaves on Y carries a critical *Hall product*

$$m \; : \; H_c^\bullet\big(\mathfrak{M}(Y), \Phi_W \otimes o\big) \otimes H_c^\bullet\big(\mathfrak{M}(Y), \Phi_W \otimes o\big) \longrightarrow H_c^\bullet\big(\mathfrak{M}(Y), \Phi_W \otimes o\big),$$

with Φ_W the perverse sheaf of vanishing cycles of the holomorphic Chern–Simons potential and o the orientation line supplied by Joyce–Upmeier orientation data. The product is defined by pull-push along the universal extension stack

$$\mathfrak{M}(Y) \xleftarrow{\;p\;} \mathfrak{Ext}^1(\mathfrak{M}(Y), \mathfrak{M}(Y)) \xrightarrow{\;q\;} \mathfrak{M}(Y),$$

where $\mathfrak{Ext}^1$ is the moduli stack of short exact sequences $0 \to A \to C \to B \to 0$, $p(0 \to A \to C \to B \to 0) = (A, B)$, and $q(0 \to A \to C \to B \to 0) = C$. The product $m = q_! \circ \mathrm{TS} \circ p^*$ is associative, where TS is the Massey–Saito–Joyce Thom–Sebastiani isomorphism on the universal extension stack [54, Theorem 5.4], [78], [12]. Its unit is the class of the empty object.

Proof. The product is the pull-push form of three standard structures. The moduli stack $\mathfrak{M}(Y)$ is locally the critical locus of the Joyce–Song holomorphic Chern–Simons potential $W : \mathrm{Crit}(W) \hookrightarrow Y_W$ where Y_W is a Darboux chart of the (-1)-shifted symplectic structure on $\mathfrak{M}(Y)$ supplied by Pantev–Toën–Vaquié–Vezzosi [95, Theorem 0.3]; the perverse sheaf Φ_W of vanishing cycles is well defined on the oriented d-critical locus by Brav–Bussi–Dupont–Joyce–Szendrői [12, Theorem 6.9]. The universal extension stack is locally a critical locus of $W_A \boxplus W_B$ on $Y_A \times Y_B$, and the Massey–Saito Thom–Sebastiani isomorphism [78, Theorem 1.1], [12, Theorem 5.10] identifies $\Phi_{W_A \boxplus W_B}$ with $\Phi_{W_A} \boxtimes \Phi_{W_B}$ on $\mathrm{Crit}(W_A) \times \mathrm{Crit}(W_B)$. The multiplicative orientation cocycle of Joyce–Upmeier [56, Theorem 4.1], [55, Theorem 1.11] trivializes the orientation gerbe on extensions, supplying the line bundle factor. When the exceptional pushforward is admissible, pulling $\Phi_W \otimes o$ along p, tensoring by Thom–Sebastiani, and pushing along $q_!$ defines m. Associativity reduces, via the two-step flag stack $\mathfrak{Flag}(\mathfrak{M}(Y), \mathfrak{M}(Y), \mathfrak{M}(Y))$, to the pentagon coherence of Thom–Sebastiani triple stratification [12, Theorem 5.10(iv)] together with the Joyce–Upmeier multiplicative cocycle on triple extensions [56, §4]. See Joyce–Song [54, §5.2, Theorem 5.5] for the original construction and Davison [26, Theorem A] for the cohomological amplification. □

$K3 \times E$ specialization. On $Y = K3 \times E$, the cosection $\sigma = \mathrm{pr}_S^* \sigma_S$ of the holomorphic two-form on the K3 factor reduces the obstruction theory through Kiem–Li (Appendix 4); the supplied moduli is $\mathfrak{M}^{\mathrm{red}}_{\mathrm{eff}}(X) \subset \mathfrak{M}(X)$, and the product is the reduced critical product

$$m^{\mathrm{red}} \;=\; q_!^{\mathrm{red}} \circ \mathrm{TS}^{\mathrm{red}} \circ p^*$$

of Proposition 0.149. Vanishing of the seven obstruction classes $\mathfrak{o}_R^{\mathrm{conf}}$, $\mathfrak{o}_R^{\mathrm{prop,LL}}$, $\mathfrak{o}_R^{\mathrm{prop,mix}}$, $\mathfrak{o}_R^{\mathrm{prop,WW}}$, $\mathfrak{o}_R^{\mathrm{conf\text{-}prop}}$, $\mathfrak{o}_{w,R}^{\mathrm{assoc}}$, and $\mathfrak{o}_{w,R}^{I,\mathrm{prod}}$, together with finite residual inertia, retained compactified flag-Quot correspondences or an explicit compact-support six-functor model, base-change, projection-formula, and quotient-after-correspondence coherences, is the data $(D0)$+$(O1)$, absorbed into Bnd, dCrit, RedOr, and Six of Definition 0.151. *The existence of a chain-level Hall product on $K3 \times E$ without those four data is open.* The resulting invariant $Y^+(X) := H_c^\bullet(\mathfrak{M}^+_{\mathrm{eff}}(X), \Phi_W \otimes o)$ is the positive-half cohomological Hall algebra (CoHA) of Kontsevich–Soibelman [64] restricted to effective dyonic class; its Drinfeld double, regulated tensor product completion, and integral form constitute the level-3 object $G(X)$, conditional on the Hall pairing and stable-envelope transport. Thus $G(X)$ arises from $Y^+(X)$ by Hall pairing, integral-form completion, and stable-envelope transport, as in the Vol III Δ_5 row [70].

D.2 The Hall coproduct via collisions

The Hall product does not by itself supply a coproduct. A collision coproduct is additional six-functor data on the reversed extension correspondence: properness or compact-support replacement for the left leg, inverse Thom–Sebastiani transport, quotient descent, vacuum unit/counit data, and the four-step flag-of-flags compatibility that compares Δm with $(m \otimes m)(1 \otimes \tau \otimes 1)(\Delta \otimes \Delta)$. The Hopf pairing and its radical quotient are separate data. When the coproduct and bialgebra compatibility are supplied, $Y^+(X)$ becomes a graded bialgebra object; after the Hopf pairing, Drinfeld doubling, completion, integral form, and stable-envelope transport, it is a candidate for the level-3 quantum group $G(X) = D_\hbar(Y^+(X))$ of Schiffmann–Vasserot type [98].

Theorem 4.2 *(Collision coproduct datum, formal conditional).* Let Y be a smooth projective Calabi–Yau threefold and suppose the Joyce–Song Hall product of Theorem 4.1 is given. Assume, in addition, that the reversed extension correspondence carries admissible compact-support six-functor data, inverse Thom–Sebastiani transport, vacuum unit/counit data, and three-step and four-step flag coherences for coassociativity and bialgebra compatibility. Then the critical Hall positive half $Y^+(Y) = H_c^\bullet(\mathfrak{M}^+_{\mathrm{eff}}(Y), \Phi_W \otimes o)$ carries a collision coproduct

$$\Delta \;:\; Y^+(Y) \longrightarrow Y^+(Y) \otimes Y^+(Y), \qquad \Delta \;=\; p^{\mathrm{coll}}_! \circ \mathrm{TS}^{-1} \circ q^{\mathrm{coll},*},$$

defined by the collision/extension correspondence

$$\mathfrak{M}^+_{\mathrm{eff}}(Y) \xleftarrow{\;q^{\mathrm{coll}}\;} \mathfrak{Ext}^{1,\mathrm{coll}}(\mathfrak{M}^+_{\mathrm{eff}}(Y)) \xrightarrow{\;p^{\mathrm{coll}}\;} \mathfrak{M}^+_{\mathrm{eff}}(Y) \times \mathfrak{M}^+_{\mathrm{eff}}(Y),$$

with $\mathfrak{Ext}^{1,\mathrm{coll}}$ the moduli of short exact sequences $0 \to A \to C \to B \to 0$ read with C on the left and (A, B) on the right, $q^{\mathrm{coll}}(0 \to A \to C \to B \to 0) = C$, $p^{\mathrm{coll}}(0 \to A \to C \to B \to 0) = (A, B)$. The coproduct Δ is graded coassociative. Together with the Hall product m of Theorem 4.1, $(Y^+(Y), m, \Delta)$ is a graded bialgebra.

Proof. The formula is the formal reverse of the Hall product. Its existence requires the extra six-functor hypotheses because the reversed leg is not automatically proper in the compact threefold problem. Given those hypotheses, coassociativity is the inverse Thom–Sebastiani pentagon on the three-step flag stack. The compatibility of m and Δ is not a consequence of the product theorem alone; it is the equality of the two pull-push functors on the four-step flag-of-flags common refinement. On quivers with potential this is the Davison–Meinhardt bialgebra theorem [27, Theorem A]; on the compact $K3 \times E$ threefold it is the finite correspondence datum of Definition 0.171 and Proposition 0.178. □

Drinfeld double. The Hall pairing $\langle \cdot, \cdot \rangle_{\mathrm{Hall}}$, when supplied and non-degenerate after the completed radical quotient, pairs Y^+ with itself; the Drinfeld double $D_\hbar(Y^+(X))$ is the quantum double in the sense of Drinfeld [58, §13], supplying the negative half $Y^-(X)$ and the Cartan generators K_γ, $\gamma \in \Gamma_{\text{eff}}$, with relations encoding the Hall pairing. The *level-3* $K3 \times E$ *quantum group* is then

$$G(X) \;:=\; D_\hbar(Y^+(X))^{\text{int}}_{\text{compl}}$$

where $(-)^{\text{int}}_{\text{compl}}$ is the integral-form completion under the Hall pairing, conditional on the stable-envelope transport of § D.6. In Schiffmann–Vasserot [98] the analogous identification on toric CY_3 ambient is established; $K3 \times E$ is not equipped here with a global toric or quiver-variety NCCR model of that kind: there is no finite fixed-point McKay presentation, no compatible product-polarized HPD self-dual model, and no Schiffmann–Vasserot stable-envelope atlas on the compact non-toric ambient. The cosection-twisted reduced obstruction theory of § D.4 is the replacement; the global $G(X)$ on the compact non-toric $K3 \times E$ is not constructed from these data without the Hall pairing, integral form, completion, and stable-envelope transport.

D.3 The Hall primitives and the formal recognition criterion

For a connected graded bialgebra $B = \bigoplus_c B_c$, the primitive elements at charge c are

$$\text{Prim}_c(B) \;=\; \{x \in B_c \mid \Delta(x) = x \otimes 1 + 1 \otimes x\},$$

equivalently the kernel of the reduced coproduct $\bar{\Delta}$. On a graded connected cocommutative Hopf algebra in characteristic zero, the Milnor–Moore theorem [75, Theorem 5.18] identifies $B \simeq U(\text{Prim}_\bullet(B))$ as Hopf algebras, where $\text{Prim}_\bullet(B)$ is the Lie algebra of primitives under the bracket $[x, y] = xy - (-1)^{|x|\,|y|}yx$.

Theorem 4.3 *(Hall primitives, formal level; β, formal).* Let $\mathfrak{M}$ carry a critical Hall positive half $Y^+(Y) = (B, m, \Delta)$ of Theorem 4.2. The graded subspace

$$\text{Prim}_\bullet(Y^+(Y)) \;=\; \bigoplus_{c \in \Gamma_{\text{eff}}} \text{Prim}_c(Y^+(Y))$$

is a graded Lie subalgebra of $Y^+(Y)$ under the antisymmetrised product $[\,\cdot\,,\,\cdot\,]$. At a charge c such that the moduli $\mathfrak{M}^+_c(Y)$ is connected and the C-component of Δ on $\mathfrak{M}^+_c(Y)$ factors through extensions whose graded pieces lie in strictly lower charges, the inclusion $\text{Prim}_c \hookrightarrow Y^+_c$ is the kernel of $\bar{\Delta}_c$. When $Y^+(Y)$ is graded connected and cocommutative, Milnor–Moore identifies $Y^+(Y) \simeq U(\text{Prim}_\bullet(Y^+(Y)))$ as graded Hopf algebras.

Proof. Primitivity of $\text{Prim}_\bullet$ under the antisymmetrised product is verified by direct computation of $\Delta([x, y])$ using the cocommutative-up-to-twist axiom and the multiplicativity of Δ; the resulting expression collapses to $[x, y] \otimes 1 + 1 \otimes [x, y]$. On a connected graded cocommutative Hopf algebra in characteristic zero, the Milnor–Moore identification is [75, §21.1]; on graded connected non-cocommutative Hopf algebras, the Cartier–Quillen–Milnor–Moore theorem [75, Theorem 21.2.1] requires either cocommutativity or a chain-level lift through bar / cobar, which is exactly the chiral Koszul separation theorem of Vol I [35]. □

Recognition. The recognition theorem of Theorem 0.215 identifies the completed formal primitive Lie superalgebra with the imported Borcherds–Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ of Theorem 9.1, conditional on the finite compact-source datum $(S1)$–$(S10)$: the completed PBW filtration, the Hopf pairing radical equality, and the parity dimension recovery. In particular, on $K3 \times E$ the formal primitive subspace

$\mathrm{Prim}_\bullet(Y^+(X))$ is the *candidate* primitive Lie superalgebra; its identification with the imported BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ is the primitive recognition part of View ③. The formal Hopf algebra structure alone gives no recognition theorem; recognition acts on it.

D.4 The cosection-twist on $K3 \times E$: Kiem–Li localisation

On $K3 \times E$, the holomorphic two-form on the K3 factor supplies the semiregularity cosection on the retained nonzero K3-class branches of the obstruction theory of $\mathfrak{M}(X)$. The Kiem–Li [59] localisation theorem then localises the virtual fundamental class to the zero-locus of the cosection when the surjectivity hypothesis holds. The localisation is the source of the *reduced* virtual class on which the Hall product of § D.1 carries chain-level coherence.

Definition 4.4 *(Cosection).* Let $\mathfrak{M}$ be a Deligne–Mumford stack equipped with a perfect obstruction theory $\phi : \mathbb{E} \to \mathbb{L}_{\mathfrak{M}}$ of amplitude $[-1, 0]$, and let $\mathrm{Ob}(\mathfrak{M}) = h^1(\mathbb{E}^\vee)$ be the coherent obstruction sheaf. A *cosection* of the obstruction theory is an $\mathcal{O}_{\mathfrak{M}}$-linear morphism

$$\sigma \ : \ \mathrm{Ob}(\mathfrak{M}) \longrightarrow \mathcal{O}_{\mathfrak{M}};$$

its *vanishing locus* is the closed substack $\mathfrak{M}^{\mathrm{red}} = (\sigma = 0)$ on which the obstruction sheaf restricts to the kernel $\ker \sigma \subset \mathrm{Ob}(\mathfrak{M})$.

Theorem 4.5 *(Kiem–Li cosection localisation;* [59] *Theorem 5.0.1; γ, proved).* Let $\mathfrak{M}$ be a Deligne–Mumford stack with perfect obstruction theory ϕ and surjective cosection $\sigma : \mathrm{Ob}(\mathfrak{M}) \twoheadrightarrow \mathcal{O}_{\mathfrak{M}}$ on $\mathfrak{M} \setminus \mathfrak{M}^{\mathrm{red}}$, where $\mathfrak{M}^{\mathrm{red}} = (\sigma = 0)$. The virtual fundamental class $[\mathfrak{M}]^{\mathrm{vir}} \in A_{\mathrm{vd}}(\mathfrak{M})$ lifts to a cosection-localised class

$$[\mathfrak{M}]^{\mathrm{vir}}_{\mathrm{loc}} \ \in \ A_{\mathrm{vd}}(\mathfrak{M}^{\mathrm{red}})$$

satisfying $\iota_*[\mathfrak{M}]^{\mathrm{vir}}_{\mathrm{loc}} = [\mathfrak{M}]^{\mathrm{vir}}$, where $\iota : \mathfrak{M}^{\mathrm{red}} \hookrightarrow \mathfrak{M}$ is the inclusion. If the cosection is used separately to quotient a trivial obstruction direction and form a reduced obstruction theory, that reduced theory has virtual dimension $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd} + 1$; this dimension shift is not the Kiem–Li localized cycle itself.

Proof. The cosection-kernel cone $C(\sigma) = \ker(\sigma : \mathrm{Ob} \to \mathcal{O})$ is a perfect obstruction cone of amplitude $[-1, 0]$ on $\mathfrak{M}^{\mathrm{red}}$ with $\mathrm{rk}\, C(\sigma) = \mathrm{rk}\,\mathrm{Ob} - 1$. The intrinsic normal cone $\mathfrak{C}$ of $\mathfrak{M}$ sits in the total obstruction bundle on $\mathfrak{M} \setminus \mathfrak{M}^{\mathrm{red}}$, where σ is surjective and the cone has codimension one in Ob; on $\mathfrak{M}^{\mathrm{red}}$, the cone embeds into $C(\sigma)$ by [59, Proposition 4.3]. Intersection with the zero section of $C(\sigma)$ on $\mathfrak{M}^{\mathrm{red}}$, pushed forward to $\mathfrak{M}$ through ι, yields $[\mathfrak{M}]^{\mathrm{vir}}$ by [59, Theorem 5.0.1]. The localized Gysin map preserves the original virtual dimension. The shift $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd} + 1$ enters only after a separate reduced obstruction theory is formed by removing the trivial obstruction summand. □

$K3 \times E$ cosection. On $X = K3 \times E$, the cosection of the obstruction theory of $\mathfrak{M}(X)$ is induced by the pullback $\sigma = \mathrm{pr}_S^* \sigma_S$ of the K3 holomorphic two-form. Concretely, for a perfect complex F on X, the chain-level cosection acts on $\mathrm{Ext}^2(F, F)$ by the semiregularity map of Buchweitz–Flenner / Bloch [81],

$$\mathrm{cs}_F \ : \ \mathrm{Ext}^2(F, F) \longrightarrow \mathbb{C}, \qquad \mathrm{cs}_F(\xi) \ = \ \int_X \sigma \wedge \mathrm{Tr}\big((\xi \otimes 1_{\Omega^1_X}) \circ \mathrm{At}(F)\big),$$

where $\mathrm{At}(F) \in \mathrm{Ext}^1(F, F \otimes \Omega^1_X)$ is the Atiyah class. The cosection vanishes precisely on the *reduced* moduli $\mathfrak{M}^{\mathrm{red}}(X) \subset \mathfrak{M}(X)$ consisting of sheaves whose Atiyah class is annihilated by σ; equivalently, sheaves whose deformation space along the K3 direction is constrained. See Maulik–Toda [81, Lemma 2.6, Theorem 2.7] and Oberdieck [90, Theorem 1, Theorem 3] for the explicit identification on the DT/PT side. Surjectivity is an additional row-level condition: the retained stratum must be identified with the nonzero K3-class branch and must record a cokernel-rank-zero semiregularity row. On the $s = 0$ Hall cusp the retained charge can lose the K3 component that makes the semiregularity map surjective; the holomorphic two-form σ_S does not vanish. The localisation then passes to the D0-degeneration limit named at obstruction $(D0)$ of § 9.

Theorem 4.6 *(Reduced virtual class on $K3 \times E$; primitive stable-pair branch).* On the stable-pair or ideal-sheaf Deligne–Mumford branch carrying a perfect obstruction theory and a surjective K3 semiregularity cosection, Kiem–Li localization gives

$$[\mathfrak{M}_c(X)]^{\mathrm{vir}}_{\mathrm{loc}} \in A_{\mathrm{vd}}(\mathfrak{M}_c(X)^{\mathrm{red}}).$$

The reduced obstruction theory obtained by quotienting the trivial obstruction direction gives a distinct reduced class

$$[\mathfrak{M}_c(X)]^{\mathrm{red}} \in A_{\mathrm{vd}+1}(\mathfrak{M}_c(X)^{\mathrm{red}}).$$

The reduced DT/PT theory of the primitive nonzero K3-class branch uses this reduced class and gives the Oberdieck–Pixton scalar $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ [90, Theorem 3]. The semistable Hall stack requires the retained compact-support six-functor and d-critical Hall datum of §0.3. On the $s = 0$ cusp the cosection degenerates; *the cosection-reduced theory there is the* $(D0)$ *degeneration limit named at § 9 and controlled by the retained D0-HN criterion of Theorem 7.7.* $[\partial, (D0)]$

D.5 BBDJS reduced d-critical orientation

The Hall product of § D.1 requires not just a virtual fundamental class but a perverse-sheaf-of-vanishing-cycles refinement and an orientation gerbe; the source is the Brav–Bussi–Dupont–Joyce–Szendrői [12] extension of the Joyce [54] d-critical-locus framework.

Theorem 4.7 *(BBDJS perverse-sheaf-of-vanishing-cycles; oriented* CY_3 *version of* [12] *Theorem 6.9; γ, proved).* Let $\mathfrak{M}$ be a derived Artin stack with a (-1)-shifted symplectic structure (PTVV [95]) whose classical truncation is a d-critical locus in the sense of Joyce [54, Definition 2.5]. Suppose $\mathfrak{M}$ carries an *orientation*, i.e. a square root o of the Joyce determinant line $K^{\mathrm{vir}}_{\mathfrak{M}} = (\det \mathbb{L}_{\mathfrak{M}})^{1/2}$, in the sense of [12, Theorem 6.9]. There is a perverse sheaf

$$\Phi_{\mathfrak{M}} \;\in\; \mathrm{Perv}(\mathfrak{M}),$$

the *perverse sheaf of vanishing cycles* of the d-critical locus; locally on a Joyce–Song Darboux chart $\mathrm{Crit}(W)$ inside a smooth ambient Y_W, $\Phi_{\mathfrak{M}}|_{\mathrm{Crit}(W)} \simeq \Phi_W[d_{Y_W}]$ is the standard perverse sheaf of vanishing cycles of the holomorphic potential $W : Y_W \to \mathbb{C}$, shifted by the ambient dimension. The chart-comparison cocycle is canonical up to the orientation choice; the orientation determines a multiplicative cocycle of Joyce–Upmeier [56, Theorem 4.1] on the moduli of objects.

Proof. Three layers stack. PTVV [95, Theorem 0.3] produces the (-1)-shifted symplectic form on the derived moduli of CY_3 coherent sheaves; Bussi–Brav–Joyce [12, §5] produce the d-critical-locus chart structure on the classical truncation; Bussi–Brav–Dupont–Joyce–Szendrői [12, Theorem 6.9] produce the perverse sheaf of vanishing cycles with its chart cocycle. The orientation is the datum needed to unify chart-local Φ_W into a global perverse sheaf $\Phi_{\mathfrak{M}}$; without it, the chart-cocycle has a sign ambiguity that obstructs the descent to a perverse sheaf [12, §6.4]. Joyce–Upmeier [56, Theorem 1.11] construct the multiplicative orientation cocycle on the universal extension stack; this is the CY_3 projective theorem for the retained Hall orientation. □

Reduced refinement on $K3 \times E$. On $X = K3 \times E$, the (-1)-shifted symplectic structure $\omega_{\mathfrak{M}}$ on $\mathfrak{M}(X)$ does not restrict to a (-1)-shifted symplectic structure on $\mathfrak{M}^{\mathrm{red}}(X)$; the cosection-twist of § D.4 contracts $\omega_{\mathfrak{M}}$ by σ, and the *reduced* (-1)-shifted form $\omega^{\mathrm{red}}_{\mathfrak{M}}$ is non-degenerate only after restriction to the cosection vanishing locus [81, §2.6]. The corresponding reduced d-critical-locus structure is supplied by the Maulik–Toda [81, Theorem 2.7] extension; the reduced perverse sheaf of vanishing cycles $\Phi^{\mathrm{red}}_{\mathfrak{M}}$ and the reduced orientation line o^{red}_C are exactly the data $(O1)$ of § 9. In the notation of (P4), the reduced orientation is twisted by a parity-flip line Π generated by the cosection cokernel,

$$o^{\mathrm{red}}_C \;=\; o^{\mathrm{std}}_C \otimes \Pi,$$

absorbing the dimension shift $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ into a $\mathbb{Z}/2$-grading shift on the oriented BBDJS line.

Proposition 4.8 *(BBDJS reduced d-critical orientation on $K3 \times E$; ∂, $(O1) + (O1)^+$).* Let $\mathfrak{M}^{\mathrm{red},+}_c(X)$ be the cosection-reduced moduli of semistable sheaves of effective dyonic class c on $X = K3 \times E$. The hypotheses *strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle (O1) and Weyl-equivariant transport along $W^{(2)}(\Lambda^{2,1}_{II})$ reflections (O1)$^+$* of § 9 are exactly the following. There exist a multiplicative orientation cocycle of Joyce–Upmeier type extending equivariantly across the reduced E-quotient, the σ_S-cosection-induced parity flip line Π, and a connected BE-equivariant trivialisation of the small-orbit gerbe class $\overline{\beta}^E_C$ on the finite-stabilizer strata, such that

$$\Phi^{\mathrm{red}}_{\mathfrak{M}^{\mathrm{red},+}_c(X)} \otimes o^{\mathrm{red}}_c$$

descends through the free E-translation quotient and trivializes the Pin^c-lift gerbe class of the reduced derived self-Ext frame bundle. The strong reduced orientation $(O1)$ is established from the retained quotient-orientation datum. Joyce–Upmeier [56, Theorem 1.11] supplies the projective Calabi–Yau-threefold orientation technology used here; the quasi-projective and quotient descent belongs to the retained $(O1_{\mathrm{quot}})$ datum. The Cao–Kool–Monavari CY_4 theorem [17, Theorem 1.6] is a fourfold analogue, not an ingredient of the $K3 \times E$ CY_3 reduction. The free E-quotient, finite-stabilizer, and Weyl-transport theorem on $K3 \times E$ follows only relative to the retained quotient-orientation datum and the chosen Weyl lifts of Theorems 7.13 and 7.14.

Proof. The construction of the reduced perverse sheaf $\Phi^{\mathrm{red}}_{\mathfrak{M}^{\mathrm{red},+}_c(X)}$ on the cosection-reduced moduli is by Maulik–Toda [81, Theorem 2.7]; the orientation gerbe lifts to the reduced Joyce d-critical locus by the BBDJS [12, §6.4] chart-cocycle calculation. The multiplicative *equivariant* orientation cocycle across the E-quotient and across the finite-stabilizer locus $BE[N]$-strata is the joint vanishing of $(O1)$ and $(O1)^+$: on the connected BE-equivariant trivialisation of $\alpha^E_C \in H^2_E(\mathfrak{M}^{\mathrm{red}}_C; \mathbb{F}_2)$ of Definition 0.109 it suffices to trivialize the small-orbit gerbe class $\overline{\beta}^E_C$ on $BE[N]$-strata, governed by the Klein-four detection criterion of Lemma 7.10. The Weyl-equivariant transport across $W^{(2)}(\Lambda^{2,1}_{II})$ reflections is the orientation character calculation of Theorem 0.253 on the three type-II walls. Theorems 7.13 and 7.14 supply these transports on $K3 \times E$; the Pin^c-lift theory of [55, 56] supplies the analytic technology in the Calabi–Yau-threefold setting. □

D.6 Stable-envelope transport: from Hall to Yangian

The level-3 quantum-group structure on $G(X) = D_\hbar(Y^+(X))$ of § D.2 is, on toric ambient or quiver varieties, identified by Maulik–Okounkov [79] with the geometric R-matrix of the *stable envelope*, providing the link from the Hall side to the Yangian / quantum-group side.

Definition 4.9 *(Stable envelope).* Let $\mathfrak{M}$ be a smooth symplectic variety with a torus T-action and a finite T-fixed locus $\mathfrak{M}^T$; let $C \subset \mathrm{Lie}(T)$ be a chamber of the T-equivariant cohomology of $\mathfrak{M}$. The *stable envelope* [79, §3] is the T-equivariant cohomological correspondence

$$\mathrm{Stab}_C \ : \ H_T^\bullet(\mathfrak{M}^T) \longrightarrow H_T^\bullet(\mathfrak{M}),$$

characterised by support, normalization, and degree axioms [79, Theorem 3.3.4]. The geometric R-matrix is the comparison

$$R^{\mathrm{MO}}_{C,C'} \ = \ \mathrm{Stab}_{C'}^{-1} \circ \mathrm{Stab}_C \ \in \ \mathrm{End}(H_T^\bullet(\mathfrak{M}^T))$$

across two chambers C, C'; it satisfies the Yang–Baxter equation by [79, Theorem 4.6.1].

THEOREM 4.10 *(Maulik–Okounkov;* [79] *Theorem 10.2.1; β, proved on quiver varieties; conditional on $K3 \times E$ compact non-toric).* Let $\mathfrak{M} = \mathfrak{M}_{Q,\mathbf{v},\mathbf{w}}$ be a Nakajima quiver variety [85, 86] attached to a finite quiver Q with dimension vectors $\mathbf{v}, \mathbf{w}$. The stable envelope Stab_C makes $H_T^\bullet(\mathfrak{M}_{Q,\mathbf{v},\mathbf{w}})$ a module over a Yangian $Y_\hbar(\mathfrak{g}_Q)$ attached to Q, with $\mathfrak{g}_Q$ the Kac–Moody Lie algebra of the quiver, and the geometric R-matrix R^{MO} is the rational R-matrix of $Y_\hbar(\mathfrak{g}_Q)$.

Proof. The Maulik–Okounkov construction proceeds through: (i) the existence and uniqueness of the stable envelope (axioms of support, polarisation, normalization), [79, §3.3]; (ii) the Yang–Baxter equation as a chamber-wall cocycle, [79, §4]; (iii) the identification of the Bethe algebra generated by the matrix elements of Stab with a Yangian, [79, §5–10]. The Schiffmann–Vasserot identification $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$ of [98, Theorem A] is the affine A_0 specialization; $\mathcal{W}_{1+\infty}$ arises *only after* Drinfeld doubling and Fock-evaluation. The three-arrow chain $\mathrm{CoHA} = Y^+ \hookrightarrow Y \xrightarrow{\mathrm{ev}_\lambda} \mathrm{End}(\mathcal{W}_{1+\infty}[\lambda]\text{-vac})$ of [64], [98] traverses the level-2 → level-3 → level-3-evaluated passage; the named identity on each arrow measures the structural distance from $\mathrm{CoHA}(\mathbb{C}^3)$ to $\mathcal{W}_{1+\infty}$. □

$K3 \times E$ specialization: open. On $X = K3 \times E$, the moduli $\mathfrak{M}^+_{\mathrm{eff}}(X)$ is not a Nakajima quiver variety, the ambient is compact non-toric, and the global stable envelope is unconstructed. The five obstructions to a global toric/quiver-variety NCCR model of the Schiffmann–Vasserot type obstruct the literal [79] construction. The cosection-twist of § D.4 supplies the *Serre-equivariant reduced substitute*: the cosection-reduced obstruction theory of Theorem 4.6 restores a CY_3-symmetric reduced obstruction theory on the $s \neq 0$ sector, and the Hall-to-Yangian comparison is then expected at the level of the reduced theory and conditional on the stable-envelope transport, after the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum of Appendix 7.

The Hall–Yangian comparison on $K3 \times E$ is therefore conjectural:

$$G(X) \ \stackrel{?}{=} \ D_\hbar(\mathrm{CoHA}^{\mathrm{red},+}_{K3\times E})^{\mathrm{int}}_{\mathrm{compl}} \ \stackrel{?}{=} \ Y_\hbar(\widehat{\mathfrak{g}}_{\Delta_5})$$

where $\widehat{\mathfrak{g}}_{\Delta_5}$ is the central extension of the imported BKM Lie superalgebra of Theorem 9.1, and the equality is conditional on a $K3 \times E$ stable-envelope transport. The Hall–Drinfeld double $\mathcal{D}_\hbar(\mathrm{CoHA}_{K3\times E})$ of Kontsevich–Soibelman [64] places this identification at the same level-3 quantum group seen from the geometric side; it is the open compact non-toric instance of the chain-fusion conjecture.

D.7 Level separations

The Beilinson cut separates the following objects.

(i) $Y^+(X)$ is the cohomological-Hall positive half $H^\bullet_c(\mathfrak{M}^+_{\mathrm{eff}}(X), \Phi_W \otimes o)$ with the Joyce–Song product/coproduct of Theorems 4.1, 4.2; $G(X) = D_\hbar(Y^+(X))^{\mathrm{int}}_{\mathrm{compl}}$ is the Drinfeld double after Hall pairing, completion, integral form, stable-envelope transport, and descent.

(ii) $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$ [98, Theorem A]; $\mathcal{W}_{1+\infty}$ appears only after Drinfeld doubling and Fock evaluation.

(iii) No global Schiffmann–Vasserot/Nakajima atlas is known for compact non-toric $K3 \times E$. The cosection-twist of § D.4 supplies the Serre-equivariant reduced substitute used here.

(iv) $\mathrm{Prim}_\bullet(Y^+(X))$ is the formal primitive Lie superalgebra of $Y^+(X)$; its identification with $\mathfrak{g}_{\Delta_5}$ is the primitive-recognition theorem of Theorem 0.215, conditional on the finite compact-source datum $(S1)$–$(S10)$.

(v) BBDJS [12, Theorem 6.9] requires an *oriented* d-critical locus; the orientation is the square root of the Joyce determinant line, and the multiplicative cocycle is Joyce–Upmeier [56, Theorem 4.1].

(vi) Kiem–Li [59, Theorem 5.0.1] requires the cosection to be surjective off the localised locus; on $K3 \times E$, the cosection degenerates on the $s = 0$ cusp, which is the $(D0)$ degeneration limit controlled by the retained D0-HN criterion of Theorem 7.7.

(vii) The cosection contracts the (-1)-shifted symplectic form; ω^{red} is non-degenerate only after restriction to the cosection vanishing locus [81, §2.6], and the orientation line carries the $\mathbb{Z}/2$-parity flip Π absorbing the dimension shift $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$.

(viii) $K3 \times E$ is a trivial-canonical threefold. The scalar-trace index is $n = \chi(\mathcal{O}_Y)$ on the closed target, whereas $\chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3})\chi(\mathcal{O}_E) = 2 \cdot 0 = 0$, and the BBDJS / Joyce–Upmeier machinery applies in the CY_3 form [12, §6], not in the CY_4 Borisov–Joyce [11] or Oh–Thomas [93, 94] form.

D.8 Four local obstruction calculations

Four local calculations enter the Hall correspondence.

(F1) Hall associativity. The Joyce–Song associativity of Theorem 4.1 requires the pentagon coherence of Thom–Sebastiani triple stratification [12, Theorem 5.10(iv)]. The chart-cocycle of the d-critical locus carries the orientation sign whose pentagon coherence is measured by the BBDJS orientation gerbe. That gerbe [12, §6.4] trivializes exactly this ambiguity; the multiplicative cocycle of Joyce–Upmeier [56, §4] extends the trivialisation to the universal extension stack. On $K3 \times E$, the gerbe descends through the free E-quotient and trivializes the small-orbit class by Theorems 7.13 and 7.14.

(F2) Cosection surjectivity. Kiem–Li [59, Theorem 5.0.1] requires $\sigma : \mathrm{Ob} \twoheadrightarrow \mathcal{O}$ on $\mathfrak{M} \setminus \mathfrak{M}^{\mathrm{red}}$. On the $s = 0$ cusp of the dyonic charge lattice of $K3 \times E$, the K3-class component needed for semiregularity can disappear and surjectivity fails, while the K3 holomorphic two-form itself remains nonzero. The smooth primitive nonzero K3-class branch satisfies surjectivity by Maulik–Toda [81, §2]; the cusp branch is the $(D0)$ degeneration limit named at obstruction $(D0)$ of § 9, controlled by the retained D0-HN criterion of Theorem 7.7. The Oberdieck–Pixton scalar $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is proved on the $s \neq 0$ branch [90, Theorem 3]; on the cusp the chain-level theory degenerates to the Borcherds imaginary-simple-root contribution.

(F3) BBDJS orientation existence on compact non-toric. The Joyce–Upmeier multiplicative cocycle [56, Theorem 4.1] is established on the CY_3 projective ambient; its extension across the cosection-reduced quotient ambient is part of the retained $(O1_{\mathrm{quot}})$ datum. On the $K3 \times E$ quotient by free E-translation, descent of the orientation line requires the small-orbit gerbe class on $BE[N]$-strata to trivialize. The joint vanishing of the connected BE-equivariant degree-two obstruction $\alpha^E_C \in H^2_E(\mathfrak{M}^{\mathrm{red}}_C; \mathbb{F}_2)$ the finite-stabilizer Cech–Borel gerbe classes, and the residual degree-one stabilizer characters is exactly

$(O1)$; Theorem 7.13 of Appendix 7 is the quotient-orientation criterion relative to this retained datum. The remaining open data here are the retained quotient orientation and the compact-source recognition rows.

(F4) **(-1)-shifted vs (-2)-shifted symplectic.** $K3 \times E$ is a trivial-canonical threefold; the moduli $\mathfrak{M}(K3 \times E)$ carries a (-1)-shifted symplectic structure by PTVV [95, Theorem 0.3]. The Borisov–Joyce [11] and Oh–Thomas [93, 94] theories are CY_4 theories. The Calabi–Yau dimension of $X = K3 \times E$ is $\dim_{\mathbb{C}}(K3) + \dim_{\mathbb{C}}(E) = 2 + 1 = 3$, the holomorphic three-form is $\sigma_S \wedge dz_E$, and the shifted symplectic is $[-1]$-shifted by [95, §2.1]. The cosection of § D.4 removes one obstruction direction and increases the reduced virtual dimension by one, but does not reduce the shifted-symplectic degree; ω^{red} is still (-1)-shifted, on the cosection vanishing locus, by the Maulik–Toda [81, Theorem 2.7] extension. The cosection-twist of § D.4 is the Serre-equivariant reduced substitute for the missing global toric/quiver-variety atlas.

The four calculations above are the local Hall forms of $(D0)$, $(O1)$, $(O1)^+$, and $(O2)$. Their vanishing is the hypothesis of the conditional cross-level identity ②↔③. The Hall correspondences above supply the compact-support functorial maps; the recognition $\mathrm{Prim}_\bullet(Y^+(X)) \simeq \mathfrak{g}_{\Delta_5}$ of Theorem 0.215 remains conditional on $(S1)$–$(S10)$. The protected Pfaffian theorem compares the View ③ Pfaffian section with $\mathcal{D}_X = \Delta_5$; the level-Z scalar trace is a later trace operation.

5 Pfaffian wall normal form on type-II walls

The three local Pfaffian wall computations supply the sign data used in Chapter 7 for the orientation-character match

$$\epsilon_o = \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II}),$$

after the global orientation transport of Theorems 7.13, 7.14, and 7.19. Every numerical identity of §5.1 and §5.6 is independently verified by exact rational lattice arithmetic and exact rational expansion of the Borcherds product.

5.1 *The three simple type-II walls*

The hyperbolic root lattice of signature $(2, 1)$ is $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus \langle 2 \rangle$, with basis (f_2, f_3, f_{-2}) and bilinear form $(f_2, f_{-2}) = -1$, $(f_3, f_3) = 2$. Its index-four type-II sublattice is

$$\Lambda^{2,1}_{II} = \{ m f_2 + l f_3 + n f_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2 \},$$

abstractly $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2 \rangle$ with discriminant -32; the reflection-group index $[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{II})] = 6$ tracks the orbit data on the chamber-walls and is independent of the sublattice index. The three simple type-II roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with pairwise pairings forming the type-II Cartan matrix

$$A = \big((\delta_i, \delta_j)\big)_{1 \le i,j \le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}, \qquad \det A = -32.$$

The matrix is symmetric generalized Cartan, hyperbolic (one negative eigenvalue, two positive), and the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ generated by $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ is the Coxeter group with relations $s^2_{\delta_i} = 1$ and no braid relation between distinct generators; the off-diagonal value -2 is the infinite Coxeter exponent. It acts on the hyperbolic plane $\mathbb{R}\mathcal{P}_{II}$. The Weyl vector ρ determined by $(\rho, \delta_i) = -1$ for $i = 1, 2, 3$ is

$$\rho = \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

The Cartan matrix, the Weyl vector, the reflection identity $s_{\delta_i}(\delta_i) = -\delta_i$, and the preservation of the form under s_{δ_i} are all direct computations in the BKM Gram.

Remark 5.1 *(Type-II versus type-I).* The lattice $\Lambda^{2,1}$ admits both type-I roots (divisibility $(\delta, \Lambda^{2,1}) = \mathbb{Z}$, simple system $\{f_{-2} - f_2, f_2 - f_3, f_{-2} - f_3\}$, trivial multiplier) and type-II roots (divisibility $2\mathbb{Z}$, simple system $\{\delta_1, \delta_2, \delta_3\}$ above, multiplier -1 on each generator). The Borcherds–Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$ is the type-II algebra, with denominator $64^{-1}\Delta_5(2Z)$ on $\Lambda^{2,1}_{II}$; the doubling $2Z$ is the passage from the type-I to the type-II lattice, and the constant $64 = 2^6$ is the theta-normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5$. The Pfaffian wall calculation is type-II throughout.

5.2 The rank-one type-II Pfaffian crossing

For each simple type-II wall $\delta \in \{\delta_1, \delta_2, \delta_3\}$, the retained rank-one chart in the datum $(O2_{\mathrm{atlas}})$ has local model

$$U_\delta = T_\delta \times (-\varepsilon, \varepsilon), \qquad s_\delta(t, u) = (t, -u),$$

with wall locus $T_\delta \times \{0\}$ coinciding with the fixed locus of s_δ. The reduced normal self-Ext factor on this retained chart is the rank-one odd Pfaffian crossing

$$K_\delta^{\mathrm{cross}} = [\mathbb{R} \xrightarrow{u} \mathbb{R}], \qquad D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad \mathrm{Pf}(D_u) = u.$$

Choose an oriented tangent Pfaffian line $\mathcal{L}_\delta^{\tan}$ on T_δ with s_δ-invariant unit v_δ:

$$s_\delta(v_\delta) = v_\delta.$$

The total local Pfaffian section is

$$\mathrm{pf}_\delta = v_\delta\, u.$$

Proposition 5.2 *(Local Pfaffian sign; conditional on the retained rank-one wall chart).* For the local rank-one type-II Pfaffian crossing $(U_\delta, \mathrm{pf}_\delta)$ of §5.2,

$$s_\delta^* \,\mathrm{pf}_\delta = -\mathrm{pf}_\delta, \qquad s_\delta^*\,(\mathrm{pf}_\delta^2) = \mathrm{pf}_\delta^2.$$

Equivalently, the local Hall/Pfaffian sign is

$$\epsilon_o^{\mathrm{loc}}(s_\delta) = (-1)^{N_\delta^{\mathrm{Pf}}} = -1, \qquad N_\delta^{\mathrm{Pf}} = 1.$$

Proof. The fixed locus of $s_\delta(t, u) = (t, -u)$ is $\{u = 0\}$, and the wall coincides with the tangent fixed locus, so the determinant factor is split: $\mathcal{L}_\delta^{\tan}$ on T_δ is the tangent factor and $K_\delta^{\mathrm{cross}}$ is the normal factor. The unit v_δ is invariant by construction; the skew matrix D_u has Pfaffian u; and s_δ sends $u \mapsto -u$ while fixing v_δ. Therefore

$$s_\delta^*(v_\delta\, u) = v_\delta\,(-u) = -v_\delta\, u = -\mathrm{pf}_\delta,$$

and squaring gives the second equality. The Pfaffian rank exponent $N_\delta^{\mathrm{Pf}} = 1$ follows because the normal factor is the single odd crossing $[\mathbb{R} \to \mathbb{R}]$. The displayed sign is $(-1)^1 = -1$. □

5.3 Match with the Maass multiplier

The Maass multiplier ν_{Δ_5} of the weight-5 Igusa cusp form is the multiplier system of the section $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ of the automorphic line bundle $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on the genus-two Siegel space $\mathbf{Sp}(2,\mathbb{Z})\backslash\mathbb{H}_2$. Its values on the type-II reflections in $\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1})$ are computed in [46, Proposition 2.1] (citing Maass [76, (1.3)–(1.5)]) via the exterior-square identification $\mathbf{PSp}_4(\mathbb{Z}) \simeq \Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$:

Lemma 5.3 *(Maass multiplier on simple type-II reflections).* The Maass character of [46, Proposition 2.1], in the normalization of [76, (1.3)–(1.5)], evaluates on the simple type-II reflections as

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

and trivially on the three $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$:

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_{-2}-f_3}) = +1.$$

The character is invariant under rescaling Δ_5 by a nonzero constant: the theta-normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ and the monic $D_5 = 64^{-1}\Delta_5$ give the same multiplier values on generators.

Derivation. Each s_{δ_i} lifts under the exterior-square isomorphism $\mathbf{PSp}_4(\mathbb{Z}) \xrightarrow{\sim} \Gamma_{3,2}$ of Theorem 1.5 to a paramodular involution acting on the genus-two Siegel space; Δ_5 transforms by the Maass character on these involutions, valued -1 on each type-II generator (Maass [76, (1.3)–(1.5)], Gritsenko–Nikulin [46, Proposition 2.1]). The trivial S_3-reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$ lift to inner Weyl permutations on the level-one paramodular and act with multiplier $+1$.

Combining Proposition 5.2 and Lemma 5.3:

Theorem 5.4 *(Local orientation match on type-II generators).* Composing Proposition 5.2 and Lemma 5.3, on the three simple type-II reflections,

$$\epsilon_o^{\mathrm{loc}}(s_{\delta_i}) = -1 = \nu_{\Delta_5}(s_{\delta_i}), \qquad i = 1, 2, 3.$$

The local Pfaffian sign and the Maass multiplier coincide on generators of the type-II reflection group, relative to the retained rank-one wall charts.

The match $\epsilon_o^{\mathrm{loc}} = \nu_{\Delta_5}$ on the three type-II simple reflections is the local seed of the global orientation-character identity. Its extension to the whole type-II Weyl group requires global Hall-stack transport of the local crossing, the $(O2)$ obstruction.

5.4 Extension to the type-II Weyl group

5.4.1 Generation and a presentation

The type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ is generated by $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ (Nikulin's reflection theorem for $\Lambda^{(1,1)} \oplus \langle 2\rangle$, [46, Theorem 4.1]). The generator pairings $(\delta_i, \delta_j) = -2$ for $i \neq j$ are obtuse and equal to -2; the corresponding Coxeter exponent m_{ij} is the order of $s_{\delta_i}s_{\delta_j}$, which is infinite because the rotation $s_{\delta_i}s_{\delta_j}$ acts on the hyperbolic plane by a hyperbolic translation. Thus

$$W^{(2)}(\Lambda^{2,1}_{II}) = \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s_{\delta_i}^2 = 1\rangle$$

is the free product of three involutions, equivalently the (∞,∞,∞)-triangle reflection group on the hyperbolic plane.

5.4.2 The determinant character

The determinant character $\det : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ is the unique homomorphism with $\det(s_{\partial_i}) = -1$ on each generator. The relations $s^2_{\partial_i} = 1$ are respected since $(-1)^2 = 1$, and there are no further relations, so det extends unambiguously. The Maass multiplier $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$ is also a $\{\pm 1\}$-character that equals -1 on each generator (Lemma 5.3); it therefore equals det:

$$\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det = (\text{sign character}).$$

This is the global Maass character on the type-II Weyl group.

5.4.3 The orientation character is a $\{\pm 1\}$-character

The orientation character ϵ_O on the Hall-stack moduli is, by construction, a homomorphism $W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ provided the orientation cocycle on the half-Hilbert orbit trivializes consistently with the local rank-one crossings of §5.2 across all generators. Under that trivialisation, the global orientation character is determined by its values on $s_{\partial_1}, s_{\partial_2}, s_{\partial_3}$, each equal to -1 by Theorem 5.4. Hence

$$\epsilon_O = \det = \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II}).$$

This is the global orientation-character match relative to the retained quotient orientation and the chosen Weyl transport. Its proof in this appendix establishes the $(O2)$ wall-atlas part: the trivialisation of the orientation cocycle on the half-Hilbert orbit.

5.5 *The (O2) obstruction: half-Hilbert orbit transport*

5.5.1 Statement of (O2)

Definition 5.5 *((O2) local Pfaffian wall normal form, global form).* The obstruction $(O2)$ is the assertion that, for every retained HN height R and every retained type-II wall transport $s_\partial \in W^{(2)}(\Lambda^{2,1}_{II})$, the local rank-one crossing of §5.2, defined on the generic wall chart $U_{\partial,R}$, extends to a strictly compatible system of charts on the half-Hilbert orbit $\mathfrak{H}^{\text{half}}_R$, with the orientation cocycle on this orbit trivialized by an explicit cochain compatible with the local Pfaffian sign on each component, the boundary of each component, and the overlap of every adjacent pair of components.

5.5.2 Local computation and global transport

Theorem 5.4 proves the local sign identity on each of the three simple type-II generators by a single chart computation: *three local sign computations, one per generator*. The global statement of §5.4, $\epsilon_O = \nu_{\Delta_5}$ on the entire Weyl group, requires that these three local seeds extend to a strictly coherent system on the half-Hilbert orbit, namely the entire orbit of the half-Hilbert moduli stack under the $W^{(2)}$-action. The component, boundary, and overlap compatibility data of Definition 5.5 are *not* produced by the local rank-one crossings alone. They are the content of the retained $(O2_{\text{atlas}})$ datum transported by Theorem 7.19.

5.5.3 Two appearances of the constant 4096

The integer 4096 appears in two structures, and the equality is scalar only. On the orientation side, the half-Hilbert orbit of the type-II reflection group on the cosection-reduced self-Ext stack carries 12 binary signs, so the orientation-cocycle support carries $2^{12} = 4096$ sign assignments; this is the reading on which the $(O2_{\text{atlas}})$ transport of Appendix 7, Theorem 7.19, operates. On the scalar side, the Oberdieck–Pixton normalization

$$Z^X_{\text{OP}}(P_{\text{OP}}, Q_{\text{OP}}, T_{\text{OP}}) = -\left(\chi^{\text{OP}}_{10}\right)^{-1} = -4096\,\Delta_5(Z)^{-2}, \qquad \chi^{\text{OP}}_{10} = D_5^2 = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2,$$

exhibits $4096 = 64^2 = ([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ as the squared theta-leading coefficient, with $-4096 = (-1)\cdot 64^2$ the OP scalar branch convention times the squared theta-leading ([91, 92]; arithmetic check by the product expansion).

The two values coincide because $2^{12} = 64^2$: the half-Hilbert orbit sign count and the squared theta-leading are the same integer after orientation has been forgotten. Theorem 7.17 of Appendix 7 makes the scalar separation precise: the orientation character $\epsilon_o = \nu_{\Delta_5}$ is fixed, relative to the retained orientation and Weyl-transport data, by wall transport of the sign-count, while the OP scalar magnitude 4096 enters $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ through the squared theta-leading.

5.5.4 The retained (O2) wall-atlas route

The retained $(O2_{\mathrm{atlas}})$ datum consists of:

(O2.a) A strictly compatible system of charts $U_{\delta,R}$, $\delta \in W^{(2)}(\Lambda^{2,1}_{II})$, on the half-Hilbert orbit at height R, each presenting the rank-one type-II Pfaffian crossing of §5.2.

(O2.b) An orientation cocycle $c_o \in Z^2(W^{(2)}(\Lambda^{2,1}_{II});\mathbb{F}_2)$ on the orbit, with explicit cochain trivialisation $\eta_o \in C^1(W^{(2)}(\Lambda^{2,1}_{II});\mathbb{F}_2)$ satisfying $\delta\eta_o = c_o$, compatible with the chosen charts.

(O2.c) Strict compatibility of the trivialisation with the component decomposition, the boundary of each component, and the overlap of every adjacent pair of components — equivalently, the component, boundary, and overlap residual entries of the retained wall-atlas datum vanish with explicit null-trivialisations.

Once (O2.a)–(O2.c), the retained quotient orientation, and the chosen Weyl transport are fixed, Theorem 5.4 extends to the global Weyl-group character match $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$.

5.6 First-timelike parity split as a recognition obligation

The orientation-character match of §5.4 is the conditional *multiplier* comparison. The recognition theorem $\mathcal{P}_X \cong \mathfrak{g}_{\Delta_5}$ of Chapter 7 requires further data, namely full parity dimensions per root degree, not just signed dimensions. The first nontrivial test occurs at the imaginary simple root $\delta_{123} = \delta_1 + \delta_2 + \delta_3$:

LEMMA 5.6 *(First-timelike parity split).* The signed root multiplicity at δ_{123} is

$$\operatorname{smult}\ \mathfrak{g}_{\Delta_5,\alpha(\delta_{123})} = f(1,1) = -64,$$

and the full target parity computed from the Gritsenko–Nikulin presentation is

$$\big(\mathrm{mult}_{\bar{0}}(\delta_{123}),\ \mathrm{mult}_{\bar{1}}(\delta_{123})\big) = (29, 93), \qquad 29 - 93 = -64.$$

The decomposition of the even part is

$$29 = 2 + 3\cdot 9,$$

where $2 = \frac{1}{3}\,\mu_{\mathrm{Mob}}(1)\,3!$ is the Witt dimension of the free Lie algebra on three generators in multidegree $(1,1,1)$ and $3\cdot 9$ is the three real–isotropic mixings each contributing the eta-9 multiplicity $\tau = 9$ of the doubled isotropic ray. The odd part 93 is the absolute value of the timelike Borcherds correction $m(\delta_{123})$, with

$$m(\delta_{123}) = -\big[\,q^1r^1s^1\,\big]\big((qrs)^{-1/2}64^{-1}\Delta_5\big) = -93.$$

Proof. The signed multiplicity is $f(1,1) = -64$ by direct expansion of $\phi_{0,1}$ in the Eichler–Zagier normalization [34, §2.2]. The full parity decomposition is the Gritsenko–Nikulin presentation reading: the even part counts free Lie words in three real-root letters, the odd part counts simple imaginary timelike roots. Witt's dimension formula $W_{\text{multidegree}}(1,1,1) = \frac{1}{n}\sum_{d|\gcd}\mu_{\text{Mob}}(d)\,(n/d)!/\prod(n_i/d)!$ with $n=3$, $\gcd(1,1,1)=1$ gives $\frac{1}{3}\cdot 1\cdot 6 = 2$. The three real–isotropic mixings each contribute $\tau = 9$ on the primitive isotropic rays (the eta-9 expansion $\prod_{k\geq 1}(1-q^k)^9$ of [48, Theorem 2.1, (5.1)]); summing over the three independent isotropic directions gives $3\cdot 9 = 27$. Hence $29 = 2+27$. The odd part is $-m(\partial_{123}) = 93$ by the Gritsenko–Nikulin imaginary simple-root formula. □

Corollary 5.7 *(Signed dimension is not recognition).* A graded super vector space with parity dimensions $(\dim_{\bar{0}}, \dim_{\bar{1}}) = (29, 93)$ and zero bracket has the same signed dimension -64 as the $\mathfrak{g}_{\Delta_5,\alpha(\partial_{123})}$ component but the wrong Lie superalgebra structure. Recognition of $\mathcal{P}_X$ with $\mathfrak{g}_{\Delta_5}$ at degree ∂_{123} requires both parity dimensions 29 and 93 and the Borcherds–Kac–Moody bracket; the determinant identity alone does not imply this recognition statement.

5.6.1 Higher-degree consistency

The same Borcherds–Kac arithmetic gives higher-degree signed checks and the finite rows where full target parity is known:

- *Doubled isotropic.* For $\partial = 2a_{ij}$, $i\neq j$,

  $$(\text{smult},\ \tau,\ \text{gap}) = (10,\ 9,\ 1),$$

  the signed multiplicity 10 splits as eta-9 multiplicity 9 plus the Kac null-root residual 1 of the rank-two affine subroot system, for each of the three pairs (i,j).
- *Height-four timelike.* For the three rows

  $$C_{k,2} = a_{ij} + 2\partial_k, \qquad \{i,j,k\} = \{1,2,3\},$$

  namely coefficient vectors $(2,1,1)$, $(1,2,1)$, $(1,1,2)$,

  $$(\text{smult},\ m,\ \text{gap}) = (108,\ 90,\ 18),$$

  each with the same triple of integers. These rows are not Weyl translates of ∂_{123}; rather

  $$s_{\partial_k}(\partial_{123}) = a_{ij} + 3\partial_k = C_{k,3}.$$

  Thus $C_{k,2}$ is a signed target row. The integer $18 = 108-90$ is the signed lower-bracket residual at height four, not a full parity block.
- *Real-string exponents.* The Borcherds–Kac real-root strings through ∂_{ij}-style mixed degrees give (real-real exponent) $= 3$ and (complement-isotropic exponent) $= 5$; the timelike string through ∂_{123} has exponent 3. All three are pairwise consistent under Weyl reflection.

These integers are the degree-wise Borcherds–Kac arithmetic used in the recognition criterion.

5.7 Local Pfaffian sign computation

Let $\partial\in\{\partial_1,\partial_2,\partial_3\}\subset\Lambda^{2,1}$, with $(f_2,f_{-2}) = -1$ and $(f_3,f_3) = 2$. On a normal chart

$$U_\partial = T_\partial\times(-\epsilon,\epsilon), \qquad s_\partial(t,u) = (t,-u), \qquad \mathbb{H}_\partial\cap U_\partial = T_\partial\times\{0\},$$

choose an oriented tangent Pfaffian unit v_∂ satisfying $s_\partial^* v_\partial = v_\partial$. The rank-one normal complex is

$$K_\partial^{\text{nor}} = [\mathbb{R}\xrightarrow{u}\mathbb{R}], \qquad D_u = \begin{pmatrix} 0 & u\\ -u & 0\end{pmatrix}, \qquad \text{Pf}(D_u) = u.$$

Thus
$$\mathrm{pf}_{\partial} = v_{\partial} u, \qquad s_{\partial}^{*}\,\mathrm{pf}_{\partial} = -\,\mathrm{pf}_{\partial}, \qquad s_{\partial}^{*}\,\mathrm{pf}_{\partial}^{2} = \mathrm{pf}_{\partial}^{2}.$$
The normal Pfaffian exponent is $N_{\partial}^{\mathrm{Pf}} = 1$, hence
$$\epsilon_{o}^{\mathrm{loc}}(s_{\partial}) = (-1)^{N_{\partial}^{\mathrm{Pf}}} = -1 = \nu_{\Delta_5}(s_{\partial})$$
for the three simple type-II reflections, by the Maass-character calculation of Gritsenko–Nikulin [46, Proposition 2.1].

The $(O2_{\mathrm{atlas}})$-transport formulas of Appendix 6 extends this local computation to the global half-Hilbert orbit by component, boundary, and overlap compatibility.

5.8 O2 relative summary

- **(O2.a) Strictly compatible chart system on the half-Hilbert orbit.** The local rank-one crossing is fixed at each simple type-II generator (§5.2); strict compatibility across all elements of $W^{(2)}(\Lambda_{II}^{2,1})$ is the half-Hilbert orbit chart system retained in $(O2_{\mathrm{atlas}})$ and transported by Theorem 7.19.
- **(O2.b) Orientation cocycle trivialisation.** $c_o \in Z^2(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2)$ is the orientation cocycle; its trivializing cochain $\eta_o \in C^1(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2)$ is part of the chosen Weyl-transport datum of Theorem 7.14 and is transported in Theorem 7.19.
- **(O2.c) Component, boundary, overlap compatibility.** The residual entries $\mathfrak{o}_{\partial,R}^{\mathrm{chart}}, \mathfrak{o}_{\partial,R}^{\mathrm{wall}}, \mathfrak{o}_{\partial,R}^{\mathrm{split}}, \mathfrak{o}_{\partial,R}^{\mathrm{unit}}, \mathfrak{o}_{\partial,R}^{\mathrm{rank}}, \mathfrak{o}_{\partial,R^+R}^{\mathrm{tr}}$ of Proposition 0.205 are fixed at finite height R for the local rank-one models; the global compatibility across the half-Hilbert orbit is retained in $(O2_{\mathrm{atlas}})$ and transported by Theorem 7.19.
- $(R1)$–$(R2)$ **scalar separation.** Separated in Appendix 7, Theorem 7.17: the orientation-cocycle support on the half-Hilbert orbit has size 2^{12} (R1), and this integer is equal to 64^2, the theta-leading square on the scalar side (R2). This is not an identification of the two data.

Relative to the chosen Weyl lifts and $(O2_{\mathrm{atlas}})$, Theorems 7.14, 7.19, and 7.17 extend Theorem 5.4 from the three local generators to the global character match $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$.

6 Eight finite constructions for the Dirac–Igusa pro-object

The Dirac–Igusa pro-object
$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_{R}\big(\mathcal{A}_{X,R}^{E_3}, \mathcal{F}_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, \mathcal{H}_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}, \mathrm{Rec}_R\big)$$
of Theorem 0.252 and Definition 0.86 is the finite source datum. Its protected Pfaffian and orientation character, after the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum, together with the finite geometric Pfaffian datum (P_{fin}), of Appendix 7, give the Igusa section and Maass character,
$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5, \qquad \epsilon_o = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The primitive identification $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the separate finite compact-source recognition datum. The finite Harder–Narasimhan height R is relative to the standing hypotheses [K3×E-fin], [K3×E-atlas], [ML], [E_3-HolFA], and [K3→E-sp]. These finite data constitute the datum $\mathfrak{K}_X = (L_X, H_X, O_X, \mathsf{Desc}_X, \mathsf{Rec}_X)$ of Theorem 0.252. The charge, moduli, and hybrid data give

(L_X) and (H_X); the reduced orientation data give the local content of (O_X); the Hall correspondence object gives the primitive product; the Pfaffian–Dirac line carries the orientation character; the Koszul source coalgebra and primitive-recognition data give (Desc_X) and (Rec_X). The compact E_3-source and specialization inputs themselves are the finite table system

formal_target_charts, field_complexes, e3_operations, bv_qme, anomaly_framing, compact_support, factorization_descen

They record the formal $K3 \times E$ source before the hybrid specialization; a target current envelope, scalar Pfaffian product, or protected trace is not an entry of this source datum. The current finite E_3-source packet records this datum only by the obstruction ledger `E3_SOURCE_OBSTRUCTION_LEDGER_VERIFIED`, with `e3_source_certification=false`. The missing rows are the formal-target charts, field complexes, E_3-operations, BV-QME, anomaly, compact-support, descent, specialization, transition, and scalar-firewall rows.

Throughout, $X = S \times E$ with S a smooth complex K3 surface and E a smooth elliptic curve. $\sigma = \sigma_{s,t}$ denotes the chosen Liu product-stability datum on the Abramovich–Polishchuk heart in $\mathsf{D}^b(\mathrm{Coh}(X))$ [1,66]. The HN sector (σ, S) is fixed and the cofinal subsystem $\mathcal{R}_{\mathrm{HN}} = \{R_0 < R_1 < \cdots\}$ is the one of Definition 0.86, governed by [K3×E-fin], [K3×E-atlas], and [ML]. The stability-heart packet `certificates/moduli/stability_heart` records this retained heart as a datum and verifies `STABILITY_HEART_DATUM_VERIFIED`. It does not prove the noetherian, HN, boundedness, finite-moduli, or derived-enhancement rows used below. The Liu product-stability packet `certificates/moduli/liu_product_stability` records the source-imported construction of $\sigma_{s,t}$ and verifies `LIU_PRODUCT_STABILITY_DEFINED`: the AP heart $\mathcal{A}_E^{\mathrm{AP}}$, the linear polynomial $L_M(n) = a(M)n + b(M) + i(c(M)n + d(M))$, the weak charge Z_t, the torsion-pair tilt $\mathcal{A}_E^t$, and the central charge $Z_E^{s,t}$. It does not prove the separate AP-compatibility row, finite-type semistable substacks, derived-enhancement rows, compact Hall stages, or protected traces. The AP–Liu compatibility packet `certificates/moduli/ap_liu_compatibility` records Proposition 1.7 and verifies `AP_LIU_COMPATIBILITY_VERIFIED`: the retained heart $\mathcal{A}_X^{\mathrm{AP/Liu}}$ is the Liu torsion-pair tilt $\mathcal{A}_E^t = \langle \mathcal{T}_t, \mathcal{F}_t[1] \rangle$ of the Abramovich–Polishchuk heart $\mathcal{A}_E^{\mathrm{AP}}$. It does not prove noetherianity, HN filtrations, bounded HN types, finite-type semistable substacks, derived-enhancement rows, compact Hall stages, or protected traces. The retained-window noetherian packet `certificates/moduli/retained_window_noetherian` records the finite ACC step and verifies `RETAINED_WINDOW_NOETHERIAN_VERIFIED`: a finite subobject-closed and quotient-closed exact window with a strict length rank is noetherian. It does not prove global noetherianity of the Abramovich–Polishchuk/Liu heart, HN filtrations, bounded HN types, finite-type moduli, or derived-enhancement rows. The retained-window HN-filtration packet `certificates/moduli/retained_window_hn_filtration` records the finite exact HN step and verifies `RETAINED_WINDOW_HN_FILTRATION_VERIFIED`: every object in a supplied finite retained window has an exact filtration by semistable quotients with strictly decreasing phases. It does not prove global HN filtrations in the Abramovich–Polishchuk/Liu heart, bounded HN types, finite-type moduli, or derived-enhancement rows. The retained HN type-bound packet `certificates/moduli/retained_hn_type_bounds` records the finite active-window type bound and verifies `RETAINED_HN_TYPE_BOUNDS_VERIFIED`: the semistable factor type set is finite, HN words have length bounded by the retained length rank, and the HN type rows mirrored into `certificates/moduli/k3e_finite_moduli/hn_type_bounds.csv` are exactly the certified rows. It does not prove finite-type semistable substacks, derived-enhancement rows, compact Hall stages, or protected traces. The retained semistable-substack packet `certificates/moduli/retained_semistable_substacks` records Proposition 1.16 and verifies `RETAINED_SEMISTABLE_SUBSTACKS_VERIFIED`: each retained semistable HN factor type has a specialization-closed finite-type substack inside a bounded Quot/Postnikov ambient, with zero semistability and boundedness defect ranks. It does not prove quasi-smooth derived enhancements, universal complexes, scalar rigidifications, finite residual inertia, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained derived-enhancement packet `certificates/moduli/retained_derived_enhancements` records Proposition 1.18 and verifies `RETAINED_DERIVED_ENHANCEMENTS_VERIFIED`: each retained finite-type semistable substack has a quasi-smooth derived Artin enhancement with cotangent amplitude $[-1, 0]$, zero Tor-amplitude defect, and restricted PTVV (-1)-shifted symplectic form with zero symplectic defect. It does not prove universal complexes, scalar rigidifications, finite residual inertia, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained rigidification-inertia packet `certificates/moduli/retained_rigidification_inertia` records Proposition 1.20 and verifies `RETAINED_RIGIDIFICATION_INERTIA_VERIFIED`: the scalar $\mathbb{G}_m$ is removed on every retained derived substack, the residual inertia group is finite, and the rigidification and inertia defects vanish. It does not prove universal complexes, finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained class-bound packet `certificates/moduli/retained_class_bounds` records Proposition 1.22 and verifies `RETAINED_CLASS_BOUNDS_VERIFIED`: C_R is finite, each retained class carries finite Hilbert-polynomial labels $P_{c,i}$, a finite cohomological-amplitude interval $[a_c, b_c]$, and a finite regularity bound N_c. It does not prove universal complexes, finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained universal-complex packet `certificates/moduli/retained_universal_complexes` records Proposition 1.24 and verifies `RETAINED_UNIVERSAL_COMPLEXES_VERIFIED`: each retained scalar-rigidified finite substack carries a universal perfect complex with base-change compatibility, finite Tor amplitude, zero descent defect, and zero perfection defect. It does not prove finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained E-translation-rigidification packet `certificates/moduli/retained_e_translation_rigidifications` records Proposition 1.26 and verifies `RETAINED_E_TRANSLATION_RIGIDIFICATIONS_VERIFIED`: the elliptic-translation action is removed on every retained finite row with zero translation-rigidification defect. It does not prove finite closed inertia stratifications, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained closed-substack packet `certificates/moduli/retained_closed_substacks` records Proposition 1.28 and verifies `RETAINED_CLOSED_SUBSTACKS_VERIFIED`: each retained finite row has a finite closed cover with finite residual inertia and zero coverage and inertia defects. It does not prove closure under extensions, closure under HN factors, closure under duals, extension or flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained extension-closure packet `certificates/moduli/retained_extension_closure` records Proposition 1.30 and verifies `RETAINED_EXTENSION_CLOSURE_VERIFIED`: each supplied binary extension row has middle term in the retained finite HN word set and zero extension-closure defect. It does not construct extension stacks, two-step flag stacks, closure under HN factors, closure under duals, cosection atlases, transitions, compact Hall stages, or protected traces. The retained HN-factor-closure packet `certificates/moduli/retained_hn_factor_closure` records Proposition 1.32 and verifies `RETAINED_HN_FACTOR_CLOSURE_VERIFIED`: each HN factor occurrence of a retained finite HN word is represented by a retained class, a finite-type semistable substack, and a retained closed substack.

It is not the dual-closure packet and does not prove extension or flag stacks, subquotient closure in flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces. The retained dual-closure packet `certificates/moduli/retained_dual_closure` records Proposition 1.34 and verifies `RETAINED_DUAL_CLOSURE_VERIFIED`: the involution $s_3 \leftrightarrow s_1, s_2 \mapsto s_2$, together with HN-word reversal, preserves the retained finite HN word set with zero defect. It does not construct extension or flag stacks, subquotient closure in flag stacks, cosection atlases, transitions, compact Hall stages, or protected traces.

The ordering charge→moduli→hybrid→orientation→Hall →Dirac+Pfaffian→Koszul→recognition is forced by the mathematics: a Pfaffian line needs the orientation datum, the orientation datum needs the quotient Hall atlas, the Hall bracket needs the hybrid correspondences, and primitive recognition needs the normal-ordered Hall primitive layer.

The four obstructions are attached to the finite data: (D_0) is the cofinal-trivialization datum at every finite R, with components in the charge, moduli, hybrid, orientation, Hall, and Pfaffian data; $(O1)$ and $(O1)^+$ live in the reduced orientation data, with Coxeter coherence feeding the Pfaffian transport; $(O2)$ lives in the Pfaffian wall normal form. Primitive recognition inherits these obstruction classes and adds the exact-completion residual (R8) of Definition 0.217. The retained D0-HN datum is not the Hilbert-scheme scalar specialization. At finite stage it is the table system

degeneration_families, retained_substacks, hn_transitions, derived_enhancements, semiregularity_cosections, vanishing_cy

of Definition 7.6.

6.1 Charge window

6.1.0.1 Hypotheses. HN height $R \in \mathcal{R}_{\mathrm{HN}}$; the formal dyonic Mukai lattice Γ_X^{form} of Definition 2.1; the Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ and the cocycle B of Lemma 2.4; the active support $\Gamma_{\mathrm{act}} = \{\gamma \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}$ of the Igusa Jacobi form $\phi_{0,1}$; the cofinal subsystem $\mathcal{R}_{\mathrm{HN}}$.

6.1.0.2 Charge-window datum. A finite cusp-positive active window

$$A_R \subset \Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{gram}}, \qquad A_R \text{ downward saturated for the cofinal target-window order,}$$

together with the normal-ordered extension

$$\widehat{\Gamma}_R = \{(c, T) \mid c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\} \subset \widehat{\Gamma}_X, \qquad \Gamma_R^{\Pi} = \overline{\Pi}_X(\widehat{\Gamma}_R) \subset \Gamma_{\mathrm{gram}},$$

the lift $L : \Gamma_{\mathrm{gram}} \to \Gamma_X^{\mathrm{form}}$ of Lemma 2.8, and the finite test window Γ_R^{test} of Definition 0.249.

6.1.0.3 Construction.

(i) Compute the active support cone

$$\Gamma_{\mathrm{act}} = \{(n, l, m) \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}$$

from the Fourier expansion of the index-one weak Jacobi form $\phi_{0,1}(\tau, z) = \sum_{n,l} c(n, l) q^n r^l$ of [34], with the index substitution $f(nm, l) := c(nm, l)$ tabulated in Appendix 8 for the $N = 1$ Igusa boundary.

(ii) Set

$$A_R^\circ = \{\gamma = (n, l, m) \in \Gamma_{\mathrm{act}} \mid n + |l| + m \le R\}.$$

Form its downward saturation

$$A_R = \{\gamma' \in \Gamma_{\mathrm{act}} \mid \exists \gamma \in A_R^\circ,\ 0 < \gamma' \le \gamma,\ \gamma - \gamma' \in Q_+\},$$

in the sense of Definition 0.217.

(iii) Normal-ordered lift. For each $c \in \Gamma_R^{HN}$ compute the B-translate orbit

$$\mathcal{T}_R(c) = \Big\{ \sum_{i<j} B(c_i, c_j) \ \Big|\ c = \sum_i c_i,\ c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \le R \Big\}.$$

Set $\widehat{\Gamma}_R = \{(c, T) : c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\}$.

(iv) Finite target test window: $\Gamma_R^{\mathrm{test}} = \overline{\Pi}_X(\widehat{\Gamma}_R)$, compatibly with $R \to R'$ by inclusion.

(v) Obtain $(A_R, \widehat{\Gamma}_R, \Gamma_R^{\Pi}, \Gamma_R^{\mathrm{test}}, L)$.

6.1.0.4 Identities. A_R is finite and downward saturated; $\overline{\Pi}_X(\widehat{\Gamma}_R) = \Gamma_R^{\Pi}$ lies in Γ_{gram}; $\Gamma_R^{HN} \subset F_{\sigma,S}(R)$ is finite by [K3×E-fin]; the cocycle identity $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ is an axiom of the cocycle. At table level the finite charge-window datum would be

active_support, window_saturation, liu_hn_window, gram_map, cocycle_identities, normal_ordered_lifts, fi

The finite-test-window definition packet

`certificates/charge/finite_test_window`

verifies `FINITE_TEST_WINDOW_DEFINITION_VERIFIED`: the definition

$$\Gamma_R^{\mathrm{test}} = \overline{\Pi}_X(\widehat{\Gamma}_R) = \{\Pi_X(c) + T \mid c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\} \subset \Gamma_{\mathrm{gram}}$$

is fixed, while the HN charge rows, normal-ordered translate orbits, image-membership rows, finiteness witnesses, transition rows, compact source compatibility rows, and relation-closed target-exhaustion rows are not supplied. The relation-closed target finite-R exhaustion packet

`certificates/charge/relation_closed_target_finite_R_exhaustion`

verifies `RELATION_CLOSED_TARGET_FINITE_R_EXHAUSTION_OBSTRUCTION_VERIFIED`: the finite target-degree-to-R rows, HN charge witnesses, normal-ordered translate witnesses, symbolic tail R_N-formulae, transition rows, and zero coverage-defect row are not supplied. The target packets `wrel_degree_closure` and `wrel_tail_staircase` are imported only as target-side evidence; they do not realize any degree inside a finite Γ_R^{test}. The current packet `certificates/charge/k3e_charge_window` is empty-blocked: its verifier reports missing active-support, downward-saturation, Liu-height, charge-window, normal-ordered lift, target-window, primitive-lift, transition, and scalar-firewall rows. Thus it is not used as evidence for a finite HN charge window. The separate formal packet `certificates/charge/mukai_gram_cocycle` records only

charges, cocycle_pairs, normal_ordering, linear_trivialization, formal_relations, scalar_firewall.

It verifies Mukai–Gram integrality on supplied formal charges, the symmetric bilinear cocycle B, $B = -\delta\Pi_X$, the polarization identity $B(c,c) = 2\Pi_X(c)$, normal-ordered additivity, raw-placement non-additivity, and the impossibility of an additive linear primitive for B. It remains formal chart arithmetic and does not supply the finite charge-window rows listed above. The further formal packet `certificates/charge/formal_primitive_lift` records

universal_formula, sample_lifts, root_grading, formal_relations, scalar_firewall.

It verifies the pointwise primitive lift $Q = e_1 + nf_1,\ P = lf_1 + e_2 + mf_2$, the e_1, e_2 minor equal to 1, wedge torsion one, and the grading $\alpha(\Pi_X(Q,P)) = 2nf_2 - lf_3 + 2mf_{-2}$. It is separate from the finite charge-window primitive-lift rows: it supplies no algebraic effectivity, no finite HN boundedness, no compact Hall support, no source Hall bracket, and no Pfaffian or wall-atlas datum. The formal Gram-fibre packet `certificates/charge/gram_fibre_kernel` records

formal_charges, same_gram_sums, kernel_rows, kernel_nonadditivity, formal_relations, scalar_firewall.

It verifies that displayed charges with the same Gram label are summed in parity blocks, that $K_{\Pi,0} = \Pi_X^{-1}(0,0,0)$ is the zero Gram fibre, and that this fibre is not an additive subgroup. It also records finite rows with parity profile $(1,1)$ and signed superdimension 0, so signed dimension is not used as a replacement for parity blocks. This is formal bookkeeping: it supplies no source parity theorem, Hopf radical ideal/coideal property, or exact finite-window pushforward. The finite exactness packet `certificates/charge/fibre_pushforward_exactness` records

finite_degrees, exact_sequences, pushforward_sums, zero_fibre_radicals, formal_relations, scalar_firewall.

It verifies, over $\mathbb{Q}$, that the displayed finite Gram/parity-block sequences remain exact after fibre-summed Gram pushforward and that the zero-fibre radical is the kernel of a displayed finite pairing matrix. This is not the Hall exactness theorem: it supplies no compact source, no Hall bracket, no Hopf ideal/coideal closure, and no bar-construction commutation. The finite bar-pushforward packet `certificates/charge/bar_pushforward_commutation` records

generators, bar_words, deconcatenation, bar_differential, formal_relations, scalar_firewall.

It verifies that finite reduced tensor-coalgebra words, their deconcatenation coproduct, and displayed bar-differential products have the same Γ_{gram}-degree whether one first forms the finite bar construction and then pushes forward by the additive normal-ordered grading, or first pushes forward the letters and then forms the finite bar construction. This is not the raw quadratic Π_X-pushforward and not the chiral Koszul quasi-isomorphism. The finite bracket-pairing pushforward packet `certificates/charge/hall_pairing_pushforward_compatibility` records

basis_degrees, bracket_rows, pairing_rows, formal_relations, scalar_firewall.

It verifies that supplied bracket rows are degree-additive after normal-ordered Gram pushforward and that supplied nonzero pairing rows have total Gram degree 0. This is finite algebraic compatibility, not construction of compact Hall correspondences, primitive closure, Hopf adjointness, Frobenius cyclicity, quotient non-degeneracy, Pfaffian orientation, or protected trace. The finite parity-pushforward packet `certificates/charge/parity_pushforward` records

source_parity_rows, pushforward_parity_sums, signed_shadow, formal_relations, scalar_firewall.

It verifies that the additive normal-ordered Gram pushforward sums the even and odd finite blocks separately and that signed superdimension is derived from the two parity ranks. The zero-degree row has profile (1, 1) and signed superdimension 0, so scalar cancellation is not used as a zero-object test. This is not the source parity theorem, the Gritsenko–Nikulin/Kac target parity equality, compact source construction, primitive recognition, Pfaffian orientation, or protected trace. The finite PBW-pushforward packet `certificates/charge/pbw_pushforward` records

source_filtered_blocks, pushforward_filtered_sums, associated_graded_rows, formal_relations, scalar_firewall.

It verifies that a supplied finite PBW filtration remains filtered after additive normal-ordered Gram pushforward and that the associated-graded rows are successive filtration quotients. This is not the compact-source PBW comparison, target PBW comparison, no-extra-relations theorem, primitive recognition, Pfaffian orientation, or protected trace. The finite triangular-pushforward packet `certificates/charge/triangular_pushforward` records

source_triangular_rows, pushforward_triangular_sums, bracket_sign_rows, formal_relations, scalar_firewall.

It verifies that a supplied integer height separates positive, Cartan, and negative finite blocks, that additive normal-ordered Gram pushforward keeps the three direct-sum parts separate, and that the supplied bracket sign rows land in the triangular part determined by the pushed height. This is not exact triangular completion, target triangular completion, compact-source primitive recognition, no-extra-relations, PBW comparison, Pfaffian orientation, or protected trace.

For the first base terminal and saturation-extension target rows the finite coefficient table

window_degrees, jacobi_coefficients, discriminant_orbits, target_comparisons, transitions, scalar_firewall

in `k3e_first_relation_window_coefficients` computes $f(nm, l)$ from the theta-series expansion of $\phi_{0,1}$ and compares it with the signed dimensions of the A071 target presentation. It is target arithmetic only: it supplies neither compact source representatives nor the even/odd parity, pairing, radical, PBW, or primitive-recognition rows. The A071 target-presentation verifier then checks the finite target reference packet itself:

target_degrees, target_simple_generators, target_hall_lie_basis, target_relation_rows, target_pairing_blocks, targ

It verifies the supplied target parity rows and verifies that $C_{k,2}$, $D_k = \delta_k + 2a_{ij}$, and $2\delta_{123}$ remain signed-only rows with no target basis, pairing, radical, or PBW block. The preceding first-window target packet `wle3_target_parity` records

target_degrees, decomposition, coefficient_comparison, real_string_checks, scalar_firewall.

It verifies the table $\delta_i : 1|0$, $a_{ij} : 10|0$, $2\delta_i + \delta_j : 1|0$, $\delta_{123} : 29|93$, including the isotropic norm of a_{ij}, the real-string Serre exponent 3, and the decomposition $29 = 2 + 27$. It remains target-only. The δ_{123} presentation-split packet `delta123_presentation_split` records

delta123_degree, real_real_real_words, mixed_real_isotropic_words, odd_imaginary_generators, split_relations,

It refines the single $29|93$ row by listing the two-dimensional real-real-real quotient of T_1, T_2, T_3, the $27 = 3 \cdot 9$ mixed words $[e_k, u_{ij,r}]$, and the 93-element odd imaginary-simple fibre. It cross-checks the

W≤ 3 target packet and the type-II chamber-isotropic packet; it remains target-only and does not provide compact source representatives, source Hall brackets, pairing radicals, PBW data, Pfaffian orientations, $O2$ wall atlases, mirror discriminants, or protected traces. The target relation-closure packet `wrel_degree_closure` records

target_window_rows, closure_generators, terminal_relation_checks, downward_saturation, saturation_extension,

It checks the 35-row target degree set

$$W^+_{\mathrm{rel}} = W^{act}_{\leq 3} \cup R_4 \cup I_4 \cup C_{\leq 7} \cup \{2\delta_{123}\},$$

including the six real-Serre terminal zero rows and the three complementary-string terminal zero rows. The downward-saturation audit records the exact defect. The three degrees

$$D_k = \delta_k + 2a_{ij}, \qquad \{i, j, k\} = \{1, 2, 3\},$$

are the only nonzero proper subdegrees below the displayed rows that are absent from W^+_{rel}, and $\mathrm{smult}(D_k) = -513$. It has no compact-source representatives, source relation matrices, source pairing radical, or source PBW row. The saturation-extension row verifies that

$$W^+_{\mathrm{sat}} = W^+_{\mathrm{rel}} \cup \{D_1, D_2, D_3\}$$

has zero remaining nonzero proper-subdegree defect; the D_k rows are still signed-only in the A071 target-presentation packet. The post-saturation real-string obligation table verifies that A071's simple-generator rows force 21 terminal codomains outside W^+_{sat}, with 10 nonzero signed terminal coefficients. Hence W^+_{sat} is not the full finite relation closure. The post-saturation terminal-layer table groups these codomains into 21 distinct terminal degrees; adjoining them creates 114 nonzero subdegree-defect incidences across 26 distinct subdegrees. The terminal-layer saturation-extension table records exactly these 26 rows. Adjoining them gives an 85-row degree set with zero remaining nonzero proper-subdegree defect. This closes downward saturation for the displayed terminal layer; relation closure remains a further finite target computation. The terminal-layer saturation real-string table records the first conditional obstruction to that computation: if the 26 saturation rows are promoted to target-generator rows, the real roots force 55 terminal checks, 54 terminal codomains outside the 85-row set, and 12 missing codomains with nonzero signed multiplicity. The defect-summary table records the next downward-saturation audit: adjoining those 55 terminal codomains would create 546 nonzero proper-subdegree defect incidences across 98 distinct subdegrees. The exact defect-layer table records those 98 rows; adjoining them gives a 237-row target degree set with zero remaining nonzero proper-subdegree defect for that layer. The next real-string summary records that promoting this 98-row layer to target-generator rows forces 206 terminal checks, 182 new terminal degrees, and 24 missing nonzero terminal degrees. The exact terminal-layer table records those 206 codomains, and the exact saturation-extension table records 50 distinct nonzero proper subdegrees with 624 incidences. Adjoining both layers gives a 469-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the 50-row layer forces 96 real-string terminal checks, 90 new terminal degrees, no missing nonzero terminal degree, and a 12-row saturation extension with 44 incidences. Adjoining these rows gives a 571-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. Promoting the 12-row layer forces 20 zero terminal checks and a 6-row saturation extension with 8 incidences. Adjoining both gives a 597-row target degree set with zero remaining nonzero proper-subdegree defects for that layer; promoting the newly adjoined 6-row layer forces 10 zero terminal checks and a 6-row saturation extension with 18 incidences. Adjoining both gives a 613-row target degree set with zero remaining nonzero proper-subdegree defects for that layer; promoting the next newly adjoined 6-row layer forces 10 zero terminal checks and a 6-row saturation extension with 8 incidences. Adjoining both gives a 629-row target degree set with zero remaining nonzero proper-subdegree defects for that layer; promoting the next newly adjoined 6-row layer forces 10 zero terminal checks and a 6-row saturation extension with 18 incidences. Adjoining both gives a 645-row target degree set with zero remaining nonzero proper-subdegree defects for that layer. The symbolic `wrel_tail_staircase` certificate proves

$$A_N \Longrightarrow B_N, \qquad B_N \Longrightarrow A_{N+2}$$

for every even $N \geq 14$, and its finite-anchor table checks A_{14} inside the `wrel_degree_closure` packet. Thus the target-degree audit has an infinite six-row tail. Its `unbounded_tail.csv` table records a strictly increasing coordinate subsequence, so a compact-source proof must use finite presentation data rather than finite target-degree enumeration.

6.1.0.5 Obstruction classes. None; this part is purely lattice combinatorics from the imported Gritsenko–Nikulin Borcherds product [8, 9, 46–48]. Downward saturation is the input requirement of (R0) in Definition 0.217. The 38-row extension discharges only the target-degree saturation condition; it does not supply the remaining real-string terminal codomains, compact-source labels, target presentation blocks, or source comparison maps required by (R0)–(R8). The new table

`post_saturation_terminal_layer` and `terminal_layer_saturation_extension` and `terminal_layer_saturati`

record the first terminal layer, the finite saturation extension forced by its proper-subdegree defects, the next real-string terminal obligations, the next induced saturation defect, and the exact finite layer that closes that downward-saturation audit, together with the associated real-string summary forced by that finite layer.

6.2 Finite moduli

6.2.0.1 Hypotheses. The retained heart $\mathcal{A}_X^{\mathrm{AP/Liu}}$ of Definition 1.8; the active window A_R and lift L of the charge window; the retained noetherian window $\mathcal{A}_{X,R}^{\mathrm{ret}}$ of Definition 1.9; the retained finite Harder–Narasimhan filtration datum of Definition 1.11; the retained bounded HN type datum of Definition 1.13; Liu's product stability condition $\sigma_{s,t}$ of Definition 1.6; the standing hypotheses [K3×E-fin] and [K3×E-atlas].

6.2.0.2 Finite-moduli datum. A finite retained Liu–Hilbert type set C_R, whose elements are tuples

$$c = (\widehat{c}, [a_c, b_c], (P_{c,i}), N_c),$$

and a lift $\ell_R : \Gamma_R^{\Pi} \to C_R$ with $\Pi_X(\widehat{c}_{\ell_R(\gamma)}) = \gamma$; the family of finite-type quasi-smooth derived Artin substacks

$$\mathfrak{M}_{R,c}^{ss} = \mathfrak{M}_{\sigma_{s,t}}^{ss}(X, c)_{\leq R} \subset \mathrm{Perf}(X), \qquad c \in C_R,$$

of $\sigma_{s,t}$-semistable objects of retained numerical class $\widehat{c}$, fixed cohomological amplitude $[a_c, b_c]$, fixed Hilbert polynomials $(P_{c,i})$, regularity bound N_c, and HN height $\leq R$; the scalar-rigidified truncation $\mathfrak{M}_{R,c}^{\mathrm{sc}}$; the universal perfect complex $\mathcal{A}_{R,c}$ on $X \times \mathfrak{M}_{R,c}^{\mathrm{sc}}$; the finite stratification by retained closed substacks of finite residual inertia.

6.2.0.3 Construction.

(i) Form C_R from the lifts $L\gamma$, $\gamma \in A_R$, by choosing the amplitude, Hilbert-polynomial, and regularity bounds above. Close C_R under retained extension sums, subobjects, quotients, duals, and intermediate flag classes.

(ii) For each $c \in C_R$ cut the substack of $\sigma_{s,t}$-semistable objects of retained HN type c by HN height $\leq R$:

$$\mathfrak{M}^{ss}_{R,c} = \big\{[A] \in \mathrm{Perf}(X) \mid \sigma_{s,t}\text{-semistable},\ \mathrm{hn}(A) \leq R,\ \nu(A) = \widehat{c},\ A \text{ has HN type } ([a_c, b_c], (P_{c,i}), N_c)\big\}.$$

The retained substack is required to be specialization-closed in the bounded Quot/Postnikov ambient stack.

(iii) Verify finite-typeness: [K3×E-fin] gives bounded Quot/Postnikov charts for the retained HN types. Liu product stability alone is not the finiteness assertion. PTVV [95] on the trivial-canonical threefold X gives the (-1)-shifted symplectic structure on $\mathrm{Perf}(X)$, restricted to the retained semistable sector.

(iv) Quasi-smoothness: the universal perfect complex $\mathcal{A}_{R,c}$ supplies a relative cotangent complex of bounded Tor-amplitude on the retained charts. Tangent at A is $T_A\mathfrak{M}^{ss}_{R,c} \simeq \mathrm{RHom}_X(A, A)[1]$.

(v) Scalar rigidification: form $\mathfrak{M}^{\mathrm{sc}}_{R,c}$ by killing the scalar $\mathbb{G}_m$-automorphism of c-stable factors.

(vi) Two-step flag stack $\mathfrak{F}^{(2)}_{R,c,c',c''}$ and extension stack $\mathfrak{E}_{R,c,c'}$ are the retained closed flag-Quot/filtration stacks over the finite HN-type set, with induced subobject and quotient HN types. A raw stack of exact triangles is only formal until this compactified model, or an explicitly admissible compact-support $q_!$, is fixed. In the compactified model the stacks are finite type and quasi-smooth in the retained sector by Proposition 0.149.

(vii) Obtain $(C_R, \ell_R, \{\mathfrak{M}^{ss}_{R,c}\}, \{\mathfrak{E}_{R,c,c'}\}, \{\mathfrak{F}^{(2)}_{R,c,c',c''}\})$ with universal perfect complexes and the finite stratification.

6.2.0.4 Identities. $\mathfrak{M}^{ss}_{R,c}$ is a finite-type quasi-smooth derived Artin substack of $\mathrm{Perf}(X)$; the scalar rigidification leaves only the finite residual inertia group $H_{R,c}$ recorded in Proposition 1.20; PTVV (-1)-shifted symplectic structure restricts.

6.2.0.5 Obstruction classes. [K3×E-fin] is invoked here in its product-boundedness form and supplies the finite-type assertion not in the cited literature for the K3×E threefold; it is part of the standing datum. [K3×E-atlas] supplies the quasi-smooth d-critical charts. These enter (D_0) at the level of the finite Hall atlas of Theorem 0.252. The finite proof tables are

hn_type_bounds, semistable_substacks, derived_enhancements, universal_complexes, scalar_rigidifications, stra

They require zero boundedness, Tor-amplitude, symplectic, descent, rigidification, inertia, subquotient-closure, cosection, orientation, composition, and $R^1\varprojlim$ defects. The $R^1\varprojlim$ rows for retained moduli cohomology are the image-stabilization rows of Definition 0.247. Liu stability by itself, the formal charge window, a target window, a Hilbert-scheme scalar test, or a Pfaffian product is not a finite-moduli construction. The current finite-moduli packet records the HN-bound, semistable-substack, derived-enhancement, and rigidification-inertia rows, together with the finite class-bound and universal-complex rows, the E-translation-rigidification rows, and the retained closed-substack rows, the retained extension-closure rows, and the retained HN-factor-closure rows, and the retained dual-closure rows. The packet verifies the remaining obstruction ledger `MODULI_OBSTRUCTION_LEDGER_VERIFIED`, with `moduli_certification=false`. It records the missing flag stacks, d-critical/cosection atlas, transitions, and scalar firewalls; it is not a construction of the finite Hall stage. The separate transition-geometry packet `certificates/moduli/finite_moduli_transition_geometry` verifies `FINITE_MODULI_TRANSITION_GEOMETRY_OBSTRUCTION_VERIFIED`: the object-stack, extension-stack, and two-step flag-stack transition morphisms, the proper-or-closed component rows, the finite-type source-target rows, the closed image or graph rows, and the base-change, retained-locus, composition, orientation-domain, and vanishing-cycle-domain zero-defect rows are not yet supplied. The moduli-cohomology $R^1 \varprojlim$ packet

`certificates/moduli/moduli_cohomology_lim1_vanishing`

verifies `MODULI_COHOMOLOGY_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: coefficient-system rows, cohomology tower rows, cohomology transition maps, functoriality rows, image-stabilization witnesses, Mittag–Leffler status rows, and zero $R^1 \varprojlim$ rows are not yet supplied.

6.3 Hybrid wrapped factorization carrier

6.3.0.1 Hypotheses. The finite moduli stacks $\{\mathfrak{M}^{\mathrm{sc}}_{R,c}\}$ of the finite moduli datum; the projection-finite/wrapped split by elliptic degree $b^{\mathrm{geom}}_R : \Gamma^{HN}_{\sigma,S}(R) \to \mathbb{Z}_{\geq 0}$ of Definition 0.10; a fixed origin $o_E \in E$; the Abramovich–Polishchuk/Liu wrapped anchor candidate.

6.3.0.2 Hybrid factorization datum. The finite hybrid local/wrapped factorization datum

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} = (C_{\sigma,S,\leq R}, \mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E), b^{\mathrm{geom}}_R, \mathcal{F}^{\mathrm{loc}}_R, \mathcal{G}^{\mathrm{wr}}_R, \mathfrak{E}^{\mathrm{loc}}_R, \mathfrak{E}^{\mathrm{hyb}}_R, \mathfrak{E}^{\mathrm{wr}}_R, \mathfrak{Flag}^{\mathrm{hyb}}_R, \mathsf{Corr}^{E,\mathrm{hyb}}_R, \lambda_R, Q_{E,R}, \theta^Q_R, o_R, I^{\mathrm{prot}}_R)$$

of Definition 0.69, together with the eight-word two-step binary flag atlas $\{w \in \{\mathrm{L},\mathrm{W}\}^3\}$, and the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$. The finite proof tables are

hybrid_ran_prestack_definition, local_stratum_bo_prestack_definition, geometric_elliptic_degree_map, geometric_

They require the hybrid prestack definition, the $b = 0$ local subprestack, the $b > 0$ wrapped subprestack, a populated geometric degree map, zero elliptic-degree additivity, geometric/Borcherds degree separation, positive-degree projection, source/target anchor memory before quotient, determinant anchors, determinant-anchor weights, the $\chi = 0$ determinant-anchor degeneracy, the degree-zero Jordan–Hoelder extra-anchor datum, support-locality, anchor-loss, Thom–Sebastiani, admissibility, associativity, pentagon, higher-coloured, quotient, integration, and $R^1 \varprojlim$ defects. Ordinary Ran locality, determinant anchors alone, quotient-first products, Fock traces, Borcherds s-degree alone, and scalar Pfaffian products are not hybrid-carrier data. The geometric elliptic-degree map packet

`certificates/hybrid/geometric_elliptic_degree_map`

verifies `GEOMETRIC_ELLIPTIC_DEGREE_MAP_OBSTRUCTION_VERIFIED`: the definition

$$(p_E)_* \operatorname{cyc}_1(\gamma) = b^{\mathrm{geom}}_R(\gamma)[E]$$

is fixed, while retained HN domain rows, one-cycle class rows, pushforward rows, local/wrapped split rows, transition rows, and row-202 additivity rows are not supplied. The geometric elliptic-degree additivity packet

`certificates/hybrid/geometric_elliptic_degree_additivity`

verifies `GEOMETRIC_ELLIPTIC_DEGREE_ADDITIVITY_OBSTRUCTION_VERIFIED`: Definition 0.11 and Proposition 0.12 give the proved finite-stage criterion; retained extension-pair rows, one-cycle additivity rows, pushed-forward additivity rows, integer degree additivity rows, and transition rows are still absent. The geometric/Borcherds separation packet

`certificates/hybrid/geometric_borcherds_degree_separation`

verifies `GEOMETRIC_BORCHERDS_DEGREE_SEPARATION_OBSTRUCTION_VERIFIED`: the support degree b_R^{geom} and the trace coordinate $m_R^{\mathrm{Bch}} = \mathrm{pr}_3\, \overline{\Pi}_X$ are distinct structure maps; branch-comparison rows are still absent. The positive elliptic-degree projection packet

`certificates/hybrid/positive_elliptic_degree_projection`

verifies `POSITIVE_ELLIPTIC_DEGREE_PROJECTION_VERIFIED`: effective one-cycles with positive elliptic degree project onto all of E; retained object support-realization rows are still absent. The hybrid Ran prestack definition packet

`certificates/hybrid/hybrid_ran_prestack_definition`

verifies `HYBRID_RAN_PRESTACK_DEFINITION_VERIFIED`: the carrier is defined as the prestack $\mathrm{Ran}^{\mathrm{hyb}}_{R,\mathrm{pre}}(E)$; component, descent, stackification, and transition rows are still absent. The local $b = 0$ stratum packet

`certificates/hybrid/local_stratum_b0_prestack_definition`

verifies `LOCAL_STRATUM_B0_PRESTACK_DEFINITION_VERIFIED`: $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ is the $J = \varnothing$ subprestack with local colours in Γ_R^{loc}; closed-configuration, local sheaf, descent, and transition rows are still absent. The wrapped $b > 0$ stratum packet

`certificates/hybrid/wrapped_stratum_bpositive_prestack_definition`

verifies `WRAPPED_STRATUM_BPOSITIVE_PRESTACK_DEFINITION_VERIFIED`: $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$ is the $I = \varnothing$ subprestack with wrapped colours in Γ_R^{wr} and anchor-retaining wrapped families; the E-equivariant prequotient definition is separated in the next packet, while anchor, support-realization, descent, and transition rows are still absent. The E-equivariant wrapped prequotient packet

`certificates/hybrid/equivariant_wrapped_prequotient_definition`

verifies `E_EQUIVARIANT_WRAPPED_PREQUOTIENT_DEFINITION_VERIFIED`: $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ is defined before the reduced E-quotient, with the E-translation action and origin 0_E; finite type is separated in the next packet, while properness, anchor population, anchor losslessness, quotient descent, aggregate hybrid-carrier population, and transition rows are still absent. The wrapped-prequotient finite-type packet

`certificates/hybrid/wrapped_prequotient_finite_type` verifies `WRAPPED_PREQUOTIENT_FINITE_TYPE_VERIFIED`: Proposition 0.8 proves finite type for $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ at fixed R from the retained finite class set, finite-type semistable substacks, and scalar rigidification; properness, anchor population, quotient descent, aggregate hybrid-carrier population, and transition rows are still absent. The local/local correspondence packet

`certificates/hybrid/local_local_correspondence_definition`

verifies `LOCAL_LOCAL_CORRESPONDENCE_DEFINITION_VERIFIED`: $\mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K}$ is the ordered projection-finite local/local diagram; extension-stack, source/target-map, descent, properness, Thom–Sebastiani, collision, and transition rows are still absent from that definition packet. The local/local target-properness packet

`certificates/hybrid/local_local_extension_properness`

verifies `LOCAL_LOCAL_TARGET_PROPERNESS_VERIFIED`: Proposition 0.20 realizes the LL extension stack as a closed relative Quot locus and proves properness of the target map q^{LL}. It does not assert properness of p^{LL}, and descent, collision compatibility, Thom–Sebastiani transport, transition rows, and aggregate hybrid-carrier population are still absent. The mixed local/wrapped correspondence packet

`certificates/hybrid/mixed_local_wrapped_correspondence_definition`

verifies `MIXED_LOCAL_WRAPPED_CORRESPONDENCE_DEFINITION_VERIFIED`: $\mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I}$ and $\mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I}$ are the one-sided unreduced mixed diagrams; extension-stack, source/target-map, populated anchor-memory, admissibility, base-change, projection-formula, Thom–Sebastiani, and transition rows are still absent from that definition packet. The mixed local/wrapped target-admissibility packet

`certificates/hybrid/mixed_local_wrapped_extension_admissibility`

verifies `MIXED_LOCAL_WRAPPED_ADMISSIBILITY_VERIFIED`: Proposition 0.22 realizes the LW and WL extension stacks as closed relative Quot loci over the wrapped target prequotient and proves properness of q^{LW} and q^{WL}. It does not assert properness of p^{LW} or p^{WL}, and compact-support pushforward, populated anchor-memory rows, base change, projection formula, Thom–Sebastiani transport, quotient descent, transition rows, and aggregate hybrid-carrier population are still absent. The wrapped/wrapped correspondence packet

`certificates/hybrid/wrapped_wrapped_correspondence_definition`

verifies `WRAPPED_WRAPPED_CORRESPONDENCE_DEFINITION_VERIFIED`: $\mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R}$ is the ordered unreduced wrapped/wrapped diagram; extension-stack, source/target-map, populated anchor-memory, admissibility, Thom–Sebastiani, symmetric-descent, and transition rows are still absent from that definition packet. The wrapped/wrapped target-admissibility packet

`certificates/hybrid/wrapped_wrapped_extension_admissibility`

verifies `WRAPPED_WRAPPED_ADMISSIBILITY_VERIFIED`: Proposition 0.24 realizes the WW extension stack as a closed relative Quot locus over the wrapped target prequotient and proves properness of q^{WW}. It does not assert properness of p^{WW}, and compact-support pushforward, populated anchor-memory rows, symmetric descent, Thom–Sebastiani transport, quotient descent, transition rows, and aggregate hybrid-carrier population are still absent. The compact-support exceptional pushforward model packet

`certificates/hybrid/compact_support_exceptional_pushforward_model` verifies COMPACT_SUPPORT_EXCEPTIONAL_PUSHFORWARD_MODEL_DEFINED: Definition 0.25 defines a chosen compactification $\mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_f \to Z$ and the formula $f_!^{\mathrm{cs}} K = \overline{f}_* j_{f,!} K$, with the proper-map specialization $f_!^{\mathrm{cs}} = f_*$. It does not prove choice independence, base change, projection formula, Thom–Sebastiani transport, symmetric descent, quotient descent, transition compatibility, associativity, or aggregate hybrid-carrier population. The mixed-correspondence base-change packet

`certificates/hybrid/mixed_correspondence_base_change`

verifies MIXED_CORRESPONDENCE_BASE_CHANGE_VERIFIED: Proposition 0.26 proves the Cartesian compact-support base-change isomorphisms for the LW and WL target maps. It does not prove compactification-choice independence, projection formula, Thom–Sebastiani transport, quotient descent, transition compatibility, associativity, or aggregate hybrid-carrier population. The mixed-correspondence projection-formula packet

`certificates/hybrid/mixed_correspondence_projection_formula`

verifies MIXED_CORRESPONDENCE_PROJECTION_FORMULA_VERIFIED: Proposition 0.27 proves the compact-support projection formula for the LW and WL target maps with retained finite-Tor target coefficients. It does not prove compactification-choice independence, Thom–Sebastiani transport, quotient descent, transition compatibility, associativity, or aggregate hybrid-carrier population. The mixed-correspondence Thom–Sebastiani packet

`certificates/hybrid/mixed_correspondence_thom_sebastiani`

verifies MIXED_THOM_SEBASTIANI_TRANSPORT_CONDITIONAL_VERIFIED: Proposition 0.28 proves the binary LW/WL reduced Thom–Sebastiani transport from supplied additive d-critical charts and zero-defect orientation transport. It does not prove $(O1)$, quotient descent, transition compatibility, two-step flag associativity, or aggregate hybrid-carrier population. The eight-word binary vocabulary packet

`certificates/hybrid/eight_word_binary_vocabulary`

verifies EIGHT_WORD_BINARY_VOCABULARY_DEFINED: Definition 0.29 fixes the alphabet {L, W}, the binary type product, the eight length-three words, and their two parenthesization symbols. It does not construct the two-step flag stacks or prove associativity. The eight-word two-step flag-stack packet

`certificates/hybrid/eight_word_two_step_flag_stacks`

verifies EIGHT_WORD_TWO_STEP_FLAG_STACKS_CONSTRUCTED: Proposition 0.30 constructs the retained two-step flag stacks and the two comparison maps for all eight words. It does not prove associativity, pentagon coherence, quotient descent, transition compatibility, or aggregate population. The LLL associativity packet

`certificates/hybrid/lll_associativity`

verifies LLL_ASSOCIATIVITY_CONDITIONAL_VERIFIED: Proposition 0.31 proves the pure local associativity defect vanishes on the LLL two-step flag stack, under the stated local base-change, projection-formula, and reduced Thom–Sebastiani coefficient rows. It does not prove the seven mixed or wrapped word associativity rows or the four-input pentagon. The LLW associativity packet

`certificates/hybrid/llw_associativity`

verifies `LLW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.32 proves the LLW associativity defect vanishes on the local/local–mixed two-step flag stack, under the stated local and mixed functorial coefficient rows. It does not prove the six remaining word associativity rows or the four-input pentagon. The LWL associativity packet

`certificates/hybrid/lwl_associativity`

verifies `LWL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.33 proves the LWL associativity defect vanishes on the ordered mixed–mixed two-step flag stack, under the stated mixed functorial coefficient rows. It does not prove the five remaining word associativity rows or the four-input pentagon. The WLL associativity packet

`certificates/hybrid/wll_associativity`

verifies `WLL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.34 proves the WLL associativity defect vanishes on the mixed–local/local two-step flag stack, under the stated local and mixed functorial coefficient rows. It does not prove the four WW-dependent word associativity rows or the four-input pentagon. The LWW associativity packet

`certificates/hybrid/lww_associativity`

verifies `LWW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.35 proves the LWW associativity defect vanishes on the mixed–wrapped/wrapped two-step flag stack, under the stated mixed functorial rows and wrapped/wrapped functorial hypotheses. It does not prove the three remaining WW-dependent word associativity rows or the four-input pentagon. The WLW associativity packet

`certificates/hybrid/wlw_associativity`

verifies `WLW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.36 proves the WLW associativity defect vanishes on the mixed–wrapped/wrapped two-step flag stack, under the stated mixed functorial rows and wrapped/wrapped functorial hypotheses. It does not prove the two remaining WW-dependent word associativity rows or the four-input pentagon. The WWL associativity packet

`certificates/hybrid/wwl_associativity`

verifies `WWL_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.37 proves the WWL associativity defect vanishes on the wrapped/wrapped–mixed two-step flag stack, under the stated mixed functorial rows and wrapped/wrapped functorial hypotheses. It does not prove the remaining WWW word associativity row or the four-input pentagon. The WWW associativity packet

`certificates/hybrid/www_associativity`

verifies `WWW_ASSOCIATIVITY_CONDITIONAL_VERIFIED`: Proposition 0.38 proves the WWW associativity defect vanishes on the wrapped/wrapped two-step flag stack, under the stated wrapped/wrapped functorial hypotheses. The eight length-three wordwise associativity defects are then zero under the stated functorial rows. The four-input pentagon remains separate. The four-input pentagon packet

`certificates/hybrid/four_input_pentagon_coherence` verifies `FOUR_INPUT_PENTAGON_CONDITIONAL_VERIFIED`: Proposition 0.39 constructs the retained four-step flag stack for every word in $\{\mathrm{L}, \mathrm{W}\}^4$ and proves the Mac Lane pentagon for the binary hybrid product on that common refinement, under the stated functorial hypotheses. It does not define the higher-coloured tree category or prove quotient descent, transition compatibility, or aggregate population. The higher-coloured tree-category packet

`certificates/hybrid/higher_coloured_tree_category_definition`

verifies `HIGHER_COLOURED_TREE_CATEGORY_DEFINED`: Definition 0.40 defines the nonunital higher-coloured hybrid tree category and the six components of $\mathfrak{o}_R^{\mathrm{col}}$. It does not prove $\mathfrak{o}_R^{\mathrm{col}} = 0$, unit compatibility, symmetric descent, wrapped order conventions, quotient descent, transition compatibility, or aggregate population. The higher-coloured residual criterion packet

`certificates/hybrid/higher_coloured_residual_vanishing_criterion`

verifies `HIGHER_COLOURED_RESIDUAL_CONDITIONAL_CRITERION_VERIFIED`: Proposition 0.41 proves the finite-stage implication from the six named component rows to $\mathfrak{o}_R^{\mathrm{col}} = 0$. The tree component is discharged by the eight associativity rows and the four-input pentagon. The unit and symmetric-relabelling rows are supplied by separate packets. The full non-tree tuple remains incomplete: refinement, ordered-chart descent, overlap, quotient-descent, transition, and aggregate population rows are not supplied. The hybrid unit packet

`certificates/hybrid/hybrid_unit_object_and_vacuum_correspondences`

verifies `HYBRID_UNIT_OBJECT_AND_VACUUM_CORRESPONDENCES_DEFINED`: Definition 0.42 defines the local zero unit object and the four split vacuum correspondences for local and wrapped inputs. It does not prove the local or wrapped unit identity laws, quotient-descent, transition compatibility, or aggregate population. The local unit compatibility packet

`certificates/hybrid/local_unit_compatibility`

verifies `LOCAL_UNIT_COMPATIBILITY_VERIFIED`: Proposition 0.43 proves that the left and right LL split vacuum correspondences act as identity pull-push maps on local coefficients. It does not prove wrapped unit compatibility, quotient-descent, transition compatibility, or aggregate population. The wrapped unit compatibility packet

`certificates/hybrid/wrapped_unit_compatibility`

verifies `WRAPPED_UNIT_COMPATIBILITY_VERIFIED`: Proposition 0.44 proves that the left LW and right WL split vacuum correspondences act as identity pull-push maps on wrapped coefficients, with wrapped anchor memory unchanged before the reduced E-quotient. It does not prove quotient-descent, transition compatibility, or aggregate population. The local configuration symmetric descent packet

`certificates/hybrid/local_configuration_symmetric_descent`

verifies `LOCAL_CONFIGURATION_SYMMETRIC_DESCENT_VERIFIED`: Proposition 0.45 proves finite symmetric-group descent for projection-finite closed local configuration charts and their reduced coefficients before the reduced E-quotient. It does not prove collision compatibility, closed-chart refinement, ordered-chart descent to the symmetric finite-colour colimit, quotient-descent, transition compatibility, or aggregate population. The wrapped insertion order convention packet

`certificates/hybrid/wrapped_insertion_order_conventions`

verifies `WRAPPED_INSERTION_ORDER_CONVENTIONS_VERIFIED`: Proposition 0.46 fixes the planar wrapped source order, the WW quotient/subobject convention, and the order preservation of the eight two-step flag comparison maps before the reduced E-quotient. It does not prove unordered wrapped-pair symmetric descent, ordered-chart descent, overlap compatibility, quotient-descent, transition compatibility, or aggregate population. The ordered two-sided mixed correspondence packet

`certificates/hybrid/two_sided_mixed_correspondence_definition`

verifies `TWO_SIDED_MIXED_CORRESPONDENCE_DEFINITION_VERIFIED`: $\mathfrak{C}^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+}$ is the ordered local–wrapped–local diagram; flag-stack, source/target-map, populated anchor-memory, admissibility, base-change, projection-formula, Thom–Sebastiani, and transition rows are still absent. The source/target anchor-memory packet

`certificates/hybrid/source_target_anchor_memory_definition`

verifies `SOURCE_TARGET_ANCHOR_MEMORY_DEFINITION_VERIFIED`: the LW, WL, WW, and LWL unreduced correspondence types carry ordered source and target anchor-memory maps before the E-quotient. The determinant anchor is constructed by the next packet and its translation weight by the following packet. The $\chi = 0$ degeneracy is isolated by the chi-zero packet; the degree-zero Jordan–Hoelder extra-anchor datum is recorded by the extra-anchor-data packet; anchor-residual, full losslessness, quotient-descent, stabilizer linearization, populated finite-stack, and transition rows are still absent from the source/target-memory packet. The determinant-anchor construction packet

`certificates/hybrid/determinant_anchor_construction`

verifies `DETERMINANT_ANCHOR_CONSTRUCTION_VERIFIED`: Definition 4.7 constructs $\lambda^{\det}_{\eta,R}$ as the normalized determinant of cohomology, checks that it lands in $\mathrm{Pic}^0(E) \simeq E$, and records determinant-functor base-change; unit-weight, $\chi = 0$ degeneracy, Jordan–Hoelder extra-anchor data, anchor-residual, losslessness, quotient-descent, populated wrapped-prequotient, and transition rows are still absent. The determinant-anchor translation-weight packet

`certificates/hybrid/determinant_anchor_translation_weight`

verifies `DETERMINANT_ANCHOR_TRANSLATION_WEIGHT_VERIFIED` and computes $\lambda^{\det}_{\eta,R}(a \cdot \mathcal{F}) = \lambda^{\det}_{\eta,R}(\mathcal{F}) + \chi_\eta a$. The determinant-anchor chi-zero degeneracy packet

`certificates/hybrid/determinant_anchor_chi_zero_degeneracy`

verifies `DETERMINANT_ANCHOR_CHI_ZERO_DEGENERACY_VERIFIED`: when $\chi_\eta = 0$, the determinant anchor is constant on the full retained E-translation orbit and has rank-one separation defect. The determinant-anchor extra-anchor-data packet

`certificates/hybrid/determinant_anchor_extra_anchor_data` verifies `DETERMINANT_ANCHOR_EXTRA_ANCHOR_DATA_VERIFIED`: the degree-zero Jordan–Hoelder divisor refines the determinant anchor and separates the rank-two pair $O_E^{\oplus 2}$ and $L \oplus L^{-1}$, $L \neq O_E$. Losslessness, anchor-residual, quotient-descent, stabilizer linearization, populated wrapped-prequotient, and transition rows are still absent. The quotient-after-correspondence pseudofunctor definition packet

`certificates/hybrid/quotient_after_correspondence_pseudofunctor_definition`

verifies `QUOTIENT_AFTER_CORRESPONDENCE_PSEUDOFUNCTOR_DEFINED`: Definition 0.50 defines the object assignment $Y \mapsto [Y/E]$, the generating-arrow assignment $(Y_0 \leftarrow \mathfrak{E}_e \rightarrow Y_1) \mapsto ([Y_0/E] \leftarrow [\mathfrak{E}_e/E] \rightarrow [Y_1/E])$, the coefficient-descent interface, and the quotient residual components. The composition component is supplied by

`certificates/hybrid/quotient_after_correspondence_composition`.

Proposition 0.51 proves preservation of finite-height composition of retained spans. The base-change square component is supplied by

`certificates/hybrid/quotient_after_correspondence_base_change`.

Proposition 0.52 proves preservation of retained finite-height cartesian squares after quotient. The Thom–Sebastiani descent component is supplied by

`certificates/hybrid/quotient_after_correspondence_thom_sebastiani`.

Proposition 0.53 proves descent of equivariant reduced Thom–Sebastiani transport through the quotient correspondence. The quotient-first exclusion component is supplied by

`certificates/hybrid/quotient_after_correspondence_quotient_first_exclusion`.

Proposition 0.54 proves that every reduced generating arrow used by $Q^{\mathrm{corr}}_{E,R}$ is the quotient of a preformed unreduced E-equivariant correspondence. The Borel–Moore chain functor component is supplied by

`certificates/hybrid/quotient_after_correspondence_bm_chain_functor`.

Proposition 0.55 constructs $Q_{E,R}$ on quotient-presented Borel–Moore chain objects. The operation-comparison maps are supplied by

`certificates/hybrid/quotient_after_correspondence_theta_mu_comparison`.

Proposition 0.56 constructs $\theta^{Q}_{\mu,R}$ for quotient-presented operation rows. The quotient two-step flag associativity component is supplied by Proposition 0.57. The quotient product-coproduct compatibility component is supplied by Proposition 0.58. The quotient primitive-projection compatibility component is supplied by Proposition 0.59. The HN-transition compatibility of $Q_{E,R}$ is supplied by Proposition 0.60. Compact-support pushforward base change, protected-integration transition compatibility, and aggregate population are not supplied. Hall product, Hall coproduct, and primitive-subspace transition preservation are not supplied by row 256. The definition of the protected integration map is supplied separately by Proposition 0.62. Its product compatibility is supplied separately by Proposition 0.63. Its coproduct compatibility is supplied separately by Proposition 0.64. Its primitive-projection compatibility is supplied separately by Proposition 0.65. Its Gram degree is supplied separately by Proposition 0.66. Its normal-ordered label comparison is supplied separately by Proposition 0.67.

6.3.0.3 Construction.

(i) *Elliptic-degree split.* Decompose $\Gamma^{HN}_{\sigma,S}(R) = \Gamma^{\mathrm{loc}}_R \sqcup \Gamma^{\mathrm{wr}}_R$ where $\Gamma^{\mathrm{loc}}_R = \{b^{\mathrm{geom}}_R = 0\}$ and $\Gamma^{\mathrm{wr}}_R = \{b^{\mathrm{geom}}_R > 0\}$. Supply the datum of Definition 0.11; then Proposition 0.12 gives additivity of b^{geom}_R on retained extensions remaining in the quotient. Keep this split separate from m^{Bch}_R as in Proposition 0.14.

(ii) *Local carrier.* Form $\mathrm{Ran}^{\mathrm{loc}}_{R,\mathrm{pre}}(E)$ by imposing $J = \varnothing$ in the hybrid prestack and allowing only colour maps $\alpha : I \to \Gamma^{\mathrm{loc}}_R$.

(iii) *Wrapped carrier.* Form $\mathrm{Ran}^{\mathrm{wr}}_{R,\mathrm{pre}}(E)$ by imposing $I = \varnothing$ in the hybrid prestack and allowing only colour maps $\eta : J \to \Gamma^{\mathrm{wr}}_R$, with wrapped prequotient families and their anchors retained.

(iv) *Projection-finite local sector.* For $\alpha \in \Gamma^{\mathrm{loc}}_R$ and finite index set I, take the closed-support incidence stack over E^I:

$$\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} = \left\{(A, \mathbf{e}) \in \mathfrak{M}^{\mathrm{sc}}_{R,L\alpha} \times E^I \;\middle|\; p_E(\operatorname{Supp} A) \subset \bigcup_{i\in I} S \times \{e_i\}\right\}.$$

Form $\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I} = (\pi_{\alpha,R,I})_!(\Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I})$ and the open-support sheaf $\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U)$ over $U \subset E$ by restricted Borel–Moore configuration sums.

(v) *Local/local correspondences.* For local colour maps $\alpha : I \to \Gamma^{\mathrm{loc}}_R$, $\beta : J \to \Gamma^{\mathrm{loc}}_R$, and a retained local target $\gamma = |\alpha| + |\beta|$, form the projection-finite diagram $\mathfrak{E}^{\mathrm{LL}}_{\alpha,\beta;\gamma,R,I,J,K}$, where $K = I \sqcup J$ modulo ordinary Ran collisions. Its target map q^{LL} is proper by Proposition 0.20; source properness is not claimed.

(vi) *Wrapped prequotient sector.* For $\eta \in \Gamma^{\mathrm{wr}}_R$, form the rigidified prequotient $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \to \mathcal{M}^{\mathrm{wr}}_{\eta,R}$ of Definition 0.6, with its E-translation action before the reduced quotient is taken. The finite-type assertion is Proposition 0.8. Then attach the determinant anchor $\lambda^{\det}_{\eta,R}$ of Definition 4.7. Its translation weight, computed in Proposition 4.8, is χ_η. On the $\chi_\eta = 0$ branch, Proposition 4.9 gives a rank-one determinant-anchor separation defect. Attach the degree-zero Jordan–Hoelder divisor of Proposition 4.10 as extra anchor data; it separates the determinant-zero rank-two test pair but is not full anchor losslessness. Unit-weight normalization, losslessness, quotient descent, stabilizer linearization, transition compatibility, and finite-stage population remain separate rows. The finite anchor residual $\mathfrak{o}^{\lambda}_R$ is Definition 0.49; its vanishing is a later row.

(vii) *Mixed local/wrapped correspondences.* For a finite local colour map $\alpha : I \to \Gamma^{\mathrm{loc}}_R$, a wrapped colour $\eta \in \Gamma^{\mathrm{wr}}_R$, and a retained wrapped target $\zeta = |\alpha| + \eta$ or $\eta + |\alpha|$, form the one-sided unreduced diagrams $\mathfrak{E}^{\mathrm{LW}}_{\alpha,\eta;\zeta,R,I}$ and $\mathfrak{E}^{\mathrm{WL}}_{\eta,\alpha;\zeta,R,I}$ before the reduced E-quotient. Their target maps are proper by Proposition 0.22; source properness is not claimed.

(viii) *Wrapped/wrapped correspondences.* For ordered wrapped colours $\eta_1, \eta_2 \in \Gamma^{\mathrm{wr}}_R$ and a retained wrapped target $\zeta = \eta_1 + \eta_2$, form the unreduced diagram $\mathfrak{E}^{\mathrm{WW}}_{\eta_1,\eta_2;\zeta,R}$ before the reduced E-quotient. Its target map is proper by Proposition 0.24; source properness is not claimed.

(ix) *Compact-support pushforward model.* For any retained finite correspondence map $f : \mathfrak{E} \to Z$, retain the compactification datum of Definition 0.25 and define $f^{\mathrm{cs}}_! K = \overline{f}_* j_{f,!} K$. Proper maps use the identity compactification. Choice independence, base change, projection formula, Thom–Sebastiani, symmetric descent, quotient-descent, transition, and associativity rows remain separate obligations.

(x) *Mixed base change.* For the LW/WL target maps, retain the Cartesian compact-support base-change isomorphisms of Proposition 0.26. Projection formula, Thom–Sebastiani, quotient-descent, transition, associativity, and aggregate population rows remain separate obligations.

(xi) *Mixed projection formula.* For the LW/WL target maps, retain the compact-support projection formula of Proposition 0.27 for retained finite-Tor target coefficients. Thom–Sebastiani, quotient-descent, transition, associativity, and aggregate population rows remain separate obligations.

(xii) *Mixed Thom–Sebastiani transport.* For the LW/WL source maps, retain the binary reduced Thom–Sebastiani transport of Proposition 0.28 on supplied additive d-critical charts with zero-defect Joyce–Upmeier orientation transport. Quotient-descent, transition, two-step flag associativity, and aggregate population rows remain separate obligations.

(xiii) *Eight-word vocabulary.* Retain the binary local/wrapped vocabulary of Definition 0.29: {LLL, LLW, LWL, WLL, LWW, WLW, WWL, WWW}, together with the L/W type product and the two parenthesization symbols. The two-step flag stacks, associativity defects, pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xiv) *Eight two-step flag stacks.* Retain the flag stacks $\mathfrak{F}^{(2),w}_{\xi_1,\xi_2,\xi_3,R}$ and their left/right comparison maps from Proposition 0.30. Associativity defects, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xv) *LLL associativity.* Retain the pure local associativity identity of Proposition 0.31. The seven remaining word associativity rows, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xvi) *LLW associativity.* Retain the local/local–mixed associativity identity of Proposition 0.32. The six remaining word associativity rows, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xvii) *LWL associativity.* Retain the ordered mixed–mixed associativity identity of Proposition 0.33. The five remaining word associativity rows, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xviii) *WLL associativity.* Retain the mixed–local/local associativity identity of Proposition 0.34. The four remaining WW-dependent word associativity rows, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xix) *LWW associativity.* Retain the mixed–wrapped/wrapped associativity identity of Proposition 0.35. The three remaining WW-dependent word associativity rows, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xx) *WLW associativity.* Retain the mixed–wrapped/wrapped associativity identity of Proposition 0.36. The two remaining WW-dependent word associativity rows, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxi) *WWL associativity.* Retain the wrapped/wrapped–mixed associativity identity of Proposition 0.37. The remaining WWW word associativity row, the four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxii) *WWW associativity.* Retain the wrapped/wrapped associativity identity of Proposition 0.38. This completes the length-three wordwise associativity list. The four-input pentagon, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxiii) *Four-input pentagon.* Retain the Mac Lane pentagon coherence of Proposition 0.39. The higher-coloured tree category, remaining unit compatibilities, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxiv) *Higher-coloured tree category.* Retain the nonunital higher-coloured tree category of Definition 0.40. The vanishing of $\mathfrak{o}_R^{\mathrm{col}}$, remaining unit compatibilities, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxv) *Higher-coloured residual criterion.* Retain the conditional criterion of Proposition 0.41. The tree component is discharged by associativity and pentagon coherence. The unit and symmetric-relabelling rows are supplied by separate proof packets. Refinement, ordered-chart descent, overlap, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxvi) *Hybrid unit datum.* Retain the local zero unit object and the split vacuum correspondences of Definition 0.42. The local and wrapped unit identity laws are supplied by the next two proof rows; quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxvii) *Local unit compatibility.* Retain Proposition 0.43. The wrapped unit identity law, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxviii) *Wrapped unit compatibility.* Retain Proposition 0.44. The unit component of the higher-coloured residual is now discharged at fixed height before quotient. Quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxix) *Local configuration symmetric descent.* Retain Proposition 0.45. The symmetry component of the higher-coloured residual is now discharged for projection-finite local configuration charts at fixed height before quotient. Collision compatibility, refinement, ordered-chart descent, overlap, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxx) *Wrapped insertion order conventions.* Retain Proposition 0.46. The wrapped order input row of the higher-coloured residual criterion is now supplied at fixed height before quotient. Unordered wrapped-pair symmetric descent, ordered-chart descent, overlap, quotient-descent, transition, and aggregate population rows remain separate obligations.

(xxxi) *Two-sided mixed correspondences.* For local colour maps $\alpha_- : I_- \longrightarrow \Gamma_R^{\mathrm{loc}}$, $\beta_+ : I_+ \longrightarrow \Gamma_R^{\mathrm{loc}}$, a wrapped colour $\eta \in \Gamma_R^{\mathrm{wr}}$, and a retained wrapped target $\zeta = |\alpha_-| + \eta + |\beta_+|$, form the ordered unreduced diagram $\mathfrak{C}^{\mathrm{LWL}}_{\alpha_-;\eta;\beta_+,R,I_-,I_+}$ before the reduced E-quotient.

(xxxii) *Source/target anchor memory.* Retain the anchor-memory maps a^{LW}, a^{WL}, a^{WW}, and a^{LWL} of Definition 0.48 on the unreduced mixed and wrapped correspondence types. This step assumes the abstract wrapped anchors $\lambda_{\eta,R}$. The determinant-anchor construction and translation-weight computation are separate packets, and the $\chi = 0$ determinant-anchor degeneracy is handled by its own packet. The degree-zero Jordan–Hoelder extra-anchor datum and the residual definition o_R^{λ} are separate packets; residual vanishing, losslessness, quotient descent, and transition rows remain separate obligations.

(xxxiii) *Quotient-after-correspondence pseudofunctor.* Retain Definition 0.50. This supplies the object, arrow, and coefficient-descent interface rows before Borel–Moore chains. Preservation of composition, stack-level base-change squares, and equivariant Thom–Sebastiani transport are supplied by Propositions 0.51, 0.52, and 0.53. Proposition 0.54 excludes independent quotient-first Hall products. Proposition 0.55 constructs $Q_{E,R}$ on Borel–Moore chains. Proposition 0.56 constructs $\theta^{Q}_{\mu,R}$ for quotient-presented operation rows. Proposition 0.57 supplies the quotient two-step flag associativity component. Proposition 0.58 supplies the quotient product-coproduct compatibility component. Proposition 0.59 supplies the quotient primitive-projection compatibility component. Proposition 0.60 supplies the HN-transition compatibility of $Q_{E,R}$. Protected integration and aggregate population remain separate obligations. Hall product, Hall coproduct, and primitive-subspace transition preservation remain outside row 256. Proposition 0.62 supplies the definition of I_R^{prot}. Proposition 0.63 supplies product compatibility. Proposition 0.64 supplies coproduct compatibility. Proposition 0.65 supplies primitive-projection compatibility. Proposition 0.66 supplies Gram degree. Proposition 0.67 supplies the normal-ordered label comparison. Transition compatibility remains separate.

(xxxiv) *Eight-word two-step flag atlas.* Retain the word set $\mathcal{W}_{\mathrm{hyb}}^{(3)}$ of Definition 0.29 and the flag stacks $\mathfrak{F}_R^{(2),w}$ of Proposition 0.30. The LLL associativity defect is zero by Proposition 0.31; the LLW associativity defect is zero by Proposition 0.32; the LWL associativity defect is zero by Proposition 0.33; and the WLL associativity defect is zero by Proposition 0.34. The LWW associativity defect is zero by Proposition 0.35. The WLW associativity defect is zero by Proposition 0.36. The WWL associativity defect is zero by Proposition 0.37; and the WWW associativity defect is zero by Proposition 0.38. The four-input pentagon coherence is supplied by Proposition 0.39. Quotient-after-correspondence descent is a separate row. This is binary/two-step data; unit maps, refinements, symmetry conventions, and higher-coloured trees remain in $\mathfrak{o}_R^{\mathrm{col}}$.

(xxxv) *Hybrid correspondence-module functor.* Let $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ be the correspondence category, and define $\mathcal{B}_R^{E,\mathrm{hyb}} : \mathsf{Corr}_R^{E,\mathrm{hyb}} \to \mathsf{Ch}_{\mathbb{C}}$ by $\mathcal{B}_R^{E,\mathrm{hyb}}(e) = q_! \,\mathrm{TS}_e^{\mathrm{red}}\, p^*$ for every generating arrow $e = (Y \xleftarrow{p} \mathfrak{E}_e \xrightarrow{q} Z)$.

(xxxvi) *Quotient pseudofunctor $Q_{E,R}$.* Use the chosen quotient-after-correspondence pseudofunctor definition, then supply the composition comparison of Proposition 0.51. Also supply the base-change square comparison of Proposition 0.52. Also supply the Thom–Sebastiani descent comparison of Proposition 0.53. Also supply the quotient-first exclusion of Proposition 0.54. Also supply the Borel–Moore chain functor of Proposition 0.55. Also supply the operation-comparison maps of Proposition 0.56. Also supply the associativity compatibility of Proposition 0.57. Also supply the coproduct compatibility of Proposition 0.58. Also supply the primitive-projection compatibility of Proposition 0.59. Also supply the HN-transition compatibility of $Q_{E,R}$ from Proposition 0.60. Also define I_R^{prot} by Proposition 0.62. Also supply product compatibility by Proposition 0.63. Also supply coproduct compatibility by Proposition 0.64. Also supply primitive-projection compatibility by Proposition 0.65. Then supply the separate coefficient, orientation, compact-support pushforward base-change, stabilizer, anchor, Hall product transition, Hall coproduct transition, primitive-subspace transition, and remaining protected integration compatibility coherences. Applying compactly supported vanishing-cycle chains to the completed datum gives the reduced object

$$\mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{geom}} = Q_{E,R} \circ \mathcal{B}_R^{E,\mathrm{hyb}}.$$

(xxxvii) Obtain the hybrid datum with all coherences and the protected integration map I_R^{prot}, with homogeneous s-degree the Borcherds trace exponent $m_R^{\mathrm{Bch}} = \mathrm{pr}_3\, \overline{\Pi}_X$.

6.3.0.4 Identities. The carrier $\mathcal{F}_{X,\sigma,S,\leq R}^{\mathrm{hyb}}$ is a hybrid correspondence-module functor on $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ with values in $\mathsf{Ch}_{\mathbb{C}}$ only after the finite proof rows below are supplied: the four-input pentagon follows from the eight-word two-step flag atlas, and the required anchor residual identity is $\mathfrak{o}_R^{\lambda} = (\mathfrak{o}_R^{\lambda,\mathrm{ex}}, \mathfrak{o}_R^{\lambda,\mathrm{unit}}, \mathfrak{o}_R^{\lambda,\mathrm{loss}}, \mathfrak{o}_R^{\lambda,\mathrm{multi}}, \mathfrak{o}_{R'R}^{\lambda,\mathrm{tr}}) = 0$. Definition 0.49 defines this tuple. Its vanishing, the hybrid residual identity $\mathfrak{O}_{\mathrm{hyb},R} = 0$, and compatibility with transition maps are separate finite proof rows.

6.3.0.5 Obstruction classes. The middle type-II wall lies in $\partial_2 \leftrightarrow (0,1,1)$. Its wall-branch elliptic degree is $d = m + 1 = 2$ by Proposition 0.15; it cannot be represented by a projection-finite local stack and demands the wrapped prequotient. The existence of compact Mukai-chart representatives for the three wall-charge triples is Proposition 3.16; those representatives do not supply the retained wrapped wall objects, anchors, or quotient orientation data. The existence of the wrapped representative $x_{\partial_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ with $\overline{\Pi}_X(x_{\partial_2,R}) = (0,1,1)$ and its compatible anchor $\lambda_{\eta,R}$ — is precisely the hybrid input to $(O2)$ at the middle wall. The three simple type-II wall profiles themselves are recovered at the hybrid operation-type level by Proposition 0.16: each has positive wall-branch elliptic degree and hence a singleton wrapped colour profile, while its local insertions are typed by LW, WL, and LWL mixed correspondences before the E-quotient. This does not populate the aggregate hybrid carrier or the O2 wall atlas. Their quotient-after-correspondence profile survival is Proposition 0.17; it is not an object-level nonvanishing, coefficient-descent, or O2-atlas statement. The type-I chamber walls are excluded from this wall-profile inlet by Proposition 0.18; this is not global colour exhaustion for the aggregate hybrid carrier.

6.4 Reduced d-critical orientation and Joyce–Upmeier transport

6.4.0.1 Hypotheses. The hybrid carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$ of the preceding section; the K3 holomorphic 2-form σ_S on S; BBDJS d-critical chart machinery [12]; the Joyce–Upmeier multiplicative orientation formalism [55, 56]; the cosection localization machinery of Kiem–Li and the cosection-reduced obstruction theory; the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ generated by $s_{\partial_1}, s_{\partial_2}, s_{\partial_3}$.

6.4.0.2 PTVV finite-substack input. Proposition 0.72 supplies the PTVV (-1)-shifted symplectic form on the retained finite derived substacks. The packet `certificates/orientation/ptvv_finite_substack_symplectic` verifies shifted degree -1, cotangent amplitude $[-1, 0]$, and zero symplectic defect on the three retained rows. It does not supply the Joyce d-critical truncation, K3 cosection, reduced self-Ext complex, determinant line, orientation square root, quotient orientation, or protected integration.

6.4.0.3 Joyce d-critical truncation input. Proposition 0.73 supplies the Joyce d-critical structure on the classical truncations of the retained finite derived substacks. The packet `certificates/orientation/joyce_dcritical_truncation` verifies the BBDJS source theorem, the three truncation rows, and zero d-critical defect. It does not supply vanishing cycles, the K3 cosection, cosection surjectivity, reduced self-Ext complexes, determinant lines, square-root orientations, quotient orientation, transition compatibility, or protected integration.

6.4.0.4 K3 semiregularity cosection input. Proposition 0.74 supplies the $\mathcal{O}$-linear semiregularity cosection from the obstruction sheaf of each retained finite classical truncation to the structure sheaf. The packet `certificates/orientation/k3_semiregularity_cosection` verifies the Atiyah-class formula, the three cosection rows, and zero construction defect. It does not supply cosection surjectivity, filtered Hall additivity, reduced self-Ext complexes, determinant lines, square-root orientations, quotient orientation, transition compatibility, vanishing cycles, or protected integration.

6.4.0.5 Cosection surjectivity obstruction. Proposition 0.75 identifies the missing row: a retained stratum must carry a nonzero K3-class witness $\beta_{R,c} \neq 0$, a branch-identification row to the reduced DT/PT semiregularity theorem, and a cokernel-rank-zero row. The packet `certificates/orientation/k3_cosection_surjectivity` verifies that these rows are not yet supplied. It is an obstruction ledger, not a proof of row 274.

6.4.0.6 Reduced self-Ext cone input. Definition 0.76 supplies the cone formula

$$\mathrm{RHom}_{\mathrm{red}}(A, A) = \mathrm{Cone}\left(\mathrm{RHom}_X(A, A) \xrightarrow{(\mathrm{tr}, \mathrm{CS}_A)} R\Gamma(X, \mathcal{O}_X) \oplus H^2(S, \mathcal{O}_S)[-1]\right)[-1].$$

The packet `certificates/orientation/rhomred_definition` verifies the three retained cone-definition rows and zero definition defect. It does not construct a square-root orientation.

6.4.0.7 Reduced self-Ext perfectness input. Proposition 0.77 proves that the cone $\mathrm{RHom}_{\mathrm{red}}(A, A)$ is perfect on each retained finite row. The proof uses the universal perfect complex of Proposition 1.24, proper perfect pushforward along $X \times \mathfrak{M}^{\mathrm{rig}}_{R,c} \to \mathfrak{M}^{\mathrm{rig}}_{R,c}$, and closure of perfect complexes under fibres. The packet `certificates/orientation/rhomred_perfectness` verifies the three perfectness rows and zero perfectness defect. It does not define $\det \mathrm{RHom}_{\mathrm{red}}(A, A)$, does not construct a square-root orientation, and does not supply quotient orientation, transition compatibility, vanishing cycles, or protected integration.

6.4.0.8 Reduced determinant-line input. Definition 0.78 defines

$$\det \mathrm{RHom}_{\mathrm{red}}(A, A)$$

by applying the Knudsen–Mumford determinant functor to the perfect complex supplied by Proposition 0.77. The packet `certificates/orientation/rhomred_determinant` verifies the three determinant-line rows and zero determinant defect. It does not construct a square-root orientation, does not compute w_2, and does not supply quotient orientation, transition compatibility, vanishing cycles, or protected integration.

6.4.0.9 Reduced square-root criterion. Proposition 0.79 gives the exact square-root criterion: a row must supply a line $o_{R,c}$ and an isomorphism

$$o_{R,c}^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A, A).$$

The packet `certificates/orientation/rhomred_square_root` verifies that no such square-root row is present in the current reduced-orientation tables. It is an obstruction ledger for row 278, not an orientation construction.

6.4.0.10 Reduced w_2-formula and obstruction. Proposition 0.80 proves the exact topological formula

$$w_2\left((\mathcal{L}^{\det}_{R,c})_{\mathbb{R}}\right) = c_1(\mathcal{L}^{\det}_{R,c}) \bmod 2$$

and identifies the required determinant-of-cohomology input: an integral c_1-row, or an equivalent GRR expansion, followed by mod-2 reduction on every retained substack. The packet `certificates/orientation/rhomred_w2` verifies that the current determinant and square-root packets supply no such c_1, GRR, or w_2-value rows. It is an obstruction ledger for row 279, not a vanishing theorem.

6.4.0.11 Reduced gerbe vanishing criterion. Proposition 0.81 identifies row 280. It must supply the two summands

$$w_2(\det K^{\mathrm{red}}_{R,c}), \qquad \mathrm{CS}^*_{R,c}\, w_2(\det \operatorname{coker} \mathrm{CS}_{R,c})$$

and the zero-sum equation defining $\alpha^{\mathrm{red}}_{R,c} = 0$. If the cosection cokernel is proved zero, the second summand vanishes; otherwise it is a separate characteristic-class computation. The packet `certificates/orientation/rhomred_alpha_red` verifies that the current tables supply neither the determinant w_2-value, nor the cokernel rank-zero row, nor the cokernel w_2-value, nor the zero-sum row. It is an obstruction ledger for row 280, not a null-homotopy and not an orientation descent theorem.

6.4.0.12 Reduced gerbe null-homotopy criterion. Proposition 0.82 identifies row 281. After choosing a Cech–Borel cocycle

$$a^{\mathrm{red}}_{R,c}$$

representing $\alpha^{\mathrm{red}}_{R,c}$, a null-homotopy is a 1-cochain $h^{\mathrm{red}}_{R,c}$ satisfying

$$\delta h^{\mathrm{red}}_{R,c} = a^{\mathrm{red}}_{R,c}.$$

The packet `certificates/orientation/rhomred_alpha_red_nullhomotopy` verifies that the current tables supply no cocycle representative, no zero-class row, no 1-cochain, and no coboundary-defect-zero row. It is an obstruction ledger for row 281, not quotient orientation.

6.4.0.13 Reduced orientation datum. The pulled-back two-form $\sigma = p_S^*\sigma_S$; the semiregularity cosection $\mathrm{CS}_{R,c}$ of Proposition 0.74; the reduced tangent complex

$$\mathrm{RHom}_{\mathrm{red}}(A,A) = \mathrm{Cone}\Big(\mathrm{RHom}_X(A,A) \xrightarrow{(\mathrm{tr},\mathrm{cs})} R\Gamma(X,\mathcal{O}_X) \oplus H^2(S,\mathcal{O}_S)[-1]\Big)[-1];$$

the required BBDJS vanishing-cycle complex $\Phi_{R,c}$; a required reduced orientation line

$$o_{R,c}, \qquad o_{R,c}^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A,A);$$

the required Joyce–Upmeier multiplicative transport $o_{c\oplus c'} \simeq o_c \otimes o_{c'}$ compatible with the reduced extension correspondences; the quotient orientation $\mathfrak{Q}^{or}_R$ of Theorem 7.13; the Weyl-equivariant lifts $\tau_{i,R,S} : o_{R,S} \to s^*_{\partial_i} o_{R,S}$ for $i = 1,2,3$ of $(O1)^+$; the local generic wall-sign data $\epsilon^{\mathrm{loc}}_{o,R}$. The current row 278 obstruction packet records that the line $o_{R,c}$ and its square isomorphism are not supplied. The current row 279 obstruction packet records that the c_1-row and mod-2 Stiefel–Whitney value are not supplied. The current row 280 obstruction packet records that the reduced gerbe zero-sum equation is not supplied. The current row 281 obstruction packet records that the reduced gerbe null-homotopy is not supplied.

6.4.0.14 Construction.

(i) *Cosection.* Proposition 0.74 supplies the semiregularity cosection $\mathrm{CS}_{R,c} : \mathrm{Ob}_{R,c} \to \mathcal{O}_{\mathfrak{M}^{\mathrm{cl}}_{R,c}}$ induced by $\sigma = p_S^*\sigma_S$. Proposition 0.75 gives the exact surjectivity criterion and records that the present finite tables lack the nonzero K3-class witness. Additivity in exact triangles is a later row.

(ii) *Reduced obstruction.* Definition 0.76 defines the reduced self-Ext cone by the displayed formula above. Proposition 0.72 supplies the unreduced PTVV (-1)-shifted symplectic input. Proposition 0.77 proves that this cone is perfect on the retained finite rows. Definition 0.78 defines its determinant line.

(iii) *BBDJS d-critical chart.* Proposition 0.73 supplies the d-critical structure on each retained classical truncation. The BBDJS vanishing-cycle complex $\Phi_{R,c}$, reduced Thom–Sebastiani transport, and two-step flag coherences are later rows.

(iv) *Reduced orientation line.* Proposition 0.79 requires an actual line $o_{R,c}$ and an isomorphism

$$o_{R,c}^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A, A).$$

The present finite tables do not supply that row. A formal expression in the reduced Ext groups is not an orientation until the Knudsen–Mumford sign, Picard-groupoid square root, and transition compatibility are supplied.

(v) *Stiefel–Whitney class.* Proposition 0.80 reduces row 279 to computing $c_1(\mathcal{L}_{R,c}^{\det})$ and reducing it modulo 2. The present finite tables do not supply the integral c_1-class, the GRR expansion of the reduced kernel, or the mod-2 value. A zero w_2-claim is therefore not part of the current orientation datum.

(vi) *Reduced gerbe class.* Proposition 0.81 reduces row 280 to the equality

$$w_2(\det K_{R,c}^{\mathrm{red}}) + \mathrm{CS}_{R,c}^{*}\, w_2(\det \operatorname{coker} \mathrm{CS}_{R,c}) = 0.$$

The present finite tables do not supply either summand or the zero-sum row. A null-homotopy of this zero class is a later row.

(vii) *Reduced gerbe null-homotopy.* Proposition 0.82 requires a Cech–Borel cocycle $a_{R,c}^{\mathrm{red}}$, a 1-cochain $h_{R,c}^{\mathrm{red}}$, and the equation

$$\partial h_{R,c}^{\mathrm{red}} = a_{R,c}^{\mathrm{red}}.$$

The present finite tables do not supply these rows. An empty quotient-Borel table is not a null-trivialization.

(viii) *Connected free E-Borel class.* Proposition 0.83 identifies row 282. Since

$$H^2(BE; \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2,$$

the free connected class has the form

$$\alpha_{R,c}^{E,\mathrm{free}} = a_1(R,c)u_1 + a_2(R,c)u_2.$$

The row must supply the equivariant Borel representative, the edge calculation in bidegree $(2,0)$, and the coefficients a_1, a_2. The packet `certificates/orientation/rhomred_free_e_borel` verifies that the present finite tables supply the free E-action but no `free_E` quotient-Borel row and no a_1, a_2 coefficients.

(ix) *Connected free E-Borel vanishing.* Proposition 0.84 identifies row 283. Once row 282 supplies

$$\alpha_{R,c}^{E,\mathrm{free}} = a_1(R,c)u_1 + a_2(R,c)u_2,$$

vanishing is exactly the pair of equations

$$a_1(R,c) = 0, \qquad a_2(R,c) = 0$$

in $\mathbb{F}_2$. The packet `certificates/orientation/rhomred_free_e_vanishing` verifies that the present finite tables supply no coefficient row and no zero-coefficient row. The null-trivialization of this zero class is row 284, not part of the row 283 criterion.

(x) *Connected free E-Borel null-trivialization.* Proposition 0.85 identifies row 284. For a chosen Cech–Borel representative

$$a_{R,c}^{E,\mathrm{free}} \in Z^2(BE;\mathbb{F}_2)$$

of the connected free E-Borel class, the row must supply a 1-cochain

$$h_{R,c}^{E,\mathrm{free}} \in C^1(BE;\mathbb{F}_2)$$

with

$$\delta h_{R,c}^{E,\mathrm{free}} = a_{R,c}^{E,\mathrm{free}}.$$

The equality $H^1(BE;\mathbb{F}_2) = 0$ removes the connected H^1-torsor ambiguity; it does not supply the cocycle, the cochain, or the defect-zero coboundary row. The packet `certificates/orientation/rhomred_free_e_nulltrivialization` verifies that the present finite tables supply none of these rows.

(xi) *Finite-stabilizer Borel class.* Proposition 0.87 identifies row 285. For a finite stabilizer $H \subset E_{\mathrm{tors}}$, the row first supplies

$$\widetilde{\beta}_{R,S}^{H} = w_2^H(\mathscr{D}_{R,S}) \in H_H^2(S;\mathbb{F}_2)$$

and an edge-reduction calculation. The resulting class $\beta_{R,S}^H \in H^2(BH;\mathbb{F}_2)$ is then recorded in the odd, Klein-four, or two-primary basis:

$$\beta_{R,S}^{E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad \beta_{R,S}^{E[N]} = A_1^{(N)}y_1 + A_{12}^{(N)}x_1x_2 + A_2^{(N)}y_2.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_beta` verifies that the present finite tables supply residual finite groups but no finite-stabilizer orientation row, no edge-reduction row, and no β^H-coefficient row. Row 286 is the later vanishing assertion.

(xii) *Finite-stabilizer Borel vanishing.* Proposition 0.88 identifies row 286. Once row 285 supplies $\beta_{R,S}^H \in H^2(BH;\mathbb{F}_2)$, vanishing is the coordinate condition

$$H = 1 \text{ or } |H| \text{ odd}: \ H^2(BH;\mathbb{F}_2) = 0, \qquad H = E[2]: \ b_{20} = b_{11} = b_{02} = 0,$$

and, for $H_{(2)} \simeq (\mathbb{Z}/2^a)^2$, $a \geq 2$,

$$A_1^{(H)} = A_{12}^{(H)} = A_2^{(H)} = 0.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_beta_vanishing` verifies that the present finite tables supply no computed β^H-value and no zero-coordinate row. Row 287 is the later choice of compatible null-trivializations.

(xiii) *Finite-stabilizer Borel null-trivialization.* Proposition 0.89 identifies row 287. After rows 285–286 supply a Cech–Borel representative

$$a_{R,S}^H \in Z^2(BH;\mathbb{F}_2)$$

and verify that its class vanishes, row 287 must supply

$$h_{R,S}^H \in C^1(BH;\mathbb{F}_2), \qquad \delta h_{R,S}^H = a_{R,S}^H.$$

For every retained subgroup inclusion $\iota\colon K \hookrightarrow H$, it must verify $\operatorname{res}_\iota a^H_{R,S} = a^K_{R,S}$ and supply a comparison 0-cochain $q^\iota_{R,S}$ with

$$\operatorname{res}_\iota h^H_{R,S} - h^K_{R,S} = \delta q^\iota_{R,S},$$

together with zero defect on chains of subgroup inclusions. The packet `certificates/orientation/rhomred_finite_s` verifies that the present finite tables supply no cocycle representative, no zero row, no 1-cochain, no subgroup-inclusion table, no comparison 0-cochain, and no chain-coherence row. Rows 288–289 are the separate linearization-character calculation.

(xiv) *Finite-stabilizer linearization character.* Proposition 0.90 identifies row 288. After the finite-stabilizer gerbe is killed, the remaining orientation descent datum is the sign character

$$\lambda^H_{R,S} \in H^1(BH;\mathbb{F}_2) = \operatorname{Hom}(H,\mathbb{F}_2).$$

Computing it requires the action of H on the trivialized orientation line, generator-basis provenance for the two-primary stabilizer, and the character values on those generators. The coordinate forms are

$$H = 1 \text{ or } |H| \text{ odd}:\ H^1(BH;\mathbb{F}_2) = 0,$$

$$H = E[2]:\quad \lambda^H_{R,S} = \lambda_1 x_1 + \lambda_2 x_2, \qquad \rho_1 = \lambda_1,\ \rho_2 = \lambda_2,\ \rho_3 = \lambda_1 + \lambda_2,$$

and, for $H_{(2)} \simeq (\mathbb{Z}/2^a)^2$, $a \geq 2$,

$$\lambda^H_{R,S} = \lambda_1^{(H)} x_1 + \lambda_2^{(H)} x_2.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_linearization_character` verifies that the present finite tables supply no orientation-line action, no generator-basis row, no character values, and no computed H^1-coordinates. Row 289 is the later vanishing assertion for the computed character.

(xv) *Finite-stabilizer zero linearization.* Proposition 0.91 identifies row 289. Once row 288 computes

$$\lambda^H_{R,S} \in H^1(BH;\mathbb{F}_2),$$

the zero-linearization row is the coordinate condition

$$H = 1 \text{ or } |H| \text{ odd}:\ H^1(BH;\mathbb{F}_2) = 0,$$

$$H = E[2]:\quad \lambda_1 = \lambda_2 = 0 \quad \text{equivalently} \quad \rho_1 = \rho_2 = \rho_3 = 0,$$

and, for $H_{(2)} \simeq (\mathbb{Z}/2^a)^2$, $a \geq 2$,

$$\lambda_1^{(H)} = \lambda_2^{(H)} = 0.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_linearization_vanishing` verifies that the present finite tables supply no computed $\lambda^H_{R,S}$-value and no zero-coordinate row. The degree-two gerbe vanishing and the row 287 null-trivialization do not imply this degree-one vanishing.

(xvi) *Gerbe-linearization degree separation.* Proposition 0.92 identifies row 290. The finite-stabilizer gerbe coordinates live in $H^2(BH;\mathbb{F}_2)$, while the residual linearization character lives in $H^1(BH;\mathbb{F}_2)$. For $H = E[2]$, the values

$$b_{20} = b_{11} = b_{02} = 0, \qquad (\lambda_1, \lambda_2) = (1, 0)$$

give $\beta^H_{R,S} = 0$ and $\lambda^H_{R,S} \neq 0$. For $H_{(2)} \simeq (\mathbb{Z}/2^a)^2$, $a \geq 2$, the values

$$A_1^{(H)} = A_{12}^{(H)} = A_2^{(H)} = 0, \qquad (\lambda_1^{(H)}, \lambda_2^{(H)}) = (1, 0)$$

give the same counterexample. The packet `certificates/orientation/rhomred_finite_stabilizer_gerbe_lineari` verifies the firewall: row 286 gerbe vanishing and row 287 null-trivialization are not row 289 zero-linearization data.

(xvii) $E[2]$ *Klein-four class.* Proposition 0.93 identifies row 291. For $E[2] \simeq (\mathbb{Z}/2)^2$ and dual basis $x_1, x_2 \in H^1(BE[2];\mathbb{F}_2)$,

$$H^2(BE[2];\mathbb{F}_2) = \mathbb{F}_2\langle x_1^2, x_1x_2, x_2^2\rangle.$$

An edge-reduced finite-stabilizer Borel class is therefore uniquely

$$\beta^{E,2}_{R,S} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_klein_four_class` verifies this local classifying-space computation and records that the present finite tables still supply no retained $E[2]$-edge row and no coefficient values. Rows 292–294 derive b_{20}, b_{11}, b_{02} from the cyclic restrictions r_1, r_2, r_3.

(xviii) *First cyclic restriction in the Klein-four class.* Proposition 0.94 identifies row 292. For $\iota_1\colon \langle e_1\rangle \hookrightarrow E[2]$ and $t \in H^1(B\langle e_1\rangle;\mathbb{F}_2)$,

$$\iota_1^*(x_1) = t, \qquad \iota_1^*(x_2) = 0.$$

Thus

$$\iota_1^*\big(b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2\big) = b_{20}t^2.$$

Writing this first cyclic restriction as r_1t^2 gives

$$b_{20} = r_1.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_klein_four_b20` verifies this local restriction computation and records that the present finite tables still supply no retained first cyclic restriction row and no value of r_1. Rows 293–294 compute the remaining coefficients.

(xix) *Diagonal cyclic restriction in the Klein-four class.* Proposition 0.95 identifies row 293. For

$$\iota_2\colon \langle e_2\rangle \hookrightarrow E[2], \qquad \iota_3\colon \langle e_1 + e_2\rangle \hookrightarrow E[2],$$

the restrictions are

$$\iota_2^*(x_1) = 0, \qquad \iota_2^*(x_2) = t,$$

and

$$\iota_3^*(x_1) = t, \qquad \iota_3^*(x_2) = t.$$

Hence

$$\iota_2^*\beta_{C,S}^{E,2} = b_{02}t^2, \qquad \iota_3^*\beta_{C,S}^{E,2} = (b_{20} + b_{11} + b_{02})t^2.$$

Writing the three cyclic restrictions as r_1t^2, r_2t^2, r_3t^2 gives

$$b_{11} = r_1 + r_2 + r_3.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_klein_four_b11` verifies this local mixed-coefficient computation and records that the present finite tables still supply no retained cyclic restriction rows and no values of r_1, r_2, r_3. Row 294 records the standalone second-square extraction $b_{02} = r_2$.

(xx) *Second cyclic restriction in the Klein-four class.* Proposition 0.96 identifies row 294. For $\iota_2 : \langle e_2\rangle \hookrightarrow E[2]$ and $t \in H^1(B\langle e_2\rangle; \mathbb{F}_2)$,

$$\iota_2^*(x_1) = 0, \qquad \iota_2^*(x_2) = t.$$

Thus

$$\iota_2^*\big(b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2\big) = b_{02}t^2.$$

Writing this second cyclic restriction as r_2t^2 gives

$$b_{02} = r_2.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_klein_four_b02` verifies this local restriction computation and records that the present finite tables still supply no retained second cyclic restriction row and no value of r_2. Row 295 is the separate vanishing step for r_1, r_2, r_3.

(xxi) *Klein-four zero-restriction criterion.* Proposition 0.97 identifies row 295. Rows 292–294 give

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3.$$

Therefore

$$r_1 = r_2 = r_3 = 0 \quad\Longleftrightarrow\quad b_{20} = b_{11} = b_{02} = 0 \quad\Longleftrightarrow\quad \beta_{C,S}^{E,2} = 0$$

for the classifying-space edge on $BE[2]$. The packet `certificates/orientation/rhomred_finite_stabilizer_klein_` verifies this algebraic criterion and records that the present finite tables still supply no retained cyclic zero restriction rows and no values of r_1, r_2, r_3. Hence row 295 is not yet a geometric vanishing proof on the retained $K3 \times E$ strata.

(xxii) *Even two-primary class.* Proposition 0.98 identifies row 296. For $2^a \parallel N$, $a \geq 2$,

$$H^*(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2,$$

and

$$H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle.$$

Thus an edge-reduced two-primary finite-stabilizer Borel class has the unique form

$$\beta^{E,N}_{C,S} = A^{(N)}_1 y_1 + A^{(N)}_{12} x_1 x_2 + A^{(N)}_2 y_2.$$

The packet `certificates/orientation/rhomred_finite_stabilizer_two_primary_class` verifies this local classifying-space computation and records that the present finite tables still supply no retained two-primary edge row and no values of $A^{(N)}_1$, $A^{(N)}_{12}$, $A^{(N)}_2$. Row 297 is the separate check that the mixed term $A^{(N)}_{12} x_1 x_2$ cannot be detected from cyclic restrictions.

(xxiii) *Detection of the two-primary mixed term.* Proposition 0.99 identifies row 297. The coefficient functional

$$\pi_{12}(y_1) = 0, \qquad \pi_{12}(x_1 x_2) = 1, \qquad \pi_{12}(y_2) = 0$$

gives

$$\pi_{12}(\beta^{E,N}_{C,S}) = A^{(N)}_{12}.$$

For every cyclic subgroup $C \subset (\mathbb{Z}/2^a)^2$, $a \geq 2$,

$$\operatorname{res}^{(\mathbb{Z}/2^a)^2}_C (x_1 x_2) = 0.$$

Indeed, order-two subgroups lie in $2(\mathbb{Z}/2^a)^2$, where the mod-2 characters x_i vanish, while cyclic subgroups of order at least four have degree-one generator u with $u^2 = 0$. The packet `certificates/orientation/rhomred_finite_stabilizer_two_primary_mixed_term_detection` verifies this local detector and records that the present finite tables still supply no retained $A^{(N)}_{12}$-value.

(xxiv) *Odd-order transfer.* Proposition 0.100 identifies row 298. If H has odd order, then

$$H^i(BH; \mathbb{F}_2) = 0, \qquad i > 0,$$

because transfer is multiplication by $|H|$, which is invertible in $\mathbb{F}_2$. Hence any edge-reduced finite-stabilizer Borel class in $H^2(BH; \mathbb{F}_2)$ and any edge-reduced residual linearization class in $H^1(BH; \mathbb{F}_2)$ vanish. The packet `certificates/orientation/rhomred_finite_stabilizer_odd_transfer` verifies this transfer calculation and records that the present finite tables still supply no retained odd-order edge-reduction row and no all-retained-stabilizer coverage row.

(xxv) *Direct-sum orientation multiplicativity.* Proposition 0.102 identifies row 299. The determinant identity for $C \oplus \mathcal{D}$ contains the off-diagonal reduced Ext factors

$$\det K^{\mathrm{red}}_{C,\mathcal{D}} \otimes \det K^{\mathrm{red}}_{\mathcal{D},C};$$

these factors become orientation data only after a chosen Calabi–Yau hyperbolic square root $q_{C,\mathcal{D}}$ and a Picard-groupoid isomorphism

$$o_{C\oplus\mathcal{D}} \simeq o_C \otimes o_{\mathcal{D}} \otimes q_{C,\mathcal{D}}$$

with symmetric, pullback, and pentagon coherence. The packet `certificates/orientation/rhomred_orientation_di` records this criterion and verifies that the current orientation tables supply no square-root row, no direct-sum Thom–Sebastiani row, and no cross-term trivialization.

(xxvi) *Extension orientation multiplicativity.* Proposition 0.103 identifies row 300. For a retained extension correspondence

$$\mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}'} \xleftarrow{p} \overline{\mathfrak{E}}^{\mathrm{red}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q} \mathfrak{M}^{\mathrm{red}}_{R,\widehat{c}+\widehat{c}'},$$

the row is the isomorphism

$$p_1^* o_{R,\widehat{c}} \otimes p_2^* o_{R,\widehat{c}'} \simeq q^* o_{R,\widehat{c}+\widehat{c}'}$$

whose square is the reduced determinant comparison and whose tensor with BBDJS Thom–Sebastiani gives the coefficient-system transport used by the Hall product. The packet `certificates/orientation/rhomred_orientation_extension_multiplicativity` verifies the criterion and records that retained extension closure and conditional mixed Thom–Sebastiani transport do not supply the missing orientation row.

(xxvii) *Thom–Sebastiani compatibility for orientation lines.* Proposition 0.104 identifies row 301. The row is the commutative square saying that the square of the orientation isomorphism $\mathrm{TS}^{o}_{\mathfrak{E}}$ is the reduced determinant Thom–Sebastiani comparison, together with the identity

$$\mathrm{TS}^{\Phi\otimes o}_{\mathfrak{E}} = \mathrm{TS}^{\Phi,\mathrm{red}}_{\mathfrak{E}} \otimes \mathrm{TS}^{o}_{\mathfrak{E}}$$

on the oriented Hall coefficient system. The packet `certificates/orientation/rhomred_orientation_thom_sebasti` verifies this criterion and records that the existing BBDJS and hybrid Thom–Sebastiani packets treat orientation compatibility as a hypothesis, not as a supplied orientation-line row.

(xxviii) *Quotient-orientation descent on the K3×E-quotient self-Ext frame bundle.* Proposition 0.105 identifies row 302. Compute the four Cech–Borel obstruction classes for every retained stratum S:

$$\alpha^{\mathrm{red}}_{R,S} = w_2(\mathscr{D}_{R,S}), \quad \alpha^{E,\mathrm{free}}_{R,S} = \mathrm{edge}_{2,0}(w_2^E(\mathscr{D}_{R,S})), \quad \widetilde{\beta}^H_{R,S} = w_2^G(\mathscr{D}_{R,S}), \quad \lambda^H_{R,S} \in H^1_G(S;\mathbb{F}_2),$$

and verify simultaneous vanishing with compatible null-trivializations on overlaps, on Hall correspondence pullbacks, on two-step flag stacks, and under successor maps. The full calculation proceeds primary stabilizer by stabilizer; cyclic restrictions alone do not see the rank-two mixed coefficient $A^{(N)}_{12}$, so the two-primary stabilizer check of Proposition 0.101 is part of the datum. If these rows are supplied, they determine the descended quotient orientation system $\mathfrak{Q}^{or}_R$. The packet `certificates/orientation/rhomred_orientation_e_quotient_descent` verifies the descent criterion and records that the current orientation_lines, quotient_borel, and finite_stabilizers tables supply no quotient-orientation rows.

(xxix) *Orientation descent and the Hall product.* Proposition 0.106 identifies row 303. For every retained compactified extension correspondence

$$\mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'} \xleftarrow{p} \overline{\mathfrak{E}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{q} \mathfrak{M}_{R,\widehat{c}+\widehat{c}'},$$

supply the quotient-orientation cocycle comparison

$$p^*(\mathfrak{q}_{R,\widehat{c}} \boxtimes \mathfrak{q}_{R,\widehat{c}'}) = q^*\mathfrak{q}_{R,\widehat{c}+\widehat{c}'}$$

in the Picard groupoid, the oriented Thom–Sebastiani row $\mathrm{TS}^{\Phi\otimes o}$, external-product descent for $\Phi^{\mathrm{red}} \otimes o$, compact-support descent for $q_!$, and the quotient-after-correspondence comparison $\theta^{Q}_{\mu,R}$. Verify that these rows commute with the two-step flag pentagon. The packet `certificates/orientation/rhomred_orientation_hall_product_descent` verifies this product-descent criterion and records that the current quotient-orientation, oriented Thom–Sebastiani, and compact Hall product rows are not supplied.

(xxx) *Orientation descent and the Hall coproduct.* Proposition 0.107 identifies row 304. For every retained splitting correspondence

$$\mathfrak{M}_{R,\widehat{c}+\widehat{c}'} \xleftarrow{q} \overline{\mathfrak{E}}_{R,\widehat{c},\widehat{c}'} \xrightarrow{p} \mathfrak{M}_{R,\widehat{c}} \times \mathfrak{M}_{R,\widehat{c}'},$$

supply the quotient-orientation cocycle comparison

$$q^* \mathfrak{q}_{R,\widehat{c}+\widehat{c}'} = p^*(\mathfrak{q}_{R,\widehat{c}} \boxtimes \mathfrak{q}_{R,\widehat{c}'})$$

in the Picard groupoid, the inverse oriented Thom–Sebastiani row, external-product descent for $\Phi^{\mathrm{red}} \otimes o$, compact-support descent for $p_!$, and the quotient-after-correspondence comparison $\theta^Q_{\Delta,R}$. Verify coassociativity, counit, and bialgebra compatibility on the retained flag and zero-object rows. The packet `certificates/orientation/rhomred_orientation_hall_coproduct_descent` verifies this coproduct-descent criterion and records that the current quotient-orientation, oriented Thom–Sebastiani, compact Hall coproduct, and counit rows are not supplied.

(xxxi) *Orientation descent and primitive projection.* Proposition 0.108 identifies row 305. For every retained finite Hall object with oriented coefficient system $K_R = \Phi^{\mathrm{red}}_R \otimes o_R$, supply the quotient-orientation descent rows for the ambient Hall object, the zero-object stratum, and the splitting correspondences entering the reduced coproduct

$$\widetilde{\Delta}_R = \Delta_R - \eta_R \epsilon_R \boxtimes \mathrm{id} - \mathrm{id} \boxtimes \eta_R \epsilon_R.$$

Also supply the compact Hall coproduct and counit matrices, primitive kernel bases, primitive idempotents, the quotient primitive comparison $\theta^{Q,\mathrm{Prim}}_R$, and the projection-commutator identity

$$\overline{\pi}^{\mathrm{Prim}}_R \theta^Q_{H,R} = \theta^{Q,\mathrm{Prim}}_R Q_{E,R} \pi^{\mathrm{Prim}}_R.$$

Transition-compatible primitive matrices and zero composition defects are required before the finite square becomes a pro-object statement. The packet `certificates/orientation/rhomred_orientation_primitive` verifies this primitive-projection descent criterion and records that the current compact Hall source, primitive-kernel, primitive-projection, and transition rows are not supplied.

(xxxii) *Weyl-equivariant lifts* $(O1)^+$. For each $i \in \{1,2,3\}$, Proposition 0.110 identifies row 306: construct the lift $\tau_{i,R,S} : o_{R,S} \longrightarrow s^*_{\partial_i} o_{R,s_{\partial_i}S}$ from source and target orientation-line rows, a retained type-II wall correspondence, determinant-square compatibility, and transport of the full quotient-orientation cocycle. The packet `certificates/orientation/rhomred_weyl_wall_transport_tau` verifies this transport criterion and records that the current orientation-line, retained wall-object, and τ_i-transport rows are not supplied. Proposition 0.111 identifies row 307: after the two lifts $\tau_{i,R,S}$ and $\tau_{i,R,s_{\partial_i}S}$ have been supplied, verify the double-reflection source loop, determinant-square compatibility, quotient-cocycle transport around the two-edge loop, and the finite τ^2-defect row

$$(s^*_{\partial_i} \tau_{i,R,s_{\partial_i}S}) \circ \tau_{i,R,S} = \mathrm{id}_{o_{R,S}}.$$

The packet `certificates/orientation/rhomred_weyl_tau_square` verifies this involutivity criterion and records that the current τ_i-transport and τ_i^2-defect rows are not supplied. Definition 0.112 identifies row 308: define the finite partial action groupoid $\mathcal{G}_R$ by object rows for retained quotient-orientation strata, generating type-II wall arrows, source and target maps, identities, inverses, and partial composition rows with zero associativity defects. The packet `certificates/orientation/rhomred_weyl_partial_action_groupoid` verifies this groupoid-datum criterion and records that the current object, arrow, inverse, and composition rows for $\mathcal{G}_R$ are not supplied. Definition 0.113 identifies row 309: for each composable pair

$$S \xrightarrow{g} T \xrightarrow{h} U$$

in $\mathcal{G}_R$, define $c_{o,R}(h, g; S)$ by

$$g^* \tau_h \circ \tau_g = (-1)^{c_{o,R}(h,g;S)} \tau_{hg}.$$

The packet `certificates/orientation/rhomred_weyl_orientation_cocycle` verifies this cocycle-definition criterion and records that the current groupoid, lift, composable-pair, normalization, and cocycle-defect rows are not supplied. Proposition 0.114 identifies row 310: after $c_{o,R}$ is a supplied closed finite 2-cocycle with coefficients in the orientation local system, choose bases for

$$C^1(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o}) \quad \text{and} \quad C^2(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o}),$$

write the coboundary matrix d^1_R, and prove

$$\operatorname{rank}_{\mathbb{F}_2}(d^1_R) = \operatorname{rank}_{\mathbb{F}_2}[d^1_R \mid v(c_{o,R})].$$

The packet `certificates/orientation/rhomred_weyl_orientation_cocycle_class_vanishing` verifies this vanishing criterion and records that the current cocycle-value, closedness, cochain-complex, coboundary-matrix, and rank rows are not supplied. Proposition 0.115 identifies row 311: after the row 310 rank certificate, choose a vector

$$b_R \in C^1(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o})$$

and verify

$$d^1_R b_R = v(c_{o,R}).$$

The packet `certificates/orientation/rhomred_weyl_orientation_cocycle_cochain_trivialization` verifies this cochain-trivialization criterion and records that the current cochain-vector, gauge-normalization, and coboundary-defect rows are not supplied. Proposition 0.116 identifies row 312: for every successor $R' \leq R$, construct transition cochain maps

$$\rho^i_{RR'} : C^i(\mathcal{G}_{R'}; \underline{\mathbb{F}}_{2,o}) \to C^i(\mathcal{G}_R; \underline{\mathbb{F}}_{2,o}) \quad (i = 1, 2)$$

with

$$d^1_R \rho^1_{RR'} = \rho^2_{RR'} d^1_{R'}, \qquad \rho^2_{RR'} v(c_{o,R'}) = v(c_{o,R}), \qquad \rho^1_{RR'}(b_{R'}) = b_R.$$

The packet `certificates/orientation/rhomred_weyl_orientation_cocycle_cochain_transition` verifies this transition-compatibility criterion and records that the current cochain-map, cocycle-pullback, b_R-defect, gauge, and two-step coherence rows are not supplied. Proposition 0.117 identifies row 313: for every retained S and type-II reflection s_{∂_i}, transport the quotient-orientation cochains by $\tau_{i,R,S}$, form

$$\Delta^{\omega}_{i,R,S} = \Theta^1_{i,R,S}(\eta_{R,S}) - \eta_{R,s_{\partial_i}S},$$

and record the class

$$\omega_{i,R,S} = [\Delta^{\omega}_{i,R,S}] \in H^1(\mathfrak{M}^{\mathrm{red}}_{R,s_{\partial_i}S};\mathbb{F}_2) \oplus \bigoplus_H H^1(BH;\mathbb{F}_2).$$

The packet `certificates/orientation/rhomred_weyl_orientation_torsor_defects` verifies this computation criterion and records that the current quotient-orientation representative, transport, H^1-basis, and defect-vector rows are not supplied. Proposition 0.118 identifies row 314: for every retained S and type-II reflection s_{∂_i}, use the row 313 coordinates and verify

$$d\Delta^{\omega}_{i,R,S} = 0, \qquad v(\omega_{i,R,S}) = 0, \qquad \mathrm{rank}(v(\omega_{i,R,S})) = 0$$

with complete retained-wall coverage. The packet `certificates/orientation/rhomred_weyl_orientation_torsor_de` verifies this vanishing criterion and records that the current computed defect-vector, closedness, zero-certificate, and retained coverage rows are not supplied. Proposition 0.119 identifies row 315: for every retained S and type-II reflection s_{∂_i}, transport the Cech–Borel reduced-gerbe cocycle and form

$$\Delta^{\mathrm{red}}_{i,R,S} = \Theta^{2,\mathrm{red}}_{i,R,S}(\alpha^{\mathrm{red}}_{R,S}) - \alpha^{\mathrm{red}}_{R,s_{\partial_i}S}.$$

Then prove

$$[\Delta^{\mathrm{red}}_{i,R,S}] = 0 \quad \text{in} \quad H^2(\mathcal{B}_{R,s_{\partial_i}S};\mathbb{F}_2)$$

by a finite coboundary witness, with complete retained-wall coverage. The packet `certificates/orientation/rhomred_weyl_alpha_red_preservation` verifies this preservation criterion and records that the current source-target α^{red}-cocycles, Weyl cochain maps, coboundary witnesses, zero class-rank certificates, and retained coverage rows are not supplied. Proposition 0.120 identifies row 316: for every retained free E-quotient stratum and type-II reflection, compute the induced map

$$M^E_{i,R,S}\colon H^2(BE;\mathbb{F}_2) \to H^2(BE;\mathbb{F}_2)$$

in the basis u_1, u_2, and verify

$$M^E_{i,R,S}\begin{pmatrix} a_1(R,S) \\ a_2(R,S) \end{pmatrix} - \begin{pmatrix} a_1(R,s_{\partial_i}S) \\ a_2(R,s_{\partial_i}S) \end{pmatrix} = 0.$$

The packet `certificates/orientation/rhomred_weyl_alpha_e_free_preservation` verifies this preservation criterion and records that the current source-target a_1, a_2 rows, induced $H^2(BE)$-matrices, zero-vector certificates, cochain-transport defects, and retained coverage rows are not supplied. Proposition 0.121 identifies row 317: for every retained finite-stabilizer stratum and type-II reflection, construct the stabilizer isomorphism

$$\varphi_{i,R,S}\colon H_{R,S} \xrightarrow{\sim} H_{R,s_{\partial_i}S},$$

compute the induced matrix

$$M^H_{i,R,S}\colon H^2(BH_{R,S};\mathbb{F}_2) \to H^2(BH_{R,s_{\partial_i}S};\mathbb{F}_2),$$

and verify

$$M^H_{i,R,S}v(\beta^{H_{R,S}}_{R,S}) - v(\beta^{H_{R,s_{\partial_i}S}}_{R,s_{\partial_i}S}) = 0$$

with edge-reduction compatibility and complete retained-wall coverage. The packet `certificates/orientation/rhomred_weyl_finite_stabilizer_beta_preservation` verifies this preservation criterion and records that the current stabilizer isomorphisms, source-target β^H-coordinates, $H^2(BH)$-matrices, edge-transport defects, zero-vector certificates, and retained coverage rows are not supplied. Proposition 0.122 identifies row 318: for every retained finite-stabilizer stratum and type-II reflection, use the same stabilizer isomorphism

$$\varphi_{i,R,S}\colon H_{R,S} \xrightarrow{\sim} H_{R,s_{\partial_i}S}$$

and compute the induced degree-one matrix

$$N^H_{i,R,S}\colon H^1(BH_{R,S};\mathbb{F}_2) \to H^1(BH_{R,s_{\partial_i}S};\mathbb{F}_2).$$

After rows 288 and 289 have supplied the source and target λ^H-values and zero rows, verify

$$N^H_{i,R,S}v(\lambda^{H_{R,S}}_{R,S}) - v(\lambda^{H_{R,s_{\partial_i}S}}_{R,s_{\partial_i}S}) = 0$$

with orientation-line action transport and complete retained-wall coverage. The packet `certificates/orientation/rhomred_weyl_finite_stabilizer_lambda_zero_preservation` verifies this preservation criterion and records that the current source-target λ^H-coordinates, stabilizer isomorphisms, $H^1(BH)$-matrices, orientation-line action transport defects, rank-zero rows, and retained coverage rows are not supplied. Definition 0.123 identifies row 319: after the strictified Weyl action, Coxeter cochain compatibility, torsor-defect vanishing, and rows 315–318 have supplied the complete quotient-oriented transport, record the residual simple-reflection signs

$$\sigma^o_{i,R} \in \{\pm 1\}, \qquad i = 1, 2, 3,$$

prove their stratum and transition independence, and define

$$\epsilon_{o,R}(s_{\partial_{i_1}}\cdots s_{\partial_{i_m}}) = \prod_{a=1}^{m} \sigma^o_{i_a,R}.$$

The no-braid Coxeter presentation of $W^{(2)}(\Lambda^{2,1}_{II})$ makes this formula a homomorphism once the finite generator signs are supplied. The packet `certificates/orientation/rhomred_weyl_orientation_character_de` verifies this definition criterion and records that the current strict Weyl action rows, generator-sign rows, stratum-independence rows, transition-compatible character rows, and inverse-limit character rows are not supplied. The automorphic Maass character packet is imported only as a firewall: ν_{Δ_5} is the comparison target, not the definition of ϵ_o. Proposition 0.124 identifies row 320: for each simple type-II wall, supply the retained $(O2_{\mathrm{atlas}})$ wall chart with

$$K^{\mathrm{nor}}_{\partial_i,R,S} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}], \qquad s_{\partial_i}(u_i) = -u_i, \qquad s_{\partial_i}(v_{i,R,S}) = v_{i,R,S},$$

and compute

$$\sigma^o_{i,R,S} = \frac{s^*_{\partial_i}(v_{i,R,S}u_i)}{v_{i,R,S}u_i} = -1.$$

After stratum and transition independence are proved, this gives $\epsilon_{o,R}(s_{\partial_i}) = -1$, hence $\epsilon_o(s_{\partial_i}) = -1$ in the inverse limit. The packet `certificates/orientation/rhomred_weyl_orientation_simple_sign_computation` verifies this local sign criterion and records that the current wall-object, rank-one chart, invariant-unit, coordinate-flip, local-sign, generator-sign, stratum-independence, and transition-compatibility rows are not supplied. Proposition 0.125 identifies row 321: after row 319 supplies a source-side character and row 320 supplies the three source generator values $\epsilon_{o,R}(s_{\delta_i}) = -1$, the determinant character $\det_W$ has the same generator values, and the no-braid type-II Coxeter presentation forces

$$\epsilon_{o,R} = \det_W \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II}).$$

The packet `certificates/orientation/rhomred_weyl_orientation_determinant_comparison` verifies this determinant criterion and records that the source character, source generator signs, equality-on-generators rows, transition compatibility rows, and inverse-limit comparison rows are not supplied. Proposition 0.126 identifies row 322: once row 321 proves $\epsilon_o = \det_W$ on $W^{(2)}(\Lambda^{2,1}_{II})$, the automorphic packet supplies

$$\nu_{\Delta_5}\big|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det_W,$$

and transitivity gives

$$\epsilon_o = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda^{2,1}_{II})}.$$

The packet `certificates/orientation/rhomred_weyl_orientation_maass_comparison` verifies this comparison criterion and records that the source determinant comparison and inverse-limit orientation character remain unsupplied. Proposition 0.127 identifies row 323. The three type-I chamber transpositions

$$s_{f_{-2}-f_2}, \qquad s_{f_2-f_3}, \qquad s_{f_{-2}-f_3}$$

generate $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$, and the Maass packet records

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_{-2}-f_3}) = +1.$$

Thus ν_{Δ_5} is trivial on the chamber automorphism group. The packet `certificates/automorphic/delta5_s3_chamber_m` verifies this automorphic statement and records that it does not extend ϵ_o to $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes S_3$. Proposition 0.128 identifies row 324. Since S_3 permutes the three type-II simple reflections and the row 320 source values, once supplied, are all equal to -1, the abstract character $\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ is S_3-invariant. Thus it admits two abstract extensions to the semidirect product, according to the two characters of S_3:

$$\widetilde{\epsilon}_\chi(w,\sigma) = \epsilon_o(w)\chi(\sigma), \qquad \chi \in \{1, \mathrm{sgn}\}.$$

The Maass-compatible choice is $\chi = 1$, because row 323 proves $\nu_{\Delta_5}|_{S_3} = 1$. The packet `certificates/orientation/rhomred_weyl_orientation_s3_extension_decision` verifies this abstract decision and records that the source-side semidirect Hall lifts, orientation-line transport, square-root compatibility, quotient-orientation transport, and semidirect relation rows are not supplied. Row 325 constructs those lifts if the compact-source extension is claimed. Proposition 0.129 identifies row 325. A geometric extension is not a new scalar character; it is an action of $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes S_3$ on the retained compact Hall correspondence package. For each chamber transposition σ it requires Hall correspondences

$$\mathcal{M}_{R,S} \longleftarrow \mathcal{Z}_{\sigma,R,S} \longrightarrow \mathcal{M}_{R,\sigma S},$$

compatibility with Hall product, coproduct, unit, counit, primitives, pairing, radicals, and PBW filtrations, a Picard-groupoid orientation transport

$$\theta^{o}_{\sigma,R,S} : o_{R,S} \xrightarrow{\sim} \sigma^{*} o_{R,\sigma S},$$

determinant square-root compatibility, quotient-orientation transport, the S_3 relations, the semidirect conjugation relations with the type-II lifts τ_i, and transition-compatible inverse-limit rows. The packet `certificates/orientation/rhomred_weyl_semidirect_hall_lifts` verifies this construction criterion and records that the present compact Hall packet, type-II partial-action groupoid, τ_i transport packet, and transition-orientation packet do not supply the required rows. Proposition 0.130 identifies row 326. The Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ is generated by the real simple reflections $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$. The imaginary simple roots of $\mathfrak{g}_{\Delta_5}$ are Borcherds correction data: they contribute multiplicities $m(a)$ and $\tau(a)$ to the denominator formula, but they do not define reflection generators of $W^{(2)}(\Lambda^{2,1}_{II})$. Consequently an as-yet-undefined local monodromy around an imaginary divisor component is not a Weyl value $\epsilon_o(w)$. The packet `certificates/orientation/rhomred_weyl_imaginary_divisor_firewall` verifies this firewall and records that row 327 must define local divisor monodromy as a separate datum. Definition 0.131 identifies row 327. A local divisor monodromy datum is a source-side compact Hall datum: a retained divisor component $\mathcal{D}_{R,S,\eta} \subset \mathcal{M}_{R,S}$, an analytic or formal punctured normal tube $N^{\times}_{R,S,\eta}$, a meridian $\gamma_{R,S,\eta}$, the restricted oriented reduced coefficient system $K_R = \Phi^{\mathrm{red}}_R \otimes o_R$, the meridian operator

$$T_{R,S,\eta} = \rho_{K_R}(\gamma_{R,S,\eta}),$$

and transition compatibility for these rows. Its domain is the set of supplied compact Hall divisor rows, not the Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$. The packet `certificates/orientation/rhomred_local_divisor_monodromy_c` verifies this definition criterion and records that no divisor component, normal tube, meridian, reduced local-system, Pfaffian-line, monodromy-operator, or transition row is presently supplied. Proposition 0.132 identifies row 328. The Maass character ν_{Δ_5} is an automorphic character evaluated on $W^{(2)}(\Lambda^{2,1}_{II})$ and on the chamber automorphism group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$; local divisor monodromy is a meridian operator $T_{R,S,\eta}$ acting on the local compact Hall coefficient system over $N^{\times}_{R,S,\eta}$. There is no defined expression $\nu_{\Delta_5}(\gamma_{R,S,\eta})$ without a separate meridian-to-automorphic comparison map. The packet `certificates/orientation/rhomred_local_divisor_monodromy_maass_irrelevance` verifies this domain separation and keeps the local monodromy operator rows and the row 329 simple-wall orientation computations outside the Maass multiplier computation. Proposition 0.133 identifies row 329. On a supplied rank-one chart over each simple type-II wall $\mathbb{H}_{\delta_i}$, with

$$K^{\mathrm{cross}}_{i,R,S} = [\mathbb{R} \xrightarrow{u_i} \mathbb{R}], \qquad \mathrm{pf}^{\mathrm{loc}}_{i,R,S} = v_{i,R,S} u_i,$$

$s_{\delta_i}(u_i) = -u_i$, and $s_{\delta_i}(v_{i,R,S}) = v_{i,R,S}$, the chart orientation sign is -1. Thus the three supplied chart rows would compute the orientation vector $(-1,-1,-1)$. The packet `certificates/orientation/rhomred_three_simple_wall_chart_orientation` verifies this row 329 computation criterion and records that the three retained wall objects, wall charts, invariant units, quotient-orientation rows, local sign rows, and transition rows are not presently supplied. Proposition 0.134 identifies row 330. For the primitive isotropic boundary rays

$$a_{12} = \delta_1 + \delta_2, \qquad a_{13} = \delta_1 + \delta_3, \qquad a_{23} = \delta_2 + \delta_3,$$

the target type-II chamber and $W_{\leq 3}$ packets record a_{ij} : 10|0 and 10 = 1 + 9. A supplied compact Hall isotropic chart realizing ten even and zero odd oriented summands would therefore have local chart sign +1, so the three supplied rows would compute (+1, +1, +1). The packet `certificates/orientation/rhomred_isotropic_aij_chart_orientation` verifies this row 330 criterion and records that source isotropic chart rows, quotient orientations, determinant-square rows, source parity rows, and transition rows are not presently supplied. Proposition 0.135 identifies row 331. For the first timelike degree

$$\delta_{123} = \delta_1 + \delta_2 + \delta_3,$$

the target presentation packets record δ_{123} : 29|93, 29 = 2 + 27, and 29 − 93 = −64. A supplied compact Hall chart realizing this source parity decomposition would therefore have local chart sign $(-1)^{93} = -1$. The packet `certificates/orientation/rhomred_delta123_chart_orientation` verifies this row 331 criterion and records that the compact source δ_{123} chart, source parity rows, source real–real–real and mixed real–isotropic rows, source odd imaginary-simple rows, quotient orientation, determinant-square row, and transition row are not presently supplied. Proposition 0.136 identifies row 332. Compatibility of the orientation line with the 29|93 split is the commutative Picard-groupoid square

$$o^{\otimes 2}_{123,R,S} \simeq \det K^{\mathrm{red}}_{123,R,S} \simeq \det K_{\bar{0},123,R,S} \otimes (\det K_{\bar{1},123,R,S})^{-1},$$

together with a source parity involution, source comparison rows realizing 29 = 2 + 27 and 93, quotient-orientation null-trivialisations, parity-preserving transitions, and strict orientation transport. The packet `certificates/orientation/rhomred_delta123_parity_compatibility` verifies this row 332 criterion and records that the source parity block, parity involution, source-to-target comparison, orientation square-root, determinant-square, quotient orientation, parity-transition, and Picard-groupoid transition rows are not presently supplied. Proposition 0.137 identifies row 333. Compatibility with negative roots requires a compact-source duality lift $\mathcal{U}_{\gamma,R,S} \to \mathcal{U}_{-\gamma,R,S}$, parity decompositions on both signs, a source positive–negative Hopf-pairing matrix with off-parity blocks zero, radical quotient rows, and a commuting Picard-groupoid square

$$\det K^{\mathrm{red}}_{-\gamma,R,S} \simeq (\det K^{\mathrm{red}}_{\gamma,R,S})^{-1} \otimes \varepsilon^{\mathrm{CY}_3}_{\gamma,R,S}$$

lifted to the orientation square roots. The packet `certificates/orientation/rhomred_negative_root_orientation_` verifies this row 333 criterion and records that the retained dual-closure rows and target formal positive–negative pairing blocks do not supply compact Hall negative-root charts, source pairing matrices, Hopf-pairing identities, radical rows, orientation square roots, Serre-duality sign lines, quotient-orientation transport, or Picard-groupoid transition rows. Proposition 0.138 identifies row 334. A Chevalley anti-involution on orientation data is an even involutive source anti-automorphism of the reduced compact Hall primitive quotient, sending $e_i \leftrightarrow f_i$, fixing the Cartan part, preserving parity, satisfying the source bracket identity $\vartheta([x, y]) = [\vartheta(y), \vartheta(x)]$, stabilizing the radical, and lifting to Picard-groupoid isomorphisms

$$o_{\gamma,R,S} \simeq o^{\vee}_{-\gamma,R,S} \otimes \varepsilon^{\mathrm{CY}_3}_{\gamma,R,S}$$

compatible with quotient orientations and transitions. The packet `certificates/orientation/rhomred_orientation_` verifies this row 334 criterion and records that target Chevalley relations, retained dual closure, formal pairing pushforward, and negative-root compatibility do not supply source simple representatives, source bracket or Chevalley relation matrices, source Hopf-pairing matrices, radical quotient rows, orientation square roots, Serre sign lines, or Picard-groupoid transition rows. Proposition 0.139 identifies row 335. The orientation pairing is Frobenius only after the compact Hall source supplies product and coproduct matrices, unit/counit and bialgebra identities, a source Hopf-pairing matrix, Hopf-adjointness rows, primitive Frobenius-cyclicity rows, radical kernel and quotient-splitting rows, radical ideal/coideal stability, orientation cyclic-sign squares, and transition plus Mittag–Leffler rows for the pairing kernels. The packet `certificates/orientation/rhomred_orientation_frobenius_pairing` verifies this row 335 criterion and records that formal Hall-pairing degree compatibility, target pairing blocks, scalar traces, and the row 334 Chevalley criterion do not prove Frobenius cyclicity or quotient nondegeneracy for the source orientation pairing. Proposition 0.140 identifies row 336. The Frobenius trace has degree -3 only after the compact source supplies a trace functional on the reduced radical quotient, a source grading convention, a trace-degree row with defect zero, trace–pairing rows $\langle x, y\rangle^o = \mathrm{tr}^o(xy)$, trace-cyclicity rows, orientation trace-line squares, the Calabi–Yau Serre shift, and transition plus Mittag–Leffler control. The packet `certificates/orientation/rhomred_orientation_frobenius_trace_degree` verifies this row 336 criterion and records that protected integration Gram s-degree, level-Z protected trace scaffolds, formal degree-zero pairings, and the row 335 Frobenius criterion do not compute the source trace degree -3. Proposition 0.141 identifies row 337. The orientation signs agree with Calabi–Yau threefold Serre duality only after the compact source supplies the reduced Serre-duality isomorphism

$$K^{\mathrm{red}}_{\gamma,R,S} \simeq \left(K^{\mathrm{red}}_{-\gamma,R,S}\right)^{\vee}[-3],$$

the source grading and determinant-ordering convention, the Calabi–Yau sign-exponent row, the Picard-groupoid Serre sign line, the determinant-duality square, sign comparisons with Thom–Sebastiani gluing, Chevalley adjunction, Frobenius cyclicity, and trace cyclicity, plus radical, quotient-orientation, transition, and Mittag–Leffler control. The packet `certificates/orientation/rhomred_orientation_cy3_serre_signs` verifies this row 337 criterion and records that PTVV shifted symplectic input, target negative-dual blocks, formal pairing degree compatibility, and the row 333–336 criteria do not compute the source Calabi–Yau Serre signs. Proposition 0.142 identifies row 338. The reduced Calabi–Yau threefold pairing is nondegenerate modulo the radical only after the compact source supplies the Hopf-pairing matrix $G^{\mathrm{CY}_3}_{\gamma,R,S}$, the left and right kernel rows, the identification of these kernels with the recorded radical, quotient splitting matrices $Q_{\gamma,R,S}$ and $Q_{-\gamma,R,S}$, a full-rank quotient matrix

$$Q_{\gamma,R,S}\, G^{\mathrm{CY}_3}_{\gamma,R,S}\, Q^{t}_{-\gamma,R,S},$$

radical ideal/coideal rows, transition-radical rows, and pairing-kernel Mittag–Leffler rows. The packet `certificates/orientation/rhomred_orientation_cy3_pairing_nondegeneracy` verifies this row 338 criterion and records that formal degree-zero pairings, target pairing blocks, the row 333, 335, and 337 criteria, transition-radical obstruction ledgers, and pairing-kernel obstruction ledgers do not prove quotient nondegeneracy.

(xxxiii) *Transition preservation of orientations.* For every successor $R' \leq R$, construct the Picard-groupoid isomorphism $(t_{RR'})^* o_{R'} \simeq o_R$; verify the determinant square-root diagram, quotient Cech–Borel null-trivialization transport, finite-stabilizer and zero-linearization transport, Thom–Sebastiani transition compatibility, Weyl-lift transition compatibility, Coxeter-cochain transition compatibility, strict composition, and $R^1 \varprojlim = 0$.

(xxxiv) *Transition preservation of reduced vanishing cycles.* For every successor $R' \leq R$, construct the transition graph of the oriented d-critical charts; verify potential compatibility up to the recorded non-degenerate quadratic stabilization, K3 semiregularity cosection pullback, reduced obstruction-complex compatibility, the graph-pullback isomorphism $p_R^* \Phi_R^{\mathrm{red}} \simeq p_{R'}^* \Phi_{R'}^{\mathrm{red}}$, BBDJS perverse normalization, Thom–Sebastiani vanishing-cycle compatibility, proper base change, strict composition, and $R^1 \varprojlim = 0$.

(xxxv) *Local wall-sign datum.* At each generic type-II wall point $\mathbb{H}_{\partial_i}$, with the rank-one local model $K_{\partial_i}^{\mathrm{cross}} = [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$, record the local generic sign $\epsilon_{o,R}^{\mathrm{loc}}(s_{\partial_i}) = -1$ of Proposition 0.205.

(xxxvi) *Finite orientation proof tables.* Populate the $(O1)$ and $(O1)^+$ tables of §0.2:

orientation_lines, ts_multiplicativity, quotient_borel, finite_stabilizers, weyl_lifts, coxeter_cocycles, transitions, scala

The finite-stabilizer table must include the rank-two mixed coefficient $A_{12}^{(N)}$ for two-primary rank-two stabilizers, not only cyclic restrictions. The scalar firewall must exclude scalar trace, squared determinant, OP scalar branch, and Maass-character values as orientation data.

(xxxvii) Obtain $(o_R, \mathfrak{Q}_R^{or}, \{\tau_{i,R,S}\}, \epsilon_{o,R}^{\mathrm{loc}})$.

6.4.0.15 Identities. o_R is a Picard-line section squaring to the reduced determinant; the four Cech–Borel classes vanish with compatible null-trivializations; the lifts $\tau_{i,R}$ satisfy $\tau_{i,R}^2 = 1$ and the type-II Coxeter relations; multiplicative transport agrees on flag stacks.

6.4.0.16 Obstruction classes. This datum is the entire site of $(O1)$ (strong reduced orientation on the K3×E-quotient self-Ext frame bundle) and of $(O1)^+$ (Weyl-equivariant transport along $W^{(2)}(\Lambda_{II}^{2,1})$ reflections). Discharging $(O1)$ requires the simultaneous vanishing and compatible null-trivialization of the four Cech–Borel classes on every retained stratum; discharging $(O1)^+$ requires $[c_{o,R}] = 0$, the Coxeter coherence, and the vanishing of all torsor defects $\omega_{i,C}$. The local wall-sign datum is the input to $(O2)$ in the Pfaffian wall normal form. The transition-orientation packet `certificates/orientation/transition_orientation_preservation` verifies `TRANSITION_ORIENTATION_PRESERVATION_OBSTRUCTION_VERIFIED`: the stack-transition input, orientation-line pullback isomorphisms, determinant square compatibility, quotient Cech–Borel null-trivialization transport, finite-stabilizer and zero-linearization transport, Thom–Sebastiani transition compatibility, Weyl-lift transition compatibility, Coxeter-cochain transition compatibility, strict Picard-groupoid transition, and orientation Mittag–Leffler rows are not yet supplied. Proposition 0.145 identifies row 339. Orientation classes survive the inverse limit only after the compact orientation source supplies finite-stage orientation rows

$$o_R^{\otimes 2} \simeq \det K_R^{\mathrm{red}}, \qquad \mathrm{cl}(o_R) = 0,$$

Picard pullback isomorphisms $\theta_{RR'}^o : (t_{RR'})^* o_{R'} \simeq o_R$, determinant-square, quotient-orientation, Thom–Sebastiani, Weyl, and Coxeter transition defect rows, strict identity and composition rows, an orientation Mittag–Leffler row with $R^1 \varprojlim = 0$, and a limit class row $o_\infty = \varprojlim_R (o_R, \theta_{RR'}^o)$. The packet `certificates/orientation/rhomred_orientation_class_limit_survival` verifies this row 339 criterion and records that the reduced orientation ledger, square-root obstruction packet, transition orientation obstruction ledger, and row 338 nondegeneracy criterion do not prove inverse-limit survival of orientation classes. Proposition 0.146 identifies row 340. No new orientation class appears at the limit only after the compact orientation source supplies the pro-Picard comparison

$$\Xi : \mathrm{Pic}^{or}_{\infty} \to \varprojlim_{R} \mathrm{Pic}^{or}_{R},$$

finite-stage effectivity rows for the determinant square, quotient-orientation, Thom–Sebastiani, Weyl, and Coxeter conditions, separatedness rows excluding pro-classes with trivial finite restrictions, automorphism and object-class Mittag–Leffler rows with zero $R^1 \varprojlim$, and a zero-defect comparison

$$\pi_0 \, \mathrm{Pic}^{or}_{\infty} \simeq \varprojlim_{R} \pi_0 \, \mathrm{Pic}^{or}_{R} .$$

The packet `certificates/orientation/rhomred_orientation_no_new_limit_class` verifies this row 340 criterion and records that row 339 survival, the reduced-orientation ledger, the square-root obstruction packet, and the transition-orientation obstruction ledger do not prove pro-Picard separatedness or exclude phantom orientation classes. The transition-vanishing-cycle packet `certificates/vanishing_cycles/transition_vanishing_cycle_preservation` verifies `TRANSITION_VANISHING_CYCLE_PRESERVATION_OBSTRUCTION_VERIFIED`: the stack-transition input, orientation-transition input, d-critical chart transitions, potential compatibility, quadratic stabilization, cosection pullback, reduced obstruction-complex maps, graph-pullback vanishing-cycle isomorphisms, perverse normalization, support and base-change rows, Thom–Sebastiani vanishing-cycle transition rows, strict composition, and vanishing-cycle Mittag–Leffler rows are not yet supplied.

6.5 Hall correspondences and primitive part

6.5.0.1 Hypotheses. The oriented hybrid carrier from the reduced orientation datum; the extension stack $\mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'}$; the two-step flag stack $\mathfrak{F}^{(2),\mathrm{geom}}_{R,\widehat{c},\widehat{c}',\widehat{c}''}$; the BBDJS reduced Thom–Sebastiani transport; six-functor admissibility of p^*, $q_!$, $p_!$ from the finite moduli datum. The input includes compactified retained filtration/flag-Quot correspondences or an explicitly admissible exceptional compact-support model, finite inertia, base-change and projection-formula witnesses, quotient-after-correspondence pseudofunctoriality, and HN-transition compatibility.

6.5.0.2 Hall correspondence datum. The finite Hall object

$$\mathcal{H}^{\mathrm{geom}}_{R} = \bigoplus_{\widehat{c}\in\widehat{\Gamma}_R} H^{\bullet}_{c}\big(\mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}}, \Phi_{R,c} \otimes o_{R,c}\big);$$

the Hall product $m_R = q_! \, \mathrm{TS}^{\mathrm{red}} \, p^*$ and Hall coproduct $\Delta_R = p_! \, (\mathrm{TS}^{\mathrm{red}})^{-1} \, q^*$; the finite-stage Hall bialgebra object $(\mathcal{H}^{\mathrm{geom}}_{R}, m_R, \Delta_R, \eta_R, \epsilon_R)$; the primitive part

$$P^{\mathrm{geom}}_{R} = \ker\big(\bar{\Delta}_R : \ker \epsilon_R \longrightarrow \ker \epsilon_R \otimes \ker \epsilon_R\big);$$

the normal-ordered Gram descent $P^{\Pi,\mathrm{geom}}_{R} = \overline{\Pi}^{\Theta_R}_{X,*} P^{\mathrm{geom}}_{R}$; the Γ^{Π}_{R}-graded super-Lie bracket $[\,\bar{x}, \bar{y}\,]_{\mathrm{red}}$ of Theorem 4.3; the Hopf pairing matrix $G_{\gamma,ab}$ and radical K_{γ}. The product m_R alone is not a primitive datum: Proposition 0.165 requires the counit, coproduct, and reduced-coproduct kernel rows before P^{geom}_{R} is a well-defined finite source subspace. Hopf pairing, radical, quotient, and transition data are subsequent recognition data on that primitive subspace. Definition 0.150 identifies row 341. The finite compact Hall object is the $\widehat{\Gamma}_R$-graded direct sum

$$\mathcal{H}_R^{\mathrm{geom}} = \bigoplus_{\widehat{c}\in\widehat{\Gamma}_R} H_*^{\mathrm{BM}}\Big(\mathfrak{M}_{R,\widehat{c}}^{\mathrm{red}}, \Phi_{R,\widehat{c}}^{\mathrm{red}} \otimes o_{R,\widehat{c}}\Big),$$

formed from retained rigidified finite-type substacks, reduced vanishing-cycle complexes, Joyce–Upmeier orientation lines, and an admissible compact-support Borel–Moore model. The packet `certificates/hall/finite_geometric_hall_object` verifies this definition and records that the current finite-moduli, orientation, vanishing-cycle, and compact Hall source packets do not yet supply the object-component rows, orientation-line rows, vanishing-cycle coefficient rows, source-degree rows, homogeneous basis provenance rows, or finite-rank rows needed to populate $\mathcal{H}_R^{\mathrm{geom}}$. Product, coproduct, Hopf pairing, radical quotient, PBW, primitive, and transition data begin only after this object has been supplied.

Definition 0.151 identifies row 342. The finite Hall product operator is the formal pull–Thom–Sebastiani–push operator

$$m_R^{\mathrm{red}} = q_! \,\mathrm{TS}^{\mathrm{red}}\, p^*$$

on the retained compactified extension correspondence. The packet `certificates/hall/finite_hall_product_operator` verifies the definition and records that the current rows do not yet prove existence of p^*, existence of $q_!$, independence of the compact-support compactification, product matrices, associativity, base-change, projection formula, reduced Thom–Sebastiani transport on the populated Hall correspondence, or transition compatibility. Thus row 342 names the operator; rows 343–345 and the later Hall-bialgebra rows supply the functorial and matrix data required for a populated product.

Proposition 0.152 identifies row 343. For a supplied retained extension morphism

$$p_{c,c'} : \overline{\mathfrak{E}}_{R,c,c'}^{\mathrm{ret}} \to \mathfrak{M}_{R,c}^{\mathrm{red}} \times \mathfrak{M}_{R,c'}^{\mathrm{red}},$$

finite type in the chosen constructible six-functor category, and for a supplied bounded constructible input coefficient

$$(\Phi_{R,c}^{\mathrm{red}} \otimes o_{R,c}) \boxtimes (\Phi_{R,c'}^{\mathrm{red}} \otimes o_{R,c'}),$$

the functor $p_{c,c'}^*$ exists and returns a bounded constructible coefficient on the extension stack. The packet `certificates/hall/finite_hall_product_pullback_functor` verifies this criterion and records that the current finite-moduli ledger has no extension-stack, finite-type flag, or source-map rows; the row 341 object is not populated; the row 342 product operator has no p^*-rows; and the compact-support model does not populate a pullback leg. This row proves the criterion for a supplied p-leg, not the existence of the actual retained $K3 \times E$ Hall product pullback.

Proposition 0.153 identifies row 344. For a supplied retained target morphism

$$q_{c,c'} : \overline{\mathfrak{E}}_{R,c,c'}^{\mathrm{ret}} \to \mathfrak{M}_{R,c+c'}^{\mathrm{red}},$$

and for a supplied bounded constructible coefficient $K_{\mathfrak{E}}$ after pullback and reduced Thom–Sebastiani transport, the exceptional pushforward leg exists if $q_{c,c'}$ is proper, or if a compact-support model

$$j_q : \overline{\mathfrak{E}}_{R,c,c'}^{\mathrm{ret}} \hookrightarrow \overline{\mathfrak{E}}_q, \qquad \overline{q} : \overline{\mathfrak{E}}_q \to \mathfrak{M}_{R,c+c'}^{\mathrm{red}}$$

is supplied in the chosen constructible six-functor formalism. The resulting functor is

$$q^{\mathrm{cs}}_{c,c',!}K_{\mathfrak{E}} = \overline{q}_* j_{q,!}K_{\mathfrak{E}},$$

and in the proper case it specializes to $q_{c,c',*}K_{\mathfrak{E}}$. The packet `certificates/hall/finite_hall_product_pushforward_functo` verifies this criterion and records that the current finite-moduli ledger has no extension-flag stack, retained q-map, properness, or coefficient row; the compact-support model has no map-population or choice-independence rows; row 342 supplies notation only; row 343 supplies only the pullback criterion; and the compact Hall source ledger has no source product matrix entries. This row proves the criterion for a supplied q-leg, not the existence of the actual retained $K3 \times E$ Hall product pushforward, and not the compactification-independence statement of row 345.

Proposition 0.154 identifies row 345. For a supplied retained Hall target map $q : \mathfrak{E} \to Z$, a supplied bounded constructible coefficient $K_{\mathfrak{E}}$, and two supplied compact-support models

$$j_i : \mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_i, \qquad \overline{q}_i : \overline{\mathfrak{E}}_i \to Z, \qquad q = \overline{q}_i \circ j_i, \qquad i = 1, 2,$$

the two compact-support pushforwards are identified once a common proper refinement

$$j_{12} : \mathfrak{E} \hookrightarrow \overline{\mathfrak{E}}_{12}, \qquad r_i : \overline{\mathfrak{E}}_{12} \to \overline{\mathfrak{E}}_i, \qquad \overline{q}_i \circ r_i = \overline{q}_{12}$$

and the comparison isomorphisms

$$r_{i,*} j_{12,!}K_{\mathfrak{E}} \simeq j_{i,!}K_{\mathfrak{E}}$$

are supplied. Proper functoriality gives

$$\overline{q}_{i,*} j_{i,!}K_{\mathfrak{E}} \simeq \overline{q}_{12,*} j_{12,!}K_{\mathfrak{E}},$$

hence the two choices of compactification give the same $q^{cs}_!$ for that pair. The packet `certificates/hall/finite_hall_produc` verifies this common-refinement criterion and records that the current compact-support model has no choice-independence rows; the finite-moduli ledger has no retained q-map, compactification-pair, or common-refinement row; row 344 proves only existence for a supplied compactification; and the compact Hall source ledger has no source product matrix entries. This row proves the independence criterion for a supplied compactification pair, not independence for the actual retained $K3 \times E$ Hall product.

Proposition 0.155 identifies row 346. For three retained charges c_1, c_2, c_3, associativity of the compact Hall product follows once the four binary product legs

$$(c_1, c_2), \qquad (c_1 + c_2, c_3), \qquad (c_2, c_3), \qquad (c_1, c_2 + c_3)$$

are populated and choice-independent, and once a retained two-step extension flag stack

$$\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$$

maps to the two iterated correspondence fibre products by comparison maps λ and ρ. Base-change, projection-formula, and reduced Thom–Sebastiani comparison rows identify both parenthesizations on this common flag stack; compactification independence identifies the two final compact-support pushforwards to $\mathfrak{M}^{\mathrm{red}}_{R,c_1+c_2+c_3}$. The resulting isomorphism is

$$m^{\mathrm{red}}_{c_1+c_2,c_3}\big(m^{\mathrm{red}}_{c_1,c_2}(K_1, K_2), K_3\big) \cong m^{\mathrm{red}}_{c_1,c_2+c_3}\big(K_1, m^{\mathrm{red}}_{c_2,c_3}(K_2, K_3)\big).$$

The packet `certificates/hall/finite_hall_product_associativity_two_step_flags` verifies this two-step-flag criterion and records that the current compact Hall product rows, compact Hall two-step flag rows, base-change/projection/Thom–Sebastiani rows, quotient-descent rows, transition rows, product matrices, and Hall-bialgebra associativity rows are not supplied. The hybrid two-step flag and wordwise associativity packets remain evidence for the method, not populated compact Hall m_R-associativity.

Definition 0.157 identifies row 347. For the retained compactified extension correspondence

$$\mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'} \xleftarrow{p_{c,c'}} \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \xrightarrow{q_{c,c'}} \mathfrak{M}^{\mathrm{red}}_{R,c+c'},$$

the finite compact Hall coproduct operator is the formal pull–inverse-Thom–Sebastiani–push expression

$$\Delta^{\mathrm{red}}_R = p^{\mathrm{cs}}_{c,c',!}\,(\mathrm{TS}^{\mathrm{red}}_{c,c'})^{-1}\,q^*_{c,c'}.$$

After Borel–Moore Kunneth identification it is read as a map

$$\mathcal{H}^{\mathrm{geom}}_{R,c+c'} \to \mathcal{H}^{\mathrm{geom}}_{R,c} \otimes \mathcal{H}^{\mathrm{geom}}_{R,c'}.$$

The packet `certificates/hall/finite_hall_coproduct_operator` verifies the definition and records that the current rows do not yet prove existence of q^*, existence of $p_!$, independence of the compact-support compactification, inverse Thom–Sebastiani transport, coproduct matrices, counit rows, coassociativity, bialgebra compatibility, or transition compatibility. Thus row 347 names the coproduct operator; rows 348–350 and the later Hall-bialgebra rows supply the functorial and matrix data required for a populated coalgebra.

Proposition 0.158 identifies row 348. For a supplied retained splitting morphism

$$p_{c,c'} : \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \to \mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'},$$

and for a supplied bounded constructible coefficient $K_{\mathfrak{E}}$ after $q^*_{c,c'}$ and inverse reduced Thom–Sebastiani transport, the exceptional pushforward leg exists if $p_{c,c'}$ is proper, or if a compact-support model

$$j_p : \overline{\mathfrak{E}}^{\mathrm{ret}}_{R,c,c'} \hookrightarrow \overline{\mathfrak{E}}_p, \qquad \overline{p} : \overline{\mathfrak{E}}_p \to \mathfrak{M}^{\mathrm{red}}_{R,c} \times \mathfrak{M}^{\mathrm{red}}_{R,c'}$$

is supplied in the chosen constructible six-functor formalism. The resulting functor is

$$p^{\mathrm{cs}}_{c,c',!}K_{\mathfrak{E}} = \overline{p}_* j_{p,!}K_{\mathfrak{E}},$$

and in the proper case it specializes to $p_{c,c',*}K_{\mathfrak{E}}$. The packet `certificates/hall/finite_hall_coproduct_pushforward_func` verifies this criterion and records that the current finite-moduli ledger has no retained splitting stack, retained p-map, properness, or coefficient row; the compact-support model has no map-population or choice-independence rows; row 347 supplies notation only; the compact Hall source ledger has no source coproduct matrix entries; and the transition-Hall-coproduct ledger has no coproduct-transport rows. This row proves the criterion for a supplied p-leg, not the existence of the actual retained $K3 \times E$ Hall coproduct pushforward, and not coassociativity or bialgebra compatibility.

Proposition 0.159 identifies row 349. For three retained charges c_1, c_2, c_3, coassociativity of the compact Hall coproduct follows once the four binary coproduct legs

$$(c_1 + c_2, c_3), \qquad (c_1, c_2), \qquad (c_1, c_2 + c_3), \qquad (c_2, c_3)$$

are populated and choice-independent, and once a retained two-step splitting flag stack

$$\mathfrak{F}^{\mathrm{ret}}_{R;c_1,c_2,c_3}$$

maps to the two iterated splitting-correspondence fibre products by comparison maps λ and ρ. Base-change, projection-formula, and inverse reduced Thom–Sebastiani comparison rows identify both parenthesizations on this common flag stack; compactification independence identifies the two compact-support pushforwards to

$$\mathfrak{M}^{\mathrm{red}}_{R,c_1} \times \mathfrak{M}^{\mathrm{red}}_{R,c_2} \times \mathfrak{M}^{\mathrm{red}}_{R,c_3}.$$

The resulting isomorphism is

$$(\Delta^{\mathrm{red}}_{c_1,c_2} \otimes 1)\Delta^{\mathrm{red}}_{c_1+c_2,c_3}(K) \cong (1 \otimes \Delta^{\mathrm{red}}_{c_2,c_3})\Delta^{\mathrm{red}}_{c_1,c_2+c_3}(K).$$

The packet `certificates/hall/finite_hall_coproduct_coassociativity_two_step_flags` verifies this two-step-splitting-flag criterion and records that the current compact Hall coproduct rows, compact Hall two-step splitting flag rows, base-change/projection/inverse-Thom–Sebastiani rows, compactification-independence rows, quotient-descent rows, transition rows, coproduct matrices, and Hall-bialgebra coassociativity rows are not supplied. The hybrid two-step flag packet remains evidence for the method before quotient and compact Hall population, not populated compact Hall Δ_R-coassociativity.

Proposition 0.160 identifies row 350. For retained charges a, b, u, v with $u + v = a + b$, set

$$\mathsf{S}(a, b; u, v) = \{(a_1, a_2, b_1, b_2) \mid a = a_1 + a_2,\ b = b_1 + b_2,\ u = a_1 + b_1,\ v = a_2 + b_2\}.$$

The bialgebra compatibility of the compact Hall product and coproduct follows once the product legs $m_{a,b}$, m_{a_1,b_1}, m_{a_2,b_2}, the coproduct legs $\Delta_{u,v}$, Δ_{a_1,a_2}, Δ_{b_1,b_2}, a retained product–splitting square stack

$$\mathfrak{Q}^{\mathrm{ret}\ \ u,v}_{R;a,b},$$

base-change, projection-formula, product–coproduct Thom–Sebastiani, Koszul-braiding, and compact-support choice-independence rows are supplied. The resulting identity is

$$\Delta^{\mathrm{red}}_{u,v} m^{\mathrm{red}}_{a,b}(K_a, K_b) \cong \bigoplus_{\mathsf{S}(a,b;u,v)} (m^{\mathrm{red}}_{a_1,b_1} \otimes m^{\mathrm{red}}_{a_2,b_2})(1 \otimes \tau \otimes 1)(\Delta^{\mathrm{red}}_{a_1,a_2} K_a \otimes \Delta^{\mathrm{red}}_{b_1,b_2} K_b),$$

and in homogeneous compact-source bases it becomes

$$D^{a+b;u,v}_R M^{a,b}_R = \sum_{\mathsf{S}(a,b;u,v)} (M^{a_1,b_1}_R \otimes M^{a_2,b_2}_R)(1 \otimes \tau \otimes 1)(D^{a;a_1,a_2}_R \otimes D^{b;b_1,b_2}_R).$$

The packet `certificates/hall/finite_hall_bialgebra_compatibility` verifies this product–coproduct square criterion and records that the current compact Hall product rows, compact Hall coproduct rows, product–splitting square rows, base-change/projection/Thom–Sebastiani rows, compactification-independence rows, transition rows, M- and D-matrices, and Hall-bialgebra compatibility rows are not supplied. Thus row 350 proves the criterion for supplied finite data, not the actual retained $K3 \times E$ Hall-bialgebra identity.

Definition 0.161 identifies row 351. If a finite Hall Hopf algebra is claimed at height R, one must supply either homogeneous antipode matrices

$$S_{R,\gamma} : \mathcal{H}_{R,\gamma} \to \mathcal{H}_{R,\iota(\gamma)}$$

for a retained charge involution ι, or a connected conilpotent filtration for which

$$S_R(1) = 1, \qquad S_R(x) = -x - \sum S_R(x')x'', \qquad \widetilde{\Delta}_R x = \sum x' \otimes x''$$

terminates in every retained homogeneous block. In both presentations the two convolution identities

$$m_R(S_R \otimes \mathrm{id})\Delta_R = \eta_R\epsilon_R, \qquad m_R(\mathrm{id} \otimes S_R)\Delta_R = \eta_R\epsilon_R$$

must be zero-defect rows, and S_R must be compatible with the HN transition maps. The packet `certificates/hall/finite_hall_antipode_datum` verifies this definition and records that the current compact Hall source has no antipode matrix rows, charge-involution rows, connected-conilpotent-recursion rows, unit/counit rows, convolution-identity rows, or transition-antipode rows. Hopf-pairing rows, Drinfeld-double comparison data, primitive closure, scalar traces, and signed multiplicities are not substitutes for the antipode.

Proposition 0.162 identifies row 352. A retained finite Hall stage with product, coproduct, unit, counit, and bialgebra identities is still not a finite Hall Hopf algebra unless the antipode datum of Definition 0.161 is supplied. The packet `certificates/hall/finite_hall_nonhopf_boundary` verifies this source-data boundary. It imports the row 351 antipode-datum packet, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger; it records that the current source has no antipode rows, no convolution identity rows, and no finite Hall Hopf-algebra claim; and it records that the manuscript does not prove non-existence of possible future antipodes. Thus the current compact Hall source may enter later arguments only as bialgebra-level data until the antipode rows and convolution identities are supplied.

Definition 0.163 identifies row 353. Given a finite counital Hall coalgebra stage, set $I_R = \ker \epsilon_R$ and define

$$\bar{\Delta}_R : I_R \to I_R \otimes I_R, \qquad \bar{\Delta}_R(x) = \Delta_R x - \eta_R(1) \otimes x - x \otimes \eta_R(1).$$

Then

$$P_R = \ker \bar{\Delta}_R$$

is the finite primitive Hall subspace. The packet `certificates/hall/finite_hall_primitive_subspace_definition` verifies this definition and records that the current compact source has no coproduct matrix, unit/counit rows, augmentation-ideal basis, reduced-coproduct matrix, primitive kernel rows, primitive basis, inclusion/projection rows, or degree–parity split. Product rows, bialgebra-compatibility criteria, Hopf-pairing rows, target root spaces, bar-coalgebra primitives, scalar traces, signed multiplicities, and rank-only data are not substitutes for the finite kernel computation.

Proposition 0.167 identifies row 354. Once a supplied finite stage is a counital graded bialgebra and $P_R = \ker \bar{\Delta}_R$ has been computed, the standard bialgebra calculation gives

$$\Delta_R([x, y]) = [x, y] \otimes 1 + 1 \otimes [x, y], \qquad \epsilon_R([x, y]) = 0$$

for homogeneous $x, y \in P_R$. The packet `certificates/hall/finite_hall_primitive_commutator_closure` verifies this relative closure criterion and records that the current compact source has no product matrix, coproduct matrix, unit/counit rows, primitive kernel rows, bialgebra identity rows, primitive-closure defect rows, or supercommutator matrix rows. Thus row 354 is a proved finite-bialgebra theorem, not a populated compact-source bracket for the present packet.

Proposition 0.168 identifies row 355. In any associative $\mathbb{Z}/2$-graded algebra,

$$[x, y] = xy - (-1)^{|x||y|} yx$$

satisfies

$$(-1)^{|x||z|}[x,[y,z]] + (-1)^{|y||x|}[y,[z,x]] + (-1)^{|z||y|}[z,[x,y]] = 0.$$

The packet `certificates/hall/finite_hall_graded_jacobi` verifies this relative theorem and records that the current compact source has no product matrix, primitive bracket matrix, associativity row, primitive-closure row, Jacobi defect row, parity row, or normal-ordered bracket matrix. Thus row 355 is the graded associative-algebra identity for supplied Hall data, not a proof that the present compact source already carries a computed Lie bracket.

Proposition 0.169 identifies row 356. If

$$[P_{R,\widehat{c}}, P_{R,\widehat{c}'}] \subset P_{R,\widehat{c}+\widehat{c}'}$$

and $\overline{\Pi}_X : \widehat{\Gamma}_R \to \Gamma_R^{\Pi}$ is the additive normal-ordered Gram homomorphism, then the fibre-summed pieces

$$P_{R,\gamma}^{\Pi} = \bigoplus_{\overline{\Pi}_X(\widehat{c})=\gamma} P_{R,\widehat{c}}$$

satisfy

$$[P_{R,\gamma}^{\Pi}, P_{R,\gamma'}^{\Pi}] \subset P_{R,\gamma+\gamma'}^{\Pi}.$$

The packet `certificates/hall/finite_hall_normal_ordered_gram_degree` verifies this relative degree theorem by importing the finite normal-ordered bracket-degree packet and records that the current compact source has no source degree rows, normal-ordered lift rows, Hall product rows, primitive bracket rows, or bracket-degree defect rows. Thus row 356 proves degree preservation for supplied homogeneous normal-ordered Hall data; it does not construct the compact $K3 \times E$ primitive bracket and does not discharge the separate parity row 357.

Proposition 0.170 identifies row 357. If $A = A_{\bar{0}} \oplus A_{\bar{1}}$ has even multiplication, then for homogeneous $x \in A_{\bar{p}}$ and $y \in A_{\bar{q}}$,

$$[x,y] = xy - (-1)^{\bar{p}\bar{q}} yx$$

has parity $\bar{p} + \bar{q}$. Combined with row 356, supplied normal-ordered Hall rows must therefore satisfy

$$[P_{R,\gamma,\bar{p}}^{\Pi}, P_{R,\gamma',\bar{q}}^{\Pi}] \subset P_{R,\gamma+\gamma',\bar{p}+\bar{q}}^{\Pi}.$$

The packet `certificates/hall/finite_hall_bracket_parity` verifies this relative parity theorem by importing the finite formal bracket-parity rows and the parity-pushforward packet. It records that the current compact source has no parity blocks, homogeneous product rows, primitive bracket rows, or bracket-parity defect rows. Thus row 357 proves parity compatibility for supplied homogeneous Hall data; it does not prove the full $K3 \times E$ source parity split and does not replace the later GN–Kac parity comparison.

Definition 0.180 identifies row 358. For a finite normal-ordered primitive Hall stage, a source degree-zero trace and compact positive-negative pairing correspondence define

$$\langle e_{\gamma,\bar{p},a}, e_{-\gamma,\bar{p},b}\rangle_R = G_{\gamma,\bar{p};ab} = \mathrm{tr}_{0,R}\big(m_R^{\Theta}(e_{\gamma,\bar{p},a}, e_{-\gamma,\bar{p},b})\big).$$

The row is only the definition of the source G-matrix. It is not the homogeneity theorem, not graded supersymmetry, not Frobenius invariance, not coproduct adjointness, and not quotient non-degeneracy. The packet `certificates/hall/finite_hall_positive_negative_pairing` imports the formal degree-zero pairing rows and records that the current compact source has no G-entries, degree-zero trace row, positive-negative pairing correspondences, orientation transport rows, or Hopf-pairing identity rows. Thus row 358 defines the finite positive-negative Hall pairing matrix; it does not discharge rows 359–362.

Proposition 0.181 identifies row 359. If the positive-negative pairing of row 358 is supplied from a homogeneous Hall product and a degree-zero trace, then its extension by zero satisfies

$$\langle P^{\Pi}_{R,\gamma,\bar{p}}, P^{\Pi}_{R,\gamma',\bar{q}}\rangle_R = 0 \quad \text{unless} \quad \gamma + \gamma' = 0 \text{ and } \bar{p} = \bar{q}.$$

The packet `certificates/hall/finite_hall_pairing_homogeneity` imports the formal degree-zero pairing rows and records that the current compact source has no G-entries, trace row, positive-negative pairing correspondences, or homogeneity defect rows. Thus row 359 proves the homogeneity criterion for supplied source data; it does not prove supersymmetry, Frobenius invariance, coproduct adjointness, or quotient non-degeneracy.

Proposition 0.182 identifies row 360. If the row 358 finite positive-negative pairing is homogeneous as in row 359, and if signed-transpose rows

$$G_{\gamma,\bar{p};ab} = (-1)^{\bar{p}} G_{-\gamma,\bar{p};ba}$$

are supplied for every retained opposite-degree parity block, then

$$\langle x, y\rangle_R = (-1)^{|x||y|}\langle y, x\rangle_R$$

for homogeneous finite primitive vectors. The packet `certificates/hall/finite_hall_pairing_supersymmetry` imports the row 359 homogeneity packet, the row 358 positive-negative pairing definition, the reduced Calabi–Yau threefold Serre-sign criterion packet, and the compact Hall source obstruction ledger. It records that the current compact source has no G-entries on both signs, parity blocks, signed-transpose rows, Serre-sign rows, or orientation-transport rows. Thus row 360 proves the supersymmetry criterion for supplied source data; it does not prove Frobenius invariance, coproduct adjointness, or quotient non-degeneracy.

Proposition 0.184 identifies row 361. If the finite primitive Hall bracket is the graded-skew supercommutator and is parity-compatible, the finite pairing is supersymmetric as in row 360, and the Frobenius cyclic defect rows

$$\mathfrak{o}^{\mathrm{Frob}}_R(x, y, z) = \langle [x, y]_\Theta, z\rangle_R + (-1)^{|x||y|}\langle y, [x, z]_\Theta\rangle_R$$

vanish for every retained homogeneous triple in the domain of the Hopf-pairing tuple, then

$$\langle [x, y]_\Theta, z\rangle_R = \langle x, [y, z]_\Theta\rangle_R.$$

The packet `certificates/hall/finite_hall_pairing_invariance` imports the row 360 supersymmetry packet, the row 357 bracket-parity packet, the row 355 graded-Jacobi packet, the reduced Calabi–Yau threefold Serre-sign criterion packet, and the compact Hall source obstruction ledger. It records that the current compact source has no primitive bracket rows, G-entries, parity blocks, Frobenius defect rows, cyclic three-point pairing correspondences, Serre-sign rows, or orientation-transport rows. Thus row 361 proves the invariance criterion for supplied source data; it does not prove coproduct adjointness or quotient non-degeneracy.

Proposition 0.185 identifies row 362. If the finite product, coproduct, and positive-negative pairing are supplied in homogeneous finite bases, and if the Hopf-adjointness defect rows

$$\mathfrak{o}^{\mathrm{adj}}_R(x, y, z) = \langle m^{\Theta}_R(x, y), z\rangle_R - \langle x \otimes y, \Delta^{\Theta}_R z\rangle_{R\otimes R}$$

vanish for every retained homogeneous triple in the domain of the Hopf-pairing tuple, then

$$\langle m_R^{\Theta}(x, y), z\rangle_R = \langle x \otimes y, \Delta_R^{\Theta} z\rangle_{R\otimes R}.$$

In finite coordinates this is

$$\sum_c M_{ab}^c G_{cd} - \sum_{u,v} D_d^{uv} (-1)^{|e_b||e_u|} G_{au} G_{bv} = 0.$$

The packet `certificates/hall/finite_hall_pairing_coproduct_adjointness` imports the row 361 invariance packet, the row 342 product operator packet, the row 347 coproduct operator packet, the row 350 bialgebra-compatibility packet, and the compact Hall source obstruction ledger. It records that the current compact source has no M-entries, D-entries, G-entries, tensor-pairing sign rows, Hopf-adjointness defect rows, product/coproduct correspondence rows, or orientation-transport rows. Thus row 362 proves the coproduct-adjointness criterion for supplied source data; it does not prove quotient non-degeneracy.

Proposition 0.186 identifies row 363. If the source positive-negative pairing matrix $G_{\gamma,\bar{p}}$, left and right radical bases $K_{\gamma,\bar{p}}$ and $K_{-\gamma,\bar{p}}$, and quotient splittings $Q_{\gamma,\bar{p}}$, $Q_{-\gamma,\bar{p}}$ are supplied, then the induced quotient matrix

$$\overline{G}_{\gamma,\bar{p}} = Q_{\gamma,\bar{p}}^t G_{\gamma,\bar{p}} Q_{-\gamma,\bar{p}}$$

is non-degenerate precisely when its rank equals both quotient dimensions:

$$\operatorname{rank} \overline{G}_{\gamma,\bar{p}} = \dim L_{\gamma,\bar{p}} = \dim L_{-\gamma,\bar{p}}.$$

The packet `certificates/hall/finite_hall_quotient_pairing_nondegeneracy` imports the row 362 coproduct-adjointness packet, the row 360 supersymmetry packet, the row 358 positive-negative pairing definition, the pairing-kernel $\varprojlim^1$ obstruction packet, the transition-radical obstruction packet, and the compact Hall source obstruction ledger. It records that the current compact source has no G-entries, K-entries, Q-entries, quotient-pairing matrices, quotient-rank rows, determinant rows, or radical-identification rows. Thus row 363 proves the quotient non-degeneracy criterion for supplied source data; it does not prove transition preservation of radicals or $\varprojlim^1$-vanishing for the radical tower.

Definition 0.187 identifies row 364. The compact geometric Hopf-pairing datum is the source-level package consisting of a degree-zero source trace, positive-negative pairing correspondences, orientation and Thom–Sebastiani transport, G-matrices, K-radical matrices, Q-quotient splittings, cyclic three-point correspondences, Hopf-adjointness witnesses, Frobenius cyclic witnesses, and quotient non-degeneracy witnesses. The packet `certificates/hall/finite_hall_compact_geometric_hopf_pairing_datum` imports the row 363 quotient non-degeneracy packet, the row 362 coproduct-adjointness packet, the row 361 Frobenius invariance packet, the reduced Calabi–Yau threefold Serre-sign criterion packet, the level-`Z` protected-trace obstruction ledger as a firewall, and the compact Hall source obstruction ledger. It records that the current compact source has no source degree-zero trace, positive-negative pairing correspondences, orientation transport, G-entries, K-entries, Q-entries, cyclic three-point correspondences, Hopf-adjointness rows, Frobenius cyclic rows, or quotient non-degeneracy witness rows. Thus row 364 records the datum required for the later Hopf-pairing defect theorem; it does not construct the datum, and the level-`Z` protected trace does not substitute for the source trace.

Proposition 0.190 identifies row 365. The proposition is the relative implication from supplied compact geometric Hall bialgebra data and supplied compact geometric Hopf-pairing data to the vanishing of $\mathfrak{D}_{\mathrm{Hpair},R}$. The packet `certificates/hall/finite_hall_hopf_pairing_defect_rows` imports the row 364 datum packet, the row 350 Hall bialgebra compatibility packet, the row 362 Hopf-adjointness criterion, the row 361 Frobenius cyclic criterion, the row 363 quotient non-degeneracy criterion, and the compact Hall source obstruction ledger. It records that the current compact source has no Hopf-adjointness rows, Frobenius cyclic rows, quotient non-degeneracy rows, M-, D-, G-, K-, or Q-entries, radical ideal/coideal rows, degree-zero source trace, positive-negative pairing correspondences, or cyclic three-point coherences. Thus row 365 certifies the theorem schema and the source/target firewall; it does not construct the compact Hopf-pairing datum or prove Hopf-radical descent for the current compact source.

Proposition 0.191 identifies row 366. The proposition is the finite-stage firewall separating a bilinear matrix calculation from a Hopf-radical quotient: G, $K = \ker G$, and Q record a pairing matrix, its linear radical, and a quotient splitting, but they do not prove Hopf adjointness, Frobenius cyclicity, quotient non-degeneracy, coideal stability, or Lie-ideal stability. The packet `certificates/hall/finite_hall_pairing_matrix_firewall` imports the row 365 Hopf-pairing defect packet, the row 362 Hopf-adjointness criterion, the row 361 Frobenius cyclic criterion, the row 363 quotient non-degeneracy criterion, and the compact Hall source obstruction ledger. It records that the current compact source has no G-, K-, or Q-entries, Hopf-adjointness rows, Frobenius cyclic rows, quotient non-degeneracy rows, radical coideal rows, radical Lie-ideal rows, or quotient Hopf-pairing rows. Thus row 366 certifies only the insufficiency theorem and the exact extra finite identities required for Hopf-radical descent.

Lemma 0.192 identifies row 367. The lemma is the finite algebraic descent criterion: Hopf adjointness and quotient tensor non-degeneracy force the radical to be a coproduct coideal; the Frobenius cyclic identity forces it to be a Lie ideal. The packet `certificates/hall/finite_hall_hopf_radical_ideal_coideal` imports the row 366 pairing-matrix firewall, the row 365 Hopf-pairing defect packet, the row 362 Hopf-adjointness criterion, the row 361 Frobenius cyclic criterion, the row 363 quotient non-degeneracy criterion, and the compact Hall source obstruction ledger. It records that the current compact source has no Hopf-adjointness rows, Frobenius cyclic rows, quotient tensor non-degeneracy rows, radical coideal rows, radical Lie-ideal rows, or G-, K-, and Q-entries. Thus row 367 proves the conditional ideal/coideal criterion for supplied finite Hopf-pairing data; it does not supply the radical-stability rows for the compact $K3 \times E$ source.

Definition 0.193 and Proposition 0.194 identify row 368. The row records the finite source datum needed before the manuscript may pass from the compact Hall heart to an integral form, a separated and complete completion, a compact Drinfeld double, stable-envelope transport, a Cartan lattice identification, or a Calabi–Yau quantum-group comparison. The packet `certificates/hall/finite_hall_integral_form_stable_envelope_transport` imports the row 367 Hopf-radical criterion, the compact Hall source obstruction ledger, the Do-HN obstruction ledger, and the $(O2)$ wall-atlas obstruction ledger. It verifies `FINITE_HALL_INTEGRAL_FORM_STABLE_ENVELOPE_TRANSPORT_OBSTRUCTI` by recording the scope of the Maulik–Okounkov, Schiffmann–Vasserot, and Kontsevich–Soibelman inputs and by keeping the current compact source integral-form rows, product-stability rows, coproduct-stability rows, pairing-stability rows, completion rows, Drinfeld-double rows, cross-relation rows, stable-envelope rows, Cartan rows, compact non-toric replacement theorem, and Calabi–Yau quantum-group comparison rows empty. Thus row 368 is a scope firewall and a datum statement; it does not construct the compact $K3 \times E$ integral form, compact Drinfeld double, or stable-envelope transport.

Definition 0.195 and Proposition 0.196 identify row 369. The row records the finite source datum needed before $C_{X,R}$ may be used as a constructed conilpotent chiral coalgebra and before $C_X =$

$\varprojlim_R C_{X,R}$ may be formed as a source coalgebra. The packet `certificates/hall/finite_hall_conilpotent_source_coalgel` imports the row 368 transport firewall, the hybrid carrier ledger, the compact Hall source obstruction ledger, the Do-HN obstruction ledger, the Hall product and coproduct transition ledgers, and the primitive-space and pairing-kernel $\varprojlim^1$ ledgers. It verifies `FINITE_HALL_CONILPOTENT_SOURCE_COALGEBRA_OBSTRUCTION_VERIF` by keeping the current compact source coalgebra rows, coaugmentation rows, counit rows, finite filtration rows, collision-coproduct rows, coalgebra-identity rows, conilpotence rows, quotient coalgebra identity rows, transition coalgebra rows, inverse-limit Mittag–Leffler rows, and bar-cobar comparison rows empty. Thus row 369 is a source-data boundary; it does not construct the compact $K3 \times E$ conilpotent source coalgebra.

6.5.0.3 Construction.

(i) *Pullback.* For $\widehat{c}, \widehat{c}' \in \widehat{\Gamma}_R$, pull back along $p : \mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'} \to \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}'}$; apply reduced Thom–Sebastiani $\mathrm{TS}^{\mathrm{red}}$ to combine vanishing cycle data [12]; pushforward exceptionally along $q : \mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'} \to \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}\star\widehat{c}'}$ to obtain $m_R(x \otimes y) = q_! \,\mathrm{TS}^{\mathrm{red}}\, p^*(x \boxtimes y)$. This operation is defined only in the retained compactified or exceptional compact-support model specified in the input.

(ii) *Coproduct.* Dually pull back along q, apply inverse reduced Thom–Sebastiani, pushforward along p to obtain $\Delta_R(z) = p_! \,(\mathrm{TS}^{\mathrm{red}})^{-1}\, q^*(z)$. The compact geometric coproduct correspondence $\mathfrak{C}^{\mathrm{geom}}_{\rho;\mu,\nu}$ of Theorem 4.3 is the support data for splitting.

(iii) *Bialgebra check.* Verify associativity of m_R and coassociativity of Δ_R on the two-step flag stack $\mathfrak{F}^{(2),\mathrm{geom}}_{R,\widehat{c},\widehat{c}',\widehat{c}''}$ by base change and Thom–Sebastiani coherence. Write the unit and counit matrices for the vacuum object and the augmentation. Verify the left and right unit identities, left and right counit identities, and the bialgebra identity

$$\Delta_R m_R = (m_R \otimes m_R)(1 \otimes \tau \otimes 1)(\Delta_R \otimes \Delta_R)$$

with the Koszul sign braiding τ, via the Massey–Saito Joyce–Upmeier multiplicative orientation transport. In finite coordinates this is the Hall bialgebra identity table: every displayed defect matrix has rank zero.

(iv) *Transition preservation of the Hall product.* For every successor $R' \leq R$, construct the transition cube for retained extension correspondences; verify coefficient-system transport for $\Phi^{\mathrm{red}} \otimes o$, Thom–Sebastiani product transport, proper base change, the projection formula, compact-support pushforward compatibility, product-matrix intertwining

$$T_{\alpha+\beta} M^{\alpha,\beta}_R = M^{\alpha,\beta}_{R'}(T_\alpha \otimes T_\beta),$$

associativity compatibility, strict composition, and $R^1 \varprojlim = 0$ for the product maps.

(v) *Transition preservation of the Hall coproduct.* For every successor $R' \leq R$, read the retained extension correspondence in the splitting direction; verify the transition cube for q-source and p-two-output squares, coefficient-system transport for $\Phi^{\mathrm{red}} \otimes o$, inverse Thom–Sebastiani coproduct transport, proper base change, the projection formula, compact-support pushforward compatibility, coproduct-matrix intertwining

$$(T_\mu \otimes T_\nu) D^{\rho;\mu,\nu}_R = D^{\rho;\mu,\nu}_{R'} T_\rho, \qquad \rho = \mu + \nu,$$

coassociativity and counit compatibility, strict composition, and $R^1 \varprojlim = 0$ for the coproduct maps.

(vi) *Primitive extraction.* Put $I_R = \ker \epsilon_R$, form the reduced coproduct

$$\bar{\Delta}_R : I_R \to I_R \otimes I_R, \qquad \bar{\Delta}_R(x) = \Delta_R x - \eta_R(1) \otimes x - x \otimes \eta_R(1),$$

and compute $P_R^{\mathrm{geom}} = \ker \bar{\Delta}_R$. Since $\bar{\Delta}_R$ is degree-graded on $\widehat{\Gamma}_R$, this is a finite linear computation in each charge once the coproduct, counit, augmentation-basis, and reduced-coproduct matrices are supplied.

(vii) *Transition preservation of primitive subspaces.* Form the reduced coproduct matrix

$$\widetilde{\Delta}_R = \Delta_R - \eta_R \epsilon_R \otimes \mathrm{id} - \mathrm{id} \otimes \eta_R \epsilon_R$$

and verify

$$\widetilde{\Delta}_{R'} T_R = (T_R \otimes T_R) \widetilde{\Delta}_R, \qquad \epsilon_{R'} T_R = \epsilon_R.$$

Compute primitive kernel rows for P_R and $P_{R'}$; restrict the ambient transition matrix to $T_R^{\mathrm{Prim}} : P_R \to P_{R'}$; verify zero primitive-image, projection-commutator, and composition defect ranks.

(viii) *Normal-ordered descent.* Apply $\overline{\Pi}_{X,*}^{\Theta_R}$ of Theorem 4.1 to obtain the Γ_R^{Π}-graded $P_R^{\Pi,\mathrm{geom}}$; verify additivity $\overline{\Pi}_X(\widehat{c} \star \widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}')$.

(ix) *Bracket and pairing matrices.* Compute the matrices

$$M_{\alpha,a;\beta,b}^{\alpha+\beta,c} = \int_{\mathfrak{E}_{\alpha,\beta;\alpha+\beta}^{\mathrm{geom}}} q^!(e_{\alpha+\beta,c}^{\vee}) \cap \mathrm{TS}^{\mathrm{red}}(p^*(e_{\alpha,a} \boxtimes e_{\beta,b})), \quad B_{\alpha,\beta}(x \otimes y) = M_{\alpha,\beta}(x \otimes y) - (-1)^{|x||y|} M_{\beta,\alpha} \mathsf{sw}(x \otimes y),$$

$$G_{\gamma,ab} = \mathrm{tr}_0(q_!\, \mathrm{TS}^{\mathrm{red}}\, p^*(e_{\gamma,a} \boxtimes e_{-\gamma,b})) \quad \text{on} \quad \mathfrak{P}_{\gamma,-\gamma;0}^{\mathrm{geom}}.$$

These entries are compact-source rows only together with the compact geometric Hopf-pairing datum of Definition 0.187: the protected degree-zero trace, the positive-negative pairing correspondences, the cyclic three-point flag coherences, and the quotient non-degeneracy witnesses. Form the radical $K_{\gamma,\bar{p}} = \ker G_{\gamma,\bar{p}} \cap \ker G_{-\gamma,\bar{p}}^t$ and the quotient $L_{\gamma,\bar{p}}$; fix splittings. Verify the finite Hopf-pairing identity rows of Definition 0.183: Hopf adjointness, the primitive Frobenius cyclic identity, and non-degeneracy of the induced quotient pairing. These are rank-zero defect checks, not consequences of the displayed G-matrix.

(x) *Transition preservation of radicals.* For every successor $R' \le R$, verify the primitive transition $T_R^{\mathrm{Prim}} : P_R^{\Pi} \to P_{R'}^{\Pi}$, source and target Hopf-pairing matrices, source and target radical kernels, and opposite-primitive lift rows. Check

$$\langle T_R^{\mathrm{Prim},\gamma} x, y' \rangle_{R'} = \langle x, L_{R'R}^{-\gamma} y' \rangle_R$$

for all retained opposite primitives y'. Restrict the transition to $T_R^{\mathrm{Rad}} : K_{R,\gamma,\bar{p}} \to K_{R',\gamma,\bar{p}}$, descend it to the quotient $L_{\gamma,\bar{p}}$, and verify zero radical-image, quotient well-definedness, splitting-compatibility, and composition defect ranks.

(xi) Obtain $(\mathcal{H}_R^{\mathrm{geom}}, m_R, \Delta_R, P_R^{\mathrm{geom}}, P_R^{\Pi,\mathrm{geom}}, \{B_{\alpha,\beta}\}, \{G_{\gamma,ab}\}, \{K_\gamma\}, \{L_\gamma\})$.

6.5.0.4 Identities. $\mathcal{H}_R^{\mathrm{geom}}$ is a finite $\widehat{\Gamma}_R$-graded Hall bialgebra; m_R is associative on the two-step flag stack; Δ_R is coassociative; η_R and ϵ_R satisfy the unit and counit identities; Δ_R is multiplicative for m_R; the supercommutator of primitives is primitive; the descended bracket is Γ_R^{Π}-graded after $\overline{\Pi}_{X,*}^{\Theta_R}$; finite-dimensional in each homogeneous parity piece by finite-typeness; Hopf adjointness and the Frobenius cyclic identity hold on the primitive quotient; the quotient pairing is non-degenerate in every retained parity block.

6.5.0.5 Obstruction classes. The global Do-orientation follows from the retained Do-HN datum through Theorem 7.7. This finite construction produces the finite Hall datum $(\mathcal{H}_R^{\mathrm{geom}}, m_R, \Delta_R, P_R)$. The seven obstruction classes $\mathfrak{o}_\Theta^{\mathrm{top}}, \mathfrak{o}_\Theta^{\mathrm{Hoch}}, \mathfrak{o}_\Theta^{\mathrm{tri}}, \mathfrak{o}_\Theta^{\mathrm{Jac}}, \mathfrak{o}_F, \mathfrak{o}_\Theta^{\mathrm{cyc}}, \mathfrak{o}_\Theta^{\mathrm{rad}}$ of Theorem 4.1 are required to vanish on this finite Hall stage and compatibly in the inverse limit. The transition-Hall-product packet `certificates/hall/transition_hall_product_preservation` verifies `TRANSITION_HALL_PRODUCT_PRESERVATION_OBSTRU` the transition-geometry input, orientation-transition input, vanishing-cycle transition input, source product matrices, extension correspondence transition squares, input and output product-stack squares, coefficient-system transport, Thom–Sebastiani product transport, proper base change, projection formula, compact-support pushforward, product-matrix intertwining, associativity-transition, and Hall-product Mittag–Leffler rows are not yet supplied. The transition-Hall-coproduct packet `certificates/hall/transition_hall_coproduct_preservation` verifies `TRANSITION_HALL_COPRODUCT_PRESERVATION_OBSTRUCTION_VERIFIED`: the transition-geometry input, orientation-transition input, vanishing-cycle transition input, source coproduct matrices, source counit maps, splitting correspondence transition squares, source object and target product stack squares, coefficient-system transport, inverse Thom–Sebastiani coproduct transport, proper base change, projection formula, compact-support pushforward, coproduct-matrix intertwining, coassociativity-transition, counit-transition, and Hall-coproduct Mittag–Leffler rows are not yet supplied. The transition-primitive-subspace packet `certificates/hall/transition_primitive_subspace_preservation` verifies `TRANSITION_PRIMITIVE_SUBSPACE_PRESERVATION_OBSTRUCTION_VERIFIED`: the source and target coproduct matrices, source and target counit maps, reduced-coproduct matrices, primitive kernel rows, ambient Hall transition matrices, Hall-coproduct transition input, counit-transition compatibility, kernel-intertwining rows, primitive projection matrices, restricted primitive transition matrices, and primitive-transition composition rows are not yet supplied. The primitive-space $R^1 \varprojlim$ packet

`certificates/hall/primitive_space_lim1_vanishing`

verifies `PRIMITIVE_SPACE_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal root-window sequence, primitive-space rows, primitive bases, finite-rank rows, parity and sign coverage rows, restricted transition matrices, transition-functoriality rows, image subspace rows, image-stabilization witnesses, Mittag–Leffler status rows, zero $R^1 \varprojlim$ rows, and comparison-tower compatibility rows are not yet supplied. The pairing-kernel $R^1 \varprojlim$ packet

`certificates/hall/pairing_kernel_lim1_vanishing`

verifies `PAIRING_KERNEL_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal root-window sequence, primitive-space input, Hopf-pairing matrix rows, Hopf-pairing identity rows, left and right kernel rows, kernel bases, finite-rank rows, pairing-transition input, restricted kernel transition matrices, transition-functoriality rows, image subspace rows, image-stabilization witnesses, Mittag–Leffler status rows, zero $R^1 \varprojlim$ rows, and comparison-kernel compatibility rows are not yet supplied. The transition-radical packet `certificates/hall/transition_radical_preservation` verifies `TRANSITION_RADICAL_PRESERVATION_OBSTRUCTION_VERIFIED`: the primitive-transition input, source and target Hopf-pairing matrices, source and target Hopf-pairing identity rows, source and target radical kernels, opposite-primitive lift rows, pairing-transition identities, radical-image identities, restricted radical transition matrices, source and target quotient splittings, descended quotient transition matrices, and radical-transition composition rows are not yet supplied. The transition-PBW packet `certificates/hall/transition_pbw_filtration_preservation` verifies

`TRANSITION_PBW_FILTRATION_PRESERVATION_OBSTRUCTION_VERIFIED`: the radical-quotient transition input, source and target PBW filtration rows, ordered word bases, relation-rewrite systems, filtered transition matrices, associated-graded transition matrices, PBW-symbol identities, A_β-PBW transition squares, and PBW-transition composition rows are not yet supplied. The PBW associated-graded $R^1 \varprojlim$ packet

`certificates/hall/pbw_associated_graded_lim1_vanishing`

verifies `PBW_ASSOCIATED_GRADED_LIM1_VANISHING_OBSTRUCTION_VERIFIED`: the cofinal root-window sequence, radical-quotient input, PBW-transition input, PBW filtration rows, ordered word bases, relation-rewrite rows, associated-graded piece rows, graded bases, associated-graded rank rows, associated-graded transition maps, PBW-symbol intertwining rows, A_β-PBW squares, transition-functoriality rows, image subspace rows, image-stabilization witnesses, Mittag–Leffler status rows, zero $R^1 \varprojlim$ rows, and comparison-graded compatibility rows are not yet supplied. The transition-parity packet `certificates/hall/transition_parity_decomposition_preservation` verifies `TRANSITION_PARITY_DECOMPOSITION_PRESERVATION_OBSTRUCTION_VERIFIED`: the parity-homogeneous bases, fermion-parity involutions, ambient transition matrices, parity-commutator identities, off-diagonal zero identities, restricted even and odd transition matrices, negative-root parity transport rows, comparison parity squares, and parity-transition composition rows are not yet supplied.

6.5.0.6 Comparison with Vol III. The datum $(\mathcal{H}_R^{\mathrm{geom}}, m_R, \Delta_R, P_R)$ is the finite protected Hall positive-half candidate attached to the K3-side Hall–Drinfeld–Borcherds problem of Vol III [70]. In the Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S}$, P_R^{geom} is the finite primitive Hall layer which can identify with the positive half $Y^+(X)$ only after the compact-source recognition datum, the Hopf pairing, completion, integral form, stable-envelope transport, and strict inverse-limit exactness have been fixed. The Drinfeld double $G(X) = D(Y^+(X))$ is therefore not obtained from $(\mathcal{H}_R^{\mathrm{geom}}, m_R)$ alone. The κ-tuple coordinate $\kappa_{\mathrm{BKM}} = 5$ is the BKM/automorphic weight of the denominator section; it is not a level-P invariant of the Hall carrier itself.

6.6 First-order Pfaffian block and Pfaffian line

6.6.0.1 Hypotheses. The oriented Hall datum $(\mathcal{H}_R^{\mathrm{geom}}, o_R, \{\tau_{i,R,S}\})$ of the reduced orientation and Hall correspondence data; the local wall-sign datum $\epsilon_{o,R}^{\mathrm{loc}}$ of the reduced orientation datum; the rank-one type-II Pfaffian crossing datum of Definition 0.204 and Appendix 5.2; the half-Hilbert orbit under $W^{(2)}(\Lambda_{II}^{2,1})$ of size $2^{12} = 4096$.

6.6.0.2 Pfaffian datum. The finite skew Pfaffian block on the orientation gerbe $\mathfrak{D}_R^{\mathrm{Pf}}$; the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R} = \det \mathrm{Ext}^1_{\mathrm{red}}(A, A)^{-1}$ (squaring to the reduced determinant); the protected Pfaffian section $\mathrm{pf}_{X,R}$; the global wall-sign character $\epsilon_{o,R} : W^{(2)}(\Lambda_{II}^{2,1}) \to \{\pm 1\}$; the Pfaffian section asymptotic; and the finite-stage line comparison

$$\iota_{\mathrm{aut},R} : \mathscr{L}_{\mathrm{Pf},R} \xrightarrow{\sim} (\mathcal{L}^5 \otimes \nu_{\Delta_5})|_R$$

compatible with cusp trivialization, squaring, and successor maps. The asymptotic is

$$\mathrm{pf}_{X,R} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in A_R} (1 - q^n r^l s^m)^{f(nm,l)} + \text{higher-window correction},$$

which exhausts to Δ_5 as $R \to \infty$.

6.6.0.3 Construction.

(i) *First-order Pfaffian block.* On each Darboux chart of $\mathfrak{M}^{\mathrm{sc}}_{R,c}$, with reduced normal self-Ext splitting $K^{\mathrm{nor}}_{R,c}$, form the finite skew block $\mathfrak{D}^{\mathrm{Pf}}_{R,c}$ representing the Pfaffian of the reduced determinant. At a generic type-II wall point the rank-one normal summand is

$$J_{\partial_i,R} = \left(\begin{smallmatrix} 0 & u_i \\ -u_i & 0 \end{smallmatrix}\right), \qquad \mathrm{Pf}(J_{\partial_i,R}) = u_i$$

(Definition 0.204).

(ii) *Pfaffian line.* Set $\mathscr{L}_{\mathrm{Pf},R} = \det \mathrm{Ext}^1_{\mathrm{red}}(A,A)^{-1}$; verify $\mathscr{L}^{\otimes 2}_{\mathrm{Pf},R} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A,A)$ by 3-Calabi–Yau Serre duality.

(iii) *Local wall sign.* At each generic point of $\mathbb{H}_{\partial_i}$ with normal model $K^{\mathrm{nor}}_{\partial_i,R} = [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$, compute the monodromy of the Pfaffian section $\mathrm{pf}_{\partial_i,R} = v_i u_i$ along the wall-reflection loop $\ell_{\partial_i,R,S}$ closed by the $(O1)^+$ lift $\tau_{i,R,S}$:

$$M_{\ell_{\partial_i,R,S}}(v_i u_i) = v_i(-u_i) = -v_i u_i, \qquad \epsilon^{\mathrm{loc}}_{o,R}(s_{\partial_i}) = (-1)^{N^{\mathrm{Pf}}_{\partial_i,R}} = -1.$$

(iv) *Half-Hilbert orbit transport.* The reflection s_{∂_i} acts on the retained half-Hilbert orbit $\mathfrak{H}^{\mathrm{half}}_R$, whose wall-atlas support has twelve binary Pfaffian parities and cardinality $2^{12} = 4096$. The quotient $\Lambda^{2,1}_{II}/2\Lambda^{2,1}_{II}$ has only eight elements and is not the source of this count. The Weyl lift transports the rank-one sign to a character value on each component:

$$\epsilon_{o,R}(s_{\partial_i}) = \epsilon^{\mathrm{loc}}_{o,R}(s_{\partial_i}; x) = -1, \qquad x \in \mathfrak{H}^{\mathrm{half}}_R.$$

The orbit support enters the protected integration datum; it is not multiplied to define the value of a simple reflection. The transported datum is the global character $\epsilon_{o,R} : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$.

(v) *Compatibility datum.* Verify the component, boundary, overlap, quotient, and protected integration compatibilities of Proposition 5.2 on every retained type-II wall stratum.

(vi) *Finite O2 wall-atlas proof tables.* Populate the $(O2_{\mathrm{atlas}})$ tables of Definition 7.1:

wall_objects, wall_charts, overlaps, orbit_transport, half_hilbert_orbit, compatibilities, scalar_separatio

The rows must record full wall-object charge match, semistability, self-Ext splitting, rank-one normal blocks, invariant units, quotient orientations, Cech/cochain overlap cancellation, Weyl transport, the twelve binary half-Hilbert coordinates, component and boundary coverage, and the Thom–Sebastiani, HN-transition, quotient-orientation, protected-integration, and scalar-separation identities. In particular, $2^{12} = 64^2 = 4096$ is recorded only as a scalar-shadow equality after the wall atlas has been supplied. The separate normalization table `delta5_theta_leading` records

normalizations, scalar_relations, form_conventions, form_relations, branch_separation, scalar_firewall,

namely the arithmetic identities $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$, $[qrs]\Delta_{10} = 64^2 = 4096$, $D_5 = 64^{-1}\Delta_5$, and $\chi^{\mathrm{OP}}_{10} = 4096^{-1}\Delta_{10}$, together with the separate convention rows for Δ_{10}, χ_{10}, Φ^{un}_{10}, and the inverse scalar branches. These rows do not replace the O2 wall charts, orientation transport, or Pfaffian-line comparison. The separate automorphic character packet `delta5_maass_character` records

character_values, character_relations, automorphic_line, scalar_firewall, namely the identities $\nu_{\Delta_5}(s_{\delta_i}) = -1$, $\nu_{\Delta_5}|_{W^{(2)}} = \det$, $\nu_{\Delta_5}|_{\mathrm{Aut}(\mathcal{P}_{II})} = 1$, $\nu_{\Delta_5}^2 = 1$, and $\Delta_5 \in H^0(\mathbb{H}_2, \mathcal{L}^5 \otimes \nu_{\Delta_5})$. These rows place the Maass multiplier in the automorphic line; they do not construct the Pfaffian orientation or the Weyl transport of reduced determinant lines. The automorphic divisor packet `delta5_humbert_divisor` records

support_components, divisor_relations, support_multiplicity_separation, scalar_firewall,

namely $\mathrm{div}(\Delta_5) = H_{\mathrm{Hum}}$ with multiplicity 1, the pole order 2 of Δ_5^{-2} on the same support, and the firewall separating this automorphic divisor data from Pfaffian walls, $O2$ atlases, mirror discriminants, and protected traces. The type-II lattice packet `type_ii_weyl_vector` records

simple_roots, cartan_matrix, isometry_basis, weyl_vector, lattice_relations, scalar_firewall,

namely the target-side identities $\Lambda_{II}^{2,1} \simeq U(4) \oplus \langle 2\rangle$, $\det(\Lambda_{II}^{2,1}) = -32$, $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$, $(\rho, \delta_i) = -1$, $(\rho, \rho) = -3/2$, and $\rho \in (\Lambda_{II}^{2,1})^\vee$. These rows do not construct a compact source, Pfaffian orientation, $O2$ wall atlas, mirror discriminant, or protected trace. The product-chamber root-map packet `product_chamber_root_map` records

chamber_sectors, closure_rules, root_map_rows, root_map_relations, denominator_conventions, scalar_

namely the three sectors of Γ_{eff}, their additive closure, the rows selected by $f(nm, l)$, the formula $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$, the decomposition $\alpha = n\delta_1 + m\delta_2 + (n + m - l)\delta_3$, active nonnegativity, inactive zero exponents, and the $Z \mapsto 2Z$ denominator-character conversion. These rows are target-side lattice arithmetic; they do not construct compact $K3 \times E$ source data, Pfaffian orientations, $O2$ wall charts, mirror discriminants, or protected traces. The type-II chamber-isotropic packet `type_ii_chamber_isotropic` records

root_divisibility, type_ii_generators, chamber_automorphisms, type_ii_coxeter_pairs, isotropic_rays, isot

namely the type-I/type-II square-two divisibility split, the three type-II simple-reflection generators, the absence of finite braid relations among distinct type-II simple generators, the S_3 chamber transpositions and their fixed Weyl vector, the three primitive isotropic boundary rays $a_{ij} = \delta_i + \delta_j$, the coefficient $f(nm, l) = 10$ on those rays, the split $10 = 1 + 9$, and the nine Gritsenko–Nikulin target isotropic directions $u_{ij,r}$ on each ray. These rows remain target-side BKM arithmetic; they do not construct compact source representatives, Pfaffian orientations, $O2$ wall charts, mirror discriminants, protected traces, or parity decompositions.

(vii) *Pfaffian section.* Define $\mathrm{pf}_{X,R}$ as the protected integration of the oriented Hall product over $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$:

$$\mathrm{pf}_{X,R} = 64q^{1/2}r^{1/2}s^{1/2}I_R^{\mathrm{prot}}\left(\bigotimes_{\gamma\in A_R} \mathrm{pf}_{\gamma,R}\right), \qquad \mathrm{pf}_{\gamma,R} = (1 - q^n r^l s^m)^{f(nm,l)}.$$

(viii) *Line comparison and scalar firewall.* Equip $\mathrm{pf}_{X,R}$ with the comparison $\iota_{\mathrm{aut},R}$ and verify compatibility with the squaring map to $\mathcal{L}^{10}$, with the order-two character $\epsilon_{o,R}$, and with the transition $\rho^{\mathrm{Pf}}_{R^+R}$. Proposition 0.201 is the negative check: the displayed scalar product, its square, and the OP scalar normalization do not define the Pfaffian section or its orientation character without this line datum.

(ix) *Finite Pfaffian proof tables.* Populate the (P_{fin}) tables of Definition 0.200:

strata, skew_complexes, pfaffian_lines, pfaffian_sections, wall_charts, automorphic_comparisons, transitions, scalar_

The scalar firewall must explicitly exclude the Borcherds product, the squared determinant, the OP scalar trace, and the exponent table as entries of (P_{fin}).

(x) Obtain $(\mathfrak{D}_R^{\mathrm{Pf}}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}, \iota_{\mathrm{aut},R})$, with the protected integration and line-comparison compatibility data.

6.6.0.4 Identities. $\mathscr{L}_{\mathrm{Pf},R}$ is a Picard-line whose square isomorphism follows from reduced Serre duality; $\mathrm{pf}_{X,R}$ is a section of $\mathscr{L}_{\mathrm{Pf},R}$ with the displayed cusp expansion and comparison $\iota_{\mathrm{aut},R}$ to $(\mathcal{L}^5 \otimes \nu_{\Delta_5})|_R$; the character $\epsilon_{o,R}$ takes values in $\{\pm 1\}$ and satisfies the Coxeter relations of $W^{(2)}(\Lambda^{2,1}_{II})$; on each simple type-II reflection $\epsilon_{o,R}(s_{\partial_i}) = -1$ by the rank-one local model.

6.6.0.5 Obstruction classes. This datum is the entire site of $(O2)$ (local Pfaffian wall normal form on type-II walls; the $2^{12} = 4096$ sign sum on the half-Hilbert orbit transport). The local rank-one Pfaffian crossing of Appendix 5.2 supplies the generic chart sign; the global half-Hilbert orbit transport is defined only relative to $(O2_{\mathrm{atlas}})$. The character identity $\epsilon_o = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda^{2,1}_{II})}$ becomes a theorem only after the retained Do-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and $(O2_{\mathrm{atlas}})$ are compatible in R.

6.7 Chiral Koszul source coalgebra

6.7.0.1 Hypotheses. The oriented Hall primitives $P_R^{\Pi,\mathrm{geom}}$ of the Hall correspondence datum; the hybrid carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$; the augmentation $\varepsilon^{\mathrm{hyb}}_{X,R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \longrightarrow k\mathbf{1}$; the hybrid residual $\mathfrak{D}_{\mathrm{hyb},R} = 0$; the bounded length N_R (positivity of HN height on every non-vacuum colour); the Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar formalism [7, 35]; the conilpotent completed domain on the elliptic curve E; the wrapped bar-cobar domain datum for the wrapped summands and mixed local/wrapped words.

6.7.0.2 Koszul coalgebra datum. The conilpotent chiral coalgebra on E:

$$C^{\mathrm{geom}}_{X,R} = \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R}) = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_E(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R});$$

the bar-length filtration $F^{\mathrm{co}}_{R,p}$; the chiral collision coproduct Δ^{ch}_R; the bar comparison $b_{X,R} = \mathrm{id}_{\bar{B}^{\mathrm{ch}}_E(\cdot)}$; conditional on the wrapped bar-cobar domain and Francis–Gaitsgory hypotheses, the source cobar counit $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_{X,R} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$; after a separate source-to-target quasi-isomorphism $\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R}$, the cobar comparison $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$; the finite comparison datum $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ of Definition 0.211, recording Weyl, Pfaffian-orientation, Hall-pairing, radical-quotient, PBW, and transition compatibility.

6.7.0.3 Construction.

(i) *Augmentation kernel.* Set $\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R} = \ker(\varepsilon^{\mathrm{hyb}}_{X,R})$; verify the bounded-length hypothesis (positivity of HN height on every non-vacuum surviving colour, hence at most N_R factors in any non-zero word).

(ii) *Bar coalgebra.* Form the reduced chiral bar

$$C^{\mathrm{geom}}_{X,R} = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_{E}(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R}),$$

length p summand generated by ordered p-fold inputs from the augmentation ideal of total HN height $\le R$ (height $> R$ terms zero); coaugmentation by the inclusion of the vacuum summand; counit by projection.

(iii) *Bar-length filtration.*

$$F^{\mathrm{co}}_{R,p} C^{\mathrm{geom}}_{X,R} = k\mathbf{1} \oplus \bigoplus_{1\le j\le p} \bar{B}^{\mathrm{ch},j}_{E}(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R}), \qquad 0 \le p \le N_R.$$

(iv) *Collision coproduct.* For $x = [x_1|\cdots|x_p]$, define the cut-map coproduct

$$\Delta^{\mathrm{ch}}_{R}(x) = \mathbf{1}\otimes x + x\otimes \mathbf{1} + \sum_{i=1}^{p-1} [x_1|\cdots|x_i] \otimes [x_{i+1}|\cdots|x_p],$$

with each cut realized by the corresponding finite collision/flag pull-push map in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R,\mathrm{geom}}$ of the hybrid carrier. Verify coassociativity by the four-input pentagon and the eight-word flag coherences; verify counit identities via vacuum collision flags.

(v) *Conilpotence.* Verify $(\overline{\Delta}^{\mathrm{bar}}_{R})^{N_R+1} = 0$: each reduced cut strictly lowers bar length on each branch.

(vi) *Bar comparison.* Take $b_{X,R}$ to be the identity of $\bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R})$, eliminating the coalgebra-side finite Koszul residuals $\mathfrak{o}^{C}_{R}, \mathfrak{o}^{\mathrm{fil}}_{R}, \mathfrak{o}^{\mathrm{conil}}_{R}, \mathfrak{o}^{\Delta,\mathrm{ch}}_{R}$ of Definition 0.208.

(vii) *Source cobar counit.* Verify that the projection-finite summands, wrapped summands, and mixed words all lie in the chosen conilpotent completed Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar domain. Under that wrapped Koszul datum, apply [35, Theorem 6.2.2] to obtain $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_{E} C^{\mathrm{geom}}_{X,R} \xrightarrow{\simeq} \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$; verify it is a source-internal quasi-isomorphism of chiral algebras on E, not a homotopy inverse of the target counit.

(viii) *Source-to-target identification.* Construct ϑ_R from the geometric primitive representatives, Hall products, coproducts, pairings, and HN transition compatibility; primitive restriction alone is insufficient by Proposition 0.213. Verify acyclicity of the full cone in every HN weight, bar length, local/wrapped word, and parity block; then verify the residual $\mathfrak{o}^{\vartheta}_{R}$ of Corollary 0.210 vanishes; set $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$.

(ix) *Finite comparison compatibility.* Verify $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$: the Weyl transport square for the chosen $(O1)^{+}$ lifts, the Pfaffian-orientation sign square for the $(O2_{\mathrm{atlas}})$ wall datum, the Hall product/coproduct and Hopf-pairing square, the radical quotient square, the PBW associated-graded square, and the transition homotopies. The residual $\mathfrak{O}_{\mathrm{Kcmp},R}$ must vanish.

(x) *Inverse limit.* Verify the Koszul Mittag–Leffler condition $R^1\varprojlim = 0$ for the inverse systems of source coalgebras, completed chiral bar/cobar constructions, target truncations, and the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$, and for the Weyl, Pfaffian, Hall-pairing, radical, PBW, and transition defect complexes in $\mathfrak{K}^{\mathrm{cmp}}_{X,R}$ [106, §3.5]. Form

$$C^{\mathrm{geom}}_{X} = \varprojlim_{R} C^{\mathrm{geom}}_{X,R}, \qquad \Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_{E} C^{\mathrm{geom}}_{X} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

(xi) Obtain $(C^{\mathrm{geom}}_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_{R}, b_{X,R}, \Theta_{\mathrm{Kos},R}, \mathfrak{K}^{\mathrm{cmp}}_{X,R})$ at every R, with successor compatibility.

6.7.0.4 Identities. $C^{\mathrm{geom}}_{X,R}$ is a coaugmented conilpotent chiral coalgebra on E with bounded bar-length filtration of length at most N_R; the coalgebra identities for Δ^{ch}_R are part of the chosen higher-coloured, unit, refinement, quotient, and transition coherences on the chosen flag stacks; the bar-cobar inversion follows from Francis–Gaitsgory chiral Koszul only after the wrapped bar-cobar domain datum is fixed; the cobar comparison $\Theta_{\mathrm{Kos},R}$ is source-internal, not the target bar-cobar counit (the source/target separation of Lemma 0.207).

6.7.0.5 Obstruction classes. The Koszul Mittag–Leffler exactness hypothesis is the separate inverse-system hypothesis for the chiral Koszul lane; it is not a consequence of the reduced-orientation (Do) endpoint. The total finite Koszul residual $\mathfrak{O}_{\mathrm{Kos},R} = (\mathfrak{o}^C_R, \mathfrak{o}^{\mathrm{fil}}_R, \mathfrak{o}^{\mathrm{conil}}_R, \mathfrak{o}^{\Delta,\mathrm{ch}}_R, \{\mathfrak{o}^{C,\mathrm{tr}}_{R'R}\}, \mathfrak{o}^b_R, \mathfrak{o}^\Omega_R, \mathfrak{o}^{\mathrm{wrKos}}_R, \mathfrak{o}^{\mathrm{hyb}}_R, \mathfrak{o}^\Pi_R, \mathfrak{o}^{\mathrm{sep}}_R, \mathfrak{o}^{\mathrm{Prim}}_R)$ has its coalgebra terms killed by the Koszul source coalgebra; the wrapped Koszul term, $\mathfrak{o}^\Omega_R$, the source-to-target map, and the ML exactness remain conditions until the displayed identities hold. The finite comparison residual $\mathfrak{O}_{\mathrm{Kcmp},R}$ is the additional obstruction to using $\Theta_{\mathrm{Kos},R}$ inside primitive recognition: it records compatibility with the Weyl action, Pfaffian orientation character, Hall pairing, radical quotient, PBW filtration, and transition maps.

6.7.0.6 Source-target separation. The bar coalgebra $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,R})$ is a *twisting / coupling* coalgebra (Maurer–Cartan classification), *not* the bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_X) = \mathrm{ChirHoch}^\bullet(\mathcal{F}^{\mathrm{hyb}}_X, \mathcal{F}^{\mathrm{hyb}}_X)$; cf. Vol II [73] (Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z}$). The chiral Koszul comparison $\Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_X \longrightarrow \mathsf{Den}_{\Delta_5,E}$ is a comparison between two algebras of the same level. It is an equivalence only under the Koszul source datum and the finite Koszul comparison datum. The homotopy inverse to the target-internal counit $\Omega^{\mathrm{ch}}_E \bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E}) \to \mathsf{Den}_{\Delta_5,E}$ is not used in either direction.

6.8 Primitive recognition

6.8.0.1 Hypotheses. The descended source primitive Lie superalgebra $P^{\Pi,\mathrm{red}}_X$ of the Hall correspondence datum (after the radical quotient $\overline{\mathrm{Rad}\langle\cdot,\cdot\rangle_X}$); the imported target denominator algebra $\mathfrak{g}_{\Delta_5} = G(\mathscr{B}_{\Delta_5})$ on $\Lambda^{2,1}_{II}$ of Definition 3.2 and Definition 4.2; the increasing cofinal height-window sequence of finite downward-saturated windows

$$W_\nu = \{\beta \in \mathcal{R}_+ \mid \mathrm{ht}\,\beta \le \nu\}, \qquad W_1 \subset W_2 \subset \cdots, \qquad \bigcup_\nu W_\nu = \mathcal{R}_+,$$

and

$$\Gamma(W_\nu) = \{(b_1, b_1 + b_2 - b_3, b_2) \mid b_1\delta_1 + b_2\delta_2 + b_3\delta_3 \in W_\nu\};$$

the Borcherds–Kac–Moody data $(H, I, p, \alpha_i, (\ ,\ ))$ of Theorem 0.215. The packet `certificates/lattice/downward_root_w` verifies this target-side convention with status `DOWNWARD_ROOT_WINDOWS_VERIFIED`. It supplies no compact source representatives, relation matrices, Hopf pairing, radical, PBW row, Pfaffian orientation, or protected trace. The packet `certificates/lattice/active_root_window_exhaustion` verifies $\bigcup_\nu W_\nu = \mathcal{R}_+$ with status `ACTIVE_ROOT_WINDOW_EXHAUSTION_VERIFIED`, by the height witness $\beta \in W_{\mathrm{ht}(\beta)}$. This is target-root exhaustion, not compact-source support exhaustion. The packet `certificates/sources/source_window_compact_support` verifies only the obstruction ledger for source-window compact support, with status `SOURCE_WINDOW_COMPACT_SUPPORT_OBSTRUCTION_VERIFIED`. It records that the compact support index set, source height function, finite source windows, compact provenance rows, cofinality witness, source-window transitions, and target-source compatibility rows are not yet supplied. The packet `certificates/sources/target_source_window_compatibility` verifies only the obstruction ledger for target/source window compatibility, with status `TARGET_SOURCE_WINDOW_COMPATIBILITY_OBSTRUCTION_VERIFIED`. It records that the source degree maps into $\Gamma(W_\nu)$, target-window matching rows, compact-provenance compatibility rows, cofinal-index compatibility rows, transition compatibility rows, and zero compatibility-defect rows are not yet supplied.

6.8.0.2 Conditional primitive-recognition datum. The cofinal finite-window primitive-recognition datum {(R0),(R1),(R2),(R3),(R4),(R5),(R6),(R7),(R8)} of Definition 0.217 for every W_ν, equivalently the finite source proof tables of Definition 0.236, compatible under $W_\nu \subset W_{\nu+1}$; the source representatives e_i^X, f_i^X, h_i^X; the source presentation map $\pi_W : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})_W \to P_X^{\Pi,\mathrm{red}}|_W$; the kernel equality $\ker \pi_W = (\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}})_W$; the conditional isomorphism

$$\boxed{P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}, \quad \Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}.}$$

This construction is a finite criterion. The isomorphism is asserted only after the compact-source rows (R0)–(R8) are fixed on a cofinal relation-closed system. A relation-closed target degree set containing W_{rel}^+ is not yet such a system: Proposition 0.239 shows that the signed rows $C_{k,2}$, D_k, and $2\delta_{123}$ lack the full target parity, pairing, radical, and PBW blocks required before a compact source comparison can be formulated. Proposition 0.251 separates a second requirement: even a relation-closed target degree must be realized by a finite HN stage R, a charge $c \in \Gamma_R^{HN}$, and a normal-ordering translate $T \in \mathcal{T}_R(c)$ before it lies in Γ_R^{test}. The compact Hall source packet records this absence by the verified obstruction ledger `blocked_obligations.csv` in `certificates/sources/k3e_compact_hall`. The ledger requires the Hall product, coproduct, unit, counit, bialgebra compatibility, primitive closure, Hopf pairing, radical quotient, relation, no-extra, generation, PBW, transition, A_β-comparison, and Koszul comparison rows; it records `compact_source_recognition=false`. The first relation-closed compact-source theorem is represented by the additional finite packet

window_closure, target_source_representatives, chevalley_serre_matrices, hall_boundary_complex, spectral_seque

It is the finite computation of source representatives, relation matrices, Hall boundary, kernel equality, PBW associated gradeds, first-window theorem status, and strict transitions. Target labels, signed multiplicities, scalar traces, Pfaffian products, denominator products, target PBW rows, arbitrary matrices, and status-only rows are not entries of this packet. The packet `certificates/first_window/k3e_relation_closed_window` currently verifies only the obstruction ledger `FIRST_WINDOW_OBSTRUCTION_LEDGER_VERIFIED`, with `first_window_certification=false`. The ledger records the missing window, representative, relation, Hall-boundary, spectral, kernel, PBW, theorem, and transition rows; it is not the first-window primitive-recognition theorem.

6.8.0.3 Construction. For every cofinal W_ν:

(R0) *Compact source labels.* Verify every finite source basis vector of degree in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$, where

$$\Gamma(W_\nu) := \{\gamma \in \Gamma_{\mathrm{gram}} \mid \alpha(\gamma) \in W_\nu\},$$

equivalently $\gamma = (b_1, b_1 + b_2 - b_3, b_2)$ for $\alpha(\gamma) = b_1\delta_1 + b_2\delta_2 + b_3\delta_3 \in W_\nu$, satisfies the compact source-admissibility condition of Theorem 0.252.

(R1) *Representatives and parities.* Supply parity-homogeneous primitive representatives $e_i^X \in P^{\Pi,\mathrm{red}}_{X,\gamma_i}$, $f_i^X \in P^{\Pi,\mathrm{red}}_{X,-\gamma_i}$, $h_i^X = \iota_H^{-1}(\alpha_i)$ for every simple-root index i with $\alpha_i \in W_\nu$ and $\alpha(\gamma_i) = \alpha_i$, with the Gritsenko–Nikulin parity fibres $\mu_{\bar 0}(a)$, $\mu_{\bar 1}(a)$ for imaginary simple roots and even real representatives for δ_1, δ_2, δ_3. Equivalently, write the finite simple-representative table naming the source basis vector, parity, Cartan representative, and target reference for every retained simple-root representative.

(R2) *Relations.* Verify in the protected Hall super-bracket the Cartan, Chevalley, real Serre, Borcherds orthogonality, and super sign relations for every relation whose displayed source and target degrees lie in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$ and $W_\nu \cup (-W_\nu) \cup \{0\}$, respectively. In the real rank-three subsystem the Cartan matrix is $((\delta_i, \delta_j)) = \mathrm{diag}(2) - 2(J - \mathrm{diag}(1))$ with real Serre exponent 3.

(R3) *Full finite parity dimensions.* For every $\gamma \in \Gamma(W_\nu)$ verify $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\bar p} = \mathrm{mult}^{\mathrm{GN}}_{\bar p}(\alpha(\gamma))$ for both parities $p = 0, 1$, and verify preservation of the two summands under $W_{\nu+1} \to W_\nu$ by the parity-transition datum of Definition 0.227. In the first timelike channel $\gamma = (1,1,1)$, whose target root is $\alpha(\gamma) = \delta_{123}$, Proposition 6.7 requires $(d^{\mathrm{GN}}_{(1,1,1),\bar 0}, d^{\mathrm{GN}}_{(1,1,1),\bar 1}) = (29, 93)$. Determinant information $29 - 93 = -64 = f(1,1)$ is consistent but not the recognition.

(R4) *Hall bialgebra, Hopf pairing, and radical.* Verify the Hall unit/counit rows and the Hall bialgebra identity rows of Definition 0.236; in particular the finite defects for associativity, coassociativity, unit, counit, product-coproduct compatibility, and primitive closure all have rank zero. Compute the positive-negative pairing matrices in parity blocks; verify the Hopf-pairing identity rows $\mathfrak{D}_{\mathrm{Hpair},R} = 0$ of Definitions 0.187 and 0.183; verify the kernel agrees with the Gritsenko–Nikulin/Kac invariant-pairing kernel; verify Lie-ideal and coideal stability under all finite brackets; verify carrying under HN transitions to the next window.

(R5) *No extra relations.* Verify the kernel equality $\ker \pi_\nu = (\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}})_{\le W_\nu}$ on the finite triangular span generated by $W_\nu \cup (-W_\nu) \cup \{0\}$.

(R6) *Generation.* Verify that every source primitive root space in $W_\nu \cup (-W_\nu)$, after the finite radical quotient, is generated by the Cartan and the chosen simple-root representatives through brackets whose intermediate positive degrees stay in W_ν. Equivalently, write the finite generation table whose rows give, for every quotient basis vector, a bracket-word span matrix with equal source and span ranks.

(R7) *Completed PBW compatibility.* Verify the PBW-graded isomorphism $\mathrm{gr}_{\mathrm{PBW}} U(P^{\Pi,\mathrm{red}}_X)_{\le W_\nu} \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}} U(\mathfrak{g}_{\Delta_5})_{\le W_\nu}$; verify strict preservation under $W_{\nu+1} \to W_\nu$ transition maps by the PBW-transition datum of Definition 0.225.

(R8) *Exact triangular completion.* Verify strictness of the transition maps and closedness of images on every finite window, so that passage to the inverse limit commutes with the finite kernels, cokernels, radicals, PBW associated gradeds, and parity pieces. The primitive-space part is the Mittag–Leffler datum of Definition 0.229; equivalently the relevant $\lim^1$ terms vanish for these triangular finite-window systems, exactly as in Propositions 0.230, 0.232, 0.234, and 0.235.

(i) If the preceding rows hold on a cofinal relation-closed system, obtain the global isomorphism $P^{\Pi,\mathrm{red}}_X \cong \mathfrak{g}_{\Delta_5}$ by passage to the inverse limit (compatible cofinal datum); together with the chiral Koszul comparison with the Koszul source coalgebra this gives $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$. The protected Pfaffian identification follows separately from Theorem 2.1.

6.8.0.4 Identities. The source presentation map $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$ is continuous and degree-preserving; (R5) gives injectivity after the completed radical quotient; (R6) gives surjectivity; (R7) gives PBW compatibility; (R8) gives exactness in the inverse-limit passage. No claim on signed superdimensions sdim $= f(nm, l)$ substitutes for the two-integer parity dimensions of (R3).

6.8.0.5 Obstruction classes. Primitive recognition inherits the obstruction tower carried by the charge, moduli, hybrid, orientation, Hall, Pfaffian, and Koszul data and adds the cofinal exact-completion residual (R8). The Beilinson–Drinfeld bar-cobar separation is respected throughout: (R5) is a source kernel-equality theorem, not an inversion of the target bar-cobar counit; (R7) is a PBW graded comparison of source and target enveloping algebras, not a graph-equality argument from the scalar square.

6.8.0.6 Comparison with Vol II. The Vol II [73] operator-algebra realization problem separates into two assertions for $K3 \times E$. The retained Pfaffian and orientation data give

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5, \qquad \epsilon_o = \nu_{\Delta_5},$$

relative to (D_0), $(O1)$, $(O1)^+$, $(O2)$. The primitive Lie-superalgebra assertion $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the cofinal compact-source datum (R0)–(R8). The Drinfeld doubling $Y^+(X) \to G(X) = D(Y^+(X))$ of Vol III [70] requires the Hopf pairing of (R4), completion by (R8), integral form, and stable-envelope transport. The κ-tuple coordinate $\kappa_{\mathrm{BKM}} = 5$ is the weight of Δ_5 and lives at level 4 of the universal stage chain; it is a consequence of the construction, not a hypothesis.

6.9 Finite HN dependencies

The finite HN system has a strict ordered sequence of mathematical dependencies. The pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ is determined by the following data for every $R \in \mathcal{R}_{\mathrm{HN}}$:

(i) the lattice combinatorics of the charge window;
(ii) the finite holomorphic E_3-source datum and the $K3$-to-E chain specialization;
(iii) the finite-type quasi-smooth derived stacks of the finite moduli datum;
(iv) the hybrid local/wrapped factorization datum and the eight-word two-step flag atlas of the hybrid carrier;
(v) the four Cech–Borel orientation classes and the $(O1)^+$ Coxeter coherence of the reduced orientation datum;
(vi) the finite-stage Hall bialgebra object and primitive matrices of the Hall correspondence datum;
(vii) the rank-one Pfaffian crossing and the half-Hilbert orbit transport of the Pfaffian line;
(viii) the bar coalgebra, conilpotent collision coproduct, and the Francis–Gaitsgory cobar quasi-isomorphism of the Koszul source coalgebra;
(ix) the cofinal datum (R0)–(R8) of primitive recognition;

together with the compatible vanishing, at every height R, of

$$(\mathfrak{D}_R^{E_3-\mathrm{src}}, \mathfrak{D}_{\mathrm{hyb},R}, \mathfrak{D}_{\mathrm{Kos},R}, \mathfrak{D}_R^{\mathrm{O1}}, \mathfrak{D}_R^{\mathrm{O1}^+}, \mathfrak{D}_R^{\mathrm{O2}}, \mathfrak{D}_R^{D_0}),$$

including the E_3-source obstruction $\mathfrak{D}_R^{E_3-\mathrm{src}}$. The current finite E_3-source packet verifies only the absence ledger `E3_SOURCE_OBSTRUCTION_LEDGER_VERIFIED`, with `e3_source_certification=false`. the finite-stage morphisms $\rho_{RR'} : \mathfrak{D}_{X,R}^{\mathrm{DI}} \to \mathfrak{D}_{X,R'}^{\mathrm{DI}}$ of Definition 0.244, and the Mittag–Leffler exactness condition [ML] of Definition 0.86 on the cofinal HN inverse system. The inverse limit is a pro-object only after these morphisms satisfy the composition law on the cofinal HN subsystem. The finite presentation has the universal property

$$\mathrm{Hom}_{\mathrm{Pro}(\mathsf{DI}_X^{\mathrm{fin}})}(\mathfrak{E}, \mathfrak{D}_X^{\mathrm{DI}}) \cong \varprojlim_R \varinjlim_S \mathrm{Hom}_{\mathsf{DI}_X^{\mathrm{fin}}}(\mathfrak{E}_S, \mathfrak{D}_{X,R}^{\mathrm{DI}}),$$

so cofinal changes of the finite HN presentation do not change the Dirac–Igusa object; this is Proposition 0.246. The finite morphism rows are cofinal_subsystems, stage_objects, stage_morphisms, component_maps, composition_laws, ml_exactness, pro_isomorphisms, and scalar_firewall. Scalar Pfaffian or denominator identities do not define the pro-object transition maps. Each obstruction is attached to the unique finite datum in which its components live.

The theorem-dependency ledger is a separate finite packet:

beilinson_levels, objects, morphisms, theorem_dependencies, dependency_edges, forbidden_edges, status_sep

It records Beilinson-level sites, typed objects, morphisms in their categories, allowed proof edges, forbidden proof edges, claim-status separation, the ambient category of each load-bearing equality, and strict transition of the dependency graph. The beilinson_levels rows fix `P` as primitive compact-source and finite HN pro-object data, `C` as chiral Koszul comparison data, `S` as Pfaffian and orientation Picard-groupoid data, `Z` as protected trace data, and `A` as automorphic, period, and open acting-object data. The equality_sites rows separate pro-category identities, K_0-class identities, Picard-groupoid equalities, derived quasi-isomorphisms, completed Lie-superalgebra equalities, automorphic line-bundle equalities, and identities that exist only after protected scalar trace. The definition_separation rows record the non-circular construction order: Pfaffian line data precede the automorphic comparison and the desired equality, primitive Hall data precede target multiplicity comparison, orientation data precede scalar signs, Koszul source data precede the target counit comparison, and the Dirac–Igusa pro-object precedes section and trace comparisons. The forbidden rows exclude the mirror conjecture, the gravity-line residual, the scalar trace, the OP scalar branch, the target counit, target labels, and automorphic-section data from proof roles in the Pfaffian, orientation, Koszul, primitive-recognition, and construction theorems unless the corresponding theorem explicitly lives at that level. The packet `certificates/dependencies/k3e_theorem_dependencies` is verified as `SCHEMA_COMPLETE_SCHEMA_ONLY`. This is a theorem-architecture certificate: it checks typed rows, zero defect ranks, required theorem types, required forbidden edges, status separation, equality-site coverage, definition separation, obstruction independence, and scalar-firewall coverage. It is not a mathematical certificate for any retained construction packet. The obstruction_independence rows record the finite-row witnesses of Proposition 0.2: each named obstruction datum has its own deleted-row residue, and no vanishing predicate is inferred from another one.

6.9.0.1 Order rigidity. Each of the seven adjacent dependencies fails under exchange:

(i) charge window $\to$ finite moduli: Λ_{ample} and the retained (H, σ)-chamber parametrise the HN strata; without the window, the moduli stack has no finite cofinal subsystem.

(ii) finite moduli $\to$ hybrid carrier: $\mathrm{Ran}^{\mathrm{hyb}}(E)$ needs the finite Hall stack as factor on which to graft the wrapped/local atlas.

(iii) hybrid carrier $\to$ reduced orientation: the cosection-twisted Joyce–Upmeier line bundle is defined on the carrier, not on the raw moduli (the cosection lives on $\mathrm{Ran}^{\mathrm{hyb}}$).

(iv) reduced orientation $\to$ Hall correspondences: the Pfaffian line bundle on the Hall stack is the orientation-twisted sign data; without orientation, the Hall product is unsigned.

(v) Hall correspondences $\to$ Pfaffian line + first-order block: the primitive Hall bracket $[-,-]_X$ defines the Hall–Drinfeld correspondence whose Pfaffian section is the Dirac–Igusa Pfaffian.

(vi) Pfaffian line $\to$ chiral Koszul source coalgebra: the source C_X is the bar of the augmented binary/two-step hybrid correspondence algebra pulled back along the Pfaffian comparison ι_{aut}; it becomes the bar of a hybrid factorization algebra only after the higher-coloured residual vanishes.

(vii) chiral Koszul source $\to$ primitive recognition: the primitive part $P_X^{\Pi,\mathrm{red}}$ is the Frobenius–Serre primitive filtration on the source; the recognition theorem matches it to $\mathfrak{g}_{\Delta_5}$ only after the finite compact-source datum is fixed.

Base-change in S and E, chamber dependence on the K3 polarization, and the elliptic-origin character correction are part of the same input order. The admissible parameter datum $\mathfrak{p} = (S, H, \sigma, E, o_E, \tau_X, \mathcal{R}_{\mathrm{HN}})$ of Definition 1.36 fixes the finite HN system before the pro-object is formed. A change of K3 polarization or stability datum requires an explicit wall-crossing system; a change of elliptic origin from o_E to a twists the wrapped anchor by $\mathcal{O}_E(\chi(\mathcal{F})(o_E - a))$ and changes the finite stabilizer linearization by the corresponding character $\vartheta^H_{a,\mathcal{F}}$ of Definition 1.38; the resulting pro-isomorphism is Proposition 1.39. The order of the constructions is forced. If the orientation datum is placed before the hybrid correspondence datum, the middle type-II wall $\delta_2 \leftrightarrow (0,1,1)$ has no projection-finite local representative by Proposition 0.15. If primitive recognition is placed before the Koszul cobar comparison, the morphism $\Omega^{\mathrm{ch}}_E C_X \to \mathtt{Den}_{\Delta_5,E}$ is absent. The same dependency obstruction applies to the remaining adjacent transpositions.

6.9.0.2 Conditional theorem. With the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum of Appendix 7, Theorem 0.253 gives

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5, \qquad \epsilon_0 = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The primitive isomorphism $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source recognition datum. The scalar square $Z_{\mathrm{BPS}}^{K3\times E} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$ is the level-$\mathtt{Z}$ protected scalar trace. Its finite trace packet is

trace_categories, protected_trace_functors, trace_operators, scalar_normalizations, trace_identities, forgetful_m

These rows place the trace in the level-$\mathtt{Z}$ category, fix the OP and unscaled normalizations, prove the inverse-square scalar identity, and record that orientation, bracket, Hopf pairing, PBW, parity, and source Koszul data lie in the kernel of the scalar trace. The gravity-residual rows must leave the level-$\mathtt{A}$ Hall–Borcherds operator algebra open and unused in the trace proof. A three-dimensional gravity-line interpretation requires the level-$\mathtt{A}$ residual; it is not asserted here. The current finite trace packet supplies only the obstruction ledger `TRACE_OBSTRUCTION_LEDGER_VERIFIED`, with `protected_trace_certification=false`. It records missing level-$\mathtt{Z}$ trace rows; it is not a construction of (Z_{HB}).

7 The D0 criterion and the retained orientation data of the Pfaffian–Dirac theorem

The Pfaffian part of the cross-level identity ②↔③ of Chapter 7 is conditional on four obstruction classes of §1.2, $(D0)$, $(O1)$, $(O1)^+$, $(O2)$, and on the finite geometric Pfaffian datum (P_{fin}). The retained D0-HN orientation datum belongs to the trivial-canonical threefold $X = K3 \times E$; the retained quotient-orientation datum belongs to $(O1)$; the $(O1)^+$ transport is relative to that datum; the retained wall-atlas datum belongs to $(O2)$. The primitive-recognition datum and the level-A gravity-line problem remain separate. The retained Do datum, the orientation data, and the finite Pfaffian sections combine Joyce–Upmeier orientation theory [56], Kiem–Li cosection localisation [59], the perverse-sheaf chart cocycle of Brav–Bussi–Dupont–Joyce–Szendrői [12], and the K3-side reduced theory of Maulik–Pandharipande [80], with $K3 \times E$-specific finite-stabilizer criteria at the Klein-four edge, the full two-primary $E[N]$-edge, half-Hilbert orbit transport, and the $(R1)$–$(R2)$ scalar separation.

7.1 $K3 \times E$ datum and criterion theorem

The standing data are: S a smooth complex projective $K3$ surface, E a smooth elliptic curve, $X = S \times E$ the trivial-canonical threefold, $\sigma_S \in H^0(S, K_S)$ the holomorphic two-form trivialising K_S, $\sigma = \mathrm{pr}_S^* \sigma_S \in H^0(X, p_S^* K_S)$ the inflated K3 two-form defining the semiregularity cosection, and $\mathfrak{M}_c^{\mathrm{red},+}(X)$ the cosection-reduced moduli of effective dyonic class $c \in \Gamma_{\mathrm{eff}}$ on X in the chosen stability heart fixed in Chapter 2. The Do sector is the zero-dimensional sector on the threefold X; the Hilbert–Chow morphism $\nu_n : \mathrm{Hilb}^n(K3) \to \mathrm{Sym}^n(K3)$ furnishes only the K3-fibre scalar test for the reduced theory. The Mukai pairing of Definition 1.3 is symmetric of signature $(4, 20)$.

Definition 7.1 (Retained type-II wall-atlas datum). The datum $(O2_{\mathrm{atlas}})$ consists, for the three simple type-II walls $\partial_1, \partial_2, \partial_3$, of retained wall objects with full $\widehat{\Gamma}_X$-charge match, semistability on the wall, reduced self-Ext splitting, rank-one normal Pfaffian block, invariant Pfaffian unit, quotient orientation, and overlap compatibility on the cofinal HN system. Concretely, for each retained HN height R it is a system of wall charts $(U_{\partial,R}, u_{\partial,R}, \mathscr{L}^{\tan}_{\partial,R}, v_{\partial,R})$ with $s_\partial(u_{\partial,R}) = -u_{\partial,R}$,

$$\mathrm{pf}_{\partial,R} = v_{\partial,R} u_{\partial,R},$$

transition isomorphisms on double overlaps, a Cech 2-cocycle $c^{\mathrm{wall}}_{\alpha\beta\gamma} \in \{\pm 1\}$, and a 1-cochain $b^{\mathrm{wall}}_{\alpha\beta}$ satisfying $\partial b^{\mathrm{wall}} = c^{\mathrm{wall}}$. These data are required to commute with Thom–Sebastiani products, HN transition maps, the chosen Weyl lifts, and the half-Hilbert orbit chart system carrying the twelve binary sign coordinates used in the scalar wall count.

7.1.0.1 Finite proof tables for $(O2_{\mathrm{atlas}})$.

At finite HN height R, the retained wall atlas is supplied by the tables

wall_objects, wall_charts, overlaps, orbit_transport, half_hilbert_orbit, compatibilities, scalar_separation.

The table wall_objects records full charge match, wall semistability, reduced self-Ext splitting, the rank-one normal block, the invariant unit, and the quotient orientation. The table wall_charts records normal rank 1, divisor order 1, reflection $u \mapsto -u$, unit invariance, and local sign -1. The tables overlaps and orbit_transport record vanishing Cech/cochain, unit-overlap, orientation-overlap, Pfaffian-transport, rank-preservation, and sign-preservation defects. The table half_hilbert_orbit records twelve binary coordinates, orbit cardinality $2^{12} = 4096$, component and boundary coverage, and zero coverage defect. The table compatibilities records the Thom–Sebastiani, HN-transition, Weyl-lift, quotient-orientation, protected-integration, boundary, and component identities. The table scalar_separation records the integer equality $2^{12} = 64^2 = 4096$ only as a scalar-shadow equality and excludes the OP scalar branch, the Maass character value, and the squared theta-leading constant as entries of the wall atlas. A single rank-one chart does not supply $(O2_{\text{atlas}})$. In the present wall-atlas packet these rows are not supplied. The finite obstruction ledger

`certificates/wall_atlas/k3e_o2_atlas/blocked_obligations.csv`

records the missing wall-object, wall-chart, overlap, Weyl-transport, half-Hilbert-orbit, compatibility, type-I-exclusion, D0-limit, and scalar-separation rows. Its verifier returns `O2_OBSTRUCTION_LEDGER_VERIFIED`, with `o2_certification=false` and `mathematical_certification=false`. Thus the packet is a finite absence ledger for $(O2_{\text{atlas}})$, not a construction of $(O2_{\text{atlas}})$.

Definition 7.2 *(Retained quotient-orientation datum).* The datum $(O1_{\text{quot}})$ consists, on every retained object, extension, wrapped, mixed, two-step flag, overlap, and HN-transition stratum, of null-trivialisations of the ordinary reduced orientation gerbe, the connected BE-edge class, the finite-stabilizer Cech–Borel classes, the zero degree-one stabilizer linearizations, and the Thom–Sebastiani compatibilities of these choices.

Definition 7.3 *(Retained finite geometric Pfaffian datum).* The datum (P_{fin}) is a cofinal HN system carrying the finite geometric Pfaffian datum of Definition 0.200: cosection-reduced skew perfect complexes, Pfaffian lines, finite Pfaffian sections, type-II wall divisor orders, automorphic line comparisons, support and parity rows, normalized leading coefficients, and Pfaffian-line Mittag–Leffler compatibilities on the cofinal HN system. In the present certificate packet these rows are not supplied. The finite obstruction ledger

`certificates/pfaffian/k3e_finite_pfaffian/blocked_obligations.csv`

records the missing retained-stratum, skew-complex, Pfaffian-line, Pfaffian-section, wall-chart, automorphic-comparison, transition, and scalar-firewall rows. Its verifier returns `PFAFFIAN_OBSTRUCTION_LEDGER_VERIFIED`, with `pfaffian_certification=false` and `mathematical_certification=false`. Thus it is a finite absence certificate for (P_{fin}), not a construction of (P_{fin}).

Definition 7.4 *(Retained Pfaffian–Dirac obstruction datum).* The retained obstruction datum on $X = K3 \times E$ is the tuple

$$\mathfrak{R}_X^{\text{DI}} = \big(D0_{\text{HN}},\ O1_{\text{quot}},\ \mathcal{T}_W,\ O2_{\text{atlas}},\ P_{\text{fin}},\ Z_{\text{HB}}\big).$$

Here $D0_{\text{HN}}$ is the retained D0-HN orientation datum of Definition 7.6; $O1_{\text{quot}}$ is the quotient-orientation datum of Definition 7.2; $\mathcal{T}_W$ is the torsor of Weyl determinant-line lifts $\{\tau_i\}_{i=1}^3$ of Definition 0.109, together with a chosen 1-cochain killing the projective cocycle c_o, the equations $\tau_i^2 = 1$, and the vanishing torsor defects $\omega_{i,C} = 0$ on every retained orbit; $O2_{\text{atlas}}$ is the retained type-II wall atlas of Definition 7.1; P_{fin} is the retained finite geometric Pfaffian datum of Definition 7.3; and Z_{HB} is the level-Z Hall–Borcherds trace comparison datum of Definition 7.20.

The projections of $\mathfrak{R}_X^{\text{DI}}$ are part of the definition. The Pfaffian and orientation statements use the first five components. The level-Z protected trace uses Z_{HB}. The primitive-recognition theorem uses neither Z_{HB} nor the mirror discriminant; it requires the finite compact-source recognition datum $(S1)$–$(S10)$. A scalar product, signed denominator table, protected trace, or mirror period identity is not a component of $\mathfrak{R}_X^{\text{DI}}$.

Theorem 7.5 *(Relative Pfaffian-orientation criterion on $K3 \times E$).* On the rank-one and finite-stabiliser strata of the moduli stack $\mathfrak{M}_c^{\text{red},+}(X)$, $X = K3 \times E$, in the stability heart fixed in Chapter 2, let $\mathfrak{R}_X^{\text{DI}}$ be the retained obstruction datum of Definition 7.4. The obstruction class $(D0)$ is represented by its $D0_{\text{HN}}$-component. The class $(O1)$ is represented by its $O1_{\text{quot}}$-component, $(O1)^+$ by its Weyl-lift torsor $\mathcal{T}_W$, and $(O2)$ by its $O2_{\mathrm{atlas}}$-component. If the Cech, Borel, torsor, and wall-atlas equations in these components vanish with the stated null-trivialisations, then the Pfaffian–Dirac theorem 2.1 and the orientation-character match 3.1 hold on $K3 \times E$ relative to $\mathfrak{R}_X^{\mathrm{DI}}$ through its first five components. The primitive recognition theorem 4.1 remains conditional on the finite compact-source recognition datum $(S1)$–$(S10)$.

The four clauses are separated below according to their retained data. The level-Z Hall–Borcherds trace comparison is distinct from the level-A gravity-line residual.

7.2 (Do) Do-degeneration limit of the cosection-reduced d-critical orientation theory

The first obstruction concerns the inverse-limit assembly of the Joyce–Upmeier orientation cocycle along the HN system $\mathcal{R}_{\mathrm{HN}}$ when the chosen stability heart degenerates to the Do-supported subcategory of zero-dimensional sheaves on $X = K3 \times E$. The Do-charge is the threefold length n. Projection to the K3 factor gives the reduced K3 scalar test; it does not replace the threefold Do moduli. The Do-degeneration limit must commute with cosection-reduction and with the orientation extension.

Definition 7.6 (Retained Do-HN orientation datum). [definitional] For each $R \in \mathcal{R}_{\mathrm{HN}}$, let $\mathfrak{M}_R$ be the retained finite-type HN substack of objects of height $\le R$ in the chosen stability heart, and let $\mathfrak{M}_R^{D\mathrm{o}} \subset \mathfrak{M}_R$ be the retained zero-dimensional threefold sector, with Do-charge equal to length on $X = K3 \times E$. The datum $(D\mathrm{o}_{\mathrm{HN}})$ consists of:

(Do.1) A flat one-parameter degeneration $\mathfrak{M}_R \to \mathbb{A}^1$ whose special fibre contains $\mathfrak{M}_R^{D\mathrm{o}}$, together with compatible object, extension, mixed, wrapped, and two-step flag substacks over $\mathbb{A}^1$.

(Do.2) Proper or closed HN transition morphisms

$$\rho^{\mathrm{stk}}_{RR'} : \mathfrak{M}_R|_{\le R'} \longrightarrow \mathfrak{M}_{R'} \qquad (R' \le R)$$

on the object, extension, and flag stacks, satisfying the composition law and preserving the Do-special fibre.

(Do.3) Quasi-smooth derived enhancements with PTVV (-1)-shifted symplectic forms, classical d-critical truncations, and perfect obstruction theories on every retained stratum of the degeneration family.

(Do.4) The K3 semiregularity cosection

$$\mathrm{CS}_{C,R} : h^1\Big((E^\bullet_{C,R})^\vee\Big) \longrightarrow \mathcal{O}_{\mathfrak{M}_R}$$

obtained by contracting the obstruction theory with $\mathrm{pr}_S^* \sigma_S$, surjective on every retained reduced stratum, compatible with the Do-special fibre, and additive on extension and flag stacks:

$$q^* \mathrm{CS}_{C,R} = p_1^* \mathrm{CS}_{A,R} + p_2^* \mathrm{CS}_{B,R} \, .$$

(Do.5) Cosection-localized vanishing-cycle complexes Φ_R^{red}, with specialization isomorphisms from the generic fibre to the Do-special fibre and transition isomorphisms under $\rho^{\mathrm{stk}}_{RR'}$.

(Do.6) Joyce–Upmeier orientation lines o_R on the reduced d-critical charts, with square-root isomorphisms for the reduced determinant complex, Thom–Sebastiani multiplicativity on retained extensions, and transition isomorphisms carrying o_R to $o_{R'}$.

(Do.7) Pfaffian-line and protected-integration compatibility: the Pfaffian lines, finite Pfaffian sections, quotient-after-correspondence maps, compact-support six-functor operations, and protected integration maps extend across the Do-special fibre and commute with $\rho^{\mathrm{stk}}_{RR'}$.

(D0.8) Strict Mittag–Leffler exactness for the inverse systems of retained moduli cohomology, reduced vanishing cycles, orientation gerbes, Pfaffian lines, compact-support operations, Hall product/coproduct maps, and the corresponding entries of the finite-stage morphism $\rho_{RR'}$ of Definition 0.244.

The D0-degeneration limit is the pro-object obtained from (D0.1)–(D0.8). The scalar Hilbert-scheme specialization on $K3$ is only a test of the reduced scalar theory; it is not an entry of $(D0_{\mathrm{HN}})$ and does not construct the transition morphisms above.

After finite presentations of the HN stacks and their correspondence atlases, $(D0_{\mathrm{HN}})$ is recorded by the D0 proof tables

degeneration_families, retained_substacks, hn_transitions, derived_enhancements, semiregularity_cosections, va

These rows are finite data: flatness defects, composition defects, obstruction-theory defects, cosection cokernels, additivity defects, specialization cones, transition defects, and $R^1 \varprojlim$-ranks must vanish. The retained-substack rows must include object, extension, mixed, wrapped, and two-step flag substacks. The Mittag–Leffler rows must include retained moduli cohomology, vanishing cycles, orientation gerbes, Pfaffian lines, compact-support operations, Hall product, Hall coproduct, and finite-stage morphisms. The scalar firewall excludes Hilbert-scheme scalar specialization, the K3 scalar test, and scalar trace as entries of $(D0_{\mathrm{HN}})$. In the present certificate packet the D0-HN rows are not supplied. The finite obstruction ledger

```
certificates/d0/k3e_d0_hn/blocked_obligations.csv
```

records the missing flat-family, D0-sector, retained-substack, HN-transition, derived-enhancement, semiregularity-cosection, vanishing-cycle, orientation, Pfaffian, protected-integration, Mittag–Leffler, and scalar-firewall rows. Its verifier returns `D0_OBSTRUCTION_LEDGER_VERIFIED`, with `d0_certification=false` and `mathematical_certification=false`. Thus it is a finite absence certificate for $(D0_{\mathrm{HN}})$, not a proof of $(D0_{\mathrm{HN}})$.

THEOREM 7.7 *((D0) criterion).* Granted $(D0_{\mathrm{HN}})$, the cosection-reduced d-critical orientation theory extends compatibly along the HN inclusions ι_R and has a D0-degeneration pro-limit on $\mathfrak{M}^{\mathrm{red}}_{\infty}$.

Proof. PTVV supplies the ambient (-1)-shifted symplectic source on the trivial-canonical threefold moduli [95, Theorem 0.3]; BBDJS constructs vanishing cycles after an oriented d-critical atlas is present [12, Theorem 6.9]; Kiem–Li gives the cosection localisation formalism [59]; Joyce–Upmeier gives the orientation-line and multiplicativity technology [55, 56]. These theorems do not by themselves produce the retained HN-compatible reduced orientation system. Clauses (D0.1)–(D0.3) supply the degeneration family and finite derived/d-critical atlases; clause (D0.4) supplies the surjective additive K3 semiregularity cosection; clauses (D0.5)–(D0.7) supply the vanishing-cycle, orientation, Pfaffian, Hall, quotient, and protected-integration specializations; and clause (D0.8) supplies the strict Mittag–Leffler exactness needed to pass to the pro-limit. The pro-limit therefore inherits the reduced vanishing-cycle complexes, orientation lines, Thom–Sebastiani transports, compact-support operations, and Pfaffian lines. Thus the retained HN datum gives the $(D0)$ vanishing condition on the pro-limit. □

Remark 7.8 *(The role of the K3 elliptic genus).* The K3 elliptic-genus formula $\chi_y(\mathrm{Hilb}^n(K3))$ [80, §4.1] tests the scalar D0 specialization. It is not a substitute for Mittag–Leffler exactness of the orientation gerbes, vanishing-cycle complexes, compact-support functors, or Pfaffian lines.

7.3 (O1) strong reduced orientation on the K3×E-quotient self-Ext frame bundle

The second obstruction is the strong reduced orientation: the multiplicative orientation cocycle of $(D0)$ must be equivariant under the free E-translation quotient and under the finite-stabilizer strata $BE[N]$ on which the E-action degenerates.

Definition 7.9 (Strong reduced orientation on the K3×E-quotient frame). [definitional] The *strong reduced orientation* on $\mathfrak{M}_c^{\mathrm{red},+}(X)$ is a multiplicative orientation cocycle o_c^{strong} on the principal Spin^c-bundle $\mathcal{F}_c^{\mathrm{red}}$ of the reduced derived self-Ext frame bundle on $\mathfrak{M}_c^{\mathrm{red},+}(X)$, satisfying:

(O1.a) *Free E-translation equivariance.* On the generic E-translation orbit of codimension 0, o_c^{strong} descends through the free quotient $\mathfrak{M}_c^{\mathrm{red},+}(X)/E$.

(O1.b) *Klein-four edge and linearization condition.* Across the $BE[2]$-strata, the Cech–Borel class reduces to the edge class of Lemma 7.10, that class vanishes, and the residual degree-one $E[2]$-linearization character is zero.

(O1.c) *Full two-primary finite-stabilizer condition.* Across the $BE[N]$-strata for $N \geq 3$, the two-primary edge class and the residual stabilizer character satisfy the criterion of Lemma 7.11.

Lemma 7.10 *(Klein-four edge criterion on rank-2 E-strata).* Assume that the finite-stabilizer Cech–Borel class on a retained $BE[2]$-stratum has no ordinary, mixed, or differential term and therefore has an edge representative in $H^2(BE[2], \mathbb{F}_2)$. For $E[2] \simeq (\mathbb{Z}/2)^2$,

$$H^*(BE[2], \mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \qquad |x_i| = 1,$$

and the edge class has the form

$$\beta^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2.$$

If r_i denotes its restriction to the three non-zero cyclic subgroups of $E[2]$, then

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3.$$

Hence $\beta^{E,2} = 0$ is equivalent to $r_1 = r_2 = r_3 = 0$ for this edge class. After this degree-two class is killed, the remaining linearization character is

$$\lambda^{E,2} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BE[2], \mathbb{F}_2),$$

and its restrictions satisfy

$$\rho_1 = \lambda_1, \qquad \rho_2 = \lambda_2, \qquad \rho_3 = \lambda_1 + \lambda_2.$$

The vanishing of r_i is therefore not the vanishing of ρ_i; quotient orientation requires both sets of equations.

Proof. The cohomology ring is the polynomial ring displayed above. Restricting to $\langle e_1\rangle$, $\langle e_2\rangle$, and $\langle e_1 + e_2\rangle$ sends the quadratic form respectively to $b_{20}t^2$, $b_{02}t^2$, and $(b_{20} + b_{11} + b_{02})t^2$. Over $\mathbb{F}_2$ this gives the displayed invertible linear change of coordinates. The last assertion is the standard statement that, after a degree-two gerbe is trivialised, linearizations form an $H^1(BE[2], \mathbb{F}_2)$-torsor. □

Lemma 7.11 *(Full two-primary finite-stabilizer criterion).* For a complex elliptic curve,

$$E[N] \simeq (\mathbb{Z}/N)^2.$$

If N is odd, the mod-2 finite-stabilizer orientation class restricts trivially by transfer. If $2^a \parallel N$, then the mod-2 calculation factors through

$$E[N]_{(2)} \simeq (\mathbb{Z}/2^a)^2.$$

For $a = 1$ one obtains Lemma 7.10. For $a \geq 2$,

$$H^*\big(B(\mathbb{Z}/2^a)^2, \mathbb{F}_2\big) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2,$$

and every degree-two edge class has the form

$$\beta^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1 x_2 + A_2^{(N)} y_2.$$

The residual stabilizer character is

$$\lambda^{E,N} = \lambda_1^{(N)} x_1 + \lambda_2^{(N)} x_2.$$

Thus quotient orientation at the N-stratum requires

$$A_1^{(N)} = A_{12}^{(N)} = A_2^{(N)} = \lambda_1^{(N)} = \lambda_2^{(N)} = 0.$$

The coefficient $A_{12}^{(N)}$ is not detected by cyclic order-two restrictions.

Proof. The group of N-torsion points of a complex elliptic curve is $(\mathbb{Z}/N)^2$. For N odd the group order is invertible in $\mathbb{F}_2$, hence $H^{>0}(BE[N], \mathbb{F}_2) = 0$ by transfer. The even-primary reduction is the primary decomposition of $E[N]$. The cohomology of $B(\mathbb{Z}/2^a)$ is $\Lambda(x) \otimes \mathbb{F}_2[y]$, with $|x| = 1$ and $|y| = 2$; the Künneth formula gives the displayed degree-two basis for the square. A cyclic order-two subgroup pulls back $x_1 x_2$ to zero, so it cannot see the mixed coefficient. □

Remark 7.12 *(What the finite-stabilizer criteria prove).* The two preceding lemmas are detection statements inside the retained datum $(O1_{\mathrm{quot}})$. They do not construct the Cech–Borel null-trivialisations, do not kill mixed Borel terms, and do not remove the residual H^1-linearization characters. They say which coordinates must vanish after the finite-stabilizer problem has been reduced to the classifying-space edge. The finite obstruction ledger `certificates/orientation/k3e_reduced_orientation/blocked_obligations.csv` records the missing orientation rows before this criterion can be used as an unconditional theorem: square-root, Thom–Sebastiani, quotient Borel, finite-stabilizer, Weyl-lift, Coxeter, transition, and scalar-firewall rows. Its verification status is `ORIENTATION_OBSTRUCTION_LEDGER_VERIFIED`, not orientation certification.

Theorem 7.13 *((O1) quotient-orientation criterion).* Granted the retained quotient-orientation datum $(O1_{\mathrm{quot}})$ of Definition 7.2, the reduced orientation of Theorem 7.7 descends to the strong reduced orientation o_c^{strong} of Definition 7.9 on $\mathfrak{M}_c^{\mathrm{red},+}(X)$.

Proof. The reduced orientation o_∞^{red} of Theorem 7.7 and the datum $(O1_{\mathrm{quot}})$ determine $(O1.a)$, $(O1.b)$, and $(O1.c)$.

$(O1.a)$ *Free E-equivariance.* On the free translation locus the connected BE-edge class $\alpha^{E,\mathrm{free}} \in H^2(BE, \mathbb{F}_2)$ is one of the null-trivialised classes in $(O1_{\mathrm{quot}})$. The chosen trivialisation, together with Joyce–Upmeier multiplicativity [56, §4], gives descent through the open quotient.

$(O1.b)$ *Klein-four extension.* Lemma 7.10 identifies the $BE[2]$-edge equations. The datum $(O1_{\mathrm{quot}})$ supplies their vanishing and the zero $H^1(BE[2], \mathbb{F}_2)$-linearization character.

$(O1.c)$ *Higher-rank extension.* Lemma 7.11 identifies the full two-primary finite-stabilizer equations for $BE[N]$, $N \geq 3$. The same datum supplies their vanishing and the compatible zero stabilizer characters on overlaps, Hall correspondences, flags, and HN transitions.

The combined cocycle is multiplicative and Pin^c-equivariant by the multiplicativity of the supplied Thom–Sebastiani transports; this is o_c^{strong}. By the multiplicativity, every product $F_1 \to E \to F_2$ of effective dyonic objects has $o_c^{\mathrm{strong}}(F_1 \to E \to F_2) = o_c^{\mathrm{strong}}(F_1) \otimes o_c^{\mathrm{strong}}(F_2)$. This is the strong reduced orientation. □

7.4 $(O1)^+$ Weyl-equivariant transport along $W^{(2)}(\Lambda_{II}^{2,1})$

The third obstruction is the Weyl-equivariance of the strong reduced orientation along the type-II Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$. The Weyl group is the 3-generator Coxeter group generated by the three simple type-II reflections $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$. Since $(\delta_i, \delta_j) = -2$ for $i \neq j$, the Coxeter exponents are $m_{ij} = \infty$; there are no braid relations among distinct generators. Hence $(W^{(2)})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^{\oplus 3}$.

Theorem 7.14 *($(O1)^+$ Weyl transport criterion and type-II character comparison).* The strong reduced orientation o_c^{strong} of Theorem 7.13, together with the chosen lifts of Definition 0.109 with vanishing projective cocycle and zero quotient-torsor defects, is $W^{(2)}(\Lambda_{II}^{2,1})$-equivariant. If, in addition, the rank-one type-II wall signs of $(O2_{\mathrm{atlas}})$ are fixed on the three simple walls, then its character on $W^{(2)}$ is

$$\epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on } W^{(2)}(\Lambda_{II}^{2,1}),$$

identical to the Maass multiplier system of the Igusa cusp form.

Proof. The proof compares the equivariant orientation line with the Maass character on the type-II Coxeter generators.

Equivariant determinant line. The vanishing of the projective cocycle c_o makes the chosen determinant-line lifts into an honest action of $W^{(2)}(\Lambda_{II}^{2,1})$ on o_c^{strong}. The vanishing of the torsor defects $\omega_{i,C}$ says that the quotient-orientation cocycle is transported with the line, not only after forgetting the finite-stabilizer data. This proves the $(O1)^+$ equivariance assertion.

Generator values from $(O2)$. Under $(O2_{\mathrm{atlas}})$, Proposition 5.2 gives the local Pfaffian sign at each simple type-II wall:

$$\epsilon_o^{\mathrm{loc}}(s_{\delta_i}) \;=\; (-1)^{N_{\delta_i}^{\mathrm{Pf}}} \;=\; -1, \qquad i = 1, 2, 3,$$

with $N_{\delta_i}^{\mathrm{Pf}} = 1$ the rank-one Pfaffian rank exponent. By Lemma 5.3, the Maass character on these generators is $\nu_{\Delta_5}(s_{\delta_i}) = -1$ for $i = 1, 2, 3$. Hence $\epsilon_o(s_{\delta_i}) = \nu_{\Delta_5}(s_{\delta_i}) = -1$ for the three generators.

The abelianisation $(\mathbb{Z}/2)^{\oplus 3}$. The type-II Weyl group has the Coxeter presentation

$$W^{(2)}(\Lambda_{II}^{2,1}) = \langle s_1, s_2, s_3 \mid s_1^2 = s_2^2 = s_3^2 = 1 \rangle,$$

with no relation on $s_i s_j$ for $i \neq j$. This is the Coxeter presentation attached to the symmetric matrix (δ_i, δ_j), because $(\delta_i, \delta_j) = -2$ gives $m_{ij} = \infty$. Abelianising only imposes commutativity and $s_i^2 = 1$, hence

$$(W^{(2)})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^{\oplus 3}.$$

Extension to the full Weyl group. A character of $W^{(2)}$ is determined by its values on the abelianisation, hence by its values on the three generators. The extension principle is standard: given $\chi : \{s_1, s_2, s_3\} \to \{\pm 1\}$, define $\chi(w) = \prod_{j=1}^{k} \chi(s_{i_j})$ for any reduced word $w = s_{i_1} \cdots s_{i_k}$. The only defining relations are $s_i^2 = 1$; hence the formula is well-defined.

Character comparison. The group $H^1(W^{(2)}, \{\pm 1\})$ is $\mathrm{Hom}(W^{(2)}, \{\pm 1\}) \simeq (\mathbb{Z}/2)^{\oplus 3}$, not the obstruction group. The obstruction is represented by the projective cocycle c_o and torsor defects $\omega_{i,C}$ of Definition 0.109. When these classes vanish and the three rank-one $(O2)$ signs are fixed, both ϵ_o and ν_{Δ_5} are $\{\pm 1\}$-valued characters of $W^{(2)}$ with the same values on the three generators by the local $(O2)$ computation; hence they are equal: $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$. □

Remark 7.15 *(Why the type-II Coxeter group reduces to its abelianisation).* The reduction to the abelianisation is forced by the order-two structure of the $\{\pm 1\}$-character group: any group homomorphism into a group of exponent two factors through the abelianisation, since exponent-two characters automatically annihilate the commutator subgroup. There is no order-three braid relation in the type-II Coxeter group; the off-diagonal value -2 is the infinite Coxeter exponent. Thus no order-three obstruction class is present in this comparison.

7.5 (O2) Local Pfaffian wall normal form and the $(R1)$–$(R2)$ scalar separation

The fourth obstruction is the type-II wall normal form: the local Pfaffian section on each type-II wall is the rank-one odd crossing of Appendix E, and the $2^{12} = 4096$ sign-sum on the half-Hilbert orbit remains separate from the OP normalization until the scalar shadow is taken.

Theorem 7.16 *((O2) wall-atlas consequence: type-II wall normal form).* Assume $(O2_{\mathrm{atlas}})$ and the Weyl-transport datum of Theorem 7.14. On each type-II wall $\delta \in W^{(2)}(\Lambda_{II}^{2,1}) \cdot \{\delta_1, \delta_2, \delta_3\}$ of $\mathfrak{M}_c^{\mathrm{red},+}(X)$, the local Pfaffian section pf_δ is the rank-one odd crossing $[\mathbb{R} \xrightarrow{u} \mathbb{R}]$ with Pfaffian rank exponent $N_\delta^{\mathrm{Pf}} = 1$ and local sign $\epsilon_o^{\mathrm{loc}}(s_\delta) = -1$.

Proof. The proof reduces higher-rank type-II walls to the rank-one normal form of Appendix E by Weyl-orbit transport.

Rank-one walls. For $\delta \in \{\delta_1, \delta_2, \delta_3\}$, the rank-one normal form is Proposition 5.2, supplying $N_\delta^{\mathrm{Pf}} = 1$ and $\epsilon_o^{\mathrm{loc}}(s_\delta) = -1$.

Orbit-stabiliser theorem. For higher-rank walls $\delta = w \cdot \delta_i$ with $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and $i \in \{1, 2, 3\}$, the local geometric chart $U_\delta = T_\delta \times (-\varepsilon, \varepsilon)$ is identified in the retained wall atlas with the w-transport of the rank-one chart $U_{\delta_i} = T_{\delta_i} \times (-\varepsilon, \varepsilon)$. By Theorem 7.14 (Weyl-equivariance of the strong reduced orientation), the local Pfaffian section transports as

$$\mathrm{pf}_\delta \;=\; w_* \,\mathrm{pf}_{\delta_i} \cdot \epsilon_o(w),$$

with $\epsilon_o(w) \in \{\pm 1\}$ the Weyl character value.

Rank exponent invariance. The Pfaffian rank exponent $N_\delta^{\mathrm{Pf}} = 1$ is preserved under the supplied Weyl-orbit chart identifications, since it is the rank of the normal factor in the reduced d-critical chart.

Local sign. The local sign $\epsilon_o^{\mathrm{loc}}(s_{\tilde{\partial}}) = \epsilon_o^{\mathrm{loc}}(ws_{\partial_i}w^{-1})$. Since the orientation sign is a character after Theorem 7.14,

$$\epsilon_o(ws_{\partial_i}w^{-1}) = \epsilon_o(w)\epsilon_o(s_{\partial_i})\epsilon_o(w)^{-1} = \epsilon_o(s_{\partial_i}) = -1.$$

Hence $\epsilon_o^{\mathrm{loc}}(s_{\tilde{\partial}}) = -1$ for every Weyl-translate. □

THEOREM 7.17 *((R1)–(R2) scalar separation: $4096 = 2^{12} = 64^2$).* Two independent normalizations have the same integer value:

(R1) (operator-level) $N_{\mathrm{wall}} = 2^{12}$, the count of sign assignments in the half-Hilbert wall atlas of (O2);
(R2) (automorphic-level) $N_{\mathrm{OP}} = 64^2 = c_\Delta(1,1,1)^2$, the squared theta-leading constant of the type-II Borcherds product.

The equality $N_{\mathrm{wall}} = N_{\mathrm{OP}} = 4096$ is an equality of integers after passing to a scalar shadow. It is not an identification of the orientation character with the OP scalar normalization.

Proof. The (O2) atlas supplies 12 binary wall signs, hence

$$N_{\mathrm{wall}} = 2^{12} = 4096.$$

The leading Fourier coefficient of Δ_5 at the type-II vacuum is

$$c_\Delta(1,1,1) = [q^{1/2}r^{1/2}s^{1/2}]\,\Delta_5 = 64$$

by Lemma 5.2. Squaring,

$$N_{\mathrm{OP}} = c_\Delta(1,1,1)^2 = 64^2 = 4096.$$

Thus

$$N_{\mathrm{wall}} = N_{\mathrm{OP}}.$$

No step identifies the wall orientation line with the OP scalar branch. □

Remark 7.18 *((R1) and (R2) live on different levels).* The wall count belongs to the Pfaffian orientation atlas. The factor 64^2 belongs to the automorphic normalization $D_5 = 64^{-1}\Delta_5$ and to the OP scalar convention. Their common integer is used only after orientation has been forgotten by squaring.

THEOREM 7.19 *((O2) wall-atlas transport on the half-Hilbert orbit).* Assume $(O2_{\mathrm{atlas}})$ and the Weyl-transport datum of Theorem 7.14. The orientation cocycle on the half-Hilbert orbit on $\mathfrak{M}_c^{\mathrm{red},+}(X)$ extends from the rank-one local data of Theorem 7.16 to a globally consistent $(\mathbb{Z}/2)$-valued system on the entire half-Hilbert orbit, with the total sign count 2^{12} realised by the twelve binary wall coordinates of the $(O2_{\mathrm{atlas}})$ datum. Equivalently, the type-II Pfaffian-wall normal form is reduced to $(O2_{\mathrm{atlas}})$.

Proof. Theorem 7.16 supplies the rank-one normal form on every type-II wall relative to the wall atlas. Theorem 7.14 supplies the $W^{(2)}$-equivariance. Theorem 7.17 identifies the integer $2^{12} = 4096$ with the squared theta-leading normalization, after passing to the scalar shadow. The half-Hilbert orbit transport is the following Weyl-orbit calculation. A base point on the orbit determines every other orbit point by Weyl transport, and the orientation sign is the product of generator signs along the chosen path. The result is independent of path by the chosen Weyl lifts and zero torsor defects of Theorem 7.14, and the total count is the $2^{12} = 4096$ count supplied by the wall atlas. This is the $(O2)$ transport theorem relative to $(O2_{\mathrm{atlas}})$. □

7.6 Vol II Hall–Borcherds trace comparison at level Z: chain-level chiral-Hochschild trace

The retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained type-II wall atlas give the relative Pfaffian and orientation hypotheses of the Pfaffian–Dirac theorem on $K3 \times E$. A separate trace comparison relates the level-n protected scalar shadow $Z_{\mathrm{BPS}}^{K3\times E} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$ of Chapter 9 to a chain-level chiral-bulk trace on the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$ on $K3 \times E$. The level-A Hall–Borcherds residual is different: it asks for the gravity-line operator algebra acting on the boundary and remains open in Vol II.

Definition 7.20 *(Vol II Hall–Borcherds trace comparison at level* Z*).* [definitional] The *level-*Z *Hall–Borcherds trace comparison* on $K3 \times E$ is the assertion identifying the protected graded trace

$$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F q^{L_0} r^{J_0} s^{\widetilde{L}_0}\big) = Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$$

of the retained E_3-prefactorization algebra $\mathcal{A}_{K3\times E}$ to the level-Z trace pairing of the chain-level chiral derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$ on $(X = K3 \times E, D, \tau)$ at the boundary vacuum b; the identification $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X) = \mathrm{ChirHoch}^\bullet(C_X, C_X)$ is the chiral-Hochschild concentration theorem of Vol I (Theorem H, [72]) on the Koszul locus, arranged by the level-Z clause (i) of the cyclic-Hochschild Beilinson stratification of Vol II [73, chiral-Hochschild Beilinson stratification]; the trace pairing of degree -3 is the CY-orientation pairing clause (ii) of the same theorem under $\mathsf{Eff}_{\mathrm{Pf,or}}$. The comparison is a class in

$$H^*_{\mathrm{Hall}}(\mathcal{A}_{K3\times E}) \;\to\; H^*_{\mathrm{Borcherds}}\big(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}_{K3\times E})\big),$$

governing whether the chiral-shadow Hall integral Δ_5^{-2} is the shadow of a chain-level chiral-Hochschild Borcherds integral on the target.

Remark 7.21 *(Distinguishing the level-*Z *trace from the level-*A *gravity line).* The level-Z Hall–Borcherds trace comparison of Definition 7.20 has trace-level scope, whereas level A asks for a gravity-line operator algebra acting on the boundary chiral algebra of the $K3 \times E$ holomorphic-topological theory, with Pentagon-face determinant section $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ and reciprocal partition character Δ_5^{-2}. This residual remains open in Vol II; its construction requires the operator-valued completion that the present level-Z trace identity does not supply. The level-Z facet is a trace pairing on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$; the level-A facet is an acting algebra on a gravity-line module. No theorem constructs that acting algebra.

7.7 The level-Z Hall–Borcherds trace comparison on $K3 \times E$

The level-Z Hall–Borcherds trace comparison of Definition 7.20 is a criterion relative to a retained trace datum on $K3 \times E$. Vol I Theorem H [72] and Vol II's chiral-Hochschild Beilinson stratification [73, chiral-Hochschild Beilinson stratification] apply only after that datum has placed the compact source in their domain. The level-A operator-algebra facet remains open in Vol II (Remark 7.21).

Definition 7.22 *(Retained level-*Z *Hall–Borcherds trace datum).* [definitional] The datum (Z_{HB}) consists of the following finite and pro-compatible data:

(Z1) a boundary chart algebra A_b in the Vol I/Vol II chiral-Koszul domain, together with a filtered quasi-isomorphism $A_b \simeq C_X$;

(Z2) a class-M weight completion of $\mathrm{Bar}(A_b)$, compatible with the HN inverse system and the finite Koszul comparison and Mittag–Leffler data;

(Z3) the Vol I chiral-Hochschild concentration hypothesis on A_b and the Vol II stratification hypotheses at level Z;

(Z4) a degree -3 cyclic trace pairing on $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$, compatible with the Calabi–Yau three-form, the Pfaffian line, and HN transition;

(Z5) compatibility of the trace normalization with the type-II Borcherds product of Δ_5, the OP chamber scalar branch, and the protected trace convention of Chapter 9.

(Z_{HB}) has trace-level scope: it supplies a trace pairing on the centre, not an operator algebra acting on a gravity-line module. In the present trace packet these rows are not supplied. The finite obstruction ledger

`certificates/trace/k3e_protected_trace/blocked_obligations.csv`

records the missing trace-category, protected-trace-functor, trace-operator, scalar-normalization, trace-identity, forgetful-map, gravity-residual, transition, and scalar-firewall rows. Its verifier returns `TRACE_OBSTRUCTION_LEDGER_VERIFIED`, with `protected_trace_certification=false` and `mathematical_certification=false`. Thus the packet is a finite absence ledger for (Z_{HB}), not a construction of the level-Z protected trace.

Theorem 7.23 *(Level-Z Hall–Borcherds trace criterion on $K3 \times E$).* Granted the retained level-Z trace datum (Z_{HB}), Vol I Theorem H [72], and Vol II's chiral-Hochschild Beilinson stratification [73] as established theorems, the level-Z Hall–Borcherds trace comparison of Definition 7.20 holds on $K3 \times E$. Equivalently, the level-n protected scalar trace $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ of Chapter 9 is identified with the chain-level CY-orientation trace of the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$ on $K3 \times E$:

$$\boxed{\mathrm{Tr}^{\mathrm{bulk}}_n\big((-1)^F \cdot \mathrm{id}\big)_{Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)} \;=\; \Delta_5^{-2}}\,.$$

The OP chamber branch is the normalized scalar representative $-4096\,\mathrm{Tr}^{\mathrm{bulk}}_n$, not the trace itself. The comparison is at level Z in the Beilinson chain of Vol II; the level-A gravity-line operator algebra is not constructed here and remains open in Vol II (Remark 7.21).

Proof. The retained datum (Z_{HB}) supplies the chart algebra A_b, its comparison with the compact source C_X, the class-M completion, the cyclic trace pairing, and the normalization comparison with the type-II Borcherds product. Clause (Z3), together with Vol I Theorem H and the level-Z clause of Vol II, identifies the chiral derived centre with chiral Hochschild cohomology on the retained Koszul locus:

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \;=\; \mathrm{ChirHoch}^{\bullet}(A_b, A_b),$$

and clause (Z4) supplies the degree--3 trace pairing

$$\mathrm{Tr}_C \colon Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \otimes Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \;\longrightarrow\; \mathbb{C}[-3]$$

at level Z. Clause (Z5), the Pfaffian normalization of Theorem 2.1, and the OP/Igusa scalar normalization of Theorem 1.2 identify this trace with the protected scalar:

$$\mathrm{Tr}^{\mathrm{bulk}}_n = \Delta_5^{-2}.$$

The OP branch convention of Convention 1.1 gives the separate scalar representative $Z^X_{\mathrm{OP}} = -4096\,\mathrm{Tr}^{\mathrm{bulk}}_n$. This proves exactly the level-Z criterion: once the retained trace datum exists with the stated normalization compatibilities, the trace comparison follows. The $\mathsf{SC}^{\mathrm{ch,top}}$-brace, the E_3-topological factorisation-algebra structure on $X \times \mathbb{R}$, and the gravity-line operator algebra are not consequences of this trace datum. □

Theorem 7.24 *(Pfaffian–Dirac theorem and level-Z Z_{BPS} trace identity on $K3 \times E$, relative to retained data).* Granted Theorems 7.7, 7.13, 7.14, 7.19, the retained finite geometric Pfaffian datum (P_{fin}), and 7.23, the last itself granted (Z_{HB}), Vol I Theorem H [72] and Vol II's chiral-Hochschild Beilinson stratification [73] as established theorems, the Pfaffian–Dirac theorem 2.1, the orientation-character match 3.1, and the level-Z trace identity for the protected scalar $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ to a chain-level CY-orientation trace on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$ all hold on $K3 \times E$. The primitive recognition theorem 4.1 is a separate conditional theorem: it requires the compact-source representatives, relations, pairings, radicals, primitive-space and pairing-kernel Mittag–Leffler exactness, PBW comparison with PBW associated-graded Mittag–Leffler exactness, and full parity dimensions of $(S1)$–$(S10)$. The level-A gravity-line operator algebra remains open in Vol II (Remark 7.21).

Proof. $(D0)$ follows from the retained D0-HN datum through Theorem 7.7; $(O1)$ follows from the retained quotient-orientation datum through Theorem 7.13; the equivariance part of $(O1)^+$ follows from the chosen Weyl lifts through Theorem 7.14; and $(O2)$ is transported from the retained wall-atlas datum by Theorem 7.19. The finite geometric Pfaffian data are the datum (P_{fin}) of Definition 7.3. The level-Z Hall–Borcherds trace comparison follows from the retained (Z_{HB}) datum through Theorem 7.23. The Pfaffian and orientation theorems of Chapter 7 then hold on $K3 \times E$ relative to these retained data. The primitive-recognition theorem asks for the additional compact-source datum $(S1)$–$(S10)$, and Appendix G gives the Pfaffian-orientation lane, not the compact-source Lie-superalgebra recognition lane. The level-Z trace identity of Chapter 9 reads $\mathrm{Tr}_n^{\mathrm{bulk}} = \Delta_5^{-2}$. The OP scalar branch is $-4096\,\mathrm{Tr}_n^{\mathrm{bulk}}$, by Convention 1.1. □

Remark 7.25 *(The conditioning is the retained level-Z datum and Vol I/Vol II chain-level theorems).* After the $K3 \times E$ retained D0-HN datum, the retained $(O1_{\mathrm{quot}})$ datum, the chosen Weyl lifts, the retained $(O2_{\mathrm{atlas}})$ datum, and the trace comparison of §7.6, the remaining chain-level conditioning for the level-Z $\mathrm{Z}_{\mathrm{BPS}}$ trace identity is the retained datum (Z_{HB}), Vol I Theorem H [72] (chiral Hochschild concentration on the Koszul locus), and Vol II [73, chiral-Hochschild Beilinson stratification] clauses (i) and (ii) (chain-level bulk identification and CY-orientation trace pairing) as correct theorems. The K3$\times E$-specific trace obstruction is precisely (Z_{HB}). The theorem is conditional on this retained datum and on the Vol I/Vol II chain-level constructions. The finite packet `certificates/trace/k3e_protected_trace` presently verifies only the absence ledger `TRACE_OBSTRUCTION_LEDGER_VERIFIED`, with `protected_trace_certification=false`; it does not supply (Z_{HB}). The level-A gravity-line operator algebra sits outside this conditioning and remains open in Vol II. Primitive recognition is independent of the retained orientation data: it is exactly the finite compact-source recognition problem of Theorem 4.1.

7.8 Compatibility of the retained data

The retained D0-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and the retained $(O2_{\mathrm{atlas}})$ datum convert the Pfaffian and orientation parts of the Pfaffian–Dirac theorem to statements on $K3 \times E$ at the Pfaffian and orientation level. The primitive Lie-superalgebra recognition remains conditional on $(S1)$–$(S10)$. The proof uses the retained data in a fixed implication chain. The retained D0-HN datum gives Theorem 7.7. The retained quotient-orientation datum gives Theorem 7.13. The Weyl lifts and simple-wall signs give Theorem 7.14. The retained $(O2_{\mathrm{atlas}})$ datum together with these Weyl lifts gives Theorem 7.19. The scalar separation $(R1)$–$(R2)$ is Theorem 7.17. The Hall–Borcherds trace comparison is Theorem 7.23. The gravity-line algebra is a separate Vol II construction.

The criteria use the following theorems, with the fourfold analogue separated from the $K3 \times E$ argument:

- Joyce–Upmeier orientation theorem [56, Theorem 1.11];
- Joyce–Tanaka–Upmeier quasi-projective extension [55, Theorem 1.4];
- the Cao–Kool–Monavari CY_4 cosection-twist extension [17, Theorem 1.6] is a fourfold analogue, not a theorem used in the $K3 \times E$ CY_3 reduction above;
- Kiem–Li cosection localisation [59, Theorem 5.0.1];
- Brav–Bussi–Dupont–Joyce–Szendrői d-critical chart cocycle [12, Theorem 6.9];
- Pantev–Toën–Vaquié–Vezzosi (-1)-shifted symplectic structure [95, Theorem 0.3];
- Maulik–Pandharipande K3 reduced theory [80, Theorem 1];
- Oberdieck–Pandharipande $K3 \times E$ Igusa cusp form factorisation [89, Theorem 0.2], [92, Theorem 1].

The $K3 \times E$-specific finite-stabilizer criteria are Lemma 7.10 (Klein-four detection on rank-2 E-strata), Lemma 7.11 (full two-primary $E[N]$-edge detection on finite-stabilizer strata), and Theorem 7.17 ($(R1)$–$(R2)$ scalar separation). The foundational citations give the chart and orientation technology; Mittag–Leffler exactness is a condition in the retained datum $(D0_{\mathrm{HN}})$.

7.9 The relative theorem on K3×E

The Pfaffian theorem 2.1 and the orientation-character theorem 3.1 hold on $K3 \times E$ at the Pfaffian and orientation level relative to $(D0_{\mathrm{HN}})$, $(O1_{\mathrm{quot}})$, the Weyl lifts, $(O2_{\mathrm{atlas}})$, and (P_{fin}). The obstruction $(D0)$ of Chapter 6 is reduced to the retained D0-HN datum by Theorem 7.7; $(O1)$ and $(O1)^+$ are reduced through Theorems 7.13 and 7.14 relative to their retained data; $(O2)$ is transported from $(O2_{\mathrm{atlas}})$ by Theorem 7.19. The $(R1)$–$(R2)$ scalar separation $4096 = 2^{12} = 64^2$ is Theorem 7.17. The level-Z Hall–Borcherds trace identity $\mathrm{Tr}_n^{\mathrm{bulk}} = \Delta_5^{-2}$ on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$ is Theorem 7.23, granted (Z_{HB}), Vol I Theorem H [72], and Vol II's chiral-Hochschild Beilinson stratification [73] as established theorems; no appeal is made to the Vol II Universal Holography master theorem, whose standard-landscape scope (affine Kac–Moody at non-critical level, principal $W_{N,c}$, Vir_c, Schellekens $c = 24$, Monster $V^\natural$, VSKR+BGG-tempered irrational cosets on a smooth projective curve) does not include the BKM-derived hybrid factorisation chart C_X on $K3 \times E$. The Saito–Kurokawa eigenline arithmetic of Proposition 2.1 and the κ-universality identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ across the CHL frame $N \in \{1, 2, 3, 4, 6\}$ of Theorem 8.1 are inscribed by direct computation. Throughout, the cited foundational results (Joyce–Upmeier, BBDJS, Maulik–Pandharipande, Oberdieck–Pandharipande, Kiem–Li, Pantev–Toën–Vaquié–Vezzosi) are applied in their published form. The Cao–Kool–Monavari CY_4 theorem is a separate fourfold analogue.

The theorem-level automorphic, BKM-denominator, and singular-theta readings of Δ_5 close at the automorphic level, and the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is identified with Δ_5 relative to the retained data. The Lie-superalgebra recognition $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source datum $(S1)$–$(S10)$. The mirror discriminant of Chapter 8 is the remaining view: the automorphic calculation gives $[\Delta_5^{-2}] = 2[\mathfrak{H}_{\mathrm{II}}]$, while the identification of this Heegner divisor with the reduced mirror discriminant and the global trivialisation of the comparison line bundle are conjectural and refined into four period conditions in Conjecture 1.1.

The unresolved data are the finite compact-source recognition datum $(S1)$–$(S10)$, the mirror period comparison of Conjecture 1.1, the explicit OPE structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$, the Type II sigma-model on $K3 \times E$ with worldsheet BPS partition function Δ_5^{-2}, the level-A gravity-line operator algebra, and the Mathieu–umbral twined denominator system. For the order-seven and order-fifteen conjugacy classes

$\{7A, 7B, 15A, 15B\}$ of M_{24}, Conjecture 1.1 of Chapter 10 refines the Cheng–Duncan–Harvey–Gannon umbral moonshine conjecture to four assertions on $\Theta(\phi_{0,1}^{(g),\mathrm{Weil}})|_{\mathbb{H}_2}$, and identifies the arithmetic obstruction in the parity of $c_g(0)$. The level-A gravity-line problem demands an operator algebra acting on the boundary chiral algebra of the $K3 \times E$ holomorphic-topological theory with Pentagon-face determinant section $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ and reciprocal partition character Δ_5^{-2}, an operator-valued Hall–Borcherds completion, and the 6d holomorphic Chern–Simons quartic anomaly vanishing on $K3 \times E$ of Vol III. The level-Z trace criterion of Theorem 7.23 is a trace pairing on the centre, not an acting algebra on a gravity-line module.

8 $N = 1$ Boundary-Compatibility Conditions

Let a chambered Hall-descent datum supply the compact comparison data of Theorem 0.252, so that the criterion of Theorem 0.215 applies. The smooth compact-realization standing datum is fixed as follows: S is a smooth complex K3 surface and E is a smooth elliptic curve. Singular K3 limits, nodal or cuspidal elliptic limits, imprimitive multiple-cover contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures require separate boundary-degeneration data. Its $N = 1$ decategorified boundary would then be constrained by the Igusa datum

$$\begin{aligned}\Gamma_{\mathrm{eff}} = \{(n, l, m) \in \mathbb{Z}^3 \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\}\\ \cup \{(n, l, 0) \in \mathbb{Z}^3 \mid n > 0,\ l \in \mathbb{Z}\}\\ \cup \{(0, l, 0) \in \mathbb{Z}^3 \mid l < 0\} \subset \mathbb{Z}^3.\end{aligned}$$

comparison Hall descent	Igusa boundary
Gram index lattice	$\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$
sector completion	Γ_{eff}
active denominator support	$\Gamma_{\mathrm{act}} = \{\gamma \in \Gamma_{\mathrm{eff}} \mid f(nm, l) \neq 0\}$
Lorentzian degree	$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$
orientation character	required geometric monodromy comparison $\epsilon_o\|_{W^{(2)}} = \nu_{\Delta_5}\|_{W^{(2)}}$
automorphic boundary	$\mathrm{Borch}_\theta(\phi_{0,1}^{\mathrm{Weil}}) = \Delta_5$
BKM denominator	$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$
denominator algebra	$\mathcal{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}$
formal current envelope	$\mathsf{Den}_{\Delta_5,E} = U_E^{\mathrm{ch}}(\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E)$
scalar square	$Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -4096\Delta_5^{-2}$ (proved for the smooth primitive nonzero $K3$-class branch).

The degree map satisfies

$$(\alpha(\gamma), \alpha(\eta)) = -2\langle\gamma, \eta\rangle_{\mathrm{ind}}, \qquad (\alpha(n, l, m), \alpha(n, l, m)) = -2(4nm - l^2).$$

The real walls are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

These equations give the denominator polarisation of the same Gram-index lattice whose cusp polarisation gives the Fock determinant. The cone is generated by the active support. Triples in Γ_{eff} with zero

Jacobi coefficient fix ordering and contribute no visible product factors. The product alone does not exclude invisible zero-superdimension root spaces.

The comparison is the $N = 1$ normalization

$$\mathrm{Borch}_\theta(\phi_{0,1}^{\mathrm{Weil}}) = \Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \varkappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = \mathrm{wt}(\Delta_5) = 5.$$

It contains no denominator theorem beyond the $N = 1$ theorem and no row map

$$\Phi_j^{\mathrm{cmp}} \longleftrightarrow F_i.$$

The protected scalar has the form

$$Z_\square^X = C_\square \Delta_5^{-2}.$$

The Oberdieck–Pixton and reduced Hilbert-scheme DT branches have constant −4096 in the quotient-by-E normalization of Theorem 5.3. The chiral/denominator half is Δ_5.

A finite radical-quotient Hall–Drinfeld double realizes the windowed protected bracket from the compactification only after the finite-first oriented Hall descent, Serre-dual opposite half, protected integration, charge-additive Hall primitives, Euler–Hall pairing, and radical quotient have been fixed. Its recognition as the compact Δ_5 current object still requires the comparison datum of Theorem 0.252 and Theorem 0.215: Borcherds relations inside the Hall super-commutator, PBW/no-extra-relations, parity, centre, associator, and the completed Hopf pairing with orientation character required to restrict to ν_{Δ_5}. Homotopy descent must produce the positive half, its completion data, and the Hall-side domain on which the comparison character is defined. This orientation character is independent of rescaling Δ_5; it comes from monodromy of the chosen square root of the Hall determinant line. The universal recognition envelope may be formed as the finite quotient killing the radical, Serre, coproduct, centre, and associator defects; it is the unquotiented compact Hall object only when compact source matrices satisfy those identities and the source-faithfulness theorem gives injectivity at every height. The Igusa boundary identifies the denominator algebra $\mathfrak{g}_{\Delta_5}$ and its protected decategorified multiplicities before those comparison data are proved.

For the Cléry–Gritsenko rows F_5, F_6, F_7, F_8, this table contains no nonabelian chamber, no simple real roots, no reflection character, and no Weyl-sum side. Rows F_5, F_6, F_7, F_8 use the separate Govindarajan CHL/WKB denominator theorem when a nonabelian bracket is needed. For F_7 this is a theorem on the μ_4-multiplier cover determined by the Cléry–Gritsenko order-four multiplier. None of these denominator rows supplies a comparison row map or a CY/DT/chiral boundary host.

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